

Euclidean Condensate Theory (ECT): Conditional Routes from a Euclidean Condensate to Lorentzian Kinematics, Quantum Structures, and Gravity

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Abstract

Euclidean Condensate Theory (ECT) asks whether physical time, Lorentzian propagation, quantum structure and gravity can arise as effective sectors of one medium defined initially on a four-dimensional Euclidean carrier. Its minimal concrete model contains a real scalar field. The present formulation then adopts—rather than dynamically derives—a physically relevant state with a nonzero scalar gradient. The state is a physically motivated structural postulate whose stationary action, stability and formation history remain Open. Its nonzero direction has exact $O(3)$ stabiliser. Given independently supplied coefficients, the scalar principal tensor

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$$

is Lorentzian exactly when $\beta(\beta - \alpha) < 0$, elliptic when the product is positive, and degenerate at $\beta = 0$ or $\alpha = \beta$. On the retained branch $\beta > 0$, the Lorentzian condition is $\alpha > \beta > 0$ and defines a dimensionless scalar characteristic slope. This is exact mathematics inside the declared effective model. It does not derive the stationary ordered state, a physical clock, a universal metric, or common matter, photon and tensor cones.

The rest of the preprint develops sector-specific completion programmes while keeping four claim types distinct: Level A is exact mathematics in a stated model and assumptions with an explicit reproducible derivation; checks verify but do not replace that derivation. Level B is an adopted structural input or conditional closure with declared missing physical steps; Level C is a phenomenological fit, benchmark or supplied application model; Open denotes a missing action, state, vertex or observable map. Level A does not by itself establish descent from the minimal scalar postulates or empirical support for ECT.

In gravity and cosmology, the displayed tangent-director map has neither a linear transverse-traceless nor a static-density seed at fixed coefficients and linear order. Separately supplied scalar-tensor and orientation models provide controlled response algebra, a direction-independent determinant identity, a classically shift-covariant trace-free completion, and a regular two-slope late-background proof of class. These results do not yet supply the physical tensor action, matter/photon metric, Newton matching, observed late-time state or a joint perturbation likelihood. Hyperbolic-response-cap (HRC) galaxy closures give exact internal response algebra and a conditional slope-four baryonic Tully–Fisher relation after a separate gravity bridge; the bridge, origin of the acceleration scale, environment law and predictive population likelihood remain Open. Current rotation-curve and ultra-diffuse-galaxy calculations are Level-C stress tests, not ECT-specific predictions.

In the quantum programme, supplied Euclidean kernels, compact winding, Osterwalder–Schrader reconstruction, canonical structures and influence-functional methods yield conditional mathematics. A declared bare strict-pullback scalar covariance fails a scoped two-time reflection-positivity gate; this excludes that covariance, not every ECT measure. Candidate gravitational records still lack an ECT

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vertex, state-dependent noise kernel and absolute rate; the Diósi–Penrose functional is an external comparator. A bounded mediator-channel audit of the enumerated candidate class found no physical gravitational mass-density record channel, but is not a no-go theorem for future action-owned vertices. The response–persistence/record formulation of the Principle of Euclidean Stationarity (PES-R) is a Level-B organisation; physical/global PES remains Open. Exchange-topology results require a supplied fermionic canonical-anticommutation-relation (CAR) completion, and the physical Born measure, unique outcome and update law are not derived. Validated Bell–Clauser–Horne–Shimony–Holt (CHSH) violations $|S| > 2$ exclude the Bell-local setting-independent common-measure premise bundle, not hidden variables as such; ECT has no derived Bell forward law. Conditional shell and boundary models identify black-hole-entropy inputs without deriving a physical ECT horizon or area law.

For gauge structure, compact winding and the $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$ identity motivate conditional Abelian and weak routes, but do not derive physical gauge fields or matter. As a reverse-engineered structural input, the colour programme separately adopts a rank-three complex bundle equipped with a positive-definite Hermitian form and a compatible unit-norm nowhere-vanishing complex volume form. These supplied data reduce its structure group to $SU(3)$ as exact conditional bundle mathematics. The inequality $\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$ excludes only a direct injective embedding in the minimal displayed algebra; physical colour connection, action, representations, confinement, strong CP and the mass gap remain Open.

ECT is therefore not yet a completed ultraviolet theory or a derivation of all known physics from one scalar. It is a status-separated common-origin architecture with one exact scalar-signature theorem, several controlled proofs of class and scoped negative results, and explicit missing physical owners. Present quantitative items are internal outputs, calibrations, external benchmarks or phenomenological fits; none is yet an owner-complete, ECT-specific numerical prediction.

Keywords: Euclidean Condensate Theory; spacetime-completion programme; conditional scalar Lorentzian signature; emergent gravity; problem of time; spontaneous symmetry breaking; $O(4) \rightarrow O(3)$ symmetry breaking; ordered condensate; generalised Einstein equations; quantum foundations; reflection positivity; Hilbert-space reconstruction; decoherence; gravitational decoherence; gravity-induced entanglement; Born rule; arrow of time; black-hole information problem; holographic entropy bound; cosmology without inflation; Hubble tension; dark energy; dark matter phenomenology; galaxy rotation curves; baryonic Tully–Fisher relation; radial acceleration relation; cosmological constant problem; conditional trace-free completion; neutrino masses; seesaw mechanism; Lorentz invariance violation; charge quantisation; conditional $SU(3)$ -structured rank-three complex bundle; Yang–Mills mass gap problem; strong CP problem; fine-structure constant

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List of Symbols

Table 1: Principal symbols. Foundational symbols (P1–P6 level) are defined in Sections 2–6. Derived and matching-level symbols are introduced in the corresponding sections.

Symbol	Definition	Value / Units
<i>Microscopic level — bare micro-action (P3)</i>		
Φ	Real scalar field of the minimal P1–P6 sector on $\mathcal{M}^4$; P7 adds a reverse-engineered internal colour module	—
μ^2	Mass parameter: $V(\Phi) = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4$, $\mu^2 > 0$	GeV ²
λ	Self-coupling in bare potential	dimensionless
ϕ_0	Bare potential minimum: $\phi_0 = \sqrt{\mu^2/\lambda}$; $[\phi_0] = \text{GeV}$	GeV
m_σ	P3 homogeneous-Gaussian radial mass: $m_\sigma \equiv m_{\Phi,\text{rad}} = \sqrt{2}\mu$, hence $m_\sigma^2 = 2\mu^2$; it is not a P4 pole, transition or cutoff owner	GeV
<i>Supplied P4 datum and conditional ordered-sector EFT; physical vacuum owner Open</i>		
u_0	Magnitude of the supplied nonzero-gradient datum: $u_0 = \langle \partial_A \Phi \rangle $; no stationary P4 action presently owns it; $[u_0] = \text{GeV}^2$	GeV ²
n^A	Orientation coordinate of the adopted P4 branch: $\langle n^A \rangle = \delta^A_w$; not an independent field in P3	—
w	Euclidean direction selected in the adopted P4 postulate; physical clock and dynamical orientation-selection owner	m
<i>Broken-phase EFT parameters</i>		
α	Anisotropy coupling: $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$; full Lorentzian condition $\beta(\beta - \alpha) < 0$, reducing to $\alpha > \beta > 0$ on the healthy $\beta > 0$ branch; $\alpha/\beta = 2$ only under the Level-B unit-slope benchmark	dimensionless; $\alpha = 2$ only after also choosing $\beta = 1$
β	Isotropic stiffness; the remaining common coefficient scale may conventionally be set to $\beta = 1$ after the ratio is supplied	dimensionless; 1 (normalisation convention)
Δ_K	Scalar stiffness gap $\Delta_K \equiv \alpha - \beta$; distinct from the longitudinal scalar coordinate χ and from the black-hole toy field χ_{sh}	dimensionless in the displayed normalisation
χ	Longitudinal scalar coordinate defined at linear order by $\delta Q_A = \partial_A \chi$; the strict coordinate reduction is Level A, while a physical one-mode interpretation requires a supplied reduced action, domain and projector	same field units as Φ
K^{AB}	Supplied scalar principal tensor: $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$; Lorentzian iff $\beta(\beta - \alpha) < 0$, with healthy shorthand $\alpha > \beta > 0$; a universal physical metric remains Open	—
$\hat{c}_*$	Dimensionless scalar characteristic slope in common-length coordinates: $\hat{c}_*^2 = \beta/(\alpha - \beta)$, Level A for the supplied broken-phase EFT; $\hat{c}_* = 1$ is a Level-B unit-slope coefficient benchmark, not a field-rescaling identity	1 (benchmark)
c_u	Dimensional conversion speed between the Euclidean common-length coordinate and a physical clock after a state, congruence and lapse have been supplied; not fixed by the principal symbol	m s ^{−1}

Table 1 continued

Symbol	Definition	Value / Units
c_{char}	Physical scalar characteristic speed. In the aligned constant-lapse special case, $c_{\text{char}} = c_{\text{u}} \hat{c}_* / \mathcal{N}_{\Phi}$; $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration, while equality to the photon/tensor cones remains Open	m s^{-1}
v_2	Electroweak matching scale: $v_2 = (\sqrt{2}G_F)^{-1/2}$ (eq. 5.7); intermediate condensate scale, not derived from P1–P6 (OP-EW-scale, Sec. 5.1)	$\approx 246.22 \text{ GeV}$
$\mathcal{J}_{\text{EW}}$	Logarithmic measure of the electroweak hierarchy: $\mathcal{J}_{\text{EW}} \equiv \ln(\phi_0/v_2)$; central reduction quantity for the OP-EW-scale closure programme (eq. 5.9, Sec. 5.1)	≈ 36.8 (dimensionless)
m_{eff}^2	EFT fluctuation mass parameter; $m_{\text{eff}}^2 \neq m_{\sigma}^2$ in general	GeV^2
$A_{L,\phi}$	Independent positive action-valued magnitude multiplying the dimensionless reduced Lorentzian scalar functional; neither its value nor equality to A_{P3} , a phase scale, S_0 or $\hbar$ is fixed by the cone/sign test	$[\hbar]$
A_P, ζ_P	Positive action magnitude and local Lorentzian orientation of a supplied pure- $P(X)$ completion, $S_P^{\text{phys}} = A_P \zeta_P \mathcal{J}_P$; on a nondegenerate Lorentzian component $\zeta_P = -\text{sgn } P_X = \text{sgn } Q'$	$[\hbar]$, sign
$A_{n,E}$	Independent positive physical-action magnitude multiplying the supplied dimensionless reduced Euclidean director functional $\mathcal{J}_{n,E}$; distinct from the missing covariant $S_n[g,n]$ and from the loop functional $\mathcal{J}_n(L)$	$[\hbar]$
ξ_{P4}	P4 pole/correlation length after an action-owned stationary background and constrained Hessian; equality to the P3 radial length is not derived	Open
ξ_{cond}	Generic local label for a separately declared correlation or coarse-graining scale; its owner must be stated at each use and no universal identity with ξ_{P3} , ξ_{P4} or ℓ_{P1} is assumed	owner-dependent
<i>Supplied smooth coherent-sector closure (BR1); winding theorem internal to that closure</i>		
$\Phi_{\text{eff}} = \rho e^{i\theta}$	Effective order parameter of coherent IR sector	—
θ	Phase of Φ_{eff} ; winding condition $\oint \partial_A \theta dX^A = 2\pi n$	rad
K_{θ}^{eff}	Supplied positive phase stiffness of the retained coherent-branch Gaussian closure, $K_{\theta}^{\text{eff}} > 0$; zero is degenerate and a negative value does not own the positive Euclidean functional used here; its magnitude is expected dimensionally to scale as $\sim \phi_0^2$ up to EFT matching coefficients	GeV^2
$\mathcal{J}_{\text{ord},E}$	Dimensionless reduced Euclidean collective functional of the supplied ordered branch; coefficients and physical measure owner remain Open	dimensionless
$A_{\text{ord},E}$	Independent positive physical-action magnitude in $S_{\text{ord},E}^{\text{phys}} = A_{\text{ord},E} \mathcal{J}_{\text{ord},E}$; its relation to A_{χ} , P3 or quantum-sector scales is Open	$[\hbar]$
ι_0^{EFT}	Dimensionless fixed-core coefficient of the supplied reduced phase model: $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}}$; not a global variational minimum	dimensionless

Table 1 continued

Symbol	Definition	Value / Units
$\mathfrak{l}_{\text{elem}}^{(\text{fixed core})}$	Dimensionless phase-only loop functional evaluated at the supplied L_{core} ; distinct from the tension-balanced minimum	dimensionless
L_{core}	Characteristic condensate core scale entering the fixed-length loop bound; explicit first-principles derivation open (OP-Q2)	m
$\mathcal{T}_{\text{loop}}$	Positive reduced line coefficient in the optional one-dimensional loop model; must be derived with $K_{\theta}^{1\text{D}}$ from one profile/action owner	energy
$\mathfrak{l}_{0,\text{bal}}^{\text{eff}}$	Dimensionless balanced-loop coefficient $\sqrt{2K_{\theta}^{1\text{D}} \mathcal{T}_{\text{loop}}}$ in the supplied reduced model; distinct from $\mathfrak{l}_0^{\text{EFT}}$	dimensionless
$K_{\vartheta}^{1\text{D}}$	Candidate reduced metric of a future compact normalisable carrier modulus ϑ after the full profile and physical quotient; not identified with the supplied BR1 phase stiffness	energy ⁻¹
$\mathfrak{l}_{0,\text{bal}}^{(\vartheta)}$	Dimensionless modulus-route coefficient $\sqrt{2K_{\vartheta}^{1\text{D}} \mathcal{T}_{\text{loop}}}$; distinct from the phase route until an owner proves equality	dimensionless
<i>Gravitational matching — Macroscopic Physics (Level B)</i>		
$\mathcal{C}_n$	Supplied orientation-operator Wilson coefficient, $\kappa_n = \mathcal{C}_n u_0^2$. An orientation-sector heavy-field form $\hat{a}_n / (16\pi^2 M_{n,\text{heavy}}^2)$ is PARAMETRIC ONLY / MISSING VERTEX; neither $M_{n,\text{heavy}}$ nor its coupling is owned by P3–P4, and no equality to a neutrino, Higgs or Yukawa-completion scale is implied	GeV ⁻²
κ_n	Orientation stiffness $\kappa_n = \mathcal{C}_n u_0^2$; $[\kappa_n] = \text{GeV}^2$. Equality to a physical tensor normalisation is Open; equal dimensions are not a derivation	GeV ²
M_G	Conditional tensor/Einstein-EFT normalisation; $M_G = \bar{M}_{\text{Pl}}$ only by weak-field matching. The dimensionally consistent bridge $\kappa_n \rightarrow M_G^2$ is Open	GeV
G_N	Observed Newton response. After a physical tensor, source vertex, clock/light map and any scalar response are matched, $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ and $G_N^{\text{SI}} = \hbar c^5 / [8\pi (E_{M_G}[\text{J}])^2]$, where $E_{M_G}[\text{J}] \equiv M_G[\text{GeV}](1 \text{ GeV})_{\text{J}}$ is the same response-matched natural-unit energy scale converted from GeV to joules; this symbol is distinct from the Diósi–Penrose self-energy E_G ; external CODATA-2022 input [1, 2]	$6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
G_*	Natural-unit scalar-action normalisation, $G_* \equiv G_*^{\text{nat}} = (8\pi \bar{M}_{\text{Pl}}^2)^{-1}$, used only with natural-unit $T_{\mu\nu}$ and U . The separate conversion target is $G_*^{\text{SI}} = \hbar c^5 / [8\pi (\bar{M}_{\text{Pl}}[\text{J}])^2]$; neither is automatically the measured G_N	GeV ⁻²

Table 1 continued

Symbol	Definition	Value / Units
$\bar{M}_{\text{Pl}}$	Reduced-Planck energy calibration target, $(8\pi G_N^{\text{nat}})^{-1/2} \approx 2.4353 \times 10^{18}$ GeV after physical response matching; equivalently $[\hbar c^5 / (8\pi G_N^{\text{SI}})]^{1/2}$ in joules. Using it in the scalar action is a matching assumption, not an ECT result [1, 2]	GeV
$g_{\mu\nu}^{\text{eff}}$	Scalar characteristic-metric candidate; the universal physical matter/photon/tensor metric is Open	—
<i>Quantum sector (Section 20)</i>		
A_{P3}	Finite positive physical-action prefactor multiplying the dimensionless reduced P3 functional; not fixed by the printed natural-unit integral	$[\hbar]$
$\Sigma_{\text{PI},\text{P3}}$	Independent finite positive action-dimension denominator of the bare P3 Euclidean measure; not identified with A_{P3} or a quantum-sector scale	$[\hbar]$
κ_{P3}	Positive dimensionless measure ratio $A_{\text{P3}}/\Sigma_{\text{PI},\text{P3}}$; after the reduced functional is fixed, this ratio controls P3 Gaussian fluctuations while the two action scales are separately unidentifiable from that measure	—
A_χ	Positive physical-action magnitude multiplying the dimensionless reduced primordial χ functional; not identified with S_0 , $\hbar$, $A_{\text{ord},\text{E}}$ or another sector scale	$[\hbar]$
$\Sigma_{\text{PI},\chi}$	Independent positive action denominator of the supplied primordial χ Euclidean measure	$[\hbar]$
κ_χ	Positive dimensionless primordial measure ratio $A_\chi/\Sigma_{\text{PI},\chi}$ after the reduced kernel and χ field normalisation are fixed; its inverse, not S_0 , normalises the named Gaussian covariance	dimensionless
Σ_θ	Finite positive owner-specific action-dimension normalisation converting the reduced BR1 phase functional to a physical action; not derived or universal	$[\hbar]$
$\Sigma_{\text{PI},\theta}$	Independent positive action-dimension denominator required only if a Euclidean phase-loop measure is supplied; the ratio $\Sigma_\theta/\Sigma_{\text{PI},\theta}$, not either scale separately, controls that reduced exponent	$[\hbar]$
Σ_ϑ	Finite positive owner-specific action-dimension normalisation for the separate CDE- Φ modulus route; equality $\Sigma_\vartheta = \Sigma_\theta$ is an Open compatibility condition	$[\hbar]$
S_0	Operational physical action-unit slot; a route can match it only through an action-valued product such as $\Sigma_\theta t_0$ or $\Sigma_\vartheta t_{0,\text{bal}}^{(\vartheta)}$, and neither product has a derived universal owner	$[\hbar]$
$\hbar$	Reduced Planck constant, $\hbar = h_{\text{P}}/(2\pi)$; target of the explicit Level B calibration $S_0 = \hbar$, whose absolute derivation, physical owner and universality are Open and are not supplied by P1–P6 or T1	1.055×10^{-34} J s
h_{P}	Ordinary Planck constant $h_{\text{P}} = 2\pi\hbar$; the subscript avoids collision with metric and Hermitian-form notation	6.626×10^{-34} J s
<i>Spacetime</i>		

Table 1 continued

Symbol	Definition	Value / Units
$\mathcal{M}^4$	Four-dimensional Euclidean manifold (P1)	—
$X^A = (x, y, z, w)$	Euclidean coordinates	m
δ_{AB}	Euclidean metric	—
x^0	Ordered-branch time-length coordinate, $x^0 = w$; it has the same length dimension as the Euclidean coordinates and is not by itself an SI clock reading	m
t, τ	Natural-unit clock parameter and physical proper time, respectively; seconds arise only after the congruence/lapse and dimensional calibration are supplied	s after SI matching
ℓ_{Pl}	Planck length: $\ell_{\text{Pl}} = \sqrt{\hbar G_N / c^3} \approx 1.616 \times 10^{-35}$ m; any P4-pole matching $\xi_{\text{P4}} \sim \ell_{\text{Pl}}$ is Open	m
<i>Decoherence (Section 26.1)</i>		
$\Xi_{ab}^{\text{full}}(T)$	Full log-modulus diagnostic $-\ln \mathcal{F}[a, b; T] $ for a named preparation and protocol; it need not be nonnegative for a generic correlated initial preparation	dimensionless
$\Phi_{ab}(T)$	Nonnegative pairwise record exponent, defined as Ξ_{ab}^{full} only after the declared full reduced construction establishes $ \mathcal{F} \leq 1$; the positive bulk Gaussian contribution is Φ_{ab}^{bulk}	dimensionless
$\Delta\mathcal{O}_{ab}(t)$	Difference of the physical record operators for alternatives a, b , after the declared resolved/unresolved split and any spatial projection	operator-dependent
$d_{ab,T}(\omega)$	Unnormalised finite-window transform $\int_0^T dt e^{i\omega t} \Delta\mathcal{O}_{ab}(t)$; no hidden division by T	$[\Delta\mathcal{O}]s$
D_{ab}^R	Retarded response kernel, $D_{ab}^R(t) = -(i/S_0)\theta(t)\langle[F_a(t), F_b(0)]\rangle$, in the passive $e^{+i\omega t}$ Fourier convention; source, state and operator dependent, and distinct from the equation-normalised scalar G_{ret}	kernel-dependent
N_{ab}	Action-normalised Hermitian time-domain noise kernel, fixed so $\int \Delta\mathcal{O} N \Delta\mathcal{O}$ has units of action	kernel-dependent
$N_{ab}^{(S)}(\omega)$	Spectral noise kernel, $N^{(S)}(\omega) := \tilde{N}(\omega)$, in the stationary convention of Eq. (29.8); not an occupation number	spectral-noise units
$S_H(\omega)$	Action-scaled spectral record kernel, $S_H(\omega) = S_0 N^{(S)}(\omega)$ in Eq. (29.9); not a retarded kernel	spectral-kernel units
$\Gamma_{\text{tr}}[q; \eta]$	Translation/deformation susceptibility for a declared dimensionless deformation amplitude when the associated quadratic-form limit exists and is finite; otherwise only extended lower/upper diagnostics are defined; optional robustness diagnostic, not a mean-force term	dimensionless
γ_a	Pole, Liouvillian, or Floquet decay rate of physical mode a after constraint/gauge reduction	s^{-1}
$\Gamma_{E,a} = S_0 \gamma_a$	Action-unit energy width; its identification with the observed quantum width uses the separate $S_0 = \hbar$ matching	energy
$\Sigma[\gamma]$	Protocol entropy production $\ln[dP_F/d(P_R^\Theta)](\gamma)$ on one common path space for named forward/reverse measures; the reciprocal identity used below requires mutual absolute continuity	dimensionless

Table 1 continued

Symbol	Definition	Value / Units
N_{eff}	Effective environmental degrees of freedom that actually couple in a declared source model; not a universal ECT mode count	—
τ_{ab}	Operational order-unity record time $\inf\{T > 0 : \Phi_{ab}(T) \geq 1\}$ only for a channel whose nonnegative Φ_{ab} has been established, when the set is non-empty; threshold and protocol dependent	s
τ_{dec}	Model-dependent decoherence-time shorthand; no general ECT identity $\tau_{\text{dec}} = 1/(N_{\text{eff}}\gamma)$ is assumed	s
$\mathfrak{P}_{\varepsilon}^{\text{ret}}(T)$	Protocol-level set of individually retained physical modes or spectral blocks after constraint/gauge reduction; the applicable finite-window diagnostics and tolerances are type specific	—
$R_{ab}(T)$	Pair-retention flag: $R_{ab} = 1$ iff both a and b belong to $\mathfrak{P}_{\varepsilon}^{\text{ret}}(T)$; it is distinct from the pairwise record exponent	binary
$\mathcal{Q}_{\Phi}(T), \mathcal{L}_{\Phi}(T)$	Sets of record-quiet and record-loud pairs defined by preregistered thresholds on an established nonnegative $\Phi_{ab}(T)$; pairs between them are intermediate	—
$\varepsilon_{\text{quiet}}, \varepsilon_{\text{loud}}$	Protocol thresholds satisfying $0 \leq \varepsilon_{\text{quiet}} < \varepsilon_{\text{loud}}$; they do not define a universal pointer or outcome criterion	dimensionless
$\mathcal{G}_a(T)$	Maximum finite-window operator-norm gain of the declared constraint-reduced physical propagator on spectral block a	dimensionless
$\mathcal{S}_a(t; \rho_a)$	State- and positive-effect-declared finite-time survival probability for an initially supported state in block a under the physical reduced map; the full curve on $0 \leq t \leq T$, not one universal scalar, enters cut/Jordan diagnostics	dimensionless
γ_L^{μ}	Lorentzian matrices in the mostly-plus house package: $\{\gamma_L^{\mu}, \gamma_L^{\nu}\} = -2\eta^{\mu\nu}$, $(\gamma_L^0)^{\dagger} = \gamma_L^0$, $(\gamma_L^i)^{\dagger} = -\gamma_L^i$; $\bar{\psi} = \psi^{\dagger} \gamma_L^0$	—
γ_A^E	Hermitian Euclidean Dirac matrices, $\{\gamma_A^E, \gamma_B^E\} = 2\delta_{AB}$; with $x_L^0 = -i\omega_E$ the house map is $\gamma_L^0 = \gamma_0^E$, $\gamma_L^i = i\gamma_i^E$ (eqs. 23.4–23.5)	—
D_E, S_E	Euclidean Dirac operator $D_E = \gamma_A \partial_A + m$ and two-point kernel $S_E = D_E^{-1}$ (eq. 23.6)	—
<i>Macroscopic Physics (Sections 11–14.1)</i>		
$\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$	Candidate anisotropic tensor form; not an ECT stress tensor	—
$T_{\mu\nu}^{(n)}$	Orientation stress obtained only from a specified covariant $S_n[g, n, \lambda_n]$; presently Open	—
$\Theta_{\mu\nu}$	Reserved only for a stress after such a variation; absent from the displayed scalar-only equation	—
$\mathcal{H}$	Hilbert space; subscripted: $\mathcal{H}_{\text{ext}}, \mathcal{H}_{\text{shell}}, \mathcal{H}_{\text{int}}$ (BH decomposition)	—
ρ_{ext}	Reduced state in a hypothetical black-hole Hilbert-factor completion, $\rho_{\text{ext}} = \text{Tr}_{\text{sh, int}} \Psi\rangle\langle\Psi $; the split and its ECT owner are Open	—

Table 1 continued

Symbol	Definition	Value / Units
$G_{\text{div}}(X)$	Natural-unit divided-equation matter coefficient $G_*/F(X)$ after every term in the natural-unit scalar metric equation has been divided by F ; $G_{\text{div}}^{\text{SI}} = G_*^{\text{SI}}/F$ is a separate conversion and neither is by itself a measured response	GeV^{-2}
G_{Cav}	Natural-unit weak-body response after the scalar range, canonical coupling, and body charges are specified; in the unscreened light-scalar limit $G_{\text{Cav}} = G_{\text{div}}(1 + \alpha_0^2)$. Its SI value is written $G_{\text{Cav}}^{\text{SI}}$	GeV^{-2}
ϕ_u	Ordered-patch amplitude coordinate $\beta_\phi^{-1} \ln(u/u_\infty)$, defined for $u/u_\infty > 0$	dimensionless
$\bar{\phi}$	Constant background value of the dimensionless amplitude coordinate; the declared reference is $\bar{\phi} = 0, F = 1$; distinct from the GeV-scale bare minimum ϕ_0	dimensionless
$\phi_{\text{bg},0}$	Present-epoch boundary value of the dimensionless late-time scalar background; implementation key phi0; not the microscopic ϕ_0	dimensionless
$\hat{\mu}_{\text{bg}}$	Dimensionless late-time mass ratio m_ϕ/H_* ; implementation key mu; not the bare P3 mass parameter μ	dimensionless
Φ_{gal}	Effective gravitational potential used only in the conditional HRC static functional G; not the fundamental ECT field Φ and not identified with ϕ_u	$\text{m}^2 \text{s}^{-2}$
ψ_K	Anisotropy/stiffness coordinate $\ln(\Delta_K/\Delta_{K,\text{vac}})$, where $\Delta_K \equiv \alpha - \beta$; no $u \leftrightarrow \Delta_K$ bridge derived	dimensionless
q	Generic scalar coordinate used in the coordinate-invariant scalar audit; not an identification among u, Δ_K and the strict scalar coordinate χ	various
$a(q)$	Scalar-audit derivative $f_{,q}/f$; distinct from the FRW scale factor and proxy coefficients	inverse units of q
$\mathcal{A}(q)$	Scalar-audit kinetic combination $K + 3f_{,q}^2/(2f)$; distinct from the candidate orientation tensor	kinetic units
$\mathcal{C}_R(\phi)$	Coefficient of a schematic candidate orientation–curvature operator; not present in the displayed scalar action and presently Open	model-dependent
$\mathcal{E}(q, \rho_{\text{m}})$	$V_{,q} - 2aV - (a/2)\rho_{\text{m}}$ in the trace convention	source units
ρ_{m}	Matter energy density in the scalar audit; $\rho_{\text{m}} = \rho c^2$ for SI mass density ρ	energy/volume
ρ_b	Baryonic mass density used in the static galactic functional G; distinct from ρ_{m}	kg m^{-3}
$W_0(s), \hat{\mu}_0(s)$	Exact HRC-0 primitive and internal response in the augmented ERP- Φ /inverse-DtN model; microscopic P1–P6 ownership Open	dimensionless
$W_3(s), \hat{\mu}_3(s)$	Conditional HRC-3 primitive and internal response in the supplied equal-three-channel rank-one model	dimensionless
s, Y_g	Internal ERP control coordinate and separate gravitational invariant $ \nabla\Phi_{\text{gal}} ^2/a_M^2$	dimensionless
$\mathcal{H}_\Phi, \mathcal{H}_R$	Constrained ordered-medium response Hessian and positive reference stiffness after physical-bundle reduction; same operator units; stationary P4 owner and microscopic map Open	model-dependent, same units

Table 1 continued

Symbol	Definition	Value / Units
$\hat{\Lambda}, \mathcal{Q}, P_s, C$	Dimensionless normalised response operator, bounded HRC occupation operator, released-channel projector, and inverse-DtN compliance in the ERP- Φ idealisation	dimensionless
D_E, Z_R, m_R	Positive massive half-space Dirichlet-to-Neumann operator, declared response normalisation, and inverse response length in the augmented model; physical interface and P1–P6 ownership Open	$[m_R] = L^{-1}$; $[D_E] = [Z_R]L^{-1}$
τ_V	Positive semifinite trace density; normalised by response volume and the declared power of m_R ; normalisation supplied and microscopically Open	dimensionless after declared normalisation
$n_{\text{ch}}, r_{\text{ch}}$	HRC-3 response-channel count and commuting projector rank, distinct from the global tensor-to-scalar ratio r ; supplied equal-channel model uses $(n_{\text{ch}}, r_{\text{ch}}) = (3, 1)$	integers / supplied model
$f(Y_g), A_g$	Open internal-to-gravity control map and normalisation; the HRC registry uses the declared identity bridge $f(Y_g) = Y_g, A_g = 1$	dimensionless
a_M	HRC gravity-bridge acceleration scale; $a_{M0} = cH_0/(2\pi)$ is matching, not a medium postulate or derivation	m s^{-2}
$\mu_{g,0}(x^2)$	$x/\sqrt{1+x^2}$ force-dimensionality diagnostic; identical to the HRC-0 shape under the identity bridge	dimensionless
$v_{\text{cl}}(y), \Sigma_{\text{cl}}$	Inverse boost paired exactly with the conditional HRC-0 point-source response $\mu_{g,0}$ and its local projected surface-density proxy; distinct from the full HRC field problem and not ECT-derived	dimensionless; kg m^{-2}
$\mathcal{R}_{\text{dyn}}$	Local UDG dynamical-to-baryonic ratio g_{obs}/g_N used only in the inverse diagnostic; distinct from radius and tensor-to-scalar r	dimensionless
$\mu_{g,k}(Y_g)$	HRC gravitational response after the separately declared bridge, $k \in \{0, 3\}$	dimensionless
g_{HRC}	Positive acceleration solving $\mu_{g,k}(g^2/a_M^2)g = g_N$ in the algebraic diagnostic	m s^{-2}
Ξ, Ξ_{alg}	Open physical environment/history factor and unconstrained UDG algebraic inverse; $\Xi_{\text{alg}} < 0$ denotes no physical $\Xi \geq 0$ solution	dimensionless
m_{loc}	Conditional local homogeneous response mass: defined on a regular branch with $\mathcal{E}_{,q} \neq 0$; real positive Yukawa response additionally requires $\mathcal{A} > 0$ and $\mathcal{E}_{,q} > 0$; not a finite-body screening mass	inverse length
$\Lambda_{\text{div}}(X)$	Natural-unit divided-equation potential coefficient $U(X)/[\bar{M}_{\text{Pl}}^2 F(X)]$; it has energy-squared units. After expressing energies in joules, $\Lambda_{\text{div}}^{\text{SI}, \text{m}^{-2}} = \Lambda_{\text{div}}/(\hbar c)^2$; neither coefficient is by itself the observed evolving cosmological response	GeV^2
n_s	Scalar spectral index (ECT-native derivation: open)	—
r	Tensor-to-scalar ratio	—
N_e	Inflationary e-folds (external; not used in ECT)	—
$w(a)$	Physical late-time equation-of-state history (if derived)	—

Table 1 continued

Symbol	Definition	Value / Units
$L_{\text{gal}} \equiv r_*$	Conditional HRC matching length $\sqrt{G_N M_{\text{bar}}/a_M}$ defined by $g_N = a_M$; not generally the $x_g = 1$ radius; microscopic origin Open	kpc
$\lambda_{\text{loc}}(\rho)$	Candidate local scalar response length m_{loc}^{-1} ; density dependence and composite-body screening are Open pending a one-coordinate finite-body calculation	m
$a_{M,\text{cl}}$	System-specific input of the exploratory cluster map; values are disclosed only to define the exact order-preservation audit, with no galactic/environment transfer law	m s^{-2}
<i>Director–matter completions (Section 7)</i>		
μ_5	Dimensionful coefficient of the homogeneous vector-current operator; its physical value may vanish	GeV
β_5	Optional dimensionless ratio μ_5/m_f when $m_f \neq 0$; not a derived species law, coefficient benchmark or body-force coupling	dimensionless
$a_\mu^{(V)}$	Covector coefficient $\mu_5 u_\mu$ of the homogeneous vector-current term	GeV
$b_\mu^{(5)}$	Independent axial-background coefficient of a separately supplied spin-dependent completion	GeV
β_ϕ	Dimensionless amplitude–curvature coupling of the ϕ -first closure, $F(\phi) = e^{\beta_\phi \phi}$ (distinct from the kinetic β)	dimensionless
ω_5	Spin-precession frequency of a separately supplied spin-dependent completion; no such frequency is generated by the homogeneous vector-density term	rad s^{-1}
η	Eötvös parameter of a separately supplied source/body-force completion; none is derived from the fixed spurion	dimensionless
<i>Neutrino sector (Section 8.9)</i>		
M_R, Y_V, C_5, Λ	Supplied seesaw/Weinberg inputs; their values, flavour structure and ECT owners are Open	model dependent
<i>ECT cosmological background and observables (§16.5)</i>		
$H_{\text{ECT}}(a)$	Conditional two-slope history computed; microscopic completion and physical map Open	inverse time
$\Sigma_{\text{ord}}, a_{\text{ord}}$	Ordered-side branch-formation front and its scale factor	—
U^μ, s^A	Physical matter-clock congruence and formation-front normal; neither is the director n^A	—
$\mathcal{N}_\Phi$	Clock/lapse map between Euclidean arc length and matter proper time	dimensionless
$d\ell_E, L_W$	Euclidean line element and homogeneous normal-depth special case	length
F_s, K_s	Non-minimal coupling and scalar kinetic coefficient of the supplied two-slope completion	$\text{mass}^2, \text{mass}^2$
a_s, b_s, c_s, κ_s	Two-slope action coefficients; supplied completion inputs, not P1–P6 predictions	dimensionless
$V_\pm$	Positive amplitudes of the two exponential potential terms	energy density
D_s, Δ_{H_V}	$D_s = \kappa_s + 3a_s^2/2$ and the reduced $(\dot{H}, \dot{\sigma})$ response determinant	dimensionless, mass^4

Table 1 continued

Symbol	Definition	Value / Units
$\omega_{\text{BD}}, \gamma_{\text{PPN}}$	Exact unscreened light-scalar Brans–Dicke parameter and PPN space-curvature coefficient	dimensionless
Ξ_{body}	Dimensionless finite-body source parameter $[(A - B)/(3A - B)](s - 1)(m_{\text{out}}R)^2$; distinct from the environment/history law Ξ	dimensionless
$D(z)$	Restricted quasistatic two-slope diagnostic computed; gauge-complete result Open	—
ζ_{ER}	Owner-specific constant slope used only in the declared early-response envelope; distinct from all BBN/PES tolerances, not the withdrawn universal ECT ε or the named two-slope orbit	dimensionless
$f\sigma_8(z)$	Redshift-space-distortion observable, conditional on the derived growth and normalisation	—
r_s	Conditional imported-recombination diagnostic computed; ECT recombination owner Open	Mpc
Φ_{W}	Weyl potential of the physical photon metric used for lensing and ISW	dimensionless
$Q_{\text{W}}^{\text{proxy}}, S_{\text{ISW}}^{\text{proxy}}$	Restricted instantaneous Weyl-amplitude and conformal-derivative carriers; not projected observables	dimensionless
t_0^{ECT}	Conditional age of a named action/state after one dimensional scale calibration; physical congruence/depth-lapse map Open	time
<i>QG/decoherence upgrade (Part III)</i>		
Δn_A	Orientation shift; its map to physical gravitational memory is Open (Section 35.4)	—
$S_{\text{gen}}(\Sigma)$	Generalised (holographic) entropy $\text{Area}(\partial\Sigma)/4G_N + S_{\text{bulk}}(\Sigma)$; Section 36.3	—
$\partial\Sigma$	Quantum extremal surface boundary in holographic entanglement computations; Section 36.3	—
<i>Vacuum offsets and conditional late-time completions (Section 15)</i>		
C_I	Global integration density in the separately supplied trace-free completion; a boundary/state datum whose ECT action and observable map remain Open	GeV^4
$c_{\Lambda}^{\text{GRref}}$	Reference normalization $c_{\Lambda}^{\text{GRref}} \equiv \rho_{\text{late},0}^{(\text{GRref})}/(\bar{M}_{\text{Pl}}^2 H_0^2) = 3\Omega_{\text{late},0}^{(\text{GRref})}$ after the physical metric and Planck calibration are fixed; not a model-native scalar–tensor fraction or an ECT determination	dimensionless
$V_{\pm}, a_s, b_s, c_s, \kappa_s$	Inputs of the conditional two-slope scalar–tensor proof of class, $V_s = V_- e^{c_s \sigma} + V_+ e^{b_s \sigma}$, $0 < c_s < 2a_s < b_s$	various
$\sigma(t)$	Scalar coordinate of the named two-slope completion; distinct in role from the microscopic radial fluctuation also denoted σ ; its map to the Φ medium is Open	dimensionless
ρ_{late}	Physical late-time source after one action, state, matter/photon metric and observable map are specified	GeV^4
<i>Adopted P7 SU(3) structural postulate (Part IV, §40–§47)</i>		
$\mathcal{C}_x$	Supplied local complex rank-three module in P7; its identification with a physical colour bundle is Open	—
h	Supplied positive-definite Hermitian form on $\mathcal{C}_x$	—

Table 1 continued

Symbol	Definition	Value / Units
Ω	Supplied nonvanishing complex volume form in $\Lambda^3 \mathcal{C}_x^*$; preserving (h, Ω) gives an $SU(3)$ frame redundancy, not by itself a physical gauge field	—
$Z(\phi)$	Kinetic coefficient in the declared leading colour-EFT truncation (eq. 41.2); positivity is branch and background dependent	dimensionless
$Y(\phi)$	Independent anisotropic colour-EFT coefficient; P7 and a shared director tensor provide neither its value nor a bound	dimensionless
$\theta_{\text{gauge}}(\phi)$	Gauge-theta angle in the convention $D_\mu = \partial_\mu - ig_s A_\mu^a T^a$; experiment constrains $\bar{\theta} \equiv \theta_{\text{gauge}} + \arg \det M_q \pmod{2\pi}$, not this angle alone	dimensionless
κ_n^{col}	Ratio Y_0/Z_0 in the declared constant-background truncation; its realised value and RG flow are Open	dimensionless
$\Lambda_{\text{col}}^{\text{src}}$	Formal one-loop transmutation scale only on the supplied $Y_0 = 0$, CP-even, matter-free constant-background source-model branch (eq. 41.5); not an ECT prediction	GeV
M_*	Supplied matching scale of the colour-EFT candidate; no ECT-specific identification is established (O1 in §47)	GeV
$S_{\text{rec}}(\omega)$	Record spectrum used in the dephasing representation; Appendix BF	—
J_W, J_{bare}	Weiss vs bare record spectral density, $J_W = J_{\text{bare}}/(2\omega)$; Appendix BE	—
$\sigma_{\text{dec}} \equiv \sigma_{\text{pair}}$	Pair-convention decoherence pressure on a phase line; Appendix BE	—
ζ_0^{pair}	Pair-normalised line-overlap constant; Appendix BE	2
c_{line}	Bessel line coefficient of the pair overlap; Appendix BG	1
ξ_{line}	Phase-line core scale; Appendix BG	—
c_{GP}	Ginzburg–Pitaevskii core constant, $c_{\text{GP}}/\pi = 0.3811$; Appendix BG	≈ 1.20
λ_n^{rec}	Gaussian width-record monitor scale; Appendix BG	—
κ_b, ℓ_w	Tilt stiffness and Airy width, $\ell_w \propto (\kappa_b \sigma_{\text{dec}})^{-1/3}$; Appendix BE	—
T_{eff}	Scalar detailed-balance record thermometer when it exists; otherwise use $S_{\pm}(\omega)$; Appendix BE	—
$\eta_{ab}^{\text{rec}}, \eta_r^{\text{leak}}$	Cross-spectrum positivity ratio; eigenchannel leakage ratio; Appendix BJ	dimensionless
$R_\Phi^{\text{oc}}, R_\Omega^{\text{oc}}$	Over-closure residuals of the P26 loop; Appendix BJ	dimensionless
$P_{\text{M3}}, D_{\text{M3-3}}$	M3 reporting packet; M3-3 data vector; Appendix BK	—
Z_b, Z_{cont}	Bound-line and continuum spectral weights, $Z_b + Z_{\text{cont}} \simeq 1$ in M3 bookkeeping; Appendix BH	dimensionless
R_{off}	Crossover offset of the M3-2 flatness transforms; Appendix BK	—

List of Abbreviations

Table 2: Principal and recurrent abbreviations used in this paper.

Abbreviation	Meaning
AQUAL	Aquadratic Lagrangian formulation; here the supplied conditional HRC static functional (algebra A; identity bridge B/Open; data application C)
BH	Black hole
CMB	Cosmic microwave background
CDT	Causal dynamical triangulations
DESI	Dark Energy Spectroscopic Instrument
DM	Dark matter
ECT	Euclidean Condensate Theory (this work)
ERP- Φ	Euclidean relaxation and threshold release principle for the Φ -medium; candidate constitutive extension
EFE	External-field effect (galactic/MOND diagnostic)
HRC	Hyperbolic response cap; exact in stated augmented response models, microscopic ECT ownership Open
UDG	Ultra-diffuse galaxy
EOM	Equation of motion
GR	General relativity
GW	Gravitational wave
JWST	James Webb Space Telescope
KG	Klein–Gordon
LIV	Lorentz invariance violation
MOND	Modified Newtonian dynamics
PT	Phase transition
QFT	Quantum field theory
RAR	Radial acceleration relation
SM	Standard Model of particle physics
SPARC	Spitzer Photometry and Accurate Rotation Curves
SSB	Spontaneous symmetry breaking
ToC	Table of contents
VEV	Vacuum expectation value
WEP	Weak equivalence principle
FDU	Fulling–Davies–Unruh (effect)
UDW	Unruh–DeWitt (detector model)
EFT	Effective field theory
FP	Fierz–Pauli (linearized gravity Lagrangian)
CKN	Cohen–Kaplan–Nelson (holographic-type effective-field-theory bound relating UV cutoff to IR length scale)
IR	Infrared (low-energy regime)
UV	Ultraviolet (high-energy regime)
DE	Dark energy
LV	Lorentz violation (generic; cf. LIV for invariance violation)
BAO	Baryon acoustic oscillations
BBN	Big Bang nucleosynthesis
BR1	Branch rule 1 (supplied coherent-sector closure)
BTFR	Baryonic Tully–Fisher relation
BVP	Boundary-value problem
CDM	Cold dark matter
CHSH	Clauser–Horne–Shimony–Holt (Bell-correlation test)
CKM	Cabibbo–Kobayashi–Maskawa (quark-mixing matrix)

Table 2 continued

Abbreviation	Meaning
CPT	Charge conjugation, parity and time reversal
CTA	Cherenkov Telescope Array
DP	Dynamical principle (the Diósi–Penrose benchmark is written out as “D–P” in Part III)
D–P	Diósi–Penrose external collapse/decoherence comparator; not an ECT rate
FLRW	Friedmann–Lemaître–Robertson–Walker (metric / cosmology)
GEE	Generalised Einstein equations
LISA	Laser Interferometer Space Antenna
LiteBIRD	Lite satellite for the studies of B-mode polarization and Inflation from cosmic background Radiation Detection
MICROSCOPE	Micro-Satellite à traînée Compensée pour l’Observation du Principe d’Équivalence (satellite WEP test)
NFW	Navarro–Frenk–White (dark-matter halo profile)
NDA	Naive dimensional analysis
NLO	Next-to-leading order
PES	Principle of Euclidean Stationarity (programme name; PES-R is the active repair direction)
PES-R	Response–persistence/record formulation of PES; Level-B taxonomy with ECT-specific split, vertices, state, and kernels Open
PPN	Parametrised post-Newtonian (formalism)
PTA	Pulsar timing array
QS	Quantum sector (Part III of this paper)
RG	Renormalisation group
SN	Supernova (type Ia unless stated otherwise)
MWI	Many-worlds interpretation
ABJ	Adler–Bell–Jackiw (anomaly; Section 9.5)
WZW	Wess–Zumino–Witten (flavour-anomaly term/sector)
CAR	Canonical anticommutation relations (Section 9.9)
OS	Osterwalder–Schrader (axioms / reconstruction theorem; Section 23.1)
RP	Reflection positivity
KMS	Kubo–Martin–Schwinger (thermal-state/detailed-balance condition)
PMNS	Pontecorvo–Maki–Nakagawa–Sakata (lepton-mixing matrix)
POVM	Positive-operator-valued measure
QED	Quantum electrodynamics
SME	Standard-Model Extension (Lorentz-violation coefficient framework)
CP	Charge conjugation and parity (combined discrete symmetry)
EW	Electroweak
FDT	Fluctuation–dissipation theorem/relation; its use requires the stated response, noise, and state hypotheses
GUT	Grand unified theory
LIGO	Laser Interferometer Gravitational-Wave Observatory
LLR	Lunar laser ranging
LSST	Legacy Survey of Space and Time at the Vera C. Rubin Observatory
MSW	Mikheyev–Smirnov–Wolfenstein (matter effect for neutrino oscillations)
OP	Open problem; an explicitly unclosed owner, derivation, matching, or observable gate
PDE	Partial differential equation
QCD	Quantum chromodynamics
QM	Quantum mechanics

Table 2 continued

Abbreviation	Meaning
WKB	Wentzel–Kramers–Brillouin asymptotic approximation
CC	Cosmic chronometers (differential-ages $H(z)$ probe)
ISW	Integrated Sachs–Wolfe (effect / cross-correlation with large-scale tracers)
RSD	Redshift-space distortions
SHOES	Supernovae and H_0 for the Equation of State (local Hubble-constant programme)
NLEE	Non-linear Einstein equations (or non-linear extensions of generalised Einstein equations)
H_Λ	Spectral shorthand retained only for auditing the full reduced Hessian; it does not derive a physical fixed-measure action or a vacuum non-renormalization theorem (Sec. 15.6).
BMV	Bose–Mazumdar–Morley / Marletto–Vedral (gravity-induced entanglement proposal, Section 32)
BMS	Bondi–Metzner–Sachs (asymptotic symmetry group, Section 35.3)
GIE	Gravity-Induced Entanglement (Section 32)
HPS	Hawking–Perry–Strominger (soft-hair proposal, Section 35.6)
LOCC	Local Operations and Classical Communication
PCCG	Post-quantum Classical Gravity (Oppenheim hybrid framework)
QES	Quantum Extremal Surface (Section 36.3)
WV	Witten–Veneziano (relation between η' mass and topological susceptibility, Part IV, §45)
OP-EW	Layered electroweak open-problem programme: OP-EW-scale ($\mathcal{A}_{EW} \approx 36.8$, four candidate mechanisms), OP-EW-locking (compact $U(1)_\theta$ origin and Hopf-fibered diagonal locking, App. N), OP-EW-gauge (chirality, hypercharge, g, g' , Weinberg angle), OP-EW-naturalness (radial-mode protection), OP-EW-matter (chiral fermions, Yukawa hierarchy)

1 Introduction

Modern theoretical physics rests on two highly successful but conceptually distinct frameworks: General Relativity and quantum field theory. General Relativity describes gravity as the curvature of spacetime geometry, while the Standard Model of particle physics describes fundamental interactions within a fixed spacetime background. Despite their empirical success, these two frameworks remain difficult to reconcile within a single consistent theoretical structure.

A conceptual peculiarity common to both theories is the distinguished status of time in the spacetime interval

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

In this expression one coordinate is fundamentally different from the others, possessing an opposite sign in the metric. While this Lorentzian structure has been extraordinarily successful phenomenologically, its physical origin remains unclear. In particular, it raises the question whether the Lorentzian signature of spacetime is a fundamental property of nature or whether it may instead arise as an emergent feature of a deeper underlying structure. Similar questions have appeared in various approaches to quantum gravity and emergent spacetime, where the possibility has been considered that the Lorentzian metric may arise dynamically from a more symmetric underlying state.

This observation motivates the exploration of scenarios in which the fundamental description is Euclidean, with all four directions initially equivalent. Euclidean Condensate Theory asks whether a real scalar medium on that carrier can own a physical ordered state and, ultimately, the clocks, fields and observable geometry of a Lorentzian world. The Euclidean arena, $O(4)$ covariance and minimal real-scalar dynamics are the first three structural inputs (P1–P3). ECT then separately adopts a physically

relevant state with a nonzero scalar gradient as its fourth structural postulate (P4). The minimal scalar action does not derive a stationary representative, stability domain, amplitude, or formation and selection history for this state; those constructive owners remain Open.

The relevant group-theoretic and principal-symbol statements are more restricted. The groups $O(4)$ and $O(1,3)$ both have six generators; they are distinct compact and non-compact real forms and neither is a lower-symmetry subgroup of the other. The supplied nonzero Euclidean vector has stabiliser $O(3)$ and abstract orbit $O(4)/O(3)$. This group geometry does not establish a physical spontaneous-symmetry-breaking event. Separately, for independently supplied $\alpha > \beta > 0$, the scalar principal tensor has one eigenvalue of sign opposite to the other three and is therefore hyperbolic. This is a change of the effective principal symbol, not spontaneous breaking $O(4) \rightarrow O(1,3)$. Formal analytic continuation relates suitable complexifications and reality conditions, but does not by itself derive a physical Lorentzian metric, state or clock. The scientific question is thus whether one owner-complete Euclidean medium can dynamically select the ordered state and connect its scalar characteristic structure to matter, photons and tensor gravity.

A closely related line of work has shown that Lorentzian dynamics can arise effectively from a fundamentally Euclidean (Riemannian) field theory. Girelli, Liberati and Sindoni [3] showed that a scalar field with a nonvanishing gradient on a Riemannian manifold can generate an effective Lorentzian metric for perturbations, albeit with scalar (Nordström-type) gravity. Mukohyama and Uzan [4] developed a more complete classical clock-field construction in which scalars, vectors, and Dirac spinors propagate in an effective Minkowski metric on a 4-dimensional Riemannian manifold. ECT shares this architectural core and reaches closely analogous conditional scalar conclusions, but differs in how the reference direction enters: here the physically relevant ordered-gradient branch is adopted as P4 rather than derived from an imposed clock-field background. Its complete stationary action, state, stability domain and dynamical $O(4) \rightarrow O(3)$ formation/selection mechanism remain Open. The two frameworks diverge further in the gravitational sector and in the extension to quantum and phenomenological questions.

For calculations, the supplied ordered direction is represented by a unit field n_A satisfying

$$n_A n^A = 1.$$

Such a constraint can arise in a separately supplied effective condensate sector, for example through a Lagrange multiplier or through a potential that fixes the magnitude of the orientation field while allowing its direction to fluctuate. Deriving that stationary sector from the minimal scalar action is the ordered-gradient completion problem (OP-grad).

Given this orientation field, the propagation of a declared scalar perturbation may be described at low energies by a supplied characteristic operator. Its two-derivative quadratic form, reduced Lorentzian functional and physical action must be kept distinct:

$$\mathcal{Q}_{2,\psi} = \frac{1}{2} \int d^4X K^{AB} \partial_A \psi \partial_B \psi, \quad \mathcal{J}_{L,\psi}^{(2)} = \varsigma_{L,\psi} \mathcal{Q}_{2,\psi}, \quad S_{L,\psi}^{(2),\text{phys}} = A_{L,\psi} \mathcal{J}_{L,\psi}^{(2)},$$

with finite, spacetime- and field-independent $A_{L,\psi} > 0$. On the retained mostly-plus, positive-energy scalar component $\varsigma_{L,\psi} = -1$. This ψ -sector scale is not identified with $A_{L,\phi}$ or any other sector scale without a matching theorem. The cone fixes neither $A_{L,\psi}$ nor any omitted higher-gradient completion. Here ψ denotes the scalar excitation governed by this supplied operator, not a generic photon, matter or tensor mode. In the ordered branch, the effective kinetic tensor is controlled by two coefficients α and β (Section 3.4). In the declared unit-slope coefficient benchmark used in the relevant subsectors of this paper, $(\alpha, \beta) = (2, 1)$ after the separate normalisation convention $\beta = 1$, so that the corresponding effective characteristic tensor reduces to

$$K^{AB} = \delta^{AB} - 2n^A n^B.$$

For independently supplied $\alpha > \beta > 0$, this tensor has one negative and three positive eigenvalues. That is an exact Lorentzian classification of the scalar principal symbol inside the stated EFT. It is not yet a universal physical spacetime metric, and it does not show that photon, matter or tensor modes share the same cone.

In the present framework the coordinate w is one of the four Euclidean coordinates and must not be identified with physical time without an independently owned clock map. The supplied direction defined by n_A labels the negative characteristic direction of the retained scalar operator when $\alpha > \beta > 0$; deriving a physical clock, rods and observable metric remains Open.

These distinctions control every later use of the scalar cone. Its signature, dispersion and dimensionless characteristic slope are derived conditional on the adopted ordered-gradient state; identifying its velocity with c is a separate scalar-clock calibration (§4.7). Physical matter, photon, gauge and tensor sectors still require their own actions and matching. Likewise, compact loops do not by themselves derive the quantum operator algebra, probability measure, outcomes or a universal action scale.

Conceptually, ECT is related to a broader class of approaches in which spacetime and gravity emerge from more fundamental microscopic degrees of freedom. Examples include Sakharov's induced gravity [5], condensed matter analogues of spacetime, and various emergent-gravity scenarios. In Sakharov's induced gravity the Einstein–Hilbert action arises from quantum fluctuations of matter fields in a fixed background spacetime. In contrast, the present framework investigates whether physical spacetime geometry can emerge from a Euclidean condensate; this has not yet been completed beyond the scalar characteristic model. Nevertheless, both approaches share the idea that gravitational dynamics may arise as an effective description of more fundamental microscopic degrees of freedom. However, the present framework differs in that it begins with a Euclidean medium possessing an $O(4)$ symmetry and investigates the conditional consequences of a supplied ordered-gradient chart. Whether a physical spontaneous-symmetry-breaking process generates that chart is Open.

A key working idea is that a P4-supplied ordered branch may support an effective Lorentzian description. Whether physical gravity and quantum theory are two sectors of that same medium is a common-origin hypothesis with separate missing actions, states and observable maps, not an established consequence of the scalar background.

It is important to emphasise that ECT is proposed as a *framework* rather than a complete fundamental theory. The aim of the present work is to explore whether a Euclidean condensate vacuum can provide a consistent setting in which spacetime geometry, quantum dynamics, and gravitational interactions may emerge from a common underlying structure.

A central question for any condensate-based theory of spacetime is: *what precisely condenses?* In ECT the answer is not a scalar vacuum expectation value (which would be $O(4)$ -invariant and produce no symmetry breaking) but the candidate *gradient field* $\partial_A \Phi$, an $O(4)$ vector whose nonzero coarse-grained reference value $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$ is supplied by P4. Here the brackets denote the supplied analysis datum, not a vacuum expectation value derived from an owned state. The strict pullback $\partial_A \Phi = (u_0 + \sigma)n_A$ contains only one longitudinal scalar coordinate at linear order; it does not furnish the independent physical orientation, tensor, gauge or quantum response owners used by later completions (Sec. 2.1).

ECT proposes a deeper Euclidean-medium programme rather than merely an interpretation of known quantum equations. What is presently established is more limited: selected Gaussian/coherent constructions reproduce standard mathematical structures after additional premises are supplied. The origin of probability, outcomes, the physical operator algebra, universal $\hbar$, and the quantum–classical transition remains Open. Likewise the physical tensor/gravity sector is not generated by the scalar action. Figure 1 is therefore a navigation map: Parts II and III develop separate completion programmes whose common microscopic owner and backreaction remain to be derived.

This difference is reflected already at the level of the equations to be solved. Standard quantum mechanics is formulated as an initial-value theory for a wavefunction, for example

$$i\hbar \partial_t \psi = H \psi,$$

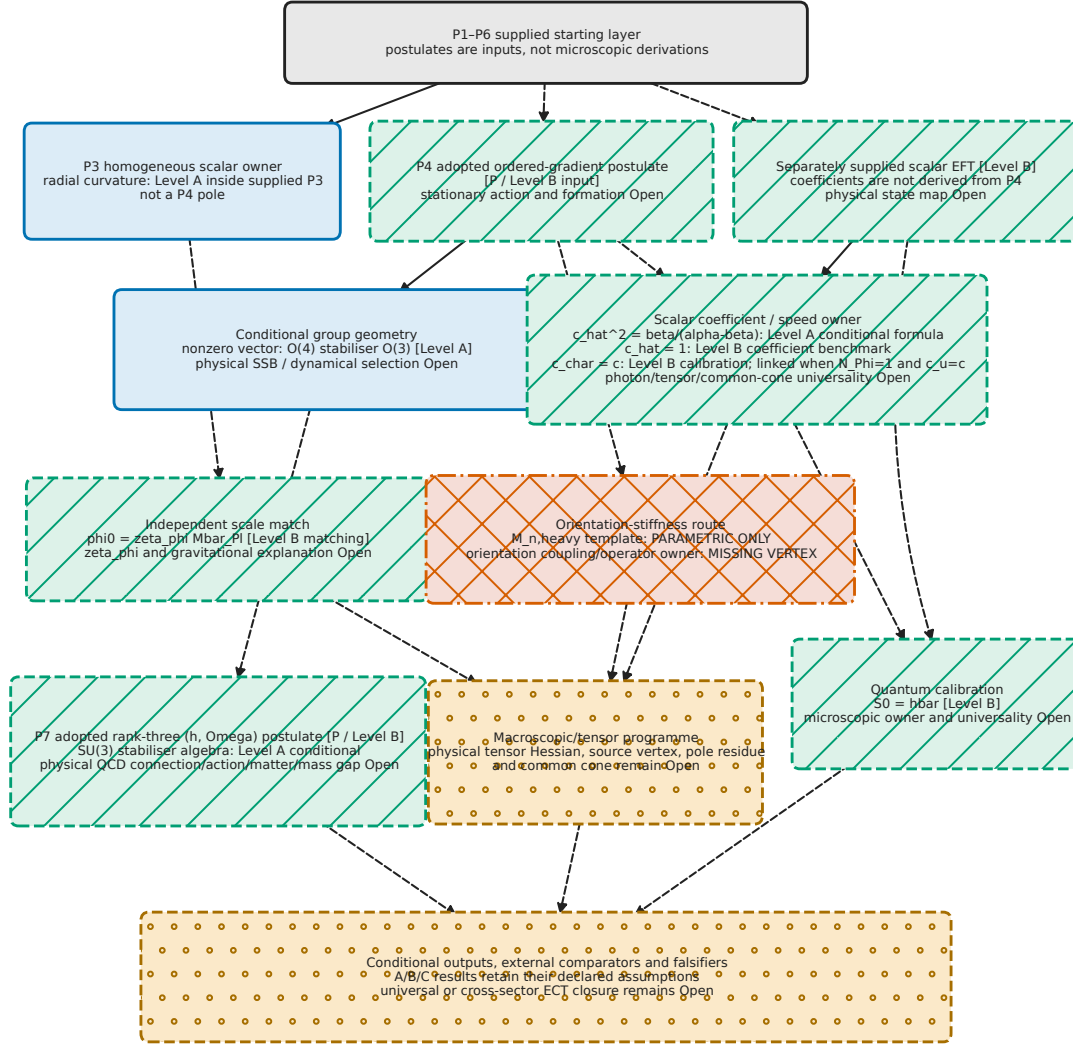
where $\psi(\mathbf{x}, t)$ is a probability amplitude and $|\psi(\mathbf{x}, t)|^2$ is interpreted statistically through the Born rule. ECT begins instead from a Euclidean field equation of the form

$$\delta^{AB} \partial_A \partial_B \Phi - V'(\Phi) = 0,$$

whose solution $\Phi(X)$ is interpreted as a physical condensate configuration in a four-dimensional Euclidean arena. The fundamental problem is therefore a boundary-value problem rather than a Cauchy evolution

ECT programme architecture

Postulates, conditional algebra, calibrations and Open physical owners remain distinct.



Solid near-black arrow: exact conditional algebra inside supplied assumptions. Dashed: conditional/Open dependency.

Status is literal and redundant with fill luminance, hatch, border and arrow style.

Figure 1: Schematic programme map of Euclidean Condensate Theory. P4 adopts the physically relevant nonzero-gradient branch and supplies its analysis chart (Level-B structural input); its stabiliser is $O(3)$ (Level-A conditional group theory). A stationary owner, stability domain and spontaneous formation/selection are not derived. The scalar characteristic theory is then separated from proposed macroscopic/tensor and coherent/quantum completions. The present action does not derive a common physical gravity–quantum backreaction vertex, so the two boxes organise targets rather than consequences of one closed action. Every arrow is a dependency or candidate route whose status is controlled by the main text and status tables. The benchmark identification $\alpha = 2\beta$ used elsewhere in the paper is not the generic branching condition; the full scalar-principal condition is $\beta(\beta - \alpha) < 0$; on the declared healthy $\beta > 0$ branch this is $\alpha > \beta > 0$.

problem: boundary data specify a Euclidean boundary-value problem rather than initial data propagated only forward in time. They fix one solution only after existence, uniqueness and stability/well-posedness have been established for the stated nonlinear equation, domain and boundary class; these global properties remain Open for ECT. In this framework probability is not postulated at the bare Euclidean level, but coarse-graining or decoherence alone does not create a probability measure or select an outcome. Those ingredients require an independently derived state, measure and update rule and remain Open.

The resulting hierarchy is one of the central structural outputs of ECT. At the most fundamental level one has the Euclidean condensate equation. In the ordered broken phase this reduces to an effective equation with kinetic tensor

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B,$$

whose retained-component Lorentzian admissibility requires $\alpha > \beta > 0$. Writing the ordered direction as the common-length coordinate $x^0 = w$ gives the hyperbolic Klein–Gordon operator. This is not yet an identification of w with a reading in seconds: the physical clock requires the state-dependent congruence/lapse map of Eq. (16.14) and a dimensional calibration. After a positive-frequency/state choice, complex envelope, operator map and S_0 normalisation are supplied, a non-relativistic Schrödinger-type equation can be recovered conditionally. This is a Level-B/Open reduction, not yet a unique descendant of P1–P6.

Terminological note. The terms “dark matter” and “dark energy” are retained as names for observational phenomena and standard comparison sectors. The Level-C HRC data diagnostic uses no particle-halo term, but this does not derive the fit from ECT, exclude particle dark matter, or determine ECT ontology. The late-time background closure is separate and has its own declared status. Fundamental particle content and the cosmological fraction budget remain Open.

Throughout this paper we distinguish four status classes: Level A results derived inside stated assumptions; Level B conditional structures with declared missing matching inputs; Level C phenomenological fits, external benchmarks or supplied application models; and Open questions for which the required owner has not been derived or identified. A summary is given in Table 70.

The paper is organised into five parts.

Part I (Foundations) develops the ordered-branch effective theory of the condensate and the conditional Lorentzian classification of one supplied scalar principal tensor. It records one scalar characteristic cone and conditional matchings for c_{char} , G_N and S_0 ; physical Dirac/CAR, gauge and tensor owners remain Open. It also inventories candidate gauge/matter completion routes and preferred-direction operators; their local bundles, representations, coefficients and observable maps remain Open. The neutrino section inventories supplied seesaw/Weinberg benchmarks; no physical chirality, sterile state, mass operator, PMNS matrix or Dirac/Majorana preference is derived.

Part II (Macroscopic Physics) develops the macroscopic branch and a scalar–tensor closure whose metric and scalar equations follow from the displayed Level-B ansatz. It studies cosmological and galactic phenomenology at their separately declared closure levels. The ordered branch also permits candidate anisotropic tensor structures, but the present field equation contains no orientation stress. Such a stress and any lensing, growth, or cluster observable attributed to it remain Open pending a covariant $S_n[g, n]$, constraints, variation, on-shell identity, and physical projection.

Part III (Quantum Sector) develops the coherent quantum branch: a conditional Osterwalder–Schrader Hilbert-space reconstruction, with reflection positivity as one required hypothesis, canonical structure, exchange topology, decoherence, and the Principle of Euclidean Stationarity. In this sector ECT aims to show that large parts of quantum formalism appear as effective ordered-branch structures rather than primitive postulates. Inside a separately supplied local first-order fermionic/CAR effective theory, a smooth strictly positive temporal kinetic factor can be removed by an invertible canonical normalisation; this conditionally excludes Pauli violation generated solely by that factor. The physical ECT fermion, CAR and spin–statistics owners, as well as the Born measure, measurement update and unique-outcome mechanism, remain Open. Part III is further extended by a dedicated upgrade layer on the gravity–matter interface: a reduced-state reformulation of the hybrid-consistency problem; an architectural

operational criterion for gravity-induced-entanglement experiments; a bounded mediator-channel audit of the enumerated candidate class, in which the tested fixed- α, β tangent-director linear route has a missing vertex and the full gravitational mediator is not identifiable; an infrared section retaining BMS, memory and soft gravitons only as external completion tests; and a structural mapping of the critical-shell black-hole picture onto the modern islands and quantum-extremal-surface language, together with a compact cross-programme diagnostic summary.

Part IV (Reverse Analysis, the P7 $SU(3)$ Structural Postulate, and Physical-Colour Constraints) is a conditional completion programme above the forward work of Parts I–III. It records P7 as an adopted Level-B global rank-three bundle with (h, Ω) ; the resulting $SU(3)$ frame group is Level-A mathematics conditional on that postulate. It does not derive a physical colour bundle, connection, action, matter representation or spectrum. The exact scoped results are the stabiliser of a nonzero fundamental vector and the absence of an injective $\mathfrak{su}(3) \hookrightarrow \mathfrak{so}(4)$ embedding; reducible, emergent/nonlinear and kernel-bundle routes remain admissible. The part then displays one leading colour-EFT truncation, restates standard anomaly conditions, and inventories every physical owner as conditional or Open. No framework choice about the Clay continuum object is promoted to a Level-A ontology or theorem about nature.

Part V (Summary, Comparisons, and Outlook) consolidates the framework through its constants map, completion tests, failure gates, open problems, and the derivational summary of structures usually treated as independent postulates or empirical inputs in standard physics. It keeps the standard conditional gauge-normalisation identities separate from the Open ECT-specific gauge/matter construction.

The four claim levels defined above govern the verbs used throughout the preprint: *derived* refers to exact mathematics in the declared model and assumptions with an explicit reproducible derivation; checks verify but do not replace it; *conditional* retains its supplied assumptions; *phenomenological* marks matching or stress testing; and *Open* marks a missing physical owner. This is part of the scientific claim, not a separate review ledger.

Later sections keep the detailed sector diagnostics with their owning derivations. Section 25.1 treats Casimir, Rindler/KMS and particle-production calculations as external targets for a future ECT vacuum completion, not as limits already derived from one ECT correlator. The galactic programme in Section 17.1 and Appendix A records two negative scalar routes and then studies a separately supplied HRC closure. Its scalar coupling, stationary-response equation, conditional baryonic Tully–Fisher algebra and exploratory environment law are derived or defined in those owning sections; the physical coordinate map, gravity bridge, extended-body response and the origin, sign and amplitude of any environment dependence remain Open.

The present formulation should be understood primarily as an effective low-energy description of condensate excitations. The analysis focuses on long-wavelength perturbations around the supplied ordered-sector chart. A supplied $\alpha > \beta > 0$ scalar principal tensor then has Lorentzian characteristics; promotion of that scalar cone to the physical clock, metric and cross-sector causal structure remains Open. The microscopic dynamics of the condensate at very high energies is also Open.

Later sections separate exact conditional calculations, external source models, data diagnostics and Open observable maps. These inventories test what an owner-complete ECT calculation would require; they are not evidence that condensate dynamics is already connected to the cited phenomena.¹

¹A shorter companion article providing an accessible overview of the main ideas, results, and open problems of ECT is available at <https://doi.org/10.5281/zenodo.19430795>.

Part I

Foundations: Postulates, Symmetry Breaking, and Emergent Laws

2 Postulates of Euclidean Condensate Theory

This section states the foundational symmetry and condensate postulates of ECT. It establishes the minimal bare framework from which later Level A/B/C results are developed; it is not itself a phenomenological or EFT-derivation section.

Euclidean Condensate Theory (ECT) starts not from already formed spacetime, matter, or energy in their familiar late-physics sense, but from a more primitive Euclidean Φ -medium. The role of the foundational postulates is not to impose the observed macroscopic world by hand, but to formalise the minimal structural properties that such a medium must possess if causal structure, geometry, and gravitational dynamics are to emerge from it.

The foundational construction is organised in three levels. First, the ontological properties of the Φ -medium are stated explicitly (Section 2.2). Second, the working postulates P1–P6 specify the concrete ECT realisation of that medium. Third, supplied effective branch rules such as BR1 define special coherent analysis sectors on the ordered chart. The manuscript then investigates a geometric macroscopic sector and a coherent sector; their common microscopic descent from one physical condensate state remains Open. The smooth coherent sector is not introduced as a fundamental postulate. It is a supplied effective branch closure: once a smooth non-vanishing single-valued complex collective field is admitted, integer winding is a topological theorem. P3 plus P4 do not themselves derive that complex field, its stiffness or the action normalisation S_0 . The later analysis of kinetic tensors and signature classes, including the healthy Lorentzian component (Section 3.4), is likewise not an extra postulate but a conditional principal-symbol analysis on the supplied chart. Subsequent sections distinguish exact consequences of stated premises from supplied effective completions, phenomenological benchmarks and Open physical maps; they do not all follow from P1–P6.

2.1 Glossary of terms

Before stating the postulates we fix the meaning of terms that in the literature are sometimes used interchangeably but carry distinct meanings in ECT.

Field Φ — the unique fundamental microscopic degree of freedom: a real scalar defined on $\mathcal{M}^4$.

Vacuum — a stationary, physically admissible background branch of the fundamental Φ -medium. P3 owns only its homogeneous stationary minima. P4 independently adopts a physically relevant nonzero-gradient sector; the completed stationary action, state and boundary/domain problem realising that postulate remain Open.

Condensate — shorthand for the adopted P4 ordered-sector description of the Φ -medium, here characterised by $\langle \partial_A \Phi \rangle \neq 0$. It is not an additional object; its stationary action/state/domain owner and formation history remain Open.

Order parameter — a supplied collective infrared variable of the BR1 coherent-sector closure, $\Phi_{\text{eff}} = \rho e^{i\theta}$, used to describe candidate low-energy excitations on the ordered chart. It is introduced only at the effective level; neither its descent from the bare micro-action nor physical symmetry breaking is established.

Orientation variable n_A — a collective coordinate on the supplied nonzero-gradient chart describing its local direction. It does not appear as an independent field in the bare micro-action, and its physical state/action owner remains Open.

Electroweak bookkeeping field Φ_{ew} and canonical doublet H — the embedded low-energy scalar variables of the secondary electroweak sector, related by $H \equiv \Phi_{\text{ew}}/\sqrt{2}$. The field H , not Φ_{ew} , is used in canonically normalised Yukawa operators. Both are distinct from the fundamental bare condensate field Φ of P3 and appear only in the electroweak-like embedding (Section 7.4).

2.2 Ontological properties of the Φ -medium

The Φ -medium is a pre-physical entity: it exists prior to spacetime, quantum mechanics, and any emergent physics. Its description must therefore not rely on the language of energy, quantum fluctuations, or effective Lagrangians. The following structural properties characterise the Φ -medium in purely geometric and configurational terms, prior to any ordering or symmetry breaking.

Layer S : Properties of the medium.

- S1 — Connected Euclidean carrier.** The Φ -medium exists on a connected four-dimensional Euclidean carrier $\mathcal{M}^4$ with a flat Euclidean metric.
- S2 — Directional isotropy.** Prior to ordering, all four directions on $\mathcal{M}^4$ are equivalent: the medium is $O(4)$ -isotropic.
- S3 — Point homogeneity.** Prior to ordering, all points of $\mathcal{M}^4$ are equivalent; ordering may break this homogeneity locally.
- S4 — Regularity.** Configurations $\Phi(X)$ are locally regular where they contain no defect cores; configurations with localised defect cores on submanifolds of lower dimension are admissible.
- S5 — Locality.** The local state of the Φ -medium at X is determined by the local configuration in a neighbourhood of X , not by non-local rules on all of $\mathcal{M}^4$.
- S6 — Multiplicity of configurations.** The configuration space is non-trivial: no configuration is privileged a priori; globally and locally distinct configurations are admissible.
- S7 — Adopted ordered sector.** ECT adopts through P4 a physically relevant ordered-gradient sector with $\langle \partial_A \Phi \rangle \neq 0$. The complete stationary action, state, stability domain, physical transition and dynamical selection mechanism realising this input are not supplied.
- S8 — Deformability and relational response.** A regular ordered configuration can respond through local deformation and reconfiguration. Physical distinctions are encoded by invariant strains, gradients, defects, holonomies and response to boundary or material probes, rather than by a passive relabelling of the same configuration. S8 does not identify arbitrary pointwise changes of the compact phase or spacetime director as gauge redundancies; a local bundle and its quotient must be supplied separately.
- S9 — Inhomogeneity of ordering.** Ordering need not be globally uniform; ordered domains, transition zones, and local deviations from a uniform background are admissible.
- S10 — Topological completeness.** Once a physical configuration space, target, boundary class and action are supplied, all compatible topological sectors are admitted rather than excluded a priori. S10 does not identify the abstract P4 orbit with the complete physical vacuum manifold.
- S11a (conditional theorem) — Infinitesimal local deformability.** Subject to fixed admissible boundary data, regularity and a fixed topological sector, a locally regular configuration admits sufficiently small compactly supported deformations. This is the tangent-space structure of the regular scalar configuration space, not a new dynamical postulate.

S11b (candidate) — Finite branch reconfigurability. Some ordered branches may be connected by finite-action admissible transition configurations. Changing the topological sector still requires an allowed defect or transition configuration. Path existence does not imply that the dynamics traverses the path, overcomes a barrier, or relaxes reversibly. S11b contains no relaxation law, channel spectrum or gravitational response.

Layer DP : Dynamical principle.

DP — Variational stationarity. Physically realisable configurations satisfy the stationarity condition for a local functional $S_E[\Phi]$ compatible with the basic symmetries of the medium, and satisfy conditions ensuring the well-posedness of the variational problem. Stationarity supplies equations of motion but does not choose one extremum/history, establish stability, assign a probability, or provide an outcome/record-selection rule.

The working postulates P1–P6 of ECT constitute the present concrete realisation of this framework within the minimal real-scalar field theory. They are adopted defining inputs, denoted $[P]$; a postulate is not demoted to Open merely because it is not derived from an earlier postulate. Open status instead attaches to a missing constructive owner, state/selection mechanism, microscopic descent or observable map. A theorem derived from named inputs is Level A conditional on them; an adopted physical matching or structural closure is Level B; a fit, toy or benchmark is Level C. In that realisation: P1 makes explicit S1–S3; P2 formalises S2 at the level of the bare symmetry; P3 realises S4, S5, and provides the minimal local variational realisation associated with DP; P4 supplies the S7 ordered-gradient datum; P5 realises S8; P6 makes explicit S6 and S9. The broader topological completeness property S10 is partially reflected, within the supplied smooth coherent branch, by the conditional winding classification of BR1. S11a is conditional configuration-space mathematics following from the regular scalar field space once its boundary and topology guards are stated. S11b is a candidate structural property: finite branch reconfigurability has not been derived from the frozen minimal P1–P6 action. Its narrower threshold-relaxation specialisation is a further independent constitutive extension. Any branch-EFT hypothesis about a localised carrier, its action cost or its elastic coefficients belongs to a later post-order constitutive layer, not to this pre-physical property list. Such a hypothesis cannot follow from deformability alone and is not inserted into the minimal P1–P6 action by this classification. The deformability/relational-response property S8 is realised by the working postulate P5, which is logically independent of P1–P4 and P6. It asserts that the ordered branch is a responsive medium whose physical state is read through invariant deformation and relational observables. It does not promote a global symmetry or the physical director to a local gauge orbit. Together, P1–P6, the additional effective closure BR1, and DP provide the present layered structural framework of ECT.

2.3 List of postulates

P1 — Four-dimensional Euclidean arena. The fundamental arena is a connected four-dimensional flat Euclidean carrier $\mathcal{M}^4$ with metric

$$ds_E^2 = \delta_{AB} dX^A dX^B, \quad A, B \in \{1, 2, 3, 4\}. \quad (2.1)$$

The local theory is formulated in flat charts. Whenever global equivalence of points and the full Euclidean group are used, P1 additionally adopts the homogeneous boundary/topology class (the canonical choice is $\mathcal{M}^4 \simeq \mathbb{R}^4$); a generic flat quotient or a manifold with a boundary would not have that global symmetry. No direction is distinguished and no time coordinate is fundamental.

P2 — $O(4)$ -invariant action. The fundamental action is invariant under global $O(4)$ rotations:

$$S_E[\Phi] = S_E[\Lambda \cdot \Phi], \quad \Lambda \in O(4), \quad (\Lambda \cdot \Phi)(X) := \Phi(\Lambda^{-1}X), \quad (2.2)$$

where Φ denotes the fundamental microscopic field of the minimal scalar realisation developed in this work. On the homogeneous flat carrier, translation invariance follows separately from the absence of explicit X -dependence; together translations and rotations form $E(4) = \mathbb{R}^4 \rtimes O(4)$. Thus $O(4)$ is the linear isotropy group, not the full isometry group. In possible future extensions with additional microscopic fields, P2 should be understood as the corresponding $O(4)$ -covariance requirement on the full microscopic action. This expresses the equivalence of all four directions in the unbroken phase.

P3 — Minimal homogeneous scalar sector. The following local real-scalar model is a deliberately minimal supplied sector on $\mathcal{M}^4$:

$$\mathcal{S}_E^{\text{P3}}[\Phi] := \int d^4X \left\{ \frac{1}{2} (\partial_A \Phi)^2 + V(\Phi) \right\}, \quad S_E^{\text{P3,phys}}[\Phi] := A_{\text{P3}} \mathcal{S}_E^{\text{P3}}[\Phi], \quad A_{\text{P3}} > 0, \quad (2.3)$$

$$V(\Phi) = -\frac{\mu^2}{2} \Phi^2 + \frac{\lambda}{4} \Phi^4, \quad \mu^2 > 0, \quad \lambda > 0. \quad (2.4)$$

In the printed natural-unit mass convention the integral $\mathcal{S}_E^{\text{P3}}$ is dimensionless. The independent constant A_{P3} restores a physical-action normalisation; it is not supplied by the displayed integral. A Euclidean path-integral measure needs a second positive owner,

$$d\mu_{\text{P3}}[\Phi] \propto \mathcal{D}\Phi \exp[-\kappa_{\text{P3}} \mathcal{S}_E^{\text{P3}}[\Phi]] = \mathcal{D}\Phi \exp\left[-\frac{S_E^{\text{P3,phys}}[\Phi]}{\Sigma_{\text{PI,P3}}}\right], \quad \kappa_{\text{P3}} := \frac{A_{\text{P3}}}{\Sigma_{\text{PI,P3}}} > 0. \quad (2.5)$$

Here $\Sigma_{\text{PI,P3}} > 0$ has action dimension and controls the measure relative to the classical action prefactor. Setting the same symbol in both places would force $\kappa_{\text{P3}} = 1$ and would erase rather than derive the fluctuation strength. Neither A_{P3} , $\Sigma_{\text{PI,P3}}$, κ_{P3} nor a relation to Σ_θ , Σ_ϑ , S_0 or $\hbar$ is derived. For constant $A_{\text{P3}} > 0$, $\delta S_E^{\text{P3,phys}} = 0$ iff $\delta \mathcal{S}_E^{\text{P3}} = 0$; the stationary set therefore survives the correction, while the measure and fluctuation amplitudes remain separate inputs. If $\mathcal{K}_{\text{P3}}$ is the Hessian of the reduced functional at a supplied stable saddle, the Gaussian covariance is $(\kappa_{\text{P3}} \mathcal{K}_{\text{P3}})^{-1}$ and the formal one-loop factor is $\det(\kappa_{\text{P3}} \mathcal{K}_{\text{P3}})^{-1/2}$, subject to the contour, zero/negative modes, domain, measure and regularisation. The printed symmetric quartic has exactly degenerate minima $\Phi = \pm \mu/\sqrt{\lambda}$, hence no false-vacuum energy splitting. It owns a domain-wall sector but, by itself, no finite false-vacuum bounce: in the standard thin-wall bookkeeping a supplied bias $\varepsilon = \Delta V > 0$ and wall tension $\sigma > 0$ give $R = 3\sigma/\varepsilon$ and $B = 27\pi^2 \sigma^4/(2\varepsilon^3)$; both diverge as $\varepsilon \rightarrow 0^+$. A finite P3 decay saddle would therefore require an explicitly supplied bias/deformation, finite boundary or protocol, or another channel. Only for such a named completion is a positive vacuum-subtracted difference weighted by $\exp[-\kappa_{\text{P3}} \Delta \mathcal{S}_E^{\text{P3}}]$; genuine real suppression requires $\Delta \mathcal{S}_E^{\text{P3}} > 0$. These measure statements identify only κ_{P3} once the normalisation of $\mathcal{S}_E^{\text{P3}}$ is fixed, not A_{P3} and $\Sigma_{\text{PI,P3}}$ separately. It owns the classical homogeneous P3 stationary minima, their Gaussian curvature and the on-shell Noether identities of this model. It is not the full DP action for the P4 gradient datum. The action and two-well potential are invariant under the internal reflection $\Phi \mapsto -\Phi$ and have two classical minima exchanged by that symmetry. Selecting one physical quantum vacuum and spontaneous $\mathbb{Z}_2$ breaking require a state and the appropriate infinite-volume or boundary/source limit; they are not asserted here. P3 creates neither a continuous vacuum manifold, compact coherent phase nor stationary nonzero-gradient branch. Gradient-sensitive $P(X)$ or other completion terms may be investigated, but none is presently adopted as the full owner (OP-grad).

P4 — Adopted ordered-gradient structural postulate. ECT postulates that its physically relevant sector is an ordered-gradient branch with

$$\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}, \quad u_0 > 0. \quad (2.6)$$

A convenient integrable representative is

$$\Phi_{\text{bg}}(X) = u_0 w + \Phi_*, \quad \Phi_* = \text{const.} \quad (2.7)$$

P4 is a foundational structural input $[P]$ (Level-B input in the claim ledger), not a stationary solution derived from the displayed P3 scalar action. Conditional on the nonzero vector, its stabiliser under $O(4)$ is exactly $O(3)$ (Level-A mathematics). The representative profile is not stationary for $\mathcal{S}_E^{P3}$, equivalently for $S_E^{P3, \text{phys}}$ with constant $A_{P3} > 0$. A complete $O(4)$ -invariant stationary action, state, constrained stability domain, physical transition and dynamical formation/selection mechanism realising P4 remain Open. Subsequent “ordered branch”, “broken-phase” and “gradient condensate” language refers to the adopted P4 sector; it does not claim that P3 derives its formation history.

P3 and P4 introduce independent quantities:

Symbol	Definition	Level
$\phi_0 = \sqrt{\mu^2/\lambda}$	homogeneous minimum of the supplied P3 potential	P3 conditional
$u_0 = \langle \partial_A \Phi \rangle $	magnitude of the supplied P4 gradient datum	P4 structural input

Their dynamical relation and the complete stationary owner that realises the adopted P4 postulate remain missing (OP-grad).

BR1 — Smooth coherent topological sector (supplied effective closure; hereafter referred to simply as the coherent branch). In a smooth infrared coherent sector appended to the real-scalar P3–P4 model, low-energy excitations are assumed to admit a collective description through an effective order parameter

$$\Phi_{\text{eff}} = \rho e^{i\theta}, \quad \rho > 0.$$

If Φ_{eff} is single-valued on a closed contour $\mathcal{C}$ that does not intersect a defect core, then the phase defines a map $\mathcal{C} \rightarrow S^1$, and therefore the winding number is necessarily integer:

$$\oint_{\mathcal{C}} \partial_A \theta dX^A = 2\pi n, \quad n \in \mathbb{Z}. \quad (2.8)$$

Thus integer winding is a topological consequence *conditional on* the supplied non-vanishing complex order parameter. The existence of that compact phase sector is not derived from the bare real scalar P3.

Configurations with zeros of ρ , discontinuities of n_A , or other non-smooth field profiles are not excluded by the full theory; however, within the supplied orientation target $O(4)/O(3) \simeq S^3$ they do not carry π_0 -, π_1 - or π_2 -protection (Section 8.10). This S^3 target treats n_A as a polar vector, so n_A and $-n_A$ are distinct. If a future physical completion instead identifies a headless director $n_A \sim -n_A$, its target is the quotient $\mathbb{RP}^3$ and the homotopy inventory must be recomputed; P4 does not choose that quotient. Their status must therefore be analysed dynamically rather than inferred from topology alone. This is consistent with the broader topological completeness property S10 of the Φ -medium. The physical target, admissible exact-gradient configurations, boundary class, action and transition owner remain Open.

Within a supplied fixed-core compact-phase EFT, the elementary coherent-loop representative is defined by

$$\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}}. \quad (2.9)$$

The equality is a definition/evaluation at the supplied characteristic core scale (Section 5.6), not a topological derivation of a nonzero global minimum. A distinct optional positive-tension 1D model has the conditional minimum $\mathcal{S}_{1, \text{min}} = 2\pi \iota_{0, \text{bal}}^{\text{eff}}$. Neither coefficient is automatically the operational S_0 . In the bending-free elementary branch, the motivated matching chain $\Sigma_\theta \iota_{0, \text{bal}}^{\text{eff}} \rightarrow S_0 \leftrightarrow \hbar$ would imply $\Sigma_\theta \mathcal{S}_{1, \text{min}} = 2\pi \hbar = h_P$; the owner arrow remains Open and the $S_0 = \hbar$ arrow is an explicit Level B calibration rather than a derivation of the measured action constant.

P5 — Deformability and relational response of the ordered branch. The ordered Φ -medium is deformable and reconfigurable: neighbouring regular configurations can be distinguished operationally by

invariant strain, gradient, defect, holonomy and response data relative to material or boundary probes. A passive relabelling of one configuration does not create a new physical state.

This statement does not equate a passive relabelling with an arbitrary local field change. The compact phase has a global shift redundancy only when its supplied action has that symmetry; a transformation $\theta(X) \rightarrow \theta(X) + \alpha(X)$ is a local gauge redundancy only after a bundle, connection and quotient have been supplied. The director $n_A = \partial_A \Phi / |\partial \Phi|$ is a constrained carrier-space orientation and can be observable relative to matter or another field; $n_A(X) \rightarrow R_A^B(X) n_B(X)$ is not a gauge orbit of the minimal scalar theory.

This postulate realises property S8 of the Φ -medium within the concrete ordered-branch framework of ECT. It is logically independent of P1–P4 and P6: symmetry, locality and branch existence do not by themselves specify the constitutive response of a deformable continuum medium.

P5 is motivated by the ordinary behaviour of media: a crystal responds to strain, a superfluid to phase differences and circulation, and a liquid crystal to director distortions relative to boundaries and probes. In each case response and deformation are physical, while a passive change of labels is not. The precise variables and symmetries differ between these systems; the analogy therefore motivates medium character but does not manufacture a gauge field.

What P5 does not assert. P5 does not exclude nonlocal observables such as winding numbers or, after a local bundle has been supplied, holonomies and Wilson-loop data. P5 does not by itself determine the gauge coupling constants or the fermion representation content, and it does not derive a physical $SU(3)$ colour connection, action or matter map. The inequality $\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$ excludes only a direct injective Lie-subalgebra/residual-factor embedding in the displayed minimal algebra. Emergent/nonlinear, reducible/singlet, independently derived bundle and defect-kernel routes are not excluded; P7 is the adopted Level-B structural completion used in this manuscript, not a theorem that it is the necessary or unique microscopic origin of physical colour.

Gauge-completion programme. P5 motivates asking which coordinate descriptions become redundant after reduction, but it does not make the physical spacetime director $n_A(X)$ a gauge orbit or localise a global compact phase. The exact inputs presently available are global compact-phase algebra in a supplied BR1 sector and the Lie-algebra identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$. Local bundles, connections, gauge-volume quotients, physical state spaces and a chiral $SU(2)_L \times U(1)_Y$ identification remain Open completion data; see Sections 7.2 and 7.3.

P6 — Multiplicity of configurations and admissibility of non-uniform ordering. The configuration space of the Φ -medium is non-trivial: no single configuration is singled out a priori as the unique physical realisation (S6). In particular, P6 admits spatially varying backgrounds, domain structures and transition configurations in the ordered-sector configuration space (S9). Their stationarity, physical occurrence, profiles and dynamics are not fixed by this admissibility postulate and remain Open. This postulate does not prescribe the magnitude or functional form of such inhomogeneities; it states only that the ECT framework is not restricted to a single homogeneous ordered configuration.

2.4 Operational principle of relationality

In the absence of a fundamental time coordinate, as specified by the arena P1, the relational approach provides a natural and operationally consistent way to define observables: physical observables are ratios between physical subsystems. This is motivated by P1 but does not follow from it as a theorem.

2.5 Independence of postulates

The six working postulates P1–P6 are logically independent. In particular, the medium-character postulate P5 is independent of P1–P4 and P6: it cannot be derived from action symmetry, locality, field content, ordering, or configuration multiplicity alone.

- P1 does not follow from P2–P6: neither $O(4)$ invariance, nor the choice of field content, nor the admissibility of an ordered branch, nor the multiplicity of admissible configurations fixes the

existence of a connected four-dimensional Euclidean arena.

- P2 does not follow from P1 and P3–P6: the existence of a Euclidean carrier, a local scalar field, an ordered branch, and a non-trivial configuration space does not by itself enforce global $O(4)$ invariance of the bare description.
- P3 does not follow from P1, P2, P4, and P6: the specific real-scalar field content and the minimal local micro-action must be stated separately.
- P4 does not follow from P1–P3 and P6: its nonzero ordered-gradient datum is an additional supplied EFT input beyond the minimal bare micro-action and the mere admissibility of non-trivial or non-uniform configurations. Physical realisation and selection remain Open.
- P6 does not follow from P1–P4: the non-triviality of the configuration space and the admissibility of inhomogeneous ordering are additional structural properties not entailed by the minimal scalar micro-action or the ordered branch specification alone.

BR1 is an adopted Level-B effective closure, not a P1–P6 theorem: it supplies a nonvanishing single-valued complex collective field on its declared infrared domain. Given that input and a closed contour avoiding zeros, integer winding is a Level-A topological consequence. Microscopic descent, formation and selection of the BR1 domain remain Open.

2.6 Discussion of the postulates

P1 — Euclidean arena

Physical meaning. Space and time are not co-equal primary structures of nature. At the fundamental level there exists only a four-dimensional geometric arena in which all directions are indistinguishable. The notions of “before” and “after”, “spatial” and “temporal” are not built into the foundation of the theory: they may emerge from it under certain conditions, but are not postulated from the outset.

This fundamentally distinguishes ECT from all theories that start with a Lorentzian spacetime $(-, +, +, +)$: in those theories the distinction between time and space is encoded from the very beginning in the sign of the metric.

Physical motivation. Euclidean geometry as a starting structure appears in physics in several independent contexts. In quantum field theory the Wick rotation $t \rightarrow -i\tau$ (Matsubara formalism [6]) can relate Lorentzian and Euclidean correlation functions when the required analyticity, contour, state and measure hypotheses hold; it does not automatically make every path integral well-defined. In the Hartle–Hawking no-boundary proposal [7] the candidate cosmological state is defined through a path integral (or semiclassical saddles) over regular compact positive-definite or complex four-geometries. In instanton calculations [8] tunnelling configurations are solutions of Euclidean equations of motion. These established Euclidean constructions have different mathematical and interpretive roles and do not imply ECT’s ontological postulate. In ECT a Euclidean scalar medium is instead proposed as physically primary.

On dimensionality. P1 adopts $d = 4$. One motivation is the exact dimension count $\dim[O(d)/O(d-1)] = d-1$, so $d = 4$ leaves three directions transverse to a nonzero ordered vector. Identifying those directions with all observed physical space still requires the clock, metric and matter maps and is not a derivation of $d = 4$ from observations. See §16.8 for the distinction between topological and force-law dimensionality. The deeper origin of the adopted dimensionality remains Open (OP0).

What P1 forbids. Lorentzian spacetime $\mathbb{R}^{1,3}$ is not an admissible fundamental arena in ECT. It may arise as an effective structure, but is not built into the foundation.

P2 — $O(4)$ -invariant action

Physical meaning. The laws of physics in the symmetric phase are identical under any rotation in $\mathcal{M}^4$. No one of the four directions is dynamically distinguished.

Physical motivation. $O(4)$ is the maximal linear group preserving the positive-definite tensor δ_{AB} ; on the homogeneous flat carrier the full isometry group is $E(4) = \mathbb{R}^4 \rtimes O(4)$. P2 adopts the $O(4)$ isotropy of the action, while point homogeneity and translations are separate domain/action statements. A smaller linear isotropy such as $O(3)$ would explicitly single out a direction. The assertion that a physical state chooses an $O(3)$ stabiliser is instead P4 input, while its stationary dynamical owner and formation remain Open. No claim that $O(4)$ is the largest possible internal or spacetime symmetry is needed.

Consequence for the effective sector. Within the supplied P4 chart, the nonzero vector datum defines a preferred direction. The most general symmetric rank-2 tensor compatible with the residual $O(3)$ -symmetry is constructed from δ^{AB} and $n^A n^B$. Its explicit form and uniqueness are established as Theorem 3.2 in Section 3.3.

What P2 forbids. Any action that explicitly singles out one of the four directions of $\mathcal{M}^4$ at the P2 microdynamic level. The present anisotropic EFT analysis is conditioned on the direction supplied by P4; it has not been derived as spontaneous ordering. Any future action that owns that physical selection must remain $O(4)$ -invariant before its state and boundary limit are specified.

P3 — Minimal scalar microdynamics

Physical meaning. The minimal P1–P6 dynamical sector contains one real scalar Φ . Within that deliberately minimal sector, effective variables, collective modes and geometric structures are constructed from this field. The reverse-engineered P7 extension adds an internal colour module and therefore must not be described as part of the scalar-only microscopic basis.

Why a scalar. A single real scalar is the minimal one-component choice compatible with P2: it carries no spacetime index and introduces no preferred direction. This is an adopted economical field content, not a uniqueness theorem; isotropic multiplets or additional fields are possible in named completions.

Form of the action. The action (2.3) at the level of P3 contains only an isotropic kinetic term and a potential. An independent director does not appear in it. The composite coordinate $n_A = \partial_A \Phi / |\partial \Phi|$ introduces no second local degree of freedom where $|\partial \Phi| > 0$, but is singular at zero gradient and changes the operator/constraint bookkeeping. Treating n_A as an independent dynamical field would enlarge P3 and requires a separately supplied action.

On the potential. The double-well ϕ^4 potential is the standard mechanism for spontaneous $\mathbb{Z}_2$ -symmetry breaking $\Phi \rightarrow -\Phi$. Importantly, this potential does not by itself create a continuous vacuum manifold nor generate the coherent phase structure later isolated as BR1; the latter belongs to the effective collective sector. The potential specifies the minimal local microdynamics of Φ and defines a local minimum at finite field amplitude $\phi_0 = \sqrt{\mu^2/\lambda}$ in the microscopic potential. However, $V(\Phi)$ alone is not sufficient to produce a stationary gradient-ordered vacuum: the nonzero-gradient analysis datum is supplied separately by P4. The value ϕ_0 is a technical characteristic of the micro-potential; the ordered-chart quantity used downstream is the supplied gradient scale u_0 (P4), whose nonzero vector has $O(3)$ stabiliser. Its physical realisation remains Open.

What P3 forbids. At the fundamental level: any additional fields. There are no fundamental spinors, vectors, or tensors. If such objects appear, they do so only as effective excitations or collective variables at a higher level of description.

P4 — Adopted ordered-gradient branch and its EFT chart

Physical meaning. P4 adopts the physically relevant sector of ECT as one in which the mean gradient is nonzero and directed along one of the four Euclidean directions. This is a defining structural input and makes the subsequent broken-phase EFT analysis nontrivial; it is not a theorem that the displayed P3 action owns a stationary vacuum of that form.

Why the gradient is the relevant order parameter. The expectation value $\langle \Phi \rangle = \phi_0 \neq 0$ is $O(4)$ -invariant: a scalar does not distinguish directions and therefore does not by itself break the Euclidean symmetry. By contrast, $\partial_A \Phi$ transforms as a vector under $O(4)$. A non-zero expectation value $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$ defines the direction w and has stabiliser $O(3)$ inside $O(4)$. Within the present minimal single-field ECT framework, $\partial_A \Phi$ is the lowest-canonical-dimension local *polar-vector* order parameter at first derivative order. It can produce directional ordering without adding a fundamental field or explicit anisotropy. This is not a uniqueness theorem for all one-scalar order parameters: a nematic composite such as $\langle \partial_A \Phi \partial_B \Phi - \delta_{AB} (\partial \Phi)^2 / 4 \rangle$ could select an unoriented axis while $\langle \partial_A \Phi \rangle = 0$, with a different target and stabiliser.

On the status of P4. In the minimal bare action P3 the standard ϕ^4 potential reaches its minimum on homogeneous configurations with $\partial_A \Phi = 0$. Therefore P4 is not a theorem of the bare micro-action. It is instead the adopted structural postulate that a physically relevant ordered-gradient sector exists. It introduces no explicit anisotropy into P3, but P3 does not derive its stationary action, amplitude, stability, formation or selection. Constructing a more general $O(4)$ -invariant gradient-sensitive functional $P(X)$, $X \equiv (\partial_A \Phi)^2$, with a stable stationary branch at $X = u_*^2 > 0$ is an Open foundational owner problem (OP-grad), not a condition for acknowledging P4 as a postulate.

On notation. The expectation values $\langle \Phi \rangle = \phi_0$ and $|\langle \partial_A \Phi \rangle| = u_0$ have different physical meanings and different dimensions: $[\phi_0] = \text{GeV}$ and $[u_0] = \text{GeV}^2$ in natural units. They should not be identified. In the scale inventory of Section 5.1, the radial symmetry-breaking amplitude $\phi_0 \sim \bar{M}_{\text{Pl}}$ is the primary energy-scale candidate (Section 5.1).

The orientation coordinate. On the supplied nonzero-gradient P4 chart, n_A is the normalised local direction, with reference map $n_A = \delta_{Aw}$. The relation $n_A = \partial_A \Phi / |\partial \Phi|$ is a coordinate construction only where $|\partial \Phi| > 0$; it is neither a selected physical vacuum nor a symmetry-breaking event, and it is not a pointwise substitution in the bare microscopic P3 action.

BR1 — Smooth coherent topological sector

Physical meaning. Within the chosen smooth BR1 effective sector, admissible configurations have a globally consistent nonvanishing compact phase on the declared contour and therefore integer winding. This restriction defines that supplied sector; it neither excludes defect/noncoherent sectors from the full configuration space nor selects which sector is physically realised.

Origin of the order parameter. In the separately supplied coherent IR closure BR1, and only within its declared low-energy domain, collective excitations are represented by an effective order parameter $\Phi_{\text{eff}} = \rho e^{i\theta}$. This is not a fundamental variable of the displayed microdynamics but an IR collective object introduced by the adopted Level-B closure. Its microscopic descent and physical selection remain Open. The phase θ belongs to this collective object, not to the real field Φ .

Topological content. The integer winding condition is determined by the homotopy group $\pi_1(S^1) = \mathbb{Z}$ [9] and is independent of contour shape or dynamics. The homotopy classification of the supplied orientation target $O(4)/O(3) \simeq S^3$, including $\pi_0 = \pi_1 = \pi_2 = 0$ and the abstract texture/skyrmion classes $\pi_3(S^3) = \mathbb{Z}$, is developed in Section 8.10. This is Level-A mathematics conditional on that target, not yet a physical defect or relic prediction. Analogous conditions underlie the quantisation of circulation in superfluid helium $\kappa = h/m$ [10] and of magnetic flux in superconductors $\Phi_0 = h/2e$ [11]. In those physical systems the circulation or flux quantum also uses a separately owned stiffness, charge or mass normalisation; integer topology alone does not determine its action scale. ECT applies the same winding classification to its separately supplied collective phase in four Euclidean dimensions; the physical normalisation, state and action remain independently owned.

What BR1 excludes from the present reduction. Within the coherent sector: configurations with fractional winding numbers of the phase, with discontinuities in n_A , or with zeros of the order parameter. All three classes are mathematically admissible within P1–P6 but do not belong to the declared smooth BR1 reduction. BR1 does not determine which sector is physically realised.

P5 / S8 — Deformability and relational response

Physical meaning. Property S8, realised by the working postulate P5, states that the Φ -medium is a deformable and reconfigurable continuum rather than a rigid structure. Its physical state is probed through invariant deformation, defect, gradient, boundary-response and, where a bundle exists, holonomy data. This is a constitutive statement about response. It is not a declaration that every pointwise field value is unobservable or that every local change is a gauge transformation.

Why S8 is not a repetition of P2 or P3. P2 specifies a global symmetry of the bare action and P3 its locality. S8 instead specifies the class of constitutive behaviour: admissible regular configurations can deform, and physical readout compares their invariant response to neighbouring configurations, matter and boundaries. Neither a global symmetry nor deformability alone creates a local redundancy.

For the supplied compact phase, a constant shift can be redundant when the action depends only on derivatives. A point-dependent shift requires an independently supplied local bundle and connection. The director $n_A = \partial_A \Phi / |\partial \Phi|$ is a physical constrained orientation of the scalar gradient, not an internal label that can be rotated arbitrarily at each point. Only passive coordinate/frame relabelling of the complete set of fields is automatically redundant.

Consequences. S8 has structural implications, but it does not create a physical gauge field:

- (i) a global phase shift can identify equivalent descriptions in the supplied derivative-only compact-phase sector, while local phase covariance is an additional completion;
- (ii) physical director distortions, gradients and defects are observable relationally and must not be quotiented as local gauge directions;
- (iii) a connection may be introduced only after an independently supplied local bundle-covariance hypothesis. Neither P5 nor S8 supplies that connection, its action, its state or its matter representation;
- (iv) the algebraic isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ does not identify the physical chiral weak group $SU(2)_L$. Combining the phase and orientation labels therefore does not derive a physical $SU(2)_L \times U(1)_Y$ gauge theory;
- (v) a continuity equation for the coherent density follows only from a supplied action with the corresponding continuous symmetry and its Noether current; medium character alone is insufficient;
- (vi) the ordered-branch gauge-topological structure provides a candidate architecture for later completions, but baryon violation additionally requires a physical bundle, anomaly structure, CP violation and a nonequilibrium state (Section 16.3).

Nonlocal observables are not excluded. S8 permits relational local and nonlocal observables. Winding numbers are available in a supplied compact phase; holonomies and Wilson loops become defined only after a local connection is supplied. Director gradients and defects are physical response data, not remnants of quotienting the director.

What S8 does not assert. S8 does not by itself determine the gauge coupling constants g, g' ; it does not derive the electroweak symmetry-breaking pattern $SU(2) \times U(1) \rightarrow U(1)_{em}$; it does not derive the fermion representation content; and it does not produce the $SU(3)$ colour gauge structure, which requires additional internal degrees of freedom beyond the minimal scalar realisation of P3.

Why S8 is natural for a medium. The proposed ontology of the minimal P1–P6 sector is that of a *medium*: a single scalar field Φ fills the Euclidean arena, while P4 supplies an ordered-gradient chart. Its formation as an effective physical vacuum is Open. Crystals respond through strain, superfluids through phase differences and circulation, and nematics through director distortions relative to boundaries. P5 adopts the common, simple property behind these examples—deformability with relational readout—without identifying their distinct microscopic variables or postulating gauge redundancy. ERP- Φ later supplies a much narrower ideal threshold-release specialisation; it is not contained in P5 by definition.

P6 — Multiplicity and inhomogeneity

Physical meaning. The Φ -medium is not restricted to a single global configuration. Ordered realisations may include spatially varying condensate backgrounds, domain structures, and transition regions. No globally uniform ordered state is imposed a priori as the unique physical realisation.

Consequences. P6 permits spatially varying backgrounds in a future owner-complete ordered sector. This makes later macroscopic constructions with branch-dependent effective quantities admissible programme elements, although their actions, states and dynamical laws must be derived separately. Domain structures and transition regions are conditional candidate configurations, not established constituents of a physical phase. Likewise, residual scatter in effective phenomenological scales (such as a_M in the galactic sector) is not excluded a priori, but requires separate dynamical analysis.

What P6 does not assert. P6 does not prescribe the magnitude, profile, or dynamical law of spatial variations in the ordered branch. It does not by itself derive a physical gravitational-response map $G_{resp}(\mathbf{x})$, $u_0(\mathbf{x})$, or galaxy-by-galaxy scatter relations. In particular, P6 does not identify the divided scalar-equation coefficient with a Cavendish or galactic response. It states only that such non-uniform ordered configurations are structurally admissible within the Φ -medium.

2.7 Three statement types

The following sections use three types of assertions:

- *Theorem* — a strict consequence of the explicitly stated model, including any supplied effective branch rule and its assumptions;
- *EFT consequence* — a result valid within the low-energy expansion about the supplied ordered chart;
- *Additional assumption* — a statement used to obtain explicit formulae but not a strict consequence of the foundational assumptions and explicitly stated branch conditions (always marked explicitly).

In this foundational section: the ordered-gradient analysis chart is supplied by P4; the coherent branch is introduced through the supplied effective closure BR1; the form of K^{AB} is an EFT consequence (Theorem 3.2); the optional inequality $\alpha < 4\beta$ is not a Lorentzian boundary and requires a separately supplied arithmetic isotropic-average prescription.

Relation to the Level A/B/C classification. The later sections of this work classify results using a finer-grained Level A / Level B / Level C / Open scheme. The correspondence is approximately: *Theorem* corresponds to Level A; *EFT consequence* corresponds to Level A when the result has been explicitly derived within a declared supplied branch framework and its stated assumptions, and to Level B when it is structurally reconstructed but not yet fully closed from bare P3 alone; *Additional assumption* corresponds to Level B or Level C. Unresolved items remain Open. This mapping is not a rigid one-to-one identification but a guide to the epistemic status of each result.

Conditional numerical benchmarks. A fully declared parameter point may be quoted as Level C arithmetic when all selected inputs, conventions and missing owners are printed, the point lies inside the declared mathematical or EFT validity domain, and all stated internal constraints are satisfied. Such a number is not thereby an ECT prediction, explanation, evidence, confirmation or whole-theory falsifier. An owner-complete action/state-to-observable and uncertainty/likelihood chain is required for those stronger claims, not for the mere admissibility of a transparent conditional benchmark. Conversely, an arbitrary observed value does not become a useful postulate or benchmark merely by being relabelled as an assumption.

Displayed digits are deterministic rounding of the stated formula evaluated at the declared inputs and conventions. They establish reproducibility of that arithmetic only; they are neither an uncertainty statement nor a claim of physical precision. Physical inference additionally requires input uncertainties and covariances, together with an owner-complete action/state-to-observable matching or likelihood chain.

3 Supplied Ordered-Gradient Structure and Conditional Scalar Signature

This section separates what the ordered-gradient datum (P4) supplies from what the scalar EFT and later response constructions add. P4 supplies a direction and residual $O(3)$ stabiliser. Conditional on independently specified coefficients, the scalar principal tensor is Lorentzian iff $\beta(\beta - \alpha) < 0$. On the healthy transverse-kinetic branch $\beta > 0$ used below, this is the shorthand $\alpha > \beta > 0$. This classification does not yet identify a physical clock, universal metric, common matter/photon/tensor cone or thermodynamic arrow. An ECT thermodynamic second law remains Open: Markovianity alone is insufficient, and a declared reduced channel, state, entropy functional, and unitality/detailed-balance/contractivity conditions are required.

The working postulates P1–P6, together with the supplied effective coherent-sector closure BR1, do not yet constitute the full physical theory. This section develops the conditional geometric consequences of the supplied P4 chart in three stages. First, we recall a conditional dynamical motivation for gradient condensation and introduce collective infrared variables on the supplied ordered chart (Sections 3.1–3.2). Second, we derive the unique local algebraic scalar kinetic tensor built from the supplied direction and classify all possible principal-symbol signatures admitted by it (Sections 3.3–3.5). Third, for the supplied hyperbolic class we derive standard scalar retarded support and then state the extra clock, state, metric and open-system owners needed for a physical time orientation and thermodynamic arrow (Sections 3.6–3.8).

3.1 Dynamic motivation for gradient condensation

The minimal bare micro-action P3 does not itself select a gradient-ordered branch: its ϕ^4 potential is minimised by homogeneous configurations with $\partial_A \Phi = 0$. P4 independently adopts the physically relevant ordered-gradient branch and supplies its EFT chart; the stationary action, state, stability and formation/selection owner remain Open.

A deeper motivation arises from one deliberately minimal separable $O(4)$ -invariant extension compatible with P1–P2. A general local first-derivative scalar density may depend on both Φ and $X = (\partial\Phi)^2$ through $F(\Phi, X)$; it need not split as $P(X) + V(\Phi)$. The lowest-dimensional gradient invariant is

$$X \equiv \partial_A \Phi \partial_A \Phi. \quad (3.1)$$

(We use X for the kinetic invariant and X^A for Euclidean coordinates.) The restricted separable choice $F(\Phi, X) = P(X) + V(\Phi)$ gives the reduced gradient functional

$$\mathcal{J}_P^{\text{grad}}[\Phi] := \int d^4X P(\partial_A \Phi \partial_A \Phi), \quad S_P^{\text{phys}} = A_P \zeta_P \mathcal{J}_P, \quad 0 < A_P < \infty, \quad \zeta_P \in \{+1, -1\}, \quad (3.2)$$

where $\mathcal{J}_P = \int d^4X [P(X) + V(\Phi)]$ when the potential is included. The finite positive magnitude A_P and sign ζ_P are constant and field-independent in the local analysis. They do not affect the stationary equation or principal cone; ζ_P is fixed only after a Lorentzian component and positive-energy orientation are declared. This is the minimal $O(4)$ -invariant extension that couples the reduced functional to the gradient magnitude rather than solely to the field amplitude; it is not by itself a positive Euclidean statistical weight.

For the pure-gradient functional $\mathcal{J}_P^{\text{grad}}$, suppose $P(X)$ has a stationary point at $X = u_*^2 > 0$,

$$P_X(u_*^2) = 0, \quad P_{XX}(u_*^2) > 0, \quad (3.3)$$

then affine configurations satisfying

$$\partial_A \Phi \partial_A \Phi = u_*^2. \quad (3.4)$$

are stationary solutions of the pure-gradient Euler–Lagrange equation. A representative integrable member of this pure-gradient solution family matches the local profile used by P4:

$$\Phi_0(X) = u_A X^A + \Phi_c, \quad u_A u_A = u_*^2. \quad (3.5)$$

For the pure-gradient functional, the Euler–Lagrange expression $-2\partial_A(P_X \partial_A \Phi)$ vanishes on this affine family. If the potential is included, however, the equation becomes

$$-2\partial_A(P_X \partial_A \Phi) + V'(\Phi) = 0,$$

so one additionally needs $V'(u_A X^A + \Phi_c) = 0$ pointwise. The nonconstant affine profile does not satisfy that condition for the P3 quartic. Thus the displayed $P(X)$ construction is a conditional pure-gradient toy owner, not yet a stationary P4 completion with the P3 potential.

The affine statement is local to the bulk Euler–Lagrange equation. On a finite domain it is a stationary variational configuration only for compatible boundary data (or after the corresponding natural boundary term is satisfied); on noncompact $\mathbb{R}^4$ a nonzero constant-gradient configuration has an extensive action unless an action-density or subtracted-state prescription is declared. Thus the bulk solution family does not by itself establish a global vacuum or an admissible physical state.

Since this conditional pure-gradient functional depends only on $u_A u_A$ and not on the direction of u_A , the displayed stationary solution family forms a three-sphere. Conditioning on one member leaves the $O(3)$ stabiliser of its nonzero vector; physical spontaneous selection additionally requires a state, boundary/infinite-volume limit and stability analysis.

It is natural to decompose the gradient into amplitude and orientation,

$$\partial_A \Phi = u n_A, \quad u \equiv \sqrt{\partial_B \Phi \partial_B \Phi} \geq 0, \quad n_A n_A = 1. \quad (3.6)$$

The scalar field $u(x)$ describes the *amplitude* of the gradient condensate; the unit vector $n_A(x)$ describes its local *orientation*. At the background level $u = u_*$ and $n_A = \delta_{Aw}$.

Status of this section. The $P(X)$ -extension of the bare action and the gradient-branch condition (3.3) are presented here as the minimal dynamical motivation for P4, not as a derived consequence of P1–P3. The specific functional form of $P(X)$, the numerical value of u_* , and the relation between u_* and the bare parameters (μ, λ) remain open problems of the ECT foundational program (OP-grad, Level Open). The subsequent broken-phase analysis (Sections 3.2–3.8) does not require the microscopic form of $P(X)$ to be fixed completely. It depends only on the existence of the ordered branch, its collective infrared variables, and the broken-phase EFT parameters (α, β) . The admissibility of an ordered branch is supplied by P4 and organised by the stated Φ -medium properties; its dynamical realisation from P1–P3, state selection and coefficient values remain Open. Non-uniform ordered backgrounds are separately admitted by S9/P6 rather than derived from the bare scalar action.

Three distinct notions (recorded for OP-grad/OP3.1a). For the pure gradient sector $\mathcal{J}_P^{\text{grad}} = \int d^4X P(X)$ the local field equation is $\partial_A(P_X(X) \partial_A \Phi) = 0$; a constant-gradient profile has $X = \text{const}$ and solves it for any constant value of P_X . The recorded condition (3.3) is a different statement: it selects the magnitude u_*^2 as a stationary-density point and, under $P_{XX} > 0$, a local minimum among homogeneous gradients. A third, again separate, question is whether the fluctuation tensor around the ordered background matches the broken-phase EFT tensor.

Computed mini-lemma (pure- P fluctuation tensor). Writing $\Phi = \bar{\Phi} + \chi$ with $\partial_A \bar{\Phi} = u_* n_A$, so that $\delta X = 2u_* n \cdot \partial \chi + (\partial \chi)^2$, the quadratic reduced functional of the pure- P sector is

$$\mathcal{J}_P^{(2)} = \int d^4X \left[P_X(X_0) (\partial \chi)^2 + 2P_{XX}(X_0) u_*^2 (n \cdot \partial \chi)^2 \right], \quad X_0 = u_*^2,$$

i.e. a fluctuation tensor $\mathcal{K}^{AB} \propto P_X(X_0) \delta^{AB} + 2P_{XX}(X_0) u_*^2 n^A n^B$, with longitudinal coefficient $P_X(X_0) + 2P_{XX}(X_0) u_*^2$ and transverse coefficient $P_X(X_0)$. At the stationary-density point $P_X(X_0) = 0$, $P_{XX}(X_0) > 0$, the pure- P sector supplies positive longitudinal stiffness but leaves the transverse orientation/Goldstone directions flat at this order. The recorded orientation stiffness $\mathcal{C}_n u^2 (\partial n)^2$ is therefore a natural EFT-level supplier of the transverse kinetic term, but deriving it from the bare bridge remains open.

Matching question M-(α, β) (a question, not an inconsistency). Direct formal matching of $\mathcal{K}^{AB}$ to the broken-phase kinetic tensor $\beta \delta^{AB} - \alpha n^A n^B$ with $\alpha > \beta > 0$ requires $P_X(X_0) > 0$ and $Q'(X_0) = P_X(X_0) + 2X_0 P_{XX}(X_0) < 0$, equivalently $P_{XX}(X_0) < -P_X(X_0)/(2X_0)$ for $X_0 > 0$. Merely requiring $P_{XX} < 0$ is insufficient. This retained matching component is not the whole Lorentzian domain and is not the recorded stationary-density point. A constant-gradient profile may still solve the local pure- P field equation, but the mechanism selecting its magnitude and matching its fluctuation tensor to the broken-phase EFT tensor is not supplied by the recorded $P_X = 0$ minimum alone. Possible bridges include mixed $P(X, \Phi)$ dependence, coupling to the potential sector, orientation stiffness at EFT level, or an explicit distinction between bare and EFT fluctuation tensors. The two tensors may legitimately live at different levels; the task recorded here is to exhibit the bridge explicitly. OP-grad/OP3.1a sharpened; completion open.

Potential-sector constraint (computed; profile reduction only). For $\mathcal{J}_P = \int d^4X [P(X) + V(\Phi)]$ in the profile reduction $\Phi = \Phi(s)$, $X = \Phi^2$, the field equation $2 \frac{d}{ds} (P_X \Phi') = V'(\Phi)$ has the first integral

$$Q(X) = E + V(\Phi), \quad Q(X) := 2X P_X(X) - P(X),$$

with the structural identity $Q'(X) = P_X(X) + 2X P_{XX}(X)$ — exactly the longitudinal fluctuation coefficient above; the gradient magnitude is slaved to the potential through $dX/d\Phi = V'(\Phi)/Q'(X)$ wherever $Q' \neq 0$ and locally $\Phi' \neq 0$. Two computed consequences follow. First, the recorded V excludes an exact global linear-gradient solution of the simple $P+V$ profile equation: away from $\Phi \in \{0, \pm\phi_0\}$ one has $V' \neq 0$, while the linear profile makes the left-hand side vanish; quasi-homogeneous gradient patches then have finite profile extent (a hedged P3 dimensional estimate of order $m_{\Phi, \text{rad}}^{-1}$, not a theorem and not a P4 scale). Second, the potential sector contributes only the position-dependent mass term $\frac{1}{2} V''(\bar{\Phi}) \chi^2$ and the background deformation — no derivative kinetic terms; mixed $P(X, \Phi)$ cross-terms likewise reduce, upon integration by parts, to background/mass-sector contributions. Hence the bridge candidate “coupling to the potential sector” is reclassified: it is a constraint and a mass-sector structure, not the kinetic bridge. The matching question correspondingly sharpens. The exact nondegenerate Lorentzian principal-symbol condition is

$$P_X(\bar{X}) Q'(\bar{X}) < 0, \tag{3.7}$$

with two components: $P_X > 0, Q' < 0$ and $P_X < 0, Q' > 0$. With $\beta = P_X(\bar{X})$ and $\beta - \alpha = Q'(\bar{X})$, the manuscript’s retained $\beta > 0$ component obeys

$$\alpha > \beta > 0 \iff P_X(\bar{X}) > 0, \quad Q'(\bar{X}) < 0.$$

For the physical candidate $S_P^{\text{phys}} = A_P \zeta_P \mathcal{I}_P$, local positive quadratic kinetic and gradient energy fixes $\zeta_P = -\text{sgn } P_X = \text{sgn } Q'$ on either Lorentzian component. This local orientation test does not prove boundedness of the full nonlinear Hamiltonian or the sign of lower-order terms. The retained component requires an operative descending branch of Q , which the simple $P+V$ profile mechanics does not itself stabilise. No inconsistency is asserted; the bridge can live at the orientation/EFT, constraint, composite-variable, or bare-versus-EFT level. The stationary-density point $P_X = 0$ remains degenerate. OP-grad/OP3.1a and the owner of A_P remain open.

Orientation sector and the matching question (computed; strict integrable regime). The orientation-sector operators of the ordered-branch EFT recorded in (3.9) (§3.2) do not alter this conclusion at the two-derivative level. Inserting the integrability reduction $\sigma = \partial_w \chi$, $\pi_i = \partial_i \chi / u_0$ (eq. (4.7), §4.3) into those operators gives

$$\frac{Z_u}{2} (\partial_A \sigma)^2 = \frac{Z_u}{2} (\partial_A \partial_w \chi)^2, \quad \frac{\mathcal{C}_n}{2} u_0^2 (\partial_A \pi_i)^2 = \frac{\mathcal{C}_n}{2} (\partial_A \partial_i \chi)^2,$$

where the background factor u_0^2 cancels exactly against the $1/u_0^2$ from $\pi_i = \partial_i \chi / u_0$. Both recorded operators pull back to fourth-order operators in χ — higher-derivative/ k^4 -type corrections — and do not modify the two-derivative tensor K^{AB} . Hence $\mathcal{C}_n$ is not a two-derivative transverse supplier in the strict integrable single-field pullback; that reading belongs to a coarse-grained level at which the exact constraint (4.5) is relaxed and n_A acquires effectively independent long-wavelength dynamics. OP-grad/OP3.1a is localised to the constraint-relaxation/coarse-graining bridge; completion remains open.

3.2 Collective variables and the ordered-branch infrared EFT

The gradient-sensitive motivation above clarifies why an ordered branch with $\partial_A \Phi \neq 0$ is natural, but it does not yet specify the appropriate long-wavelength variables of that branch. For the subsequent development of both the geometric and coherent sectors, it is useful to separate the ordered condensate into its amplitude and orientation components:

$$\partial_A \Phi = u n_A, \quad u \geq 0, \quad n_A n_A = 1. \quad (3.8)$$

Here $u(x)$ measures the local amplitude of the gradient condensate, while $n_A(x)$ specifies its local orientation in the underlying Euclidean medium.

At wavelengths large compared with the microscopic condensate scale, the ordered branch admits a local derivative expansion in terms of these collective variables. Before any geometric interpretation is imposed, the minimal Euclidean infrared functional of the ordered branch may be written in the schematic form

$$\mathcal{J}_{\text{ord,E}}[u, n] = \int d^4 X \left[\frac{Z_u}{2} (\partial_A u)(\partial_A u) + \frac{\mathcal{C}_n}{2} u^2 (\partial_A n_B)(\partial_A n_B) + W(u) + \Lambda_n (n_A n_A - 1) + \dots \right]. \quad (3.9)$$

Here $\mathcal{J}_{\text{ord,E}}$ is the dimensionless reduced functional in the displayed natural-unit convention, $W(u)$ is the effective branch potential for the condensate amplitude, Λ_n enforces $n_A n_A = 1$, and the ellipsis denotes suppressed higher-derivative or higher-order operators. A physical action requires the separate supplied magnitude

$$S_{\text{ord,E}}^{\text{phys}} = A_{\text{ord,E}} \mathcal{J}_{\text{ord,E}}, \quad 0 < A_{\text{ord,E}} < \infty, \quad (3.10)$$

taken constant and field-independent within this truncation. Neither a Euclidean measure denominator nor equality of $A_{\text{ord,E}}$ to the primordial A_χ or any quantum-sector action scale is implied.

Several comments are important.

(i) This is not yet the geometric branch action. Equation (3.9) is a pre-geometric infrared description of the ordered condensate branch itself. It does not yet assume the existence of an effective metric, and it does not yet identify gravity with any particular operator. Rather, it provides the minimal collective-variable layer that should exist between the microscopic Euclidean Φ -medium and the later macroscopic geometric realisation.

(ii) Why these variables are natural. Once the branch condition $\partial_A \Phi \neq 0$ holds, the decomposition into amplitude u and orientation n_A is not an extra postulate but a natural rewriting of the ordered phase. This makes explicit that two qualitatively different kinds of long-wavelength excitations may exist: amplitude fluctuations of the condensate and orientational distortions of the preferred direction.

(iii) Relation to the subsequent signature analysis. For a uniform ordered background

$$u = u_*, \quad n_A = \delta_{Aw},$$

the infrared ordered-branch EFT reduces to the background on which the broken-phase kinetic tensor is classified in the following subsections. The symmetry-based uniqueness theorem for K^{AB} and the Lorentzian viability analysis therefore do not depend on the detailed microscopic form of $W(u)$, but only on the existence of the ordered branch and its collective orientation variable.

(iv) Status. At the present stage (3.9) should be read as the minimal infrared organisational principle of the ordered branch, not yet as a fully derived theorem from the bare microscopic action P3. Its detailed coefficients ($Z_u, \mathcal{C}_n$), the precise form of $W(u)$, the physical magnitude $A_{\text{ord,E}}$, a measure denominator, and its matching to the later macroscopic gravitational closure remain part of the open foundational programme. Nevertheless, introducing this intermediate ordered-branch EFT removes an important conceptual gap: the theory no longer jumps directly from the bare Euclidean scalar action to macroscopic geometry, but passes through a well-defined collective condensate layer.

Theorem 3.1 (A pure $P(X)$ closure does not generate orientation stiffness). *Let the Euclidean action depend only on the gradient invariant $X \equiv \partial_A \Phi \partial^A \Phi$:*

$$S_E[\Phi] = \int d^4X P(X).$$

Writing the gradient in polar form $\partial_A \Phi = u n_A$, $n_A n^A = 1$, one finds $X = u^2$. Therefore $P(X)$ depends only on the modulus u but neither on the orientation n_A nor on its derivatives $\partial_B n_A$. A functional of the form $P(X)$ cannot generate a local orientation-stiffness term $\propto (\partial_B n_A)(\partial^B n^A)$.

Proof. Since $n_A n^A = 1$, one has $X = (\partial_A \Phi)(\partial^A \Phi) = u^2 n_A n^A = u^2$. The function $P(u^2)$ contains n_A only through the algebraic constraint $n_A n^A = 1$; no derivative of n_A appears. $\square$

Structural consequence. This theorem says only that the displayed bare local functional $\int P(X)$ contains no independent n_A or $(\partial n)^2$ term after $X = u^2$ is used. It does not forbid derivative operators in a separately supplied/effective orientation action, and it does not derive the P4 ordered branch or a gravitational sector. Next-to-leading derivative operators or a controlled integration over additional modes are candidate routes, whose coefficients, signs, constraint reduction and physical tensor matching remain Open. If a specific UV integration generates such an operator, dimensional analysis may involve the radial scale, but that scaling is not a consequence of this theorem. Sakharov induced gravity is therefore only a comparison programme [5], not an existing first-principles ECT loop derivation.

Two-level description of the orientation sector. There are two distinct but compatible levels of description of the orientation dynamics. At the χ -level, the integrability-reduced scalar fluctuation of the supplied broken-phase EFT propagates with the anisotropic principal tensor K^{AB} (Universality Corollary, Section 4.3). This is A-math conditional for one constrained scalar coordinate and its characteristic cone. It is not an independent n_A orientation action, a physical universal cone, or a gauge/tensor propagation theorem. At the n_A -level, however, one writes an effective action directly for the local orientation field n_A . The corresponding stiffness term $\kappa_n (\partial n)^2$ belongs to the NLO/induced sector, because n_A -gradients correspond to second derivatives of Φ . Therefore the scalar principal-symbol result and any separately supplied n_A stiffness/gravitational normalisation are logically distinct questions; no coefficient or physical bridge follows from the former.

3.3 Uniqueness of the scalar kinetic-tensor form on the supplied P4 chart

Theorem 3.2 (Uniqueness of K^{AB}). *Any symmetric rank-2 tensor that is (i) symmetric in A, B ; (ii) $O(3)$ -invariant under the residual symmetry; (iii) built locally and algebraically from $\{\delta^{AB}, n^A\}$ without additional derivatives must take the form*

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad \alpha, \beta \in \mathbb{R}. \quad (3.11)$$

Proof. A local algebraic rank-2 tensor compatible with the supplied nonzero-vector chart and its residual $O(3)$ may depend only on δ^{AB} and n^A . Symmetry in A, B excludes antisymmetric structures. Without additional derivatives or pseudo-tensors (Levi-Civita is antisymmetric and excluded), the only symmetric rank-2 tensors built from $\{\delta^{AB}, n^A\}$ are δ^{AB} and $n^A n^B$. Any such tensor is therefore a linear combination $K^{AB} = a \delta^{AB} + b n^A n^B$, which gives (3.11) with $a = \beta$, $b = -\alpha$. Full proof via $O(3)$ representation theory is in Appendix C. $\square$

This characteristic tensor owns the scalar ordered-branch cone. It does not by itself own a physical graviton or a universal matter metric; those require the completion tests in Section 8.5.

The supplied-chart object fixed by the principal tensor is the characteristic quadratic form

$$\begin{aligned} \mathcal{Q}_{2,\varphi}[\varphi] &= \int d^4X \left\{ \frac{\beta}{2} (\partial_A \varphi)^2 - \frac{\alpha}{2} (n^A \partial_A \varphi)^2 + \frac{m_{\text{eff}}^2}{2} \varphi^2 \right\}, \\ \mathcal{J}_{L,\varphi}^{(2)} &\equiv \varsigma_L \mathcal{Q}_{2,\varphi}, \quad \varsigma_L \in \{+1, -1\}, \\ S_{L,\varphi}^{(2),\text{phys}} &\equiv A_{L,\varphi} \mathcal{J}_{L,\varphi}^{(2)}, \quad 0 < A_{L,\varphi} < \infty, \quad \partial_A A_{L,\varphi} = 0, \quad \frac{\delta A_{L,\varphi}}{\delta \varphi} = 0. \end{aligned} \quad (3.12)$$

Here m_{eff}^2 is the EFT fluctuation mass parameter, not identified a priori with $V''(\phi_0)$ of the bare potential. In the printed natural-unit convention both $\mathcal{Q}_{2,\varphi}$ and $\mathcal{J}_{L,\varphi}^{(2)}$ are dimensionless. The characteristic cone fixes neither the overall Lorentzian orientation ς_L nor the positive, spacetime- and field-independent action-valued magnitude $A_{L,\varphi}$. Writing $x^0 = w$ and $\Delta_K \equiv \alpha - \beta$, direct Legendre transformation of the reduced functional gives

$$\pi_\varphi^{\text{red}} = -\varsigma_L \Delta_K \partial_0 \varphi, \quad \mathcal{H}_{L,\varphi}^{\text{red}} = -\frac{\varsigma_L}{2} [\Delta_K (\partial_0 \varphi)^2 + \beta (\nabla \varphi)^2 + m_{\text{eff}}^2 \varphi^2]. \quad (3.13)$$

The corresponding physical variables are $\Pi_\varphi^{\text{phys}} = A_{L,\varphi} \pi_\varphi^{\text{red}}$ and $\mathcal{H}_{L,\varphi}^{\text{phys}} = A_{L,\varphi} \mathcal{H}_{L,\varphi}^{\text{red}}$. On the retained branch $\beta > 0$, $\Delta_K > 0$, positive kinetic and gradient energy fixes $\varsigma_L = -1$. With $\tilde{\varphi} = \sqrt{\Delta_K} \varphi$, $\hat{c}_*^2 = \beta / \Delta_K$ and $\hat{M}_{\text{eff}}^2 = m_{\text{eff}}^2 / \Delta_K$, the reduced Lorentzian functional and Hamiltonian are

$$\mathcal{J}_{L,\varphi}^{(2)} = \int dx^0 d^3x \left[\frac{1}{2} (\partial_0 \tilde{\varphi})^2 - \frac{\hat{c}_*^2}{2} (\nabla \tilde{\varphi})^2 - \frac{\hat{M}_{\text{eff}}^2}{2} \tilde{\varphi}^2 \right], \quad (3.14)$$

$$H_{L,\varphi}^{\text{red}} = \int d^3x \left[\frac{1}{2} (\pi_{\tilde{\varphi}}^{\text{red}})^2 + \frac{\hat{c}_*^2}{2} (\nabla \tilde{\varphi})^2 + \frac{\hat{M}_{\text{eff}}^2}{2} \tilde{\varphi}^2 \right]. \quad (3.15)$$

The physical common-length generator is $H_{L,\varphi}^{\text{phys}} = A_{L,\varphi} H_{L,\varphi}^{\text{red}}$. In the aligned constant-lapse patch define $a_\tau \equiv dx^0 / d\tau = c_u / \mathcal{N}_\Phi > 0$ and $\tilde{\varphi}_\tau = a_\tau^{-1/2} \tilde{\varphi}$. The physical-clock canonical generator is then

$$H_\tau^{\text{phys}} = A_{L,\varphi} a_\tau H_{L,\varphi}^{\text{red}} = A_{L,\varphi} \int d^3x \left[\frac{1}{2} (\pi_\tau^{\text{red}})^2 + \frac{c_{\text{char}}^2}{2} (\nabla \tilde{\varphi}_\tau)^2 + \frac{M_{\text{eff}}^2}{2} \tilde{\varphi}_\tau^2 \right]. \quad (3.16)$$

The positive constants $A_{L,\varphi}$ and a_τ preserve the energy sign; a variable lapse requires a separate local canonical/constraint analysis. On a declared self-adjoint spatial domain the exact quadratic condition is $\inf \text{spec}[\beta(-\nabla^2) + m_{\text{eff}}^2] \geq 0$. Periodic, Dirichlet and Neumann realisations are standard examples once their domains are declared. A general Robin realisation carries the corresponding boundary contribution

in the variational quadratic form; its spectrum must not be paired with a volume-only Hamiltonian lacking that boundary owner. Thus $m_{\text{eff}}^2 \geq 0$ is a simple sufficient condition when $-\nabla^2 \geq 0$, and is also necessary when the spectral infimum of $-\nabla^2$ is zero, but it is not necessary on every bounded domain. The sign test is Level-A Legendre mathematics inside the supplied form; $A_{L,\varphi}$ and its relation to A_{P3} , Σ_θ , Σ_ϑ , S_0 or $\hbar$ remain Open matching inputs. The indefinite $\mathcal{Q}_{2,\varphi}$ is not a positive Euclidean statistical weight.

3.4 Eigenvalue analysis and signature classification

The supplied P4 datum defines one direction $n_A = \delta_{Aw}$, but it does not by itself determine the kinetic properties of fluctuations. The kinetic tensor K^{AB} is fixed only by the additional EFT parameters (α, β) . For different values of these parameters the ordered-chart kinetic operator can be elliptic, degenerate, or hyperbolic. We now classify all three possibilities and state the conditional inequality that defines the Lorentzian/hyperbolic class. Physical admissibility and identification with observed time require independently supplied clock, state, and matter/photon/tensor maps.

With $n_A = \delta_{Aw}$ (supplied P4 direction), the tensor (3.11) becomes diagonal:

$$K^{AB} = \text{diag}(\beta, \beta, \beta, \beta - \alpha).$$

Three signature classes:

Condition	Signature	Propagation
$\beta(\beta - \alpha) > 0$	nondegenerate elliptic	all eigenvalues have the same sign
$\beta = 0$ or $\alpha = \beta$	degenerate	at least one principal eigenvalue vanishes
$\beta(\beta - \alpha) < 0$	Lorentzian/hyperbolic	one sign is opposite to the other three

Inside the adopted P4 chart EFT, once the direction and coefficients are given, Lorentzian signature follows exactly when $\beta(\beta - \alpha) < 0$. This classification neither derives P4 nor needs to: P4 is the adopted ordered-gradient postulate. Its stationary action/state, dynamical formation and selection, and the realised coefficient values remain Open. For the rest of the scalar Hamiltonian and every later shorthand use of the ‘‘Lorentzian window’’ we restrict explicitly to the healthy transverse-kinetic branch $\beta > 0$; there the condition is $\alpha > \beta > 0$. The stronger relation $\alpha = 2\beta$ is not needed for the existence of that healthy Lorentzian branch. It is the canonical common-length unit-slope coefficient benchmark $\hat{c}_* = 1$, equivalently $\alpha = 2\beta$; it is not generated by a common field rescaling and is not a derivation of the photon speed. Separately, ECT adopts the scalar-clock calibration $c_{\text{char}} = c$ after fixing the conversion speed and lapse. Equality with the photon and tensor cones remains Open.

Signature classification, Lorentzian admissibility, and the ECT branch. On the full real coefficient plane the three regimes are determined by the product $\beta(\beta - \alpha)$, not by the ordering of α and β alone. Negative product gives a nondegenerate hyperbolic principal symbol, positive product gives a nondegenerate elliptic one, and $\beta = 0$ or $\alpha = \beta$ is degenerate. On the explicitly selected healthy branch $\beta > 0$, these reduce respectively to $\alpha > \beta > 0$, $\beta > \alpha$, and the boundary $\alpha = \beta$ (with $\beta = 0$ separately excluded). Only the first of these supplied healthy-branch cases admits the scalar hyperbolic initial-value problem. No claim about a universal physical causal sector follows without the independent clock, state and cross-sector maps.

The second conclusion concerns normalisation and physical matching. The principal symbol fixes

only the dimensionless common-length slope

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta},$$

whereas the physical scalar speed in an aligned constant-lapse patch is

$$c_{\text{char}} = \frac{c_u \hat{c}_*}{\mathcal{N}_\Phi}.$$

The adopted Level-B scalar-clock target is

$$c_{\text{char}} = c, \quad \frac{c_u}{\mathcal{N}_\Phi} = \frac{c}{\hat{c}_*}.$$

Its physical owner is distinct from the Level-B coefficient benchmark $\hat{c}_* = 1$ because c_u and $\mathcal{N}_\Phi$ are separate clock-map data. They are not algebraically independent after additionally selecting the common-unit slice $\mathcal{N}_\Phi = 1$ and $c_u = c$: on that slice $c_{\text{char}} = c$ if and only if $\hat{c}_* = 1$, and hence $\alpha = 2\beta$. Neither statement proves that the compact phase owns an unbroken photon or that matter, photon and tensor clocks share the scalar cone.

This removes the circular reading. We do not argue that $\alpha > \beta$ because Lorentzian physics is observed. Rather, the supplied-chart EFT itself classifies which parameter region is capable of supporting a Lorentzian scalar branch. The point $\alpha = 2\beta$ is selected only when the Level-B unit-slope coefficient benchmark $\hat{c}_* = 1$ is adopted.

On the explicitly declared healthy transverse-kinetic branch $\beta > 0$, the Lorentzian admissibility condition is $\alpha > \beta > 0$. On the full real coefficient plane the exact condition is $\beta(\beta - \alpha) < 0$; its second mathematical component $\beta < 0, \alpha < \beta$ has the opposite overall principal-tensor sign and is excluded by the healthy-kinetic convention used here. The stationary P4 owner, a microscopic derivation of the adopted clock/lapse calibration, the physical photon owner and cross-sector cone equality remain separate Open problems. The following section keeps the exact hyperbolic component separate from an optional arithmetic isotropic-average benchmark. The latter is neither a homogenisation theorem nor a physical upper bound on α .

3.5 Healthy scalar Lorentzian component and optional arithmetic average

Variation of any finite nonzero constant multiple of the supplied characteristic form gives the same homogeneous equation; in standard Klein–Gordon sign orientation it is

$$(\alpha - \beta) \partial_w^2 \varphi - \beta \partial_t^2 \varphi + m_{\text{eff}}^2 \varphi = 0. \quad (3.17)$$

Dispersion relation:

$$(\alpha - \beta) k_w^2 = \beta \mathbf{k}^2 + m_{\text{eff}}^2. \quad (3.18)$$

The effective propagation speed is:

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta}. \quad (3.19)$$

The healthy scalar Lorentzian component, including the separately supplied transverse-kinetic sign, is

$$0 < \beta < \alpha. \quad (3.20)$$

Its logical ingredients must be kept distinct:

- the exact full-plane hyperbolicity criterion is $\beta(\beta - \alpha) < 0$ (Level A mathematics inside the supplied scalar EFT);
- after the healthy transverse branch $\beta > 0$ is supplied, that criterion reduces to the lower condition $\alpha > \beta > 0$;

- the optional additional condition $\alpha < 4\beta$ assumes a declared ensemble/cell, scale ℓ_{iso} , arithmetic isotropic orientation average, and positivity of that averaged coefficient form. It is derived only as that benchmark in Appendix D; it is not a condition for Lorentzian signature or a derived homogenised operator.

P4 supplies none of these coefficient signs or numerical values.

Physical speed matching and its present status. At Level A conditional mathematics, the supplied second-order reduced scalar operator has one principal cone (conditional one-operator corollary, §4.3). Integrability fixes one scalar coordinate but does not by itself fix the derivative order or number of characteristic factors. The physical value additionally requires c_u , the clock lapse and a physical sector-specific action. In particular, a compact phase does not by itself derive a local gauge redundancy, an unbroken massless photon, or its kinetic tensor. Thus $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration, whereas $c_\gamma \stackrel{?}{=} c_{\text{char}}$ and the tensor-cone comparison remain Open. The GW170817/GRB 170817A interval is an external kill gate for any completed photon–tensor metric, not evidence that the present scalar cone already owns the gravitational-wave speed [12]. The coefficient choice $(\beta, \alpha) = (1, 2)$ realises the separate Level-B unit-slope benchmark $\hat{c}_* = 1$; a common scalar field rescaling does not select this ratio.

Photon-sector Lorentz-violation status. The supplied scalar/director structure does not derive a physical photon action or higher-dimension propagation operator. An Open parametrisation template for a future named photon completion may, only after its gauge-invariant operator, derivative order, CPT/parity and birefringence class are fixed, be written schematically as

$$\omega^2 = k^2 \left[1 + \xi_n \left(\frac{k}{\Lambda_\gamma} \right)^n + \dots \right], \quad n \in \mathbb{N}, \quad \Lambda_\gamma > 0, \quad \frac{k}{\Lambda_\gamma} \ll 1. \quad (3.21)$$

The branch domain must also keep $\omega^2 \geq 0$. Here ξ_n , Λ_γ , the helicity/birefringence assignment, photon clock, source-lag model and detector likelihood are supplied completion data. Until the operator is named this equation is a template, not a defined model. No unit coefficient, numerical delay or ECT interpretation of high-energy timing data follows. Fermi-LAT and other propagation tests are external gates on such a named completion (Appendix E).

Emergent special relativity. The possibility that an anisotropic kinetic structure on a Euclidean background can yield effective Lorentzian propagation has also been demonstrated in closely related constructions. Girelli, Liberati and Sindoni [3] showed this in a scalar-gradient setting with scalar gravity, and Mukohyama and Uzan [4] later developed a classical clock-field model in which scalar, vector, and spinor sectors propagate in an effective Minkowski metric. In that sense, ECT is not unique in proposing emergent Lorentzian dynamics from a Euclidean substrate. Its distinctive claim is instead that the preferred direction is supplied by the P4 $O(4) \rightarrow O(3)$ ordered condensate-gradient branch, rather than imposed as a fixed coordinate of the Euclidean arena. Its dynamical formation from P1–P3 is Open.

Given P4, the supplied broken-phase coefficients and the displayed second-order scalar principal operator, one derives its Lorentzian classification and one scalar cone; this does not yet derive physical special relativity for all rods, clocks and fields. Residual $O(3)$ symmetry restricts the scalar principal tensor to K^{AB} (Theorem 3.2). In the homogeneous ordered phase, the scalar Lorentzian branch may be parametrised by the ordered-branch common-length coordinate $x^0 = w$, so that the scalar characteristic metric reduces locally to the Minkowski form $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ at $\hat{c}_* = 1$, and the isometry group is the Poincaré group. (The Wick/Picard–Lefschetz continuation belongs to the path-integral justification discussed elsewhere and should not be conflated with this common-length scalar parametrisation.) The displayed second-order scalar operator possesses one causal cone (conditional one-operator corollary, §4.3). The phase-sector continuation is controlled by the same ordered-phase tensor (§4.7). Gauge and fermionic sectors inherit the same causal structure only within their respective reconstruction routes (Level B,

§4.7, §9). Any higher-order Lorentz-violating corrections would have to be matched and sufficiently suppressed; neither their operator, suppression scale nor coefficients follows from the current condensate action. Existing observations constrain such deviations very strongly [12, 13, 14, 15]. Thus Minkowski-form scalar kinematics is recovered conditionally as the homogeneous infrared limit of this supplied second-order ordered-branch operator; physical special relativity for rods, clocks and all fields remains an Open matching problem.

3.6 Effective Lorentzian metric

At this stage Lorentzian signature has already appeared in the ordered-branch kinetic tensor. We write the ordered common-length coordinate as

$$x^0 \equiv w, \quad \hat{c}_*^2 \equiv \frac{\beta}{\alpha - \beta}. \quad (3.22)$$

Thus x^0 has the same length dimension as (x, y, z) and $\hat{c}_*$ is a dimensionless characteristic slope. Neither quantity is by itself a reading in seconds. A physical clock requires a conversion speed c_u , a matter congruence and the state-dependent lapse $\mathcal{N}_\Phi$. In an aligned constant-lapse patch,

$$d\tau = \frac{\mathcal{N}_\Phi}{c_u} dx^0, \quad c_{\text{char}} = \frac{c_u \hat{c}_*}{\mathcal{N}_\Phi}. \quad (3.23)$$

Only the aligned constant-lapse choice $\mathcal{N}_\Phi = 1$, $c_u = c$ gives $\tau = w/c$. The additional, logically separate scalar-cone choice $\hat{c}_* = 1$ gives $c_{\text{char}} = c$ and still requires photon matching. This convention contains no assumption about a spherical boundary; the general congruence integral is given in Eq. (16.14).

Equation (3.22) is the step used to write the hyperbolic field equation. A separate formal Euclidean–Lorentzian continuation belongs only to the amplitude/path-integral language discussed later in the coherent sector. Restoring the physical clock by Eq. (3.23) gives, in a locally constant patch,

$$\begin{aligned} \partial_\tau^2 \varphi - c_{\text{char}}^2 \nabla^2 \varphi + M_{\text{eff}}^2 \varphi &= 0, \\ M_{\text{eff}}^2 &= \frac{c_u^2}{\mathcal{N}_\Phi^2} \hat{M}_{\text{eff}}^2 = \frac{m_{\text{eff}}^2 c_{\text{char}}^2}{\beta}. \end{aligned} \quad (3.24)$$

Fluctuations see the effective Lorentzian metric:

$$g_{\mu\nu}^{\text{eff}} dx^\mu dx^\nu = -c_{\text{char}}^2 d\tau^2 + \delta_{ij} dx^i dx^j. \quad (3.25)$$

In common-length coordinates the equivalent local form is $ds_{\text{eff}}^2 = -\hat{c}_*^2 (dx^0)^2 + d\mathbf{x}^2$. At $(\alpha, \beta) = (2, 1)$ one has $\hat{c}_* = 1$; the further matched special case above gives $c_{\text{char}} = c$.

Free-sector consistency. For supplied $\alpha > \beta > 0$, the cone and dispersion follow from the characteristic form independently of its overall sign. The positive-energy Hamiltonian (3.15) additionally uses the explicit orientation $\varsigma_L = -1$ and a positive action magnitude $A_{L,\varphi}$. On the declared self-adjoint domain its exact spectral condition is $\text{inf spec}[\beta(-\nabla^2) + m_{\text{eff}}^2] \geq 0$; $m_{\text{eff}}^2 \geq 0$ is a simple sufficient condition when $-\nabla^2 \geq 0$. Those mass/domain inputs, the relative sign and the action-valued magnitude are not fixed by the principal tensor. This is *not* yet a full OS-reconstruction of the interacting theory, but it is a necessary internal consistency check for the supplied free Lorentzian sector.

3.7 Physical picture and its present scope

The mathematical results admit a concrete physical interpretation.

A supplied direction as a prerequisite, not a birth theorem. P4 supplies the distinguished-direction datum needed by every later temporal construction; it does not derive an $O(4) \rightarrow O(3)$ transition. In the symmetric phase all four directions of $\mathcal{M}^4$ are equivalent, so P1–P3 alone select neither a temporal orientation nor a preferred scalar principal direction. After P4 is supplied, one direction and an $O(3)$ stabiliser exist. A scalar causal cone becomes definable only after the additional coefficient condition $\beta(\beta - \alpha) < 0$; physical clocks, rods, matter/photon/tensor cones, particles and an entropy arrow require their own actions, states and observable maps. Thus ordering is a necessary architectural prerequisite for the proposed physical world, not by itself a derivation of causality, time, particles or geometry.

What condenses. In superfluid helium-II below the Bose–Einstein condensation temperature atoms enter a collective ordered state; the condensate carries all characteristic properties of superfluidity [10]. In ECT the supplied analogue is the ordered-gradient datum $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$. It defines a conditional EFT chart rather than proving that Φ dynamically freezes into a physical state. It is also not a condensate in the strict quantum-mechanical sense (QM has not yet been derived at this stage).

Origin of time. In the isotropic reference description ($u_0 = 0$) all four directions of $\mathcal{M}^4$ are equivalent. On the supplied P4 chart one direction (w) is distinguished. This makes w a candidate for the role of “time”. By itself, however, supplying a direction does not yet yield causality or the arrow of time—those require the kinetic operator to be hyperbolic ($\alpha > \beta > 0$ on the retained component) and a declared initial-boundary problem. P4 supplies the ordered chart and BR1 a separate smooth coherent-sector closure; neither selects a physical branch.

Condensed-matter analogy. In a superconductor the condensate gives electromagnetic modes a Meissner mass and defines a material rest frame [16]. This is an analogy for ordered-medium response, not an example in which fundamental Lorentz symmetry is spontaneously broken. On the supplied ECT chart, the preferred direction w similarly permits anisotropic response, but its physical formation and the response action remain Open. Note that in the scalar-only ECT basis, the orientation variable n_A associated with the supplied nonzero-gradient chart is not an independent field but a derived collective variable, subject to integrability constraints (Section 4.3).

Problem of time in ECT. ECT does not introduce external physical time at the P1 level. P4 supplies a preferred Euclidean direction, but this direction is only a candidate clock coordinate. A scalar temporal classification requires independently supplied $\beta(\beta - \alpha) < 0$ (equivalently $\alpha > \beta > 0$ on the retained $\beta > 0$ component); a physical time observable additionally requires a realised branch, clock/lapse and matter/metric map. Classical temporality and an arrow of time require still further state, record and entropy owners. ECT therefore organises a route toward the problem of time rather than deriving its complete solution [17, 18]; the single-outcome and thermodynamic-arrow problems remain Open.

3.8 Conditional temporal orientation and the Open arrow-of-time programme

The stabiliser and abstract orbit of the P4-supplied vector follow as Level-A conditional mathematics. Interpreting that datum as a physical temporal orientation is a separate Open step. In a supplied $\beta(\beta - \alpha) < 0$ scalar model with a chosen initial-boundary orientation, standard hyperbolic theory supplies a retarded scalar problem. A physical clock/lapse, universal cone and thermodynamic second law require independent owners.

Physical picture: what is and is not derived. P4 supplies the direction and residual $O(3)$ stabiliser. The conditional scalar model supplies retarded support only after its coefficients and orientation are declared. Neither fact proves low initial entropy, monotonic entropy production or physical time. The thermodynamic arrow is an Open ECT programme requiring a reduced map, state, entropy functional and appropriate contractivity or detailed-balance hypotheses; see Section 3.9.

Starting point: no external time. ECT assumes no external physical time at the foundational level. The present section therefore records a hierarchy of completion requirements rather than taking an emergent physical Lorentzian structure as established.

Result A1: geometry of the supplied direction. Before ordering, the Euclidean Φ -medium possesses no distinguished direction: all four directions of $\mathcal{M}^4$ are equivalent. After the nonzero P4 datum is supplied, its vector defines a preferred direction w through $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw} \neq 0$, providing the directional input for the conditional scalar-signature classification (Section 3.6). The asymmetry is not explicit in P3; its physical formation and selection remain Open. Given P4, the geometry of the supplied preferred direction and its residual $O(3)$ stabiliser follow at Level A. P1–P3 do not derive that ordered gradient branch, its amplitude or its state; the OP-grad dynamical owner remains Open.

Result A2: conditional scalar Lorentzian signature. The effective principal tensor for perturbations on the supplied chart,

$$G_{\text{eff}}^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad (3.26)$$

has Lorentzian signature when $\alpha > \beta > 0$ (Level A mathematics conditional on the supplied EFT coefficients; Section 3.4). On the retained component $\beta > 0$, once the supplied coefficients satisfy $\alpha > \beta > 0$, the distinguished coordinate w is the negative characteristic direction of this scalar tensor. The common-length parametrisation $x^0 = w$, with

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta}.$$

This does not yet define a clock reading in seconds. The realised physical branch, time orientation, lapse and matter/photon/tensor metric remain Open. Only after those owners are supplied may the negative scalar characteristic direction be interpreted as physical time.

Result A3: retarded support in the supplied scalar problem. The supplied constant-coefficient scalar EFT admits the explicit symmetric-hyperbolic reduction of Appendix G. Equivalently, its linear retarded fundamental solution obeys

$$\boxed{G_{\text{ret}}(t, \mathbf{x}) = 0 \quad \text{for } t < 0.} \quad (3.27)$$

For the declared orientation and retarded boundary prescription, scalar perturbations have no support before the chosen scalar-time origin. This is Level A operator mathematics inside the supplied quadratic model. A separately supplied quasilinear $P(X)$ completion has the exact nondegenerate sign test $P_X(P_X + 2XP_{XX}) < 0$. The retained component that matches $\beta > 0, \alpha > \beta$ is $P_X > 0$ and $P_X + 2XP_{XX} < 0$; the opposite-sign component is Lorentzian as well. Local nonlinear domain of dependence further requires uniform hyperbolicity, an owned timelike foliation and regular initial–boundary data. A nonlinear Green function is not defined. Neither statement selects the physical arrow, clock, universal cone or thermodynamic orientation.

Result A4: passive linear response in a declared source model. Let $\mathcal{O} = \mathcal{O}^\dagger$ be a named physical operator, let the drive enter as $H_{\text{drive}}(t) = +j(t)\mathcal{O}$, and define

$$D_{\mathcal{O}\mathcal{O}}^R(t) := -\frac{i}{S_0} \theta(t) \langle [\mathcal{O}(t), \mathcal{O}(0)] \rangle. \quad (3.28)$$

With the Fourier convention $\tilde{D}(\omega) = \int dt e^{+i\omega t} D(t)$, a stationary passive state has $\text{Im} D_{\mathcal{O}\mathcal{O}}^R(\omega) \leq 0$ for $\omega > 0$. For $j(t) = \text{Re}[j_\omega e^{-i\omega t}]$, the cycle-averaged work put into the system is therefore

$$\bar{P}_{\text{abs}}(\omega) = -\frac{\omega}{2} \text{Im} D_{\mathcal{O}\mathcal{O}}^R(\omega) |j_\omega|^2 \geq 0. \quad (3.29)$$

The factor $1/2$ is absent if $|j|^2$ is defined as an RMS rather than peak phasor. This D^R is not the equation-normalised scalar fundamental solution G_{ret} of Result A3. Retarded support alone fixes neither the absorptive sign nor the response magnitude: the operator, source vertex, state, action normalisation and Fourier/source convention are premises. Identifying this passive kernel in ECT remains Open.

Result A5: positive semidefinite pairwise noise quadratic. Fix a named joint initial preparation and closed-time-path contour, and centre the unresolved-sector one-point function or absorb its contribution into the declared system force. For a compatible factorised preparation, or for a correlated preparation whose initial-boundary contribution has been retained explicitly and separated from the following bulk term, let $d = q - q'$ denote the physical record difference and let $N = N^\dagger \succeq 0$ be the action-normalised bulk time-domain noise kernel. The bulk Gaussian Feynman–Vernon functional may be written as

$$\mathcal{F}_{\text{bulk}}[q, q'] = \exp \left\{ \frac{i}{S_0} \Gamma_R[q, q'] - \frac{1}{S_0} \Gamma_{\text{irr}}[d] \right\}. \quad (3.30)$$

The associated non-negative functional and dimensionless exponent are

$$\Gamma_{\text{irr}}[d] := \frac{1}{2} \langle d, Nd \rangle \geq 0, \quad \Phi_{qq'}^{\text{bulk}} = \frac{\Gamma_{\text{irr}}[d]}{S_0}, \quad \Gamma_{\text{irr}}[0] = 0. \quad (3.31)$$

Here Γ_{irr} has units of action and $\Phi_{qq'}^{\text{bulk}}$ is the dimensionless nonnegative exponent of the declared bulk Gaussian functional. It must not be identified with the full $\Phi_{qq'}$ unless the complete preparation-dependent construction is also shown to be contractive. Therefore

$$|\mathcal{F}_{\text{bulk}}[q, q']| \leq 1, \quad |\mathcal{F}_{\text{bulk}}[q, q']| = 1 \iff N^{1/2}d = 0 \quad \text{almost everywhere.} \quad (3.32)$$

Thus $q = q'$ is sufficient but not necessary for equality: an unmonitored or decoherence-free difference can lie in the null space of N . Distinct histories are suppressed only to the extent that their physical record difference has support outside that null space. The spectral kernel $S_H = S_0 N^{(S)}$ and the finite-window transform are fixed explicitly in Eqs. (29.8)–(29.10). This is Level A algebra in the declared quadratic source model; the ECT split, record operator, state, and kernel remain Open. For a correlated joint preparation the full effective action may also contain an initial-boundary term S_{ic} , or its equivalent on an extended contour. The full $-\ln |\mathcal{F}|$ can therefore receive preparation-dependent contributions not contained in $\Gamma_{\text{irr}}[d]$. Result A5 proves positivity of the displayed bulk noise quadratic; it does not prove that initial-correlation terms are absent.

What A5 does and does not establish. Positive semidefiniteness prevents amplification by the modulus of the declared bulk Gaussian influence functional. It does not imply that every pair of mathematically distinct histories is distinguishable to the environment, that the dephasing exponent grows monotonically with elapsed time, or that a von Neumann/coarse-grained entropy is monotone. Decoherence-free directions and finite-time recoherence are permitted. An entropy theorem additionally requires the reduced channel, state, entropy functional, and hypotheses such as unitality, detailed balance, or contractivity; Markovianity alone is insufficient (Section 3.9).

Status of the causal kernel. Retarded support $K(\tau < 0) = 0$ is established at Level A for the supplied quadratic scalar operator (Appendices F and G). The same appendix gives the exact principal tensor and conditional local hyperbolicity gate for a separately supplied $P(X)$ completion; it does not establish a full nonlinear ECT kernel. What is not yet derived is the full first-principles identification of the open-system response kernel entering the influence functional directly from Euclidean microdynamics, together with the auxiliary bath closures (ohmicity, Markov limit, environment modelling) used in the macroscopic irreversibility estimates. Results A4 and A5 are therefore conditional on those bath-closure assumptions, not on the retarded-support property itself.

Result A6: branch selection as the basis for a low-entropy initial sector. Postulate P4 adopts the physically relevant ordered-gradient branch $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$ as a Level-B structural input; it neither selects a unique realised configuration nor supplies an ensemble measure or initial state. Calling that branch low entropy additionally requires macrovariables, coarse-graining, state counting and a preparation/selection law, none of which is presently derived. The low-entropy interpretation is therefore a Level-C/Open programme, not an alternative derivation of the Past Hypothesis [19].

Combined statement. P4 supplies a preferred Euclidean direction, not a physical clock. For an independently supplied scalar quadratic operator with $\beta(\beta - \alpha) < 0$ (equivalently $\alpha > \beta > 0$ on the retained $\beta > 0$ component), A1–A3 give one hyperbolic scalar characteristic direction and retarded support only after an initial–boundary orientation is supplied. This conditional result does not derive a universal observable metric, physical time or a thermodynamic arrow. In a declared passive quadratic response model, A4 gives nonnegative cycle-averaged absorbed power; in a declared Gaussian record channel, A5 gives a positive-semidefinite pairwise noise quadratic. Their identification with physical ECT open-system kernels and states remains Open. The low-entropy reading of a selected ordered realisation (A6) is a Level-C/Open programme rather than a consequence of A1–A5.

- (a) P4 supplies a preferred Euclidean direction; a scalar temporal parametrisation additionally requires the stated hyperbolic operator and a physical clock/lapse map remains Open (A1+A2);
- (b) retarded scalar support follows only for that supplied operator after an initial–boundary orientation; it is not a forward-only theorem for all Φ -medium physics (A3);
- (c) a declared passive stationary quadratic model absorbs nonnegative cycle-averaged power from the named drive (A4);
- (d) a declared positive Gaussian noise kernel does not amplify record differences and suppresses only components outside its null space (A5);
- (e) selecting one ordered realisation as a low-entropy sector requires an ensemble, macrovariables, coarse-graining and a preparation law (A6).

Why this is not yet the full second law. The above results establish directionality but not strict monotonicity of entropy. For a generic derivative-coupled super-ohmic bath ($J_{\text{bare}}(\omega) \propto \omega^4$, hence $J_W(\omega) \propto \omega^3$ in the declared Weiss convention; Appendix I), temporary partial recoherences are not excluded. The step to strict monotonicity requires additional closure assumptions (Section 3.9).

Compatibility with variational dynamics. There is no contradiction between emergent irreversibility and the variational principle of the underlying medium. At the fundamental level, classical solutions of the Φ -medium remain extremals of $S_E[\Phi]$ (Section 6.6). A variational microscopic model is compatible with irreversible reduced dynamics only after a system/environment split, coarse-graining map, state, time orientation and reduced channel are supplied. Coarse graining alone does not establish irreversibility or entropy growth in ECT; monotonicity requires additional channel hypotheses, and recoherence or entropy backflow is not excluded by the present architecture. An ECT-native reduced channel and arrow theorem therefore remain Open.

Comparison with other approaches. *Penrose-type low-entropy cosmology* [19]: The ECT programme is compatible with using a physically realised ordered state as a candidate low-entropy sector. This is a Level-C/Open interpretation: neither P4 nor A6 derives the state, its entropy, its preparation, or a transition tying low entropy to condensate ordering.

Spontaneous symmetry breaking in condensed matter [20]: The supplied P4 chart motivates the Level-C/Open programme that an owner-complete ordering process might participate in the origin of

effective spacetime structure. The analogy does not derive a stationary ordered state, spontaneous selection, physical clock, or common metric.

Cosmological arrow without external time [21]: P4 supplies a Euclidean orientation. A temporal orientation exists only for an independently supplied hyperbolic scalar operator together with an initial–boundary choice; a physical clock/lapse and thermodynamic arrow remain Open. A declared pairwise record channel can suppress specified coherences, but A5 alone is neither directional in the thermodynamic sense nor an entropy theorem.

Caldeira–Leggett open-system theory [22]: Ohmic and Markov limits can yield a local reduced generator in a declared source model, but do not by themselves prove strict entropy growth. The positive quadratic Γ_{irr} used here is meaningful for any nonnegative declared spectral density; converting it into an entropy theorem still requires the channel, state, entropy functional, and unitality/detailed-balance/contractivity hypotheses.

Status summary.

Statement	Status	Comment
A1: Supplied nonzero P4 vector defines w and has stabiliser $O(3)$	Level A conditional kinematics / physical branch Open	Exact stabiliser result inside the supplied P4 datum; no dynamical vacuum selection
A2: Supplied scalar principal tensor is Lorentzian iff $\beta(\beta - \alpha) < 0$	A-math conditional / physical branch Open	Healthy shorthand $\alpha > \beta > 0$; realised time and metric require separate maps; §3.6
A3: Retarded propagation of Φ , $G_{\text{ret}}(t < 0) = 0$	A-math in the supplied hyperbolic model / physical branch Open	Leray–Choquet–Bruhat; App. G
A4: Passive linear-response absorption	A in declared source model / ECT Open	Requires $H_{\text{drive}} = +j\mathcal{O}$, $D^R = -(i/S_0)\theta\langle[\mathcal{O}, \mathcal{O}]\rangle$, the $e^{+i\omega t}$ Fourier convention, a stationary passive state and the peak/RMS phasor convention; this D^R is not the scalar G_{ret} ; Eq. (3.29)
A5: Positive semidefinite pairwise noise quadratic	A in declared Gaussian source model / ECT Open	Equality includes the noise-kernel null space; ECT record operator, state, and kernel Open; App. H
A6: Physical realisation and selection of a single ordered branch	Open	Requires a stationary action, state, stability domain and dynamical selection mechanism
Low-entropy interpretation of initial branch	Level C/Open	Requires explicit owners for the macrovariables, coarse-graining, state-counting and preparation/selection law; none is presently derived

Statement	Status	Comment
Retarded structure of ordered Φ -branch	A in supplied quadratic operator / conditional $P(X)$ gate	Physical clock, nonlinear action, foliation and boundary owner Open; App. G
Open-system bath-kernel identification from Euclidean microdynamics	C (open)	OP1, OP5
Strict entropy monotonicity for an ECT reduced channel	Open	Markovianity is insufficient; declare the map, state, entropy functional, and unitality/detailed-balance/contractivity hypotheses; §3.9
Full microscopic H-theorem	C (open)	OP1, OP5, OP-Q17

3.9 Reduced-state entropy production: conditional programme

Status: Level A for retarded support of the supplied quadratic scalar operator and positivity of the declared quadratic noise kernel; Open for an ECT-specific entropy-production theorem. Markovianity and a positive noise kernel alone do not imply monotone von Neumann entropy.

For reference, an effective ohmic spectral ansatz and a short-memory condition may be written

$$J(\omega) \simeq N_{\text{eff}} \gamma \omega, \quad (3.33)$$

$$\tau_{\text{env}} \ll \tau_{\text{dyn}}. \quad (3.34)$$

These are useful model assumptions, but they are not by themselves an entropy theorem [22].

The second law as a Level B extension of the arrow of time. *Status:* Open for ECT. Positivity of a declared pairwise noise quadratic gives dephasing in the monitored directions, not a second law. A thermodynamic statement additionally requires a physical reduced map, state, entropy functional, and contractivity or detailed balance. The ordered branch supplies a direction for causal response, but a thermodynamic second law additionally requires a declared reduced channel, coarse graining, entropy functional, initial state, and conditions such as unitality, detailed balance, or contractivity relative to a stationary state. For example, a unital finite-dimensional dynamical semigroup makes the von Neumann entropy non-decreasing, whereas an amplitude-damping channel may purify a state. Thus the earlier scalar equality $\dot{S}_{\text{ent}} = N_{\text{eff}} \gamma (\dot{q})^2$ is not a consequence of the Gaussian–Markov influence functional.

The Boltzmann analogy. The Boltzmann analogy is limited to a separation between reversible microscopic dynamics and a reduced macroscopic description. It does not supply molecular chaos, a Past Hypothesis, or monotone entropy for ECT; those ingredients must be stated and tested independently.

Statistical speciality of the ordered branch (Level B). Selection of one ordered orientation is not a proof that its statistical state has low entropy. Low-entropy initial data are therefore an additional cosmological/state question, not a consequence of P4 or of a single realised branch.

Entropy growth from inhomogeneous ordering (Level B). Inhomogeneous ordering and domain formation can enlarge the number of coarse-grained configurations, but this kinematic multiplicity has no model-independent sign for an entropy rate. Its conversion into entropy production remains conditional on the specified dynamics and coarse graining.

Conditional defect and domain production programme (Level B/Open). If a wider physical configuration space, action and inhomogeneous ordered state are supplied, defect and domain configurations may become legitimate excitations. Their existence, abundances, lifetimes and contribution to any entropy budget must be computed from the owning field equations and state; mode counting alone is neither an existence proof nor an entropy theorem.

Two Level B closure assumptions. Two logically independent closure inputs are required: a derived system–environment/reduced-state map and a declared thermodynamic contractivity condition. Neither follows from the existence of many condensate modes.

Conditional entropy theorem. The applicable statement is the data-processing inequality $dD(\rho_w||\rho_*)/dw \leq 0$ for a named contractive channel with stationary ρ_* ; whether ECT supplies such a channel is Open. A valid conditional statement may be recorded as

$$\frac{d}{dw}D(\rho_w||\rho_*) \leq 0 \quad \text{for a declared contractive channel with stationary } \rho_*, \quad (3.35)$$

where D is quantum relative entropy. This is standard open-system mathematics; its ECT application remains Open until the physical reduced map is derived.

Statistical suppression of time-reversed trajectories. Suppression of a time-reversed trajectory is meaningful only for normalised forward and reverse protocols with a specified reversal map. Pairwise loss of interference is not a trajectory-probability ratio and cannot replace microscopic reversibility or local detailed balance.

Three regimes. Three regimes must be kept distinct: causal retarded propagation, pairwise reduced-state record formation, and thermodynamic irreversibility. ECT has structural owners for the first and conditional Gaussian models for the second; the third remains Open.

What remains open. The open inputs are the physical resolved/unresolved split, record operators, state-dependent kernels, reduced map, entropy functional, and forward/reverse protocols. Until they are derived, no universal ECT second law or fluctuation theorem is claimed.

Status-separated arrow/entropy inventory.

Statement	Status	Scope
Causal retarded response	Level A in owning sectors	Does not determine entropy sign
Positive pairwise noise quadratic	Level A in declared Gaus- sian kernel	Suppresses selected coherences
Ohmic/Markov assumptions	Model inputs	Insufficient for monotone entropy by themselves
ECT monotone-entropy theorem	Open	Reduced map, state, entropy func- tional and contractivity required
ECT Crooks relation	Open	Forward/reverse protocols and local detailed balance required

Likewise, a reverse/forward comparison exists only after protocols and path measures are defined on compatible measurable spaces. Let $(\Omega, \mathcal{F})$ be the common forward path space and let $\Theta\gamma = \tilde{\gamma}$ be a

measurable time-reversal involution. Transport the normalised reverse measure to this space by

$$Q(A) \equiv P_R^\Theta(A) := P_R(\Theta A), \quad A \in \mathcal{F}.$$

If $P_F \ll Q$, the Radon–Nikodym theorem permits the definition

$$\Sigma[\gamma] := \ln \left(\frac{dP_F}{dQ}(\gamma) \right), \quad \frac{dQ}{dP_F}(\gamma) = e^{-\Sigma[\gamma]} \quad \text{when } P_F \sim Q. \quad (3.36)$$

The first definition is P_F -almost sure and requires only $P_F \ll Q$. The displayed reciprocal, and consequently $\langle e^{-\Sigma} \rangle_F = 1$, additionally require $Q \ll P_F$. If the transported reverse measure has a component singular with respect to P_F , the unit identity is replaced by the corresponding absolute-irreversibility correction. A pointwise shorthand $\ln(P_F[\gamma]/P_R[\tilde{\gamma}])$ is legitimate only when common densities with respect to a declared dominating measure have been supplied. A Crooks theorem still requires microscopic reversibility or local detailed balance and the appropriate initial ensembles; it does not follow from this measure-theoretic definition or from a positive decoherence exponent. Deriving the ECT reduced channel, entropy functional, protocols and path measures remains OP-Q17.

3.10 Comparison with other theories

Einstein–aether theory [23]. In Einstein–aether theory the anisotropy is introduced by a postulated aether vector field u^A ; it does not arise spontaneously. In ECT the orientation variable n_A is a collective variable of the supplied nonzero-gradient chart—it enters the kinetic EFT tensor, not the bare action P3. P4 supplies rather than dynamically selects that direction; a stationary state and spontaneous selection, the coefficient sign, and a physical Lorentzian metric remain Open. A systematic comparison of emergent-gravity approaches is given in Section 8.5.

Hořava–Lifshitz gravity. This theory starts from Lorentzian spacetime and introduces explicit high-energy anisotropy between space and time. ECT instead starts from Euclidean $O(4)$ invariance and P4 supplies a nonzero-vector direction whose stabiliser is $O(3)$; a physical ordered state and spontaneous selection are Open, and no pre-existing Lorentz symmetry is being dynamically broken here. Lorentzian/hyperbolic signature is the conditional scalar classification for supplied $\beta(\beta - \alpha) < 0$, equivalently $\alpha > \beta > 0$ on the retained $\beta > 0$ component.

CDT (Causal Dynamical Triangulations) [24]. In CDT the Lorentzian structure is imposed as a constraint on the lattice geometry. In ECT the scalar characteristic tensor is conditional on the supplied ordered-background EFT; a physical spacetime metric remains Open.

Hartle–Hawking no-boundary proposal [7]. The no-boundary proposal defines a candidate cosmological state through a path integral (or semiclassical saddles) over regular compact positive-definite or complex four-geometries. It does not posit ECT’s Euclidean scalar medium.

Summary. Residual $O(3)$ symmetry fixes the form of the supplied scalar principal tensor. For supplied P4 background and EFT coefficients it is Lorentzian iff $\beta(\beta - \alpha) < 0$; on the declared healthy $\beta > 0$ branch this is $\alpha > \beta > 0$; this classification is Level-A mathematics inside that conditional model. P4 adopts the ordered-gradient sector but does not fix the sign of $\beta - \alpha$, supply its stationary action/state or dynamical selection, or derive a physical clock, universal metric or common photon/matter/tensor cone; those constructive owners remain Open. The arrow-of-time programme likewise requires the independently declared record dynamics of Section 26.1. Unlike Einstein–aether [23] and Horava–Lifshitz [25], ECT does not begin with a Lorentzian metric plus a preferred vector or foliation. P4 adopts the physically relevant ordered-gradient sector and its Euclidean direction; the stationary/dynamical owner and physical metric remain completion targets. Unlike CDT, that metric is not a lattice input. The ECT Euclidean-medium ontology is a separate postulate, not a consequence of the Hartle–Hawking construction.

Because the ordered quantity $Q_A = \partial_A \Phi$ is an exact one-form rather than an independent multiplet, the orientation sector is subject to an integrability constraint that reduces the naive $O(4)/O(3)$ Goldstone content (Appendix J). Within the stated model this differs from treating the order parameter as an unconstrained independent multiplet; no literature-wide uniqueness claim follows.

Uniqueness of the scalar characteristic cone. The integrability constraint has a direct consequence for the uniqueness within the stated linear scalar subsector. In Einstein–aether theory [23, 26], the aether vector is an independent dynamical field with four free kinetic coefficients; different spin sectors depend on different coefficient combinations, and equality of their cones imposes model-dependent constraints. In the SME [27, 28], each sector can carry sector-dependent Lorentz-violation parameters. In the He-3 analogy [20], different quasiparticle species can see different cones. In the supplied scalar truncation, the fundamental field is one (Φ), the principal tensor is one (K^{AB}), and integrability expresses all soft-coordinate perturbations of the strict scalar map as derivatives of one scalar χ . There are no free sector-dependent Lorentz-violation coefficients inside that one displayed scalar operator (Universality Corollary, §4.3). This is a theorem only for the leading linear scalar soft sector of the supplied model. Gauge, fermion, matter and tensor sectors require independent actions and principal symbols; equality with the scalar cone is an Open physical matching test, not a reconstruction mandate. A detailed comparison of speed structures is given in Appendix K.

Forward connection to the Principle of Euclidean Stationarity. The signature analysis of the broken-phase EFT provides the structural precondition for the response–persistence formulation introduced in Section 29 of Part III. PES-R neither determines the full sign choice $\beta(\beta - \alpha) < 0$ (or $\alpha > \beta > 0$ on the retained $\beta > 0$ component) nor supplies the clock/photon matching needed for $c_{\text{char}} = c$. Once the Lorentzian ordered branch and its physical modes have been established by their owning equations, PES-R can only classify their finite-time retention and pairwise record distinguishability for a declared protocol. It does not impose phase compactness on sectors that do not possess it, minimise one universal environmental scalar, or select realised outcomes. The $\alpha \rightarrow \beta$ singular boundary remains a property of the kinetic tensor; no additional PES-R transition law is inferred from it.

4 Spectrum and Effective Dynamics of the Broken Phase

This section organises the excitation spectrum of the broken phase: amplitude, orientation, and macroscopic branches; the integrability constraint on the soft sector; and the resulting hierarchy of effective modes. The scalar-coordinate reduction is Level A kinematics, whereas the interpretation of that coordinate as one physical propagating mode is Level B inside a supplied reduced EFT. The physical χ^2 operator, its stationary-action owner, its sign and magnitude, and any Ward or protection mechanism remain Open.

Given the supplied P4 gradient background and the independently supplied BR1 coherent-sector hypothesis, one may classify candidate coordinates and conditional excitations about that EFT chart. Neither input selects a stationary physical vacuum or determines its spectrum.

4.1 Polar decomposition and collective variables

An affine representative of the supplied P4 gradient datum is

$$\Phi_{\text{bg}}(X) = u_0 w + \Phi_*, \quad \partial_A \Phi_{\text{bg}} = u_0 \delta_{Aw}. \quad (4.1)$$

It is not thereby a stationary background. The ordered-chart decomposition

$$\partial_A \Phi = (u_0 + \sigma) n_A \quad (4.2)$$

defines the amplitude coordinate $\sigma(X)$ and orientation coordinate $n_A(X)$ about the supplied reference value $n_A = \delta_{Aw}$.

Status of n_A . The orientation variable is an ordered-branch EFT coordinate, not a second fundamental microscopic field of P3 and not a selected vacuum state. This decomposition defines a P4 amplitude coordinate σ . Its quadratic operator, pole, residue and particle interpretation are Open; the distinct homogeneous P3 curvature is discussed in Section 8.3.

4.2 Sector map: amplitude, soft orientation, and macroscopic ϕ -sector

Three qualitatively distinct structures must be kept separate throughout the rest of the theory.

P4 amplitude coordinate and Open physical mode. The decomposition $\partial_A \Phi = (u_0 + \sigma)n_A$ defines an amplitude coordinate about the supplied P4 gradient background. It does not supply that background's stationary action or quadratic operator. Therefore the P4 pole, residue, correlation length, decoupling scale, particle interpretation and EFT threshold are Open. Separately, the homogeneous P3 action has the conditional Gaussian curvature $m_{\Phi, \text{rad}}^2 = V''(\phi_0) = 2\mu^2$. Equating the two is an Open P3–P4 matching hypothesis, not an identity. Neither coordinate is the Standard-Model Higgs boson.

Orientation coordinate sector. On the supplied nonzero P4 chart one may write

$$Q_A \equiv \partial_A \Phi = u n_A, \quad u = |Q|, \quad n_A n^A = 1. \quad (4.3)$$

A supplied nonzero direction has stabiliser $O(3)$ and abstract orbit $O(4)/O(3) \simeq S^3$. This is conditional group geometry, not a derived physical vacuum manifold or symmetry-breaking event. The orbit has dimension three, but in the scalar-only exact-gradient basis its independent coordinate content is reduced by integrability, as shown below. A physical mode count requires a stationary reduced action, domain and projector. However, in the present scalar-only ECT basis Q_A is not an arbitrary vector field but an exact one-form, $Q_A = \partial_A \Phi$. Therefore the orientation sector is subject to the integrability constraint

$$\partial_A Q_B - \partial_B Q_A = 0. \quad (4.4)$$

This sharply reduces the naive Goldstone counting; see Section 4.3 and Appendix J. The conditional particle-spectrum interpretation of the scalar coordinate χ inside a supplied reduced EFT is discussed in Section 8.2.

Macroscopic scalar ansatz and galactic response. The amplitude coordinate ϕ_u enters a Level-B scalar ansatz. ERP- Φ and HRC are a separate augmented-model constitutive programme; their source/force/metric match to ϕ_u remains Open. The preserved SPARC/RAR/BTFR benchmarks have been recomputed with the HRC registry as Level-C diagnostics and are not an ECT explanation of particle ontology. The soft sector must therefore be assessed on its own dynamics rather than added or excluded by the fit result.

4.3 Integrability and the status of the orientation sector

The crucial point is that in the current ECT basis the ordered quantity is $Q_A = \partial_A \Phi$, so the decomposition $Q_A = u n_A$ does *not* define independent fields u and n_A . Instead they must satisfy the exact integrability condition

$$\partial_A (u n_B) - \partial_B (u n_A) = 0. \quad (4.5)$$

Linearising around the ordered background $Q_A^{(0)} = u_0 \delta_{Aw}$ and writing

$$u = u_0 + \sigma, \quad n_A = (\pi_i, 1) + \mathcal{O}(\pi^2), \quad (4.6)$$

one finds $\delta Q_i = u_0 \pi_i$, $\delta Q_w = \sigma$. Since δQ_A must itself be a gradient, there exists a scalar fluctuation χ such that $\delta Q_A = \partial_A \chi$, hence

$$u_0 \pi_i = \partial_i \chi, \quad \sigma = \partial_w \chi. \quad (4.7)$$

Therefore the apparent orientation triplet is not a set of three independent scalar coordinates: at the linear scalar-only kinematic level all of its components are parametrised by *one longitudinal scalar coordinate* χ . Equivalently, $\varepsilon_{ijk} \partial_j \pi_k = 0$, so the transverse/vortical part of the naive orientation coordinate is eliminated. This exact coordinate reduction does not by itself determine the constrained physical mode count, kinetic normalisation, spectrum, or mass; those require a supplied reduced action, domain and projector.

Interpretation: inverse-Higgs-type reduction. This is an inverse-Higgs-type reduction [29, 30] characteristic of spacetime symmetry breaking: the broken object $Q_A = \partial_A \Phi$ is a geometric gradient rather than an internal multiplet. One must therefore distinguish two statements:

1. the nonzero vector supplied by P4 has $O(3)$ stabiliser and admits a minimal abstract $O(4)/O(3) \simeq S^3$ orientation description;
2. in the strict scalar-only map, all linear soft variations are represented by one longitudinal scalar coordinate χ ; interpreting it as one physical propagating mode is a further Level-B reduced-EFT claim.

The full technical derivation is given in Appendix J.

Corollary: one characteristic cone of the supplied second-order scalar operator. The integrability result has a further consequence that distinguishes ECT from multi-field Lorentz-violation frameworks [23, 26, 25, 27, 20].

Corollary (conditional one-operator scalar cone). In the scalar-only ECT basis, $Q_A = \partial_A \Phi$ is an exact one-form, so Eq. (4.7) expresses the linear soft coordinates (σ, π_i) as derivatives of one scalar coordinate χ . Suppose in addition that the supplied constrained quadratic scalar EFT has the second-order principal part $K^{AB} \partial_A \chi \partial_B \chi$, with no additional independent scalar principal operator. Then that displayed operator has the single principal polynomial $K^{AB} \xi_A \xi_B$ and hence one scalar characteristic cone with common-length slope $\hat{c}_*^2 = \beta/(\alpha - \beta)$ (3.19). This is Level A conditional mathematics for the supplied operator and Level B as a physical ECT mode; it is not a P1–P4-only theorem or a universal physical cone for gauge, fermion, matter or tensor sectors.

Proof. From (4.7), $\sigma = \partial_w \chi$ and $\pi_i = \partial_i \chi / u_0$. By the additional second-order reduced-EFT assumption, the quadratic principal object is the characteristic form $\mathcal{Q}_{2,\chi} = \frac{1}{2} \int d^4 X K^{AB} \partial_A \chi \partial_B \chi$, which gives one equation for one field. On the retained positive-energy component its Lorentzian reduced functional is $\mathcal{J}_{L,\chi}^{(2)} = -\mathcal{Q}_{2,\chi}$; this overall sign changes neither the principal polynomial nor the one-cone conclusion. A physical action would additionally require an independent finite, positive, spacetime- and field-independent action-valued factor $A_{L,\chi}$; no equality to $A_{L,\varphi}$ is assumed. Its characteristics satisfy $\beta \mathbf{k}^2 - (\alpha - \beta) k_w^2 = 0$, defining a single cone with speed $\hat{c}_*^2 = \beta/(\alpha - \beta)$. By construction no second principal symbol is present in this supplied operator. $\square$

Scope and limitations. This corollary establishes one principal symbol only inside the displayed second-order quadratic scalar branch. Integrability alone supplies the coordinate reduction, not the derivative order, physical Hessian, projector or spectrum. A single coordinate can carry a higher-order characteristic polynomial with multiple factors: for example, the formal substitution $p_w = i\omega$ in the pure quartic kernel of Appendix AA gives $K_4(i\omega, k) = \frac{1}{2} (k^2 - \omega^2) (\mathcal{C}_n k^2 - Z_u \omega^2)$, with two distinct slopes when $Z_u \neq \mathcal{C}_n$. That kernel is used here only for Euclidean seed statistics, not asserted as a physical Lorentzian equation, but it is an explicit counterexample to the inference “one coordinate $\Rightarrow$ one cone.” At $Z_u = \mathcal{C}_n$ the factors coincide into a double characteristic; this still does not prove a healthy one-mode or unitary continuation. The distinct reflection-positivity analysis for the named bare constant-numerator K_4 covariance is given in Appendix AA: the coincident case is handled by the double-pole witness, while Eq. (AA.10) gives the unequal-pole gate. These tests concern Euclidean transfer positivity, not the conditional second-order cone statement above. The extension to phase, gauge, and fermionic sectors requires additional construction steps and is not part of the present conditional Level A mathematical statement: conditional gauge completion (§7), spinorial reconstruction (§9). Those extensions are discussed in §4.7 and remain conditional/Open until their actions are supplied.

Contrast with competing frameworks. In Einstein–aether theory [23, 26] the unit vector n^μ is an independent dynamical field with four free kinetic coefficients $c_1 \dots c_4$. Different spin sectors generically propagate at different speeds; universality of the causal cone requires fine-tuning of the c_i . In Hořava–Lifshitz gravity [25] different sectors may acquire distinct UV dispersion relations. In the Standard Model Extension (SME) [27, 28] each sector carries an independent set of Lorentz-violation coefficients; universality is not a structural consequence. In the He-3 analogy of Volovik [20] different quasiparticle species can see different effective light cones. Inside the supplied second-order scalar operator there is only the one declared principal tensor K^{AB} , so it has one cone. The fact that all strict-map soft coordinates are derivatives of one scalar χ does not independently prove this operator assumption and does not establish cross-sector physical universality. A detailed comparison of speed structures is given in Appendix K.

4.4 Minimal soft orientation EFT and phenomenological status

Although the strict scalar-only microscopic content is reduced by integrability, it remains useful to organise the supplied P4 chart by a minimal orientation EFT. In that effective description the supplied orientation target $O(4)/O(3) \simeq S^3$ supports a three-component orientation map, but this is an *effective parametrisation*, not a derived physical vacuum manifold or a theorem about three independent physical Goldstone particles.

Quadratic soft-sector EFT. If the broken-phase EFT contains the local orientation-elastic term $\frac{\mathcal{C}_n(u)}{2} u^2 (\partial_B n_A) (\partial^B n^A)$, then at uniform background $u = u_0$ the effective stiffness is

$$\kappa_n = \mathcal{C}_n u_0^2. \quad (4.8)$$

This is a Level B EFT identity. The coefficient $\mathcal{C}_n$ carries dimension mass^{-2} (so that $[\kappa_n] = \text{mass}^2$; this dimension alone does not provide a spin-2 normalisation scale) and is not yet derived from bare postulates P1–P6. A pure $P(X)$ closure cannot generate this term (Theorem 3.1). The displayed local n -gradient operator is next-to-leading relative to that narrow pure- $P(X)$ truncation. This does not exclude separate fields/actions, constraints, nonlocal response or coarse-grained owners; the origin and coefficient of orientation stiffness remain Open (Section 5.1). The nonlinear sigma-model form, unit constraint, and scalar-integrability caveat are collected in Appendix AB.

If the supplied reduced EFT is derivative-only and its constrained quadratic operator contains no χ^2 term, the χ coordinate has the conditional linear dispersion relation

$$\omega^2 = c_\chi^2 k^2, \quad c_\chi^2 = \frac{K_s}{K_w} c_{\text{char}}^2, \quad (4.9)$$

where K_w, K_s are EFT stiffness coefficients along w and spatial directions respectively. The stronger statement $c_\chi = c_{\text{char}}$ requires $K_s = K_w$, which is not yet established. Neither integrability nor the broken-generator count provides an owned internal shift symmetry or Ward identity that forbids a mass term. Indeed, the potential-sector expansion of Section 3.1 can contribute $\frac{1}{2} V''(\bar{\Phi}) \chi^2$ in a profile reduction. The physical mass, its sign and its action-to-EFT matching therefore remain Open.

What remains true, and what does not. Without the medium-character postulate P5, there is no local $O(3)$ gauge structure at the level of P1–P4 and P6 alone, hence no Higgs-type eating mechanism in the foundational sector [31]. Any electroweak interpretation requires an additional scale $v_2 \sim 246 \text{ GeV}$ and a second dynamical step (OP-EW).

The Level-C HRC algebraic diagnostic uses neither a soft-orientation component nor a particle-halo term. This fit property does not show that ECT already explains the corresponding phenomena, and it neither requires nor excludes a particle or collective dark sector. The phenomenological role of any physical mode represented by the longitudinal coordinate remains Open.

4.5 EFT quadratic action for fluctuations

Status: EFT consequence (Theorem 3.2 applied to the broken-phase EFT).

The displayed quadratic action for φ is a supplied broken-phase EFT ansatz with tensor (3.11). It cannot presently be advertised as the result of integrating out the P3 homogeneous radial mode: no stationary P4 action, Hessian or matching map to that pole has been derived. Appendix L is therefore a conditional reduction template, not an owner proof. The background profile and the status of its conditional fluctuation expansion are recorded explicitly in Appendix B.

4.6 Mode hierarchy and the branching structure

Coordinate/mode sector	Scale	Role
P3 radial coordinate	$m_{\Phi,\text{rad}}^2 = 2\mu^2$ about a classical homogeneous P3 minimum	Does not own a P4 pole or EFT breakdown scale; P4 Hessian/matching Open
Orientation δn_i	Declared long-wavelength P4 domain; physical pole/cutoff Open	Geometric sector; integrability parametrises the strict scalar-only map by one longitudinal coordinate χ (Level A kinematics). One physical soft mode follows only inside a supplied reduced action, domain/quotient and projector (Level B; see Section 4.3)
Scalar φ	Declared low-energy scalar EFT; P4 threshold Open	Conditional scalar excitations of the supplied Lorentzian principal sector
Phase θ	Topological sector	Coherent IR; compact winding exact; fixed-core and positive-tension loop coefficients are distinct conditional candidates, and the operational action-unit owner is Open

Fixed-map status of the geometric tensor. The symmetric object $g_{AB}^{\text{eff}}(n)$ is a characteristic-metric candidate for the scalar ordered branch. Calling it rank 2 does not make its linear variation a helicity-2 particle. At fixed α, β and on the homogeneous ordered background, a tangent director perturbation gives only h_{iw} , while

$$h_{ij} = h_{ww} = h_{ij}^{\text{TT}} = 0.$$

Under strict integrability the surviving h_{iw} is longitudinal shift-type data. Therefore, at fixed α, β and to linear order in a tangent director perturbation, the current director map supplies neither a linear TT seed nor a linear static-density vertex. The exact scoped component theorem and its assumptions are stated in Theorem 8.1; it is not a no-go theorem for nonlinear, composite, additional-field, or dynamical-parameter vertices.

The standard Fierz–Pauli field, two TT polarisations and an Einstein completion remain useful external targets. ECT ownership requires a new constrained tensor Hessian, positive residues, a nonzero matter vertex and a universal physical metric; none follows from the vector character of δn_i or from dimensional equality of stiffness coefficients.

4.7 Causal cone inheritance by the phase and gauge sectors

The supplied second-order scalar operator has one principal cone; integrability provides only its scalar-coordinate reduction (§4.3). This subsection records conditional phase/gauge matching routes; it does not

prove inheritance by physical sectors.

Physical significance: why a single cone matters. A decisive test of any emergent-spacetime framework is whether different low-energy sectors inherit a single causal structure rather than sector-dependent propagation cones. In many modified-gravity and Lorentz-violating frameworks, different fields can have different limiting speeds, destroying universal relativistic kinematics. In the displayed scalar/phase ansatz the two fields share a kinetic tensor K^{AB} by construction. Independent gauge, fermion and tensor sectors require their own principal-symbol checks (Theorem 3.2). The joint GW170817/GRB 170817A result constrains the relative signal speed to $-3 \times 10^{-15} \leq (c_{\text{GW}} - c_\gamma)/c_\gamma \leq +7 \times 10^{-16}$ under the stated association and emission-time assumptions [12]; this is an external gate for a future tensor completion, and the scalar cone alone does not predict it. This is in contrast to, e.g., Einstein-aether theory [23], where different sectors may generically have different speeds.

Phase sector (Level A/B bridge). The phase θ enters ECT through the polar decomposition $\Phi_{\text{eff}} = \rho e^{i\theta}$. At wavelengths large compared to the radial-mode scale, the supplied phase characteristic form is (21.2):

$$\mathcal{Q}_{\theta,K} = \frac{K_\theta^{\text{eff}}}{2} \int d^4X \hat{K}^{AB} \partial_A \theta \partial_B \theta, \quad \hat{K}^{AB} \equiv K^{AB}/\beta,$$

on the declared $\beta > 0$ branch, with supplied $K_\theta^{\text{eff}} > 0$. The positive-energy reduced Lorentzian functional in this two-derivative truncation is $\mathcal{S}_{\theta,L} = -\mathcal{Q}_{\theta,K}$. Thus the phase-sector closure inherits the same cone, while the positive rescaling by $1/\beta$ keeps the independently supplied stiffness normalisation explicit. This is a conditional shared-tensor ansatz, not a circular speed matching: symmetry restricts the form of K^{AB} , but neither the compact phase sector, its stiffness, physical action, higher operators nor map from bare P3 is derived. Unspecified higher-gradient terms are not oriented by the displayed minus sign and require their own pole, constraint and reduced-Hessian tests. Status: A-math conditional for the tensor/cone algebra and Level-B/Open for the physical phase owner.

Gauge-sector completion route (Open). P5 motivates a redundancy interpretation but does not create a local bundle, connection, gauge-volume quotient or transverse photon. Conditionally, after those structures are supplied:

1. On an ordered background the leading parity-even two-derivative quadratic basis has two independent invariants, $I_1 = F_{AB}F^{AB}$ and $I_2 = (n^A F_{AB})(n_C F^{CB})$ (Proposition 3, §7). Maxwell form is the special isotropic choice $Z_2 = 0$, not a consequence of P5.
2. The anisotropic coefficient and physical reduced Hessian must be supplied or derived independently; the ordered director permits rather than excludes this second gauge kinetic structure. This contrasts with Einstein-aether [23], where the gauge sector may have its own kinetic coefficients, and with the SME [27], where each sector carries independent Lorentz-violation parameters.
3. The scalar-clock calibration and photon-cone target are distinct and neither follows from P5:

$$c_{\text{char}} = c \quad (\text{adopted Level-B calibration}), \quad c_\gamma \stackrel{?}{=} c_{\text{char}} \quad (\text{Open photon matching}). \quad (4.10)$$

Caveat: independent gauge-sector extension. If the gauge sector were allowed an independent EFT extension beyond the current candidate gauge-EFT route compatible with P5, the anisotropic operator $n^\mu F_{\mu\nu} n^\alpha F_\alpha{}^\nu$ might acquire a free coefficient, and the universality argument would no longer hold. P5 alone does not prefer one Wilson coefficient or exclude this anisotropic operator.

Summary of the cone-matching programme. The hierarchy of derivation levels is:

1. Level A conditional mathematics: one causal cone of the supplied second-order scalar principal operator. Integrability supplies the coordinate reduction but not the operator order or physical projector.
2. Level A/B bridge: inside the separately supplied phase closure, the phase sector is assigned the same tensor structure at leading order; this is not a microscopic-descent theorem.
3. Open: construct an unbroken transverse photon and show that its principal cone satisfies $c_\gamma = c_{\text{char}} = c$.

The relation $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration. The equalities $c_\gamma = c_{\text{char}}$ and $c_{\text{tensor}} = c_{\text{char}}$ remain Open; GW170817/GRB170817A constrains a completed photon–tensor construction. None of these physical statements is derived by the dimensionless scalar principal symbol.

Sector-by-sector cone ownership. The phrase “one condensate” is not a principal-symbol theorem for every effective sector. The current owner statements are instead:

- (i) the supplied scalar principal tensor has the dimensionless common-length slope $\hat{c}_*$ as Level-A conditional mathematics; c_{char} is defined only after a separately supplied local clock/conversion-speed/lapse map, and its physical value remains Open;
- (ii) the compact-phase field shares that cone only inside the explicitly written phase closure using the same tensor and its supplied Lorentzian orientation (Level A algebra in that model, with its coefficient and action matching still conditional);
- (iii) a physical photon has $c_\gamma = c_{\text{char}}$ only after the compact-phase construction is matched to a transverse Maxwell field and independent anisotropic kinetic operators are excluded;
- (iv) the present fixed director map supplies no physical tensor principal symbol, so c_{GW} is Open.

Accordingly the valid statement is the owner ledger

$$c_{\text{scalar}} = c_{\text{char}}, \quad c_{\text{phase}} = c_{\text{char}} \text{ (within the supplied phase characteristic closure)}, \quad c_\gamma \stackrel{?}{=} c_{\text{char}}, \quad c_{\text{GW}} \text{ Open.} \quad (4.11)$$

What GW170817 tests. The measured photon–GW arrival-time agreement is an external kill gate for the principal symbol of a future tensor completion. It is not a present ECT consistency check because the tensor action and c_{GW} have not been identified. Likewise, a Planck-suppressed dispersion formula is only a conditional operator benchmark until its tensor operator and coefficient are derived.

Falsifiable completion statement. Once a photon or tensor action is supplied, each sector must independently pass hyperbolicity, residue, low-energy cone and high-energy-dispersion tests. An unsuppressed speed difference would reject that completion. Agreement would establish consistency of the completed operators, not retroactively derive them from the scalar cone. Thus the robust structural result is the single scalar-branch cone; cross-sector universality remains a matching and completion programme.

4.8 Status of results in this section

Table 8: Status of results in this section.

Statement	Status	Basis
Polar decomposition $\partial_A \Phi = (u_0 + \sigma)n_A$	Level A	P4 + Theorem 3.2
Characteristic form, reduced Lorentzian functional and physical-action factor (3.12)	A-math conditional / Open	K^{AB} fixes the supplied principal form; positive energy on $\alpha > \beta > 0$ fixes $\zeta_L = -1$, while $A_{L,\varphi} > 0$ and its cross-sector matching remain Open
Equation of motion (3.17)	Level A (EFT)	Variation is invariant under any finite nonzero constant overall multiplier
Lorentzian dispersion (3.18) for the healthy branch $\alpha > \beta > 0$	Level A (EFT)	Theorem 3.2 plus the supplied healthy-branch signs
Klein–Gordon equation and mass map (3.24)	Level A (EFT)	Real parametrisation $x^0 = w$ plus the supplied clock/lapse map
Free Hamiltonian $H \geq 0$ (3.15)	A-math conditional / Open	Requires $\zeta_L = -1$, $A_{L,\varphi} > 0$, $\alpha > \beta > 0$ and $\inf \text{spec}[\beta(-\nabla^2) + m_{\text{eff}}^2] \geq 0$ on the declared self-adjoint domain; magnitude and cross-sector ownership remain Open
Identification of m_{eff}^2 with $V''(\phi_0)$	Open (OP-grad)	Requires explicit EFT matching
Integrability of $Q_A = \partial_A \Phi$	Level A	Scalar-only basis of P3
Linear reduction $\pi_i = \partial_i \chi / u_0$, $\sigma = \partial_w \chi$	Level A	Integrability around the ordered background
Unique principal cone of the supplied linear scalar branch	A-math conditional / physical Open	Universality Corollary (§4.3): one scalar coordinate plus the supplied quadratic tensor $\Rightarrow$ one principal symbol
Cone assignment in the supplied phase characteristic form	A-math conditional / Open	Same K^{AB} is assumed; positive-energy action further requires $K_\theta^{\text{eff}} > 0$ and the supplied orientation, while the physical phase owner remains Open
$c_\gamma = c_{\text{char}}$ (gauge-completion target)	Open	P5 does not derive a transverse photon or its principal symbol (§4.7)

Continued on next page

Table 8 continued

Statement	Status	Basis
Cross-sector cone universality (4.11)	Open	Requires independent gauge, fermion and tensor actions
Cross-sector action-scale universality (§5.6)	B/Open	Common normalisation is a proposed matching programme
Cross-sector variational-descent programme (§22.2)	Open	Euclidean DP, supplied reduced actions, macroscopic completions, and Lorentzian stationary phase have distinct owners; no universal descent chain is derived
Four structural universality programmes (§11.1)	B/Open	Separate closure tests, not established universalities
Radial-mode EFT breakdown threshold (§8.3)	A-math conditional	Scalar mass/correlation scale only; not a UV regulator or microscopic completion
Minimal 3-component orientation EFT on $O(4)/O(3) \simeq S^3$	Level B (effective description)	Broken-branch parametrisation only
Strict scalar-only map: one longitudinal scalar coordinate χ ; physical one-mode interpretation conditional	A kinematics / B physical EFT	Inverse-Higgs-type coordinate reduction; physical count needs action, quotient/domain and projector [29, 30]
Pure $P(X)$ closure does not generate $(\partial n)^2$	Level A	Theorem 3.1
Orientation-sector stiffness $\kappa_n = \mathcal{C}_n u_0^2$; $[\mathcal{C}_n] = \text{mass}^{-2}$	Level B (EFT)	Algebraic form exact within NLO EFT; full coefficient and normalisation not derived at this level
Candidate one-loop induced structure and minimal leaf-conformal kernel for $\mathcal{C}_n$	Level B conditional	Conditional partial progress: an NLO/one-loop route and a conditional foliation kernel are identified; the finite Wilson coefficient, normalisation, trace-sector reduction and spin-2 map remain open
Derivation of $\mathcal{C}_n$ from bare P1–P6	Open	OP-Planck; NLO + spin-2 normalisation
Linear soft dispersion $\omega^2 = c_\chi^2 k^2$	A-math conditional / Level B ECT	Supplied derivative-only reduced EFT; a physical χ^2 operator is not excluded

Continued on next page

Table 8 continued

Statement	Status	Basis
Equality $c_\chi = c_{\text{char}}$	Open	Requires $K_s = K_w$; not yet derived
Three independent physical Goldstone particles	Not established	Not represented by the strict scalar coordinate map; a larger physical bundle is not derived
No Higgs-type eating without a separately supplied local gauge completion	Open	P5 supplies no local $O(3)$ gauge redundancy, connection or physical Higgs map
Physical m_χ^2 and any protection or pseudo-Goldstone mechanism	Open (OP-GM)	Stationary owner, constrained Hessian/projector, regulator-compatible Ward identity and counterterms are missing
Relation to electroweak gauge/Higgs sector	Open (OP-EW)	Requires a dynamically generated second condensate scale and an explicit mechanism for the second ordering step; derivation open
Variational stationarity of realised Φ -configurations	Postulate	DP supplies a necessary on-shell condition, not a selector among extrema
Emergent extremal dynamics in reduced effective sectors	Level B/Open	Equations follow from each owning supplied or derived reduced action; descent from DP additionally requires an explicit variation-compatible restriction or controlled elimination, and no unique-history selection follows
Candidate loop scales and operational action unit	Level B/Open	l_0^{EFT} and $l_{0,\text{bal}}^{\text{eff}}$ are dimensionless coefficients of distinct supplied loop closures; either product with the independent action normalisation Σ_θ may be mapped to operational S_0 only through an Open owner map, while $S_0 = \hbar$ is an explicit Level B physical calibration rather than a derivation

Continued on next page

Table 8 continued

Statement	Status	Basis
Cross-sector action-scale universality	Open	Common use of S_0 is a programme; independent reduced-sector normalisations and Born probability ownership are not derived (§5.6)
Strict identification $S_0 = \hbar$	Level B matching / Open owner	Adopted as a physical calibration; microscopic ownership, absolute derivation and cross-sector universality require full quantum-sector reconstruction
Induced effective-metric tensorial sector from orientation fluctuations	Level B/Open	Scalar characteristic geometry is established; physical tensor content is not produced by projection alone; see Section 8.5

What is intentionally not done here. This section does not introduce $\hbar$, canonical commutation relations, or the Born rule. Only the result that broken-phase excitations satisfy a Lorentzian wave equation with relativistic dispersion is obtained. The quantum regime is developed in Section 20 after deriving the corresponding formalism from the condensate structure.

5 Fundamental Scales, Matching Relations, and the Status of Constants

This section separates the energy-dimension quantities ϕ_0 and v_2 from the conditional HRC matching length L_{gal} . It also records c_{char} , the matching for G_N , radial-mode scales, and S_0 . Most entries are matching relations or EFT-level consistency statements. The gradient-condensate mechanism and the ECT derivation of L_{gal} remain Open.

After establishing the postulates and the broken-phase dynamics, we systematically distinguish: which parameters are microscopic, which are effective, which are fixed by observational matching, and which are not yet derived.

Physical significance: conditional matching inventory. In standard physics, c , $\hbar$, and G are independent measured constants. ECT currently supplies only a scalar common-length slope $\hat{c}_*$ in one broken-phase EFT. The dimensional clock conversion and lapse used by the scalar calibration, the physical photon speed, a nonzero TT normalisation M_G , and the microscopic owner of the operational action slot are independent sectors. The displayed G_N relation and the candidate-to- S_0 maps remain conditional/Open; $c_{\text{char}} = c$ and $S_0 = \hbar$ are separately adopted Level-B calibrations. They are not three projections already derived from one medium, and photon/tensor cone universality remains Open. The explicit matching formulas are collected in Table 109.

5.1 Two energy/matching scales and one conditional HRC matching length

The present organisation distinguishes a UV stiffness scale, a low-energy matching scale, and a conditional HRC matching scale. The last is not yet derived as a condensate scale. They are *not* three independent VEVs of three different fields.

What is established, what is matched, and what remains open in the hierarchy programme. The ECT scale hierarchy should be read in separate owner layers. First, the homogeneous P3 model supplies the amplitude scale ϕ_0 and the conditional curvature $m_{\Phi,\text{rad}}^2 = 2\mu^2$. It does not supply a radial pole or threshold for the distinct P4 gradient branch. The condensate quantities define an orientation-stiffness scale $\kappa_n = \mathcal{C}_n u_0^2$. Their relation to a physical tensor coefficient M_G is Open (§5.1). Independently, ECT adopts the physically motivated Level-B scale matching $\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}$ with dimensionless $\zeta_\phi > 0$, $\zeta_\phi = O(1)$ and $\bar{M}_{\text{Pl}} = (8\pi G_N)^{-1/2}$. The displayed tensor bridge contains no ϕ_0 and therefore does not derive this ansatz. Third, what remains open is the completed hierarchy problem: the first-principles derivation of the induced coefficient $\mathcal{C}_n$, the emergence of the second electroweak scale v_2 , and the explicit dynamical origin of the large ratio $v_2/\phi_0 \sim 10^{-16}$.

Scale	Value	Physical origin in ECT	Status	Observable anchor
ϕ_0	$\zeta_\phi \bar{M}_{\text{Pl}}$, with the canonical benchmark $\zeta_\phi = 1$ giving $2.435 \times 10^{18} \text{ GeV}$	Independent physically motivated scale ansatz; not derived by the displayed tensor bridge	Adopted Level B match / microscopic explanation Open	G_N
v_2	$\approx 246.22 \text{ GeV}$	Low-energy matching to electroweak EFT; structural candidate for second transition	Constraint / Level B	G_F
$L_{\text{gal}} \equiv r_*$	Conditional HRC matching length $\sqrt{G_N M_{\text{bar}}/a_M}$	Identity/source bridge and microscopic origin Open	Bridge B/Open; data use C	SPARC/RAR

These three entries are not quantities of the same dimension or status. The supplied P4 gradient amplitude u_0 has mass dimension two in the adopted conventions; it is not itself a UV energy scale. Only an action-owned pole or cutoff with a dimensional matching map could define such a scale. The intermediate scale v_2 is, in the present formulation, a low-energy matching constraint if a second transition exists. The galactic scale is a conditional HRC matching/crossover length of the supplied gravity layer, not a second microscopic condensate VEV. Its derivation from the scalar ansatz is Open.

The multiplicity S6 and topological completeness S10 of the Φ -medium make a multi-scale structure structurally admissible, but do not by themselves explain the specific hierarchy between these scales.

Caveat on dimensions and status. The pair (ϕ_0, v_2) contains energy-dimension quantities. By contrast, $L_{\text{gal}} \equiv r_*$ is the conditional HRC matching length $\sqrt{G_N M_{\text{bar}}/a_M}$ used in Level-C diagnostics. It is neither a third VEV nor a derived condensate scale, and it must not be ordered numerically against energies.

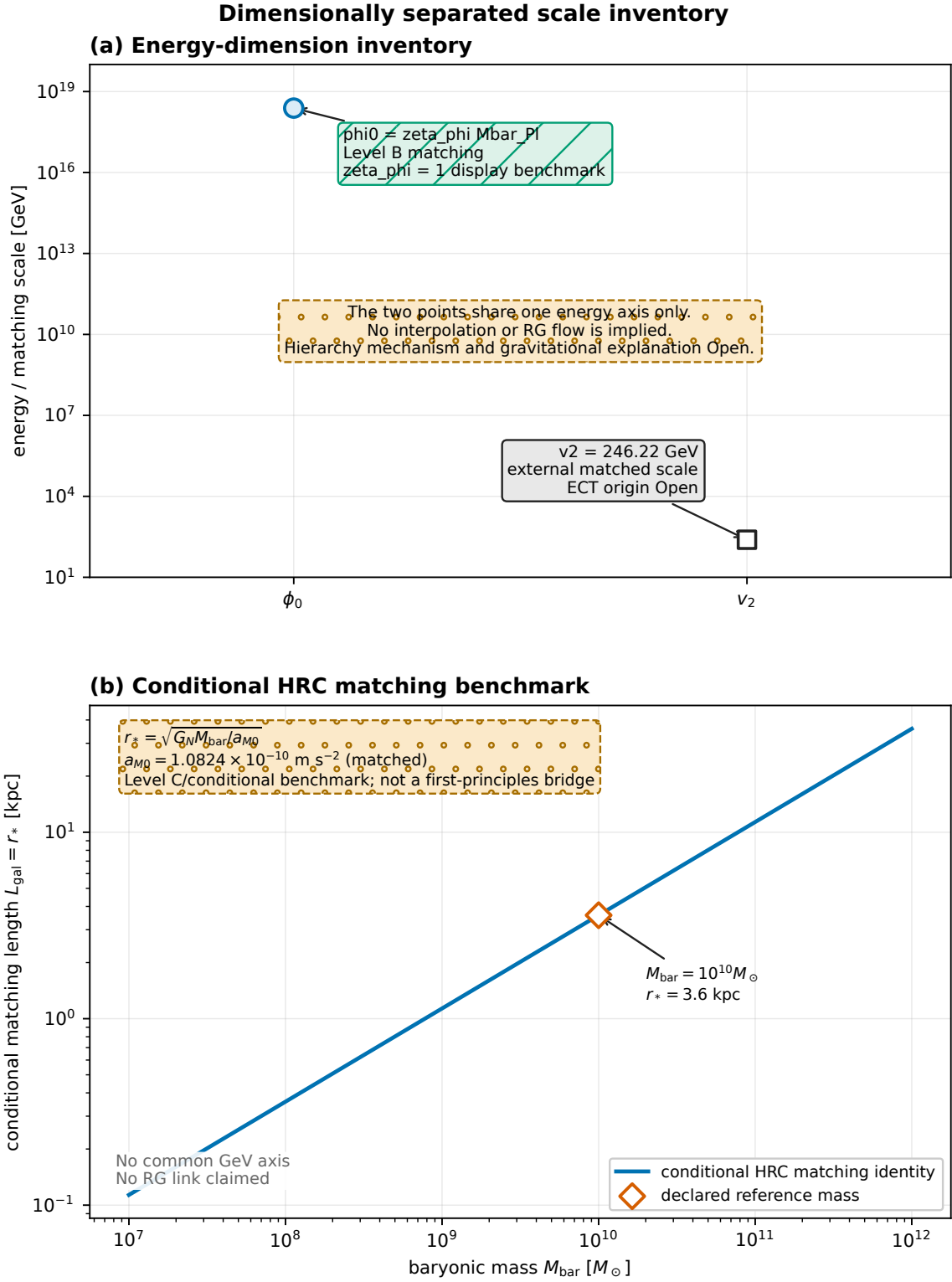


Figure 2: Dimensionally separated scale inventory. Panel (a) compares only energy-dimension quantities; the origin of the matched electroweak scale and the hierarchy mechanism remain Open. Its two entries are independent inventory points; no connecting curve or interpolation is implied. Panel (b) shows the conditional HRC matching identity at the declared matched scale; it supplies no common GeV axis or RG flow. Colours encode structural, external, conditional and Open roles and remain ordered by luminance in grayscale.

Gravitational normalisation and the Planck scale. The parameter u_0 sets only the magnitude of the supplied P4 nonzero-gradient datum. Neither an orientation stiffness nor dynamical $O(4) \rightarrow O(3)$ breaking follows from it. Moreover, u_0 cannot be identified directly with the reduced Planck mass: the two quantities have different dimensions ($[u_0] = \text{GeV}^2$ in the present convention, $[\bar{M}_{\text{Pl}}] = \text{GeV}$).

In the supplied geometric-sector EFT, orientation variables δn_i are assigned a kinetic term with effective stiffness $\kappa_n = \mathcal{C}_n u_0^2$ (Section 4.4), where $\mathcal{C}_n$ is an effective coefficient with dimension mass^{-2} . A pure $P(X)$ closure of the ordered branch cannot generate this term (Theorem 3.1); orientation stiffness therefore requires additional structure. Candidate possibilities include mixed radial–orientation dependence, an independent stiffness operator, constraint-sector matching, higher derivatives, induced contributions, or nonlocal response. The two examples below are not exhaustive and no owner is presently selected by P1–P6. A generic orientation-sector heavy-field matching template may instead be written

$$\mathcal{C}_n = \frac{\hat{a}_n}{16\pi^2 M_{n,\text{heavy}}^2}, \quad \kappa_n = \mathcal{C}_n u_0^2, \quad (5.1)$$

where $M_{n,\text{heavy}}$, the vertex that couples that field to $(\partial n)^2$, the regulator, subtraction and finite coefficient $\hat{a}_n$ must all be supplied by one stationary completion. The current P4 model owns none of them, so this is PARAMETRIC ONLY / MISSING VERTEX; the P3 curvature $m_{\Phi,\text{rad}}$ cannot be inserted as $M_{n,\text{heavy}}$ by notation, and no equality to another sector’s heavy scale is implied. At equal mass dimension one may pose only the independent Open tensor bridge

$$\boxed{M_G^2 \stackrel{?}{=} c_M \kappa_n} \quad (\text{Open representation and pole-residue map}). \quad (5.2)$$

Separately from the Open tensor bridge, ECT adopts

$$\boxed{\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}}, \quad \zeta_\phi > 0, \quad \zeta_\phi = O(1) \quad (\text{Level-B physical scale ansatz}). \quad (5.3)$$

The canonical benchmark $\zeta_\phi = 1$ uses $\bar{M}_{\text{Pl}} = (8\pi G_N)^{-1/2} \approx 2.435 \times 10^{18} \text{ GeV}$. A first-principles derivation of ζ_ϕ , of $\mathcal{C}_n$, or of an equation containing both ϕ_0 and the tensor residue remains Open (OP-Planck, Section 38.1).

Planck-scale candidate: conditional dimensional route. The NLO estimate concerns orientation stiffness, not a graviton residue. Only after a physical constrained tensor Hessian, matter vertex and positive pole residue have been derived can c_M be calculated and one ask whether the adopted coefficient ζ_ϕ is explained by the same owner. At present $\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}$ is an explicit matching input, not an induced-gravity result and not evidence that the Planck scale has been derived from the condensate.

Common-medium hierarchy hypothesis. ECT adopts as a Level-B programme hypothesis that the P3/P4, electroweak, infrared and HRC scales should admit a common-medium matching in a completed theory. No relation among ϕ_0 , u_0 , v_2 , a_M and m_χ follows at present unless it is displayed as a separate named closure. The P3 length $\xi_{\text{P3}} = 1/(\sqrt{2\lambda} \phi_0)$ and a future P4 pole length must remain distinct; matching either to a Planck length requires an explicit Planck convention and an independent dimensionless coefficient.

Connection to the universal structures already established. The hierarchy programme is not isolated from the rest of ECT. The P3 amplitude ϕ_0 , a possible P4 orientation stiffness and the coherent-sector action-scale programme (§5.6) may share a future microscopic completion, but no common matching owner is currently derived. Whether it also owns a physical tensor normalisation is precisely the Open bridge above; numerical proximity to $\bar{M}_{\text{Pl}}$ cannot establish that ownership. A successful completion of the hierarchy problem must therefore be compatible with all of those earlier universalities simultaneously.

The supplied gradient datum and the open problem OP-grad. Postulate P4 introduces u_0 as the ordered-chart gradient scale $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$, independently of the bare potential minimum $\phi_0 = \sqrt{\mu^2/\lambda}$ (P3). P4 deliberately specifies the gradient datum independently of the P3 field minimum. Whether that datum belongs to a stationary physical vacuum remains Open. One may nevertheless write the following Level-C dimensional ansatz by combining the homogeneous P3 amplitude $\phi_0 = \mu/\sqrt{\lambda}$ with its P3 Gaussian mass $m_{\Phi, \text{rad}} = \sqrt{2}\mu$; neither quantity is a P4 correlation owner:

$$u_0 \sim \frac{\phi_0}{\xi_{\text{P3}}} \sim \phi_0 m_{\Phi, \text{rad}} \sim \frac{\mu}{\sqrt{\lambda}} \cdot \sqrt{2}\mu \sim \frac{\mu^2}{\sqrt{\lambda}}. \quad (5.4)$$

This is only a dimensional self-consistency estimate, not a derivation. The action/state mechanism that could physically realise this gradient datum remains an open foundational problem (OP-grad). Until OP-grad is closed, u_0 should be treated as an independent Level 2 parameter (Section 5.2).

Galactic HRC matching scale: conditional definition. The galactic sector introduces no second microscopic VEV. Its present-epoch matching value is $a_{M0} = cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m s}^{-2}$ for $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This is a declared matching condition, not a Φ -medium postulate or a derived coefficient. For a declared a_M and baryonic mass define

$$r_* \equiv \left(\frac{G_N M_{\text{bar}}}{a_M} \right)^{1/2}; \quad (5.5)$$

for $M_{\text{bar}} = 10^{10} M_\odot$ and the matched value this is about 3.6 kpc. It is conditional HRC algebra: the internal response, gravity bridge, source/metric map and microscopic owner of a_M remain distinct questions.

L_{gal} is a conditional HRC matching length; its ECT origin is Open.

(5.6)

Why $a_M \sim cH_0$. In the present implementation, $a_{M0} \equiv cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m s}^{-2}$ is a declared Level-C matching convention. Its numerical proximity is therefore built into the definition and is neither an ECT prediction nor evidence for a common galactic–cosmological origin. Such an origin would require one microscopic action, source/force map, physical background solution and observable matching calculation. Those owners, and hence any derivation of a_M from condensate dynamics, remain Open (OP-galactic; Section 17.1).

Intermediate scale v_2 and the electroweak matching. If the ECT condensate undergoes a second ordering step, the intermediate scale is fully fixed by low-energy matching:

$$v_2 = \left(\sqrt{2} G_F \right)^{-1/2} \approx 246.22 \text{ GeV}, \quad (5.7)$$

where $G_F = 1.1663785(6) \times 10^{-5} \text{ GeV}^{-2}$ is the current PDG-2026 electroweak-review input [32]. The resulting $v_2 = 246.21967 \text{ GeV}$ is standard electroweak matching, not an ECT prediction. This is not a free parameter: any deviation would give wrong W and Z masses. A structural candidate mechanism for the second transition $O(3) \rightarrow O(2)$ is discussed in Appendix M; it remains an open problem of ECT.

Energy hierarchy and separate HRC matching length. Only the two energy-dimension quantities enter the hierarchy

$$\phi_0 \sim \bar{M}_{\text{Pl}} \gg v_2, \quad \frac{v_2}{\phi_0} \approx 10^{-16}. \quad (5.8)$$

Separately, $L_{\text{gal}} \equiv r_* = \sqrt{G_N M_{\text{bar}}/a_M}$ is a conditional HRC matching/crossover length; its fitted use is Level C, while the gravity bridge and origin of a_M remain Open. No dimensionally meaningful ordering between L_{gal} and the energies above is implied.

The hierarchy problem in the ECT context. The ratio $v_2/\phi_0 \approx 10^{-16}$ is the ECT counterpart of the standard gauge hierarchy problem. ECT does not solve this problem in the current formulation. Under the adopted common-medium Level-B programme hypothesis, v_2 is treated as the matching scale of a possible second ordering step within the same condensate rather than as a fundamental VEV of an independent field (Appendix M). The existence, action and dynamics of that second step remain Open. If it has a dynamical origin—for instance through RG running of a supplied effective orientation coupling from ϕ_0 down to v_2 (Fig. 2)—the large hierarchy could in principle be accommodated logarithmically rather than by fine-tuning. This is a structural possibility, not a derived result: the explicit RG calculation within ECT has not been performed and remains part of OP-EW. Any claim that ECT resolves the hierarchy problem would be premature.

Structural status of v_2 and its open-problem dependencies. In the present formulation, v_2 is a matched intermediate scale, not a first-principles condensate prediction. What remains open is not the low-energy value $v_2 = (\sqrt{2}G_F)^{-1/2}$ itself, but the dynamical origin of the second ordering step within the same condensate. A dimensional-analysis argument by itself cannot decide which mechanism, if any, owns the hierarchy $v_2/\phi_0 \sim 10^{-16}$. Polynomial or rational combinations of the primary ECT parameters ($\alpha, \beta, \lambda, \mu, u_0, \phi_0$) may merely relocate the hierarchy into a small dimensionless coefficient unless a symmetry, protection theorem or parameter measure is supplied; conversely, such additional structure is a counterexample to any blanket polynomial no-go claim. The present work supplies no naturalness measure and proves no uniqueness theorem. Furthermore, any renormalisation-group-based derivation would require β -functions of any supplied gauge completion, whose fields and normalisations are not yet derived from condensate parameters. In that sense, progress on OP-EW is structurally tied to the larger open programme of constructing a local gauge sector (OP-SM). The phase–orientation alignment scenario of Appendix M is one candidate route, not an exhaustive or selected mechanism; its realisation remains Open.

Exact reparameterisation and unresolved OP-EW-scale. The present parameter list does not select $v_2/\phi_0 \sim 10^{-16}$. Possible future mechanisms include protected small parameters, dimensional transmutation, non-polynomial dynamics and phase–orientation constructions; all remain Open.

We do not solve OP-EW here or reduce it uniquely to one mechanism. The observed hierarchy can be written, as an exact matched reparametrisation, as

$$v_2 = \phi_0 e^{-\mathcal{J}_{EW}}, \quad \mathcal{J}_{EW} \equiv \ln(\phi_0/v_2) \approx 36.8, \quad (5.9)$$

where the numerical value follows from $\phi_0 \simeq \bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and $v_2 = (\sqrt{2}G_F)^{-1/2} \approx 246.22$ GeV (5.7), taken in their present matched form.

Equation (5.9) is only a logarithmic change of variables. It neither reduces the hierarchy problem nor identifies a natural, preferred, or order-one microscopic input. A physical derivation would have to provide a protected secondary operator, its complete local gauge and matter representations, the relevant beta functions or nonperturbative dynamics, threshold matching, and a stability/radiative closure. None of these owners follows from P1–P7. Accordingly no particular scale-generation mechanism, including the candidate phase–orientation constructions discussed later, is selected here.

Positioning relative to existing approaches. Dimensional transmutation as a route to the electroweak hierarchy has a long history: classical scale invariance with radiative breaking [33], conformal extensions of the Standard Model [34], walking technicolor [35, 36], and composite Higgs from coset structures [37]. The novelty of the ECT route, *if* it can be closed, is that the running/critical sector is not introduced as an external technicolor or conformal sector but emerges from the phase–orientation structure of the Euclidean condensate. The closest condensed-matter analogue is the $A \leftrightarrow B$ transition between ordered phases of superfluid ^3He [20], where two distinct broken-symmetry patterns coexist within a single underlying condensate.

Discriminant with respect to the standard hierarchy problem. ECT reframes the hierarchy problem in a way that differs sharply from the Standard Model. In the Standard Model, the electroweak scale is the vacuum expectation value of an independent Higgs field, while the Planck scale is an external gravitational input. In the current programme, ϕ_0 and m_σ belong to the real-scalar model, while S_0 , the TT gravitational scale and the electroweak scale belong to separately supplied or matched closures. A common microscopic origin is a hypothesis to be tested, not a derived fact. Thus the primary hierarchy question is no longer “why are gravity and electroweak physics unrelated scales?” but rather “how does one condensate generate a secondary low-energy ordering step $v_2 \ll \phi_0$?” This is still open, but it is a different and structurally narrower problem.

The energy hierarchy and the separate HRC matching programme lead to three distinct open problems:

OP-grad (gradient condensate). What is the dynamical mechanism by which the gradient condensate $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$ emerges from the bare action P3? Dimensional analysis (5.4) is consistent, but a first-principles derivation remains open. A three-notion sharpening (local field equation; stationary-density magnitude selection; fluctuation-tensor matching) and the matching question $M(\alpha, \beta)$ are recorded in the dynamic-motivation subsection; completion remains open.

OP-EW (electroweak hierarchy and locking programme). Why does a second condensate scale arise at $v_2 \ll \phi_0$, and how does the reduced orientation-sector transition extend to the full electroweak-like generator structure? The old single OP-EW problem is decomposed below into OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, and OP-EW-matter. The reduced $O(3) \rightarrow O(2)$ candidate is described in Appendix M; the Hopf-fibered diagonal-locking upgrade is described in Appendix N. Both remain structural programmes, not derived results.

OP-galactic (IR closure). What determines a_M and the corresponding L_{gal} from a microscopic ECT calculation? The relation $a_M \sim cH_0/(2\pi)$ is a Level-C matched comparison/input. Its coefficient, scalar-to-AQUAL source/force map, and condensate origin are Open.

Falsifier for the ECT hierarchy programme. The proposed common-origin hierarchy programme would fail if any of the following were established: (i) the effective gravitational scale required a microscopic amplitude parametrically unrelated to ϕ_0 ; (ii) a named future completion predicted a P4 pole or validity scale and that derived scale failed its prespecified matching test; (iii) the electroweak scale v_2 could only be introduced as the vacuum expectation value of an entirely independent sector unrelated to the original condensate; or (iv) the galactic scale and the cosmological background scale required a second independent IR medium rather than branch variation of the same condensate. Conversely, an actual derivation tying $\phi_0, m_\sigma, S_0, M_G$ and secondary matching scales to one reduced action would support the one-condensate hypothesis; no such derivation currently exists.

Programme-level hypotheses of the hierarchy architecture. The present ECT hierarchy picture lists four programme-level possibilities, not current predictions or preferred expectations. First, the P3 homogeneous radial curvature, a future action-owned P4 pole/validity scale, the physical tensor normalisation and the Planck energy are distinct owners. A completed reduction may test a matching among them; the current framework establishes no such correlation. Second, any successful electroweak completion should generate v_2 as a secondary ordering or matching scale of the same medium, not as a fundamental second vacuum unrelated to ϕ_0 . Third, any future derivation of the HRC matching scale should connect it to the same medium rather than introduce a third fundamental vacuum expectation value; that connection is presently Open. Fourth, a completed hierarchy solution inside ECT should reduce the number of independent dimensionful inputs relative to the Standard Model plus GR, not increase them.

Consistency programme for gravitational and quantum scales. ECT supplies a one-medium research architecture in which the scalar characteristic speed is controlled by the ordered branch, while the two reduced-loop action candidates and G_N are conditional matching programmes. Mapping either $\Sigma_\theta t_0^{\text{EFT}}$

or $\Sigma_\theta t_{0,\text{bal}}^{\text{eff}}$ to the operational S_0 , and then matching $S_0 \leftrightarrow \hbar$, requires a universality theorem, and G_N exists here only through weak-field matching inside a separately supplied gravity completion (Appendix O); the condensate-to-tensor vertex and normalisation are Open. At the current stage, however, their explicit parametrisations are not yet fully reduced to an identical set of microscopic variables, because the relation between the gradient condensate scale u_0 and the bare potential data remains open (OP-grad). The relevant matching relations are:

$$G_N^{\text{nat}} = \frac{1}{8\pi M_G^2}, \quad t_0^{\text{EFT}} = \frac{c_\perp K_\theta}{2m_\phi^2}, \quad (5.10)$$

where M_G , K_θ , and m_ϕ belong to different completion/matching layers. Here the gravitational relation is in natural units. Only after the physical tensor residue and source vertex, clock/light map and any scalar response are fixed may it be restored as

$$G_N^{\text{SI}} = \frac{\hbar c^5}{8\pi [E_{M_G}(\text{J})]^2}, \quad E_{M_G}[\text{J}] \equiv M_G[\text{GeV}](1\text{GeV})_{\text{J}}. \quad (5.11)$$

The c in this SI conversion is the measured conversion speed; it is not an unmatched use of c_{char} . The symbol E_{M_G} is reserved for this response-matching scale and must not be confused with the Diósi–Penrose self-energy E_G used in Sec. 26.1. In the simplest supplied phase EFT one estimates $K_\theta \sim \phi_0^2$ and $m_\phi = \sqrt{2\lambda} \phi_0$, where $\phi_0 = \sqrt{\mu^2/\lambda}$ is the bare field VEV; Section 5.6). Neither G_N nor either loop coefficient is derived from the other. The operational S_0 , either candidate-owner map and its identification with $\hbar$ remain parametric/Open, while M_G needs an independent nonzero TT completion before the first relation becomes physical. Their common microscopic origin is therefore an architectural hypothesis to be tested, not an established consistency relation.

Status summary:

Statement	Status	Comment
Supplied P4 gradient magnitude u_0	Input; UV/cutoff role Open	P4 owns no pole, correlation length or EFT breakdown scale
$P(X) \not\equiv (\partial n)^2$	A	Theorem 3.1
$\kappa_n = \mathcal{C}_n u_0^2; M_G^2 \stackrel{?}{=} c_M \kappa_n$	A for orienta- tion coefficient / Open tensor bridge	Equal dimensions do not derive a tensor owner
$M_G = \bar{M}_{\text{Pl}}$ (weak-field matching)	B	Section 5.4
$\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}, \zeta_\phi > 0, \zeta_\phi = O(1)$	Adopted Level B scale ansatz / micro- scopic explana- tion Open	Independent of the still-Open tensor bridge
$u_0 \sim \phi_0 \cdot m_\phi$: dimensional self-consistency	B	Eq. (5.4); OP-grad
$v_2 = (\sqrt{2}G_F)^{-1/2}$ as IR matching constraint	Constraint	If second transition exists
Galactic HRC response and scale	A/B/C/Open by layer	Response and BTFR slope algebra are A inside supplied models; BTFR is B after the declared bridge; bridge and a_M ownership Open; data comparison C
$a_M \sim cH_0/(2\pi)$: matched comparison/input	C / origin Open	Coefficient and microscopic map Open (OP-galactic)

Statement	Status	Comment
G_N and the two loop candidates share a proposed one-medium matching programme	Open ownership	Eq. (5.10) displays only the fixed-core branch; the balanced branch additionally requires $\mathcal{T}_{\text{loop}}$. No common owner, universality map or physical tensor vertex is derived
$\mathfrak{so}(3) \cong \mathfrak{su}(2)$: structural compatibility with EW-like second transition	A	Algebra only
Reduced $O(3) \rightarrow O(2)$ transition as orientation-sector analogue	B/C	Two broken directions only; not full EW counting
Gradient condensate from bare action	Open (OP-grad)	
Hopf-fibered diagonal locking $SU(2)_{\text{orient}} \times U(1)_{\theta} \rightarrow U(1)_{\text{diag}}$	Candidate geometry	Three broken directions at pre-gauge/global level; dynamics not derived (App. N)
Origin of compact $U(1)_{\theta}$ as EW-locking participant	Open (OP-EW-locking.0)	Effective phase sector exists; EW role not derived
First-principles derivation of v_2	Open (OP-EW-scale)	The logarithmic identity $\mathcal{J}_{\text{EW}} = \ln(\phi_0/v_2)$ supplies no generating dynamics
Local chiral gauge completion, hypercharge, g, g' , Weinberg angle	Open (OP-EW-gauge)	Not implied by Hopf geometry alone
Radiative protection of σ_2	Open (OP-EW-naturalness)	Goldstone protection does not protect radial mode
First-principles derivation of a_M	Open (OP-galactic)	

5.2 Parameter hierarchy by level of description

Level 1. Bare micro-parameters (P3). The potential $V(\Phi)$ contains two parameters: $\mu^2 > 0, \lambda > 0$, with bare radial stiffness:

$$m_{\sigma}^2 = V''(\phi_0) = 2\mu^2, \quad \phi_0 = \sqrt{\mu^2/\lambda}. \quad (5.12)$$

These characterise the microscopic field dynamics only and do not derive the P4 gradient datum or a physical ordered vacuum.

Level 2. Supplied ordered-chart parameters (P4). Postulate P4 introduces $u_0 > 0$, the gradient scale $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$. This is an independent parameter; its relation to ϕ_0 and λ from the bare theory requires additional matching (OP-grad).

Level 3. Conditional ordered-chart EFT parameters. The effective kinetic tensor contains $\alpha > \beta > 0$ and m_{eff}^2 , where $m_{\text{eff}}^2 \neq m_{\sigma}^2$ in general.

Level 4. Coherent-sector parameters (BR1). Within a supplied compact-phase EFT (BR1), winding supplies only the integer label. A fixed-core route defines ι_0^{EFT} by evaluating the phase-gradient functional at a supplied L_{core} ; an alternative positive-tension route defines $\iota_{0,\text{bal}}^{\text{eff}}$ by a conditional minimum (Section 5.6). The 4D owners, route selection, cross-sector universality and map to operational S_0 remain Open. The separate calibration $S_0 \leftrightarrow \hbar$ is retained as a Level B physical matching, not as a microscopic derivation.

5.3 Dimensionless scalar slope and physical-speed map

Status: Level A mathematics for $\hat{c}_$ in the supplied scalar EFT; $\hat{c}_* = 1$ is a Level-B unit-slope coefficient benchmark; $c_{\text{char}} = c$ is an adopted Level-B scalar-clock calibration; photon and tensor cone matching remains Open.*

From the dispersion relation (3.18):

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta}. \quad (5.13)$$

This is a strict EFT consequence; it requires neither quantum formalism nor a gravitational sector.

Physical clock and observational matching. Equation (5.13) fixes only the common-length slope. After an independent conversion speed c_u and clock lapse $\mathcal{N}_\Phi$ are supplied, $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$. The equality $c_{\text{char}} = c$ is adopted as the scalar-clock calibration. The further equality $c_\gamma = c_{\text{char}}$ requires the missing photon owner, while GW170817 constrains an independently missing tensor principal symbol [12]. The benchmark coefficient choice $(\beta, \alpha) = (1, 2)$ combines the Level-B unit-slope benchmark $\hat{c}_* = 1$ with the convention $\beta = 1$; a common field rescaling cannot select the ratio α/β . The parameters (α, β) are EFT parameters not derivable from the bare micro-action P3 in the current formulation.

5.4 Conditional tensor scale and Newton's constant

The orientation coefficient $\kappa_n = \mathcal{C}_n u_0^2$ has mass dimension two. The fixed-map theorem shows that it does not normalise a helicity-2 field in the currently displayed director channel. Introduce instead M_T as the coefficient of a *separately supplied* healthy tensor action. In that conditional EFT the standard weak-field matching is

$$G_N^{\text{nat}} = \frac{1}{8\pi M_T^2}. \quad (5.14)$$

This relation defines the matched M_T once the physical metric and vertex are specified. A microscopic bridge $\kappa_n \rightarrow M_T^2$ is Open; neither dimensional analysis nor an induced orientation term supplies it.

5.5 Radial-mode scales and the correlation length

The homogeneous P3 potential owns the local curvature

$$m_{\Phi, \text{rad}}^2 := V''(\phi_0) = 2\mu^2. \quad (5.15)$$

For that homogeneous P3 fluctuation, the Euclidean propagator has the large-distance behaviour $G_{\Phi, \text{rad}}(r) \propto e^{-m_{\Phi, \text{rad}} r} / r^{3/2}$, so its correlation length is $\xi_{\text{P3}} = m_{\Phi, \text{rad}}^{-1}$ within P3 (Appendix P). This result does not own a fluctuation about the independently postulated P4 gradient background.

A P4 pole and correlation length require a stationary P4 action, boundary and constraint data, and the constrained Hessian

$$\mathcal{H}_{\text{P4}} = \left. \frac{\delta^2 S_{\text{P4}}}{\delta \Phi \delta \Phi} \right|_{\bar{\Phi}_{\text{P4}}}, \quad \xi_{\text{P4}}^{-1} := m_{\text{P4}, \text{pole}}. \quad (5.16)$$

None is presently supplied. Thus $m_{\text{P4}, \text{pole}} = m_{\Phi, \text{rad}}$ and $\xi_{\text{P4}} = \xi_{\text{P3}}$ are Open matching hypotheses, not identities.

The exact P3 relation $m_{\Phi, \text{rad}} = \sqrt{2\lambda} \phi_0$ does not fix a Planck-scale mass or Planck-length correlation scale because λ is an independent owner. It neither establishes a Planck-heavy P4 particle nor follows from the reduced phase-loop coefficient. The latter is dimensionless; its physical match is $\Sigma_\theta \iota_0^{\text{EFT}} = \hbar$ with independent, Open Σ_θ . Consequently Eq. (5.26) alone fixes neither λ nor either radial pole. The separate P3, P4 and secondary electroweak radial statuses are developed in Section 8.3.

5.6 Dimensionless loop coefficients and the Open physical action normalisation

Status: three-layered with two explicitly distinct Layer-2 closures. Layer 1 is Level-A variational mathematics for a dimensionless reduced functional in a supplied $U(1)$ -extended phase model. Layer 2a defines the dimensionless fixed-core coefficient ι_0^{EFT} after a loop length is supplied. Layer 2b defines the distinct dimensionless balanced coefficient $\iota_{0,\text{bal}}^{\text{eff}}$ after a positive line tension is supplied. At $n = 1$ and $L_{\text{core}} = L_{,1}$ they differ by a factor of two and must not be conflated. The compact phase, carrier, profile, tension and common four-dimensional owner remain Level-B/Open. Layer 3 introduces a separate action-valued normalisation Σ_θ ; only products such as $\Sigma_\theta \iota_0$ may be matched to the operational S_0 and then to $\hbar$. This is calibration, not a parameter-free derivation.*

A central question for ECT is whether the physical universal quantum of action can be derived from the coherent branch rather than supplied. The present framework does not settle that question. It instead exhibits two dimensionless reduced-model coefficients: the phase-only fixed-core ι_0^{EFT} and the tension-balanced $\iota_{0,\text{bal}}^{\text{eff}}$. Neither is an action. A physical Euclidean phase-loop action requires $S_{\theta,E}^{\text{phys}} = \Sigma_\theta \mathcal{J}_{\theta,E}$ with a finite positive owner-specific action-dimension constant $\Sigma_\theta > 0$. Positivity preserves the reduced stationary points and minima; it does not by itself define a statistical suppression factor. If a positive Euclidean loop measure is later supplied, it needs an independent denominator $\Sigma_{\text{PI},\theta} > 0$ and the dimensionless exponent $\kappa_\theta \mathcal{J}_{\theta,E}$, where $\kappa_\theta = \Sigma_\theta / \Sigma_{\text{PI},\theta}$. Neither this denominator nor its equality to S_0 is derived. Reusing Σ_θ in the Lorentzian phase action is a separate same-sector continuation/matching input, not a consequence of winding. Only then can $\Sigma_\theta \iota_0$ be compared with the operational S_0 and the observed $\hbar$. What follows separates the exact reduced mathematics from these Open owner and matching maps. Appendix Q collects the periodicity, fixed-core, and action-normalisation chain with the same status separation.

Phase single-valuedness fixes integer winding at fixed loop data, but it does not supply the compact phase sector, transverse profile, core length, line tension, operational action unit or universal quantum of action.

Layer 1: compact phase and the variational bound (strict within the reduced $U(1)$ -extended phase-loop EFT). *Coherent phase variable.* In the smooth coherent branch (BR1), the collective state is characterised not only by an amplitude ρ but also by a phase-like variable θ . Because the coherent state is physically single-valued, this phase is defined only modulo a full winding:

$$\theta \sim \theta + 2\pi. \quad (5.17)$$

It should be noted that the real scalar field Φ of P3 does not by itself carry a compact $U(1)$ phase variable. The extension to $\Phi_{\text{eff}} = \rho e^{i\theta}$ is an additional modelling step for the coherent charged sector. Accordingly, the periodicity $\theta \sim \theta + 2\pi$ and the associated winding number are strict statements within the $U(1)$ -extended phase EFT, not direct topological consequences of the supplied director orbit $O(4)/O(3) \simeq S^3$ (which has $\pi_1(S^3) = 0$).

Dimensionless reduced phase functional. Conditional on a supplied phase sector and amplitude freeze-out, write

$$\mathcal{J}_{\theta,E}[\theta] = \frac{K_\theta}{2} \int d^4X \delta^{AB} \partial_A \theta \partial_B \theta, \quad S_{\theta,E}^{\text{phys}} := \Sigma_\theta \mathcal{J}_{\theta,E}, \quad (5.18)$$

where $\mathcal{J}_{\theta,E}$ is the positive Euclidean reduced loop functional, dimensionless in the printed natural-unit normalisation, and Σ_θ is an independent action-valued coefficient. The retained Gaussian phase closure assumes the same supplied stiffness $K_\theta = K_\theta^{\text{eff}} > 0$; zero is degenerate, while a negative value does not define the positive Euclidean functional used in this layer. In the simplest quartic bare-potential realisation one expects $K_\theta \sim \phi_0^2$, where $\phi_0 = \sqrt{\mu^2/\lambda}$. Throughout this section and Appendix AS, K_θ is an abbreviation for the same supplied coherent-branch stiffness denoted K_θ^{eff} in Part III; it is not an independently adjustable bare coefficient. The owner map used below is therefore $K_\theta^{\text{ID}} = K_\theta^{\text{eff}} \mathcal{A}_\perp$. If a microscopic reduction produces a different coefficient, the fixed-core and balanced-loop constructions must be treated as a different model rather than silently identified with the Part III branch.

One-dimensional reduction and the transverse three-volume weight. For a supplied closed one-dimensional Euclidean carrier the four-dimensional phase action reduces to an effective one-dimensional functional only after integrating its localised profile over the three transverse coordinates. This geometry is a codimension-three curve or worldline; it is not, by this notation alone, a spatial vortex tube. We encode this reduction in the effective one-dimensional stiffness

$$K_\theta^{\text{1D}} \equiv K_\theta \mathcal{A}_\perp, \quad \mathcal{A}_\perp = c_\perp \xi_{\text{core}}^3, \quad [K_\theta^{\text{1D}}] = \text{GeV}^{-1}, \quad (5.19)$$

where $\xi_{\text{core}} \sim m_\phi^{-1}$ is the radial-core scale and $c_\perp = \mathcal{O}(1)$ is a profile-dependent geometric coefficient of the transverse three-volume weight that is *not* fixed at the present EFT level (Appendix AS). This step is a dimensionally consistent reduced-loop closure, not a first-principles derivation of the transverse profile (Level B).

Sharp lower bound on the dimensionless carrier functional (Cauchy–Schwarz inequality). Consider a supplied closed one-dimensional carrier γ of proper length L in the phase EFT with winding number $n[\gamma] = \frac{1}{2\pi} \oint_\gamma \partial_\ell \theta d\ell$. The dimensionless reduced 1D functional along the loop, built on the reduced one-dimensional stiffness (5.19), is

$$\mathcal{J}_{\text{loop}}[\theta] = \frac{K_\theta^{\text{1D}}}{2} \oint_\gamma d\ell (\partial_\ell \theta)^2. \quad (5.20)$$

By the Cauchy–Schwarz inequality applied to the constant function $f = 1$ and $g = \partial_\ell \theta$:

$$\oint d\ell (\partial_\ell \theta)^2 \geq \frac{1}{L} \left(\oint d\ell \partial_\ell \theta \right)^2 = \frac{(2\pi n)^2}{L}, \quad (5.21)$$

with equality if and only if $\partial_\ell \theta = \text{const}$ along the loop (uniform winding). Therefore:

$$\boxed{\mathcal{J}_{\text{loop}}[\theta] \geq \frac{2\pi^2 K_\theta^{\text{1D}} n^2}{L}, \quad \text{equality at uniform winding.}} \quad (5.22)$$

This is a strict variational result inside the supplied reduced model: the minimiser of the dimensionless loop functional at fixed length and winding is the uniform-gradient configuration, and its minimum scales as n^2/L .

Important structural remark. Equation (5.22) provides a sharp lower bound for each fixed L , but because the bound falls as $1/L$, $\mathcal{J}_{\text{loop}} \rightarrow 0$ as $L \rightarrow \infty$. The pure phase topology alone does *not* yield a nonzero global minimum of the reduced functional; additional physical input is needed even to define a distinguished dimensionless coefficient, and Σ_θ is still required to obtain an action.

Layer 2: supplied loop scales and dimensionless coefficients (Level A mathematics / Level B–Open ownership). *Why the fixed- L bound is not yet a nonzero global minimum.* Equation (5.22) gives the minimum of the dimensionless reduced functional in a fixed length-and-winding sector, but because the bound falls as $1/L$, it does not define a nonzero global reduced minimum when the loop length is allowed to vary. A physical action requires the independent positive factor Σ_θ .

Supplied carrier-length benchmark. To obtain a distinguished fixed-core dimensionless EFT coefficient, one must supply the proper length or period of an elementary closed carrier. A core-sized benchmark may be written

$$L_{\text{core}} \sim 2\pi \xi_{\text{core}} = \frac{2\pi}{m_\phi}. \quad (5.23)$$

This is a Level B/Open convention, not a consequence of topology or of the straight-vortex profile. The local correlation length motivates a transverse scale but does not select a spatial circumference, Euclidean proper period or worldline ensemble. In particular, a localised-carrier expansion does not fail merely because $L \gg \xi_{\text{core}}$; curvature and profile corrections instead become important when the radius of curvature approaches the core width. A physical interpretation of L therefore requires a separate embedding, ensemble and stabilisation mechanism.

Core-width calculation and carrier guard. The estimate $\xi_{\text{core}} \sim m_\phi^{-1}$ follows in a supplied straight global-vortex problem for the (ρ, θ) system. This is a codimension-two profile analogy; without an explicit embedding and dimensional reduction it does not own the codimension-three d^3y worldline coefficient used above. For a rectilinear vortex with unit winding, the radial profile $\rho(r)$ satisfies $\rho(0) = 0$ and $\rho(r) \rightarrow \phi_0$ at $r \gg m_\phi^{-1}$, with the core size fixed by the balance between gradient energy and potential energy at $\xi_{\text{core}} \sim m_\phi^{-1}$. For a closed loop of circumference $L \sim 2\pi\xi_{\text{core}}$, curvature corrections are $O(1)$ and do not change the scaling estimate. It does not identify the circumference of an elementary closed loop or a nonzero global action minimum. The order statement $L_{\text{core}} = O(\xi_{\text{core}})$, with $L_{\text{core}} = 2\pi\xi_{\text{core}}$ as the explicit benchmark used above, remains a Level-B/Open closure choice pending the full 4D closed-loop variational problem.

Defined fixed-core dimensionless coefficient. Evaluating the fixed- L lower bound (5.22) at the supplied elementary-loop size with $n = 1$ gives

$$l_{\text{elem}} := \frac{2\pi^2 K_\theta^{\text{1D}}}{L_{\text{core}}} = \pi c_\perp \frac{K_\theta}{m_\phi^2}, \quad S_{\text{elem}}^{\text{phys}} := \Sigma_\theta l_{\text{elem}}. \quad (5.24)$$

Define

$$l_0^{\text{EFT}} \equiv \frac{l_{\text{elem}}}{2\pi} = \frac{c_\perp K_\theta}{2m_\phi^2}. \quad (5.25)$$

Dimensionless form and parameter compression. In the simplest quartic realisation (in the convention $V(\Phi) = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4$ of P3, so that $K_\theta \sim \phi_0^2$ and $m_\phi^2 = 2\lambda\phi_0^2$) the elementary-loop scale becomes

$$l_0^{\text{EFT}} \sim \frac{c_\perp}{4\lambda}, \quad (5.26)$$

independent of ϕ_0 inside this supplied profile and normalisation. It is not an ECT prediction because K_θ , $c_\perp$, the loop length and the real-to-complex reduction are not derived. Consequently the conditional physical matching $\Sigma_\theta l_0^{\text{EFT}} = \hbar$ constrains only the product $\Sigma_\theta c_\perp / (4\lambda)$. Because Σ_θ and $c_\perp$ are not independently derived, it neither fixes λ nor constitutes a parameter-free prediction. This is a candidate dimensionless EFT coefficient after the fixed-core closure is supplied, not a topologically selected or globally minimal physical action scale. Only $\Sigma_\theta l_0^{\text{EFT}}$ is action-valued (Level A reduced algebra / Level B route / Open owner).

Notation. The dimensionless fixed-core representative is $l_{\text{elem}}^{(\text{fixed core})} = 2\pi l_0^{\text{EFT}}$, while its physical action would be $S_{\text{elem}}^{\text{phys}} = \Sigma_\theta l_{\text{elem}}^{(\text{fixed core})}$. The supplied length and coefficient are not fixed by topology, and no global variational minimum over arbitrary loop lengths is asserted.

Optional tension-balanced closure route (additional supplied physics). A different, constructive way to remove the $L \rightarrow \infty$ zero-infimum is to supply the same reduced one-dimensional carrier with a positive reduced inverse-length coefficient $\mathcal{T}_{\text{loop}}$. For $K_\theta^{\text{1D}} > 0$, $\mathcal{T}_{\text{loop}} > 0$ and nonzero winding n , the supplied dimensionless 1D model is

$$\mathcal{J}_n(L) = \mathcal{T}_{\text{loop}}L + \frac{2\pi^2 K_\theta^{\text{1D}} n^2}{L}, \quad L > 0, \quad [K_\theta^{\text{1D}}] = L, \quad [\mathcal{T}_{\text{loop}}] = L^{-1}. \quad (5.27)$$

In the phase route the physical coefficients are $K_{\theta, \text{phys}}^{\text{1D}} = \Sigma_\theta K_\theta^{\text{1D}}$ and $\mathcal{T}_{\theta, \text{phys}} = \Sigma_\theta \mathcal{T}_{\text{loop}}$, with dimensions action times length and action per length, respectively. The reduced coefficients do not supply the action unit. It is strictly convex for every $L > 0$ and diverges at both ends of the allowed interval. Hence, at fixed nonzero winding and within this one-coordinate ansatz, it has a unique global minimum with respect to L :

$$L_{*,n} = \pi|n| \sqrt{\frac{2K_\theta^{\text{1D}}}{\mathcal{T}_{\text{loop}}}}, \quad \mathcal{J}_{n,\text{min}} = 2\pi|n| \sqrt{2K_\theta^{\text{1D}} \mathcal{T}_{\text{loop}}}, \quad (5.28)$$

with equal tension and phase-gradient contributions at $L_{*,n}$. For $n = 0$, however, $\mathcal{J}_{n=0}(L) = \mathcal{T}_{\text{loop}}L$ has infimum zero as $L \rightarrow 0^+$ and no finite interior minimum. It is useful to define the balanced-model scale

$$l_{0,\text{bal}}^{\text{eff}} \equiv \frac{\mathcal{J}_{1,\text{min}}}{2\pi} = \sqrt{2K_\theta^{\text{1D}} \mathcal{T}_{\text{loop}}}. \quad (5.29)$$

This symbol is deliberately distinct from the phase-only fixed-core ι_0^{EFT} . For the elementary sector $n = 1$, if one chooses $L_{\text{core}} = L_{*,1}$ and evaluates only the phase-gradient term used in the fixed-core definition, then

$$\iota_0^{\text{EFT}}|_{L_{\text{core}}=L_{*,1}} = \frac{1}{2}\iota_{0,\text{bal}}^{\text{eff}}. \quad (5.30)$$

Thus, at the same elementary $n = 1$ length, the balanced total coefficient is twice the phase-only coefficient. This factor is a normalisation consequence of the two equal contributions and is not, by itself, evidence of double counting.

The minimisation in Eqs. (5.27)–(5.30) is Level A mathematics inside the supplied 1D model. Its ECT interpretation is only a Level B/Open route: the compact $U(1)$ sector, transverse profile, positive length-independent tension, the correct worldline or worldsheet embedding, declared ensemble, shape modes, curvature, collapse, splitting and self-intersection terms, unwinding barriers and backreaction must be obtained from one 4D reduced action. In particular, $\mathcal{T}_{\text{loop}}$ and $K_{\vartheta}^{\text{1D}}$ cannot be fitted as unrelated knobs, and no equation above derives $\hbar$ from P1–P6. The full ownership and falsification conditions are collected in Appendix AS.

Post-order candidate constitutive hypothesis CDE- Φ : cohesive defect elasticity. CDE- Φ is not a pre-physical property, a new frozen postulate or an action owner. It is a branch-EFT acceptance criterion: after an ordered branch and its dynamics have already been declared, at least one stationary localised profile must heal to the ambient state, have a positive reduced line coefficient together with a finite positive physical-action prefactor, and have a positive projected metric on each retained physical normalisable modulus. In plain language, a completed branch must be able to hold a localised disturbance together, and both extending it and varying a genuine internal coordinate must cost positive action. This criterion mentions neither gravity nor quantisation and supplies no compact modulus, topology, rigidity, measure or action quantum.

The actual owner is the full tuple consisting of an explicit four-dimensional completion action, its physical state and boundary data, a stationary profile, one fixed vacuum subtraction and regulator, and the constraint/gauge/frame/zero-mode quotient used to define physical fluctuations. Suppose that tuple supplies completion fields Ψ^I (which may include an independent orientation or other collective field and are not silently identified with the single real P3 scalar Φ), a stationary localised profile $\Psi_*^I(y; \vartheta)$ around a one-dimensional Euclidean carrier $X^A(s)$, and a genuine compact normalisable collective coordinate $\vartheta \sim \vartheta + 2\pi$ after the physical quotient. A controlled collective-coordinate reduction [38] can then have the dimensionless reduced form

$$\begin{aligned} \mathcal{S}_{E,\text{red}}^{\text{sub}} &= \int ds \left[\mathcal{T}_{\text{loop}} + \frac{K_{\vartheta}^{\text{1D}}}{2} (D_s \vartheta)^2 + \frac{\kappa_b}{2} \mathcal{K}^2 + \dots \right], \\ S_{E,\text{phys}}^{\text{sub}} &= \Sigma_{\vartheta} \mathcal{S}_{E,\text{red}}^{\text{sub}}, \quad \Sigma_{\vartheta} > 0. \end{aligned} \quad (5.31)$$

where $\mathcal{K}$ is the carrier curvature, D_s contains any derived normal-bundle or internal connection, and the ellipsis is deliberately non-exhaustive: twist, torsion, mixed curvature–modulus, nonlocal and higher-derivative terms must be retained whenever the owner permits them. Before profile relaxation, the two leading coefficients are schematically

$$\mathcal{T}_{\text{loop}} = \int d^3y [\mathcal{L}_E(\Psi_*; \mathfrak{s}) - \mathcal{L}_E(\Psi_{\text{vac}}; \mathfrak{s})], \quad (5.32)$$

$$K_{\vartheta}^{\text{1D}} = \int d^3y G_{IJ}(\Psi_*; \mathfrak{s}) \partial_{\vartheta} \Psi_*^I \partial_{\vartheta} \Psi_*^J, \quad (5.33)$$

where $\mathfrak{s}$ records the state and boundary data. The physical coefficient is obtained only after constrained profile relaxation, gauge reduction and the corresponding Schur complement. Equations (5.32) and (5.33) show how the two parameters could be computed from one action rather than postulated independently. They are a derivation template, not evidence that the minimal real-scalar P3 action has the required profile or compact modulus.

The transverse d^3y measure is crucial. It describes a codimension-three curve in four-dimensional Euclidean space—most naturally the worldline of a point-like texture on a selected three-dimensional

slice. A spatial $U(1)$ vortex loop persisting along the selected w direction instead traces a two-dimensional worldsheet, has a transverse d^2y reduction, and would make the bracketed expression an energy or an action per unit w -duration. These two embeddings are not interchangeable. Here $L = \int_{\Gamma} ds$ is the proper length or Euclidean period of the closed worldline, not the spatial radius of the texture. Varying L is meaningful only in a declared closed-worldline/proper-time ensemble with its einbein or parametrisation convention, boundary period and measure fixed. For propagation over a prescribed duration, or when L is a gauge modulus, L is external and the stationary-length calculation is not a physical size prediction. Current-carrying string constructions provide an external example in which the string carrier and a longitudinal internal mode are separate ingredients [39]; they do not derive either ingredient in ECT.

The best-matched ECT-native owner candidate is therefore a stabilised three-dimensional orientation texture whose worldline carries a compact, normalisable internal modulus. The carrier charge and the longitudinal winding n are then distinct: the texture degree labels the spatial sector and obstructs smooth unwinding only while the target, nonzero-order-parameter domain, regularity and boundary assumptions hold, whereas n labels winding of an independently established internal modulus along a closed worldline. Existence, finite size, lifetime and a barrier remain separate gates. This route remains Open. A two-derivative texture collapses by Derrick scaling for a static positive-energy profile with the stated regular boundary conditions and no additional charge, the exact-gradient ordered branch can restrict the orientation image to a contractible hemisphere, and a stabilising higher-gradient or independent-orientation completion has not been derived. The familiar stable-texture models [40, 41, 42, 43, 44] are mathematical analogies and existence proofs in other actions, not evidence that P1–P6 owns this branch.

The modulus route defines a separate dimensionless conditional coefficient

$$\iota_{0,\text{bal}}^{(\vartheta)} \equiv \sqrt{2K_{\vartheta}^{\text{1D}} \mathcal{T}_{\text{loop}}}. \quad (5.34)$$

It may be identified with the earlier BR1 phase coefficient $\iota_{0,\text{bal}}^{\text{eff}}$ only if the owner calculation proves $\vartheta = \theta|_{\Gamma}$, $K_{\vartheta}^{\text{1D}} = K_{\theta}^{\text{1D}}$ and compatible subtraction and quotient conventions. No such map is presently derived. Neither coefficient has dimensions of action.

What follows without additional matching. After additionally freezing the proper-time ensemble, discarding the ellipsis in Eq. (5.31), setting $\kappa_p = 0$, taking constant positive coefficients and working in a trivialised connection sector with $\oint_{\Gamma} D_s \vartheta ds = 2\pi n$, the modulus route has the dimensionless two-term truncation

$$\mathcal{J}_n^{(\vartheta)}(L) = \mathcal{T}_{\text{loop}} L + \frac{2\pi^2 K_{\vartheta}^{\text{1D}} n^2}{L}. \quad (5.35)$$

Inside that supplied truncation,

$$L_{*,n} = |n| L_{*,1}, \quad \mathcal{J}_{n,\text{min}} = |n| \mathcal{J}_{1,\text{min}}, \quad D_s \vartheta|_* = \text{sgn}(n) \sqrt{\frac{2\mathcal{T}_{\text{loop}}}{K_{\vartheta}^{\text{1D}}}}, \quad \mathcal{J}_n''(L_{*,n}) = \frac{2\mathcal{T}_{\text{loop}}}{L_{*,n}}. \quad (5.36)$$

Thus the optimal modulus gradient is independent of $|n|$, while the length and reduced minimum scale linearly with $|n|$. The same linearity makes one $|n|$ loop exactly marginal against $|n|$ separated unit-winding loops in this truncation; it does not predict which channel is populated.

Conditional physical matching, not a derivation. An independent finite positive owner-specific action-valued constant $\Sigma_{\vartheta} > 0$ converts this reduced functional to a candidate physical action,

$$S_{n,\text{phys}}^{(\vartheta)} = \Sigma_{\vartheta} \mathcal{J}_n^{(\vartheta)}. \quad (5.37)$$

Its equality to the phase-sector coefficient Σ_{θ} is an additional Open compatibility condition, like the identification $\vartheta = \theta|_{\Gamma}$. Writing $h_p \equiv 2\pi\hbar$, the bending-free elementary matching programme is therefore

$$\Sigma_{\vartheta} \iota_{0,\text{bal}}^{(\vartheta)} \xrightarrow{\mathcal{M}_{\text{bal}}} S_0 \xleftrightarrow{\mathcal{M}_{\hbar}} \hbar \implies \Sigma_{\vartheta} \mathcal{J}_{1,\text{min}} = 2\pi\hbar = h_p, \quad 2K_{\vartheta}^{\text{1D}} \mathcal{T}_{\text{loop}} = \left(\frac{\hbar}{\Sigma_{\vartheta}} \right)^2. \quad (5.38)$$

The reduced minimisation is Level A mathematics inside the supplied T1 truncation. The coefficient owner, the map $\mathcal{M}_{\text{bal}}$ and the identification of the modulus route with the phase route remain Open; the

empirical calibration $\mathcal{M}_{\hbar}$ is adopted at Level B. The equation is restricted to the bending-free $n = 1$ branch: a material positive κ_b shifts the minimum and requires a separately declared matching.

If the same reduction also yields $\kappa_b > 0$, a circular carrier is selected at fixed length by the curvature bound. For a circle,

$$\begin{aligned}\mathcal{J}_n^{\text{circ}}(L) &= \mathcal{T}_{\text{loop}}L + \frac{2\pi^2(K_{\vartheta}^{\text{1D}}n^2 + \kappa_b)}{L}, \\ L_{*,n}^{\text{circ}} &= \pi\sqrt{\frac{2(K_{\vartheta}^{\text{1D}}n^2 + \kappa_b)}{\mathcal{T}_{\text{loop}}}}, \\ \mathcal{J}_{n,\min}^{\text{circ}} &= 2\pi\sqrt{2\mathcal{T}_{\text{loop}}(K_{\vartheta}^{\text{1D}}n^2 + \kappa_b)}.\end{aligned}\tag{5.39}$$

Hence bending rigidity can stabilise shape but also shifts the original scale; it cannot be invoked while retaining the two-term numerical value unchanged. In the restricted space of nonempty smooth one-component closed curves, the $n = 0$ branch has a round-circle minimum $L_{*,0}^{\text{circ}} = \pi\sqrt{2\kappa_b/\mathcal{T}_{\text{loop}}}$ and $\mathcal{J}_{n=0,\min}^{\text{circ}} = 2\pi\sqrt{2\kappa_b\mathcal{T}_{\text{loop}}}$. The empty vacuum and core-fill-in paths are absent from that curve space, so this is not a proof of a stable or metastable physical loop. The minimum is no longer linear in $|n|$. With one identical positive per-carrier κ_b and all omitted interactions set to zero, the connected-circle truncation obeys $\mathcal{J}_{n,\min}^{\text{circ}} < n\mathcal{J}_{1,\min}^{\text{circ}}$ for $n > 1$. This is exact same-core energetic subadditivity in the supplied curve model, not a physical binding verdict: carrier existence, core relaxation, $V_{\text{sep}}(D)$, reconnection, splitting and the decay barrier remain Open.

PES-R enters only after a candidate carrier, Lorentzian continuation, physical modes, state, noise and record protocol have been supplied. It can then test dynamical admissibility and persistence. It does not select $\mathcal{T}_{\text{loop}}/K_{\vartheta}^{\text{1D}}$, prove marginality, map the action-valued product $\Sigma_{\theta}l_{0,\text{bal}}^{\text{eff}}$ to $\hbar$, or produce a universal $e^{-2\pi|n|}$ occupation law. Such a law would additionally require a nucleation saddle, determinant, zero-mode measure, state and an independently derived measure scale.

External calibration, not a gap equation. Equation (5.25) defines a dimensionless reduced coefficient. Only the action-valued product $\Sigma_{\theta}l_0^{\text{EFT}}$ can be compared with $\hbar$. Imposing $\Sigma_{\theta}c_{\perp}K_{\theta}/(2m_{\varphi}^2) = \hbar$ is an external calibration after the inputs are supplied; it is not a self-consistency or BCS-like gap equation because no internal stationary equation determines these parameters. The content of this programme is not that $\hbar$ or G_N has been computed from the minimal medium. Rather, the same ordered-medium architecture contains conditional matching routes for an EFT action scale and for the gravitational coupling; their physical ownership remains Open. At the current stage, however, their explicit parametrisations are not yet fully reduced to an identical set of microscopic variables, because the relation between the gradient condensate scale u_0 and the bare potential data remains open (OP-grad). The relevant matching relations are:

$$l_0^{\text{EFT}} = \frac{c_{\perp}K_{\theta}}{2m_{\varphi}^2}, \quad G_N^{\text{nat}} = \frac{1}{8\pi M_G^2},\tag{5.40}$$

where K_{θ} and m_{φ} are parameters of the supplied phase EFT, while M_G and the physical c_{char} require independent tensor/clock matching. They are not all derived functions of $(\phi_0, \lambda, \alpha, \beta)$. ECT therefore poses a common-origin consistency test between gravitational and quantum completions. The displayed relations alone do not show that they originate in the same ordered condensate.

Common-origin hypothesis and its obstruction. P3 contains one real microscopic field Φ , the reduced functional $\mathcal{J}_E^{\text{P3}}$, the separately normalised physical action $S_E^{\text{P3,phys}}$, and a possible independently normalised measure. The compact phase, physical fermion/gauge sectors and nonzero TT gravitational sector used later are additional effective closures, not decompositions already derived from that field.

The bare P3 measure, if supplied, is

$$Z_{\text{P3}} = \int \mathcal{D}\Phi \exp[-\kappa_{\text{P3}}\mathcal{J}_E^{\text{P3}}[\Phi]] = \int \mathcal{D}\Phi \exp\left[-\frac{S_E^{\text{P3,phys}}[\Phi]}{\Sigma_{\text{PI,P3}}}\right], \quad \kappa_{\text{P3}} = \frac{A_{\text{P3}}}{\Sigma_{\text{PI,P3}}} > 0.\tag{5.41}$$

The dimensionless reduced functional, the physical classical-action prefactor A_{P3} and the path-integral denominator $\Sigma_{PI,P3}$ are distinct owners. The displayed classical action does not determine the measure, and the measure does not prove equality with independently renormalised effective-sector normalisations. None of A_{P3} , $\Sigma_{PI,P3}$ or κ_{P3} is identified with Σ_θ , Σ_ϑ , the operational S_0 , $\hbar$, or the protocol entropy production $\Sigma[\gamma]$. Such relations would require a derived real-to-complex reduction, canonical normalisation and coarse-graining/RG matching; those steps remain Open.

Formal analytic-continuation bookkeeping (conditional). The formal analytic continuation $w_E \rightarrow \pm i x_L^0$ (with $x^0 = w$ the ordered-branch time-length coordinate; physical SI units enter only through the separate map (3.23)) relates Euclidean coherent weights to Lorentzian phase evolution. It should not be confused with the real parametrisation $x^0 = w$, which is the primary route to the hyperbolic field equations. At the present stage the underlying ordered-branch hyperbolicity has already been fixed independently by $\alpha > \beta > 0$.

Parameter-counting diagnostic (conditional). In its simplest bare-potential realisation, ECT has three structurally independent bare/EFT parameters $(\phi_0, \lambda, \alpha)$ at $\beta = 1$. Several observational calibration relations would be needed to fix them, and they are not all owned by the present action:

- the Level-B unit-slope coefficient benchmark $\hat{c}_* = 1$ fixes $\alpha = 2\beta$, while $c_{\text{char}} = c$ is the separate adopted scalar-clock calibration after the clock/lapse convention is fixed; physical photon/tensor owners and cone equality remain Open (Section 3.5);
- in natural units the response-matched tensor scale obeys $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$. Only after the physical tensor residue/source, clock/light and scalar-response matchings are supplied may one convert it to $G_N^{\text{SI}} = \hbar c^5 / [8\pi (E_{M_G}[\text{J}])^2]$, where $E_{M_G}[\text{J}] = M_G[\text{GeV}](1 \text{ GeV})_{\text{J}}$ and c is the measured light speed. This is a Level B/Open matching route, not an identification of M_G with an orientation stiffness or a mixed-unit formula involving an unmatched c_{char} (Section 5.4);
- the route-specific test $\Sigma_\theta \iota_0^{\text{EFT}} \rightarrow S_0 = \hbar$ constrains the dimensionless product $(\Sigma_\theta/\hbar)c_\perp/(4\lambda)$ (Eq. (5.26)); it does not fix $c_\perp/\lambda$ without the independent action normalisation. The balanced route instead tests the independent combination $\Sigma_\theta \sqrt{2K_\theta^{\text{ID}} \mathcal{T}_{\text{loop}}} \rightarrow S_0 = \hbar$. They are alternative diagnostics unless a compatibility map is derived.

This list is therefore a bookkeeping map, not a closed three-equation derivation of the observed constants. The matching $\Sigma_\theta \iota_0^{\text{EFT}} = \hbar$ would constrain

$$\Sigma_\theta \frac{c_\perp K_\theta}{2m_\phi^2} \approx \hbar.$$

In the quartic realisation this is a constraint on $\Sigma_\theta c_\perp/(4\lambda)$, not on λ alone. Since neither Σ_θ nor $c_\perp$ is derived, no parameter-free determination or fine-tuning conclusion follows.

Layer 3: adopted calibration to the reduced Planck constant $\hbar$ (Level B; candidate-owner maps Open). The preceding relations define two distinct reduced-model candidates. Their possible relation to the operational action unit and to the observed reduced Planck constant is the three-map programme

$$\Sigma_\theta \iota_0^{\text{EFT}} \xrightarrow{\mathcal{M}_{\text{fc}}} S_0, \quad \Sigma_\theta \iota_{0,\text{bal}}^{\text{eff}} \xrightarrow{\mathcal{M}_{\text{bal}}} S_0, \quad S_0 \xleftarrow{\mathcal{M}_{\hbar}} \hbar. \quad (5.42)$$

The two candidate-owner arrows remain Open; the final $S_0 = \hbar$ arrow is an explicit Level B empirical calibration rather than a definition or a parameter-free theorem. In particular, at $L_{\text{core}} = L_{*,1}$ the two candidates differ by a factor of two, so they cannot both be identified with one physical S_0 without an additional change of normalisation or dynamics.

In ordinary quantum theory, $\hbar$ is the quantity that renders the action dimensionless in the phase factor $\mathcal{A} \sim e^{iS/\hbar}$. The two supplied reduced models assign analogous candidate roles to distinct coefficients:

$$S_0^{\text{candidate}} \in \{\Sigma_\theta \iota_0^{\text{EFT}}, \Sigma_\theta \iota_{0,\text{bal}}^{\text{eff}}\} \xrightarrow{\mathcal{M}_{\text{owner}}} S_0 \xleftrightarrow{\mathcal{M}_{\hbar}} \hbar. \quad (5.43)$$

This equation names two proposed upstream candidates and two matching stages; it derives neither a nonzero coefficient nor physical universality.

Clarification of loop terminology. The supplied fixed-core evaluation is denoted $\iota_{\text{elem}}^{(\text{fixed core})}$; it is not called $S_{\text{loop,min}}$ because the tension-balanced model now has the distinct, genuine one-coordinate minimum $\mathcal{J}_{n,\text{min}}$ of Eq. (5.28). The fixed-core object is only the dimensionless reduced phase-gradient functional evaluated on an elementary winding representative at a supplied length; only $\Sigma_\theta \iota_{\text{elem}}^{(\text{fixed core})}$ has action dimension. It is not a global minimum or an arbitrary closed condensate configuration. The proposed $\hbar$ matching refers to a declared candidate coefficient and is not derived by phase topology.

Why this is not yet Level A. To promote the identification to Level A, one would need to derive from ECT not just a number with units of action, but the full structural role of $\hbar$:

- (i) derive one candidate-owner map in Eq. (5.42) and then a universal phase weighting $\mathcal{A} \sim e^{iS/S_0}$ for arbitrary histories, not only special loops;
- (ii) a composition law for coherent amplitudes;
- (iii) canonical commutators normalised by the same operational S_0 ;
- (iv) ultimately, reconstruction of the Hilbert-space structure and the Born rule.

This is the programme of the Quantum Sector (Section 20), not a one-line identification. Within that programme, partial results are already available (Sections 21.1–23.3), but the full reconstruction is not yet complete.

Cross-sector action-scale programme (Open). The later conditional quantum and response constructions use an operational action-unit slot denoted S_0 . This unadorned symbol is not a third loop derivation and is not shorthand for either reduced-loop candidate. The proposed owner maps $\Sigma_\theta \iota_0^{\text{EFT}} \leftrightarrow S_0$ and $\Sigma_\theta \iota_{0,\text{bal}}^{\text{eff}} \leftrightarrow S_0$ are separate Open matchings. The operational symbol is reused in:

- (i) the coherent-history phase weight $\mathcal{A} \sim e^{i\Sigma_\theta \mathcal{J}_{\theta,L}/S_0}$ (Section 22.1);
- (ii) the Schrödinger-type wave reconstruction (22.7);
- (iii) the natural canonical commutator candidate (23.18);
- (iv) the decoherence architecture (Section 26.1) and the later Born/measurement programme (Section 30.4).

This reuse is bookkeeping, not a universality theorem. The real-scalar, phase, canonical, decoherence and probability sectors have not been derived as one reduced measure with one renormalised normalisation; Born weights in particular require an independently admitted probability measure. Equality of the operational S_0 across those conditional sectors and either reduced-loop-to-operational owner map are independent Open matching tests. The final equality $S_0 = \hbar$ is instead the adopted Level-B physical calibration; its microscopic derivation and cross-sector universality remain Open. Likewise only the scalar cone is owned; a common photon/matter/fermion/tensor cone is Open.

Falsifiable content of action-scale universality. If the loop-sector quantization scale, the wave-sector action normalisation, the canonical commutator normalisation, and the Born/decoherence-sector normalisation were found to require *mutually inconsistent* fundamental action scales rather than a single coherent-branch scale S_0 , then the ECT programme of action-scale universality would fail. Such an outcome would falsify this proposed common-origin completion. The empirical universality of $\hbar$ is the target to reproduce, not evidence that the present ECT reductions already share one microscopic normalisation.

Relation to the classical limit and variational dynamics. The reduced-loop candidates are compatible with the supplied variational phase models, but a generic Lorentzian phase weight uses the operational S_0 only after one of the Open owner maps $\mathcal{M}_{fc}$ or $\mathcal{M}_{bal}$ has been supplied. At the fundamental level, realised configurations are required by DP to be stationary points of $S_E[\Phi]$ (Section 6.6); DP does not select which stationary point is realised. Inside a supplied Lorentzian oscillatory functional integral, with its measure, contour, state and stationary-phase hypotheses fixed, $|\Delta S| \gg S_0$ makes relative phases rapidly oscillatory and can yield saddle dominance. At $|\Delta S| \sim S_0$ interference can remain relevant. This is conditional asymptotic mathematics, not a derivation of classicality from the Euclidean variational principle; actual decoherence requires the separate open-system owners stated below.

Coarse graining and emergent classicality. A supplied coherent phase weight can remain interference-sensitive at resolution comparable to the operational S_0 after an explicit owner map. Coarse graining or $N_{eff} \gg 1$ does not by itself wash out phases. Decoherence requires an identified subsystem–environment coupling, environment state, influence/noise kernel, partial trace, and a timescale on which off-diagonal terms are suppressed. Until those owners are specified, loss of access to S_0 -sensitive information is a Level B/Open mechanism proposal rather than a derived consequence of multiplicity.

Path-integral hint. The most natural amplitude assignment suggested by the coherent branch structure is

$$\mathcal{A} \sim e^{iS/S_0}. \quad (5.44)$$

Equation (5.44) uses the operational action slot only after an explicit candidate-to- S_0 owner map has been supplied. It should be read as a conditional path-integral phase ansatz, not as a derivation of the quantum measure, the Born rule, or the physical value of S_0 .

Comparison with familiar quantisation ideas. *Bohr–Sommerfeld quantisation* [45]: in semiclassical mechanics one has $\oint p dq = 2\pi n \hbar$. The compact phase supplies the integer winding label n only. A Bohr–Sommerfeld action integral requires a separately owned canonical pair or action–angle map; it does not follow from the winding condition alone.

Flux quantisation and superfluid winding [20]: this is perhaps the closest analogy. In superfluids, compact phase plus single-valuedness forces quantised circulation. ECT uses the same structural logic, applied to the action-phase relation of the coherent branch rather than to a material superfluid.

Path-integral phase assignment [46]: standard quantum theory uses $e^{iS/\hbar}$ and takes $\hbar$ as primitive. The supplied fixed-core phase EFT contains a distinguished coefficient ι_0^{EFT} ; it does not derive its nonzero value or cross-sector universality, and it only motivates testing its identification with $\hbar$.

Status summary.

Statement	Status	Comment
<i>Layer 1: reduced $U(1)$-extended phase-loop EFT</i>		
Coherent compact phase variable in the $U(1)$ -extended phase EFT	B	Modelling extension of real Φ

Statement	Status	Comment
Single-valuedness on closed loops	A	Topological consistency within that EFT
Nontrivial winding sectors	A	Integer winding number
Sharp lower bound on loop action at fixed length	A	Eq. (5.22); Cauchy–Schwarz
Uniform winding minimises the fixed- L action	A	Equality case of (5.21)
<i>Layer 2: elementary-loop scale</i>		
Elementary loop size $L_{\text{core}} \sim 2\pi/m_\phi$	B/Open	Supplied closed-loop length; straight-vortex core width does not fix it
$\iota_0^{\text{EFT}} = c_\perp K_\theta / (2m_\phi^2)$	B	Reduced-loop closure; transverse coefficient $c_\perp$ open
Tension-balanced minimum $\mathcal{J}_{n,\min} = 2\pi n \sqrt{2K_\theta^{\text{1D}}\mathcal{T}_{\text{loop}}}$	A/B/Open	Exact minimum in the supplied positive two-term 1D model; common 4D owner, carrier/ensemble stability and ECT ownership Open
ι_0^{EFT} and G_N common-origin hypothesis	Open	Phase and TT/gravity owners are independent in the current model
External calibration $\Sigma_\theta \iota_0^{\text{EFT}} \approx \hbar$ constrains only the supplied parameter product	B/Open	Not a gap equation; Σ_θ and the physical owner map are Open
<i>Structural bridges</i>		
Single-field-to-effective-sector normalisation	Open	P3 does not fix independently renormalised phase/gauge/tensor sectors
Formal continuation of a supplied phase normalisation	B/Open	Physical state, RP/OS and clock map required
Parameter-counting diagnostic for $\Sigma_\theta \iota_0^{\text{EFT}} = \hbar$	B/Open	Matching input, not a necessary final calibration
<i>Layer 3: identification with $\hbar$</i>		
The physical ratio $ \Delta S /S_0$ conditionally separates saddle- and phase-sensitive regimes	B/Open	Requires an owner map $\Delta S = \Sigma_\theta \Delta \mathcal{J}$, a supplied Lorentzian oscillatory measure and the operational S_0 ; decoherence separately requires coupling, state, kernel, trace and timescale
Phase factor $\mathcal{A} \sim e^{iS/S_0}$ after an explicit candidate-to-operational owner map	B/Open	Conditional path-integral hint; owner map, measure/state and physical normalisation Open
$\Sigma_\theta \iota_0^{\text{EFT}}$ as a candidate physical action for matching to $\hbar$	B/Open	Reduced coefficient and action normalisation have separate Open owners

Statement	Status	Comment
Route-dependent owner maps $\Sigma_\theta \iota_0^{\text{EFT}}$ or $\Sigma_\theta \iota_{0,\text{bal}}^{\text{eff}} \rightarrow S_0$, followed by $S_0 = \hbar$	Open owner maps / Level-B calibration	The two candidates differ in general; no candidate-to-operational map is parameter-free and their simultaneous use requires an explicit compatibility condition. The final equality is the separately adopted calibration
Universal amplitude composition / Born rule	C (open)	Later quantum programme

5.7 Summary table: parameters and constants

Parameters and constants.			
Quantity	Physical meaning	Status	Basis / note
μ^2, λ	Bare micro-parameters P3	Micro-parameters	P3; not directly linked to P4
u_0	Gradient condensate amplitude	P4 (postulated)	OP-grad for derivation from P3
α, β	Kinetic tensor parameters	EFT parameters	Theorem 3.2
$\hat{c}_*$	Dimensionless scalar principal slope in common-length coordinates	Level A (supplied EFT)	Eq. (5.13)
$c_u, \mathcal{N}_\Phi$	Conversion speed and physical clock lapse	Open/matched	Eq. (3.23)
c_{char}	Physical scalar characteristic speed after the clock map	Conditional	$c_u \hat{c}_* / \mathcal{N}_\Phi$
$\alpha = 2\beta$	Scalar/phase cone relation; photon matching conditional and tensor cone Open	Level A/B/Open	Eq. (4.11); §4.7
Four structural universality programmes (§11.1)	B/Open	Independent cone, scale, variational and conservation closure tests	
Physical identification $c_{\text{char}} = c$	Level B (matching)	§4.7 + [12]	
$m_\sigma^2 = 2\mu^2$	Bare radial stiffness	Level A in P3	$V''(\phi_0)$ from P3
m_{eff}^2	EFT fluctuation mass	EFT parameter	$\neq m_\sigma^2$ in general

Quantity	Physical meaning	Status	Basis / note
$\xi_{P3} = m_{\Phi, \text{rad}}^{-1}$	Homogeneous P3 Gaussian correlation length	Level A in the supplied free P3 model	Does not determine a P4 pole, correlation length or cutoff
M_G	Gravitational stiffness	Level B (matching)	$G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$
G_N	Newton's constant	Matched	Observational input
$\bar{M}_{\text{Pl}}$	Reduced Planck mass	Level B (matching)	$\bar{M}_{\text{Pl}}^2 = M_G^2$
v_2	Electroweak matching scale	Constraint / Level B	$v_2 = (\sqrt{2} G_F)^{-1/2} \approx 246 \text{ GeV}$; if second transition exists
$L_{\text{gal}} \equiv r_*$	Conditional HRC matching/crossover length	Definition A after supplied source/scale; bridge Open; fitted use C	$\sqrt{G_N M_{\text{bar}}/a_M}$; ECT origin of a_M Open
S_0	Operational action-unit slot for conditional wave, canonical and response closures	B/Open owner map (§5.6)	Fixed-core ι_0^{EFT} and balanced $\iota_{0, \text{bal}}^{\text{eff}}$ are distinct upstream candidates; their owner maps, absolute derivation and cross-sector universality remain Open, while $S_0 = \hbar$ is the explicit Level B calibration
$\hbar$	Reduced Planck constant used by the adopted action-unit calibration	Level B calibration / microscopic owner Open (§5.6 and 21.2)	$S_0 = \hbar$ is adopted; its absolute microscopic derivation and cross-sector universality remain Open

Similarly, in the electroweak and fermionic sections Φ_{ew} denotes the embedded low-energy bookkeeping doublet and $H \equiv \Phi_{\text{ew}}/\sqrt{2}$ its canonically normalised form. Both are distinct from the fundamental bare condensate field Φ of the postulate layer. Yukawa operators and the vacuum convention $H^0 = (v_2 + h)/\sqrt{2}$ are written through H .

6 Branching Structure of ECT

This section describes the branching structure of ECT into the geometric macroscopic branch and the coherent branch, establishes the variational and Noether backbone of the framework, and collects the corresponding foundational consequences. The underlying symmetry and variational structures are predominantly Level A, while the explicit reduction to branch-specific effective descriptions and some physical interpretations remain Level B or open.

The six working postulates P1–P6, together with the P4-supplied ordered chart, candidate excitation inventory, scale hierarchy, and the supplied smooth coherent-sector closure BR1, define a candidate condensate framework for two distinct but potentially compatible effective-branch programmes. Their stationary physical states, selection and common reduction remain Open.

6.1 Two classes of observables

In the minimal P1–P6 sector ECT contains a single fundamental field Φ (P3). The separately supplied P7 internal module is an extension rather than part of this scalar-only basis. Postulate P4 introduces the ordered branch and, through the broken-phase EFT analysis, the orientation variable n_A and the kinetic tensor (3.11), producing the geometric causal sector. The supplied smooth coherent-sector closure BR1 introduces the collective phase variable θ with the winding condition (2.8).

These two steps are structurally distinct and produce two qualitatively different classes of observables:

1. *Phase-insensitive observables*: probe the coarse-grained geometric and stiffness response of the vacuum. Primary variables: n_A , K^{AB} and the scalar characteristic g_{AB}^{eff} ; a physical tensor coefficient M_G belongs only to a supplied completion. These constitute the *geometric branch*.
2. *Phase-sensitive observables*: probe coherence, winding structure, and non-trivial loop configurations. Primary variables: Φ_{eff} , θ , winding numbers, S_0 . These constitute the *coherent branch*.

6.2 The geometric branch (Macroscopic Physics)

Primary variables: n_A , K^{AB} , the scalar characteristic g_{AB}^{eff} and long-wavelength orientation-stiffness fields. A physical tensor normalisation M_G is an Open completion variable.

This branch contains in embryo: the scalar Lorentzian common-length principal slope $\hat{c}_*$ (Level A inside the supplied scalar EFT) and a conditional clock map to c_{char} , plus a programme for a physical tensor completion and G_N matching (Open); the long-wavelength sector from which gravity, cosmology, and astrophysical phenomenology are developed. The mode content and effective dynamics of this geometric branch are developed in Section 4, while the physical-gravity completion programme is discussed in Section 8.5 and the conditional macroscopic development in Section 11.

6.3 The coherent branch (Quantum Sector)

Primary variables: Φ_{eff} , θ , winding condition (2.8), S_0 .

This branch will later develop: identification of S_0 with observable $\hbar$; canonical quantisation relations; connection between phase single-valuedness, loop quantisation, and wave dynamics; decoherence, probabilistic interpretation, and the quantum limit of ECT. Its effective phase dynamics and action-scale structure are developed in Sections 5.6 and 20.

6.4 Why branching is necessary

The branching is an effective organisational proposal:

1. P4 supplies the ordered-gradient chart and orientation coordinate used by the geometric programme.
2. BR1 supplies the smooth coherent phase closure with integer winding.
3. A common microscopic descent of both effective programmes from one physical condensate state remains Open.
4. Neither branch reduces to the other without loss of physical content.

Criterion. In the present organisation an effective sector is defined by the supplied closure and retained observables; this is not a dynamical physical selection.

6.5 Correlation, coherent interaction, and record formation

P4 supplies a preferred Euclidean direction; it does not by itself derive Lorentzian causal structure. Within an independently supplied hyperbolic scalar problem, coherent interactions may occur without a large record exponent. Common-medium correlations likewise do not by themselves imply entanglement, Bell violation or a detector outcome.

Interaction after a supplied physical completion. A time-directed interaction event requires a supplied physical metric/clock, matter vertices and state. A Euclidean correlation or boundary-value dependence is not itself an interaction, measurement event or stochastic transition.

Common-medium correlation in the coherent branch. A common medium is compatible with a separately supplied non-factorisable coherent amplitude, but common support neither derives nor forces one. This is an architectural possibility; the physical amplitude, state, apparatus map and operational correlators remain Open.

The conditional reconstruction. For a supplied scalar principal operator, independently supplied coefficients satisfying $\beta(\beta - \alpha) < 0$ give a hyperbolic classification. A physical metric, clock and realised branch remain Open. Environmental records may dephase alternatives after their coupling is specified, but they neither select the signature nor turn a Euclidean orientation into a physical arrow.

Comparison with standard quantum mechanics. Standard quantum theory remains the quantitative owner of the laboratory amplitudes discussed here. ECT presently supplies a Euclidean- first interpretation and candidate microscopic programme, not a replacement calculation for those experiments. Decoherence is a separate reduced-state process in which unresolved degrees of freedom acquire records of alternatives. A pairwise exponent can suppress off-diagonal terms for a specified channel, but it neither changes the metric signature nor selects one classical event. The ECT-specific interaction and record vertices remain sector-dependent/Open.

6.6 Variational principle and Noether structure

Status: layered. Variational stationarity is the defining DP input. Noether structure from action symmetries is Level A. A branch-restricted action inherits stationarity and Noether structure only under an explicitly derived variation- and symmetry-compatible restriction or controlled elimination. Separately supplied branch EFTs own their own equations and currents at Level B/Open rather than inheriting them from DP. Compatibility with the arrow of time and with the classical/quantum divide is Level B/Open. Full derivation of effective branch functionals and late-time charge reconstruction remains Open.

The variational principle is the mathematical device that supplies on-shell field equations. Within the ontology of the Φ -medium, DP states a necessary condition for a physically realised configuration; it does not select one solution from the stationary set. The action principle belongs to the same foundational *dynamical* layer as DP, not to the ontological layer of properties S1–S10, conditional property S11a or candidate property S11b. Post-order constitutive hypotheses for a supplied branch EFT are later inputs and must not be read back into that ontological layer.

From admissibility to realisation. The structural properties S1–S10, conditional S11a and candidate S11b specify what kinds of configurations are admissible: regular or defect-adjacent, homogeneous or inhomogeneous, coherent or topologically nontrivial. The dynamical principle DP then states that physically realised configurations satisfy stationarity of a local action functional compatible with the basic symmetries of the medium:

$$\delta S_E[\Phi] \Big|_{\text{on-shell}} = 0. \quad (6.1)$$

ECT thus separates two logically distinct notions:

- *admissibility* — governed by the ontology of the medium;
- *on-shell dynamics* — constrained by variational stationarity; realisation among multiple extrema remains an additional state/boundary or selection question.

The same stationarity structure can reappear in a reduced sector only after its effective action and boundary/state reduction have been constructed. Stationary-phase dominance in a supplied coherent amplitude is separate from choosing one ontically realised history (§22.2).

Noether structure. For the supplied translation-invariant reduced P3 functional, Noether's theorem [47] gives the off-shell identity

$$\tau_{P3}^{AB} := \partial^A \Phi \partial^B \Phi - \delta^{AB} \mathcal{L}_\Phi, \quad \partial_A \tau_{P3}^{AB} = -\frac{\delta \mathcal{J}_E^{P3}}{\delta \Phi} \partial^B \Phi. \quad (6.2)$$

The corresponding physically normalised current is

$$T_{P3,phys}^{AB} = A_{P3} \tau_{P3}^{AB}, \quad \partial_A T_{P3,phys}^{AB} = -\frac{\delta S_E^{P3,phys}}{\delta \Phi} \partial^B \Phi. \quad (6.3)$$

Both divergences vanish on the same P3 shell for finite constant $A_{P3} > 0$. Their normalisation is not fixed by the independent measure ratio κ_{P3} . Admissible boundary conditions are additionally required for differentiability of the generators and conservation of integrated charges. $O(4)$ symmetry similarly produces angular currents whose local divergence vanishes on the owning shell. These statements belong to P3 and do not supply a conserved P4 source or curved-space stress tensor. The full derivation, including the orientation-field sector, is given in Appendix R.

Off-shell P3 tensor diagnostic on the supplied P4 profile. The P3 canonical tensor may be evaluated algebraically on the affine datum $\partial_A \Phi_{bg} = u_0 \delta_{Aw}$:

$$\tau_{P3}^{ww}[\Phi_{bg}] = \frac{u_0^2}{2} - V(\Phi_{bg}) \quad (6.4)$$

$$\tau_{P3}^{ij}[\Phi_{bg}] = -\delta^{ij} \left[\frac{u_0^2}{2} + V(\Phi_{bg}) \right] \quad (6.5)$$

$$\tau_{P3}^{wi}[\Phi_{bg}] = 0. \quad (6.6)$$

These are components of the reduced Noether current. The physically normalised components are $T_{P3,phys}^{AB} = A_{P3} \tau_{P3}^{AB}$; the path-integral denominator $\Sigma_{P1,P3}$ does not replace that factor. and hence, pointwise and for each i with no sum,

$$\tau_{P3}^{ww}[\Phi_{bg}] - \tau_{P3}^{ii}[\Phi_{bg}] = u_0^2, \quad T_{P3,phys}^{ww} - T_{P3,phys}^{ii} = A_{P3} u_0^2. \quad (6.7)$$

This is only an off-shell substitution identity. The affine P4 profile is not a solution of the printed P3 action: $V(\Phi_{bg})$ varies and

$$\partial_A \tau_{P3}^{AB} = -\frac{\delta \mathcal{J}_E^{P3}}{\delta \Phi} \partial^B \Phi_{bg} \neq 0, \quad \partial_A T_{P3,phys}^{AB} = A_{P3} \partial_A \tau_{P3}^{AB} \neq 0$$

generically. Therefore neither the reduced components nor their A_{P3} -scaled physical normalisation is a conserved P4 background stress, vacuum source, energy density or pressure. The independent measure ratio κ_{P3} does not alter this off-shell conclusion. The former ρ_{bg}^{eff} , p_{bg}^{eff} and w_{bg}^{eff} readings are withdrawn. A physical P4 stress tensor, equation of state, Lorentzian/covariant source and bridge to gravity require a stationary P4 action and its metric/source variation and remain Open.

Preserved algebraic component proxies. For comparison with the former notation one may retain the pointwise off-shell abbreviations

$$\rho_{P3}^{proxy} := \frac{u_0^2}{2} - V(\Phi_{bg}), \quad p_{P3}^{proxy} := -\left(\frac{u_0^2}{2} + V(\Phi_{bg}) \right). \quad (6.8)$$

Where the denominator is nonzero their algebraic ratio is

$$w_{P3}^{\text{proxy}} := \frac{p_{P3}^{\text{proxy}}}{\rho_{P3}^{\text{proxy}}} = -\frac{u_0^2/2 + V(\Phi_{\text{bg}})}{u_0^2/2 - V(\Phi_{\text{bg}})}. \quad (6.9)$$

These formulae preserve the exact reduced component bookkeeping. Multiplication by A_{P3} supplies the corresponding P3 physical-current normalisation, but neither object is a conserved P4 stress tensor, physical P4 energy density, pressure or equation of state. Their downstream physical use requires the missing stationary P4 action, metric variation, state and matching map; $\Sigma_{P1,P3}$ instead owns the Euclidean measure.

Ward identities and the quantum programme. The Noether conservation laws suggest that Ward-type identities should exist in the coherent quantum sector as well. In a completed quantum reconstruction of ECT, such identities may help constrain the effective action and its symmetry structure. At present, however, neither the explicit form of these identities nor their relation to effective general covariance has been derived. This remains part of the open quantum programme.

Constrained structure of possible Lorentz violation. The ordered-branch construction suggests that Lorentz-violating effects, if present, should exhibit correlated rather than arbitrary structure: they are expected to reflect the condensate anisotropy associated with the preferred branch direction. In that sense ECT points toward a more constrained pattern than a fully general SME-style parameterisation. A precise mapping of the effective LV operator basis remains open.

Sector dependence of variational owners and physical interpretation. The fundamental real-scalar action owns its translational and rotational Noether currents. A reduced sector inherits stationarity or a current only when an explicit restriction, elimination, or effective-action construction preserves the relevant variations and symmetry. Independently supplied phase, gauge, gravity, cosmology, HRC, and open-system actions instead own their displayed equations conditionally; they do not descend from DP merely because they use variational notation. Both the operator owner and the later physical interpretation of a current or charge must therefore be identified sector by sector.

Conditional inheritance under an explicitly derived reduction. The supplied nonzero P4 vector and its $O(3)$ stabiliser permit reduced variables, but neither coarse graining nor EFT matching alone preserves the stationary set. In general,

$$\delta S_E[\Phi] = 0 \not\Rightarrow \delta S_{\text{eff}} = 0 \quad \text{without a derived, variation-compatible reduction map.} \quad (6.10)$$

Inheritance can be proved for a specified consistent restriction of the fundamental action, or for an effective action obtained by a controlled elimination whose variations and boundary data are tracked. The phase, gravity, cosmology, HRC, and other EFT actions supplied independently in this article therefore remain separate conditional owners until such a map is constructed. This distinction is Level A variational mathematics and an Open ECT descent problem (Appendix S).

Relation to the arrow of time. The fine-grained variational principle is compatible with a separately derived reduced irreversible dynamics, but compatibility is not a theorem of entropy growth. Multiplicity (S6), inhomogeneous ordering (S9), and coarse graining motivate a Level-B programme; the sign of an entropy rate still requires the physical reduced map, state, entropy functional, and the relevant contractivity or detailed-balance hypotheses (Section 3.8).

Relation to S_0 and the classical/quantum divide. Euclidean DP and a supplied Lorentzian coherent weighting are different owners. In the latter, S_0 controls the phase-difference expansion: $|\Delta S| \gg S_0$ yields the usual stationary-phase asymptotic, whereas $|\Delta S| \sim S_0$ retains interference in the supplied integral. This identifies neighborhoods of declared saddles in an integrated amplitude; it does not select an actual history, establish decoherence, or derive classicality. The Lorentzian measure, contour, state, and S_0 matching are Level B/Open inputs. The Euclidean variational principle alone derives none of them (Section 5.6).

Coarse graining and effective extremality. Coarse graining does not eliminate the fundamental variational principle; it changes the level of description. Many fine-grained configurations may map to one macrostate, which permits—but does not prove—an irreversible reduced description. A macroscopic entropy theorem must still be derived from the named channel. Some effective sectors can be written in extremal form even when their exact reduction from the bare action remains a Level-B closure.

Defect and topological sectors. Because S9 and S10 allow inhomogeneous ordering and topological sectors, defects, domain walls, and defect-adjacent configurations must also be regarded as admissible members of the overall variational framework. What changes is not whether they are governed by extremality, but whether a simple smooth effective reduction is available. Defect sectors are therefore allowed as separately supplied completions but are not derived or selected by the present smooth low-energy EFT.

Comparison with other variational frameworks. *Ordinary classical field theory [48]:* the action principle is usually taken as the formal starting point of dynamics on a pre-existing spacetime. In ECT the action defines stationary candidate configurations of a pre-spacetime medium. It does not by itself select which candidate is realised; stability, state, record and update dynamics are separate owners.

Condensed-matter effective actions [20]: in controlled condensed-matter examples, collective actions can be obtained from a microscopic substrate by an explicit restriction or elimination. ECT tests whether analogous controlled maps can be constructed from its medium; the separately supplied EFTs in this manuscript are not evidence that such inheritance has already been derived.

Stationary phase and path-integral logic [46]: for a supplied Lorentzian oscillatory path integral, S_0 controls the stationary-phase expansion around declared saddles. This is standard conditional mathematics, not an ECT derivation of the Lorentzian integral from DP. Realised-history selection, probability, and decoherence do not follow from stationarity.

Status summary.

Statement	Status	Comment
Stationarity required by DP for candidate on-shell configurations	Postulate	Necessary condition within the fundamental action; not a unique-history or state selector
P3 Noether T^{AB} : off-shell identity and conservation on P3 shell	A	P1 + P3 + DP
Noether $J^{C,AB}$: existence and conservation	A	P1 + P2 + DP
Symmetry-compatible action implies Noether structure	A	Standard variational logic
Lorentzian interpretation of translation currents as energy–momentum	B	Ordered-branch Lorentzian EFT interpretation
Background anisotropy $T_{bg}^{ww} - T_{bg}^{ii} = u_0^2$	A	Direct computation; independent of V_{bg}
Effective vacuum equation-of-state interpretation	B	Requires Lorentzian reading + background subtraction/matching

Statement	Status	Comment
Consistent restriction of a variational problem preserves extremality	A-math conditional	Requires an explicit variation-compatible restriction and boundary data
Explicit reduced branch functional S_{eff}	B/Open	Must be derived by controlled elimination or supplied as a separate EFT owner
Coherent-branch variational equation	B/Open	Owned by an explicit supplied/derived branch action; no generic descent from DP
Macroscopic source-tensor matching programme	B/Open	Requires an explicit coarse-graining/EFT map and observable matching; no controlled descent from DP is yet derived; §13.3
Constrained LV structure from condensate anisotropy	B	Structural expectation; explicit basis open
Ward identities in the coherent quantum sector	C (open)	Quantum reconstruction not complete
Conditional defect inventory compatible with a supplied variational framework	B/Open	Physical configuration space, action and admissibility rules remain to be supplied
Irreversible reduced dynamics compatible with microscopic extremality	B/Open	Requires a system/environment split, state, time orientation, coarse-graining map and reduced channel; coarse graining alone is insufficient (§3.8)
Conditional Lorentzian saddle dominance ($ \Delta S \gg S_0$)	B/Open	Requires supplied oscillatory measure, state and an explicit loop-candidate-to- S_0 owner map; not the Euclidean Laplace criterion
Full derivation of all effective branch functionals	Open	Not yet completed
Late-time charge reconstruction in all sectors	Open	Depends on gauge/quantum programme

6.7 Remark on defect sectors beyond the smooth coherent sector

The supplied coherent-sector closure BR1 restricts the present section to smooth fields on declared contours or domains where $\rho > 0$ and the phase is well defined. Integer winding can occur on a contour surrounding an excluded core or in a noncontractible domain, so “defect-free” is not implied by the winding theorem. S10 conditionally admits all sectors compatible with a separately declared physical target, configuration space and action; it does not establish those owners or identify the supplied orbit $O(4)/O(3) \simeq S^3$ with the complete physical target. Candidate configurations outside the smooth reduction include:

- *Vortex cores*: configurations with $\rho = 0$ where phase θ is undefined;
- *Domain-wall-like configurations*: discontinuous or sharply varying profiles of n_A , not protected as topologically stable walls by the π_0 of the supplied S^3 target; physical wall status remains Open;
- *Instantons*: non-trivial topological configurations in Euclidean space describing tunnelling between different vacuum sectors [8].

The smooth coherent-sector restriction selects a calculational sector and leaves Open whether an owner-complete ECT configuration space contains this richer candidate state space. For the supplied polar-vector S^3 target, $\pi_0 = \pi_1 = \pi_2 = 0$ and $\pi_3(S^3) = \mathbb{Z}$ are Level-A conditional homotopy statements; they are not physical defect or relic anti-predictions. The conditional inventory is summarised in Section 8.10.

6.8 Foundational consequences and branch-selection rules

For later use it is convenient to collect the main consequences of the core foundational framework (Sections 2–6).

1. **Ordered-branch rule.** P4 supplies an ordered-gradient analysis chart with non-zero gradient datum, $\langle \partial_A \Phi \rangle \neq 0$, whose nonzero vector has $O(3)$ stabiliser. Physical vacuum realisation and spontaneous selection remain Open.
2. **Amplitude–orientation decomposition.** On the ordered branch the condensate gradient admits the decomposition

$$\partial_A \Phi = u n_A, \quad u \geq 0, \quad n_A n_A = 1.$$

Because $Q_A = \partial_A \Phi$ is an exact one-form, u and n_A are not independent: they satisfy the integrability condition $\partial_A(u n_B) - \partial_B(u n_A) = 0$, which reduces the orientation sector to one longitudinal soft scalar at linear order (Level B; Appendix J).

3. **Uniqueness of the broken-phase kinetic tensor.** Local symmetric rank-2 tensors compatible with the residual $O(3)$ symmetry take the form

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B.$$

Here n_A is the derived ordered-branch orientation variable, not an independent microscopic field (Section 4.2).

4. **Signature classification rule.** Depending on (α, β) , the ordered-chart kinetic operator has Euclidean, degenerate, or Lorentzian signature.
5. **Macroscopic viability rule.** Only a hyperbolic branch ($\beta(\beta - \alpha) < 0$; retained component $\alpha > \beta > 0$) can serve as a Lorentzian macroscopic causal sector. Inside the displayed scalar EFT, the Level-B unit-slope coefficient benchmark $\hat{c}_* = 1$ fixes $\alpha = 2\beta$. The physical characteristic speed is instead $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$; $c_{\text{char}} = c$ is the separate adopted scalar-clock calibration and cannot by itself select $\alpha = 2\beta$. The theory identifies the Lorentzian admissibility region internally; photon/tensor cone ownership and universality remain Open.
6. **Smooth coherent-sector rule (BR1).** On a declared closed contour in the coherent infrared branch, with non-vanishing amplitude and single-valued collective field, the winding number is necessarily integer:

$$\oint_{\mathcal{C}} \partial_A \theta dX^A = 2\pi n, \quad n \in \mathbb{Z}.$$

7. **Two-branch rule.** The ordered condensate framework supports two effective developments: the geometric macroscopic branch and the coherent branch. They are organised as two candidate effective programmes within the same ordered-medium hypothesis, but retain distinct supplied closures and observables; their common microscopic descent remains Open.

These statements will be used repeatedly in the subsequent development of the gravitational, cosmological, astrophysical, gauge, particle, fermionic, and quantum sectors.

7 Gauge Symmetries, Matter Couplings, and the ECT-Specific Fifth Coupling

This section separates exact global compact-phase algebra from conditional local gauge completions and from the embedded matter-sector operator benchmark. P5 supplies deformability and relational response, while compact phase supplies global phase structure; neither establishes arbitrary local gauge

redundancy. A local $U(1)$ or non-Abelian theory additionally requires a supplied bundle or reduced quotient, connection and action, state and matter representations. The second half develops an allowed director–matter deformation and its observational status. Global algebra is Level A in its supplied model; local gauge, electroweak and matter-completion claims are conditional/Open.

7.1 Internal structures of the condensate relevant to interactions

Status: mixed structural A/B/C. The decomposition itself and the existence of the ordered-branch direction n_A are structural Level A results. This subsection does not derive the physical Standard Model gauge group. What it establishes is the structural decomposition of the ordered branch relevant to the interaction sector: (i) the compact phase sector, (ii) multiplet/non-Abelian internal structure, (iii) colour-like extensions, and (iv) the preferred condensate direction n_A as an ordered-branch geometric ingredient. The compact $U(1)$ phase route is then developed below to near-theorem level within the extended phase EFT, whereas the non-Abelian and electroweak sectors remain Level B/C embeddings.

The ordered condensate branch established in Sections 3–6 introduces not only an effective Lorentzian spacetime structure but also several distinct structures relevant to interactions. Some of them are genuine internal structures of the condensate description (phase, multiplet organisation, colour-like extension), while others arise from the geometry of the ordered branch itself. To avoid later confusion, these structures should be sharply separated from one another.

Proposition: interaction-relevant structural decomposition. Within the present ECT formulation, the ordered branch supports four conceptually distinct structures relevant to the interaction sector:

1. a *compact phase sector* of the coherent EFT,
2. *multiplet/non-Abelian internal structure*,
3. *colour-like extensions* of such internal structure,
4. a *preferred condensate direction* $n_A = \partial_A \Phi / |\partial \Phi|$.

Proof. The first three ingredients are distinguished by the type of field-space structure carried by the effective condensate description: compact phase, internal multiplet indices, and colour-like triplet organisation. The fourth ingredient is of a different nature: it is not an internal symmetry but the normalised gradient orientation defined on the P4-supplied nonzero-gradient chart. P4 supplies the preferred direction; it does not derive a physical selection event or broken-phase state. Since these four ingredients act at distinct structural levels, they must be treated separately in the interaction analysis. $\square$

Compact phase sector. As a structural low-energy representation of the coherent branch, consider the complex condensate field

$$\Phi_{\text{eff}}(X) = \rho(X) e^{i\theta(X)}. \quad (7.1)$$

Here $\theta(X)$ is the compact phase variable of the coherent branch. In the emergent complex ordered sector (Section 21.1; the minimal ECT field is real, and the phase language belongs to the two-component extension), a constant phase shift

$$\Phi_{\text{eff}} \rightarrow e^{i\alpha} \Phi_{\text{eff}} \quad (7.2)$$

leaves the effective action invariant. This compact phase route provides the cleanest and most developed entry point from condensate structure to gauge-like interactions. It is developed in Section 7.2.

Multiplet/non-Abelian structure. If the condensate is organised into internal multiplets, non-Abelian internal structures become available. In particular, the ordered branch admits the algebraic bridge $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, which makes electroweak-like embeddings structurally available. This is an algebraic and EFT-level availability statement, not yet a first-principles derivation of the physical non-Abelian gauge sector. It is developed in Sections 7.3 and 7.4.

Colour-like extension. If the condensate carries suitable triplet structure in an internal sector, an $SU(3)$ -type colour-like extension can be accommodated at the level of model building (see §40 for a candidate completion). At the present stage this remains a structural possibility rather than a derivation of QCD or confinement from the bare condensate action. This is discussed in Section 7.3.

Preferred condensate direction. The P4 chart supplies a preferred direction

$$n_A = \frac{\partial_A \Phi}{|\partial \Phi|}. \quad (7.3)$$

This quantity is not an internal gauge degree of freedom and not a multiplet index. It is the normalised-gradient orientational variable on the supplied chart. The nonzero vector has $O(3)$ stabiliser; a physical ordered state and its spontaneous selection remain Open. Later, it enters the matter sector through the ECT-specific fermion–condensate coupling. This ingredient therefore belongs to the interaction architecture, but not to the internal gauge-symmetry side of that architecture.

What is strictly established here. The present subsection does *not* derive the Standard Model gauge group. What it does establish is the structural decomposition of the interaction sector into: (i) a compact phase route, (ii) non-Abelian multiplet embeddings, (iii) colour-like extensions, and (iv) the ECT-specific preferred-direction ingredient n_A . Among these, the compact phase route is the most developed and is the only one carried below to near-theorem level.

Distinction from spacetime symmetry breaking. The supplied nonzero P4 vector carries a Euclidean coordinate index A and has $O(3)$ stabiliser inside $O(4)$. This group statement neither produces a physical ordered branch nor establishes Lorentzian structure. The internal structures discussed here act at a different level: phase, multiplet organisation, and colour-like structure belong to the effective condensate description, while n_A is a geometric datum of the supplied ordered chart. This separation between spacetime-level ordering and interaction-level structure is essential for the logic of the gauge sector. Its relation to the Coleman–Mandula logic is discussed further in Section 7.11.

Roadmap. The remainder of this section proceeds in four steps. Section 7.2 develops the compact $U(1)$ phase route. Sections 7.3 and 7.4 consider the non-Abelian and electroweak-like embeddings. The later fifth-coupling block treats the ECT-specific matter-sector deformation associated with the preferred condensate direction n_A .

Status summary.

Structure	Status	Comment
Compact phase sector	A/B	Structural $U(1)$ route in the coherent phase EFT
Non-Abelian multiplet structure	B/C	Algebraic/EFT embedding, not derivation
Colour-like extension	C	Model-level possibility only
Preferred condensate direction n_A	A	Ordered-branch geometric ingredient
Standard Model gauge group fully derived	Open	Not established here

7.2 Compact phase and conditional local $U(1)$ completion

Status: layered. Layer 1 is Level A mathematics within a supplied compact coherent-phase EFT: once the ordered branch is described by the compact phase variable θ , the Noether current, its conservation, and the homogeneous limit follow strictly. Layer 2 is conditional: P5 motivates a local-redundancy completion, but neither global compact phase nor medium character alone creates a local gauge bundle. Once a local $U(1)$ bundle and connection are supplied, bundle covariance fixes the standard compensating derivative and the two-invariant ordered-background kinetic basis; Maxwell form is the special $Z_2 = 0$ case. Layer 3 is Open for the physical $U(1)_{\text{em}}$ owner, an unbroken massless photon, the gauge-reduced state space, matter charges and vertices, and QED matching. A charged condensate can instead Higgs the candidate connection.

Connection to the coherent branch. The phase field $\theta(X)$ appearing in this subsection is the same compact variable introduced in the coherent branch (BR1). Its compactness supplies the integer winding classification used by the fixed-core construction, while the finite coefficient ι_0^{EFT} also requires the supplied core length, profile, and normalisation (Section 5.6). Its dynamics is developed further in the Quantum Sector (Section 21.1); compact winding alone does not derive a universal action quantum. The present subsection separates exact global-phase consequences from a conditional local gauge completion of that compact phase structure.

Layer 1: global $U(1)$ symmetry and the Noether current (Level A within the compact phase EFT).

Assumption for Layer 1. The coherent branch admits an effective compact phase representation $\Phi_{\text{eff}}(X) = \rho(X)e^{i\theta(X)}$, with global phase symmetry (7.2). Within that phase EFT, the following statements are strict.

Theorem 1: Noether current. For the global phase symmetry (7.2), the conserved current is

$$j^A \equiv \frac{-i}{2} (\Phi_{\text{eff}}^* \partial^A \Phi_{\text{eff}} - \Phi_{\text{eff}} \partial^A \Phi_{\text{eff}}^*). \quad (7.4)$$

Using $\Phi_{\text{eff}} = \rho e^{i\theta}$, this reduces to

$$j^A = \rho^2 \partial^A \theta. \quad (7.5)$$

Proof. Under an infinitesimal global phase shift, $\delta \Phi_{\text{eff}} = i\alpha \Phi_{\text{eff}}$, $\delta \Phi_{\text{eff}}^* = -i\alpha \Phi_{\text{eff}}^*$. Noether's theorem applied to the canonical complex kinetic structure gives (7.4). Substituting the polar form yields (7.5). The full derivation is given in Appendix T. $\square$

Corollary 1: current conservation. The Noether current satisfies $\partial_A j^A = 0$.

Proof. Direct Noether consequence of the exact global phase symmetry of the compact phase EFT. $\square$

Corollary 2: homogeneous limit. For a homogeneous phase configuration, $\partial_A \theta = 0 \implies j^A = 0$.

Proof. Immediate from (7.5). $\square$

Physical interpretation. The current (7.5) is the phase gradient weighted by the condensate density. In the ECT interpretation, this suggests that charge-carrying excitations are associated with phase-topological or phase-gradient structures of the ordered medium. This is structurally analogous to supercurrents in superfluids and superconductors [20], though not physically identical to a material condensate.

Layer 2: local $U(1)$ completion and gauge connection. The following three conditions are separate completion assumptions. P5 motivates their investigation but does not derive them:

(L1) locality of the effective description — compatible with P3, but a local effective bundle must still be supplied;

- (L2) only phase gradients, not the absolute phase, enter local observables — global shift invariance from P2 does not imply invariance under arbitrary local rephasings;
- (L3) local phase redefinitions are treated as redundancies of the effective description — this is an additional local-bundle hypothesis, not the direct content of P5.

Conditional on L1–L3, the standard gauge-completion construction below proceeds.

Proposition 2: local phase redundancy requires a compensating connection. Under the local phase transformation $\Phi_{\text{eff}}(X) \rightarrow e^{i\alpha(X)}\Phi_{\text{eff}}(X)$, the ordinary derivative is not covariant:

$$\partial_A \Phi_{\text{eff}} \rightarrow e^{i\alpha(X)}(\partial_A \Phi_{\text{eff}} + i(\partial_A \alpha)\Phi_{\text{eff}}). \quad (7.6)$$

To restore covariance one must introduce a compensating field A_A and define

$$D_A = \partial_A - ieA_A, \quad (7.7)$$

together with the transformation law

$$A_A \rightarrow A_A + \frac{1}{e}\partial_A \alpha. \quad (7.8)$$

Then $D_A \Phi_{\text{eff}} \rightarrow e^{i\alpha(X)}D_A \Phi_{\text{eff}}$.

Proof. This is the unique first-order local compensating completion of the ordinary derivative under local phase redefinition. Any compensating field must remove the inhomogeneous term $i(\partial_A \alpha)\Phi_{\text{eff}}$. At leading derivative order this requires a connection transforming as (7.8), which then yields the covariant derivative (7.7). See Appendix T for details. $\square$

Remark on status. Proposition 2 is an exact conditional statement about a supplied local $U(1)$ bundle. Proposition 3 classifies the leading parity-even operator after that bundle, a nondegenerate gauge-reduced Hessian and its kinetic coefficient are supplied. Neither proposition derives the bundle, its physical reduced state space, or the electromagnetic identification from P1–P6.

Proposition 3: leading ordered-background gauge kinetic basis. Define the gauge-invariant field strength

$$F_{AB} = \partial_A A_B - \partial_B A_A. \quad (7.9)$$

On a supplied ordered background with director n^A , the local parity-even gauge-invariant terms quadratic in the field strength and containing at most two derivatives span two independent invariants:

$$I_1 = F_{AB}F^{AB}, \quad I_2 = (n^A F_{AB})(n_C F^{CB}), \quad S_{\text{gauge}} = -\frac{1}{4} \int d^4X [Z_1 I_1 + 2Z_2 I_2]. \quad (7.10)$$

Proof. Gauge invariance excludes any mass term $A_A A^A$. At two-derivative order, antisymmetry of F_{AB} and residual $O(3)$ symmetry reduce the parity-even quadratic contractions to I_1 and I_2 up to total derivatives. They are independent whenever the director is present. Maxwell form is recovered in the isotropic/unordered limit or if an additional symmetry or matching condition enforces $Z_2 = 0$. $\square$

Status guard. The two-invariant classification is Level-A mathematics conditional on a supplied local bundle and ordered director. P5 fixes neither Z_1, Z_2 nor their signs, the gauge-reduced Hessian, the unbroken physical photon, or its map to the measured electromagnetic cone; those owners remain B/Open.

Coupling constant. The gauge coupling e is not fixed by the above argument. It remains a free EFT matching parameter.

Layer 3: compact holonomy, flux quantisation, and photon matching.

Proposition 4: conditional fluxoid relation in a supplied local phase EFT. For compact phase $\theta \sim \theta + 2\pi$, define the gauge-invariant phase gradient

$$\mathcal{D}_A \theta \equiv \partial_A \theta - e A_A. \quad (7.11)$$

Single-valuedness implies $\oint_\gamma \mathcal{D}_A \theta dX^A = 2\pi n$ and therefore the exact conditional identity

$$\oint_\gamma \mathcal{D}_A \theta dX^A = 2\pi n - e \oint_\gamma A_A dX^A, \quad n \in \mathbb{Z}, \quad (7.12)$$

and only with the additional zero-circulation condition $\oint_\gamma \mathcal{D}_A \theta dX^A = 0$ does this reduce to

$$e \oint_\gamma A_A dX^A = 2\pi n \quad (7.13)$$

up to sign convention.

Proof. Compactness of θ gives the winding integral. Subtracting the supplied connection gives (7.12); Eq. (7.13) requires the stated extra equilibrium/fluxoid condition. $\square$

Photon matching and the Higgs guard. If a supplied local $U(1)$ is unbroken and its Maxwell coefficient has a healthy sign, its transverse connection can be massless. Compactness alone does not establish this: a charged condensate generically permits a Higgs or London mass. ECT has not yet derived the physical unbroken $U(1)_{\text{em}}$, its gauge-volume quotient, transverse photon state, coupling to matter, or observable metric. Photon identification therefore remains Open rather than a completed matching result.

Superfluid analogy. The present $U(1)$ construction is structurally analogous to the phase–current and phase–holonomy logic of superfluids and superconductors [20]. The analogy is structural, not an exact physical identification. The ECT condensate is not a material medium; it is the ordered background underlying the effective spacetime and interaction sectors.

Interpretive example: the Aharonov–Bohm effect (Level B). The Aharonov–Bohm (AB) effect—in which an observable phase shift occurs even in regions where the local field strength vanishes—is a standard external target for any completed ECT gauge sector. Because the compact phase $\theta(X)$ and a local connection A_A are both supplied in this completion, the AB phase shift

$$\Delta\phi = \frac{e}{S_0} \oint A_\mu dx^\mu \quad (7.14)$$

is the standard holonomy of that conditional $U(1)$ connection around a closed loop. This would connect local phase symmetry, loop/winding bookkeeping and the physical gauge potential only after the missing local bundle, charge and observable owners are supplied. The standard AB phase formula is accommodated after the local bundle, charged representation and action-scale matching are supplied; it is not an ECT derivation. The effect illustrates why the gauge potential, rather than only the field strength, carries physical information when phase is a primary condensate variable.

What is strict and what remains Open. Within the compact phase EFT, the Noether current, current conservation, and the homogeneous limit are strict results. After a local $U(1)$ bundle is supplied, the compensating connection, covariant derivative and the two-invariant ordered-background kinetic basis are standard conditional mathematics. P5 does not supply that bundle. Compact phase winding constrains holonomy only after the phase, connection and charged representation are consistently identified. The physical unbroken photon, charge lattice, matter vertex, coupling and phenomenology remain Open.

What has been established and what remains open. The controlled hierarchy is:

- (i) BR1 and the compact phase EFT are additional effective closure inputs; they are not derived from the minimal real scalar P3;
- (ii) the global Noether current is exact inside that supplied EFT;
- (iii) local phase redundancy is an additional bundle hypothesis;
- (iv) conditional on it, A_A is the standard first-order compensating connection and (I_1, I_2) form the leading parity-even ordered-background kinetic basis; Maxwell form requires $Z_2 = 0$.

The following steps remain open:

- (v) the physical reduced Hessian, positive transverse residues and unbroken-photon condition are not derived;
- (vi) the charge representations, gauge coupling e , matter vertex, physical photon metric and QED phenomenology are not derived.

Thus the compact phase supplies a conditional gauge-completion route, not a derived electromagnetic sector.

Status summary.

Statement	Status	Comment
<i>Layer 1: compact phase EFT</i>		
Global phase symmetry in compact phase EFT	A/B	EFT structure admitted; consequences below strict
Noether current $j^A = \rho^2 \partial^A \theta$	A	Strict within compact phase EFT
Current conservation $\partial_A j^A = 0$	A	Noether theorem
Homogeneous limit gives $j^A = 0$	A	Direct corollary
<i>Layer 2: supplied local-bundle completion</i>		
Compensating connection A_A	A-math conditional	Requires supplied local $U(1)$ bundle
Covariant derivative $D_A = \partial_A - ieA_A$	A-math conditional	Same
Field strength F_{AB}	A-math conditional	Same
Ordered-background kinetic basis (I_1, I_2)	A-math conditional / Open	Basis classification is strict after bundle/director input; coefficients, signs and physical Hessian are Open
Compact holonomy / flux quantisation	A-math conditional	Requires a consistent compact representation and connection
Gauge coupling e	C	Free matching parameter
<i>Layer 3: physical matching</i>		
Massless $U(1)$ boson structurally consistent with photon	Open	Unbroken owner, transverse state and matter map missing; Higgs phase possible
QED fully derived from P1–P6	Open	

Structural route to the fine-structure constant

If a separately supplied local $U(1)$ bundle and connection are matched to the compact-phase sector, then a Maxwell-type term may be written with a normalisation coefficient Z_A . In the mostly-plus Lorentzian house package used below, one consistent Abelian convention is $D_\mu = \partial_\mu - iq_A A_\mu$, $\psi \mapsto e^{iq_A \alpha} \psi$ and $A_\mu \mapsto A_\mu + \partial_\mu \alpha$. Its value is calculable only after the physical reduced Hessian and state are supplied:

$$\mathcal{L}_{\text{eff}} = -\frac{Z_A}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} i \gamma_L^\mu D_\mu \psi + \dots \quad (7.15)$$

Here q_A is the charge entering this Abelian covariant derivative; it is not the integer hypercharge character weight introduced later. After canonical normalisation $A_\mu^{(c)} = \sqrt{Z_A} A_\mu$, the observed electromagnetic coupling becomes

$$e = \frac{q_A}{\sqrt{Z_A}}, \quad \alpha_{\text{fs}} = \frac{e^2}{4\pi} = \frac{q_A^2}{4\pi Z_A}. \quad (7.16)$$

Conditional on the supplied local $U(1)$ completion, this reduces the problem of the fine-structure constant to determining its gauge-field normalisation Z_A ; it does not derive that completion from P1–P6.

Required magnitude. For $q_A = 1$ the observed value $\alpha_{\text{fs}}^{-1} \approx 137$ requires

$$Z_A^{\text{IR}} = \frac{1}{4\pi \alpha_{\text{fs}}} \approx 10.9. \quad (7.17)$$

This is a moderate coefficient of order ten, not an extreme number.

Electroweak structure of the ultraviolet problem. The low-energy relation $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$ is defined in terms of the photon normalisation Z_A . However, above the electroweak scale M_W the photon is not a propagating eigenstate: the physically relevant gauge basis is $U(1)_Y \times SU(2)_L$, with independent electroweak gauge-kinetic normalisations for the hypercharge field B_μ and the weak isospin field W_μ^a . The low-energy photon kinetic term emerges only after electroweak symmetry breaking through the Weinberg mixing:

$$\frac{1}{e^2} = \frac{1}{g'^2} + \frac{1}{g^2}, \quad \alpha_{\text{fs}}^{-1} = \alpha_Y^{-1} + \alpha_2^{-1}. \quad (7.18)$$

Consequently, the ultraviolet problem for α_{fs} is not the determination of a single coefficient $Z_A^{(0)}$ at the condensate scale, but of the pair $(Z_B^{(0)}, Z_W^{(0)})$ and the electroweak mixing angle θ_W that relates them to the low-energy photon.

Electroweak target with colour ownership still Open (Level B/Open; benchmark Level C). The P7-supplied rank-three module with (h, Ω) defines only an $SU(3)$ -structured frame candidate (§40). It supplies no physical quark sections, chiral matter representations, hypercharges, multiplicities, connection or action. Consequently it removes none of the earlier matter-content ambiguity in the fine-structure matching problem. The observed Standard-Model matter content used below remains an external benchmark. Outcome: MISSING PHYSICAL MATTER-REPRESENTATION OWNER.

Using the observed Standard-Model electroweak couplings and the observed three-generation matter content only as an external benchmark (not as an ECT derivation), and employing the standard hypercharge normalisation with one-loop beta coefficients $b_Y = 41/6$ and $b_2 = -19/6$, one finds at the electroweak scale $\alpha_Y^{-1}(M_Z) \approx 98.4$ and $\alpha_2^{-1}(M_Z) \approx 29.6$. In particular, the observed three-generation Standard-Model matter content is used here only as benchmark input, not as a quantity already derived within ECT. Running to the primary condensate scale $\phi_0 \approx 2.435 \times 10^{18}$ GeV yields

$$\alpha_Y^{-1}(\phi_0) \approx 57, \quad \alpha_2^{-1}(\phi_0) \approx 49, \quad (7.19)$$

corresponding to benchmark ultraviolet gauge-kinetic coefficients $Z_B^{\text{bench}} \approx 4.6$ and $Z_W^{\text{bench}} \approx 3.9$. Within the specified external one-loop Standard-Model extrapolation, these benchmark numbers are of order unity. That numerical observation is Level C/source-model bookkeeping only; it is not a derivation from P7 or from the ECT action.

Open matching interpretation (Level B/Open; benchmark Level C). The quantities $Z_B^{(0)}$ and $Z_W^{(0)}$ are external matching targets for a future electroweak completion. The possible identifications $Z_B^{(0)} \leftrightarrow K_\theta^{\text{eff}}$ and $Z_W^{(0)} \leftrightarrow \kappa_n$ are Open hypotheses. They require a microscopic local $U(1)_Y \times SU(2)_L$ bundle and action, a physical charged spectrum, a gauge-reduced response calculation, and a stated regulator, renormalisation scheme and threshold prescription. Neither P7 nor the shared scale ϕ_0 supplies those owners. In particular, no present result shows that the coefficients are naturally comparable, predicts their ratio, or favours $\sin^2 \theta_W(\phi_0) = \mathcal{O}(1/2)$.

For reference, the same external benchmark arithmetic gives

$$\sin^2 \theta_W^{\text{bench}}(\phi_0) = \frac{49}{57 + 49} \approx 0.46. \quad (7.20)$$

This is not an ECT prediction and provides no evidence for a condensate matching map.

Whether purely induced electroweak kinetic terms, microscopic bare terms, or a mixture can reproduce the matching targets is Open. Deciding this requires the UV charged spectrum and multiplicities, its couplings, regulator and thresholds, and an explicit induced-Z calculation. Different allowed multiplicities and logarithmic windows are counterexamples to any inference from the displayed Standard-Model running alone; that benchmark excludes neither a purely induced nor a mixed mechanism. Outcome: PARAMETRIC ONLY; MISSING ELECTROWEAK ACTION/RESPONSE OWNER.

Technical details are collected in Appendix AA.

The Abelian construction provides the cleanest example of how gauge structure can emerge from condensate variables. The next subsections ask to what extent this logic can be extended to the non-Abelian and electroweak sectors.

7.3 Non-Abelian gauge structure and its status

Status: conditional/Open. The algebraic identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ is exact. It neither turns the carrier-space orientation triplet into an internal weak bundle nor promotes a global rotation to local gauge redundancy. P5 describes a medium but does not supply an independent local multiplet fibre, gauge-volume quotient, connection, kinetic coefficient, representations or matter vertex. Once such data are supplied, the usual covariant derivative, field strength and Yang–Mills operator follow by standard EFT completion. A physical $SU(2)_L \times U(1)_Y$, chirality, colour and electroweak matching remain Open/model-building inputs.

General non-Abelian framework. Suppose that the effective condensate description contains internal multiplet degrees of freedom

$$\Phi = (\Phi_1, \Phi_2, \dots, \Phi_N)^T, \quad (7.21)$$

and that the effective action is invariant under global $SU(N)$ transformations

$$\Phi \rightarrow U\Phi, \quad U \in SU(N). \quad (7.22)$$

If one further admits local multiplet redundancy $U \rightarrow U(X)$, covariance requires the introduction of gauge connections. The corresponding covariant derivative is

$$D_A = \partial_A - ig A_A^a T^a, \quad (7.23)$$

where T^a are the generators of the Lie algebra and A_A^a are the associated gauge fields.

Field strength and ordered-background kinetic basis. The gauge-covariant field strength is

$$F_{AB}^a = \partial_A A_B^a - \partial_B A_A^a + g f^{abc} A_A^b A_B^c, \quad (7.24)$$

where f^{abc} are the structure constants of the group. On a supplied ordered background, local parity-even terms quadratic in the field strength and containing at most two derivatives span two independent invariants,

$$I_1^{\text{NA}} = F_{AB}^a F^{aAB}, \quad I_2^{\text{NA}} = (n^A F_{AB}^a)(n_C F^{aCB}), \quad S_{\text{gauge}} = -\frac{1}{4} \int d^4 X [Z_1^{\text{NA}} I_1^{\text{NA}} + 2Z_2^{\text{NA}} I_2^{\text{NA}}]. \quad (7.25)$$

The usual Yang–Mills kinetic term is the special choice $Z_2^{\text{NA}} = 0$; residual $O(3)$ symmetry alone does not enforce it.

Conditional theorem: local non-Abelian EFT completion. Once an independent local multiplet redundancy and its bundle are specified, the covariant derivative (7.23), field strength (7.24), and the two-invariant ordered-background kinetic basis (7.25) follow by the same EFT logic as in the Abelian case. The isotropic Yang–Mills form follows only after the additional condition $Z_2^{\text{NA}} = 0$. This proves compatibility of a supplied completion, not its descent from the minimal Φ action.

Proof sketch. Local multiplet covariance requires a compensating connection valued in the Lie algebra. Closure of the local algebra forces the gauge field to transform in the adjoint and introduces the commutator term in the field strength. Antisymmetry of F_{AB}^a and the ordered director reduce the local parity-even quadratic contractions to I_1^{NA} and I_2^{NA} , up to total derivatives. A stronger symmetry or matching condition is required to remove the director-sensitive invariant. $\square$

Appendix reference. The explicit structural local-completion steps and the $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ bridge are written out in Appendix U.

Status after P5. P5 does not make arbitrary pointwise rotations of the physical director $n_A(X)$ a redundancy: n_A carries carrier-space orientation information, and its variations are constrained by $Q_A = \partial_A \Phi$. Treating $n_A \rightarrow R n_A$ as a gauge orbit would require a separately supplied internal fibre and reduced Hessian. Conditional on that completion, a connection valued in the chosen Lie algebra, its field strength and the two-invariant ordered-background kinetic basis follow by standard EFT logic. The special Yang–Mills choice $Z_2^{\text{NA}} = 0$, the bundle, representation content, kinetic coefficients, physical state space and coupling g remain open matching parameters.

Conditional gauge-completion architecture. The supplied compact-phase sector has global $U(1)$ algebra, while the connected carrier-orientation stabiliser supplies the Lie-algebra identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$. These facts motivate, but do not derive, a local internal electroweak bundle:

$$U(1)_{\text{global}} \oplus [\mathfrak{so}(3) \cong \mathfrak{su}(2)] \not\cong SU(2)_L \times U(1)_Y. \quad (7.26)$$

Supplying distinct local bundles, connections, gauge-volume quotients, positive reduced Hessians, chiral matter representations and an unbroken photon defines a B/Open completion route. None of these physical owners follows from P2–P5.

The algebraic bridge: $\mathfrak{so}(3) \cong \mathfrak{su}(2)$. For the nonzero vector supplied by P4, the stabiliser algebra is

$$\mathfrak{so}(3) \cong \mathfrak{su}(2). \quad (7.27)$$

This is a strict Lie-algebra isomorphism and therefore a genuine Level A statement. It means that among the simple Standard Model gauge factors, $SU(2)$ has a direct algebraic analogue in the P4 stabiliser. This does not derive a physical SSB event or an electroweak gauge owner.

Global-group caveat. The statement $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ does *not* identify the global groups $SO(3)$ and $SU(2)$. Their representation content differs: $SU(2)$ admits doublets, whereas $SO(3)$ does not. Thus the algebraic bridge is significant but not yet sufficient to derive the physical weak-interaction sector. Its relation to chirality and the possible selection of $SU(2)_L$ is discussed separately in Section 9.3.

Accommodation of Standard Model gauge groups. Within this framework, the Standard Model gauge factors define three separate completion targets:

- $U(1)$: the coherent-branch compact phase sector (Section 7.2); a local bundle, physical photon and charge map remain Open.
- $SU(2)$: algebraically distinguished by the isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$; a structural algebraic analogy for an electroweak-like sector if an independent internal bundle and second transition are supplied (Section 7.4).
- $SU(3)$: direct injection into the displayed $O(4)$ algebra is excluded by dimension (conditional A). P7 adopts a positive-definite Hermitian rank-three bundle (Level-B structural input), whose $SU(3)$ frame group is conditional A. Its physical-colour map, connection/action and dynamics remain Open; emergent/nonlinear, reducible/singlet and defect-kernel derivation routes are not excluded; see §40). Massless physical gluons and confinement are not derived.

Coupling constants. The gauge couplings g , g' , and g_s are free EFT matching parameters in the present formulation. Neither their numerical values nor their running are derived from the condensate data. Whether condensate structure could in principle constrain them remains an open problem of the ECT quantitative-unification programme (OP-GUT4 in Table 115).

Confinement as an open problem. The present ECT formulation shows how an $SU(3)$ -like gauge structure may be structurally accommodated in the low-energy sector, but it does not derive confinement or replace the standard QCD account of the infrared strong-interaction regime. A full derivation of confinement within ECT would require a genuinely nonperturbative analysis going beyond the perturbative EFT embedding used here. Whether condensate structure can provide an ECT-specific route to confinement remains open (OP-GUT1 in Table 115).

What is strict and what is embedded. The Lie-algebra statement $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ is strict. The standard non-Abelian local completion is also structurally strong once the local multiplet EFT assumptions are admitted. By contrast, the identification of the physical Standard Model gauge group, the actual matter representation content, the electroweak sector, the strong-interaction infrared regime, and the observed coupling constants are not derived here. These remain embeddings, matchings, or open problems.

What would be needed for full Level A. To promote the non-Abelian sector to full Level A from P1–P6, ECT would need to show:

- (i) how internal condensate multiplets arise dynamically from the bare scalar condensate description;
- (ii) why local non-Abelian redundancy is selected rather than merely admitted as an EFT completion;
- (iii) how the physical gauge group is fixed;
- (iv) how matter representations (doublets, triplets, singlets) are selected;
- (v) how the gauge couplings are determined;
- (vi) how electroweak breaking and confinement emerge from the same condensate logic.

None of these steps has yet been completed.

Status summary.

Statement	Status	Comment
<i>Exact algebra and Open localisation</i>		
$\mathfrak{so}(3) \cong \mathfrak{su}(2)$	A	Strict Lie-algebra fact
Local internal orientation bundle	Open	The spacetime director is physical/constrained, not a gauge orbit
Physical $SU(2)_L \times U(1)_Y$ gauge architecture	Open	Not implied by P2–P5
<i>Local EFT gauge completion</i>		
Non-Abelian covariant derivative / field strength	A/B	EFT completion once multiplet structure specified
Ordered-background non-Abelian kinetic basis	A/B conditional	$I_1^{\text{NA}}, I_2^{\text{NA}}$ allowed after bundle supply; $Z_2^{\text{NA}} = 0$ not derived
<i>Physical electroweak identification</i>		
$SU(2)_L \times U(1)_Y$ electroweak-like embedding	B/Open	Conditional model-building route, not derivation
$SU(3)$ -like colour sector	C	Model-level extension
Gauge couplings (g, g', g_s)	Open/matched	Free EFT matching parameters
Confinement	Open	Not derived
Full SM gauge group from P1–P6	Open	

Among the non-Abelian sectors, the $SU(2)$ -like case is algebraically distinguished by the residual rotation algebra of the ordered branch. The next subsection develops this connection into the electroweak-like embedding.

7.4 Electroweak-like embedding and its current status

Status: layered structural embedding. The generator counting associated with a candidate second transition $O(3) \rightarrow O(2)$ is a strict structural statement once such a transition is admitted. The electroweak-like doublet EFT, the attempted identification with the physical $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$ pattern, the scale matching $v_2 \simeq 246 \text{ GeV}$, and the Higgs-like radial mode are Level B/C embedding or matching statements. A first-principles derivation from the minimal scalar ECT postulates P1–P6 is not yet available. Connection: Section 5.1, Appendix M.

Two-layer completion route. The geometric layer supplies the exact residual rotation algebra and the global compact-phase algebra, but neither a local internal bundle nor a preferred chiral $SU(2)_L$ factor. The phenomenological layer supplies a candidate doublet EFT matched at $v_2 \simeq 246 \text{ GeV}$. Its standard mass relations are conditional on that input. Thus the two layers organise an Open electroweak completion; they do not explain why the physical gauge group exists or derive how it breaks from P1–P6.

Why the electroweak-like sector is structurally distinguished. Among the non-Abelian sectors, the $SU(2)$ -like case is singled out by the Lie-algebra bridge $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ established in Section 7.3. This gives the ordered branch a structural relation to a weak-like sector that is stronger than for other simple gauge factors, but still insufficient for a derivation of the physical electroweak sector. If, in addition to the primary ordering step $O(4) \rightarrow O(3)$, the condensate undergoes a second effective ordering step

$$O(3) \rightarrow O(2), \tag{7.28}$$

then two generators are broken and one remains unbroken. This is a strict counting statement and provides the closest reduced structural analogue within ECT to a weak-like symmetry-breaking pattern. By itself, however, it is only an algebraic/combinatorial analogy: it does not reproduce the full Standard Model electroweak generator counting, does not derive the physical $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$ mechanism, and does not derive the observed $W^\pm/Z/\gamma$ spectrum.

Candidate second-transition mechanism. A structural candidate mechanism for such a second transition is outlined in Appendix M. At present it requires additional assumptions beyond P1–P6 (in particular, phase–orientation coupling and potential alignment). Accordingly, the second transition is not yet a theorem of the bare condensate action. A concise summary of what this appendix establishes and what remains open is given in Appendix M.

Electroweak-like embedding model. A convenient effective embedding starts from the bookkeeping doublet

$$\Phi_{\text{ew}} = \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix}, \quad H \equiv \frac{\Phi_{\text{ew}}}{\sqrt{2}}, \quad (7.29)$$

and uses H as the canonically normalised electroweak doublet. This is a field redefinition, not a new degree of freedom. In particular, the two forms of the supplied action are exactly equivalent:

$$\begin{aligned} S_{\text{ew}} &= \int d^4X \left[\frac{1}{2} |D_A \Phi_{\text{ew}}|^2 + \frac{\lambda_2}{4} (\Phi_{\text{ew}}^\dagger \Phi_{\text{ew}} - v_2^2)^2 \right] \\ &= \int d^4X \left[|D_A H|^2 + \lambda_2 \left(H^\dagger H - \frac{v_2^2}{2} \right)^2 \right]. \end{aligned} \quad (7.30)$$

The equality follows term by term from $\Phi_{\text{ew}} = \sqrt{2}H$: the kinetic prefactor and both powers in the potential transform together. Thus $|\Phi_{\text{ew}}| = v_2$ is the same vacuum as $\langle H^\dagger H \rangle = v_2^2/2$. In unitary gauge the neutral component is fixed throughout this manuscript as

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_2 + h \end{pmatrix}, \quad H^0 = \frac{v_2 + h}{\sqrt{2}}, \quad m_f = \frac{y_f v_2}{\sqrt{2}}. \quad (7.31)$$

Every term in either action has mass dimension four in four dimensions, with $[H] = [\Phi_{\text{ew}}] = 1$, $[v_2] = 1$, and $[\lambda_2] = 0$. The gauge-covariant derivative is

$$D_A = \partial_A - ig \frac{\tau^a}{2} W_A^a - ig' Y B_A. \quad (7.32)$$

Here τ^a are the Pauli matrices, W_A^a the $SU(2)$ gauge fields, B_A the hypercharge gauge field, and g, g' the corresponding couplings. The Y used here is the small Standard-Model hypercharge: $Q_{\text{em}}/e = T_3 + Y$. The integer compact- $U(1)$ character used in global quotients is a separate weight $q_f = 6Y_f$ for each multiplet.

This construction is not derived from the minimal scalar ECT postulates P1–P6. Rather, it provides a low-energy structural embedding showing how an electroweak-like EFT could be accommodated if a second condensate transition exists. The displayed equality is Level A algebra inside that supplied EFT; its physical ECT ownership remains Level B/C/Open.

Scale matching and the hierarchy problem. If such an intermediate transition is realised, the associated scale is fixed by low-energy matching:

$$v_2 = (\sqrt{2}G_F)^{-1/2} \approx 246.22 \text{ GeV}, \quad (7.33)$$

equivalently by the observed relations

$$m_W = \frac{gv_2}{2}, \quad m_Z = \frac{\sqrt{g^2 + g'^2}}{2} v_2. \quad (7.34)$$

Thus 246 GeV is not a free choice once the electroweak EFT is matched. The ratio

$$\frac{v_2}{\phi_0} \sim 10^{-16} \quad (7.35)$$

is the ECT counterpart of the standard gauge hierarchy problem and is discussed in Section 5.1.

Higgs-like radial mode. In this embedding the radial excitation of the electroweak condensate plays the structural role of a Higgs-like scalar. Writing the doublet in polar form around the broken minimum, the radial mode σ_2 inherits its mass from the local curvature of the effective potential at $|\Phi_{\text{ew}}| = v_2$:

$$m_{\sigma_2}^2 = 2\lambda_2 v_2^2. \quad (7.36)$$

The observed Higgs mass $m_H \simeq 125 \text{ GeV}$ therefore constrains λ_2 through matching. This is a matching relation, not a prediction from bare condensate microphysics. In ECT the point is structural: the Higgs-like scalar is interpreted as the radial mode of a *second condensate sector*, not as a completely independent fundamental scalar unrelated to the condensate architecture.

ECT interpretation: candidate second phase of the same medium. If the second transition is realised, ECT interprets the electroweak condensate as a possible second ordered phase of the same underlying Φ -medium. The first transition supplies only a preferred Euclidean direction and, in an independently supplied scalar EFT with $\alpha > \beta > 0$ on the retained $\beta > 0$ component, a conditional scalar characteristic cone. A physical spacetime metric, gravity action/source map and electroweak sector remain separate completion problems. The common-medium reading is therefore an interpretive unification proposal, not a completed derivation.

What is strict and what is embedded. The generator counting associated with a candidate $O(3) \rightarrow O(2)$ step is strict once such a step is admitted. The algebraic bridge $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ is also strict. By contrast, the existence of the second transition itself, the doublet structure Φ_{ew} , the identification with the physical electroweak symmetry-breaking pattern, the values of g, g' , the matching scale v_2 , the Higgs-like radial mode σ_2 , and the reconstruction of the full Standard Model electroweak spectrum are Level B/C embedding or matching statements.

What would be needed for full Level A. To promote the electroweak-like sector from structural embedding to a first-principles derivation, ECT would need to:

- (i) derive the second condensate transition $O(3) \rightarrow O(2)$ dynamically from the bare action;
- (ii) show that the resulting breaking pattern reproduces the physical $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$ structure;
- (iii) derive the doublet representation content rather than admit it as an EFT embedding;
- (iv) derive the gauge couplings g, g' ;
- (v) derive the Yukawa structure and fermion mass generation within the same mechanism;
- (vi) derive the hierarchy ratio $v_2/\phi_0 \sim 10^{-16}$ from condensate dynamics.

None of these steps has yet been completed. The electroweak-like sector is therefore one of the places where the gap between structural embedding and first-principles derivation is currently largest.

Status summary.

Statement	Status	Comment
$O(3) \rightarrow O(2)$ generator counting	A	Strict combinatorial statement
$\mathfrak{so}(3) \cong \mathfrak{su}(2)$	A	Strict Lie-algebra fact
Second condensate transition exists	B/C	Candidate mechanism only
Electroweak doublet embedding Φ_{ew}	B/C	Structural EFT embedding
Matching scale $v_2 \simeq 246 \text{ GeV}$	Constraint	Fixed once EW EFT is matched
Weak-like gauge-boson sector	B/C	EFT embedding only; full SM electroweak spectrum not reproduced from the current $O(3) \rightarrow O(2)$ counting
Higgs-like radial mode σ_2	B/C	Structural role, mass matched
Gauge couplings g, g'	C	Free EFT parameters
Hierarchy ratio v_2/ϕ_0	Open	Not derived
Full electroweak sector from P1–P6	Open	

Status of the electroweak scale v_2 in the present formulation. In the present formulation of ECT, the scale $v_2 \approx 246 \text{ GeV}$ should be read as the electroweak-scale secondary condensate scale required by phenomenological matching, not as a first-principles derivation from the primary condensate dynamics. The two-energy-scale hierarchy and separate HRC matching inventory in Section 5.1, together with the structural candidate for a second transition described in Appendix M, suggests a possible hierarchical organisation of scales within one condensate framework, but the actual origin of v_2 , the dynamical mechanism that would generate the second transition, and the corresponding threshold and matching structure remain open. In particular, the present article does not derive $v_2/\phi_0 \sim 10^{-16}$, the gauge couplings g, g' , or the Yukawa hierarchy from condensate dynamics alone.

It should also be noted that v_2 already enters the present formulation as an input in several sectors beyond the hierarchy discussion: the electroweak boson mass formulae $m_W = gv_2/2$ and $m_Z = \sqrt{g^2 + g'^2} v_2/2$ of the present subsection; the fermion mass matching $m_f = y_f v_2/\sqrt{2}$ used in the matter-coupling discussion (§7.5); the unselected neutrino mass operators and scales, whose owners remain Open (§8.9); and the independently unselected coefficient of the director operator (§7.6). A future derivation of v_2 would not by itself fix that coefficient without an explicit cross-sector matching theorem.

This inventories a conditional gauge-completion scaffold. It does not derive a physical bundle, gauge field, matter representation or Standard-Model interaction sector. The next subsection records standard matter couplings only after those data are supplied.

7.5 Matter couplings and the Standard Model interaction pattern

Status: conditional bridge subsection. The previous subsections inventoried a gauge-completion scaffold rather than deriving physical gauge sectors. This subsection records how separately supplied local gauge completions would couple to supplied matter representations. All content here is Level B/C/Open embedding or matching, except where noted otherwise.

General framework. Once gauge fields are admitted in the low-energy EFT—whether Abelian, non-Abelian, or electroweak-like—their action on matter follows the standard covariant-derivative structure. In this sense, the observed Standard Model interaction pattern is *reproduced by embedding* rather than derived directly from the bare condensate postulates P1–P6. The gauge-matter coupling is therefore structurally correct at the EFT level, but not yet established as a necessary consequence of the condensate architecture.

Strong sector. Within the non-Abelian embedding (Section 7.3), the $SU(3)$ -like colour sector is taken to remain in the unbroken (symmetric) phase, which is structurally consistent with massless gluons and the absence of a colour-Higgs mechanism. A full derivation of confinement within ECT would require a genuinely nonperturbative analysis going beyond the perturbative EFT embedding used in this section; confinement remains an open problem (OP-GUT1 in Table 115) and is not claimed as an ECT result.

Fermion masses and Yukawa couplings. At the embedded EFT level, fermion masses arise through Yukawa couplings to the electroweak condensate sector:

$$m_f = y_f v_2 / \sqrt{2}, \quad (7.37)$$

where y_f is the Yukawa coupling constant and $v_2 \approx 246 \text{ GeV}$ is the electroweak matching scale. The Yukawa coefficients y_f are Standard-Model-like matching inputs, not derived from the bare condensate action. The detailed development of the fermionic sector, including spin structure and chirality, is deferred to the fermion sections of this paper.

Gauge coupling constants. At the present stage, the coupling constants e , g , g' , and g_s remain free EFT matching parameters. Neither their numerical values nor their running are derived from condensate microphysics. Whether the condensate structure could in principle constrain them—for example, through topological arguments, RG flow from the UV condensate scale, symmetry or state-selection mechanisms—remains a central open question of the ECT quantitative-unification programme (OP-GUT4 in Table 115).

Anomaly cancellation. The anomaly-cancellation structure of the Standard Model is likewise not derived in the present ECT embedding. In the Standard Model, anomaly cancellation depends on the specific fermion representation content and hypercharge assignments, and its fulfilment is a nontrivial consistency condition. Any full Level A completion of the gauge and matter sectors within ECT would need to reproduce the observed anomaly cancellations explicitly. At the present stage, this remains part of the open quantitative-unification programme (OP-GUT6a, OP-GUT6b in Table 115). A systematic anomaly audit for any supplied chiral gauge completion, including the standard electroweak consistency constraints, the anomaly-transparency of the director coupling $\mathcal{L}_5$, and the open status of the colour-sector completion, is developed in §9.5.

Separation between embedded SM structure and ECT-specific deformation. The content of the present subsection and the four preceding ones describes conditional Standard-Model-like completions that can be organised alongside the condensate architecture of ECT. P1–P6 do not derive the required local bundles, connections, kinetic operators or matter representations, so this organisation is a model-building programme rather than an embedding obtained from the minimal action.

The remainder of this section inventories an ECT-motivated operator allowed when a preferred direction and a fermion EFT are separately supplied. Its coefficient may vanish, so it is not a derived interaction or prediction.

7.6 ECT-motivated fermion–director operator

Status: conditional EFT operator inventory; physical channel Open. P4 supplies a preferred background direction, but neither an independent dynamical director field nor a completed fermion sector. If fermionic excitations and the restricted parity-even, zeroth-derivative EFT basis are supplied, the vector bilinear $n_A \bar{\Psi} \gamma^A \Psi$ is an allowed leading operator. Its Wilson coefficient may vanish, it is not unique once axial or derivative operators are admitted, and its observable map is Open. The conditional chain is:

- (i) *the P4 branch supplies n_A — postulated branch input;*
- (ii) *fermionic excitations exist — Ph (empirical fact);*

- (iii) within the restricted basis, an allowed leading operator coupling a spinorial excitation to n_A without derivatives is uniquely $n_A \bar{\Psi} \gamma^A \Psi$ — conditional operator classification;
- (iv) a nonzero coupling $\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi$ is an additional EFT matching choice, not a consequence of P1–P6.

Its coefficient μ_5 is not derived from P1–P6, may vanish, and has no retained numerical benchmark. A first-principles derivation from condensate microdynamics remains Open.

Terminology: director coupling. By analogy with the director field of nematic liquid crystals and the anchoring energy in those systems, the coupling $\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi$ is henceforth referred to as the *director coupling*. The order parameter n_A plays the role of the liquid-crystal director; the coupling describes how fermionic excitations interact with the orientation of the ordered medium.

Structural origin. As discussed in Section 7.1, the P4 chart supplies a preferred direction

$$n_A \equiv \frac{\partial_A \Phi}{|\partial \Phi|}. \quad (7.38)$$

This is not a gauge degree of freedom and not an internal multiplet index; it is a kinematic orientational variable of the ordered branch supplied by P4. Its independent physical dynamics is not derived by the bare real-scalar action.

Leading parity-even operator. Consider the low-energy fermionic EFT on the ordered branch and impose the following restrictions: (i) locality, (ii) linear dependence on the preferred direction n_A , (iii) no derivatives on the fermion field, (iv) parity-even operator basis. Under these assumptions, the leading ECT-specific fermion bilinear is

$$\Delta \mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi, \quad (7.39)$$

where μ_5 has mass dimension one.

This statement should be read carefully. It is *not* a theorem of the bare scalar condensate action. It is the leading operator in the specified low-energy EFT basis. If the parity-even restriction is dropped, an axial structure $n_A \bar{\Psi} \gamma^A \gamma^5 \Psi$ would also be available. The formal EFT operator classification is recorded in Appendix V.

Why this operator is ECT-motivated. The operator (7.39) has no counterpart in the minimal Standard Model fermion sector, because the SM contains no derived preferred spacetime direction analogous to (7.38). Its availability reflects the supplied preferred direction and restricted operator basis, not a nonzero interaction derived from ECT. With the phenomenological input that fermionic excitations are observed in nature, $\mathcal{L}_5$ remains an admissible member of the restricted EFT basis. Observation of fermions does not force its coefficient to be nonzero. The coefficient μ_5 , its sign, species dependence and physical observability remain Open matching data.

Lorentzian-branch form and sign convention. In a supplied Lorentzian matter completion compatible with the $\alpha > \beta > 0$ retained scalar branch (more generally $\beta(\beta - \alpha) < 0$), let the future-oriented unit vector in its local rest frame be $u^\mu = (1, 0, 0, 0)$. With $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ this fixes $u_\mu = (-1, 0, 0, 0)$. Writing the interaction with this lowered vector gives

$$\Delta \mathcal{L}_5 = \mu_5 u_\mu \bar{\Psi} \gamma_L^\mu \Psi = -\mu_5 j^0 \quad \text{in the local rest frame,} \quad (7.40)$$

where $j^\mu = \bar{\Psi} \gamma_L^\mu \Psi$ and $j^0 = \Psi^\dagger \Psi \geq 0$. Equivalently one may define the independently oriented vector-current coefficient $a_\mu^{(V)} \equiv \mu_5 u_\mu$; changing its orientation or replacing μ_5 by $-\mu_5$ changes the displayed rest-frame sign. ECT has not derived that physical sign. This is a preferred-direction deformation of

the fermion sector, not an additional internal gauge interaction. In this homogeneous rest frame it is proportional to the number density and to the identity in spin space. Consequently

$$[H_{\mu_5}, \mathbf{S}] = 0, \quad \Delta E_{\uparrow\downarrow} = 0, \quad \omega_{\text{prec}}^{(\mu_5 \text{ vector})} = 0 \quad (7.41)$$

for this term by itself. A spin-precession observable requires a separately specified spin-dependent completion—for example an independent axial background $b_\mu^{(5)} \bar{\Psi} \gamma^\mu \gamma^5 \Psi$, a derivative/Pauli operator, or an inhomogeneous vector completion whose explicit nonrelativistic reduction contains spin–orbit, curl or Pauli structure—and its detector Hamiltonian. ECT currently derives neither such a completion nor a precession-frequency window.

Discrete-symmetry classification of $\mathcal{L}_5$ (Level B). The coupling (7.40) selects a preferred timelike direction in that supplied Lorentzian matter completion and therefore introduces a native ordered-medium asymmetry source absent in the minimal Standard Model fermion sector. Using the standard discrete-transformation conventions for the Dirac vector current, one finds that j^μ is C-odd, while j^0 is P-even and T-even. Multiplication by the real scalar coefficient and the fixed rest-frame sign does not change those transformation parities. Accordingly, $\mathcal{L}_5$ is *C-odd, P-even, T-even, CP-odd, and CPT-odd* (Level B). Reality of μ_5 ensures hermiticity of the operator (required for a real action), but does *not* select its physical sign or restore CPT invariance. The systematic classification, its relation to the conditional CPT route for reconstructible local sectors, and the re-interpretation of $\mathcal{L}_5$ as a branch-sensitive Lorentz- and CPT-violating phenomenological operator are developed in Section 9.4. Its possible role in Sakharov condition (ii) is only a conditional completion (Section 16.3): a nonzero coefficient, physical matter/charge map, branch state and transport must be supplied. The conditional CPT backbone of Section 9.4 is not affected because $\mathcal{L}_5$ explicitly lies outside the reconstructible OS setup as a branch-selecting extension. As a vector bilinear, $\mathcal{L}_5$ does not modify the anomaly coefficients of a supplied chiral gauge completion; the systematic anomaly architecture is discussed in Section 9.5.

Coefficient dimension and missing matching owner. Since the bilinear $\bar{\Psi} \gamma^A n_A \Psi$ has mass dimension three, the coefficient μ_5 must have mass dimension one. It is therefore convenient to parameterise

$$\mu_5 \equiv \beta_5 m_f, \quad (7.42)$$

where β_5 is dimensionless. Thus the genuinely dimensionful coefficient is μ_5 , while β_5 is only a convenient ratio to an externally supplied fermion mass. Dimensional analysis does not select a nonzero coefficient, its sign, species dependence, or a suppression scale. Coefficients proportional to ϕ_0 , v_2 , mixed scales, an independent Wilson scale, or exactly zero are all unselected until a physical matching calculation is supplied.

Relation to the Standard-Model Extension viewpoint. In the Lorentzian branch, the operator $u_\mu \bar{\Psi} \gamma^\mu \Psi$ is structurally analogous to a vector background deformation in the sense of Lorentz-violating EFT. For a single fermion species in a strictly constant flat background, such a term can be partly absorbed by field redefinitions and may fail to produce directly observable effects. Physical effects arise when at least one of the following is true:

- the coupling differs between fermion species;
- the preferred direction varies across spacetime or environment;
- the background is curved or inhomogeneous;
- the coupling enters relative observables rather than an isolated single-species dispersion relation.

For ECT this means that the phenomenological relevance of (7.40) is tied not merely to the existence of the operator, but also to the ordered-branch environment and the relation between different matter sectors. A fixed homogeneous spurion does not by itself define a source-generated fifth force or an Eötvös parameter. That calculation requires a dynamical field equation or an explicitly prescribed inhomogeneous profile, source and test-body charges, propagator and range, microscopic-to-composite composition map, and any screening/environment law. None is supplied by the vector operator classification.

Preferred-direction effects. On a separately supplied physical metric/tetrad and fermion completion, the operator (7.39) parametrises preferred-direction corrections to the corresponding Lorentz-covariant fermion EFT. It parametrises residual sensitivity of the low-energy matter sector to the ordered-branch orientation. This preferred-direction structure is absent in the exact Lorentz-symmetric limit. It becomes an ECT discriminant only if a nonzero Wilson coefficient, species reduction and observable bridge are derived.

What is strict and what is embedded. The existence of the preferred direction n_A after $O(4) \rightarrow O(3)$ is a strict structural consequence of the ordered branch (Level A). The statement that $n_A \bar{\Psi} \gamma^A \Psi$ is the leading parity-even zeroth-derivative operator is a statement about the chosen low-energy EFT basis (Level A/B conditional). Dimensions require a dimension-one coefficient but do not select a nonzero coefficient, its sign, suppression scale, mass dependence or species law. The fermion masses m_f are embedded SM inputs (Level C). A first-principles derivation of the coefficient from condensate microphysics has not been completed.

What would be needed for full Level A. To promote this coupling from a structural EFT deformation to a first-principles result, ECT would need to:

- (i) derive the operator directly from the bare condensate action or from a controlled integration-out procedure;
- (ii) derive its coefficient from condensate microdynamics rather than dimensional analysis;
- (iii) determine whether the vector operator is uniquely selected, or whether axial and derivative competitors survive;
- (iv) establish under what conditions the coupling is physically observable rather than removable by field redefinitions in the low-energy Lorentzian limit.

None of these steps has yet been completed.

Status summary.

Statement	Status	Comment
Preferred direction n_A in the ordered branch	P4 input	Supplied by the ordered-branch postulate
Leading parity-even zeroth-derivative operator $n_A \bar{\Psi} \gamma^A \Psi$	A-math conditional	Allowed within the stated restricted EFT basis
Operator is absent from the minimal Lorentz-invariant SM basis	B	Conditional preferred-background comparison
Dimensionful coupling μ_5 required	A	Follows from mass dimension counting
Suppression scale and species law	Open	Not selected by dimensions or the operator inventory
Fermion masses m_f as inputs	C	SM-matched, not ECT-derived

Statement	Status	Comment
Coefficient derived from condensate micro-dynamics	Open	Not completed
Physical observability in the Lorentzian limit	Open	Depends on species dependence / gradients / environment

7.7 Strength, range, and the three-layer distinction

Status: the microscopic preferred-direction operator M and the scalar response ansatz S are distinct by variable/operator content. This does not derive the map from S to the conditional HRC gravity functional G .

M — microscopic matter operator. The local fermion deformation is

$$\Delta\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi, \quad \mu_5 = \beta_5 m_f, \quad (7.43)$$

where the second equality is only a dimensionless parameterisation. Its operator classification is A/B, while the coefficient, its possible vanishing and its observable multi-species reduction are Open.

S — scalar response ansatz. The separate amplitude coordinate

$$\phi_u = \beta_\phi^{-1} \ln(u/u_\infty) \quad (7.44)$$

enters the Level-B scalar ansatz through $F(\phi_u) = u/u_\infty$. The corrected equations give conditional local response identities, not a finite-body profile, screening solution, or galactic source/force law.

G — conditional HRC gravity functional. After the declared identity bridge, the supplied static HRC functional gives $\mu_{g,k}(g^2/a_M^2)g = g_N$ in spherical symmetry. This is Level A algebra inside that functional and a Level B conditional corollary of the bridge; galaxy-by-galaxy use is Level C. The physical coordinate, source/force normalisation, ownership of a_M , and mapping $S \rightarrow G$ remain Open. Therefore $M \neq S$ is a structural nonidentity; it supplies no proof of $S \rightarrow G$. Environmental modulation likewise remains Open pending matched field functions, a body solution, and an observable projection.

Species dependence and the route to observability. Section 7.6 already noted that in a strictly constant flat background, a vector term of the form $u_\mu \bar{\Psi} \gamma^\mu \Psi$ may be partly removable by field redefinitions for a single fermion species. Observable effects therefore require at least one of: species dependence, background gradients, curvature, or relative multi-sector observables.

Species dependence is not implied by the operator. It requires a flavour-dependent matching calculation; a universal coefficient, a non-universal coefficient and a vanishing coefficient are all currently allowed. The precise multi-species reduction has not been carried out, so no species hierarchy or numerical plot is retained.

Relation to other fifth-force frameworks. The preferred-direction operator M is a local fermion EFT deformation. The scalar ansatz S is a separate gravitational closure, while G is a separately supplied conditional HRC gravity functional. Saying that all are studied within the ECT programme is organisational and does not establish a common derivation.

What would be needed for a first-principles bridge. One must derive the coefficient and multi-species reduction of M , the matched functions and body solution of S , and an explicit source/force/normalisation map from S to G . None of these steps is supplied by dimensional analysis.

Status summary.

Statement	Status	Comment
Coefficient scale and species dependence	C/Open	No matching calculation or numerical/predictive benchmark; the optional H_{5m} free- ζ_{5f} ansatz is non-identifiable
M: local n_A matter operator	A/B	Coefficient and observability not derived
$M \neq S$: n_A operator versus ϕ_u scalar ansatz	A/B	Structural nonidentity only
S: $F(\phi_u) = u/u_\infty$ scalar response ansatz	B	Local identities conditional; body solution Open
$S \rightarrow G$: scalar ansatz to the HRC source/force map	Open	Physical coordinate and normalisation missing
G: supplied HRC static functional, bridge, and diagnostics	A / Open / C	In-model algebra / bridge / data layers
Environmental modulation and screening	Open	Sign, profile, charge and observable map missing

7.8 Observability conditions, consistency checks, and falsification logic

Status: conditional observability inventory. The present subsection does not provide precision phenomenology. It identifies which classes of observables can in principle be sensitive to the supplied fermion–director operator, which effects are structurally excluded, and what would be required for quantitative predictions. Possible observable channels are Level-C/Open benchmarks; the physical director, coefficient, species law, composite-body charge and measurement map remain Open.

What is established, what is benchmarked, and what remains open. At the present stage, the ordered-branch fifth-interaction programme contains three distinct layers. First, the vector operator $\Delta\mathcal{L}_5 = \mu_5 \bar{\Psi}\gamma^A n_A \Psi$ is the leading term only in the explicitly restricted parity-even, zeroth-derivative basis; broader bases contain competitors and its coefficient may vanish. Second, spin-precession, weak-equivalence-principle and compact-star observables require distinct spin/body/source completions in addition to a coefficient; adopting a number for μ_5 alone does not produce them. Third, the microscopic normalization of μ_5 , its flavour structure, background profile, composite-body reduction and global multi-sector fit remain Open.

Structural operator versus calibrated coupling. ECT classifies an allowed operator once fermions and the restricted basis are admitted, but it fixes neither its physical presence nor the coefficient μ_5 . Accordingly, the operator inventory is weaker than a phenomenology-level statement: the former is a conditional mathematical classification, whereas the latter remains benchmark-level until a controlled microscopic matching is achieved.

From operator existence to physical observability. Sections 7.6–7.7 showed that a supplied fermion EFT and restricted basis admit the ECT-motivated local operator

$$\Delta\mathcal{L}_5 = \mu_5 \bar{\Psi}\gamma^A n_A \Psi,$$

as the leading parity-even zeroth-derivative deformation of the fermion sector in the stated EFT basis. The existence of the operator, however, is not yet the same as a directly observable laboratory signal. In the

Lorentzian branch, a vector background coupling may be partly removable for a single fermion species in a strictly constant flat background. Observable consequences therefore require additional structure.

Conditions under which the coupling can become observable. The preferred-direction coupling can contribute to physical observables only when at least one of the following is present:

- (i) non-universal couplings between different fermion species;
- (ii) gradients or inhomogeneities of the ordered background;
- (iii) curved or environment-dependent Lorentzian branch geometry;
- (iv) relative observables comparing distinct matter sectors.

These conditions do not by themselves produce a signal, but they define the structural routes through which the ECT-specific coupling could survive field redefinitions and become measurable.

External constraints on distinct completions. Two experimental classes are relevant only after their separate observable owners are supplied. Weak-equivalence-principle tests constrain source-generated, composition-dependent accelerations. The MICROSCOPE final result gives $\eta(\text{Ti, Pt}) = [-1.5 \pm 2.3 (\text{stat}) \pm 1.5 (\text{syst})] \times 10^{-15}$ [49]. Mapping this external result to ECT requires a field equation or inhomogeneous profile, source and test charges, range, body-composition map and screening/environment law. The homogeneous vector-density spurion supplies none of them and therefore predicts no Eötvös window.

Precision spin searches constrain axial, derivative/Pauli or other explicitly spin-dependent Hamiltonians in comagnetometer, torsion-balance and spectroscopy experiments [50]. The homogeneous timelike vector-density term has no spin splitting, so those searches do not constrain it as a spin interaction. No common coefficient or numerical window relates the force and spin classes until a declared completion derives such a relation.

Observable classes and their owners. The admissible operator inventory does not identify one universal “fifth-force” observable. Distinct classes require distinct completions:

- **Vector-current relative observables:** the homogeneous $a_\mu^{(V)} \bar{\Psi} \gamma^\mu \Psi$ term can become observable only through a supplied non-universal multi-species map, gradients, curvature or relative-sector measurement. It is neither a spin interaction nor a source-generated body force by itself.
- **Spin-sensitive observables:** an independent axial $b_\mu^{(5)}$, derivative/Pauli or inhomogeneous spin operator must supply its coefficient and nonrelativistic detector Hamiltonian.
- **Composition-dependent forces:** a dynamical or prescribed inhomogeneous background must supply source and test charges, propagator/range, body-composition map and screening law. No species or body-charge scaling is supplied by the operator alone.
- **Propagation or environmental effects:** a declared kinetic operator and branch profile must derive any anisotropy or ambient dependence rather than insert it phenomenologically.

At present, this is a classification of observable *types*, not a list of precision predictions. The homogeneous dimension-three vector coefficient $a_\mu^{(V)}$ belongs to relative energy/phase and CPT-odd particle–antiparticle or interspecies observables; it is spin independent. Axial spin/precession observables and mediator force/WEP observables have independent owners. No vector-to-axial or vector-to-force inference is made. Because no coefficient or species law is derived, no numerical comparison with other interaction strengths is retained.

What this operator does not control. The microscopic operator M is not the scalar ansatz S and cannot by itself generate its gravitational response. Separately, S has not been matched to the supplied conditional HRC gravity functional G . Thus the operator does not account for the fitted galactic $O(1)$ modification, while the existence of that fit does not establish any microscopic fifth-force signal. This subsection concerns matter-sector observability, not a derivation of the galactic law.

Consistency checks. Even without precision predictions, the present structure already imposes nontrivial consistency requirements:

- the operator must remain sufficiently suppressed in the Lorentzian low-energy regime;
- any observable effect must respect the micro/macro sector distinction established in Section 7.7;
- non-universal matter effects, if present, must arise through the multi-species structure of the EFT rather than through a universal unsuppressed background shift;
- environmental dependence, if invoked, must ultimately be derived from the ordered-medium field equations rather than inserted ad hoc.

Falsification logic. Operator admissibility is not a nonzero-signal prediction. In particular, $\mu_5 = 0$ and a constant universal vector-current term removable from all relative observables are admissible limits; either outcome closes that signal route but does not falsify the restricted-basis classification or ECT as a whole. Tests become falsifiers only after a named completion supplies its operator, nonzero coefficient, background, source/body or detector map and uncertainty model:

- (i) a vector-current completion is rejected if its derived relative-sector observable is incompatible with the corresponding data;
- (ii) a spin-dependent completion is rejected if its derived Hamiltonian and signal fail spin-sensitive bounds or a positive-signal consistency test;
- (iii) a source-generated force completion is rejected if its derived body accelerations fail WEP or fifth-force data;
- (iv) an environment-dependent completion is rejected if its action-owned profile necessarily produces an excluded signal.

Failure of one completion does not transfer to a different operator or observable map. Programme-level exclusion would require ECT to select a unique nonzero completion whose derived predictions are ruled out.

No unified numerical falsifier without a common owner. Composition-dependent free fall and spin-sensitive searches probe distinct source/body-force and detector-Hamiltonian completions. The homogeneous vector-density operator supplies neither map, so no single effective coupling is presently required to fit both data classes. A joint numerical falsifier exists only if a future action-owned completion derives a shared coefficient and both observable projections; failure would then reject that named common completion, not the independent operator classes or all ECT.

What is strict and what is open. What is established is only that the vector operator is allowed in the restricted supplied basis. No species sensitivity follows from P1–P6 or from the operator alone. What remains open is the full mapping from the operator to concrete measurement channels. No precision observable is derived at this stage.

What would be needed for quantitative predictions. To turn the present observability logic into a predictive framework, ECT would need to:

- (i) derive the coefficient μ_5 from condensate microphysics rather than dimensional EFT scaling;
- (ii) derive the environmental dependence of the ordered branch in laboratory, astrophysical, and cosmological settings;
- (iii) perform the full multi-species Lorentzian EFT reduction and determine which parts of the operator survive field redefinitions;
- (iv) map the resulting effective couplings onto standard Lorentz-violation and equivalence-principle test frameworks;
- (v) identify which observable classes remain genuinely distinct from the macroscopic ϕ -sector signals.

Table 20: Status map for the ECT fifth-interaction programme.

Claim	Status	Comment
Admissibility of the operator $\Delta\mathcal{L}_5$	A-math conditional	Restricted EFT basis; its coefficient may vanish
Uniqueness within the restricted operator basis	A-math conditional	Leading parity-even zeroth-derivative ordered-branch fermion coupling
CP-asymmetric dynamical route	C/Open	Requires a nonzero physical coefficient and completed matter map; not yet full CP phenomenology
Coefficient scale and species law	Open	No first-principles matching
Source-generated WEP / Eötvös completion	Open	Requires field/profile, source and test charges, body-composition map, range and screening law
Spin-sensitive operator completion	Open	Requires an independent axial, derivative/Pauli or inhomogeneous operator and detector Hamiltonian
Homogeneous vector-current relative observable	B conditional / Open	Operator classification is conditional; coefficient, non-universal map and measurable projection are Open
Compact-star response completion	Open	Requires a dense-matter action, EOS, background profile and observable map
Environmental modulation	Open	Sign, profile, coefficient and observable map not derived
Full flavour structure	Open	Not yet derived from condensate microphysics
Common spin–force coefficient and global fit	Not assumed / Open	No common owner relates the distinct observables; a joint fit is meaningful only after one is derived

7.9 Structural constraints on the coupling coefficient

Status: operator-dimension bookkeeping and Open matching analysis. This subsection does not derive the coefficient of the fermion–director coefficient from first principles. Its purpose is to state what dimensions fix and to expose what they do not fix. No condensate scale, species law or numerical coefficient is

selected.

What has already been established. Section 7.6 established that the preferred- direction operator can be written as

$$\Delta\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi, \quad (7.45)$$

with $[\mu_5] = 1$. Dimensions neither require a nonzero coefficient nor select its sign, mass dependence, suppression scale or flavour structure.

Dimensional scale assignment in the present EFT closure. Three ingredients enter the present argument.

First, the operator $\bar{\Psi} \gamma^A n_A \Psi$ has mass dimension three, so its coefficient must have mass dimension one. Second, the preferred direction n_A is supplied by the P4 chart, whose nonzero vector has $O(3)$ stabiliser, and is associated with the scale u_0 . Its physical formation through an $O(4) \rightarrow O(3)$ transition is Open. Third, the radial/Planck-scale parameter ϕ_0 is one available UV scale in the supplied ordered-branch closure; its association with the first transition does not make it the coefficient owner.

The scale ϕ_0 is therefore not adopted as a numerical benchmark in the present manuscript. A compact statement of the missing matching inputs is given in Appendix X.

No scale-ordering inference from the existence of n_A . The first ordered branch supplies only the tensor n_A . It does not fix when the operator coefficient is generated. A later symmetry may generate or forbid the operator, so ϕ_0 -, v_2 -, mixed-sector and independent Wilson coefficients, as well as $\mu_5 = 0$, are equally Open.

Admissible corrections. Once the coefficient is treated as an EFT object rather than a first-principles result, several classes of corrections become possible in principle:

- (i) multiplicative $O(1)$ matching factors;
- (ii) additional suppression by ratios involving lower scales, such as v_2/ϕ_0 , if the second transition participates in the operator completion;
- (iii) background-dependent corrections induced by the ordered-medium environment;
- (iv) loop corrections from the fermion sector or from collective condensate modes.

At present, none of these corrections has been computed in a controlled way. No numerical scaling law is retained.

Radiative stability: what can and cannot be said. The present work does not contain a loop analysis of the fermion–condensate operator. Accordingly, radiative stability has not been proved. A controlled one-loop calculation in the ordered-branch EFT would be required to determine whether the coefficient is radiatively stable, whether additional operator mixing occurs, and whether lower-scale corrections enter systematically. Until that action-owned calculation is performed, even the sign and size of radiative corrections are Open; the tree-level dimensional suppression is not a radiative-stability argument.

Could integrating out heavy condensate modes generate the operator? A possible but unselected completion is that the preferred-direction coupling is generated, or renormalised, after integrating out heavy condensate degrees of freedom such as the radial mode. At present this remains a structural possibility only. The article does not yet contain an explicit matching calculation from the heavy-mode sector to the low-energy fermion EFT. Therefore it is legitimate to note this as a possible origin channel, but not to assign it a definite formula. Any quantitative claim about heavy-mode induced corrections would be premature without a controlled derivation.

The rôle of the second transition. If the weak-like $O(3) \rightarrow O(2)$ embedding is realised, it may still modify the preferred-direction coupling indirectly. Two broad possibilities exist:

- the preferred-direction operator is already present at the u_0 scale and the second transition only renormalises its coefficient;
- the second transition contributes additional operators or mixing terms that alter the low-energy form of the coupling.

The present article does not decide between these possibilities. The existence of n_A belongs to the supplied first branch, but this makes no statement about the origin of μ_5 ; the coefficient could arise only after a later transition or vanish identically.

What is strict and what is open. Strictly speaking, the present section establishes only that the preferred-direction operator requires a mass-dimension-one coefficient. What remains open is the actual microphysical derivation of the coefficient, the size and sign of loop corrections, the rôle of heavy condensate modes, and the operator basis across the two successive transitions.

What would be needed for full Level A. To close the question of the coefficient at Level A, ECT would need:

- a controlled derivation of the operator coefficient from the ordered-branch microdynamics;
- an explicit operator-basis analysis across the successive transitions $O(4) \rightarrow O(3) \rightarrow O(2)$;
- a loop calculation establishing whether the coefficient is radiatively stable and whether lower-scale corrections enter;
- a matching analysis showing whether heavy condensate modes generate, renormalise, or leave unchanged the leading operator.

None of these steps has yet been completed.

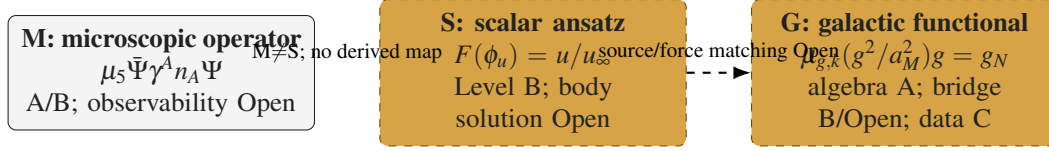
Status summary.

Statement	Status	Comment
Coefficient must have mass dimension one	A	Follows from operator dimension counting
Suppression scale and species law	Open	No matching calculation selects either
ϕ_0 -, v_2 -, mixed- or independent-scale coefficient	Open	All remain unselected; the coefficient may also vanish
Heavy-mode induced corrections	B/C	Structurally admissible, not derived
Radiative stability established	Open	No loop analysis yet
Full coefficient derived from microphysics	Open	Not completed

7.10 Connection to the galactic weak-field sector

Status: bookkeeping bridge only. $M \neq S$ is structural; $S \rightarrow G$ is Open.

The microscopic operator M , scalar ansatz S , and galactic functional G must not be compressed into one “macroscopic ϕ mechanism.” Their current logical relation is shown in Fig. 3.



One research programme does not imply one derived mechanism.

Figure 3: Status-sensitive separation of the fifth-force and galactic objects. $M \neq S$ follows from variable/operator content. The dashed $S \rightarrow G$ arrow is the missing physical-coordinate, source, force and normalisation match. The Level-C HRC algebraic diagnostic is not derived by the microscopic operator or the Level-B scalar identities. Gray denotes an external/supplied operator and orange dashed boxes denote Open matching routes; the text and border styles preserve the distinction in grayscale.

Open completion. Higher-order terms may connect M and S, and a future body/source calculation may connect S to G. Those are independent Open gates.

7.11 Gauge-sector summary, structural synthesis, and the open programme

Status separated. P5 is a medium/redundancy motivation, not a derivation of local $U(1)$ or $SU(2)$ gauge dynamics. Global compact-phase Noether algebra and $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ are exact in their supplied models. Local bundles, gauge-volume reduction, connections, kinetic coefficients, an unbroken photon, physical matter representations, charges, couplings, colour and electroweak identification remain conditional/Open. The director-fermion operators are allowed basis terms only; their Wilson coefficients may vanish and their observable map remains Open.

What Section 7 has established. After the introduction of the medium-character postulate P5, Section 7 establishes the following hierarchy of results:

- (i) The compact-phase $U(1)$ construction (Section 7.2): a global Noether current is exact inside a supplied compact-phase EFT. A local bundle, compensating connection and two-invariant ordered-background kinetic basis form a conditional completion; they do not follow from P5 alone.
- (ii) The non-Abelian orientation sector (Section 7.3): the ordered-branch orientation algebra gives the exact identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, but the spacetime director is not an internal weak gauge field. A local chiral bundle and connection require independent completion data.
- (iii) The electroweak-like embedding (Section 7.4): a candidate $O(3) \rightarrow O(2)$ step gives a structural analogue of the weak-sector breaking pattern, but not a completed derivation of the physical electroweak sector (Level B/C).
- (iv) The matter-coupling architecture (Section 7.5): Standard-Model-like interaction pattern reproduced at the EFT matching level.
- (v) The candidate preferred-direction coupling (Sections 7.6–7.10): the displayed vector bilinear is allowed only in the stated restricted parity-even fermion EFT basis. Its nonzero coefficient, sign, species dependence and observable bridge remain Open.

Separation of spacetime and internal symmetries. The P4 direction datum carries a coordinate index of the Euclidean arena, whereas internal gauge symmetries act on multiplet indices of condensate components. These sectors are therefore structurally distinct from the outset. Section 7.2 records the additional assumptions needed to complete the compact phase into a local $U(1)$ theory. The construction is conditional: the compact phase alone does not derive the bundle, reduced Hessian, kinetic coefficients or physical matter map.

The Coleman–Mandula theorem [51] constrains the structure of exact symmetries of a nontrivial relativistic S-matrix under specific assumptions. The present ECT discussion is formulated at the level of the condensate action, the ordered branch, and the corresponding low-energy EFT structures, prior to any full S-matrix reconstruction. For that reason, the theorem does not by itself settle the viability of the present embedding programme. This should not be read as a circumvention of Coleman–Mandula, but as a statement about the current level of analysis.

Connection to chirality and the matter sector. The exact decomposition $Spin(4) \simeq SU(2)_L \times SU(2)_R$ does not by itself select a weak factor. Fixing the ordered vector leaves the diagonal stabiliser $SU(2)_{\text{diag}}$, so the broken-phase geometry does not derive a left/right matter asymmetry. A physical chiral weak bundle, one-factor vertex, coupling and matter representations must be supplied independently.

What P5 motivates and what remains open. P5 motivates looking for a local-redundancy completion but closes neither the localisation step nor a physical gauge architecture. The candidate $SU(2) \times U(1)$ is therefore B/Open model building. The open steps are:

- (i) derive the local phase and internal-multiplet bundles, their gauge-volume quotients, connections and positive kinetic operators;
- (ii) identify an unbroken physical $U(1)_{\text{em}}$ rather than a Higgsed connection, and derive its transverse state and metric;
- (iii) for $SU(3)$ colour, only the direct injective embedding in the displayed minimal algebra is excluded; P7 is an adopted Level-B structural postulate whose physical-colour map remains C/Open, while emergent/nonlinear, reducible/singlet, independent-bundle and defect-kernel routes and all physical colour owners remain Open;
- (iv) the coupling constants (e, g, g', g_s) are not derived from condensate parameters;
- (v) the candidate second transition $O(3) \rightarrow O(2)$ has not been derived dynamically from the bare action;
- (vi) the full Standard Model gauge sector, including anomaly cancellation and the correct matter representations, has not been reproduced.

The corresponding Level A programme for the ECT-specific preferred- direction matter coupling is given separately in Sections 7.6–7.9.

Status summary.

Statement	Status	Comment
<i>Abelian compact-phase route</i>		
Global Noether current and current conservation	A	Strict within the compact phase EFT
Local phase redundancy	Open input	Not implied by global phase symmetry or P5
Compensating connection and covariant derivative	A-math conditional	After a local bundle is supplied
Ordered-background kinetic basis (I_1, I_2)	A-math conditional / Open	Basis classification after bundle/director input; coefficients and physical Hessian Open
Compact holonomy and flux quantisation	A-math conditional	Requires consistent compact representation and connection

Statement	Status	Comment
Photon identification	Open	Unbroken photon, charge/matter map and QED not derived
<i>Non-Abelian completion route</i>		
$\mathfrak{so}(3) \cong \mathfrak{su}(2)$	A	Strict Lie-algebra identity
Local internal multiplet redundancy	Open input	Carrier-space orientation triplet is not a gauge field
Ordered-branch $SU(2) \times U(1)$ gauge architecture	B/Open	Supplied model-building route, not derived
Non-Abelian EFT gauge completion	A-math conditional	Once the bundle and multiplet are supplied
$SU(3)$ direct-injective embedding no-go; P7 structural postulate	A restricted / Level-B input	Dimension excludes only direct injection; $SU(3)$ frame reduction is conditional A, while the physical-colour map and alternative microscopic origins remain Open
$O(3) \rightarrow O(2)$ as candidate weak-like step	B	Reduced algebraic/counting analogue only
v_2 as electroweak matching scale	Constraint	If a second condensate transition exists
<i>Matter couplings and interaction architecture</i>		
Gauge couplings (e, g, g', g_s)	Matched	EFT matching parameters, not derived
Yukawa/matter-coupling structure	B/C	Embedded at the EFT level
Chiral selection of $SU(2)_L$	Open	Vector stabiliser is diagonal; independent one-factor vertex required
<i>ECT-specific matter-sector deformation</i>		
Preferred-direction operator $n_A \bar{\Psi} \gamma^A \Psi$	A-math conditional / physical Open	Basis classification only; axial competitors depend on assumptions
Dimensionful coupling μ_5	A-math/Open	Dimension fixed; value, sign, scale and species law Open
M: n_A operator $\neq$ S: ϕ_u scalar ansatz	A/B	Structural nonidentity only
S scalar ansatz $\rightarrow$ G HRC gravity map	Open	Source/force/normalisation matching missing
G supplied HRC static functional, bridge, and diagnostics	A / Open / C	In-model algebra / bridge / data layers
Coefficient derived from condensate micro-physics	Open	Not yet completed

Section 7 therefore establishes a layered inventory rather than a single derivation. P5 motivates deformable relational response but neither creates local gauge redundancy nor closes the bundle, connection, action, state or matter-representation gates.

Interaction map of ECT at the present stage. At the present stage, ECT organises three interaction-completion programmes; none is yet a physical interaction derived from P1–P6:

- (i) **Electromagnetic route:** the compact coherent-phase sector supplies global compact-phase algebra. A local redundancy, ordered-background gauge kinetic coefficients, matter charge and an unbroken physical photon require added completion data.
- (ii) **Weak-interaction route:** the ordered-orientation sector supplies a rotation algebra, not a local chiral $SU(2) \times U(1)$ gauge theory. The weak bundle, fields, representations and coefficients remain Open.
- (iii) **Gravity completion route:** the scalar characteristic geometry is derived in its stated model, but at fixed α, β and tangent-director linear order about the homogeneous background the fixed director map has no linear TT/static-density vertex. A supplied scalar–tensor/HRC action provides conditional response algebra; a physical tensor, universal metric and nonlinear gravity owner remain Open.

By contrast, a colour $SU(3)$ sector is not derived within the present minimal condensate framework; its absence sharply localises the missing internal structure required for a complete interaction unification.

Native derivation criterion. A fundamental interaction is *native* in ECT if: (a) it arises from an ordered-branch redundancy or ordered-branch tensor mode; (b) its leading low-energy EFT closure is fixed internally by the postulate framework, up to explicit matching parameters and clearly identified unresolved higher-order sectors; (c) it does not require importing an external gauge group or an external background structure. By this criterion no physical electromagnetic, weak, colour or gravitational interaction is yet native in the strong derivational sense. The compact phase, rotation algebra and scalar principal tensor are conditional starting points; all local gauge, chiral, charge/matter and tensor/source completions remain Open.

8 Elementary Particles and Field Excitations in ECT

Status guide for this chapter. The material below spans three levels of rigour, organised into three categories of excitations (Section 8.1):

- (i) **Condensate-coordinate and candidate-excitation sectors (Level A/B/Open).** Physical graviton: Open completion target; at fixed α, β and tangent-director linear order the fixed director image has no linear TT/static-density seed (Section 8.5); nonlinear, composite or additional-field vertices are not excluded. One longitudinal orientation coordinate χ (Level A kinematics), with a one-mode physical interpretation only in a supplied reduced EFT (Level B; Section 8.2). Homogeneous P3 radial coordinate σ , with $m_{\Phi, \text{rad}}^2 = 2\mu^2$ inside P3 (Level A conditional; Section 8.3); a P4 amplitude pole or particle remains Open. The P3 radial coordinate and the strict orientation-coordinate map have different owners; physical P4 amplitude and χ modes and a graviton require additional actions and reductions.
- (ii) **Conditionally supplied SM-like EFT embeddings (Level B/C/Open).** Candidate gauge fields in separately supplied local-bundle completions (Section 8.6; full development in Section 7). Higgs boson reinterpreted as secondary radial mode (Section 8.7). Fermions: structural reconstruction on the ordered branch (Section 8.8; full development in Section 9). Neutrino sector: seesaw-type embedding (Section 8.9). These assume Standard-Model structure as external input; the SM gauge group, Yukawa couplings, and generation structure are not derived from the minimal bosonic condensate action.
- (iii) **Topological and conditional sectors (A-math conditional / physical Open).** Abstract texture/skyrmion classes of the supplied target are classified by $\pi_3(S^3) = \mathbb{Z}$, with $\pi_0 = \pi_1 = \pi_2 = 0$; conditional secondary defects (Section 8.10). Physical finite-action configurations, stability, masses and abundances require a configuration/action/transition owner and remain Open.

One candidate fermion deformation allowed in the restricted ordered-branch EFT is the director operator $\mu_5 \bar{\Psi} \gamma^A n_A \Psi$ (Section 9.8). A condensed overview with status table is given in the chapter summary (Section 8.11).

8.1 General Structure of Particle Excitations

Status: structural roadmap for the particle chapter. This subsection does not yet derive the full particle content of nature. Its purpose is to classify which coordinates or candidate excitations occur in the supplied ordered-gradient chart, which sectors are only structurally embedded at the EFT level, and which states are topological or conditional. The strongest outputs of this chapter are therefore not the full Standard Model particle spectrum, but the clear separation between exact coordinate content, conditional condensate modes, embedded matter sectors, and conditional topological states.

In the scalar-only ECT basis the fundamental degree of freedom is the condensate field Φ . The orientation field n_A is introduced only as a derived ordered-branch variable through $Q_A = \partial_A \Phi = u n_A$ and is not an independent microscopic field in the present formulation (Section 4.2). Candidate particles and field excitations are organised as condensate coordinates, fields in separately supplied internal/spinorial structures, or maps into declared topological targets. Their physical poles, state spaces and configuration owners must be established separately.

Why a particle chapter is justified at all. The transition from Part I to the present chapter is not a change of subject, but a change of level of description. Part I analysed the supplied ordered chart, its conditional sector decomposition, its scale inventory, and gauge-side structural embeddings. Section 8 asks what these declared structures permit when read at the level of particle-like excitations. The chapter is therefore not an additional postulate about particles, but the excitation-level inventory of the proposed condensate architecture.

What is meant here by a “particle”. In the present ECT framework, “particle” does not always mean a fundamental microscopic field inserted into the bare action. Depending on the sector, it may instead refer to: (i) a required fluctuation mode of the ordered condensate, (ii) an EFT field structurally accommodated by internal symmetries of the condensate background, or (iii) a map into a supplied topological target, with physical configuration status still Open. This distinction is essential for interpreting the status labels used throughout this chapter.

Three categories of excitations. The particle content discussed in this chapter falls into three categories with different derivation status:

(i) *Condensate coordinates and candidate excitations.* These statements have different status and must not be conflated:

- homogeneous P3 amplitude coordinate with $m_{\Phi, \text{rad}}^2 = 2\mu^2 = 2\lambda\phi_0^2$; its numerical mass and a physical P4 amplitude pole are Open (Section 8.3);
- one longitudinal scalar coordinate χ from the integrability-constrained orientation map; its physical one-mode reduced EFT is conditional (Section 8.2);
- physical graviton as an Open tensor-completion target; the displayed director map is not a helicity-2 owner (Section 8.5).

(ii) *Conditionally supplied Standard-Model-like EFT embeddings.* These may be accommodated on the ordered background but are not derived from P1–P6:

- candidate gauge fields in supplied local bundles motivated by compact-phase and rotation algebras (Section 8.6);
- electroweak-like gauge embedding and the associated secondary condensate sector (Section 8.7);

- fermions as spinorial excitations compatible with the ordered background (Section 8.8);
- neutrino sector (Section 8.9).

(iii) *Topological sectors and other conditional states.* These are abstract target-space classifications or depend on a possible second transition:

- abstract texture/skyrmion classes from $\pi_3(S^3) = \mathbb{Z}$ and vanishing lower homotopy groups $\pi_0 = \pi_1 = \pi_2 = 0$ for the supplied target;
- (conditional) secondary defects from $\pi_2(S^2)$ if $O(3) \rightarrow O(2)$ exists;
- the Higgs-like secondary radial mode h as the *conditional particle-level interpretation* of the secondary condensate sector, if the weak-like $O(3) \rightarrow O(2)$ transition is realised.

Topological sectors are developed in Section 8.10, while the conditional Higgs-like reinterpretation is discussed primarily in Section 8.7.

Classification criterion. The division into the three categories above is not merely descriptive. It reflects three distinct logical relations to the ECT foundation:

- a coordinate is *kinematically required* if it follows from the supplied parametrisation itself; physical propagation still requires an owned action, domain and state;
- a state is *embedded* if it can be consistently realised in the ECT low-energy framework but is not derived from P1–P6;
- a state is *conditional or topological* if it depends on the topology of a supplied target or on an additional transition not yet established from first principles.

These three classes therefore correspond not merely to different kinds of excitations, but to different logical relations to the foundation of ECT: strict consequence, EFT embedding, and conditional/topological realisation.

Physical interpretation and condensed-matter analogy. The three categories have direct analogues in condensed-matter physics. Category (i) coordinates are analogous to the variables used to describe phonons in a crystal, but the analogy does not prove an ECT pole: an ordered physical medium, its quadratic action and state would first have to exist. Category (ii) fields are analogous to *impurity atoms in a crystal lattice*: they are compatible with the medium and respect its symmetries, but they are not derived from the lattice structure alone. Category (iii) states are analogous to *topological defects* (dislocations, vortices): their possible topological protection is classified by the homotopy of a declared target, while physical existence, finite action and abundance depend on dynamics and boundary conditions.

Comparison with quantum field theory. In standard QFT, particles are postulated as quanta of independent fields $(\phi, A_\mu, \psi, \dots)$ defined on a fixed spacetime background. Each field carries its own Fock space. The architecture does not place all candidate sectors on the same logical footing. Category (i) contains condensate coordinates and fluctuations that arise in the supplied broken-branch parametrisation; their physical poles, states and particle interpretation still require the corresponding actions, domains and projectors. Category (ii) fields are structurally embedded in the condensate symmetry architecture but are not derived from the minimal scalar action. Category (iii) consists of maps into the supplied target S^3 ; topology fixes abstract sector labels and non-protection statements, not stable particles, masses or abundances. The vanishing groups π_2 , π_0 and π_1 are exact for that target. Turning them into physical exclusions of monopoles, walls or strings requires identifying it with the complete physical vacuum/configuration space, which remains Open.

Particle scales versus the galactic fit length. The particle discussion contains the exact P3 relation $m_{\Phi, \text{rad}} = \sqrt{2\lambda}\phi_0$, an Open and distinct P4 pole, the conditional secondary scale $m_h \sim v_2 \approx 246 \text{ GeV}$, and the separately classified longitudinal coordinate χ with its conditional reduced-EFT mode interpretation. The kpc quantity L_{gal} is not a particle or condensate mass scale; it is a conditional HRC matching length whose ECT and scalar/AQUAL matching remain Open.

Dependence on earlier results. The present classification relies on four earlier structural results of Part I: the broken-phase sector map of Section 4, the dimensionally separated scale inventory of Section 5.1, the branching architecture of Section 6, and the gauge-sector embedding analysed in Section 7. Section 8 should therefore be read not as an independent particle postulate, but as the particle-level unpacking of those earlier structural ingredients.

What this chapter does not claim. The present chapter does not claim that the full Standard Model particle spectrum has been derived from the minimal scalar condensate action. In particular, the gauge group, Yukawa hierarchy, generation structure, and the detailed fermion spectrum remain beyond the present Level A closure. What is established here is a structured map of which sectors already follow from the ordered branch, which are admissible EFT embeddings, and which remain conditional or open.

Dependence on later parts of the paper. Section 8 identifies particle-like sectors and their derivational status, but it does not by itself complete the full phenomenological or cosmological role of those sectors. Questions of macroscopic closure, dark-sector behaviour, and late-time observational consequences are deferred to the later parts of the paper. This prevents the particle classification from being overinterpreted as a complete physical model already at the level of Part I.

The purpose of the following subsections is therefore not merely to list possible excitations, but to determine which particle-like sectors already belong to the broken condensate branch itself, which require additional EFT embedding, and which remain conditional on topology or a secondary transition. In this sense, the chapter is a derivation-status map of the ECT particle sector, not a claim of complete particle unification. Its organising principle is not phenomenological completeness, but logical dependence on the ordered-branch foundation.

Multi-particle description: why it is necessary and how ECT differs from QM and QFT. Ordinary non-relativistic quantum mechanics operates on a Hilbert space $\mathcal{H}_N$ of fixed particle number N . Processes with variable N (photon emission, pair creation, annihilation) require quantum field theory, where the Fock space $\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N$ and creation/annihilation operators are introduced as additional postulates. In ECT the architecture is different: fundamental are not “individual particles” but configurations of a single condensate field Φ . Changing particle number is a reconfiguration of the same medium, analogous to the appearance and disappearance of vortices in a superfluid.

Multi-particle description remains necessary for coherent and decohering regimes alike. Coherent many-body interaction occurs within the ordered Lorentzian dynamics; it does not require a large record exponent or a change of metric signature. Four independent reasons are:

- (a) *Entanglement*: a supplied nonfactorising state/channel requires a many-body description; a common condensate alone does not force it (Section 31);
- (b) *Exchange statistics*: for the separately supplied defect-free $\mathbb{R}^3$ identical-point configuration space, the topology $(\pi_1(\mathcal{M}_2) = \mathbb{Z}_2)$ supplies two exchange homotopy classes; the physical spatial slice, excitation map, representation and spin–statistics map remain Open (Section 24.4);
- (c) *Decoherence channels*: a rate is protocol-, state-, coupling- and spectrum-dependent; no universal $1/(N_{\text{eff}}\gamma)$ law follows;

- (d) *Variable excitation number:* pair creation, annihilation and radiation require a Fock/sector completion. Particle production at the $O(4) \rightarrow O(3)$ ordering event additionally requires a nonequilibrium transition history and is not yet derived.

Status of Fock space. ECT does not require Fock space as a fundamental postulate. However, an explicit emergent sector decomposition with variable excitation number—especially for the fermionic many-body sectors—remains to be constructed from condensate dynamics. This is a nontrivial reconstruction problem rather than a foundational gap (Open; OP-Q11).

8.2 Soft Orientation Sector of the Condensate

*Status: minimal orientation EFT is Level B. Three independent physical Goldstone particles are **not** claimed in the present scalar-only basis; see Section 4.3 and Appendix J.*

Why this coordinate sector is considered. A supplied nonzero P4 vector has stabiliser $O(3)$ and abstract orientation orbit $O(4)/O(3) \simeq S^3$. This conditional group geometry motivates an orientation-coordinate sector, but does not prove a physical vacuum manifold or propagating mode. What is nontrivial in scalar-only ECT is that the strict gradient map has much smaller independent coordinate content than naive broken-generator counting would suggest. The number of physical propagating modes follows only after a reduced action, constraint quotient/domain and projector are supplied. For this reason, the soft orientation sector belongs to the class of condensate-coordinate and candidate-excitation sectors introduced in Section 8.1; its physical particle content is not fixed by integrability alone.

Minimal conditional chain. The orientation-coordinate analysis uses three successive facts: (i) a supplied nonzero vector has $O(3)$ stabiliser and abstract orbit $O(4)/O(3) \simeq S^3$; (ii) naive broken-generator counting therefore suggests three soft directions; (iii) in the strict scalar-only basis, the order parameter is generated by a single exact one-form $Q_A = \partial_A \Phi$, so the kinematically independent soft-coordinate content is reduced to one longitudinal scalar χ . The orbit geometry is exact conditional mathematics, whereas its identification with a physical sector and its particle-like content remain Open/conditional on supplied reduced dynamics.

A supplied nonzero vector has $O(3)$ stabiliser, so its orientation orbit is

$$O(4)/O(3) \simeq S^3. \quad (8.1)$$

In a generic internal-symmetry breaking this would yield three independent Goldstone bosons. However, in the present scalar-only ECT basis the ordered quantity is $Q_A = \partial_A \Phi$, which is an exact one-form. The integrability constraint $\partial_A Q_B - \partial_B Q_A = 0$ implies that the orientation triplet reduces kinematically to one longitudinal scalar coordinate χ at linear order [29, 30]:

$$\pi_i = \frac{1}{u_0} \partial_i \chi, \quad \sigma = \partial_w \chi. \quad (8.2)$$

Minimal orientation EFT. A three-component orientation parametrisation of S^3 remains a useful effective language for the supplied chart, but it should be understood as a minimal EFT organisation, not as three independent propagating particles. The broken-generator counting $\dim O(4) - \dim O(3) = 3$ is a geometric statement; the strict scalar map has one independent longitudinal coordinate at linear order. A supplied reduced EFT may then realise $N_G^{\text{phys}} = 1$ (Level B), but integrability alone does not prove that physical count.

Three levels of statement. It is important to distinguish:

- (i) the *geometric* statement that the supplied nonzero vector has an S^3 orientation orbit;

- (ii) the *group-theoretic* statement that $\dim O(4) - \dim O(3) = 3$;
- (iii) the *conditional physical* statement that a supplied reduced EFT with the required domain and projector promotes the one longitudinal coordinate to one independent propagating soft mode at linear order.

The first two statements and the coordinate reduction are strict conditional mathematics; the third is Level B and supplies a particle interpretation only inside its declared reduced EFT. Physical SSB and identification of the orbit with the complete vacuum manifold remain Open. Confusion between these three levels is precisely what leads to the incorrect “three physical Goldstones” reading: the present scalar-only map does not contain three independent coordinates, while a larger physical mode bundle is not derived.

Mass status of the longitudinal scalar coordinate. The supplied derivative-only reduced EFT used for the linear soft-dispersion benchmark contains no χ^2 term and therefore has $m_\chi = 0$ inside that declared truncation. This is a conditional property of the displayed kernel, not a deduction of a physical zero from P1–P6. The spacetime/gradient integrability reduction is kinematic: it supplies neither an internal shift symmetry nor a regulator-compatible Ward identity that protects the physical mass. Potential, constraint, radiative, boundary or additional-field terms may change the reduced Hessian. Conversely, an exact zero would become protected only after an owned stationary action, domain, constrained projector and counterterm-closed identity are given. None is presently available, so the physical m_χ^2 , its sign and its radiative stability are NOT IDENTIFIABLE / Open (OP-GM). A pseudo-Goldstone mass is one possible conditional deformation, not the only logically admissible mass owner.

Named optional soft-scalar hypotheses. One C/Open completion may supply a shift-compatible derivative operator whose dimensionless momentum-space coupling is

$$g_\chi(k) = c_\chi \frac{k}{f_\chi}, \quad f_\chi > 0, \quad (8.3)$$

with independent coefficient c_χ and scale f_χ ; the optional match $f_\chi = \phi_0$ is a further input. A distinct named scale hypothesis is

$$\begin{aligned} H_0^{(\text{freq})} > 0, \quad H_{0,E} &:= \hbar H_0^{(\text{freq})}, \quad \zeta_H > 0, \\ m_{\chi,E} &:= m_\chi c^2 = \zeta_H H_{0,E}, \quad \xi_\chi = \frac{\hbar c}{m_{\chi,E}} = \frac{c}{\zeta_H H_0^{(\text{freq})}}. \end{aligned} \quad (8.4)$$

where ζ_H is a supplied positive coefficient and may differ from unity. The separate boundary choice $\zeta_H = 0$ is a massless branch with $\xi_\chi = \infty$ only as a limit; the displayed quotient is not used there. Neither branch is selected by P1–P6. They derive no abundance, dark-matter/dark-energy role, HRC map or laboratory signal without the action, state, source and observable owners.

For scale orientation only, the explicit point $\zeta_H = \zeta_\phi = 1$, $\phi_0^{\text{bench}} = \bar{M}_{\text{Pl}} = 2.4353 \times 10^{18} \text{ GeV}$ and $H_0^{(\text{freq})} = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ gives

$$\begin{aligned} m_{\chi,E}^{\text{bench}} &= 1.493 \times 10^{-33} \text{ eV}, \quad \xi_\chi^{\text{bench}} = 1.322 \times 10^{26} \text{ m}, \\ \frac{m_{\chi,E}^{\text{bench}}}{\phi_0^{\text{bench}}} &= 6.131 \times 10^{-61}. \end{aligned} \quad (8.5)$$

Here $\xi_\chi^{\text{bench}} = c/H_0^{(\text{freq})}$ is a reduced-Compton proxy; a physical correlation length still requires an owned propagator. These values are Level C consequences of the displayed parameter point, not ECT predictions, evidence, or an identification of χ with a dark-sector component. Nor do these scale conversions establish a numerical decoupling conclusion.

Soft-sector role versus the HRC no-halo algebraic diagnostic. The Level-C HRC algebraic diagnostic is evaluated on selected benchmarks without a particle-halo or Goldstone component in that diagnostic model. This is a phenomenological choice, not an architectural proof that the soft orientation sector or particle dark matter is absent. The mass, couplings, abundance and independent cosmological role of any physical mode represented by χ remain Open.

Role in the candidate-state inventory. The scalar coordinate χ , together with its conditional one-mode reduced EFT interpretation, is listed as item (ii) in the inventory beyond the observed SM spectrum (Section 8.10). It is not a derived ECT particle; its mass, couplings, stability and possible cosmological abundance remain Open.

This concludes the softest condensate sector of the particle chapter. The next subsections turn to heavier or more structurally elaborate excitations, beginning with the radial condensate mode.

Status summary.

Statement	Status	Comment
Orientation-coordinate sector motivated by the supplied orbit	A-math/B/Open	Orbit geometry and scalar-coordinate reduction are conditional mathematics; physical propagation needs a supplied reduced action/domain/projector
Supplied nonzero vector has $O(3)$ stabiliser and S^3 orbit	A-math conditional / physical SSB Open	Group geometry only
Naive Goldstone counting: $\dim O(4) - \dim O(3) = 3$	A	Broken-generator counting
Integrability gives one longitudinal scalar coordinate; a supplied reduced EFT may have $N_G^{\text{phys}} = 1$	A kinematics / B physical	Physical count additionally requires the action, quotient/domain and projector; see Appendix J
Supplied derivative-only χ kernel has $m_\chi = 0$	A-math conditional / B as ECT model	Property of the declared truncation, not of P1–P6 alone
Optional H_χ scale $m_{\chi,E} = \zeta_H H_{0,E}$	C/Open input	Supplied positive ζ_H ; the $\zeta_H = 0$ boundary is massless; no abundance, dark-sector role or signal follows
Physical m_χ^2 and protection mechanism	Open (OP-GM)	Stationary owner, projector, Ward identity and radiative closure missing
Physical soft-mode mass, operator and abundance	Open	No numerical infrared scaling or cosmological identification is retained
Soft-mode dark-matter interpretation	Not claimed	The HRC diagnostic omits a soft-mode component; its abundance, galactic role and $S \rightarrow G$ matching remain Open

8.3 Radial Condensate Mode

Status: three distinct owners. The homogeneous P3 potential has a Level-A conditional amplitude coordinate with curvature $m_{\Phi,\text{rad}}^2 = 2\mu^2$. P4 instead postulates a gradient-ordered background; its amplitude coordinate, stationary action, constrained Hessian, pole and correlation length are Open. Equality of a future P4 pole with the P3 curvature is only a matching hypothesis. A separately supplied electroweak-like second condensate may contain a Higgs-like radial mode at Level C/Open and is neither the P3 coordinate nor a derived P4 particle.

What is required and what is not. For the homogeneous P3 extrema, decomposing fluctuations of the scalar amplitude is standard and unavoidable; the local curvature defines the P3 radial coordinate. This statement does not transfer to the independent P4 gradient datum. A physical P4 amplitude excitation exists only if a stationary P4 action and its physical fluctuation reduction produce a pole. It is therefore currently an Open candidate-excitation sector in the classification of Section 8.1.

Contrast with the soft orientation sector. The scalar coordinate χ of Section 8.2 parametrises a supplied linear orientation map. A normal P4 amplitude coordinate would be distinct from χ , but neither its kinetic term nor its pole is present in the displayed P4 data. The P3 homogeneous amplitude is a different owner again. Thus no current result identifies a heavy P4 mode or the breakdown scale of the broken-phase P4 EFT; each physical spectrum requires its declared action and Hessian.

Minimal owner chain. P3 supplies a scalar action, homogeneous extrema and their local Hessian, so its amplitude curvature is calculable. A tangential/normal decomposition about P4 would first require the P4 configuration to be stationary for an owned constrained action. The postulated gradient profile is generally not stationary for the P3 action. Therefore a P4 radial pole, its physical scale and any role as a P4 ultraviolet threshold remain Open rather than following from the existence of a nonzero ordered datum.

Tangential versus normal coordinates. For a supplied stationary broken-phase action, tangential and normal coordinates are geometrically inequivalent. Geometry alone, however, does not determine their kinetic operators, pole residues or masses. In the current P4 programme the orientation coordinate is a conditional reduction and the normal amplitude pole is Open; calling the latter heavy would assume the missing Hessian.

First owner: homogeneous P3 amplitude fluctuation. For the P3 scalar action, write $\Phi = \phi_0 + \delta\Phi$ about either homogeneous minimum. Its local curvature is exactly

$$m_{\Phi,\text{rad}}^2 := V''(\phi_0) = 2\mu^2 = 2\lambda\phi_0^2. \quad (8.6)$$

This is a classical homogeneous-P3-minimum result. It is not the pole of the P4 gradient amplitude in $\partial_A\Phi = (u_0 + \sigma)n_A$: that pole needs a stationary P4 action, constraints and Hessian. The exact P3 mass is

$$m_{\Phi,\text{rad}} = \sqrt{2\lambda}\phi_0. \quad (8.7)$$

The value of λ is not selected, so matching ϕ_0 alone does not fix a Planck-scale mass. This is not a derived P4 particle or a P4 EFT threshold. It is also distinct from the supplied secondary electroweak Higgs-like candidate.

The positivity $V''(\phi_0) > 0$ gives tree-level local stability of the homogeneous P3 amplitude coordinate within the stated potential. It does not establish stability of the P4 gradient background. The one-loop vacuum stability analysis, the question of false vacuum and Coleman vacuum decay, and the comparison with Standard-Model electroweak metastability are developed in Section 8.4.

Spin and propagation. The P3 amplitude coordinate is scalar under the Euclidean rotations of its own action. A Lorentzian P4 particle interpretation, pole, residue and propagation cone are Open until the P4 action and continuation are supplied. Equality with matter, photon and tensor cones is likewise not derived.

Why the P3 radial coordinate is distinct from the soft orientation coordinate. The homogeneous-P3 radial coordinate σ must also be distinguished from the longitudinal scalar coordinate χ of the orientation sector (Section 4.3). This distinction is ECT-specific and follows from the integrability of the gradient condensate: $Q_A = \partial_A \Phi$ is an exact one-form, so amplitude and orientation fluctuations are not independent generic fields. At coordinate level, a putative σ would vary $|Q_A| = |\partial_A \Phi|$, whereas χ parametrises the supplied longitudinal orientation variation. They are distinct coordinates, but coordinate separation is not a two-pole theorem. Their physical mode content requires separate reduced actions; the χ model is conditional and the P4 amplitude pole remains Open.

Conditional higher-dimension expansion. If a positive isolated P4 pole is later derived and couples locally to the retained fields, standard EFT matching may generate

$$\Delta \mathcal{L}_{\text{EFT}} \sim \sum_{n \geq 1} \frac{\mathcal{O}_{2n+4}}{m_{\text{P4,pole}}^{2n}}. \quad (8.8)$$

This is a conditional decoupling template, not a current P4 result. The P3 homogeneous curvature may analogously set a threshold only for a supplied EFT built about that chosen classical P3 minimum; it cannot be imported into P4 without matching. A physical quantum vacuum still requires its state and infinite-volume or boundary/source-limit owner.

What is strict and what is not. Strictly, the homogeneous P3 action has the curvature $m_{\Phi,\text{rad}}^2 = 2\mu^2$. A P4 radial pole and ultraviolet threshold do not follow from P4's nonzero gradient datum. The secondary mode at v_2 is separately conditional on the supplied weak-like transition.

Conditional bridge to the candidate action coefficient. The supplied fixed-core phase closure contains

$$t_0^{\text{EFT}} = \frac{c_{\perp} K_{\theta}}{2m_{\varphi}^2}. \quad (8.9)$$

The displayed t_0^{EFT} formula is dimensionless and exact inside the supplied reduced closure. Identifying its core parameter with the P3 mass, $m_{\varphi} = m_{\Phi,\text{rad}}$, or with a future P4 pole is an additional matching ansatz. A physical action further requires $S_{0,\text{phys}} = \Sigma_{\theta} t_0^{\text{EFT}}$. Neither the loop functional nor P3 derives a P4 threshold, the fixed-core data or a common physical owner; the bridge remains Level-B/Open.

P3 radial correlation and the Open P4 threshold. For the homogeneous P3 action, the Gaussian amplitude coordinate has $m_{\Phi,\text{rad}}^2 = 2\mu^2$. Its Euclidean propagator has the large-distance form

$$G_{\Phi,\text{rad}}(r) \propto \frac{e^{-m_{\Phi,\text{rad}} r}}{r^{3/2}} \quad (r \rightarrow \infty), \quad (8.10)$$

so $\xi_{\text{P3}} = m_{\Phi,\text{rad}}^{-1}$ is the P3 correlation length. This result does not provide a propagating pole, range or cutoff for the independently postulated P4 gradient background.

P4 EFT-threshold programme. A physical P4 breakdown threshold requires a stationary P4 action, its constrained Hessian and a positive isolated pole $m_{\text{P4,pole}}$. Those owners are not presently supplied, so $m_{\text{P4,pole}}$, $\xi_{\text{P4}} = m_{\text{P4,pole}}^{-1}$ and any P4 decoupling threshold are Open. If such a pole is later derived, standard EFT matching may use it as a conditional threshold. Identifying it with $m_{\Phi,\text{rad}}$ is a separate matching hypothesis, not an identity. Neither statement supplies a microscopic ultraviolet regulator or proves that the full theory is ultraviolet complete.

Connection to the quantum-gravity programme. The P3 amplitude scale, a future P4 pole and the physical action-unit programme are distinct owners. No current theorem turns the P3 correlation length into a gravity cutoff or a universal action quantum. A completed P4 action could later provide a pole and a controlled EFT domain; that result would still require independent tensor, state and action-normalisation maps.

Status of the primary and secondary amplitude coordinates. A homogeneous P3 amplitude curvature exists conditionally in the supplied P3 action. The P4 decomposition supplies only an amplitude coordinate until its stationary action and pole are derived. Therefore no physical “primary radial particle” is required merely by P4 ordering. A secondary Higgs-like mode is even more conditional: it exists only inside a separately supplied electroweak-like transition and field completion. The two coordinates, their actions and any mixing vertex must be independently owned.

Second radial mode: candidate from the electroweak-like transition. If the second condensate transition $O(3) \rightarrow O(2)$ is realised at scale v_2 (Appendix M, assumptions A1–A2), the effective broken-phase description at that scale admits a radial fluctuation around the second condensate vacuum. In an effective field theory language (*not* a second independent fundamental field, but an effective condensate sector at scale v_2):

$$\phi_{\text{ew}} = v_2 + h, \quad m_h^2 = V''_{\text{ew}}(v_2) = 2\lambda_2 v_2^2, \quad (8.11)$$

where λ_2 is the quartic coupling of the second-condensate effective sector at scale v_2 .

This mode is *structurally analogous* to the Standard Model Higgs boson in the following sense:

- it is the radial excitation of the second-condensate VEV (formally analogous to the Higgs field);
- the spontaneous breaking of the internal symmetry at v_2 would generate masses for the vector bosons of the embedded electroweak-like gauge sector, if a local gauge realisation exists (formally analogous to the Higgs mechanism);
- the relation $m_h = \sqrt{2\lambda_2} v_2$ has the same structure as the SM Higgs mass formula.

Honest status of the Higgs-like mode. Three levels must be distinguished:

1. **Level B/C (structural):** if the second transition exists and if the corresponding $O(3)$ -sector admits a local gauge realisation (not implied by P1–P6; part of OP-EW), a radial mode h at scale $v_2 \approx 246$ GeV is structurally present.
2. **Matching constraint:** the PDG-2025 review quotes the ATLAS Run-1+Run-2 diphoton-plus-four-lepton combination $m_H = 125.11 \pm 0.09$ (stat) ± 0.06 (syst) GeV [52]. Consistency with this external measurement would require $\lambda_2(v_2) \approx 0.13$. This value is not derived from ECT postulates P1–P6; it is a matching input if one demands consistency with the measured Higgs mass.
3. **Open (OP-EW):** the dynamical mechanism producing v_2 , the origin of λ_2 , and the localisation of the $O(3)$ symmetry at scale v_2 are not derived from the minimal scalar condensate and remain open problems.

Relation to the scale inventory. The P3 radial sector has the exact supplied-model relation $m_\sigma = \sqrt{2\lambda}\phi_0$; its numerical mass and the distinct P4 pole remain Open. Conditionally, the secondary sector has $m_h \sim v_2$. The distinct kpc quantity L_{gal} is a conditional HRC matching length used in Level-C diagnostics; assigning it to the amplitude sector requires the still-Open scalar-to-galactic map.

Higgs self-coupling and testability. If the secondary condensate potential is quartic near v_2 ,

$$V_{\text{ew}}(\phi_{\text{ew}}) = \frac{\lambda_2}{4} (\phi_{\text{ew}}^2 - v_2^2)^2, \quad (8.12)$$

then the Higgs mass satisfies

$$m_h^2 = 2\lambda_2 v_2^2, \quad \lambda_2 = \frac{m_h^2}{2v_2^2}. \quad (8.13)$$

The corresponding cubic self-interaction is

$$V_{\text{ew}} \supset \lambda_2 v_2 h^3, \quad (8.14)$$

or equivalently

$$V_{\text{ew}}'''(v_2) = 6\lambda_2 v_2 = \frac{3m_h^2}{v_2}. \quad (8.15)$$

Different collider conventions for the trilinear Higgs coupling differ by combinatorial factors; the expression above refers to the third derivative of the effective potential at the minimum. Deviations from this quartic relation would signal non-quartic corrections to the effective potential at scale v_2 —for instance, induced by the primary condensate sector or by the dynamics of the second transition itself. Deviations from this quartic relation would indicate that the effective secondary potential is not approximately quartic near its minimum, or that additional condensate-sector corrections enter already at that scale. The present framework does not determine the size of such deviations.

Reframing the hierarchy problem. The usual SM hierarchy problem asks: why is the Higgs mass so small compared to the Planck scale [53]? ECT recasts this as a condensate-spectrum question: why does a second condensate branch appear at $v_2 \sim 10^{-16} \phi_0$? This is still an open problem (OP-EW), but the reframing is itself meaningful: instead of a fine-tuning of one scalar mass, the challenge becomes explaining the dynamical origin of a second ordered phase at an intermediate scale. The updated OP-EW-scale programme treats this hierarchy as a non-polynomial scale-generation problem, $v_2 = \phi_0 e^{-\mathcal{J}_{\text{EW}}}$ with $\mathcal{J}_{\text{EW}} \approx 36.8$ (Section 5.1). Candidate mechanisms include alignment running, walking/near-conformal dynamics, Miransky/BKT scaling, and topological/Hopf-like routes (Appendix N). None is derived from P1–P6, and none by itself solves the radiative stability problem of the secondary radial mode.

Radiative stability: what ECT does and does not solve. The reframing of the hierarchy problem as a condensate-spectrum question does not by itself resolve the radiative stability issue. If the secondary radial mode h couples to other fields (gauge bosons, fermions) with generic EFT couplings, its mass receives quadratic radiative corrections of order

$$\delta m_h^2 \sim \frac{g^2}{16\pi^2} \Lambda_{\text{sec}}^2, \quad (8.16)$$

where Λ_{sec} is an independently supplied regulator or heavy threshold of the secondary EFT. This is a schematic Wilsonian sensitivity; its coefficient and even its explicit quadratic form depend on the regulator, couplings and field content. Choosing $\Lambda_{\text{sec}} \sim \bar{M}_{\text{Pl}}$ is a benchmark, not a result: neither the P3 mass m_σ nor an Open P4 pole is automatically the secondary cutoff. The resulting fine-tuning is of the same order as in the Standard Model. ECT does not claim to solve this problem. What it offers is a different *physical context*: the fine-tuning question becomes whether the dynamics of the second condensate transition naturally selects $v_2 \ll \phi_0$. A possible mechanism is dimensional transmutation via the RG running of the effective quartic coupling from ϕ_0 to v_2 (Section 5.1, Appendix M), but this remains Level C and has not been demonstrated within ECT. Absent an additional symmetry, compositeness mechanism, or dynamical selection principle, the secondary scalar remains quadratically sensitive to the primary UV scale. Any claim that ECT resolves the naturalness problem would be premature.

Comparison with other Higgs frameworks. *Standard Model Higgs:* in the SM the Higgs is an elementary scalar doublet introduced phenomenologically [54, 55]. In ECT the P3 homogeneous radial coordinate is present in the supplied P3 model, while a physical P4 radial pole and the Higgs-like secondary mode both require additional action owners; the latter is also conditional on the second transition.

Condensed-matter amplitude (Higgs) modes: in superfluids and in appropriate broken-phase condensed-matter systems, one encounters a phase mode together with a massive amplitude (Higgs-like) mode [56, 20]. The homogeneous P3 model has an amplitude coordinate and Gaussian curvature analogous at the level of its supplied quadratic model. A physical P4 amplitude particle, however, requires its own stationary action, pole and observable map and is Open.

EFT hierarchy and decoupling: in standard EFT, heavy particles decouple at low energies and generate suppressed higher-dimension operators. A future P4 completion with a positive isolated heavy pole could play that EFT role after its vertex and scale separation are derived. The P3 curvature does not presently license an integration-out step for P4.

A further useful comparison is with frameworks in which the Higgs is itself a non-elementary excitation of an underlying sector.

Composite Higgs models [57, 37]: in these models the Higgs is not elementary but a pseudo-Nambu–Goldstone boson of a new strongly coupled sector. The ECT secondary radial mode shares the feature of being a *condensate excitation* rather than a fundamental scalar, but differs in mechanism: it is the amplitude fluctuation of a second ordered phase, not a pseudo-NGB of a confining gauge theory. Unlike pseudo-Nambu–Goldstone Higgs scenarios, the ECT secondary radial mode is not protected by an approximate shift symmetry in the present formulation. The two approaches face the same hierarchy challenge; ECT recasts it as a condensate-dynamics question rather than a compositeness question.

Status summary.

Statement	Status	Comment
P3 homogeneous radial curvature/pole $m_\sigma^2 = 2\mu^2$	A conditional in P3	Gaussian expansion about either homogeneous P3 minimum; no P4 owner is implied
P4 amplitude coordinate σ	A-math kinematics	Coordinate in a supplied ordered-branch parametrisation, not yet a physical excitation
Stationary P4 radial Hessian, pole and positive residue	Open	Requires a stationary P4 action, state, constrained Hessian, domain and projector
Primary scalar sector is distinct from the SM Higgs	A by definition	P3 and the conditionally supplied secondary electroweak sector have different owners
Radial and longitudinal scalar coordinates separate at linear order	A-math conditional	Supplied broken-branch parametrisation and integrability of Q_A (Section 4.3)
$m_\sigma = \sqrt{2\lambda}\phi_0$	A inside supplied P3 / numerical value Open	Exact P3 curvature relation; neither λ nor a P4 mass is selected
P4 EFT validity/breakdown threshold	Open	Requires an action-owned heavy pole and matching calculation; the P3 correlation mass is not a P4 cutoff

Statement	Status	Comment
Candidate $m_{\sigma-t_0^{\text{EFT}}}$ matching	Open ansatz	Requires a derived common action, vertex and normalisation; dimensional proximity is insufficient
Secondary Higgs-like radial mode h	B/C	Conditional on a realised $O(3) \rightarrow O(2)$ transition
Secondary mode is structurally analogous to SM Higgs	B/C	Formal radial analogue; not a derivation
$m_h = \sqrt{2\lambda_2} v_2$ with v_2 from G_F matching	B/C	Matching input, not ECT prediction
$\lambda_2(v_2) \approx 0.13$ if matched to $m_H \approx 125$ GeV	Constraint	External SM matching, not an ECT derivation
Radiative stability of m_h	Open	Quadratic sensitivity to the primary UV scale remains
Quartic relation for m_h and the cubic interaction near the minimum	B/C	Holds if the secondary potential is nearly quartic
Yukawa / gauge mass generation fully derived	Open	Not completed; part of OP-EW

8.4 Vacuum Stability, False Vacuum, and Vacuum Decay

8.4.1 Why the question matters

In standard quantum field theory, the vacuum state of a scalar field can be metastable: the observed vacuum may be only a local, not a global, minimum of the effective potential. If a deeper minimum exists, the field can undergo a non-perturbative transition — Coleman vacuum decay [58] — in which a bubble of the “true vacuum” nucleates inside the “false vacuum” and subsequently expands, converting the entire spacetime. For the Standard-Model Higgs field, radiative corrections driven primarily by the large top-Yukawa coupling are known to make the electroweak vacuum metastable at high field values [59, 60]: the running quartic coupling $\lambda_{\text{SM}}(\mu)$ crosses zero near $\mu \sim 10^{10}$ GeV, and the effective potential turns downward at large Higgs-field amplitudes.

Whether the primary ECT condensate suffers from an analogous instability is therefore a central self-consistency question. This subsection analyses the problem from first principles: the structure of the ECT potential, its critical points, the one-loop running of the quartic coupling, the absence of fermion-driven destabilisation, the nature of the cosmological ordering transition, and the role of subleading sectors.

8.4.2 Structure of the minimal ECT potential

The bare potential of the primary condensate (P3) is

$$V(\Phi) = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4, \quad \mu^2 > 0, \quad \lambda > 0. \quad (8.17)$$

Its critical points are:

- $\Phi = 0$: symmetric stationary point, with $V''(0) = -\mu^2 < 0$. This is an *unstable hilltop*, not a metastable minimum. Its P3 second variation has a negative homogeneous direction; a real-time roll or spinodal law requires a separately supplied Lorentzian dynamics and state.
- $\Phi = \pm\phi_0$ with $\phi_0 = \sqrt{\mu^2/\lambda}$: the two global minima, with $V''(\phi_0) = 2\mu^2 > 0$ (tree-level local stability) and $V(\phi_0) = -\mu^4/(4\lambda) < 0$. These two minima are exactly degenerate by the $\mathbb{Z}_2$ symmetry $\Phi \rightarrow -\Phi$ of the potential.

The minimal ECT potential therefore contains *no metastable local minimum* separated from a deeper minimum by a barrier. Both minima are global, and the symmetric point is a maximum, not a false vacuum. This is a crucial structural difference from any double-well configuration that would be needed for Coleman vacuum decay.

8.4.3 Inapplicability of Coleman vacuum decay

Coleman’s semiclassical theory of false-vacuum decay [58, 61] requires two specific ingredients: (i) a *metastable* minimum (the “false vacuum”), in which the field can sit for a long time, and (ii) a *deeper* minimum (the “true vacuum”), separated from the first by a potential barrier. The decay rate per unit volume is then

$$\Gamma/V \sim A e^{-B/\hbar}, \quad (8.18)$$

where $B = S_E[\Phi_{\text{bounce}}] - S_E[\Phi_{\text{false}}]$ is the difference in Euclidean action between the bounce configuration and the homogeneous false vacuum.

In the minimal ECT potential (8.17), *neither* ingredient is present:

1. $\Phi = 0$ is a hilltop ($V''(0) < 0$), not a metastable minimum. A field configuration at $\Phi = 0$ does not “sit” there; it rolls away immediately (spinodal regime, not metastable trapping).
2. The two minima $\pm\phi_0$ are exactly degenerate ($\Delta V = 0$). They therefore do not furnish a false-to-lower-vacuum Coleman channel. Degenerate-domain walls, finite-volume tunnelling or other Euclidean saddles are different questions and are not excluded by this observation.

Therefore, the standard **metastable-to-lower-vacuum Coleman decay hypotheses are absent from the homogeneous tree-level P3 potential**. The formula (8.18) has no such false-vacuum subject in that bounded model. This does not prove the absence of every finite-action saddle or decay channel in an owner-complete P4/quantum theory.

8.4.4 The Open cosmological-ordering owner is not fixed by the P3 false-vacuum test

P4 supplies the gradient-ordered chart used by the conditional scalar-signature construction (Section 6.8); it does not derive a cosmological $O(4) \rightarrow O(3)$ transition. No current owner identifies a pre-ordered P4 regime with the homogeneous P3 coordinate $\Phi \simeq 0$. Consequently the P3 hilltop algebra neither proves that physical ordering is spinodal nor determines a nucleation/bounce exponent. Spinodal ordering and barrier-mediated nucleation remain distinct conditional completion classes, each requiring a real-time or state/transition owner and compatible boundary data.

8.4.5 One-loop stability of the primary amplitude sector

Even though no Coleman false vacuum exists at tree level, one must still ask whether radiative corrections could generate new minima at large field values, as happens in the Standard Model. This is the question of *RG-driven metastability*.

The running coupling in strict scalar closure. The one-loop β -function for a single real scalar in the minimal ECT action (Eq. (8.29), Section 5.1) is

$$\beta_\lambda \equiv \frac{d\lambda}{d\ln\mu} = \frac{9\lambda^2}{8\pi^2} > 0, \quad (8.19)$$

which is *strictly positive* for any $\lambda > 0$. This is the standard one-loop result in the manuscript’s $V = \lambda\phi^4/4$ normalisation. Relative to $V = g_4\phi^4/4!$, one has $g_4 = 6\lambda$.

With the boundary condition $\lambda(M_{\text{Pl}}) \approx 10^{-3}$ (Section 5.1), the running coupling has the explicit solution

$$\lambda(\mu) = \frac{\lambda_0}{1 - \frac{9\lambda_0}{8\pi^2} \ln \frac{\mu}{M_{\text{Pl}}}}, \quad \lambda_0 \equiv \lambda(M_{\text{Pl}}). \quad (8.20)$$

Consequences for vacuum structure. Since $\beta_\lambda > 0$, the coupling $\lambda(\mu)$ grows with the renormalisation scale μ in this strict one-real-scalar, one-loop leading-log truncation. Three scoped statements follow:

1. **No zero crossing occurs in the displayed running below its pole.** For $\lambda_0 > 0$, the denominator in Eq. (8.20) stays positive for $\mu < \mu_{\text{LP}}$, where $\mu_{\text{LP}} \approx M_{\text{Pl}} e^{8.77 \times 10^3}$. This is a statement about the displayed truncation, not about scales beyond its EFT domain.
2. **The Standard-Model-like negative-running route is absent at this order.** The leading-log expression $V_{\text{LL}}(\phi) \sim \lambda(\phi)\phi^4/4$ does not turn downward through a zero of λ . Positivity of β_λ alone does *not* prove that the full effective potential has no additional extrema. Threshold matching, other fields, higher loops, higher-dimension operators and non-perturbative effects remain outside this calculation.
3. **The formal pole lies outside the stated EFT range.** The displayed truncation has a separately declared domain, while its physical P4 cutoff is not derived. Accordingly a formal pole beyond the controlled domain is not a physical prediction. The boundary value $g(M_{\text{Pl}}) = \lambda_0/(8\pi) \approx 4 \times 10^{-5}$ is small within the declared benchmark.

Why Standard-Model metastability does not transfer. The Standard-Model running of the Higgs quartic differs qualitatively. At one loop, the schematic structure is

$$\beta_{\lambda_{\text{SM}}} = \frac{1}{16\pi^2} [24\lambda_{\text{SM}}^2 + \lambda_{\text{SM}}(12y_t^2 - 9g^2 - 3g'^2) - 6y_t^4 + \text{gauge quartic terms}], \quad (8.21)$$

so that the large top-Yukawa contribution drives $\lambda_{\text{SM}}(\mu)$ downward and, in the observed parameter range, eventually through zero near $\mu \sim 10^{10}$ GeV [59, 60]. This is the origin of the familiar electroweak metastability story.

In the strict isolated primary-scalar truncation, the one-loop β -function contains only the positive self-coupling term $+9\lambda^2/(8\pi^2)$ and no top-Yukawa-like negative term. This excludes that particular negative-running mechanism at this order and is a structural difference between the two stated truncations (Level B conditional). It is not a proof of the full primary-sector effective potential after threshold, matter, higher-loop or non-perturbative completion.

8.4.6 Analysis of subleading contributions

Soft orientation sector. In the supplied derivative-only reduced benchmark, the physical-mode interpretation of the longitudinal coordinate χ (Section 4.3) is parametrically softer than the homogeneous-P3 radial coordinate and provides no identified destabilising channel for the primary amplitude potential in that minimal closure. The integrability constraint sharply reduces the independent soft sector relative to a generic multi-field broken-symmetry system, and no analogue of the Standard-Model top-Yukawa destabilisation mechanism is present here. A full constrained fluctuation-determinant analysis for the coupled amplitude–orientation system has not yet been carried out, so the present statement should be read as follows: within the current minimal closure, the conditional soft sector does not generate an identified mechanism capable of overturning the one-loop amplitude-sector stability result. This is not a derived physical χ loop or a proof of full branch stability.

Conditional target-space sectors. The supplied orientation target $\mathcal{T}_{\text{P4}} = O(4)/O(3) \simeq S^3$ has map classes classified by $\pi_3(S^3) = \mathbb{Z}$ (Section 8.10). Within a separately supplied positive two-derivative S^3 sigma model, nonuniform representatives carry additional gradient action $\int (\partial n)^2 d^4X \geq 0$ relative to a uniform map. This conditional calculation does not establish that such maps are admissible exact-gradient

ECT configurations or tunnelling endpoints. Likewise $\pi_0(S^3) = 0$ says that this target has no disconnected components; a physical domain-wall statement requires the complete vacuum/configuration owner and remains Open.

Embedded electroweak sector. If the conditional secondary transition (Section 8.7) realises a Standard-Model-like Higgs sector at the scale $v_2 \approx 246 \text{ GeV}$, the question of electroweak vacuum metastability belongs to that *embedded* sector, not to the primary condensate. The primary amplitude sector does not inherit the SM-type instability mechanism (which requires fermionic couplings absent at this level), but the full threshold and matching analysis between primary and secondary sectors has not been completed. The corresponding threshold and matching problem remains part of the second-transition programme (OP-EW).

8.4.7 Taxonomy of vacuum-decay scenarios

Table 25: Applicability of vacuum-decay scenarios to the primary ECT condensate.

Scenario	Applies?	Reason
Coleman false $\rightarrow$ true	No	No metastable minimum in the minimal potential; $\Phi = 0$ is a hilltop
SM-type negative-running route	Absent in the strict one-loop scalar truncation	$\beta_\lambda > 0$ below the formal pole; full effective-potential stability is Open
Thermal first-order transition	No	Article explicitly distinguishes the ordering transition from thermal bubble nucleation
Ordering / spinodal-like instability of the symmetric hilltop	Structural analogue	The cosmological ordering transition is closer to this class than to Coleman nucleation, but its precise non-equilibrium mechanism remains open
Branch-to-branch (π_3 target class)	Open in ECT	A supplied positive sigma model raises the gradient action, but physical configurations, endpoints and transition action are not owned
Embedded EW metastability	Separate question	Belongs to the secondary sector, not to the primary condensate

8.4.8 Summary: derivation status

Table 26: Vacuum-stability results and their derivation level.

ID	Level	Result
R1	A	Local tree-level stability: $V''(\phi_0) = 2\mu^2 > 0$
R2	A-math conditional / physical Open	For the supplied S^3 target, $\pi_0 = \pi_1 = \pi_2 = 0$; physical wall, string and monopole conclusions require the complete vacuum/configuration space and transition owner
R3	B conditional	In the strict scalar one-loop leading-log closure, $\beta_\lambda > 0$ keeps the displayed $\lambda(\mu)$ positive below its pole and excludes the Standard-Model-like negative-running route at this order; additional extrema and full effective-potential stability remain Open

ID	Level	Result
R4	B / Open	No identified destabilising channel from the integrability-reduced χ sector in the minimal closure; a full constrained amplitude–orientation determinant analysis remains open
R5	A-math conditional / physical Open	$\Phi = 0$ is a hilltop ($V''(0) < 0$), not a metastable minimum; the standard false-to-lower-vacuum Coleman hypotheses are absent from the homogeneous tree-level P3 potential, while other saddles and the P4 transition owner remain Open
O1	Open	Full branch-architecture stability (nonlinear amplitude–orientation couplings, non-perturbative channels, matching to extended sectors)
O2	Open	Electroweak metastability: belongs to the embedded secondary sector (OP-EW)

8.4.9 Conclusion: vacuum decay in ECT

In the printed tree-level minimal potential, the ordered minima are not Coleman false vacua: the symmetric point is a hilltop and there is no second metastable local minimum. Separately, the strict one-real-scalar one-loop leading-log running excludes the Standard-Model-like quartic zero-crossing mechanism at that order. Neither statement establishes the global structure of a completed ECT effective potential. Threshold and matter sectors, the constrained amplitude–orientation determinant, higher-loop and non-perturbative effects remain Open.

The P3 hilltop can motivate a spinodal analogy, but it does not own the cosmological P4 ordering transition or identify its pre-ordered state. No metastable-to-lower-vacuum channel has been identified inside the stated homogeneous P3 truncation; this bounded negative result is not a proof of absolute, all-saddle or all-branch stability.

8.4.10 Euclidean formulation and the status of bounce configurations

A conceptual remark is in order regarding the role of the Euclidean formulation. In the standard semiclassical decay construction, a Euclidean bounce is used after the relevant analytic continuation and boundary conditions are specified; this does not assign ECT’s medium ontology to the Euclidean variables. In ECT the situation is reversed: the four-dimensional Euclidean manifold is *fundamental* (P1), while the present theory only conditionally reconstructs a Lorentzian scalar characteristic sector; physical spacetime remains an Open completion. If a bounce-like saddle configuration of the condensate field existed — connecting, say, two different ordered sectors — it would be a *native* object of the theory, not an analytic-continuation artefact.

This observation has two consequences. First, it means that the absence of Coleman vacuum decay established above is not an artefact of a Lorentzian-to-Euclidean continuation: the ECT potential genuinely lacks the required metastable-minimum structure on its native Euclidean arena. Second, it constrains any future ordering completion: if a physical $O(4) \rightarrow O(3)$ transition is derived, it must be owned by a stationary Euclidean action/state and its saddle or real-time completion, rather than assumed to be tunnelling in a pre-existing Lorentzian spacetime. The present P3 action does not supply that transition. Non-perturbative Euclidean transitions (branch changes, texture formation, defect-mediated processes) are possible future completion classes, but none is derived or identified with a Coleman-type decay of the supplied P4 chart (see Table 25); their physical realisations remain Open.

Ontological reading: alternative configurations versus time-evolving decay. The statements above concern the minimal P3 potential and the separately supplied P4 chart. They should also be read with the correct ontology in mind. In standard semiclassical language, false-vacuum decay is often described as a Lorentzian-time process in which a metastable field configuration tunnels into a lower-energy

configuration through a Coleman bounce. ECT does not take such a Lorentzian time-history as primitive. At the primitive level the object is a four-dimensional Euclidean condensate configuration, and there is no external time parameter with respect to which one complete global configuration literally becomes another. The Coleman formalism remains the correct effective-field-theory tool where its assumptions are satisfied; what changes in ECT is the interpretation. Without a supplied Euclidean measure, contour, state and continuation, a saddle is only a stationary configuration and controls neither a relative weight nor a transition amplitude. In the minimal P3 potential, the required false-vacuum structure is absent. In more general extensions, possible vacuum-transition questions must therefore be posed first as questions about Euclidean saddle structure and boundary data. Only after the missing measure and observable owners are supplied may they be translated into an effective Lorentzian decay-rate language.

8.4.11 Comparison with other frameworks

The vacuum-stability question takes different forms in different frameworks.

In the Standard Model, the electroweak vacuum is metastable because the large top-Yukawa coupling drives the Higgs quartic downward at high scale, eventually sending $\lambda_{\text{SM}}(\mu)$ through zero. In string-landscape settings, by contrast, vacuum decay is tied to the existence of many distinct local minima and transition channels between them.

The minimal P3 scalar model differs structurally from both pictures. Within the strict scalar closure, there is no fermionic destabilising channel analogous to the Standard-Model top-Yukawa mechanism, and the minimal primary potential does not exhibit a landscape of competing isolated minima. The resulting completion problem is therefore different: P3 has its stated homogeneous minima, whereas stability of any physical nonzero-gradient P4 branch and of its low-energy amplitude sector remains Open.

8.5 Fixed-map tensor no-go and the physical-gravity completion programme

Status. Level A for the scalar characteristic tensor and for the negative theorem below, within the displayed fixed- (α, β) map. Open for a physical helicity-2 field, its constrained action, canonical residues, universal matter vertex, nonlinear completion and quantum state. Fierz–Pauli, Einstein gravity, BMS/memory and gravitational-wave formulae are retained only as standard completion targets or external tests.

The separately supplied scalar quadratic model uses the principal tensor

$$G_{\text{eff}}^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad n_A n^A = 1, \quad (8.22)$$

or, using $P_{\parallel}^{AB} = n^A n^B$ and $P_{\perp}^{AB} = \delta^{AB} - n^A n^B$, $G_{\text{eff}}^{AB} = \beta P_{\perp}^{AB} - (\alpha - \beta) P_{\parallel}^{AB}$. For $\alpha > \beta > 0$ this tensor has one hyperbolic direction. This is a statement about a characteristic geometry. It does not by itself identify the physical metric seen by matter, photons and a possible tensor mode.

The inverse of the displayed tensor is

$$g_{AB}^{\text{eff}} = \frac{1}{\beta} P_{AB}^{\perp} - \frac{1}{\alpha - \beta} P_{AB}^{\parallel}, \quad G_{\text{eff}}^{AC} g_{CB}^{\text{eff}} = \delta^A_B. \quad (8.23)$$

At the homogeneous point $\bar{n}^A = \delta_w^A$ and for a tangent variation $\bar{n} \cdot \delta n = 0$ at fixed α, β ,

$$\delta G_{\text{eff}}^{AB} = -\alpha (\bar{n}^A \delta n^B + \delta n^A \bar{n}^B), \quad (8.24)$$

and variation of the inverse relation gives

$$h_{AB} \equiv \delta g_{AB}^{\text{eff}} = -\frac{\alpha}{\beta(\alpha - \beta)} (\bar{n}_A \delta n_B + \delta n_A \bar{n}_B). \quad (8.25)$$

Theorem 8.1 (Fixed-map linear tensor/static-density no-go). *For the map above, with fixed α, β , tangent director fluctuations and the condensate time direction w , the only nonzero linear components are*

$$h_{iw} = -\frac{\alpha}{\beta(\alpha - \beta)} \delta n_i, \quad h_{ij} = h_{ww} = 0. \quad (8.26)$$

Consequently $h_{ij}^{\text{TT}} = 0$ and a static nonrelativistic density has $T^{AB}h_{AB} = \rho h_{ww} = 0$. Moreover $\delta \ln |\det G| = g_{AB}\delta G^{BA} = 0$. Under strict integrability, $\delta n_i = \partial_i \chi / u_0$, the surviving component is longitudinal shift-type data, not an independent TT graviton.

Proof. At $\bar{n}^A = \delta_w^A$, tangent fluctuations have $\delta n_w = 0$. Substitution in Eq. (8.25) gives the component statement directly. The TT projector annihilates the zero spatial block, while a static $T^{ww} = \rho$ contracts only with h_{ww} . The determinant result follows by tracing Eq. (8.24) with the inverse metric. $\square$

The theorem excludes the specific live chain $\delta n_i \rightarrow h_{AB}[n] \rightarrow (h_{ij}^{\text{TT}}, h_{ww})$; it is not a no-go theorem for all possible ECT gravities. A viable completion may contain an independent constrained tensor, dynamic α or β , an additional amplitude/composite variable, or a nonlocal reduction map. Such a completion must supply a physical projector, positive residues, the matter/photon metric, a source vertex and, for PES applications, a state and noise kernel.

Conditional Fierz–Pauli target. If an independent healthy symmetric tensor $H_{\mu\nu}$ is supplied, its standard massless quadratic target is the Fierz–Pauli action [62]

$$S_{\text{FP}}^{(2)} = -\frac{M_T^2}{8} \int d^4x [\partial_\lambda H_{\mu\nu} \partial^\lambda H^{\mu\nu} - 2\partial_\mu H^{\mu\nu} \partial^\lambda H_{\lambda\nu} + 2\partial_\mu H^{\mu\nu} \partial_\nu H - \partial_\lambda H \partial^\lambda H]. \quad (8.27)$$

After the usual constraints and gauge quotient this conditional theory has exactly two transverse-traceless polarisations and no ghost. These are standard Fierz–Pauli facts, not consequences of Eq. (8.25). Any proposal in which the physical spin-2 carrier is composite must also state which assumptions of the Weinberg–Witten obstruction do or do not apply [63]; the present fixed-map negative result is not itself an application of that theorem. The Deser bootstrap [64] is a possible nonlinear route only after a physical FP seed and universal $H_{\mu\nu}T^{\mu\nu}/2$ vertex exist.

Normalisation firewall. The orientation coefficient $\kappa_n = \mathcal{C}_n u_0^2$ and the squared conditional tensor scale M_T^2 have the same mass dimension but are not thereby equal. Inside a separately supplied Einstein/FP completion one may match

$$G_N^{\text{nat}} = \frac{1}{8\pi M_T^2}, \quad (8.28)$$

but $\kappa_n \rightarrow M_T^2 \rightarrow G_N$ remains Open. The former scalar-RG expression

$$g_{\text{sc}}(\mu) \equiv \frac{\lambda(\mu)}{8\pi} \quad (8.29)$$

and the scalar Landau-pole estimate

$$\mu_{\text{LP}} = M_{\text{Pl}} \exp\left(\frac{8\pi^2}{9\lambda(M_{\text{Pl}})}\right) \quad (8.30)$$

remain conditional scalar/matching algebra; they are not graviton-loop renormalisation results.

External theorem and observation gates. The Weinberg–Witten assumptions must be evaluated only after a physical Lorentzian tensor completion is specified; the Euclidean starting point alone does not construct an evasion. Likewise GW170817/GRB 170817A is a stringent external gate on the principal symbol of any future tensor completion. The current scalar characteristic cone does not determine c_{GW} .

Completion falsifiers. A proposed tensor completion fails if its reduced spectrum contains a ghost, cannot produce a nonzero projected source vertex, violates universal matter coupling, predicts an excluded tensor speed or polarisation content, or cannot be joined consistently to its nonlinear equations. Until those tests pass, ECT possesses a tensor-completion programme rather than a derived graviton.

Tensor-owner status.

Statement	Status	Comment
Scalar characteristic tensor and Lorentzian signature	Level A	Principal-symbol result in the stated scalar model
Fixed-map component theorem $h_{iw} \neq 0$, $h_{ij}^{\text{TT}} = h_{ww} = 0$	Level A negative	Exact at fixed α, β and linear tangent order
Physical matter/photon metric and density vertex	Open	Not fixed by a scalar characteristic cone
Fierz–Pauli action and two TT polarisations	Conditional comparison	Standard if a healthy independent tensor is supplied
$\kappa_n \rightarrow M_T^2 \rightarrow G_N$	Open	Dimensional equality is not a derivation
c_{GW} and tensor LIV operators	Open / external gate	Require the tensor principal symbol and coefficients
Nonlinear Einstein/Deser completion	Open	Requires a physical FP seed and universal source

Detailed tensor-completion audit

Why a physical tensor sector would be required. The scalar characteristic geometry determines a causal cone for the stated ordered-branch scalar equations, but it does not by itself provide the dynamical field that carries the radiative, tidal and static-potential observables attributed to gravity. A complete theory must identify the physical variable before assigning it helicity, count its independent modes after all constraints, and derive the metric to which matter and photons couple. The fixed-map theorem above shows why those steps cannot be skipped: the only linear image of a tangent director fluctuation lies in h_{iw} , whereas both the spatial TT block and the static h_{ww} contact vanish.

Separation from the radial mode. A hypothetical P4 amplitude pole and the director fluctuation would be distinct operators. If one common stationary completion supplied such a pole and a vertex to the orientation operator, integrating it out could renormalise $\kappa_n = \mathcal{C}_n u_0^2$. Neither pole nor vertex is presently owned, and a coefficient correction would not change the representation content of the surviving linear field. Conversely, keeping σ as an explicit scalar can produce a trace coupling in a supplied scalar–tensor completion, but it does not generate the two helicity-2 polarisations. Any proposed mixed route must give the full constrained Hessian in the $(\sigma, \delta n)$ basis and demonstrate which projected pole carries the gravitational residue.

Conditional Fierz–Pauli calculation. For an independently supplied symmetric field $H_{\mu\nu}$, the quadratic Fierz–Pauli action displayed above is invariant under $H_{\mu\nu} \mapsto H_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. The lapse- and shift-like components impose constraints; after quotienting the gauge directions, two transverse-traceless modes remain. This calculation is standard and remains in the manuscript as the target that an ECT tensor owner must reproduce. It is conditional because neither $H_{\mu\nu}$ nor its gauge redundancy follows from the current director image. In particular, even a separately supplied internal $SU(2) \times U(1)$ redundancy would not imply linearised diffeomorphism redundancy.

Source and normalisation requirements. The physical source test is operator-level. A candidate must yield a nonzero term of the form

$$\frac{1}{2} \int d^4x H_{\mu\nu} T^{\mu\nu}$$

or a fully specified alternative with the same weak-field observables. Noether conservation of $T^{\mu\nu}$ is necessary for a massless tensor completion but does not create this vertex. Likewise, the equality of mass dimensions $[\kappa_n] = [M_T^2]$ does not identify the two coefficients. The chain $\kappa_n \rightarrow M_T^2 \rightarrow G_N$ therefore contains an Open first arrow and a standard weak-field matching second arrow.

Weinberg–Witten and emergent-medium comparisons. The Weinberg–Witten assumptions apply only after one specifies the Lorentzian Hilbert space, the covariant conserved stress tensor and the particle pole to which the theorem is being applied. A pre-geometric Euclidean medium may evade one or more assumptions, but an evasion is not a construction of the missing pole. Analogue-gravity systems provide useful examples of characteristic metrics without a dynamical Einstein tensor. Sakharov-type induced gravity can generate an Einstein–Hilbert coefficient after a field content, regulator and state are fixed, but it likewise does not turn the present fixed director image into a TT field. These comparisons delimit completion routes rather than establish ECT ownership.

Nonlinear completion. The Deser bootstrap and Lovelock uniqueness results are retained in their standard conditional form. Starting from a healthy universal massless spin-2 seed, iterative self-coupling leads to an Einstein-type nonlinear theory under the usual locality and derivative assumptions. The theorems do not manufacture the seed, its positive residue, or its universal matter vertex. If ECT supplies a different constrained spectrum, the nonlinear completion must be redone rather than inferred by analogy.

Cone and propagation gates. The dimensionless scalar slope $\hat{c}_*$ is derived in the supplied scalar model; c_{char} additionally requires the clock map. A physical tensor completion has its own principal symbol and speed c_T . Only after the tensor Hessian is known can one compare c_T with c_{char} and with the GW170817 bound. Similarly, an expression such as $\omega^2 = c_T^2 k^2 [1 + \zeta_2 (k/M)^2 + \dots]$ is a conditional operator benchmark until M and ζ_2 are derived. Agreement with observations would test the completion; it cannot be counted as a result of the fixed map.

Polarisation and mass tests. A completed massless Fierz–Pauli branch predicts two TT polarisations. A massive, scalar–tensor, vector–tensor or nonlocal completion may predict a different spectrum. Therefore two-polarisation and graviton-mass statements are not present predictions of ECT; they are discriminants among future completion candidates. Each candidate must be tested for ghosts, tachyons, gradient instabilities, strong coupling and compatibility with binary-pulsar, GW-propagation and Solar-System data.

Ultraviolet bookkeeping. The dimensionless expression $g_{\text{sc}}(\mu) = \lambda(\mu)/(8\pi)$ and the associated Landau-pole estimate are scalar self-coupling diagnostics in the displayed parametrisation. The notation carries no Newton coupling and is not a graviton-loop beta function. A genuine UV claim requires the canonical tensor propagator, vertices, gauge fixing, counterterms and a regulator compatible with the constrained physical subspace.

Application protocol. Before any tensor result is used in cosmology, BMV/GIE, memory, soft theorems, black-hole radiation or PES, the same ordered checklist must be passed:

- (a) identify the reduced physical tensor bundle and projectors;
- (b) derive its poles and positive residues;
- (c) derive the matter/photon vertex and the physical metric;
- (d) specify the nonlinear background and boundary conditions;
- (e) for open-system applications, specify a state, retarded kernel, noise kernel and resolved/unresolved split.

Until then the manuscript retains standard formulas, external bounds and conditional routes as a developed completion programme, not as derived ECT gravity.

8.6 Gauge Bosons from Condensate Phase Symmetries

Status: global compact-phase algebra is exact inside supplied BR1; local gauge bundle and physical photon Open. P5 motivates a redundancy programme but does not localise the phase, supply a gauge quotient, or protect a massless transverse state.

Status after P5. P5 does not turn a global phase symmetry into a local redundancy. A local $U(1)$ description becomes available only after a bundle and gauge-volume quotient are supplied. The connection and two-invariant ordered-background kinetic basis are then standard conditional mathematics. An unbroken photon, e , charged matter and the QED vertex remain Open.

Particle-level reading of the $U(1)$ route. At the level of a supplied compact-phase EFT and a separately supplied local bundle, one may write the standard Abelian completion. For a complex condensate

$$\Phi = \rho e^{i\theta} \quad (8.31)$$

the conditional local completion is invariant under

$$\theta \rightarrow \theta + \Lambda(X). \quad (8.32)$$

Introducing the covariant derivative

$$D_A = \partial_A - ieA_A \quad (8.33)$$

permits the ordered-background gauge kinetic basis

$$S \supset -\frac{1}{4} \int [Z_1 F_{AB} F^{AB} + 2Z_2 (n^A F_{AB})(n_C F^{CB})], d^4X. \quad (8.34)$$

Neither gauge covariance nor P5 sets $Z_2 = 0$; Maxwell form requires an additional isotropic-limit symmetry or matching condition. If this $U(1)$ is unbroken and the reduced Hessian is positive, its transverse mode is massless,

$$m_\gamma = 0, \quad (8.35)$$

as a property of the supplied completion. A charged condensate can instead Higgs the connection, so compactness alone gives no photon theorem.

What is strict and what is not. What follows strictly is only the operator algebra inside the locally completed model under the stated unbroken-phase assumptions. What does *not* follow from first principles is the full identification of this sector with physical quantum electrodynamics. In particular, the present subsection does not derive the electric charge e , the full QED matter coupling structure, radiative corrections, or the complete Standard Model electromagnetic sector from P1–P6 alone.

Prediction and non-prediction. The positive output of the present construction is structural rather than phenomenological: If a local compact- $U(1)$ bundle, healthy kinetic coefficients, a positive reduced Hessian and an unbroken phase are supplied, that completion can have a massless transverse Abelian mode. Compactness and P5 alone do not supply those owners, so ECT presently makes neither a photon-mass prediction nor a quantitative electromagnetic prediction from P1–P6.

Dependence on earlier results. The detailed derivation-status analysis of the Abelian gauge sector was already given in Section 7.2. The present subsection should therefore be read as the particle-level placement of that result within the classification scheme of Section 8, not as an independent second derivation.

Status summary.

Statement	Status	Comment
Compact phase supplies global winding data	A	It motivates, but does not derive, a local $U(1)$ bundle
Local $U(1)$ completion and ordered kinetic basis	Open	Bundle, coefficients, quotient/reduced Hessian and physical owner are missing
Massless Abelian mode in a supplied unbroken completion	A conditional / Open ECT	Standard property of that completion, not a P1–P6 theorem
Electric charge e derived from condensate microphysics	Open	Not completed
Full QED matter coupling structure	Open	Not derived from P1–P6
Full physical photon identification	Open	Unbroken $U(1)_{\text{em}}$, charge, matter and observable maps are missing

This identifies the conditional Abelian completion and its Open physical owners. The next subsection treats the still more conditional electroweak-like model.

8.7 Electroweak Sector and the Higgs Interpretation in ECT

Status: three-layer electroweak completion programme. Level A only for the Lie-algebra identity in the supplied branch; B/Open for the local internal bundle and chiral weak completion; B/C for the supplied secondary doublet scaffold matched at v_2 . Open for the first-principles derivation of v_2 , the quantitative electroweak parameter set, custodial protection, and Yukawa hierarchies. This subsection consolidates the radial-mode analysis (Section 8.3), the electroweak-like gauge embedding (Section 7.4), and the secondary-transition scaffold (Appendix M) and the candidate Hopf-fibered locking geometry (Appendix N).

What is established, what is matched, and what remains open in the electroweak programme. The electroweak programme should be read in three layers. First, the residual $O(3)$ after $O(4) \rightarrow O(3)$ provides the strict Lie-algebra identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, but no local internal bundle or chiral factor. These must be supplied in a candidate weak-like route (Sections 7.3–7.4, 9). Second, a lower transition scale v_2 can be introduced as the closure scale of a secondary ordered-branch step $O(3) \rightarrow O(2)$, providing a matched scaffold for the $W^\pm$, Z , and Higgs-like mass pattern. Third, what remains open is the actual first-principles derivation of $v_2 \approx 246 \text{ GeV}$, the quantitative electroweak mass spectrum, custodial protection, and Yukawa hierarchies. ECT therefore organises a conditional electroweak model, not a derived physical sector.

Why this sector is conditional rather than required. Unlike the P3 radial fluctuation and the strict longitudinal scalar-coordinate map on the supplied chart, neither the electroweak-like sector nor a physical graviton is forced by P4. It can arise only if the condensate admits a second effective ordering step beyond $O(4) \rightarrow O(3)$. For that reason, this sector belongs to the conditional/embedded part of the Section 8 classification rather than to the class of required condensate excitations.

The Higgs discussion in ECT has two complementary sides: the gauge-embedding side, where the electroweak-like EFT is introduced (Section 7.4), and the radial-mode side, where the secondary amplitude excitation is analysed as a condensate mode (Section 8.3). The present subsection brings these two viewpoints together.

Minimal derivation chain. The present electroweak-like construction rests on four steps: (i) the supplied nonzero P4 vector has $O(3)$ stabiliser; (ii) this gives the strict algebraic bridge $\mathfrak{so}(3) \cong \mathfrak{su}(2)$;

(iii) if a second transition $O(3) \rightarrow O(2)$ is admitted, a reduced weak-like symmetry-breaking pattern becomes structurally available; (iv) only after that step can one embed a doublet EFT with a Higgs-like radial mode at the scale v_2 . The first two steps are exact group mathematics conditional on the P4 datum; the latter two are conditional, while physical P4 realisation remains Open.

The Higgs boson as a second condensate radial mode. In the Standard Model, the Higgs boson is an elementary scalar field introduced independently of the spacetime metric [54, 55]. In the supplied secondary-doublet completion, the Higgs can be represented by the radial excitation h of a second condensate vacuum at scale $v_2 \approx 246 \text{ GeV}$ (Section 8.3):

$$\phi_{\text{ew}} = v_2 + h, \quad m_h^2 = 2\lambda_2 v_2^2. \quad (8.36)$$

This is an interpretation inside that supplied model, not a derivation that the observed Higgs is a mode of the primary Φ field. The distinction is conceptual: the SM Higgs and the spacetime metric are unrelated structures; in ECT they would both arise from a single underlying medium. This reduced analogy should not be overstated: it provides a structural route toward a weak-like sector, not a derivation of the full electroweak Standard Model.

Coupling pattern. If the electroweak embedding is realised at the level of the SM gauge structure (Section 7.4), the Higgs couplings to $W^\pm$, Z , and fermions follow the Standard Model pattern at leading order. This is a direct consequence of the embedding: the low-energy EFT is *designed* to reproduce the SM structure, and the Higgs coupling pattern is therefore an input, not a prediction. Accordingly, agreement with the observed tree-level Higgs coupling pattern is a consistency requirement of the embedding, not an emergent prediction of minimal ECT.

Possible non-minimal corrections. Beyond the minimal embedding, additional condensate-sector corrections to the Higgs-like state are not excluded. These may in principle include mixing with the homogeneous-P3 radial coordinate or with a future action-owned P4 amplitude pole, departures from an approximately quartic effective potential near the secondary minimum, or higher-dimension corrections to the Yukawa sector. At present none of these effects is derived quantitatively within ECT. They should therefore be treated only as logically admissible non-minimal possibilities, not as predictions of the present formulation.

Property	Standard Model	ECT
Ontological status	Elementary scalar field	Candidate radial interpretation in a supplied secondary doublet
Relation to spacetime	Independent of metric	Common-medium ownership is an Open completion hypothesis
$v_2 = 246 \text{ GeV}$	Fundamental VEV	Matching constraint (if second transition exists)
$m_h = 125 \text{ GeV}$	Determined by λ and v	Determined by λ_2 and v_2 (same formula)
Coupling pattern	Predicted by SM gauge structure	Reproduced by embedding (input, not prediction)
Cubic Higgs interaction near the minimum	$V'''(v) = 3m_h^2/v$	Same if V_{ew} is quartic; deviations possible

Property	Standard Model	ECT
Radiative sensitivity (schematic)	Cutoff estimate $\delta m_h^2 \sim \Lambda^2$, with its regulator and renormalisation interpretation to be specified	A portal correction would scale schematically as $\delta m_h^2 \sim g^2 \Lambda_{\text{sec}}^2 / (16\pi^2)$; the portal, sector cutoff, symmetry protection and renormalisation prescription are Open, and m_σ is not by itself the cutoff
Radial partner	None	P3 relation $m_{\Phi, \text{rad}} = \sqrt{2\lambda} \phi_0$; no numerical mass or P4 pole is derived
Mixing with primary condensate radial mode	No analogue in the minimal SM	Possible σ - h mixing

Comparison: SM Higgs versus ECT Higgs.

What ECT adds and what it does not. ECT does not derive the Higgs boson from first principles. What it adds is a *structural context*: the Higgs is reinterpreted as a conditional secondary radial mode of the proposed medium. The distinct homogeneous-P3 radial coordinate has $m_{\Phi, \text{rad}} = \sqrt{2\lambda} \phi_0$; its numerical mass is not selected, it is not a derived P4 particle, and it does not define the ultraviolet threshold of the P4 effective theory. This reinterpretation does not currently lead to a quantitative prediction that differs robustly from the Standard Model, but it changes the conceptual status of the hierarchy problem (Section 8.3): instead of asking why an elementary scalar is light, one asks why the condensate undergoes a second ordering step at $v_2 \ll \phi_0$. In the language of Section 8, this makes the Higgs-like state a *conditional particle-level interpretation* of the secondary condensate sector, not a required excitation of the primary ordered branch. Its positive content is therefore conceptual rather than derivational: if the second transition exists, then the Higgs-like scalar and the Lorentzian/gravitational sector are interpreted as belonging to different ordered phases of the same underlying medium.

The electroweak sector remains a structural embedding at Level B/C, not a completed derivation. Its full first-principles realisation within ECT requires closing the layered OP-EW programme: OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, and OP-EW-matter (Section 5.1, Appendices M and N).

Discriminant with respect to Standard-Model and purely matched electroweak sectors. The electroweak sector can appear in three sharply distinct ways. In the Standard Model, $SU(2) \times U(1)$ and the Higgs scale are fundamental inputs. In the present ECT programme, only the algebraic analogy is singled out; the local gauge architecture and lower scale v_2 remain supplied/matched. In a completed ECT electroweak programme, both the architecture and the secondary scale would arise from one condensate logic. The present text establishes conditional model arithmetic, not a physical electroweak derivation.

Observational anchors for the electroweak programme. The observed electroweak parameters serve as matched consistency targets, not derived fits: $m_W \approx 80.4 \text{ GeV}$, $m_Z \approx 91.2 \text{ GeV}$, $m_H \approx 125 \text{ GeV}$, the tree-level ρ -parameter near unity, left-handed weak charged currents, and the absence of observed right-handed weak neutrino currents. Within the present embedding, these quantities are used as consistency-level targets for the secondary closure scaffold; they would become genuine predictions only if v_2 and the electroweak coupling constants were derived from the condensate dynamics.

Conditional completion targets of the electroweak-like scaffold. A supplied completion would have to pass four tests. First, if v_2 is genuinely a secondary ordered-branch closure scale, a completed action must determine which scalar and vector mass operators share that scale. Weak chirality is a separate

representation-and-vertex problem and is not implied by their common numerical scale. Second, the custodial-like near-degeneracy $m_W/(m_Z \cos \theta_W) \approx 1$ should eventually be constrained by the closure geometry rather than left as a free ratio. Third, the proposed secondary-radial interpretation is a model choice to be tested; it is not selected by P1–P6. Fourth, if ν_2 cannot eventually be connected to condensate running or a secondary transition mechanism, the ECT electroweak programme remains structurally incomplete.

Rejection criteria for that supplied scaffold. Failure of any listed condition would reject the proposed electroweak-like completion, not the minimal scalar ordered branch. Because the physical weak bundle, chirality, representations and Higgs owner are not derived, these are completion tests rather than ECT predictions or falsifiers of P1–P6.

Prediction and non-prediction. The positive output of the present subsection is structural rather than precision-phenomenological: ECT admits a weak-like embedded sector with a Higgs-like radial mode if a second condensate transition is realised. What it does not yet predict is the quantitative electroweak parameter set of the Standard Model. In particular, the values of g , g' , λ_2 , and the full $W^\pm/Z/\gamma$ spectrum remain matters of EFT matching rather than first-principles derivation.

The same incompleteness propagates directly into the neutrino sector: the mass/chirality problem discussed in Section 9 remains tied to the still-open electroweak closure problem rather than forming an independent unresolved sector. In that sense, the neutrino programme is a downstream diagnostic of the secondary-transition completion, not a detached add-on.

Status summary.

Statement	Status	Comment
Conditional status of the electroweak-like sector	A/B	Requires a separately supplied second-transition completion beyond the P4 ordered-gradient analysis chart; neither physical transition is derived by the present action
Higgs as radial excitation of second condensate phase	B/C	Conditional particle-level interpretation of the secondary condensate sector
SM coupling pattern reproduced by embedding	B/C	Input, not prediction
Possible non-minimal condensate-sector corrections to the Higgs-like state	C	Logically admissible, but not derived quantitatively
Higgs and spacetime associated with the same underlying medium	B/C	Conceptual reinterpretation, if the second transition exists
Radiative naturalness of the Higgs-like mode	Open	Same quadratic sensitivity issue as in the SM
Full EW sector derived from P1–P6	Open	Decomposes into OP-EW-scale, OP-EW-locking, OP-EW-gauge, OP-EW-naturalness, and OP-EW-matter

8.8 Fermions (Level B: structural route)

Status: Level B structural reconstruction. Spinorial excitations are compatible with the ordered condensate background but are not derived from the bare bosonic action. Full development: Section 9.

Within the classification of Section 8.1, fermions belong to category (ii): structurally embedded Standard Model fields.

The stabiliser of the supplied nonzero P4 vector is $O(3)$. Its identity component is $SO(3)$, whose spin double cover is $\text{Spin}(3) \simeq SU(2)$ and admits half-integer representations. Identifying this $SO(3)$ with the physical spatial-rotation group and realising stable spinorial excitations remain Open. Only after a physical Lorentzian metric/tetrad and spin structure are separately supplied does the standard $SL(2, \mathbb{C})$ Lorentz-spinor reading become available. Dirac spinors may then be represented as two Weyl components $\Psi = (\psi_L, \psi_R)^T$. This is a structural compatibility statement, not a derivation of spinors as collective condensate excitations.

The fermionic sector is developed in Section 9, which covers:

- spinor representations of the ordered branch (Section 9.1);
- structural reconstruction of the Dirac equation (Section 9.2);
- chirality and the selection of $SU(2)_L$ (Section 9.3);
- Yukawa couplings and fermion masses (Section 9.6);
- allowed fermion–director operator and its Open microscopic coefficient (Section 9.8);
- spin-statistics structural route (Section 9.9).

A symmetry-allowed fermion–director operator is $\Delta\mathcal{L} = \mu_5 \bar{\Psi} \gamma^A n_A \Psi$. The restricted operator-basis statement does not require a nonzero coefficient; $\mu_5 = \beta_5 m_f$ is only a parameterisation, and no coefficient benchmark is retained. The operator form is introduced in Section 9.8, while its detailed phenomenology, observational constraints, and falsification tests are developed in Section 7.

In the present chapter fermions are treated only at the level of structural compatibility and effective embedding. The detailed fermionic programme is deferred to Section 9, where spinor structure, the Dirac form, chirality, Yukawa couplings, the fermion–condensate coupling, and the spin-statistics route are discussed separately. This subsection therefore serves only as the particle-level bridge to the dedicated fermionic programme of Section 9.

8.9 Neutrino Sector

Status: layered. Observation already excludes a strictly massless active-neutrino completion and forces any viable ECT completion to accommodate nonzero neutrino masses and flavour mixing (A given Ph). The present ordered-branch and $SU(2) \times U(1)$ discussions do not derive a chiral weak bundle or a neutrino representation. ECT therefore does not yet constrain the viable neutrino completion more strongly than a generic EFT and does not derive the number of generations, the flavour textures, the PMNS angles, the CP phases, the mass ordering, or the absolute neutrino masses. Those latter issues remain completion-dependent and are not derived from the minimal condensate action.

Structural setting and scope. In the current formulation of ECT, the neutrino sector is not yet a closed flavour theory. The framework does not derive the number of generations, the flavour textures, the PMNS angles, the CP phases, the mass ordering, or the absolute neutrino masses. Nor does it yet constrain the type of viable neutrino completion more strongly than a generic EFT. The Euclidean identity $\text{Spin}(4) \simeq SU(2)_L \times SU(2)_R$ supplies two algebraic factors, while the ordered-vector stabiliser is diagonal and selects neither physical weak chirality nor an internal weak factor. A physical $SU(2)_L \times U(1)_Y$ bundle, one-factor vertex and matter representations must be supplied independently. The neutrino question is therefore which supplied completion, if any, is compatible with the ordered background; no geometrically one-chiral weak sector has been derived.

Canonical notation. Throughout the neutrino discussion the anisotropic kinetic structure is written in the canonical form

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad \alpha > \beta > 0, \quad \hat{c}_*^2 = \frac{\beta}{\alpha - \beta}. \quad (8.37)$$

Any shorthand using $\alpha - 1$ is valid only after an explicit normalisation $\beta = 1$. This point matters not only locally for the neutrino section but also for consistency with all related sections discussing the Lorentzian branch, the causal cone, and cosmological redshift corrections.

What is ECT-specific and what is borrowed. Type-I seesaw algebra, the Weinberg operator, the PMNS/MSW formalism, leptogenesis, and neutrinoless double-beta decay are standard neutrino-model-building ingredients. They are not derived from the current ECT postulates P1–P6. The ECT-specific research inputs lie elsewhere: first, in the proposed but Open route to a chiral weak sector; second, in a possible interpretation of separately supplied opposite-chirality states as sterile fields; third, in the possibility of a dimensional intermediate anchor scale associated with the condensate hierarchy; fourth, in an allowed preferred-direction operator whose coefficient and observable map are Open. This separation should be kept explicit to avoid presenting borrowed machinery as an ECT derivation.

Three logically distinct mass-generation routes. At the present stage, three neutrino-completion routes remain open.

(i) *Pure Dirac route.* A right-handed singlet is added with

$$\mathcal{L}_D = -\bar{L} Y_V \tilde{H} N_R + \text{h.c.}, \quad (m_D)_{\text{entry}} = y_V v_2 / \sqrt{2}, \quad m_V^{\text{phys}} = |y_V| v_2 / \sqrt{2}. \quad (8.38)$$

The route is algebraically consistent but requires a separately supplied small Yukawa coupling and remains structurally unowned in the absence of a derivation of flavour hierarchies. No numerical coupling is retained without a declared scale, scheme and source. Status: standard conditional algebra / Open ECT.

(ii) *Pure Planck-suppressed Majorana/Weinberg route.* For definiteness, this subsection declares the one-flavour operator convention for the imported Weinberg interaction [65]

$$\mathcal{L}_5 = -\frac{C_5}{2\Lambda} (\bar{L}^c \tilde{H}^*) (\tilde{H}^\dagger L) + \text{h.c.}. \quad (8.39)$$

With $H^0 = v_2 / \sqrt{2}$ this convention gives

$$(M_V)_{\text{entry}} = \frac{C_5 v_2^2}{2\Lambda}, \quad m_V^{\text{phys}} = \frac{|C_5| v_2^2}{2\Lambda}. \quad (8.40)$$

The first quantity is the generally complex one-flavour Majorana mass-matrix entry; the physical Takagi singular value contains $|C_5|$. The numerical factor $1/2$ is tied to the explicit operator and Majorana-mass normalisation above; it is not universal, because another convention can absorb it into C_5 . Identifying Λ with any condensate scale and fixing C_5 numerically are additional model choices, not consequences of P1–P6. Status: Level A algebra inside the declared operator convention; the Wilson coefficient, scale identification and ECT mass-operator ownership remain Open.

(iii) *Embedded type-I seesaw route.* If right-handed neutrinos are admitted as additional supplied fields, a type-I seesaw mechanism can be placed on top of the ECT low-energy sector:

$$\mathcal{L}_V = -\bar{L} Y_V \tilde{H} N_R - \frac{1}{2} N_R^T C M_R N_R + \text{h.c.} \quad (8.41)$$

In the basis (v_L, N_R^c) the mass matrix reads

$$\mathcal{M}_V = \begin{pmatrix} 0 & m_D \\ m_D^T & M_R \end{pmatrix}, \quad m_D = Y_V v_2 / \sqrt{2}. \quad (8.42)$$

For several flavours, let the complex-symmetric heavy block M_R be invertible and define

$$M_H \equiv \|M_R\|_2, \quad \mu_H \equiv \sigma_{\min}(M_R), \quad \kappa_2(M_R) \equiv M_H/\mu_H, \quad \Theta_{\text{alg}} \equiv m_D M_R^{-1}.$$

The controlled theorem domain used here is

$$\kappa_2(M_R) \leq \kappa_0, \quad \mu_H/M_H \geq g_0 > 0, \quad \varepsilon_2 \equiv \|\Theta_{\text{alg}}\|_2, \quad \kappa_0 \varepsilon_2 \leq \varepsilon_0 < \frac{1}{2}, \quad (8.43)$$

together with a declared heavy–light Takagi-singular-value gap. Under these hypotheses a perturbative unitary Takagi congruence exists and, in a declared block-diagonal basis, gives

$$M_{\nu}^{\text{light}} = -m_D M_R^{-1} m_D^T + R_{\text{light}}, \quad M_{\nu}^{\text{heavy}} = M_R + R_{\text{heavy}}. \quad (8.44)$$

The physical masses are the Takagi singular values of these blocks. The algebraic matrix $\Theta_{\text{alg}} = m_D M_R^{-1}$ is a convenient mixing proxy; in the standard unitary-block convention the displayed off-diagonal Takagi rotation obeys $\Theta^* = \Theta_{\text{alg}} + R_{\Theta}$, while a formula without the complex conjugation requires an explicitly named alternative convention. Degenerate Takagi vectors are basis-dependent, so only their invariant subspaces/projectors are controlled there.

For fixed $\kappa_0, g_0, \varepsilon_0$ and block convention, constants $C_l, C_h, C_{\Theta} < \infty$ exist such that

$$\|R_{\text{light}}\|_2 \leq C_l M_H \varepsilon_2^4, \quad \|R_{\text{heavy}}\|_2 \leq C_h M_H \varepsilon_2^2, \quad \|R_{\Theta}\|_2 \leq C_{\Theta} \varepsilon_2^3.$$

These are norm bounds with hypothesis-dependent constants, not universal matrix coefficients. The light matrix identifies only the combination $m_D M_R^{-1} m_D^T$; flavour bases, heavy degeneracies and the usual Casas–Ibarra-type inverse freedom prevent recovery of m_D and M_R separately from low-energy masses alone. For one generation, use the available phase convention to set $M_R > 0$ and define the actual expansion parameter

$$\varepsilon \equiv \frac{|m_D|}{M_R} \ll 1. \quad (8.45)$$

The physical light Takagi singular value and its controlled expansion are

$$\begin{aligned} m_{\nu} &= \frac{\sqrt{M_R^2 + 4|m_D|^2} - M_R}{2} \\ &= \frac{|m_D|^2}{M_R} - \frac{|m_D|^4}{M_R^3} + \mathcal{O}\left(\frac{|m_D|^6}{M_R^5}\right) \\ &= \frac{s_y^2 v_2^2}{2M_R} - \frac{s_y^4 v_2^4}{4M_R^3} + \mathcal{O}\left(\frac{s_y^6 v_2^6}{M_R^5}\right), \quad s_y \equiv |y_{\nu}|, \quad |m_D| = \frac{s_y v_2}{\sqrt{2}}. \end{aligned} \quad (8.46)$$

Writing y_{ν}^2 instead of the squared Yukawa singular value $s_y^2 = |y_{\nu}|^2$ is valid only after the additional positive-real one-flavour phase choice used for the plotted scalar benchmarks. The singular-value formula is phase safe. This algebra is standard and imported; it is Level A inside the supplied type-I model, not an ECT first-principles result [66]. Its possible relevance to ECT is only model-building: a right-handed singlet and its Majorana mass have not been derived from the ordered branch. The embedded seesaw route is a supplied completion option whose physical ECT owner remains Open, not a preferred prediction.

Minimum number of heavy sterile states in the minimal type-I-only route. If the displayed type-I seesaw is the sole source of light-neutrino masses, $\text{rank } M_{\nu}^{\text{light}} \leq \text{rank } m_D \leq n_R$. Reproducing two independent nonzero oscillation splittings therefore requires at least two heavy sterile states in that restricted route, while a generic full-rank 3×3 type-I completion uses three. This lower bound is not model-independent once a Weinberg operator, radiative mass term or another light-mass owner is also admitted. It is standard imported neutrino physics, not an ECT-specific result.

Optional neutrino scale hypotheses. Dimensional consistency alone does not select a unique scale, but named completion hypotheses are admissible. Two useful C/Open branches are

$$M_H := \|M_R\|_2 = c_R \sqrt{\phi_0 v_2}, \quad c_R > 0, \quad \Lambda = c_\Lambda \phi_0, \quad c_\Lambda > 0, \quad (8.47)$$

with independent supplied dimensionless coefficients c_R and c_Λ . The first relation fixes only a characteristic heavy singular-value scale, not the general complex-symmetric flavour matrix. Equivalently a full matrix branch would require a supplied dimensionless complex-symmetric matrix $\hat{C}_R$ in $M_R = \sqrt{\phi_0 v_2} \hat{C}_R$. All light masses and mixings remain conditional on Y_ν , C_5 , flavour structure, RG convention and the ultraviolet operator. Inverting the seesaw formula against an externally supplied light-mass target fits a Yukawa; it is not an ECT prediction. Likewise $C_5 = 1$ and $c_\Lambda = 1$ define a unit-Wilson benchmark, not a preferred mass. No leptogenesis or phenomenological preference follows without its separate owner chain.

Declared unit-coefficient orientation points. Using the separately adopted scale point $\zeta_\phi = 1$, together with $c_R = 1$, gives the characteristic heavy scale

$$M_H^{\text{unit}} = \sqrt{\bar{M}_{\text{Pl}} v_2} = 2.4487 \times 10^{10} \text{ GeV}. \quad (8.48)$$

On the leading one-flavour seesaw branch, inversion at the illustrative external targets $m_\nu = 0.050 \text{ eV}$ and 0.009 eV , motivated only as the characteristic square roots of the observed oscillation splittings [67], gives, respectively,

$$\begin{aligned} s_y &= 6.355 \times 10^{-3}, & |m_D| &= 1.1065 \text{ GeV}, & \varepsilon &= 4.519 \times 10^{-11}, \\ s_y &= 2.696 \times 10^{-3}, & |m_D| &= 0.4695 \text{ GeV}, & \varepsilon &= 1.917 \times 10^{-11}. \end{aligned} \quad (8.49)$$

These are inverse fits inside the supplied one-flavour model, not predictions of the absolute neutrino masses, Yukawa matrix or mixing. On the independent unit-Wilson point $C_5 = c_\Lambda = \zeta_\phi = 1$,

$$m_\nu^{(5),\text{unit}} = \frac{v_2^2}{2\bar{M}_{\text{Pl}}} = 1.2447 \times 10^{-5} \text{ eV}. \quad (8.50)$$

This last number is about 4.0×10^3 below the 0.050 eV orientation target. In the same one-flavour normalisation, reproducing that supplied orientation scale requires

$$\frac{|C_{5,\text{eff}}|}{c_\Lambda \zeta_\phi} \simeq 4.02 \times 10^3, \quad (8.51)$$

where $C_{5,\text{eff}}$ denotes the relevant low-energy Takagi direction after the still-Open flavour, threshold and RG matching. This required coefficient/scale combination is not a selected Wilson coefficient, UV completion, data fit or ECT prediction. The quoted unit point diagnoses only the named Planck-suppressed branch: changing $|C_{5,\text{eff}}|/(c_\Lambda \zeta_\phi)$, lowering the matching scale, or taking $C_{5,\text{eff}} = 0$ changes or removes it. Without a common UV owner, the relation between the type-I and Weinberg contributions is NOT IDENTIFIABLE, so they must not be added by default. If C_5 is obtained by matching the same N_R fields, $C_5/\Lambda = -Y_\nu M_R^{-1} Y_\nu^T$ in the declared operator/sign convention and simultaneous inclusion double counts unless a subtraction prescription is supplied. A separately owned residual C_5 from another UV sector may instead add coherently. None of the orientation points is ECT evidence or a selected neutrino completion.

Required recovery of standard oscillation phenomenology. At leading order the low-energy oscillation Hamiltonian remains the standard one:

$$H_{\text{osc}} = \frac{1}{2E} U_{\text{PMNS}} \text{diag}(m_1^2, m_2^2, m_3^2) U_{\text{PMNS}}^\dagger + V_{\text{MSW}}, \quad U_{\text{PMNS}} = U_e^\dagger U_\nu. \quad (8.52)$$

An ECT completion would have to reproduce this successful low-energy oscillation kinematics; the present action does not yet derive the origin of the neutrino mass matrix. The present framework supplies the standard PMNS/MSW Hamiltonian as the target low-energy closure; it does not derive the two mass matrices or their relative diagonalisation.

Preferred-direction operators: precise domain and conventions. The following is only a broken-phase, low-energy, free-propagation EFT on a flat Lorentzian patch. Above electroweak breaking the gauge-covariant operators act on the full lepton doublets L_a :

$$\delta\mathcal{L}_L = -(a_L)_{\mu}^{ab} \bar{L}_a \gamma_L^{\mu} L_b - \frac{i}{2} (c_L)_{\mu\nu}^{ab} \bar{L}_a \gamma_L^{\mu} \overleftrightarrow{D}^{\nu} L_b, \quad [(a_L)_{\mu}] = 1, \quad [(c_L)_{\mu\nu}] = 0. \quad (8.53)$$

The neutrino-only form follows after electroweak symmetry breaking by projecting L_a to ν_{La} and, for free propagation, replacing D by ∂ . Before symmetry breaking gauge covariance ties the same doublet operator to the charged-lepton partners; a neutrino-only a_L/c_L tensor is not an independent gauge-invariant isolated coefficient. The plot below is only the declared post-EWSB projection and is not a global SME coefficient fit. The coefficients are assumed constant and homogeneous. The spatial Dirac operator is taken on a self-adjoint domain: for example the standard $H^1(\mathbb{R}^3)$ domain with adequate falloff, or a periodic H^1 domain on a three-torus. Smooth compactly supported fields may be used as a core, but are not substituted for the completed self-adjoint domain. The associated boundary form, and hence the surface term from integration by parts, vanishes. Curved backgrounds, interfaces, spacetime-dependent coefficients, absorption and decay are outside this benchmark. These operator classes and their CPT assignments are standard external SME results [68, 69], not ECT evidence.

Hermiticity in flavour space requires

$$(a_L)_{\mu}^{ab} = [(a_L)_{\mu}^{ba}]^*, \quad (c_L)_{\mu\nu}^{ab} = [(c_L)_{\mu\nu}^{ba}]^*. \quad (8.54)$$

Use the mostly-plus metric $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$ and the positive-frequency convention $v(x) = u(p)e^{ip_{\mu}x^{\mu}}$, with

$$p^{\mu} = (E, \mathbf{p}), \quad p_{\mu} = (-E, \mathbf{p}), \quad \hat{p}^{\mu} = (1, \hat{\mathbf{p}}), \quad \hat{p}_{\mu} = (-1, \hat{\mathbf{p}}). \quad (8.55)$$

The leading-order Hamiltonian reduction additionally assumes that the coefficient of $i\partial_0$ is admissible. After multiplying the field equation by γ_L^0 , its time-kinetic form is

$$\mathcal{K} = \gamma_L^0 [\gamma_L^0 + (c_L)_{\mu 0} \gamma_L^{\mu}] \equiv \mathbf{1} + \delta\mathcal{K}. \quad (8.56)$$

The declared sufficient perturbative guard is $\delta\mathcal{K} = \delta\mathcal{K}^{\dagger}$ and $\|\delta\mathcal{K}\|_{\text{op}} < 1$. It makes $\mathcal{K}$ positive definite and nonsingular on spinor-flavour space, so that the local constant field redefinition $\nu_L = \mathcal{K}^{-1/2} \chi_L$ exists and gives a canonical Schrödinger form; the positive square root and its inverse are unique. Flavour Hermiticity makes this finite-dimensional time-kinetic perturbation Hermitian, but does not by itself impose the norm bound. The norm bound is sufficient, not necessary. For Hermitian $\mathcal{K}$, positive definiteness is necessary and sufficient for this positive congruence canonicalisation; nonsingularity alone is insufficient. A self-adjoint spatial domain remains a separate condition on the full Hamiltonian generator. All formulas below retain only terms linear in a_L and c_L (and leading m^2/E). At this order define, with the sign fixed by the displayed derivative and plane-wave conventions,

$$\begin{aligned} A_{ab}(\hat{p}) &= (a_L)_{ab}^{\mu} \hat{p}_{\mu}, & C_{ab}(\hat{p}) &= -(c_L)_{ab}^{\mu\nu} \hat{p}_{\mu} \hat{p}_{\nu}, \\ H_{\text{vac}} &= \frac{1}{2E} U_{\text{PMNS}} \text{diag}(m_1^2, m_2^2, m_3^2) U_{\text{PMNS}}^{\dagger}, \\ V_{\text{MSW}} &= 0, & H_{\text{eff}}^{\nu} &= H_{\text{vac}} + A + EC. \end{aligned} \quad (8.57)$$

For the displayed minus-sign operator, the massless plane-wave equation is $\gamma_L^{\mu} q_{\mu} u = 0$ with $q_{\mu} = p_{\mu} + (a_L)_{\mu} - (c_L)_{\mu\nu} p^{\nu}$ at first order. For example, if the only nonzero one-flavour coefficient is $(c_L)_{00} = \varepsilon$ and $a_L = 0$, then $q_0 = -(1 + \varepsilon)E$. The null condition $q^2 = 0$ therefore gives $E = |\mathbf{p}|/(1 + \varepsilon) = |\mathbf{p}|(1 - \varepsilon) + O(\varepsilon^2)$ and $\delta E = -\varepsilon|\mathbf{p}| = EC$ at first order. This derives the minus sign in the definition of C rather than choosing it by analogy. It also exposes the excluded counterexample $\varepsilon = -1$, for which $\mathcal{K}$ is singular even though the one-flavour coefficient is Hermitian.

The cited SME sources do not literally use the house display above. To make the comparison unambiguous, denote their mostly-minus convention by S : $g_{\mu\nu}^S = \text{diag}(+, -, -, -)$, $L(x) = u(p)e^{-ip_\mu^S x^\mu}$, and a source derivative operator $+(i/2)(c_L^S)_{\mu\nu}\bar{L}\gamma_S^\mu\overleftrightarrow{D}_S^\nu L$. The exact dictionary used here is

$$\eta_{\mu\nu}^H = -g_{\mu\nu}^S, \quad \gamma_H^\mu = \gamma_S^\mu, \quad p_\mu^H = -p_\mu^S, \quad D_H^\nu = -D_S^\nu, \quad (a_L^H)_\mu = (a_L^S)_\mu, \quad (c_L^H)_{\mu\nu} = (c_L^S)_{\mu\nu}. \quad (8.58)$$

The source plus-sign derivative term then becomes the displayed house minus-sign term. Raising one a_L index introduces one further minus sign, whereas raising both c_L indices introduces none. Consequently

$$A_H = (a_L^H)^\mu \hat{p}_\mu^H = (a_L^S)^\mu \hat{p}_\mu^S = A_S, \quad C_H = -(c_L^H)^{\mu\nu} \hat{p}_\mu^H \hat{p}_\nu^H = -(c_L^S)^{\mu\nu} \hat{p}_\mu^S \hat{p}_\nu^S = C_S. \quad (8.59)$$

Thus the effective A and C used in the source comparison are invariant under the declared translation even though the literal derivative-term sign changes. No numerical coefficient table is imported in this benchmark.

For every real direction, flavour Hermiticity makes A and C Hermitian. Together with the self-adjoint spatial domain and the norm-bounded, positive-nonsingular $\mathcal{H}$ assumptions above, the canonically reduced H_{eff}^ν is Hermitian and $U_\nu(L) = \exp[-iH_{\text{eff}}^\nu L]$ is unitary on this vacuum free-propagation domain. The earlier general PMNS/MSW target remains valid, but the present figure explicitly sets the matter potential to zero. Dropping flavour Hermiticity can permit complex eigenvalues; violating the norm/positivity guard can obstruct the Hamiltonian reduction. Real eigenvalues alone would still be insufficient: the non-normal Jordan matrix $H_J = \begin{pmatrix} 0 & g \\ 0 & 0 \end{pmatrix}$, $g \neq 0$, has real spectrum but $e^{-iH_J L} = \mathbf{1} - iH_J L$ is not unitary and can grow norms. This is a counterexample outside the Hermitian, self-adjoint benchmark, not an allowed propagation Hamiltonian.

With the same momentum convention, the corresponding antineutrino Hamiltonian is

$$H_{\text{eff}}^{\bar{\nu}} = H_{\text{vac}}^* - A^* + EC^*. \quad (8.60)$$

Thus the CPT-odd a_L contribution changes sign with the appropriate flavour complex conjugation, whereas the CPT-even c_L contribution retains its sign and uses the conjugate flavour matrix. A matter potential would also reverse appropriately for antineutrinos, but it is absent here by the declared $V_{\text{MSW}} = 0$ vacuum benchmark.

Oscillations are blind to a flavour-identity contribution. Therefore the relevant matrix for either $X = A, C$ is

$$X_{\text{osc}} = X - \frac{\text{tr} X}{3} \mathbf{1}_3. \quad (8.61)$$

In particular, $C = c_0 \mathbf{1}_3$ changes the common phase but gives $C_{\text{osc}} = 0$ and no oscillation signal. A norm of the unprojected coefficient is therefore not by itself an oscillation observable.

Oscillation-sensitive $d = 4$ operator class; coefficient Open. At a fixed propagation direction, the CPT-even contribution is the traceless Hermitian matrix $EC_{\text{osc}}(\hat{p})$. Its oscillation effect depends on eigenvalue differences and on its orientation relative to H_{vac} ; a flavour-identity component is unobservable. Because H_{vac} and C_{osc} need not commute, equality between separate norms or eigenspaces would not establish an eigenvalue crossing of the full Hamiltonian $H_{\text{vac}} + EC$. The operator classification is external SME physics [68, 69]. ECT currently derives no nonzero coefficient, selected direction, flavour map, energy scale or microscopic vertex, so no numerical curve, equality scale, prediction, fit or constraint is retained.

CPT and observable ownership. The $d = 3$ a_L and $d = 4$ c_L operators belong to distinct CPT classes and must not be identified. Unitary free propagation is established here under the declared Hermiticity, self-adjoint-domain, perturbative and time-kinetic-admissibility assumptions above. Oscillations depend on eigenvalue differences and mixing of A_{osc} or C_{osc} , not on an identity shift. Under antineutrino conjugation the CPT-odd term changes sign, $A \mapsto -A^*$, while the CPT-even term retains its sign, $C \mapsto C^*$.

A measurable effect additionally requires a state, baseline, selected direction, matter profile and detector observable. The operator inventory uses the vacuum case, not a matter profile. ECT currently supplies none of the microscopic coefficients or flavour data; both routes remain an external operator inventory with Open physical ownership.

Connection to leptogenesis. A heavy sterile sector can simultaneously support seesaw masses and leptogenesis. This is standard model-building, not an ECT result. The present action derives neither a heavy sterile spectrum nor the baryon current, CP-violating vertex, real-time state, production history, transport or washout dynamics. Accordingly no numerical scale comparison is retained, and no coincidence is evidence for the mechanism. The entire connection remains Open for ECT.

Cosmological neutrino-mass bounds. Published neutrino mass-sum limits are likelihood results conditional on a named cosmological model and data combination. Standard base- Λ CDM analyses provide an external reference [70]; they are not a model-independent consequence of the supplied mass-operator inventory above. Conversely, the present ECT inventory has no Boltzmann hierarchy, recombination history or joint likelihood and therefore supplies no quantitative weakening of those limits. The divided coefficient $G_{\text{div}}(z)$ alone fixes neither the sign nor magnitude of a shift. Any ECT-specific bound requires an action-owned scalar–metric background and perturbations, neutrino free-streaming, growth, expansion and parameter degeneracies to be solved jointly. That analysis remains Open.

Relation to the dark sector. In the supplied heavy-sterile benchmark, sterile neutrinos are assigned to the seesaw/leptogenesis sector rather than to the dark sector. Any dark-sector role of ultralight condensate excitations is discussed separately in Section 8.10 and remains open.

Comparison with supplied completion classes. In the minimal Standard Model there is no neutrino mass mechanism at all. In a pure Dirac extension, neutrino masses are allowed but require an exceptionally tiny Yukawa coupling. Type-I seesaw completions supply heavy sterile singlets and their mass and coupling matrices. Left-right symmetric models instead supply an explicit $SU(2)_R$ gauge extension and its representation content. The present ECT inventory supplies neither completion and has not selected among Dirac, Majorana/seesaw or left-right routes. The Euclidean spinor identity does not select a physical weak bundle or sterile state; until the vertices and representations are supplied, the neutrino completion is no more constrained than a generic EFT.

Table 31: Comparison of neutrino frameworks.

Framework	Mass origin	Chirality origin	Sterile origin	$0\nu\beta\beta$	Leptogenesis	Flavour closure
SM	—	postulated	—	—	—	N/A
Dirac ext.	$y_V v_2/\sqrt{2}$	postulated	added	no	no	no
Type-I seesaw	seesaw	postulated	added (ad hoc)	yes	yes	no
Left–right sym.	seesaw	gauge $SU(2)_R$	gauge	yes	yes	partial
ECT candidate	supplied seesaw	weak chirality Open	supplied singlet	conditional	conditional	open

*conditional = only on a supplied Majorana/seesaw branch; no mass, mixing or leptogenesis scale is selected by ECT.

ECT neutrino model-building inventory:

1. Chirality: geometric comparison only; physical weak map Open.

2. Seesaw route: standard imported algebra; M_R , Y_ν and sterile content are supplied and Open for ECT. No condensate-scale expression is selected.
3. Preferred-direction operators: independent flavour-Hermitian $d = 3$ CPT-odd A_{osc} and $d = 4$ CPT-even C_{osc} channels on the declared perturbative self-adjoint vacuum/free-propagation domain with norm-bounded positive-nonsingular time kinetic form; $C = -(c_L)^{\mu\nu} \hat{p}_\mu \hat{p}_\nu$ for the displayed minus-sign derivative operator, and antineutrinos use $A \mapsto -A^*$ and $C \mapsto C^*$. Coefficients, direction/flavour map and physical ECT vertex remain Open; no numerical benchmark is retained.
4. Heavy sterile neutrinos belong to the seesaw/leptogenesis sector, not to the minimal neutrino-oscillation sector.

Neutrinoless double-beta decay in supplied Majorana completions. If a Majorana/seesaw branch is supplied, neutrinoless double-beta decay is a generic test of Majorana physics. The effective mass

$$m_{\beta\beta} = \left| \sum_i U_{ei}^2 m_i \right| \quad (8.62)$$

is generically nonzero, although strongly texture-, ordering-, and phase-dependent. Because the present ECT framework does not yet derive flavour textures, it cannot predict the exact value of $m_{\beta\beta}$. The Dirac/Majorana distinction is not selected by ECT. A future positive signal would support Majorana neutrino physics but would not discriminate ECT without an ECT-specific derived mass/vertex prediction. A deep null result would constrain Majorana completions generically.

Possible future routes. Two speculative but conceptually interesting future directions deserve brief mention. First, a completed matter action must identify the physical source of PMNS phases and test whether it has any dependence on the ordered background. Second, Majorana lepton-number violation requires a derived mass operator and state. The vector stabiliser supplies neither weak chirality nor an “opposite branch” whose decoupling could generate that operator.

What would be needed for full Level A. To elevate the neutrino sector beyond speculative embedding, ECT would need to:

- (i) derive whether right-handed neutrinos are required at all;
- (ii) derive whether neutrinos are Dirac or Majorana in the ordered branch;
- (iii) derive the origin and scale of the Majorana mass M_R , if present;
- (iv) derive the neutrino Yukawa couplings y_ν ;
- (v) derive the observed neutrino mass splittings and mixing pattern.

None of these steps has yet been completed.

Table 32: Status map for the ECT neutrino programme.

Claim	Status	Comment
Exactly massless active-neutrino completion	Excluded	Incompatible with oscillation data
Observed neutrino masses require some completion	A given Ph	Phenomenological requirement, not internal derivation
Weak chiral branch selected by the ordered background	Not derived	The exact connected orientation stabiliser is $SU(2)_{\text{diag}}$ and selects no internal weak factor

Continued on next page

Table 32 continued

Claim	Status	Comment
Identification with the active weak sector	Open	Requires an independent chiral bundle, vertex and representations
Pure Dirac branch	Open	Requires a supplied Yukawa operator and coefficient; the present theory neither selects nor disfavors this route
Dimension-5 Weinberg normalisation	A algebra in supplied operator / Open ECT owner	$m_v^{\text{phys}} = C_5 v_2^2/(2\Lambda)$ in the declared convention; C_5 , Λ and their ECT ownership are not derived
Embedded seesaw completion	C	Imported mechanism and supplied N_R ; not preferred by P1–P6
Neutrino mass-scale selection	Open	No heavy scale, Wilson coefficient, Yukawa, mixing or phenomenological consequence is selected by P1–P6
Standard PMNS/MSW Hamiltonian supplied as the required target	C/Open	Mass matrices and preferred-direction coefficient un-derived
$0\nu\beta\beta$ in a supplied Majorana branch	C/Open	Generic Majorana test; no ECT-specific prediction
Seesaw–leptogenesis connection	Open	Requires a supplied heavy spectrum, baryon current, CP vertex, real-time state, production history, transport and washout
Cosmological $\sum m_\nu$ bound model-dependent under a completed scalar–metric background and perturbation response	Open	$G_{\text{div}}(z)$ alone fixes neither sign nor magnitude; quantitative bound requires Boltzmann/MCMC analysis
Full flavour closure (generations, textures, PMNS, ordering)	Open	Systematised architecturally in §9.7; quantitative derivation Open under OP-Yukawa/OP-generations

This closes the neutrino subsection at the level currently justified by ECT: an Open route to a chiral weak sector, a conditional assessment of three imported mass-generation routes, no ECT-owned mass-scale ansatz, and several still-open completion routes for the full flavour programme. The next subsection turns away from imported Standard-Model extensions and returns to sectors that are more directly tied to the topology of the ordered condensate itself.

8.10 Conditional Topological Classification and Candidate New States

Status: layered. The homotopy groups of the supplied orientation target $\mathcal{T}_{P4} = O(4)/O(3) \simeq S^3$ are Level-A conditional mathematics. Identifying this orbit with the complete physical vacuum and configuration space requires an owned stationary action, state, exact-gradient admissibility, boundary class and transition history; those inputs are Open. The Derrick result is exact inside a separately supplied positive two-derivative S^3 sigma model, but physical texture existence, stabilisation, mass, abundance and production remain Open.

What is exact and what is not. Unlike several embedded sectors discussed earlier in the chapter, the homotopy calculation does not borrow Standard-Model structure. It follows exactly once the target $\mathcal{T}_{P4} \simeq S^3$ is supplied. It classifies maps into that target and the target-space protection that they could carry. It does not establish that this orbit is the physical vacuum manifold, that all maps are admissible exact-gradient ECT configurations, or that physical defects, particles or relics exist or are absent.

The supplied architecture inventories three qualitatively distinct classes of candidate or structural sectors beyond the observed Standard Model spectrum: smooth condensate coordinates/candidate excitations, maps into a supplied orientation target, and (conditionally) defect configurations from a possible second transition. This inventory is not a claim that every entry is an identified physical ECT particle. This subsection classifies each class and specifies its derivation status.

The spectrum discussed here combines three strands developed earlier in the chapter: the smooth radial-mode sector (Section 8.3), the electroweak-like conditional sector (Section 8.7), and the conditional topological classification of supplied targets (Appendix AB and Section 34).

Minimal conditional chain. The logic is: (i) P4 supplies a nonzero vector whose stabiliser and orbit are $O(3)$ and $\mathcal{T}_{P4} = O(4)/O(3) \simeq S^3$; (ii) the homotopy groups of S^3 are computed exactly; (iii) those groups classify abstract target-space sectors and non-protection statements; (iv) physical defect claims require identification of the target with the full vacuum/configuration space plus an action, boundary class and transition history. Steps (i)–(iii) are Level-A mathematics conditional on P4; step (iv) is Open.

Class 1: smooth condensate excitations. (a) *P3 radial coordinate and Open P4 pole.* The homogeneous P3 quadratic fluctuation operator has mass parameter $m_{\Phi, \text{rad}}^2 = 2\mu^2$ and the corresponding Euclidean Gaussian propagator pole. Its identification with a physical amplitude mode, mass or breakdown threshold of the P4 gradient-ordered branch requires a stationary P4 action and Hessian and is Open. The former Planck-heavy P4 particle statement is withdrawn.

(b) *Longitudinal scalar coordinate χ and its conditional mode.* In the scalar-only ECT basis, the integrability constraint $\partial_A Q_B - \partial_B Q_A = 0$ reduces the naive three-component Goldstone-coordinate map to a single longitudinal scalar coordinate χ (Section 4.3, Appendix J). A supplied reduced EFT may promote this coordinate to one physical soft mode (Level B), but its mass, couplings and stability are Open. Whether such a mode exists physically or plays a cosmological role (e.g. as a dark-sector candidate) remains Open (Section 18.1).

Class 2: maps into the supplied orientation target. The P4-supplied nonzero vector has abstract orientation target $\mathcal{T}_{P4} = O(4)/O(3) \simeq S^3$. Its homotopy groups give the following exact target-space classification [9] (Level-A conditional mathematics):

Homotopy group	Value	Consequence
$\pi_0(S^3)$	0	No π_0 -protected walls within this target
$\pi_1(S^3)$	0	No π_1 -protected strings within this target
$\pi_2(S^3)$	0	No π_2 -protected monopoles within this target
$\pi_3(S^3)$	$\mathbb{Z}$	Integer classes of maps into this target

Conditional non-protection statements. The vanishing of π_0 , π_1 , and π_2 means only that maps into the supplied S^3 target do not possess those three homotopy protections. A physical absence claim does not follow until this target is shown to be the complete physical vacuum/configuration space of an owned transition. Hence no ECT relic-dilution or monopole conclusion is derived at this stage.

Texture/skyrmion target classes (A-math conditional; physical Open). The non-trivial homotopy group $\pi_3(S^3) = \mathbb{Z}$ guarantees integer classes of maps into the supplied target. Representatives may be written as texture- or skyrmion-like maps that wrap S^3 . Whether they are admissible exact-gradient ECT configurations with finite action is Open.

The existence of nontrivial homotopy classes does not by itself imply the existence of stable finite-size particle-like solitons. Whether the corresponding configurations are dynamically stabilised against collapse or unwinding depends on the nonlinear terms of the effective action, possible higher-derivative contributions, gauge effects, or other UV-completion data. Accordingly, topology guarantees target-map labels, but not physical ECT configuration sectors or stable isolated particles (Open; Section 34).

Sector existence versus particle existence. The statement $\pi_3(S^3) = \mathbb{Z}$ guarantees distinct homotopy classes of maps into the supplied target. It does *not* by itself guarantee distinct sectors of the physical ECT configuration space or stable, finite-size, particle-like solitons. That stronger claim requires a dynamical analysis of the condensate energy functional and, in particular, a mechanism preventing shrinkage or unwinding. This distinction is the central reason why the topological classification is Level-A conditional mathematics while the physical sector and particle interpretation remain Open.

Derrick theorem and the Open mass owner. In a separately supplied three-spatial-dimensional positive two-derivative S^3 sigma model, let a texture representative of characteristic size R be governed only by the orientation energy. Under spatial rescaling its energy obeys

$$E_2(R) \propto R. \quad (8.63)$$

Hence E_2 decreases as $R \rightarrow 0$: that supplied two-derivative model has no stable finite-size soliton. The integer $Q \in \pi_3(S^3) = \mathbb{Z}$ labels configuration sectors but does not determine a spectrum $m(Q)$. A stable object would require additional higher-derivative, gauge, potential or boundary terms whose scaling balances E_2 . If such terms are generated solely by the primary high-stiffness sector, dimensional analysis places the resulting mass at the primary scale up to unknown profile and Wilson coefficients, not at a separately chosen low-energy scale. A low-energy topological mass instead requires a separately derived secondary action. Its existence, stabilisation, profile, mass and production history are all Open. In particular, the sigma-model target and action are not derived by P3/P4, so no numerical topological mass spectrum is published here.

Class 3: defect configurations from a possible second transition (conditional). If the second condensate transition $O(3) \rightarrow O(2)$ is realised (Appendix M), the secondary vacuum manifold is

$$\mathcal{V}_2 = O(3)/O(2) \simeq S^2. \quad (8.64)$$

The symbol $\mathcal{V}_2$ is reserved for this conditional secondary vacuum manifold and is distinct from the two-particle exchange space $\mathcal{M}_2$ used in Section 24.4. Its homotopy groups are:

Homotopy group	Value	Consequence
$\pi_0(S^2)$	0	No π_0 -protected map class in the supplied target; physical walls Open
$\pi_1(S^2)$	0	No π_1 -protected map class in the supplied target; physical strings Open
$\pi_2(S^2)$	$\mathbb{Z}$	Protected map classes exist in the supplied target; physical monopole-like configurations remain conditional/Open

Thus if the second transition exists, ECT would admit a sector of $\pi_2(S^2) = \mathbb{Z}$ topological configurations. Their interpretation as global hedgehogs or monopole-like states depends on whether the secondary symmetry is global or gauged and on the details of the corresponding EFT completion. For a global secondary symmetry this would correspond to hedgehog-type (global-monopole) configurations; in a gauged realisation one may obtain monopole-like configurations, but their detailed status depends on the full gauge embedding and cannot be inferred from homotopy alone. This is conditional on OP-EW and is Level B/C (Appendix M).

Candidate states and structural sectors beyond the observed Standard Model spectrum. Summarising the three classes above, the current inventory is:

- (i) the homogeneous P3 radial coordinate σ , with $m_{\Phi, \text{rad}}^2 = 2\mu^2 = 2\lambda\phi_0^2$ inside P3; its numerical mass, a P4 pole, particle and UV threshold remain Open;
- (ii) one longitudinal scalar coordinate χ in the constrained orientation map (Level A kinematics), with a physical one-mode interpretation only in a supplied reduced EFT (Level B; existence, mass and role Open);
- (iii) nontrivial texture/skyrmion target-map classes labelled by $\pi_3(S^3) = \mathbb{Z}$ (A-math conditional; physical configurations and dynamics Open);
- (iv) (conditional) secondary monopole-like defects from $\pi_2(S^2) = \mathbb{Z}$ if the second transition exists (Level B/C).

The radial coordinate and the χ coordinate reduction, together with the topological classification in item (iii), are statements about supplied coordinates and targets. Their physical configuration/particle interpretation, masses and stability do not follow from those statements. Item (iv) is conditional on OP-EW.

The Higgs-like secondary radial mode h discussed in Section 8.7 is not a new state beyond the observed SM spectrum; rather, it is an ECT reinterpretation of the observed Higgs boson, conditional on the second transition being realised.

Relation to the dark sector. The Level-C HRC algebraic diagnostic contains no particle-halo term. This does not show that ECT particle or collective sectors are unnecessary. No relic abundance, stability theorem, or production mechanism has been established for the states listed here, and their possible cosmological role remains Open.

Observational signatures and falsifiability. The present falsification handle is structural rather than a numerical mass-spectrum target:

- **Finite primary texture:** observation of a stable primary texture would require a stabilising operator beyond the two-derivative action. A claimed ECT mass prediction must first derive that operator, its profile and its Wilson coefficients; no collider mass target follows from topology alone.
- **Target-space discriminator:** if a future owner-complete P4 theory identifies its complete vacuum/configuration target with S^3 , a defect protected specifically by π_0 , π_1 or π_2 of that same target would contradict the identification. Present ECT has not established the antecedent, so this is not yet a physical falsifier.

Conditional classification and non-prediction. The strongest result of this subsection is the exact homotopy classification of the supplied S^3 target. Present ECT does not yet predict the physical absence of walls, strings or monopoles, nor the physical existence of a π_3 texture sector. All such claims require the missing physical vacuum/configuration and action owners; masses, abundances, production rates and observational signatures require further dynamics.

Status summary.

Statement	Status	Comment
<i>Smooth condensate excitations</i>		
Primary P3 radial coordinate σ ; exact curvature relation only	A inside supplied P3 / physical Open	Numerical mass and physical P4 pole are Open
One longitudinal scalar coordinate χ ; physical mode conditional	A kinematics / B physical	Action/domain/projector, mass and role Open
<i>Conditional topology of the supplied target</i>		
$\pi_0 = \pi_1 = \pi_2 = 0$ for S^3	A-math conditional / physical Open	No corresponding homotopy protection inside the supplied target; physical defects not classified
$\pi_3(S^3) = \mathbb{Z}$: integer target-map classes	A-math conditional / physical Open	Physical ECT configuration sectors require an owner
Target-space classification $\neq$ physical sector or stable particle	A-math/Open	Homotopy fixes map classes; admissibility and dynamics are Open
Dynamical stability of localised texture/skyrmion states	Open	Requires nonlinear condensate energy functional and profile solutions
Integer $Q = n$ labels maps into the supplied target	A-math conditional	Follows from $\pi_3(S^3) = \mathbb{Z}$
Two-derivative supplied S^3 sigma model, $E_2(R) \propto R$	A-math within that supplied model / physical Open	Collapses under Derrick scaling; no physical ECT texture owner follows
Stabilised topological mass spectrum	Open	Requires an additional action, profile and Wilson coefficients; a secondary-sector owner is not derived
<i>Conditional: second transition</i>		
$\pi_2(S^2) = \mathbb{Z}$: monopole-like defects	B/C	If $O(3) \rightarrow O(2)$ exists; nature depends on local vs global realisation
<i>Connection to dark sector</i>		
Galactic mass-discrepancy benchmark without a particle halo	C	Property of the conditional HRC algebraic diagnostic; ECT matching Open
Topological states as possible DM candidates	C	Role not established
<i>Reinterpretation of observed states</i>		
Higgs-like secondary radial mode h	B/C	Reinterpretation of the observed Higgs, conditional on OP-EW

8.11 Summary of the particle and excitation spectrum

Status: synthetic chapter summary. Section 8 does not derive the full Standard Model particle spectrum from the minimal scalar condensate action. Its strongest outputs are instead: (i) the separation of exact condensate-coordinate content from conditional candidate excitation sectors organised around the supplied ordered chart, (ii) the structural embedding of selected Standard-Model-like sectors at the EFT level, and (iii) conditional homotopy statements tied to a supplied orientation target, with physical topology Open. The purpose of the present subsection is to collect those results in a single disciplined map.

What Section 8 has established. The chapter organises three classes of candidate particle-like content around the supplied ordered chart and its possible completions. First, the homogeneous P3 action supplies a radial coordinate with known local curvature, while the supplied P4 orientation map has an exact longitudinal coordinate reduction. These are different owners: their promotion to physical P4 spectra requires declared actions and reductions, and a physical-graviton sector is not presently derived. Second, it inventories conditionally supplied low-energy fields that do not follow from P1–P6 alone but may be placed consistently on the ordered background, including local-bundle gauge completions, the weak-like/Higgs-like sector, and fermionic and neutrino embeddings. Third, it records nontrivial target-map classes and vanishing lower homotopy groups of the supplied S^3 orbit. Their promotion to physical sectors or defect exclusions remains Open.

(i) Condensate coordinates and candidate excitations.

- **Physical graviton:** an Open tensor-completion target. At fixed α, β and tangent-director linear order about homogeneous $n = w$, the fixed orientation map has $h_{ij}^{\text{TT}} = h_{ww} = 0$; this does not exclude separately guarded quadratic, nonlinear, composite or additional-field routes. A constrained tensor Hessian, source vertex and canonical normalisation remain to be supplied (Section 8.5).
- **Soft orientation coordinate χ :** the exact longitudinal coordinate of the constrained scalar-only orientation map; one physical soft mode is conditional on a supplied reduced action, domain and projector (Section 8.2).
- **P3 radial coordinate σ :** the homogeneous P3 Gaussian mode has $m_\sigma^2 = 2\mu^2$. The analogous P4 amplitude coordinate is kinematic only: its stationary Hessian, pole, residue and any Planck-scale identification remain Open (Section 8.3).

(ii) Conditionally supplied Standard-Model-like EFT embeddings.

- **Gauge fields:** conditional $U(1)$, electroweak-like, and non-Abelian local-bundle completions motivated by global compact-phase and rotation algebras (Section 8.6; full development in Sections 7.2 and 7.11).
- **Higgs sector:** the observed Higgs boson is reinterpreted as the radial excitation of a secondary electroweak condensate phase, if the second transition exists (Section 8.7).
- **Fermions:** spinorial fields compatible with the ordered branch, with a conditional Dirac-form reconstruction after a physical Lorentzian matter metric is supplied (Section 8.8; full development in Section 9).
- **Neutrinos:** a seesaw-type embedded mechanism with $m_\nu \sim |y_\nu|^2 v_2^2 / (2M_R)$ as standard supplied algebra. The heavy scale, Yukawa matrix and mass operator are not selected; no numerical neutrino conclusion is attributed to ECT (Open; Section 8.9).

(iii) **Topological and conditional sectors.**

- **Supplied-target topology:** the orientation orbit $O(4)/O(3) \simeq S^3$ has map classes classified by $\pi_3(S^3) = \mathbb{Z}$ and vanishing lower homotopy groups. Physical texture sectors and wall/string/monopole conclusions remain Open (Section 8.10).
- **Conditional secondary defects:** if the second transition $O(3) \rightarrow O(2)$ exists, monopole-like defects associated with $\pi_2(S^2) = \mathbb{Z}$ may arise (Section 8.10).

Structural statements and non-predictions. The strongest exact statements of this chapter are conditional and structural. At the present level, the homogeneous P3 model contains a radial Gaussian coordinate; its identification with a physical P4 amplitude pole remains Open. Separately, scalar integrability gives an exact kinematic reduction of the naive soft-orientation sector to one longitudinal scalar coordinate in the strict linear map. Treating that coordinate as the sole physical soft mode additionally requires a reduced action, constraint quotient/domain and physical projector (Level B/Open). The topological classification of the supplied orientation target, including $\pi_0 = \pi_1 = \pi_2 = 0$ and $\pi_3(S^3) = \mathbb{Z}$, is established as conditional mathematics. A physical vacuum manifold and physical defect sectors are not established. By contrast, the full Standard-Model particle spectrum, the gauge group, the fermion generations, the quantitative electroweak parameters, and the neutrino masses are not predicted here. Those sectors remain embedded, matched, or open.

Status summary.

Statement	Status	Comment
<i>Condensate coordinates and candidate excitations</i>		
P3 radial coordinate σ / Open P4 pole	A mathematics in P3 / Open physical P4 pole	Homogeneous P3 Gaussian radial coordinate; its continuation to an action-owned stationary P4 background and physical amplitude excitation is not derived
Strict soft-orientation map has one scalar coordinate; one physical χ mode is conditional	A kinematics / B physical	Scalar integrability plus supplied reduced action/domain/projector
Physical graviton sector	Open	No linear TT/static-density seed at fixed α, β and tangent-director order about homogeneous $n = w$; quadratic/nonlinear alternatives remain parametric/Open; Fierz–Pauli is supplied
<i>Structurally embedded sectors</i>		
Massless Abelian gauge boson	A conditional / Open ECT	Standard output only after an unbroken local $U(1)$, healthy ordered-background kinetic coefficients and reduced Hessian are supplied
Electroweak-like embedded sector	B/C	Conditional on secondary $O(3) \rightarrow O(2)$ transition
Higgs-like secondary radial mode	B/C	Conditional particle-level interpretation of the secondary sector

Statement	Status	Comment
Fermionic sector	A conditional / Open ECT	A supplied free spinor kernel admits the standard OS/Dirac reconstruction only under its complete applicable corrected OS-II package; the physical interacting owner is missing
Neutrino sector	C/Open	Standard supplied seesaw benchmarks; no ECT chirality or mass owner
<i>Topological and conditional sectors</i>		
$\pi_0 = \pi_1 = \pi_2 = 0$ for the supplied S^3 target	A-math conditional / physical Open	Target-space non-protection only
$\pi_3(S^3) = \mathbb{Z}$: integer target-map classes	A-math conditional / physical Open	Physical configuration sectors require an owner
Stable localised texture/skyrmion-like states	Open	Requires nonlinear condensate dynamics
Secondary monopole-like defects	B/C	Only if the secondary transition exists
<i>Not derived in Section 8</i>		
Full Standard Model particle spectrum	Open	Not derived from P1–P6
Gauge/Yukawa parameters	Open	External matching inputs
Topological mass spectrum	C/Open	Phenomenological ansatz only

Section 8 therefore establishes the bosonic, topological, and particle-classification side of the ordered-branch spectrum, but not the full fermionic reconstruction. The next section turns to the dedicated fermionic programme, where the spinorial structure, Yukawa couplings, and the ECT-specific fermion–condensate interaction are treated explicitly.

9 Fermionic Sector

Status guide. This section has three distinct layers:

Structural reconstruction (Level B). The Dirac form on the ordered Lorentzian branch, given additional representation-theoretic and first-order-minimality inputs. Fermions are *not* derived as collective excitations of the bare bosonic condensate; they are introduced as additional fields compatible with the condensate symmetry.

Phenomenological embedding (Level C). Yukawa couplings, fermion masses, neutrino sector. These assume Standard-Model structure as external input.

Open problems. Three fermion generations, Yukawa hierarchy, fermionic statistics from condensate structure, right-handed neutrino scale.

Connection to the fundamental ECT action. The fermionic sector is introduced consistently with the full ECT action (2.3). The scalar condensate Φ couples to fermions via supplied Yukawa interactions $y\bar{\Psi}\Phi\Psi$. Global compact-phase and rotation algebras motivate conditional gauge completions, but P5 does not derive local gauge fields or the physical $SU(2)_L \times U(1)_Y$ group (Sections 7–7). The *structural recon-*

struction of the Dirac form below is written as an Euler–Lagrange equation of $S_{\text{ECT}+f}$; the fermion field Ψ and its couplings are additional inputs not contained in the bare scalar condensate action (Level B/C):

1. The ordered-branch vacuum is characterised by $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$, where u_0 is the gradient condensate amplitude (Section 5.1), with the induced gravitational scale $\phi_0 \sim \bar{M}_{\text{Pl}}$ (Section 5.1; note that $n_A = \partial_A \Phi / |\partial \Phi|$ is a derived variable, not an independent field).
2. Fermions Ψ are introduced as additional fields in a supplied spinorial representation of $\text{Spin}(3)$, the double cover of the connected $SO(3)$ subgroup of the $O(3)$ stabiliser. After a physical Lorentzian metric/tetrad and spin structure are separately supplied, a corresponding $SL(2, \mathbb{C})$ representation may be used. Neither physical identification follows from P4 alone.
3. Euclidean and Lorentzian actions are kept as separate owners. Before continuation the supplied extension has the schematic form

$$S_E = S_{\text{ECT}}^E + S_f^E,$$

where S_f^E is a Euclidean spinor action such as the free kernel fixed in Section 23.1. Only after a continuation, physical Lorentzian metric, tetrad and spinor bundle are supplied do we write

$$S_f^L = \int d^4x e(x) \left[\frac{i}{2} Z \bar{\Psi} \gamma_L^\mu e_a^\mu \overleftrightarrow{D}_\mu \Psi - M \bar{\Psi} \Psi + \mu_5 u_\mu \bar{\Psi} \gamma_L^\mu \Psi + \dots \right], \quad e = \sqrt{-\det g_L}. \quad (9.1)$$

Here $Z > 0$, M , the connection in D_μ , the preferred covector and all matching coefficients belong to the supplied completion. In particular $u_\mu j^\mu = -j^0$ for the future-oriented rest vector used above, while the physical sign of μ_5 remains Open. The Euclidean condensate functional and this Lorentzian action are not terms of one uncontinued integral.

Once these additional fermionic inputs are in place, the Dirac equation (Section 9.2) is the Euler–Lagrange equation of S_f^L with respect to $\bar{\Psi}$, evaluated in the supplied condensate background (structural reconstruction, Level B).

Fermionic degrees of freedom can be introduced consistently with the spinor representations of $\text{Spin}(4) \simeq SU(2)_L \times SU(2)_R$, the double cover of the underlying Euclidean rotation group. This section performs a *structural reconstruction* of the Dirac form. Once (i) a supplied physical Lorentzian metric/tetrad completion and (ii) spinorial fields and representations are admitted as Level B inputs, the Dirac form can be constructed algebraically. Neither input is derived from the bare scalar condensate. The section does *not* derive fermions from the bare bosonic condensate action.

Honest framing. The symmetry group structure $\text{Spin}(4) \simeq SU(2)_L \times SU(2)_R$ admits spinor representations, and under the standard generator continuation the relevant group becomes $SL(2, \mathbb{C})$ the standard Weyl and Dirac spinors are obtained by Clifford linearisation. However, two important points should be noted. First, the spinor structure of the rotation group by itself does not imply fermionic *statistics* (anti-commutation relations): the spin-statistics connection requires additional input. Second, the present reconstruction obtains the Dirac equation as an effective equation for fields on a supplied Lorentzian background — it does not microscopically derive fermions as collective excitations of the condensate. The latter would require a separate construction. This section reconstructs the Dirac equation structurally (Level B) and discusses chirality in the ECT framework.

The fermionic programme developed here remains at the level of structural reconstruction on the ordered branch: it does not derive fermions from the bare bosonic condensate, but shows how spinorial fields, Dirac kinematics, chirality, and fermion couplings can be consistently organised once a physical Lorentzian metric/tetrad background is supplied. Quantum-statistical questions are addressed only at the level of a structural route and belong ultimately to the broader quantum programme.

Section roadmap. The logic of this section is layered. Section 9.1 establishes the group-theoretic compatibility of spinorial representations with the ordered branch. Section 9.2 reconstructs the Dirac form on that supplied Lorentzian background. Section 9.3 discusses the structural route to chiral asymmetry.

Sections 9.6 and 9.8 treat, respectively, the Standard-Model-like Yukawa embedding and the ECT-specific fermion–condensate coupling. Section 9.9 outlines the structural route toward spin-statistics, and Section 9.10 summarises the open programme. The Euclidean counterpart of the present conventions (hermitian gammas, the Euclidean Dirac kernel, and the graded OS reflection structure) is fixed in §23.1 and used in the spin–statistics analysis of Section 9.9 and Appendix AU.

9.1 Spinor structure in the condensate background

Status: Level A for the group-theoretic statements ($Spin(4) \simeq SU(2)_L \times SU(2)_R$, $\pi_1(SO(3)) = \mathbb{Z}_2$). $\pi_1(SO(3)) = \mathbb{Z}_2$ is a Level A topological fact about the P4-vector stabiliser group; its relevance as a structural motivation for half-integer representations is Level B. Fermions are not derived as collective condensate excitations; they are introduced as additional fields compatible with the condensate symmetry.

Section 8.8 provided the brief particle-spectrum roadmap; the present section develops that route in detail.

Spinors as compatible excitations (Level B/Open physical). In conventional quantum field theory, spinor fields transform under a spin cover of the relevant physical Lorentz or rotation group. Within the ECT architecture, the exact conditional group fact is that the connected $SO(3)$ subgroup of the P4-vector stabiliser has double cover $Spin(3) \simeq SU(2)$. Spinorial representations are therefore algebraically available. Identifying this subgroup with physical spatial rotations, supplying a physical metric/tetrad and spin structure, and realising stable spinorial excitations remain Open. Clifford linearisation yields a Dirac form only after those additional owners are supplied; it is not a derivation from the bare scalar condensate.

Euclidean rotation group and its spinorial content. At the Euclidean-operator level the undeformed arena is invariant under rotations in four dimensions, $O(4)$. The connected Euclidean spin group satisfies $Spin(4) \simeq SU(2)_L \times SU(2)_R$, while at the Lie-algebra level $\mathfrak{so}(4) \simeq \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (Level A). Rotations in four dimensions thus naturally split into two independent $SU(2)$ sectors. The fundamental representation of each $SU(2)$ factor is two-dimensional; two-component spinors are among the irreducible representations naturally available in this decomposition.

Important caveat. The group-theoretic structure *admits* spinorial representations; it does not guarantee that spinorial excitations are dynamically present as stable quasiparticles of the bare bosonic condensate. The step from “spinor representations exist” to “spinorial particles appear in the spectrum” requires introducing additional fermionic degrees of freedom as external inputs.

With that proviso, if spinorial excitations are admitted, they can be written as

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (9.2)$$

where ψ_L and ψ_R transform under the two $SU(2)$ factors of $Spin(4)$.

Topological availability of half-integer representations. The continuous stabiliser of the supplied nonzero P4 vector is $SO(3)$, whose fundamental group is $\pi_1(SO(3)) = \mathbb{Z}_2$ (Level A). This topological fact means that the connected rotation group has a nontrivial double cover. Accordingly, spinorial representations become structurally available: a 2π rotation, trivial in $SO(3)$ itself, can act as -1 in the spinorial representation lifted to the double cover. It does not by itself imply that dynamical fermionic excitations must occur. What remains an additional input is the physical assumption that such representations are *realised* by dynamical excitations of the ordered branch.

From Euclidean to Lorentzian spinors. After P4 supplies a preferred direction and a supplied scalar EFT satisfies $\alpha > \beta > 0$ on the retained $\beta > 0$ component, that scalar principal tensor is hyperbolic. The algebraic automorphism group of its Lorentzian canonical form is conjugate to $O(1, 3)$; identifying its

connected component with a common physical Lorentz symmetry requires a supplied metric/tetrad, reality structure and cross-sector map. It is not the residual subgroup of the Euclidean $O(4)$ breaking. If, in addition, a spin structure and spinor action are supplied, $Spin^+(1,3) \simeq SL(2, \mathbb{C})$ covers $SO^+(1,3)$ and the pair (ψ_L, ψ_R) is compatible with a Dirac spinor completion (Level B). None of these compatibility steps derives dynamical fermionic quasiparticles from the bare bosonic action.

Why spinors: scoped Clifford factorisation. For a four-complex-component irreducible Clifford module carrying $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ under the connected Lorentz group, require local, linear, constant-coefficient Lorentz-covariant dynamics and the exact factorisation

$$(\Gamma \cdot k + mc_{\text{char}})(\Gamma \cdot k - mc_{\text{char}}) = (-k^2 - m^2 c_{\text{char}}^2) \mathbf{1}, \quad \{\Gamma^\mu, \Gamma^\nu\} = -2\eta^{\mu\nu}. \quad (9.3)$$

The irreducible complex Clifford module, and hence the matrices Γ^μ , is then unique up to similarity and convention choices. This is a uniqueness theorem only inside that factorisation class. Reducible direct sums, Duffin–Kemmer–Petiau or Maxwell systems, constrained or gauge systems, additional background tensors and non-Clifford first-order systems are outside its hypotheses. The theorem is standard mathematics; it neither derives a physical ECT spinor sector nor proves that every first-order system is Dirac. Its ECT relevance begins only after the common physical metric, spin structure, field content and action are supplied, as discussed in Section 9.2.

Comparison with other emergent-fermion approaches. In superfluid $^3\text{He-A}$ [20], Weyl fermions emerge as low-energy excitations near topological Fermi points in momentum space: the fermionic character is a property of the quasiparticle spectrum, not an external input. In ECT the situation is different: fermions are introduced as additional fields compatible with the ordered branch. ECT currently provides a structural compatibility route and a Lorentzian-spinorial reconstruction, but not yet a microscopic derivation of fermions as quasiparticles of the bare bosonic medium.

Status summary.

Statement	Status	Comment
$Spin(4) \simeq SU(2)_L \times SU(2)_R$	A	Double cover of $SO(4)$
Spinorial representations available through the double cover of the connected rotation group	A	Group-theoretic availability; dynamical realisation not implied
$\pi_1(SO(3)) = \mathbb{Z}_2$: topological availability of half-integer representations	A	Topological fact of residual rotation group
Half-integer representations as structural motivation for fermionic excitations	B	Physical interpretation of topological availability
Dirac spinors compatible with ordered Lorentzian branch	B	Structural compatibility
Dirac operator unique in the irreducible four-component Clifford-factorisation class	B	Conditional standard theorem after a common physical metric and spinor sector are supplied
Fermions as collective condensate excitations	Open	Not derived from bare bosonic action

Physical interpretation of spin (Level B: interpretive). The exact group-theoretic input is the availability of spinorial representations of $Spin(4) \simeq SU(2)_L \times SU(2)_R$ and their 4π return property. Interpreting those representations as physical fermions requires a separately supplied physical metric/tetrad, spinor

action and matter state; P4 alone does not provide the $SL(2, \mathbb{C})$ Lorentzian spinor sector. A magnetic-moment calculation requires, in addition, an independently supplied unbroken local $U(1)$ completion, charge representation and photon state. The compact global phase and Euclidean covering-group algebra are candidate antecedents only; the physical fermion and electromagnetic owners remain Open.

Limitation. The present formulation does not provide a first-principles derivation of the electron g -factor, hydrogenic fine structure, or the numerical value of α_{fs} . These remain open targets of the fermionic programme (Section 9.10).

9.2 Structural Reconstruction of the Dirac Form

Status: Level B structural reconstruction. If one admits (i) a physical Lorentzian metric compatible with the conditional scalar-signature branch, (ii) spinorial representations compatible with the broken $O(4) \rightarrow O(3)$ background, and (iii) first-order Clifford linearisation as the minimal relativistic option, then a Dirac-type first-order equation can be constructed algebraically. Each of these inputs goes beyond the minimal scalar condensate postulate; the reconstruction is therefore Level B. The ECT-specific programme is to obtain the Clifford algebra and physical metric from the ordered medium rather than introduce them on a fundamental Lorentzian spacetime. That descent is not yet derived. A more structural representation-theoretic route to the same Dirac form is developed later in Section 24.5, where the equation is written first in terms of the pre-identification action scale S_0 and only subsequently matched to $\hbar$.

Metric, Fourier covector and mass shell. The $O(3)$ stabiliser of the P4 vector does not supply a Lorentzian matter metric, nor does the scalar principal tensor alone. Conditional on supplying its flat mostly-plus limit, we fix the complete house convention

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1), \quad x^0 = c_{\text{char}} t.$$

For a free plane wave we define

$$\Psi_k(x) = u(k) \exp \left[-\frac{i}{\hbar} (Et - \mathbf{p} \cdot \mathbf{x}) \right], \quad i\hbar \partial_\mu \Psi_k = k_\mu \Psi_k,$$

so that the Fourier eigen-covector is

$$k_\mu = (E/c_{\text{char}}, -\mathbf{p}), \quad k^2 = -E^2/c_{\text{char}}^2 + \mathbf{p}^2 = -m^2 c_{\text{char}}^2.$$

This definition avoids identifying k_μ with a contravariant physical momentum or with a metric-lowered vector under a different Fourier sign.

Clifford, adjoint and current package. Introduce Lorentzian matrices obeying

$$\{\gamma_L^\mu, \gamma_L^\nu\} = -2\eta^{\mu\nu} \mathbf{1}, \quad (\gamma_L^0)^\dagger = \gamma_L^0, \quad (\gamma_L^i)^\dagger = -\gamma_L^i.$$

With

$$\bar{\Psi} = \Psi^\dagger \gamma_L^0, \quad (\gamma_L^\mu)^\dagger = \gamma_L^0 \gamma_L^\mu \gamma_L^0,$$

the vector current is $j^\mu = \bar{\Psi} \gamma_L^\mu \Psi$ and $j^0 = \Psi^\dagger \Psi \geq 0$. These are standard conditional identities for the supplied representation; the scalar action does not derive the physical spinor bundle.

Clifford factorisation and Dirac equation. Defining $\not{k} = \gamma_L^\mu k_\mu$ gives

$$\not{k}^2 = -k^2, \quad (\not{k} + mc_{\text{char}})(\not{k} - mc_{\text{char}})u = (-k^2 - m^2 c_{\text{char}}^2)u = 0.$$

The first-order equation and its coordinate form are

$$(\not{k} - mc_{\text{char}})u = 0,$$

$$\boxed{(i\hbar\gamma_L^\mu D_\mu - mc_{\text{char}})\Psi = 0.} \quad (9.4)$$

For an Abelian connection our convention is $D_\mu = \partial_\mu - iq_A A_\mu$, $\Psi \mapsto e^{iq_A \alpha} \Psi$ and $A_\mu \mapsto A_\mu + \partial_\mu \alpha$. In the free case,

$$(i\hbar\gamma_L \cdot \partial + mc_{\text{char}})(i\hbar\gamma_L \cdot \partial - mc_{\text{char}}) = \hbar^2 \square - m^2 c_{\text{char}}^2, \quad \square = \eta^{\mu\nu} \partial_\mu \partial_\nu.$$

The corresponding equal-time Hamiltonian is

$$i\hbar\partial_t \Psi = [-i\hbar c_{\text{char}} \alpha \cdot \nabla + \beta_D m c_{\text{char}}^2] \Psi, \quad \alpha^i = \gamma_L^0 \gamma_L^i, \quad \beta_D = \gamma_L^0,$$

which is Hermitian in the supplied equal-time inner product, subject to its operator domain and boundary conditions.

The factorisation is Level A standard conditional mathematics once the metric, representation, domain and first-order equation are supplied. Its use as an ECT fermion sector is Level B/Open: the physical metric, spinor field, CAR/Fock structure, mass and action-scale matching remain un-derived. The equality $c_{\text{char}} = c$ used here is the adopted scalar-clock calibration, not a result of this fermion factorisation; fermion-cone equality remains Open.

Connection to the supplied Lorentzian action. The Dirac-type equation (9.4) is the free, constant-coefficient limit of the separately supplied Lorentzian fermion action S_f^L (9.1). Varying that action with respect to $\bar{\Psi}$ gives the Dirac operator together with any retained preferred-direction and interaction terms discussed separately in Section 9.8. This is variational consistency inside the supplied extension. It neither adds a Lorentzian fermion integrand directly to the Euclidean condensate action nor derives the fermionic sector from the bare bosonic medium.

Weyl decomposition. In the chiral representation the Dirac spinor decomposes into two Weyl spinors

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (9.5)$$

transforming under the two inequivalent two-component representations of $SL(2, \mathbb{C})$. This connects directly to the ECT observation that the Euclidean spin group admits the decomposition $Spin(4) \simeq SU(2)_L \times SU(2)_R$ (Section 9.1).

Curved-background target. If a physical mostly-plus metric/tetrad completion is supplied independently, define

$$\Gamma^\mu = e_a^\mu \gamma_L^a, \quad \{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}.$$

The standard curved-background target is then

$$(i\hbar\Gamma^\mu \nabla_\mu - mc_{\text{char}})\Psi = 0, \quad (9.6)$$

where the tetrad, spin structure and spin connection belong to that completion. The fixed condensate director and its gradients do not by themselves derive a vierbein, spin connection or universal physical metric. Their descent from a constrained Φ -medium action remains Open; the displayed equation is a standard conditional target, not an ECT result.

ECT-specific fermion–condensate coupling. Beyond the Dirac form itself, the ordered branch admits an additional fermion–condensate coupling $\Delta\mathcal{L} = \mu_5 \bar{\Psi} \gamma^A n_A \Psi$. Its status within the fermionic reconstruction is discussed in Section 9.8, while its phenomenology is developed separately in Section 7.

What is and is not reconstructed here. The present subsection reconstructs the *Dirac form* only conditionally: it is the minimal first-order spinorial equation after a physical Lorentzian metric/tetrad and spinorial field have been supplied. It does not yet derive the existence of fermions from the bare bosonic condensate, nor does it derive fermionic quantisation, anticommutation relations, or the full spin-statistics theorem. Nor does the present subsection derive the observed fermion spectrum, generation structure, or Yukawa hierarchy. Those questions belong to the broader fermionic and quantum programmes developed in the following subsections and in Section 24.4.

Status summary.

Statement	Status	Comment
Scalar quadratic invariant in the supplied ordered-branch EFT	A-math conditional	Applies to that scalar principal operator only
Clifford algebra on a supplied physical metric/tetrad	B/Open	Standard algebra after the metric/tetrad and spinor bundle are admitted; their ECT descent is Open
Dirac-type first-order equation via Clifford linearisation	B	Structural reconstruction once the spinorial and Lorentzian inputs are admitted
Connection to $S_{\text{ECT}+f}$ variational structure	B	Euler–Lagrange equation of the extended action
Weyl decomposition $\leftrightarrow \text{Spin}(4) \simeq SU(2)_L \times SU(2)_R$	A/B	Group-theoretic (A); physical relevance (B)
Curved-background Dirac generalisation on a supplied metric/vierbein/spin-connection completion	B/Open	Standard conditional target; the Φ -medium descent remains Open

9.3 Chirality and fermion interactions

Status: exact group-theory obstruction and Open physical sector. The decomposition $\text{Spin}(4) \simeq SU(2)_+ \times SU(2)_-$ is Level A mathematics. A supplied nonzero vector has diagonal stabiliser $SU(2)_{\text{diag}}$ and does not select either factor as the physical weak $SU(2)_L$. A local internal weak bundle, chiral matter representations and parity-violating vertex remain Open.

Exact stabiliser calculation. Represent a Euclidean vector by a Hermitian two-by-two matrix N on which $(g_+, g_-) \in SU(2)_+ \times SU(2)_-$ acts as $N \mapsto g_+ N g_-^{-1}$. At a reference ordered direction one may take $N \propto \mathbf{1}$. Its stabiliser satisfies

$$g_+ \mathbf{1} g_-^{-1} = \mathbf{1} \iff g_+ = g_-, \quad (9.7)$$

and is therefore $SU(2)_{\text{diag}}$. Thus the ordered vector identifies the two Euclidean factors; it does not privilege $SU(2)_+$ or $SU(2)_-$. The earlier-looking inference “preferred orientation $\Rightarrow$ one chiral weak factor” is not a consequence of the stabiliser.

Euclidean and Lorentzian chirality. For a supplied spinor kernel, $\text{Spin}(4)$ Weyl modules and their continuation to the Lorentzian $(1/2, 0)$ and $(0, 1/2)$ modules are standard algebra. An orientation convention can label the two modules, but relabelling is not a dynamical energy splitting and supplies no interaction vertex. A physical weak theory additionally needs a local internal bundle, a gauge-reduced kinetic operator, an unbroken or broken state, matter representations and a vertex that couples only to the observed left-handed doublets. None follows from the vector stabiliser or from P1–P6.

Director operator does not select weak chirality. The conditional operator $\mathcal{L}_5 = \mu_5 n_\mu \bar{\Psi} \gamma^\mu \Psi$ couples to a vector current. In a Weyl decomposition its temporal component contains the number densities of both chiralities with the same vector-current assignment; it is not the Standard-Model projector $P_L = (1 - \gamma^5)/2$. An axial operator could distinguish chiralities, but it is an independent EFT operator and is not selected by the stated parity-even restricted basis. Therefore the director programme and the weak-chirality programme do not share a derived vertex.

Consequences for neutrinos and $SU(2)_R$. The observed left-handed active-neutrino representation cannot be inferred from the ordered direction. Likewise, a “fate of $SU(2)_R$ ” is not yet a dynamical ECT question: no physical $SU(2)_L \times SU(2)_R$ internal gauge bundle has been derived. Left–right models may be supplied as completion benchmarks, but the current action selects none.

Closure test. A physical derivation would have to construct the constrained local weak bundle and kinetic action, derive a chiral projector in the matter vertex, obtain the observed representations, and pass local and Witten anomaly conditions. The anomaly audit of Section 9.5 tests a supplied completion; it cannot create the missing bundle or representations.

Status summary.

Statement	Status	Comment
$Spin(4) \simeq SU(2)_+ \times SU(2)_-$	A	Exact double-cover identity
Stabiliser of the ordered vector	A	$SU(2)_{\text{diag}}$; neither factor is selected
Continuation of supplied free spinor modules	A conditional	Standard representation theory; physical ECT spinor owner Open
Physical local weak bundle and chiral projector	Open	Not supplied by orientation or P5
Observed fermion and neutrino representations	Open	Matter map and mass operators missing
Anomaly cancellation	A conditional / Open ECT	Standard check after representations are supplied
Full derivation of Standard-Model chirality	Open (OP-chirality)	Requires bundle, action, state and matter vertex

9.4 Discrete symmetries of the emergent matter sector: P, T, C, and conditional CPT

Status: layered. The Euclidean reflection identities follow from the $O(4)$ structure postulated in P2 (Level A mathematics). Their identification with physical parity or time reversal requires an ordered-branch clock/rod map and a completed matter sector and is therefore conditional. Charge conjugation has a conditional, internal status tied to the fermionic/charged sector (Level B). A conditional CPT route is identified for those local sectors where the complete applicable corrected OS-II reconstruction package enumerated in paragraph B3 below is available: supplied Euclidean correlators/measure, observable algebra and time reflection, temperedness plus the required regularity/growth condition, Euclidean covariance, reflection positivity, permutation/graded symmetry, and applicable locality/clustering properties. This is a conditional standard-QFT route for supplied free/quadratic local sectors and remains Open for the full nonlinear interacting theory. The phenomenological preferred-direction operator $\mathcal{L}_5$ introduced in Section 7.6 is re-classified here as a CPT-odd branch-sensitive operator (Level B), complementing but not replacing the exact symmetry backbone. The present subsection should be read as a structural identification of the discrete-symmetry architecture of ECT, not as a complete closed proof of CPT for the full nonlinear interacting theory.

Transition from the chirality sector. The previous subsection (§9.3) showed that the ordered vector leaves $SU(2)_{\text{diag}}$ and therefore does not select either factor of $Spin(4) \simeq SU(2)_+ \times SU(2)_-$ as a physical weak chirality. We now separate exact Euclidean reflection identities from conditional physical parity, time-reversal and matter-sector maps. The key point is that the Euclidean theory already contains exact reflections at the level of $O(4)$, whereas the operator-level notions of T , C , and CPT emerge only conditionally in those local sectors for which the Euclidean quantum reconstruction succeeds. The discussion is organised into three structurally distinct layers: (A) the geometric discrete backbone of the Euclidean manifold; (B) the conditional operator-level theorem structure that emerges after Osterwalder–Schrader reconstruction; (C) the branch-sensitive phenomenology, including the re-classification of $\mathcal{L}_5$ and the branch-pair symmetry architecture that is characteristic of ECT.

Layer A: Geometric discrete backbone of ECT

A1. Euclidean reflections in $O(4)$. On the Euclidean manifold $\mathcal{M}^4$ with metric δ_{AB} the $O(4)$ group acts in its full form $O(4) = SO(4) \rtimes \mathbb{Z}_2$, where the $\mathbb{Z}_2$ factor is generated by improper reflections (determinant -1 transformations). Postulate P2 asserts invariance of the Euclidean action S_E under $O(4)$, *not* merely under $SO(4)$; hence improper reflections are automatically exact Euclidean symmetries of the supplied classical action. This exact identity does not by itself identify a physical parity, an anti-unitary time-reversal operation or CPT. The three natural generators of the improper part are:

$$R_s : (x^1, x^2, x^3, w) \mapsto (-x^1, -x^2, -x^3, w), \quad (9.8)$$

$$R_w : (x^1, x^2, x^3, w) \mapsto (x^1, x^2, x^3, -w), \quad (9.9)$$

$$I_4 = R_s \circ R_w : (x^1, x^2, x^3, w) \mapsto (-x^1, -x^2, -x^3, -w). \quad (9.10)$$

All three are exact isometries of the Euclidean metric and, by P2, exact symmetries of S_E (Level A).

A2. Candidate geometric antecedent of Lorentzian parity from R_s . After the retained Lorentzian component $\alpha > \beta > 0$ is supplied and the real parametrisation $x^0 = w$ is introduced (§3.5), the spatial reflection R_s preserves the ordered-branch time coordinate and acts only on spatial indices. At the level of coordinates it is therefore a candidate geometric precursor of Lorentzian parity $P : (t, \mathbf{x}) \mapsto (t, -\mathbf{x})$. The Euclidean reflection identity is Level A mathematics; the physical identification is conditional on the branch clock/rod and matter maps.

The spinorial content of R_s is fixed by the double cover $Spin(4) \simeq SU(2)_L \times SU(2)_R$. The improper elements $O(4) \setminus SO(4)$ exchange the two $SU(2)$ factors. Consequently R_s acts on a Dirac-like doublet $\Psi = (\psi_L, \psi_R)^\top$ by swapping the two chiral components, up to the standard phase conventions of Euclidean spinor QFT [71, 72]. These phases are inherited from the group-theoretic structure and introduce no additional postulates.

Conditional consequence. An independently supplied vertex that couples to only one Lorentzian Weyl module would violate parity. Neither the ordered vector nor its diagonal stabiliser supplies that vertex or selects a module. Reproducing the maximally parity-violating Standard-Model weak sector therefore remains an Open action-and-representation problem (OP-P1, §9.10):

- Euclidean improper element exchanges chiral factors: Level A (group theory);
- one-factor chiral coupling: conditional property of an independently supplied matter vertex, not an ordered-branch result;
- exact maximal weak-sector pattern: Open (future derivation, OP-P1).

A3. Geometric antecedent of time reversal from R_w . The reflection R_w inverts the preferred Euclidean axis $w \mapsto -w$. Under the real parametrisation $x^0 = w$ it reverses the common-length coordinate x^0 . Interpreting this as $t \mapsto -t$ additionally uses the ordered-branch clock map of Eq. (3.23); it is therefore only a candidate geometric antecedent of Lorentzian time reversal.

We emphasise that this classical coordinate map is *not*, by itself, the anti-unitary time-reversal operator of a conditionally reconstructed Lorentzian quantum sector. Anti-unitarity is a property of an operator on a Hilbert space, not of a coordinate transformation on a manifold, and it requires a reconstruction step to which we now turn (Layer B).

Layer B: Conditional operator-level theorem structure

B1. Complex conjugation from reflection positivity. The fundamental ECT field Φ is a real scalar; there is no fundamental complex structure in the bare action. In a Euclidean subsector satisfying all Osterwalder–Schrader hypotheses, a complex Hilbert space and the conjugation required for anti-unitary operators can be reconstructed conditionally [73, 74]. Those hypotheses and the identification of the reconstructed parameter with a physical clock are not derived for the full interacting Φ theory.

The relevant construction proceeds as follows. For each local observable A with support in a half-space of the Euclidean manifold, reflection positivity asserts

$$\langle \Theta(A)A \rangle_E \geq 0, \quad (9.11)$$

where Θ is the Euclidean reflection operator (for the w -axis one takes $\Theta = \Theta_w$ associated with R_w). Reflection positivity permits the definition of a positive-semidefinite sesquilinear form

$$\langle \Psi_A, \Psi_B \rangle_{\mathcal{H}} \equiv \langle \Theta(A^*)B \rangle_E, \quad (9.12)$$

on the positive-time observable algebra. Quotienting by its null space and then completing gives the OS Hilbert space. A self-adjoint nonnegative Hamiltonian follows only when the complete applicable corrected OS-II package and time-translation structure are supplied; reflection positivity alone does not provide it. Within such a completed reconstruction, complex conjugation on $\mathcal{H}$ is an emergent structure rather than a fundamental ingredient. An anti-unitary time-reversal operator T on $\mathcal{H}$ is then built as a combination of the Euclidean reflection Θ_w with complex conjugation on $\mathcal{H}$; anti-unitarity follows by construction.

The Euclidean reflection R_w supplies the geometric reversal of the preferred axis. An anti-unitary realisation of time reversal in a conditionally reconstructed Lorentzian quantum sector requires the full OS hypotheses, the reconstructed Hilbert-space structure and a separately justified physical clock/matter interpretation. In this sense, the geometric Euclidean reversal and the anti-unitary quantum operator T are related but not identical objects.

B2. Charge conjugation as an internal operation. Charge conjugation C does *not* arise from $O(4)$ spacetime reflections. The fundamental ECT scalar Φ carries no phase, so the $\theta \rightarrow -\theta$ construction of charge conjugation in a compact-phase sector is available only after a compact phase has been supplied and matched to an independently admitted local $U(1)$ bundle, connection and charged matter representation (see §21.1 and the conditional local completion in §7.2).

For the Dirac sector a group-theoretic basis for C is nonetheless available: on $Spin(4)$ representations the charge-conjugation matrix C is determined (up to convention) by the relations

$$C\gamma^A C^{-1} = -(\gamma^A)^\top, \quad C^\top = -C, \quad (9.13)$$

and such a matrix exists group-theoretically. In this sense C has a structural basis in the spinor bundle, although the basis is internal (spinor/charge structure) rather than geometric (spacetime reflection). Its status is therefore Level B, conditional on the existence of the relevant charged/fermionic sector.

A complete derivation of the action of C on all emergent charged sectors of ECT, including the interplay with the compact-phase $U(1)$ structure, remains open (OP-CPT2, §9.10).

B3. Conditional CPT route via Osterwalder–Schrader. There are two standard routes to the CPT theorem: the Minkowski-side proof of Jost [75, 76] and the Euclidean-side route through Osterwalder–Schrader reconstruction [73, 74]. Because ECT is formulated on a Euclidean manifold and already employs the corrected OS-II reconstruction route in Part III, with reflection positivity as one required hypothesis (§23.1), the OS route is the natural one.

Conditional CPT route. Let $\mathcal{S}$ denote a local sector of ECT defined by an operator algebra generated by observables supported in finite spatial regions of the Euclidean manifold. Whenever the applicable reconstruction hypotheses are supplied, including

- (a) Euclidean Schwinger functions or a measure, an observable algebra and a specified time reflection with the required OS-II temperedness and regularity/growth hypotheses,
- (b) Euclidean covariance of the correlators is established (action-level $O(4)$ invariance from P2 alone is not sufficient),
- (c) reflection positivity is verified for the algebra of observables of $\mathcal{S}$,
- (d) permutation/graded symmetry and the applicable locality and cluster decomposition properties hold for $\mathcal{S}$, and
- (e) the Osterwalder–Schrader reconstruction closes for $\mathcal{S}$,

the associated emergent Lorentzian theory obtained through OS reconstruction inherits the standard CPT structure: there exists an anti-unitary operator Θ_{CPT} on the reconstructed Hilbert space satisfying the usual CPT transformation rules for bosonic and fermionic fields.

The important point is the formulation: in the present article we *identify* this route and its precise prerequisites; we do not claim a complete closed proof of the full nonlinear case. The current status is:

- supplied free and quadratic local sectors for which conditions (a)–(d) are actually checked: conditional standard OS/CPT consistency, not a derivation of physical ECT matter;
- full nonlinear interacting ECT: Open (OP-CPT1), pending the broader programme on low-energy QFT architecture (locality and cluster structure in nonlinear ECT) together with OP-OS1.

B4. Comparison with Minkowski-side proofs. The standard Minkowski-side CPT route is associated with the Jost / Pauli–Lüders line of argument in axiomatic Lorentzian QFT [75, 77, 76], where CPT follows from the usual locality and covariance assumptions. The route identified here is the Euclidean Osterwalder–Schrader one: once the complete applicable reconstruction hypotheses listed above hold and OS reconstruction applies, the reconstructed Lorentzian theory inherits the standard CPT structure. We emphasise again that the present subsection identifies these prerequisites explicitly and specifies where they are already met and where they remain open within ECT.

Layer C: Branch-sensitive phenomenology

C1. Conditional branch-pair symmetry architecture. The supplied orientation orbit is S^3 (§34); for every direction n_A it contains the antipodal point $-n_A$. This exact target geometry motivates a three-fold conditional classification:

- (i) *exact symmetries of the full Euclidean theory* that act on the entire ensemble of ordered branches;
- (ii) *branch-exchange operations* that exchange n_A and $-n_A$ (equivalently, map an ordered branch to its antipodal partner), examples of which include R_w and I_4 in the orientation coordinate;
- (iii) *effective symmetries of one selected branch*, which form a subgroup of (i) and may be broken by branch selection.

The antipodal structure is exact for the supplied orbit. Its identification with physical vacuum branches and its dynamical realisation as a symmetry of cosmological solutions remain Open. It is therefore a candidate symmetry layer, not an established property of physical ECT vacua.

A simple structural observation follows from the homotopy of S^3 : since $\pi_0(S^3) = 0$ (§34), antipodal points lie in the same connected target. This does not exclude a physical barrier or domain wall until the full vacuum/configuration space and action are owned.

Possible cosmological interpretation. One may speculate that our observable matter-dominated branch is paired with an antipodal antimatter-dominated partner, with the Euclidean pair being CPT-symmetric while branch-level asymmetries are local to each partner. Such a cosmological reading is reminiscent of Boyle–Turok-type CPT-symmetric cosmologies [78], but native to the ECT $O(4) \rightarrow O(3)$ structure. It is recorded here as a possible future direction (Level C, OP-BRANCH1, §9.10), not as an established result.

C2. Re-classification of the phenomenological operator $\mathcal{L}_5$. The preferred-direction fermion–condensate coupling introduced in §7.6,

$$\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma_A^F n_A \Psi \quad (\text{Euclidean notation}), \quad (9.14)$$

admits a precise discrete-symmetry classification after a Lorentzian completion is supplied. There we use $j^\mu = \bar{\Psi} \gamma_L^\mu \Psi$ and the future-oriented rest vector $u^\mu = (1, 0, 0, 0)$, hence $u_\mu = (-1, 0, 0, 0)$. The Lorentzian operator is

$$\mathcal{L}_5 = \mu_5 u_\mu j^\mu = -\mu_5 \bar{\Psi} \gamma_L^0 \Psi = -\mu_5 \Psi^\dagger \Psi. \quad (9.15)$$

The minus sign follows only from lowering the declared vector in the mostly-plus metric; it may be absorbed into the Open coefficient and is not a derived physical sign.

The current is C-odd, while j^0 is P-even and T-even. Therefore the operator is *C-odd, P-even, T-even, CP-odd, and CPT-odd*; its overall real sign does not change that classification. Reality of μ_5 ensures hermiticity but does not restore CPT invariance. Consequently $\mathcal{L}_5$ is a branch-sensitive Lorentz- and CPT-violating phenomenological operator of the ordered-branch EFT, not a foundation of any exact symmetry theorem. Its possible roles in a supplied completion are:

- (i) to enter a candidate Sakharov CP-bias term only after a nonzero coefficient, physical baryon current, state and transport are supplied (§16.3);
- (ii) to parameterise a possible branch-sensitive CP-asymmetric source, not to prove that the realised state activates one;
- (iii) to serve as a phenomenological channel for testable Lorentz/CPT-violation signatures, particularly in the neutrino sector (§8.9).

The conditional CPT route of Layer B for reconstructible local sectors is not affected by $\mathcal{L}_5$: the operator explicitly lies outside the reconstructible OS setup as a branch-selecting extension of the ordered-branch EFT, and therefore does not conflict with the exact CPT backbone of those sectors. Compatibility of $\mathcal{L}_5$ with standard anomaly cancellation, following from its vector-operator nature, is systematically analysed in §9.5.

Comparison with SME-type constructions. A separate line of research, the Standard-Model Extension (SME) of Kostelecký and collaborators [27, 28, 15], systematically parametrises Lorentz and CPT violation at the level of effective Lorentzian QFT. The ECT operator $\mathcal{L}_5$ fits into the general SME operator class at the effective level. In the present ECT candidate the preferred direction is a supplied P4 datum; its dynamical selection from an initially Euclidean $O(4)$ -symmetric phase is not derived. SME-type constructions typically introduce Lorentz-violating tensors directly at the level of the effective Lorentzian theory. Thus a supplied ECT direction permits operators in an SME-like class but does not yet provide their pre-geometric physical origin.

C3. Neutrino sector: corrected operator interpretation. The preferred-direction inventory contains two independent possibilities. On the declared constant-coefficient, perturbative post-EWSB vacuum/free-propagation domain with $V_{\text{MSW}} = 0$, a self-adjoint spatial domain and norm-bounded positive-nonsingular time kinetic form permit canonical Hamiltonian reduction; flavour Hermiticity alone does not. At leading order $C_{ab}(\hat{p}) = -(c_L)_{ab}^{\mu\nu} \hat{p}_\mu \hat{p}_\nu$ has the minus sign derived from the displayed action and plane-wave conventions. Together with kinetic admissibility, flavour-Hermitian a_L and c_L define Hermitian $A(\hat{p})$ and $C(\hat{p})$ and hence unitary propagation. Only the traceless projections A_{osc} and C_{osc} can affect oscillations. Antineutrinos use $A \mapsto -A^*$ for the CPT-odd term and $C \mapsto C^*$ for the CPT-even term. For an eigensplitting ΔA in the dimension-three CPT-odd route, the vacuum-scale ratio is $2\Delta A E_\nu / \Delta m^2$. A flavour-identity coefficient is a counterexample to treating a raw coefficient magnitude as an oscillation signal. For the separate $d = 4$ CPT-even class, no nonzero tensor coefficient, direction/flavour map, above-EWSB completion or ECT vertex is presently derived, and no numerical comparison is retained. Because H_{vac} and C_{osc} need not commute, a comparison of their separate eigenspaces would not by itself locate a crossing of the full Hamiltonian. Above EWSB gauge covariance also ties these doublet operators to charged-lepton partners, so the post-EWSB neutrino-only operator is not an independent gauge-invariant coefficient or a global fit. Neutrinoless double-beta decay is an external discriminator of Majorana character. If observed, it would require a supplied Majorana mass owner; neither preferred-direction operator nor the ordered branch selects Dirac versus Majorana character. Its absence constrains only a declared Majorana completion.

C4. Baryogenesis completion inventory. The present subsection supplies only candidate ingredients for a future baryogenesis completion (§16.3). The Sakharov owners would have to be realised as follows:

- (i) *Baryon number violation:* a physical gauge/matter map and vertex relating a defect transition to baryon current; topology alone is insufficient (§22.3, §16.3).
- (ii) *C and CP violation:* the phenomenological CPT-odd and CP-odd operator $\mathcal{L}_5$ (Layer C of this subsection), combined with complex flavour phases (OP-Yukawa; §9.6). The conditional CPT route of Layer B for reconstructible sectors remains intact; $\mathcal{L}_5$ lies outside that reconstructible setup as a branch-sensitive phenomenological channel.
- (iii) *Departure from thermal equilibrium:* a derived finite-rate ordering history, clock, thermal state and species rates. P4 supplies only the ordered branch, not this process.

The branch-pair architecture of C1 provides an additional, still speculative, possibility in which the observed matter-antimatter asymmetry is a branch-local property of our ordered partner, the full Euclidean pair being CPT-symmetric. This is recorded as OP-BRANCH1, not as an established result.

Three distinct notions of “time-reversal”

It is useful to disentangle three concepts that are logically distinct in ECT and that are sometimes conflated in the discrete-symmetry literature:

- (i) *Fundamental anti-unitary T* on the reconstructed Hilbert space, obtained through the OS route of Layer B. Level B (free/quadratic local sectors); Open for the full nonlinear theory (OP-CPT1).
- (ii) *Effective branch-level T-violation* carried by $\mathcal{L}_5$ at the phenomenological level of Layer C. Level B; independent of the fundamental- T statement.
- (iii) *Thermodynamic arrow of time* (Open for ECT). Monotonicity in a supplied reduced protocol requires its map, state or fixed point, entropy functional and contractivity/detailed-balance hypotheses; retarded support and dephasing do not derive the physical arrow. This question is independent of both (i) and (ii).

The conditional CPT route may hold in reconstructible sectors and $\mathcal{L}_5$ may parametrise branch-level symmetry violation in a supplied EFT. Neither statement establishes a thermodynamic arrow; that remains a separate Open reduced-dynamics problem.

Locality/cluster input for CPT closure. The conditional CPT route identified in Layer B depends not only on reflection positivity but also on the locality and cluster decomposition properties required by OS reconstruction. Their status in the full nonlinear interacting theory remains part of the broader low-energy QFT architecture programme (OP-Loc1, §9.10), of which the CPT route is one element. This open-problem dependency is explicitly recognised here to avoid misinterpretation of the conditional CPT statement as a standalone closed theorem.

Connection to quantum statistics. The discrete-symmetry architecture developed here complements the topological structure of the fermionic sector discussed in §9.9 (quantum statistics). Three distinct $\mathbb{Z}_2$ facts enter the fermionic sector architecture, and it is important to keep them apart. Their coexistence does not by itself establish the full relativistic spin-statistics theorem; that linkage remains part of the broader open programme:

- the $\mathbb{Z}_2$ factor of $O(4) = SO(4) \rtimes \mathbb{Z}_2$, generated by improper reflections (present subsection);
- the topological fact $\pi_1(SO(3)) = \mathbb{Z}_2$, underlying the group-theoretic availability of half-integer representations (§9.1);
- the exchange topology $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$, which supplies two exchange-loop classes in the declared point model; the characters ± 1 require an additional one-dimensional unitary representation, and physical statistics remain Open (§9.9).

Table 40: Discrete-symmetry architecture of ECT: status summary.

Statement	Status	Basis / open problem
Euclidean reflections $R_s, R_w, I_4 \in O(4)$	A	From $O(4)$ -invariance in P2
Candidate Lorentzian P precursor from R_s	B/Open	Requires the physical branch clock/rod and matter maps
R_s exchanges $SU(2)_L \leftrightarrow SU(2)_R$ in $Spin(4)$	A	Group theory of double cover
Physical parity asymmetry in a weak vertex	Open	The vector stabiliser is $SU(2)_{\text{diag}}$ and selects no chirality (§9.3)
Exact maximal weak-sector parity pattern	Open	OP-P1 (§9.10)
Candidate geometric antecedent of T from R_w	B/Open	The $O(4)$ identity is exact; its physical identification is not
Anti-unitary T in quadratic local sectors after OS	B	RP + complex conjugation from OS formula
Full nonlinear anti-unitary T closure	Open	OP-CPT1, depends on OP-OS1 and OP-Loc1
Charge conjugation C in charged/Dirac sector	B	Internal; Dirac-operator structure
Complete C transformation on all emergent charged sectors	Open	OP-CPT2
Conditional CPT route for reconstructed local sectors	B/Open	Identified via OS prerequisites (a)–(d)
Full interacting CPT closure in nonlinear ECT	Open	OP-CPT1

Statement	Status	Basis / open problem
Branch-pair geometry of the supplied orientation orbit	A-math conditional / physical Open	Antipodal structure of S^3 ; physical vacuum branches are not derived
Cosmological CPT-symmetric branch-pair realisation	C	OP-BRANCH1
$\mathcal{L}_5$ as CPT-odd branch-sensitive phenomenological operator	B	Discrete-symmetry classification of the vector bilinear
Locality/cluster prerequisite for full CPT closure	Open	OP-Loc1
Full CKM/PMNS-scale flavour closure	Open	Systematised in §9.7; quantitative derivation remains under OP-Yukawa / OP-generations

This architectural contribution is recorded in the global Level-4 self-consistency checklist (§19.3).

9.5 Anomaly audit for a supplied chiral gauge completion

Status: consistency architecture. This subsection systematises standard consistency conditions inherited by any supplied reconstructible chiral gauge completion and records what ECT still lacks before those conditions can be applied to a physical sector. It is a consistency architecture, not a first-principles derivation of Standard-Model representation content. Layers A and B correspond to group-theoretic inputs and inherited anomaly consistency in reconstructible local sectors; Layer C isolates the full Standard-Model completion and the ECT-native interpretation of the mixed $U(1)_Y$ -gravitational condition in a future physical metric/tensor completion.

Transition from the discrete-symmetry subsection. The previous subsection (§9.4) established the discrete-symmetry backbone of the emergent matter sector: geometric precursors of P and T from $O(4)$ reflections, a group-theoretic basis for C , and a conditional CPT route for reconstructible local sectors. With these ingredients in place, the natural next consistency question is whether a future supplied chiral gauge completion is quantum-mechanically consistent at the loop level. Anomaly cancellation is a *non-negotiable self-consistency requirement* already recorded in the gauge-sector discussion (§49.14, “Anomaly cancellation as a self-consistency requirement”): any acceptable fermionic completion of a supplied gauge sector must satisfy it; failure would reject that completion, not contradict the existence of the scalar ordered branch. The present subsection organises the resulting architecture into three layers and attaches each outcome to the already-existing unification programme; the associated open problems are recorded in the existing OP-GUT unification programme, in particular OP-GUT3b and the OP-GUT6 consistency family (Table 115).

Layer A: Topological and group-theoretic inputs

A1. Load-bearing homotopy facts. The relevant topological inputs for the anomaly-architecture discussion are:

- $\pi_4(SU(2)) = \mathbb{Z}_2$ — Witten global-anomaly topology [79];
- $\pi_3(SU(N)) = \mathbb{Z}$ for $N \geq 2$ — Chern–Pontryagin density / instanton background relevant to the ABJ structure;

- $\pi_1(U(1)) = \mathbb{Z}$ — integer winding in a supplied compact group; a physical electric-charge lattice additionally requires the gauge and matter map (§49.14).

All three are Level A: group theory.

A2. Conditional chirality bookkeeping. $Spin(4) \simeq SU(2)_L \times SU(2)_R$ supplies two algebraic factors, but the carrier-space orientation director does not dynamically select an internal weak bundle or the observed chiral representations. Labelling one factor L is conventional; the physical weak/chiral owner remains Open (§9.3 and §9.4).

Layer B: Inherited anomaly consistency of a supplied reconstructible chiral gauge EFT

B1. Osterwalder–Schrader reconstruction enables standard one-loop analysis. Once the Osterwalder–Schrader reconstruction closes for the local chiral gauge EFT (prerequisite inherited from §9.4, Layer B), the standard Adler–Bell–Jackiw one-loop analysis applies directly to the reconstructed Lorentzian theory [76, 71, 72]. ECT inherits the framework; it does not formulate a new ECT-specific anomaly theorem. This is a Level B inheritance, conditional on OS closure.

B2. Anomaly consistency conditions in the unified left-handed Weyl convention. We adopt the unified left-handed Weyl convention throughout this subsection. Fermions are expressed as left-handed Weyl fields Q, u^c, d^c, L, e^c (and optionally ν^c). The Standard-Model hypercharge assignments in this convention are:

$$Y_Q = +\frac{1}{6}, \quad Y_{u^c} = -\frac{2}{3}, \quad Y_{d^c} = +\frac{1}{3}, \quad Y_L = -\frac{1}{2}, \quad Y_{e^c} = +1, \quad Y_{\nu^c} = 0. \quad (9.16)$$

The four local anomaly cancellation conditions are then:

$$[SU(3)]^2 U(1)_Y : \quad 2Y_Q + Y_{u^c} + Y_{d^c} = 0, \quad (9.17)$$

$$[SU(2)]^2 U(1)_Y : \quad 3Y_Q + Y_L = 0, \quad (9.18)$$

$$[U(1)_Y]^3 : \quad 6Y_Q^3 + 3Y_{u^c}^3 + 3Y_{d^c}^3 + 2Y_L^3 + Y_{e^c}^3 (+Y_{\nu^c}^3) = 0, \quad (9.19)$$

$$U(1)_Y\text{-grav}^2 : \quad 6Y_Q + 3Y_{u^c} + 3Y_{d^c} + 2Y_L + Y_{e^c} (+Y_{\nu^c}) = 0. \quad (9.20)$$

Scope labels: all four equations are standard consistency conditions only after a local chiral gauge bundle and its representations are supplied. Equation (9.18) then tests the supplied electroweak representations; Eq. (9.20) additionally needs the physical metric/gravity completion; Eqs. (9.17) and (9.19) need the full colour and hypercharge representation content. None is yet a constraint on a gauge sector derived from P1–P6.

B2'. Explicit verification for the inherited assignments. Substituting the assignments (9.16) into the four conditions (9.17)–(9.20) gives, per generation,

$$[SU(3)]^2 U(1)_Y : \quad 2 \cdot \frac{1}{6} - \frac{2}{3} + \frac{1}{3} = 0,$$

$$[SU(2)]^2 U(1)_Y : \quad 3 \cdot \frac{1}{6} - \frac{1}{2} = 0,$$

$$U(1)_Y\text{-grav}^2 : \quad 6 \cdot \frac{1}{6} - 3 \cdot \frac{2}{3} + 3 \cdot \frac{1}{3} - 2 \cdot \frac{1}{2} + 1 (+0) = 1 - 2 + 1 - 1 + 1 = 0,$$

$$U(1)_Y^3 : \quad 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1^3 (+0) = \frac{1}{36} - \frac{32}{36} + \frac{4}{36} - \frac{9}{36} + \frac{36}{36} = 0,$$

where the parenthetical $(+0)$ marks the optional hypercharge-neutral ν^c contribution. Witten $SU(2)$ anomaly: per generation there are 3 colour copies of Q_L plus one L_L , hence $4N_{\text{gen}}$ left-handed $SU(2)$ doublets, which is even. This is Level A arithmetic/algebra inside supplied inherited charge assignments; it is not an ECT derivation of those assignments or the physical gauge sector; the scope classification of B3 below (which conditions are *meaningful* in current ECT) is unchanged by it.

B3. Hard disclaimer. Current P1–P6 supply no physical local $SU(2)_L \times U(1)_Y$ bundle, chiral matter representations or charge assignments. Consequently none of Eqs. (9.17)–(9.20) is presently an ECT-derived anomaly condition. Once a named local completion is supplied, $[SU(2)]^2 U(1)_Y$ tests its electroweak representations; the mixed gravitational condition also awaits the physical metric/tensor owner; and the colour and cubic-hypercharge conditions await the full matter completion (OP-GUT1, OP-GUT2, OP-GUT3b). F4 therefore records an inherited verification protocol, not anomaly closure of an existing ECT gauge sector.

B4. The preferred-direction operator $\mathcal{L}_5$ is anomaly-transparent. The preferred-direction operator

$$\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^\mu n_\mu \Psi, \quad (9.21)$$

re-classified in §9.4, Layer C as a CPT-odd, CP-odd, C-odd, P-even, T-even branch operator, does *not* alter the chiral representation content of the fermions, their gauge charges, or the local gauge-anomaly coefficients computed from that chiral representation data. It is a background-vector bilinear insertion in the ordered-branch EFT, not a change in gauge representation data. Therefore it can affect branch-level phenomenology, dispersion relations, and discrete-symmetry properties, but it does not modify the local gauge-anomaly consistency constraints of the supplied gauge completion at the level of representation data.

Lemma (anomaly transparency at the local anomaly-coefficient level). In a supplied reconstructible ordered-branch chiral gauge EFT, the deformation $\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^\mu n_\mu \Psi$ does not change the local gauge-anomaly coefficients computed from the chiral representation data, and does not change the Witten $SU(2)$ doublet-counting class. *Proof (representation-data).* Anomaly coefficients are determined by the chiral fermion representations and charges. Since $\mathcal{L}_5$ changes neither the gauge representation nor the chirality assignments — it is the coupling of the background vector $B_\mu = \mu_5 n_\mu$ to the gauge-singlet vector current $\bar{\Psi} \gamma^\mu \Psi$ — the anomaly coefficients are unchanged; the $SU(2)$ doublet count is likewise unchanged. *Secondary check (Fujikawa).* Equivalently, in a Fujikawa description this vector background shifts the Dirac operator without changing the chiral representation trace that defines the anomaly coefficient. *Scope.* Local gauge-anomaly coefficients and the Witten count only; flavour-diagonal $\mathcal{L}_5$ as defined; possible mixed gravitational/global or 't Hooft-type bookkeeping involving the background direction is outside the lemma; the ECT application inherits the B1 OS-closure conditionality. Status: candidate Level A at the stated level.

The anomaly coefficients are determined by the chiral representation content and gauge charges, not by background-vector bilinear insertions that leave this representation data unchanged. In this sense the anomaly structure is insensitive to the preferred-direction deformation $\mathcal{L}_5$: the operator modifies branch-level phenomenology, but not the anomaly coefficients entering the standard consistency conditions.

In this sense the F3 reclassification of $\mathcal{L}_5$ has a concrete F4 payoff: despite being CPT-odd at the branch level, $\mathcal{L}_5$ is *anomaly-transparent* with respect to the standard gauge-anomaly consistency conditions. This is a clean structural bridge from §9.4 to the present subsection. Flavour structure is similarly anomaly-decoupled, as discussed in §9.7, Layer B5.

B5. Witten global anomaly. The standard Witten requirement is that the number of left-handed $SU(2)$ doublets be even. Per SM-like generation there are three coloured quark doublets Q plus one lepton doublet L , giving four doublets per generation; for an SM-like generation structure this gives $4N_{\text{gen}}$, which is even for any integer $N_{\text{gen}} \geq 1$. In ECT this is inherited as a consistency test once the fermion spectrum is fixed; it is automatically satisfied for an SM-like generation structure, but is not yet a first-principles derivation of that structure (OP-GUT5).

B6. Connection to the instanton sector. The triangle anomaly is structurally related to instanton-induced chiral charge violation through the Chern–Pontryagin density $\text{tr} F \wedge F$. In this sense the anomaly architecture discussed here is naturally connected to the instanton sector already explored elsewhere in the manuscript. In particular, the standard relation between the ABJ anomaly and nonperturbative $SU(2)$

topological transitions carries over, on the same conditional basis as in B1 above, to the reconstructible ordered-branch sector of ECT.

Layer C: Full Standard-Model completion and the emergent-graviton aspect

C1. Specific hypercharge derivation remains open. The specific values in (9.16) remain Standard-Model input; their derivation from condensate dynamics is recorded as OP-GUT3b and remains Open.

C2. Scope of full Standard-Model anomaly closure. The $[SU(3)]^2 U(1)_Y$ condition (9.17) is directly conditional on the colour-sector completion (OP-GUT1). The $[U(1)_Y]^3$ condition (9.19) depends on the full chiral matter content and the observed hypercharge assignments, and therefore remains conditional on the combined completion of the matter-representation problem (OP-GUT2) and the observed charge-assignment problem (OP-GUT3b). Thus the full Standard-Model anomaly closure remains completion-dependent in current ECT.

C3. Mixed $U(1)_Y$ -gravitational anomaly in the emergent-graviton setting. In ordinary 3+1-dimensional chiral gauge theory the relevant local gravitational consistency condition is the mixed $U(1)_Y$ -gravitational anomaly constraint

$$\sum_i Y_i = 0 \quad (9.22)$$

over left-handed Weyl fermions. A local pure gravitational anomaly is not present in the ordinary 4D setting. The ECT-specific question is therefore not whether a new 4D pure gravitational anomaly appears, but how the standard mixed $U(1)_Y$ -gravitational consistency condition should be interpreted when the graviton itself is emergent from condensate collective modes rather than fundamental at the pre-emergent level.

Since the ordered branch distinguishes a preferred direction n^A , the detailed ECT interpretation of gravitational consistency conditions should eventually be formulated in the language appropriate to the emergent condensate background rather than assumed from the outset. However, the present work does not establish any modification of the standard mixed $U(1)_Y$ -gravitational condition. The conservative and technically correct statement is that ECT is expected to reproduce the usual long-wavelength condition (9.22), while its precise pre-emergent interpretation remains open (OP-GUT6c).

C4. 't Hooft matching: outlook. A more ambitious future question is whether exact global symmetries of the ordered-branch pre-SM sector admit a meaningful 't Hooft anomaly-matching analysis with the infrared effective theory. Since the relevant exact global symmetries of the current ECT UV architecture have not yet been fully identified, this lies beyond the scope of the present subsection.

Solution structure of the one-generation anomaly equations

Status: candidate Level A algebraic classification under the inherited one-generation chiral content $\{Q, u^c, d^c, L, e^c\}$ (all left-handed Weyl). This is a classification of the solution set of the consistency conditions, not a derivation of the observed assignment; OP-GUT3b remains Open.

The linear conditions (9.17), (9.18) and (9.20) give $Y_L = -3Y_Q$, $Y_{e^c} = 6Y_Q$, $Y_{u^c} + Y_{d^c} = -2Y_Q$; writing $Y_{u^c} = -Y_Q + \delta$, $Y_{d^c} = -Y_Q - \delta$, the cubic condition (9.19) reduces to

$$18Y_Q(9Y_Q^2 - \delta^2) = 0. \quad (9.23)$$

Apart from the trivial all-zero charge assignment, the solution set, up to overall normalisation, consists of two nontrivial assignments: (a) the *non-degenerate branch* $Y_Q \neq 0$, $\delta = \pm 3Y_Q$, where the two sign choices exchange the labels $u^c \leftrightarrow d^c$ and reproduce the Standard-Model assignment (9.16); (b) the *degenerate branch* $Y_Q = 0$ ($Y_L = Y_{e^c} = 0$, $Y_{u^c} = -Y_{d^c}$), in which hypercharge acts only on u^c, d^c and the lepton sector is hypercharge-neutral. This two-assignment structure matches the published classification [80, 81], where branch (b) corresponds to the non-SM assignment sometimes called “bizarre” in that literature.

Exclusion of the degenerate branch. Branch (b) is excluded by the inherited Standard-Model charge-assignment requirement for the charged-lepton sector — equivalently by the inherited identification of e^c as the charged-lepton singlet with nonzero hypercharge. Anomaly cancellation alone does not exclude it; this exclusion is recorded explicitly as inherited input. The corresponding uniqueness statement in §42 is scoped accordingly.

The ν^c freedom and the $B-L$ flat direction. Starting from the non-degenerate SM-like branch, if a singlet ν^c with *free* hypercharge is admitted, the shifted assignment $Y \rightarrow Y + t(B-L)$ is anomaly-free for all t . Here $B-L$ is understood in the left-handed conjugate convention: $Q_L : +\frac{1}{3}, u^c, d^c : -\frac{1}{3}, L_L : -1, e^c, \nu^c : +1$. Hence the anomaly system alone no longer fixes hypercharge. If ν^c is admitted but constrained from the outset to be a hypercharge-neutral singlet, $Y_{\nu^c} = 0$, this particular $B-L$ flat direction is removed.

What compactness and the supplied centre interlock add. Compactness alone gives integer characters only after a local bundle and representation are supplied. In the small- Y convention, the full colour-weak interlock of §49.14 uses $q_f = 6Y_f$ and $2t_f + 3d_f + q_f \equiv 0 \pmod{6}$; its $\mathbb{Z}_2$ subcondition is $d_f + q_f \equiv 0 \pmod{2}$. Applied to the $B-L$ flat direction this discretises but does not eliminate it. The observed assignment and the physical global quotient therefore remain OP-GUT3b/Open inputs.

The N_c -dependent classification and what it does not select. *Status: candidate Level A algebraic classification under the stated inherited content at general N_c ; a constraint statement, not a derivation of the colour rank.* The classification extends to a general number of colours $N_c \geq 2$, with $Q \in (\mathbf{N}_c, \mathbf{2})_{Y_Q}$, $u^c, d^c \in (\bar{\mathbf{N}}_c, \mathbf{1})$, $L \in (\mathbf{1}, \mathbf{2})_{Y_L}$, $e^c \in (\mathbf{1}, \mathbf{1})_{Y_e}$. The $[SU(N_c)]^3$ condition is satisfied identically (vector-like colour content); the linear conditions give $Y_L = -N_c Y_Q$, $Y_{e^c} = 2N_c Y_Q$, $Y_{u^c} + Y_{d^c} = -2Y_Q$; and with $Y_{u^c} = -Y_Q + \delta$, $Y_{d^c} = -Y_Q - \delta$ the cubic condition reduces to

$$6N_c Y_Q (N_c^2 Y_Q^2 - \delta^2) = 0, \quad (9.24)$$

recovering (9.23) at $N_c = 3$. Apart from the all-zero and degenerate branches (structure as above), the local anomaly equations admit an SM-like hypercharge family for *every* $N_c \geq 2$: $\{Y_{u^c}, Y_{d^c}\} = \{-(N_c+1)Y_Q, (N_c-1)Y_Q\}$. For $N_c \geq 2$, the electroweak Witten condition ($N_c + 1$ left-handed doublets per generation) imposes N_c *odd* — not $N_c = 3$. With the charged-lepton normalisation $Q_e = -1$ the quark charges become N_c -dependent,

$$Q_u = \frac{N_c + 1}{2N_c}, \quad Q_d = -\frac{N_c - 1}{2N_c}, \quad (9.25)$$

an N_c -quark colour-singlet baryon with k up-type quarks has integer charge $Q_B = k - \frac{N_c-1}{2}$ for any odd N_c (a unit-charge proton analogue at $k = \frac{N_c+1}{2}$), and the anomalous π^0 coefficient is N_c -independent, $N_c(Q_u^2 - Q_d^2) = 1$ [82, 83]. Hence charge integrality, atom neutrality, and the two-flavour pion-photon sector are all blind to N_c ; phenomenological information sensitive to N_c exists in the $N_f \geq 3$ Wess–Zumino–Witten (WZW)/flavour-anomaly sector (for example through explicit N_c -dependent terms in $\eta \rightarrow \pi^+ \pi^- \gamma$) and in the R -ratio class [82]. *No-overclaim statement.* Anomaly cancellation (including the Witten condition) constrains the N_c -dependent charge assignments — N_c odd, hypercharge pattern fixed up to normalisation and the $u^c \leftrightarrow d^c$ relabelling, degenerate branch as in the $N_c = 3$ analysis — but does not by itself select $N_c = 3$; identifying $N_c = 3$ requires additional input, such as inherited observed-charge assignments, phenomenological information sensitive to N_c (for example in the $N_f \geq 3$ WZW/flavour-anomaly sector or the R -ratio class), or an ECT-native rank-selection mechanism (the zero-mode programme of OP-GUT1, §40).

Anomaly cancellation does not explain the generation count. Anomaly cancellation does not explain the existence of exactly three generations. That question remains part of the broader flavour/generation problem (OP-GUT5) and lies outside the present anomaly-architecture result.

Disentanglement of three notions of “anomaly”. It is useful to keep three logically distinct notions separate: (i) local ABJ triangle anomalies in reconstructible local sectors (Level B, conditional on OS); (ii) global topological Witten anomaly tied to $\pi_4(SU(2))$ (Level A for topology, Level B for numerical check once the fermion spectrum is fixed); (iii) ’t Hooft anomaly matching between UV and IR descriptions, treated here only as a longer-term outlook.

Consistency bridge. The present subsection should therefore be read as a consistency bridge: it records the standard anomaly template that any reconstructible ordered-branch chiral gauge completion must pass, while leaving the physical bundle, representations and full Standard-Model completion explicitly Open.

Table 41: Anomaly audit for a supplied chiral gauge completion: status summary.

Statement	Status	Comment
Integer representations of a supplied compact local $U(1)$	A conditional / Open ECT	Theorem 23.1 assumes the local bundle and matter map (§49.14)
Ordered-vector stabiliser and physical weak chirality	A / Open	$SU(2)_{\text{diag}}$ is exact; local chiral weak sector is not derived (§9.3)
$\pi_4(SU(2)) = \mathbb{Z}_2$ Witten input	A	Group theory
Standard one-loop anomaly analysis in reconstructible local sectors	B	Conditional on OS / local EFT closure
$[SU(2)]^2 U(1)_Y$ condition	A arithmetic / B conditional	Standard check of a supplied local weak/hypercharge bundle and matter data; not an ECT weak-sector derivation
Mixed $U(1)_Y$ -gravitational condition	A arithmetic / B/Open physical	Additionally requires a physical metric/tensor completion
$[SU(3)]^2 U(1)_Y$ and $[U(1)_Y]^3$ conditions	Conditional / Open	Depend on colour-sector and full matter completion (OP-GUT1, OP-GUT2, OP-GUT3b)
Explicit verification of the four conditions for the inherited assignments	A	Arithmetic of inherited input (B2’); verification, not derivation
$\mathcal{L}_5$ does not modify anomaly coefficients	A cand. / B	Representation-data lemma at the local anomaly-coefficient level (B4); ECT application inherits B1 conditionality
Witten anomaly test for a supplied chiral $SU(2)$ sector	A conditional / Open ECT	Standard consistency check once the bundle and fermion spectrum are supplied
Solution structure of the one-generation anomaly system	A cand.	Two nontrivial assignments up to normalisation and $u^c \leftrightarrow d^c$ relabelling; degenerate branch excluded by inherited charged-lepton input

Statement	Status	Comment
N_c -dependent classification at general N_c	A cand.	Local conditions admit an SM-like hypercharge family for every $N_c \geq 2$ (9.24); Witten imposes N_c odd; the colour rank is not selected by anomalies alone
$B-L$ flat direction for a free-hypercharge singlet ν^c	A cand.	Anomaly system alone does not fix Y ; a supplied $\mathbb{Z}_6$ interlock discretises but does not eliminate the residual family
Specific observed hypercharge assignments from condensate dynamics	Open	OP-GUT3b
ECT interpretation of mixed $U(1)_Y$ -gravitational condition with emergent graviton	Open	Extension OP-GUT6c
$N_{\text{gen}} = 3$ from anomaly cancellation	No	Not constrained by anomalies alone (OP-GUT5)

Its whole-framework consistency status is recorded in the global Level-4 self-consistency checklist (§19.3).

Hypercharge normalisation: distinct notions and exact centre interlock. Four notions must be kept separate: the rational hypercharge pattern within a generation (OP-GUT3b); the charge-lattice/centre structure (OP-GUT3a); the coupling magnitude (g' , $Z_B^{(0)}$, $\sin^2 \theta_W$); and the normalisation convention. The assignments (9.16) use small hypercharge, so throughout the manuscript

$$Q_{\text{em}}/e = T_3 + Y.$$

For this global-quotient paragraph define the integer compact- $U(1)$ character of multiplet f by $q_f \equiv 6Y_f \in \mathbb{Z}$. This character is not the coupling g' .

Let $t_f \in \mathbb{Z}_3$ be colour triality ($\mathbf{3} \rightarrow 1$, $\bar{\mathbf{3}} \rightarrow -1$, singlet $\rightarrow 0$) and $d_f = 2T_f \pmod{2}$ the weak-isospin parity. The central element

$$z_6 = (\omega_3 \mathbf{1}_3, -\mathbf{1}_2, e^{i\pi/3}), \quad \omega_3 = e^{2\pi i/3},$$

acts trivially exactly when

$$\frac{t_f}{3} + \frac{d_f}{2} + \frac{q_f}{6} \in \mathbb{Z} \iff 2t_f + 3d_f + q_f \equiv 0 \pmod{6}. \quad (9.26)$$

For $(Q, u^c, d^c, L, e^c, \nu^c)$ the tuples (t, d, q) are respectively

$$(1, 1, 1), (-1, 0, -4), (-1, 0, 2), (0, 1, -3), (0, 0, 6), (0, 0, 0),$$

and the six values of $2t + 3d + q$ are $6, -6, 0, 0, 6, 0$. Thus the supplied representations are compatible with the $\mathbb{Z}_6$ quotient. The $\mathbb{Z}_3$ and $\mathbb{Z}_2$ subquotient conditions are respectively

$$2t_f + q_f \equiv 0 \pmod{3}, \quad d_f + q_f \equiv 0 \pmod{2}.$$

Compatibility does not select the physical quotient among $\mathbb{Z}_6/\mathbb{Z}_3/\mathbb{Z}_2/\text{trivial}$ possibilities; that requires the complete field, operator and line-operator content [84].

Since an $SU(2)$ weight satisfies $T_3 \equiv d/2 \pmod{1}$, the same supplied interlock implies the conditional standard correlation

$$Q_{\text{em}}/e \equiv -\frac{t_f}{3} \pmod{1}.$$

This is quotient arithmetic for admitted representations, not an ECT derivation of the observed charges or of the colour module.

Exact residual lattice. For the left-handed assignment $Y'_f = \lambda Y_{\text{SM},f} + c(B-L)_f$ including v^c , the v^c condition first gives $c \in \mathbb{Z}$, while the quark-doublet condition gives $\lambda + 2c \equiv 1 \pmod{6}$; the remaining multiplets then follow. Hence the necessary-and-sufficient family is

$$\boxed{c \in \mathbb{Z}, \quad \lambda + 2c \equiv 1 \pmod{6}}, \quad c = m, \quad \lambda = 1 - 2m + 6n, \quad m, n \in \mathbb{Z}.$$

Consequently $c = 0$ gives $\lambda \equiv 1 \pmod{6}$, while $\lambda = 1$ requires $c \in 3\mathbb{Z}$ (not $6\mathbb{Z}$). The observed point $(1, 0)$ passes. The point $(-1, 1)$ also passes and relabels $u^c \leftrightarrow d^c$ together with $e^c \leftrightarrow v^c$. By contrast $(-1, 2)$ fails: its three quark interlock residues are $2, 4, 4 \pmod{6}$.

Thus the centre condition discretises but does not eliminate the $B-L$ flat direction. OP-GUT3a is sharpened at the level of standard quotient arithmetic; OP-GUT3b and ECT selection of the observed representative, representations and physical global gauge group remain Open.

P7-restricted matter audit (OP-GUT2): bounded first pass

Status: candidate Level A algebra inside the restricted class for the computed sectors; Level B conditional as ECT application, since the class itself is structurally imposed; OP-GUT2 remains Open.

Definition (P7-restricted admissible matter class). Left-handed Weyl fields valued only in tensor products of the structures already assumed in this framework: the colour module E or E^* ($\mathbf{3}$ or $\bar{\mathbf{3}}$), the weak doublet/singlet structure, and a compact- $U(1)$ charge-lattice label. This is a structural restriction defining a test class, not a derivation of matter representations; the exclusion of higher colour tensors ($\mathbf{6}, \mathbf{8}, \dots$) and higher weak isospin is a minimality/available-module assumption, not a consequence of anomaly cancellation or of condensate dynamics.

Minimality convention. At most one copy of each chiral type (R_c, R_w, Y) ; vector-like pairs are not counted as part of the minimal chiral core; no sterile singlets unless an explicit v^c branch is analysed. This is a classification convention, not a dynamical selection rule.

Baseline SM-pattern sector. For $\mathcal{P}_{\text{SM}} = \{(\mathbf{3}, \mathbf{2})_{Y_Q}, (\bar{\mathbf{3}}, \mathbf{1})_{Y_u}, (\bar{\mathbf{3}}, \mathbf{1})_{Y_d}, (\mathbf{1}, \mathbf{2})_{Y_L}, (\mathbf{1}, \mathbf{1})_{Y_e}\}$ the cubic colour condition closes as $2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 2 - 1 - 1 = 0$, the colour-weighted Witten count is $3 + 1 = 4$ (even), and the remaining hypercharge conditions are exactly the one-generation system whose solution structure is recorded above (9.23). The v^c branches are: (a) no v^c (baseline); (b) hypercharge-neutral v^c , $Y_{v^c} = 0$, leaving all conditions unchanged; (c) free-hypercharge v^c , opening the $B-L$ flat direction recorded there.

Small-sector audit. Vector-like pairs contribute zero to all five local anomaly conditions and an even number to the Witten count, so anomaly conditions do not constrain vector-like additions. Pure-singlet sets with up to four nonzero charges admit no chiral solution: after removing vector-like $(y, -y)$ pairs and neutral singlets, $\sum_i y_i = 0$ and $\sum_i y_i^3 = 0$ force zeros or $\pm$ pairs. With no weak doublets and at most two nonzero coloured weak-singlet multiplets, the conditions force a single vector-like pair (and $(\mathbf{3}, \mathbf{1})_0 \oplus (\bar{\mathbf{3}}, \mathbf{1})_0$ is vector-like under colour already); the colourless doublet sector likewise closes at minimal type-count. With no weak doublets, $[SU(3)]^3$ requires equal numbers of $\mathbf{3}$ and $\bar{\mathbf{3}}$ multiplets, excluding all odd coloured multiplet counts. A four-multiplet coloured weak-singlet sector, however, already gives an explicit non-SM chiral anomaly-free family,

$$\{(\mathbf{3}, \mathbf{1})_a, (\mathbf{3}, \mathbf{1})_{-a}, (\bar{\mathbf{3}}, \mathbf{1})_b, (\bar{\mathbf{3}}, \mathbf{1})_{-b}\}, \quad a, b \neq 0, \quad b \neq \pm a,$$

anomaly-free by direct substitution and containing no gauge-invariant mass bilinear (invariance of $\mathbf{3}_y \otimes \bar{\mathbf{3}}_{y'}$ requires $y + y' = 0$, which is excluded): $U(1)$ -protected, $SU(2)$ -singlet coloured chiral quartets. For example, $a = 1, b = 2$ is an explicit integer-charge witness. Larger chiral patterns remain open; this is a bounded first-pass audit, not a classification of all P7-restricted matter contents.

No-overclaim lemma (with witness). Anomaly cancellation, Witten parity, and lattice quantisation do not by themselves select the Standard-Model matter content: the restricted class demonstrably admits non-SM chiral anomaly-free content (the quartet family above) alongside the SM-pattern solution branches. Within the class, the SM pattern is distinguished by the minimality convention and by carrying the full colour–weak–hypercharge chiral structure, not by anomaly consistency alone; its dynamical derivation remains the open problem OP-GUT2.

9.6 Yukawa couplings and fermion masses

Status: Level C phenomenological embedding. Yukawa couplings and fermion masses are not derived from the ECT condensate action; they are imposed as Standard-Model input. In the canonical electroweak convention $H \equiv \Phi_{\text{ew}}/\sqrt{2}$ and $H^0 = (v_2 + h)/\sqrt{2}$, the mass relation is exactly $m_f = y_f v_2/\sqrt{2}$ inside the supplied EFT. Possible condensate-sector corrections are discussed in Section 8.7.

Once fermionic fields Ψ are admitted as additional spinorial degrees of freedom compatible with the ordered branch (Section 9.1), their gauge-invariant Yukawa operators are written through the canonical doublet H (or $\tilde{H} = i\tau^2 H^*$), for example

$$\mathcal{L}_{\text{Yukawa}}^{\text{eff}} = -y_d \bar{Q}_L H d_R - y_u \bar{Q}_L \tilde{H} u_R - y_e \bar{L}_L H e_R + \text{h.c.} \implies m_f = \frac{y_f v_2}{\sqrt{2}}. \quad (9.27)$$

Here the schematic flavour matrices and gauge contractions are supplied Standard-Model data. The bookkeeping field $\Phi_{\text{ew}} = \sqrt{2}H$ may be used in the equivalent scalar action, but inserting it unchanged into the canonical Yukawa operator would double-count the field normalisation. The factor $1/\sqrt{2}$ is therefore fixed within this manuscript; changing the field convention requires simultaneous redefinition of the Yukawa coefficients and vacuum parameter. This is Level A action algebra inside the supplied EFT. The values and microscopic origin of the Yukawa matrices remain Level C/Open.

Chiral structure of the Yukawa coupling. In the Standard Model, Yukawa interactions couple left- and right-chiral fermion sectors through H or $\tilde{H}$, schematically as $y_f \bar{\psi}_L H \psi_R + \text{h.c.}$ with the representation-appropriate doublet. Within ECT, the distinction between ψ_L and ψ_R has a structural background in the $\text{Spin}(4) \simeq \text{SU}(2)_L \times \text{SU}(2)_R$ decomposition and in the chirality discussion of Section 9.3. This makes a left–right Yukawa bilinear compatible with the ECT chirality framework, but neither the physical electroweak representations nor the numerical Yukawa matrices are derived from condensate dynamics. The Yukawa sector remains an external low-energy embedding.

Yukawa hierarchy: structural route. The observed fermion spectrum is strongly hierarchical. Numerical Yukawa values and ratios depend on a declared renormalisation scale, scheme and external source; none is imported here as an ECT result. If the three fermion generations correspond to distinct normalizable fermionic modes ψ_i in the post-transition condensate background (see below), then a supplied localisation model would parametrise the Yukawa matrix by overlap integrals:

$$Y_{ij} \sim \int d^3x \psi_i^\dagger(x) \mathcal{H}(x) \psi_j(x), \quad (9.28)$$

where $\mathcal{H}(x)$ is the effective Higgs/condensate coupling profile. If the modes are quasi-localised with characteristic inverse localisation lengths κ_i , a supplied tail model must also provide an effective separation or penetration length d_i . Its reduced overlap exponent is the dimensionless quantity $\Xi_i^{\text{ov}} := \kappa_i d_i$. More generally a WKB owner may write $\Xi_i^{\text{ov}} = B_i/\Sigma_{\text{ov}}^{\text{act}}$ with a finite positive action unit $\Sigma_{\text{ov}}^{\text{act}}$. The parametrisation $Y_i \sim Y_{0i} e^{-\Xi_i^{\text{ov}}}$ suppresses magnitudes only when $\text{Re} \Xi_i^{\text{ov}} \geq 0$; phases and prefactors remain part of the supplied overlap model. In general, $\ln |Y_a/Y_b| = \ln |Y_{0a}/Y_{0b}| - \text{Re} \Xi_a^{\text{ov}} + \text{Re} \Xi_b^{\text{ov}}$, so the exponent differences are not identifiable from Yukawa ratios without a prefactor model. They equal suppression-exponent differences only under an explicitly supplied common-prefactor-magnitude convention. This identity states only what a fitted overlap model must reproduce; the prefactors, profiles, d_i or B_i , action unit and

their ECT owner are not derived, so this is not an ECT naturalness calculation. A chosen near-rank-1 ansatz may be written $Y \approx Y^{(0)} + \delta Y$ with $\text{rank}(Y^{(0)}) = 1$ and $\|\delta Y\| \ll \|Y^{(0)}\|$. Singular-value perturbation bounds then leave at most one singular value of order $\|Y^{(0)}\|$, while the other two are bounded by $\|\delta Y\|$ and may include zeros. Exponential scale separation follows only after the supplied overlap model additionally establishes, for example, $\|\delta Y\|/\|Y^{(0)}\| = \mathcal{O}(e^{-\text{Re}\Xi})$ with $\text{Re}\Xi \gg 1$.

This route is Level C/Open: the profiles, boundary conditions, physical fermion action, normalisation and computation of Y_{ij} from condensate microphysics remain unsupplied (OP-Yukawa). Details are in Appendix AA. The flavour architecture connecting the Yukawa overlap mechanism to CKM mixing and the seesaw embedding to PMNS is systematised in §9.7.

Three fermion generations: spectral route. The Standard Model contains exactly three generations of quarks and leptons. One conditional model-building route to this number uses the spectral structure of a separately supplied post-transition background. If fermions experience an effective potential from the condensate profile (e.g. of kink/domain-wall type), the number of normalizable bound states is controlled by the dimensionless ratio

$$\nu = \kappa_{\text{PT}} \frac{|y|}{\sqrt{2\lambda_{\text{eff}}}}, \quad \kappa_{\text{PT}} > 0, \quad \lambda_{\text{eff}} > 0, \quad (9.29)$$

where y is an effective Yukawa coupling, λ_{eff} controls the local post-transition profile, and κ_{PT} is a supplied profile/operator matching coefficient in this toy spectral model. It should not be identified automatically with the fundamental bare quartic of the full ECT condensate action. For the conventional attractive Pöschl–Teller operator $H_\nu = -\partial_x^2 - \nu(\nu+1)\text{sech}^2 x$ on $L^2(\mathbb{R}, dx)$, with dimensionless x and self-adjoint domain $H^2(\mathbb{R})$ (equivalently the Friedrichs realisation of the $C_c^\infty(\mathbb{R})$ restriction), the negative discrete levels are indexed by integers $n \geq 0$ with $n < \nu$. Hence, for $\nu > 0$, their number is $N_- = \lceil \nu \rceil$; when ν is an integer, the $n = \nu$ threshold solution is not an additional negative normalisable state. If these negative levels are identified with toy light fermionic modes, three modes require exactly $2 < \nu \leq 3$, hence the conditional wedge

$$\frac{2\sqrt{2\lambda_{\text{eff}}}}{\kappa_{\text{PT}}} < |y| \leq \frac{3\sqrt{2\lambda_{\text{eff}}}}{\kappa_{\text{PT}}}. \quad (9.30)$$

This is exact algebra inside the named toy after κ_{PT} is supplied. It is a wedge, not a universal rectangular Yukawa window, and the physical fermion operator/profile map is unowned. It cannot be used as a probability, selection argument, naturalness measure or derivation of three generations.

A future topological route could use an index only after specifying an elliptic chiral Dirac operator, bundle and connection, matter representation, domain/boundary conditions and admissible background. Under those supplied hypotheses an index theorem may constrain $N_L - N_R$; P1–P6 currently supply none of those owners, so no family multiplicity follows.

Both routes define concrete conditional spectral tasks, but neither the fermion operator nor its background, boundary data and representation map are outputs of P1–P6. The count $N_- = \lceil \nu \rceil$ and the wedge $2 < \nu \leq 3$ are Level A conditional mathematics inside the named self-adjoint toy operator. Matching that operator and its levels to physical fermion modes is Level C/Open, and exactly three physical generations remain Open (OP-generations). Details are in Appendix AA.

Possible condensate-sector corrections. In the energy-scale pair relevant here (Section 5.1), the primary condensate scale $\phi_0 \sim \bar{M}_{\text{Pl}}$ is separated from the electroweak scale $v_2 \approx 246 \text{ GeV}$ by many orders of magnitude. If the two condensate sectors are not perfectly decoupled, higher-dimension operators suppressed by a separately supplied physical Yukawa-completion scale $M_{Y,\text{heavy}} > 0$ could induce corrections to the low-energy Yukawa relations organised as powers of $v_2/M_{Y,\text{heavy}}$. Only under the additional named matching $M_{Y,\text{heavy}} = c_Y \phi_0$, $c_Y > 0$, does this become an expansion in $v_2/(c_Y \phi_0)$. No equality to the orientation or neutrino heavy scales is implied. This possibility is discussed qualitatively in Section 8.7; at the present stage ECT does not predict the size of such effects (Level C).

Connection to the neutrino sector. In the present ECT formulation, neutrino masses are treated through a separate embedded mechanism beyond the minimal charged-fermion Yukawa structure; the seesaw-type discussion is given in Section 8.9.

What is and is not derived here. This subsection imposes the Standard-Model Yukawa structure as external input at Level C, but identifies structural routes toward derivation at Level B/C. The supplied completion provides: (i) standard spinor and chiral representations in which a Yukawa coupling can be written, while ECT’s physical spinor and weak-bundle owners remain Open (Sections 9.1–9.3); (ii) a spectral route to the generation number through fermionic bound-state counting in the condensate background (§9.6); (iii) an overlap/tunnelling mechanism for hierarchical Yukawa couplings (§9.6). The numerical values of y_f and the exact generation number remain open calculational targets (Appendix AA).

Status summary.

Statement	Status	Comment
Mass formula $m_f = y_f v_2 / \sqrt{2}$	C	SM embedding; y_f not derived
Chiral structure of a supplied Yukawa coupling	C/Open	Standard-model input; ECT representation content, bundle and couplings are not derived
Yukawa hierarchy	B/C	Structural route via overlap/tunnelling (§9.6); explicit computation open (OP-Yukawa)
Three fermion generations	C/Open	Supplied spectral toy via bound-state counting (§9.6); strict derivation open (OP-generations)
Condensate-sector corrections to y_f	C	Possible in principle; size not predicted
Neutrino mass mechanism	C	Seesaw-type embedding; Section 8.9

9.7 Flavour architecture of the emergent matter sector: CKM and PMNS in ECT

Status: systematisation, not derivation. Sections 9.4 and 9.5 established the discrete-symmetry and anomaly bookkeeping for a supplied chiral gauge completion. With three fermion generations and the Yukawa structure taken as external input (§9.6, OP-Yukawa, OP-generations), the natural next question is whether the observed flavour structure—the CKM quark-mixing matrix and the PMNS lepton-mixing matrix—admits any ECT-native framing. This subsection is a consistency architecture, not a first-principles derivation. Layer A records group-theoretic inputs; Layer B organises structural routes already present in the preprint; Layer C identifies open targets and concluding remarks. The three concrete derivation targets (CKM hierarchy, PMNS angles and CP phases, absolute neutrino spectrum) all fall under the existing flavour open-programme labels OP-Yukawa and OP-generations, with broader unification-level bookkeeping recorded under OP-GUT5 in Table 115. It should therefore be read as a unifying flavour bridge across the already existing Yukawa and neutrino subsections, not as a replacement for either of them.

Transition from the anomaly subsection. The previous subsection (§9.5) showed the standard anomaly-cancellation framework that a supplied electroweak chiral completion must satisfy, with the preferred-direction operator $\mathcal{L}_5$ being transparent at the representation-data level. A natural next question is how the off-diagonal flavour structure—the CKM and PMNS mixing matrices, the associated

CP phases, and the absolute neutrino spectrum—relates to the already-established Yukawa-overlap mechanism of §9.6 and the seesaw embedding of §8.9.

Layer A: Group-theoretic and structural inputs

A1. Flavour-group bookkeeping. In an SM-like three-generation chiral matter sector without explicit sterile-neutrino fields, the maximal flavour symmetry of the kinetic terms is

$$G_F^{\text{SM}} = U(3)_Q \times U(3)_{u^c} \times U(3)_{d^c} \times U(3)_L \times U(3)_{e^c}. \quad (9.31)$$

If a heavy sterile-neutrino sector is supplied in a seesaw completion, this is enlarged by an additional factor $U(3)_N$ before the Majorana mass term is introduced; the Majorana sector then breaks part of this symmetry further. In the present subsection this distinction is used only as group-theoretic bookkeeping; no new flavour symmetry is derived from the condensate microdynamics.

A2. Parameter counting. For three generations the physical-parameter count is the standard one:

- CKM: 3 angles + 1 CP phase = 4 parameters;
- PMNS (Dirac case): 3 angles + 1 Dirac CP phase = 4 parameters;
- PMNS (Majorana case, conditional on a supplied Majorana/seesaw completion in §8.9): + 2 Majorana phases = 6 parameters.

A3. Three generations as external input. The number of fermion generations $N_{\text{gen}} = 3$ remains external input at the present stage (OP-generations; §9.6, §9.6), addressed via the spectral route of the Yukawa subsection rather than within the flavour-architecture discussion presented here.

Layer B: Structural routes already present, now unified

B1. CKM as basis misalignment. The CKM matrix

$$V_{\text{CKM}} = U_u^\dagger U_d, \quad (9.32)$$

arising from the misalignment between up-type and down-type Yukawa diagonalising rotations, is a standard feature of chiral gauge theories with Yukawa couplings. This is a Level A structural input; its specific numerical entries are not an ECT output.

B2. Supplied overlap ansatz as an Open model-building route. In a separately supplied overlap/localisation model, chosen generation profiles can produce small off-diagonal matrix elements and therefore can be fitted to a hierarchical quark-mixing matrix. This generic flexibility is not an ECT derivation or explanation of the Wolfenstein-like hierarchy

$$|V_{us}| \gg |V_{cb}| \gg |V_{ub}|, \quad (9.33)$$

with V_{CKM} close to the identity. The hierarchy and representative elements are external global-fit results with their declared three-generation/unitarity assumptions [85]. The physical profiles, fermion action, normalised Yukawa operators, renormalisation scale/scheme and generation owner are missing. Thus even qualitative ECT ownership remains Open; the specific Wolfenstein parameters are not derived.

B3. External “CKM small / PMNS large” pattern and Open route. The observed difference between quark and lepton mixing may be represented inside a supplied overlap-plus-seesaw model by choosing different charged- and neutral-lepton matrices. This is a generic model-building possibility, not a consequence of ECT: neither the overlap profiles nor the seesaw/Weinberg operator and flavour owner is derived. All mixing angles, phases and the qualitative pattern remain Open under OP-Yukawa/OP-generations.

B4. Standard location of CP phases in supplied flavour matrices. In any supplied complex Yukawa and Majorana/seesaw completion, physical CP phases reside in the corresponding effective flavour matrices after unphysical rephasings are removed. This is standard flavour bookkeeping, not an ECT result. The current framework derives neither those matrices nor the observed values of δ_{CKM} , δ_{PMNS} or Majorana phases; their ECT owner remains Open.

B5. Flavour structure is anomaly-decoupled. The flavour structure (Y_u, Y_d, Y_e, Y_ν matrices) does not enter the anomaly coefficients of the chiral gauge sector; those coefficients depend only on the chiral representation content and the gauge charges (§9.5, Layer B4). Flavour is therefore absent from those anomaly coefficients once the representations and charges are supplied, just as the displayed $\mathcal{L}_5$ insertion is anomaly-transparent. This is standard conditional anomaly algebra, not a derived flavour mechanism.

B6. Near-rank-one matrices do not by themselves imply alignment. A leading rank-one form $Y \approx Y^{(0)} + \delta Y$ does not ensure that the up- and down-sector diagonalising rotations are nearly aligned: two rank-one matrices can point in different flavour directions, and their null subspaces admit arbitrary rotations before the perturbations are specified. Relative profiles and normalised perturbations must be derived before CKM or FCNC suppression can be inferred. The standard GIM mechanism remains external to the missing ECT flavour completion.

B7. PMNS/MSW remains an external target. Because no physical ECT neutrino propagation correction has been derived, the standard leading-order PMNS/MSW Hamiltonian remains an external target rather than an ECT stability result. Whether a future flavour completion preserves or modifies it is Open.

Layer C: Open targets and concluding remarks

C1. Open programme under existing umbrella labels. No new formal OP-IDs are introduced by F8. The three concrete derivation targets recorded here fall under the existing flavour open-programme labels:

- CKM mixing angles and Wolfenstein hierarchy (OP-Yukawa);
- PMNS angles and Dirac/Majorana CP phases (OP-Yukawa, with the neutrino-sector discussion in §8.9);
- absolute neutrino spectrum and ordering (OP-Yukawa / neutrino programme).

Broader unification-level bookkeeping is recorded under OP-GUT5 in Table 115, while the detailed local flavour bookkeeping remains under OP-Yukawa and OP-generations in the fermionic-sector discussion.

C2. Absolute neutrino scale Open. A heavy scale can be inserted into a supplied seesaw embedding, but the ECT action does not select it or pin the light-neutrino spectrum to it (§8.9). The absolute scale, internal hierarchy, ordering, phase structure, sterile content and mass operator all remain Open. No scale-setting ECT input or partial derivation of the spectrum is claimed.

C3. Baryogenesis connection. The flavour architecture supplies only a bookkeeping location for possible CP-odd coefficients and complex flavour phases. A nonzero physical coefficient for $\mathcal{L}_5$, a baryon-number current and violating vertex, a real-time transition state, and transport/washout dynamics are not derived. Consequently no numerical leptogenesis benchmark is retained as part of the ECT argument, and F8 predicts neither the sign nor the magnitude of the observed baryon asymmetry.

C4. Strong CP problem: longer-term outlook. A longer-term question is how the flavour sector, the instanton/ θ -sector, and the open colour completion combine in the ECT framing of the strong CP problem; this remains outside the scope of the present subsection.

Disentanglement of three flavour-related notions. It is useful to keep three notions separate:

- (i) flavour mixing itself (CKM/PMNS matrices as basis misalignment);
- (ii) CP-violating flavour phases in the effective quark/lepton mass structures;
- (iii) baryogenesis, for which the physical current, violating vertex, CP-odd coefficient, state and transport system remain Open (§16.3).

The manuscript provides only a candidate bookkeeping location for these distinct objects. Their joint physical realisation and consistency are not derived.

Table 43: Flavour architecture of the emergent matter sector: status summary.

Statement	Status	Comment
Standard flavour-group bookkeeping for three-generation matter sectors	A	Optional sterile-sector extension for seesaw
CKM as basis misalignment between up- and down-sector diagonalisation	A	Standard field-theory structure
Supplied overlap model can be fitted to hierarchical CKM mixing	C/Open	Profiles and physical fermion action not derived
A supplied overlap-plus-seesaw model can parametrise “CKM small / PMNS large”	C/Open	Neither sector is ECT-derived
Physical CP phases reside in supplied complex Yukawa / seesaw matrices	A standard bookkeeping / Open ECT	Matrices and ECT owner not derived
Flavour structure is anomaly-decoupled	B	Extends F4
Near-rank-one Yukawa matrices do not by themselves imply near-alignment	A counterexample / Open ECT	Relative profiles and perturbations are required before CKM or FCNC claims
Standard PMNS/MSW Hamiltonian supplied as target phenomenology	C/Open	Mass matrices, mixing and physical ECT correction un-derived
Exact CKM/PMNS parameters, CP phases, and neutrino spectrum from condensate dynamics	Open	Under OP-Yukawa / OP-generations
$N_{\text{gen}} = 3$ from ECT	Open	Outside F8; OP-generations

The resulting flavour-architecture status is recorded in the global Level-4 self-consistency checklist (§19.3).

9.8 ECT-specific fermion–condensate coupling

Status: conditional EFT bridge; physical channel Open. The preferred direction n_A is supplied by the P4 ordered branch and is not generated dynamically by P1–P3. Within a parity-even, zeroth-derivative fermionic EFT, the operator $n_A \bar{\Psi} \gamma^A \Psi$ is the leading ECT-specific fermion bilinear (A-math conditional on the restricted EFT basis). Its coefficient is not derived from condensate microphysics. The full operator analysis is given in Section 7.6 and Appendix V; full phenomenology is developed in Section 7.

This subsection gives the fermionic-sector inventory for a candidate preferred-direction operator. Its detailed operator classification, dimensional analysis, Lorentzian reduction, and observability caveats are developed in Section 7.6 and Appendix V. Its detailed phenomenology, observational constraints, and

falsification tests are developed separately in Section 7.

The P4 chart supplies a preferred condensate direction

$$n_A \equiv \frac{\partial_A \Phi}{|\partial \Phi|}, \quad (9.34)$$

which is a kinematic orientational variable of the supplied ordered branch. Within the parity-even, zeroth-derivative low-energy fermionic EFT, the leading ECT-specific deformation of the Dirac operator is

$$\Delta \mathcal{L} = \mu_5 \bar{\Psi} \gamma^A n_A \Psi, \quad (9.35)$$

where μ_5 has mass dimension one.

The coefficient μ_5 is an independent dimension-one EFT coefficient. For $m_f \neq 0$ on the supplied $\phi_0 > 0$ branch, define the dimensionless ratio $\beta_{5f} := \mu_{5f}/m_f$. A named optional additional hypothesis H_{5m} assumes that this coefficient is generated by the same fermion-mass spurion, is analytic in $x_f := m_f/\phi_0$ near $x_f = 0$, and has a vanishing constant term. Its leading-order C/Open truncation is

$$\beta_{5f} = \zeta_{5f} \frac{m_f}{\phi_0}, \quad \mu_{5f} = \beta_{5f} m_f = \zeta_{5f} \frac{m_f^2}{\phi_0}, \quad (9.36)$$

where the dimensionless $\zeta_{5f} \in \mathbb{R}$ are independent matching inputs and may vanish; their reality is the diagonal Hermiticity condition. The vector current itself is chirally even, so neither the mass-spurion owner, analyticity nor the vanishing constant term follows from chiral symmetry or from P1–P7. Higher analytic powers of x_f are allowed. Thus H_{5m} is a physically motivated ansatz, not a dimensional theorem or an ECT prediction. This is a diagonal ansatz in a declared fermion mass-eigenstate basis at a separately supplied fixed scale $\mu_{\text{match}} > 0$; off-diagonal flavour coefficients require a separate Hermitian matrix owner and are not represented by f . With one unrestricted diagonal coefficient per nonzero-mass species this form is NOT IDENTIFIABLE: any such diagonal coefficient set can be reproduced by choosing $\zeta_{5f} = \mu_{5f} \phi_0 / m_f^2$. It becomes a cross-species scaling hypothesis only after a future completion supplies a nontrivial restriction or correlation among the ζ_{5f} . Dimensional analysis alone selects neither such a correlation nor a universal coefficient. The operator remains only an allowed preferred-direction class in a supplied fermion completion; the ansatz is not evidence for residual Lorentz violation and by itself implies no fifth force, spin precession or equivalence-principle violation.

Its main structural features are:

- if $\zeta_{5f} \neq 0$ and the resulting operator survives the relevant field redefinitions and relative-observable reduction, it parametrises preferred-direction corrections through the condensate direction n_A ;
- it is absent in the minimal Lorentz-invariant Standard Model fermion basis and represents a candidate preferred-medium deformation;
- its coefficient, suppression scale, sign and any correlation among species are Open, and every coefficient may vanish.

As discussed in Section 9.3, this vector-current operator does not select a weak chirality. The preferred direction supplies its spacetime tensor, while a chiral gauge projector would require an independent internal bundle and matter vertex.

What is and is not derived here. What is structurally established is the existence of the preferred direction n_A and, within the stated EFT basis, the admissibility of the operator $n_A \bar{\Psi} \gamma^A \Psi$ as the leading parity-even zeroth-derivative deformation of the fermion sector. What is not derived is the microscopic coefficient of this operator. Accordingly, no numerical coefficient or predictive cross-species correlation is retained. The physical observability of the coupling in the Lorentzian low-energy limit depends on additional issues such as species dependence, background gradients, and environment; these are analysed in Section 7.6.

Status summary.

Statement	Status	Comment
Preferred direction n_A in the ordered branch	P4 input	Ordered-branch postulate
Leading parity-even zeroth-derivative operator $n_A \bar{\Psi} \gamma^A \Psi$	A-math conditional	Allowed within the stated restricted EFT basis
Dimensionful coefficient μ_5	A	Required by mass dimension counting
Optional parametrisation $\mu_{5f} = \zeta_{5f} m_f^2 / \phi_0$	C/Open / non-identifiable	Independent supplied ζ_{5f} , including zero, fit any nonzero-mass species pattern; a testable scaling law requires an additional correlation owner
Physical value, nonredundancy and species reduction	Open	Requires matching and a relative-observable or gradient completion
Relation to weak chirality	Open / negative constraint	The vector current contains no derived P_L projector
Coefficient derived from condensate dynamics	Open	Section 9.10

9.9 Spin-statistics: structural route and open problem

Status: Level A topology conditional on a separately supplied defect-free $\mathbb{R}^3$ point-excitation configuration space / Open physical ECT identification. For two supplied identical unlabeled points, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$. This does not derive the physical P4 spatial slice, particle-identity map, representation choice, CAR or the spin-statistics theorem. Full quantum-sector development: Section 24.4.

Why spin-statistics enters the fermionic section. Sections 9.1–9.2 established that spinorial representations are compatible with the ordered branch and that a Dirac-type equation can be constructed. However, the existence of a spinorial field representation does not by itself fix the quantum exchange law of the corresponding excitations. The question of why half-integer-spin fields are associated with antisymmetric exchange belongs to the broader quantum programme, but it is stated here because it directly concerns the physical interpretation of the fermionic fields introduced in this section.

Two distinct $\mathbb{Z}_2$ structures. It is important to distinguish two different $\mathbb{Z}_2$ facts that arise in different contexts:

- $\pi_1(SO(3)) = \mathbb{Z}_2$ (Section 9.1): this is the topological fact that makes spinorial representations available through the double cover of the connected rotation group;
- $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ (this subsection): this is the topology of exchange classes for two identical excitations in three spatial dimensions.

The first concerns *single-particle representation theory*; the second concerns *many-body exchange topology*. Spinorial transformation law and fermionic exchange law are distinct structures, even though they are linked in relativistic quantum theory. The spin-statistics problem is precisely the problem of linking these two structures.

Exchange topology. Here $\mathcal{M}_2$ denotes the reduced configuration space of two identical excitations in three spatial dimensions, obtained after removing coincidence points and quotienting by particle exchange. Its fundamental group is

$$\pi_1(\mathcal{M}_2) = \mathbb{Z}_2. \quad (9.37)$$

This is a statement about two homotopy classes of exchange loops. If one additionally restricts to one-dimensional unitary representations, the two characters are $+1$ and -1 , conventionally called the bosonic and fermionic exchange sectors. Topology alone neither chooses a character nor constructs a quantum state, CAR, or a spin–statistics link. For $N \geq 3$ identical point excitations the corresponding group is S_N , which also has higher-dimensional representations; whether those sectors are physically equivalent to ordinary statistics with an internal multiplicity requires the additional locality and relativistic-QFT hypotheses.

The genuine three-dimensional conclusion is narrower: in the separately supplied defect-free $\mathbb{R}^3$ point model, the two-dimensional braid group and its continuous Abelian anyonic phases are not the exchange topology of the supplied points. This is not yet a theorem that the physical P4 branch is $3+1$; the physical spatial slice and excitation map remain Open. Effective anyonic quasiparticles in reduced-dimensional media remain possible. Pairing integer-spin tensor sectors with the $+1$ character and half-integer spinor sectors with the -1 character is a conditional compatibility route only after the physical field content, state space and local quantum dynamics are supplied. The fuller quantum programme is given in Section 24.4.

What ECT does not yet derive. ECT does not yet derive:

- (i) the full relativistic spin-statistics theorem;
- (ii) the full operator anti-commutation algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$;
- (iii) an explicit emergent many-excitation sector decomposition with variable excitation number from condensate dynamics (ECT does not require Fock space as a fundamental postulate, but the effective reconstruction remains open).

Within the fermionic sector, these questions belong to the open programme summarised in Section 9.10 (Open; OP-Q11).

Decomposition of OP-Q11 and the Euclidean OS route. For status bookkeeping OP-Q11 is decomposed, analogously to OP3, into: *Q11.1a* — the OS spin–statistics test for a free minimal Euclidean $Spin(4)$ Dirac kernel (Level A mathematical result for the free kernel; conventions verified at page level against the original Osterwalder–Schrader and Lüscher constructions, Appendix AU; below); *Q11.1b* — its application to the ECT spinor quadratic subsector (Level B conditional); *Q11.2* — descent into the ordered branch (Level B conditional); *Q11.3* — CAR/Fock reconstruction (Level B conditional); *Q11.4* — the full interacting fermionic sector (open). ECT identifies a native Euclidean route to the spin–statistics connection: in a free minimal Euclidean $Spin(4)$ Dirac kernel, OS positivity is compatible with the Grassmann/antisymmetric quantisation and excludes the corresponding commuting/symmetric half-integer spinor quantisation under the assumptions stated below. Applying this route to ECT fermions remains conditional on deriving or verifying the ECT spinor quadratic kernel and its reflection-positive structure. The present subsection thereby retains its structural/topological content unchanged; the new material upgrades only the free spinor-kernel statement, not the full fermionic sector.

Proposition Q11.1a: OS spin–statistics test for a free minimal Euclidean $Spin(4)$ Dirac kernel (Level A; mathematical statement). For a local, minimal, first-order Euclidean Dirac kernel transforming in a half-integer representation of $Spin(4) \simeq SU(2)_L \times SU(2)_R$, with the OS reflection data fixed in §23.1 (eqs. 23.4–23.8) and a positive mass gap $m > 0$, the Grassmann Gaussian construction gives a reflection-positive reconstruction in the convention fixed there. Conversely, within the same convention and for the same kernel, the corresponding commuting/symmetric quantisation fails reflection positivity.

This is a convention-fixed free-kernel test; the page-level comparison with the Euclidean Fermi-field OS literature has been completed and is recorded in Appendix AU. The statistics assignment itself is not among the assumptions: the comparison is between the Grassmann/antisymmetric and the commuting/symmetric construction for the *same* quadratic spinor kernel. *Scope restriction.* For the minimal Dirac kernel considered here, the leading local half-integer quadratic operator has the first-order $\gamma_A \partial_A + m$ structure; more general local higher-derivative spinor kernels would require a separate RP analysis and are not covered by Q11.1a. *Proof.* Appendix AU. Sketch: the half-space contour evaluation reduces the graded OS form, at fixed spatial momentum, to spinor-space matrices; for the Grassmann construction both elementary quadratic families yield orthogonal projectors (hence non-negative forms), while for the commuting construction the statistics-dependent reordering sign leaves a matrix with spectrum $\{0, -1\}$ and an explicit negative-norm test function exists. Status: Level A mathematical result for the free minimal Euclidean Dirac kernel; the conventions have been verified at page level against the original Euclidean Fermi-field constructions, as recorded in Appendix AU.

Q11.1b: conditional ECT application (Level B). Q11.1a is a Level A mathematical result for the free minimal Euclidean Dirac kernel (Appendix AU), but Q11.1b is not Level A unless the ECT spinor quadratic kernel itself is obtained from the condensate construction and shown to satisfy the stated OS assumptions. The Dirac form of §9.2 currently enters with Level B inputs, and the established reflection-positivity results cover the free quadratic scalar subsectors. The composite/effective character of the ECT fermionic fields does not invalidate the route, but it precludes claiming a first-principles Level A ECT derivation of fermionic statistics at this stage.

Q11.2: descent to the ordered branch (Level B conditional). Restricting supplied $Spin(4)$ modules to the $Spin(3)$ covering the connected $SO(3)$ stabiliser makes a half-integer representation route algebraically available. Carrying that content into a physical ordered Lorentzian branch is conditional and is not by itself a spin–statistics theorem. The descent additionally requires:

- (i) compatibility of the OS reflection Θ with the ordered-branch time $x^0 = w$;
- (ii) continuity of the half-integer representation content across the breaking pattern $Spin(4) \rightarrow Spin(3)$;
- (iii) a separately supplied physical three-dimensional spatial slice and its declared point-excitation exchange topology, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$, with no effective anyonic sector for fundamental exchange in that declared defect-free point model;
- (iv) for the pointlike primary-branch excitations considered in the spin–statistics statement, no topological-defect, skyrmion, or texture sector changes the exchange class; extended/solitonic sectors are outside Q11.2 and remain separately classified;
- (v) inherited locality/domain-of-dependence and positive-Hilbert-space reconstruction in the ordered branch;
- (vi) the proof is performed in the symmetric Euclidean/OS sector and transported into the ordered branch; it is not derived from the broken $O(3)$ structure alone.

Q11.3: CAR algebra and effective Fock reconstruction (Level B conditional). Given Q11.1b and Q11.2, the OS reconstruction applied to antisymmetric Schwinger functions yields the CAR on the reconstructed one-particle space; in an orthonormal mode basis this becomes $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$. The result is conditional on locality, cluster decomposition, and positive-Hilbert-space reconstruction, and addresses items (ii)–(iii) of the “what ECT does not yet derive” list above at conditional Level B, not as a standalone result.

Status summary.

Statement	Status	Comment
$\pi_1(SO(3)) = \mathbb{Z}_2$: spinorial representations available through the double cover	A	Single-particle representation topology
$\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$: two homotopy classes for supplied point excitations	A conditional	Supplied defect-free $\mathbb{R}^3$ point model; physical spatial/excitation map Open
Characters $+1$ and -1	A representation theory	Only after selecting a one-dimensional unitary representation
Physical bosonic/fermionic assignment	Open ECT	Species representation, state space, CAR and spin-statistics owner are not derived
Continuous Abelian braid phases of the 2D anyon type unavailable for fundamental point-particle exchange	A topology conditional	Does not remove higher-dimensional S_N representations or derive physical statistics
Full spin-statistics theorem	Open (OP-Q11; decomposed Q11.1a–Q11.4)	Euclidean OS route above; developed further in Section 24.4
Q11.1a: OS spin-statistics test, free minimal Euclidean $Spin(4)$ Dirac kernel	A (math., free kernel)	Converse calculation completed; conventions verified at page level against the original OS/Lüscher constructions (Appendix AU); $m > 0$, minimal first-order kernel
Q11.1b: application to the ECT spinor quadratic subsector	B conditional	Requires derivation/verification of the ECT spinor kernel and spinorial reflection positivity; composite/effective fermions prevent Level-A ECT status
Q11.2: descent $Spin(4) \rightarrow Spin(3)$ into the ordered branch	B conditional	Pointlike primary-branch excitations only; conditions (i)–(vi) above
Q11.3: CAR/Fock reconstruction via OS	B conditional	Conditional on Q11.1b + Q11.2
Q11.4: full interacting fermionic sector	Open	Inherits the Level B/open status of interacting reflection positivity
Operator anti-commutation algebra	Open (OP-Q11.3 route identified)	Not yet derived; conditional OS-reconstruction route, see Q11.3 above

9.10 Open programme and status summary for the fermionic sector

Status: this subsection collects the open problems and the joint status summary for the fermionic sector (Sections 9.1–9.9). The fermionic sector is among the most heavily embedded parts of ECT: its content is introduced through structural reconstruction on the ordered branch rather than derived from the bare bosonic condensate.

Open programme. To promote the fermionic sector from structural reconstruction to first-principles derivation, the following programme would need to be completed:

- (i) derive the dynamical origin of spinorial fields from the bare bosonic condensate, including their fully induced vierbein/spin-connection coupling on non-uniform condensate backgrounds (currently: external input; Section 9.1);
- (ii) derive fermionic statistics beyond the structural $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ exchange-topology route (currently: Level B compatibility plus the Euclidean OS route Q11.1a–Q11.4 of Section 9.9 and Appendix AU: the free-kernel test is a Level A mathematical result, the ECT application remains conditional);
- (iii) derive the Yukawa coupling pattern and the fermion mass hierarchy (currently: SM input, Level C; Section 9.6);
- (iv) derive the coefficient of the fermion–condensate coupling beyond the dimensional EFT estimate, equivalently derive μ_5 (or β_5 in the parameterisation $\mu_5 = \beta_5 m_f$) from condensate microdynamics (currently: Open for the physical coefficient; Section 7.6, Appendix V);
- (v) derive a physical local weak bundle, a chiral matter projector and the observed $SU(2)_L$ representations dynamically (currently Open; the vector stabiliser is $SU(2)_{\text{diag}}$ and selects neither chirality; Section 9.3);
- (vi) derive the origin of exactly three fermion generations (currently: no mechanism identified; Section 9.6);
- (vii) derive quantitative atomic consequences of the fermionic reconstruction—in particular the hydrogenic fine structure, the microscopic origin of the fermionic g -factor, and possible tiny orientation-sensitive corrections to atomic energy levels associated with the preferred-direction/fifth-interaction sector (Section 7.6); this belongs to the interface between the fermionic programme and the phenomenological discriminant programme;
- (viii) **OP-CPT1:** close the conditional CPT route of Section 9.4 for the full nonlinear interacting theory, including the Osterwalder–Schrader reconstruction of interacting local sectors and the associated anti-unitary time-reversal operator (currently: Level B for free/quadratic local sectors; depends on OP-OS1 and OP-Loc1);
- (ix) **OP-CPT2:** derive the complete action of charge conjugation on all emergent charged sectors of ECT, including the compact-phase $U(1)$ structure and the full Dirac content beyond the group-theoretic basis of Section 9.4, Layer B;
- (x) **OP-P1:** derive the exactly maximal parity-violation pattern of the Standard-Model weak sector from a completed action and matter representation; the ordered vector alone supplies no chiral selection (Section 9.4, Layer A);
- (xi) **OP-BRANCH1:** investigate the physical realisation of the branch-pair symmetry architecture of Section 9.4, Layer C, as a CPT-symmetric cosmological pair, including the identification of observational discriminators; currently Level C speculation;
- (xii) **OP-Loc1:** establish the locality and cluster decomposition properties required for the Osterwalder–Schrader reconstruction of full nonlinear interacting ECT, on which OP-CPT1 depends; this is an element of the broader low-energy QFT architecture programme (Section 9.4, Layer B).

None of these steps has been completed. The fermionic sector is a structural reconstruction programme, not a completed derivation from the bare scalar condensate.

What would Level A look like? A hypothetical Level A fermionic sector would mean that spinorial excitations emerge as dynamical quasiparticles of the bare bosonic condensate, with their statistics, mass spectrum, and gauge quantum numbers determined by condensate microphysics. An instructive analogy is provided by emergent fermionic quasiparticles in systems such as superfluid $^3\text{He-A}$ [20], although ECT does not currently provide a comparable microscopic mechanism. The structural route developed in this section should be understood as the present limit of what the framework can establish for fermions. A genuine Level A completion would require qualitatively new input beyond what is currently available in ECT.

Connections to the broader ECT programme. Several fermionic open problems are shared with other sectors of the theory:

- the spin-statistics question (ii) connects to the quantum sector (Section 24.4; OP-Q11);
- the fifth-force phenomenology (iv) connects to the phenomenological and macroscopic sectors (Section 7);
- the Open weak-bundle and chiral-projector problem connects to the electroweak completion (Section 7.4);
- the generation puzzle (vi) remains without a connection to any other part of ECT at the present stage.

Joint status summary. The table below gives a condensed overview. Detailed status tables are provided in each subsection.

Subsection	Highest level	Summary
9.1 Spinor structure	A (group theory) / B (physical motivation)	$Spin(4) \simeq SU(2)_L \times SU(2)_R$; spinorial reps available; fermions not derived from bare condensate
9.2 Dirac form	B/Open	Conditional Clifford linearisation on a supplied physical metric/tetrad and spinor bundle; their ECT descent remains Open
9.3 Chirality	A obstruction / Open physical	The ordered-vector stabiliser is $SU(2)_{\text{diag}}$ and selects no weak chirality; the local bundle, projector and representations are Open
9.6 Yukawa / masses	C	SM embedding; hierarchy and three generations open (OP-Yukawa, OP-generations)
9.8 Fermion–condensate coupling	B/Open operator inventory; coefficient Open	$\mu_5 \bar{\Psi} \gamma^A n_A \Psi$ is symmetry-allowed in the restricted basis but its coefficient may vanish; no scale or species law is derived in Section 7
9.9 Spin-statistics	A topology conditional / B compatibility	$\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ for a separately supplied defect-free three-dimensional point-configuration space; this gives two loop classes, while the physical slice, excitation map, representation assignment and full spin–statistics theorem remain conditional/Open (OP-Q11)
<i>Overall</i>		

Subsection	Highest level	Summary
Fermions as collective condensate excitations	Open	Not derived from bare bosonic action
Full first-principles fermionic sector from P1–P6	Open	Structural reconstruction only

Spin discreteness and Stern–Gerlach measurements. The discreteness of spin projections does not arise in ECT from a new dynamical mechanism. It follows, as in standard quantum theory, from the compact representation theory of the rotation group and its spinorial double cover. What the present architecture adds is a candidate route: the P4-supplied nonzero vector has an $O(3)$ stabiliser, and its connected $SO(3)$ subgroup has the double cover $\text{Spin}(3) \simeq SU(2)$. Identifying this subgroup with the physical spatial-rotation group and realising its spinorial representations remain Open. In a Stern–Gerlach-type measurement, the role of the macroscopic field gradient and detection chain is then to amplify one projection channel into a record-bearing Lorentzian outcome. ECT therefore does not alter the mathematical discreteness of spin. It supplies an ordered-branch notation in which the residual rotation group can be tracked conditionally; the physical state and matter map remain Open. Environmental coupling may amplify and stabilise a Stern–Gerlach record, but selection of one realised outcome and its update rule remain Open (OP-Q19).

Overall assessment. At the present stage, the fermionic sector of ECT is best viewed as a coherent structural programme rather than a completed derivation. It organises conditional spinor, Dirac, chirality and candidate-operator routes under shared ordered-branch notation, but does not yet derive the physical fermionic sector or establish an advantage over a supplied Standard-Model embedding. At the same time, none of the decisive microscopic steps—fermion emergence, statistics, Yukawa pattern, or generation structure—has yet been derived from the bare condensate action.

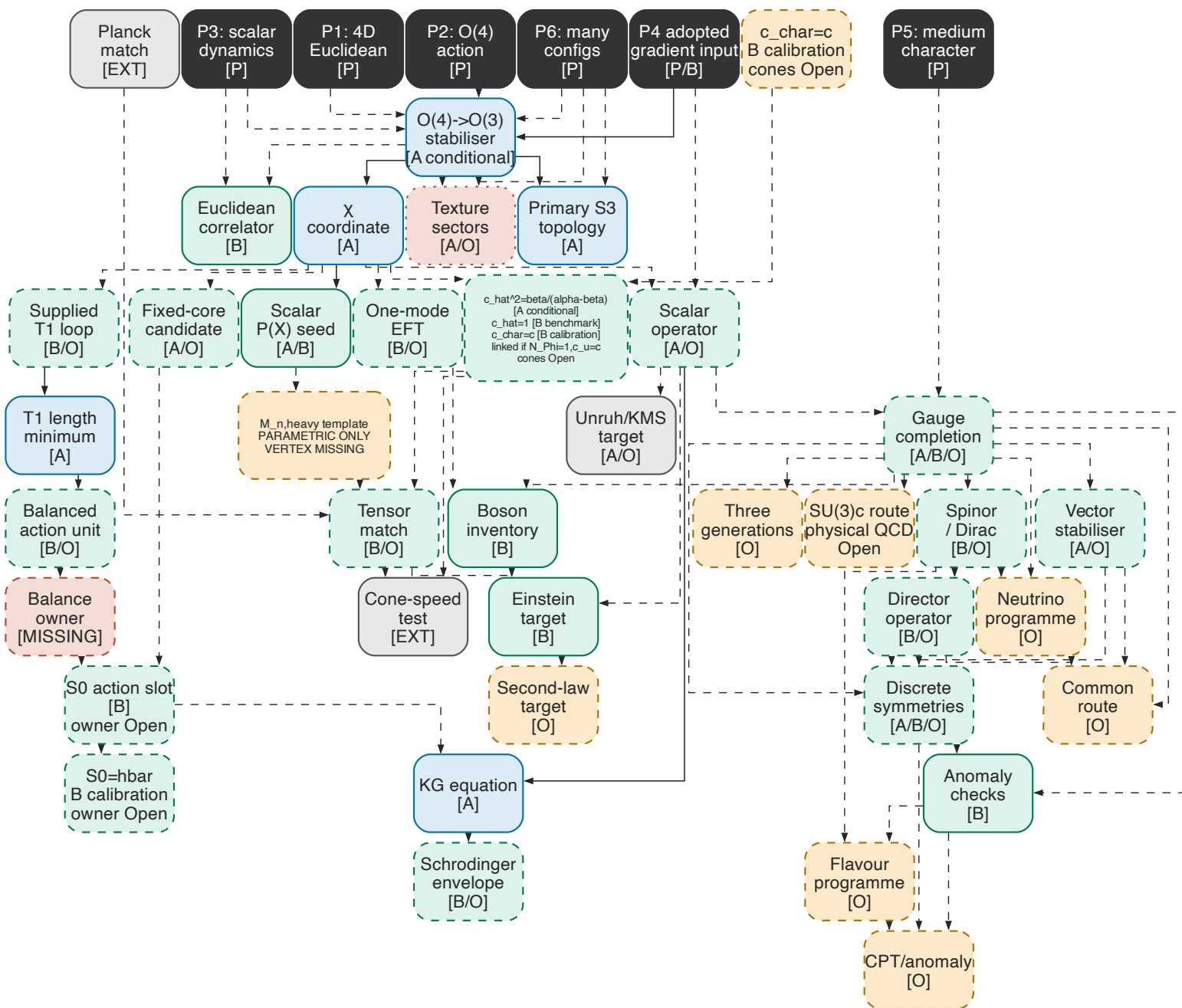
10 Conclusions of Part I: Foundations

This section summarises the foundational construction of ECT established in Part I (Sections 2–9). It collects only the outputs that belong to the foundational branch of the theory, before the macroscopic and quantum-sector developments of Parts II–III. All items are classified by the Level A/B/C/Open discipline maintained throughout the paper. The status and dependency logic of the foundational construction is shown schematically in Figure 4.

Selected status and dependency map. Figure 4 is a connected Part I status and dependency graph. It displays the selected dependency topology on one A4 reader map rather than replacing it by disconnected reader fragments. The diagram separates the causal/ordered-branch structure from the phase/winding structure and includes the conditional T1 route without turning it into a derivation of the operational action unit or of $\hbar$. Solid arrows denote declared dependency or organisation only; dashed arrows denote conditional, matching or Open links. Literal node text is authoritative; the deterministic companion key in the reproducibility package preserves every full source label and edge record.

Selected Part I status and dependency map

Status focus: P4 adopted P/B; O(4)->O(3) stabiliser A conditional; $c_{\text{hat}}^2 = \beta / (\alpha - \beta)$ A conditional; $c_{\text{hat}} = 1$ B coefficient benchmark; $c_{\text{char}} = c$ B calibration, linked when $N_{\text{Phi}} = 1$ and $c_u = c$; cone universality Open; M_n heavy route PARAMETRIC ONLY / VERTEX MISSING.



P postulate/input | A exact in stated model | B conditional/calibration | C fit/toy | O Open | MISSING owner absent. Solid: exact declared step; dashed/dotted: conditional or Open dependency. Literal node status controls.

How to read Figure 4. Starting from P1–P6, DP and the supplied coherent-branch closure BR1, P4 supplies a preferred direction datum but does not fix the kinetic signs. A separately supplied $\alpha > \beta > 0$ selects the retained Lorentzian scalar-PDE regime, while clock, photon, matter and tensor matching remain separate. The compact-phase closure supplies winding sectors; the fixed-core and tension-balanced action-scale candidates remain distinct from the operational S_0 . CDE- Φ and the texture-worldline reduction specify an acceptance criterion and a generic route for what the actual 4D action/state/profile/quotient owner would have to calculate, but the carrier, compact modulus, stability and owner map are Open. Gauge, charge, chirality and fermion completion likewise retain their stated owners. In particular, the allowed homogeneous timelike vector-current operator is spin independent and supplies no source-generated composition force; any spin or Eötvös signal belongs to a separately declared completion.

The calm colourblind-safe palette uses blue for results exact in their declared model and assumptions, green for conditional constructions, pale orange for fits or benchmarks, orange dashed borders for Open routes, neutral gray for external inputs and charcoal for structural inputs. These roles remain distinguishable in grayscale through luminance, border/arrow form and direct status labels; no hatch texture is required. The vector poster is the reader-facing topology authority. Exact node/edge records and source hashes remain in the repository machine-topology package rather than as a printed multi-page edge ledger.

Foundational outcome of Part I. Part I establishes the minimal ECT core: an $O(4)$ -invariant scalar medium, a supplied gradient-ordered broken phase, a two-branch architecture, and a layered low-energy sector decomposition. Within the supplied P4/coefficient EFT it classifies the scalar principal tensor. Compact-phase winding and the $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$ algebraic correspondence are exact only in their declared domains. A physical clock and metric, local gauge bundles and actions, an unbroken photon, chiral $SU(2)_L$ representations, charges and matter assignments remain Open completion owners. The coherent branch rule BR1 supplies conditional action-scale logic, and the preferred-direction matter operator is distinct from the later scalar ansatz S. Other main open fronts include the microphysical origin of the gradient condensate, nonlinear gravity, physical colour and fermionic completion, matching of the scalar-background ansatz, and the separate map from S to the conditional HRC gravity functional G.

The four cross-sector universalities isolated later in this chapter (Section 11.1) should be read as the capstone compression of this foundational logic.

10.1 Core axiomatic framework (Sections 2–6)

The foundational layer of ECT establishes:

1. A connected four-dimensional Euclidean arena with initially equivalent directions and points (P1; ontology layer S1–S3).
2. $O(4)$ -invariant scalar microdynamics with the minimal variational realisation of the Φ -medium (P2–P3, DP).
3. A supplied nonzero-gradient chart whose vector datum has $O(3)$ stabiliser and defines the orientation variable n_A (P4); a stationary physical vacuum and spontaneous selection remain Open.
4. The unique kinetic tensor $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$ of the broken phase (Level A; Theorem 3.2).
5. Lorentzian signature as an ordered-branch admissibility condition: the broken-phase kinetic tensor $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$ admits Euclidean, degenerate, and Lorentzian regimes. It is hyperbolic exactly when $\beta(\beta - \alpha) < 0$; on the retained healthy component this is $\alpha > \beta > 0$ (Level A conditional tensor algebra). This is not an observational selection rule and not an added postulate, but a mathematical admissibility property of the ordered-branch EFT.
6. A two-step cone determination: first, independently supplied coefficients satisfying $\beta(\beta - \alpha) < 0$ classify the scalar principal tensor as hyperbolic and give one scalar characteristic cone (Level A

mathematics in that EFT); it does not select the realised physical branch or a clock map. Second, $\hat{c}_*^2 = \beta/(\alpha - \beta)$ and the Level-B unit-slope coefficient benchmark $\hat{c}_* = 1$ give $\alpha = 2\beta$ inside that EFT. The physical relations to clocks, photons, matter and tensor cones remain separate Open matchings.

7. A scale-and-cone research architecture: the supplied scalar principal tensor fixes a common-length slope $\hat{c}_*$ (Level A in that EFT), while physical c_{char} requires clock matching. A future stationary P4 completion could test a heavy-pole matching ansatz $\kappa_n \sim u_0^2/m_{\text{P4,pole}}^2$; neither that pole nor its vertex is presently derived. The further $\kappa_n \rightarrow M_G^2$ remains an Open tensor bridge. Independently, $\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}$, $\zeta_\phi > 0$ and $\zeta_\phi = O(1)$, is the adopted Level-B scale ansatz; its microscopic explanation remains Open. The coherent branch supplies two distinct loop-scale candidates and later conditional quantum closures reuse an operational action-unit slot S_0 . The owner map, nonzero universal value and cross-sector universality remain Open, while the physical identification $S_0 = \hbar$ is Level B matching (Sections 4.7, 5.1, 5.6).
8. A layered arrow-of-time architecture with six nested levels (A1–A6). Retarded response and positive pairwise noise are retained; an ECT thermodynamic second law is Open because ohmicity and Markovianity alone are insufficient (Section 3.8).
9. Clear sector separation: homogeneous P3 amplitude curvature, Open P4 amplitude-pole programme, and the strict soft-orientation coordinate χ reduced by integrability to one longitudinal scalar coordinate, and the macroscopic ϕ -sector (Section 4.2).
10. A hierarchy programme in which the primary microscopic amplitude ϕ_0 , a possible P4 pole, the orientation stiffness and the coherent action-scale programme are separate matching owners. Their common microscopic descent and physical gravitational normalisation remain Open. The present open problem is no longer “why are gravity and electroweak physics unrelated scales?”, but rather how one condensate generates a secondary low-energy matching scale $v_2 \ll \phi_0$ while retaining the same medium origin (Section 5.1).
11. The coherent branch BR1 with compact phase θ , integer winding sectors, fixed-core representative $\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi\iota_0^{\text{EFT}}$ in the supplied reduced closure, together with the programme of testing whether one common action scale consistently normalises amplitudes, wave dynamics and canonical structure. Born weights require an independently admitted probability measure and are not fixed by S_0 .
12. The supplied real-scalar action, together with its declared symmetries and DP, has conserved translational and rotational Noether currents. Their Lorentzian energy–momentum reading is a separate ordered-branch interpretation; it does not supply the later EFT actions (Section 6.6).
13. A two-branch structure — geometric macroscopic branch and coherent branch — organised as candidate programmes of the same ordered medium. P4 supplies the geometric chart and BR1 separately supplies the coherent closure; their common microscopic descent, back-reaction and equality of characteristic cones remain architectural hypotheses (Section 6).
14. Admissibility of non-uniform ordered backgrounds (P6), treated here as an auxiliary structural admissibility assumption rather than a core dynamical postulate. Its rôle is to make branch-dependent inhomogeneous ordered configurations conceptually available for the later macroscopic ϕ -sector development, not to derive that sector by itself.

Capstone of the foundational layer. Taken together, these results organise a one-medium research architecture, not a completed descent theorem. The supplied scalar branch conditionally fixes one scalar principal cone while physical clocks and cross-sector cones remain Open; (ϕ_0, v_2) is kept distinct from the conditional HRC matching length; and the coherent branch organises an action-scale programme.

The macroscopic and quantum sections then introduce separately owned effective actions, states, and observable maps. Connecting those owners back to the fundamental Euclidean action is an explicit Open programme, so later sections must be read as conditional completions rather than automatic consequences of one variational backbone.

10.2 Gauge-sector architecture, particles, and fermions (Sections 7–9)

Building on the core framework, the sector-development chapters establish:

15. The $U(1)$ phase–gauge correspondence: within the compact phase EFT, the Noether current and current conservation follow; after a local bundle/director are supplied, the two-invariant kinetic basis follows at Level A/B with clearly stated assumptions (Section 7.2). This is the strongest gauge-sector result of Part I.
16. Structural embedding of non-Abelian gauge sectors: $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ provides the cleanest algebraic bridge (Level A). For $SU(3)$, dimension excludes only a direct injective Lie-subalgebra/residual-factor embedding in the displayed minimal algebra (Level A conditional); P7 supplies the adopted positive-definite Hermitian rank-three bundle (Level-B), whose $SU(3)$ frame reduction is exact conditional mathematics. Compact winding and the restricted Lie-algebra no-go are exact under their assumptions; local Abelian/chiral- $SU(2)$ sectors, alternative emergent/nonlinear, reducible/singlet, independent-bundle and defect-kernel colour routes, and physical colour remain Open (Sections 7.3–7.4; see also Section 49.14).
17. A three-category particle and excitation spectrum: (i) kinematically identified or conditionally modelled condensate excitations (χ, σ) , with the physical χ mass and mode projector Open, and a physical graviton only as an Open completion target; (ii) structurally embedded SM fields; (iii) topological and conditional sectors (Section 8.11).
18. A conditional free-spinor reconstruction: the $\text{Spin}(4) \rightarrow \text{Spin}(3)$ representation bookkeeping and a Dirac-form low-energy operator can be written once the corresponding spinor completion is supplied. This does not identify the physical fermion multiplets, a chiral weak bundle, its representations, or the Yukawa sector. In particular, the vector order-parameter stabiliser is $SU(2)_{\text{diag}}$ and selects neither $SU(2)_L$ nor $SU(2)_R$; the physical origin of weak chirality therefore remains Open (Section 9).
19. The standard compact- $SU(2)$ origin of discrete spin projections, together with a conditional ECT route to its relevance: the supplied $O(4) \rightarrow O(3)$ vector chart has connected stabiliser $SO(3)$, whose spinorial double cover is $\text{Spin}(3) \simeq SU(2)$ (Level-A conditional group theory). Identifying this stabiliser with the physical spatial-rotation group, and realising spinorial excitations on a physical three-dimensional slice, require separately supplied branch, metric/tetrad, spin-structure and matter-sector owners and remain Open. Conditional on a declared spin–apparatus–environment channel, Stern–Gerlach-type record formation may be analysed through the environmental-locking logic used elsewhere. This supplies neither selection of one realised outcome nor its update rule; both remain Open.
20. The ECT-specific fermion–condensate coupling $\Delta\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi$: a candidate preferred-medium matter-sector deformation (operator inventory A-math conditional; coefficient, species law and physical map Open; Section 7.6).
21. Three-layer status separation: the microscopic n_A operator M is not the Level-B scalar ansatz S, while the map from S to the supplied conditional HRC gravity functional G is Open (Section 7.7; Appendix W; Figure 3).
22. Observability conditions and falsification logic for the ECT-specific coupling: four classes of observable channels identified (spin-sensitive, composition-sensitive, propagation, environment-dependent), together with explicit conditions under which the present ECT closure could be refuted (Section 7.8). No precision predictions are available at this stage.

23. Structural constraints on the coupling coefficient: neither the coefficient nor the suppression scale, species dependence, v_2 corrections or radiative stability is selected by the present action (Section 7.9).
24. A neutrino sector via type-I seesaw embedding with two independently supplied mass operators and scales. The action derives no heavy scale, Yukawa, mixing or leptogenesis consequence (Section 8.9).

10.3 Status-separated structural statements and forward completion tests

Part I contains exact conditional statements, structural routes, external gates, and Open completion tests. None receives a common prediction status; each is classified here by its own derivation level and owner.

Conditional homotopy classification of the supplied orientation target. The supplied target $O(4)/O(3) \simeq S^3$ has vanishing lower homotopy groups: $\pi_0(S^3) = \pi_1(S^3) = \pi_2(S^3) = 0$. This is Level-A mathematics conditional on the target. It becomes a physical wall/string/monopole statement only after the target is identified with the complete physical vacuum/configuration space and a transition owner is supplied; those steps remain Open (Section 8.10).

Texture/skymion target classes (A-math conditional; physical Open). The non-trivial $\pi_3(S^3) = \mathbb{Z}$ guarantees integer classes of maps into the supplied target. In the separately supplied positive two-derivative S^3 sigma model, $E_2(R) \propto R$ and Derrick collapse follows. Admissible exact-gradient ECT configurations, stabilising action, profile, mass spectrum and production remain Open (Section 8.10).

Ordered-branch signature classification and the supplied second-order scalar cone (Level A conditional / B physical). Part I fixes the signature logic of the ordered branch in three distinct steps. First, on the full real coefficient plane the exact Level-A classification inside the supplied scalar EFT is hyperbolic iff $\beta(\beta - \alpha) < 0$, elliptic iff the product is positive, and degenerate for $\beta = 0$ or $\alpha = \beta$. The hyperbolic set has two mathematical components: $\beta > 0, \alpha > \beta$ and $\beta < 0, \alpha < \beta$. Second, the separately supplied healthy transverse-kinetic convention retains the first component, hence $\alpha > \beta > 0$; P4 does not select it. Third, $\hat{c}_*^2 = \beta/(\alpha - \beta)$ and the adopted Level-B unit-slope coefficient benchmark $\hat{c}_* = 1$ give $\alpha = 2\beta$ inside that supplied branch; a common field rescaling cannot select this ratio. The the separate equality $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration after fixing the conversion/lapse convention; a microscopic explanation of that calibration and equality with photon and tensor cones remain Open. This reconstructed chain is a central non-circular output of Part I and stays conceptually separate from later decoherence- or PES-based arguments.

Lorentz-invariance violation: missing physical operators (Open). Part I supplies no photon, neutrino or tensor LIV operator, coefficient, common suppression scale or observable map. External propagation data constrain future named completions; no unit-coefficient benchmark or cross-sector unification is retained (Section 3.6).

Matching relations for emergent fundamental constants (Level A/B). Part I collects separate candidate matching relations for the effective propagation scale, gravitational normalisation, and coherent action coefficient. Their common one-medium origin is a matching programme, not a derived common control by one parameter set. At the ordered-branch level one has the separate conditional relations

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta}, \quad G_N^{\text{nat}} = \frac{1}{8\pi M_G^2}, \quad \iota_0^{\text{EFT}} = \frac{c_{\perp} K_{\theta}}{2m_{\phi}^2}, \quad \iota_{0,\text{bal}}^{\text{eff}} = \sqrt{2K_{\theta}^{\text{1D}} \mathcal{T}_{\text{loop}}}.$$

The first relation is the dimensionless scalar slope at the EFT level; $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_{\Phi}$ requires the separate clock map. The second is the geometric matching relation that defines the effective gravitational scale

M_G of the ordered branch; the explicit microscopic relation between M_G and the condensate parameters remains an open derivational target. The third and fourth are distinct coherent-sector loop-scale candidates. Mapping either one to the operational S_0 remains Open; the subsequent $S_0 = \hbar$ calibration is an adopted Level B physical input rather than a derivation. Likewise $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration; its microscopic explanation and cross-sector cone universality remain Open. These are independent matching inputs/programmes and do not yet reduce the number of physical inputs.

Species dependence remains Open. The operator alone assigns no species law. A multi-species signal requires an explicit flavour-dependent matching and observable reduction (Sections 7.6 and 7.8).

Longitudinal soft coordinate and conditional scalar mode. The integrability constraint reduces the naive three-component Goldstone-coordinate map to one longitudinal scalar coordinate χ (Level A kinematics). A supplied reduced EFT may realise one physical soft mode (Level B), but its existence, mass, stability and cosmological role are Open. It is therefore only a possible dark-sector candidate, not a completed dark-sector prediction (Section 8.2).

Neutrino mass operators and scales (C/Open). The present ECT action selects no heavy-neutrino scale, Weinberg coefficient or Yukawa matrix. Consequently no ECT-selected numerical mass comparison, Yukawa determination, active–sterile mixing, or leptogenesis inference follows. Fully declared unit-coefficient orientation points and inverse fits remain admissible Level-C benchmarks, not predictions. The theory does not determine the absolute mass scale, the mass ordering, the PMNS pattern, or the Dirac/Majorana character (Section 8.9).

Conceptual outputs of Part I. Beyond the formal structural results, Part I produces the following conceptual outputs that distinguish ECT from standard approaches: (i) *Three logical classes of excitations:* ECT does not place all candidate particle sectors on the same footing—condensate coordinates and conditional modes, structurally embedded SM fields, and topologically selected sectors occupy different logical tiers, unlike QFT where all fields are equally fundamental postulates (§8.1). (ii) *Correlation versus interaction:* a common medium permits product, classically correlated or entangled states; it does not by itself derive a physical interaction, factorisation or exchange vertex. Decoherence does not create Lorentzian interactions. (iii) *Open chiral completion:* the exact $Spin(4) \simeq SU(2)_L \times SU(2)_R$ decomposition survives, but fixing the ordered vector leaves $SU(2)_{\text{diag}}$ and selects neither weak factor. Physical chirality, CAR/spinor states and the weak vertex remain Open. (iv) *Multi-excitation structure as a reconstruction target:* ECT does not require Fock space as a foundational postulate; variable particle number should emerge as a sector decomposition of one condensate medium (OP-Q11). (v) *Conditional 3+1 bookkeeping:* after P1 supplies four Euclidean dimensions and P4 supplies one ordered direction, three transverse coordinates remain. This does not derive $D = 4$, the ordered branch, a physical clock, or why Nature has $3 + 1$ spacetime dimensions. (vi) *Tensor completion:* at fixed α, β and tangent-director linear order about homogeneous $n = w$, the fixed director map has $h_{ij}^{\text{TT}} = h_{ww} = 0$. The scoped theorem does not exclude quadratic/nonlinear/composite routes; a massless Fierz–Pauli sector is retained only as a conditional external completion target (Open).

10.4 What Part I derives and what it does not

Exact or conditional mathematics retained. For supplied P4 and coefficients, the scalar principal tensor has the stated form/sign classification and one scalar characteristic cone; the integrability reduction is scoped in §4.3. The named scalar initial-boundary problem admits the retarded solution of Appendix G. Standard Euler–Lagrange/Noether identities, $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, and integer winding after supplied BR1 also survive in their declared domains. None of these derives a physical light speed, universal matter/photon/tensor cone, metric, branch actualisation, gauge sector, common gravity/quantum origin, or quantum action scale from P1–P3.

Conditional closures, matchings and Open bridges. G_N , $S_0 = \hbar$ and $c_{\text{char}} = c$ are comparison/matching inputs, not derived constants. Schrödinger, Einstein, local $U(1)$, Dirac, entropy and spin-label constructions reproduce standard forms only after their respective fields, states, operators, boundary conditions, gauge reductions or channel hypotheses are supplied. The director coefficient/species law, weak sector, BR1 origin, gravitational/electroweak hierarchy and HRC gravity bridge remain Open or Level-C benchmarks.

Supplied gauge/matter modules (conditional/Open). A compact phase gives global Noether algebra; bundle covariance is standard mathematics only after a local $U(1)$ bundle is supplied. The spacetime orientation triplet is not an internal gauge field, and the vector stabiliser selects no chiral factor. Physical Abelian/weak/colour gauge fields, couplings, representations, charges, anomaly cancellation, fermion masses, Yukawa/flavour data and three generations remain supplied/Open rather than derived from the minimal scalar.

Explicitly open. The dynamical origin of the gradient condensate (OP-grad), the energy-hierarchy mechanism and separate galactic matching (OP-EW, OP-galactic), the full graviton sector (OP1), the full SM gauge completion including colour derivation, matter representations, and anomaly cancellation (OP-GUT1, OP-GUT2, OP-GUT3b, OP-GUT6 family in Table 115), the full fermionic sector from first principles (Section 9.10), the microscopic derivation of the fifth-coupling coefficient, radiative stability of β_5 , the full quantum reconstruction from S_0 (Section 20), and the microscopic H-theorem.

10.5 What remains open at the end of Part I

The following list reformulates the open points above as concrete research targets for Parts II–III and for future work, rather than repeating the derived/open classification itself. The key open problems are:

- Gradient condensate from bare action: the dynamical mechanism generating $\langle \partial_A \Phi \rangle \neq 0$ from P3 (OP-grad).
- Energy hierarchy: mechanism generating $v_2 \ll \phi_0$ (OP-EW); separately, derivation of the HRC matching length, a_M , and scalar-to-HRC gravity bridge (OP-galactic).
- **OP-Planck:** Derive the full NLO orientation-stiffness coefficient $\mathcal{C}_n$ from first principles, including the induced contribution from integrating out the radial sector, the explicit map $\kappa_n \rightarrow M_G^2$ through the spin-2 normalisation, and a complete dimension-and-normalisation audit (Appendix O).
- Full derivation of the graviton sector: Fierz–Pauli closure, nonperturbative spectrum, gauge reduction (OP1).
- Gauge/matter derivation: local $U(1)$ and chiral $SU(2)_L$ are not selected by the global phase/vector stabiliser, and the $SU(3)_c$ module is supplied. Derive the bundles, gauge-volume reduction, kinetic terms, representations, charges, anomaly cancellation and couplings (OP-GUT1, OP-GUT2, OP-GUT3b, OP-GUT6 family; see also Sections 49.14 and 9.5).
- Full first-principles fermionic sector: dynamical origin of spinorial excitations, chiral gauge selection, Yukawa hierarchy, neutrino-mass mechanism, three generations (Section 9.10).
- Microscopic derivation of the fifth-coupling coefficient μ_5 from condensate microdynamics, and establishment of radiative stability.
- Derivation of an unbroken physical $U(1)_{\text{em}}$, its transverse photon state, positive reduced Hessian, matter vertex and physical metric (OP-gauge); only then can the Open relation $c_\gamma = c_{\text{char}}$ be tested. The scalar equality $c_{\text{char}} = c$ is already the adopted clock calibration.

- Full mapping of the ECT-specific operator onto standard Lorentz-violation and equivalence-principle test frameworks.
- Full quantum reconstruction from S_0 : amplitude composition, Born rule, operator algebra, Hilbert-space structure (Section 20).
- Microscopic H-theorem: first-principles derivation of the decoherence kernel and irreversible thermodynamics.
- Nonlinear closure of each effective branch.
- Strict identification $\Sigma_\theta \iota_0^{\text{EFT}} = \hbar$ beyond Level B matching.
- Full subsystem-level reconstruction of quantum dynamics from the descriptive Euclidean condensate equation: in particular, the bridge from the deterministic boundary-value formulation of the coherent sector to the standard probabilistic subsystem equations beyond the presently established Schrödinger/Klein–Gordon reconstruction.

Additional open consistency problems of the emergent matter sector are now organised explicitly in the structured OP-GUT family (Table 115) and the fermionic-architecture open programme (§9.10), including the conditional CPT closure problem (OP-CPT1), the anomaly-consistency family (OP-GUT6a, OP-GUT6b, OP-GUT6c), and the exact weak-sector parity pattern (OP-P1).

10.6 Roadmap to Parts II–III

Table 47: Structural branching of ECT after the foundational construction.

Branch	Primary variables	Developed in
Geometric (Macroscopic Physics)	$g_{AB}^{\text{eff}}, n_A, \phi(X)$, and long-wavelength background variations of the ordered branch	Gravity, cosmology, galactic phenomenology, observational tests
Coherent (Quantum Sector)	$\Phi_{\text{eff}}, \theta, S_0$, coherent winding sectors, and coherent correlators	Action-scale quantisation, wave dynamics, quantum kinematics, decoherence, probability

Part II (Section 11 onwards) develops the geometric macroscopic branch: the homogeneous ordered-phase limit, the emergent gravitational sector, the ordered-branch cosmological programme, galactic and cluster-scale phenomenology, dark-sector interpretation, and observational tests. Part II distinguishes a Level-B scalar ansatz from separately supplied conditional HRC galactic phenomenology. Part I establishes only that the microscopic n_A operator is a different object; it does not supply the Open scalar-to-AQUAL source/force map (Section 7.10).

Part III (Section 20 onwards) develops the phase-sensitive coherent branch across eleven thematic steps: coherent phase dynamics and the pre-quantum action scale, wave kinematics and conservation laws, the Hilbert-space bridge (reflection positivity, unitarity, canonical structure, uncertainty), exchange sectors and Dirac structure, unified vacuum response, decoherence and the arrow of time, the quantum–classical boundary and Born-type probability, entanglement, the wider topological/dark-sector ontology, black-hole thermodynamics and the information problem, and the analogue-laboratory programme.

Part II

Macroscopic Physics: Gravity, Cosmology, and Observational Tests

11 Scope, Bridge, and Approximation Hierarchy

Part II develops the macroscopic physics of the ordered condensate branch. Before proceeding to the specific derivations—local special-relativistic kinematics (§12), the gravitational sector (§13), cosmology, and galactic phenomenology—the present section fixes the scope, states the approximation regime, and records explicitly which results of Part I are used as structural inputs below.

The purpose of this section is threefold. First, to define the branch of the theory developed in Part II. Second, to make explicit which approximations are used, where they enter, and what their derivation status is. Third, to draw a sharp boundary between what Part II imports from Part I and what it does not import from the Quantum Sector of Part III.

Throughout this chapter we distinguish three levels of status: strictly derived structural inputs from Part I (Level A), structural but conditional results inherited from controlled reconstruction routes (Level B), and closure-dependent macroscopic specialisations used in phenomenology.

11.1 Macroscopic Physics as the development of the geometric branch

Connection to ECT Foundations (Part I): Sections 6, 6.2, and the summary rules collected in Section 6.8, especially the amplitude–orientation decomposition rule and the macroscopic viability rule. Status: conceptual bridge; all claims structural.

After completing ECT Foundations (Sections 2–6), ECT is no longer an undifferentiated scheme. As organised in the branching section, P4 supplies an ordered-gradient chart and BR1 separately supplies a coherent-sector closure. The manuscript then studies phase-insensitive geometric and phase-sensitive coherent programmes; no physical branch selection or common microscopic descent is derived. The present section follows the first of these developments. Throughout Part II we work with the derived scalar cone and, wherever SI units are displayed, use the adopted Level-B scalar-clock calibration $c_{\text{char}} = c$. Part I does not supply the reduced gauge bundle, unbroken photon or physical photon principal operator required to establish the separate equality $c_\gamma = c_{\text{char}}$ (§4.7).

Two conditional programmes organised around one medium. The manuscript organises a *geometric/macroscopic programme* (§6.2) around phase-insensitive long-wavelength variables and a *coherent/quantum-reconstruction programme* (§6.3) around a separately supplied complex closure, winding sectors, reduced-state kernels and amplitudes. This organisation is the one-medium architecture being tested; it is not a derivation that one frozen scalar action already owns both physical sectors. Each programme requires its own action, state, constraint reduction and observable map. The present section develops the first programme exclusively. The Level-B scalar ansatz of Part II is an amplitude/background closure within this programme. The supplied conditional HRC gravity functional is a separate response construction whose connection to that ansatz remains Open; neither should be confused with the coherent phase θ .

Definition of macroscopic programme variables. ECT Foundations supplies an ordered-gradient chart and separately defines a BR1 coherent closure. The subsequent analysis is organised into a *tensorial, phase-insensitive programme* and a *coherent, phase-sensitive programme*; their physical common owner is Open. The geometric branch is defined precisely by the former. The primary variables and completion symbols of this programme are:

- the scalar characteristic tensor K^{AB} and its convenient inverse metric-like notation; a universal physical metric remains Open;
- the orientation collective variable n_A (P4);
- long-wavelength orientation coordinates δn_i around the supplied chart (Section 4.3);
- action-specific Noether stress and angular-current structures (Section 6.6, Appendix R);
- the conditional tensor coefficient M_G in a supplied gravity completion; the bridge from orientation stiffness to M_G and then to the observed Newton constant is Open.

Some precursor symbols appear in ECT Foundations, but their physical metric, tensor, source and matching ownership is not thereby established.

The conditional coarse-graining criterion. A macroscopic P4 programme may be organised only after a P4 pole is supplied. Define conditionally

$$L \gg \xi_{\text{P4}} := m_{\text{P4,pole}}^{-1}, \quad E \ll m_{\text{P4,pole}}. \quad (11.1)$$

The P3 value $m_{\Phi,\text{rad}}^2 = 2\mu^2$ does not own $m_{\text{P4,pole}}$. Thus Eq. (11.1) is a conditional EFT regime, not a derived numerical cutoff, until the stationary P4 action and Hessian are supplied. Coarse graining also does not prove the retained orientation/background inventory to be complete.

Why the phase drops out. The macroscopic branch is defined not only by scale but by the loss of sensitivity to local phase information. In geometric-branch observables the phase θ of the coherent order parameter Φ_{eff} does not appear explicitly; what remains are effective metric relations, source couplings, causal propagation, and large-scale background structure. This is the bookkeeping reason why the present section studies conditional gravity, cosmology and galactic completions, while wave amplitudes, operator algebra, decoherence and probability owners are analysed separately. It is not a theorem that coarse graining alone produces either completed sector.

What is already established in ECT Foundations. From ECT Foundations the following conditional inventory is available for the geometric programme: the supplied chart is parametrised by collective infrared variables (Section 3.2), and the declared local algebraic scalar operator inventory fixes its two-tensor basis. A physical coarse-grained tensor owner remains Open.

- (i) a constrained soft orientation sector (integrability gives one longitudinal scalar coordinate χ in the strict scalar map; its one-mode physical interpretation is a Level B minimal-EFT statement; see Section 4.3 and Appendix J);
- (ii) conditional scalar Lorentzian component $\alpha > \beta > 0$ once the healthy sign, P4 chart and EFT coefficients are supplied; the optional arithmetic-average benchmark $\alpha < 4\beta$ requires its own declared ensemble/cell and is not a homogenised or physical viability bound (Section 3.5);
- (iii) action-specific conserved T^{AB} from translational invariance and angular currents $J^{C,AB}$ from $O(4)$ invariance (Section 6.6, Appendix R);
- (iv) a conditional tensor-normalisation symbol M_G is introduced for a supplied gravity completion; its microscopic owner and the bridge from orientation stiffness are Open, while standard matching to G_N is reviewed in Section 13.4;
- (v) a conditional Deser-type self-coupling route, applicable only after a healthy universal spin-2 seed and source vertex are independently derived (Section 4.6);

- (vi) the scalar-cone result: integrability of $Q_A = \partial_A \Phi$ forces the linear scalar ordered branch into a single principal slope $\hat{c}_*$ in common-length coordinates (Level A inside the supplied scalar EFT); its physical speed requires c_u and $\mathcal{N}_\Phi$;
- (vii) restricted cross-sector cone programme: the scalar cone is derived, and the coherent-phase and Abelian-gauge routes share it under their stated matching assumptions. The physical tensor principal symbol and c_{GW} remain Open (§4.7, Eq. (4.11));
- (viii) the physical identification $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration; the photon and tensor cone equalities remain independent Open kill gates.

Four structural common-origin programmes of ECT. The single-medium architecture (P3) motivates four cross-sector universality programmes with different closure status:

- (I) **Common-cone programme:** the scalar characteristic cone is derived; phase and Abelian-gauge inheritance is conditional on their stated maps, while the physical tensor cone is Open (§4.7, Eq. (4.11)).
- (II) **Common action-scale programme:** a common parameter S_0 is used across supplied coherent-branch constructions. Its universality and identification to $\hbar$ remain matching problems (§5.6).
- (III) **Common extremal-backbone programme:** stationary or variational principles occur in the declared bare and reduced models, but a universal one-history physical identity across macroscopic, open-system and quantum sectors is not derived (§22.2).
- (IV) **Common conservation-owner programme:** translational/ $O(4)$ Noether currents of the real scalar action, the phase current and winding of the supplied BR1 closure, and the later probability-current reading are distinct structures. A single reduced action/state owner connecting all of them remains Open (§22.3).

These are conceptually distinct research programmes probing causal propagation, action normalisation, variational organisation and conservation structure. They are not evidence that one medium already owns all their operators, states or measured normalisations. The observed single values of c and $\hbar$ are matching targets and cannot validate common microscopic ownership before the corresponding bridges are derived.

Combined falsifiable content of the universalities. If *any* of the four proposed common-origin links were shown to be impossible—for example if completed native sectors required incompatible causal cones, action scales or conservation owners—then the corresponding ECT single-medium programme would fail. Such an outcome would show that the native low-energy sectors of the theory are not controlled by one and the same condensate structure. Conversely, the present incompleteness of the bridges is not positive evidence for common ownership. Each link must be tested only after the relevant physical actions, states and observable maps are supplied.

Programme-level failure-gate map. The programme developed in Parts I–III admits a compact status map in which structural routes are paired with the conditions that would reject a specified completion and with external observational constraints relevant after that completion is supplied. The current Open routes are not thereby promoted to predictions or already-operational falsifiers. Table 48 collects thirteen programme-level failure gates.

Table 48: Status-separated failure gates of the current ECT programme.

Programme statement	Failure condition for the named completion	External constraint or future test window
Common-cone programme (§4.7)	Sector-dependent limiting speeds	$ c_{\text{GW}}/c - 1 \lesssim 10^{-15}$ [12]
Common action-scale programme (§5.6)	Mutually inconsistent quantum-sector normalisations	Single $\hbar$ across all established quantum phenomena
One extremal backbone (§22.2)	Incompatible sector-dependent extremal/classical crossover criteria in native ECT sectors	Interferometric, atomic, and macroscopic regimes are all consistent with action-controlled crossover logic, while the precise universal threshold remains a structural programme-level claim rather than an established measured law
One conservation origin (§22.3)	Independent non-condensate conservation structures for native sectors	Charge stability bounds such as $\tau(e \rightarrow \gamma \nu) > 10^{26}$ yr [86]; the broader baryogenesis connection is developed separately in Section 16.3
Candidate electroweak completion (separately supplied; §7)	No ECT-specific falsifier exists until a named local $SU(2)_L \times U(1)_Y$ bundle, action, chirality and matter representations are derived or supplied	Once supplied, test anomaly cancellation, charge assignments, the unbroken photon and the mass pattern; the observed Standard-Model organisation is an external target, not verification of an ECT route
Open entanglement-from-medium reconstruction programme (§31)	After a physical state/channel is derived, no reconstruction from the medium is possible without importing the full tensor-product structure as an independent primitive	Bell-inequality violation well established; loophole-free tests confirm non-classical correlations
Tensor-closure graviton programme (§8.5)	Induced metric sector cannot close to FP/TT, or tensor cone differs from ordered branch	GW speed $ c_{\text{GW}}/c - 1 \lesssim 10^{-15}$; no extra polarisations detected
Analogue source-model comparison programme (§37)	A frozen analogue–ECT bridge fails after its source action, state, operator, observable, uncertainty and discriminator guards pass	Quantised circulation, dynamical Casimir, acoustic mode mixing and tunable decoherence are external source-model results; they are neither evidence for ECT nor a generic ECT falsifier
Quantum-sector one-medium unification (§20)	Different quantum substructures requiring mutually unrelated action scales or principles	All established QM phenomena remain consistent with a single $\hbar$, one effective probabilistic layer of Born type, and one Schrödinger dynamics
Singularity as branch boundary (§14.1)	Ordered branch remains valid to formal GR singularity with no condensate breakdown	No observed sub-Planckian singular physics; BH exterior consistent with EFT description

Continued on next page

Table 48 continued

Programme statement	Failure condition	External constraint / future window
P4 radial-mode EFT-threshold programme (§8.3)	A stationary P4 action fails to produce a healthy isolated amplitude pole, or its derived threshold conflicts with observation	No P4 pole or threshold is presently predicted; this row is a future theory-and-data gate, not a passed test
Einstein nonlinear closure (§13.5)	Non-Einstein two-derivative completion required, or non-universal matter-metric coupling	Classical GR weak-field success; binary-pulsar radiative consistency; absence of compulsory extra low-energy tensor polarisations. External sensitivity footprint of $\mathcal{R}_{\mu\nu}^{\text{cand}}[n]$ inventoried across five sectors (Table 51); no ECT contribution is identified before S_n , projection, sign, and coefficients are derived (OP-c1)
Galactic dark-sector fit scope (§18)	Failure of the HRC no-halo algebraic diagnostic on held-out galactic data	Level-C SPARC fits and conditional BTFR/RAR algebra use the matched a_M benchmark; scalar-to-galactic matching and ECT particle content remain Open

The same map yields conditional programme-level hypotheses:

- (i) No native low-energy sector should require its own independent fundamental action unit: all coherent-branch normalisations should reduce to one operational S_0 , after an explicit owner map rather than by identifying either loop candidate by notation.
- (ii) No native low-energy sector should require its own independent extremality law: all effective equations of motion should descend from one dynamical principle DP.
- (iii) Any future tensor completion must pass the external constraint $|c_{\text{GW}}/c_\gamma - 1| \lesssim 10^{-15}$; the present scalar action predicts neither equality nor Planck-suppressed departures (§13.8).
- (iv) If the ordered-medium electroweak route is correct, future deeper closure should *reduce* external gauge input rather than increase it.
- (v) Cosmological, decoherence, and vacuum-response developments should reinforce cross-sector unification rather than fragment into unrelated sector-specific constants.

These are programme-level hypotheses: each is falsifiable in principle but tests the overall architecture rather than a single formula. Their collective consistency forms a major part of the present structural case for the ECT single-medium programme.

What does not follow automatically. Items (i)–(viii) above do not automatically give: the full nonlinear gravitational closure; standard Einstein equations in final form; complete matching of all phenomenological constants; or any quantum-mechanical statement. These are developed step by step below.

Logical order of Part II. First: the homogeneous ordered-phase limit and emergent special relativity (§12). Then: the macroscopic gravitational sector, including the effective metric, source structure, gravitational stiffness matching, nonlinear closure, weak-field limit, post-Newtonian regime, and gravitational waves (§§13.1–13.8). Finally: cosmology, galactic phenomenology, dark-sector interpretation, and observational tests (§§14–19.1). The conditional HRC gravity functional G used later is separately supplied. Its response algebra is Level A inside the augmented model, the map from the ordered phase or scalar ansatz S is Level B/Open, and its data applications are Level C.

What Part II does not import from Part III. The macroscopic branch is constructed without using the reduced Planck constant $\hbar$, the Born rule, wavefunction collapse, canonical commutation relations, decoherence functionals, or instanton/loop amplitudes. These belong exclusively to the Quantum Sector developed in Part III. The role of Part II is purely geometric and macroscopic. Accordingly, all macroscopic equations in Part II are organised without taking $\hbar$ as a dynamical input of the geometric branch itself. In particular, the action-scale logic associated with S_0 and its matching to $\hbar$ is not used as an input to the geometric branch, even when c and G_N are already employed as matched macroscopic constants.

Branch structure at the coarse-grained level. At the coarse-grained level relevant for the present chapter, observables separate into a phase-insensitive geometric development and a phase-sensitive coherent development. Part II develops only the former. This distinction is sufficient for the present scope and does not require a stronger theorem-level claim of exhaustive RG classification.

Physical picture. If ECT Foundations answers: *what is the ordered medium and how does it branch?*, then Macroscopic Physics answers: *how does the geometric branch of that medium appear to a large-scale observer insensitive to coherent phase data?* Gravity is not introduced as an independent external sector. Rather, the working hypothesis of Macroscopic Physics is that the large-scale causal and geometric response of the ordered condensate admits an effective description which, in the viable Lorentzian branch, takes geometric form.

11.2 Approximation hierarchy used in Part II

The derivations of Part II use a nested sequence of approximation regimes. The table below records the approximation levels explicitly used in the chapter and indicates where each of them enters. The purpose is organisational clarity rather than a theorem-level claim of formal completeness.

Table 49: Approximation levels used in Part II.

Approximation	Validity condition	Where used	Origin / status
Conditional retained scalar hyperbolic component	$\beta > 0, \alpha > \beta$	All of Part II	Level A mathematics inside the supplied scalar operators; physical macroscopic viability and a common metric remain Open (§3.5)
Optional arithmetic-average benchmark	$\alpha < 4\beta$ only after a declared ensemble/cell, scale ℓ_{iso} and arithmetic isotropic averaging prescription	Only where separately invoked	Conditional coefficient-average positivity; not a Lorentzian boundary, homogenised operator or derived ultraviolet window
Conditional P4 macroscopic EFT regime	$L \gg \xi_{\text{P4}}, E \ll m_{\text{P4,pole}}$	Only owner-declared P4-EFT formulae	Open until a stationary P4 Hessian, positive pole and scale separation are derived

Approximation (cont.)	Condition	Where used	Status
Homogeneous supplied ordered chart	$\nabla n_A = 0, \alpha, \beta = \text{const}$	§12 (SR)	Level A conditional kinematics; stationary physical phase and selection Open
Linear perturbative regime	$ \delta n_i \ll 1$ (equivalently $ \partial_i \chi /u_0 \ll 1$ in the declared linear map)	§13.2 (linearised gravity)	Level A conditional expansion; physical linearised-gravity owner Open
Minimal nonlinear closure	A1–A4, amplitude-dominated reduction	§13.5 (generalised EE)	Level B
Algebraic frozen- F limit	$F = \text{const}$ and scalar derivatives vanish; no independent orientation source included	§13.7 (conditional GR reduction)	Level A algebra inside the Level-B ansatz; dynamical reachability Open
Weak-field, quasi-static	$ h_{\mu\nu} \ll 1, v \ll c_{\text{char}}, \partial_t \ll c_{\text{char}} \nabla $	§13.6 (Newtonian limit)	Level B within the adopted closure
FRW background specialisation	$n_A \rightarrow \bar{n}_A, \phi = \phi(t)$, isotropic	§14 (cosmology)	Level B background ansatz
Critical IR galactic regime	deep-acceleration branch with non-analytic IR closure	§17.1 (galaxies)	Closure-dependent

These approximations should be understood as the explicit approximation map used in the present chapter. They are introduced here in order to make the logical dependencies of the later derivations transparent.

11.3 Bridge from Part I: structural inputs used in Part II

The list above states the conceptual content in prose; the table below recasts the same inherited results as an operational bridge map for the later macroscopic derivations. The following table records the items retained from Part I that are used as structural inputs in Part II. It is intended as a bridge map: for each inherited item, we indicate its status and what role it plays in the macroscopic development.

Table 50: Structural inputs from Part I used in Part II.

Part I item	Level	Section	Role in Part II
P4-supplied nonzero vector n_A with $O(3)$ stabiliser	Supplied datum / A-math; physical SSB Open	§3	Conditional scalar orientation chart; no physical causal structure by itself
Unique $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$	A-math conditional	§3.3	Supplied scalar operator basis; physical metric Open
Retained scalar Lorentzian component $\beta > 0, \alpha > \beta$	A-math conditional / physical Open	§3.5	Exact scalar hyperbolicity on the supplied healthy component; the optional arithmetic-average condition is a separate assumption, not part of this row

Part I item (cont.)	Level	Section	Role
Supplied second-order scalar operator: one principal cone	A-math conditional / B physical	§4.3	Integrability fixes the coordinate map, not operator order; no GW-speed owner
$\hat{c}_*^2 = \beta/(\alpha - \beta)$	A-math conditional	§5.3	Dimensionless scalar characteristic ratio on the retained component
$c_{\text{char}} = c$ (scalar-clock calibration)	Level B adopted	§4.7	Uses a declared clock/lapse convention; microscopic owner Open, while photon/tensor cone equality requires separate actions
Ordered-branch collective variables	A-math/B/Open	§3.2	Conditional amplitude/ ϕ coordinates and orientation reduction; physical modes require their actions and projectors
On-shell Noether currents T^{AB} , $j^{C,AB}$	A-math in their owning actions	§6.6	Current identities only; no gravitational source vertex or Bianchi identity follows
Phase-to-gauge completion route (L1–L3)	Open	§7	transverse photon and $c_\gamma = c_{\text{char}}$ require independent owners

Immediate matching step in Part II. In addition to the inherited inputs listed above, Part II performs the gravitational stiffness matching

$$G_N^{\text{nat}} = \frac{1}{8\pi M_G^2},$$

which is not an input from Part I but an early macroscopic matching step carried out within the present chapter (§13.4).

Characteristic scales of the macroscopic regime. Conditional on a stationary P4 action and pole, the geometric EFT may be organised at

$$L \gg \xi_{\text{P4}} = m_{\text{P4,pole}}^{-1}, \quad E \ll m_{\text{P4,pole}}.$$

Neither quantity is inherited from the P3 homogeneous radial pole. The dimensionless scalar slope $\hat{c}_*$, the physical clock/photon speed and the tensor normalisation M_G likewise have distinct owners. Therefore P3, P4, photon and tensor scales must be matched rather than identified by notation.

12 Special Relativity as the Homogeneous Ordered-Phase Limit

Before allowing the ordered condensate background to vary across spacetime, it is useful to isolate the homogeneous limit in which the effective geometry is flat. This limit provides the local special-relativistic kinematics on which the subsequent macroscopic gravitational construction is built.

The logic is the following. Conditional on the supplied P4 ordered branch, residual $O(3)$ covariance permits the displayed scalar kinetic tensor $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$ (Theorem 3.2), while integrability of $Q_A = \partial_A \Phi$ reduces the strict linear soft map to one scalar coordinate (§4.3). The use of a single second-order physical scalar principal operator is an additional supplied-EFT assumption. A compact coherent phase and a local gauge bundle are additional completion inputs; their principal tensors, the physical photon and cone equality are Open.

The purpose of the present section is therefore not to postulate special relativity anew, but to state explicitly which parts of local relativistic kinematics follow from the ordered branch and how the Minkowski interval arises.

12.1 Status and scope

The conclusions of this section are intentionally stratified.

- **Level A conditional mathematics:** the supplied second-order linear scalar principal operator has one causal cone with dimensionless common-length slope $\hat{c}_*$; integrability alone does not fix the operator order (conditional one-operator corollary, §4.3). The dimensionful c_{char} additionally requires a supplied clock, conversion speed and lapse map.
- **Level A mathematics:** in the supplied homogeneous scalar model, the principal tensor has flat Lorentzian form and Poincaré-form isometries in common-length coordinates.
- **Adopted scalar matching / Open universality:** $c_{\text{char}} = c$ is the Level-B scalar-clock calibration; rods, a transverse photon and the equalities $c_\gamma = c_{\text{char}}$ and $c_{\text{tensor}} = c_{\text{char}}$ require independent owners (§4.7).
- **Not claimed here:** a theorem-level derivation of the full gauge structure, nor a complete derivation of all fermionic sectors from the microscopic Euclidean theory.

12.2 Homogeneous ordered phase and the supplied scalar cone

Consider the homogeneous ordered branch with constant $n^A = \delta_w^A$ and constant coefficients α, β . The ordered-branch operator (3.11) reduces to

$$K^{AB} \partial_A \partial_B = \beta \delta^{ij} \partial_i \partial_j - \Delta_K \partial_w^2, \quad \Delta_K \equiv \alpha - \beta. \quad (12.1)$$

On the manuscript's retained $\beta > 0$ component, $\Delta_K > 0$ makes this supplied second-order operator hyperbolic. Globally the exact nondegenerate condition is $\beta \Delta_K > 0$. In the retained component the operator has the single scalar principal cone

$$\beta \mathbf{k}^2 - \Delta_K k_w^2 = 0 \quad \implies \quad \hat{c}_*^2 = \frac{\beta}{\Delta_K}. \quad (12.2)$$

This is the one cone of the explicitly supplied second-order operator $K^{AB} \partial_A \partial_B$. Integrability of $Q_A = \partial_A \Phi$ expresses the strict-map soft coordinates as derivatives of one scalar χ , but does not by itself exclude higher-order or additional physical principal factors. The conditional one-operator result and its counterexample guard are given in Appendix K.

12.3 Minkowski form of the local effective geometry

In the homogeneous branch the effective quadratic form (13.8) becomes

$$g_{\text{eff}}^{AB} = \text{diag}(\beta, \beta, \beta, -\Delta_K). \quad (12.3)$$

Introducing the ordered-branch time-length coordinate $x^0 = w$ (slope $\hat{c}_*$; cf. (12.2)), the common-length line element is

$$ds_*^2 = -\hat{c}_*^2 (dx^0)^2 + d\mathbf{x}^2 = -c_{\text{char}}^2 d\tau^2 + d\mathbf{x}^2 \quad (12.4)$$

where the second equality is restricted to the locally aligned constant-lapse map (3.23). At $\hat{c}_* = 1$ in common-length units this is $ds_*^2 = -(dx^0)^2 + d\mathbf{x}^2 = \eta_{\mu\nu} dx^\mu dx^\nu$, the Minkowski metric. Within the independently supplied scalar EFT, $\alpha > \beta > 0$ on the retained $\beta > 0$ component classifies this principal tensor as hyperbolic (the full nondegenerate criterion is $\beta(\beta - \alpha) < 0$). P4 does not generate those coefficients or select the realised physical branch, clock or universal metric (§3.6).

12.4 Lorentz transformations from cone invariance

The admissible linear transformations between inertial coordinate systems of the homogeneous ordered phase are those preserving the interval (12.4). Imposing linearity, spatial isotropy of the residual $O(3)$, and—after the physical clock map—invariance of the null cone $x = \pm c_{\text{char}} \tau$, one obtains the Lorentz boost along x with $t \equiv \tau$ below denoting that local physical clock (not the common-length coordinate $x^0 = w$):

$$x' = \gamma_*(x - vt), \quad t' = \gamma_* \left(t - \frac{v}{c_{\text{char}}^2} x \right), \quad \gamma_* = \frac{1}{\sqrt{1 - v^2/c_{\text{char}}^2}}. \quad (12.5)$$

All standard kinematic consequences follow immediately: time dilation $\Delta t = \gamma_* \Delta \tau$, length contraction $L = L_0/\gamma_*$, and velocity addition $u \oplus v = (u + v)/(1 + uv/c_{\text{char}}^2)$.

Equivalently, these transformations are precisely the linear isometries of the flat ordered-branch metric (12.4); the usual Lorentz subgroup of the Poincaré group is therefore realised as the kinematic symmetry of the homogeneous ordered phase.

12.5 Relativistic dispersion, energy, and momentum

The scalar ordered-branch equation (3.24) in the homogeneous phase gives $\omega^2 = c_{\text{char}}^2 \mathbf{k}^2 + M_{\text{eff}}^2$. Its positive-energy interpretation uses the separately displayed $\varsigma_L = -1$ Lorentzian orientation and $A_{L,\varphi} > 0$ in Eq. (3.14); the cone fixes neither the energy sign nor its action-valued magnitude. When m denotes the physical mass used below, the standard quantum matching is $M_{\text{eff}} = mc_{\text{char}}^2/\hbar$, and the dispersion is

$$\omega^2 = c_{\text{char}}^2 \mathbf{k}^2 + \frac{m^2 c_{\text{char}}^4}{\hbar^2}. \quad (12.6)$$

Upon the standard quantum identification $E = \hbar\omega$, $\mathbf{p} = \hbar\mathbf{k}$, one obtains the familiar relativistic energy–momentum relation

$$E^2 = p^2 c_{\text{char}}^2 + m^2 c_{\text{char}}^4. \quad (12.7)$$

For massless modes of this supplied scalar operator $E = pc_{\text{char}}$, so their group and phase velocities coincide at c_{char} . These are scalar-wave characteristic kinematics only; a physical particle interpretation and the photon, matter, fermion and tensor cones require independent owners.

The identification $E = \hbar\omega$, $\mathbf{p} = \hbar\mathbf{k}$ is used here as the standard quantum matching between wave kinematics in the ordered medium and particle energy–momentum. In ECT this matching is not treated as an independent fundamental postulate: its structural route is the Schrödinger-type coherent-branch reconstruction $iS_0 \partial_t \Psi = \hat{H}_{\text{coh}} \Psi + \dots$ (Eq. (22.7)). In the coherent stationary sector, where the decoherence and higher-gradient corrections denoted by the ellipsis are neglected and $\hat{H}_{\text{coh}} \Psi = E\Psi$, a phase $\Psi \sim e^{-i\omega t}$ gives $E = S_0 \omega$. The final identification with the observed reduced Planck constant uses the adopted Level-B calibration $S_0 = \hbar$; its microscopic owner and universality remain Open, as discussed in Sections 5.6, 21.2, and 22.1.

12.6 Signal-speed bound of the supplied scalar operator

Scalar bound from the characteristic cone. For a supplied quasilinear scalar equation (G.2), the exact nondegenerate condition $P_X(P_X + 2XP_{XX}) < 0$, with both factors bounded away from zero, defines a local scalar characteristic cone. The manuscript’s retained $\beta > 0$ matching component is $P_X > 0$ and $P_X + 2XP_{XX} < 0$; the opposite-sign component is not excluded by signature alone. Under the foliation, regularity and initial–boundary orientation assumptions of Appendix G, the Leray–Choquet–Bruhat theorem [87, 88, 89] gives finite domain of dependence and retarded support for that scalar problem. This is neither a branch-wide fundamental speed theorem nor a physical clock/metric derivation. Photon, matter, fermion and tensor principal symbols and their equality to this scalar cone remain independently owned Open gates.

Group and phase velocities of linear modes. For linear plane waves, the dispersion relation (12.6) makes the causal bound explicit. A massive mode ($m \neq 0$) has the group velocity

$$v_{\text{gr}} = \frac{\partial \omega}{\partial k} = \frac{c_{\text{char}}^2 k}{\sqrt{c_{\text{char}}^2 k^2 + m^2 c_{\text{char}}^4 / \hbar^2}} < c_{\text{char}}, \quad (12.8)$$

For $m \neq 0$ and finite k , $v_{\text{gr}} < c_{\text{char}}$; $v_{\text{gr}} \rightarrow c_{\text{char}}$ as $k \rightarrow \infty$, while for $m = 0$ and $k > 0$ one has $v_{\text{gr}} = c_{\text{char}}$. Here $k \equiv |\mathbf{k}|$. Thus the centre of a narrow-band massive wave packet is sub-characteristic. Its causal front is not: because the mass term is lower order, $v_{\text{front}} = \lim_{k \rightarrow \infty} \omega(k)/k = c_{\text{char}}$. In flat $3 + 1$ dimensions the massive retarded Green function has a distributional term on the scalar characteristic cone and a tail in its interior, whereas the massless retarded fundamental solution is supported on the cone. Both are confined to the closed scalar causal cone.

The phase velocity of a massive mode, by contrast, is

$$v_{\text{ph}} = \frac{\omega}{k} = c_{\text{char}} \sqrt{1 + \frac{m^2 c_{\text{char}}^2}{\hbar^2 k^2}} > c_{\text{char}}, \quad (12.9)$$

and formally exceeds the cone speed. This does not violate causality: the phase velocity does not transport information or causal influence. The causal bound is set by the characteristic cone of the principal symbol—equivalently, by the front speed—not by the phase velocity. Within this supplied scalar operator, the speed bound is a bound on scalar causal propagation, not on every quantity carrying the name “velocity”. Its promotion to a universal physical speed bound remains an independent cross-sector Open gate.

Consistency check: velocity addition. The velocity-addition law derived from cone invariance (12.5), $u \oplus v = (u + v)/(1 + uv/c_{\text{char}}^2)$, preserves the interval $|u|, |v| < c_{\text{char}} \Rightarrow |u \oplus v| < c_{\text{char}}$. This confirms the internal closure of the sub-cone velocity domain and is consistent with the causal bound following from the principal symbol, but is not its independent source.

Scalar signal speed and the physical-cone gap. The front speed and domain-of-dependence boundary of the supplied scalar operator are determined by its characteristic cone. No physical photon or universal matter/tensor signal speed follows from that statement. A local Abelian connection and an unbroken massless photon require independent owners; a charged condensate can Higgs the connection, and its kinetic tensor need not equal the scalar one. Photon, matter and tensor cone equality is therefore a future consistency target, with GW170817/GRB 170817A an external kill gate, not a present ECT prediction.

Universality across inertial frames. Coordinate changes preserving the scalar null cone give the usual Lorentz group with invariant parameter c_{char} inside this conditional subsector. This establishes scalar cone kinematics only. Universality for rods, clocks, photons, fermions and tensors requires their physical actions and a common observable metric and remains Open.

Status and caveats. *Theorem-level inside the supplied operator:* retarded propagation of the fixed quadratic scalar EFT; conditional local domain-of-dependence confinement for a separately supplied uniformly hyperbolic $P(X)$ branch (Appendix G). For $m \neq 0$ and finite k , the centre of a narrow-band packet has sub-characteristic group velocity, while the wavefront remains on the scalar characteristic cone. In flat $3 + 1$ dimensions the massive retarded kernel has both a cone-boundary term and an interior tail; in the massless case the flat- $3 + 1$ retarded fundamental solution is supported on the cone. *Conditional/Open:* photon cone matching requires the physical gauge owner, while the gravitational-wave cone is Open until a tensor action is supplied. In slowly varying backgrounds the statement is local: massive WKB packet centres are locally timelike where that approximation is valid, whereas geometric-optics wavefronts follow the local null characteristics. No global Huygens statement is implied. Extension to strongly nonlinear,

solitonic, defect or fully interacting sectors is not proved here and requires its own action, branch and boundary analysis.

ECT adopts $c_{\text{char}} = c$ as a scalar-clock calibration. This does not identify the observed photon as an excitation of the scalar owner: the photon action and the equality $c_\gamma = c_{\text{char}}$ remain Open.

12.7 The status of the identification $c_{\text{char}} = c$

Up to this point, the derivation is entirely internal to the homogeneous ordered branch. It establishes a unique invariant speed c_{char} for the linear scalar sector. To identify this speed with the observed speed of light, one further step is required.

The compact phase can be assigned the same tensor inside a supplied phase EFT. A physical transverse photon, however, requires an independent local gauge bundle, gauge-volume reduction, unbroken $U(1)_{\text{em}}$ and principal operator. P5 does not exclude anisotropic gauge kinetic terms. Consequently the adopted scalar calibration and the Open photon match must be written separately:

$$c_{\text{char}} = c \quad (\text{adopted Level-B scalar-clock calibration}), \quad c_\gamma \stackrel{?}{=} c_{\text{char}} \quad (\text{Open photon-sector matching}). \quad (12.10)$$

Neither statement is implied by the dimensionless scalar relation $\hat{c}_*^2 = \beta/(\alpha - \beta)$. The first fixes a physical unit/lapse calibration; the second additionally needs the photon action. Tensor and matter cone universality remains Open.

12.8 From the local scalar characteristic model to a macroscopic completion

The present section describes a homogeneous scalar ordered-phase model with a constant characteristic ratio. A physical locally Minkowskian metric and special-relativity interpretation additionally require clock, rod, matter and photon maps. A candidate macroscopic completion allows the background to vary slowly:

$$g_{\text{eff}}^{\mu\nu} \rightarrow g_{\text{eff}}^{\mu\nu}(X).$$

The displayed flat scalar principal tensor is then replaced by a slowly varying candidate geometry. This is an ansatz for the Part II programme, not a derivation of the equivalence principle or a physical metric from P1–P6.

An important clarification concerns the status of c_{char} in the macroscopic regime. The EFT parameters (α, β) are independently supplied coefficients of the local ordered-chart scalar model; P4 does not derive them as physical vacuum-state parameters. In the homogeneous benchmark they are held position-independent by assumption. The candidate gravity completion associates slow variation of $n^A(X)$ and amplitude with a varying effective tensor $g_{\text{eff}}^{\mu\nu}(X)$. Whether this tensor is the physical metric and whether its local cone is shared by matter, photons and tensors are separate Open gates. If certain phenomenological closures are expressed using X -dependent effective ordered-branch parameters, this describes the varying geometry, not a varying vacuum light speed. The equivalence principle is not derived by this scalar tangent-space construction; it is a separate metric/matter-vertex gate.

12.9 What is proved and what remains open

Strictly established inside the supplied scalar model (Level A mathematics). Residual $O(3)$ fixes the form of an independently supplied scalar principal tensor. When its independently supplied coefficients satisfy $\beta(\beta - \alpha) < 0$, that tensor is hyperbolic and defines one scalar characteristic cone. The corresponding flat scalar equation has Poincaré-form algebra, and after a clock/lapse map is supplied its dispersion may be written with $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$. Retarded support for the nonlinear scalar problem additionally requires the stated regularity and initial–boundary orientation. These results do not derive the realised physical branch, a universal observable metric or the photon, matter, fermion and tensor cones.

Conditional model algebra. The supplied compact-phase characteristic form may use the same ordered-branch tensor, with a separately supplied positive-energy Lorentzian orientation. This does not derive a physical phase action, the physical photon or $c_\gamma = c_{\text{char}}$.

Still open. A theorem-level derivation of the full gauge sector from the microscopic Euclidean theory. A theorem-level derivation of the precise suppression pattern of non-minimal gauge-sector operators beyond the minimal route. A complete analysis of higher-order Lorentz-violating corrections beyond the homogeneous limit.

In this sense, the supplied scalar model reproduces Lorentzian kinematics conditionally. Physical special relativity additionally requires clocks, rods, matter states and all observable sector cones.

Summary.

$$\begin{aligned} \text{homogeneous ordered phase} &\implies \text{unique scalar-branch cone } c_{\text{char}} \\ &\implies \text{conditional scalar Poincaré-form kinematics,} \end{aligned} \tag{12.11}$$

with the separate physical completion target

$$c_\gamma = c_{\text{char}} = c. \tag{12.12}$$

Having established the homogeneous scalar principal-operator result, we now turn to the separately supplied macroscopic gravity completion and its missing physical owners.

13 Conditional macroscopic gravity completion

Scope and status. The ordered branch supplies a scalar characteristic geometry, while Theorem 8.1 shows that the displayed fixed director map does not supply a linear TT or static-density vertex. The covariant scalar–tensor action used below is therefore a supplied Level-B completion ansatz. Its field equations are Level A algebra inside that ansatz; its identification as the physical ECT metric, its matter/photon universality and its tensor completion are Open.

The macroscopic problem is to identify a physical metric and action whose weak-field, radiative and cosmological sectors share one consistent source. Residual $O(3)$ symmetry fixes the scalar principal tensor but does not select a unique nonlinear metric completion. In particular, Noether conservation $\partial_A T^{AB} = 0$ does not create an interaction vertex, and the fixed-map orientation perturbation has no linear T^{ww} contact.

The action developed in this section is retained because it provides an explicit, testable scalar–tensor model. When F is constant and scalar derivatives vanish, its equations reduce algebraically to the Einstein form. That conditional limit does not prove that ECT dynamically reaches it, does not derive a tensor mode and does not choose the physical metric among the admissible completion families. Galactic HRC response is another augmented completion and must not be reused as a PES bath without a declared Wilsonian split and subtraction.

Status layers.

Level A: scalar characteristic tensor; exact variations of each explicitly supplied action; fixed-map negative theorem.

Level B: conditional scalar–tensor and HRC completion algebra, with all matching inputs declared.

Level C: fits or imported phenomenological benchmarks.

Open: physical metric, tensor action, $\kappa_n \rightarrow M_T^2 \rightarrow G_N$, finite-body scalar charge, and the ECT record state/noise kernel.

13.1 Effective metric and the macroscopic limit of the condensate

Status: Level A for the residual-symmetry form and signature classification of the supplied scalar principal tensor (Theorem 3.2). Its interpretation as a physical macroscopic metric is Level B/Open and requires a matter/clock/tensor completion. The position-dependent divided-equation coefficient introduced below belongs to the separate ϕ -first closure and is Level B; a measured gravitational response still requires its own transfer.

The purpose of this subsection is to identify the minimal rank-2 tensorial object that governs the causal coarse-grained branch of ECT. Given the supplied P4 ordered-sector chart and a declared long-wavelength domain, a future coarse-grained route may retain slow orientation variables after integrating out a positive isolated P4 amplitude pole, if such a pole and a controlled scale separation are derived. Their constrained physical content and projector are not fixed by this coarse-graining ansatz. However, this by itself does not yet determine whether the resulting coarse-grained description admits a geometric interpretation, nor which tensorial object should play that role. We now classify a restricted tensor basis. Given only a supplied director and the local algebraic symmetric rank-2 basis $\{\delta^{AB}, \tilde{n}^A \tilde{n}^B\}$, with no derivatives or additional tensors, residual symmetry leaves one two-coefficient form. In its conditional Lorentzian region this is a scalar characteristic-tensor candidate; it is not a unique coarse-grained or physical metric.

Conditional coarse-graining and the macroscopic regime

Let $m_{\text{P4,pole}}$ and $\xi_{\text{P4}} = m_{\text{P4,pole}}^{-1}$ denote quantities to be extracted from a future stationary P4 action and constrained Hessian. They are not fixed by the P3 homogeneous radial pole. Conditional on those inputs, choose

$$\xi_{\text{P4}} \ll \ell \ll L, \quad E \ll m_{\text{P4,pole}}. \quad (13.1)$$

For a supplied window W_ℓ , define the coarse-grained director

$$\bar{n}_A(X) = \frac{1}{\mathcal{N}_\ell} \int_{B_\ell(X)} d^4Y W_\ell(X - Y) n_A(Y). \quad (13.2)$$

where W_ℓ is a smooth window function localised at scale ℓ and $\mathcal{N}_\ell$ is the normalisation factor. Even when $n_A n^A = 1$ pointwise, its coarse average need not be unit. Define

$$\rho_\ell = (\bar{n}_A \bar{n}^A)^{1/2}, \quad \tilde{n}_A = \frac{\bar{n}_A}{\rho_\ell}, \quad \tilde{n}_A \tilde{n}^A = 1, \quad (\rho_\ell > 0). \quad (13.3)$$

Thus $\bar{n}_A \bar{n}_B$ is not an idempotent rank-one projector unless $\rho_\ell = 1$. The normalised director $\tilde{n}_A$ owns the projectors below; ρ_ℓ is a separate coarse order-parameter amplitude. At $\rho_\ell = 0$ there is no preferred coarse direction and this local broken-branch chart does not apply. Conditional on a supplied P4 pole and $\rho_\ell > 0$, the slow unit-director hierarchy is

$$|\partial_B \tilde{n}_A| \ll \xi_{\text{P4}}^{-1}, \quad |\partial_C \partial_B \tilde{n}_A| \ll \xi_{\text{P4}}^{-2}. \quad (13.4)$$

This is conditional EFT organisation. It neither derives the P4 pole nor proves that a scalar characteristic tensor is the physical metric.

Uniqueness of the macroscopic rank-2 tensor

On a supplied P4 patch where $\rho_\ell > 0$, the local algebraic directional rank-2 tensors built only from the declared inventory are generated by the Euclidean tensor δ_{AB} and the unit-director projector $\tilde{n}_A \tilde{n}_B$. Scalar coefficients may separately depend on ρ_ℓ and other residual invariants. By Theorem 3.2 (Appendix C), any symmetric rank-2 tensor compatible with the residual $O(3)$ symmetry must therefore have the form:

$$K^{AB} = \beta \delta^{AB} - \alpha \tilde{n}^A \tilde{n}^B. \quad (13.5)$$

Introducing the orthogonal projectors

$$P_{\parallel}^{AB} = \tilde{n}^A \tilde{n}^B, \quad P_{\perp}^{AB} = \delta^{AB} - \tilde{n}^A \tilde{n}^B, \quad (13.6)$$

this becomes:

$$K^{AB} = \beta P_{\perp}^{AB} - (\alpha - \beta) P_{\parallel}^{AB}. \quad (13.7)$$

This is the macroscopic version of the tensor-uniqueness result established in ECT Foundations (§3.3).

At this stage one should distinguish two logically separate claims. First, the residual symmetry fixes the admissible tensorial structure uniquely. Second, only in the hyperbolic Lorentzian regime does that structure define a viable scalar causal problem. Interpreting the same tensor as the observable metric requires independent clock, matter and tensor-sector owners.

No physical metric is introduced here as an additional primitive field. Rather, residual symmetry fixes the available scalar rank-2 principal tensor. Whether a completed ECT matter/tensor theory promotes it, possibly up to a conformal or disformal map, to a common physical metric remains Open.

Geometric realisation of the viable tensor structure

Since K^{AB} (13.7) controls the principal part of the supplied long-wavelength scalar equation, it defines that equation's characteristic cone. In the viable Lorentzian coarse-grained regime the corresponding inverse characteristic-metric candidate is:

$$g_{\text{eff}}^{AB} = \beta P_{\perp}^{AB} - (\alpha - \beta) P_{\parallel}^{AB}, \quad (13.8)$$

equivalently written as:

$$g_{\text{eff}}^{AB} = \beta \delta^{AB} - \alpha \tilde{n}^A \tilde{n}^B. \quad (13.9)$$

Because $P_{\perp}$ and $P_{\parallel}$ are orthogonal projectors, the covariant inverse metric is obtained exactly:

$$g_{AB}^{\text{eff}} = \frac{1}{\beta} P_{AB}^{\perp} - \frac{1}{\alpha - \beta} P_{AB}^{\parallel}, \quad g_{\text{eff}}^{AC} g_{CB}^{\text{eff}} = \delta^A_B. \quad (13.10)$$

At this stage the scalar characteristic metric is fixed only up to an overall positive conformal factor—sufficient for its scalar cone and relative longitudinal/transverse stiffness. This does not fix the normalisation or universality of a physical metric; later matching conditions are conditional tests (§13.4).

Lorentzian signature of the ordered phase

Choose the local ordered unit frame $\tilde{n}^A = (0, 0, 0, 1)$. Equation (13.8) then gives:

$$g_{\text{eff}}^{AB} = \text{diag}(\beta, \beta, \beta, -(\alpha - \beta)). \quad (13.11)$$

The displayed tensor is Lorentzian exactly when $\beta(\beta - \alpha) < 0$. On the retained $\beta > 0$ component this is $\alpha > \beta > 0$. The macroscopic programme retains this same scalar condition. It may separately impose $\alpha < 4\beta$ as the optional arithmetic-average benchmark of Section 3.5, but that extra inequality is not a Lorentzian or macroscopic-viability theorem. At the benchmark coefficients $\alpha = 2, \beta = 1$: $g_{\text{eff}}^{AB} = \text{diag}(1, 1, 1, -1) = \eta^{AB}$ (Level A in EFT).

This is the precise scalar-model form of a Lorentzian principal direction. P4 adopts the physically relevant nonzero-vector branch and supplies its analysis chart; the bare P3 action does not generate it. A physical time coordinate further requires the clock/congruence map and is not derived by the sign change alone.

Defining $\hat{c}_*^2 = \beta/(\alpha - \beta)$ (3.19) and writing $x^0 = w$, the common-length principal hyperbolic operator is:

$$g_{\text{eff}}^{AB} \partial_A \partial_B = \beta \left(\nabla^2 - \frac{1}{\hat{c}_*^2} \partial_0^2 \right) \propto \partial_0^2 - \hat{c}_*^2 \nabla^2. \quad (13.12)$$

For the supplied scalar operator this hyperbolic structure underlies the scalar retarded Green functions of Appendix F and one scalar characteristic cone. It does not own the physical tensor sector of §13.2.

Null structure and causal cones

The effective null cones of the macroscopic branch:

$$g_{\text{eff}}^{AB} k_A k_B = 0 \implies \beta \mathbf{k}^2 - (\alpha - \beta) k_w^2 = 0 \implies k_0^2 = \hat{c}_*^2 \mathbf{k}^2, \quad \omega^2 = c_{\text{char}}^2 \mathbf{k}^2 \text{ only after Eq. (3.23).} \quad (13.13)$$

The displayed null surface is the characteristic cone of the supplied scalar principal operator. Retarded scalar response follows only after an initial–boundary orientation is supplied. P4 does not by itself reconstruct physical causal order, a clock/lapse, a universal metric or the cones of other sectors. Analogue-gravity systems provide useful conditional comparisons for such characteristic geometry [90, 20].

Phenomenological ϕ -first closure of the macroscopic branch

The uniqueness of the form in the declared local algebraic scalar principal-symbol basis $\{\mathcal{S}^{AB}, n^A n^B\}$ is a strict Level A result. It does not cover additional fields, derivatives of n^A or nonlocal operators; its interpretation as a physical effective metric remains Open. By contrast, the position-dependent quantity introduced through a macroscopic amplitude variable is the divided-equation coefficient $G_{\text{div}} = G_*/F$. It is Level A algebra inside the supplied Level-B ordered-branch closure, but it is not an effective Newton response until the scalar range, canonical coupling, body charges and observable map are solved.

To parameterise slow background variations of the ordered branch, define the logarithmic coordinate of the macroscopic amplitude u by

$$\phi_u(X) \equiv \frac{1}{\beta_\phi} \ln \frac{u(X)}{u_\infty}, \quad \phi \equiv \phi_u, \quad \phi|_{u=u_\infty} = 0. \quad (13.14)$$

This logarithmic chart is defined only on an ordered patch with $u/u_\infty > 0$; it is singular at the formation point $u = 0$, which must be described in u or another regular variable. Thus $u/u_\infty = e^{\beta_\phi \phi}$ is a coordinate identity on that patch. The further identification of this ratio with the coefficient of the macroscopic curvature term is an independent Level B matching assumption,

$$F_u(\phi_u) = e^{\beta_\phi \phi_u}, \quad G_{\text{div}}(\phi_u) \equiv \frac{G_*}{F_u(\phi_u)} = G_* e^{-\beta_\phi \phi_u}. \quad (13.15)$$

Here $u(X)$ is the ordered-branch amplitude introduced in Section 3.2; u_∞ is its reference value; and β_ϕ is a dimensionless closure parameter, distinct from the kinetic coefficient β in K^{AB} and from the fifth-force coupling β_5 .

The quantity $\Delta_K \equiv \alpha - \beta$ used in the exploratory galactic appendix is instead an anisotropy/stiffness coordinate. Its logarithmic coordinate will be denoted $\psi_K \equiv \ln(\Delta_K/\Delta_{K,\text{vac}})$. No map $u \leftrightarrow \Delta_K$, and hence no map $\phi_u \leftrightarrow \psi_K$, has been derived from the present action. Formulas written in these two coordinates are therefore not interchangeable. When a statement is independent of that unresolved identification, a generic scalar–tensor coordinate q will be used.

Equations (13.14)–(13.15) define the current amplitude-dominated macroscopic closure; they are not a first-principles theorem from bare P3. The matching $F_u = e^{\beta_\phi \phi_u}$ and hence $G_{\text{div}} = G_*/F_u$ are closure assumptions. The latter is the matter coefficient of the completely divided scalar metric equation; neither a Cavendish nor a cosmological response follows without its own transfer problem.

What is established and what is not

Established (Level A mathematics in the supplied broken-phase scalar EFT):

- (i) Residual $O(3)$ fixes the form of the supplied symmetric scalar principal tensor.
- (ii) For independently supplied coefficients satisfying $\beta(\beta - \alpha) < 0$, that tensor is hyperbolic and defines one scalar characteristic cone; on the retained $\beta > 0$ component the premise is $\alpha > \beta > 0$.

- (iii) The label “characteristic metric” is convenient inverse-principal- symbol notation inside this scalar problem; it is not yet a universal observable metric.

Not yet established:

- (i) P4 adopts the physically relevant ordered branch but does not select $\beta(\beta - \alpha) < 0$, supply its stationary action/state, or derive a clock/lapse map.
- (ii) A physical universal metric, conformal normalisation, matter coupling universality and the photon, fermion and tensor cones require independent actions and matching.
- (iii) $G_{\text{div}}(X) = G_* e^{-\beta_\phi \phi_u}$ is a closure-level divided coefficient, not a first-principles theorem or an observable gravitational response.

The Level A statement is therefore the conditional algebraic classification of a supplied scalar principal tensor; physical metric and causal structure remain Open.

Physical interpretation

P4 distinguishes one Euclidean direction. If a physical clock, metric and matter-response completion is later derived, the scalar characteristic tensor may become one ingredient of its long-wavelength description. The current fixed-director construction does not itself establish that identification.

Summary

The supplied scalar principal tensor can be written

$$g_{\text{eff}}^{AB} = \beta (\delta^{AB} - \tilde{n}^A \tilde{n}^B) - (\alpha - \beta) \tilde{n}^A \tilde{n}^B. \quad (13.16)$$

Theorem 3.2 fixes this residual- $O(3)$ form. For $\beta(\beta - \alpha) < 0$ it is hyperbolic and defines one scalar characteristic cone, up to normalisation. The subsequent subsections keep four logically separate tasks: a physical-tensor completion, the conserved-current inventory versus the missing interaction vertex, weak-field normalisation inside a supplied gravity action, and nonlinear/phenomenological tests. None follows from this scalar characteristic tensor alone.

13.2 Conditional tensor completion and propagation tests

Status: Open for ECT ownership. The director equation below is a massless scalar/longitudinal orientation equation. The Fierz–Pauli equation is retained as a standard target for a separately derived physical tensor.

The minimal orientation EFT may contain gapless variables obeying

$$\partial_t^2 \delta n_i - c_n^2 \nabla^2 \delta n_i = 0, \quad \omega^2 = c_n^2 k^2. \quad (13.17)$$

Gaplessness and the index i do not imply helicity 2. Under the current integrability constraint the director fluctuation is longitudinal, and Theorem 8.1 gives $h_{ij}^{\text{TT}} = h_{\text{ww}} = 0$.

For comparison, if a new physical symmetric tensor $H_{\mu\nu}$ is supplied, the standard massless quadratic target can be written

$$S_{\text{FP}}[H] = -\frac{M_T^2}{8} \int d^4x [(\partial_\lambda H_{\mu\nu})^2 - 2\partial_\mu H^{\mu\nu} \partial^\lambda H_{\lambda\nu} + 2\partial_\mu H^{\mu\nu} \partial_\nu H - (\partial_\lambda H)^2]. \quad (13.18)$$

Its gauge quotient has two TT modes, but neither $H_{\mu\nu}$ nor this action has been obtained from the fixed director map. The scale M_T is therefore a conditional tensor-EFT parameter and the possible coefficient relation $M_T^2 \stackrel{?}{=} c_M^{(n)} \kappa_n$ is Open (the dimensionless matching coefficient $c_M^{(n)}$ is unrelated to the tensor propagation speed below).

Even a separately supplied internal $SU(2) \times U(1)$ redundancy would not imply linearised diffeomorphism redundancy. Likewise, the Deser/Lovelock route states that a healthy universal massless spin-2 seed has an Einstein-type nonlinear completion under its standard assumptions; it does not manufacture the seed, source vertex or universal metric. These remain explicit closure gates.

Physical completion checklist. A candidate must provide: (i) the constrained tensor Hessian and positive residues; (ii) a nonzero projected $H_{\mu\nu}T^{\mu\nu}/2$ vertex; (iii) two TT modes or a fully tested alternative spectrum; (iv) the tensor principal cone; (v) nonlinear Bianchi consistency and universal matter coupling. Only after these steps may M_T be matched to $\bar{M}_{PI}$ and the completion compared with GW/PPN observations.

Detailed linear-completion protocol

Director equation versus helicity. The wave equation for δn_i establishes only that the reduced orientation EFT can possess gapless disturbances. A spatial index is not a helicity label. Under the strict pullback relation $u_0\delta n_i = \partial_i\chi$, the disturbance is generated by one scalar potential; its Fourier polarisation is parallel to $\mathbf{k}$. The fixed metric map then produces a longitudinal mixed component h_{iw} and no h_{ij}^{TT} . This is the precise reason that the former inference “gapless director $\Rightarrow$ massless graviton” fails.

What P5 does and does not add. P5 motivates, but does not itself create, a local compact-phase bundle. A separately supplied local compact-phase redundancy can organise a gauge description of the phase sector. It does not create the linearised diffeomorphism redundancy of a symmetric tensor, does not add the missing TT projector and does not turn the orientation triplet into a gauge field. Any tensor gauge redundancy must be derived from the constrained tensor action itself.

Comparison with Einstein–aether models. Einstein–aether theory begins with a Lorentzian metric, a unit timelike vector and a covariant action containing all allowed two-derivative operators. Its spin-2, spin-1 and spin-0 speeds follow only after the complete quadratic action is diagonalised. ECT currently has a scalar characteristic tensor and a director variable but no analogous covariant orientation action. The comparison therefore supplies a calculation template, not numerical ECT speeds or residues.

Conditional FP constraints. For the supplied $H_{\mu\nu}$ target, variation with respect to the temporal components gives the standard constraints. Together with the gauge quotient these leave two TT modes and a positive quadratic Hamiltonian when the sign of M_T^2 is positive. A candidate ECT completion must reproduce this result or state explicitly which alternative degrees of freedom remain and why they are healthy. Merely matching its dispersion relation to $\omega^2 = c_{\text{char}}^2 k^2$ is insufficient.

Deser and Lovelock route. If the preceding physical seed and a universal conserved vertex are derived, the Deser self-coupling argument provides a controlled route from the linearised tensor theory to an Einstein-like nonlinear action. Lovelock uniqueness further restricts local second-order metric equations in four dimensions. Both statements are conditional on assumptions that the present director map has not met. They remain developed closure routes rather than a derivation from P1–P6.

Linear falsifiers. The first inexpensive gates for any proposed tensor owner are: a nonzero static-density projection; a nonzero TT residue; absence of negative-norm and gradient-unstable poles; a principal cone compatible with GW170817; and a matter/photon metric compatible with weak-equivalence and PPN tests. Failure of one gate rejects that proposed owner without rejecting the already derived scalar characteristic branch.

13.3 Matter coupling and the Noether source structure

Status: the P3 canonical translation identity is Level A mathematics and becomes a conservation law only on solutions of its owning P3 action. The P4 gradient datum is generally not such a solution. A physical covariant metric stress, P4 source law, coarse-grained conservation equation and universal matter coupling require their own action, equations, projection and matching and remain Open (OP2, OP-c1, OP-matter). Migrated and expanded from: sec:noether, Appendix R, and material deferred from Part I.

The previous subsections established a scalar characteristic structure and stated a conditional tensor-completion target; they did not derive a physical tensorial propagation sector. The question now is: *what plays the role of the gravitational source in ECT?* In general relativity the answer is the stress-energy tensor, which is postulated to appear on the right-hand side of the Einstein equations. In ECT the scalar ordered branch is not postulated from the outset as a physical spacetime theory. It must therefore be shown separately that a physical metric/tensor completion exists and that its source structure arises *variationally* from an owned physical action. Its descent from the same microscopic condensate is an Open common-origin hypothesis.

The variational origin of conserved source currents

At the microscopic level the ECT action is local and does not depend explicitly on the coordinates X^A of the Euclidean arena (P1, P3). By Noether's theorem applied to translational invariance [47], this yields a conserved rank-2 current for the scalar sector:

$$\tau_{P3}^{AB} = \frac{\partial \mathcal{L}_\Phi}{\partial (\partial_A \Phi)} \partial^B \Phi - \delta^{AB} \mathcal{L}_\Phi, \quad \partial_A \tau_{P3}^{AB} = - \frac{\delta \mathcal{J}_E^{P3}}{\delta \Phi} \partial^B \Phi, \quad T_{P3, \text{phys}}^{AB} = A_{P3} \tau_{P3}^{AB}. \quad (13.19)$$

Both reduced and physical divergences vanish only on the same P3 shell for constant $A_{P3} > 0$; their normalisation is independent of the path-integral ratio κ_{P3} . The complete form including translational and rotational currents is derived in Appendix R. The P3 identity does not become $\nabla_\mu T^{\mu\nu} = 0$ merely by renaming w as a time coordinate. That covariant law requires a diffeomorphism-owned Lorentzian action, its metric stress tensor and the complete on-shell equations. These are not provided by the current P4 datum. Thus only the scoped P3 on-shell Noether identity is Level A conditional mathematics; the physical P4/macroscale covariant source law is Open.

Caveat: time-dependent backgrounds. A time-dependent solution need not be invariant under time translations, but that fact alone does not break a time-translation symmetry of its owning action or destroy the corresponding Noether charge. Absence of a conserved global energy requires an additional obstruction—for example, a completed Lorentzian geometry with no timelike Killing vector, explicit time dependence in the owner, nonconserving boundary flux, or an open subsystem. Whether an ECT cosmological completion has such an obstruction is Open; it cannot be inferred from the P4 datum or from background time dependence alone [91].

From microscopic current to macroscopic source

The tensor (13.19) is defined at the microscopic level. The macroscale branch requires a coarse-grained version. Applying the same window function (13.2):

$$\bar{\tau}_{P3}^{AB}(X) = \frac{1}{\mathcal{N}_\ell} \int_{B_\ell(X)} d^4Y W_\ell(X-Y) \tau_{P3}^{AB}(Y), \quad \bar{T}_{P3, \text{phys}}^{AB} = A_{P3} \bar{\tau}_{P3}^{AB}. \quad (13.20)$$

For a translation-invariant window, averaging the reduced P3 identity gives

$$\partial_A \bar{\tau}_{P3}^{AB} = - \frac{\delta \mathcal{J}_E^{P3}}{\delta \Phi} \partial^B \Phi \quad \text{plus declared boundary/window terms.} \quad (13.21)$$

It vanishes only for a supplied P3 on-shell ensemble and compatible boundary conditions. This does not establish conservation of a physical P4 macroscopic source or a covariant gravitational stress tensor.

Physically, the macroscopic observer does not measure microscopic gradients of Φ directly. What is reconstructed are long-wavelength energy densities, momentum fluxes, and stress-like components — the hydrodynamic analogue of the microscopic Noether current.

Candidate orientation sector and the missing macroscopic stress

The reduced flat-background sigma-model EFT contains orientation variables, but it does not by itself supply a covariant macroscopic stress tensor. For bookkeeping one may retain the schematic reduced-EFT expression

$$T_{AB}^{(n),\text{EFTcand}} = K_n [(\nabla_A n_C)(\nabla_B n^C) - \frac{1}{2} g_{AB}(\nabla_C n_D)(\nabla^C n^D)] + \lambda_n n_A n_B + \Delta_{AB}^{(n)}. \quad (13.22)$$

It is a candidate reduced-EFT object, not the ECT gravitational source. Establishing a macroscopic orientation stress requires a generally covariant $S_n[g, n, \lambda_n]$, its metric and orientation variation, the integrability/unit constraints, and an on-shell projection:

$$T_{\mu\nu}^{(n)} = -\frac{2}{\sqrt{-g}} \frac{\delta S_n}{\delta g^{\mu\nu}}.$$

Those inputs are Open (OP2, OP-c1).

Rotational currents and the role of $O(4)$ symmetry

From $O(4)$ -invariance (P2), Noether's theorem gives rotational currents:

$$J^{C,AB} = X^A T^{CB} - X^B T^{CA} + S^{C,AB}, \quad \partial_C J^{C,AB} = \text{Euler-Lagrange terms}, \quad \partial_C J^{C,AB} = 0 \quad \text{on the owning shell}, \quad (13.23)$$

Here $S^{C,AB}$ is the spin part (zero for the P3 scalar). Admissible boundary conditions are additionally needed for differentiable generators and conserved integrated angular charges. A separately supplied orientation action can have its own rotational current; P2 and the P4 datum do not create one or a macroscopic stress without that owner.

The distinction between translational and rotational Noether structures must be kept precise. The conservation $\partial_A T^{AB} = 0$ comes from *translation* invariance, not from $O(4)$ rotations. The $O(4)$ symmetry constrains the tensorial form of the ordered phase and permits candidate anisotropic operators. It does not by itself produce a covariant macroscopic orientation stress.

Candidate anisotropic source corrections

A covariant orientation extension may contain anisotropic curvature operators. For later bookkeeping define the candidate tensor structure

$$\mathcal{X}_{\mu\nu}^{\text{cand}}[n] \equiv c_1 \left(n_\mu n^\lambda R_{\lambda\nu} + n_\nu n^\lambda R_{\lambda\mu} - \frac{1}{2} g_{\mu\nu} n^\alpha n^\beta R_{\alpha\beta} \right). \quad (13.24)$$

This candidate tensor form is not the orientation stress tensor, nor does a Riemann-commutator manipulation by itself derive its coefficient or completion. A consistent use requires a specified generally covariant action $S_n[g, n]$, the unit/integrability constraints, and

$$T_{\mu\nu}^{(n)} = -\frac{2}{\sqrt{-g}} \frac{\delta S_n}{\delta g^{\mu\nu}}.$$

The existence of the macroscopic orientation action, the complete tensor, its projection, and the matching of c_1 are therefore Open (OP2, OP-c1). The scalar-only field equation below must not contain $\mathcal{X}_{\mu\nu}^{\text{cand}}$ unless such an S_n is added and varied.

Why conservation does not supply a gravitational vertex

The coarse-grained Noether current is conditionally conserved bookkeeping data of the translated P3 theory only under the stated on-shell ensemble, window and boundary assumptions. Conservation alone does not determine an interaction term. A gravitational source requires an explicit physical field $H_{\mu\nu}$ and an action containing, at linear order, a nonzero projected vertex such as $H_{\mu\nu}T^{\mu\nu}/2$ (or a declared alternative), together with its constraints and normalisation.

For the displayed fixed director map, at fixed α, β and first order in a tangent perturbation about homogeneous $n = w$, the candidate static contraction vanishes:

$$T^{AB}h_{AB} = \rho h_{ww} = 0 + O(\delta n^2).$$

Thus only the current linear director route terminates at MISSING VERTEX. The exact quadratic δg_{ww} and its source-convention dependence are recorded in Section 33.3; they are PARAMETRIC ONLY, not a linear vertex or a derived gravity owner. The existence of T^{AB} , its long-wavelength conservation and the existence of a rank-2 characteristic tensor do not close that gap. In a separately supplied diffeomorphism-invariant scalar–tensor or tensor completion, variation of the declared matter action can provide a covariantly conserved source; that is conditional completion algebra, not a consequence of Euclidean translation invariance alone.

Universality of matter coupling: current status

A stronger claim is that *all* slow matter sectors couple universally to the same effective metric $g_{\mu\nu}^{\text{eff}}$. This is the analogue of the weak equivalence principle. Within the present formulation it is a working assumption (A4):

Within a separately supplied universal matter completion and its declared domain, all included species couple to one physical metric $g_{\mu\nu}^{\text{phys}}$.

Coarse-graining away phase information does not imply that all retained sectors share one metric: different species can inherit different principal symbols, vertices or disformal factors. Universality is therefore an explicit completion assumption and an equivalence-principle test, not a consequence of coarse graining. Deriving it from an ECT action for specific matter sectors remains Open (OP-matter).

What is established is a Noether tensor of the supplied action. Its coupling as the source of a physical TT metric, and universality for arbitrary matter sectors, are Open.

Physical interpretation

The Noether construction motivates the adopted Level-B/Open programme hypothesis $H_{N \rightarrow G}$: a future macroscopic source tensor should arise by a declared metric-response and coarse-graining map from a stationary full ordered-sector action, with background subtraction, state and matching fixed. The displayed P3 Noether tensor is not itself that universal matter-to-gravity vertex, and the withdrawn off-shell P3 density/pressure proxies are not physical cosmological sources. At fixed α, β and tangent-director linear order about homogeneous $n = w$, the present fixed map has no linear TT/static-density seed; quadratic, nonlinear, composite and additional-field routes remain parametric/Open.

A macroscopic observer reconstructs the familiar language of gravity: metric perturbations respond to stress-energy; causal propagation is governed by the effective Lorentzian metric; matter distributions shape the geometry of low-energy probes. In ECT this is a completion target constrained by variational and symmetry properties, not a reconstruction already completed from P1–P6.

What is established and what is not

Established (Level A):

- (i) The P3 off-shell translation identity and vanishing local divergence on P3 shell, Eq. (13.19); integrated charges additionally require admissible boundary data.
- (ii) A coarse-grained P3 current is conserved only for a P3 on-shell ensemble with compatible translation-invariant window and boundary terms, Eq. (13.21).
- (iii) Conservation supplies a rank-2 Noether current, but no gravitational interaction vertex follows; the fixed- α, β tangent-director linear route about homogeneous $n = w$ has a MISSING VERTEX, while quadratic/nonlinear/composite routes remain parametric/Open.
- (iv) Rotational currents $J^{C,AB}$ exist. Whether they induce a covariant macroscopic anisotropic stress is Open and requires $S_n[g, n]$.

Open or conditional:

- (i) Full universality of matter coupling is Level B (OP-matter).
- (ii) Explicit nonlinear source structure beyond linear EFT: Section 13.5.
- (iii) Coefficient c_1 not yet derived from first principles (OP-c1). In particular, determine whether the completed tensor/orientation closure yields phenomenologically viable functions $\mu(k, z)$, $\eta(k, z)$, and $\Sigma(k, z)$ on cosmological linear scales, and whether their sign and scale dependence can be simultaneously reconciled with weak lensing, redshift-space distortions, CMB lensing, cluster lensing, and the Hubble-sector benchmark.
- (iv) Matching of K_n, λ_n to bare P3 condensate parameters (OP2).

Summary

The supplied P3 action provides off-shell translational and rotational Noether identities whose local divergences vanish on its equations of motion. Differentiable generators and conserved integrated charges additionally require admissible boundary or fall-off data. Coarse graining preserves the corresponding local or integrated statement only for a declared on-shell ensemble and compatible window/boundary terms. The P4 profile is not on the P3 shell, and neither a physical macroscopic source nor an anisotropic stress follows until a generally covariant P4/orientation action is supplied and varied. Thus this section inventories conditional Noether currents; the source vertex for physical gravity and the Newton normalisation remain Open matching problems.

13.4 Conditional tensor normalisation and Newton matching

The ordered scalar/orientation EFT contains a coefficient with mass dimension two,

$$\kappa_n = \mathcal{C}_n u_0^2. \quad (13.25)$$

It controls the operator in which it is actually derived. Because the fixed director image has no linear TT/static-density seed, κ_n cannot be identified with a gravitational Planck coefficient by dimension alone.

For a separately supplied tensor completion define

$$M_G^2 \equiv M_T^2 \quad (\text{conditional tensor-action coefficient}), \quad (13.26)$$

and make the standard weak-field match

$$G_N^{\text{nat}} = \frac{1}{8\pi M_G^2}. \quad (13.27)$$

This equation is valid inside that supplied EFT; it is not a P1–P6 derivation of G_N .

The orientation estimate

$$\kappa_n = \mathcal{C}_n u_0^2 \quad (13.28)$$

and the often-used dimensional candidate

$$M_G^2 \stackrel{?}{=} c_M \kappa_n \quad (13.29)$$

must therefore remain an Open matching hypothesis. The actual dependency is

$$(u_0, \alpha, \beta, \dots) \longrightarrow \kappa_n, \quad \boxed{\kappa_n \longrightarrow M_G^2 \text{ Open}}, \quad M_G \xrightarrow{\text{match}} G_N. \quad (13.30)$$

Likewise the hierarchy

$$m_{\Phi, \text{rad}}, \quad m_{\text{P4, pole}}, \quad \sqrt{u_0}, \quad \kappa_n^{1/2}, \quad M_G \quad (13.31)$$

contains quantities with comparable mass dimension but distinct operator owners. Here the P3 mass is owned, whereas $m_{\text{P4, pole}}$ is Open; no constrained tensor Hessian or residue calculation presently relates them.

Detailed normalisation and matching audit

Operator ownership. The coefficient κ_n belongs to the orientation operator from which it was derived. A tensor Planck coefficient belongs to the kinetic term of a physical symmetric tensor after constraints and canonical normalisation. Although both coefficients have mass dimension two, their equality is an additional dynamical statement. It can be tested only by deriving the physical pole and its residue from one common reduced Hessian.

Definition versus prediction. Writing $M_G^2 = M_T^2$ defines the normalisation of a supplied tensor EFT. The weak-field convention

$$G_N^{\text{nat}} = \frac{1}{8\pi M_G^2}$$

then follows by comparing that action with the Einstein–Hilbert normalisation. This is exact matching algebra inside the supplied theory, not a prediction of G_N from condensate parameters. In particular, using the measured G_N to infer $M_G = \bar{M}_{\text{Pl}}$ cannot subsequently be counted as an independent ECT explanation of the Planck scale.

Candidate microscopic chain. A genuine derivation would have to execute

$$(u_0, \alpha, \beta, Z_u, \mathcal{C}_n, \dots) \longrightarrow \mathcal{H}_{\text{constrained}} \longrightarrow (H_{\mu\nu}^{\text{phys}}, Z_T, M_T) \longrightarrow G_N.$$

The first arrow includes solving the unit and integrability constraints; the second includes mode projection and canonical normalisation; the third is the standard weak-field match. At present only the orientation coefficient and the last conditional match are available. The middle tensor-owner arrow is Open.

Radial-mode contributions. Eliminating a supplied heavy field that has an explicit $(\partial n)^2$ vertex can generate local operators and shift $\mathcal{C}_n$; the present P4 model supplies neither ingredient. The resulting Wilson coefficient must be evaluated with a declared regulator and subtraction scheme. Even a finite positive shift of κ_n would still require a representation map to a TT tensor residue. Conversely, retaining the radial mode explicitly produces a scalar pole whose coupling is trace-like; it cannot be relabelled as M_G without double counting the mode already absorbed into the low-energy coefficients.

Scale hierarchy. The symbols m_σ , u_0 , $\kappa_n^{1/2}$ and M_G denote different objects: respectively the P3 homogeneous Gaussian mass, the supplied P4 directional-datum amplitude, an orientation stiffness and a conditional tensor-action coefficient. A physical P4 radial pole is a separate Open quantity. Numerical proximity or dimensional similarity is not an identification. Each equality in a proposed hierarchy must state whether it is a definition, a fitted match, a loop result or an Open hypothesis.

Falsification and closure data. To close the normalisation chain one must provide the constrained quadratic action, the physical tensor projector, the pole residue, the matter vertex and the map to the metric measured by rods, clocks and photons. Solar-System PPN coefficients, binary-pulsar damping and GW propagation then test the same normalisation. A coefficient that fits Newton’s law but fails one of those linked gates is not a viable gravitational owner.

13.5 Nonlinear closure and the generalised field equations

Status: conditional completion / Open ECT ownership. A nonlinear metric theory becomes necessary only after a physical metric, propagating tensor Hessian and matter source vertex have been supplied. The fixed-director map derives none of those three owners: it supplies only scalar principal-tensor algebra and has no linear transverse–traceless/static-density vertex. The ϕ reparameterisation and the functions $F(\phi)$, $\omega(\phi)$ and $U(\phi)$ belong to a separately supplied scalar–tensor completion. Lovelock uniqueness is Level A mathematics only under assumptions A1–A5; descent of the full gravity action from bare P3 remains Open (OP2, OP3, OP-matter).

Sections 13.1–13.4 therefore provide an inventory of three completion targets—a physical metric, propagating tensor sector and conserved matter vertex—rather than an ECT derivation of them. The central question of this subsection is therefore: *what is the strongest nonlinear gravitational structure currently justifiable within the ECT framework?*

Conditional nonlinear-completion programme

A nonlinear metric completion is not forced by the supplied P4 datum. If a future theory independently supplies (i) a physical metric, (ii) a healthy massless or otherwise viable tensor sector, (iii) a universal matter source vertex and (iv) locality/gauge assumptions that require self-coupling, then finite-amplitude consistency becomes a nonlinear completion problem. Standard Deser/Lovelock routes may constrain that named completion under their own hypotheses. This is conditional Level B/Open programme logic, not a Level A consequence of P1–P6 or of a geometric branch whose owners are still missing.

Why the closure is not taken to be pure GR immediately

The geometric branch is not purely metric in origin. It emerges from an ordered condensate that carries additional collective information at the macroscopic level:

- the orientation field n_A ,
- the amplitude-derived order variable $\phi = \frac{1}{\beta_\phi} \ln(u/u_\infty)$ describing slow variations of the degree of condensate ordering,
- candidate anisotropic tensor structures tied to n_A ; an orientation stress requires a separately specified covariant $S_n[g, n]$ and is Open.

One supplied scalar–tensor ansatz, optionally enlarged by an action-owned orientation sector, is a useful proof-of-class completion. It is not the most general local or nonlocal closure: higher derivatives, additional fields, different constraints and nonlocal response remain possible. The ϕ -first model is therefore a named conditional ansatz, not a class selected or narrowed uniquely by P1–P6.

Why the same variational principle yields different equations

Hilbert’s variational derivation of the Einstein equations was mathematically complete for the Einstein–Hilbert action: varying $S_{EH} = \frac{1}{16\pi G} \int \sqrt{-g} R d^4x + S_{\text{matter}}$ yields exactly the Einstein field equations, because the action contains only the metric and minimally coupled matter fields. In the present ECT manuscript, the displayed macroscopic action is the amplitude-dominated scalar action. Varying it gives eq. (13.38). A possible orientation completion would require a separate covariant $S_n[g, n]$ and would

add the stress from its own variation; it is not included in eq. (13.38). The difference from pure GR in the displayed closure is therefore scalar field content and non-minimal coupling. GR is recovered only algebraically when the scalar is frozen; dynamical reachability is Open.

From the ordered-branch amplitude to the ϕ -field

The ordered-branch infrared EFT introduced in Section 3.2 already identifies two distinct collective variables of the condensate: the amplitude $u(x)$ and the orientation field $n_A(x)$. At macroscopic scales these variables need not enter symmetrically. The causal tensor structure of the geometric branch is controlled primarily by the ordered orientation sector, whereas slow departures from homogeneous gravitational response are naturally carried by the condensate amplitude.

This motivates the following organisational step. Let u_∞ denote the declared asymptotic reference value of the ordered branch in the homogeneous large-scale limit. It is then natural to introduce a dimensionless amplitude variable

$$\phi(x) \equiv \frac{1}{\beta_\phi} \ln \frac{u(x)}{u_\infty}. \quad (13.32)$$

On an ordered patch with $u/u_\infty > 0$, construction gives

$$\phi = 0$$

at the frozen reference configuration, while $\phi \neq 0$ measures a local departure of the condensate amplitude from that background. The chart does not contain the formation point $u = 0$; any derivation through formation must return to u or another regular coordinate.

Several points are important.

(i) ϕ is not a new fundamental field. It is a macroscopic reparameterisation of the ordered-branch amplitude. Thus the ϕ -first closure is not introducing an extra scalar by hand; it is choosing a convenient infrared variable for the amplitude sector already present in the ordered condensate EFT.

(ii) Why a logarithmic variable is natural. The logarithmic form makes the frozen-reference limit multiplicative rather than additive, and it turns local relative changes of the condensate response into additive fluctuations of ϕ . It also makes exponential factors such as $e^{\beta_\phi \phi} = u/u_\infty$ appear naturally in the later macroscopic closure.

(iii) Candidate scalar carrier for a macroscopic ansatz. The ordered-branch EFT contains an amplitude coordinate $u(x)$, while other scalar coordinates or composites can arise in supplied reductions. Declaring u (or its logarithm) as the carrier of a scalar–tensor closure is a model-building ansatz; no coarse-graining, continuation or matching theorem shows that it modulates physical gravity.

(iv) Status. The existence of an ordered-branch amplitude variable is part of the collective EFT structure of Section 3.2. The specific nonlinear macroscopic closure written in terms of ϕ remains Level B. This subsection supplies only a change of variable and a declared candidate carrier. It does not provide the missing physical bridge between the Euclidean amplitude sector and gravity.

Scalar-only macroscopic reduction

The current macroscopic closure retains the condensate-amplitude coordinate as its scalar variable:

$$S_{\text{ord,E}}^{\text{phys}}[u, n] = A_{\text{ord,E}} \mathcal{J}_{\text{ord,E}}[u, n] \longrightarrow S_{\text{macro}}^{\text{scalar}}[g, \phi], \quad \phi = \frac{1}{\beta_\phi} \ln \frac{u}{u_\infty}. \quad (13.33)$$

This is a declared scalar reduction. It neither freezes nor absorbs a macroscopic orientation stress. The ordered branch supplies a kinematic preferred direction, while a gravitational orientation sector is a separate Open extension requiring $S_n[g, n]$ and its full variation.

What is being assumed and what is not. This reduction does not yet assume the explicit ϕ -first action (13.35). It only declares that the amplitude sector is retained as the scalar coordinate in this first macroscopic ansatz. The detailed scalar functions $F(\phi)$, $\omega(\phi)$, and $U(\phi)$ remain Level-B matching inputs. A covariant coupling of n_A to the macroscopic field equation is Open.

Consequence. The nonlinear macroscopic closure used below is a scalar–tensor ansatz motivated by the amplitude sector of the ordered branch. Its detailed matching to the microscopic condensate remains part of the Level-B closure.

From ordered-branch EFT coefficients to closure templates

The amplitude-dominated reduction identifies the ordered-branch amplitude as a candidate scalar carrier for a first macroscopic closure. The exact result available at this stage is narrower: a change of variable inside the supplied Euclidean amplitude subsector produces Euclidean kinetic and potential templates. It does not derive the Lorentzian scalar–tensor coefficient functions

$$F(\phi), \quad \omega(\phi), \quad U(\phi).$$

Those functions require a coarse-graining, continuation and matching map that is not owned by the displayed microscopic action.

The starting point is the Euclidean ordered-branch functional (3.9),

$$\mathcal{J}_{\text{ord,E}}[u, n] = \int d^4X \left[\frac{Z_u}{2} (\partial u)^2 + \frac{\mathcal{C}_n}{2} u^2 (\partial n)^2 + W(u) + \Lambda_n (n_A n_A - 1) + \dots \right].$$

After amplitude-dominated coarse-graining and the logarithmic reparameterisation

$$\phi = \frac{1}{\beta_\phi} \ln(u/u_\infty), \quad u = u_\infty e^{\beta_\phi \phi},$$

three structural correspondences become natural.

(i) Candidate gravitational response matching. The amplitude ratio is a natural dimensionless candidate, so the present closure *supplies* the matching

$$F(\phi) \equiv \frac{f(\phi)}{\bar{M}_{\text{Pl}}^2} = \frac{u}{u_\infty} = e^{\beta_\phi \phi}.$$

Only the last equality is a coordinate identity. Neither the logarithmic reparameterisation nor the displayed Euclidean amplitude functional derives the coefficient $f(\phi)$ of a Lorentzian curvature term. That action-level descent and its normalisation remain Level B/Open matching inputs.

(ii) Exact Euclidean kinetic template. The amplitude-gradient term in the ordered-branch EFT transforms as

$$\frac{Z_u}{2} (\partial u)^2 = \frac{Z_u \beta_\phi^2 u_\infty^2}{2} e^{2\beta_\phi \phi} (\partial \phi)^2.$$

This is an exact coordinate identity inside the supplied Euclidean amplitude functional. It defines a candidate Euclidean coefficient template, but it is not the normalised Lorentzian function $\omega(\phi)$. Coarse-graining, operator mixing, field normalisation and Euclidean-to-Lorentzian action descent can change that coefficient; their matching remains Level B/Open.

(iii) Exact Euclidean potential template. The same coordinate change rewrites the supplied branch potential as

$$W_\phi^{(E)}(\phi) \equiv W(u_\infty e^{\beta_\phi \phi}).$$

On the regular logarithmic chart $u/u_\infty > 0$ with $\beta_\phi \neq 0$ it also obeys the exact chain-rule identity

$$\frac{dW_\phi^{(E)}}{d\phi} = \beta_\phi u W_{,u}.$$

These equalities define only the reparameterised Euclidean potential. The macroscopic Lorentzian $U(\phi)$ may use it as a matching template, but the identification $U = W_\phi^{(E)}$, its normalisation and any coarse-graining shifts are not derived here.

What is and is not fixed here. The variable change alone derives none of the three macroscopic closure functions. It records a candidate matching programme:

$$F(\phi) \stackrel{\text{supplied}}{=} u/u_\infty,$$

$\omega(\phi)$ is to be matched from renormalised amplitude stiffness,

$U(\phi)$ is to be matched from the coarse-grained branch potential.

The equality for F is a Level B ansatz; the normalised ω and U and the Euclidean-to-Lorentzian action descent remain unowned. The utility of one common amplitude coordinate therefore organises the closure but does not constitute its derivation. The amplitude coordinate supplies a common organising language for later cosmological and galactic constructions. The cosmological benchmark remains a phenomenological closure choice. The HRC response algebra is Level A only inside its supplied augmented model, while the scalar-to-HRC source/force bridge and cosmological matching remain Open.

Minimal generally covariant operator basis

The amplitude-dominated macroscopic reduction identifies the minimal field content of the geometric branch:

- the effective metric $g_{\mu\nu}$,
- the condensate amplitude variable ϕ (reparameterising the ordered-branch amplitude u/u_∞),
- the orientation structure n^μ inherited from the broken phase (a condensate-derived collective variable, not an independent aether-like field),
- matter fields Ψ minimally coupled to $g_{\mu\nu}$.

At two-derivative order, a representative local diffeomorphism-invariant operator inventory may be written as

$$S_{\text{macro}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} f(\phi) R - \frac{1}{2} Z_\phi(\phi) (\nabla \phi)^2 - U(\phi) + \mathcal{C}_R(\phi) n^\mu n^\nu R_{\mu\nu} + \mathcal{B}(\phi) (\nabla_\mu n_\nu) (\nabla^\mu n^\nu) + \mathcal{L}_{\text{mat}} \right]. \quad (13.34)$$

For dimensionless ϕ and n^μ ,

$$[f] = [Z_\phi] = [\mathcal{C}_R] = [\mathcal{B}] = \text{mass}^2, \quad [U] = [\mathcal{L}_{\text{mat}}] = \text{mass}^4, \quad F \equiv f/\bar{M}_{\text{Pl}}^2.$$

Equation (13.34) is an illustrative inventory, not a complete orientation-action basis or a microscopic derivation of its coefficients. A full two-derivative orientation completion can contain additional independent contractions and its constraint sector must be varied with the metric. Higher-derivative operators are omitted under A2; parity-odd terms are not used in the stated residual- $O(3)$ branch.

The ϕ -first closure adopted below retains only the scalar terms $f(\phi)$, $Z_\phi(\phi)$ and $U(\phi)$, written equivalently through the dimensionless $F = f/\bar{M}_{\text{Pl}}^2$ and $\omega = Z_\phi/\bar{M}_{\text{Pl}}^2$. It does not absorb the orientation inventory into Eq. (13.38); a covariant orientation action, constraints and varied stress remain a separate Open extension.

Working assumptions of the ϕ -first macroscopic closure

The assumptions below should be read on top of the amplitude-dominated macroscopic reduction introduced above. They do not create the ϕ -sector from nothing; they specify the first nonlinear closure built on that reduction.

- A1 Conditional Lorentzian scalar branch.** Supply coefficients in the domain $\beta(\beta - \alpha) < 0$; on the retained healthy component $\beta > 0$ this premise is $\alpha > \beta > 0$. Given that supplied premise, diagonalisation proves Lorentzian signature as Level A conditional tensor algebra inside the supplied scalar EFT. P4 adopts the physically relevant ordered branch but does not select the coefficient signs, supply its stationary owner, or derive the clock or universal metric.
- A2 IR locality.** Macroscopic dynamics is restricted to a separately declared completion domain and truncated at local, two-derivative order. This is a closure assumption; P4 does not currently own the corresponding cutoff.
- A3 Four-dimensional geometry.** The macroscopic ansatz retains the four-dimensional arena postulated in P1. Four-dimensionality is inherited input, not a derived macroscopic result.
- A4 Metric universality for slow matter.** Slow matter sectors are assumed to couple to $g_{\mu\nu}^{\text{eff}}$ (Level B/Open, OP-matter).
- A5 Pure-metric reduction.** When ϕ and n_A are frozen, the closure is assumed to reduce to a pure-metric two-derivative theory for the Lovelock step only.

A1 is conditional Level-A mathematics inside the supplied scalar principal-tensor model; A2 and A5 are closure assumptions; A3 is inherited from postulate P1; A4 and the physical metric/branch ownership remain Open.

The ϕ -first effective IR action

Under A1–A4 and the amplitude-dominated closure, consider the scalar-only Jordan-frame ansatz

$$S_{\text{IR}} = \int d^4x \sqrt{-g} \left\{ \frac{\bar{M}_{\text{Pl}}^2}{2} F(\phi) R - \frac{\bar{M}_{\text{Pl}}^2}{2} \omega(\phi) (\partial_\mu \phi)^2 - U(\phi) + \mathcal{L}_{\text{mat}}[g_{\mu\nu}, \Psi] \right\}, \quad F(\phi) = e^{\beta_\phi \phi}. \quad (13.35)$$

Equation (13.35) is written in natural units $\hbar = c = 1$, with $\bar{M}_{\text{Pl}}$ an energy. Its constant coefficient is

$$G_*^{\text{nat}} \equiv \frac{1}{8\pi \bar{M}_{\text{Pl}}^2}. \quad (13.36)$$

All field equations in this section use the same natural-unit convention: $T_{\mu\nu}$ has energy-density units and the unsuperscripted symbol means $G_* \equiv G_*^{\text{nat}}$. SI restoration is a separate dictionary,

$$G_*^{\text{SI}} = \frac{\hbar c^5}{8\pi [\bar{M}_{\text{Pl}}(\text{J})]^2}, \quad (13.37)$$

which is not substituted into the natural-unit field equation below. The measured conversion speed is used here only through the explicitly adopted scalar-clock calibration $c_{\text{char}} = c$; this does not assume or derive photon/tensor cone equality. Neither G_* nor G_*^{SI} is automatically the local Newton response. In the unscreened light-scalar weak-body limit $G_{N,\text{obs}}^{\text{nat}} = G_* F_0^{-1} (1 + \alpha_0^2)$; a finite-range or screened regime has its own response factor. Thus inserting the observed reduced Planck energy in the scalar action is itself a local-response matching assumption. The functions F , ω , and U are macroscopic closure inputs. In particular, $F = u/u_\infty$ is the amplitude-dominated matching hypothesis, whereas $\omega(\phi)$ and $U(\phi)$ have not been derived in full from bare P3. Equation (13.35) contains no independent orientation field n_μ ; orientation stress may be appended only after a separate covariant action $S_n[g, n]$ has been specified. The displayed scalar-only action is Level B as an ECT closure, while its variational identities below are exact inside that stated model. The assumptions, variations, and limits internal to this supplied closure are collected in Appendix Y; that appendix does not supply the missing P1–P6 descent or physical metric map.

Scalar-only metric equation

Writing $X_\phi \equiv (\partial\phi)^2$, variation of Eq. (13.35) with respect to $g^{\mu\nu}$ gives

$$F G_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) F - \omega (\partial_\mu \phi \partial_\nu \phi - \tfrac{1}{2} g_{\mu\nu} X_\phi) + \frac{U}{\bar{M}_{\text{Pl}}^2} g_{\mu\nu} = 8\pi G_* T_{\mu\nu}^{\text{mat}}. \quad (13.38)$$

All functions in this equation are evaluated at ϕ . The coefficient on the right is the constant fixed by the normalisation of the action. Only after dividing the *complete* equation by $F \neq 0$ may one define

$$G_{\text{div}}(\phi) \equiv \frac{G_*}{F(\phi)}, \quad \Lambda_{\text{div}}(\phi) \equiv \frac{U(\phi)}{\bar{M}_{\text{Pl}}^2 F(\phi)}. \quad (13.39)$$

These are exact algebraic coefficients of the divided equation, not response observables. Here every coefficient is natural-unit: $G_{\text{div}} = G_*/F$ and $G_{\text{Cav}} = G_{\text{div}}(1 + \alpha_0^2)$ in the stated unscreened light-scalar limit, while $\Lambda_{\text{div}} = U/(\bar{M}_{\text{Pl}}^2 F)$ has energy-squared units. Only in a separate conversion step may one write $G_{\text{div}}^{\text{SI}} = G_*^{\text{SI}}/F$ and $\Lambda_{\text{div}}^{\text{SI}, \text{m}^{-2}} = \Lambda_{\text{div}}^{\text{nat}}/(\hbar c)^2$, with all energies first expressed in joules. Retaining $F G_{\mu\nu}$ on the left while also replacing $G_* \rightarrow G_*/F$ on the right would double-count F^{-1} . The unit dictionary is Eq. (13.36); matching either coefficient to the observed Newton response remains response dependent. A constant background does not remove a light scalar force, so $G_{\text{div}} = G_{\text{Cav}}$ requires a separately established range/coupling/body-response limit. Away from a constant background, scalar derivative and kinetic terms likewise prevent one local coefficient from encoding the full gravitational response.

If a covariant orientation action is later added, its variational tensor $-2(\sqrt{-g})^{-1} \delta S_n / \delta g^{\mu\nu}$ must be added consistently, together with the n_μ equation and constraint. It is not generated by Eq. (13.35).

Bianchi consistency and matter conservation

Because $\mathcal{L}_{\text{mat}}$ is minimally coupled to the Jordan-frame metric, diffeomorphism invariance gives ordinary matter conservation on the matter equations of motion. The scalar variational identity then yields

$$\nabla^\mu T_{\mu\nu}^{\text{mat}} = 0, \quad \nabla^\mu [\text{left-hand side of Eq. (13.38)}] = 0 \quad \text{on the } \phi \text{ equation of motion.} \quad (13.40)$$

Thus a varying ϕ does not by itself imply exchange with, or non-conservation of, the minimally coupled matter tensor. In a future orientation extension the corresponding total identity also uses the n_μ equation; it still does not alter Jordan-frame matter conservation.

Background reduction of the macroscopic closure

For cosmological applications the present calculation uses the scalar-only isotropic reduction

$$\phi = \phi(\tau), \quad ds^2 = -c_{\text{char}}^2 d\tau^2 + a^2(\tau) \delta_{ij} dx^i dx^j,$$

No orientation field, action, or stress is included in this background system. FRW symmetry can constrain a separately specified covariant $S_n[g, n]$, but it cannot absorb or determine a missing orientation sector. The resulting background and numerical outputs are therefore scalar-closure benchmarks only.

Under this isotropic background reduction, the ϕ -first action (13.35) yields the schematic parent coupled system for the scale factor $a(t)$ and the macroscopic amplitude variable $\phi(t)$. One explicit conditional scalar–tensor completion is recorded in Appendix AE. Its action, state and globally regular two-slope orbit own the numerical results used below. This is a Level A result inside that supplied completion and a Level B proof of class; P1–P6 do not yet select it as the unique cosmology. Writing

$$F(\phi) \equiv e^{\beta_\phi \phi},$$

the background equations take the schematic scalar–tensor form

$$3F(\phi)H^2 = \frac{\rho_m + \rho_r + U(\phi)}{\bar{M}_{\text{Pl}}^2} + \frac{1}{2} \omega(\phi) \dot{\phi}^2 - 3H\dot{F}, \quad (13.41)$$

$$-2F(\phi)\dot{H} = \frac{\rho_m + \frac{4}{3}\rho_r}{\bar{M}_{\text{Pl}}^2} + \omega(\phi)\dot{\phi}^2 + \ddot{F} - H\dot{F}, \quad (13.42)$$

together with the scalar background equation

$$\omega(\phi)(\ddot{\phi} + 3H\dot{\phi}) + \frac{1}{2}\omega_{,\phi}\dot{\phi}^2 + \frac{1}{\bar{M}_{\text{Pl}}^2}U_{,\phi} - 3(2H^2 + \dot{H})F_{,\phi} = 0. \quad (13.43)$$

Several remarks are important.

(i) Status inside the supplied closure. These equations are exact isotropic reductions of the supplied macroscopic closure. This does not derive that closure from P1–P6 or identify it as the physical ECT gravity sector. The closure functions $F(\phi) = e^{\beta_\phi\phi}$, $\omega(\phi)$ and $U(\phi)$, together with their physical action/state/metric ownership, remain Level B/Open.

(ii) Scope of the orientation sector. The displayed scalar background reduction contains no independently varied orientation action or stress. A future covariant $S_n[g, n]$ could define an additional sector, but its FRW contribution cannot be inferred by saying that it is frozen, absorbed, or subleading. That contribution remains Open.

(iii) Relation to late-time cosmology. The late-time background apparatus used in §16.5 and in Appendix AE should be read as a benchmark specialisation of Eqs. (13.41)–(13.43) to the present amplitude-first closure. Thus the cosmological sector is no longer an isolated add-on: it is the background branch of the same macroscopic system used in the geometric theory.

(iv) Conditional observational outputs. Once a background solution $H(z), \phi(z)$ and a physical metric/clock map are supplied, the conditional cosmological observables used later in the paper follow through the standard FRW integral relations:

$$D_L(z), \quad t_{\text{lookback}}(z), \quad t_U(z), \quad t_{\text{gal}}(z_{\text{obs}}; z_{\text{form}}).$$

These are derived outputs inside that supplied background and observable map, not independent phenomenological inserts; they are not predictions of P1–P6.

Conditional frozen- F limit

If $\phi = \bar{\phi}$ and hence $F(\bar{\phi}) = \bar{F}$ are constant, scalar derivatives vanish, and no separately specified orientation source is included, Eq. (13.38) reduces algebraically to

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda_{\text{div}}(\bar{\phi})g_{\mu\nu} = 8\pi G_{\text{div}}(\bar{\phi})T_{\mu\nu}^{\text{mat}}. \quad (13.44)$$

For the reference normalisation $\bar{F} = 1$, $G_{\text{div}} = G_*$. This is an exact GR-form *background* equation inside the Level-B scalar ansatz. It is not yet a physical GR limit: the constant branch must solve the scalar equation and be stable, and scalar perturbations or finite-body charge must decouple, be short-ranged, or be screened. It proves neither that response condition nor dynamical approach to the branch.

Lovelock uniqueness of the pure-metric limit

Under A1–A5, Lovelock’s theorem [92] applies: in four dimensions, the unique local diffeomorphism-invariant two-derivative pure-metric field equations are those of Einstein–Hilbert gravity with cosmological constant.

Status: Level A *under* A1–A5. It is a theorem that the pure-metric frozen-order reduction cannot be anything other than Einstein–Hilbert + Λ . It is *not* a theorem that the full ECT geometric branch is always pure GR.

Comparison with neighbouring theories

General relativity. GR is recovered algebraically in the conditional frozen- F scalar limit. No dynamical statement about ϕ or an orientation field becoming invisible has been derived.

Brans–Dicke scalar-tensor theories. ECT shares an effective scalar modulating gravitational response. But ϕ is not a new fundamental field; it is the logarithmic macroscopic amplitude variable of the ordered condensate branch.

Einstein–aether theory [23]. ECT shares a preferred-direction collective variable and permits candidate anisotropic tensor structures. No covariant ECT orientation stress has yet been derived from it. The variable is not fundamental in bare action P3; it belongs to the ordered condensate phase. This is the structural difference with Einstein-aether theories, which postulate such a field from the outset.

Hořava–Lifshitz gravity. ECT also has an emergent preferred structure, but does not begin by explicitly breaking Lorentz symmetry in the microscopic action. In the present model the preferred direction is supplied by P4; its macroscopic realisation through a stationary state and spontaneous ordering of the Euclidean medium remains an Open completion target.

Physical interpretation

The displayed nonlinear closure is scalar–tensor: it describes how the amplitude coordinate and its non-minimal response factor modify the metric equation inside the declared ansatz. Its pure-metric GR equation follows only in the algebraic frozen- F condition. A separate orientation stress, finite-body screening, and the dynamical route to that condition are Open.

Two scalar departures and one separate candidate extension. The scalar equation (13.38) contains two closure-dependent departures from standard Einstein gravity. A possible orientation response is listed separately and is not a term in that equation:

(i) *Divided potential coefficient* $\Lambda_{\text{div}}(\phi)$. The ratio $U/(\tilde{M}_{\text{Pl}}^2 F)$ is the exact potential coefficient of the divided metric equation. It equals a GR-form cosmological constant only on a dynamically available stable constant-scalar branch with the physical metric, clock, vacuum offset, and branch fixed. On an evolving branch, $\dot{F}$, $\ddot{F}$, scalar kinetic energy, and the state also enter H , so Λ_{div} is not by itself an observed cosmological response.

(ii) *Divided matter coefficient* $G_{\text{div}}(\phi) = G_* e^{-\beta_\phi \phi}$. Changing u changes this exact coefficient inside the scalar ansatz. A Cavendish, finite-body, orbital, or cosmological response additionally requires the canonically normalised scalar coupling, its range, the source and test-body charges, and the relevant metric/clock map. No ECT-specific transfer to those observables or to the separately supplied conditional HRC gravity response has yet been derived.

(iii) *Candidate orientation response.* The scalar action (13.35) contains no independent n_μ and therefore does not produce an orientation stress on variation. Einstein–aether-like tensors, including $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$, may be studied as Level-C external proxies, but an ECT orientation stress remains Open until a covariant $S_n[g, n]$, its constraints, and its on-shell stress tensor are derived.

Derivation-level summary

Strict (Level A): local Lorentzian branch; necessity of a covariant macroscopic completion; GR recovery when additional branch variables freeze; Lovelock uniqueness of the pure-metric limit under A1–A5.

Structural (Level B): generalised nonlinear equations from the ϕ -first closure and the exact divided coefficients G_{div} and Λ_{div} inside that ansatz; their physical response maps, an orientation extension and dynamical PPN reachability remain Open.

Closure-dependent: the supplied functions $F(\phi)$, $\omega(\phi)$ and $U(\phi)$, their cosmological benchmarks, and the post-Newtonian completion. Once F, U are supplied, the divided coefficients are algebraic; detailed galactic fits belong to G, not to those coefficients.

What is established and what is not (detailed)

Established or structurally fixed:

- (i) A nonlinear closure is *required* once metric structure, tensor propagation, and source coupling all exist (Level A).
- (ii) The pure-metric two-derivative frozen-order limit is uniquely Einstein–Hilbert + Λ in $d = 4$ (Level A under A1–A5).
- (iii) The current best-developed closure is the ϕ -first one, Eq. (13.35) (Level B).
- (iv) The macroscopic scalar variable ϕ is interpreted as a reparameterisation of the ordered-branch amplitude u/u_∞ , not as an extra fundamental field (bridge principle; structural).
- (v) The first macroscopic nonlinear reduction retained here is scalar-only: the amplitude coordinate is reparameterised by $\phi = \frac{1}{\beta_\phi} \ln(u/u_\infty)$. A macroscopic orientation stress is a separate Open extension.
- (vi) The FRW system used later is the scalar-only isotropic reduction. Symmetry does not freeze, absorb, or determine an unspecified orientation action or stress.
- (vii) The ordered-branch amplitude supplies exact Euclidean coordinate templates for its kinetic coefficient and branch potential. The curvature matching $F \stackrel{\text{supplied}}{=} u/u_\infty$ and the Lorentzian functions $\omega(\phi)$ and $U(\phi)$ remain separate Level B/Open matching inputs; no action-level descent follows from the logarithmic coordinate alone.
- (viii) The displayed scalar action gives exact conditional FLRW equations, but P1–P6 do not select its closure functions, physical state or boundary/initial data.
- (ix) A physical calculation must evolve one varied action continuously from a derived ordering event; no external high-redshift continuation is admissible.
- (x) The article–code contract is the exact solver protocol of Appendix AR: constraints, physical response determinant, kinetic/gradient stability and two-method reproducibility are mandatory outputs.
- (xi) Background tables, distance/time quadratures, perturbations and observable likelihood products may be published only after that common solution passes its owning gates.

Not yet established as first-principles theorems:

- (i) Full derivation of the nonlinear closure from bare P3 (OP2).
- (ii) Microscopic derivation of $U(\phi)$ (OP3).
- (iii) Universality of matter coupling (OP-matter).
- (iv) Exact nonlinear role of $\mathcal{R}_{\mu\nu}^{\text{cand}}[n]$ in all regimes (OP-c1).
- (v) Whether any supplied $K(\phi)$ and $V(\phi)$ define a globally regular, stable cosmological solution. The conditional two-slope action in Appendix AE passes its homogeneous gates, but a physical completion must additionally derive the ordering state, metric, perturbations and finite-body limit from the same action.

Observational role of the full nonlinear equations across the ECT programme

The summary below is a conditional proxy inventory. The displayed scalar-only action does not produce $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$. Each orientation entry therefore assumes a future separately specified covariant n^A action whose variation yields the candidate operator and whose on-shell identities are consistent. The coefficient, sign, and even the surviving projected channel are Open; existing numbers illustrate sensitivity only.

Table 51: Conditional proxy footprint of a candidate orientation-sector extension; these entries are not derived from the displayed scalar-only action.

Sector	Benchmark / present disposition	NLEE channel	Role	Separability	Appendix
Hubble tension (§16.5)	no ECT-native recombination-to-distance-ladder result	no independently computed orientation contribution	Scalar-only result	Orientation channel Open	AR
Universe age (§16.4)	scalar-background benchmark only	no independently computed orientation contribution	Scalar-only result	Orientation channel Open	AR
JWST / growth (§16.5)	background, perturbation and galaxy forward model not jointly derived	external orientation-tensor sensitivity proxy	Not included in scalar result	Level C proxy / ECT Open	AH
Galactic / RAR / BTFR (§17.1)	Closure eq.; slope 4	Degenerate with $a_{M,\text{eff}}$; scatter	Degenerate with effective-scale calibration	Not separately extractable within present closure	AK.5
Cluster projected proxy (§17.2)	Exact order/set-valued-argmax preservation for declared baryonic maps	No cosmological-drift, amplitude or orientation-slip correction retained	No morphology/lensing or observed mass-ratio inference	A algebra / physical collision, action and lensing map Open	AI

The table identifies where a completed orientation extension could have the largest observational leverage: sub-horizon growth and cluster-merger lensing. It does not establish that the candidate $\mathcal{X}_{\mu\nu}^{\text{cand}}$ is realised physically. The first priority is a covariant action, physical projection, and on-shell consistency; only then can response and lensing observables be calculated.

Phenomena beyond the scope of the present work

A future covariant orientation extension could generate the phenomenological channels listed below. At present they are scenario targets, not omitted predictions of the scalar–tensor equation. Their assessment requires the missing orientation action and projection in addition to Boltzmann, 3D, and lensing calculations.

Table 52: NLEE effects in phenomena not examined quantitatively in the present work.

Phenomenon	Expected NLEE channel	Qualitative expectation
Cosmic shear / weak lensing (S_8 / lensing–growth sector)	external candidate-tensor sensitivity proxy	No ECT correction presently predicted; a future S_n and Boltzmann calculation would be required
Void–galaxy correlations	external environment-sensitive proxy with unspecified sign	Open until an orientation action and coefficient are derived
Integrated Sachs–Wolfe effect	external candidate-tensor proxy after a future S_n derivation	No ECT correction presently predicted
Strong lensing in cluster cores	external slip sensitivity proxy	No ECT correction presently predicted
Tidal / asymmetric environments	external candidate-tensor proxy after a future S_n derivation	No ECT contribution presently predicted

S_8 / **lensing–growth proxy programme (Level C/Open).** A future covariant orientation action might supply a lensing–growth channel. The scalar-only generalised Einstein equation does not contain $\mathcal{R}_{\mu\nu}^{\text{cand}}[n]$, so the present μ , η , and Σ discussion is an external proxy rather than an ECT-derived modification. For bookkeeping, use the standard observational definitions

$$-k^2\Psi = 4\pi G_N a^2 \mu(k, z) \rho \Delta, \quad \eta(k, z) \equiv \frac{\Phi}{\Psi}, \quad (13.45)$$

$$-k^2(\Phi + \Psi) = 8\pi G_N a^2 \Sigma(k, z) \rho \Delta, \quad \Sigma(k, z) = \frac{\mu(k, z)[1 + \eta(k, z)]}{2}. \quad (13.46)$$

An Einstein–aether-like tensor can be used to explore these functions only after it is explicitly declared as an external model. Whether a covariant $S_n[g, n]$ exists, which operator survives projection, and its coefficients, state, and FRW response are all Open. Consequently, no resolved S_8 value or ECT orientation correction is predicted here.

Structural comparison with neighbouring frameworks

Table 53 summarises how the generalised field equations of ECT compare structurally with GR, Brans–Dicke, and Einstein-aether theories.

Table 53: Structural comparison of generalised Einstein equations. “✓” = present; “*post.*” = postulated; “*deriv.*” = derived from action; “—” = absent; “candidate / Open” = an allowed tensor form whose ECT action-level source has not been supplied. A constant scalar gives a GR-form background equation, but a physical GR response additionally requires the scalar perturbation and finite-body charge to decouple, be short-ranged, or be screened. No orientation-sector reduction is claimed without a separately specified $S_n[g, n]$.

Term	Einstein-aether [23]	Brans–Dicke [93]	$f(R)$	Horava–Lifshitz [25]	ECT
$R_{AB} - \frac{1}{2}g_{AB}R$	✓	✓	✓	✓	✓

Term	Einstein-aether	Brans–Dicke	$f(R)$	Horava–Lifshitz	ECT
$\Lambda_{\text{div}}(X) g_{AB}$ (divided potential coefficient)	—	—	$\checkmark^*$	—	coefficient derived / response Open
$n_A n^C R_{CB}$ (anisotropic)	<i>post.</i>	—	—	$\checkmark^{**}$	candidate / Open
$G_{\text{div}}(X)$ (divided matter coefficient)	—	$\checkmark$	—	—	coefficient derived / response Open
Condensate back-reaction $T_{AB}^{(n)}$	$\checkmark^{***}$	—	—	—	missing S_n / Open

* In $f(R)$ gravity an analogous divided potential coefficient arises from the scalar representation of the curvature sector, not from a condensate; its cosmological response likewise requires the full solution.

** In Horava–Lifshitz, anisotropy is between space and time, not tied to a condensate direction.

*** Einstein-aether back-reaction involves the aether EMT, which is structurally similar to $T_{AB}^{(n)}$ but with free coupling constants $c_{1..4}$.

Summary

The geometric branch of ECT requires a nonlinear completion. The most developed currently available amplitude-dominated completion is the ϕ -first scalar-tensor closure (13.35)–(13.38), whose homogeneous frozen-order limit reduces to standard Einstein gravity with cosmological constant. The full first-principles derivation of this nonlinear sector from microscopic condensate dynamics remains open, but the EFT logic already constrains the closure strongly. This prepares the next stage: the weak-field limit and the emergence of Newtonian and post-Newtonian gravity from the nonlinear geometric branch.

13.6 Weak-field and Newtonian limit

Status: Level A/B for the algebraic Newtonian limit of the declared Level-B scalar–tensor ansatz when F and the scalar are frozen and any separately specified orientation correction vanishes. The dynamical reachability of this limit for an extended Solar-System source, the composite-body scalar charge, and the full PPN profile are Open. The modified weak-field and galactic equations below are closure-level parametrisations, not a derivation of density screening.

Weak-field expansion of the scalar–tensor ansatz

We expand about a declared vacuum-reference background,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1, \quad |q - \bar{q}| \ll 1, \quad v \ll c_{\text{char}}, \quad (13.47)$$

and use the quasistatic nonrelativistic ordering

$$|\partial_t| \ll c_{\text{char}} |\nabla|, \quad T_{00} \gg T_{0i}, T_{ij}. \quad (13.48)$$

In an explicit SI Newtonian-gauge convention, let Ψ_N and Φ_N be dimensional potentials in $\text{m}^2 \text{s}^{-2}$:

$$ds^2 = - \left(1 + \frac{2\Psi_N}{c^2} \right) c^2 dt^2 + \left(1 - \frac{2\Phi_N}{c^2} \right) \delta_{ij} dx^i dx^j. \quad (13.49)$$

Conditional frozen-order limit

If $F = 1$ is constant, $\nabla_\mu q = 0$, and any independently defined orientation-sector correction vanishes, the corrected metric equation reduces algebraically to GR. Its nonrelativistic limit is

$$\nabla^2 \Psi_N = 4\pi G_N^{\text{SI}} \rho_{\text{bar}}, \quad \Phi_N = \Psi_N, \quad (13.50)$$

and hence

$$\gamma_{\text{PPN}} \equiv \frac{\Phi_N}{\Psi_N} = 1. \quad (13.51)$$

These equations prove the conditional limit, not that the scalar profile of a real body reaches it. Establishing that step requires the canonical Einstein-frame Hessian, an interior/exterior body solution, its scalar charge, and the associated metric potentials.

Weak-field closure away from the frozen reference

For a slowly varying scalar background, write the weak-field equation in one explicit SI convention, with Ψ_N in $\text{m}^2 \text{s}^{-2}$ and ρ_{bar} in kg m^{-3} :

$$\nabla^2 \Psi_N(X) \simeq 4\pi G_{\text{div}}^{\text{SI}}(X) \rho_{\text{bar}}(X) + \mathcal{D}_q^{\text{SI}}(X) + \mathcal{D}_n^{\text{SI}}(X). \quad (13.52)$$

Every term in this equation has units s^{-2} . Here $\mathcal{D}_q^{\text{SI}}$ denotes the consistently converted scalar derivative terms obtained from the corrected equation. $\mathcal{D}_n^{\text{SI}}$ is only a placeholder for a separately specified covariant orientation extension; it does not follow from the displayed scalar-only action. Neither term currently supplies a solved galactic matching or Solar-System screening profile. This SI weak-field equation is a separate observable dictionary; it is not substituted into the natural-unit covariant field equation.

Local variation of the divided coefficient and the LLR transfer. For the main coordinate convention $F = e^{\beta_\phi \phi_u}$,

$$\frac{\dot{G}_{\text{div}}}{G_{\text{div}}} = -\beta_\phi \dot{\phi}_u, \quad G_{\text{div}} = \frac{G_*}{F}. \quad (13.53)$$

Because F is dimensionless and the conversion factor is constant, the same logarithmic identity holds for $G_{\text{div}}^{\text{SI}} = G_*^{\text{SI}}/F$. It is algebraic; neither version is yet the quantity fitted by lunar ranging. For a universally coupled scalar that is unscreened and light on the experimental scale, define

$$a \equiv \frac{f_{,q}}{f}, \quad K_E \equiv \frac{K}{f} + \frac{3}{2}a^2, \quad \alpha_0^2 \equiv \frac{a_0^2}{2K_{E,0}}. \quad (13.54)$$

Then the weak-body Cavendish response is

$$G_{\text{Cav}} = G_{\text{div}}(1 + \alpha_0^2), \quad \frac{\dot{G}_{\text{Cav}}}{G_{\text{Cav}}} = \frac{\dot{G}_{\text{div}}}{G_{\text{div}}} + \frac{d}{dt} \ln(1 + \alpha_0^2). \quad (13.55)$$

A finite scalar mass and extended bodies further introduce range-dependent Yukawa factors and body charges. A 2021 primary lunar-laser-ranging analysis reports $\dot{G}/G_0 = (-5.0 \pm 9.6) \times 10^{-15} \text{ yr}^{-1}$ [94]. It constrains the fitted orbital/Cavendish response, not G_{div} alone. The ECT transfer from the action-owned scalar and finite bodies to that fitted quantity remains Open.

Local-deviation variable. Define

$$\varepsilon_{\text{div}}(X) \equiv F^{-1}(X) - 1 \approx -\beta_\phi \phi_u(X) \quad (|\beta_\phi \phi_u| \ll 1). \quad (13.56)$$

No numerical tolerance on this divided-equation coefficient is assigned. A bound requires a finite-body response map, screening solution and experiment-specific likelihood; none is presently owned.

Two phenomenological weak-field regimes

The conditional HRC response after the identity bridge distinguishes a high-acceleration Newtonian asymptote from a low-acceleration critical branch. This is useful closure-level organisation. It must not be identified with a proven density-driven scalar-screening transition: the map from the action and a finite-body solution to either branch remains Open.

Status summary

Established conditionally: the GR-form background Poisson equation follows when the scalar and any separate orientation correction are frozen; the GR PPN values additionally require the scalar perturbation and body charge to decouple, be short-ranged, or be screened. **Level B scalar ansatz:** the schematic divided weak-field equation. **Layered HRC status:** the response algebra is Level A inside the supplied augmented model, the identity/source bridge is Level B/Open, and only the data diagnostics are Level C. **Open:** dynamical reachability, composite-body charge, full PPN parameters, a covariant orientation action, and the scalar-to-galactic source/force match.

13.7 Post-Newtonian regime and Solar System tests

Status: the frozen- F GR limit is an algebraic conditional result. Solar-System reachability, finite-body sensitivity, and the complete PPN parameter set are Open.

Conditional GR value. If F is constant, the scalar gradient vanishes, any separately defined orientation extension is absent, and the linear scalar charge is zero or its force is short-ranged or screened for the bodies and scale considered, then Eq. (13.44) supplies the GR-form background and

$$\gamma_{\text{PPN}} = 1, \quad \beta_{\text{PPN}} = 1. \quad (13.57)$$

This conditional value is compatible with the primary Cassini result $\gamma_{\text{PPN}} - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [95], but consistency with that measurement is not a prediction that an extended Solar-System body realises the assumed response limit.

Constancy of the homogeneous background F alone is insufficient. A massless Brans–Dicke counterexample has $\alpha_0^2 = (2\omega_{\text{BD}} + 3)^{-1}$, $G_{\text{Cav}} = G_{\text{div}}(2\omega_{\text{BD}} + 4)/(2\omega_{\text{BD}} + 3)$, and $\gamma_{\text{PPN}} = (\omega_{\text{BD}} + 1)/(\omega_{\text{BD}} + 2)$ even though the background F is constant.

Required closure calculation. A test requires a canonical scalar normalisation, an asymptotic background, regular interior and exterior solutions, the composite-body scalar charge, and the two metric potentials entering the PPN observables. None is fixed by the positivity of the local homogeneous Hessian alone.

Current empirical status. No complete coupled finite-body PPN/WEP comparison is presently available. For the named two-slope near-GR point, the scalar small-body calculation in Appendix AE supplies no mass-screening suppression; its unscreened value satisfies only the displayed absolute γ -deviation proxy and lies 1.35σ from the primary Cassini central value. A full likelihood and the metric backreaction, β_{PPN} , preferred-frame coefficients, composition dependence and lunar-ranging observables remain Open.

13.8 Gravitational waves: completion targets and empirical gates

Status: Open for ECT. The equations below define standard conditional tensor-EFT targets; they are not derived from the fixed director map.

On a spatially flat emergent FRW interval, define the dimensionless TT metric perturbation by $g_{ij} = a^2(\delta_{ij} + H_{ij})$, with $\partial_i H_{ij} = H_{ii} = 0$. Let $\Pi_{ij}^{\text{TT,phys}}$ denote the TT projection of the physical orthonormal

spatial anisotropic stress (equivalently the mixed spatial stress on this background, with displayed spatial indices moved by δ_{ij}). A separately supplied constant- (M_T, c_T) tensor EFT may then be normalised as

$$S_T^{(2)} = \frac{M_T^2}{8} \int d\eta d^3x a^2 [(H'_{ij})^2 - c_T^2 (\partial_k H_{ij})^2] + \frac{1}{2} \int d\eta d^3x a^4 H_{ij} \Pi_{ij}^{\text{TT,phys}}. \quad (13.58)$$

Its variation assumes vanishing endpoint variations and periodic or sufficiently decaying spatial data. On curved spatial slices the displayed k^2 is replaced by the appropriate TT tensor-Laplacian eigenvalue. This is conditional standard tensor-EFT input, not an action derived from the fixed director map. Ordinary massless Fierz–Pauli theory is its flat-space Lorentz-invariant specialization with $c_T = 1$; allowing $c_T \neq 1$ means that an additional Lorentz-breaking tensor operator has been supplied. The tensor propagation speed of Eq. (13.58) is, by definition,

$$c_{\text{GW}} = c_T, \quad (13.59)$$

not automatically the scalar characteristic speed c_{char} . The observed GW170817/GRB 170817A interval is an external kill gate for c_T [12].

If the completed tensor action contains the explicitly declared operator

$$\omega^2 = c_T^2 k^2 [1 + \xi_2 (k/M_{\text{LIV}})^2 + O(k^4/M_{\text{LIV}}^4)], \quad (13.60)$$

then

$$\delta v_{\text{GW}}/c_T = O(\xi_2 E_{\text{GW}}^2/M_{\text{LIV}}^2), \quad (13.61)$$

and

$$\Delta t_{\text{GW}} = O[(D/c_T) \xi_2 E_{\text{GW}}^2/M_{\text{LIV}}^2]. \quad (13.62)$$

ECT presently predicts neither this tensor operator nor its coefficient or suppression scale. Accordingly no tensor propagation delay is assigned; a future completed action must compute the dispersion relation before numerical comparison with timing data.

If the completion is exactly massless Fierz–Pauli, the standard constraint analysis leaves two TT polarisations. The current director image does not supply that completion. Extra scalar/vector polarisations and c_T must be computed from the completed Hessian before comparison with data.

Ordering-transition tensor radiation

Varying the separately supplied action Eq. (13.58) with constant M_T and c_T gives the standard conditional sourced relation (in units $c = 1$)

$$H''_{ij} + 2\mathcal{H} H'_{ij} + c_T^2 k^2 H_{ij} = \frac{2a^2}{M_T^2} \Pi_{ij}^{\text{TT,phys}} \equiv \frac{2}{M_T^2} \Pi_{ij}^{\text{TT,conf}}, \quad \Pi_{ij}^{\text{TT,conf}} := a^2 \Pi_{ij}^{\text{TT,phys}}. \quad (13.63)$$

The GR-normalised specialization is $c_T = 1$ and $M_T^2 = (8\pi G_N^{\text{nat}})^{-1}$; a running tensor coefficient would also modify the friction term as $2\mathcal{H} \mapsto 2\mathcal{H} + (M_T^2)' / M_T^2$ in this minimal action. Equation (13.63) is valid only after the physical tensor, its TT projector, normalization and anisotropic source have been derived. The $O(4) \rightarrow O(3)$ ordering event does not by itself prove nonzero Π_{ij}^{TT} , its sign, efficiency or spectrum. A stochastic background, its relation to baryogenesis and its redshifting history therefore remain an Open completion programme. The dimension-only source/redshift scaffold for the same separately supplied tensor completion is recorded in Appendix AF; it assigns neither an ECT source nor a spectrum and is not a pre-Lorentzian production calculation.

Terminal tensor/GW status. The scalar cone, a possible tensor cone and the photon cone are distinct owners until their kinetic operators and universal couplings are derived. GW170817, polarisation tests and stochastic-background searches are external tests of future completions, not present passes of ECT.

Detailed gravitational-wave completion programme

Principal-symbol calculation. For a supplied physical tensor H_{ij}^{TT} , an isotropic quadratic low-energy equation may be organised schematically as

$$Z_T (\ddot{H}_{ij}^{\text{TT}} - c_T^2 \nabla^2 H_{ij}^{\text{TT}} + m_{\text{tens}}^2 H_{ij}^{\text{TT}}) = \Pi_{ij}^{\text{TT}}, \quad c_T^2 = K_T / Z_T.$$

Here $Z_T > 0$ and $K_T > 0$ are kinetic coefficients of mass dimension two, whereas $m_{\text{tens}}^2 \geq 0$ is the physical tensor mass squared in this minimal stable benchmark. None of $Z_T, K_T, m_{\text{tens}}$ or the physical TT projector is determined by the scalar characteristic tensor. The fixed director map has a zero spatial TT block, so it cannot be used to evaluate this equation.

GW170817 gate. The multimessenger observation constrains the low-energy tensor cone relative to the photon cone as $-3 \times 10^{-15} \leq (c_{\text{GW}} - c_\gamma)/c_\gamma \leq +7 \times 10^{-16}$, subject to the source-association and emission-time assumptions [12]. A future ECT tensor action must therefore yield a principal symbol passing this bound. The observation does not validate ECT before that symbol exists, and it does not prove equality of all scalar, gauge and tensor cones.

Higher-derivative dispersion. If the completed tensor action contains a term producing

$$\omega^2 = c_T^2 k^2 [1 + \zeta_2 (k/M_{\text{LIV}})^2 + O(k^4/M_{\text{LIV}}^4)],$$

then the group-velocity shift and accumulated delay follow from the displayed operator by expansion. Since neither the operator, coefficient nor scale has an ECT tensor owner, no numerical delay is assigned and no empirical pass can be claimed.

Polarisation content. Exactly two TT polarisations follow for a healthy massless Fierz–Pauli completion. Scalar–tensor, vector–tensor, massive or nonlocal completions can contain additional modes. ECT must derive its constrained spectrum before using polarisation counts. Observations of extra modes would reject a strict FP completion, not the scalar ordered-branch result by itself.

Ordering-transition source. An $O(4) \rightarrow O(3)$ transition provides time-dependent scalar and director backgrounds, but a stochastic GW source requires their nonzero projection onto the physical tensor anisotropic stress. The conditional sourced equation

$$H_{ij}'' + 2\mathcal{H}H_{ij}' + c_T^2 k^2 H_{ij} = \frac{2a^2}{M_T^2} \Pi_{ij}^{\text{TT,phys}} = \frac{2}{M_T^2} \Pi_{ij}^{\text{TT,conf}}$$

is standard once H_{ij}^{TT} , M_T and Π_{ij}^{TT} are owned. Here $\Pi_{ij}^{\text{TT,conf}} = a^2 \Pi_{ij}^{\text{TT,phys}}$; omitting a^2 without this rescaled-source convention would mix normalisations. The ordering event alone fixes neither the sign nor the amplitude of Π_{ij}^{TT} ; in the fixed director map the linear TT projection is zero. Consequently no stochastic spectrum $\Omega_{\text{GW}}(f)$ is presently derived.

Conditions for an appreciable ordering signal. A calculable signal requires a departure from exact homogeneity, a nonzero TT source, a specified transition duration, correlation length, wall or defect content, tensor efficiency and subsequent expansion history. If the transition is adiabatic, spatially isotropic or has only scalar/longitudinal stress, the leading tensor source vanishes or is strongly suppressed. These conditions replace the former assertion that ordering generically creates an observable background.

Relation to baryogenesis. Temporal coincidence between an ordering epoch and a baryogenesis mechanism does not fix a GW spectrum. A joint prediction would require one common background solution, a CP-violating source, a tensor anisotropic-stress kernel, and transfer functions to today. Until those owners are supplied, the possible correlation is an Open programme rather than a sign-level ECT result.

External observational anchors. The completion should be confronted with: the GW170817 speed bound; limits on the graviton mass and anomalous damping; polarisation searches; binary-pulsar energy loss; stochastic-background constraints from pulsar timing, LIGO– Virgo–KAGRA and future space interferometers; and consistency between GW luminosity distance and the matter/photon metric. These data constrain a specified tensor completion. They are not evidence for a tensor that has not yet been derived.

Terminal status. The equations in this subsection preserve the complete calculation protocol for propagation, dispersion, polarisations and an ordering source. Their status is conditional standard tensor-EFT algebra. ECT ownership remains Open until the physical tensor bundle, action, vertex, background and state are supplied and the fixed-map no-go is bypassed explicitly.

13.9 Gravitational-sector conditional results, external gates, and Open tests

Summary of conditional gravitational-sector statements, consistency conditions, external gates, and Open completion tests, together with their derivation status.

Table 54: Status-separated gravitational-sector statements, consistency conditions, and completion gates.

Statement / condition	Formula / estimate	Status	External gate / future test
Einstein–Hilbert uniqueness of the pure-metric two-derivative closure	Lovelock–Deser route in the pure-metric limit (§13.5)	Level A/B	Classical GR; binary pulsars
Conditional GR recovery when F is frozen	Eq. (13.44)	Algebraic inside scalar ansatz; dynamical reachability Open	—
Universal metric coupling	All matter sectors couple to one $g_{\mu\nu}$	Level A/B	WEP tests; MICROSCOPE
GW propagation	c_{GW} requires the physical tensor principal symbol	Open / external gate	GW170817
Ordering-transition stochastic GW route	Eq. (13.63)	Conditional standard tensor-EFT algebra / ECT source and spectrum Open	PTA, LISA
TT polarisation recovery	requires the OP-gauge reduction to the GR-like tensor sector	Level B / open reduction step	LIGO/Virgo
Solar-System scalar-screening / PPN completion	a constant background plus a decoupled, short-range or screened scalar response gives $\gamma_{\text{PPN}} = 1$	Open body-profile and response-transfer test	Cassini, MICROSCOPE, LLR
Higher-order LIV completion	photon and tensor operators, coefficients and suppression scales are not derived; external data constrain only explicitly supplied completions	Open	High-energy propagation tests
Divided potential coefficient $\Lambda_{\text{div}}(\phi)$	exact coefficient $U/(\bar{M}_{\text{Pl}}^2 F)$ in the divided supplied equation	Level A algebra in Level B ansatz / cosmological response Open	Cosmological data

Statement (cont.)	Formula	Status	Gate / future test
Divided matter coefficient $G_{\text{div}}(\phi)$	exact coefficient G_*/F in the divided supplied equation	Level A algebra in Level B ansatz / local and cosmological responses Open	Cavendish, PPN, LLR, cosmology
Force-law dimensionality diagnostic	$d_{\text{force}}: 3 \rightarrow 7/3 \rightarrow 2$ inside the conditional HRC-0 point-source diagnostic (not the full HRC field problem)	Level C conditional algebra	Data-level extraction with measured $p_N(r)$ remains Open
Candidate orientation-sector gravitational slip	conditional proxy from a separately specified n^A action	Open / proxy	Lensing–dynamics, Euclid

Section-level status summary

At the current stage, the emergent gravitational sector separates three logically different layers.

Strict structural core: a supplied ordered scalar EFT with $\beta(\beta - \alpha) < 0$ admits a Lorentzian scalar principal tensor. A separately supplied scalar–metric action has a covariantly conserved matter source, and the Lovelock–Deser assumptions single out Einstein–Hilbert form for any two-derivative pure-metric completion. These statements do not provide a linear helicity-2 mode, a static-density graviton vertex, or the physical matter/photon metric.

Level-B scalar ansatz: the corrected scalar–tensor equations, the exact divided coefficients $G_{\text{div}} = G_*/F$ and $\Lambda_{\text{div}} = U/(\bar{M}_{\text{Pl}}^2 F)$, and the algebraic constant-background GR-form equation. Those coefficient identities are Level A inside the supplied ansatz; the particular F, K, V match to the condensate is closure-dependent. A physical GR, Cavendish or PPN limit additionally requires the scalar range and finite-body response.

Open extensions and empirical gates: a covariant orientation action that produces $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$, its on-shell identities and coefficient; the finite-body scalar charge and Solar-System PPN profile; and the microphysical match from the scalar ansatz to the conditional HRC gravity response. Thus neither orientation-stress phenomenology nor density screening is presently a derived result of the displayed scalar-only action.

14 ECT and the Classical Cosmological Problems

The three classical cosmological problems—horizon, flatness, and relic/monopole—are historically resolved by inflationary cosmology through dynamical mechanisms operating inside an already existing Lorentzian spacetime. ECT explores structurally different conditional routes at the level of a supplied ordering scenario, before a Lorentzian branch is physically owned.

This section presents ECT routes and their separate statuses: the horizon route is conditional, physical FLRW flatness still needs the metric/embedding bridge, and $\pi_2(S^3) = 0$ is a closed mathematical result only for the supplied orientation target. Its physical monopole/relic interpretation remains Open. The pre-Lorentzian scenario and primordial perturbation transfer remain incomplete. The flat Euclidean ambient metric of P1 is Level A, but inheritance by the physical FLRW metric is Level B/Open. The supplied-target homotopy and the separately supplied sigma-model minimum are Level-A conditional mathematics; physical vacuum topology and cosmological transfer remain Open. Throughout this section c denotes the observed conversion speed. Whenever $c_{\text{char}} = c$ is used, it is the explicit adopted Level-B scalar-clock calibration, not a derived gauge-emergence result; $c_\gamma = c_{\text{char}}$ remains a separate Open target (§4.7).

14.1 Pre-Lorentzian cosmological scenario

Status: Level A for the distinction between the P1 Euclidean configuration class and the separately supplied P4 ordered class; Level B/Open for their physical chronology, clock/metric transition and cosmological realisation.

Bridge from the gravitational sector. Section 13 supplies a conditional generally covariant completion of an ordered branch. The present subsection asks the separate question of which configuration classes are present before any chronology is assigned to them.

Proposition (Euclidean and ordered configuration classes). P1 admits Euclidean configurations, and the separately supplied P4 branch admits configurations with a preferred direction. In the former class no supplied scalar Lorentzian principal tensor has yet been declared. This is a Level-A statement about the stated configuration classes, not a theorem that one is a physically earlier cosmological epoch or that physical spacetime, clock and metric are absent/present in a derived history.

Cosmological realisation (Level B/Open scenario). Interpreting the two classes as successive physical epochs requires a nonequilibrium state and ordering dynamics, clock/lapse, observable metric and transfer to matter/radiation. Those owners are Open. The term “pre-Lorentzian” therefore denotes a candidate branch relation rather than ordinary clock-time evolution inside an already supplied spacetime. Consequently the Big Bang as a branch boundary, the pre-Lorentzian horizon reframing and the trans-Planckian interpretation remain scenario-level claims.

Characteristic scales (Open matching). The homogeneous P3 pole $m_{\Phi,\text{rad}}^2 = 2\mu^2$, a P4 pole, the physical tensor scale and the Planck energy are distinct owners. Neither $m_{\text{P4,pole}} = m_{\Phi,\text{rad}}$ nor a Planck-scale value for either mass is derived. No earliest Lorentzian relaxation time follows until the nonequilibrium P4 action, state and clock map are supplied.

Conceptual consequences. (i) *Reinterpretation of the Big Bang.* Within this scenario, the backward boundary of a supplied Lorentzian branch may be represented by a different configuration class of the same arena. This is a candidate reinterpretation, not a derived replacement of the Big Bang or a theorem about solutions of completed nonequilibrium dynamics.

Singularities as ordered-branch boundaries: unified programme. The ECT singularity programme should be read in three layers. First, the ordered-branch Lorentzian geometric EFT is only a coarse-grained branch description, valid when the condensate remains in its ordered regime. Accordingly, divergent classical GR singularities are treated only as candidate indicators that a supplied ordered geometric branch may have left its domain of applicability. Whether and where the physical branch actually breaks down remains Open (Level B/Open). Second, in the present black-hole (§36, Appendix AZ.7) and cosmological constructions, this interpretation is implemented structurally through the critical-shell and pre-ordered-regime logic: before one reaches a putative classical singularity, the ordered branch itself enters a breakdown region in which the coarse-grained Lorentzian description is no longer trusted. Third, what remains open is the unique microscopic completion of that breakdown region: ECT does not yet provide a full solved interior state, a unique bounce, or a completed non-perturbative resolution of all singularity questions.

Breakdown criterion for the ordered geometric branch. The ordered-branch Lorentzian EFT is expected to fail when the local state of the condensate approaches the critical regime in which the ordered amplitude description ceases to be valid. The relevant indicators are: (i) the ordered-branch amplitude $\rho \rightarrow 0$; (ii) the local effective temperature $T_{\text{loc}} \rightarrow T_c$ (Appendix AZ.7); (iii) the geometric curvature radius or gradient length of the ordered branch approaching a breakdown scale derived from a specified

P4 completion. No such scale is currently owned; the homogeneous P3 correlation length cannot be substituted for it. This candidate breakdown criterion points to the Open P4 threshold programme of §8.3. No current result identifies the P3 homogeneous curvature with a P4 pole or proves that one mass controls the validity of the reduced ordered-branch action. Interpreting singular behaviour as branch failure therefore remains Open.

Open branch-boundary interpretation of a cosmological singularity. The manuscript considers, but does not derive, an interpretation in which a formal backward-extrapolated Lorentzian Big-Bang singularity signals the breakdown of the ordered-branch effective description rather than a physical first instant. A possible prior configuration is the P1 Euclidean class, but its physical connection to the ordered branch is not yet solved. The branch-boundary reading of the Big-Bang singularity is therefore a Level-B/Open scenario, not an established replacement or a completed bounce.

(ii) *Horizon problem reformulated.* The standard horizon problem may be reformulated, conditionally, as a branch-coherence question: under what conditions can an ordering transition be macroscopically coherent on a connected progenitor domain of sufficient extent? This Open route is developed in §14.3.

(iii) *Trans-Planckian issue softened.* If a completed ordering history replaces an indefinitely extended earlier Lorentzian branch, the usual trans-Planckian extrapolation may be altered. Without the state, clock/metric and perturbation transfer, no resolution is yet derived.

Comparison with other pre-Big-Bang approaches. Several other frameworks address the question of what precedes the standard Big Bang. In the Hartle–Hawking no-boundary proposal [7], a Euclidean geometry replaces the initial singularity via a path integral over compact metrics. In Vilenkin’s tunnelling-from-nothing scenario, the Universe nucleates quantum-mechanically from a state with no prior spacetime. In cyclic/ekpyrotic models, a pre-existing dynamical stage within Lorentzian/cosmological language is assumed. ECT differs at the scenario level by using its P1/P4 configuration classes as the proposed ingredients. Their chronology and physical transition remain additional completion problems rather than consequences of signature classification alone.

Status summary. *Strictly established (Level A):* the distinction between the P1 Euclidean configuration class and the separately supplied P4 ordered class. *Scenario-level (Level B/Open):* their chronological realisation, physical clock/metric transition, detailed nonequilibrium dynamics and observable relics.

The comparative cosmological timeline is shown in §16.5 (§16.7); quantitative details of Scenarios A and B are discussed there.

14.2 Post-transition ordered-branch cosmological scenarios

Status: Level B/Open for the scenario classification and its declared consistency conditions. Neither ordered-domain growth nor positive energy selects the observed expanding branch. Branch orientation (OP3.2) partially clarified: time-reversal covariance lemma (Level A) and conditional persistence criterion (Level B) below. Open for first-principles ordering dynamics (OP3.1), exact $\phi_b(z)$ trajectory, thermal completion of the hot Big-Bang history (OP3.3), and CMB-peak structure with ECT matter content and matching Open. Connection: pre-Lorentzian regime (§14.1), background reduction (Eqs. (13.41)–(13.43)), far-future Scenarios A/B (§16.1), conditional bare scalar-ratio identity (eq. (16.2); $\omega > 0$ and a positive dimensionally normalised bare denominator), second law (§3.9), arrow of time (§3.8).

Problem statement: branch onset versus cosmological history

Two questions must be sharply distinguished.

The first is why the $O(4) \rightarrow O(3)$ transition occurred at all. This is the full nonequilibrium transition problem (OP3) and is not addressed here. For status bookkeeping it is useful to decompose OP3 into four sub-problems of different difficulty: *OP3.1* — the ordering mechanism itself, including the microscopic

derivation of the closure functions from bare P3 (open; sharpened below into subproblems OP3.1a–c with recorded interface requirements; mechanism open); *OP3.2* — branch orientation: why the observed branch expands (partially clarified below by a time-reversal covariance lemma and a conditional Level B persistence criterion); *OP3.3* — thermal completion: entropy production, thermalisation, radiation domination, and BBN-compatible initial data (open; a conditional bookkeeping ledger and a late-stage Kibble–Zurek scaffold are recorded below as necessary constraints and a definition/open-problem pair; neither can be extrapolated to the Euclidean ordering process before a physical chronology is supplied); *OP3.4* — relics of the ordering event (open observational programme).

Sharpened decomposition of OP3.1 (Open; programme sharpened). The ordering problem separates into three subproblems. *OP3.1a (bare-to-ordered gap)*. The bare micro-action is explicitly recorded (eq. (2.3), with field equation (38.1)), but the gradient-ordered sector is introduced by P4 as a postulate: the current minimal bare action does not by itself exhibit the gradient-ordered sector, and the article records OP-grad as the missing bridge. Exhibiting an ECT-native bridge — an admissible $P(X)$ -type extension, a boundary-data mechanism of the recorded boundary-value formulation, or another mechanism — coincides with completing OP-grad. *OP3.1b (EFT-data derivation)*. The task is to derive the ordered-branch EFT data $Z_u, \mathcal{C}_n, W(u)$ of (3.9) by controlled expansion around the ordered configuration — hence F, ω, U within the recorded closure-function map (§13.5) and its normalisation freedoms; this subsumes the recorded matching problem OP2. Postulating $W(u)$ does not close OP3.1. *OP3.1c (selection as measure concentration)*. The task is to establish dominance of the ordered sector within the recorded S_0 -weighted Euclidean extremal-action framework (§22.2); “selection” here is a statement about the Euclidean measure, not a process in time.

Interface requirements (recorded; no mechanism is implied). Any candidate ordering mechanism must not presuppose emergent-time or causal-cone structure prior to ordering; it must formulate selection as Euclidean measure concentration rather than time evolution; it must derive the ordered sector (P4) and the EFT data rather than assume them; it must hand ρ_{onset} to the thermal-completion ledger only after an ordered Lorentzian branch exists, with no energy statements at the pre-emergent Euclidean level; and it must respect the recorded no-ghost expectations. The bare scalar-ratio identity (16.2) is a bookkeeping check after a state is supplied, not an interface condition on the observable expansion history. OP3.1 remains Open; interface requirements recorded / programme sharpened.

The second is: given that an ordered Lorentzian branch has formed, which cosmological histories are dynamically admissible, structurally natural, or excluded? This is the question addressed in the present subsection.

A further distinction is equally important: *a proposed ordered-branch onset is not identical to thermal Big-Bang completion*. The proposed onset supplies a preferred-direction background; scalar hyperbolicity additionally assumes $\beta(\beta - \alpha) < 0$ (or $\alpha > \beta > 0$ on the retained $\beta > 0$ component), while a physical metric, clock and common causal cone remain Open. Thermal completion requires further entropy-production, thermalisation, radiation-era and BBN-compatible state owners.

Thermal-completion bookkeeping ledger (Level B conditional). The following necessary or conditional bookkeeping constraints organise any thermal-completion mechanism once an FLRW reduction of the ordered branch is available; they constrain OP3.3 but do not close it, and no mechanism is implied. Notation: ρ_{onset} is the ordered-branch *effective* energy density at branch onset, defined after emergence/matching within the FLRW reduction — not an energy statement at the pre-emergent Euclidean level; T_{th} is the thermal-completion temperature at which a standard radiation-dominated bath is first established (no inflationary reheating mechanism is assumed); w_{pre} is the time-averaged effective equation of state of the dominant condensate sector between onset and thermal completion. The scalar-stress lemma (16.2) does not establish a bound for that nonequilibrium effective sector; the first-pass ledger therefore explicitly assumes the standard range $-1 < w_{\text{pre}} \leq 1$. (*L1*) *Conversion ceiling*. If at any moment a fraction of ρ_{onset} were converted into an instantly thermalised bath, its temperature

obeys $T_{\max} \leq (30\rho_{\text{onset}}/\pi^2 g_*(T_{\max}))^{1/4}$; this is a ceiling, not a prediction — if conversion is inefficient or delayed, $T_{\text{th}} \ll T_{\max}$. Its numerical size depends on the matching estimate for ρ_{onset} and on the effective relativistic degrees of freedom. *(L2) Radiation-domination floor.* Any completion must deliver $T_{\text{th}} \gtrsim T_{\text{BBN}}^{\min} \sim \text{few MeV}$, the lowest thermal-completion temperature compatible with standard BBN and neutrino decoupling; a representative decaying-particle analysis gives $T_{\text{th}} \gtrsim 4 \text{ MeV}$ at 95% C.L., with channel-dependent islands down to $\sim 1 \text{ MeV}$ if the dominant decay is directly to neutrinos [96]. *(L3) Pre-thermal expansion corridor.* With $\rho_{\text{th}} = \frac{\pi^2}{30} g_*(T_{\text{th}}) T_{\text{th}}^4$ and an approximately constant $w_{\text{pre}} > -1$, the dilution law $\rho \propto a^{-3(1+w_{\text{pre}})}$ gives

$$N_{\text{pre}} := \ln \frac{a_{\text{th}}}{a_{\text{onset}}} = \frac{1}{3(1+w_{\text{pre}})} \ln \frac{\rho_{\text{onset}}}{\rho_{\text{th}}}, \quad w_{\text{pre}} > -1;$$

the case $w_{\text{pre}} \rightarrow -1$ degenerates (quasi-de-Sitter-like behaviour) and requires separate treatment. For fixed ρ_{onset} , T_{th} , and $g_*(T_{\text{th}})$, this equation gives only a bookkeeping corridor for the duration of the pre-thermal ordered-branch epoch; it is not a prediction of w_{pre} or of N_{pre} . *(L4) BBN expansion-response condition.* A useful external proxy holds the gravitational rescaling constant during the short BBN epoch, keeps the particle scales fixed, and writes $H \propto \sqrt{G_{\text{BBN}}^{\text{proxy}}}$. In that restricted model, a 2020 BBN+CMB analysis finds $G_{\text{BBN}}^{\text{proxy}}/G_0 = 0.99^{+0.06}_{-0.05}$ at 2σ [97]. This is not a direct inequality on $G_{\text{div}} = G_*/F$. In the scalar action, $\bar{F}$, scalar kinetic energy, and U enter the expansion equation alongside the divided coefficient. Compatibility must therefore be tested by solving the complete background, propagating the physical $H(T_\gamma)$ and matter microphysics through the reaction network, and evaluating the abundance likelihood. Only if that solution reduces to the constant-rescaling proxy may the quoted interval be transferred to G_{div} . No such ECT-specific transfer or relaxation mechanism is presently derived. *(L5) Entropy compatibility.* The comoving entropy of the bath at thermal completion, $S = \frac{2\pi^2}{45} g_{*s}(T_{\text{th}}) T_{\text{th}}^3 a_{\text{th}}^3 V_c$, is, if subsequent evolution is approximately adiabatic, comparable with the observed entropy in the present observable Universe, $S_{\text{obs}} \sim 10^{88}$, after specifying the comoving-volume convention V_c ; non-adiabatic late evolution would modify this accounting. Combined with L1–L3 this is an accounting identity linking $(\rho_{\text{onset}}, w_{\text{pre}}, N_{\text{pre}}, T_{\text{th}})$; the mechanism of entropy production is not derived and is not implied. *(L6) Charge-selective routing (event-ordering condition).* Within the recorded charge assignments, the relaxing amplitude mode carries no recorded holonomy or charge label, and within the recorded induced-gauge architecture it supports no tree-level σF^2 portal, loop-induced portals reducing to the charged-sector selection rule (§16.5). Conversion of condensate relaxation energy into a gauge/radiation bath therefore proceeds through the charged matter chain, whose masses anchor to $m_f = y_f v_2 / \sqrt{2}$ with $v_2 = \phi_0 e^{-\mathcal{J}_{\text{EW}}}$. Any thermal-completion mechanism hence inherits the event ordering: gradient ordering $\rightarrow$ metric/ Φ -sector matching (so that the v_2 chain exists) $\rightarrow$ charged-spectrum population $\rightarrow$ entropy production and radiation domination (L2). This is a routing/ordering constraint on mechanisms, conditional on the recorded charge-based selectivity; no conversion efficiency, timescale, or mechanism is implied, and, as for L1–L5, no statement is made at the pre-emergent level.

Late-stage Kibble–Zurek scaffold (Open; definition recorded). *Definition (late-stage ordering window).* Validity domain: an effective Lorentzian ordered-branch description exists; coarse-grained ordering variables and an ordering time parameter are meaningful; the passage through the ordering threshold is characterised by an effective quench-like rate in the already Lorentzian ordered branch. Input data: relaxation time $\tau(\varepsilon)$ and correlation length $\xi(\varepsilon)$ of the ordering variables near threshold, and a freeze-out scale $\hat{\xi}$, if standard Kibble–Zurek matching conditions [98, 99] can be justified in the late-stage ordered-branch regime (see [100] for a review). Output target: a characteristic domain/texture correlation-volume scale — for pointlike relic counts one would estimate $n \sim \hat{\xi}^{-3}$; for the $\pi_3(S^3)$ texture sectors (Appendix AB) this is a domain/texture correlation-volume estimate, not an automatically conserved defect number. Such estimates constrain possible residual gradient/defect energy contributions, which must remain compatible with the bookkeeping ledger above; they are not an entropy source. *Open problem (OP3.3, sharpened).* Exhibit the ECT-native $\tau(\varepsilon), \xi(\varepsilon)$ from the ordered-branch EFT and justify the matching assumptions; no

exponents are computed here and no claim is made. *Firewall*. The scaffold applies only after an effective Lorentzian description exists; it is not an account of the Euclidean ordering process before a physical chronology is supplied, which belongs to OP3.1.

Admissible post-transition scenarios

The effective background dynamics of the ordered branch is governed by the scalar–tensor system Eqs. (13.41)–(13.43), developed in §13.5. Within the ϕ -first closure, the qualitative post-transition behaviour is controlled by the sign and magnitude of the residual condensate energy $U(\phi)$, the kinetic coupling $\omega(\phi)$ and its sign (no-ghost condition), the non-minimal response factor $F(\phi) = e^{\beta_\phi \phi}$, and the matter/radiation content.

The effective Friedmann equation admits both expanding and contracting solutions, $H = \pm\sqrt{\dots}$. Positive condensate energy permits both real branches and supplies no dynamical preference for the sign of H . The branch-orientation question (OP3.2) is partially clarified by the following statements.

Lemma: time-reversal covariance of the homogeneous background system (Level A). The homogeneous background equations (13.41)–(13.43) are covariant under the simultaneous reversal $t \mapsto -t$, $H \mapsto -H$, $\dot{\phi} \mapsto -\dot{\phi}$, $\dot{F} \mapsto -\dot{F}$, with $\ddot{H}$, $\ddot{\phi}$, and $\ddot{F}$ unchanged. Hence, before a thermodynamic arrow and retarded boundary conditions are specified, the sign of H alone does not define an invariant physical distinction: an expanding solution in one parametrisation is paired with a contracting solution in the reversed parametrisation. This removes a purely parametrisation-dependent part of the expansion-versus-contraction question, but it does not by itself select the observed branch. Once the thermodynamic arrow and retarded support are fixed (§3.8), the physical branch-orientation problem remains and is addressed conditionally below.

No-bounce sublemma (spatially flat positive-potential/no-ghost sector; Level A conditional). At a putative regular crossing $H = 0$, the retained Friedmann constraint gives

$$0 = \frac{\rho_m + \rho_r + U(\phi)}{\bar{M}_{\text{Pl}}^2} + \frac{1}{2} \omega(\phi) \dot{\phi}^2.$$

For $F > 0$, $\omega > 0$, $\rho_m, \rho_r \geq 0$ and a smooth $U \geq 0$ in the relevant neighbourhood, every term must vanish. At the all-zero corner, smooth non-negativity also gives $U_{,\phi} = 0$. The scalar and Raychaudhuri equations then imply

$$\omega \ddot{\phi} = 3F_{,\phi} \dot{H}, \quad -2F \dot{H} = F_{,\phi} \ddot{\phi},$$

and hence

$$\left(2F + \frac{3F_{,\phi}^2}{\omega} \right) \dot{H} = 0.$$

The coefficient is positive, so $\dot{H} = 0$: the all-zero solution is not a regular bounce. Thus a smooth solution cannot change the sign of H inside this stated sector. Spatial curvature, $U < 0$, ghosts, singular evolution or terms outside the retained closure reopen the question.

What persistence does and does not imply. The no-bounce result establishes only sign persistence of H in its stated domain. It does not fix the sign of $\dot{H}$, monotonic growth of $|H|$, total density or curvature, nor a breakdown time. For example, a contracting de Sitter solution is obtained with

$$H = -H_\Lambda, \quad \dot{H} = 0, \quad \phi = \phi_0, \quad \rho_m = \rho_r = 0,$$

$$U(\phi_0) = 3\bar{M}_{\text{Pl}}^2 F_0 H_\Lambda^2, \quad U_{,\phi}(\phi_0) = 6\bar{M}_{\text{Pl}}^2 H_\Lambda^2 F_{,\phi}(\phi_0),$$

as realised locally by $U = 3\bar{M}_{\text{Pl}}^2 H_\Lambda^2 F^2 / F_0$. It has no bounce, but its density and curvature are constant.

A narrower monotonicity statement survives for any separately conserved perfect-fluid component:

$$\dot{\rho}_i = -3H(1 + w_i)\rho_i.$$

If $H < 0$, $\rho_i > 0$ and $w_i > -1$, then $\dot{\rho}_i > 0$. Inferring total curvature growth or EFT breakdown additionally requires that component to remain conserved and dynamically relevant, control of non-minimal cancellations and energy exchange, a specified breakdown threshold, and bounds on $H(t)$ and $w_i(t)$. A timescale of order $|H|^{-1}$ follows only under further slow-variation and order-one logarithmic-distance assumptions. Consequently neither the contracting nor expanding physical branch is selected by this lemma; ordering dynamics, state and arrow-compatible boundary conditions remain Open (OP3.1).

Table 55: Qualitative post-transition cosmological branches in ECT.

Scenario	Structural conditions	ECT status
Expanding branch with a conditional late scalar source	Requires a selected action and state; a positive potential alone does not derive the physical source or metric	Proof of class inside supplied completions; ECT ownership Open
Asymptotic de Sitter in the two-slope completion	$V = V_- e^{cq} + V_+ e^{bq}$ with $0 < c < 2a < b$, positive coefficients and regular expanding initial data	Level A inside the named action; calibrated background Level C; ECT map Open
Branch degeneration (Scenario B of §16.1)	$\phi_b \rightarrow -\infty$, $u \rightarrow 0$, so G_{div} grows; the physical gravitational response is not fixed by this coefficient alone and the geometric chart eventually degenerates	Level B coefficient/chart statement; response and timescale Open
Always decelerating	$U(\phi) \rightarrow 0$ rapidly; no persistent vacuum-like component	Possible; observationally disfavoured
Contracting branch	$H < 0$ sign choice	Kinematically admissible; no-bounce fixes only sign persistence. Growth, breakdown and any timescale require a specified action/state; first-principles selection remains open (OP3.1)
Bounce / oscillatory	Turning points in ϕ dynamics; nontrivial $U(\phi)$	Excluded in the positive-potential/no-ghost sector by the Friedmann constraint (no-bounce sublemma); admissible only if $U < 0$ or scalar–tensor derivative terms dominate; not developed
Effective phantom-like history ($w_{\text{eff}} < -1$)	Determined by the solved $H(a)$ and non-minimal $F, \dot{F}, \ddot{F}$ terms	Not excluded by the bare identity (16.2); test each complete action/state through its observable map and likelihood

Why phase-ordering growth does not select the cosmological sign

Growth of ordered domains in phase-ordering systems is sometimes used as an analogy for a post-transition branch. In condensed-matter systems undergoing a symmetry-breaking transition, ordered regions generically *grow*—they expand into the surrounding disordered medium, a late-stage process often modelled by Allen–Cahn or Kibble–Zurek-type effective descriptions once an ordering time parameter is meaningful. That spatial coarsening occurs in a separately clocked dissipative source model does

not select the sign of an FRW scale factor. In ECT the required chronology, transition action and state, dissipative kernel, and map from an ordering domain to a physical metric and scale factor are not derived. Consequently the analogy supplies neither evidence nor a Level-B/C preference for an expanding branch. A sign-selection result would require the completed nonequilibrium ordering dynamics (OP3.1), branch-orientation owner (OP3.2) and thermal completion (OP3.3).

Minimal viable post-transition branch

A minimally viable post-transition ordered-branch history must satisfy the following conditions simultaneously:

- (i) *expanding ordered branch selected*;
- (ii) *early scalar background and expansion compatible with BBN*: the same solved action/state must yield a regular $H(T_\gamma)$ and microphysical BBN pipeline within the observational likelihood. A nearly constant F and a response-calibrated nearly constant G_{div} are at most useful reduction condition, not by itself BBN compatibility;
- (iii) *hot radiation epoch available*: thermal completion of the hot Big-Bang branch, including entropy production, thermalisation, and a radiation-dominated era (open—§14.1);
- (iv) *delayed effective late-component dominance*: after one explicit scalar–tensor bookkeeping convention is fixed, its action-owned effective late component must be subdominant early and comparable to matter only late; the bare $K + U$ pieces alone do not define that component;
- (v) *conditional bare scalar-ratio diagnostic*: $w_\phi^{\text{bare}} \geq -1$ follows at Level A inside the supplied stress when $\omega(\phi) > 0$ and $K_\phi^{\text{phys}} + U > 0$, but places no bound on w_{eff} ;
- (vi) *perturbation sector not grossly inconsistent*: a single adiabatic scalar mode at the structural level, with a conditional route toward a near-scale-invariant spectrum (§14.6). Restricted growth diagnostics exist, but observational compatibility with large-scale structure is Open (§AH).

A nearly frozen supplied scalar component can become relevant after matter dilutes; this is generic conditional scalar-field phenomenology, not an ECT-native trajectory or explanation. The action-owned background, initial state, physical metric/clock map and observed late-time scale remain Open.

Connection to the arrow of time

The expanding-branch sign and the thermodynamic arrow are separate questions. The ordered branch supplies causal response, while a thermodynamic entropy-production theorem requires a specified reduced channel, coarse graining, entropy functional, and detailed-balance/contractivity assumptions. No alignment of the cosmological and thermodynamic arrows follows from a large decoherence scalar alone, and the sign choice $H > 0$ remains Open.

Observational discriminants

The scenario classification connects to several observational channels already developed elsewhere in the paper. These are not new derivations but reinterpretations as *branch discriminants*:

- (i) *conditional bare scalar-ratio sign*: if $\omega(\phi) > 0$ and $K_\phi^{\text{phys}} + U > 0$, the supplied bare pieces obey $w_\phi^{\text{bare}} \geq -1$. This is an algebraic identity, not a sign prediction for the action-owned effective expansion history;
- (ii) *running $G_{\text{div}}(z)$ and response transfers*: the coefficient varies by $O(\beta_\phi \Delta\phi_b)$, but BBN constrains the full expansion/microphysics map, while lunar ranging and pulsar timing constrain distinct Cavendish, orbital and binary response maps (§16.1);

- (iii) *correlated ΔH_0 –age–JWST signatures*: a future complete action must jointly determine these through one $\phi_b(z)$ (§16.5);
- (iv) *S_8 and weak lensing*: a future conditional test of the gauge-reduced growth and photon-metric completion; no ECT growth kernel or alleviation is presently derived (§AH);
- (v) *Scenario A versus B*: a measured $\dot{G}_{\text{Cav}}$ or orbital response constrains a completed scalar/range/body-charge transfer. A zero or nonzero fitted value does not uniquely select either scenario without additional no-cancellation and branch assumptions (§16.1).

What is explained and what remains open

Established only conditionally: the homogeneous equations are time-reversal covariant; the stated spatially flat positive-potential system has the conditional no-bounce lemma; and the supplied stress has the separate bare scalar-ratio identity; and the separately supplied two-slope action has a stable late-de-Sitter proof of class. None of these statements selects the physical ECT action, state, late source, or arrow. The signs of the cosmological and thermodynamic arrows require separate dynamical owners, and their alignment is not presently explained.

Open:

- (i) the full nonequilibrium ordering dynamics that forms the Lorentzian branch itself (OP3.1);
- (ii) first-principles derivation of the assumptions entering the branch-orientation persistence criterion (OP3.2), including onset-neighbourhood positivity, controlled scalar–tensor derivative terms, and the absence of an internal bounce mechanism;
- (iii) the exact background trajectory $\phi_b(z)$;
- (iv) thermal completion of the hot Big-Bang history (entropy production, thermalisation, radiation bath);
- (v) reproduction of precision early-universe observables—in particular, the acoustic CMB peak structure with the cosmological matter content and matching left Open;
- (vi) precision determination of the observed cosmological parameters from a global ECT fit.

Status summary. *Level A conditional mathematics*: (i) the distinction, inside the supplied branch construction, between its Euclidean-side configuration domain and its Lorentzian-side chart; (ii) the distinction between a geometric branch onset and a separately supplied thermal hot-Big-Bang completion; (iii) time-reversal covariance of the supplied homogeneous background equations. The physical existence, state and formation dynamics of a pre-ordered regime remain Open.

Level A inside the supplied scalar action / ECT Open: (i) the dimensionally normalised bare scalar-ratio inequality; it is not an observable w_{eff} bound.

Level B: (i) the scalar-only FLRW closure route for the ordered branch; (ii) future Scenarios A/B; (iv) post-transition scenario classification (Table 55); (v) minimal viable post-transition branch conditions; (vi) observational discriminants reinterpreted as branch discriminants; (vii) conditional persistence criterion for the expanding branch orientation (under explicit positivity, no-ghost, and no-bounce assumptions); (viii) thermal-completion bookkeeping ledger (conditional necessary constraints L1–L5; mechanism open).

No assigned ECT status: the condensed-matter domain-growth analogy is motivation only and supplies no preference for the expanding branch.

Open: (i) first-principles ordering dynamics (OP3.1; interface requirements recorded, subproblems OP3.1a–c separated above — programme sharpened, mechanism open); the branch orientation (OP3.2) is reduced to a conditional persistence criterion whose assumptions (potential positivity at onset, controlled scalar–tensor derivative terms) remain themselves open; (ii) exact $\phi_b(z)$ trajectory; (iii) thermal completion of the hot Big-Bang branch; (iv) precision CMB acoustic structure without particle dark matter; (v) precision global cosmological fit.

14.3 Horizon problem: branch coherence

Status: Level A only for the minimum of the separately supplied sigma-model functional on a connected domain whose admissible boundary and topological class contains a constant map; Open for an ECT cosmological horizon solution, because the ordering action/state, admissible progenitor class, coherence scale, curvature transfer and CMB likelihood are not derived.

The pre-ordered configuration regime established in §14.1 provides the structural setting in which the standard horizon problem is reformulated.

The standard horizon problem is usually formulated inside an already existing Lorentzian FLRW spacetime: why do widely separated regions of the observed Universe appear so nearly homogeneous even though, in an ordinary early Lorentzian history, they would not have been in mutual causal contact?

In the Level-B/Open branch-history scenario, the observable physical Lorentzian completion is not taken as primordial. The proposed sequence is:

$$\begin{aligned} &\text{pre-Lorentzian / Euclidean stage} \longrightarrow \\ &O(4) \rightarrow O(3) \text{ ordering transition} \longrightarrow \text{candidate physical Lorentzian completion.} \end{aligned}$$

Accordingly, the horizon question may be reformulated as asking whether a candidate progenitor branch was coherently ordered on a sufficient macroscopic scale. This branch-coherence route is not yet a solution: its state, ordering dynamics, clock/metric transfer and CMB likelihood remain Open.

One conditional carrier of such coherence is a separately supplied orientation-sector action associated with the P4 $O(4) \rightarrow O(3)$ ordered branch. Let this sector be described by a unit field

$$n_A(x), \quad n_A n^A = 1,$$

encoding the orientation of the selected branch. Its natural low-gradient Euclidean ordering functional on a connected progenitor domain Ω_{br} takes the sigma-model form

$$S_{\text{ord}}[n] = \frac{\kappa}{2} \int_{\Omega_{\text{br}}} d^d x (\partial_i n_A)(\partial_i n^A), \quad \kappa > 0. \quad (14.1)$$

Lemma (boundary-compatible minimum of the Euclidean ordering functional). Let Ω_{br} be connected and let the declared admissible class contain a constant map (equivalently, no imposed boundary value, topological sector or defect insertion excludes all constant representatives). Then the absolute minimum of S_{ord} in that class is $S_{\text{ord}} = 0$, and it is attained exactly by maps that are constant almost everywhere.

Proof. Since $(\partial_i n_A)(\partial_i n^A) \geq 0$ pointwise, $S_{\text{ord}}[n] \geq 0$. The assumed constant representative attains zero. Conversely, equality implies $\partial_i n_A = 0$ almost everywhere, which on a connected domain gives one constant representative. If the boundary data or topological sector excludes constant maps, the lower bound remains true but zero need not be attainable in that restricted class. $\square$

This establishes a precise but narrow Level A mathematical statement inside the stated admissible class. It does not establish that this functional is generated by P1–P3, that the realised boundary/state class contains a constant map, that the realised state occupies the minimum, or that the progenitor domain has the required size and connectivity.

This does not mean that all non-uniform branch realizations are forbidden. What it does establish is that a macro-coherent constant branch is *admissible in the supplied functional* whenever the declared boundary and topological class contains that representative. If the observed Universe corresponds to such a coherently selected connected branch, then its large-scale homogeneity is inherited from the branch-selection event itself rather than manufactured later by causal equilibration inside an already formed spacetime. In this sense, a macro-coherent branch transition is a candidate route for reformulating the standard horizon problem, not its present solution.

The assumption that the progenitor domain is connected, and that its coherence scale is large enough to satisfy

$$L_{\text{br}} \frac{a_0}{a_{\text{br}}} \gtrsim L_{\text{hom}},$$

is at present a working hypothesis rather than a first-principles derivation. Its role here is to formulate the coherent-branch scenario precisely. A dedicated quantitative estimate of the required coherence scale is left for future work. As a purely kinematic benchmark, this condition may be rewritten as

$$L_{\text{br}} \gtrsim L_{\text{hom}} \frac{a_{\text{br}}}{a_0}.$$

Thus the required coherence scale on the progenitor branch is smaller than the present homogeneity scale by the same expansion factor that separates the branch-selection epoch from today. This does not yet solve the branch-selection dynamics, but it clarifies that the required pre-branch coherence need not be macroscopic in today's units.

At the level of classical action comparison, the coherent branch is also the natural low-gradient baseline. This stronger preference statement goes beyond the exact minimum result above and should therefore be interpreted more cautiously: large $O(1)$ twists of the orientation field across a scale L carry a positive action cost

$$S_{\text{ord}} \sim \kappa (\Delta\theta)^2 L^{d-2},$$

so macroscopically fragmented branches are not free. This preference statement, however, is made only at the level of classical action comparison; a full semiclassical treatment including the entropy of the configuration space would be needed to establish the preference rigorously. Accordingly, the preference claim should be read as a Level B statement.

The net result is a conditional coherent-branch scenario distinct from inflationary stretching, not a derived solution. The displayed minimum defines a calculable acceptance route for the horizon problem within a supplied ordering model. What remains open is the full nonequilibrium dynamics of branch selection, the semiclassical weighting of coherent versus fragmented realisations, and the quantitative mapping to the observed primordial perturbation sector. The flatness problem is addressed in Section 14.4, and the relic and monopole problem in Section 14.5.

14.4 Flatness problem: Euclidean structural input and the missing physical-metric bridge

Status: Level A for flatness of the P1 Euclidean ambient metric; Level B/Open for spatial curvature of the physical cosmological metric.

In standard FLRW cosmology the flatness question concerns the curvature of the physical spatial metric seen by matter and photons, ${}^{(3)}R[h_{\text{phys}}] = 6k/a^2$, or equivalently the observable $\Omega_K = -k/(a^2 H^2)$. Postulate P1 instead states that the pre-Lorentzian ambient configuration space has the flat Euclidean metric δ_{AB} . Those are not the same assertion: ambient flatness does not, by itself, derive the induced physical FLRW metric, its embedding, lapse/shift, matter congruence, or the rule connecting the ordered phase to photon geodesics.

There is a useful conditional construction. If a macrocoherent ordered branch is represented by a planar aligned interface with constant director n_A , if the physical spatial metric is exactly the induced metric on the orthogonal leaves, and if the subsequent homogeneous evolution is $h_{ij} = a^2(t)\delta_{ij}$, then

$${}^{(3)}R[h] = 0, \quad k = 0, \quad \Omega_K = 0. \quad (14.2)$$

This is exact inside the stated geometric completion. It is a proof-of-class for a spatially flat ECT branch, not a theorem that every physical ECT completion inherits $k = 0$ from P1.

The external Euclidean view makes the missing step explicit. We use the Riemann-sign convention $(\nabla_\mu \nabla_\nu - \nabla_\nu \nabla_\mu)V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma$. For an induced three-metric h_{AB} and a unit normal with $n^A n_A = \varepsilon$, the pointwise contracted Gauss identity is

$${}^{(3)}R[h] = h^{AC} h^{BD} R_{ABCD}^{(4)} + \varepsilon (K^2 - K_{ab} K^{ab}). \quad (14.3)$$

Here the ECT ambient Euclidean convention has $\varepsilon = +1$; for a flat ambient space the projected $R_{ABCD}^{(4)}$ term vanishes. Boundary divergences arise instead when an ambient scalar-curvature *action* is decomposed

into leaf and normal pieces; they are not terms in this pointwise Gauss identity. Consequently, if one additionally identifies an umbilic codimension-one embedding by $K_{ij} = (H/c)h_{ij}$, then $K^2 - K_{ij}K^{ij} = 6H^2/c^2$. Requiring the same leaf to have ${}^{(3)}R = 0$ in flat E^4 therefore forces $H = 0$.

This conditional calculation rules out only that simultaneous naive identification; it does not derive a P4 physical metric, foliation, lapse or shift, heavy-mode vertex, tensor residue, or gravity owner. A physical completion may require a distinct response fibre, non-umbilic embedding, higher codimension, curved effective ambient geometry, or a metric not equal to the naive induced metric. Which route is realised is Open and must follow from the same varied action and state used for age, perturbations and lensing.

The observed near-flatness bound is consequently a sharp acceptance test. The conditional planar baseline is compatible with it, but ECT has not yet derived a unique value or distribution for Ω_K . Perturbative backreaction, imperfect coherence and global topology are separate questions; local Euclidean flatness neither fixes them nor excludes compact flat topologies. Unlike inflationary dynamical suppression, the ECT route is at present a structural candidate whose decisive missing owner is the physical cosmological metric/embedding map.

14.5 Relic and monopole problem: conditional target-space classification

Status: Level-A mathematics for the homotopy of the supplied S^3 orientation target; Open for its identification with the complete physical vacuum/configuration space, the ordering transition and relic production. Later-sector relic details are likewise Open/conditional.

In many grand-unified-theory (GUT) extensions of the standard model, a hot early Universe is expected to produce an overabundance of heavy topological relics — magnetic monopoles, domain walls, cosmic strings — whose predicted density would vastly exceed observational bounds. Standard inflation addresses this through dynamical dilution: exponential expansion reduces any pre-existing relic density to negligible levels.

In ECT the relevant question is whether a future owner-complete Lorentzian branch has a vacuum/configuration target whose topology protects relics. The present P4 orbit alone does not answer that physical question; it supplies a conditional mathematical classification and a list of missing owners.

(i) Homotopy of the supplied orientation target (Level-A conditional mathematics). The nonzero P4 vector has abstract orbit

$$\mathcal{T}_{P4} = O(4)/O(3) \simeq S^3.$$

The homotopy groups of S^3 are [9]:

$$\pi_0(S^3) = 0, \quad \pi_1(S^3) = 0, \quad \pi_2(S^3) = 0, \quad \pi_3(S^3) = \mathbb{Z}.$$

For maps into this supplied target:

- $\pi_0(\mathcal{T}_{P4}) = 0$: no π_0 protection;
- $\pi_1(\mathcal{T}_{P4}) = 0$: no π_1 protection;
- $\pi_2(\mathcal{T}_{P4}) = 0$: no π_2 protection.

These statements do not establish absence of physical walls, strings or monopoles. That conclusion requires a stationary physical vacuum, its full configuration space, exact-gradient admissibility, boundary conditions and a transition history, all of which remain Open.

A nontrivial $\pi_3(S^3) = \mathbb{Z}$ gives integer classes of maps into the supplied target. Whether any such map is an admissible finite-action ECT configuration, let alone a cosmological texture, is Open. The status-separated inventory of candidate states and conditional topological sectors, including conditional secondary defects, is given in Section 8.10. The present result is therefore a conditional target-space calculation, not a structural removal of the monopole catastrophe.

(ii) Standard GUT monopole overproduction is not structurally required (Level B). Standard GUT magnetic monopoles are topological defects of the GUT gauge symmetry, arising when a unified

group G_{GUT} breaks to a subgroup containing a $U(1)$ factor with $\pi_2(G_{\text{GUT}}/H) \neq 0$. This scenario requires a hot symmetric GUT phase that then cools through a sequence of thermal gauge-symmetry breakings. ECT does not postulate such a phase as a structural necessity of its cosmological architecture. The candidate ordered cosmological branch is associated with $O(4) \rightarrow O(3)$ rather than a derived hot GUT thermal history. If no such GUT phase belongs to a completed branch history, the standard monopole overproduction scenario does not apply: *ECT does not structurally require a monopole-overproducing GUT thermal history.*

(iii) Later-sector relics: further development, not structural catastrophe (Level B). The target-space calculation above covers only the supplied S^3 orbit. Remaining questions — including monopole-like configurations that might arise in the electroweak sector, non-topological heavy relics, and metastable branch configurations — are not yet fully classified in ECT. However, none of these has been shown to be structurally inevitable or to produce a cosmologically dangerous abundance in the observed branch. No smooth, relic-poor observed branch is selected by P4 or BR1. Relic production, dilution and abundance remain sector-specific physical questions.

Further development. If the electroweak-like condensate embedding is physically realised, it would involve a transition of the form $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$ at the matched scale $v_2 = 246 \text{ GeV}$. At present this sector should be treated as a structural candidate rather than an established part of the derived ECT core. A naive global-symmetry analysis gives $\pi_2(SU(2)/U(1)) \neq 0$, which may admit monopole-like configurations at this scale; however, in gauge theory this analysis is more subtle and the precise topological status of such objects depends on the UV completion. Their cosmological production in the ECT setting requires a dedicated analysis and belongs to the broader ECT relic programme.

Comparison with inflation. Standard inflation addresses the relic problem through a dynamical dilution mechanism operating inside an already existing Lorentzian cosmology: a separate inflationary sector dilutes any pre-existing relic density exponentially. ECT does not yet supply an alternative physical solution. Its supplied S^3 target has vanishing lower homotopy groups, but the physical vacuum and transition owner are Open. It also does not postulate a hot GUT history; whether a completed ECT cosmology contains one is a model-dependent question.

14.6 Primordial perturbations: current status

Status: Level A only for the one-scalar-coordinate inventory of the stated strict map; a one-mode physical EFT is Level B. The physical scalar projector, metric/clock map, tensor action and propagation speed, and the full derivation of n_s , A_s , r are Open.

Sections 14.1 and 14.3 define a pre-ordered-to-ordered scenario. P1 supplies a flat Euclidean substrate, but spatial flatness, macrocoherence and branch formation in the physical cosmological metric require an explicit embedding, state and nonequilibrium history and remain conditional/Open. The next question is therefore what perturbation content follows inside a declared ordered-chart cosmological closure and which parts of the physical primordial spectrum remain Open. The purpose of the present subsection is structural rather than fully predictive: to identify what is already fixed by the ordered branch, and what still requires a quantitative perturbation theory.

Observational target

The observed dimensionless primordial perturbation spectrum is nearly scale-invariant,

$$\Delta_\zeta^2(k) \equiv \frac{k^3}{2\pi^2} P_\zeta(k) \approx A_s \left(\frac{k}{k_*} \right)^{n_s-1}, \quad (14.4)$$

with $n_s = 0.9649 \pm 0.0042$ from Planck 2018 and $r_{0.05} < 0.036$ at 95% C.L. from the BICEP/Keck 2018 data [70, 101]. Any cosmological theory must eventually account for these observed values. ECT does not yet derive n_s or r from first principles, and the present work makes no such claim. However, the strict scalar map already fixes one longitudinal coordinate at a structural level, and a supplied reduced EFT

identifies a conditional route toward the primordial spectrum. The complete physical perturbation field content is not fixed.

Perturbation variables, scalar-coordinate reduction and conditional one-mode structure

The condensate field Φ on the ordered branch admits a natural decomposition into fluctuation sectors (§3): orientational (Goldstone) fluctuations δn_A , amplitude fluctuations $\delta\sigma = \delta|\partial\Phi|$, and phase fluctuations $\delta\theta$. These provide a useful perturbative coordinate inventory of the condensate/branch sector; the physical quotient and mode count remain to be specified.

Proposition (single scalar coordinate in the strict ordered-branch map). At leading linear order, every soft variation of the strict scalar-only map $Q_A = \partial_A\Phi$ is parametrised by one scalar coordinate χ .

Proof. By the Universality Corollary (§4.3), the integrability condition $Q_A = \partial_A\Phi$ reduces the linear soft coordinate variations of the strict ordered-branch map to derivatives of a single scalar coordinate χ . Hence the strict map possesses one scalar coordinate at leading order. $\square$

This is the structural analogue of a single-scalar parametrisation. A physical statement about one propagating degree of freedom additionally requires the supplied reduced action, constraint quotient, boundary domain and projector; it is Level B rather than a consequence of integrability alone. The result does not derive the observed curvature spectrum and does not exclude independent phase, gauge, matter or completion-sector scalars.

Conditional isocurvature statement. Inside a supplied reduced EFT whose entire physical scalar content is the single χ coordinate, there is no second independent scalar from which to construct an intrinsic entropy mode at leading order. This is a Level-B truncation statement, not a proof that the full ECT completion has no isocurvature sector. Phase, gauge, matter, constraint or additional-field completions remain unclassified and may supply such modes.

Tensor sector

The current ECT action does not identify a physical primordial tensor variable. A completion must supply its constrained quadratic Hessian, TT projector, principal symbol, normalised primordial state and coupling to the observable metric. Consequently the tensor speed, spectrum and tensor-to-scalar ratio are Open. The primordial and late-time GW sectors must arise from the same completed owner and pass the same cone, polarisation, damping and ghost gates; that is a consistency requirement rather than a present prediction.

Seed statistics, notation, and the projection diagnostic

Before estimating the spectrum it is useful to fix and separate the field content of the ordered-branch perturbation sector: $\Phi(X)$ is the fundamental real scalar; $Q_A \equiv \partial_A\Phi$ is the *gradient order parameter* used by the supplied P4 ordered-sector chart (this is the origin of the gradient-based architecture of ECT, not a particle-like Φ excitation); $u_0 \equiv |\langle Q_A \rangle|$ with $\langle \partial_A\Phi \rangle = u_0 \delta_{Aw}$ is the magnitude of that supplied coarse-grained datum; no stationary vacuum, condensate transition or branch-selection mechanism follows from this notation; χ parametrises the one longitudinal scalar coordinate of the strict linear map, the integrability constraint $\partial_{[A}Q_{B]} = 0$ producing this kinematic reduction (Universality Corollary). Interpreting χ as the complete physical scalar mode requires a supplied reduced action, constraint domain and projector (Level B); and ζ is the curvature perturbation, generated through a displacement of the ordering hypersurface,

$$\delta\Phi \longrightarrow \delta X_{\text{transition}} \longrightarrow \delta N \longrightarrow \zeta.$$

Projection diagnostic (Level A: conditional mathematics). Let $k = |\mathbf{k}| > 0$, let $s > 0$ be real, and let the relevant fluctuation have

$$G_4(p_w, \mathbf{k}) = \mathcal{A}_s (p_w^2 + k^2)^{-s},$$

where $\mathcal{A}_s > 0$ is real and independent of p_w and k over the declared band. These restrictions are what make G_4 an ordinary nonnegative covariance; the Gamma-function expression below has an analytic continuation in s , but that continuation is not by itself a physical covariance or power spectrum. Denote the complete field-to-curvature transfer, including any background-gradient normalisation, by $\zeta_k = \mathcal{T}_{\chi \rightarrow \zeta}(k) \chi_k$, with $\mathcal{T}_{\chi \rightarrow \zeta}(k) \sim k^q$. On an infinite translation-invariant w direction the equal-slice integral converges as an ordinary covariance if and only if $s > 1/2$, and in that domain

$$\begin{aligned} P_3(k) &= \int \frac{dp_w}{2\pi} G_4(p_w, \mathbf{k}) = \mathcal{A}_s \frac{\Gamma(s - \frac{1}{2})}{2\sqrt{\pi}\Gamma(s)} k^{1-2s}, \\ \Delta_\zeta^2(k) &= \frac{k^3}{2\pi^2} P_3(k) \mathcal{T}_{\chi \rightarrow \zeta}^2(k) \propto k^{4-2s+2q}. \end{aligned} \tag{14.5}$$

At $k = 0$ no positive real value of s makes the uncut integral convergent at both the infrared and ultraviolet ends, so that mode requires a separate infrared/zero-mode prescription.

At the marginal value, a symmetric cutoff gives

$$P_3^{(\Lambda)}(k) = \frac{\mathcal{A}_{1/2}}{\pi} \operatorname{arsinh} \frac{\Lambda}{k} = \frac{\mathcal{A}_{1/2}}{\pi} \log \frac{2\Lambda}{k} + O(k^2/\Lambda^2).$$

For $0 < s < 1/2$,

$$P_3^{(\Lambda)}(k) = \frac{\mathcal{A}_s \Lambda^{1-2s}}{\pi(1-2s)} + \mathcal{A}_s \frac{\Gamma(s - \frac{1}{2})}{2\sqrt{\pi}\Gamma(s)} k^{1-2s} + O(k^2 \Lambda^{-1-2s}).$$

The first term is cutoff-local. Hence at and below $s = 1/2$ the uncut exponent identity is not a physical projected power spectrum without a declared filter and subtraction prescription.

For a scale-independent complete transfer, $s = 1 - \eta/2$ is inside the domain when the real anomalous exponent obeys $\eta < 1$ and gives $P_3 = \mathcal{A}_1/(2k)$ at $\eta = 0$ and $\Delta_\zeta^2 \propto k^{2+\eta}$, i.e. $n_s = 3 + \eta$. The isotropic Euclidean bi-Laplacian case $s = 2$ gives $P_3 = \mathcal{A}_2/(4k^3)$ and a scale-invariant seed, $n_s = 1$. These are conditional projection statements; the ECT owners of s , the state and the complete transfer remain Open. Logarithmic corrections may modulate these convergent-domain results but do not remove their leading powers.

Critical $O(4)$ fluctuations: three routes to the curvature spectrum

As a separately supplied comparison model, consider a four-dimensional $O(4)$ critical theory with S^3 target. Near criticality its order-parameter propagator has the standard form $G_4(p) \sim (p^2)^{-(1-\eta/2)}$, with $\eta = 0$ at leading order in $d = 4$ (the upper critical dimension of the $O(N)$ model). By the diagnostic (14.5) three routes to a curvature spectrum must be distinguished.

Route I (ordinary criticality; insufficient). With $s \simeq 1$ and a scale-independent complete transfer the projected three-dimensional spectrum is $\Delta^2 \propto k^2$ (blue). An ordinary $O(4)$ -critical $1/p^2$ correlator therefore does not by itself produce a scale-invariant curvature spectrum. The earlier logarithmic estimate $1 - n_s \sim \hat{p}/\ln(L/\xi_0)$ with $\hat{p} \sim 4-5$ is consequently *not* treated as a derived ECT prediction: a multiplicative logarithm cannot remove the leading k^2 factor, and $\hat{p} \sim 4-5$ is at most a diagnostic fitting parameter, not a quantity following from the ECT critical theory.

Route II (transfer-compensated; open). Scale invariance could be recovered from $s = 1$ if the complete curvature transfer carried $q = -1$, $\mathcal{T}_{\chi \rightarrow \zeta}(k) \sim 1/k$. The naive Poisson relation $k^2 \Phi_k \sim G_N \delta \rho_k$ does not deliver this (it yields $k^{\pm 2}$, depending on the density-field relation, not a flat spectrum). Whether a gauge-invariant transfer with the required scaling exists is open (Level C/Open).

Route III (Euclidean second-gradient / strict-pullback candidate). With $s = 2$ and a scale-independent complete transfer the projected spectrum is flat. This is the current conditional strict-pullback seed candidate, developed next; its action-level coefficients, state, physical projector and transfer are not yet ECT-owned, and the projection exponent does not prove uniqueness of this route.

Lifshitz-window conditional theorem

Write the dimensionless Euclidean quadratic functional of the surviving ordering mode as

$$\mathcal{J}_{\chi,E}^{(2)} = \frac{1}{2} \int d^4X \left[\kappa_4 (\Delta_E \chi)^2 + \kappa_2 (\partial_A \chi)(\partial^A \chi) + m_\chi^2 \chi^2 \right], \quad \Delta_E \equiv \delta^{AB} \partial_A \partial_B, \quad (14.6)$$

and separate its physical action from its measure denominator:

$$S_{\chi,E}^{(2),\text{phys}} = A_\chi \mathcal{J}_{\chi,E}^{(2)}, \quad 0 < A_\chi < \infty, \quad 0 < \Sigma_{\text{PI},\chi} < \infty, \\ d\mu_\chi \propto \exp \left[-\frac{S_{\chi,E}^{(2),\text{phys}}}{\Sigma_{\text{PI},\chi}} \right] \mathcal{D}\chi = \exp[-\kappa_\chi \mathcal{J}_{\chi,E}^{(2)}] \mathcal{D}\chi, \quad \kappa_\chi \equiv \frac{A_\chi}{\Sigma_{\text{PI},\chi}} > 0. \quad (14.7)$$

At a fixed renormalisation/matching scale, the finite action scales are constant and field-independent in this supplied Gaussian truncation. After the displayed reduced kernel and the χ field normalisation are fixed, only their dimensionless ratio κ_χ controls its covariance; equality to S_0 , $\hbar$, $A_{\text{ord},E}$ or another sector's ratio is not implied. Whenever $\kappa_\chi(k)$ is used below, it denotes a separately supplied RG-improved coefficient $\kappa_\chi(\mu)|_{\mu=k}$ in a stated scheme. Absent such a scheme, κ_χ is fixed and its logarithmic derivative vanishes. Its stationary owner, RG flow and cross-sector matching remain Open. The operator in (14.6) is the *full isotropic* Euclidean bi-Laplacian $(\Delta_E \chi)^2$; a spatial-only higher-gradient operator $(\partial_i^2 \chi)^2$ together with an ordinary second-order w -term would project to $\Delta^2 \propto k$ (blue) and is *not* sufficient. The isotropic form is a sufficient idealisation: Appendix AA shows that the strict integrable pullback of the recorded $Z_u, \mathcal{C}_n$ operators gives the more general kernel $\frac{1}{2}(p_w^2 + k^2)(Z_u p_w^2 + \mathcal{C}_n k^2)$. For $Z_u, \mathcal{C}_n > 0$ its covariance gives a scale-invariant equal- w local seed exactly when $\kappa_\chi \sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})$ is scale-independent over the band; separate constant $\kappa_\chi, Z_u, \mathcal{C}_n$ are sufficient but not necessary. The exact $O(4)$ bi-Laplacian requires $Z_u = \mathcal{C}_n$, but the local seed tilt does not. This is a supplied reduced Fourier kernel. Neither its lower-order coefficients nor the physical χ projector follow from integrability alone, and positivity of the Fourier kernel is not Osterwalder–Schrader reflection positivity; the exact distinction and counterexample are given in Appendix AA.

Conditional theorem. Suppose that, over the band of comoving modes relevant for cosmological observations and for every p_w in the equal-slice projection, with $p^2 = p_w^2 + k^2$, the quadratic functional reaches a *Lifshitz window*

$$\kappa_4 > 0, \quad \kappa_4 p^4 \gg |\kappa_2| p^2, \quad \kappa_4 p^4 \gg |m_\chi^2|. \quad (14.8)$$

Then, at fixed nonzero spatial momentum and with the massless zero mode regulated or removed, $G_\chi(p) = \langle \chi(p) \chi(-p) \rangle \simeq \kappa_\chi^{-1} / (\kappa_4 p^4)$, and the equal- w projection gives

$$P_\chi(k) \simeq \int \frac{dp_w}{2\pi} \frac{\kappa_\chi^{-1}}{\kappa_4 (p_w^2 + k^2)^2} = \frac{\kappa_\chi^{-1}}{4\kappa_4 k^3}. \quad (14.9)$$

If the curvature perturbation arises from a displacement of the ordering hypersurface, write

$$\zeta \simeq \delta N \simeq \frac{T_{\chi \rightarrow \zeta}(k)}{u_0(k)} \chi_* \equiv \mathcal{T}_{\chi \rightarrow \zeta}(k) \chi_*, \quad \mathcal{T}_{\chi \rightarrow \zeta} \equiv \frac{T_{\chi \rightarrow \zeta}}{u_0}.$$

Any leading scale-independence assumption concerns the complete transfer $\mathcal{T}_{\chi \rightarrow \zeta}$, not the numerator $T_{\chi \rightarrow \zeta}$ alone. Since the seed is the gradient potential $\delta Q_A = \partial_A \chi$, one has $[\chi] = \text{GeV}$ and $[u_0] = \text{GeV}^2$; hence $[\mathcal{T}_{\chi \rightarrow \zeta}] = \text{GeV}^{-1}$. The numerator factorises as $T_{\chi \rightarrow \zeta} = \Lambda_T \hat{T}_{\chi \rightarrow \zeta}$, where $[\Lambda_T] = \text{GeV}$ is supplied by the ordering/emergence map and $\hat{T}_{\chi \rightarrow \zeta}$ is dimensionless. The minimal estimate is $\hat{T}_{\chi \rightarrow \zeta} = O(1)$ (not $T_{\chi \rightarrow \zeta} \simeq 1$ as a pure number), and in general χ need not be identified with $\delta\Phi$. Both factors in the complete transfer and their running are open completion targets. Then

$$\Delta_\zeta^2(k) \simeq \frac{1}{8\pi^2} \frac{\kappa_\chi^{-1} T_{\chi \rightarrow \zeta}^2}{u_0^2 \kappa_4}. \quad (14.10)$$

The coefficient $1/(8\pi^2)$ follows from the displayed $dp_w/(2\pi)$ projection, $P_\chi = \kappa_\chi^{-1}/(4\kappa_4 k^3)$, $\Delta_\zeta^2 = k^3 P_\zeta/(2\pi^2)$ and $\zeta = \mathcal{T}_{\chi \rightarrow \zeta} \chi$. Once those definitions are fixed it is not an additional convention-dependent factor; the sign $\simeq$ records finite-window/deformation and transfer approximations. In the strict isotropic pure-quartic limit, $n_\chi = 1$ exactly when $d \ln(\kappa_\chi \kappa_4)/d \ln k = 0$; separate constant κ_χ and κ_4 are a sufficient special case. In the finite window (14.8) this is a leading approximation whose corrections are controlled by the exact deformation formulas of Appendix AA (Level A for the convergent $k > 0$, infinite-line projection mathematics; Level B/C as an ECT mechanism, conditional on the window and on the transfer map). Under that exact seed condition, $n_s = 1$ additionally requires the complete transfer $\mathcal{T}_{\chi \rightarrow \zeta} = T_{\chi \rightarrow \zeta}/u_0$ to be scale-independent. The established content of this route is narrower: a supplied positive quartic strict-pullback kernel with controlled zero mode and vanishing lower-order deformations gives $P_\chi \propto k^{-3}$ by an exact equal-slice integral. Neither $m_\chi = 0$ nor the required finite band is presently derived from a stationary ECT owner, so the leading $n_s = 1$ is a conditional kernel limit, not a generic ECT prediction. The observed red tilt $n_s \simeq 0.965$ and the amplitude A_s are *not* derived predictions: they are controlled by the open quantities in the tilt budget (transfer running, conditional normalisation drift, freeze-out map, and anisotropy-ratio running; Appendix AA, eq. (AA.2)), which converge on the single closure-level ordering/Hubble problem. In the isotropic idealisation, (14.10) gives the proportionality

$$A_s \propto \frac{\kappa_\chi^{-1} T_{\chi \rightarrow \zeta}^2}{u_0^2 \kappa_4}, \quad (14.11)$$

where the historical label is retained for cross-references but the named primordial covariance is normalised by κ_χ^{-1} , not by S_0 . The independently supplied transfer remains part of this closure; equality of κ_χ to another sector's measure ratio is an Open same-channel or cross-sector matching test. For the general strict-pullback kernel of Appendix AA, the factor κ_4^{-1} of this isotropic idealisation is replaced, up to the same convention-dependent normalisation, by $1/[\sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})]$, reducing to $1/(4\kappa_4)$ when $Z_u = \mathcal{C}_n = 2\kappa_4$; the robust statement is thus an amplitude matching condition involving $\kappa_\chi^{-1} T_{\chi \rightarrow \zeta}^2$, u_0 , Z_u and $\mathcal{C}_n$, not a parameter-free prediction. In the isotropic idealisation the tilt is then

$$\begin{aligned} n_s - 1 &= \frac{d}{d \ln k} \ln \left[\frac{\kappa_\chi^{-1}(k) T_{\chi \rightarrow \zeta}^2(k)}{u_0^2(k) \kappa_4(k)} \right] \\ &= -\frac{d \ln \kappa_\chi}{d \ln k} + 2 \frac{d \ln T_{\chi \rightarrow \zeta}}{d \ln k} - 2 \frac{d \ln u_0}{d \ln k} - \frac{d \ln \kappa_4}{d \ln k}. \end{aligned} \quad (14.12)$$

Define the complete field-to-curvature transfer $\mathcal{T}_{\chi \rightarrow \zeta}(k) \equiv T_{\chi \rightarrow \zeta}(k)/u_0(k)$, so that $\zeta_k = \mathcal{T}_{\chi \rightarrow \zeta}(k) \chi_k$. Equation (14.12) is equivalently $n_s - 1 = -d \ln(\kappa_\chi \kappa_4)/d \ln k + 2 d \ln \mathcal{T}_{\chi \rightarrow \zeta}/d \ln k$. Here $\kappa_\chi(k)$ is conditional RG-improved notation: $\kappa_\chi(\mu)|_{\mu=k}$ in a separately supplied renormalisation or matching scheme. At a fixed scale, and absent such a scheme, $d \ln \kappa_\chi/d \ln k = 0$. No microscopic running or cross-sector universality is derived. Equivalently, the tilt budget contains measure-normalisation, Lifshitz-kernel, transfer and other running terms; the pure fixed p^4 kernel contributes zero by itself. For the general strict pullback of Appendix AA,

$$n_s - 1 = \frac{d}{d \ln k} \ln \left[\frac{\kappa_\chi^{-1}(k) T_{\chi \rightarrow \zeta}^2(k)}{u_0^2(k) \sqrt{\mathcal{C}_n(k)} (\sqrt{Z_u(k)} + \sqrt{\mathcal{C}_n(k)})} \right],$$

up to convention-dependent normalisation and only after the complete $\chi \rightarrow \zeta$ transfer is specified; in the isotropic limit $Z_u = \mathcal{C}_n = 2\kappa_4$ it returns (14.12).

Guards and open conditions (Level C/Open). (i) *Euclidean-statistical, not Lorentzian-propagating.* The bi-Laplacian in (14.6) is a Euclidean correlation structure fixing seed statistics on the ordering slice; it is *not* a propagating higher-time-derivative Lorentzian equation of motion. The intended supplied Lorentzian completion is required to be second order in its physical clock; the projection of the Euclidean p^4 seed onto such a ghost-free completion remains Open. Higher- w derivatives therefore cannot simply

be declared absent from the on-shell spectrum before that projection is constructed. The blue statement applies to the displayed spatial $z = 2$ kernel with an ordinary second-order w term; it is not a no-go theorem for every space- w anisotropic operator. For example, a supplied one-pole kernel $p_w^2 + \lambda^2 k^6$ also projects as k^{-3} . That alternative is not derived by ECT either, but it shows why the strict-pullback route must not be advertised as uniquely selected by the projection exponent alone. The comparison with Hořava-type constructions [25] is therefore operator-specific. (ii) *Second-gradient structure and anisotropy*. In the simplified theorem above, the full isotropic Euclidean bi-Laplacian is a sufficient idealisation. Appendix AA shows that the strict integrable pullback of the recorded $Z_u, \mathcal{C}_n$ operators is more general: in the supplied pure-quartic limit it gives a scale-invariant local seed spectrum for positive $Z_u, \mathcal{C}_n$ when the combined normalisation $\kappa_\chi \sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})$ is scale-independent (separate constants suffice), even when the four-dimensional kernel is not exactly $O(4)$ bi-Laplacian (this holds for the projection route, and not necessarily for transfer-compensated routes). What fails is a spatial-only Lifshitz operator combined with an ordinary second-order w -term, which projects to a blue spectrum. Ordinary three-dimensional statistical anisotropy g_*^{aniso} requires subleading terms that break the residual spatial $O(3)$, not merely a mismatch between Z_u and $\mathcal{C}_n$. (iii) *Nearly $O(4)$ -symmetric seed statistics*. The flat-spectrum route assumes the seed correlations are generated in the nearly $O(4)$ -symmetric Euclidean ordering regime, before macroscopic Lorentzian anisotropy has imprinted a preferred direction on the correlator. Residual w -versus-spatial differences in the fourth-order seed kernel can change the amplitude prefactor and transfer-sector correlators, as in Appendix AA; a genuine three-dimensional g_*^{aniso} requires additional subleading operators that break the residual spatial $O(3)$ symmetry. (iv) *The physical Lifshitz window is not derived*. The integrability constraint $\partial_{[A} Q_{B]} = 0$ makes the leading physical stiffness of the ordering mode plausibly a second-gradient quantity $(\partial Q)^2 \sim (\partial^2 \Phi)^2$, the ECT-native candidate mechanism for suppressing κ_2 ; it does not by itself prove a vanishing two-derivative coefficient or mass. The fixed-coefficient equal-slice problem is now solved exactly in Appendix AA: separate dimensionless controls δ_a, δ_m and ρ determine pointwise dominance, amplitude and tilt tolerances. What remains Open is the stationary action-to-coefficient map, running, radiative stability, state and physical projection. Until that owner exists, the number of small ratios is a set of unprotected hierarchy obligations, not a theorem about the number of independent microscopic tunings. (v) *Infrared sensitivity*. In four dimensions $G_4 \sim 1/p^4$ corresponds to a logarithmically correlated field, $G(r) \sim -\ln r$; the variance per logarithmic interval is constant, while the total variance carries the usual logarithmic dependence on the box size / horizon cutoff and on the zero mode.

Amplitude. If the curvature perturbation arises through a local displacement of the ordering hypersurface, then

$$\zeta(\mathbf{x}) \simeq \delta N(\mathbf{x}) \simeq \frac{\delta Q_*^{\text{eff}}}{u_0}, \quad \delta Q_*^{\text{eff}} \equiv T_{\chi \rightarrow \zeta} \chi_*. \quad (14.13)$$

Here δN denotes a *schematic Euclidean surface-displacement map*—the relabelling of an ordering-hypersurface displacement into the observable curvature variable—and is *not* a pre-Lorentzian e-fold count nor an import of the inflationary δN formalism, which presupposes a Lorentzian expansion history absent at the Euclidean ordering level (cf. the symbol-table entry for N_e , “external; not used in ECT”). The observed amplitude $\sqrt{A_s} \approx 4.6 \times 10^{-5}$ requires

$$\frac{\delta Q_*^{\text{eff}}}{u_0} \sim 5 \times 10^{-5}, \quad [\delta Q_*^{\text{eff}}] = [u_0] = \text{GeV}^2. \quad (14.14)$$

Here δQ_*^{eff} is a mass-squared ordering-gradient composite, not a perturbation of the mass-dimension-one field Φ . In the Lifshitz-window route this is equivalently a matching condition on the combination $\kappa_\chi^{-1} T_{\chi \rightarrow \zeta}^2 / (u_0^2 \kappa_4)$ (the isotropic-limit scaling of Eq. (14.11)); it is not a parameter-free prediction (Level C). Loop-scale weak/Higgs effects may contribute to the transfer or the running, but they are not used here as the derivation of A_s .

Gauge-sector running remains unowned. Writing four real components as a complex doublet is an algebraic repackaging; it does not supply the Standard-Model gauge representations, couplings or their matching to the ordering mode. No gauge–Higgs anomalous-dimension number is therefore imported into the tilt budget. Any such contribution would require an explicitly derived gauge/matter completion and matching into the owner-specific runnings of κ_χ , $T_{\chi \rightarrow \zeta}$, u_0 or κ_4 in Eq. (14.12). The physical tilt remains Open.

Tensor-to-scalar ratio. No claim of a small r is made. If a supplied physical χ sector reaches the strict-pullback / second-gradient Lifshitz seed window while the transverse-traceless geometric sector does not, tensor power may be relatively suppressed, depending on the independently supplied tensor normalisation, state, freeze-out and transfer; no $r \ll 1$ conclusion follows without those owners. If the tensor sector also reaches such a window, tensors acquire their own near-flat spectrum; the tensor sector carries its own second-gradient stiffness coefficients $Z_T, \mathcal{C}_T$, so r depends on the scalar-versus-tensor second-gradient stiffness and on the transfer, reducing to $r \sim \kappa_S / \kappa_T \times (\text{transfer factors})$ only in the isotropic idealisation. Thus r is a diagnostic of which sectors become Lifshitz-critical, and remains open (Level C/Open).

Comparison with inflationary perturbation theory

In standard inflationary cosmology n_s and r follow from the slow-roll dynamics of a separately introduced inflaton. ECT introduces no such field. In the supplied reduced scalar truncation, integrability fixes one longitudinal scalar coordinate (Universality Corollary), while its physical one-mode interpretation remains Level B. The leading spectrum—when the Lifshitz window holds—arises from Euclidean second-gradient seed statistics projected onto the emergent three-dimensional branch (the isotropic bi-Laplacian is a sufficient idealisation, while the strict-pullback $Z_u, \mathcal{C}_n$ kernel gives the same local seed tilt when its combined measure–kernel normalisation is scale-independent under the conditions of Appendix AA), rather than from horizon exit during quasi-de Sitter expansion. The limited resemblance is only the conditional one-mode scalar truncation; it is not yet a statement about the complete ECT perturbation spectrum.

Research agenda

The concrete completion programme for the ECT primordial spectrum is:

1. derive a stationary regulated action and match its constrained quadratic operator to $a_w, Z_u, \mathcal{C}_n, m_\chi^2$ and their running; then test the exact anisotropic amplitude/tilt inequalities of Appendix AA, rather than an averaged-stiffness proxy (OP-grad);
2. establish the physical χ mode/projector and derive the gauge-invariant numerator $T_{\chi \rightarrow \zeta}$ and background factor u_0 entering the complete transfer $\mathcal{T}_{\chi \rightarrow \zeta} = T_{\chi \rightarrow \zeta} / u_0$, then determine rather than assume its running;
3. compute the amplitude (14.11) and the tilt (14.12) in terms of $\kappa_\chi, u_0, Z_u, \mathcal{C}_n$, the transfer, and the isotropic-limit κ_4 where applicable;
4. determine the tensor sector and hence r , and the level of statistical anisotropy g_*^{aniso} from subleading residual-spatial- $O(3)$ -breaking terms (not from a $Z_u \neq \mathcal{C}_n$ mismatch alone);
5. compute the non-Gaussianity ($f_{\text{NL}}, g_{\text{NL}}$) generated by the nonlinear surface-displacement map $\zeta \simeq \delta N(\chi)$;
6. cross-check against the alternative FRW-ordered-background freeze-out description, and match to the homogeneous ordered-branch cosmology of Section 16.

What is established and what remains open

At Level A, the strict scalar map fixes a single scalar coordinate and the projection diagnostic (14.5) is exact on its declared $k > 0$, infinite- w , real $s > 1/2$, $\mathcal{A}_s > 0$ covariance domain: an ordinary $1/p^2$ critical correlator projects to a blue spectrum and an isotropic Euclidean $1/p^4$ correlator projects to a

scale-invariant one. The marginal and subcritical cases $0 < s \leq 1/2$, and the $k = 0$ mode, require explicit regulator, subtraction and zero-mode prescriptions and are not included in that uncut exponent statement. The strict $Z_u, \mathcal{C}_n$ pullback of Appendix AA gives the same local seed tilt in its pure-quartic limit for $Z_u, \mathcal{C}_n > 0$ exactly when the combined measure–kernel normalisation is scale-independent (separate constants suffice), even when the four-dimensional kernel is not exactly $O(4)$ bi-Laplacian. The same appendix now gives the exact finite-deformation ratio and tilt for the supplied constant-coefficient kernel. A physical one-mode interpretation remains Level B, and isocurvature sectors beyond that truncation are Open. What remains Open is the existence of a broad observable Lifshitz band, the physical mass and coefficient owners, reflection-positive/projected state, the gauge-invariant transfer, the red tilt, the amplitude, the tensor ratio r , the non-Gaussianity, the scale-map, and the ghost-free projection of the Euclidean p^4 seed sector onto the Lorentzian ordered branch.

14.7 Summary and comparison with inflationary resolution

Conceptual difference from inflation. Inflation addresses the horizon, flatness, and relic problems as dynamical problems inside an already existing Lorentzian spacetime. ECT explores whether these questions can instead be reformulated at the level of branch emergence: a separately supplied ordering functional gives a conditional coherence route, the physical-flatness bridge remains Open, and the supplied S^3 target has vanishing π_2 , with physical vacuum/relic interpretation Open. These ingredients do not yet constitute a quantitative replacement for inflation.

Table 56: Classical cosmological problems: standard resolution vs. ECT.

Problem	Standard treatment	ECT treatment	Status
Horizon	Inflationary causal stretching inside an already existing Lorentzian spacetime	Conditional branch-coherence route: a constant map minimises the separately supplied positive ordering functional on a connected domain when the admissible boundary/topological class contains it	A-math in supplied model; ECT solution Open
Flatness	Dynamical suppression of $ \Omega_K $ by inflation	P1 supplies a flat ambient metric; $k = 0$ follows only in the declared planar induced-metric completion	Level B/Open physically
Relic / monopole	Dynamical dilution by inflation	The supplied S^3 orientation target has $\pi_2 = 0$; its identification with the complete cosmological vacuum manifold and transition owner is Open	A-math conditional / Open ECT

What is established and what remains open. At Level A mathematics inside the separately supplied ordering model, its positive functional admits a uniform-map minimum; ownership and physical selection of that model remain Open. Independently, the supplied orientation target satisfies $\pi_2(S^3) = 0$; its physical vacuum/relic interpretation is Open. P1 also fixes the ambient Euclidean metric. A planar induced-metric completion then gives $k = 0$ exactly, but ownership of that metric by the physical cosmological branch is Level B/Open. Likewise, one longitudinal scalar coordinate parametrises the stated constrained linear map; its physical mode owner, as well as the tensor and curvature-transfer owners, remains Level B/Open. What remains open is the fully dynamical realisation of branch selection, the quantitative perturbation theory across the ordering transition (including A_s , n_s , and r), and whether any physical $\pi_3(S^3)$ texture sector exists and has a relic spectrum.

15 Vacuum Offsets, Trace-Free Completion, and Late Acceleration

15.1 The problem and the four statements that must be separated

In general relativity coupled to quantum field theory, a constant term in the matter effective action has the metric dependence

$$S_\Lambda = - \int d^4x \sqrt{-g} \rho_\Lambda, \quad (15.1)$$

and therefore sources the metric equations. The observed late-time source is conventionally written

$$\rho_{\Lambda,\text{obs}} = 3\Omega_{\Lambda 0} \bar{M}_{\text{Pl}}^2 H_0^2 \simeq 10^{-47} \text{ GeV}^4, \quad (15.2)$$

whereas unprotected zero-derivative contributions to an effective action are sensitive to much higher scales. This is the standard vacuum-energy naturalness problem [102, 103, 104].

Four logically different statements are relevant in ECT and are kept separate throughout this section.

1. The orientation kinetic tensor has an exactly orientation-independent determinant at fixed amplitude and fixed coefficients. This is an algebraic result of the ordered-branch ansatz.
2. A separately supplied trace-free/fixed-measure scalar–tensor action is classically covariant under a constant shift of the scalar potential when an integration density is shifted oppositely. This is an exact result inside that declared completion.
3. The orientation determinant identity alone does not derive the trace-free metric action, its Bianchi completion, the matter/photon metric, or the integration density.
4. A regular late-accelerating scalar–tensor proof of class exists, but its action, state and dimensional calibration are not yet selected by the frozen ECT postulates. The observed late source, H_0 , and the age of our condensate are therefore matching/state questions until that map is derived.

The standard trace-free algebra is stated below as a conditional completion; it is not attributed to the orientation identity without the missing action-level descent.

15.2 Three source terms and their owners

It is useful to distinguish three quantities that are often called vacuum energy.

Microscopic baseline. A field-independent additive constant in the fundamental Euclidean Lagrangian changes the value of the Euclidean action. Its effect on Lorentzian observables depends on the derived continuation, measure and physical metric action; the absence of an independent Euclidean metric variation is not by itself a proof that the corresponding Lorentzian source vanishes.

Zero-derivative quantum term. Integrating out fluctuations can generate local terms with field-dependent prefactors. Their contribution is governed by the full constrained Hessian, measure and counterterms, not by the determinant of one kinetic block alone.

Late-time source. The source that enters the physical Friedmann and photon-redshift equations may be an integration density, a dynamical scalar contribution, another ordered-branch sector, or a combination of them. Its law and observable map must be derived from one complete action; its numerical value for our condensate may then be fixed as boundary/initial data or inferred from observations, just as other cosmological state data are.

15.3 Scope of Weinberg's no-go result

Weinberg's adjustment no-go result [102] is not evaded merely because the microscopic description is Euclidean or because a Lorentzian metric is emergent. Once a local low-energy action, a physical metric and the adjustment fields are specified, the theorem's hypotheses must be checked on that completed theory. The orientation-determinant identity does not remove those hypotheses and therefore is not a solution of the cosmological-constant problem. A separately declared trace-free action can change the local variational premises and replace a Lagrangian offset by a global integration datum, but ECT has not yet derived that action, its quantum measure or the rule selecting the datum. The two-slope ordinary completion changes the late dynamics but does not evade the additive-offset problem at all. Thus Weinberg's result remains the relevant comparison and consistency test; whether a complete ECT action lies outside one of its assumptions is Open rather than established.

15.4 Exact orientation determinant and its scope

For the fixed-coefficient orientation tensor

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad \delta_{AB} n^A n^B = 1, \quad (15.3)$$

the matrix determinant lemma gives

$$\det K^{AB} = \beta^3 (\beta - \alpha), \quad (15.4)$$

independent of the direction of n^A . If, in addition, this tensor is identified with a densitized inverse metric through

$$K^{AB} = C_K \sqrt{-g_{\text{eff}}} g_{\text{eff}}^{AB} \quad (C_K = \text{field independent}), \quad (15.5)$$

then the determinant of that particular representative is fixed,

$$\sqrt{-g_{\text{eff}}} \propto \sqrt{-\det K} = \sqrt{\beta^3 (\alpha - \beta)}, \quad (15.6)$$

and tangent orientation variations obey

$$\delta_n \sqrt{-g_{\text{eff}}} = 0. \quad (15.7)$$

Exact lemma and non-converse. Equations (15.4)–(15.7) are Level A algebra inside the fixed- α, β orientation ansatz. Their converse is false: a constant determinant does not determine the metric equations. For example,

$$g_{\mu\nu}(s) = \text{diag}(-e^{3s}, e^{-s}, e^{-s}, e^{-s}), \quad \det g = -1, \quad (15.8)$$

has the same determinant for every s but different lapse, spatial rulers, redshifts and response. Moreover, the orientation sphere supplies only three tangent variations, while a trace-free symmetric metric contains nine components before diffeomorphism constraints. Amplitude dependence, orientation–amplitude mixing and field-dependent normalization can also change the physical measure while leaving Eq. (15.4) intact.

Consequently the exact orientation lemma is not an unconditional vacuum-offset decoupling theorem. Deriving the physical constrained metric action, its constraint algebra, its Bianchi identity and its universal matter/photon coupling remains an Open action-level task.

15.5 Conditional trace-free/fixed-measure completion

The standard unimodular mechanism can nevertheless be stated exactly once a specific completion is supplied. Consider

$$S_{\text{TF}} = \int d^4x \left\{ \sqrt{-g} \left[\frac{1}{2} f(q) R - \frac{1}{2} K(q) (\partial q)^2 - V(q) + \mathcal{L}_m \right] + \lambda(x) (\sqrt{-g} - \varepsilon_0) \right\}. \quad (15.9)$$

Variation with respect to λ fixes the measure. Metric variation, the scalar equation, matter conservation and the Bianchi identity imply $\lambda = -C_I$ with $\partial_\mu C_I = 0$, and reconstruct

$$f G_{\mu\nu} = T_{\mu\nu}^m + K \left(\partial_\mu q \partial_\nu q - \frac{1}{2} g_{\mu\nu} (\partial q)^2 \right) - [V(q) + C_I] g_{\mu\nu} + \nabla_\mu \nabla_\nu f - g_{\mu\nu} \square f. \quad (15.10)$$

The scalar equation contains $V_{,q}$ but not C_I . Therefore

$$V(q) \longrightarrow V(q) + C_0, \quad C_I \longrightarrow C_I - C_0 \quad (15.11)$$

leaves the complete classical equations invariant. This is the precise conditional vacuum-offset result: only the combination $V + C_I$ enters the metric equation, while the split is conventional.

In a constant-scalar vacuum,

$$R = \frac{4[V(q) + C_I]}{f(q)}, \quad V_{,q} - 2 \frac{f_{,q}}{f} [V(q) + C_I] = 0. \quad (15.12)$$

For $f = f_* e^{aq}$ and $V = V_1 e^{bq}$, the root is

$$e^{bq_0} = \frac{2aC_I}{(b-2a)V_1}. \quad (15.13)$$

It exists as a real positive exponential precisely when $aC_I/[(b-2a)V_1] > 0$ (with nonzero denominator). Choices such as $C_I > 0$, $V_1 > 0$, $a > 0$ and $b > 2a$ are one branch, not the general existence condition.

Interpretation of C_I . C_I is a global integration/state datum, not a value determined by the local equations. This does not require a fundamental law to calculate the particular datum of our condensate: it may be selected by boundary/initial conditions and measured. What ECT still owes is the derivation of Eq. (15.9) (or another complete shift-covariant action) from the Φ medium and the map by which C_I enters physical clocks, matter and photons. Until then, the trace-free result is Level A within the named completion and Level B/Open as an ECT descent.

15.6 Why the fixed determinant does not supply the completion

The earlier spectral condition H_Λ can be represented schematically by a quadratic fluctuation operator

$$S^{(2)} = \frac{1}{2} \int d^4 x_E [\mathcal{K}^{AB}(\phi, n, \dots) \partial_A \chi \partial_B \chi + \mathcal{M}^2(\phi, n, \dots) \chi^2]. \quad (15.14)$$

Even if

$$\mathcal{K}^{AB} = C K^{AB}(n), \quad C = \text{constant}, \quad (15.15)$$

the determinant controls only that Gaussian block. It does not establish the physical constrained metric action or a radiative non-renormalization theorem. If instead $\mathcal{K}^{AB} = Z(q) K^{AB}$, then

$$\delta_q \Gamma_{\text{vac}}^{(0)} \propto \Lambda_{\text{UV}}^4 Z(q) Z_{,q}(q), \quad (15.16)$$

and a scalar naturalness problem is explicit. Conditions on the full reduced Hessian, measure, constraints and counterterms are therefore needed even after the classical trace-free action is derived.

15.7 Ordinary two-slope completion: regular dynamics, not shift invariance

A distinct constructive result uses the ordinary Jordan action

$$S_J = \int d^4x \sqrt{-g} \left[\frac{1}{2} f(q) R - \frac{1}{2} K(q) (\partial q)^2 - V(q) \right] + S_m[g, \psi], \quad f = f_* e^{aq}, \quad K = \kappa f, \quad C_I = 0, \quad (15.17)$$

with

$$V(q) = V_- e^{cq} + V_+ e^{bq}, \quad 0 < c < 2a < b, \quad V_{\pm} > 0. \quad (15.18)$$

Its Einstein-frame potential $\mathcal{U}_E \propto V_- e^{(c-2a)q} + V_+ e^{(b-2a)q}$ is coercive and has one strict minimum,

$$e^{(b-c)q_{\text{vac}}} = \frac{(2a-c)V_-}{(b-2a)V_+}, \quad \mathcal{E}_{,q}(q_{\text{vac}}) = (2a-c)(b-c)V_- e^{cq_{\text{vac}}} > 0. \quad (15.19)$$

Provided $f > 0$, $\kappa + 3a^2/2 > 0$, and the solution is on the expanding finite-energy branch, this supplies a regular late-de-Sitter proof of class and, for a frozen near-GR benchmark, a connected radiation–matter–de-Sitter numerical history. It does not make the ordinary action invariant under $V \rightarrow V + C_0$: an added constant still gravitates and changes the scalar balance. The two-slope mechanism and the trace-free offset mechanism are therefore complementary, not interchangeable.

Here $\kappa + 3a^2/2 > 0$ is equivalently the positivity condition for the Einstein-frame scalar kinetic coefficient in the displayed normalisation.

For a homogeneous scalar the bare Jordan-bookkeeping scalar pieces are

$$\rho_q = \frac{1}{2} K \dot{q}^2 + V(q), \quad (15.20)$$

$$p_q = \frac{1}{2} K \dot{q}^2 - V(q). \quad (15.21)$$

They are neither separately conserved in general nor the full effective dark-energy density and pressure: non-minimal-coupling terms proportional to $\dot{f}$, $\ddot{f}$ and the chosen metric rearrangement also enter the Friedmann bookkeeping. The ratio $p_q/\rho_q \geq -1$ follows for these bare pieces only when $K \geq 0$ and $\rho_q > 0$; it is not a prediction of the physical ECT dark-energy equation of state.

Current observational context and a conditional falsifier. DESI DR2 BAO are well described by flat Λ CDM in isolation. When combined with CMB data and, dataset-dependently, supernova compilations, w_0 – w_a fits prefer evolving dark energy, with the quoted significance depending on the combination. The extended DESI analysis finds a common low-redshift trend and a preference for phantom crossing, while non-crossing alternatives are disfavoured but not excluded [105, 106]. This is observational context, not an ECT prediction or a discovery claim. In the positive-kinetic two-slope Jordan scalar the bare ratio $w_q = (K\dot{q}^2/2 - V)/(K\dot{q}^2/2 + V)$ obeys $w_q \geq -1$ when $K \geq 0$ and $\rho_q > 0$, but the nonminimal- f terms mean that this bare ratio is not the effective $w(z)$ fitted by cosmological data. A complete action-owned ECT history robustly constrained never to cross -1 would therefore be disfavoured by that combined-data trend. The present candidate has not supplied the physical photon/perturbation likelihood and is untested by this falsifier.

15.8 Observed late source, H_0 , and age

After choosing the present physical metric and rewriting its measured background in the GR-normalised form with fixed reference $\bar{M}_{\text{Pl}}$, define

$$\rho_{\text{late},0}^{(\text{GRref})} \equiv 3\Omega_{\text{late},0}^{(\text{GRref})} \bar{M}_{\text{Pl}}^2 H_0^2. \quad (15.22)$$

This is an observational convention/definition, not the generic scalar–tensor Friedmann equation and not a dimensional derivation. In the supplied scalar–tensor action the present constraint also contains F_0 and

$-3H_0\dot{F}_0$ (and any separately supplied orientation stress); how those terms are assigned to an effective late component must be stated. Writing $\rho_{\text{late},0}^{(\text{GRref})} = c_{\Lambda}^{\text{GRref}} \bar{M}_{\text{Pl}}^2 H_0^2$ therefore defines $c_{\Lambda}^{\text{GRref}} = 3\Omega_{\text{late},0}^{(\text{GRref})}$ only in this GR-reference bookkeeping after H_0 , the physical metric and Planck calibration are fixed. It is not an ECT determination or evidence that a microscopic vacuum baseline has decoupled.

As a measured-input dimensional check, $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$ and $H_0 \simeq 1.44 \times 10^{-42} \text{ GeV}$ give $\bar{M}_{\text{Pl}}^2 H_0^2 \simeq 1.2 \times 10^{-47} \text{ GeV}^4$, of the same dimensional order as the observed late source. This arithmetic uses measured inputs and is not a derivation of either the scale or its coefficient from ECT.

The dimensional alternatives also diagnose what has and has not been explained. A density proportional to $\bar{M}_{\text{Pl}} H_0^3$ is smaller than $\bar{M}_{\text{Pl}}^2 H_0^2$ by $H_0/\bar{M}_{\text{Pl}}$, while H_0^4 is smaller by $(H_0/\bar{M}_{\text{Pl}})^2$; neither can replace the observed scale without an additional large dimensionless factor or new scale. The Cohen–Kaplan–Nelson UV/IR consistency estimate $\rho \lesssim \bar{M}_{\text{Pl}}^2/L_{\text{IR}}^2$ [107] reproduces the same dimensional order only after $L_{\text{IR}} \sim H_0^{-1}$ and near saturation are supplied. It is a useful external comparison, not an ECT derivation.

The two-slope proof-of-class yields, for one explicitly calibrated near-GR benchmark,

$$\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368677864. \quad (15.23)$$

Using measured values of H_0 as two disclosed alternatives for the single dimensional scale calibration of this already fixed orbit gives 13.98050 Gyr for $67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and 12.90095 Gyr for $73.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$. These are conditional outputs of one calibrated action and state, not unique ECT predictions. Different regular completions or values of C_I give different histories. Conversely, leaving C_I as a measured state datum is logically acceptable once its action-level and observable maps are derived.

15.9 Infrared carrier remains unidentified

The current action derives no physical soft-mode operator, mass, state or abundance that carries the observed late source. A pseudo-Goldstone interpretation would require all of those owners and a cosmological forward model. Writing $H_{0,E} := \hbar H_0^{(\text{freq})}$ so that the physical frequency and its natural-unit energy representation are explicit, the separately named C/Open scale hypothesis $m_{\chi,E} = \zeta_H H_{0,E}$ may be used only for conditional orientation; it is not selected by the action and supplies no late-source identification. No Planck-scale field excursion, numerical infrared-density scaling, abundance or cosmological-role inference is retained.

15.10 Comparison and status

The corrected result should also be compared with established external routes rather than presented in isolation:

Table 57: Compact comparison with external vacuum-energy approaches.

Approach	What it changes	What remains input	Relation to the present ECT programme
GR plus QFT	A constant term sources the metric	Renormalized residual vacuum density	Baseline naturalness problem [102]
Unimodular/trace-free gravity	Local trace equation is replaced; a global integration constant appears	Boundary/state value and quantum completion	Mathematical template for the conditional C_I branch [108]
Sequestering/global constraints	Global variables cancel selected vacuum contributions	Complete global action, measure and boundary prescription	Possible external completion class, not derived here
Dynamical scalar/quintessence	A rolling field supplies late acceleration	Potential, initial state, matter metric and radiative stability	The ordinary two-slope branch is a conditional member of this broad class
Anthropic/landscape selection	Restricts attention to vacuum values compatible with observers or structure	Ensemble, measure and selection rule	External comparison only; not an ECT derivation [109]

CKN UV/IR bound	Relates an infrared size to an allowed energy density	Choice of L_{IR} and degree of saturation	Dimensional comparison only [107]
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Table 58: Vacuum-offset and late-source status. “Conditional” always means inside the explicitly named completion, not a result of the orientation determinant alone.

Structure	Exact result	Missing owner	Status
Orientation tensor	$\det K$ is direction independent at fixed α, β	Physical metric action and measure	Level A algebra; descent Open
Explicit trace-free action	$V + C_I$ is shift covariant; C_I is a global datum	Derivation from the Φ medium and physical clock/photon map	Level A conditional; ECT map Level B/Open
Ordinary two-slope action	Coercive potential, stable vacuum and regular background proof of class	Selection of action, parameters and state; not shift invariant	Level A conditional / Level C benchmark
Observed late source	Friedmann normalization after H_0 , metric and state are fixed	One complete ECT action and state-to-observable map	Open as a unique ECT cosmology; measurable state datum allowed

15.11 Binding hierarchy and open tests

Established. The orientation determinant identity is exact in its stated ansatz. The shift covariance of $V + C_I$ is exact in the separately supplied trace-free completion. The coercive two-slope potential and its stable vacuum are exact in the separately supplied ordinary scalar–tensor completion.

Not established. The fixed orientation determinant does not derive the trace-free action; the microscopic baseline and loop contribution have not been shown to be absent from the complete physical ECT metric equations; the physical matter/photon metric is not identified by this section; and the observed late-time source is not a unique output of the frozen postulates.

15.12 Open programme

- O1.** Derive, or refute, the trace-free/fixed-measure action and its Bianchi completion from constrained Φ -medium variables. The derivation must cover amplitude as well as orientation variations.
- O2.** Derive the universal physical metric and show how the integration density or dynamical late source enters clocks, redshift, lensing and PPN observables.
- O3.** Compute the full reduced Hessian, measure and counterterms before claiming radiative protection of a zero-derivative term.
- O4.** Decide whether the two-slope completion, the trace-free completion, or a coupled version follows from the ordered Φ phase. If C_I is a state datum, identify its boundary/state-selection rule; its particular value need not be a universal constant predicted by local postulates.
- O5.** Fit one action-owned background and perturbation system jointly to expansion, growth, CMB/lensing and local-gravity data. Conditional age estimates remain valid benchmarks but are not promoted by the fit alone.

15.13 Summary

The robust result carried by the ordered orientation tensor is the fixed determinant of one fixed-coefficient kinetic block. It is not, by itself, a vacuum-energy decoupling theorem. An explicit trace-free/fixed-measure scalar–tensor completion does have an exact classical shift covariance: $V \rightarrow V + C_0$ is absorbed

by $C_I \rightarrow C_I - C_0$. The integration density is a legitimate global state/boundary datum, but ECT has not yet derived the completion or its observable map. Separately, an ordinary two-slope action provides a stable and globally regular late-accelerating proof of class, but it is not invariant under an added constant and its numerical history is a calibrated conditional result. Hence the vacuum-offset route is constructive, while the physical late source, H_0 and the age of our condensate remain to be identified by one complete action, state and metric map.

The technical derivations supporting Section 15 are collected in Appendix AD; their rank-one determinant identity and non-converse are isolated in Section AD.1.

16 Cosmology from the Ordered Branch

Boundary-value formulation and ordered-branch cosmology. The boundary-value character of ECT should not be read as a second cosmological ontology competing with the Lorentzian FLRW branch. At the microscopic level the theory is formulated by an $O(4)$ -invariant Euclidean action and, statistically, by the corresponding Euclidean functional integral over admissible Φ -configurations. At the macroscopic level P4 supplies an ordered-gradient analysis chart in which a nonzero vector datum provides the preferred direction used by the conditional Lorentzian EFT. Its physical vacuum, state and dynamical origin remain part of OP-grad, while the subsequent cosmological background closure belongs to OP3.

The global Euclidean boundary data and domain should therefore be understood as the admissible global support and boundary context of the condensate configuration, not as an external time-evolution mechanism. In an elliptic Euclidean formulation, boundary and interior data are mutually constrained components of one global solution or statistical ensemble, rather than cause and effect. The Lorentzian FLRW description is obtained only after an ordered branch is available and can be read in terms of relational clocks, rods, and subsystems.

The distinction between parameter levels is essential. The bare parameters of P3 are theoretical inputs characterising the microscopic scalar action; they are not deduced from the realised universe. The ordered-branch amplitude u_0 , broken-phase EFT parameters such as (α, β) , and cosmological closure coefficients such as ω_0, m_u, c_3, c_4 belong to higher effective levels. In the present formulation these higher-level quantities are not all derived from bare P3 or from global boundary data; their full matching to a gradient-sensitive Euclidean functional and admissible ordered-domain realisations remains part of the open closure programme.

Consequently, ECT statements have different dependence on the realised cosmological branch. Some results are structural consequences of the ordered branch or of a specified closure, while quantities such as $a(t)$, $H(z)$, $\phi_b(z)$, and the size of a finite ordered domain are realisation- and closure-dependent. The finite-domain picture and the FLRW branch are therefore complementary: the former is a possible global completion of the ordered region, while the latter is the effective interior description used to compute cosmological observables.

Having separated the conditional horizon route, the physical-flatness owner problem and the primary monopole topology in the preceding section, the present section develops a conditional post-transition cosmology of the ordered branch. It uses equations obtained by varying the supplied ϕ -first action and then taking its scalar-only homogeneous isotropic FRW reduction in §13.5 (Eqs. (13.41)–(13.43)). The applicable entries of Table 49 are the retained scalar Lorentzian component, any explicitly invoked arithmetic-average benchmark, the conditional macroscopic EFT regime, and the Level-B amplitude-dominated nonlinear closure and the FRW background specialisation; no separately numbered levels “0–1, 3, 5, 8” exist. The orientation stress is omitted rather than derived away. Status: the displayed equations are Level A algebra inside their supplied conditional action; the F, ω, U matching, ECT ownership, physical state, orientation sector and global cosmological solution remain Level B/Open.

What is established, what is benchmarked, and what remains open. At the present stage, the ordered-branch cosmology supports three different levels of statement. First, the supplied homogeneous scalar model has exact conditional equations and a dimensionally normalised inequality for its bare

scalar bookkeeping ratio when the stated kinetic and energy assumptions hold. That inequality is not a bound on the observable effective equation of state. Second, inverse reconstruction proves that unowned functions and states do not select one physical $H(a)$. Third, one named two-slope action and state supplies conditional age, background, high-redshift clock, sound-horizon, growth and Weyl/velocity-carrier diagnostics. Full Hubble inference, JWST abundance or morphology, the perturbation spectrum (A_s, n_s, r) and a global data fit remain Open targets of one action-owned completion.

16.1 Homogeneous FLRW branch and late-time evolution

Status: Level B. The FRW reduction follows from the macroscopic closure of §13.5; specific closure functions remain open (OP3). The bare scalar-ratio identity $w_\phi^{\text{bare}} \geq -1$ is Level A algebra inside the supplied action when $\omega(\phi) > 0$ and its dimensionally normalised bare denominator is positive. It is not a Level-B ECT prediction or a bound on observable w_{eff} . The exact scope of the frozen-scalar, bare-ratio, Hubble-sign, JWST, and common-owner statements is consolidated in Appendix AG.

Bridge from the gravitational sector

The full background equations governing the homogeneous ordered branch are already derived in §13.5 (Eqs. (13.41)–(13.43)) as the isotropic background reduction of the ϕ -first closure. The present subsection analyses their physical content: late-time dark-energy behaviour, evolution of the divided coefficient G_{div} , and observational constraints only after the separately solved local or cosmological response is supplied. The detailed late-time benchmark closure apparatus is recorded in Appendix AE. In what follows, it is important to distinguish between structural properties of the supplied homogeneous equations (including the bare scalar-ratio identity and the distinct full acceleration equation) and particular background parameterisations used for illustration or data contact. An inflation-like effective description of the ordering transition, if retained, is discussed in §14.1–14.3.

Conditional frozen-scalar FLRW limit

If ϕ is constant and $\dot{\phi} = 0$, the background equations reduce algebraically to the Einstein–Friedmann form with a vacuum contribution fixed by the chosen scalar reference. This is a consistency limit of the ϕ -first closure, not a derivation of dynamical density screening or of a cosmological approach to Λ CDM.

Conditional bare scalar-stress lemma; observable acceleration owner Open

The action in Eq. (13.35) fixes the physical kinetic energy density, so the dimensionally homogeneous Jordan-bookkeeping pieces are

$$K_\phi^{\text{phys}} \equiv \frac{1}{2} \bar{M}_{\text{Pl}}^2 \omega(\phi) \dot{\phi}^2, \quad \rho_\phi^{\text{bare}} = K_\phi^{\text{phys}} + U(\phi), \quad p_\phi^{\text{bare}} = K_\phi^{\text{phys}} - U(\phi). \quad (16.1)$$

Whenever $\omega(\phi) > 0$ and $\rho_\phi^{\text{bare}} > 0$, their algebraic ratio obeys

$$\boxed{w_\phi^{\text{bare}} = \frac{K_\phi^{\text{phys}} - U}{K_\phi^{\text{phys}} + U}, \quad w_\phi^{\text{bare}} + 1 = \frac{2K_\phi^{\text{phys}}}{K_\phi^{\text{phys}} + U} \geq 0.} \quad (16.2)$$

This is Level A algebra inside the supplied stress decomposition, but those pieces are not separately conserved and do not own the observable Jordan-frame equation of state. Before division by F , the non-minimal scalar contribution to the two background equations may be organised as

$$\rho_\phi^{(F)} = K_\phi^{\text{phys}} + U - 3\bar{M}_{\text{Pl}}^2 H \dot{F}, \quad p_\phi^{(F)} = K_\phi^{\text{phys}} - U + \bar{M}_{\text{Pl}}^2 (\ddot{F} + 2H\dot{F}), \quad (16.3)$$

so $\rho_\phi^{(F)} + p_\phi^{(F)} = 2K_\phi^{\text{phys}} + \bar{M}_{\text{Pl}}^2(\dot{F} - H\dot{F})$ has no sign fixed by $K_\phi^{\text{phys}} \geq 0$. Division by F and alternative moves of the varying Planck coefficient between the two sides give other equivalent bookkeeping conventions. The invariant background diagnostic is the solved $H(a)$ (for example $w_{\text{eff}} = -1 - 2\dot{H}/(3H^2)$), and it can lie below -1 even while $w_\phi^{\text{bare}} \geq -1$.

This distinction has an exact local counterexample inside the supplied equations. At one regular vacuum event choose

$$F = 1, \quad H = H_* > 0, \quad \dot{F} = \dot{\phi} = 0, \quad \ddot{F} = -H_*^2, \quad F_{,\phi} \neq 0. \quad (16.4)$$

Equation (13.41) then fixes $U = 3\bar{M}_{\text{Pl}}^2 H_*^2$, while Eq. (13.42) gives $\dot{H} = H_*^2/2$ and hence

$$w_{\text{eff}}^{\text{geom}} = -1 - \frac{2\dot{H}}{3H^2} = -\frac{4}{3}. \quad (16.5)$$

Since $\dot{\phi} = 0$, the chosen $\ddot{F}$ implies $\ddot{\phi} = -H_*^2/F_{,\phi}$. The scalar equation is satisfied at the same event by the smooth local potential slope

$$U_{,\phi} = \bar{M}_{\text{Pl}}^2 H_*^2 \left(\frac{15}{2} F_{,\phi} + \frac{\omega}{F_{,\phi}} \right). \quad (16.6)$$

Thus positive ω and the exact bare non-phantom lemma do not imply a geometric non-phantom lower bound. This is Level A local, model-internal evidence for the logical non-implication; it neither selects a global ECT history nor constructs a crossing from above to below -1 ; such a global crossing remains Open.

A chosen positive potential can parametrise a phase with $K_\phi^{\text{phys}} \ll U$ and hence $w_\phi^{\text{bare}} \simeq -1$. This establishes only a statement about that bookkeeping ratio, not acceleration; acceleration must satisfy the full equation (18.2). It does not establish that the Φ -medium supplies the observed acceleration, excludes a separate component, or predicts a present value or redshift history. The minimal P1–P6 action and this stress lemma do not identify a physical $w(a)$ or a late-time likelihood. The separately supplied two-slope completion below does provide a regular conditional history, but its microscopic selection, physical metric/state map, and owner-complete likelihood remain Open.

Evolution of the divided matter coefficient

The exact divided coefficient of the supplied scalar metric equation is

$$G_{\text{div}}(z) = G_* e^{-\beta_\phi \phi_b(z)}. \quad (16.7)$$

This is algebraic bookkeeping inside the ϕ -first closure, not a physical cosmological gravitational response. The latter is the solved $H(z)$ and perturbation/observable map, which also depend on $\dot{F}$, $\ddot{F}$, scalar kinetic and potential terms, the state, and the physical matter/photon metric. No P1–P6 theorem fixes its sign or magnitude.

BBN consistency constraint. Constancy of F is neither necessary nor sufficient for BBN compatibility. The complete action/state must determine the physical metric and clock, $H(T_\gamma)$, scalar kinetic and potential contributions and any variation of particle scales. The forward owner must then propagate the nuclear reaction network and rate covariance, neutron lifetime and weak rates, neutrino distributions and decoupling, plasma thermodynamics and finite-temperature QED corrections, baryon-to-photon ratio, initial abundances and the observational abundance likelihood. An inverse- F curve is only G_{div} bookkeeping. It becomes the constant-rescaling proxy used by an external BBN bound only after the complete solution is shown to reduce to that proxy; no such ECT-specific reduction is yet derived.

Quantitative cosmology figure status. The named regular two-slope action/state now supplies one reproducible post-ordering background orbit, so its conditional $H(z)$, supplied two-slope coordinate σ , distance and age diagnostics may be displayed. Those plots are Level A only inside that supplied action/state and Level C after the declared dimensional calibration. They are not universal P1–P6 predictions: selection of the completion, passage through the ordering front, the physical matter/photon metric, primordial transfer and joint likelihood remain Open. In particular, an inverse- F background curve is not identified with a locally measured Newton constant.

What is established and what remains open

Status-separated results: (i) the FRW equations are Level A algebra inside the supplied Level-B ϕ -first action and reduce to an Einstein–Friedmann form when the scalar is frozen; calibration to Λ CDM is additional matching; (ii) $w_\phi^{\text{bare}} \geq -1$ is Level A algebra for the dimensionally normalised bare scalar pieces, while observable w_{eff} has no such model-independent bound because $\dot{F}$ and $\ddot{F}$ contribute; (iii) the sign and magnitude of Hubble inference are not fixed by the conditional coupling formula and remain Open.

Open: (i) explicit form of $U(\phi)$ and $\omega(\phi)$ from bare P3 (OP3); (ii) a P1–P6-selected self-consistent background $\phi_b(z)$ and a derived ordering-front continuation with early-time saturation compatible with BBN; (iii) quantitative MCMC fit to CMB+BAO+SN data. The displayed conditional orbit below closes none of these three owner requirements by itself.

16.2 Candidate ordering-to-background stages: acceptance requirements

The amplitude coordinate $\phi = \beta_\phi^{-1} \ln(u/u_\infty)$ is valid only after an ordered patch with $u > 0$ exists. It can organise approach to a regular background and later evolution, but it cannot parameterise the formation point $u = 0$. Formation must be solved in u or another regular variable and matched onto the logarithmic chart. These are solver domains, not a predicted three-stage trajectory. A physical completion must derive the ordering event and then evolve the same action and state continuously; it may not interpolate the regimes by hand.

The acceptance requirements are: regular passage from the ordering event; positive physical kinetic eigenvalues and nonnegative characteristic speeds; no response-matrix degeneration; a high-redshift state compatible with BBN and recombination after its matter/photon map is derived; and a present state that satisfies local-gravity constraints. The amplitude need not be monotonic, saturate before BBN, or drift at late times unless the solved equations demonstrate those behaviours. No redshift or gigayear plot is admissible unless a named solution, dimensional calibration and observable map own its axes and its remaining conditional status is stated explicitly.

External and internal descriptions of the same history. The external–internal bridge is the state-dependent congruence construction already defined by Eqs. (16.14) and (16.15), subject to the integrability gate in Eq. (16.17). From the outside the ordered region is described by a formation front, Euclidean arc length and boundary state; from the inside matter clocks measure τ , photons define redshift and the homogeneous state defines $a(\tau)$. These relations are local along a supplied action-owned congruence. They require neither a spherical outer boundary nor a universal radius, and they do not derive the front, lapse, matter congruence or physical matter/photon metric.

Ordering stage. The ordering calculation must determine the physical branch, independent degrees of freedom, defect content, initial entropy and the state inherited by Lorentzian perturbations. An equilibrium minimum alone does not determine the non-equilibrium state or its spectrum. The output is a boundary-data functional for the cosmological initial-value problem, not a prescribed redshift.

Shape-agnostic external Euclidean view and internal cosmological history

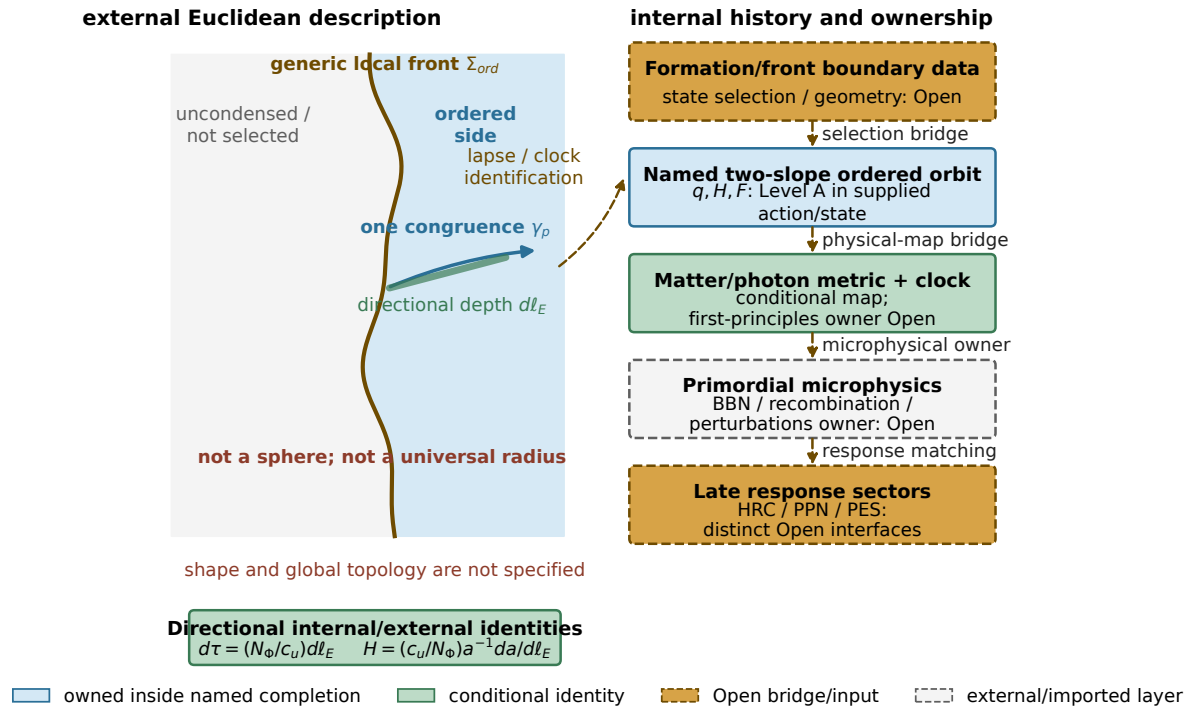


Figure 5: Shape-agnostic external/internal history map. The external panel depicts a generic local formation front and one action-owned congruence, not a spherical boundary. The internal panel separates an orbit that is Level A only inside the named supplied action/state, a Level B conditional metric/clock map, imported external microphysics, and Open interfaces. Formation/front selection, the first-principles physical metric, primordial microphysics, Boltzmann perturbations, HRC/PPN matching and PES record kernels remain Open. Colour follows the luminance-first publication status palette and is redundant with border style and literal status text.

Radiation and nucleosynthesis stage. The continuation must reach a regime in which the physical photon temperature, reaction rates, expansion rate and effective gravitational coupling are calculable. The BBN forward problem uses

$$\{H(T_\gamma), m_e(T_\gamma), \alpha_{\text{fs}}(T_\gamma), \eta_b, N_{\text{eff}}; \tau_n, \langle \sigma v \rangle_i, f_v(p, T_\gamma), \dots\} \longrightarrow \{Y_p, \text{D/H}, {}^3\text{He/H}, {}^7\text{Li/H}\}. \quad (16.8)$$

The coefficient G_{div} enters this map only through the action-owned solved background/response, unless a separate scalar dependence of the microphysical rates is derived; listing both $H(T_\gamma)$ and an independent $G_{\text{eff}}(T_\gamma)$ would otherwise double-count or leave the transfer unspecified. No late-time fit can compensate for failure of this early-state map.

Recombination and structure stage. For any proposed completion, the same orbit would have to supply the expansion and physical couplings used by recombination, while a derived gauge-reduced quadratic action and primordial state would have to supply the transfer functions. Neither owner-complete pipeline is supplied here. A claimed early-structure enhancement must be traced to a specific operator in this chain and checked against CMB spectra, lensing, growth and local gravity.

Late-time stage. Acceleration, HRC phenomenology and finite-body screening are presently distinct candidate descriptions assigned to the same putative medium; their common action ownership has not been derived. A viable completion must show how their scales and couplings coexist without assigning one field coordinate incompatible roles. The cosmological Hubble rate is a homogeneous expansion observable; the HRC argument is a quasistatic acceleration invariant; the PPN parameters are finite-source metric coefficients. Equality between any two scales is a matching hypothesis until the common action derives it.

The supplied regular two-slope action/state supports a quantitative post-ordering orbit. This removes the need for the former hand-drawn scalar history while leaving the transition through the formation front Open. The inverse- F curve below is a background diagnostic coordinate and is not identified with a locally measured Newton constant.

16.3 Matter–antimatter asymmetry in the ECT framework

Status: Level C/Open conditional completion. The discussion below is an inventory of operators and state assumptions that could realise the Sakharov conditions; P1–P6 do not yet derive baryon number, a baryon-violating vertex, a physical CP-odd coefficient, the transition/thermal state, or the transport and washout system. Consequently neither $\eta_B \neq 0$ nor its magnitude is presently an ECT prediction. Connection: Section 22.3 (defect-mediated topology change), Section 7 ($\mathcal{L}_5$ coupling), Section 7.2 (gauge structure).

In the named base- Λ CDM Planck likelihood, $\Omega_b h^2 = 0.02237 \pm 0.00015$; the standard conversion $\eta_{10} \simeq 273.9 \Omega_b h^2$ gives $\eta_B \simeq (6.13 \pm 0.04) \times 10^{-10}$ [70]. This is a model-dependent fit plus conversion, not a direct count and not ECT evidence. It corresponds approximately to one net baryon per billion early particle–antiparticle pairs. All visible matter—stars, planets, observers—consists of this surviving excess.

Any baryogenesis mechanism must satisfy the three Sakharov conditions [110]: (i) baryon number violation, (ii) C and CP violation, and (iii) departure from thermal equilibrium.

The development below records a conditional ordered-medium completion in which supplied owners could implement the Sakharov ingredients. It is not a replacement for an explicit gauge, matter and transport calculation.

Candidate Sakharov ingredients and missing owners

(i) Baryon number violation: defect-mediated topology change.

Conditional post-ordering evolution of the supplied two-slope condensate

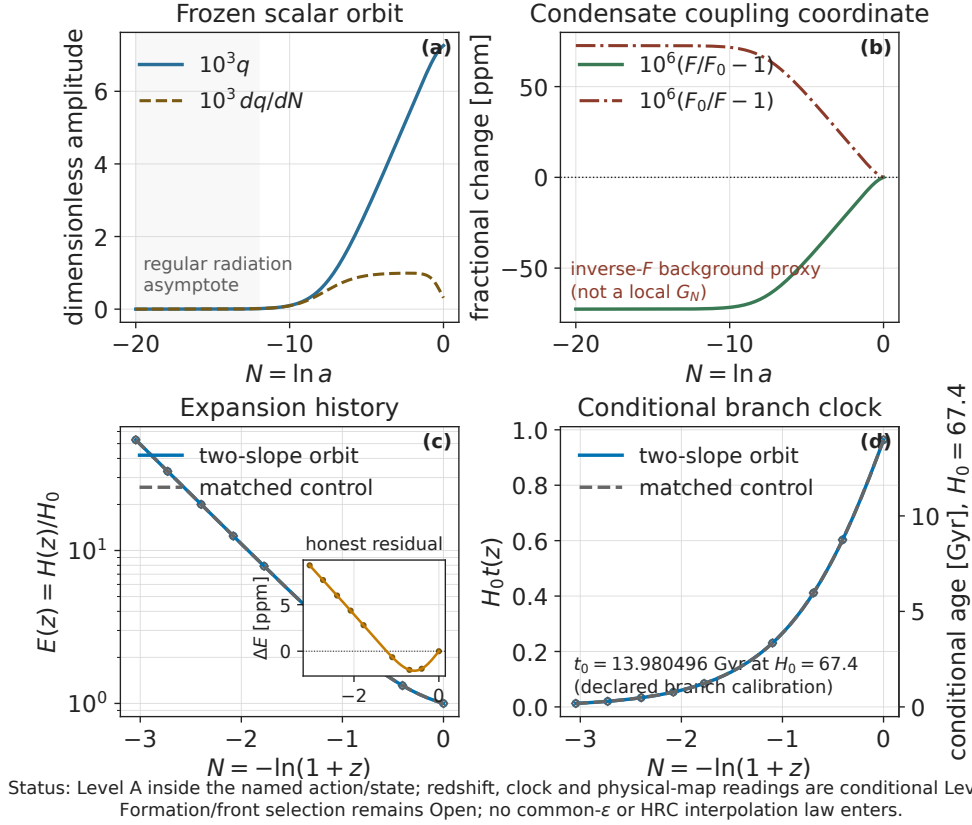


Figure 6: Conditional post-ordering orbit of the supplied two-slope scalar completion. Panel (a) shows its scalar coordinate σ and velocity $d\sigma/dN$ on the frozen regular orbit. Panel (b) shows F_s/F_{s0} and the inverse- F_s background proxy, which is not a local G_N . Panel (c) compares the action-owned expansion with the same-present-fraction matched control and exposes the ppm residual rather than visually exaggerating it. Panel (d) shows the branch clock using the externally supplied reference dimensional calibration $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This calibration is state/data input for the displayed history, not a universal P1–P6 prediction. The dynamical curves are Level A only inside the named action/state; redshift, Gyr and physical-map readings are conditional Level C. The integration begins on an already ordered regular radiation branch: no trajectory through the Open formation front is inferred. Panels (c), its inset and (d) show nine evaluated redshift nodes as unconnected markers; no intermediate interpolation is implied. No common- ε or HRC interpolation law enters this figure. *Display semantics: Panels (a,b) are dense frozen solver histories without decorative markers; panels (c,d) use solver-owned dense background traces with the nine published diagnostic nodes shown separately.*

The supplied coherent branch carries integer winding sectors (Section 22.3). A defect can change that collective-field winding when the amplitude vanishes at a core. This is a statement about the supplied order parameter, not about physical baryon number.

Only after supplying a physical gauge/fermion bundle and a map from its baryon current to this winding could a defect transition represent baryon violation (Sections 7.2 and 7.3). In an additional defect model one may then estimate rates such as

- in a separately specified thermal state with a derived finite barrier E_{def} , an Arrhenius regime may give $\Gamma_{\text{def}} \propto e^{-E_{\text{def}}/(k_B T)}$;
- outside that regime a rate requires an explicit finite-action Euclidean saddle or real-time transition calculation, prefactor and state. Topology alone fixes none of these and does not imply that a process becomes unsuppressed at any named ECT scale.

These rate forms are illustrative: the defect action, thermal state, baryon-current map and transition vertex are not derived. Condensate topology alone is therefore not an ECT analogue of a physical sphaleron.

(ii) CP violation: preferred-direction fermion coupling.

If fermionic excitations are empirically admitted (phenomenological input, analogous to $c_{\text{char}} = c$), then the unique leading renormalisable coupling of a spinorial excitation to the ordered direction field n_A is

$$\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi \quad (16.9)$$

(Level A given empirical fermions: EFT uniqueness theorem; Section 7). This operator is a candidate ingredient only after a nonzero coefficient, physical fermion map and branch-state dynamics are supplied. As systematically classified in Section 9.4, $\mathcal{L}_5$ is a branch-sensitive, CPT-odd and CP-odd phenomenological operator (C-odd, P-even, T-even, CP-odd, CPT-odd). Combined with complex phases of the fermionic mass matrix (open parameter, OP-Yukawa, Section 9.6), it yields the required CP-asymmetric dynamics. Importantly, the conditional CPT route for reconstructible local sectors identified in Section 9.4 remains intact: $\mathcal{L}_5$ explicitly lies outside that reconstructible setup as a phenomenological branch extension, so the exact symmetry backbone of the reconstructible sectors is not compromised. The branch-pair architecture of Section 9.4 additionally admits a speculative cosmological interpretation in which our observable matter-dominated branch is paired with an antipodal antimatter-dominated partner, the full Euclidean pair being CPT-symmetric; this is recorded as OP-BRANCH1 rather than as an established result. The complex flavour phases appear explicitly in the flavour-architecture subsection (§9.7), providing the flavour-sector part of the CP-violation bookkeeping. The baryogenesis estimate in the present section remains tied to the leptogenesis-style assumptions stated above and does not claim that the CKM phase alone explains the observed asymmetry. In particular, the present framework does not claim that the observed baryon asymmetry can be generated by the CKM phase alone.

(iii) Departure from thermal equilibrium: the ordering transition.

P4 supplies an ordered-gradient chart; it does not derive a real-time transition, its clock map, duration, thermal state, or coupling to physical species. If a future completion supplies a finite-rate ordering event, species whose interaction rates are too slow to track it would be out of equilibrium. If the transition completes on a timescale $\tau_{\text{tr}} \sim M_{\text{Pl}}^{-1}$, then species with $\Gamma \ll \tau_{\text{tr}}^{-1}$ would be driven out of equilibrium. Both scales in this conditional statement are presently inputs.

Candidate bias operator inventory

Given the missing owners above, one may write a spontaneous-baryogenesis candidate rather than derive a bias mechanism.

In a supplied non-equilibrium history, $n_A(X)$ and $\theta(X)$ may be time dependent. The following expression is then a candidate operator inventory, not a derived chemical potential:

$$\mu_{\text{eff}} \sim c_\theta \dot{\theta} + c_\Omega \Omega_t + c_5 \mu_5 n_0 + \cdots \quad (16.10)$$

where Ω_t is a time-dependent scalar built from the angular velocity of the ordered frame, and the coefficients c_θ , c_Ω , c_5 depend on the effective Lagrangian of the coherent branch and on the fermion–condensate coupling structure.

For μ_{eff} to act as a physical chemical potential, the completed matter action must couple it to a derived baryon current with the indicated normalisation. Only then, and only while an independently derived baryon-violating process is active, can it source an asymmetry. In a hot plasma with small μ_{eff}/T , standard equilibrium thermodynamics gives [111]:

$$n_B - n_{\bar{B}} \propto \mu_{\text{eff}} T^2. \quad (16.11)$$

After freeze-out at temperature T_D , the asymmetry is frozen:

$$\eta_B \sim C \frac{\mu_{\text{eff}}(T_D)}{T_D}, \quad (16.12)$$

where C is a model-dependent dimensionless coefficient whose sign and possible vanishing depend on the charge content, active degrees of freedom, entropy normalisation, conversion factors, transport and washout details. It is not generically of order unity.

When matter–antimatter asymmetry arises and when it does not

The conditional equations show what a completed model would have to supply; they do not make a nonzero asymmetry generic.

Necessary ingredients for a conditional $\eta_B \neq 0$. A nonzero matter–antimatter asymmetry could arise only when the following conditions are simultaneously realised by the missing action/state:

- (a) At least one term in Eq. (16.10) has an action-owned nonzero coefficient and history, producing $\mu_{\text{eff}} \neq 0$, and the completed matter action couples that bias to a derived baryon current with fixed normalisation. Time dependence alone does not guarantee a nonzero charge bias. In particular, $\mu_5 \neq 0$ is necessary only for the director term $c_5 \mu_5 n_0$, not for independent $c_\theta \dot{\theta}$ or $c_\Omega \Omega_t$ routes.
- (b) The susceptibility/conversion factor represented by C in Eq. (16.12) is nonzero after charge content, entropy, transport and washout are specified. Its value and sign are not selected.
- (c) *B-violating transitions are active* during the epoch when the bias is nonzero. No such ECT rate has yet been calculated.
- (d) *Freeze-out occurs before complete washout*: the B-violating rate drops below the expansion rate while $\eta_B \neq 0$.

Conditions under which $\eta_B = 0$. The asymmetry vanishes if any of the following conditions holds:

- (i) **Exactly homogeneous transition**: the ordering history is sufficiently symmetric that no effective bias μ_{eff} is generated.
- (ii) **Vanishing effective CP-bias source**: the relevant effective couplings are such that the ordered background induces no net CP-asymmetric bias.
- (iii) **Vanishing susceptibility or conversion**: the completed charge, entropy and transport map gives $C = 0$ even when a formal μ_{eff} is nonzero.
- (iv) **Exact dynamical cancellation**: the bias changes sign during the transition in such a way that the integrated asymmetry source cancels.
- (v) **Complete washout**: baryon-number-violating processes remain active long enough after the bias epoch to erase the produced asymmetry.

None of these is known to be exceptional in the present theory. Their relative likelihood cannot be ranked without a measure, transition state and transport solution.

Comparison with the Standard Model

In the Standard Model, electroweak baryogenesis fails at observed parameters because the electroweak phase transition is a crossover rather than a strongly first-order transition. Additional beyond-minimal ingredients are therefore required for successful electroweak baryogenesis at the observed parameters.

ECT supplies an ordered $O(4) \rightarrow O(3)$ branch, not yet the physical transition process or its temperature. The candidate completion above would add all three Sakharov owners rather than derive them from ordering alone. Whether such a completion exists and reproduces the observed asymmetry is Open.

No retained numerical baryogenesis benchmark

External baryogenesis models span different masses, CP sources, washout histories and transport systems. Importing one numerical parameter tuple would test only that external model and would not constrain the missing ECT owners. The present manuscript therefore retains no leptogenesis number as an ECT consistency check. The observed η_B is only an external target for a future owner-complete calculation.

Status summary.

- **Level A conditional mathematics:** integer winding and its possible change through a supplied vanishing-core configuration concern the collective order parameter, not baryon number.
- **Conditional EFT candidates:** the allowed restricted-basis fermion–condensate operator $\mathcal{L}_5$ can provide a CP-bias candidate only if a nonzero coefficient and physical matter map are supplied. The displayed bias and freeze-out equations are standard conditional templates.
- **Open:** the physical baryon current and violating vertex, gauge/colour/fermion map, CP-odd coefficient, clock and thermal transition state, transport, freeze-out, washout, and any ECT value or sign of η_B .

16.4 Age of the Lorentzian ordered branch: exact definition and present identifiability

ECT does not identify cosmic age with elapsed time since a fundamental singularity. The physical quantity is the duration of the Lorentzian ordered branch, beginning at a relational ordering event. On a monotonically expanding post-order branch, after the physical photon/matter redshift map has established $1+z = a_0/a$ and the conventional normalisation $a_0 = 1$ has been chosen, the duration at redshift z is the kinematic identity

$$t_{\text{ECT}}(z) = \int_{a_{\text{ord}}}^{1/(1+z)} \frac{da}{aH_{\text{ECT}}(a)}, \quad t_{0,\text{ECT}} = \int_{a_{\text{ord}}}^1 \frac{da}{aH_{\text{ECT}}(a)}. \quad (16.13)$$

Equation (16.13) is Level A kinematics after $H_{\text{ECT}}(a)$ and a_{ord} are supplied. It is not by itself a dynamical prediction of either quantity.

This internal Lorentzian duration Δt_L must be distinguished from an external-Euclidean formation or relaxation parameter. The clock-zero surface Σ_{ord} may be an internal ordering front, or the ordered-side portion of an actual finite-condensate boundary if the latter is what the completed dynamics selects. The present action has not yet identified which of these owns the realised branch start. No assumption of spherical topology or shape is made in either case. Three objects must not be identified without a derivation: the ordered-medium director $n^A = \partial^A \Phi / u$, the normal s^A to the branch-formation front Σ_{ord} , and the physical matter-clock congruence U^μ . They coincide in a planar homogeneous special case, but need not do so for a front defined, for example, by a level set of $u = |\partial \Phi|$.

For a present event p , let γ_p be the action-owned matter/comoving curve that intersects the ordered side of the formation front once. The outside description of branch age is then the congruence integral

$$\tau_{\text{age}}(p) = \int_{\gamma_p: \Sigma_{\text{ord}}^+ \rightarrow p} \frac{\mathcal{N}_\Phi[\Phi, \text{matter}](X)}{c_u} d\ell_E, \quad d\ell_E^2 = \delta_{AB} dX^A dX^B. \quad (16.14)$$

The $+$ indicates that a sharp transition is approached from the ordered side, where the supplied director exists. A physical metric, lapse and matter-clock congruence remain inputs of the completion rather than properties derived by P4. Only for an owned foliation and congruence does the same orbit also give the local internal–outside differential bridge

$$d\tau = \frac{\mathcal{N}_\Phi}{c_u} d\ell_E, \quad H_{\text{ECT}} = \frac{1}{a} \frac{da}{d\tau} = \frac{c_u}{\mathcal{N}_\Phi} \frac{1}{a} \frac{da}{d\ell_E}. \quad (16.15)$$

Thus the present Hubble rate and the integrated branch age are two observables of one state-dependent orbit and lapse map, not independent adjustable inputs. Equation (16.15) is a directional congruence identity, not a statement about a radius or a spherical boundary. Only for $\gamma_p \parallel n^A \parallel s^A$, constant $\mathcal{N}_\Phi$, and planar homogeneous leaves does this reduce to a normal-depth formula $\tau_{\text{age}} = \mathcal{N}_\Phi L_w / c_u$. No spherical Universe or contemporaneous spatial edge is involved.

A local exact director does not by itself provide a global scalar clock. If

$$\vartheta = \frac{\mathcal{N}_\Phi}{c_u u} d\Phi, \quad (16.16)$$

then a necessary local integrability condition for $\vartheta = d\tau$ is

$$d\left(\frac{\mathcal{N}_\Phi}{u}\right) \wedge d\Phi = 0. \quad (16.17)$$

Without it, age remains well-defined along the specified congruence but may be a spatial field rather than one global number. ECT need not select an absolute Euclidean origin or arbitrary front position; it must derive the front, crossing rule, congruence and clock response that make Eq. (16.14) agree with the internal FLRW integral. Those owners are presently Open.

Status of the numerical age. Universal laws need not select the integration constants and dimensional calibration of our realised condensate. Thus an observationally inferred H_0 may be declared as the single dimensional scale calibration of one fixed dimensionless orbit after ECT has supplied the physical metric, redshift map and forward likelihood. This is a pure rescaling only while no independently fixed Planck, particle, clock or potential scale has already fixed the same dimensional normalisation. What the frozen postulates and current macroscopic actions do not yet supply is one selected law-level history together with the physical congruence/depth-lapse map of Eq. (16.14). Therefore

$$\boxed{H_{\text{ECT}}(a) \text{ and } t_{0,\text{ECT}} \text{ are calculable for a named action/state, but are not uniquely selected by P1–P6.}} \quad (16.18)$$

Conditional gigayear estimates for explicitly supplied completions are given in Section AE. What is not identified at the present closure level is one unique law-level age; this is not a claim that legitimate state data or one dimensional scale calibration are defective.

Exact reason: inverse reconstruction of the scalar closure. For the scalar-only background equations

$$\begin{aligned} 3FH^2 &= \frac{\rho_m + \rho_r + U}{\bar{M}_{\text{Pl}}^2} + \frac{1}{2}\omega\dot{\phi}^2 - 3H\dot{F}, \\ -2F\dot{H} &= \frac{\rho_m + \frac{4}{3}\rho_r}{\bar{M}_{\text{Pl}}^2} + \omega\dot{\phi}^2 + \ddot{F} - H\dot{F}, \end{aligned} \quad (16.19)$$

let $H(t)$ and $F(t) > 0$ be sufficiently smooth and let ordinary matter and radiation be conserved. Define

$$\mathcal{K}(t) \equiv \omega\dot{\phi}^2 = -2F\dot{H} - \frac{\rho_m + \frac{4}{3}\rho_r}{\bar{M}_{\text{Pl}}^2} - \ddot{F} + H\dot{F}, \quad (16.20)$$

$$\frac{U(t)}{\bar{M}_{\text{Pl}}^2} = 3FH^2 + F\dot{H} + \frac{1}{2}\ddot{F} + \frac{5}{2}H\dot{F} - \frac{\rho_m}{2\bar{M}_{\text{Pl}}^2} - \frac{\rho_r}{3\bar{M}_{\text{Pl}}^2}. \quad (16.21)$$

Direct substitution makes both equations in (16.19) identities. On every patch where $F = e^{\beta_\phi \phi}$ is monotonic and $\dot{\phi} \neq 0$, one has $\phi = \beta_\phi^{-1} \ln F$ and obtains $\omega(\phi)$ and $U(\phi)$ by composition with $t(\phi)$, provided $\omega = \mathcal{K}/\dot{\phi}^2$ is finite and has the required physical sign. The scalar equation follows only with the corresponding Bianchi identity and conserved matter/radiation stress tensor; a zero of $\dot{\phi}$ requires a new patch or a separate regular-variable construction. Thus the currently unowned closure functions can support a continuum of expansion histories. This is an exact non-identifiability theorem inside the displayed scalar equations; it is not a failure of a numerical integrator.

Constructive boundary of the theorem. Non-identifiability does not mean that every nearby scalar completion is singular. Section AE gives an explicit two-slope action with exact radiation and dust limits, a stable vacuum and globally regular two-solver trajectories. Its near-GR member has $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368677864$ without a background splice. Two disclosed single-scale calibrations of the same dimensionless orbit give 13.98050 or 12.90095 Gyr. These are conditional ages of a named calibrated ECT-compatible candidate completion. Universal laws need not fix one state-specific number; what remains binding in Eq. (16.18) is the Open microscopic selection and physical metric/clock/front map, not any demand for a universal age.

Additional independent missing owners. Even after choosing the scalar functions, the physical age still requires:

- (i) a stationary ordered-phase action—the P4 gradient background is not stationary under the frozen bare P3 action;
- (ii) the covariant orientation action and its homogeneous stress tensor;
- (iii) a non-circular metric and Planck-scale matching from (u, n_A) ;
- (iv) the physical state, matter/radiation abundances, branch and boundary data;
- (v) the external-depth/lapse map that turns the ordering-front evolution into the start and duration of the Lorentzian matter clock;
- (vi) stability, Newtonian, PPN and perturbation gates for the same solution.

What the external Euclidean view does and does not add. No spherical boundary is assumed. On an arbitrary smooth interface the full flat-boundary massive Euclidean Dirichlet-to-Neumann symbol is $\sqrt{M^2 + |\xi|_h^2}$. In the ordinary classical high-frequency expansion its homogeneous principal symbol is $|\xi|_h$; the mass enters at lower order, while curvature-dependent subprincipal and lower terms depend on the second fundamental form K_{ij} . Boundary shape and state therefore cannot be replaced by one global radius. A uniform translation of a flat interface is an ambient Euclidean isometry and has $\dot{\Lambda} = 0$; it cannot simply be renamed Hubble expansion. Moreover, identifying $K_{ij} = (H/c)h_{ij}$ with a spatially flat codimension-one FLRW slice in flat E^4 violates the Gauss equation unless $H = 0$. A distinct response fibre, curved ambient completion, higher codimension, or derived director invariant is required. The outside view therefore sharpens the missing geometric owners but does not supply an age for free.

Age, lookback time, and early objects. Once a unique background exists, lookback and formation intervals follow from the same solution,

$$t_{\text{lookback}}(z) = \int_{1/(1+z)}^1 \frac{da}{aH_{\text{ECT}}(a)}, \quad (16.22)$$

$$t_{\text{gal}}(z_{\text{obs}}; z_{\text{form}}) = \int_{1/(1+z_{\text{form}})}^{1/(1+z_{\text{obs}})} \frac{da}{aH_{\text{ECT}}(a)}. \quad (16.23)$$

These formulae do not establish a JWST time budget until the same derived H_{ECT} is coupled to a perturbation, halo-formation and baryonic forward model. An external high-redshift history may not be grafted onto the solution, and the dimensional calibration may not be changed independently between observables. After the complete dimensionless ECT orbit has been solved, a measured H_0 may be declared once as a single dimensional scale calibration of that same dimensionless orbit, with that status carried through every clock and distance output.

Falsifier and completion criterion. A conditional ECT age is calculable once one complete varied action, one physical state, one ordering-front/depth-lapse map, dimensional unit data and one self-consistent background integration are supplied. It becomes an independent prediction only to the extent that some of those state data are fixed by other observables and the remaining age or Hubble quantity follows. ECT need not make the realised age a universal constant: H_0 and age may be legitimate observed descriptors of our condensate state. What is mandatory is that one orbit, one dimensional scale calibration and one lapse map impose the dimensionless combination $H_0^{\text{cal}} t_0$ after the declared calibration and the full correlated clock–distance history. Different stable closures or depth/lapse maps that pass their own assumptions but yield materially different correlations demonstrate that the dynamical closure, not the concept of state data, remains underdetermined.

16.5 ECT cosmological background: owners, observables, and closure gates

The cosmological sector is organised by physical owners. The Hubble, JWST, chronometer, growth, lensing and ISW questions are assigned to the dynamical operators that actually control them rather than to one shared empirical parameter.

The action-to-observable chain. A cosmological prediction requires the ordered sequence

$$\boxed{\mathcal{S}_{\text{ECT}}^{\text{cov}} \longrightarrow \{H(a), \phi(a), n_A(a), \rho_i(a)\} \longrightarrow \{\delta_i, \Phi_W, \mathcal{R}\} \longrightarrow \{D_A, r_s, f\sigma_8, C_\ell, \mathcal{N}_{\text{halo}}, \gamma_{\text{PPN}}, \beta_{\text{PPN}}\}.} \quad (16.24)$$

Every arrow must be owned by the same varied action and state. A static HRC response law does not replace the homogeneous background or its perturbation completion: the comoving FLRW congruence has vanishing proper acceleration, so the HRC acceleration invariant is zero on the exact homogeneous solution.

Currently controlled equations. Inside the supplied scalar-only Jordan-frame model, variation yields the background equations collected in Eqs. (13.41)–(13.43). Their algebra is exact inside that model. Their ownership by ECT remains Level B/Open because F , ω , U , the orientation action and the state are not fixed by P1–P6. The inverse-reconstruction theorem of §16.4 proves that these missing functions are sufficient to support inequivalent $H(a)$ histories.

Table 59: Owner map for the active ECT cosmology programme.

Target	Required dynamical owner	Present disposition	Status
Physical back-ground $H(a)$	Complete covariant ordered-phase action, state and initial/boundary data	Exact inverse reconstruction proves that P1–P6 do not select one history; a named two-slope completion supplies one regular conditional $H(a)$ and a nine-point parameter scan	A inside named action / B proof of class / C diagnostics / Open P1–P6 ownership

Target	Required dynamical owner	Present disposition	Status
ECT age	Named action/state, physical matter congruence, external depth/lapse map and one declared dimensional calibration	Exact quadrature; $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$ and disclosed Gyr conversions for one two-slope state; the official 15-point BC03 covariance calibrates the same orbit to $H_0 = 68.59 \pm 3.95$ and $t_0 = 13.74 \pm 0.79$ Gyr. State data and one dimensional calibration are legitimate inputs, while the physical map and microscopic completion remain Open	A/C/Open
Hubble inference	Recombination, photon metric, r_s , D_A and late distance ladder in one likelihood	Conditional $H(z)$, D_M , imported- recombination r_s and a covariance-aware 15-point chronometer scale calibration computed; no ECT-native joint early+late likelihood	C/Open
JWST abundances	Background age/volume, primordial spectrum, transfer, non-linear halo and baryonic selection model	Conditional high- z clock budgets computed at nine prespecified parameter nodes; all nine give $\Delta t(z=10) < 0$ relative to the declared control, with the most negative sampled shift -10.491167% ; no continuous-family, abundance, or morphology conclusion	C/Open
Linear growth	Gauge-complete scalar/vector/tensor perturbation action, matter coupling, slip and sound speeds	Restricted quasistatic $\mu_{QS} = F_0/F$ diagnostic computed; full kernel and likelihood Open	C/Open
ISW and lensing	Photon metric and Weyl-potential transfer	Restricted same-metric, zero-slip, quasistatic Weyl-amplitude and derivative carriers computed; projected spectra and horizon-scale ISW remain un-owned	C/Open

Target	Required dynamical owner	Present disposition	Status
Clusters and mergers	Non-local/retarded metric response plus collision and baryonic forward model	A strictly increasing local map preserves the set-valued argmax and cannot move an attained total-response maximizer away from the supplied total-baryon maximizers; one common real Debye pole plus ballistic transport is conditionally excluded; inherited gas/BCG offsets and residual extrema lie outside the local theorem	Level A theorem in supplied map / conditional Level A one-pole / physical cluster map Open
Solar System / PPN	Covariant weak-field metric completion around a finite source	Scalar cone or galactic response alone does not determine $(\gamma_{\text{PPN}}, \beta_{\text{PPN}})$	Open

Hubble tension. The observable is not a direct function of one late-time coupling. It requires the sound horizon

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H_{\text{ECT}}(z)} dz \quad (16.25)$$

together with the angular-diameter distance, recombination history, photon metric and the late distance-ladder likelihood. Until those are derived and fitted in one pipeline, ECT neither resolves nor uniquely predicts the magnitude or sign of the inferred H_0 shift. For the supplied near-GR two-slope state, Table 128 gives $H(z)$ and distance directly, while Eq. (AE.40) gives an imported-recombination diagnostic: $\Delta r_s/r_s = -0.00316\%$. This is a conditional diagnostic of that supplied action/state plus imported recombination, not an ECT prediction of H_0 ; the named state generates no material shift within those assumptions.

JWST early massive objects. An early-object prediction needs the derived time and volume elements, the primordial perturbation spectrum, transfer functions, scale-dependent growth, collapse statistics, stellar mass-to-light modelling and survey selection. Changing only a background coupling or age does not establish an abundance, morphology or formation history. The current ECT result is a programme with explicit owners plus conditional clock budgets, not a claimed resolution. At the nine prespecified nodes in Table 130, stronger sampled departures reduce $t(z=10)$ by as much as 10.491167% while increasing the homogeneous Planck drift. Hence none of those nine background nodes enlarges the sampled $t(z=10)$ age budget. This finite-grid result is neither a theorem over the continuous two-parameter family nor a JWST abundance no-go; unsampled background regions and distinct perturbative/nonlinear mechanisms remain Open and must pass the same clock and local-gravity gates.

Growth, ISW and cosmological lensing. The relevant quantities are obtained from the quadratic action for physical perturbations and the metric seen by matter and photons. A future completion must derive a Poisson kernel, gravitational slip, all physical sound speeds, anisotropic stress and initial conditions. Only then can it predict $f\sigma_8$, the Weyl potential, ISW cross-correlations and weak-lensing spectra. Importing a multiplicative growth factor is not an ECT calculation. Equation (AE.30) is a named quasistatic approximation: it gives sub- $2 \times 10^{-3}\%$ changes in the near-GR state and is not a substitute for the gauge-reduced calculation.

HRC and cosmology. HRC supplies a candidate quasistatic response in the galactic mass-discrepancy sector after a separate gravity bridge. It does not supply the stress tensor of the homogeneous condensate

or the cosmological perturbation action. A relation such as $a_M \propto cH$ is an independently testable matching hypothesis until a single action makes both sides outputs. Critical interface response may reduce HRC owner classes, but it does not by itself determine $H(a)$.

Conditional cosmology from a supplied action and state

The calculation uses a family of complete scalar actions and states whose equations are integrated from radiation domination to the present. For the calibrated slice

$$c_s = a_s, \quad b_s = 3a_s, \quad \frac{V_-}{V_+} = \frac{b_s - 2a_s}{2a_s - c_s} = 1, \quad \sigma(0) = 0, \quad H(0) = H_*, \quad (16.26)$$

the last two conditions are shooting targets, not fitted observables. The scan uses $\Omega_{m0} = 0.315$, $\Omega_{r0} = 9.19 \times 10^{-5}$ and starts at $N = -20$ with $\sigma' = 0$. Every point below was integrated independently with DOP853 and Radau. Their dimensionless ages agree within 1.4×10^{-15} ; F_s , H^2 , the Friedmann denominator and the reduced response determinant remain positive on the sampled trajectories.

Table 60: Conditional calibrated two-slope family. The sound horizon uses the explicitly imported standard photon/recombination layer $\Omega_b = 0.0493$, $\Omega_\gamma = 5.45 \times 10^{-5}$ and $z_{\text{drag}} = 1059.94$; the displayed $H_0^{(\theta_*)}$ is only the rough fixed-acoustic-angle rescaling $67.4 r_s^{\text{ctl}}/r_s^{2s}$, not an ECT likelihood result. The displayed early-to-present ratio is the divided coefficient $G_{\text{div},s}^{\text{early}}/G_{\text{div},s}^0 = F_s(0)/F_{s,\text{early}}$; it is not by itself a BBN or locally measured gravity response.

a_s	κ_s	Δt_0 [%]	Δt_{10} [%]	Δr_s [%]	$G_{\text{div},s}^{\text{early}}/G_{\text{div},s}^0$	$ \gamma_{\text{PPN}} - 1 $	$H_0^{(\theta_*)}$
0.01	0.05	0.003321	-0.19304	-0.64208	1.01502	1.9920×10^{-3}	67.836
0.03	0.05	0.029258	-1.69939	-5.54599	1.14110	1.7375×10^{-2}	71.357
0.08	0.05	0.181262	-10.49117	-30.64606	2.32553	1.0191×10^{-1}	97.183
0.01	1	0.000166	-0.00968	-0.03226	1.00075	9.9980×10^{-5}	67.422
0.03	1	0.001497	-0.08700	-0.28977	1.00674	8.9838×10^{-4}	67.596
0.08	1	0.010565	-0.61396	-2.03130	1.04864	6.3191×10^{-3}	68.797
0.01	10	0.000017	-0.00097	-0.00323	1.00007	9.9998×10^{-6}	67.402
0.03	10	0.000150	-0.00871	-0.02904	1.00067	8.9984×10^{-5}	67.420
0.08	10	0.001065	-0.06189	-0.20620	1.00479	6.3918×10^{-4}	67.539

The corresponding fixed-acoustic-angle proxy is retained in Appendix BM.

What this scan establishes. It establishes, at Level A inside each supplied action/state, that regular background trajectories exist and that the age, high-redshift clocks, sound horizon and long-range scalar response can be propagated from the same orbit. As a Level C completion scan it also exposes a tradeoff: the points with the largest reduction of r_s and largest rough upward acoustic rescaling of H_0 simultaneously have large early divided-coefficient shifts and fail the unscreened massless-scalar PPN limit. The column $G_{\text{div},s}^{\text{early}}/G_{\text{div},s}^0$ records Planck-coefficient bookkeeping, not a BBN, Cavendish or finite-body response. This is not a universal no-go for ECT, but the same action's finite-body calculation below finds no mass-screening regime for the named calibration. The strongly coupled rows therefore require a different derived response or metric completion; they cannot borrow suppression from this scalar mechanism.

Exact local-gravity diagnostic for this family. For this explicitly specified unscreened constant-exponential benchmark only—not as a universal ECT prediction—the Einstein-frame scalar coupling is

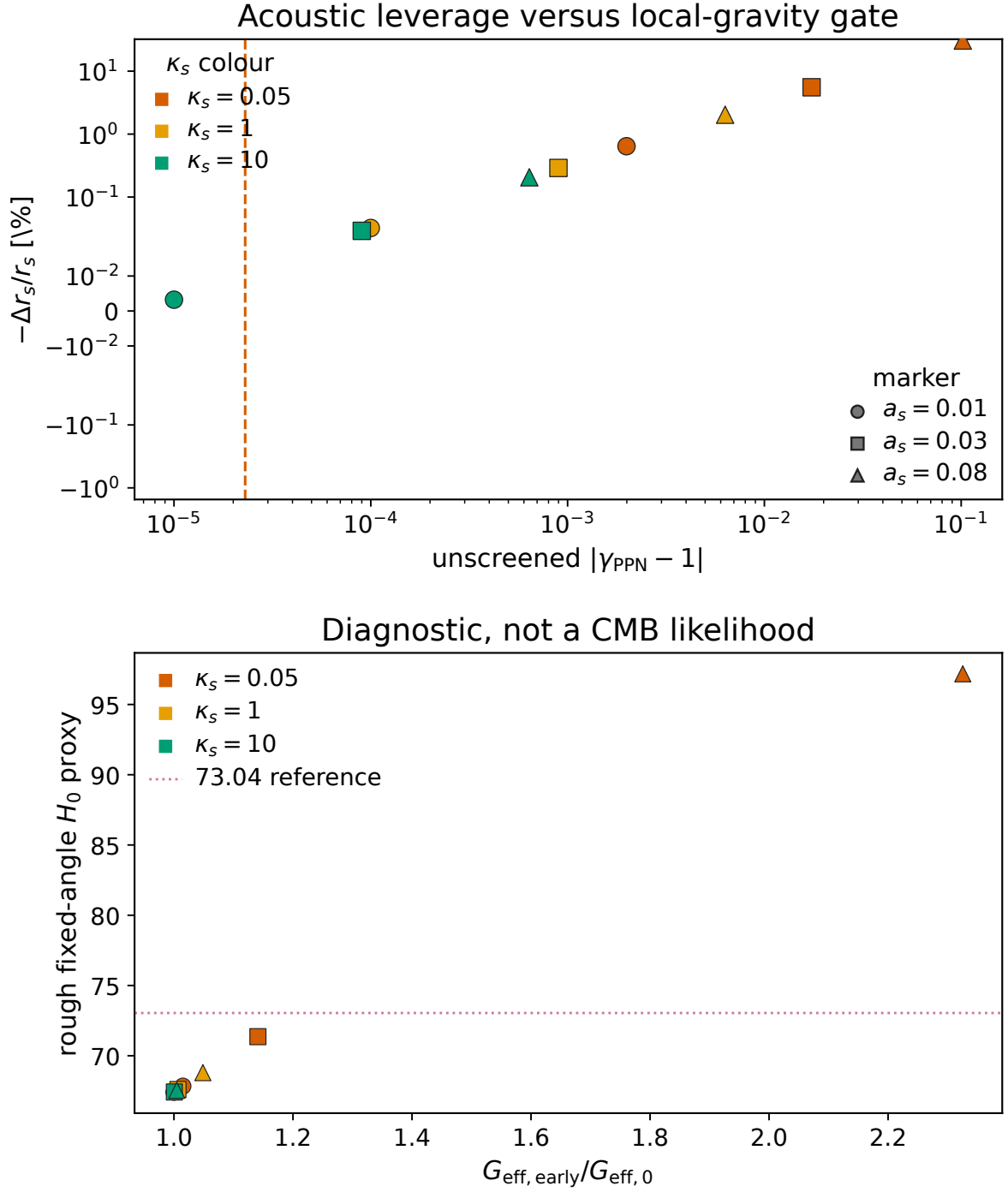


Figure 7: Conditional nine-point two-slope acoustic–PPN scan. This page shows acoustic leverage against the exact unscreened massless-scalar PPN gate of the same supplied action/state. The rough fixed-acoustic-angle diagnostic is a separate full-width atlas panel and is not a CMB or distance-ladder likelihood. The nine independently computed parameter nodes are shown without connecting segments: no fit or continuously evaluated parameter path is implied. Okabe–Ito colours and the a_s marker shapes provide redundant encodings.

constant and

$$\omega_{\text{BD}} = \frac{\kappa_s}{a_s^2}, \quad \gamma_{\text{PPN}} = \frac{\omega_{\text{BD}} + 1}{\omega_{\text{BD}} + 2} = \frac{\kappa_s + a_s^2}{\kappa_s + 2a_s^2}, \quad \beta_{\text{PPN}} = 1. \quad (16.27)$$

For this same massless, universally coupled weak-body limit,

$$\alpha_s^2 = \frac{a_s^2}{2\kappa_s + 3a_s^2}, \quad G_{\text{div},s} = \frac{1}{8\pi F_s}, \quad G_{\text{Cav},s} = G_{\text{div},s}(1 + \alpha_s^2) = \frac{1}{8\pi F_s} \frac{2\kappa_s + 4a_s^2}{2\kappa_s + 3a_s^2} \quad (c = \hbar = 1). \quad (16.28)$$

Because α_s is constant along each scan row, the early-to-present $G_{\text{Cav},s}$ ratio equals the corresponding F_s^{-1} ratio only in this explicitly unscreened massless weak-body limit. A finite scalar range, body charges or screening changes the transfer; the ratio is not thereby an LLR or BBN observable. Only $(a_s, \kappa_s) = (0.01, 10)$ satisfies the manuscript's displayed absolute proxy gate $|\gamma - 1| < 2.3 \times 10^{-5}$ in this unscreened massless limit. The primary Cassini result is $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [95]; the conditional proxy value $\gamma - 1 = -0.99998 \times 10^{-5}$ lies 1.3478σ from that central value, outside the literal one-sigma interval but inside two sigma. Thus the row satisfies only the declared absolute-deviation proxy gate, not a full Cassini likelihood or a complete ECT PPN fit.

Hubble-channel consequence. The fixed- θ_* values in Table 60 are a transparent one-number diagnostic, not a posterior. They hold the reference acoustic angle fixed while changing only the conditional sound horizon. A valid ECT inference must recompute recombination, $D_M(z_*)$, diffusion damping, the perturbation spectra and the late distance ladder. Nevertheless the diagnostic is useful: the only scan point that passes the displayed absolute γ proxy gate moves the rough value by merely $0.002 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Among the nine computed nodes, those with $H_0^{(\theta_*)} \simeq 71.4$ or 97.2 fail the displayed unscreened γ_{PPN} proxy and have large homogeneous divided-coefficient ratios, $G_{\text{div},s}^{\text{early}}/G_{\text{div},s}^0 = 1.14110$ and 2.32553 . No separately defined “early-Planck-scale gate” was evaluated; these ratios are not BBN observables, and BBN/recombination viability requires the forward calculation specified above.

Early-object consequence. None of the nine evaluated background nodes enlarges the sampled $t(z = 10)$ age budget; the largest sampled departure makes that age smaller by 10.491167% . This is an exact result for the finite protocol, not a theorem over the continuous two-parameter family and not a JWST abundance no-go. An ECT explanation of early massive objects would need either an as-yet untested viable background region or a derived change in the primordial spectrum, scale-dependent transfer/growth, collapse efficiency or baryonic assembly, all propagated through the same local and recombination gates.

Observable-by-observable calculation protocols

Chronometers and ages. For each action/state point the publication object is the ordered tuple

$$\mathcal{O}_{\text{age}} = \{H(N_i), H_0^{\text{cal}} t(N_i), H_0^{\text{cal}} t_0, \mathcal{N}_w(w), \mathcal{C}_{\text{bg}}(N_i)\}. \quad (16.29)$$

Here $\mathcal{C}_{\text{bg}}$ is the full constraint residual and $\mathcal{N}_w$ is the external-depth/lapse map of Eq. (16.14). Gyr values are secondary products obtained only after one dimensional scale calibration of the fixed state/orbit. The same $H(z)$ array must be used for chronometers, distances and early-object clocks; independent interpolants are not permitted.

Distance-ladder and BAO chain. Starting from the physical photon metric, the solver publishes $D_M, D_A, D_L, dV_c/(dz d\Omega)$ and r_{drag} on one common grid. The dimensionless likelihood inputs are

$$\frac{D_M(z)}{r_{\text{drag}}}, \quad H_\gamma(z)r_{\text{drag}}, \quad m_B - M_B - 5 \log_{10} D_L(z), \quad D_L^{\text{GW}}(z), \quad (16.30)$$

with the calibration covariance and possible matter–photon–tensor metric differences explicit. A reference r_{drag} may be used only as the labelled diagnostic in Table 60; it cannot enter an ECT likelihood.

Recombination and CMB chain. The background is coupled to a recombination solver through the physical electron mass, fine-structure constant, Thomson opacity and photon temperature. The required residual vector is

$$\mathbf{R}_{\text{CMB}} = \{\theta_*, \theta_{\text{diff}}, z_*, z_{\text{drag}}, C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}\}_{\text{ECT}} - \{\mathcal{D}\}_{\text{obs}}. \quad (16.31)$$

No single component can be minimised by changing a parameter that is kept fixed in the other components. The ECT action/state hash and the external-depth/lapse map accompany every transfer file.

Optional low-multipole completion target. An ordering/coherence scale can motivate asking whether a future completed primordial state modifies only the lowest wavenumbers; it does not determine the sign, amplitude or shape of that modification. A named Open ansatz may, for example, declare

$$\begin{aligned} \mathcal{P}_{\mathcal{R}}(k) &= \mathcal{P}_{\text{ref}}(k) [1 - A_L W(k/k_c)], \\ k_c &> 0, \quad W : [0, \infty) \rightarrow [0, 1], \\ W(0) &= 1, \quad \lim_{x \rightarrow \infty} W(x) = 0, \quad 1 - A_L W(k/k_c) \geq 0. \end{aligned} \quad (16.32)$$

where A_L , k_c and the low-pass window W are independently supplied. Unless a window family is fixed, its shape and k_c are not separately identifiable from this parametrisation. The usual transfer relation

$$C_\ell^{XY} = 4\pi \int d \ln k \mathcal{P}_{\mathcal{R}}(k) \Delta_\ell^X(k) \Delta_\ell^Y(k) \quad (16.33)$$

then becomes calculable only after a common gauge-invariant perturbation action, state, recombination/Boltzmann transfer and likelihood are supplied. $A_L > 0$ would describe suppression inside this ansatz, but P1–P7 selects neither $A_L > 0$ nor any quadrupole value. Thus the low- ℓ branch is a C/Open completion target, not an ECT prediction, explanation or evidence.

Growth and redshift-space distortions. The conditional quasistatic result in Eq. (AE.30) is retained because it is a reproducible limiting calculation, but it is not promoted into the physical growth curve. The full calculation begins from the reduced kinetic, gradient and mass matrices, evolves all adiabatic and isocurvature modes, and publishes

$$\mathcal{O}_{\text{gr}} = \{T_m(k, z), T_\theta(k, z), \mu_{\text{cos}}(k, z), \eta_{\text{slip}}(k, z), P_{\text{lin}}(k, z), P_\ell^s(k, z)\}. \quad (16.34)$$

Only a covariance-weighted fit of these quantities can establish a growth or S_8 result. Multiplying a reference growth curve by one background number is not an ECT calculation.

Lensing and ISW. The metric bridge must determine the Weyl potential $\Phi_W = (\Phi + \Psi)/2$ seen by photons. The jointly owned targets are

$$C_\ell^{\kappa\kappa}, \quad C_\ell^{g\kappa}, \quad C_\ell^{Tg}|_{\text{ISW}}, \quad C_L^{\phi\phi}, \quad D_L^{\text{GW}}/D_L^\gamma. \quad (16.35)$$

A background scalar coupling does not determine any of these without the slip and transfer system. A local HRC relation is likewise insufficient.

JWST abundance and morphology. For a frozen survey selection $\mathcal{S}$, a calculable ECT prediction is

$$\frac{dN}{dz dM_* d\Omega} = \frac{dV_c}{dz d\Omega} \int dM_h \frac{dn_h}{dM_h} P(M_* | M_h, z) \mathcal{S}(M_*, z), \quad (16.36)$$

together with image-level size and morphology likelihoods. The halo mass function must be generated from the ECT linear spectrum and a validated non-linear collapse calculation. The conditional time budgets constrain the integral's possible histories but do not replace it.

Clusters and collision lensing. The local-HRC theorem fixes a narrower baseline: a strictly increasing pointwise transformation preserves the set-valued argmax and, whenever maxima are attained, cannot displace a total-response maximizer relative to the maximizers of the supplied total projected baryon proxy. Any gas/BCG separation in the synthetic maps is inherited from hand-set inputs, not dynamically generated by HRC. The theorem does not determine residual-map extrema, a non-spherical HRC field solution or the photon metric. The physical calculation must solve a retarded response problem,

$$\mathcal{Q}(x, t) = \int_{-\infty}^t dt' \int d^3x' K_{\text{ret}}(x, t; x', t') \mathcal{S}_{\text{bar}}(x', t'), \quad (16.37)$$

derive the photon metric from $\mathcal{Q}$ and forward-model shear, X-ray gas, galaxies and collision geometry. Positivity, causality, memory time and energy balance of K_{ret} are independent gates.

One-real-pole transport no-go. Define the matching-timescale reparameterisation

$$\tau_{a_M}^{\text{match}} \equiv \frac{c}{a_M^{\text{match}}} = \frac{2\pi}{H_0}. \quad (16.38)$$

It is not a derived relaxation time. It equals 87.7664 Gyr for the explicitly declared calibration $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (and 84.1–91.2 Gyr across $H_0 = 73.04\text{--}67.4$). Only if one additionally sets a single constant real Debye pole $\tau_{\text{pole}} = \tau_{a_M}^{\text{match}}$ do ballistic merger speeds $v = 2250\text{--}3000 \text{ km s}^{-1}$ imply

$$v\tau_{a_M}^{\text{match}} = 201.96\text{--}269.28 \text{ Mpc}, \quad \tau_{70\text{--}100\text{kpc}} = 22.82\text{--}43.46 \text{ Myr}. \quad (16.39)$$

The scales differ by roughly three orders of magnitude. Thus one constant real pole with the ballistic law $\ell = v\tau$ cannot own both the static a_M matching and 70–100 kpc Bullet-like offsets. This conditional no-go does not equate a KMS imaginary-time period with a retarded pole and does not exclude a multipole, dispersive or separately owned merger-response kernel.

Solar-System closure. For every cosmologically interesting point, the same action must be solved around an extended gravitating body with its cosmological boundary state. The outputs are the scalar charge, metric coefficients, preferred-frame terms and ephemeris observables. The tested finite-body scalar mechanism has $\Xi_{\text{body}} \ll 1$ and supplies no mass screening for the named near-GR state. Therefore strongly coupled scan points that fail the unscreened diagnostic in Eq. (16.27) need a different derived response or metric completion; suppression may not be assumed.

Robustness axes. The minimum robustness programme varies, one axis at a time and jointly: (i) ordering initial data; (ii) external-depth/lapse map; (iii) scalar action coefficients; (iv) radiation and neutrino content; (v) matter and photon metric assignments; (vi) recombination microphysics; (vii) perturbation initial conditions; (viii) finite-body/environmental response; and (ix) data calibration and covariance. Each variation is rerun through all downstream observables. A point is not *robust* when only its background constraint remains small.

Completion and falsification protocol. For every observable the next accepted result must provide: (i) the varied action and constrained physical fields; (ii) state and boundary data; (iii) a stable background; (iv) the quadratic perturbation operators; (v) the map to matter and photon observables; (vi) a frozen likelihood or forward model; and (vii) counterexamples and limiting tests. Failure of one completion falsifies that completion, not unconnected ECT sectors. Conversely, agreement obtained by separately tuned operators is not evidence for one unified ECT mechanism.

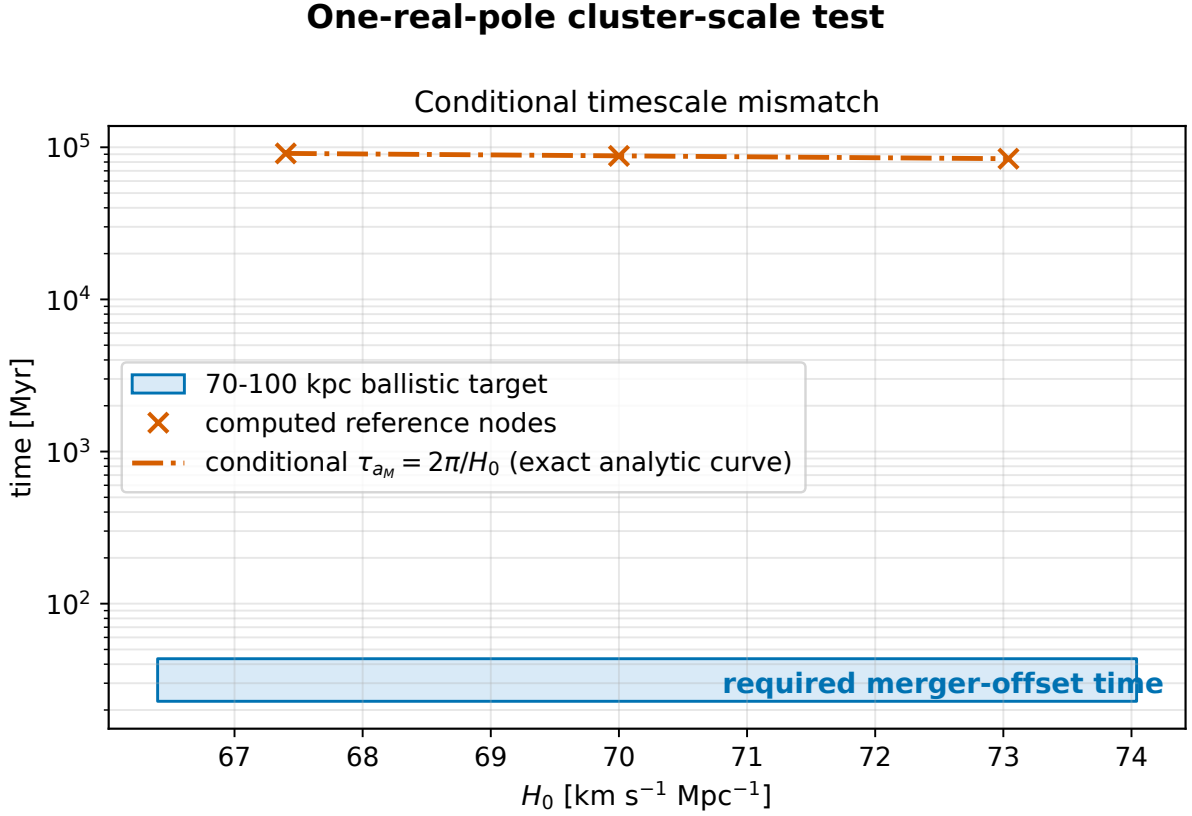


Figure 8: Restricted transport-scale audit under the joint assumptions of one constant real Debye pole, $\tau_{\text{pole}} = \tau_{aM}^{\text{match}} = 2\pi/H_0$, and ballistic propagation $\ell = v\tau_{\text{pole}}$. The resulting 201.96–269.28 Mpc transport scale is compared with the declared 70–100 kpc merger-offset target, exposing a factor 2.02×10^3 – 3.85×10^3 mismatch. This is a conditional Level-A no-go inside those assumptions, not a constraint on multipole, dispersive, non-ballistic, or separately owned causal response kernels. The displayed line is the densely sampled exact analytic relation $\tau_{\text{pole}} = 2\pi/H_0$; the three crosses are the frozen reference nodes. Colour is redundant with line style, markers, luminance contrast and direct labels. The declared target interval is repeated by its luminance-separated fill and both boundary lines, rather than by a decorative texture. Complementary full-width panels are retained in Appendix BM.

16.6 Cosmological status map: conditional outputs, benchmarks, and Open targets

The preceding subsections analysed a range of cosmological statements at different derivation levels. The purpose of the present subsection is to collect them in a single summary table, analogous to the gravitational-sector summary in §13.9, and to distinguish explicitly between strictly derived results, structural closure-level consequences, and illustrative benchmark outcomes.

Table 61: Status map for the ordered-branch cosmology programme.

Claim	Status	Comment
Late-time acceleration in a supplied scalar completion	Open / conditional	A chosen positive potential can parametrise acceleration (proof-of-class); the ECT source, state and globally regular solution are Open
Bare scalar-ratio inequality $w_\phi^{\text{bare}} \geq -1$	A inside supplied stress / ECT Open	Dimensionally normalised algebra only; $\dot{F}, \ddot{F}$ leave observable w_{eff} unconstrained
Conditional frozen-scalar Einstein–Friedmann consistency form	A algebra inside supplied action / ECT Open	Identification with calibrated Λ CDM is Level C and requires $F_0, U_0, \rho_{m0}, \rho_{r0}$, physical metric and state; reachability Open
ECT Hubble inference	Open	Sign and magnitude require one action-owned recombination-to-distance-ladder likelihood
JWST abundance and morphology	Open	Require one background, perturbation, halo, baryonic and survey-selection forward model
Cross-channel cosmological consistency	Open	Age, distance, growth and potential evolution become correlated only after one action and state derive them
Ordered-branch late-time-source route	Open / conditional	Scalar proof-of-class only; no derived ECT source or exclusion of an independent component
Background-to-galactic correlation hypothesis	Open	$a_{M,\text{bg}} = c_{\text{char}} H / (2\pi)$ is an auxiliary background proxy; no map to fitted $a_{M,\text{eff}}$ is derived
Physical late-time-source history $w(a)$ from ECT	Open	Requires solved background, metric, state and joint likelihood
Primordial observables (A_s, n_s, r) from ECT	Open	Not yet obtained from first principles in the present ECT cosmology programme

It is especially important to separate: (i) sign-level and closure-level structural consequences of the ordered branch, (ii) benchmark numerical outcomes obtained within a specific ϕ -first truncation, and (iii) genuinely open cosmological targets that still require a solved background history, perturbation closure, or full data fit. This subsection should be read as a claim-by-claim status inventory; the chapter-level synthesis of what is established, what is closure-dependent, and what remains open is deferred to Section 16.9. Quantities not yet derived from ECT—such as an ECT-native formula for n_s , a physical late-time-source history $w(a)$, or a first-principles baryogenesis asymmetry—are not listed below as ECT predictions.

Table 62: Status-separated cosmological statements, conditional consequences, and Open targets.

Statement / consequence	Formula / current disposition	Level	Section	Observational context
Spatial-flatness branch	$k = 0$ only after the physical metric/embedding is supplied	B / Open	§14.4	topology is structural; cosmological metric ownership Open

Statement (cont.)	Current disposition	Level	Sec.	Obs. context
Supplied-target monopole classification	$\pi_2(S^3) = 0$ for the supplied orientation target	A-math conditional / physical Open	§14.5	complete physical vacuum/configuration target and production history remain Open
Scalar-only perturbation inventory	one longitudinal scalar coordinate in the stated constrained map	A in map / physical Open	§14.6	complete perturbation action and metric map Open
No isocurvature in the scalar-only truncation	conditional on that one-mode inventory	B / physical Open	§14.6	orientation/matter entropy modes not yet classified
Physical tensor propagation	scalar characteristic cone does not determine it	Open	§14.6	tensor action, source, speed and damping missing
Macrocoherent branch minimum	$S_{\text{ord}} = 0$ iff $n_A = \text{const}$ in a connected, constant-compatible admissible class	A-math in supplied model / ECT Open	§14.3	boundary/topological class and physical branch selection remain Open
Conditional physical-time completion	P4 supplies a direction; $\beta(\beta - \alpha) < 0$ classifies a scalar operator; clock/lapse and chronology remain Open	A-math conditional / Open	§14.1	no physical-time theorem from ordering alone
Conditional bare scalar-ratio lemma	$w_\phi^{\text{bare}} \geq -1$ if $\omega > 0$ and $K_\phi^{\text{phys}} + U > 0$	A in supplied stress / ECT Open	§16.1	non-minimal terms leave physical $w_{\text{eff}}(a)$ unconstrained
Ordered-branch age	$t_0 = \int_{a_{\text{ord}}}^1 da/(aH)$	A kinematics / owners Open	§16.4	supplied-completion estimate exists; physical map and microscopic completion Open
Correlated observables	$t_0, t_U, t_{\text{gal}}, D_L$ from one $H(z)$	B struct.	§16.4	age / distance / JWST
ECT Hubble inference	sign and magnitude not identified without the full early+late pipeline	Open	§16.5	CMB, BAO, ladder and sirens
Cross-channel consistency	one future action and state must jointly generate $w(a), H(a)$ and observables	Open	§16.5	joint background/perturbation fits
BBN consistency	accepted background must pass an ECT-native BBN/recombination pipeline	Open requirement	§16.5	no predicted saturation or drift history
Ordering-to-background stages	solver partition, not a predicted trajectory	Open scenario	§16.2	continuous exact-action evolution required
Candidate out-of-equilibrium ordering-transition route	missing-owner scenario; P4 supplies no real-time transition or state	C/Open	§16.3	no jointly owned Sakharov completion, transport or washout

Statement (cont.)	Current disposition	Level	Sec.	Obs. context
ECT background, age, Hubble and early-object closure	one supplied two-slope state gives conditional $H(z)$, distances, τ_{2s} and high- z clocks; the physical map and microscopic completion remain Open	B/C/Open	§AE, §16.5	full perturbation, recombination, abundance and local-gravity pipeline required
Orientation determinant identity	$\det(\beta I - \alpha n n^\top) = \beta^3(\beta - \alpha)$ at fixed amplitude and coefficients	A (algebra only)	§15	does not derive a trace-free metric action or vacuum decoupling
Conditional trace-free shift covariance	$V \rightarrow V + C_0$, $C_I \rightarrow C_I - C_0$ leaves the explicitly supplied constrained scalar–tensor equations invariant	A inside named completion; ECT descent B/Open	§15.5	C_I is an allowed global state/boundary datum
Two-slope late-background proof of class	a coercive ordinary scalar–tensor potential gives one stable de-Sitter root and a regular calibrated history, but is not shift invariant	A inside named action; benchmark C; ECT map Open	§15.7	no claim that the microscopic vacuum offset decouples
Conditional bare scalar-stress ratio	$w_\phi^{\text{bare}} = (K_\phi^{\text{phys}} - U)/(K_\phi^{\text{phys}} + U) \geq -1$ under the stated sign conditions	A in supplied stress / ECT Open	§16.1	F -derivative terms determine the physical late-time expansion
Pseudo-Goldstone late-source route	Physical soft-mode operator, mass, state and abundance not identified	Open	§15.9	a C/Open Hubble-frequency orientation benchmark is retained; no late-source identification or numerical infrared-density scaling

What is explicitly not yet predicted. The present ECT cosmological programme does *not* yet provide: (i) an ECT-native formula for the scalar spectral index n_s or the tensor-to-scalar ratio r ; (ii) a physical late-time-source history $w(a)$; (iii) a first-principles derivation of the baryon asymmetry η_B . These remain open tasks identified earlier in §§14.6, 16.1, and 16.3.

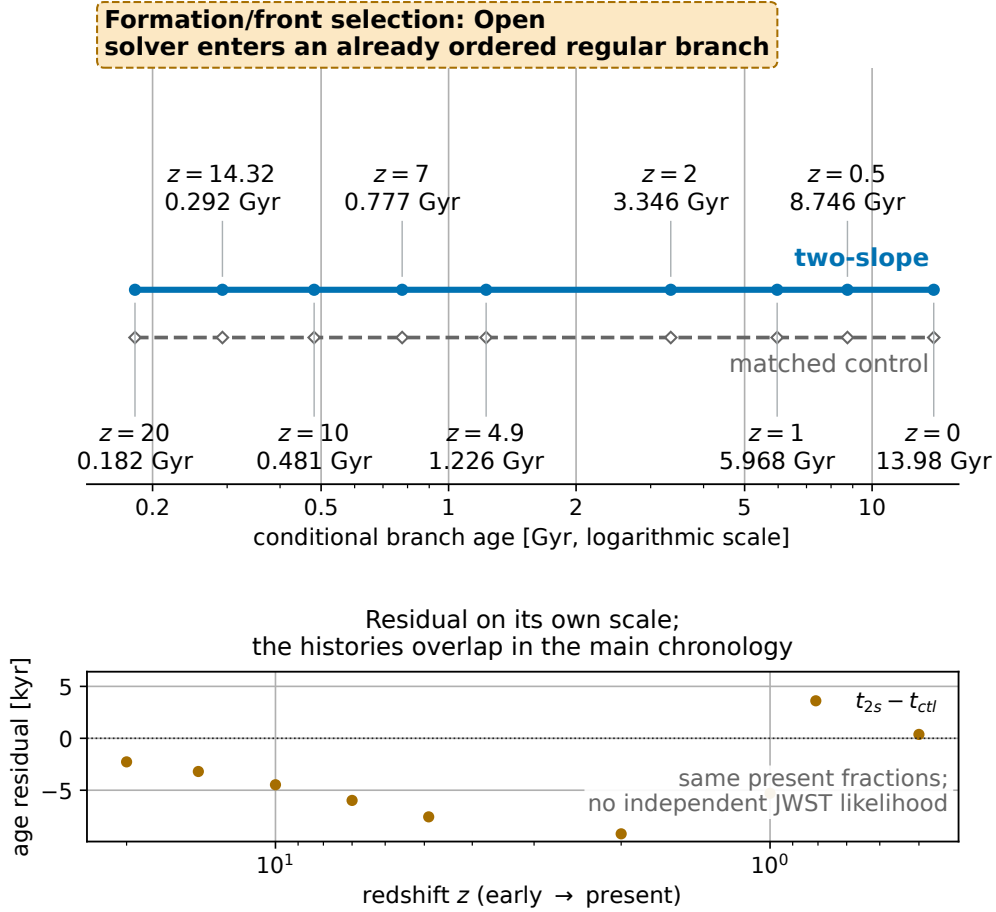
Status summary. The strongest current cosmological statements are conditional and structural, not ECT-native numerical predictions: a conditional planar spatial-flatness proof-of-class, the supplied-target $\pi_2(S^3) = 0$ classification, the constrained longitudinal scalar-coordinate inventory, and the conditional positive-kinetic scalar-stress lemma. The ordered-branch age definition is exact kinematics after its owners are supplied. One named two-slope action/state has active conditional age, $H(z)$, distance, sound-horizon and high-redshift clock benchmarks. Microscopic selection of the two-slope completion and the physical metric/clock/front map remain Open, as do full Hubble inference and JWST abundance or morphology. A fully ECT-native perturbation spectrum and a first-principles baryogenesis model remain central open problems. A chapter-level synthesis of these statuses is given in §16.9.

16.7 Cosmological evolution: ECT vs. Λ CDM

Cosmological visualisations. A universal formation-to-present P1–P6 history remains Open. The named two-slope completion nevertheless supplies one reproducible post-ordering conditional chronology, so it is displayed with its action/state, dimensional-calibration and observable-map qualifications. It must

not be read as a derived ordering transition, a universal age prediction, a Hubble tension solution or a JWST abundance calculation.

Conditional chronology of the named two-slope ordered branch



Externally supplied calibration: $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $t_0 = 13.980496 \text{ Gyr}$.
The separate BC03 clock fit ($H_0 = 68.586 \pm 3.949$) is not used here.
Neither calibration is a universal P1-P6 prediction.

Figure 9: Conditional chronology of the named two-slope ordered branch. The main track uses the externally supplied reference dimensional calibration $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, giving $t(z = 10) = 0.4814276482 \text{ Gyr}$ and $t_0 = 13.9804960874 \text{ Gyr}$. The two-slope and matched-control histories nearly overlap; their difference is therefore shown separately in kyr. The distinct BC03 clock fit $H_0 = 68.586 \pm 3.949 \text{ km s}^{-1} \text{ Mpc}^{-1}$ would give a different dimensional reading of the same orbit and is not mixed into this panel. Event ages share one externally supplied reference dimensional calibration and are not independent measurements or universal P1-P6 predictions. Formation-front selection, BBN/recombination microphysics and the physical metric/clock map remain Open. The figure is neither a universal P1-P6 age prediction, a Hubble-tension solution nor a JWST abundance calculation. The residual panel shows nine evaluated event redshifts as unconnected markers; no intermediate residual interpolation is implied.

Table 63 summarises the key differences between the standard Λ CDM picture and the ECT picture at each cosmological epoch. This comparison reflects the current status of the theory: exact kinematic identities, conditional scalar equations and one globally regular supplied two-slope $H(a)$ orbit are available, but P1-P6 do not select that action/state and no owner-complete observable pipeline has yet

been derived. The early-Universe ordering layer likewise remains a structural scenario rather than a completed first-principles nonequilibrium derivation. The horizon problem can be reformulated in ECT as a branch-coherence question. A uniform map is the exact minimum of a separately supplied positive sigma-model functional (Section 14.3), but P1–P6 do not yet derive that functional, the selected state, the required coherence scale, or the primordial transfer. This is therefore a conditional route rather than a solution; an inflation-like ordering epoch, if retained, is likewise only a conditional effective scenario.

Table 63: Cosmological evolution: ECT vs. Λ CDM.

Epoch	Λ CDM	ECT
Pre-Planckian	Undefined; initial singularity at $t = 0$ marks the boundary of the theory	Pre-Lorentzian Euclidean condensate regime (Level B scenario); $O(4)$ symmetry unbroken, no Lorentzian metric is yet established, and no macroscopic time coordinate is available
Phase transition	Not present	P4 supplies a nonzero vector with $O(3)$ stabiliser; physical $O(4) \rightarrow O(3)$ SSB remains Open. The supplied scalar principal operator admits a Lorentzian characteristic representative on the declared coefficient component. A physical clock/matter metric and nonequilibrium ordering history remain Open
Inflation	Separate inflaton field with ad-hoc potential	Possible inflation-like effective description of the ordering transition (Level B structural scenario; no derived ECT horizon-resolution mechanism is presently available; see Section 14.3); ECT-native derivation of the primordial spectrum remains open (see §14.6); not yet a full first-principles replacement for inflation
Flatness problem	Separate inflationary mechanism usually invoked to suppress $ \Omega_K $ dynamically; in its absence, extreme fine-tuning of initial conditions is typically required	P1 supplies a flat Euclidean ambient metric. A planar induced-metric completion has $k = 0$ exactly, but the physical FLRW metric/embedding is not yet derived (Level B/Open; see Section 14.4)
Relic / monopole problem	Requires dynamical dilution mechanism (inflation) to dilute overproduced GUT monopoles and topological relics from the hot early Universe	The supplied S^3 orientation target has $\pi_2 = 0$, but physical-vacuum identification, transition and relic abundance are Open. A hot GUT history is not postulated, but its absence is not derived; EW and later-sector relics require further development (see Section 14.5)

(continued on next page)

Epoch	Λ CDM	ECT
Primordial perturbations	Observed CMB spectrum: $n_s = 0.9649 \pm 0.0042$ [70], $r_{0.05} < 0.036$ at 95% C.L. [101]; standard framework: inflaton slow-roll	One scalar coordinate in the strict map (Level A kinematics); one physical scalar mode is Level B in the supplied reduced EFT; an ECT-native derivation of n_s , r remains open (OP3)
Reheating / radiation	Standard; inflaton decays into SM fields	A completed ordered branch must transfer energy from condensate excitations to the physical radiation bath; charged-spectrum population, efficiency and thermal completion remain Open (OP3.3)
Matter domination	Cold dark matter particles	Cosmological matter content and matching Open; no ECT replacement for the cold component is derived. Candidate condensate or particle sectors have uncomputed abundance and stability
Late-time acceleration	Cosmological constant $\Lambda = \text{const}$	A regular two-slope scalar–tensor proof of class and a conditional scalar stress exist; the ECT action, state, metric and physical history are Open
Galactic scales	Dark matter halo in the standard interpretation	HRC-0/HRC-3 are augmented-model candidates with recomputed algebraic diagnostics; scalar-to-gravity matching, the field-level likelihood and any environment law remain Open
Origin of cosmic time	Lorentzian time fundamental from Big Bang onward	A relational Lorentzian-clock candidate is available only on the ordered branch; the physical clock/matter-metric bridge remains Open
Perturbation field content	Model-dependent; often single inflaton + tensor sector	One longitudinal scalar coordinate occurs in the stated constrained linear map; its physical projector and the tensor field, source, cone and tensor-to-scalar ratio remain Open (§14.6)
Cosmic age interpretation	Time since Big-Bang singularity (13.8 Gyr)	Duration of Lorentzian ordered branch; conceptually different quantity (§16.4)
Hubble tension	Discrepancy between early- and late-inferred H_0	No ECT sign or magnitude is identified; requires one recombination, distance and likelihood pipeline (§16.5)
JWST early galaxies	Requires unusually rapid early assembly or revised forward modelling	ECT abundance and morphology are Open until the background, perturbation, halo, baryonic and selection owners are solved

(continued on next page)

Epoch	Λ CDM	ECT
Age–distance– Hubble–JWST re- lation	Often analysed as partly separate tensions	These channels must be generated by one future action and state; no common ECT mechanism is presently derived
Far future of the named models	Flat positive- Λ Λ CDM tends to de Sitter expansion	The supplied two-slope action tends to its stable root $\sigma \rightarrow \sigma_0$ and is future- regular (Level A only inside that ac- tion/state; P1–P6 selection and the physical map remain Open)
Alternative future completion	A phantom completion can produce a Big Rip	Recollapse is an Open acceptance case for a different explicitly supplied action/state; it is not produced by the named two-slope orbit and no ECT timescale has been derived (OP3)

Pre-Lorentzian regime. The pre-ordered / pre-Lorentzian regime, its structural status (Level A for configuration-space existence; Level B for cosmological realisation), and its conceptual consequences are developed in §14.1; the branch-coherent reformulation of the horizon problem is given in §14.3.

Condensate evolution and future branches. Ordering, a regular post-ordering background and any later evolution form a solver and acceptance partition, not a universal three-stage trajectory. The named two-slope action supplies only one regular post-ordering branch ending at its stable de Sitter root. Formation/front dynamics and any alternative recollapse branch require a different complete action/state and remain Open. Observable drift bounds and the physical expansion history are acceptance tests of any completed branch.

Dimensionally separated scale inventory. The energy/matching pair (ϕ_0, v_2) and the conditional HRC matching length L_{gal} have different dimensions and status. No common hierarchy, RG cascade, or background-to-galactic scale map is derived.

Distinctive structural features of the ECT cosmological picture. Beyond the epoch-by-epoch comparison recorded in the table, several broader distinctions separate ECT from Λ CDM. The Euclidean construction supplies a pre-ordered configuration-space regime and a scalar Lorentzian characteristic representative on the ordered branch. A physical cosmological clock/metric, globally regular background, late-time source and the distance/Hubble/JWST observables are not established consequences of that structure. Their possible common ownership by one condensate is the programme tested by the action and observable gates of Appendix AI, not a present Level B cosmological mechanism.

Internal versus comparative visualisations. §16.5 and Appendix AI are owner and acceptance maps. Figures 5, 6 and 9 now complement them with the external/internal map, conditional post-ordering orbit and direct numerical age comparison of the named supplied completion. They do not replace the missing formation, metric, perturbation and likelihood owners.

What is established and what remains open. The comparison above defines a common ECT calculation programme for flatness/relic questions, the ordered-branch clock, Hubble inference and early-object observables. It does not establish a dynamical link among those channels. The strongest current statements concern the declared configuration-space topology and the scalar characteristic operator; their

promotion to physical cosmology requires the missing metric, state and perturbation owners. P1–P6 selection of the late-time background, full Hubble inference and the JWST forward model are Open. The named two-slope action/state does supply a conditional numerical age and background orbit, but not their microscopic selection or an owner-complete observational likelihood. What remains open is a fully ECT-native perturbation spectrum (in particular n_s and r), the quantitative nonequilibrium dynamics of the ordering transition, and fully nonlinear structure formation. The epoch-by-epoch comparison above should be read together with the status synthesis of §16.9.

16.8 Universe size, boundary, and force-law effective dimensionality

Status: mixed. The finite-domain / boundary discussion is a Level B cosmological scenario for a possible large-scale completion of the ordered branch. The force-law dimensionality below is instead point-source algebra inside the conditional HRC-0 point-source response after the identity bridge, distinct from the full HRC field problem and not an action-derived ECT prediction. The first part of this subsection concerns the speculative cosmological completion; the second records a conditional HRC-0 point-source diagnostic. Its formula follows only after the declared identity bridge and source profile are supplied.

Size of the ECT domain. In one possible ECT macroscopic completion (Level B), the observable universe is a finite ordered Lorentzian domain in which the macroscopic geometric branch is active. Beyond the causal boundary of this domain the ordered-branch description itself ceases to remain reliable. As the ordered amplitude weakens toward the edge of a finite domain, the Lorentzian branch description is expected to break down and the pre-ordered / pre-Lorentzian condensate regime to take over (Level B scenario; explicit dynamics requires OP3). This should be understood not as reaching a physical wall, but as entering a regime in which the macroscopic geometric branch is no longer a valid effective description.

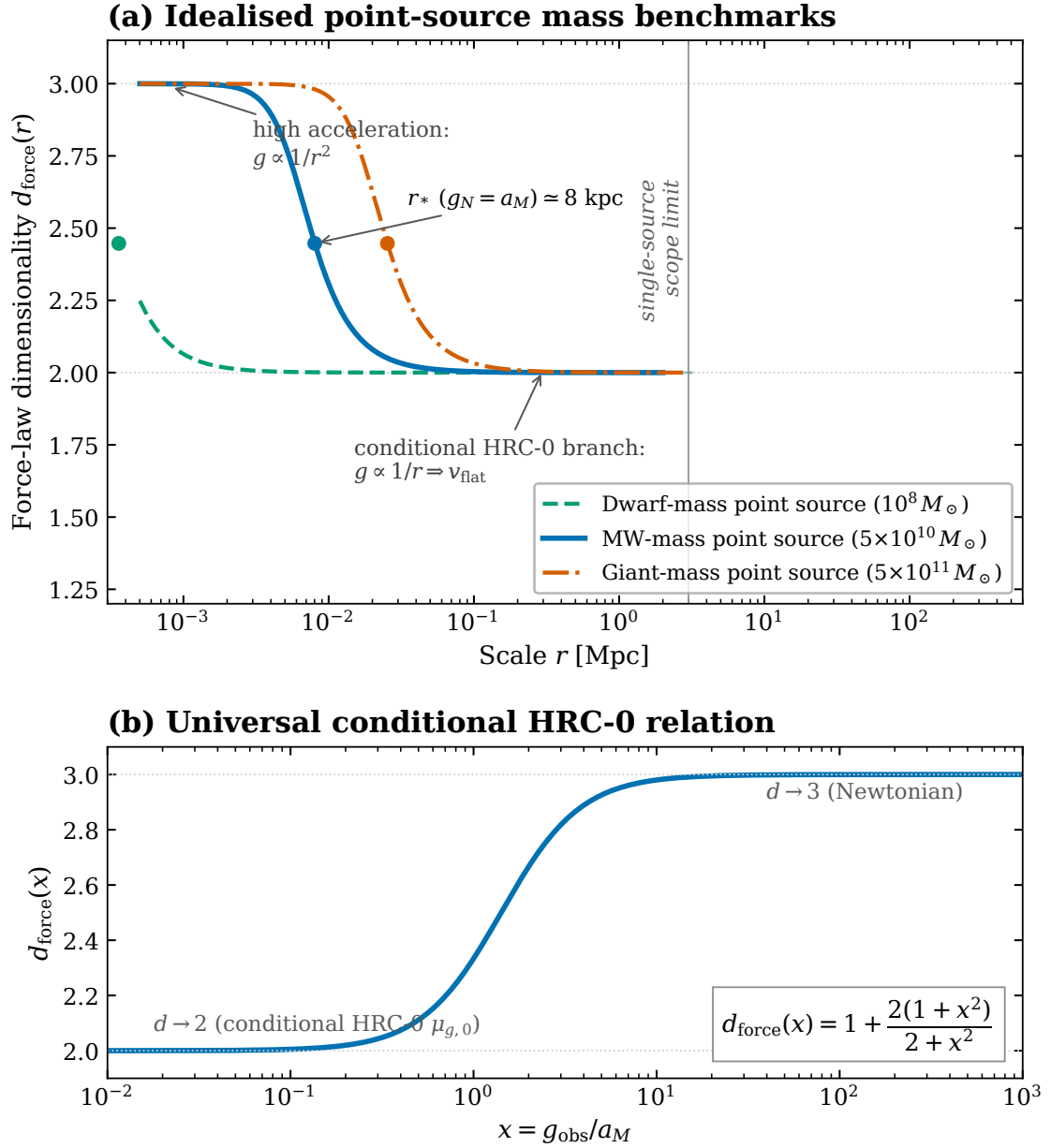
No action-owned boundary solution or observable map fixes a finite-domain size, and no inequality relating R_{domain} to c/H_0 follows from the present framework. The domain size remains Open.

This finite-domain scenario should not be read as fixing the late-time background trajectory or the cosmological closure coefficients by itself. It supplies a possible global support and boundary context for the ordered branch. The observable FLRW quantities remain interior relational variables of that branch and must ultimately be obtained consistently with the same Euclidean ordered-domain realisation.

From domain scenario to measurable force-law response. The finite-domain discussion above concerns a speculative large-scale completion of the ordered branch. The force-law dimensionality below is exact Level-A algebra inside the supplied HRC-0 point-source relation. Its identity/source bridge is Level-B/Open and its synthetic or data application is Level-C. It is independent of the finite-domain scenario and distinct from the full HRC field problem.

Topological versus force-law dimensionality. Two distinct notions of dimensionality appear in ECT. The first is the topological dimensionality of the underlying arena, which is postulated in P1 (see §2 for why $d = 4$ is special). The second is the effective force-law dimensionality extracted from the radial scaling of the observed gravitational response. These notions are different: the arena remains topologically four-dimensional, while the diagnostic computed inside the supplied HRC-0 point-source relation after a separately declared B/Open gravity bridge moves away from its near-Newtonian value in the imposed low-acceleration regime. This conditional point-source response is not the full HRC field problem. This does not mean that matter distributions become literally two-dimensional; it refers only to the scaling of the effective gravitational law.

Effective dimensionality at different scales. The conditional HRC-0 point-source relation supplies a scale-dependent force-law diagnostic. This does not imply a change of topological dimension, a property of the full HRC field problem, or a closure-independent condensate prediction.



Synthetic HRC-0 benchmark: $a_{M0} = 1.0824 \times 10^{-10} \text{ m s}^{-2}$.
 Idealised point sources; no observational data overlay.
 Faded isolation cutoffs are display-only.

Figure 10: Point-source force-law diagnostic of the HRC-0 response $\mu_{g,0}(x^2) = x/\sqrt{1+x^2}$. The shape is Level-A algebra inside the supplied augmented model; the identity/source bridge is Level-B/Open and this synthetic figure is Level-C. **(a)** The three curves differ only by the radial scale $r_* = \sqrt{G_N M}/a_M$ defined by $g_N = a_M$; r_* is not the $x = 1$ radius, which is $2^{1/4}r_*$ here. The imposed point-source relation tends from $d_{\text{force}} = 3$ to 2 and has $d_{\text{force}}(x = 1) = 7/3$. **(b)** The exact point-source collapse $d_{\text{force}}(x) = 1 + 2(1+x^2)/(2+x^2)$. Extended mass profiles instead require their local $p_N(r) = -d \ln g_N / d \ln r$ in Eq. (16.41); no observed-galaxy markers are inferred.

Figure 10 separates the imposed point-source diagnostic from the extended-profile generalisation developed below.

Force-law effective dimensionality. The conditional HRC-0 force-law diagnostic is defined by the local acceleration exponent:

$$d_{\text{force}}(r) \equiv 1 - \frac{d \ln g_{\text{obs}}}{d \ln r}. \quad (16.40)$$

In d spatial dimensions the gravitational force scales as $g \propto 1/r^{d-1}$; eq. (16.40) reads off d from the local force law.

Conditional one-variable collapse. After the supplied HRC-0 point-source relation and identity bridge are declared, d_{force} is a function only of $x = g_{\text{obs}}/a_M$ for a point-mass source. Its mass independence is therefore algebra inside this one-variable closure. It does not follow from the ECT Universality Corollary and does not establish the gravity bridge, source map, or critical scale.

General formula. *Status of the response function: the analysis uses the exact HRC-0 shape $\mu_{g,0}(x^2) = x/\sqrt{1+x^2}$. Its internal algebra is Level A in the augmented model; the identity/source bridge remains Level B/Open and the point-source application is Level C. The calculation does not derive the bridge or a_M . The limiting behaviour below is shared only by functions for which the Newtonian and low-acceleration asymptotes are imposed.*

Starting from the conditional HRC-0 point-source relation $\mu_{g,0}(x^2)g_{\text{obs}} = g_N(r)$ with $x \equiv g_{\text{obs}}/a_M$, one obtains by logarithmic differentiation:

$$d_{\text{force}}(r) = 1 + \frac{p_N(r)}{1 + \frac{d \ln \mu_{g,0}}{d \ln x}}, \quad (16.41)$$

where $p_N(r) \equiv -d \ln g_N / d \ln r$. For a point mass $p_N = 2$. The derivation is given in Appendix AJ.

Closed-form result. For HRC-0, $\mu_{g,0}(x^2) = x/\sqrt{1+x^2}$, one finds $d \ln \mu_{g,0} / d \ln x = 1/(1+x^2)$, and therefore

$$d_{\text{force}}(x) = 1 + 2 \frac{1+x^2}{2+x^2} \quad (16.42)$$

for a point-mass Newtonian source $p_N = 2$. Thus:

- in the high-acceleration Newtonian asymptote $x \gg 1$: $d_{\text{force}} \rightarrow 3$, corresponding to $g \propto 1/r^2$;
- in the imposed low-acceleration regime $x \ll 1$: $d_{\text{force}} \rightarrow 2$, corresponding to $g \propto 1/r$ and asymptotically flat rotation curves;
- at the transition point $x = 1$: $d_{\text{force}}(1) = 1 + \frac{4}{3} = \frac{7}{3} \approx 2.33$.

Thus the conditional HRC-0 point-source law yields a smooth diagnostic crossover from an effectively three-dimensional Newtonian regime at large acceleration to an effectively two-dimensional force-law regime at low acceleration.

Representative characteristic radii. Define

$$r_* = \sqrt{\frac{G_N M}{a_M}}$$

as the reference radius where $g_N = a_M$. It is not the exact $x_g = 1$ radius: $r_{x_g=1} = r_*/\sqrt{\mu_g(1)}$. Thus HRC-0 gives $r_{x_g=1} = 2^{1/4}r_*$, while conditional HRC-3 gives $r_{x_g=1} = \sqrt{3\sqrt{2}/2}r_*$. The characteristic radius is set by the independently supplied M and a_M . Because a_M is fitted or matched rather than derived, these radii are conditional coordinate outputs, not ECT predictions or evidence for an observable transition.

Observational extraction and closure test. A data-level test would require independent profiles for $g_{\text{obs}}(r)$ and baryonic $g_N(r)$ together with the local $p_N(r)$ in Eq. (16.41). The closed $p_N = 2$ curve is a synthetic point-source benchmark, not a universal extended-galaxy prediction. No such observational extraction is performed here.

Status. The displayed point-source $d_{\text{force}}(x)$ is Level-A algebra inside the supplied HRC-0 response model. The identity/source bridge is Level-B/Open and the plotted application is Level-C. It is not a theorem about microscopic topology and does not derive a finite-domain boundary.

What is established and what remains open. The finite-domain interpretation is a Level-B cosmological scenario whose large-scale completion remains Open (OP3). Separately, the force-law formula is Level-A algebra inside the supplied HRC-0 response, Level-B/Open after the declared gravity/source bridge, and Level-C when applied as a synthetic or data diagnostic. What remains open is: (i) a first-principles derivation of any finite-domain boundary from bare P3; (ii) a full data-level extraction of $d_{\text{force}}(x)$ from observed rotation-curve samples; (iii) the extension of the force-law dimensionality concept beyond the galactic branch to morphological or cosmic-web scales (OP8).

16.9 What is established and what remains open

Physical interpretation

The present cosmological programme studies a possible large-scale history built on the supplied ordered-gradient chart and its macroscopic background variable $\phi_b(z)$. It explores the conditional interpretation that a Lorentzian cosmological history begins only after a physical ordered state has formed. P4 itself does not establish such a state or transition. A pre-ordered regime therefore motivates, but does not derive, a relational start of the Lorentzian clock. A stationary state and selection mechanism, homogeneous FLRW solution, late-time source, numerical age, Hubble inference and JWST forward model are separate owner-dependent targets until one complete action and state derive them together.

This unified reading should not be overstated. Some parts of the cosmological sector are already strict structural results, while others remain closure-level consequences of the present ϕ -first macroscopic completion. The purpose of the present subsection is therefore not to introduce new cosmological claims, but to summarise which parts of the ECT cosmological programme are already established, which are benchmarked, and which remain open. It should therefore be read as a status synthesis of the chapter rather than as an additional derivational step. A detailed claim-by-claim status table is given in §16.6. The present synthesis instead groups the chapter by epistemic status and by the role each result plays in the ordered-branch cosmological programme.

Strictly established and status-separated

- (i) The programme compares a hypothetical pre-ordered Euclidean class with the ordered-gradient chart supplied by P4. P4 itself neither owns the former class nor derives a transition between them. Calling the former “pre-Lorentzian” is programme language; a realised physical Lorentzian branch, clock and chronology remain Open (§14.1).
- (ii) Spatial flatness $k = 0$ is a property of the named baseline cosmological completion, not a consequence of P4 alone (§14.4).
- (iii) The supplied orientation target satisfies $\pi_2(S^3) = 0$. Excluding a physical monopole class requires identifying it with the complete vacuum/configuration space; that step is Open (§14.5).
- (iv) Scalar integrability leaves one longitudinal scalar coordinate in the strict map; one physical mode is conditional on the supplied linear EFT, domain and projector. This does not exclude independent isocurvature modes in additional matter/completion sectors (§14.6).

- (v) For supplied $\beta(\beta - \alpha) < 0$, the scalar principal tensor defines one scalar characteristic cone. A physical tensor Hessian, metric, clock and c_{GW} remain Open completion gates (§14.6).
- (vi) In a separately supplied positive sigma-model functional, a uniform orientation is its global minimum. The ECT owner, state selection, coherence scale and cosmological transfer remain Open (§14.3).

Structural closure-level results (Level B)

- (i) The homogeneous FLRW branch of the ϕ -first closure is well-defined and supplies the cosmological background equations (§16.1).
- (ii) The supplied stress obeys the bare scalar-ratio identity $w_{\phi}^{\text{bare}} \geq -1$ when $\omega(\phi) > 0$ and $K_{\phi}^{\text{phys}} + U > 0$; this does not constrain the observable effective history (§16.1).
- (iii) The age of the Universe is reinterpreted as the duration of the Lorentzian ordered branch rather than the elapsed time since a fundamental singularity (§16.4).
- (iv) The quantities t_0 , $t_U(z)$, $t_{\text{gal}}(z_{\text{obs}}; z_{\text{form}})$, and $D_L(z)$ are all controlled by the same ordered-branch background $H(z)$ (§16.4).
- (v) One named ECT-compatible candidate completion now gives a derived conditional age and background distances after a declared state and dimensional calibration. Full Hubble inference and JWST abundance remain Open. A completed cosmology must derive their physical maps and correlations from one action/state and one consistent calibration (§16.5).
- (vi) P4 supplies an ordered-gradient chart, but not the real-time transition history or nonequilibrium state required by Sakharov baryogenesis. The baryon current, violating vertex, CP coefficient, transport and washout owners remain Open (§16.3).
- (vii) The three named regimes—ordering, a regular post-ordering background and any later evolution—are a solver/acceptance partition, not a structural three-stage history. The supplied two-slope action has one future-regular orbit; P1–P6 selection, the formation front and any alternative far-future branch remain Open (§16.2).
- (viii) The force-law dimensionality crossover $d_{\text{force}} : 3 \rightarrow 7/3 \rightarrow 2$ is conditional point-source algebra inside the conditional HRC-0 point-source diagnostic, not the full HRC field problem; data-level extraction remains Open (§16.8).

Current numerical status

A named supplied two-slope action/state now has a globally regular two-integrator orbit, the conditional value $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$, distances, high-redshift clock budgets, an imported-recombination sound-horizon diagnostic, restricted quasistatic growth, and restricted instantaneous Weyl/derivative carriers. None is a P1–P6-selected universal cosmology or a full likelihood result. The physical metric, ordering state, recombination, gauge-complete perturbations and finite-body closure remain Open. Every further numerical result must use the exact solver protocol of Appendix AR.

What remains open

- (i) A fully ECT-native derivation of the primordial spectrum, in particular n_s , A_s , and r (§14.6, OP3).
- (ii) The nonequilibrium microdynamics of the ordering transition from bare P3.
- (iii) The microscopic origin of the effective closure functions, in particular $U(\phi)$ and the full background history $\phi_b(z)$ (OP3).

- (iv) A closure-independent late-time fit of the ECT background sector to the combined age / distance / Hubble / JWST data.
- (v) A first-principles ECT derivation of the baryon asymmetry η_B ; the present manuscript contains only an operator/state owner inventory, not an established Sakharov-compatible ECT environment or numerical benchmark (§16.3, OP7).
- (vi) A fully self-consistent high-redshift completion of the background ansatz compatible with BBN and with the late-time Hubble sector.
- (vii) Fully nonlinear structure-formation calculations (N-body / hydrodynamic level) for the JWST regime.
- (viii) A first-principles derivation of any finite ordered-domain boundary and of its relation to the pre-Lorentzian regime (§16.8).

Chapter-level synthesis. The strongest ECT cosmological results are structural rather than numerical. The architecture supplies a pre-ordered regime; P1 fixes ambient Euclidean flatness, while $k = 0$ requires the declared planar induced-metric completion. The supplied S^3 orientation target has no π_2 class; the physical vacuum/relic conclusion is Open. The strict scalar map has one longitudinal perturbation coordinate. Ownership of the physical FLRW metric and of a complete one-mode scalar perturbation sector remains Level B/Open. The late-time cosmological sector is organised by explicit owners. The central gaps are a globally regular action-owned background, the ordering clock, the gauge-reduced perturbation and photon/matter-metric sectors, and the observable-specific forward models listed in Appendix AI.

17 Galactic and Astrophysical Phenomenology

Scope and status of the galactic programme. This chapter keeps two constructions separate: S, the Level-B scalar ansatz, and ERP- Φ /HRC, the augmented-model response candidate with its newly recomputed Level-C diagnostic pipeline. The physical coordinate, source/force/metric normalisation, body solution, screening and map from the live action to HRC remain Open.

What is and is not claimed here. The HRC algebraic diagnostic addresses selected BTFR, RAR and rotation-curve benchmarks without an explicit particle-halo term in that fit model; it is not yet the nonspherical field-level calculation. The exact local-response theorem proves that every strictly increasing algebraic transform preserves the ordering and maxima of the supplied total projected baryon proxy. It therefore cannot itself displace the total-response maximum, but may inherit gas/BCG separations already present in the input and does not constrain residual-map extrema. These are Level-C closure outputs, not first-principles consequences of the corrected scalar equation. Neither a signed environment law nor Solar-System screening is inferred from the fits.

The scalar audit is documented in Appendix A; the conditional HRC gravity functional and SPARC diagnostic registry in Appendices AK and AL; the cluster-merger owner protocol is in Appendix AI. Implementation algorithms are collected in Appendix Z.

17.1 Scalar audit S and conditional HRC gravity functional G

Status: the amplitude coordinate and conditional scalar identities belong to the Level-B scalar ansatz. ERP- Φ /HRC is a new augmented-model constitutive candidate with the layered status stated in Sec. 17.3; its gravity ownership is Open. The SPARC fit outputs and plotted curves below are HRC-only Level-C algebraic diagnostics with a poor absolute held-out likelihood. Their fitted a_M values are not a derivation of the scale, an environment law, or a field-level disk prediction.

For the amplitude coordinate and conditional scalar-response identities, see Appendix A (Sections AJ.1–AJ.4). The conditional HRC gravity functional and its Open identity-bridge ownership are documented below and in Appendix AK.

Amplitude coordinate of the separate scalar ansatz S

The separate Level-B scalar ansatz S defines

$$\phi \equiv \phi_u = \frac{1}{\beta_\phi} \ln \frac{u}{u_\infty}, \quad F(\phi) = \frac{u}{u_\infty} = e^{\beta_\phi \phi}. \quad (17.1)$$

This is a definition of the u -based coordinate together with the Level B matching $f = \bar{M}_{\text{Pl}}^2 F$. It is distinct from the exploratory anisotropy coordinate $\psi_K = \ln(\Delta_K/\Delta_{K,\text{vac}})$ used in Appendix A; no $u \leftrightarrow \Delta_K$ bridge has been derived. The particular functions $\omega(\phi)$ and $U(\phi)$ used in phenomenological calculations are closure choices, not inherited coefficients already fixed by bare P3.

Scalar-only local response. For a coordinate-independent audit, write the scalar closure as

$$S_q = \int d^4x \sqrt{-g} \left[\frac{1}{2} f(q) R - \frac{1}{2} K(q) (\partial q)^2 - V(q) \right] + S_m[g, \Psi], \quad a(q) \equiv \frac{d \ln f}{dq}.$$

Eliminating the Ricci scalar between the scalar equation and the trace of the metric equation gives, for non-relativistic matter $T \simeq -\rho_m$,

$$\mathcal{A}(q) \square q + \mathcal{C}_X(q) (\partial q)^2 - \mathcal{E}(q, \rho_m) = 0, \quad (17.2)$$

$$\mathcal{A}(q) = K(q) + \frac{3f_{,q}^2}{2f}, \quad (17.3)$$

$$\mathcal{C}_X(q) = \frac{1}{2} K_{,q} + \frac{1}{2} aK + \frac{3}{2} a f_{,qq}, \quad (17.4)$$

$$\mathcal{E}(q, \rho_m) = V_{,q} - 2aV - \frac{a}{2} \rho_m. \quad (17.5)$$

No universal 4π coefficient appears in this equation. Factors of $8\pi G_N$ enter only after a specified normalisation and division by f ; they cannot replace the non-minimal coupling derivative $f_{,q}$.

A homogeneous local root satisfies $\mathcal{E}(\bar{q}, \rho_m) = 0$. The derivative required by the implicit-function theorem is the partial derivative at fixed Jordan-frame matter density,

$$\mathcal{E}_{,q}|_{\rho_m} = V_{,qq} - 2a_{,q}V - 2aV_{,q} - \frac{a_{,q}}{2} \rho_m, \quad (17.6)$$

$$\mathcal{E}_{,q}|_{\mathcal{E}=0} = V_{,qq} - \left(2V + \frac{\rho_m}{2} \right) a_{,q} - a^2(4V + \rho_m). \quad (17.7)$$

Let f, K, V be sufficiently smooth, let $f > 0$, take ρ_m as an independently specified local Jordan-frame density, and fix a regular scalar coordinate together with one boundary/state branch. The implicit-function theorem requires $\mathcal{E}(\bar{q}, \rho_m) = 0$ and $\mathcal{E}_{,q}(\bar{q}, \rho_m) \neq 0$; it does not require a sign for $\mathcal{E}_{,q}$. It then gives

$$\frac{d\bar{q}}{d\rho_m} = \frac{a(\bar{q})}{2\mathcal{E}_{,q}}, \quad \frac{d \ln \bar{f}}{d\rho_m} = \frac{a(\bar{q})^2}{2\mathcal{E}_{,q}}. \quad (17.8)$$

On a stable-root branch with $\mathcal{E}_{,q} > 0$ the second expression is non-negative. The separate condition $\mathcal{A} > 0$ is a kinetic-stability guard for the supplied local scalar equation, not a hypothesis used by the implicit-function theorem. The logarithmic response is invariant under a smooth regular reparameterisation of q ; the sign of $d\bar{q}/d\rho_m$ alone is coordinate dependent.

Two dimensionally explicit negative controls make the scope fail-closed. Let q be dimensionless, $f_\star > 0$ carry mass dimension two, $V_\star > 0$ carry mass dimension four, and $\hat{\rho} \equiv \rho_m/V_\star$. First take

$$a(q) = q, \quad f(q) = f_\star e^{q^2/2}, \quad V(q) = V_\star \left[\frac{1}{4} + (q-1) + \frac{5}{4}(q-1)^2 \right].$$

At $(q, \hat{\rho}) = (1, 1)$, $\mathcal{E}/V_\star = 0$ and the full fixed-density derivative is $\mathcal{E}_{,q}/V_\star = -1/2$, whereas deleting the $a_{,q}$ terms gives the opposite sign, $+1/2$. Thus those terms can reverse the root-curvature verdict even when the independently supplied kinetic guard is healthy. Second, take

$$\mathcal{E} = V_\star \left(q - q^2 - \frac{\hat{\rho}}{2} \right).$$

The point $(q, \hat{\rho}) = (1/2, 1/2)$ has $\mathcal{E} = \mathcal{E}_{,q} = 0$, while the lower branch $q_- = [1 - \sqrt{1 - 2\hat{\rho}}]/2$ has divergent susceptibility as $\hat{\rho} \rightarrow (1/2)^-$. Division by $\mathcal{E}_{,q}$ is therefore forbidden at the fold; no continuation, screening law or finite-body charge is inferred there. For relativistic matter the full trace T replaces $-\rho_m$; scalar-dependent matter masses or correlated matter–scalar initial data also alter the differentiation. This is Level A conditional local mathematics inside the supplied closure. It establishes neither a finite-body charge, density-growing response mass, screening law, PPN limit, nor the earlier 4π , a_M or Ξ formulae.

Derived origin versus benchmark closure. The amplitude coordinate is a variable of the supplied ordered-chart EFT; existence of a physical ordered branch does not follow from that parametrisation. The identifications $f/\bar{M}_{\text{Pl}}^2 = u/u_\infty$, the functions K, V , their connection to the soft anisotropy sector, and the HRC response used in the algebraic SPARC diagnostic are separate matching inputs. The HRC response algebra is Level A inside the supplied augmented model. Its identity/microscopic gravity bridge is Level B/Open, whereas the galaxy fits are Level C; it is not a realisation of the separate scalar ansatz until those bridges are derived.

What the bare condensate-amplitude test establishes

For the supplied linear proxy equation, a compact source has an unsourced exterior Yukawa tail of range $\sqrt{\beta}/m_\sigma$. Therefore a proposed range L must satisfy the exact necessary condition $m_\sigma/\sqrt{\beta} \lesssim L^{-1}$. ECT presently supplies no owned map from the P3 curvature to the physical proxy parameters, so this theorem does not support a numerical galactic application. It also does not forbid an extended profile driven by an extended source; that response remains unquantified until the source vertex is supplied.

If a light amplitude completion is proposed instead, Solar-System and equivalence-principle tests constrain its canonically normalised matter coupling and finite-body solution. They do not imply an automatic order-unity exclusion without those inputs. The result therefore rejects only the separately specified unscreened linear Brans–Dicke benchmark; the radial-proxy branch supplies a symbolic range gate, not a numerical exclusion. It does not select ϕ_u or derive the conditional HRC functional G; its algebra is A inside the augmented model, its microscopic bridge is B/Open, and its data applications are C. Appendix A.3 records the recovered exterior-tail theorem, while Fig. 11 displays the remaining Open matching problem.

Local response mass and the open screening problem

Status: the local linearisation below is derived inside the stated scalar–tensor closure. Density-enhanced screening, finite-body charges, and compatibility with Solar-System bounds remain Open (OP-screen).

Let $\bar{q}(\rho_m)$ be a stable homogeneous root of $\mathcal{E} = 0$. Linearising $q = \bar{q} + \delta q$ and neglecting background gradients gives

$$(\nabla^2 - m_{\text{loc}}^2) \delta q = -\frac{a(\bar{q})}{2\mathcal{A}(\bar{q})} \delta \rho_m, \quad m_{\text{loc}}^2(\rho_m) = \frac{\mathcal{E}_{,q}(\bar{q}, \rho_m)}{\mathcal{A}(\bar{q})}. \quad (17.9)$$

For SI mass density ρ , $\rho_m = c^2 \rho$. Positivity of $\mathcal{A}$ and $\mathcal{E}_{,q}$ gives a locally stable Yukawa response. Along a regular root, the exact total derivative is

$$\frac{dm_{\text{loc}}^2}{d\rho_m} = \frac{1}{\mathcal{A}} \left[-\frac{a_{,q}}{2} + \frac{a \mathcal{E}_{,qq}}{2\mathcal{E}_{,q}} \right] - \frac{a \mathcal{A}_{,q}}{2\mathcal{A}^2}, \quad (17.10)$$

with all functions evaluated on that branch. Its sign is unrestricted and depends on the complete f, K, V closure. Therefore local stability does not derive density-enhanced screening.

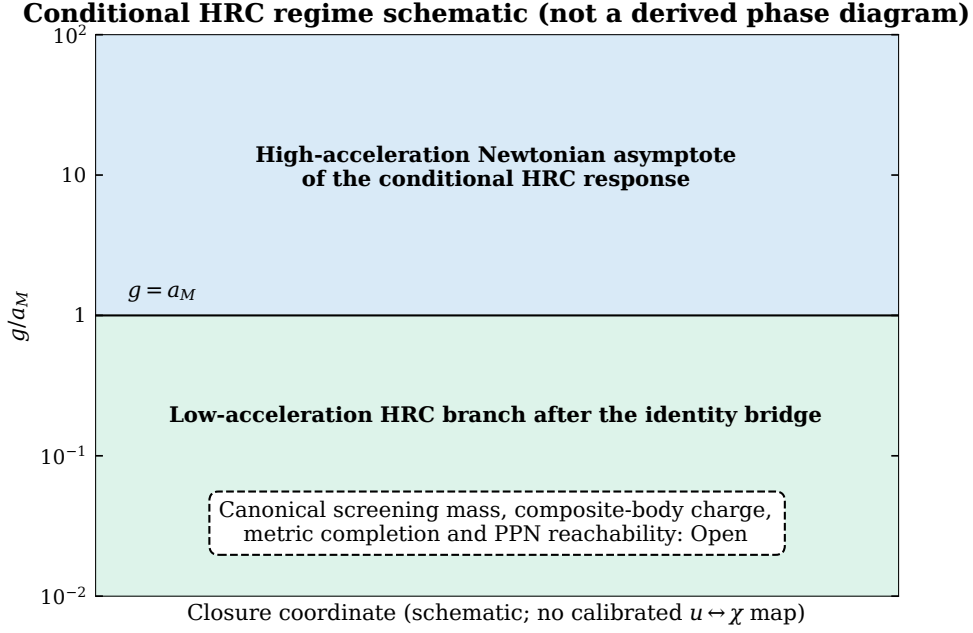


Figure 11: Status diagram for the galactic programme. The vertical axis shows only the two response regimes of the conditional HRC diagnostic. The horizontal scalar-closure axis has no calibrated physical coordinate: neither $u \leftrightarrow \Delta_K$ nor $S \rightarrow G$ has been derived. The Open gate collects the required finite-body solution, scalar charge, screening trend, PPN test, and galactic source/force normalisation.

What a screening claim additionally requires. A viable screened branch must supply (i) a specified closure for f, K, V ; (ii) regular interior and exterior solutions for an extended source; (iii) the scalar charge and matching across the body; (iv) the metric potentials entering post-Newtonian observables; and (v) a parameter region satisfying laboratory, equivalence-principle, and Cassini constraints. None of these follows from the positivity of m_{loc}^2 alone.

Benchmark length scales. The local response length $\lambda_{\text{loc}} = m_{\text{loc}}^{-1}$ is defined only where the assumptions of Eq. (17.9) hold. No calibrated sub-AU, kpc, or Mpc hierarchy follows from the present f, K, V system. Numerical response lengths require a specified closure and finite-body solution. Appendix AN records the corresponding response-length guards and finite-body owner checklist.

Relation to known screening paradigms. If a future solution yields a response mass increasing with ambient density, its phenomenology would resemble chameleon screening [112]. At present this is an analogy, not a derived ECT mechanism. Symmetron and Vainshtein comparisons likewise require the missing nonlinear action and body solutions [113, 114].

Distinct length scales and their current status

Status: the definitions are distinct; their numerical ordering in the galactic closure is not presently derived.

Four scales appear in different parts of the programme:

- (i) **Microscopic condensate scale.** The homogeneous-P3 radial correlation length $\xi_{\text{P3}} = m_{\Phi, \text{rad}}^{-1}$ is microscopic under the declared Planck-scale calibration. It is neither a P4 pole length nor the unresolved soft scalar coordinate used in galactic fits.
- (ii) **Local scalar response length.** Where Eq. (17.9) is applicable, $\lambda_{\text{loc}} = m_{\text{loc}}^{-1}$. Its density dependence and its finite-body interpretation are Open.

- (iii) **Phenomenological transition radius.** The fitted acceleration $a_{M,\text{eff}}$ defines $r_* = \sqrt{G_N M_{\text{bar}}/a_{M,\text{eff}}}$. This is an algebraic scale of the HRC algebraic diagnostic, not yet a prediction of the scalar action.
- (iv) **Cosmological reference scale.** H_0^{-1} is the cosmological length. The numerical proximity of a fitted galactic acceleration to $cH_0/(2\pi)$ is a phenomenological matching observation; its ECT coefficient and dynamical origin remain open.

The candidate ordering sometimes used for intuition is

$$\xi_{P3} \ll \lambda_{\text{loc}} \lesssim r_* \ll H_0^{-1} \quad (\text{benchmark hypothesis, not a derived hierarchy}). \quad (17.11)$$

It may be tested only after $m_{\text{loc}}(\rho)$, the body charge, and the map to a_M are fixed. Consequently this hierarchy cannot presently be used to claim simultaneous derivation of Solar-System screening and galactic critical dynamics.

Anisotropy-loop audit and the missing amplitude bridge

Status: exact Gaussian-normalisation cancellation / derivative coefficients parametric only / Open amplitude matching. Appendix A.6 shows that constant $\kappa_n = \mathcal{C}_n u_0^2$ cancels from the Δ_K -dependent determinant and that the $O(3)$ background permits distinct $Z_\perp$ and Z_w . The old $K_\perp \propto e^{-\psi_K/2}$ form survives only as a separately supplied surrogate, not a loop result, and cannot be transferred to ϕ_u without a derived $u \leftrightarrow \Delta_K$ map.

A generic amplitude-sector derivative expansion may be written as

$$\Gamma[\phi_u] = \int d^3x \left[\frac{K_u(\phi_u)}{2} (\nabla \phi_u)^2 + \frac{L_u(\phi_u)}{3} |\nabla \phi_u|^3 + \dots \right]. \quad (17.12)$$

For explicit cross-reference, the corresponding amplitude coefficient obeys

$$K_{\phi_u} \quad \text{is not determined by the } \psi_K \text{ loop,} \quad (17.13)$$

and hence

$$|\nabla \phi_u|_{\text{trans}} = K_u/L_u \quad \text{is not determined by that loop either.} \quad (17.14)$$

The present loop calculation fixes neither K_u , L_u , nor L_u/K_u . Thus a field-dependent transition gradient and the required ordering of Newtonian and nonlinear regimes are design conditions on a future infrared completion, not consequences derived by the current one-loop branch.

Critical infrared regime

Status: Level C conditional scaling candidate / ECT fixed point Open. If the infrared theory is local in three spatial dimensions, controlled by a fixed point, depends only on $Y_q = (\nabla q)^2$, and has a dimensionless field with negligible anomalous dimension, then scale counting gives $\mathcal{F} \propto Y_q^{3/2}$. The existence of that fixed point, the physical ECT coordinate, source normalisation, metric-force map, and a_M coefficient are not derived.

For bookkeeping, the conditional fixed-point functional may be written

$$F_{\text{IR}}[q] = \int d^3x \left[\Lambda^4 \mathcal{F} \left(\frac{Y_q}{Y_*} \right) + U_{\text{eff}}(q; \rho) \right], \quad \mathcal{F} \propto Y_q^{3/2} \text{ only under the stated assumptions.} \quad (17.15)$$

Under those assumptions the exponent 3/2 is exact and the BTFR exponent follows algebraically after the source and force maps are additionally assumed. These are properties of the conditional critical-AQUAL benchmark, not a result of the quadratic one-loop exterior.

Source normalisation convention. In the AQUAL-like equations below, ρ_b is the ordinary baryonic mass density and $4\pi G_N \rho_b$ is the normalisation supplied in the conditional three-dimensional HRC/AQUAL functional. It is not inherited from the corrected homogeneous scalar trace equation. In that equation the matter term is $(\kappa_F/2)\rho_m$, with $\rho_m = \rho_b c^2$ and $\kappa_F = d \ln F / dq$; a fixed 4π coefficient cannot replace this coupling derivative. Matching the AQUAL source and force variable to the scalar–tensor action is therefore Open.

Conditional HRC potential equation (not a scalar- q equation). Only after the separately declared identity bridge does the galactic layer use the HRC potential equation

$$\nabla \cdot \left[\mu_{g,k} \left(\frac{|\nabla \Phi_{\text{gal}}|^2}{a_M^2} \right) \nabla \Phi_{\text{gal}} \right] = 4\pi G_N \rho_b, \quad k \in \{0, 3\}. \quad (17.16)$$

The registered HRC branches obey

$$\mu_{g,k}(Y_g) \rightarrow 1 \quad (Y_g \gg 1), \quad \mu_{g,k}(Y_g) \rightarrow \sqrt{Y_g} \quad (Y_g \ll 1). \quad (17.17)$$

Here ρ_b is baryonic mass density. Neither $\Phi_{\text{gal}} = \phi_u$ nor $\Phi_{\text{gal}} = \psi_K$ is assumed; both identifications remain excluded until an explicit S→G map is derived.

The numerical registry may fit an effective scale per galaxy as a Level-C diagnostic. A possible environment/history dependence is recorded only as the test family

$$a_{M,\text{eff}} = a_{M0} \Xi(\rho_{\text{env}}, \bar{q}_{\text{env}}, \dots), \quad a_{M0} = \frac{cH_0}{2\pi}, \quad (17.18)$$

where Ξ is Open. Its variables, sign, amplitude and density-only reduction are not derived; scatter in algebraic fits is not by itself evidence for a physical environment law. A prespecified regression protocol and interpretation guards for this Open family are given in Appendix AO.

Status summary. The corrected scalar response remains a Level-B ansatz. HRC-0 is Level A in the augmented ERP- Φ /inverse-DtN model; HRC-3 is Level A in its supplied three-channel model and a Level-B proof of class. The identity gravity bridge, the origin of a_M , the scalar/action map, finite-body screening and the environment law remain Open. The current galaxy-by-galaxy calculations are Level-C algebraic diagnostics and do not replace the nonspherical field solution. The owner-by-owner status and falsifier ledger is given in Appendix AP.

17.2 Local-HRC argmax theorem and missing physical cluster owner

Status: the strictly increasing-map order/argmax theorem is Level A inside the supplied local map. The physical cluster source, metric, retarded response, collision dynamics and observational map remain Open. No synthetic cluster masses, offsets, widths, critical scales, reconstructed maps or finite-box amplitudes are retained as ECT phenomenology.

For any supplied positive projected baryonic field Σ_b , the inverse enhancement paired with the conditional HRC-0 point-source response can be written algebraically as

$$\mu_{g,0}(x^2) = \frac{x}{\sqrt{1+x^2}} \iff v_{\text{cl}}(y) = \sqrt{\frac{1 + \sqrt{1+4/y^2}}{2}}, \quad y = \frac{g_N}{a_{M,\text{cl}}}, \quad (17.19)$$

and the local proxy is

$$\Sigma_{\text{cl}} = v_{\text{cl}} \left(\frac{2\pi G_N \Sigma_b}{a_{M,\text{cl}}} \right) \Sigma_b. \quad (17.20)$$

This is not the nonspherical HRC field equation and has no covariant photon/lensing owner. It is used only to prove what a positive pointwise map can and cannot do.

Let D be the projected domain and define the set-valued $\arg \max_D g := \{\mathbf{x} \in D : g(\mathbf{x}) = \sup_D g\}$, which may be empty. Let a static projected response be pointwise, $\Sigma_{\text{resp}}(\mathbf{x}) = f[\Sigma_b(\mathbf{x})]$ on D , on a physical branch where f is strictly increasing on $\Sigma_b(D)$. A convenient sufficient smooth condition is $f'(u) > 0$ for every u in the interval hull of $\Sigma_b(D)$, not merely at the sampled image values. Then for any two pixels

$$\Sigma_b(\mathbf{x}_1) > \Sigma_b(\mathbf{x}_2) \iff \Sigma_{\text{resp}}(\mathbf{x}_1) > \Sigma_{\text{resp}}(\mathbf{x}_2), \quad \arg \max \Sigma_{\text{resp}} = \arg \max \Sigma_b. \quad (17.21)$$

The set identity remains valid when both argmax sets are empty. When the maximizer set is nonempty—as for a finite pixel map, or a continuous field on a compact domain—a strictly increasing positive local map can rescale contrast but cannot displace any attained total-response maximizer relative to the supplied total-baryon maximizers. It may inherit a gas/BCG separation already present in that input. The theorem is analytical and does not require a proxy-map construction. The expanded monotonicity proof and the missing physical merger/lensing owner checklist are given in Appendix AM.

For Eq. (17.20), setting $s = \Sigma_b > 0$ and $k = 2\pi G_N / a_{M,\text{cl}} > 0$ gives the stronger explicit result

$$\begin{aligned} \Sigma_{\text{cl}}^2(s) &= \frac{1}{2} \left[s^2 + \frac{s}{k} \sqrt{k^2 s^2 + 4} \right], \\ \frac{d\Sigma_{\text{cl}}^2}{ds} &= s + \frac{1}{2k} \left[\sqrt{k^2 s^2 + 4} + \frac{k^2 s^2}{\sqrt{k^2 s^2 + 4}} \right] > 0. \end{aligned} \quad (17.22)$$

With $D = \sqrt{1 + 4/y^2}$,

$$\frac{d \ln v_{\text{cl}}}{d \ln y} = -\frac{D-1}{2D}, \quad \frac{d \ln \Sigma_{\text{cl}}}{d \ln \Sigma_b} = \frac{D+1}{2D} \in \left(\frac{1}{2}, 1 \right). \quad (17.23)$$

Hence the local map preserves every positive-pixel ordering and compresses contrast; it cannot preferentially select a compact collisionless component.

The result is a no-go for the local static route, not for every ECT completion. A Bullet-like calculation requires the retarded/non-local medium state, collision dynamics, physical photon metric, hydrodynamics, collisionless transport, and an observational shear/convergence likelihood listed in Appendix AI. Residual cluster mass and morphology remain Open.

17.3 Galactic dynamics: BTFR, rotation curves, and the effective critical scale

Status overview. The present galactic programme separates four layers. The amplitude coordinate and corrected scalar variational identities remain Level-B closure input. Internal branch reconfigurability and ERP- Φ are a proposed medium property and a new candidate constitutive principle; they contain no gravity. HRC-0 is Level A in the stated augmented inverse-DtN/recruitment model, while HRC-3 is Level A in the supplied equal-three-channel matrix model and a Level B proof of class. Their microscopic P1–P6 ownership and the medium-to-gravity bridge remain Open. The SPARC/RAR/BTFR/UDG numbers have been recomputed as HRC-only Level-C diagnostics. The active calculation uses only the stated HRC map and the baseline $a_{M0} = cH_0/(2\pi)$ is matching only, and fitted scatter has no identified physical environment component until held-out correlation and microscopic matching are supplied.

Scope. This subsection starts from the effective gravitational potential Φ_{gal} and the separately supplied conditional HRC gravity functional G , then derives its Level-A in-model AQUAL/RAR/BTFR algebra and reports SPARC benchmarks. The scalar coordinate ϕ_u is not used in G unless the still-Open $S \rightarrow G$ physical-coordinate, source, force, and normalisation map is supplied. The tested scalar routes are audited separately in Appendix A.

Notation for the HRC scale. a_M denotes the gravity-bridge scale. With the declared $H_0 = 70.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the matching input is

$$a_{M0} \equiv a_M^{\text{match}} \equiv \frac{cH_0}{2\pi} = 1.0824013602391996 \times 10^{-10} \text{ m s}^{-2}. \quad (17.24)$$

It is not an ECT-derived coefficient or a second fitted scale. $a_{M,\text{eff}}$ denotes an independently fitted Level-C nuisance value. The notation does not assert a physical environment dependence; that requires a derived Ξ operator.

Local deformability, branch reconfigurability, and the HRC response programme

Scope and separation of logical layers. This subsection introduces a candidate constitutive extension of the ordered Φ -medium. The medium principle below contains no gravitational field, Newton constant, acceleration scale, matter source, or metric. Its output is only a dimensionless quasistatic response. A separate, explicitly labelled bridge later identifies that response with the weak-field galactic functional. This separation is mandatory: a property of the pre-geometric medium must not silently assume the gravitational phenomenon it is intended to help explain.

Deformability versus reconfigurability: S11a and S11b. Two claims that were previously grouped together must be separated. First, if Φ is locally regular and $\delta\Phi \in C_c^\infty$ preserves the declared boundary data, then

$$\Phi_\varepsilon = \Phi + \varepsilon \delta\Phi \quad (17.25)$$

remains regular and in the same topological sector for sufficiently small $|\varepsilon|$, provided the background stays a positive distance from excluded defects in the chosen norm. This infinitesimal local deformability, S11a, is conditional Level A configuration-space mathematics rather than a new physical postulate.

Second, finite branch reconfigurability, S11b, is a candidate property: some ordered branches may be joined by finite-action admissible transition configurations, while a change of topological sector still requires the defect or transition configurations already allowed by the topological postulates. In plain language, a regular medium configuration can always be nudged locally, but only selected ordered states are proposed to be connected by a physically admissible finite reorganisation. Even when a path exists, the dynamics need not climb a barrier or relax along it. Neither statement contains gravity, force, acceleration, metric, Hessian, a minimisation rule, or a preferred global boundary shape.

For an ordered branch, S11b permits *internal branch reconfigurability*: the state need not remain on a locally inadmissible branch when another admissible branch is accessible. It does not guarantee that every branch is connected to every other branch, and says nothing yet about a Euclidean continuation direction, the number or equality of response channels, their coupling, the sharpness of a threshold, or the response after crossing.

S11a is registered as a conditional theorem with explicit guards. S11b is the proposed foundational property and is not claimed to be realised or derived by the frozen minimal P1–P6 action. It is analogous to internal rearrangement in an ordered elastic or multistable medium, but the analogy is not a derivation. The particular ERP- Φ idealisation used here adds four restrictive clauses:

- (a) equivalent normalised channels couple to one scalar control parameter with equal strength;
- (b) quasistatic relaxation follows the minimum-energy Euclidean continuation;
- (c) during one monotonic sweep, a channel crosses its threshold at most once; in the ideal no-hysteresis limit a reversed sweep returns it at the same threshold rather than making the release irreversible;
- (d) the ideal limit is symmetric and non-hysteretic, so the response is a state function rather than a history-dependent plastic law.

These clauses are scientifically stronger than S11b. They are exposed separately so that future microscopic calculations can confirm, modify, or refute them without changing the meaning of the general property.

ERP- Φ : ideal Euclidean relaxation and threshold release. There is a prerequisite before the response Hessian can be attributed to the live theory. The representative P4 background $\Phi_{\text{bg}} = u_0 w + \Phi_*$ is not a stationary point of P3:

$$\mathcal{E}_{P3}[\Phi_{\text{bg}}] = V'(u_0 w + \Phi_*) \neq 0, \quad \mathcal{H}_{P3}^{\text{formal}} = -\Delta - \mu^2 + 3\lambda(u_0 w + \Phi_*)^2. \quad (17.26)$$

The second operator is w -dependent and, because the first variation does not vanish, is not a physical homogeneous small-oscillation Hessian about a P3 equilibrium. Consequently the ERP response Hessian below is supplied by the augmented model; it cannot be identified with the frozen P3 Hessian about P4. Closing OP-grad by one explicit stationary ordered-phase action is the first microscopic ownership gate.

Let $\mathcal{H}_\Phi$ be the constrained response Hessian of the ordered Φ -medium on the physical response bundle, after gauge and redundant directions have been removed, and let $\mathcal{K}_R > 0$ be a declared reference stiffness on that bundle. The dimensionless positive operator is

$$\hat{\Lambda} = \mathcal{K}_R^{-1/2} \mathcal{H}_\Phi(0) \mathcal{K}_R^{-1/2}, \quad \lambda \in \text{spec } \hat{\Lambda}. \quad (17.27)$$

ERP- Φ is the following candidate constitutive principle:

An imposed scalar mismatch lowers equally the remaining stability margin of equivalent normalised response channels. The ordered Φ -medium relaxes by the minimum-energy continuation into the available Euclidean direction. A channel that reaches the boundary of stability changes from the bound to the relaxed branch once on a monotonic sweep and contributes its normalised quasistatic compliance; in the ideal no-hysteresis limit, reversal returns it at the same threshold.

The formal idealisation is

$$\hat{\Lambda}_s = \hat{\Lambda} - sI, \quad P_s = \Theta(sI - \hat{\Lambda}), \quad (17.28)$$

where s is an internal dimensionless control coordinate and P_s projects onto channels released by its current value. The symbol s is deliberately not identified here with any gravitational invariant. The projector is a zero-temperature, infinitely slow and sharp-threshold limit; finite temperature, finite rate, disorder or hysteresis would replace it by a broadened occupation law and must be derived rather than inserted ad hoc.

The binary recruitment has an explicit bounded variational owner. For the positive semifinite trace density τ_V , minimise over the bounded HRC occupation operator $\mathcal{Q}$,

$$\Omega_s[\mathcal{Q}] = \tau_V[(\hat{\Lambda} - sI)\mathcal{Q}] \longrightarrow \min, \quad 0 \leq \mathcal{Q} \leq I. \quad (17.29)$$

Away from a threshold eigenspace, the minimiser is

$$\mathcal{Q}_s = P_s = \Theta(sI - \hat{\Lambda}). \quad (17.30)$$

Equations (17.29)–(17.30) select the occupied branch before its compliance is read out. This separation is an essential protected-probe assumption of the exact idealisation. Indeed, if recruitment and readout enter one generic finite-coupling functional,

$$\Omega_{s,\eta}[\mathcal{Q}] = \tau_V[(\hat{\Lambda} - sI)\mathcal{Q}] + \eta \tau_V(C\mathcal{Q}), \quad \mathcal{Q}_{s,\eta} = \Theta(sI - \hat{\Lambda} - \eta C), \quad (17.31)$$

where the last expression uses $[C, \hat{\Lambda}] = 0$, the crossing condition is $\lambda - s + \eta C(\lambda) = 0$, not $\lambda = s$. Exact P_s and exact HRC therefore require the protected-probe order of limits $\eta \rightarrow 0$ before the compliance is evaluated, or a derived Ward identity, topological protection or counterterm that cancels this threshold backreaction. No such protection has yet been obtained from P1–P6; finite-coupling ownership remains Open.

There is nevertheless a narrower finite-coupling closure route that does not shift the threshold. If the same positive compliance that reads the channel also multiplies its full branch gap, take

$$\tilde{\Omega}_s[\mathcal{Q}] = \tau_V \left[C^{1/2}(\hat{\Lambda} - sI)C^{1/2}\mathcal{Q} \right], \quad C > 0, \quad [C, \hat{\Lambda}] = 0, \quad 0 \leq \mathcal{Q} \leq I. \quad (17.32)$$

Because every spectral coefficient is $C(\lambda)(\lambda - s)$, its sign changes exactly at $s = \lambda$ and

$$\begin{aligned} \arg \min_{\mathcal{Q}} \tilde{\Omega}_s[\mathcal{Q}] &= P_s, \\ -\frac{d\tilde{\Omega}_*}{ds} &= \tau_V(CP_s), \end{aligned} \quad (17.33)$$

away from isolated threshold points; the moving-boundary term vanishes there because it is multiplied by $\lambda - s$. This supplies Level A algebra inside the stated commuting factorised model and a Level B finite-coupling closure route. It is not a P1–P6 derivation. Any positive spectral weight $h(\hat{\Lambda})$ would preserve the same crossing, so the live action must independently derive why the inverse-DtN compliance $C = (I + \hat{\Lambda})^{-1/2}$ multiplies the branch gap, why it commutes with $\hat{\Lambda}$, and why no non-factorising additive vertex remains. Moreover $\tau_V(CP_s)$ is the generalised force $-d\tilde{\Omega}_*/ds$, not automatically the macroscopic gravity-action density. The trace-to-action bridge therefore remains Open.

This is not the generic response of every condensate. For example, an ordinary quartic Landau channel with energy $\frac{1}{2}(\lambda - s)n^2 + \frac{u}{4}n^4$ gives the continuous ramp $n^2 = (s - \lambda)\Theta(s - \lambda)/u$, not a normalised binary projector. Consequently the bounded two-branch channel is an explicit new microscopic owner required by ERP- Φ ; its realisation by frozen P1–P6 remains Open.

For a positive massive Euclidean half-space response, minimum-energy continuation gives the Dirichlet-to-Neumann operator and its normalised inverse

$$D_E = Z_R m_R \sqrt{I + \hat{\Lambda}}, \quad C = Z_R m_R D_E^{-1} = (I + \hat{\Lambda})^{-1/2}. \quad (17.34)$$

Thus the candidate medium response is the compliance of the released subspace, not the number of released modes alone:

$$W_\Phi(s) = \frac{1}{2\pi} \tau_V(CP_s), \quad \hat{\mu}_\Phi(s) = \frac{dW_\Phi}{ds}. \quad (17.35)$$

Here τ_V is a declared semifinite trace density, not a finite matrix trace. In the homogeneous model one divides by response volume and by the appropriate power of m_R before taking the limit. Equal coupling and this trace-density normalisation are parts of the supplied ERP- Φ ideal model; they have not yet been derived from the frozen P1–P6 action. The statement is therefore a new constitutive extension, not a retroactive theorem of the original postulates.

Withdrawal of PHR-17a. An earlier proposed lemma claimed that the released-channel projector alone forced a uniform channel load. The positive diagonal counterexample below refutes that claim: PHR-17a and its reduction of the owner count are withdrawn. The variational projector/recruitment result itself is not affected.

PHR-19: conditional scale-equivariant load lemma and its limits. Projector/recruitment structure alone does not make the channel load uniform: a diagonal positive counterexample such as $L(\lambda) = 1 + 0.4\lambda/(1 + \lambda)$ leaves the threshold projector intact but deforms the cumulative response. Equal shift and homogeneous final load are therefore distinct owner requirements.

A narrower statement is exact. Suppose that, after all constraints and coarse graining, the normalised load is one positive scalar multiplier $L(\lambda)$ on the reduced channel bundle; it is diagonal in the same basis, independent of the gate s , and exactly, continuously homogeneous on the whole positive half-line,

$$L(a\lambda) = a^w L(\lambda), \quad a, \lambda > 0. \quad (17.36)$$

Keep the displayed density of states and compliance fixed rather than absorbing L into them. Continuity then gives $L(\lambda) = L(1)\lambda^w$. Because the baseline HRC-0 response tends to one as $s \rightarrow \infty$, imposing the pointwise Newtonian normalisation on the loaded response forces

$$w = 0, \quad L(1) = 1, \quad L(\lambda) = 1. \quad (17.37)$$

This is Level A algebra under its stated premises, not a derivation of those premises or of S12. Rotations and dilations are transitive on nonzero radial shells but their representation is reducible, so criticality alone does not give the required irreducible equivariant source. The massive compliance $C = (1 + \lambda)^{-1/2}$ contains a scale and breaks exact dilation symmetry. For a massless interface the principal DtN symbol is $q = |k|$, whereas the dimensionless variable used below is $\lambda = |k|^2/m_R^2$; they must not be identified. Formally combining a massless inverse-DtN factor with the three-dimensional radial density gives a constant spectral response, not HRC-0, and the scale m_R needed for the crossover is absent. Finite form factors, anomalous dimensions, mixtures of scaling weights, logarithmic or discrete-scale corrections, and Schur coarse graining can all spoil exact homogeneity. Hence uniform load remains an independent Open owner to be checked on the final reduced bundle.

Exact HRC-0 response inside the augmented model. For the homogeneous three-dimensional spectral measure,

$$\frac{1}{2\pi} \tau_V [f(\widehat{\Lambda})P_s] = \int_0^s \sqrt{\lambda} f(\lambda) d\lambda, \quad \frac{d}{ds} \frac{1}{2\pi} \tau_V [f(\widehat{\Lambda})P_s] = \sqrt{s} f(s). \quad (17.38)$$

This equation defines the measure convention used below. For a homogeneous isotropic three-dimensional response space, $\lambda = |k|^2/m_R^2$ and the trace density has spectral measure $2\pi\sqrt{\lambda} d\lambda$ in the present normalisation. The exponent $1/2$ is therefore a three-dimensional density-of-states result, whereas the coefficient is a declared normalisation to be matched by a microscopic calculation.

The smooth measure also contains a non-commuting infrared order of limits. For example, on a finite periodic cubic response interface of size L , used only as a nonspherical counterexample,

$$\lambda_n = \frac{4\pi^2 |n|^2}{(m_R L)^2}, \quad n \in \mathbb{Z}^3. \quad (17.39)$$

The first nonzero mode enters P_s only if

$$m_R L \sqrt{s} \geq 2\pi. \quad (17.40)$$

At fixed finite $m_R L$, taking $s \rightarrow 0$ first therefore gives a staircase or gap, not the smooth $\sqrt{s}$ law; an isolated zero mode gives a point contribution rather than the continuum density. The thermodynamic or local continuum limit must control the spectrum before the deep-IR limit, unless a separate noncompact or gapless internal response fibre is derived.

For the frozen 165-galaxy registry, the internal parameter was reconstructed row by row from $\mu = g_N/g$ rather than by imposing one matched a_M on folds that used different fitted scales. The smallest values are

$$s_{\min}^{(0)} = 0.0010949568, \quad s_{\min}^{(3)} = 0.0015006288, \quad (17.41)$$

so even the first periodic mode requires respectively $m_R L > 189.88$ and $m_R L > 162.20$. Continuum expected-count targets of 100 modes give the diagnostic, not theorem-level, requirements $m_R L \simeq 546.75$ and 467.03. Thus $m_R L = \mathcal{O}(1)$ is incompatible with a smooth HRC branch in this finite-cube realisation over the sampled range. This does not select a global shape for the Universe; it is a counterexample to universal compact-boundary infrared smoothness.

With the compliance in Eq. (17.34),

$$W_0(s) = \int_0^s \sqrt{\frac{\lambda}{1+\lambda}} d\lambda = \sqrt{s(1+s)} - \operatorname{arsinh} \sqrt{s}, \quad (17.42)$$

$$\boxed{\widehat{\mu}_0(s) = W_{0,s}} = \boxed{\sqrt{\frac{s}{1+s}}}. \quad (17.43)$$

Writing $x = \sqrt{s}$ gives

$$\mu_0(x) = \frac{x}{\sqrt{1+x^2}}, \quad \mu_0 = x - \frac{1}{2}x^3 + \mathcal{O}(x^5), \quad \mu_0 = 1 - \frac{1}{2x^2} + \mathcal{O}(x^{-4}). \quad (17.44)$$

Algebraically this is exactly the function conventionally called the MOND-standard interpolation, $\mu_{\text{standard}}(x) = x/\sqrt{1+x^2}$, within the phenomenological programme initiated by Milgrom [115]. Therefore HRC-0 has no interpolation-shape novelty at the local algebraic galaxy level. Its distinct content is the proposed augmented-medium response owner, together with the still-Open acceleration-scale and gravity bridges; HRC-3 is a separate post-hoc conditional crossover channel. The primitive and response are Level A consequences of the stated augmented spectral model. Their ownership by the microscopic Φ action remains Open because the physical constrained bundle, equal-shift Ward identity and trace normalisation have not yet been obtained from P1–P6.

Conditional three-channel crossover HRC-3. An explicitly supplied three-channel matrix response can deform the base compliance without changing either endpoint. Put a rank- r_{ch} projector P inside the capped trace and define

$$V(\widehat{\Lambda}) = I - 2 \frac{\widehat{\Lambda}}{I + \widehat{\Lambda}} P, \quad \frac{1}{n_{\text{ch}}} \text{tr}(V^\dagger V) \Big|_\lambda = 1 - \frac{4r_{\text{ch}}}{n_{\text{ch}}} \frac{\lambda}{(1+\lambda)^2}. \quad (17.45)$$

This identity assumes $[P, \widehat{\Lambda}] = 0$ and a projector of constant rank r_{ch} over the spectral support. The baseline trace weights of the three channels are equal; the rank-one vertex selects one direction only after the ordered branch is specified. Thus HRC-3 is a less symmetric deformation of the ideal ERP response. The live scalar-only action does not yet own either that selected direction or the commuting spectral vertex. In the equal-three-channel, rank-one case, $n_{\text{ch}} = 3$, $r_{\text{ch}} = 1$, and

$$\boxed{\widehat{\mu}_3(s) = \widehat{\mu}_0(s) \left[1 - \frac{4}{3} \frac{s}{(1+s)^2} \right]}. \quad (17.46)$$

The coefficient $4/3$ is fixed inside this matrix model, not fitted in the formula. HRC-3 is Level A within that supplied response model and a Level B proof of class. It is Open as a consequence of the live scalar-only P1–P6 theory, which does not yet provide the required three-channel physical response bundle or its equal weights.

The selection is sharper than a free numerical choice. For $n_{\text{ch}} = 3$ equal channels the allowed projector ranks give

$$r_{\text{ch}} = 0, 1, 2, 3 \implies A = \frac{4r_{\text{ch}}}{3} = 0, \frac{4}{3}, \frac{8}{3}, 4. \quad (17.47)$$

For the reciprocal family, positivity requires $A < 4$, monotonic response requires

$$A \leq \frac{20}{9}, \quad (17.48)$$

and the bound is transparent rather than numerical. With $z = s/(1+s) \in [0, 1)$,

$$\frac{d\widehat{\mu}_A}{ds} = \frac{(1-z)^2}{2\sqrt{z}} (1 - 3Az + 5Az^2). \quad (17.49)$$

For $A \geq 0$ the bracket has its interior minimum at $z = 3/10$, where it equals $1 - 9A/20$, proving the stated monotonicity limit. Ellipticity of the radial AQUAL eigenvalue is controlled explicitly by

$$\ell_A(s) \equiv \widehat{\mu}_A + 2s \frac{d\widehat{\mu}_A}{ds} = \sqrt{z} (2 - z - 4Az + 9Az^2 - 5Az^3) > 0. \quad (17.50)$$

The first-contact equations $\ell_A = 0$ and $\partial_z \ell_A = 0$ reduce to $10z^3 - 39z^2 + 36z - 8 = 0$, whose relevant root is $z_* = 0.3305721538\dots$. They give

$$A < A_{\text{ell}} = 3.2140953737\dots \quad (17.51)$$

Consequently $r_{\text{ch}} = 1$, hence $A = 4/3$, is the unique nonzero monotonic member of the stated equal-three-channel affine-reflection class. This is a conditional selection theorem for the supplied matrix model, not a derivation of the three-channel bundle from P1–P6.

The response has a complete quasistatic primitive, and therefore a static functional owner after the separate gravity bridge, inside the same supplied model. It is not a complete causal dynamical action for the response sector. For the reciprocal one-parameter family

$$\hat{\mu}_A(s) = \hat{\mu}_0(s) \left[1 - A \frac{s}{(1+s)^2} \right], \quad (17.52)$$

put $m = \sqrt{s/(1+s)}$. A primitive satisfying $W_A(0) = 0$ is

$$W_A(s) = W_0(s) - 2A \left[\text{artanh } m - m - \frac{m^3}{3} \right]. \quad (17.53)$$

Hence HRC-3 uses

$$W_3(s) = W_0(s) - \frac{8}{3} \left[\text{artanh } m - m - \frac{m^3}{3} \right], \quad W_{3,s} = \hat{\mu}_3(s). \quad (17.54)$$

This closes the primitive–response relation inside the matrix model; it does not close the missing microscopic ownership of that matrix response.

Separate weak-field gravity bridge. Only at this second logical layer do we introduce the gravitational invariant

$$Y_g \equiv \frac{|\nabla \Phi_{\text{gal}}|^2}{a_M^2}, \quad s = f(Y_g), \quad W_g(Y_g) = A_g W_\Phi(f(Y_g)), \quad (17.55)$$

where f and A_g belong to the medium-to-gravity bridge, not to ERP- Φ . Its response is

$$\mu_g(Y_g) = W_{g,Y_g} = A_g \hat{\mu}_\Phi(f(Y_g)) f'(Y_g). \quad (17.56)$$

The exact HRC-0 or HRC-3 gravitational law used below is obtained only under the separately declared matching conditions

$$f(Y_g) = Y_g, \quad A_g = 1. \quad (17.57)$$

These are bridge assumptions, not medium postulates. With this bridge, the supplied static functional

$$S_{\text{HRC}}^{\text{stat}} = - \int dt d^3x \left[\frac{a_M^2}{8\pi G_N} W_g(Y_g) + \rho_b \Phi_{\text{gal}} \right] \quad (17.58)$$

gives

$$\nabla \cdot [\mu_g(Y_g) \nabla \Phi_{\text{gal}}] = 4\pi G_N \rho_b. \quad (17.59)$$

For an isolated spherical source, both HRC-0 and HRC-3 have $\mu \sim g/a_M$ in the deep branch and therefore give, conditionally,

$$g^2 \simeq g_N a_M, \quad v_{\text{flat}}^4 = G_N M_b a_M. \quad (17.60)$$

These are exact consequences of the supplied static gravity bridge. They do not derive that bridge, the matter vertex, photon coupling, both metric potentials, or the acceleration scale from the microscopic medium.

Exterior Euclidean interpretation without a global shape assumption. The inverse-DtN construction is local to a smooth response interface. It does not require the condensate domain, the Universe, or its boundary to be spherical. In local boundary coordinates the principal symbol is $D_E \sim \sqrt{-\Delta_\Sigma + m_R^2}$; curvature and finite-depth terms enter as subleading geometric corrections that must be calculated for the actual interface. A global sphere is therefore neither assumed nor used to obtain Eqs. (17.38)–(17.43). Conversely, the local half-space result alone does not identify the physical interface, its boundary condition, or its embedding in the live Φ action. Those are explicit exterior-side closure tests.

Acceleration scale. The present-epoch relation

$$a_{M0} \equiv \frac{cH_0}{2\pi} \quad (17.61)$$

is retained only as an explicit matching condition. Neither ERP- Φ nor the HRC trace currently derives the coefficient $1/(2\pi)$. The stronger hypothesis $a_M(z) \propto H(z)$ is a separate falsifiable proposal and is not used as a derived result here.

HRC-only numerical diagnostic and its limitations. A controlled algebraic diagnostic on 165 SPARC galaxies, signed gas and five repeated whole-galaxy splits gave the following absolute held-out statistics. The fixed- M/L mode uses 3343 points per seed; the common- M/L transfer mode uses a validity mask frozen at $\Upsilon_d = 0.3$ and 3342 points per seed:

Response	Fixed- M/L $\langle\chi^2\rangle$	Fixed- M/L SD	Common- M/L $\langle\chi^2\rangle$	Fixed- M/L fitted $\langle a_M \rangle$
HRC-0	117598.72	3645.86	109439.95	$1.75 \times 10^{-10} \text{ ms}^{-2}$
Conditional HRC-3	99501.19	2967.18	103763.46	$1.27 \times 10^{-10} \text{ ms}^{-2}$

The corresponding fold-averaged training values of a_M in the common- M/L transfer mode are $1.2629 \times 10^{-10} \text{ ms}^{-2}$ for HRC-0 and $1.1170 \times 10^{-10} \text{ ms}^{-2}$ for HRC-3; they belong to a different stellar- M/L protocol from the final column above. Across the five frozen whole-galaxy fold seeds, the population standard deviations of the common- M/L held-out χ^2 totals are 3194.14 for HRC-0 and 4537.24 for HRC-3. The absolute values correspond to χ^2/point of order 30 and do not constitute an acceptable survey likelihood. HRC-3 was constructed after the crossover residuals were inspected and is not an independent observational prediction. HRC-0 has a demonstrated tension in the local algebraic disk approximation; a full nonspherical AQUAL disk verdict remains not identifiable from these inputs. These numbers are Level C diagnostics, not a survey likelihood. A complete calculation still requires the axisymmetric finite-thickness field equation, the curl field, boundary conditions and external field, consistent distance, inclination and stellar-population nuisance parameters, covariance and a prespecified held-out protocol.

Operational HRC registry and provenance policy. The manuscript uses two named response modes: HRC-0 as the exact base response of the augmented model and HRC-3 as the conditional principal numerical candidate. All active galactic formulae, figures and tables use this HRC-only registry and its frozen inputs, hashes and status-qualified diagnostics.

Closure tests and falsifiers. The present extension remains conditional until the following finite tests are passed:

- (i) construct one stationary, stable ordered-phase action that actually supports P4, then derive the physical constrained response bundle and its dimension from its Schur Hessian;
- (ii) derive or refute the equal scalar Hessian shift and trace weights;

- (iii) derive the quasistatic zero-temperature occupation law and quantify threshold broadening or hysteresis; then derive the commuting factorised gap of Eq. (17.32), a protected Kubo limit, another exact cancellation, or the actual finite-coupling threshold shift;
- (iv) derive the physical interface/fibre and verify the low-spectrum continuum condition, including discrete finite-size corrections rather than extrapolating a high-frequency Weyl law to $s \rightarrow 0$;
- (v) obtain the trace-to-gravity action, matter and photon vertices and both weak-field metric potentials without double counting;
- (vi) derive or independently match a_M and test any redshift evolution;
- (vii) solve the finite-disk HRC field equation and rerun the complete publication likelihood before promoting the present algebraic figures.

A non-three-dimensional physical response density, unequal channel weights, an unavoidable history dependence, a nonpositive constrained Hessian, a failed Newtonian limit, or failure of the metric/PPN gates would refute the corresponding HRC completion even if an algebraic galaxy fit remained good.

Status summary. Infinitesimal local deformability S11a is conditional configuration-space mathematics; finite branch reconfigurability S11b is the proposed weak general medium property. ERP- Φ is a further sharp, symmetric, additive, non-hysteretic quasistatic idealisation rather than a restatement of S11b. HRC-0 is Level A mathematics in the augmented inverse-DtN/recruitment model. HRC-3 is Level A in the stated equal-three-channel rank-one response model and a Level B proof of class. The gravitational bridge and the ownership of either response by frozen P1–P6 remain Open; the numerical comparisons are Level C. Apart from the explicit P3–P4 stationarity guard required to define the response operator, no quantum, PES, tensor, cluster, cosmological or PPN claim is modified by this proposal.

Claim	Status in this candidate
Infinitesimal local deformability S11a	Conditional Level A configuration-space result with boundary, regularity and topology guards; it does not imply finite accessibility or relaxation
Finite branch reconfigurability S11b	Candidate gravity-free property; it does not imply ERP's threshold, channel or no-hysteresis clauses
ERP- Φ	New constitutive principle / Level B candidate conditional on its stated ideal clauses; Open as a consequence of frozen P1–P6
P4-supporting response Hessian	Open / MISSING ACTION; P3 does not own it
Factorised finite-coupling route	Level A algebra in the supplied commuting model; Level B closure route; Open in P1–P6 and in the trace-to-action map
PHR-19 load lemma	Level A only under exact scale-equivariant scalar-load and Newtonian-normalisation premises; uniform load/S12 is an independent Open owner
Finite-interface IR gate	Compact-boundary counterexample Level A; physical interface/fibre and continuum-first limit Open
HRC-0	Level A in the augmented inverse-DtN/recruitment model; Level B candidate mechanism; Open in live ECT
HRC-3	Level A in the supplied three-channel matrix model; Level B proof of class; Level C numerical comparison; Open in live ECT
Gravity bridge and a_M	Open; $a_{M0} = cH_0/(2\pi)$ is matching only
HRC SPARC/RAR diagnostics	Level C algebraic held-out diagnostics; full finite-disk likelihood and independent prediction still Open

Conditional algebra after the HRC gravity bridge

The following chain is exact only after the separate bridge in Eqs. (17.55)–(17.57). The internal ERP coordinate s and the gravitational invariant Y_g must not be identified by notation alone: the calculation below explicitly adopts $s = f(Y_g) = Y_g$ and $A_g = 1$. Deriving or refuting that map from the live Φ action remains an Open task.

- (I) **Internal response and gravity bridge.** ERP- Φ supplies a model-conditional internal primitive $W_k(s)$ and response $\hat{\mu}_k = W_{k,s}$, with $k = 0$ or 3 . The bridge defines

$$Y_g \equiv \frac{|\nabla\Phi_{\text{gal}}|^2}{a_M^2}, \quad W_{g,k}(Y_g) = W_k(Y_g), \quad \mu_{g,k}(Y_g) = W_{g,k,Y_g}. \quad (17.62)$$

These equalities are the declared identity bridge, not a postulate about gravity at the pre-geometric medium level.

- (II) **Conditional static functional.** The supplied quasistatic weak-field action is

$$\Gamma_{\text{gal}}^{(k)} = - \int dt d^3x \left[\rho_b \Phi_{\text{gal}} + \frac{a_M^2}{8\pi G_N} W_{g,k}(Y_g) \right]. \quad (17.63)$$

It is a complete variational owner of the static HRC field equation once supplied. It is not yet a covariant metric completion and does not derive the matter or photon vertices from P1–P6.

- (III) **Deep-branch scaling.** Both registered responses satisfy

$$W_{g,k}(Y_g) = \frac{2}{3} Y_g^{3/2} + \mathcal{O}(Y_g^{5/2}), \quad \mu_{g,k}(Y_g) = \sqrt{Y_g} + \mathcal{O}(Y_g^{3/2}). \quad (17.64)$$

The exponent is exact inside either stated HRC model. Its ownership by the live action and the origin of a_M remain Open.

- (IV) **Euler–Lagrange equation.** Varying Eq. (17.63) gives

$$\nabla \cdot [\mu_{g,k}(Y_g) \nabla \Phi_{\text{gal}}] = 4\pi G_N \rho_b. \quad (17.65)$$

For $x_g = |\nabla\Phi_{\text{gal}}|/a_M$,

$$\mu_{g,k}(x_g^2) \rightarrow x_g \quad (x_g \ll 1), \quad \mu_{g,k}(x_g^2) \rightarrow 1 \quad (x_g \gg 1). \quad (17.66)$$

- (V) **Spherical reduction and RAR asymptotes.** For an isolated spherical source, integration of the divergence equation gives

$$\mu_{g,k}(g^2/a_M^2)g = g_N, \quad g_N = \frac{G_N M_b(< r)}{r^2}. \quad (17.67)$$

Hence $g \simeq g_N$ in the Newtonian branch and $g^2 \simeq g_N a_M$ in the deep branch. This is the symmetry-reduced equation used in the BTFR derivation below.

- (VI) **Nonspherical guard.** For a disk, the algebraic relation need not hold pointwise because

$$\mu_{g,k}g = g_N + \nabla \times H. \quad (17.68)$$

The full density, scale height, boundary data and external field are needed to determine H . The current radius-by-radius SPARC solver is therefore a Level-C response-shape diagnostic rather than the field-level disk solution.

- (VII) **Scale and fit layers.** The present-epoch value $a_{M0} = cH_0/(2\pi)$ is an explicit matching condition. A globally fitted a_M , any per-object effective value and any environment/history function are data-interface quantities until the medium-to-gravity map and scale owner are derived. No redshift law follows merely from the present matching.

Baryonic Tully–Fisher relation as a conditional HRC corollary

Layered status: the slope 4 is Level A algebra inside the supplied static HRC functional. It is a Level B conditional corollary of the declared identity gravity bridge and the isolated spherical deep-branch reduction. Ownership of that bridge and of a_M by P1–P6 remains Open; comparison with galaxy catalogues is Level C. No environment or redshift law is implied.

For either HRC-0 or HRC-3, $\mu_{g,k}(g^2/a_M^2) \sim g/a_M$ as $g/a_M \rightarrow 0$. In spherical symmetry Eq. (17.59) therefore gives

$$g^2 \simeq g_N a_M, \quad \boxed{v_\infty^4 = G_N M_b a_M}. \quad (17.69)$$

The exponent is not fitted. The observed normalisation, intrinsic scatter, and finite-disk corrections do not by themselves identify a microscopic Φ -medium owner for a_M .

The frozen HRC registry also evaluates an internal outer-tail proxy. For the outermost 25% of usable points (at least three), it uses the inverse-variance weighted mean observed velocity and separately defines $R_{\text{tail}} = \text{median } R_i$ and $g_{N,\text{tail}} = \text{median } g_{N,i}$. The plotted proxy is $M_{\text{bar,eff}} = R_{\text{tail}}^2 g_{N,\text{tail}} / G_N$. It is constructed from the SPARC decomposition entering the fit and is therefore not an independent catalogue mass. Figure 12 shows the conditional slope-four lines and the distribution of independently fitted scales. For 165 fixed- M/L fits the median ratios to the matched scale are

$$\frac{a_{M,\text{med}}^{(0)}}{a_{M0}} = 1.222, \quad \frac{a_{M,\text{med}}^{(3)}}{a_{M0}} = 0.921, \quad (17.70)$$

with 16–84 percentile intervals $[0.592, 2.460]$ and $[0.408, 2.112]$, respectively. Their breadth is not a derived environment effect: it also contains distance, inclination, stellar-population, bulge, noncircular-motion and algebraic-disk systematics.

Orientation-sector separation. The live scalar-only reduction supplies no independent orientation triplet in the galactic action. A future covariant orientation term could modify the response, but its source, sign and coefficient are not presently available. No such term is included in the HRC diagnostics below.

HRC algebraic SPARC diagnostic

For a symmetry-reduced field, the HRC relation used in the frozen diagnostic is implicit:

$$g_N(R) = \mu_{g,k} \left(\frac{g^2(R)}{a_M^2} \right) g(R), \quad k \in \{0, 3\}, \quad (17.71)$$

where $\mu_{g,0}$ and $\mu_{g,3}$ are obtained from Eqs. (17.43) and (17.46). This numerical registry uses the explicit identity bridge Eq. (17.57); a nontrivial $s = f(Y_g)$ would define a different gravitational response even if the internal ERP response were unchanged. The baryonic input is

$$g_N(R) = \frac{V_{\text{gas}} |V_{\text{gas}}| + \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2}{R}. \quad (17.72)$$

The signed gas term is essential. The present calculation applies the local relation (17.71) radius by radius. For a generic disk this is an approximation to the nonlinear three-dimensional equation:

$$\mu g = g_N + \nabla \times H. \quad (17.73)$$

The available SPARC table contains mid-plane component velocities rather than the full three-dimensional source, vertical scale-height priors and boundary data. It therefore cannot determine the curl field or supply a final disk verdict.

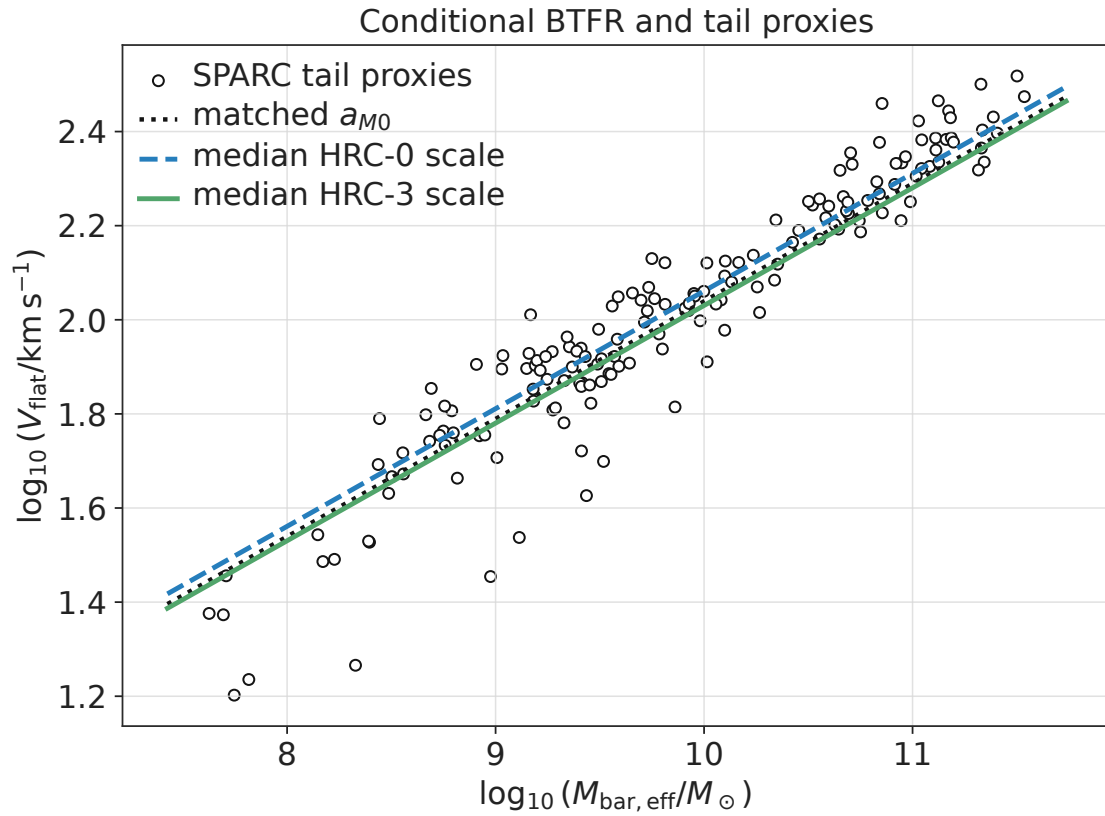


Figure 12: HRC-only BTFR diagnostic. This page shows SPARC outer-tail baryonic-mass proxies with the conditional slope-four relations evaluated at the matched scale and the two fixed- M/L median scales. Because the mass proxy comes from the same baryonic decomposition, this is an internal pipeline check rather than an independent catalogue BTFR likelihood. The per-galaxy fitted-scale distributions follow as a separate full-width continuation. Colour is redundant with line style, marker shape, luminance contrast and direct labels; no texture distinguishes the HRC models.

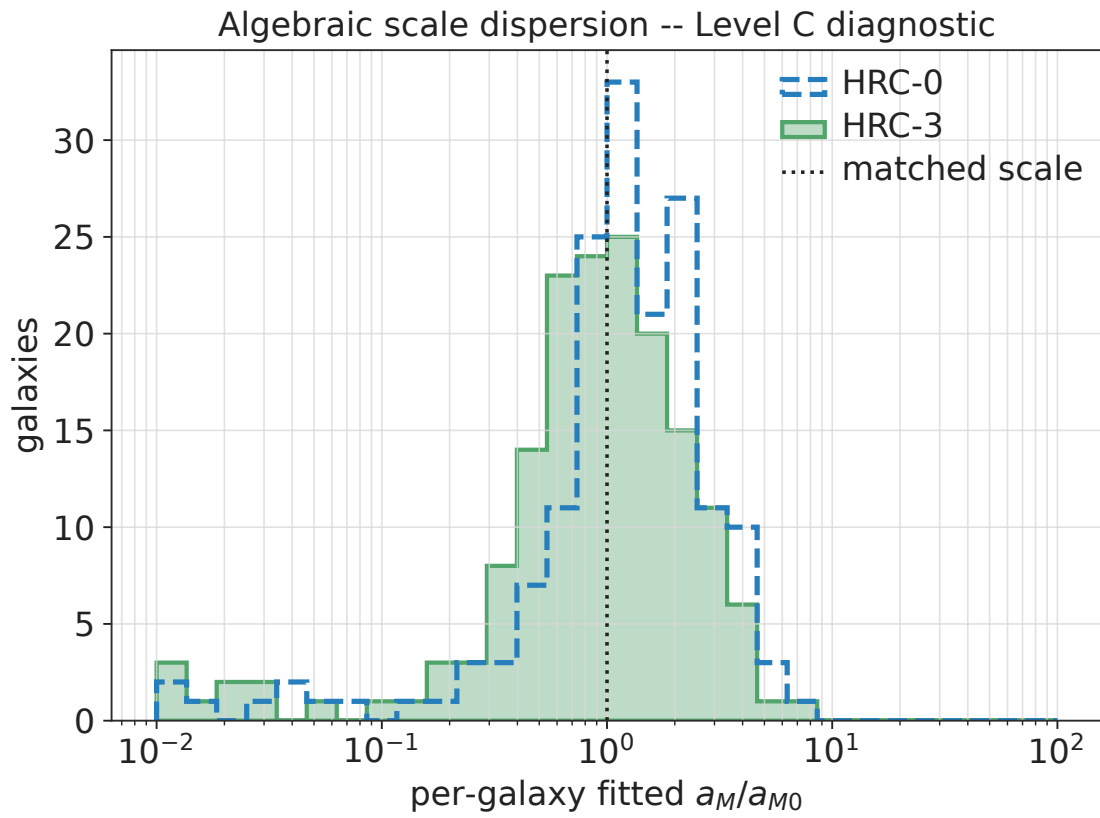


Figure 12, continued. Per-galaxy fitted-scale dispersion. Its breadth also contains distance, inclination, stellar-population, bulge, non-circular-motion and algebraic-disk systematics.

Frozen protocol. The current Level-C diagnostic contains 165 SPARC galaxies, signed gas, five whole-galaxy held-out folds repeated for five seeds, and two declared stellar- M/L modes. The fixed- M/L mode uses 3343 points per seed. The common- M/L transfer mode fixes its validity mask at the lower bound $\Upsilon_d = 0.3$ and therefore uses 3342 points per seed. Only HRC-0 and HRC-3 are evaluated in the figures and table below. The results are:

Table 64: HRC-only algebraic SPARC diagnostics. These are absolute held-out statistics of the local relation (17.71), not a full survey likelihood.

Response	Fixed- M/L $\langle\chi^2\rangle$	Fixed- M/L SD	Common- M/L $\langle\chi^2\rangle$	Fixed- M/L fitted $\langle a_M \rangle$ / m s^{-2}
HRC-0	117598.72	3645.86	109439.95	1.75×10^{-10}
HRC-3	99501.19	2967.18	103763.46	1.27×10^{-10}

The corresponding fold-averaged training values of a_M in the common- M/L transfer mode are $1.2629 \times 10^{-10} \text{ m s}^{-2}$ for HRC-0 and $1.1170 \times 10^{-10} \text{ m s}^{-2}$ for HRC-3. These values are reported separately because they belong to a different stellar- M/L protocol from the final column of the table. Across the five frozen whole-galaxy fold seeds, the population standard deviations of the common- M/L held-out χ^2 totals are 3194.14 for HRC-0 and 4537.24 for HRC-3.

The absolute scale $\chi^2/\text{point} \sim 30$ is a warning, not evidence of a precision fit. The current likelihood omits covariance, intrinsic scatter, distance and inclination marginalisation, hierarchical stellar populations, the external-field response and the full disk equation. HRC-3 was formulated after the crossover residual was inspected and is not an independent prediction.

As a separate joint nuisance test, the baseline varies only $10^{-2} \leq a_M/a_{M0} \leq 10^2$ per galaxy at fixed $\Upsilon_d = 0.5$ and $\Upsilon_b = 0.7$. The comparison jointly varies that same scale and an independent disk ratio $0.1 \leq \Upsilon_d \leq 2.5$ per galaxy while keeping $\Upsilon_b = 0.7$. It therefore probes the a_M -disk- M/L degeneracy rather than a free- M/L -only model. The median reduced χ^2 changes from 2.449 to 1.293 for HRC-0 and from 2.327 to 1.273 for HRC-3. The corresponding median scale ratios move from 1.222 to 0.912 and from 0.921 to 0.812. This improvement measures the expected a_M - M/L degeneracy, not additional support for a non-universal physical scale. A population prior and joint distance/inclination inference are still required. The fitted tails are boundary-censored: for HRC-0/HRC-3 the fixed scale reaches its lower bound in 2/3 galaxies and the free- M/L scale in 6/6; Υ_d reaches the lower bound in 21/19 and the upper bound in 14/9 galaxies. These objects are retained as stress diagnostics, not used to infer an uncensored physical scale distribution.

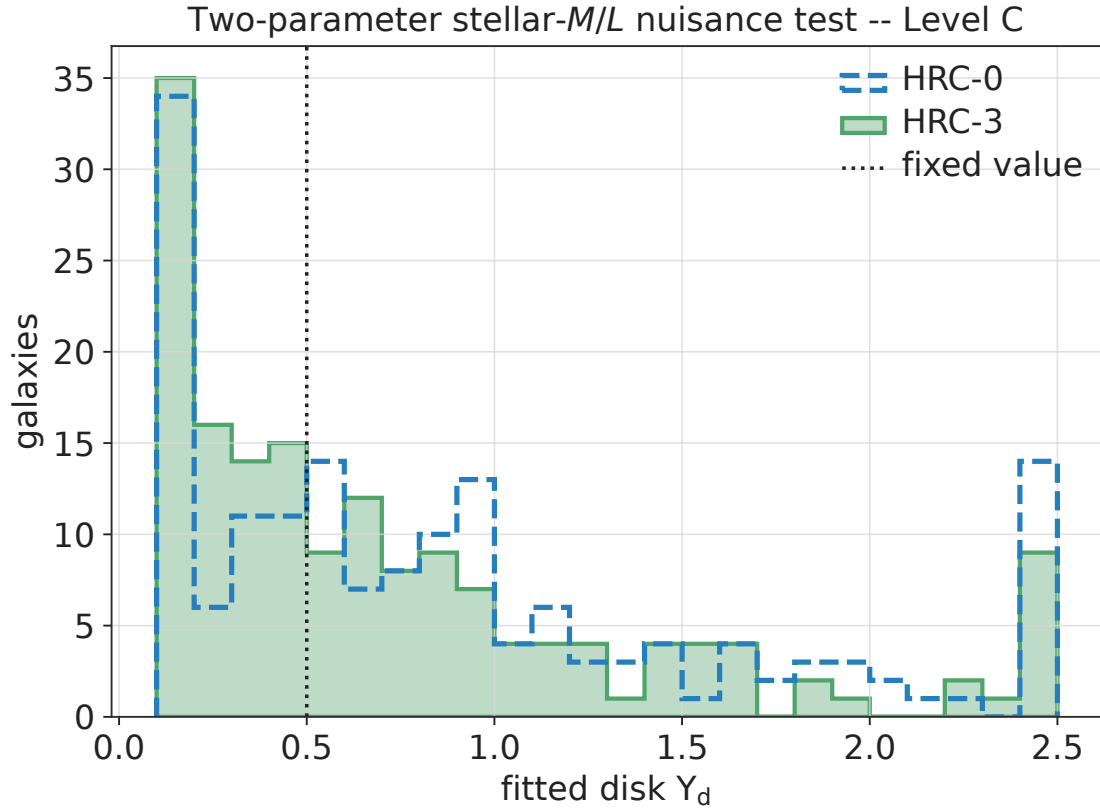


Figure 13: HRC-only stellar- M/L sensitivity. The first two panels compare fixed- and free- M/L fitted scales; the third gives the fitted disk mass-to-light distributions. The diagonal and fixed value are shown by dotted black lines. Boundary accumulation near $Y_d = 0.1$ or 2.5 , and at the disclosed $a_M/a_{M0} = 0.01$ lower bound, is a nuisance-model warning; distribution tails are censored. Complementary full-width panels are retained in Appendix BN.

Reading the response-law panel. The preceding nuisance test concerns fitted stellar inputs and the declared acceleration scale. Figure 14 now isolates the response functions themselves, before the external rotation-curve examples are introduced; no fit protocol or claim status changes at this transition.

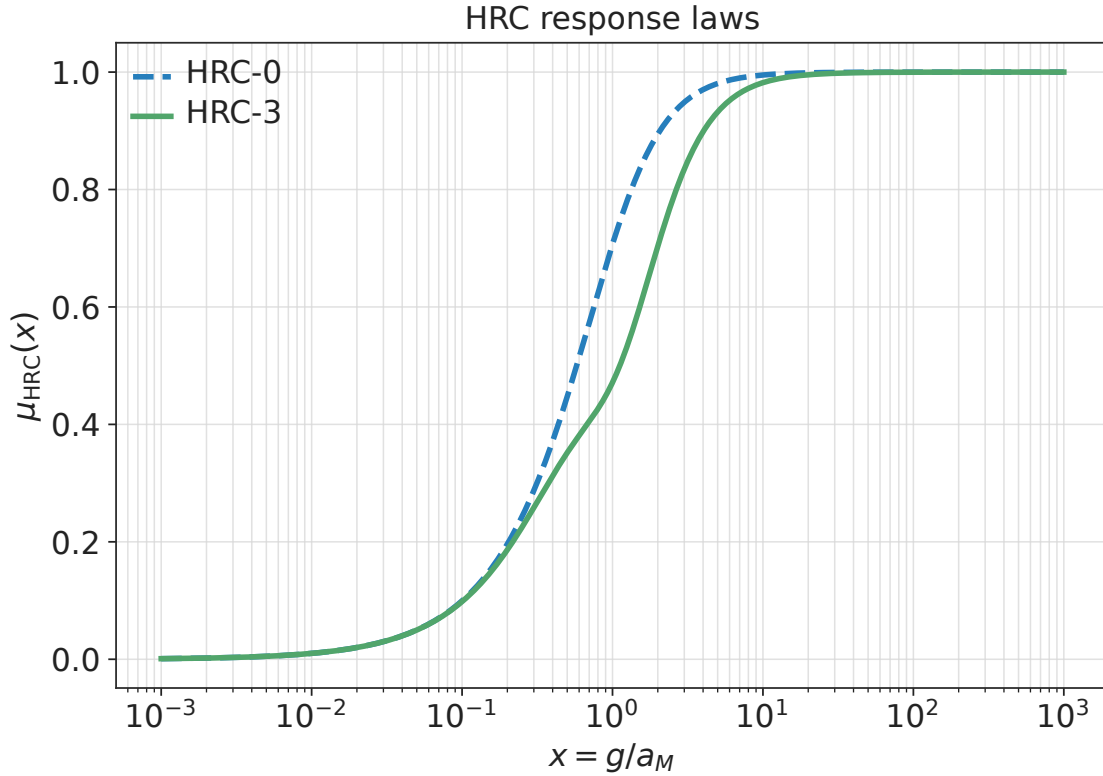


Figure 14: HRC-only response-law diagnostic. This page compares the exact HRC-0 response in the augmented inverse-DtN/recruitment model with the conditional HRC-3 response. The held-out deep/crossover/Newtonian χ^2 -regime census is a separate full-width atlas panel. Colour is redundant with line style, marker shape, border contrast and direct labels, so the figure remains legible in grayscale without decorative hatching.

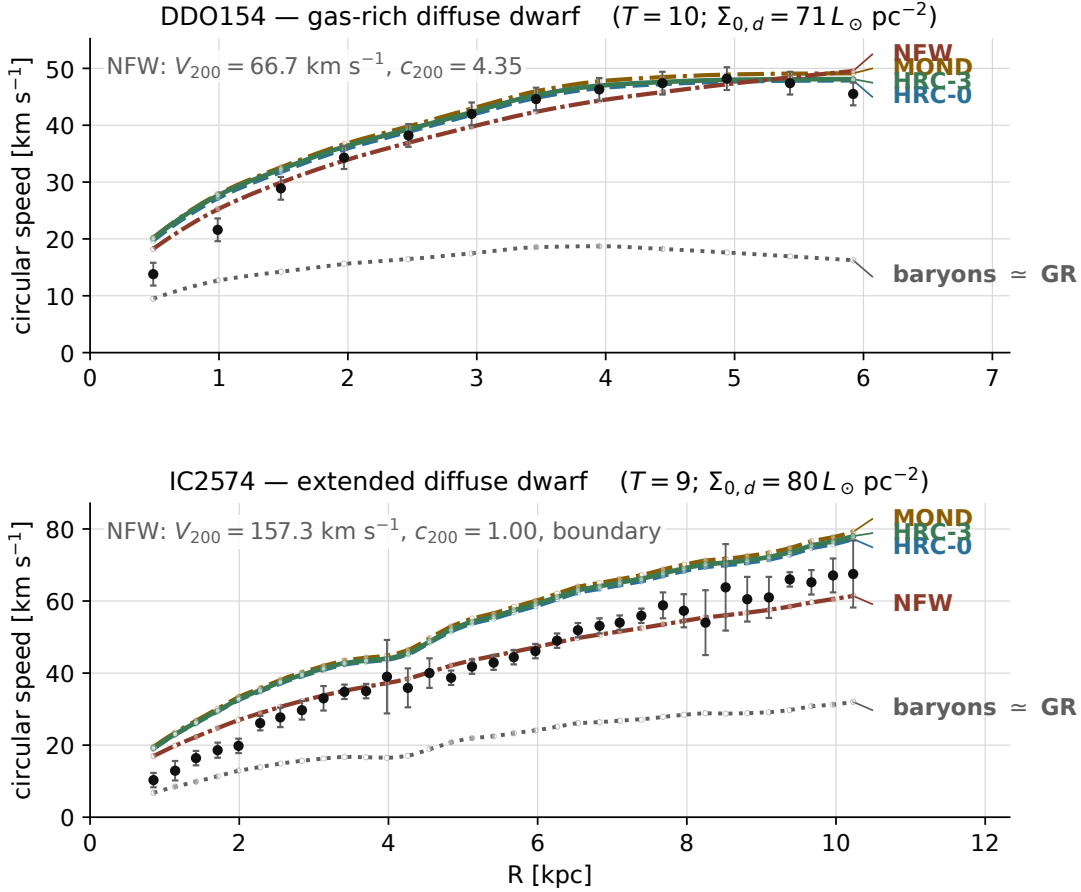
External model-comparison guard. Figures 15–17 hold the baryonic decomposition and stellar mass-to-light ratios fixed while separating a fixed-input response comparison from a two-parameter Navarro–Frenk–White (NFW) halo fit [116]. HRC-0 is not shape-distinct from the standard MOND interpolation used here: at equal acceleration scale the two curves coincide identically. Conversely, an NFW fit is not a unique Λ CDM curve; it becomes a population-level Λ CDM prediction only after halo-population, concentration–mass, abundance, feedback and nuisance priors are supplied. The six systems were fixed before the rendering pass and do not establish survey-level model selection. Every panel uses $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$, signed gas and a 2 km s^{-1} error floor. Colour is redundant with line style, markers and direct labels; no hatch is used. The comparison remains a local algebraic disk diagnostic, not a finite-thickness field solution or hierarchical likelihood. SPARC measurements are shown only as unconnected error-bar markers. Source model ordinates are retained as faint, unconnected nodes. Smooth traces are residual-gated, shape-preserving display approximations through curvature-regularised display nodes, with a maximum node-displacement tolerance constrained to $0.10\text{--}0.25 \text{ km s}^{-1}$. All fit and diagnostic metrics use the original nodes; no continuous physical owner is implied by a display trace.

SPARC comparison I: diffuse systems

Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.

Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.

Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.



Λ CDM-motivated free NFW fit; not a unique Λ CDM prediction.
All panels: $Y_d = 0.5$, $Y_b = 0.7$, signed gas and a 2 km s^{-1} error floor.

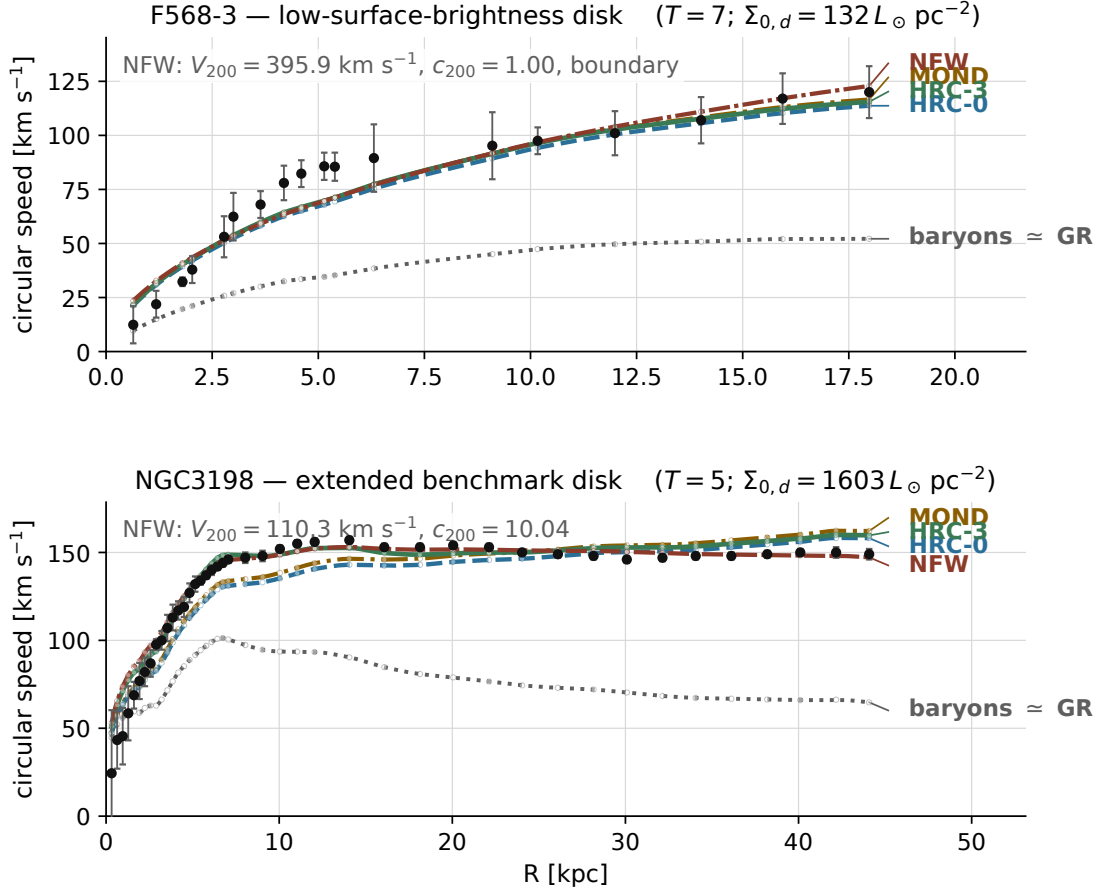
Figure 15: Representative SPARC external-model comparison, set I. DDO154 and IC2574 sample diffuse and low-surface-brightness regimes. Black points are SPARC velocities; directly labelled curves show baryon-only weak-field GR, fixed-scale MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW halo. HRC-0 and MOND-standard have the same response shape and differ here only through their declared scales. The HRC matching input is $a_M = cH_0/(2\pi)$ at the externally declared $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$; its coefficient and ECT physical owner remain Open. LambdaCDM-motivated free NFW fit; fitted to the displayed rotation curve; not a unique LambdaCDM prediction. Fixed baryonic inputs and boundary-hit flags are printed in the figure. This is a Level-C algebraic diagnostic, not a population model-selection result. The full comparison atlas in Appendix BN applies the same externally guarded comparison to held-out, stratified-gallery and post-hoc stress examples.

SPARC comparison II: low-surface-brightness and benchmark disks

Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.

Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.

Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.



Λ CDM-motivated free NFW fit; not a unique Λ CDM prediction.
All panels: $Y_d = 0.5$, $Y_b = 0.7$, signed gas and a 2 km s^{-1} error floor.

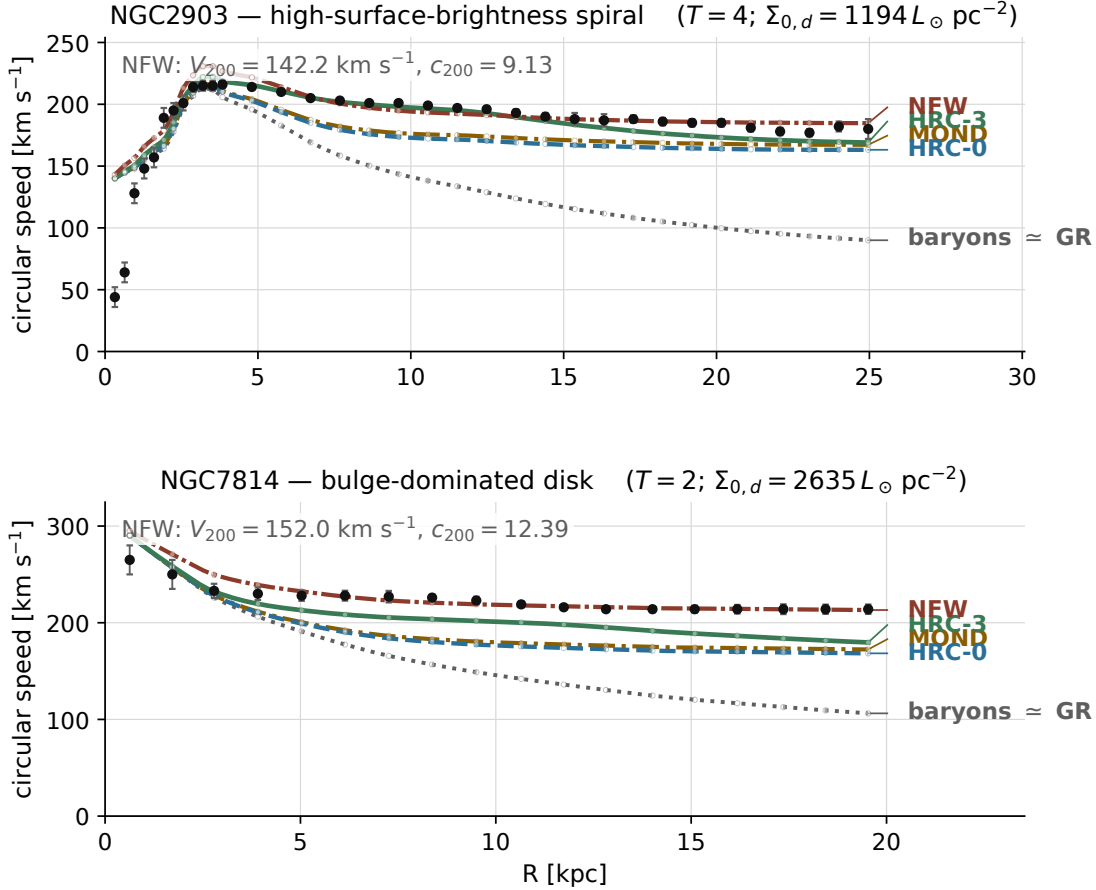
Figure 16: Representative SPARC external-model comparison, set II. F568-3 and NGC3198 provide low-surface-brightness and extended benchmark cases. Black points and directly labelled curves use the same fixed baryonic inputs as set I: baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW halo. The HRC matching input is $a_M = cH_0/(2\pi)$ at the externally declared $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$; its coefficient and ECT physical owner remain Open. Printed V_{200} and c_{200} are fitted separately. LambdaCDM-motivated free NFW fit; fitted to the displayed rotation curve; not a unique LambdaCDM prediction. The residual patterns expose where the fixed-scale HRC/MOND curves or fitted halo are stressed. This is a Level-C six-example diagnostic, not survey-level model selection.

SPARC comparison III: bright and bulge-dominated disks

Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.

Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.

Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.



Λ CDM-motivated free NFW fit; not a unique Λ CDM prediction.
All panels: $Y_d = 0.5$, $Y_b = 0.7$, signed gas and a 2 km s^{-1} error floor.

Figure 17: Representative SPARC external-model comparison, set III. NGC2903 and NGC7814 provide high-surface-brightness and bulge-dominated cases. Black points and directly labelled curves use the same fixed baryonic inputs as sets I–II: baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW halo. The HRC matching input is $a_M = cH_0/(2\pi)$ at the externally declared $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$; its coefficient and ECT physical owner remain Open. The printed V_{200} and c_{200} are fitted separately. The NFW curve is a Λ CDM-motivated free fit to the displayed rotation curve, not a unique Λ CDM prediction. This is a Level-C six-example diagnostic, not survey-level model selection.

Reader guide. The three preceding pages form one fixed-baryonic external-comparison suite, following the two locally placed nuisance and response-law checks above. They remain Level-C diagnostics, not population-level model-selection results. Appendix BN supplies a full-width comparison atlas with baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW benchmark for every displayed galaxy; the NFW curves are Λ CDM-motivated fits, not unique Λ CDM predictions.

Scale status. The present matching value is

$$a_{M0} = \frac{cH_0}{2\pi} \approx 1.08 \times 10^{-10} \text{ m s}^{-2} \quad (H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}), \quad (17.74)$$

but the coefficient is not derived. A possible environment/history response is only a test family,

$$a_{M,\text{eff}} = a_{M0} \Xi(\rho_{\text{env}}, \bar{q}_{\text{env}}, \text{history}, \dots), \quad (17.75)$$

with variables, sign and amplitude Open. No Milky-Way-specific HRC result is implied by this ansatz.

Milky Way sensitivity test

The representative HRC sensitivity calculation uses 14–25 kpc velocity points representative of the Eilers et al. reconstruction [117] and a declared exponential disk ($M_d = 5.0 \times 10^{10} M_\odot$, $R_d = 2.5$ kpc). It is a sensitivity calculation, not a modern Milky-Way posterior: gas, bulge, halo geometry, correlated systematics and baryonic-model uncertainty are absent. The algebraic best fits are

	a_M/a_{M0}	χ_{red}^2
HRC-0	2.798	1.813
HRC-3	1.703	2.287

(17.76)

for this fixed disk. The fitted displacement from a_{M0} is therefore a stress on the simplified baryonic model/universal-scale pair, not evidence for a measured environment law.

The readable HRC-only Milky-Way sensitivity curve is retained as provenance in Appendix BN; it is not a main-text external-model comparison.

17.4 Radial Acceleration Relation in the HRC diagnostic

The observed and baryonic accelerations are

$$g_{\text{obs}} = \frac{V_{\text{obs}}^2}{R}, \quad g_N = \frac{V_{\text{bar}}^2}{R}, \quad (17.77)$$

and the symmetry-reduced conditional HRC response law is

$$g_N = \mu_{g,k}(g_{\text{obs}}^2/a_M^2)g_{\text{obs}}, \quad k \in \{0, 3\}. \quad (17.78)$$

As shown in Figure 18, the deep and Newtonian asymptotes are fixed by the declared HRC laws, whereas the crossover distinguishes HRC-0 from HRC-3. A claim of an ECT RAR prediction requires both the missing medium-to-gravity bridge and a prespecified full-field calculation.

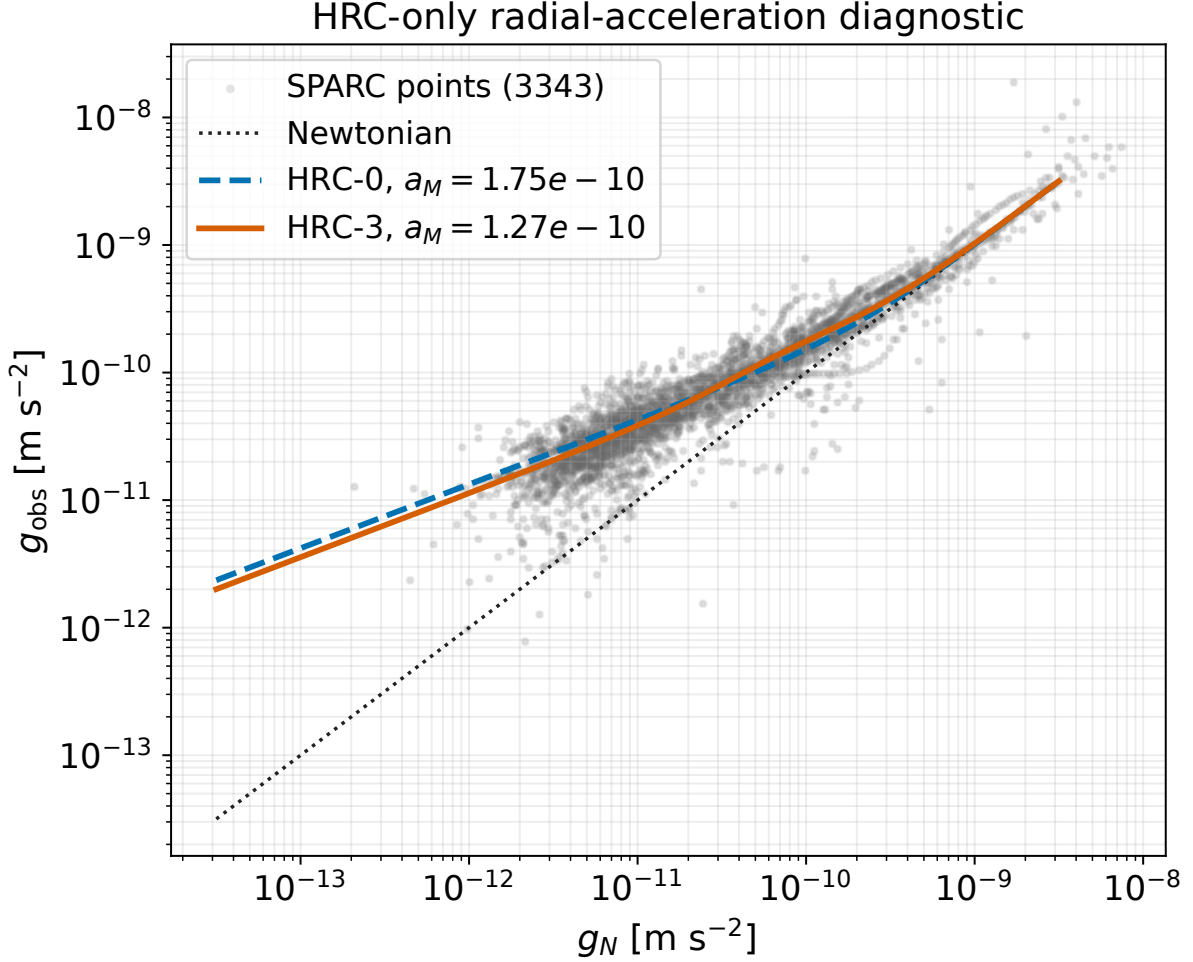


Figure 18: HRC-only RAR diagnostic. Grey points are the 3343 SPARC entries; the HRC curves use the mean held-out fitted scales of the frozen algebraic pipeline. The plot is a Level-C diagnostic and does not include the disk curl field, covariance or hierarchical nuisance model. Colour is redundant with line style.

17.5 Dark-matter-deficient ultra-diffuse galaxies as an HRC stress test

For a spherical central-value proxy, define $\mathcal{R}_{\text{dyn}} = g_{\text{obs}}/g_N$. The HRC equation admits a finite positive- a_M enhancement solution only when $\mathcal{R}_{\text{dyn}} > 1$; equality is the Newtonian boundary $a_M = 0$. For HRC-0 the required scale is obtained analytically:

$$\frac{a_M^{(0)}}{g_{\text{obs}}} = \sqrt{\mathcal{R}_{\text{dyn}}^2 - 1}. \quad (17.79)$$

For HRC-3, x_3 is the unique positive solution of $\mu_{g,3}(x_3^2) = 1/\mathcal{R}_{\text{dyn}}$ and $a_M^{(3)} = g_{\text{obs}}/x_3$.

Using $R_{1/2} = 4R_e/3$, enclosed stellar mass $M_\star/2$ and $g_{\text{obs}} = 3\sigma^2/R_{1/2}$ gives the central proxy values below. The pressure-supported inputs are drawn from the DF2/DF4, FCC 224, NGC 5846-UDG1 and Dragonfly 44 literature [118, 119, 120, 121, 122, 123, 124, 125]; heterogeneous measurement procedures mean that the rows are not a common-confidence sample. AGC 114905 is a gas-rich rotator [126] and is excluded from the spherical proxy.

Table 65: HRC-only central Wolf-proxy diagnostic. Values are not a Jeans posterior, and AGC 114905 requires a disk solver.

Object	$\mathcal{R}_{\text{dyn}}$	$a_M^{(0)}/a_{M0}$	$a_M^{(3)}/a_{M0}$	Domain/status
NGC 1052–DF4	0.787	—	—	No positive HRC enhancement solution at central inputs
FCC 224	1.229	0.0155	0.00780	Strong suppression of the matched scale required
NGC 1052–DF2	2.328–6.988	0.0476–0.470	0.0278–0.457	Dispersion bracket remains ambiguous
NGC 5846-UDG1	10.262	0.947	0.935	Close to the matched HRC scale
Dragonfly 44	31.734	4.951	4.944	Scale substantially above the matched value

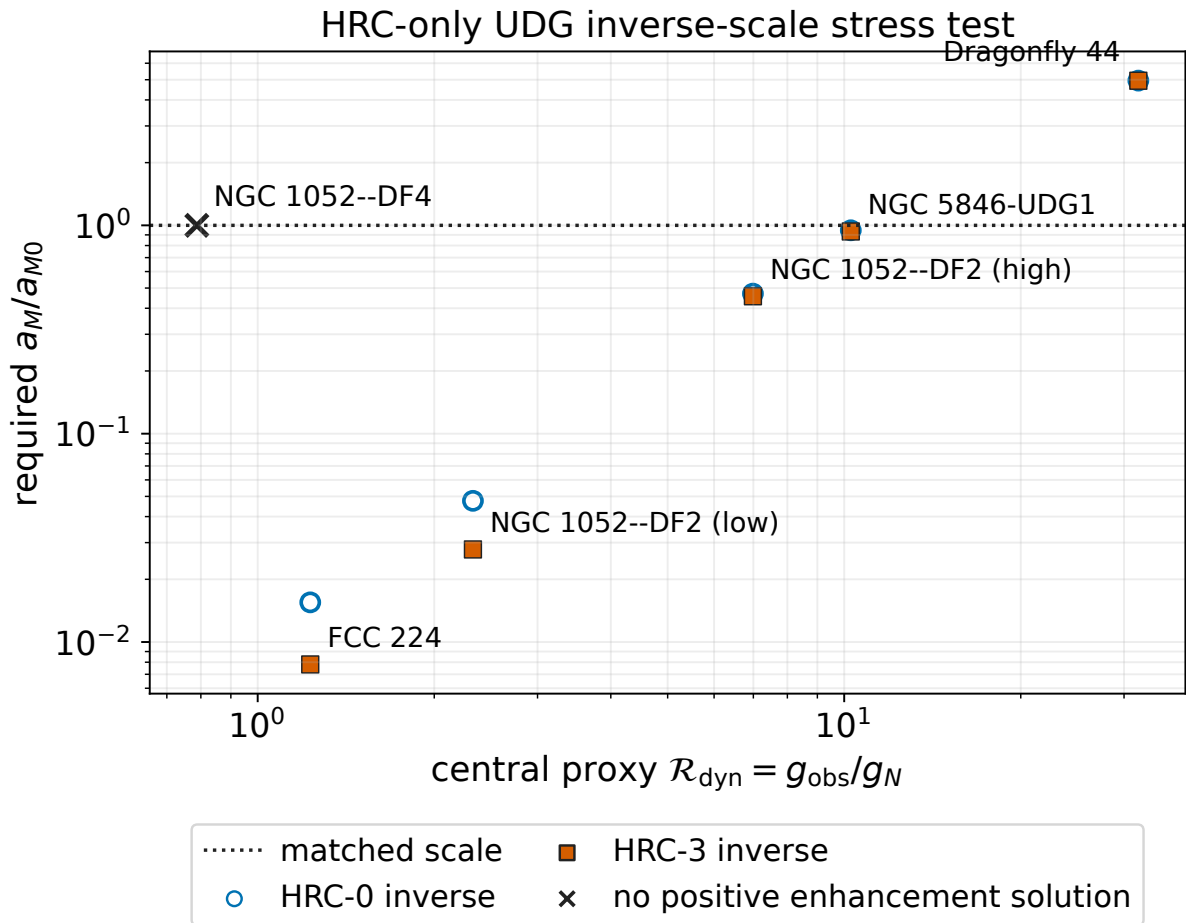


Figure 19: HRC-only inverse-scale UDG stress test. The two DF2 points are the low- and high-dispersion endpoints; the cross marks a central proxy outside the positive-enhancement domain. The matched-scale line is not an environment fit. HRC-0 and HRC-3 nearly coincide for the largest $\mathcal{R}_{\text{dyn}}$, while they differ appreciably near the crossover.

Figure 19 visualises the same central-proxy stress test: similar diffuse systems require radically different effective response scales, while DF4 is outside the positive-enhancement domain at its central proxy inputs. It does not derive an environment law, explain the incidence rate, or supersede a non-spherical

Jeans analysis. A successful completion must predict the additional variable or history dependence rather than fitting a separate a_M to each object.

Finite completion tests. The galactic numerical sector is not complete until it supplies: (i) a density-aware axisymmetric HRC solver; (ii) the external-field response in the same operator; (iii) distance, inclination, covariance, intrinsic-scatter and hierarchical- M/L marginalisation; (iv) a prespecified independent galaxy test; and (v) the covariant metric needed before any lensing, cluster, PPN or cosmological use.

17.6 Summary and status of galactic and cluster-scale phenomenology

This chapter records an HRC galactic diagnostic and a separate exact local-map cluster theorem. Its current result ledger is:

1. The amplitude coordinate ϕ_u and the conditional homogeneous scalar response are available inside the Level-B scalar ansatz. No AU–kpc–Mpc screening hierarchy is derived; canonical mass, body charge, and PPN reachability remain Open.
2. The strictly increasing-map order/argmax theorem is Level A inside the supplied local map: it preserves the complete ordering, the possibly empty set-valued argmax and every attained maximum of the supplied total projected baryon field. No synthetic replay or amplitude result is retained. The physical cluster map, collision dynamics and general causal merger kernel remain Open.
3. The 165-galaxy SPARC statistics and associated figures have been recomputed with HRC-0 and HRC-3. They are Level-C algebraic diagnostics and not a full field-level likelihood.
4. HRC-0 and HRC-3 share the deep response $\mu \sim g/a_M$, so the BTFR slope 4 follows conditionally from the separate static gravity bridge. The normalisation $a_{M0} = cH_0/(2\pi)$ remains a matching condition.
5. HRC-0 is exact in the augmented inverse-DtN/recruitment model and HRC-3 is conditional on the supplied three-channel matrix owner. Their same-pipeline comparison is Level C; full finite-disk fields and the publication cascade remain uncomputed. The force-dimensionality curve uses the algebraically identical HRC-0 shape but is not retroactively promoted to a microscopic ECT derivation. Residual scatter has no derived environment/history component, sign, or amplitude.

The HRC-only fit arithmetic is a Level C algebraic diagnostic. ERP- Φ is a new constitutive proposal; the HRC responses have their model-conditional statuses stated in Sec. 17.3. The gravity bridge, environment law Ξ , action/source matching and scalar screening remain Open. A known stress test of the HRC algebraic diagnostic is the class of dark-matter-deficient ultra-diffuse galaxies (NGC 1052–DF4, FCC 224); these are listed as entry ND-13 in Table 72 and discussed diagnostically in Appendix AQ, and the HRC-only central-proxy inversions are reported as stress diagnostics, not as field-level explanations. For the cosmological sector of the ordered-branch programme, see the summary in Section 16.9.

18 Mass-discrepancy and late-time-acceleration phenomenology in ECT

Status: mixed. The scalar–tensor ansatz and late-time background closure are Level B at their declared scope. The galactic HRC-only algebraic outputs are Level-C diagnostics. ERP- Φ /HRC is the active candidate programme: exact or conditional in its stated augmented response models, but its P1–P6 and gravity ownership remains Open. Additional condensate dark sectors and the cosmological fraction budget remain Open.

This section separates galactic mass-discrepancy fits, possible additional collective excitations, and late-time background acceleration. The absence of a particle halo is a property of the conditional HRC algebraic diagnostic, not an ECT theorem about particle content.

18.1 Scope and dark-sector logic

The present manuscript separates three model questions without claiming a derived common ontology. First, the separately supplied conditional HRC gravity functional G contains no particle-halo term and is evaluated against selected benchmarks; its scalar/ECT matching and fundamental particle content are Open. Second, the late-time scalar-background ansatz parametrises accelerated expansion; its microscopic origin, uniqueness, and cosmological matter budget are Open. Third, possible particle, collective, defect, or topological sectors are research candidates with no derived abundance or assigned cosmological role.

P4, P6, and the branch architecture do not by themselves identify the galactic fit with ϕ_u , derive a dark-energy source, or exclude hidden components. The separation below is therefore an organisational status ledger, not a postulate-grounded derivation of the dark sector.

Table 66: Three distinct dark-sector questions in the present ECT programme.

Question	Present statement	Status	Next discriminating gate
Galactic mass-discrepancy response	HRC-0/HRC-3 are active augmented-model candidates with current algebraic no-halo diagnostics; the full field solve remains Open	A/B/C by layer; live ECT Open	HRC disk likelihood; derive the medium, source and metric map
Late-time acceleration	The two-slope action supplies a regular proof of class; a trace-free completion supplies independent shift covariance through C_I	A in supplied actions / C benchmark / ECT Open	Derive the Φ -medium action and state, physical metric, perturbations, and joint likelihood
Additional sectors	Particle, collective, defect, and topological candidates have no derived abundance or assigned dark-sector role	Open	Spectrum, stability, production, abundance, coupling, lensing, and growth

18.2 HRC no-halo algebraic diagnostic and Open particle content

HRC no-halo diagnostic and Open particle content

The conditional HRC static functional contains no particle-halo term and is evaluated on selected flat-curve, RAR, and BTFR benchmarks. Its absolute held-out likelihood is poor; this is a property of the specified diagnostic, not a derivation that ECT selects the functional and not evidence that particle dark matter is absent. The amplitude coordinate is motivated by the Level-B scalar ansatz, whereas the AQUAL-type functional, its source/force map, the origin of a_M , and the environment law Ξ have not been derived from that ansatz.

Status and discriminator. The current result is therefore a Level-C no-halo fit benchmark with Open ECT matching and Open fundamental particle content. A particle species shown independently to explain the galactic discrepancy would disfavor this minimal no-halo fit as a complete description, not the ECT framework. Conversely, null direct-detection searches do not confirm the closure.

Numerical anchors from Section 17. Inside the declared functional, the BTFR slope 4 follows by exact conditional algebra. The baseline $a_{M0} = cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m s}^{-2}$ is a matched benchmark, the observed RAR scatter is $\sigma \approx 0.13 \text{ dex}$ [127, 128], and the SPARC pipeline fits one $a_{M,\text{eff}}$ per galaxy for 165 galaxies. These fit facts do not supply the missing scalar-to-galactic derivation or a cosmological particle-fraction prediction.

Collective excitations and topological sectors

Status: Level B/C structural possibilities. Relic abundance, stability, and production history not yet computed from bare P3.

The ordered branch supports a constrained soft orientation sector. Its strict scalar map has one longitudinal coordinate χ at linear order; one physical mode is a Level-B reduced-EFT interpretation. Its dark-sector role, abundance, and stability are not established. The HRC algebraic diagnostic does not use this mode or a particle-halo term, but that fit property neither requires nor excludes particle or collective dark sectors.

The topological completeness of the Φ -medium (S10) together with the coherent branch (BR1) admits the possibility that part of the dark sector is tied to the smooth coherent-branch configurations: winding sectors, vortex-like structures, or topological configurations (see Section 8.10 for the full classification). Their definition and dynamics depend on winding numbers and defect stability beyond the coherent approximation. They are noted here as a conceptual bridge to the Quantum Sector (Section 34); their detailed treatment is deferred.

No physical large-scale mass prediction. For any supplied relativistic mass parameter one may define a Compton scale $\xi_G = \hbar/(m_G c)$, but ECT derives neither a physical soft-mode mass operator nor a value of m_G . The separately named $m_{\chi,E} = \zeta_H H_{0,E}$ point is retained only as a nonpredictive C/Open orientation benchmark, with its quoted reduced-Compton proxy. It supplies no physical or predicted Hubble-scale mass, coincidence explanation, relic-abundance estimate or dark-sector inference.

The homotopy group $\pi_3(S^3) = \mathbb{Z}$ is structurally fixed for the supplied orientation target. Its identification with the complete physical configuration space, and whether corresponding admissible configurations are dynamically stabilised as particle-like or long-lived dark-sector objects, remain Open.

Ultra-light modes and topological condensate configurations remain candidate additional excitations. They are not terms in the conditional HRC algebraic diagnostic, but that omission neither excludes nor constrains their possible abundance. Any cluster or cosmological role remains Open.

Direct-detection discriminator and cosmological-fraction status

The Level-C HRC algebraic diagnostic contains no particle-halo term and is evaluated on selected rotation-curve benchmarks with poor absolute held-out likelihood. This neither derives that ECT selects the closure nor excludes particle dark matter, and it predicts no absence of a laboratory particle signal. A robust particle detection shown independently to explain the galactic discrepancy would disfavor the minimal no-halo fit as a complete description, not ECT as a whole. Conversely, a null direct-detection result does not confirm this closure.

ECT does not yet derive Ω_b , Ω_{DM} , or Ω_Λ from first principles. The galactic no-halo fit, the late-time background closure, and possible condensate or particle excitations must therefore keep separate status labels. Their abundances, stability, mutual matching, and cosmological roles remain Open.

18.3 Late-time scalar stress: conditional lemma and open acceleration owner

The amplitude coordinate in this subsection is $\phi = \phi_u = \beta_\phi^{-1} \ln(u/u_\infty)$ on an ordered patch $u > 0$, not the distinct anisotropy coordinate $\psi_K = \ln(\Delta_K/\Delta_{K,vac})$. The map between those coordinates is Open. Inside a supplied scalar completion, Eqs. (16.1)–(16.2) give the exact bare-ratio lemma

$$\omega > 0, \quad K_\phi^{\text{phys}} + U > 0 \quad \implies \quad w_\phi^{\text{bare}} = \frac{K_\phi^{\text{phys}} - U}{K_\phi^{\text{phys}} + U} \geq -1. \quad (18.1)$$

The lemma is dimensionally correct Level A algebra for the bare scalar pieces. It neither derives the potential/state nor bounds the action-owned effective history, because Eq. (16.3) also contains $\dot{F}$ and $\ddot{F}$.

Proof-of-class, not an ECT acceleration result. A selected $U(\phi) > 0$ and a state with $K_\phi^{\text{phys}} \ll U$ give a vacuum-like bare scalar ratio, but acceleration follows only from the full condition below. In a non-minimally coupled completion the actual acceleration also contains derivatives of $F(\phi)$ and must be obtained from the complete metric equations. Thus the existence of an admissible scalar parametrisation is Level B proof-of-class; ECT ownership is Open. The action does not identify a physical $w(a)$, a coincidence history, or any redshift-dependent late-time-source likelihood.

Exact acceleration condition in the supplied scalar model. Combining the two Jordan-frame background equations gives

$$\frac{\ddot{a}}{a} = -\frac{1}{6F} \left[\frac{\rho_m + 2\rho_r}{\bar{M}_{\text{Pl}}^2} + 2\omega\dot{\phi}^2 - \frac{2U}{\bar{M}_{\text{Pl}}^2} + 3(\ddot{F} + H\dot{F}) \right]. \quad (18.2)$$

Therefore $w_\phi^{\text{bare}} \simeq -1$ is neither necessary nor sufficient by itself when F evolves. A supplied trajectory accelerates precisely when the bracket in Eq. (18.2) is negative. This equality is an internal Level A test of the supplied action and state; the identification of that action with ECT remains Level B/Open.

Two-slope late-time proof-of-class. The potential

$$V_s(\sigma) = V_- e^{c_s \sigma} + V_+ e^{b_s \sigma}, \quad F_s = F_* e^{a_s \sigma}, \quad K_s = \kappa_s F_s, \quad (18.3)$$

has a finite stationary point of V_s/F_s^2 when $(c_s - 2a_s)V_- + (b_s - 2a_s)V_+ = 0$. With $c_s < 2a_s < b_s$ and positive coefficients, the effective potential is bounded on both sides and the stationary point is a minimum. This removes the zero-density degeneration of a single exponential and supplies a globally regular scalar-background example. The nine calibrated numerical nodes in Table 60 then show how the sampled potential parameters affect age, distances, r_s , Planck-scale drift and local gravity; they do not establish a continuous-family trend. It does not make the potential or its coefficients consequences of P1–P6.

Effective equation of state as a derived diagnostic. For any solved trajectory define

$$w_{\text{eff}}(N) = -1 - \frac{2}{3} \frac{H'(N)}{H(N)}, \quad q_{\text{dec}}(N) = -1 - \frac{H'(N)}{H(N)}. \quad (18.4)$$

These quantities describe the total expansion history and are the correct background diagnostics even when the scalar stress cannot be separately conserved. They must be reported together with the decomposition into matter, radiation, scalar kinetic/potential and F -derivative terms. Fitting w_0, w_a is a compression of a solved trajectory, not a substitute for solving it.

Coincidence and stability requirements. A physical late-acceleration mechanism must explain why the accelerating regime occurs near the present state without selecting that state by a terminal shooting condition. Along the complete orbit the reduced scalar mass, kinetic eigenvalues and characteristic speeds must remain physical, and the same asymptotic state must pass finite-body PPN and varying- G constraints. A potential that accelerates only on a locally excluded branch does not close the late-time sector. This is a dynamical selection question, not a prejudice against a constant contribution by itself; the distinction is usefully emphasised in Ref. [129].

Current DESI context and conditional falsifier. DESI DR2 BAO are well described by flat Λ CDM in isolation. When combined with CMB and, dataset-dependently, supernova compilations, w_0 – w_a fits prefer evolving dark energy; the quoted significance varies with the data combination [105]. The extended DESI analysis finds a common low-redshift trend and a preference for phantom crossing, while non-crossing alternatives are disfavoured but not excluded [106]. This is observational context, not an ECT prediction or a discovery claim. For the positive-kinetic two-slope Jordan scalar the bare bookkeeping ratio

$w_q = (K\dot{\sigma}^2/2 - V)/(K\dot{\sigma}^2/2 + V)$ obeys $w_q \geq -1$ when $K \geq 0$ and $\rho_q > 0$; nonminimal F terms mean that this bare ratio is not the fitted action-owned effective $w(z)$. A robust ECT completion whose observable effective history is constrained never to cross -1 would be disfavoured, but the present completion has not been run through the full DESI+CMB+SN likelihood and is therefore untested.

Required observational comparison. The background test set is

$$\{H(z), D_M(z), D_L(z), r_{\text{drag}}, t(z)\}, \quad (18.5)$$

while the perturbative test set is

$$\{f\sigma_8, C_\ell^{\kappa\kappa}, C_\ell^{Tg}, C_L^{\phi\phi}, D_L^{\text{GW}}/D_L^\gamma\}. \quad (18.6)$$

One joint likelihood is required because a non-minimal $F(\phi)$ changes background distances, growth, photon lensing and local calibration simultaneously. Agreement in the first set with failure in the second rejects the supplied completion.

Relation to the vacuum-offset result. The fixed determinant is an exact orientation-sector identity, not a decoupling theorem. Vacuum-offset shift covariance follows only in the separately supplied trace-free completion, where $V + C_I$ is invariant under opposite shifts. The ordinary two-slope action can generate a positive, stable late source but is not additive-shift invariant. Neither statement alone selects the source, magnitude or evolution observed in our condensate; the action descent and observable matching remain Open.

Completion and falsification gates. A late-time ECT result requires one varied ordered-medium action to determine F , ω , U , the orientation stress and the state; one globally regular stable background; a physical matter/photon metric; and a joint background-plus-perturbation likelihood. Only then can $w(a)$ and the sign and magnitude of accelerated expansion be compared with observations. A failure rejects that named completion. The full owner chain is given in §16.5 and Appendix AI.

18.4 Dark-sector discriminants and falsifiers of the present closure

This subsection separates closure-specific tests, imported benchmarks, and Open research programmes relevant to dark-sector interpretation. The status-tagged inventory in Table 71 must not be read as a list of uniformly derived ECT predictions.

Exploratory environment/history test (F1). The fitted scale $a_{M,\text{eff}}$ may be cross-correlated with declared environment and history covariates as a test of the Open law Ξ . The corrected scalar equation does not derive the earlier $\gamma 4\pi\bar{\rho}_m/U_0''$ combination, nor does it predict a positive sign or a ten-percent amplitude. A null result constrains the specified regression model; a detected correlation does not by itself establish the scalar source/screening chain or ECT.

Minimal-fit particle-halo scope test. The Level-C HRC algebraic diagnostic is fitted to the selected rotation-curve sample and returns model outputs without a particle-halo term; its absolute held-out likelihood is poor. This is not a derivation that the mass discrepancy in ECT must arise from that closure and is not a prediction that particle dark matter is absent. The LZ 4.2-tonne-year analysis reports a minimum observed 90% C.L. spin-independent WIMP–nucleon limit $\sigma_{\text{SI}} < 2.2 \times 10^{-48} \text{ cm}^2$ at a WIMP mass of $40 \text{ GeV}/c^2$, with no positive WIMP signal [130]. This external exclusion curve does not by itself determine whether any particle species accounts for the galactic mass discrepancy. A robust detection of a particle species specifically shown to account for the galactic mass discrepancy would disfavor the minimal no-halo fit as a complete description of that channel; it would not by itself falsify ECT. Detection of a subdominant species would not decide the status of the Level-C HRC data diagnostic or its still-Open ECT matching.

Bare scalar-ratio guard and physical falsifier. Equation (18.1) tests only the bare scalar decomposition. Because the non-minimal terms in Eq. (16.3) enter the solved expansion history, an observational preference for $w_{\text{eff}} < -1$ does not contradict that identity. It would reject a named completion only if the complete action-owned background, photon/perturbation map and likelihood independently constrained its observable $w_{\text{eff}}(a)$ never to cross -1 . Agreement with either sign would not establish ECT ownership.

Local cluster-response no-go (F10). For any positive baryonic projected field and any strictly increasing local map F , the response $\Sigma_{\text{resp}}(\mathbf{x}) = F(\Sigma_b(\mathbf{x}))$ preserves the complete pointwise ordering and the set-valued identity $\text{argmax} \Sigma_{\text{resp}} = \text{argmax} \Sigma_b$, with both sets possibly empty. Whenever the maxima are attained, it therefore cannot dynamically displace a total-response maximizer away from the supplied total-baryon maximizers or encode merger memory. This does not constrain the residual $\Sigma_{\text{resp}} - \Sigma_b$, which is generally non-monotone, or a gas/BCG offset already supplied in Σ_b . Bullet-like systems are empirical targets for the Open retarded two-potential collision problem, not outputs of the local algebraic HRC response.

Disambiguation of M, S, and G. The microscopic preferred-direction operator M, the Level-B scalar ansatz S, and the conditional HRC static functional G are three distinct objects. M is not S by operator and variable content; G is a phenomenological kpc fit, while the map $S \rightarrow G$ remains Open. None may be used as a derived surrogate for another in dark-sector comparisons.

Taken together, these items have different statuses. The bare scalar-ratio identity is conditional algebra inside its scalar closure and does not bound the physical background history; the cluster script is only an exact peak-preservation sanity audit; the rotation-curve fits are benchmarks; and the environment/history law Ξ is Open. They should not be presented as jointly derived consequences of one scalar mechanism until the missing matching and screening calculations are supplied.

18.5 Section-level status summary

The dark-sector interpretation of ECT, as developed from the ordered-branch architecture in Sections 18.1–18.4, should be read at the present stage in three layers.

Mixed status: the Level-C HRC algebraic diagnostic is fitted to selected mass-discrepancy data with poor absolute held-out likelihood and without a particle-halo term, while its ECT action/source matching remains Open. The late-time scalar background is a separate Level-B closure, and its bare scalar-ratio identity does not upgrade the galactic functional.

Structurally possible but not yet developed quantitatively: collective condensate excitations and additional coherent-branch sectors as dark components beyond the conditional HRC algebraic diagnostic. Their detailed treatment requires the coherent branch (BR1) and is deferred to Section 34 of the Quantum Sector.

Open: the relic abundance, stability, and production history of such dark-sector excitations; the first-principles derivation of the effective cosmological fractions; and the observational test of a future action-owned effective $w(z)$. The bare scalar-ratio identity is conditional algebra and does not bound that observable history. The most important near-term discriminants are summarised in Table 71, especially the prespecified environmental modulation of $a_{M,\text{eff}}$ (F1). Entry F5 is therefore a completion-specific effective-history test, not a universal no-phantom guard or an independent ECT prediction. A genuine cluster-lensing test remains an Open specification rather than a current discriminant.

The macroscopic dark-sector reading developed here should therefore be seen as a controlled Part II interpretation layer, not as a complete microscopic ontology of the dark sector. Unlike the standard Λ CDM reading, it does not begin by postulating two independent hidden ingredients for the galactic mass discrepancy and the late-time accelerated expansion. The detailed ontology of coherent-branch and collective dark-sector excitations belongs instead to the Quantum Sector.

19 Observational Tests, Falsifiers, and Chapter Conclusions

19.1 Astrophysical diagnostics, external anchors, and completion tests

Status: mixed conditional, Level C, external, and Open material; each item carries its own status and no common prediction status is implied.

Postulate grounding and scope discipline. The present subsection does not introduce a new independent sector of the theory. It collects conditional calculations, fitted diagnostics, external anchors, and Open targets of the ordered-branch programme already developed in Sections 16.5–18. Its structural basis is the ordered branch itself (P4) together with the admissibility of non-uniform or slowly evolving branch responses at the macroscopic level (P6). For clarity, the discussion is organised into three owner layers: the conditional covariant background action, its exact constraint/stability audit, and the observable-specific forward models that remain to be constructed.

Early-object observations and the ECT calculation target

JWST observations motivate a quantitative early-object forward model, but they do not select an ECT background or an effective response by themselves. The relevant data include luminosity functions, stellar-mass posteriors, spectroscopic redshifts, compact-object/host scaling relations, morphologies and survey selection. An ECT prediction requires the common pipeline in Eq. (AH.6); qualitative similarity to an early maturity channel is not a derived abundance or morphology prediction.

19.1.1 Integrated Sachs–Wolfe and lensing target

The ISW source and weak-lensing kernel depend on the time-dependent Weyl potential of the physical photon metric. Under a deliberately restricted same-metric, zero-slip, sub-horizon quasistatic closure, the named near-GR two-slope state changes the instantaneous Weyl carrier by only 0.00267–0.00292% and its conformal-derivative carrier by 0.00308–0.00813% over $0 \leq z \leq 2$. These Level C limiting numbers are not C_ℓ , lensing, or ISW predictions: the quasistatic approximation excludes horizon-scale ISW, and ECT must still derive the photon metric, gauge-reduced perturbation action, initial conditions and tracer kernels from the same completion. The full owners and acceptance tests are listed in Appendix AI.

Exploratory strong-curvature frontier: black holes and compact objects

Black-hole and compact-object implications of the ordered condensate branch are potentially important but remain exploratory (Level B/Open); they are not part of the present controlled macroscopic derivation and should be read as a forward research direction rather than as an established observational output of Part II. For GW-related LIV effects Section 13.8 supplies only a conditional operator benchmark. ECT currently identifies neither c_{GW} nor a propagation-time shift. The value of this block is therefore primarily organisational: it marks the strong-curvature observational frontier that any future ECT completion must eventually address.

Dark-sector synthesis. The present manuscript separates three closure programmes: (i) a Level-C HRC algebraic diagnostic evaluated on selected mass-discrepancy benchmarks that returns outputs without a particle-halo term; (ii) a late-time ordered-background route at its declared closure status; and (iii) the Open possibility of additional collective, topological, or particle sectors. The first is currently a fit result rather than an ECT derivation or an inference about fundamental particle content. Table 66 summarises the three channels and their discriminants.

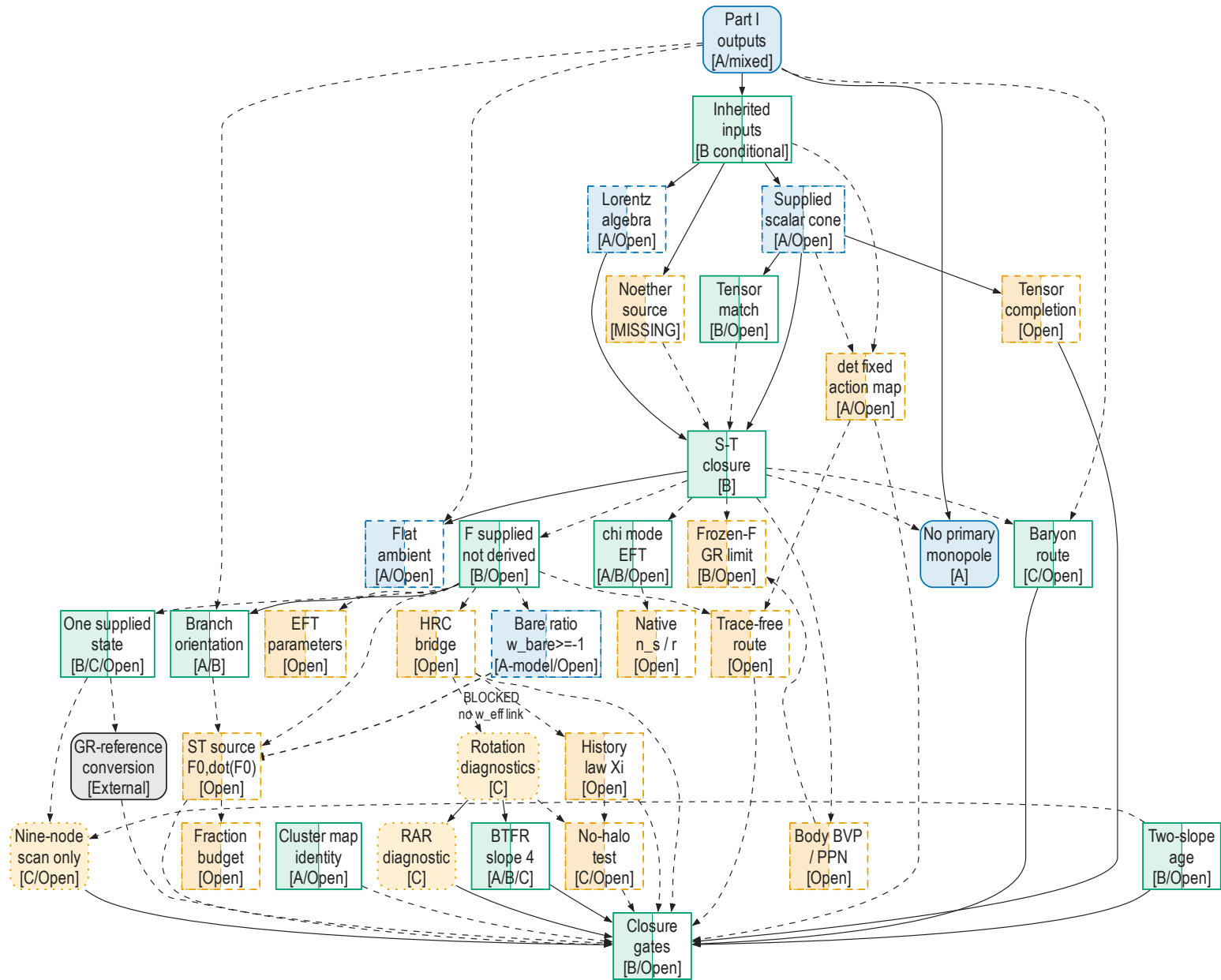
Cosmological status. The active output combines the owner map of §16.5 with one conditional two-slope background: its age, $H(z)$, distances, high- z clocks, restricted growth/ r_s , and instantaneous Weyl/derivative diagnostics are computed. P1–P6 do not select that completion, while full Hubble inference, JWST abundance/morphology, gauge-complete growth, projected ISW/lensing spectra, cluster transport and finite-body PPN await their listed owners.

19.2 Conclusions of Part II: Macroscopic Physics

This subsection summarises the macroscopic construction of ECT developed in Part II (Sections 11–18). It collects the conditional macroscopic completion programmes and Level-C diagnostics developed on the macroscopic branch: gravity and cosmology completion programmes, galactic and cluster phenomenology, and the associated dark-sector reinterpretation. All items are classified by the Level A/B/C/Open discipline maintained throughout the paper. The status and dependency logic of the macroscopic branch is shown schematically in Figure 20.

Selected status and dependency map. Figure 20 displays a selected connected Part II graph on one connected A4 reader map. It replaces the former navigation thumbnail and three route fragments; no branch is removed merely to make an A4 panel. Internal scalar algebra remains separate from the Open clock/rod, physical-metric, tensor, source, screening and finite-body owners. HRC/RAR/BTFR entries remain conditional algebraic or Level-C diagnostics, and the named background, JWST, cluster, growth and PPN routes keep their explicit conditional/Open status. The supplied amplitude function is not derived from the logarithmic coordinate; the bare scalar ratio does not bound observable w_{eff} ; the old implication is retained as a visibly blocked edge. The late source is split into a GR-reference conversion and an Open generic scalar–tensor owner, while the JWST result is limited to the nine evaluated nodes rather than the continuous family.

Selected Part II status and dependency map



A exact | B conditional | C fit/benchmark | Open/MISSING explicit. Solid border = A/external; striped = conditional/Open; dotted = C; dashed + tee + BLOCKED = no implication. Colour is redundant.

Reading discipline. Solid arrows in Figure 20 organise declared dependencies; dashed arrows expose assumptions, matches and missing owners. An arrow never upgrades a node's status. The map's direct labels and luminance-separated palette remain authoritative in colour and grayscale. The repository source and machine topology retain the exact edge inventory, which is not repeated as a reader-facing ledger.

Macroscopic outcome of Part II. Part II develops the macroscopic physics of ECT from the ordered-branch inputs inherited from Part I. Its organising postulate-level basis is the existence of the ordered branch itself (P4), together with the admissibility of non-uniform and slowly varying ordered configurations at macroscopic scales (P6). Within that inherited framework, Part II develops conditional gravity and classical-cosmology completion programmes, supplied-branch algebra, Level-C galactic and cluster diagnostics, and a status-separated dark-sector reinterpretation.

The strongest current outputs of the chapter are: the distinction between ambient Euclidean flatness and the still-open physical FLRW-curvature bridge, including a conditional planar $k = 0$ proof-of-class, the conditional $\pi_2(S^3) = 0$ classification of the supplied target, *the Level A orientation determinant identity and the distinct conditional trace-free shift covariance* (Section 15): fixed $\det K^{AB} = \beta^3(\beta - \alpha)$ holds on the fixed-coefficient orientation submanifold, but does not derive a physical constrained metric action. In the separately supplied trace-free completion, $V \rightarrow V + C_0$ and $C_I \rightarrow C_I - C_0$ leave the classical equations invariant; C_I may be a measured boundary/state datum. The ordinary two-slope completion independently supplies a regular late-de-Sitter proof of class, but is not shift invariant. The physical ECT action, state, metric and late-source map remain Open, the dimensionally normalised bare scalar-ratio inequality inside the supplied positive-kinetic stress (with no corresponding bound on observable $w(z)$), the BTFR slope 4 as Level-A algebra inside the supplied HRC functional and a Level-B/Open corollary after the declared gravity bridge, the Level-C matched input $a_{M0} \approx cH_0/(2\pi)$ with microscopic origin Open, an Open environment/history programme, and the cluster peak-preservation identity as Level-A mathematics in the supplied monotone map, with the physical cluster response and morphology remaining Open. The chapter distinguishes exact equations inside declared actions, conditional ECT ownership, exact negative constraint/stability tests, and observable-specific Open forward models. This status separation is essential for interpreting every numerical claim.

Status-separated outputs of Part II. Within the inherited ordered-branch framework, the macroscopic branch records the following conditional statements, benchmarks, and Open programmes:

1. A controlled pure-metric gravitational branch and a Level-B scalar-tensor ansatz with corrected field equations. The exact divided matter coefficient is $G_{\text{div}} = G_*/F$; its Cavendish, finite-body and cosmological response maps remain Open. The constant-background GR-form equation is conditional and is not by itself a physical GR-response limit. A covariant orientation action, any resulting $T_{\mu\nu}^{(n)}$, and the condensate matching of F, K, V remain separate Open/closure inputs rather than derived functions of the displayed scalar-only action.
2. A scalar signal-speed bound (§12.6): the characteristic cone of the supplied scalar principal symbol confines disturbances of that scalar model to its closed causal cone. For $m \neq 0$ and finite k , a narrow-band packet centre has sub-characteristic group velocity, while the front remains on the cone; in flat $3 + 1$ dimensions the massive retarded kernel has both a cone-boundary term and an interior tail. In the massless flat- $3 + 1$ case the retarded fundamental solution is supported on the cone. No physical photon, matter or tensor speed follows until its action and observable metric are supplied. The independently supplied EFT parameters (α, β) are fixed in the homogeneous benchmark; P4 does not derive them as vacuum-state parameters. A macroscopic completion may vary candidate geometry, but its physical cone and matter/photon/tensor matching remain Open.
3. P1 supplies a flat Euclidean ambient metric; a spatially flat physical branch is a conditional planar proof-of-class, not yet a derived FLRW metric. The supplied target has $\pi_2(S^3) = 0$, while a physical monopole/relic conclusion remains Open; one longitudinal scalar coordinate parametrises the stated constrained linear map; the physical one-mode reading is conditional.

4. For the supplied translation-invariant strict-pullback kernel, the exact equal- w power, sharp quartic coercivity, fixed- k and full-domain positivity conditions, finite-deformation ratio and seed tilt, pointwise dominance bounds and finite-band tolerances are derived. This is Level A conditional mathematics of the displayed fixed-coefficient kernel. Reflection positivity, a physical projector, coefficient ownership/running and the $\chi \rightarrow \zeta$ observable transfer remain Open; the result is not a CMB prediction.
5. A scalar-only ordered-branch cosmological model whose equations are exact inside the supplied action, together with a proof that the unowned closure functions and state do not identify one $H(a)$. The age quadrature is exact after $H(a)$ and a_{ord} are derived; Hubble, JWST, growth, ISW, lensing, clusters and PPN remain owner-specific Open calculations (§16.5).
6. A conditional HRC static functional whose algebraic diagnostics address rotation-curve/RAR/BTFR phenomenology without a particle halo in that test. Its identity bridge, action ownership, source map and field-level disk solution remain Open.
7. A matched critical-scale reference $a_{M0} \approx cH_0/(2\pi)$ and fitted per-galaxy $a_{M,\text{eff}}$ values; their microscopic relation and any environment/history law are Open.
8. An exact theorem that any supplied strictly increasing local transform preserves the ordering and possibly empty set-valued argmax of its input baryonic field. No cluster-specific synthetic input or amplitude is retained; physical morphology, response and observational calibration are Open.
9. A sector-by-sector inventory of external orientation proxies. No $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$ contribution is derived from the scalar action; a covariant $S_n[g, n]$, projection, and coefficients remain Open.
10. A dark-sector reinterpretation in which galactic mass discrepancy, possible coherent/topological dark sectors, and late-time dark energy are treated as distinct questions within the same condensate architecture.

What Part II inherits, what it establishes, and what remains open. Part II does not start from a blank slate. It conditionally inherits the P4 ordered branch and its scalar principal tensor from Part I; the physical metric/gravity and coherent-state sectors remain separate completion programmes rather than inherited theorems.

Within that inherited framework, Part II establishes the strongest macroscopic structural outputs: P1 provides a flat Euclidean ambient metric and a conditional planar route to $k = 0$; the physical FLRW metric remains Open. The primary supplied S^3 target has no π_2 -protected map class; whether this excludes any physical monopole channel remains Open. The stated constrained perturbative map contains one longitudinal scalar coordinate; a physical one-mode sector requires the supplied reduced EFT and projector.

At the next level, the declared scalar/cosmological and galactic closures yield conditional algebraic consequences and benchmark fits. These include the bare scalar-ratio inequality within the positive-kinetic ansatz (not an observable w_{eff} bound), the Hubble/JWST programme, the HRC-0/HRC-3 response construction with its separate Open gravity bridge, the HRC-only BTFR/RAR diagnostic record, and the cluster proxy's exact peak-preservation identity (which is not morphology evidence). The present scale is matching, and environment/history dependence of the response scale is Open.

The displayed scalar action contains no independent n_μ . Accordingly, the orientation footprint is an inventory of Level-C external proxies for a possible future $S_n[g, n]$, not a derived beyond-benchmark layer. This status must remain explicit when interpreting the numerical sensitivities.

At a weaker benchmark level, specific late-time trajectories, calibrated dark-energy reconstructions, semi-analytic maturity budgets, and Level-C galaxy-by-galaxy fits depend on closure choices and calibrated parameters. Section 19.1 organises the evidence by equation owner, constraint/stability result and observable forward model; no attached background is treated as a physical cosmological result.

What remains open is the primordial spectrum (n_s, r), the cosmological fraction budget, the full nonlinear ϕ boundary-value problem, and the detailed ontology of coherent dark-sector excitations.

Table 67: Numerical inputs, conditional outputs, external anchors, and fit diagnostics of Part II. The entries are not all of the same logical type: some are conditional algebra, some are closure-level benchmark estimates, some are external constraints, and some are comparative fit diagnostics. The final column records this status explicitly.

Item	Value / conditional output	Observation / comparison	Status
BTFR slope	4 for either HRC deep branch after the gravity bridge	3.98 ± 0.08	B / bridge Open
a_{M0} (matched HRC gravity scale)	$cH_0/(2\pi) = 1.08 \times 10^{-10} \text{ m/s}^2$	$a_0 \approx 1.2 \times 10^{-10}$ (McGaugh)	Matching; ECT origin Open
MW representative disk sensitivity	HRC-0: $\chi_v^2 = 1.813$, $a_M/a_{M0} = 2.798$	HRC-3: $\chi_v^2 = 2.287$, $a_M/a_{M0} = 1.703$	Level C; not a modern Milky-Way posterior
n_s, A_s, r (primordial spectrum)	Supplied pure-quartic strict-pullback kernel gives local $n_\chi = 1$ exactly when its combined κ_χ -kernel normalisation is scale-independent (separate constants suffice); exact finite-band bounds depend on δ_a, δ_m, ρ ; physical mass, measure/state/projector and complete transfer open	CMB scalar/tensor targets: $n_s \simeq 0.965$, A_s obs., r bounded	A-math conditional / B/Open physical
η_B (baryon asymmetry)	No ECT value	External target from a named Planck base- Λ CDM fit plus standard conversion [70]	Open; physical current, vertex, state and transport absent
GW-EM relative speed	Open: no ECT value	Joint interval $[-3, +0.7] \times 10^{-15}$ [12]	External gate
Late-time expansion history	Named near-GR two-slope orbit; official 15-point BC03 chronometer calibration gives $H_0 = 68.59 \pm 3.95 \text{ km s}^{-1} \text{ Mpc}^{-1}$	BAO, supernovae, chronometers and growth data	Level C calibration; owner-complete joint likelihood Open
ECT cosmological background and age	Named two-slope scalar-tensor action/state: $H_{2s}(0)\Delta t = 0.9636868$ and covariance-calibrated $t_0 = 13.74 \pm 0.79 \text{ Gyr}$ conditionally	Exact owner audit (§16.5)	Level A within the named completion; dimensional scale calibration Level C; P1-P6 ownership Open

Bridge to Part III. With the macroscopic branch in place, the next task is to return to the coherent and quantum structure of the condensate. Part III develops the coherent branch across eleven thematic steps: the action-scale and winding programme, wave kinematics and conservation laws, the Hilbert-space bridge through reflection positivity and unitarity, exchange-sector topology and Dirac structure, the unified vacuum-response programme, decoherence and the arrow of time, the quantum–classical boundary and Born-type probability, entanglement, the wider topological/dark-sector ontology, black-hole thermodynamics and the information-exclusion result, and the analogue source-model comparison and bridge-building programme.

Conceptual output: scalar characteristic-metric notation. ECT takes the Φ -medium, rather than a physical metric, as the primitive model entity. In the supplied scalar EFT it is convenient to denote the inverse principal tensor by “characteristic metric” because it records the propagation characteristics of that scalar operator. This notation is not a derivation of a universal observable metric. A possible gravity completion may be a long-wavelength condensate response, but the fixed director map has no linear transverse–traceless/static-density vertex; physical rods, clocks, photons, matter and nonlinear gravity require separate owners. Two further foundational issues find natural homes in the macroscopic branch: the primary condensate vacuum satisfies $V''(\phi_0) > 0$ at tree level, while the strict one-real-scalar one-loop running has $\beta_\lambda > 0$ and therefore excludes only the Standard-Model-like negative-running route at that order (§8.4.5; Level B conditional). Full effective-potential, branch-architecture and embedded-sector stability remain Open [59, 60]; §8.3); and the duration and relative density of any late-time source are not identified until one globally regular background, physical metric and state are derived. The cosmological-coincidence question therefore remains an Open dynamical problem; this is logically separate from the conditional trace-free vacuum-offset completion and its still-Open Φ -medium action descent (Section 15).

19.3 Global Level-4 self-consistency checklist

Role of the checklist. The present table is not a second derivation map and not a duplicate of the status-separated claim inventory. Its purpose is different: it provides a compact Level-4 audit of the framework as a whole, combining inherited consistency conditions from Part I, macroscopic results from Part II, and deferred or open items that extend into the Quantum Sector. It should therefore be read as a global status board rather than as a local summary of Section 19 alone.

Postulate-level organisation. At the structural level, the checklist probes four layers of the ECT architecture: the microscopic consistency of the Euclidean Φ -medium (P1–P3), the ordered macroscopic branch and its admissible non-uniform developments (P4–P6), the coherent sector (BR1, S10), and the still-open bridges required to close the full framework. The PASS / PART. / PROB. / OPEN labels should be interpreted in that same layered sense. Unlike the derivation map, which records logical dependencies, and unlike the falsifier table, which records observational tests, the present checklist answers a different question: which framework-level consistency targets are already passed, only partially controlled, structurally motivated but not rigorous, or still open.

Architectural completions from the emergent matter-sector trilogy. The discrete-symmetry architecture (§9.4), the anomaly architecture (§9.5), and the flavour architecture (§9.7) now provide an explicit conditional consistency audit for a supplied chiral gauge completion. Their Level-4 audit status is therefore recorded in the checklist below alongside the original whole-framework consistency items.

Scope legend: I = inherited from Part I (foundations); II = macroscopic branch of Part II; III = Quantum Sector / Part III; G = global whole-framework closure problem.

Table 68: Level-4 self-consistency checklist. PASS = a named mathematical/source-model test reproduced under its displayed assumptions, not automatic physical ECT closure; PART. = a partial conditional calculation with its missing owners declared; PROB. = structurally motivated, rigorous proof pending; OPEN = known open problem.

Status	Scope	Check	Outcome	Ref.
PART.	I	Condensate stability	Tree-level local stability plus a positive-running check in the strict one-loop scalar truncation; full effective-potential and branch-architecture stability remain open	§8.4.5, §3
PART.	I	Quadratic stability of supplied scalar closure	Kinetic/gradient signs are healthy only after the supplied $\alpha > \beta > 0$ component, $\zeta_L = -1$ orientation and $A_{L,\varphi} > 0$; full non-negativity is the declared spectral/domain test, whose stationary P4 owner remains Open	§3
OPEN	I	Physical tensor/ghost test	No linear TT/static-density seed at fixed α, β and tangent-director order; quadratic response is parametric and a constrained tensor Hessian/residues remain missing	§8.5
OPEN	G	Full non-perturbative ghost-freedom and exact TT closure of a candidate physical graviton completion	Completion-level question tied to gauge/gravity closure (OP3)	§8.5, §38.1
OPEN	I	Tensor propagation gate	c_{GW} and tensor LIV operators are not identified; GW170817 is an external completion test for a future action	§13.8
PART.	I	Discrete-symmetry architecture of emergent matter sector	Euclidean reflection identities are A-math; their physical P/T identification, OS/CPT reconstruction and completed matter action are conditional/Open; $\mathcal{L}_5$ is only a candidate preferred-background operator	§9.4
OPEN	I	Anomaly audit for a supplied chiral gauge completion	Standard anomaly equations and arithmetic are A conditional on supplied representations; $\mathcal{L}_5$ is representation-data transparent; the physical local bundle, chirality and matter spectrum are Open (OP-GUT6 family)	§9.5
OPEN	G	Flavour architecture (CKM/PMNS)	Basis-misalignment algebra is standard; overlap profiles, generations, representations, Yukawa matrices, CKM/PMNS entries and phases require supplied spectral/matter closures	§9.7
PASS	I	Scalar-operator causality test	Retarded propagation for the supplied hyperbolic scalar operator; positive pairwise noise quadratic only in a declared Gaussian source model, with physical matter/photon/tensor cones and the ECT record kernel Open	§3.8
OPEN	G	ECT entropy-production theorem	Ohmicity/Markovianity and positive pairwise noise are insufficient; reduced channel, state, entropy functional, and unitality/detailed-balance/contractivity hypotheses required (OP-Q17)	§3.9

Status	Scope	Check	Outcome	Ref.
OPEN	III	Born measure	A complete applicable corrected OS-II package for a supplied Gaussian model reconstructs a positive quadratic norm. Independently, ordinary projector Gleason represents a supplied normalised, noncontextual, countably additive projector measure only for a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. State/preparation, event, probability origin, outcome and update remain Open	§30.4
OPEN	II	Primordial perturbations n_s, r	The strict map has one scalar coordinate (A kinematics); a supplied reduced EFT may promote it to one physical mode (B). The curvature variable, state, transfer and CMB likelihood are Open. A conditional Lifshitz window is not a prediction of n_s or r	§14.1
OPEN	II	Baryogenesis	A supplied transition/matter completion may host Sakharov ingredients; no ECT value of η_B is calculated (OP7)	§16.3
COND.	III	Unruh temperature	Standard Rindler/KMS result only after a Lorentzian QFT state, detector coupling, analyticity and $S_0 = \hbar$ matching are supplied; ECT owner Open	§25.4
COND.	III	Orientation-sector Casimir (ideal boundaries)	Idealised boundary-value calculation; whether matter imposes those boundaries and the force magnitude/sign are Open	§25.3
OPEN	III	Casimir: lab applicability	Material boundaries $\rightarrow$ soft orientation sector: not derived	§25.3
OPEN	II	Tensor ratio r	ECT-native derivation pending; removed borrowed slow-roll formula	§14.1
OPEN	G	Renormalisability	A scalar EFT on a fixed background does not establish UV safety of a physical tensor/gauge/matter completion; its action, constraints and counterterms are missing	§38.1
OPEN	G	3 fermion generations	Three-state counts occur only in supplied spectral wells with parameter/boundary choices; physical generations, representations, masses and mixings are not derived	§9.6, §9.7
OPEN	G	α_{fs} ; SU(3)	The identity $\alpha_{fs} = q^2/(4\pi Z_A)$ is conditional on a supplied physical gauge normalisation; ECT derives neither the electroweak owners nor the colour sector (§7.2)	§7.2
COND.	II	Cosmological background and age	One supplied two-slope action/state gives conditional $H(z)$, distances and ages after one declared calibration; the official 15-point BC03 covariance gives $H_0 = 68.59 \pm 3.95$ and $t_0 = 13.74 \pm 0.79$ Gyr for that orbit. The microscopic completion and physical clock/metric map, not the legitimacy of state data, remain Open	§AE, §16.5

Status	Scope	Check	Outcome	Ref.
OPEN	II	JWST early-object forward model	Conditional clock and hybrid rare-peak sensitivities are computed; abundance and morphology still require perturbations, halo/baryonic modelling and survey selection	§19.1
PART.	II	Force-law dimensionality diagnostic	$d_{\text{force}}(x)$ crossover $3 \rightarrow 7/3 \rightarrow 2$ algebraic for the HRC-0 shape; no retroactive action derivation	§16.8
OPEN	II	Environment/history dependence of $a_{M,\text{eff}}$	An exploratory factor Ξ is defined for cross-correlation tests; its microscopic map, sign, amplitude, and density-only closure are Open	§17, §18.4
COND.	II	HRC constitutive response	HRC-0 exact in the augmented ERP- Φ /inverse-DtN model; HRC-3 conditional in the supplied three-channel model; gravity bridge and P1–P6 ownership Open	§17.3
COND.	II	Galactic dark-sector benchmark	HRC is the active candidate; HRC-only algebraic fit outputs are Level-C diagnostics, not predictions; cosmological fraction budget Open	§18.2
OPEN	II	Cluster projected-proxy sanity audit	Strictly local v_{cl} transform preserves every positive baryonic-pixel ordering exactly; no morphology evidence; action/source/lensing response and observational calibration Open	§17.2, §18.4
OPEN	II	Varying-constants protection (OP-CONST)	Selective-coupling and frozen-background constructions are conditional routes only; the physical matter map and microscopic protection theorem remain Open	§16.5

19.4 Global dependency, status, and completion-test maps

Role of the global maps. The following tables provide three complementary global views of the ECT programme. Table 69 records the logical status and dependency architecture: which sectors depend on which earlier inputs and closure assumptions. Table 70 records the epistemic status of those outputs in the Level A / Level B / Open discipline adopted throughout the paper. Table 71 records conditional calculations, external constraints and Open completion tests. An Open row is not a present ECT signature or falsifier. These three maps should therefore not be read as redundant summaries, but as answers to three different questions: what follows from what, how strongly it is established, and how it can fail.

Postulate-level organisation. At the highest structural level, all three maps are organised by the same postulate hierarchy: the microscopic Euclidean Φ -medium and its consistency conditions (P1–P3), the ordered macroscopic branch and its admissible non-uniform developments (P4–P6), and the coherent sector (BR1, S10) whose full ontology is taken up later in the Quantum Sector.

The narrative conclusions of Part II were summarised in Section 19.2, while the audit-style checklist was given in Section 19.3. The tables below complement those layers by presenting the same framework as a status-and-dependency map, a status map, and a completion-test map. Entries should be read against the same layered vocabulary used in the final status board (Section 19.5): inherited structural inputs, closure-level outputs within the present ϕ -first programme, benchmark/calibrated implementations, and open problems. For the macroscopic branch developed in Part II, the declared near-term test is the prespecified environment/history regression (F1). The bare scalar-ratio identity (F5) is a conditional internal check rather than an observable bound or ECT prediction. A physical cluster-lensing test (F10) remains Open; the current local proxy is not a discriminant.

Table 69: Status and dependency map of ECT: physical objects and their declared inputs or owners in the Level A / Level B / Open discipline. Entries: “derived” denotes Level A, exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. “Structural / effective (Level A/B)” means proven in the pure-metric limit and Level B in the ϕ -first closure; “matched (Level B)” denotes an EFT matching relation; “embedded” means introduced consistently, not derived from the bare scalar action; “open” denotes an active research direction.

Object	Origin in ECT	Status
4D Euclidean space	Postulate P1	postulate
$O(4)$ symmetry	Postulate P2	postulate
Condensate field Φ	Postulate P3	postulate
Ordered-gradient physical sector	Postulate P4	postulate; stationary owner Open
Scalar characteristic-metric candidate	Supplied P4 branch and scalar principal tensor with $\beta(\beta - \alpha) < 0$; healthy shorthand $\alpha > \beta > 0$ only for $\beta > 0$	conditional A-math; universal physical metric Open
Physical/thermodynamic arrow of time	Retarded support in a supplied scalar problem and dephasing in a supplied reduced channel are insufficient	Open: state, map, entropy functional and contractivity/detailed-balance owner required
Scalar common-length slope $\hat{c}_*$	$\hat{c}_* = \sqrt{\beta/(\alpha - \beta)}$ from the supplied scalar principal tensor	A-math conditional
Dimensionful scalar speed c_{char}	$c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$ after a supplied clock, conversion speed and lapse	$c_{\text{char}} = c$ adopted Level B calibration; photon/tensor cone equality Open
Gravitational coupling G_N	Geometric EFT matching: $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$	matched (Level B)
Ordered-branch amplitude variable ϕ	Macroscopic closure variable for slow amplitude variations of the ordered branch	effective (Level B)
Pre-quantum action scales	The phase-only fixed-core model defines ι_0^{EFT} after L_{core} is supplied; the optional tension-balanced model defines the distinct $\iota_{0,\text{bal}}^{\text{eff}} = \sqrt{2K_\theta^{1D} \mathcal{T}_{\text{loop}}}$. CDE- Φ is only a post-order acceptance criterion; the actual owner must be one explicit 4D action, state, profile, subtraction and physical quotient	A mathematics for the supplied bounds/minimum / Level-B generic-EFT route / carrier, compact modulus and ECT ownership Open

(continued on next page)

Object	Origin in ECT	Status
Operational action unit and physical reduced Planck constant	Candidate owner maps $\Sigma_\theta \iota_0^{\text{EFT}}$ or $\Sigma_\theta \iota_{0,\text{bal}}^{\text{eff}} \rightarrow S_0$, followed separately by the Level-B calibration $S_0 = \hbar$	Calibration adopted conditionally; physical owner, absolute derivation and universality remain Open and are not identified by the present action
Schrödinger-type equation	Supplied norm-preserving complex sector, Hamiltonian, domain, clock and NR limit	conditional algebra; ECT owner B/Open
Macroscopic scalar closure	Pure-metric conditional limit + Level-B ϕ_u ansatz; galactic functional separate	A/B; S→G Open
Physical graviton	$h_{ij}^{\text{TT}} = h_{ww} = 0$ only at fixed α, β and tangent-director linear order about homogeneous $n = w$; quadratic/nonlinear routes remain parametric/Open; Fierz–Pauli supplied	A-math conditional negative result / Open physical spin-2 completion
Photon	Compact phase gives global Noether algebra; a local bundle/connection, unbroken transverse photon, charge/matter map and metric require completion	conditional route / Open
$W^\pm, Z^0$	Supplied electroweak doublet and chiral gauge completion	embedded model-building / Open descent
Higgs-like boson	Radial mode of the secondary electroweak condensate doublet	structural / phenomenological matching (Level B+)
Fermions	Supplied spinor/CAR fields on the ordered background	conditional/Open physical origin
Neutrino sector	No physical chiral/mass operator derived; weak-sector, PMNS/MSW and seesaw closures may be supplied; the geometric-scale expression is an unselected historical ansatz only	C/Open
Late-time acceleration source	Regular two-slope proof of class plus a distinct conditional trace-free C_I branch; physical ECT source, state and observable map Open	A in supplied actions / C benchmark / Open ECT
Dark-sector role	No particle halo in the Level-C HRC algebraic diagnostic; particle content and cosmological fractions Open	C/Open
Flat rotation curves and RAR/BTFR phenomenology	HRC-0/HRC-3 augmented-model response plus separate gravity bridge; HRC algebraic diagnostics recomputed	A/B/C by layer / live ECT Open

(continued on next page)

Object	Origin in ECT	Status
Macroscopic entropy theorem	Conditional reduced-channel result only after the map, state, entropy functional, and contractivity/detailed-balance hypotheses are declared	Open for ECT
Universal causal cone	One scalar characteristic cone only; gauge, fermion, matter and tensor cones require independent owners	Open beyond scalar subsector
LV suppression	Higher-dimensional-operator benchmark, with coefficients and sector maps unspecified	conditional EFT benchmark
Proton stability	Three selection-rule patterns conditional on gauge/flavour completion (§44)	conditional (Level C)
Vacuum-offset and late-source status	The orientation determinant is fixed only on its stated submanifold and does not derive a metric action. An explicit trace-free completion is shift covariant through $V + C_I$; a separate ordinary two-slope completion is regular but not shift invariant. The ECT action descent, physical metric and state-to-observable map are Open	orientation algebra A; supplied completions A; ECT descent B/Open; calibrated late history C

Table 70: Structured classification of ECT statements, benchmarks, and Open targets. Level A = exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. Level B = structural/conditional derivation or closure route with a declared open matching, input, or identification step; Level C = phenomenological fit, benchmark, or candidate mechanism; Open = missing/unidentified input or future programme.

#	Statement	Level	Declared model / assumptions / owner	Section
D1	Supplied scalar principal tensor is Lorentzian iff $\beta(\beta - \alpha) < 0$; on the healthy $\beta > 0$ branch this is $\alpha > \beta > 0$; realised branch selection is Open	A-math conditional / Open	P4 does not determine the sign of $\beta - \alpha$ or the physical vacuum/clock/metric	3
D2	KG form and mass map from the supplied scalar operator plus the declared clock/lapse map; Schrödinger limit only after an additional rest-energy/low-momentum reduction	A-math conditional / B–Open	The KG algebra uses no Osterwalder–Schrader continuation; the P4 action, mass/domain matching and physical-mode ownership remain Open	3, 22.1
D3	G_N from geometric EFT matching ($G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$)	B	Matching between ordered-branch tensor EFT and observed gravity; explicit $(u_0, \alpha, \beta) \rightarrow M_G$ derivation open (OP2, OP3)	13.4

#	Statement	Level	Declared model / assumptions / owner	Section
D4a	A supplied nonzero P4 vector has stabiliser $O(3)$ and abstract orbit $O(4)/O(3) \simeq S^3$; physical SSB and vacuum-manifold identification are Open	A-math conditional / Open	Group orbit only; P4 is not a stationary vacuum owner	4.2
D4b	$Q_A = \partial_A \Phi$ implies integrability; $\pi_i = \partial_i \chi / u_0$	A	Scalar-only basis	App. J
D4c	One physical longitudinal scalar mode χ in a supplied reduced EFT	B	Requires the kinematic reduction plus action, quotient/domain and projector	App. J
D4d	Minimal 3-component orientation EFT	B	Effective parametrisation	4.4
D4e	Three independent physical Goldstone particles	Open / excluded in current scalar basis	Scalar integrability leaves one longitudinal coordinate in the strict map; a larger physical bundle is not derived	4.3
D5a	Fixed director-map tensor test: $h_{ij}^{\text{TT}} = h_{ww} = 0$	A negative result / physical graviton Open	Fixed α, β and tangent director perturbation	8
D5b	Fierz–Pauli form, TT projection, ghost-free counting at linearised level	B	TT gauge; explicit mode-counting from microscopic condensate variables open (OP3 partially resolved)	8, App. Y
D6	Dirac form (structural reconstruction once spinorial fields admitted)	B/Open	Spin(3) representation after restricting to the connected $SO(3)$ stabiliser; its physical-rotation and fermion owners are Open; not from the bare bosonic action	9, 9.2
D7	Derrick scaling in the separately supplied two-derivative S^3 sigma-model target	A-math within supplied model / physical sector and mass Open	Target-space and action assumption; ECT configuration stabilisation, profile and mass remain Open	8.10
D8	Thermodynamic entropy-production programme	Open	Retarded response and positive pairwise noise do not imply monotone von Neumann entropy; derive the reduced channel, entropy functional, and detailed-balance/contractivity hypotheses (OP-Q17)	3.9

#	Statement	Level	Declared model / assumptions / owner	Section
D9	Physical identification of an action-scale candidate with $\hbar$	Level B matching / Open owner	The supplied fixed-core and tension-balanced models define distinct candidates, related by Eq. (5.30) only in the elementary sector when $L_{\text{core}} = L_{*,1}$. CDE- Φ is an acceptance criterion, not an owner or carrier. The calibration $S_0 = \hbar$ is explicit; the profile, compact modulus, cross-sector normalisation and operational owner map remain Open (OP-Q1/OP-Q2)	5.6, 21.2, 23.3
D10	Linearised scalar/metric equation; orientation extension separate	B/Open	Scalar closure conditional; covariant S_n and its stress missing	11
D11	Primordial perturbation inventory: one scalar coordinate in the strict map	A kinematics / B physical EFT	Universality Corollary supplies the coordinate reduction; action, physical mode projector, state, n_s and r remain open (OP3)	14.6
D12	Baryogenesis completion inventory	Open	P4 supplies an ordered-gradient chart, not a derived real-time transition; baryon current/violation, CP coefficient, state, transport and washout are missing, and no ECT value of η_B is calculated (OP7)	16.3
D13	Physical tensor-sector quantisation and UV closure	Open	The orientation/effective-metric sector is a structural candidate, but physical constraints, canonical normalisation, interacting RP/OS reconstruction, and cross-sector action-scale comparison are missing; PES-R is not a quantisation mechanism	8, 38.1, 29
Q-PES	PES-R response–persistence taxonomy	B/Open	Sector dynamics first; then finite-time retention and pairwise records; topology only where owned. Same-channel Wightman/KMS tests and primordial record-kernel constraints remain Open until physical operators, states, common filters and transfer maps are derived	29, 29.13, App. BE.6
D14	Corrected scalar–tensor IR closure; candidate anisotropic tensor proxy	B/Open	Scalar variation exact in the ansatz; the projected-Riemann identity is false and orientation action/projection remain Open	11, App. AC

#	Statement	Level	Declared model / assumptions / owner	Section
D15a	Retarded support $K(\tau < 0) = 0$ for the supplied scalar operator after an initial–boundary orientation	A-math conditional	Quadratic pole proof plus symmetric-hyperbolic reduction; physical time/arrow Open	3.8, F, G
D15b	Open-system identification of bath response kernel / pre-branch origin of causal selection	C	Retarded ordered-branch dynamics provides the structural basis; full identification from Euclidean microdynamics not yet complete	3.9, 38.1
D16a	Compact phase and local $U(1)$ completion	A global / Open local	Winding is exact; bundle, reduced Hessian, photon and matter map are Open	7.2
D16b	Weak algebra and physical chiral $SU(2)$ sector	A algebra / Open physical	$\mathfrak{so}(3) \simeq \mathfrak{su}(2)$ and $SU(2)_{\text{diag}}$ stabiliser are exact; no chiral factor is selected	7.3, 9.3
D16c	$SU(3)_c$ colour sector	P7 Level-B / A conditional / physical Open	Direct injective embedding is excluded by dimension; P7 is the adopted structural bundle and its $SU(3)$ frame reduction is exact conditional mathematics, while physical colour and alternative microscopic origins remain Open. Anomaly cancellation does not select the colour rank (N_c odd only, §9.5)	40
D16d	Neutrino-sector completion	C/Open	Physical chirality, sterile content, mass operator and PMNS matrix are not derived; the historical geometric-scale expression is unselected, and no numerical c_L coefficient is retained. The a_L/c_L propagation algebra is conditional on flavour Hermiticity, identity projection, perturbative coefficients, a self-adjoint spatial domain and norm-bounded positive-nonsingular time kinetic form in the declared broken-phase constant-coefficient vacuum/free domain; $C = -c_L^{\mu\nu} \hat{p}_\mu \hat{p}_\nu$ for the displayed minus-sign derivative operator, and antineutrinos use the declared CPT conjugations	8.9

#	Statement	Level	Declared model / assumptions / owner	Section
D17	ECT entropy-production theorem	Open	Retarded support and a positive pairwise noise kernel do not fix entropy sign; derive the reduced map, state, entropy functional, and contractivity/detailed-balance conditions (OP-Q17)	3.9, 38.1
D18	Physical ordered-domain size	Open	No inequality relative to the Hub-ble radius follows without a formation history, boundary dynamics and physical metric/clock map (OP3)	16.8
D19	Conditional reduced loop coefficients and physical action normalisation in the $U(1)$ -extended EFT	A/B/Open	The fixed-length Cauchy–Schwarz bound and minimum of the supplied positive-tension 1D functional are exact dimensionless mathematics. The d^3y coefficient has the geometry of a codimension-three worldline reduction, not an ordinary spatial vortex worldsheet. CDE- Φ is an acceptance criterion; only an explicit 4D action/state/profile/subtraction/quotient can own the route. A physical action requires $\Sigma_{\vartheta} \mathcal{J}_n$; in the bending-free $n = 1$ branch the Level-B calibration is $\Sigma_{\vartheta} \mathcal{J}_{1,\min} = 2\pi\hbar = h_p$. Carrier, compact modulus, ensemble, profile, stability, Σ_{ϑ} , its possible equality to Σ_{θ} , and universality remain Open; the distinct dimensionless coefficients ι_0^{EFT} , $\iota_{0,\text{bal}}^{\text{eff}}$ and $\iota_{0,\text{bal}}^{(\vartheta)}$ are not interchangeable	5.6, AS
D20	Supplied scalar slope and physical scalar-clock calibration $\hat{c}_* = \sqrt{\beta/(\alpha - \beta)}$, $c_{\text{char}} = c$	A-math conditional / Level-B calibration / cone universality Open	The slope is exact in the supplied scalar principal EFT; $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_{\Phi}$ uses the declared clock, conversion speed and lapse, and $c_{\text{char}} = c$ is the adopted scalar-clock calibration. Photon and tensor cone equality remains Open	3

#	Statement	Level	Declared model / assumptions / owner	Section
D21	Supplied N -scalar Dirichlet Casimir boundary-value problem	A mathematics / Open ECT	The scalar formula is exact inside its supplied toy model; for separately supplied $N = 3$ its ratio to the two-polarisation QED force is arithmetically $3/2$. The physical ECT mode count, material boundary map, stress observable and force are not derived, so no ECT correction is predicted	25.3
D22	Standard Euclidean–Rindler/KMS temperature target	A external / Open ECT	The standard formula follows after a physical metric/cone, state/KMS condition, detector coupling and $S_0 = \hbar$ are supplied; P1–P6 do not derive an observable detector response	25.4
D23a	Bogoliubov transformation for a supplied time-dependent quadratic mode problem	A external / Open ECT	ECT transition action/clock/history, in/out states, particle map and rates are not derived	25.5
D23b	Primordial spectral parameters (n_s, A_s, r) : strict-pullback candidate	A-math conditional / B/Open physical	Ordinary critical projection is blue; the supplied pure- K_4 kernel gives local $n_\chi = 1$ exactly when $\kappa_\chi \sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})$ is scale-independent (separate constants suffice), while exact lower-order corrections and tolerances depend on δ_a, δ_m, ρ (App. AA). The isotropic amplitude scales as $\kappa_\chi^{-1} \mathcal{T}_{\chi \rightarrow \zeta}^2 / \kappa_4$, where $\mathcal{T}_{\chi \rightarrow \zeta} = T_{\chi \rightarrow \zeta} / u_0$. Measure/coefficient ownership, mass protection, RP-compatible physical covariance, observed n_s, A_s, r , transfer, scale-map and continuation are Open	14.1
D24	Gravitational stiffness matching: $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$; direct microscopic relation to ordered-branch parameters remains open	B	Geometric EFT matching; explicit $(u_0, \alpha, \beta) \rightarrow M_G$ derivation open (OP2, OP3)	13.4
D25	Conditional Tolman crossover $\rho_* = \hbar c / (2\pi k_B T_*)$	C/Open	T_* , the physical P4 transition, horizon metric and Tolman control owner are not derived; no universal Planck-scale thickness is predicted	36

#	Statement	Level	Declared model / assumptions / owner	Section
D26	Conditional final-state distinguishability under a supplied global isometry	A mathematics / Open ECT	Excludes only a fully state-erasing complete final channel; thermal marginals and Page curve not fixed	36
D27	Transfer amplitude and response kernel inside optional H_{sh}	C/Open	Supplied stationary passive-scattering completion; background, state, boundary data, flux normalisation and ECT shell identity Open	AZ.4, AZ.5
D27b	Conditional boundary-cell counting lemma (Theorem 36.1)	A mathematics under supplied inputs / Open ECT	Requires explicit coarse-graining, finite local capacity, a physical split and an area-law state-class assumption; m_{σ}^{-1} is not a UV cutoff	36.5
D27c	Integer representations of a supplied compact local $U(1)$ (Theorem 23.1)	A conditional mathematics / Open ECT	BR1 supplies only a compact global phase; local bundle, photon, charge unit, matter representations and physical charge map are Open	23.4
D27d	Post-transition scenario classification and conditional branch-orientation persistence criterion	B	Structural classification of expanding/contracting/accelerating branches; orientation criterion conditional (spatially flat retained closure, positivity, no-ghost, no-bounce); full transition dynamics (OP3.1) and exact background trajectory open	14.2
D28	Effective shell Hamiltonian structure (form, not coefficients)	C	GL criticality + Rindler geometry; coefficients open	AZ.5
D29	Optional H_{sh} vertex $\mathcal{L}_{\text{int}} = -\frac{g}{2}\delta\chi_{\text{sh}}(\partial\varphi)^2$	C/Open	Explicit supplied operator; g may vanish and is not selected by ECT; other vertices remain possible	AZ.5
D30	Bare scalar-ratio identity: $w_{\phi}^{\text{bare}} \geq -1$	A inside supplied stress / ECT Open	Uses $K_{\phi}^{\text{phys}} = \bar{M}_{\text{Pl}}^2 \omega \dot{\phi}^2 / 2$; $\dot{F}, \ddot{F}$ leave observable w_{eff} unconstrained	16.1

#	Statement	Level	Declared model / assumptions / owner	Section
D31	Ordered-branch age quadrature and supplied two-slope estimate	A/C/Open	Quadrature and named-action orbit are exact internally; $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368678$ for the disclosed state. Only after the explicit calibration $H_{2s}(0) = H_0^{\text{cal}}$ may this be written as $H_0^{\text{cal}}t_0$. State data and one dimensional calibration are legitimate; microscopic completion and the physical clock/metric/front map remain Open	16.4
D32	Hubble inference	C/Open	Conditional $H(z)$, distances and an imported-recombination r_s are computed for one action. The official 15-point BC03 covariance calibrates its common scale to $H_0 = 68.59 \pm 3.95$ without distinguishing it from the matched control; a complete ECT recombination-to-distance-ladder likelihood is still missing	16.5
D33	JWST time budget, abundance and morphology	C/Open	Conditional high- z ages are computed at nine prespecified nodes; all nine reduce $t(z=10)$ relative to the declared control, with the most negative sampled shift -10.491167% . This is not a continuous-family or JWST no-go; abundance/morphology require the end-to-end forward model	16.5
D34	Force-law dimensionality $d_{\text{force}}(x): 3 \rightarrow 7/3 \rightarrow 2$	C	Conditional algebra for the HRC-0 shape; old diagnostic is not a retroactive microscopic derivation	16.8
D35	Scalar-model retarded cone and domain-of-dependence bound	A in supplied scalar operator / Open physically	Symmetric hyperbolicity fixes support for the displayed scalar operator; common matter, photon, fermion and tensor cones require their independent actions and observable metric	12.6, G
D36	Conditional phase/gauge cone inheritance; physical tensor cone Open	A/B/Open	The scalar cone is derived; phase/gauge maps are conditional and the tensor principal symbol is missing	12.6, 4.7

#	Statement	Level	Declared model / assumptions / owner	Section
D37	Proton stability: three selection-rule patterns (I/II/III)	C	Conditional on gauge/flavour completion, especially colour sector and baryon-number selection rules; OP-proton	44

Reading the mixed-status tests table. The entries in Table 71 do not have one common logical status. Structural theorems, conditional closure tests, external benchmarks and Open observable programmes are labelled separately. In the cosmological sector one supplied two-slope action/state now has active conditional age, $H(z)$, distance, high-redshift clock, imported-recombination r_s , restricted growth, Weyl/ISW-carrier and peculiar-velocity diagnostics; the local cluster result is the exact monotone-map no-go only. None is a unique P1–P6 cosmology, full Hubble likelihood, JWST abundance/morphology, gauge-complete growth, projected ISW/lensing spectrum, merger model or finite-body PPN prediction. Each future completion is falsified by failure of its background, perturbation, photon/matter-metric, structure or local-gravity gate.

Table 71: Status-sensitive tests and discriminants of the ECT framework. F1 and F2 test Open closure programmes; F3 is an effective scalar–tensor consistency check; F10 is an Open future cluster-lensing specification; F12 is a conditional $\mu_{g,0}$ source-model test not yet performed.

#	Candidate test / statement	Observable	Target experiment	Verdict if falsified
F1	Exploratory environment/history dependence of $a_{M,\text{eff}}$ (Section 17.1)	$a_{M,i}$ versus declared environment and history covariates; no sign or 10% amplitude is presently predicted	SPARC $\times$ 2MRS/SDSS	a null result rejects or constrains the tested Ξ ansatz; a correlation does not by itself establish the scalar source/screening chain or ECT
F2	Canonical scalar screening and composite-body charge (Open)	PPN observables and fifth-force bounds from a solved extended-body profile	Cassini, MICROSCOPE, LLR	failure excludes the stated screening completion; the algebraic frozen- F GR limit alone is not a dynamical screening result
F3	Physical GW cone, polarisations and normalisation	Joint speed interval [12]	GW/GRB event	Open kill gate: at fixed α, β and tangent-director linear order about homogeneous $n = w$, the fixed map has $h_{ij}^{\text{TT}} = h_{ww} = 0$; quadratic/nonlinear routes remain parametric/Open, and no physical ECT tensor principal symbol or c_{GW} is yet derived
F4	Separately declared director–matter force or spin completion	Composition-dependent acceleration from a supplied source/body-force map, or spin response from an independent spin-dependent Hamiltonian	MICROSCOPE [49]; spin-sensitive searches [50]	Each dataset constrains only its named completion. The homogeneous timelike vector-density term alone supplies neither spin splitting nor a source-generated Eötvös force
F5	One action-owned background and physical metric must reproduce late-time distances, expansion and growth simultaneously	Externally inferred background and perturbation likelihoods	BAO, supernovae, chronometers, RSD and lensing	Reject the tested completion if the joint likelihood fails

#	Candidate test / state- ment	Observable	Target experi- ment	Verdict if falsified
F6	Stable primary texture requires a derived stabilising operator beyond the two-derivative action	Profile and spectrum computed from that completed action; no preset mass target	Theory-specific soliton and collider searches after owner closure	Reject the tested stabilised completion, not the π_3 sector labels
F7	Scalar unit-slope and sector-matching completion (<i>currently non-operational</i>)	No LIV observable is presently owned by $\alpha = 2\beta$: that relation fixes only $\hat{c}_* = 1$ in the supplied scalar common-length operator. An operational test requires a physical scalar excitation and action, clock/lapse, source/detector map, and an independently derived relation to the measured sector	Future sector-specific propagation tests only after owner closure	Failure would reject the named scalar/sector-matching completion. Present photon/GW bounds do not test $\alpha = 2\beta$; gauge, fermion and tensor cones remain independent Open owners
F8	ECT boundary-response completion (Open; Section 25.3)	Force and sign obtained from a derived reduced bundle, material boundary operator and renormalised stress tensor; no $3/2$ ratio is predicted	Future experiment selected only after the boundary map is derived	Rejects the tested completed boundary model, not the present P1–P6 framework
F9	Redshift dependence of fitted a_M (Open)	Test whether one matched baseline remains adequate across redshift; no map from the cosmological scalar background to the galactic fit scale is derived	JWST, SKA high- z	Constrains the tested redshift-regression family, not the scalar background or ECT as a whole
F10	Action-matched cluster-lensing response	Derive a physical, possibly nonlocal map from independently constrained baryons to convergence; the current local proxy only preserves $\arg \max \Sigma_b$	Weak-/strong-lensing plus baryonic maps	Constrains only the prespecified future response; current toy maps are not morphology evidence
F11	Graviton polarisation: two TT modes in the completed linear closure	Polarisation decomposition of stochastic GW background	LISA, PTA, 3G detectors	Observation of robust extra scalar/vector GW polarisations would disfavour a supplied massless Fierz–Pauli completion, but not the current fixed-map negative result
F12	Conditional HRC-0 point-source $d_{\text{force}}(x)$ diagnostic: $3 \rightarrow 2$	Synthetic slope identity for $\mu_{g,0}$ with $p_N = 2$; an extended-galaxy test requires measured $p_N(r)$ and has not been performed	SPARC, THINGS, next-generation kinematic surveys	Constrains the supplied point-source diagnostic only; an extended-source HRC field test remains future and is not an ECT-wide falsifier

#	Candidate test / state- ment	Observable	Target experi- ment	Verdict if falsified
F13	ECT makes no graviton-mass prediction; $m_g = 0$ belongs only to the conditional Fierz–Pauli completion	Mass-dependent dispersion of GW signal	LISA, LIGO O5+	Would disfavour only the supplied massless Fierz–Pauli completion; the live theory does not yet own a physical tensor mass operator
F14	Particle-halo scope of the Level-C HRC no-halo diagnostic	A confirmed particle species shown to account for the galactic mass discrepancy on galaxy scales	Direct detection, collider, indirect and halo-mapping evidence	Disfavours the no-particle-halo interpretation of the Level-C HRC data diagnostic; it does not by itself falsify ECT or exclude additional sectors
F15	Trace-free shift covariance in the named constrained completion	$V + C_I$ is invariant under $V \rightarrow V + C_0$, $C_I \rightarrow C_I - C_0$	Direct variation, Bianchi reconstruction and constant-shift tests	The named completion is rejected if its reduced equations are not shift covariant; ECT membership remains Open until the Φ -medium action is derived
F16	Regular two-slope background proof of class	One stable de-Sitter root and a nonsingular connected benchmark history	Independent symbolic residuals, two-solver integration, stability and physical-metric/PPN gates	The named completion is rejected by a failed regularity/stability replay; its survival would still not prove ECT ownership or vacuum-offset decoupling

Five consequential tests and completion gates (cross-reference). For orientation, the following five handles carry the most weight in discriminating ECT from neighbouring frameworks. They are not new content but pointers into the detailed candidate-test table above and into the corresponding sections.

- T1. Environment/history dependence of $a_{M,\text{eff}}$** (Open; F1). The law $a_{M,\text{eff}} = a_{M0}\Xi$ is an exploratory hypothesis; neither non-universality, systematic variation, sign, nor amplitude is derived. A null held-out correlation constrains the tested Ξ ansatz, whereas a detected correlation still requires a microscopic ECT matching calculation.
- T2. DM-deficient UDG stress test** (Level C / Open). Under the declared proxy assumptions, DF4/FCC224 stress or disfavor the tested monotonic density-only Ξ ansatz; the small, non-selection-neutral sample does not exclude the whole function class. A completion must also accommodate the matched DM-rich control NGC 5846-UDG1 and the observed $\sim 0.6\text{--}1\%$ incidence. This is a closure stress test, not a solved prediction or ECT-wide exclusion.
- T3. Correlated slip / S_8 programme** (Open / external proxy). A correlated pattern across $\mu(k, z)$, $\eta(k, z)$, $\Sigma(k, z)$, RSD, weak lensing, cluster lensing, and CMB lensing could test a future specified S_n completion. The present manuscript derives neither this channel nor its sign and amplitude; therefore a null pattern cannot falsify ECT. It would constrain only the tested proxy or the specified future completion.
- T4. Sector-specific LIV completion programme** (Open; F7/ND10). The supplied K^{AB} restricts only the leading second-order scalar operator. ECT currently derives no photon, neutrino, fermion or tensor LIV operator, coefficient, common suppression scale or observable map; no cross-sector sign or coefficient pattern is therefore predicted. A future common-descent completion could be tested for correlations only after each sector’s action and matching map are derived. Incompatible data would falsify that named completion, not P1–P6.

T5. Vacuum-offset action and radiative completion (Open ECT descent). The fixed orientation determinant is exact algebra, but neither the classical physical trace-free action nor quantum non-renormalization follows from it. The explicit constrained completion supplies a classical shift-covariance target. The full reduced Hessian, measure, constraints and counterterms must be derived before any loop decoupling statement is made.

The S_8 and weak-lensing sector. This sector is neither solved nor exempted by the current architecture. It requires the same gauge-complete perturbation and photon-metric owners as growth and lensing. Agreement or disagreement can be assessed only after that completion is specified.

Baryon-number observables. A future owner-complete model could realise exact low-energy protection, allow $\Delta B = 2$ while forbidding $\Delta B = 1$, or allow proton-decay operators. This conditional taxonomy is a Level C/Open physical-completion discriminant, not a prediction of P7 or of the present ECT framework. The current theory supplies no physical baryon current, violating vertex, matching scale or branching map; an observation would constrain the corresponding named completion (Section 44).

Candidate observational channels and completion gates. Several observational programmes expected to deliver data within the next decade provide external data that could constrain future ECT completions. Table 71 already separates conditional tests, external gates, fitted diagnostics, and Open observable programmes. Table 72 extends that status map with future observational channels. Rows marked Open or C are not present ECT predictions, and no mission-ready forecast is claimed. Five observational windows are useful completion targets: full-sky CMB polarisation (LiteBIRD [131]), precision ground-based CMB lensing and growth measurements (Simons Observatory [132], Euclid, Rubin/LSST), low-frequency space-based gravitational-wave detection (LISA [133]), mesoscopic quantum tests in levitated-nanoparticle optomechanics and related near-term gravitational-decoherence / gravity-induced- entanglement protocols (Section 32, Section 32.5), and asymptotic IR/soft channels including gravitational-wave memory (Section 35.4).

Table 72: Candidate observational channels and completion gates. Rows marked Open or C are not ECT discriminants or mission-ready forecasts; they become tests only after the named physical owners and forward maps are derived.

#	Observable	ECT statement	Level	Mission / programme	Verdict if data conflict
ND1	Primordial tensor signal (large-angle CMB B-mode polarisation)	A future owner-complete early-Universe model must supply the ordering background, scalar and tensor actions, state, freeze-out and transfer. The current ECT formulation derives neither an inflationary phase nor a quantitative primordial tensor spectrum	Open	LiteBIRD, Simons Observatory, BICEP Array	A robust tensor spectrum would test a named condensate-based early-Universe completion after that completion supplies its tensor action, state and transfer map; no result can exclude every presently Open completion without those owners

#	Observable	ECT statement	Level	Mission	Verdict if conflict
ND2	CMB lensing and growth consistency	The scalar background can be confronted with growth data. A future orientation channel is an external Level-C proxy until a covariant S_n , projection, and coefficients are derived	B / C / Open	Simons Observatory, Euclid, Rubin/LSST	Growth–lensing data constrain a declared non-GR response completion only after its covariant perturbation action, physical metric, coefficients and likelihood map are supplied. No ECT-specific precision threshold is derived here; screened, decoupled or zero-coefficient limits remain admissible until ECT selects one physical sector
ND3	GW propagation speed and possible dispersion	A future ECT tensor completion must derive its own principal symbol, physical GW cone and dispersion law before comparison with data. The present scalar cone is not a GW prediction, and no LISA-band magnitude or suppression law is derived here	Open	LISA	A clear leading-order mismatch constrains only a future tensor completion; the present scalar cone does not generate a GW prediction
ND4	GW polarisation content	At fixed α, β and tangent-director linear order about homogeneous $n = w$, the declared fixed map has no TT tensor content (Level A inside that map). A separately supplied Fierz–Pauli completion would have two TT modes only conditionally; quadratic, nonlinear, composite and additional scalar/vector routes remain Open	A in fixed map / B conditional / Open completion	LISA, PTA, 3G detectors	A confirmed extra polarisation would reject only the tested completion; the current fixed-map linear route has no physical GW sector, while untested nonlinear routes remain Open
ND5	Standard-siren determination of H_0	A future ECT completion must predict the siren distance–redshift relation from its physical metric	Open	LISA standard sirens	Any proposed completion is tested against the same siren likelihood; no shift is currently predicted

#	Observable	ECT statement	Level	Mission	Verdict if conflict
ND6	Director coupling: spin-sensitive completion	For a homogeneous time-like spurion, $\mu_5 u_\mu \bar{\Psi} \gamma^\mu \Psi$ is a number-density term and produces no spin splitting [Eq. (7.41)]. Precession requires a separately specified spin-dependent operator—axial, derivative/Pauli, or an inhomogeneous-vector completion whose nonrelativistic reduction contains spin-orbit, curl or Pauli structure—and its detector Hamiltonian. No ECT coefficient or frequency window is derived	B conditional operator classification / Open observable	Spin-polarised torsion balance, co-magnetometer	A null or positive result constrains only the declared spin-dependent completion; it does not test the homogeneous vector-density term by itself
ND7	Director coupling: composition-dependent force completion	The fixed preferred-direction spurion does not define a source-generated fifth force. A prediction requires a field equation or prescribed inhomogeneous profile, source and test charges, propagator/range, microscopic-to-body composition map and screening/environment dependence. No ECT Eötvös-parameter window is derived	Open	MICROSCOPE final result [49]	Use the measured result as an external constraint on a declared force completion. The former $\eta \sim 10^{-15}$ target is withdrawn as an output of the fixed vector spurion
ND8	Diósi–Penrose benchmark and the open ECT gravitational record channel	The imported symbolic functional is given by Eqs. (26.12)–(26.13). ECT has not yet supplied the physical source projector, state-dependent noise kernel, or absolute normalisation needed to derive this rate; the minimal free scalar vacuum is incompatible with a universal D–P linear-time law	External source-model formula / Open ECT rate	Levitated-nanoparticle optomechanics; MAQRO-class space missions [134, 135, 136]	measurements constrain the imported D–P benchmark and, after a declared spectral mapping, any future ECT kernel; they do not presently test an ECT-derived D–P rate

#	Observable	ECT statement	Level	Mission	Verdict if conflict
ND9	Gravity-induced entanglement (BMV-class)	The present action does not identify the physical gravity mediator or its record channel: the tensor vertex is missing, the scalar route is parametric, and the complete protocol map is NOT IDENTIFIABLE (Section 32, Operational criterion §32.4); inversion of an external collapse formula does not define an ECT entangling or decoherence threshold	Open	BMV-class near-term proposals [137, 138, 139, 140]	BMV data constrain a future declared mediator completion; with the current missing owner, neither a positive nor null result is an ECT-specific falsifier
ND10	LIV / superluminal sector	No photon or tensor LIV operator, coefficient, common suppression scale or time-of-flight map is owned by the present action	Open	FERMI-LAT, CTA, IceCube, LHAASO [141]	Data constrain the corresponding supplied photon/tensor operators; there is no current ECT operator unification to falsify
ND11	Gravitational-wave memory as persistent asymptotic condensate shift	One candidate interpretation of gravitational memory [142] is a persistent rearrangement of an asymptotic ordered-branch orientation, $n_A^{(0)} \rightarrow n_A^{(0)} + \Delta n_A$ (Section 35.4); the physical tensor/memory owner and amplitude are NOT IDENTIFIABLE	Open	LISA, 3G ground-based detectors, PTA memory searches	Memory data constrain only a future declared tensor/IR completion; quantitative closure is required before this row is an ECT test
ND12	Black-hole information: Page-curve-compatible evaporation	ECT's critical ordered-phase shell picture is structurally aligned with the islands/QES language (Section 36.3, Table 92); a Page-curve turnover is architecturally admissible, not derived	C (structural)	BH echo/ringdown searches [143]; analogue-lab horizon programmes	Direct or indirect evidence for fundamental information loss in BH evaporation (as opposed to reduced-state apparent mixedness) would falsify a future globally unitary strong-field completion, not P2 itself; a confirmed non-Page-curve evolution of observer-accessible radiation would also falsify it

#	Observable	ECT statement	Level	Mission	Verdict if conflict
ND13	Dark-matter-deficient ultra-diffuse galaxies (DF4, FCC 224; DF2 ambiguous)	DF4 and probably FCC 224 exhibit tracer kinematics consistent with baryon-only dynamical mass in the probed region; the current conditional HRC galactic diagnostic (eq. 17.71) at the baseline $a_{M,\text{eff}} = a_{M0}$ tends toward the supplied HRC low-acceleration branch for these systems and thereby over-predicts the observed line-of-sight dispersion by a factor of order three. The minimal density-only Open ansatz (eq. 17.18) for Ξ does not naturally accommodate both the low-dispersion stress-test objects and matched DM-rich controls such as NGC 5846-UDG1 within a single simple environmental scalar law. The presently known sample is small and not selection-neutral. A full diagnostic treatment is given in Appendix AQ. Possible theory-level directions for a future completion (an action-owned finite-body scalar-metric response built from eq. 13.38, history-dependent extensions of Ξ, and orientation-sector coupling) belong to structures already present in the broader framework, but none is presently derived for the galactic closure; addressing them is linked to OP3. ΛCDM with bullet-dwarf formation [144, 145, 146] reproduces the observed dispersions through partial dark-halo stripping and remains a viable comparator	C / Open	Archival: DF4 [119], FCC 224 [120, 121], DF2 [122, 123], NGC 5846-UDG1 [124]; future: isolated-field DM-deficient searches, IFU $\sigma(R)$ profiles, matched DM-rich controls	Observation-side: if the sample expands and the proxy-level tension persists or worsens under same-tracer same-aperture matched analysis, the tested density-only Open Ξ ansatz is disfavoured and an action-owned finite-body scalar-metric response derived from $F(\phi)$, a history-dependent Ξ, or an orientation-sector completion becomes necessary. Theory-side: a derivation that maps the scalar action, its source and body charges, and the solved metric potentials onto the conditional HRC gravity layer is required before any ECT rescue can be claimed

A stochastic gravitational-wave background is an Open target for a future tensor completion. The ordering event alone does not derive a TT source, sign or spectrum $\Omega_{\text{GW}}(f)$. A future ECT derivation of the primordial scalar spectrum could further connect this programme to constraints on n_s , running, and

non-Gaussianity.

19.5 Status table and open problems of the macroscopic branch

Role of the final status table. The present table is the final status board of the macroscopic branch. Unlike the global checklist of Section 19.3, which audits framework-wide consistency targets, and unlike the maps of Section 19.4, which record logical derivation structure and falsifier logic, the table below gives the compact present judgement on the main macroscopic outputs of Part II.

Postulate-level reading. The table should be read against the same layered architecture used throughout the paper: P1–P3 control the microscopic Euclidean medium and its EFT consistency, P4 adopts the physically relevant ordered branch and supplies its macroscopic analysis chart, while its complete stationary/dynamical owner remains Open; P6 permits the non-uniform macroscopic ordered responses needed for the cosmological and galactic ϕ -programmes. The labels “Level A”, “Level B”, “Matched”, and “Open” refer to that organised framework rather than to isolated claims.

A separate numerical/regime-level inventory is now collected in Section 27. That inventory should be read together with the present status table: the environmental-decoherence entries are closure-level consistency checks, the photonic entries are weak-locking consistency statements, and the mesoscopic gravitational entries are imported Diósi–Penrose benchmarks and targets for a future ECT record-channel calculation. They are not ECT-specific predictions until the physical ECT operator, vertex, state, and noise/response kernels have been derived.

Status table and open problems of the macroscopic branch

Statement / claim	Status	Layer	Basis / OP
<i>Metric structure and tensor sector</i>			
Conditional coarse-graining regime (11.1)	Level B/Open P4-pole owner	Definition after $m_{P4,pole}$ is supplied; its existence and value are Open	EFT
Uniqueness of the supplied scalar principal-tensor form (13.8)	Level A con- ditional math- ematics / physical met- ric Open	Conditional EFT	Thm. 3.2
Lorentzian scalar principal tensor for supplied $\beta(\beta - \alpha) < 0$ (or $\alpha > \beta > 0$ on the healthy $\beta > 0$ branch)	Level A math- ematics con- ditional on P4 / physical metric Open	Conditional EFT	P4 adopts the ordered branch, but not the coefficient sign, sta- tionary owner, clock or cross- sector metric
c_{GW}	Open / exter- nal gate	Tensor action absent	GW170817
Massless tensor propagation	Open	Conditional FP target	Tensor completion
Tensor dispersion and propagation delay	Open	Requires an explicit tensor operator, co- efficient and scale	External timing gate
Fierz–Pauli tensor sector (supplied com- pletion target)	Open	Structural/ closure	EFT coarse-graining; exact full-ECT tensor closure still open

Statement / claim (cont.)	Status	Layer	Basis / OP
Two TT polarisations	Open	Conditional FP target only	Physical tensor completion and constraint analysis
<i>Source structure and coupling</i>			
$\partial_A T^{AB} = 0$ on P3 shell (Noether)	Level A conditional	Inherited only from its owning P3 action	P1+P3
Window-averaged P3 translation identity (13.21)	Level A mathematics conditional / physical P4 conservation Open	P3 Noether identity	Divergence vanishes only for a P3 on-shell ensemble with compatible translation-invariant window, boundary terms and averaging domain; no physical P4 or covariant gravitational- source conservation follows
Physical matter source vertex	Open	MISSING VERTEX only for the fixed- α, β tangent- director linear route; quadratic/ nonlinear routes parametric/Open	bounded record-channel audit
Matter universality A4	Level B	Closure	OP-matter
Orientation EMT $T_{AB}^{(n)}$	Open	Missing co- variant action	OP2, OP-c1
<i>Gravitational stiffness and Newton constant</i>			
M_G in a supplied tensor EFT	Conditional	Standard target normal- isation	Tensor completion
$G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$: matching	Level B	Closure	Comparison with GR
$M_G = \bar{M}_{\text{Pl}}$: numerical matching	Matched	Calibration	Observational identification
$M_G^2 \sim c_M \kappa_n$	Open hypothesis	Dimensional candidate only	Derive tensor Hessian/residue
$(u_0, \alpha, \beta) \rightarrow \kappa_n \rightarrow M_G^2 \rightarrow G_N$	Open	Open	OP-grad, OP2–OP3
<i>Nonlinear closure</i>			
Closure required (structural)	Level A	Structural	Geometric branch
ϕ -first scalar–tensor action (13.35)	Level B	Closure	Scalar ansatz; orientation action separate/Open
Corrected scalar metric and field equations	Level A within Level- B ansatz	Algebra	One consistent placement of F ; minimal Jordan matter con- served
Candidate anisotropic tensor $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$; orientation stress	Open	Missing action-level input	Covariant n^A action, variation, constraints, projection and on- shell identity required
Conditional constant-background GR- form equation (13.44)	Level A within Level- B ansatz	Structural conditional	Does not prove the scalar root, dynamical reachability or a decoupled local response

Statement / claim (cont.)	Status	Layer	Basis / OP
$G_{\text{div}} = G_*/F$ in the fully divided equation	Level A within Level-B ansatz	Algebra	Exact equation coefficient, not a response observable; no orientation term is created by division
Main amplitude coordinate ϕ_u and $F = u/u_\infty$	Level B	Structural	$u \leftrightarrow \Delta_K$ bridge Open
Underlying EFT parameters (Z_u, W, β_ϕ)	Open	Open	OP-grad
<i>Weak-field and phenomenology</i>			
Frozen- F weak-field $\rightarrow$ Newtonian	Level A/B conditional	Structural	Assumed frozen background
Canonical screening mass and composite-body charge	Open	Not identifiable yet	Einstein-frame Hessian + body profile
Solar-System PPN passage of the proposed screening route	Open	Open test	Cassini/MICROSCOPE/LLR calculation
Modified Poisson closure (13.52)	A within Level B scalar ansatz	Schematic conditional equation	Body/orientation/source closure Open
<i>Cosmology and galactic</i>			
Pre-Lorentzian Euclidean cosmological scenario	Level B	Structural	Structurally motivated by P1–P4
Primordial n_s, r : ECT-native derivation	Open	Open	OP3
Conditional bare scalar-ratio identity $w_\phi^{\text{bare}} \geq -1$ for $\omega > 0$ and $K_\phi^{\text{phys}} + U > 0$; physical expansion history unconstrained by this identity	A in supplied stress / Open ECT	Conditional model	OP3
BTFR slope 4 within the supplied HRC deep branch	A in-model / B after bridge / C data	Exact augmented-model algebra; bridge corollary conditional	Existence/source/force matching Open
Baseline $a_{M0} \sim cH_0/(2\pi)$	Level C / origin Open	Matched input	Coefficient not first-principles
Environment/history law Ξ for $a_{M,\text{eff}}$	Open	Exploratory hypothesis	Sign and amplitude not derived
SPARC HRC algebraic scale diagnostic	Level C / benchmark	HRC-0 median $a_M/a_{M0} = 1.222$; HRC-3 0.921 over 165 galaxies	Not ECT-specific evidence
Sakharov-compatible baryogenesis framework (ordered-medium route; quantitative completion open)	Level C/Open	Conditional completion inventory	Physical baryon current/vertex, CP coefficient, transition state and transport owners absent; no numerical ECT result is retained

Statement / claim (cont.)	Status	Layer	Basis / OP
Conditional late-source routes	A in named completions / ECT Open	The trace-free branch carries a free state datum C_I ; the ordinary two-slope branch gives a stable late-de-Sitter proof of class	Neither the physical source nor $w(z)$ is selected by the frozen ECT action
Candidate cosmological / black-hole singularity as ordered-branch breakdown	Level B/Open	Scenario and programme	No derived breakdown surface, replacement interior, or microscopic transition
<i>Open problems (macroscopic branch)</i>			
OP2: microscopic derivation of M_G from ordered-branch parameters	Open	Open	Full EFT
OP3.1: ordering mechanism and derivation of effective ϕ -closure parameters from bare P3	Open	Open	OP-grad (functional forms derived); OP3.2 partially clarified by the conditional persistence criterion
OP-gauge: explicit two-TT-polarisation proof in full ECT language	Open	Open	OP-gauge
OP-matter: first-principles matter universality	Open	Open	OP-matter
OP-c1: covariant orientation action, constraints, stress, projection and coefficients	Open	Missing action-level input	OP-c1
OP-screen: screening mechanism from bare P3	Open	Open	OP-screen
OP-CONST: protection of dimensionless constants (α, μ) on the cosmological branch	Open	Structural consistency	Conditional selective-coupling protection route: selective ϕ -dressing; two-variable separation $u_{\cos}$ vs u_0, ϕ_0 (OP-grad); charge-neutral drift mode + frozen matching (conditional); verified anchors

Part III

Quantum Sector: Coherent Branch

The manuscript organises two candidate observational programmes around the ordered-medium hypothesis. The first class—developed in Part II (Macroscopic Physics)—concerns observables that are insensitive to the global phase of the effective order parameter: long-wavelength metric fluctuations, gravitational stiffness, cosmological dynamics, and galactic phenomenology. In that conditional programme the phase is averaged away and the proposed effective description is geometric; the physical metric, source and observable maps remain separately status-labelled.

The second class—developed here—concerns observables that *remain sensitive* to the global phase structure, winding topology, loop configurations, and coherent correlators of the ordered condensate. This defines the supplied coherent-branch programme studied in Part III; it does not by itself establish a physical coherent sector.

Origin in the branching structure. The two-branch architecture is organised in Part I (Section 6). P4 supplies the ordered-gradient chart used by the geometric programme, whereas BR1 separately supplies the compact coherent phase. They are potentially compatible effective programmes within the same ordered-medium hypothesis, but their common microscopic descent from one physical action and state remains Open. The geometric branch was developed first because it requires less structural machinery—its observables are already accessible at the level of the coarse-grained metric EFT. The coherent branch, by contrast, requires the full phase-sensitive effective description and the action-scale logic prepared in Part I (Sections 6.3 and 5.6). Its development is therefore deferred to the present chapter.

What Part I already prepared. The following structural ingredients are available at the entrance to Part III:

- (i) the ordered phase admits a genuine phase-sensitive sector, distinct from the geometric branch inside the supplied broken-phase EFT;
- (ii) topologically quantised winding sectors exist in the coherent order parameter as Level A topology inside the supplied compact-phase closure BR1;
- (iii) supplied fixed-core and positive-tension loop models define the distinct candidates t_0^{EFT} and $t_{0,\text{bal}}^{\text{eff}}$; winding alone derives neither coefficient, neither owner map to the operational S_0 , nor $S_0 = \hbar$ (Appendix AS);
- (iv) the scalar branch admits a conditional Poincaré-form principal operator via $x^0 = w$, while physical clocks, metric and quantum state remain Open; Noether identities hold for each declared action.

Part III takes these as its structural starting point and develops them into wave dynamics, canonical structure, the Hilbert-space bridge, exchange-sector topology, Dirac structure, vacuum response, decoherence, the quantum–classical boundary, Born-type probability, entanglement, topological/dark-sector ontology, black-hole thermodynamics, and the analogue-laboratory programme.

What Part III does not import. Part III does not assume:

- that $S_0 = \hbar$ has already been established (this is one of the central tasks of the present chapter);
- that canonical commutation relations or the Born rule hold (their physical operators, normalisation and probability measure remain independent reconstruction targets);
- that decoherence, measurement, or the arrow of time have been addressed (these belong to the open-system development of the coherent branch);

- any result from standard quantum mechanics or quantum field theory as an ECT prediction merely because its standard algebra is reproduced after the corresponding state/operator inputs are supplied.

Methodological standard. Throughout Part III the same Level A/B/Open classification is maintained as in Parts I and II. Level A is exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. Results conditional on supplied P4, BR1, a state, action or operator completion remain labelled as such; BR1 is not called a derived consequence of P1–P6. Level B denotes results that additionally require stated closure assumptions, effective approximations, or matching identifications. Open problems are flagged explicitly and collected in the final status map (Section 38).

The correct reading of Part III is therefore neither “full derivation of quantum mechanics from ECT postulates” nor “purely conjectural sketch”. It is a status-separated reconstruction programme: some mathematical substructures are explicit, while the physical state/operator, probability, outcome and detector owners remain conditional or Open.

Chapter outline. Part III contains nineteen sections. The eleven core themes are listed below; eight dedicated bridge and test modules treat source-model anchors (Section 27), hybrid consistency (Section 28), PES-R (Section 29), Bell correlations (Section 31), gravity-induced entanglement (Section 32), the terminal mediator taxonomy (Section 33), infrared completion targets (Section 35), and analogue tests (Section 37).

- (1) **Scope, inputs, and logical programme** (Section 20): entry into the coherent branch, structural imports from Parts I–II, and the derivation roadmap.
- (2) **Coherent phase dynamics and the pre-quantum action scale** (Section 21): phase field, loop sectors, winding classification, and the distinguished scale S_0 .
- (3) **Euclidean-to-Lorentzian wave kinematics** (Section 22): common-length hyperbolic parametrisation, wave dynamics, extremal action, and conservation laws.
- (4) **Hilbert-space structure, reflection positivity, canonical algebra, and unitarity** (Section 23): the central bridge from the Euclidean coherent structure to the quantum-mechanical formalism.
- (5) **Euclidean path-dependence, quantum nonclassicality, statistics, and Dirac structure** (Section 24): structural origins of interference, tunnelling, exchange sectors, and the spinorial route.
- (6) **Quantum vacuum structure and universal response** (Section 25): Casimir, Unruh, particle production, and the unified coherent-vacuum logic.
- (7) **Open quantum systems, decoherence, and irreversibility** (Section 26): influence functional, arrow of time, entropy production.
- (8) **Quantum–classical boundary, probability, and measurement** (Section 30): reverse $O(3) \rightarrow O(4)$ transitions, Born-type interpretation, and the status of measurement theory.
- (9) **Topological sectors, defects, and the dark sector** (Section 34): the wider state space beyond the smooth coherent branch.
- (10) **Black-hole thermodynamics and the information problem** (Section 36): horizon thermality as a coherent-vacuum phenomenon.
- (11) **Status map and open problems** (Section 38): final assessment of Part III.

20 Scope, Inputs, and Logical Programme of the Quantum Sector

Part III develops the coherent branch of the ordered condensate—the branch in which the global phase structure, winding sectors, and loop configurations of the effective order parameter remain physically active. Before proceeding to the specific constructions—phase dynamics, wave kinematics, the Hilbert-space bridge, vacuum response, decoherence, and measurement status—the present section identifies the primary variables of the coherent branch, states the main derivation targets of Part III, and records what has and has not been established at the point of entry.

Throughout Part III the same Level A/B/Open classification is maintained as in Parts I and II: Level A is exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. Results conditional on supplied P4, BR1, a state, action or operator completion remain labelled as such; BR1 is not a derived consequence of P1–P6; Level B denotes results that additionally require stated closure assumptions, effective approximations, or matching identifications.

How ECT changes the status of quantum reconstruction targets. ECT supplies structural routes to several kinematic ingredients, with distinct closure levels:

- supplied free Gaussian measures admit the standard OS Hilbert-space reconstruction only after their complete covariance, time-reflection, RP, symmetry/regularity, locality/clustering and observable-algebra package; the interacting ECT measure and physical identification remain Open;
- global unitarity follows only where the complete applicable corrected OS-II package and reconstruction are established; reflection positivity is necessary but not sufficient, and reduced channels can be effectively non-unitary;
- independently of OS-II or dephasing, ordinary projector Gleason represents a supplied normalised, noncontextual, countably additive projector measure on a complex Hilbert space only for $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses, while state/preparation, event, probability origin, outcome and update remain Open;
- decoherence can suppress interference and stabilise records, but pointer selection, one realised outcome, and the update theorem remain Open.

No universal $\Gamma_{\text{loop}} \sim 1$ signature or collapse law is claimed.

Three distinct questions about quantumness in ECT. The question “why quantum mechanics?” is not a single problem; it decomposes into three logically independent sub-questions, each addressed by a different part of the ECT programme:

- Structural reconstruction.** *Why does a quantum-theoretic structure exist at all?* ECT provides a structural reconstruction programme for quantum theory through its Euclidean functional measure together with the reflection-positivity / Osterwalder–Schrader reconstruction route (Section 23.1). For a supplied reflection-positive Euclidean measure satisfying the OS hypotheses, the OS theorem constructs a Hilbert space, Hamiltonian and semigroup/unitary continuation. The relevant positivity checks are Level A mathematics for specified free quadratic kernels; identifying those kernels with physical ECT states and extending the construction to the interacting coherent sector remain Open.
- Candidate loop scales and an operational action unit.** *Why can a finite pre-quantum action coefficient appear?* In a supplied compact-phase effective closure (Section 21), integer winding is exact and the fixed-length loop bound is calculable. A fixed-core coefficient requires a supplied loop length and profile. Alternatively, a supplied positive-tension 1D model selects a finite length and coefficient by minimisation. Neither route is derived from P1–P6 or identified as the owner of the operational S_0 .

- (iii) **Physical identification.** *Why is that action scale the observed $\hbar$?* The identification $S_0 = \hbar$ is adopted as a Level B empirical calibration (Section 21.2); the required owner map, real-to-complex reduction, absolute derivation, cross-sector normalisation and physical quantum reconstruction remain Open.

These three questions are complementary rather than competing. The first concerns the *existence* of quantum-theoretic structure; the second concerns the *origin* of the action scale that organises that structure; the third concerns the *empirical matching* of the scale to observed data. Conflating them obscures the actual logical status of each step.

What is established, what is structural, and what remains open in the quantum-sector programme.

The ECT quantum-sector programme should be read in three layers.

First, supplied coherent-branch models provide a compact-phase sector, two distinct loop-scale candidates and a separately supplied operational action unit S_0 (whose owner, nonzero value and universality remain Open; §5.6), conditional scalar wave kinematics after $x^0 = w$, continuity identities and a stationary-phase limit (§22.2). Each result inherits the assumptions of its own action/state and is summarised in Table 95.

Second, on top of that infrastructure, ECT provides structural routes towards the standard quantum architecture: Schrödinger-type dynamics (§22.1), Hilbert-space reconstruction via reflection positivity (§23.1), canonical commutation structure (§23.3), Born-type weighting (§30.4), and spin/statistics constraints (§24.4). These routes are strong, but not all of them are yet fully closed at the same logical level.

Third, what remains open is the unique first-principles completion of the whole quantum package: a fully closed measurement theory, a fully closed Born theorem without extra closure input, and a completed fermionic/spinorial sector that would settle Pauli exclusion and spin-statistics at the same level as the bosonic coherent branch.

Connection to the structural universalities. The quantum programme tests four proposed cross-sector universalities (§11.1). Its supplied models reuse a scalar cone parameter c_{char} , an action parameter S_0 , variational stationarity and conservation identities, but their physical universality across geometric, gauge and coherent sectors is not derived. The P4 threshold programme (§8.3) asks whether a stationary ordered-branch action has a healthy amplitude pole that limits the coherent EFT. No such pole or cross-Part threshold is presently derived; the homogeneous P3 curvature cannot supply it by owner transfer.

Discriminant with respect to standard and alternative quantum programmes. The ECT quantum programme differs from two familiar alternatives:

- (i) **Axiomatic QM:** ECT asks whether Schrödinger dynamics, a probability current and Born weighting can ultimately share owners, but the present conditional routes do not yet trace all three to one action/state;
- (ii) **Ad hoc hidden-variable or collapse models:** ECT does not introduce extra collapse variables by decree at the outset. It organises a supplied coherent branch, its candidate action coefficient, its continuity structure, and a conditional Hilbert-space route, and then states which measurement closures remain missing.

This defines a potentially unifying research programme, not a demonstrated replacement for the standard axiom list.

Falsifier for the quantum-sector unification programme. The present ECT quantum-sector programme would be seriously weakened if different quantum substructures ultimately required mutually unrelated normalisations or principles: for example, if Schrödinger dynamics required one action scale,

Born weighting another, canonical commutators a third, and the fermionic Pauli/spin-statistics sector a fourth independent closure principle. It would also be weakened if a completed coherent-sector analysis showed that no single ordered-branch architecture can jointly support wave kinematics, probability current, Hilbert reconstruction, and fermionic completion. Conversely, an algebraic cross-link or reproduction of a supplied source model does not support the one-medium ontology. A positive common-owner result would require one independently derived ECT action and state together with compatible preparation, composition and observable maps across the linked sectors.

20.1 Quantum Sector as the development of the coherent branch

Connection to ECT Foundations: Sections 6, 6.3, 5.6. Status: entry point to the coherent branch. ECT Foundations supplied a phase-sensitive coherent-sector closure and established integer winding and a fixed-length variational bound inside it. Neither loop-scale candidate, their ownership of the operational action-unit slot S_0 , nor a nonzero universal quantum of action was derived from P1–P6. What follows in Part III is not assumed from the outset to be a complete derivation of quantum mechanics; rather, it is a staged programme in which some structures are obtained structurally, some only conditionally, and some remain open.

This subsection does not repeat the general introduction to Part III given above. Its more specific purpose is to identify the effective variables of the coherent branch, to state the main derivation targets of the Quantum Sector, and to clarify which additional ingredients are required in order to approach the standard quantum-mechanical formalism.

Primary variables of the coherent branch. In the coherent branch the ordered phase is described by variables that retain global phase information. The primary objects are:

- the effective order parameter

$$\Phi_{\text{eff}} = \rho e^{i\theta} \quad (20.1)$$

- the phase variable θ , which is not coarse-grained away;
- the winding quantisation condition,

$$\oint_{\gamma} \partial_A \theta dX^A = 2\pi n, \quad n \in \mathbb{Z} \quad (20.2)$$

- the candidate coefficient ι_0^{EFT} of a supplied fixed-core phase EFT, represented by

$$\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}} \quad (20.3)$$

- the distinct optional tension-balanced candidate $\iota_{0,\text{bal}}^{\text{eff}}$ of Eq. (5.29);
- an operational action-unit slot S_0 for later conditional wave, canonical and response closures. Its owner is not fixed by either loop construction.

From these ingredients one later constructs coherent correlators, loop amplitudes, and the effective dynamical objects of the Quantum Sector. These structures were prepared in ECT Foundations but not yet developed into a full dynamical quantum programme. Part III begins that development.

Why the phase is not coarse-grained away here. In the geometric branch, local phase information was irrelevant because the observables under study were governed by the coarse-grained metric response and long-wavelength stiffness. In the coherent branch the relevant observables are different: global phase consistency, nontrivial loop sectors, and winding-sensitive amplitudes carry physical content. The phase field θ is therefore retained not as an aesthetic choice but because interference-sensitive, loop-sensitive, and coherence-sensitive observables would be lost if it were averaged away.

Physical meaning of the Quantum Sector. The Quantum Sector should not be understood as importing ordinary quantum mechanics into ECT from outside. Its role is the opposite: to determine how far the ordered phase, once its coherent and topological data are retained, can structurally motivate, constrain, or in some cases partially reconstruct structures that standard quantum theory usually takes as postulated.

A central issue is not merely the recovery of wave-like evolution, but the status of the bridge from Euclidean coherent dynamics to a physical Hilbert-space description with positive norm, positive spectrum, and unitary Lorentzian evolution. That bridge is examined in Section 23.

Derivation targets of Part III. ECT Foundations and Macroscopic Physics did not yet establish:

- (i) a microscopic owner and cross-sector universality for the adopted Level-B calibration $S_0 = \hbar$;
- (ii) a full Hilbert-space reconstruction of the interacting coherent branch with positive norm and global unitarity;
- (iii) a full derivation of the standard canonical commutation algebra, rather than only a structural route toward it;
- (iv) a complete derivation of Born weights;
- (v) a complete theory of measurement;
- (vi) the full dynamics and observational status of defect sectors;
- (vii) a complete nonperturbative closure of the black-hole sector.

These are the principal targets, conditional steps, and open problems of Part III.

Roadmap. The chapter proceeds from phase dynamics and the distinguished action scale (§21) to wave kinematics (§22), then to the Hilbert-space / unitarity bridge (§23), nonclassicality and statistics (§24), coherent-vacuum response (§25), open-system decoherence (§26), the quantum-classical boundary and measurement status (§30), wider topological and defect sectors (§34), and finally black-hole thermodynamics and the information problem (§36), before the concluding status map in §38. The response–persistence formulation PES-R (§29) is introduced after the open-system machinery as a status taxonomy: sector dynamics, finite-time retention, and pairwise records remain non-equivalent owners. It does not derive measurement outcomes, discrete spectra, or a universal quantum–classical boundary.

Status at entry. At the entry to Part III, ECT has established the coherent branch, its phase-sensitive variables and winding classification. The two supplied loop closures define distinct candidate coefficients, while Part III uses a separately declared operational slot S_0 . Topology alone derives neither candidate, their owner map to that slot, nor universal physical ownership. What remains is to determine how much of quantum kinematics, probability, irreversibility, and information flow can be derived from that ordered-phase framework without overclaiming closure where it has not yet been achieved.

21 Coherent Phase Dynamics, Loop Sectors, and the Pre-Quantum Action Scale

This section develops the first technical layer of the Quantum Sector. Its purpose is to identify the effective variable that survives in the smooth coherent branch, to show how nontrivial loop sectors arise in the reduced compact-phase description, and to separate the exact winding and fixed-length bound from the supplied coefficient used as a candidate action unit.

The logic is staged. Section 21.1 introduces the phase variable and separates the supplied positive Euclidean loop functional $\mathcal{J}_{\theta,E}$ from the indefinite ordered-branch characteristic form $\mathcal{Q}_{\theta,K}$ and its

conditional Lorentzian orientation, while retaining the winding-sector organisation of the coherent branch. Section 21.2 then distinguishes the two supplied loop-scale candidates from the operational slot S_0 : none is derived from P1–P6, neither loop candidate is automatically the owner of that slot, and S_0 is not automatically the observed reduced Planck constant $\hbar$. In this way, Section 21 provides the bridge from coherent loop structure to the later wave, canonical, and probability constructions of Part III.

21.1 Coherent phase dynamics and loop sectors

Status: Level A for the existence of the phase variable, winding sectors, and the compact-phase loop classification inside the coherent branch; Level B for the specific effective dynamical closure used to connect these structures to wave dynamics. Connection to ECT Foundations: BR1, Section 5.6, Appendix AS.

We begin with the phase-sensitive effective description of the smooth coherent branch. This is not yet ordinary quantum mechanics. Rather, it is the reduced compact-phase EFT in which nontrivial loop sectors are retained and organised before the later wave-dynamical and canonical constructions are attempted.

Phase as the natural variable of the coherent branch. Once the ordered condensate is written as

$$\Phi_{\text{eff}}(X) = \rho(X) e^{i\theta(X)}, \quad (21.1)$$

the phase field θ becomes physically meaningful whenever configurations with nontrivial winding are retained. In the geometric branch such information is averaged away. In the coherent branch it is exactly this global phase structure that distinguishes one sector from another. The phase is not merely a convenient reparametrisation: it is the variable that remembers the global consistency conditions of the condensate—single-valuedness, winding, and compact-phase loop structure.

Ordered-branch phase characteristic form. At wavelengths large compared to the microscopic core scale, and within the smooth coherent sector defined by the supplied closure BR1, the radial mode ρ is approximately frozen and the effective dynamics is dominated by the phase field. The quadratic phase principal form in this regime takes the generic form (see Section 5.6 and Appendix AS):

$$\begin{aligned} \mathcal{Q}_{\theta,K} &= \frac{K_{\theta}^{\text{eff}}}{2} \int d^4X \hat{K}^{AB}(X) \partial_A \theta \partial_B \theta, \\ \hat{K}^{AB} &\equiv K^{AB}/\beta \quad (\beta > 0). \end{aligned} \quad (21.2)$$

Here $K^{AB}(X)$ is the supplied ordered-branch principal tensor. The positive division by β changes neither its cone nor its signature and keeps K_{θ}^{eff} as the independently supplied phase stiffness. $K_{\theta}^{\text{eff}} > 0$ is assumed for this retained two-derivative closure; zero is degenerate and a negative value reverses its leading Hamiltonian sign. Throughout the present wave, stationary-phase, Hessian and canonical discussion, $\mathcal{Q}_{\theta,K}$ denotes only this displayed quadratic truncation. Higher-gradient and interaction operators require a separate Lorentzian continuation, relative sign, pole/constraint analysis and symplectic owner; they are not included by an ellipsis. $\mathcal{Q}_{\theta,K}$ and the canonical formulae below are evaluated only in a locally homogeneous, aligned frozen-coefficient patch. For variable $K^{AB}(X)$ the Euler–Lagrange operator is instead in divergence form, $\partial_A [\hat{K}^{AB}(X) \partial_B \theta]$, and derivatives of the coefficients cannot be discarded; that variable-coefficient completion and its boundary domain remain Open. $\mathcal{Q}_{\theta,K}$ is an indefinite characteristic form, not the positive Euclidean loop functional $\mathcal{S}_{\theta,E}$ and not yet the physical Lorentzian action. Its positive-energy Lorentzian orientation and the clock/lapse map are supplied below in Section 22.1.

Here K_{θ}^{eff} denotes the *effective phase stiffness* of the coherent branch. At the level of the present closure it is an independent effective coefficient. Relating it to the supplied P4 datum u_0 , a future action-owned P4 radial pole, or the anisotropy α requires an explicit stationary matching calculation; the P3 mass m_{σ} cannot substitute for that missing owner. Dimensional analysis in the simplest quartic realisation suggests $K_{\theta}^{\text{eff}} \sim \phi_0^2$ up to EFT matching coefficients. However, its explicit first-principles reduction to these primary

parameters is not assumed here: it requires a dedicated EFT matching step in which every omitted physical mode and its coupling are first derived and then integrated out consistently. That matching remains an open part of the coherent-sector EFT programme pending the full loop-sector calculation (Appendix AS). Equation (21.2) should therefore be understood as a Level B effective closure, not yet as a full microscopic derivation from bare P3.

Why this characteristic form is a useful closure. Equation (21.2) is the phase-sector analogue of the anisotropic EFT structures already used in ECT Foundations. Its role is to capture the lowest-order principal dynamics of smooth coherent configurations inside a declared reduced-model domain; no P4 heavy-mode integration is presently derived and the phase is retained as a supplied dynamical degree of freedom. Although $\mathcal{Q}_{\theta,K}$ is written in the original Euclidean-coordinate chart, it is indefinite and is not the positive Euclidean statistical functional $\mathcal{J}_{\theta,E}$. Its positive-energy Lorentzian orientation is supplied separately in Section 22.1; a formal Euclidean–Lorentzian amplitude continuation is a third, distinct operation. The complex phase field, coefficients and descent from the real-scalar P3 remain Level B/Open.

Winding sectors and topological classification. The coherent branch is partitioned into sectors classified by winding number (cf. (20.2)):

$$n[\gamma] = \frac{1}{2\pi} \oint_{\gamma} \partial_A \theta dX^A \in \mathbb{Z}. \quad (21.3)$$

This is a Level A structural statement within the reduced compact-phase description defined by the supplied BR1 coherent-sector closure. It should be stressed, however, that the supplied orientation target $O(4)/O(3) \simeq S^3$ has $\pi_1(S^3) = 0$, so the winding classification does not follow from the topology of the coset space itself. Rather, it arises in the $U(1)$ -extended compact-phase EFT of the coherent branch, where $\Phi_{\text{eff}} = \rho e^{i\theta}$ is treated as the relevant low-energy order parameter (see Section 5.6). The trivial sector $n = 0$ contains contractible smooth configurations; nontrivial sectors $n \neq 0$ correspond to configurations with nontrivial phase winding that cannot be eliminated by a smooth deformation within the reduced compact-phase coherent EFT.

Elementary loop action and the appearance of S_0 . Integer winding fixes a topological sector. A finite elementary-loop action is obtained only after the characteristic closed-loop length and phase-EFT normalisation are supplied. At the supplied fixed core length, the $n = 1$ representative is described by the dimensionless reduced coefficient

$$\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi \iota_0^{\text{EFT}}, \quad (21.4)$$

where the candidate coefficient ι_0^{EFT} depends on the coherent-sector EFT parameters (Appendix AS). This is not a global variational minimum over arbitrary loop lengths. At this stage ι_0^{EFT} is a Level-B/Open parameter of the declared fixed-core model; neither a nonzero global variational minimum nor its identification with $\hbar$ is established. Under the later Level B matching hypothesis that the characteristic elementary loop is core-sized, one expects this scale to be associated with microscopic circumferences of order the core length. Its possible relation to the observed value of $\hbar$ is analysed in Section 21.2; it is not assumed at the present stage.

An optional tension-balanced closure route supplies a second candidate: if a positive line tension and the phase stiffness descend together into the same 1D loop functional, its finite minimiser defines $\iota_{0,\text{bal}}^{\text{eff}} = \sqrt{2K_{\theta}^{\text{1D}} \mathcal{T}_{\text{loop}}}$. The added term makes a finite conditional minimum possible in the supplied model, but it does not derive the tension, its 4D owner or $\hbar$; nor is it the same coefficient as ι_0^{EFT} (Eqs. (5.29)–(5.30)).

Physical meaning of the loop sectors. A nontrivial loop sector represents more than a winding number. Physically, it means that the condensate admits coherent histories that cannot be continuously deformed into the trivial sector within the reduced compact-phase description of the smooth coherent branch. Loop

sectors provide a candidate setting for amplitudes and interference, but topology alone does not derive quantum dynamics, an action quantum or probabilities. This logic is structurally analogous to circulation quantisation in superfluids and flux quantisation in superconductors (see Table 74).

Smooth coherent branch versus defect sectors. The present discussion is restricted to the smooth coherent sector defined by BR1. Configurations with singular cores, vanishing ρ , or discontinuous orientation data belong to the wider defect sector and are treated in Section 34. The role of BR1 is not to deny the existence of defects, but to define the branch in which smooth phase dynamics can first be developed cleanly.

From loop sectors to wave dynamics. Once the condensate admits coherent sectors weighted by their phase action, the next question is how these sectors combine into an effective wave description. This step requires: (a) showing how the phase-sector path integral leads to a Schrödinger-like amplitude language; (b) verifying that the real parametrisation $x^0 = w$ (established for the geometric branch; Section 10.6) carries the phase action into a standard kinetic term; (c) identifying the physical owner of the operational S_0 and testing its separate match to $\hbar$. Those questions belong to Sections 21.2 and 22.1.

Status summary. *Established inside the supplied compact-phase closure* (Level A mathematics): integer winding and the fixed-length variational bound. *Not yet established:* (i) the descent of the complex phase field from P3; (ii) a physical closed-loop length and nonzero global minimum; (iii) the explicit common-action values of K_θ^{eff} , K_θ^{1D} and $\mathcal{T}_{\text{loop}}$, and the physical owner map from either loop candidate to the operational S_0 (open, Appendix AS); (iv) the microscopic explanation and universality of the separately adopted Level-B calibration $S_0 = \hbar$ (Section 21.2); (v) the full wave-dynamical superposition rule between sectors; (vi) the role of defect sectors (Section 34).

21.2 Pre-quantum action scale and the status of $\hbar$

Status: Level A mathematics for winding, the fixed-length bound, and the positive-tension minimum inside their respective supplied models; Level B/Open for both loop-scale candidates and either owner map to the operational S_0 ; $S_0 = \hbar$ is the separately adopted Level-B calibration, whose microscopic explanation and cross-sector universality remain Open. Connection to ECT Foundations: Section 5.6 and Appendix AS.

The previous subsection established that the coherent branch of ECT is organised by nontrivial loop sectors of the phase field and that, within the reduced $U(1)$ -extended phase-loop EFT, the fixed-length loop action admits a sharp lower bound. The present subsection clarifies the precise status of three distinct objects. ι_0^{EFT} is the phase-only fixed-core candidate, $\iota_{0,\text{bal}}^{\text{eff}}$ is the positive-tension balanced candidate, and the unadorned S_0 is the operational action-unit slot used by later conditional quantum and response formulae. Neither candidate is derived from P1–P6 or identified as the owner of S_0 . The final equality $S_0 = \hbar$ is adopted as a Level-B calibration; its microscopic derivation and universality are not established.

What topology fixes and what it does not. The conditional construction uses four declared ingredients, organised into two alternative closure routes:

- (i) the supplied $U(1)$ -extended EFT contains a phase variable θ ;
- (ii) inside the supplied compact-phase BR1 closure, smooth configurations are partitioned into integer winding sectors (Level A topology under that premise);
- (iii) the fixed-core route supplies a characteristic loop length and phase normalisation to define ι_0^{EFT} ;
- (iv) the balanced route separately supplies a positive line tension to define $\iota_{0,\text{bal}}^{\text{eff}}$.

Only the winding, fixed-length bound and the explicit two-term minimisation inside their declared models are theorems. They do not imply that the physical coherent branch possesses a unique finite action scale associated with nontrivial compact-phase transport. The winding structure and fixed-length bound are strict within that model (Level A mathematics). The fixed-core finite coefficient additionally requires the unowned length L_{core} . The balanced coefficient instead requires a positive $\mathcal{T}_{\text{loop}}$, a common reduction of $\mathcal{T}_{\text{loop}}$ with K_{θ}^{1D} from one 4D owner, and survival of the minimum in the full carrier/profile/ensemble variational problem (Level B/Open).

The corresponding EFT action scale, recalled from Section 5.6, is

$$\iota_0^{\text{EFT}} = \frac{c_{\perp} K_{\theta}^{\text{eff}}}{2m_{\phi}^2}, \quad (21.5)$$

where K_{θ}^{eff} is the effective phase stiffness of the coherent branch, m_{ϕ} is the inverse core-length scale entering the elementary-loop closure. Here $c_{\perp} = \mathcal{O}(1)$ is the profile-dependent coefficient of the transverse three-volume reduction of the supplied codimension-three carrier (Section 5.6, Eq. (5.19)). As emphasised in ECT Foundations, this quantity refers to the *elementary loop evaluated at the characteristic core scale*, not to a global variational minimum over arbitrary loop lengths.

What the three action-scale symbols mean physically. ι_0^{EFT} is the phase-gradient contribution evaluated at a supplied fixed-core length. $\iota_{0,\text{bal}}^{\text{eff}}$ is the total minimum of a different supplied 1D model containing a positive line tension. The unadorned S_0 is an operational input to later wave, canonical and PES formulae. Compact phase fixes only winding and the fixed-length bound; the finite loop coefficients depend on additional profiles and couplings, and neither has been shown to own the universal operational unit observed in quantum processes.

Why the adopted calibration $S_0 = \hbar$ is not a derivation. Compact topology establishes integer winding and the supplied phase EFT gives a fixed-length n^2/L bound. That phase-only bound has zero infimum when L varies freely. The supplied positive-tension model has a finite minimum, but this does not establish its descent from ECT, its stability under the full carrier, profile and proper-time ensemble dynamics, or its ownership of the operational action unit. In particular, they do not prove that:

- (i) all coherent amplitudes are weighted by the same unit S_0 ;
- (ii) canonical quantisation rules are governed by S_0 ;
- (iii) the wave-dynamical limit of the coherent branch reproduces the observed quantum kinematics with S_0 in the rôle of $\hbar$;
- (iv) the numerical value of S_0 and its universal role across the coherent-sector amplitude, wave, and canonical structures match the observed reduced Planck constant $\hbar = 1.054 \dots \times 10^{-34} \text{ J s}$.

These are additional statements; none follows merely from the structural existence of nontrivial loop sectors.

What is required to explain the calibration $S_0 = \hbar$. ECT already adopts the equality as a Level-B physical calibration. To turn that input into a microscopic derivation and universality result, the following multi-step programme is required:

- (i) derive the reduced loop action explicitly within the coherent EFT, rather than only introducing its structural form (Appendix AS);
- (ii) show that the resulting action scale controls the effective phase weight of coherent histories in the path-integral sense;

- (iii) demonstrate that the canonical and wave-dynamical limits of the coherent sector are governed by the same scale;
- (iv) perform the numerical matching to the observed $\hbar$.

This is a multi-step closure problem of the Quantum Sector, not a one-line identification.

Role of Appendix AS. Appendix AS derives both reduced-model formulae, their factor-two relation in the elementary matched-length case, and the guards needed before either can be assigned an ECT owner. The remaining task is not merely numerical evaluation: it is a common 4D reduction of profile, stiffness, tension, embedding, unwinding and backreaction, followed by an independent map to the operational S_0 .

Why the two conditional routes are useful. The fixed-core construction shows how a supplied length, profile and phase stiffness define one finite coefficient. T1 adds a different result: in a supplied positive-tension 1D functional, competition between tension and winding stiffness selects a finite loop length and total minimum rather than requiring the length to be chosen by hand. This is useful conditional mechanism-building, but neither construction derives its four-dimensional owner or explains the universal physical quantum of action.

The same operational symbol S_0 reappears across the conditional coherent-branch constructions developed in Part III: phase weights, wave reconstruction, canonical commutators and decoherence architecture (§5.6). This is a consistency ansatz, not proof that either loop candidate owns the slot or that one microscopic normalisation controls every sector. The identification $S_0 = \hbar$ is the separately adopted Level-B matching input; its microscopic explanation remains Open.

Structural bridges and the three matching relations. The action-scale bridge is a two-step, route-dependent programme rather than a direct identification of the fixed-core candidate with $\hbar$. The single-field reduction, physical analytic-continuation state and cross-sector normalisation are Open.

In the simplest bare-potential realisation, the effective theory is organised by three independent classes of matching relations:

$$\begin{aligned}
 c_{\text{char}} &= c, \\
 G_N^{\text{nat}} &= \frac{1}{8\pi M_G^2}, \\
 \iota_0^{\text{EFT}} \text{ or } \iota_{0,\text{bal}}^{\text{eff}} &\xrightarrow{\mathcal{M}_{\text{owner}}} S_0 \xleftarrow{\mathcal{M}_{\hbar}} \hbar.
 \end{aligned} \tag{21.6}$$

The first two connect the ordered branch to Lorentzian propagation and to observed gravity; the third line would connect one selected coherent-loop candidate to the operational action slot and then to the observed quantum of action. These are independent observational matchings, not a closed three-parameter derivation. For the full discussion of these bridges, see Section 5.6.

Bridge to the rest of Part III. Once a distinguished action scale exists, the natural next question is how coherent sectors combine into an effective wave description and how the ordered-branch real parametrisation $x^0 = w$ yields the standard kinetic term, while the separate formal continuation $w_E \rightarrow \pm it$ provides the amplitude/path-integral bridge for the same coherent sector. That question leads first to the wave-dynamical development of the Quantum Sector in Section 22.

Beyond wave kinematics, the same scale also enters the later quantum programme: the normalisation of canonical structures (Section 23.3), the Hilbert-space / unitarity bridge (Section 23), the open-system decoherence architecture (Section 26.1), and the probability problem (Section 30.4). In all of these later uses, however, the physical identification $S_0 = \hbar$ remains a Level B matching step rather than an already completed Level A derivation.

Status summary. *Established inside supplied models:* Level A mathematics for winding and the fixed-length phase bound, and for the one-coordinate minimum of the positive-tension functional at fixed $n \neq 0$. The phase-only class by itself still has zero infimum as $L \rightarrow \infty$. *Not established (Open):* a common four-dimensional owner of K_θ^{1D} and $\mathcal{T}_{\text{loop}}$, absence of double counting, and stability of the balanced stationary point against shape, transverse, splitting, unwinding and backreaction directions. *Open owner maps / adopted calibration:* the candidate-owner maps and cross-sector normalisation remain Open, while the last equality below is the adopted Level-B physical calibration:

$$\Sigma_\theta \iota_0^{\text{EFT}} \xrightarrow{\mathcal{M}_{\text{fc}}} S_0, \quad \Sigma_\theta \iota_{0,\text{bal}}^{\text{eff}} \xrightarrow{\mathcal{M}_{\text{bal}}} S_0, \quad \boxed{S_0 = \hbar} \quad (\text{Level-B calibration}). \quad (21.7)$$

The first two arrows are alternative route-dependent owner tests, not simultaneous identities. The final equality is used as the third empirical calibration, after $c_{\text{char}} = c$ and G_N matching; its microscopic derivation and universality remain Open. The completion of the $\hbar$ -programme in ECT—universal amplitude weighting, canonical normalisation, and Born-type probability weights—belongs to the later constructive development of the Quantum Sector.

Connection to PES-R. PES-R supplies no independent normalisation of the coherent action scale. It may compare a physical decay rate γ_a with the action-unit width $\Gamma_{E,a} = S_0 \gamma_a$ and may use the dimensionless pairwise exponent Φ_{ab} , but neither relation determines S_0 . The fixed-core $n = 1$ winding calculation remains owned by the compact-phase sector, while the identification $S_0 = \hbar$ remains a separate Level-B canonical matching. PES-R therefore neither strengthens that identification nor establishes a positive global lower bound on the elementary-loop action.

For orientation, it is useful to compare the ECT loop-sector logic with other physical systems in which a reduced circle-valued phase degree of freedom leads to quantised loop data.

Table 74: Topological or compact-phase loop quantisation in different physical systems. In each case the relevant reduced phase degree of freedom is circle-valued, so that the loop classification is governed by $\pi_1(S^1) = \mathbb{Z}$. For ECT this refers to the reduced $U(1)$ -extended coherent-phase EFT, not directly to the full vacuum coset $O(4)/O(3) \simeq S^3$.

System	Object	Topological origin	Quantum
Superfluid ^4He [10]	Quantised vortex	$\pi_1(S^1) = \mathbb{Z}$	$\kappa = h/m$
Type-II superconductor	Flux quantum	$\pi_1(S^1) = \mathbb{Z}$	$\Phi_0 = h/2e$
Early QM (de Broglie)	Bohr–Sommerfeld orbit	$\pi_1(S^1) = \mathbb{Z}$	$\oint p dq = n\hbar$
ECT coherent phase EFT	Two supplied loop closures	compact $U(1)$ phase, $\pi_1(S^1) = \mathbb{Z}$	$\iota_{\text{elem}}^{(\text{fixed core})} = 2\pi\iota_0^{\text{EFT}}$; optional $\mathcal{I}_{1,\text{min}} = 2\pi\iota_{0,\text{bal}}^{\text{eff}}$; ECT owner Open

22 From Euclidean Coherent Dynamics to Lorentzian Wave Kinematics

This section tests a conditional Lorentzian wave-kinematical completion while keeping three objects distinct: the positive Euclidean loop functional $\mathcal{I}_{\theta,E}$, the indefinite characteristic form $\mathcal{Q}_{\theta,K}$, and the supplied positive-energy Lorentzian functional $\mathcal{I}_{\theta,L}$. A Hamilton–Jacobi/continuity decomposition still requires a separately supplied norm-preserving complex law, Hamiltonian and domain; the full Hilbert-space and measurement questions are treated in subsequent subsections.

The logic remains status-separated. Section 22.1 studies a supplied coherent wave-kinematical completion while distinguishing $x^0 = w$ from a formal Euclidean–Lorentzian amplitude bridge. Section 22.2 treats stationary phase only for its separately supplied Lorentzian integral. Section 22.3 compares currents

of the bare real-scalar and supplied phase/complex actions; it does not derive one current hierarchy across them.

Section 38.3 collects the Euclidean condensate equation, ordered-branch equations, Klein–Gordon closure, and Schrödinger-type limit as separately owned equations with their stated statuses. Their ordered presentation is not a derivation of a universal descent chain.

22.1 Wave dynamics, ordered-branch hyperbolic parametrisation, and Schrödinger-type evolution

Status: Level A mathematics for the tensor/cone algebra of the supplied characteristic form (21.2); Level B/Open for the compact phase sector; $K_\theta^{\text{eff}} > 0$, the Lorentzian action owner, clock and continuation map. The Hamilton–Jacobi/continuity split is Level A only conditional on a separately supplied norm-preserving complex law, Hamiltonian and domain; the Schrödinger-type law remains Level B/Open; $S_0 = \hbar$ is the adopted Level-B calibration, while its microscopic owner and universality remain Open. Connection to ECT Foundations: Sections 5.6, 6.3; Appendix AS.

The previous two subsections established the phase field and winding sectors inside the supplied compact-phase EFT, displayed two distinct conditional loop-scale candidates, and reserved the unadorned S_0 for an operational action-unit slot whose physical owner remains Open. The question now is whether a supplied value of that slot supports a wave-dynamical description. This is the first place in the Quantum Sector where the ECT structure begins to resemble ordinary quantum theory. The resemblance must be built, not assumed.

From phase action to coherent amplitudes

The natural starting point is the ordered-branch characteristic form (21.2). On the retained $\beta > 0$ component its positive-energy Lorentzian orientation in the supplied quadratic truncation is $\mathcal{J}_{\theta,L} := -\mathcal{Q}_{\theta,K}$. Within a supplied coherent closure with an operational action-unit input, a candidate amplitude for a history $\theta(\cdot)$ is assigned the phase weight

$$\mathcal{A}[\theta] \sim \exp\left(\frac{i\Sigma_\theta \mathcal{J}_{\theta,L}[\theta]}{S_0}\right). \quad (22.1)$$

Here $S_{\theta,L}^{\text{phys}} = \Sigma_\theta \mathcal{J}_{\theta,L}$, while S_0 is the supplied operational action-unit slot. The symbol ‘ $\sim$ ’ marks a structural assignment rather than a normalised path-integral measure. For finite positive, spacetime- and field-independent constant Σ_θ , stationarity is unchanged: $\delta(\Sigma_\theta \mathcal{J}_{\theta,L}) = 0$ iff $\delta \mathcal{J}_{\theta,L} = 0$. Reusing the Euclidean loop coefficient Σ_θ here is a declared same-sector continuation/matching input; it is not derived by winding, P4 or the cone.

At this stage Eq. (22.1) is not the final quantum path integral. It records the proposed reuse of a supplied operational unit in a phase-weight closure; it does not derive its loop-sector owner, an intrinsic universal scale, or the physical quantum measure. The phase-weight assignment is Level B/Open because its state, normalisation and cross-sector ownership remain to be established. The equality $S_0 = \hbar$ used in it is the adopted Level-B calibration, not a result of this construction.

Analytic continuation versus ordered-branch hyperbolic parametrisation

Once the retained $\beta > 0$ component satisfies $\alpha > \beta > 0$, the ordered-chart tensor $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$ already has Lorentzian sign structure. Accordingly, the common-length ordered coordinate and its scalar slope are introduced by Eq. (3.22), $x^0 = w$ and $\hat{c}_*^2 = \beta/(\alpha - \beta)$. The physical time τ is introduced only through the separate clock map Eq. (3.23). In particular, the physical clock must not be obtained by dividing w by a symbol that is simultaneously being used for the dimensionless principal slope. The canonical aligned special case is $\tau = w/c$ after $c_u = c$, $\mathcal{N}_\Phi = 1$, and an aligned planar branch are supplied. The logically separate choice $\hat{c}_* = 1$ is required only if the scalar characteristic speed is also matched to c (with the physical photon owner still Open).

A distinct formal operation is the Euclidean–Lorentzian amplitude continuation

$$w_E \rightarrow \pm i x_L^0, \quad (22.2)$$

which is used only to relate Euclidean coherent weights to Lorentzian phase-amplitude language. It should not be identified with the derivation of hyperbolicity itself.

Two distinct conditional operations.

- (i) **Common-length hyperbolic parametrisation:** $x^0 = w$ may be used in a supplied hyperbolic scalar equation; physical clock readings require Eq. (3.23) and remain an independent matching problem.
- (ii) **Formal Euclidean–Lorentzian amplitude bridge:** the continuation (22.2) translates Euclidean coherent-sector weights into Lorentzian amplitude language only in sectors satisfying the required reconstruction hypotheses.

Remark: continuation versus physical dynamics. Analytic continuation, a common-length hyperbolic parametrisation and a physical real-time evolution law are three different claims. The first is an amplitude-language operation; the second belongs to the supplied scalar principal operator; the third requires a clock/lapse, state and physical action. The present coherent closure supplies no general theorem identifying all three (see also Section 6.3).

Conditional hyperbolic form of the coherent phase dynamics

With the adapted broken-phase basis $n^A = \delta_w^A$, the ordered-branch characteristic form entering Eq. (21.2) obeys

$$\hat{K}^{AB} \partial_A \theta \partial_B \theta = \delta^{ij} \partial_i \theta \partial_j \theta - \frac{1}{\hat{c}_*^2} (\partial_w \theta)^2.$$

At this stage the ordered-branch hyperbolicity and the time-length coordinate have already been fixed by $\alpha > \beta > 0$ on the retained $\beta > 0$ component and $x^0 = w$; the physical proper-time map is still separate.

With $\partial_0 \equiv \partial_w$, the reduced phase-sector action in common-length coordinates is

$$\mathcal{I}_{\theta,L} \equiv -\mathcal{Q}_{\theta,K} = \frac{K_\theta^{\text{eff}}}{2} \int dx^0 d^3x \left[\frac{1}{\hat{c}_*^2} (\partial_0 \theta)^2 - (\nabla \theta)^2 \right] \quad (22.3)$$

In a locally aligned constant-lapse patch, Eq. (3.23) rewrites the same bracket as $c_{\text{char}}^{-2} (\partial_\tau \theta)^2 - (\nabla \theta)^2$, up to the corresponding constant measure normalisation. Thus the dimensionless slope and the dimensionful clock speed are not used interchangeably. The formal Euclidean–Lorentzian continuation (22.2) is kept separate and is used only at the level of coherent amplitudes and correlator language, not in the derivation of the ordered-branch kinetic form itself.

This is again a Level B effective closure. Its importance is that it provides the correct Lorentzian kinetic structure for coherent phase evolution, with the spatial gradient and time-derivative terms separated by the effective speed c_{char} .

Hamilton–Jacobi / continuity structure

Conditional polar decomposition. Introduce the polar representation

$$\Psi(x, t) = \sqrt{\rho(x, t)} \exp\left(\frac{i \mathcal{S}(x, t)}{S_0}\right). \quad (22.4)$$

This change of variables alone supplies no evolution equation. If a norm-preserving complex evolution law, Hamiltonian, operator domain, and boundary conditions are independently supplied, its real and imaginary parts may be written as

$$\partial_t \mathcal{S} + H_{\text{coh}}[\rho, \mathcal{S}] = 0, \quad (22.5)$$

and

$$\partial_t \rho + \nabla_i J_{\text{coh}}^i[\rho, \mathcal{S}] = 0. \quad (22.6)$$

For a specified Schrödinger Hamiltonian this is the exact Madelung decomposition [147]. It is Level A mathematics conditional on that supplied law. The phase action shown here does not derive H_{coh} , its norm-preserving complex dynamics, the current, or the physical operator domain; their ECT ownership and the matching of the supplied coefficient S_0 remain Level B/Open. Accordingly neither the $S_0 \rightarrow 0$ classical limit nor a universal non-classical term is claimed before an owning evolution law is fixed.

Schrödinger-type recombination (Level B)

The two real equations above can be assembled into a single complex equation:

$$iS_0 \partial_t \Psi = \hat{H}_{\text{coh}} \Psi + \dots \quad (22.7)$$

where $\hat{H}_{\text{coh}}$ contains the spatial gradient and potential terms, and the ellipsis denotes decoherence and higher-gradient corrections.

What this recombination is and is not. The recombination into Eq. (22.7) is a structural route, not a derivation of the Schrödinger equation from first principles. To qualify as a Level A derivation, one would additionally need:

- (i) a theorem on linear complex amplitude composition for coherent histories;
- (ii) the explicit derivation of $\hat{H}_{\text{coh}}$ for the relevant coherent sector (OP-Q3);
- (iii) use of the adopted Level-B calibration $S_0 = \hbar$ (21.7); this supplies the numerical action unit but not the missing Hamiltonian or composition theorem.

For these reasons Eq. (22.7) is Level B.

Summary of what is established. Given a supplied norm-preserving complex evolution law, Hamiltonian, domain, and scalar principal/clock convention, its polar (Madelung) decomposition into Hamilton–Jacobi-type and continuity equations is exact mathematics. The phase action alone does not derive that complex evolution law. ECT ownership of the Hamiltonian/current and physical clock, and the origin, finite value, or cross-sector universality of S_0 , remain Level B/Open. The displayed Schrödinger-type recombination and the matching $S_0 = \hbar$ are supplied closure inputs, not Level A consequences of P1–P6.

Relation to bound-state and spinorial descriptions (Level B). The Schrödinger-type equation (22.7) should be understood as a low-energy envelope description. In bound systems—for example, a charged spinorial field in a Coulomb-like background belonging to a separately supplied compact local $U(1)$ completion (Section 7.2)—the full structural description is first gauge-coupled and, for fermionic matter, spinorial (Section 9.2). The Schrödinger orbital, with its familiar discrete eigenvalues and probability densities, then appears as the non-relativistic limit of that fuller spinorial structure: the regime in which velocities, binding energies, and spin-dependent relativistic corrections are all small compared with the rest scale. At the structural level, the discrete spectrum may be interpreted as arising from phase-consistency requirements on admissible bound configurations, in harmony with the winding/single-valuedness conditions of the coherent branch. This structural hierarchy—spinorial bound mode, non-relativistic envelope, Schrödinger orbital—does not add new quantitative content beyond what is already established above, but it clarifies the logical position of the Schrödinger equation within the broader ECT architecture.

Why this is already structurally nontrivial

The supplied compact-phase extension organises a conditional research architecture beyond a generic real broken-phase scalar model:

- compact-phase loop sectors with integer winding (BR1);
- fixed-core and positive-tension loop functionals defining the distinct candidates t_0^{EFT} and $t_{0,\text{bal}}^{\text{eff}}$; their four-dimensional owners, map to operational S_0 , and universal identification with $\hbar$ remain Open;
- a supplied scalar principal/clock convention rather than a derived physical time map;
- a conditional coherent wave reduction using the same coefficient.

This architecture sharpens a route toward explaining wave mechanics; it is not yet such an explanation.

Bridge to canonical structure

The conditional sequence tested in the Quantum Sector is:

$$\begin{array}{ccc} \text{supplied phase sector and } S_0 & \xrightarrow{\text{supplied complex law}} & \text{wave dynamics} \\ & \xrightarrow{\text{Open operator owner}} & \text{candidate canonical structure.} \end{array} \quad (22.8)$$

Only after a Schrödinger-type coherent law is supplied can one ask whether its variables admit a physical canonical organisation and whether its commutators share the same coefficient S_0 . That question belongs to Section 23.3.

Structural cross-check: restricted causal-cone inheritance. The coherent wave-kinematical reconstruction and the scalar geometric characteristic use the same ordered-phase tensor K^{AB} under the stated continuation, and therefore share c_{char} . This statement does not determine the principal symbol or speed of an Open physical tensor completion. In the scalar geometric branch this speed enters through the common-length ordered-branch parametrisation $x^0 = w$ and the resulting effective Lorentzian metric. In the coherent branch the formal continuation (22.2) is used only for the Euclidean–Lorentzian amplitude correspondence. The shared scalar/phase cone is a consequence of writing those two supplied actions with the same tensor, not a theorem for every physical sector. The joint GW170817/GRB 170817A interval $-3 \times 10^{-15} \leq (c_{\text{GW}} - c_\gamma)/c_\gamma \leq +7 \times 10^{-16}$ [12] shows, under the stated association and emission-time assumptions, that the observed gravitational and electromagnetic signals are consistent with a common causal cone. For ECT this is an external kill gate on the missing photon and tensor principal symbols, not a present prediction (Section 4.7).

Observational orientation (Level B). After the matched identification $S_0 = \hbar$, the coherent-wave organisation developed above becomes structurally compatible with the standard de Broglie and dispersion relations used in ordinary quantum mechanics. These relations are already confirmed by electron and neutron diffraction, atom and molecule interferometry, and Bose–Einstein condensate matter waves. ECT does not claim a completed first-principles derivation of all such phenomena in this subsection alone. Its claim here is conditional: a supplied phase EFT on the scalar ordered-branch background can be matched to standard wave kinematics. The real-scalar action does not derive the phase sector or its action scale.

Status summary. *Conditional Level A mathematics / ECT owner Open:* after a norm-preserving complex evolution law, Hamiltonian, domain and clock convention are supplied, its polar decomposition yields Hamilton–Jacobi-type and continuity equations.

Level B/Open closure: the displayed Schrödinger-type law and wave-mechanical reading require that supplied complex dynamics and the separate matching $S_0 = \hbar$; they do not follow from the real phase action alone.

Level B/open: the fully explicit coherent Hamiltonian $\hat{H}_{\text{coh}}$ in general regimes, the completed normalised quantum interpretation, and the full identification with ordinary quantum mechanics beyond the present structural route.

22.2 Principle of extremal action and saddle dominance

Status: Level A mathematics for stationary-phase / saddle-point asymptotics only after a Lorentzian oscillatory functional measure, contour, state, action, and domain are supplied; Level B/Open for an ECT physical owner and for finite- S_0 applications. Euclidean DP does not derive this Lorentzian weighting or its saddles (Section 6.6).

The Quantum Sector uses a separately supplied Lorentzian oscillatory functional integral. Within that declared model, its leading asymptotic is the standard stationary-phase expansion of the same supplied S_0 -weighted amplitude. This is not a derivation of the Lorentzian action, measure, contour, or state from Euclidean DP.

Coherent amplitude weighting

The coherent branch is weighted by amplitudes of the form (22.1):

$$\mathcal{A}[q] \sim \exp(iS[q]/S_0).$$

A separately supplied Euclidean measure instead uses its own dimensionless exponent:

$$\mathcal{W}_E[q] \sim \exp[-\mathcal{J}_E^\mu[q]]. \quad (22.9)$$

For the bare P3 source model, $\mathcal{J}_E^\mu = \kappa_{\text{P3}} \mathcal{J}_E^{\text{P3}} = S_E^{\text{P3,phys}} / \Sigma_{\text{PI,P3}}$. This measure owner is not obtained by reusing the Lorentzian canonical S_0 . The two asymptotics must not be conflated. In Lorentzian signature, $|\Delta S|/S_0 \gg 1$ produces oscillatory stationary-phase cancellation. For a real positive Euclidean weight there is no phase cancellation: Laplace/steepest-descent asymptotics selects minima or specified complex thimbles according to the sign, contour, measure, and boundary data. $|\Delta \mathcal{J}_E^\mu|$ together with the Hessian, contour, measure and boundary data controls the Euclidean asymptotic; $|\Delta S_E|/S_0$ is not an owner-correct Euclidean criterion.

Stationary-phase structure

Starting from the coherent amplitude weighting (22.1), the saddle-point structure can be exhibited explicitly. The relevant criterion is not merely that an action be “large” in an absolute sense, but that phase differences satisfy $|\Delta S|/S_0 \gg 1$, so that neighbouring histories contribute with rapidly oscillating relative phases. In that regime, the coherent functional integral is dominated by the vicinity of stationary configurations θ_{cl} satisfying

$$\left. \frac{\delta \mathcal{J}_{\theta,L}}{\delta \theta} \right|_{\theta_{\text{cl}}} = 0. \quad (22.10)$$

Expanding around such a configuration, $\theta = \theta_{\text{cl}} + \eta$, gives

$$\mathcal{J}_{\theta,L}[\theta_{\text{cl}} + \eta] = \mathcal{J}_{\theta,L}[\theta_{\text{cl}}] + \frac{1}{2} \int d^4x d^4y \eta(x) \mathcal{K}_{\text{cl}}(x,y) \eta(y). \quad (22.11)$$

For the explicitly retained quadratic functional this expansion terminates exactly at second order. A separately named interacting completion would restore higher variations, with its own Lorentzian signs, couplings and domain; none is inherited here by an ellipsis. Here

$$\mathcal{K}_{\text{cl}}(x,y) := \left. \frac{\delta^2 \mathcal{J}_{\theta,L}}{\delta \theta(x) \delta \theta(y)} \right|_{\theta_{\text{cl}}}.$$

Within this quadratic truncation, the corresponding formal Gaussian evaluation takes the form

$$Z_{\text{coh}} \approx \exp \left[i \frac{\Sigma_{\theta} \mathcal{I}_{\theta,L}[\theta_{\text{cl}}]}{S_0} \right] \cdot \left[\det \left(\frac{\Sigma_{\theta}}{S_0} \mathcal{K}_{\text{cl}} \right) \right]^{-1/2}, \quad (22.12)$$

up to the usual phase and regularisation subtleties of oscillatory functional determinants [46]. The first factor gives the leading coherent saddle weight. Because $\mathcal{K}_{\text{cl}}$ is the Hessian of the dimensionless functional, its Gaussian operator carries the same dimensionless ratio Σ_{θ}/S_0 as the exponent. The determinant remains formal: zero and negative modes, contour or thimble, measure normalisation, operator domain and regularisation must be supplied in the reduced coherent-sector EFT.

Saddle-point conditions

For a supplied Lorentzian oscillatory integral, stationary-phase histories satisfy

$$\delta S[q] = 0. \quad (22.13)$$

For a real Euclidean Laplace problem, an interior extremum obeys

$$\delta S_E[q] = 0, \quad (22.14)$$

but dominance additionally requires the correct minimum/thimble, Hessian signature, contour, measure, and boundary conditions. The Euclidean and Lorentzian conditions therefore do not follow from one common large-phase argument.

Relation to the Euclidean dynamical principle. Equation (22.13) is the standard saddle equation of the supplied Lorentzian functional integral. It resembles the stationarity equation of Euclidean DP, but the two have different actions, measures, contours, states, and boundary data. They coincide only if an explicit continuation or effective-action map is derived. Large relative phases then justify an integrated-amplitude asymptotic around the declared Lorentzian saddles; they do not recover actual classical histories from DP or select one realised path.

Status-separated variational uses (A-math conditional/B/Open). Variational notation occurs in several sectors, but that fact alone does not make their actions descendants of one Euclidean stationary problem:

- (I) **Fundamental Euclidean owner (DP):** stationarity of the declared condensate action $S_E[\Phi]$ supplies its field equation (Section 6.6); it is not a unique-history rule.
- (II) **Reduced-sector actions:** the equations of a reduced action follow from variation of that owning action. They inherit DP only if a variation-compatible restriction or controlled elimination from S_E is explicitly derived, with boundary data and discarded modes tracked. The scalar, phase/orientation, and Dirac-type EFTs displayed here are otherwise supplied conditional owners with sector-dependent status; shared variational form does not establish a common backbone.
- (III) **Macroscopic completion programmes (Part II):** Einstein/scalar–tensor, cosmological, and HRC equations follow by variation only after their respective effective actions and observable maps are supplied. The fixed director map has no linear TT or static-density vertex in its scoped theorem, so these physical actions do not presently descend from DP alone.
- (IV) **Conditional Lorentzian stationary phase (Part III):** after an oscillatory functional measure, contour, state and coefficient S_0 are supplied, $|\Delta S|/S_0 \gg 1$ gives the standard stationary-phase asymptotic near declared saddles. This does not follow from the Euclidean DP alone and is not decoherence or unique-history selection.

The four layers therefore share variational notation but not one completed physical owner: bare Euclidean stationarity, variation of separately supplied reduced actions, conditional macroscopic completions, and Lorentzian stationary-phase asymptotics have different assumptions.

Falsifiable content of the proposed operational-unit closure. Sector-dependent operational action thresholds would reject the proposed reuse of one S_0 , not the mathematical existence of either loop candidate, P1–P6 or ECT as a whole.

Conditional asymptotic content

Inside the supplied Lorentzian oscillatory integral, large phase differences can suppress non-stationary contributions in an integrated amplitude and small phase differences can retain interference. This is an asymptotic statement about that integral, not a theorem that actual trajectories become classical. Decoherence, records and outcome selection require their own coupling, state, kernel, trace and timescale. Finite- S_0 terms quantify departure from the saddle approximation only within the supplied integral. At the formal level they are encoded by the fluctuation operator $\mathcal{H}_{cl}$ and higher-order terms in the expansion around the stationary history. Their detailed interpretation in specific reduced systems remains part of the later constructive quantum programme.

Two distinct conditional ingredients associated with classical behaviour. The following asymptotic and open-system ingredients must not be conflated:

- (a) **Conditional saddle asymptotics:** when a supplied Lorentzian oscillatory integral has $|\Delta S|/S_0 \gg 1$, its leading integrated amplitude is approximated by neighborhoods of declared stationary points. This does not select an individual trajectory or establish physical classicality.
- (b) **Decoherence candidate:** off-diagonal terms are suppressed only if an identified coupling, state, influence/noise kernel, trace and finite timescale produce that result; unresolved multiplicity alone is insufficient.

The first mechanism is conditional on a supplied Lorentzian oscillatory measure. The second is a separate open-system calculation and is not yet an ECT explanation of the absence of macroscopic superpositions. The quantitative decoherence side of this transition is developed later in Sections 26.1 and 30.1.

Quantitative orientation (Level B). If one adopts the later Level B identification $S_0 = \hbar$, the stationary-phase criterion acquires its familiar scale. For a macroscopic mechanical system—for example, a kilogram-scale oscillator with centimetre-scale motion—characteristic actions exceed $\hbar$ by roughly 10^{30} – 10^{32} , so saddle dominance is extremely sharp. By contrast, in atomic-scale systems the relevant action differences are of order $\hbar$, and no such sharp suppression of neighbouring histories occurs. These estimates are standard consequences of the stationary-phase criterion; they serve only as scale orientation and do not replace the later decoherence analysis. The quoted orders of magnitude are illustrative rather than system-specific and should be read as scale estimates, not as a completed phenomenology of any particular experimental platform.

Observational orientation and ownership. Neutron, atom, and large-molecule interferometry illustrate the standard coexistence of stationary-phase approximations and observable interference. They test their owning quantum source models, not the ECT origin of the Lorentzian functional measure or of decoherence. ECT may use this separation as a target: a future completed coherent-sector action must reproduce the stationary-phase regime, while a separately derived physical environment channel must account for any loss of interference. No ECT-specific discrimination follows until both owners and a held-out correction are derived.

Conditional benchmarks for the extremal-action sector. Within a supplied Lorentzian oscillatory model the standard benchmarks are:

- (i) stationary-phase accuracy and environmental dephasing are logically distinct and can occur at different scales;

- (ii) $|\Delta S|/S_0$ controls the saddle expansion of that integral, not the absolute system size alone;
- (iii) physical macroscopic classicality additionally requires a derived system–environment coupling, state, kernel, trace, and timescale;
- (iv) a future ECT completion may test whether one physical action scale consistently normalises independently owned sectors. Born probabilities require their own measure and event owners.

The first two are standard conditional mathematics, the third is a standard open-system requirement, and the fourth is an Open ECT unification test. None is derived from P3 or Euclidean DP alone.

Status summary. *Established (Level A mathematics, conditional):* stationary-phase / saddle asymptotics for a supplied Lorentzian oscillatory integral with declared measure, contour, state, action, and domain. *Separately owned (B/Open):* reduced EFT equations and Part-II macroscopic equations follow from their own supplied or explicitly derived actions; they are not established descendants of Euclidean DP. *Open:* a universal map linking DP, reduced EFTs, macroscopic completions, and the Lorentzian coherent integral, together with the physical matching $S_0 = \hbar$ and quantitative applications.

22.3 Conservation laws in the coherent sector

Status: Level A within the supplied real-scalar action for its translational and rotational Noether currents; Level A within the supplied compact-phase EFT for its phase current $j^A = \rho^2 \partial^A \theta$ and for winding-number conservation. These are different owners. A continuity law is Level A mathematics only inside a separately supplied norm-preserving complex dynamics, while its physical probability-current reading, the Lorentzian energy–momentum interpretation, and globally defined observed charges remain Level B/Open. Connection: Section 6.6, Appendix R, Section 7.2, Section 13.3, and Section 22.1 (Eq. (22.6)).

In standard physics, conservation of energy-momentum, conservation of charge, conservation of probability, and loop/flux quantisation phenomena are introduced through distinct structural inputs: spacetime symmetry (Noether), gauge symmetry ($U(1)$), unitarity of quantum evolution, and topological or grand-unification arguments respectively.

ECT instead has owner-specific statements. Translation/rotation invariance of a declared action gives its Noether currents. A supplied compact-phase EFT gives a phase current and integer winding labels. A probability-current interpretation requires a physical complex state, inner product and operator normalisation and is not derived from the bare real scalar. These results may be tested for common ownership in a future completion, but they do not yet trace to one established conservation law.

Energy-momentum conservation

For the owning P3 action, Noether’s theorem gives

$$\partial_A \tau_{P3}^{AB} = -\frac{\delta \mathcal{J}_E^{P3}}{\delta \Phi} \partial^B \Phi, \quad T_{P3, \text{phys}}^{AB} = A_{P3} \tau_{P3}^{AB}, \quad \partial_A T_{P3, \text{phys}}^{AB} = 0 \quad \text{only on P3 shell.} \quad (22.15)$$

This is the local identity. Differentiable translation generators and conserved integrated charges additionally require admissible boundary or fall-off conditions. The curved-space law

$$\nabla_\mu T^{\mu\nu} = 0 \quad (22.16)$$

holds only inside a separately supplied diffeomorphism-invariant completion after all owning matter and metric equations are imposed. It is not a reinterpretation obtained by setting $x^0 = w$ and remains Open for P4.

When energy is not globally conserved. Equation (22.16) is a *local* conservation law. For a generally covariant completed geometry, a standard conserved global energy additionally requires appropriate asymptotics or an exact time-translation Killing vector. Two important conditional cases are:

- (a) **Cosmological expansion:** a completed expanding geometry may lack a timelike Killing vector and hence the corresponding standard global energy, as in generic GR cosmologies [91]. A time-dependent field solution on a fixed time-translation-invariant background does not by itself imply this failure. The physical ECT geometry and charge owner are Open (Section 13.3).
- (b) **Open coherent subsystem:** in a supplied system–environment model, energy can flow from the resolved subsystem to unresolved modes while the complete owning dynamics may retain its stated conservation law. This is standard reduced-system bookkeeping, not yet an ECT derivation of dissipation: the physical split, coupling, state, current, and trace remain Open (Section 26.1).

Angular momentum conservation

For a declared $O(4)$ -invariant Euclidean action, Noether’s theorem gives an off-shell rotational identity and

$$\partial_C J^{CAB} = 0 \quad \text{only on the owning equations of motion.} \quad (22.17)$$

This is the local on-shell statement. Differentiable generators and conserved integrated angular charges additionally require admissible boundary or fall-off conditions. For the supplied P4 vector datum, the generator algebra decomposes into three stabiliser generators J^{ij} and three mixed generators J^{wi} . This is conditional algebra; a physical broken vacuum and conserved orientation currents require their missing action/state owner. The standard complex generator continuation can represent iJ^{wi} as Lorentzian boosts only after a physical metric/clock and representation are supplied. That conditional algebraic correspondence is not an emergent physical Poincaré owner.

Phase-current conservation in the coherent branch

The coherent branch carries an additional conserved current beyond the translational and rotational structures. The global $U(1)$ phase symmetry of the compact-phase EFT (Section 7.2) yields:

$$j^A = \rho^2 \partial^A \theta, \quad \partial_A j^A = 0 \quad (\text{Level A within the compact-phase EFT}), \quad (22.18)$$

recalled from Eq. (7.5). The compact-phase Noether equation is Level A mathematics within the supplied phase EFT. A probability-current reading is different: it requires a supplied norm-preserving complex evolution law, state/inner product, operator domain, physical clock and the matching $S_0 = \hbar$. Only after those Level B/Open inputs are fixed may j^0 be compared with $|\Psi|^2$ and j^i with a probability current. Noether conservation alone does not derive unitarity or a physical probability interpretation.

Winding-number conservation

In addition to the differential Noether currents, the coherent branch carries a topological conservation law: the winding number

$$n[\gamma] = \frac{1}{2\pi} \oint_\gamma \partial_A \theta dX^A \in \mathbb{Z} \quad (22.19)$$

is invariant under smooth deformations of the field configuration within the supplied compact-phase BR1 closure (Level A topology under that premise).

Winding-number conservation is topological: it does not depend on the specific dynamics, only on the single-valuedness of Φ_{eff} and the smoothness of the configuration. This is a global sector-classification law, distinct from local Noether conservation. It constrains global sector changes and remains stable under smooth deformations within the compact-phase EFT.

Defect-mediated topology change

The topological protection of winding number holds only within the smooth coherent branch (BR1), where $\rho > 0$ everywhere. If the condensate amplitude vanishes at a localised defect core ($\rho \rightarrow 0$), the compact-phase description breaks down and the winding number can change discontinuously. This is the ECT structural analogue of instanton and sphaleron processes in gauge field theory.

Physical mechanism. In a superfluid, a quantised vortex can be created or annihilated only through processes in which the superfluid density drops to zero at the vortex core [10, 20]. In ECT, analogously, a change of the topological charge requires a configuration in which Φ_{eff} passes through zero. Such defect-mediated transitions are admissible configurations under S4. Their physical rate is not fixed by topology: it requires a defect action, barrier, statistical state and real-time continuation.

Conditional thermal suppression template (Level C/Open). If a supplied defect completion assigns an activation energy E_{defect} and an approximately thermal dilute state, the standard Arrhenius template is

$$\Gamma_{\text{defect}} \propto \exp\left(-\frac{E_{\text{defect}}}{k_B T}\right), \quad (22.20)$$

with a completion-dependent prefactor. The illustrative assignment $E_{\text{defect}} = \bar{M}_{\text{Pl}}$ is not retained: ECT has not derived an activation energy, thermal state or prefactor. The symbolic Arrhenius template classifies what a supplied thermal defect model would require; it does not predict defect accessibility at any temperature.

Connection to baryogenesis. These defect-mediated topology changes do not by themselves define physical baryon number, a baryon-violating vertex, a nonzero CP coefficient or a real-time nonequilibrium state. They are therefore only entries in the completion inventory of Section 16.3, not an ECT-native baryogenesis mechanism.

Observational orientation

The compact-phase conservation structure developed here is the structural counterpart of quantised circulation and flux in condensed matter: in superfluid ^4He , circulation is quantised as $\kappa = h/m_{\text{He}} \approx 9.97 \times 10^{-8} \text{ m}^2/\text{s}$; in type-II superconductors, the magnetic flux quantum is $\Phi_0 = h/2e \approx 2.07 \times 10^{-15} \text{ Wb}$. In each case the quantisation follows from the single-valuedness of a compact order-parameter phase (cf. Table 74).

Experimentally, electric-charge conservation has been tested to extraordinary precision: the best limits on electron charge non-conservation give lifetimes exceeding 10^{26} years [86]. ECT does not yet map these defect sectors to physical electric charge or derive a decay rate, so the observed bound is an external target rather than evidence for the ordered-medium architecture. The connection between defect-mediated transitions and baryogenesis is developed in Section 16.3.

ECT structural expectations and falsifiable requirements

The conservation-law architecture of the coherent sector suggests the following structurally testable expectations:

- (i) **Unified origin:** energy–momentum Noether currents, the BR1 phase current, loop-sector quantisation and the later probability-current reading are realised in different declared sectors. A common reduced action/state owner has not been derived; their common origin is a falsifiable architectural hypothesis rather than a result.

- (ii) **Topological robustness:** winding-number conservation provides robustness of the loop-sector classification under smooth perturbations within the compact-phase EFT. It is stronger than a purely local differential conservation law in that it constrains global sector changes rather than only local current flow.
- (iii) **Defect-mediated topological sector change:** transitions between different winding/topological sectors are possible only through defect processes (vanishing ρ) inside the supplied compact-phase closure. Their thermal rate, suppression scale and any lifting near a condensate scale require a supplied defect action, statistical state and real-time continuation; no Planck-temperature baryon-violation prediction follows from topology alone.
- (iv) **Cross-sector S_0 programme:** supplied coherent closures currently reuse S_0 in wave and canonical normalisations. A universality theorem is Open, and Born weights do not follow from S_0 without an independently admitted probability measure.

Conditional closure tests. Two observations would test the supplied completion rather than the bare postulates:

- (a) If the coherent wave, candidate probability-current, and gauge/charge closures required different action normalisations, that would falsify the proposed one- S_0 completion. P3 alone neither supplies nor equates those sector normalisations.
- (b) If a declared winding sector admitted a smooth transition between different integer sectors while the order parameter remained nonzero and single-valued on the contour, with no boundary flux, singularity or defect crossing, that would falsify the protection assumptions of that supplied sector.

Observed universality of $\hbar$ is an external target for item (a). Charge conservation tests a different statement and does not validate a common S_0 owner.

Status summary. *Established (Level A within the stated owner):* translational and rotational Noether currents of the supplied real-scalar action given P1–P3 + DP; the phase current $j^A = \rho^2 \partial^A \theta$ of the separately supplied compact-phase EFT; and winding-number conservation under smooth deformations within BR1. Their coexistence does not establish one common reduced action or probability-current owner. *Level B:* the Lorentzian interpretation of Noether currents as energy, momentum, and angular momentum; a continuity equation only after a norm-preserving complex law is supplied; the later probability-current reading additionally requires its state, inner product, domain, clock and $S_0 = \hbar$ matching and is not the phase Noether law itself. *Level C/Open:* the conditional Arrhenius template for thermal suppression of defect-mediated topology change, whose defect action and thermal history are supplied rather than derived. *Open:* fully explicit topological current algebra in the coherent branch; the precise defect-sector dynamics and its connection to non-perturbative baryogenesis; conserved charges associated with the full defect sector.

Bridge to the Hilbert-space programme. The conservation structures established in the present section are the last pre-Hilbert ingredients of the coherent branch. What remains is to determine under what conditions the Euclidean coherent architecture admits a positive-norm Hilbert-space reading, unitary evolution, canonical algebra, and the uncertainty principle. That bridge is the task of the next section.

23 Hilbert-Space Structure, Reflection Positivity, Canonical Algebra, and Unitarity

Inputs from previous sections: the supplied phase characteristic form (21.2) and its conditional Lorentzian orientation (22.3), the supplied operational action-unit slot S_0 and its Open loop-owner map (Section 21.2),

compact-phase topology and winding sectors (BR1), real parametrisation $x^0 = w$, and Schrödinger-type reduction (22.7). Aim: to identify the conditions under which the Euclidean coherent structure of ECT gives rise to a physical Hilbert space, unitary evolution, canonical commutation relations, and the uncertainty principle. Status: Level A mathematics for reflection positivity and the standard free-field OS reconstruction of the explicitly supplied quadratic scalar measures (massive radial mode and free massless Goldstone-coordinate measures, with their declared infrared/domain qualifications); Level A for the canonical organisation, for the topological quantisation of the global compact-phase generator, and for the Robertson–Schrödinger uncertainty inequality; Level A for the conceptual distinction between global unitarity of the full coherent description and effective reduced-sector non-unitarity; Level B/Open for the commutator normalisation candidate S_0 , whose nonzero value and uniqueness are not fixed by winding; Level B/open for the full interacting Hilbert-space reconstruction, global unitarity beyond the Gaussian regime, the completed local operator algebra, and the final physical identification $S_0 = \hbar$. Connections: P1 (Euclidean arena), P2 ($O(4)$ -invariant action), P5 (medium character), the UV-threshold programme (§8.3), and the downstream Born, entanglement, decoherence, and black-hole-information programmes.

The previous sections supplied a compact complex branch with integer winding sectors, two distinct loop-scale candidates, an operational S_0 with Open owner map, a conditional wave-dynamical reduction, and a scalar Lorentzian kinetic structure. They did not derive a nonzero universal action scale or physical quantum operator algebra. The present section addresses the next logical question: *under what conditions does this Euclidean coherent structure give rise to the full apparatus of quantum mechanics*—a physical Hilbert space with positive norm, a self-adjoint Hamiltonian, unitary time evolution, and canonical commutation relations?

This is the central bridge of the Quantum Sector. Without it, the resemblance between the coherent-branch wave dynamics and ordinary quantum mechanics would remain suggestive but formally incomplete. With it, the entire downstream quantum architecture—Born-type probabilities (§30.4), entanglement (§31), decoherence (§26.1), and the black-hole-information programme (§36)—acquires its common structural foundation.

Plan of this section. The section proceeds in four steps:

- (i) reflection positivity and the Osterwalder–Schrader route to a physical Hilbert space (Section 23.1);
- (ii) the status of unitarity in the coherent branch (Section 23.2);
- (iii) canonical structure and the status of commutation relations (Section 23.3);
- (iv) the uncertainty principle as a mathematical consequence (Section 23.5).

23.1 Reflection positivity and the route to a physical Hilbert space

Status: Level A mathematics for reflection positivity and standard OS reconstruction of the explicitly supplied free quadratic scalar measures (massive radial and free massless Goldstone-coordinate measures, with stated infrared/domain qualifications); Level B/open for the full interacting coherent-sector measure, observable algebra, and physical ECT identification. Connection: P1 (Euclidean arena), P2 ($O(4)$ -invariant action), P5 (medium character), the real parametrisation $x^0 = w$, the UV-threshold programme (§8.3), and the downstream Hilbert-space, Born, entanglement, decoherence, and black-hole-information programmes.

The standard route from a Euclidean field theory to a physical Lorentzian quantum theory passes through the *Osterwalder–Schrader (OS) reconstruction theorem*, in the form corrected in OS II [73, 74]. The theorem states that if supplied Euclidean Schwinger functions or a Euclidean measure satisfy the applicable OS package—temperedness together with a valid OS-II reconstruction regularity/growth condition (for example the $E0'$ linear-growth condition), Euclidean covariance, reflection positivity with

respect to a specified time reflection, permutation/graded symmetry, an identified observable algebra, and the required locality and clustering conditions in the chosen formulation—then it is possible to reconstruct a Lorentzian quantum field theory with a physical Hilbert space, a positive-definite norm, a self-adjoint Hamiltonian bounded below, and unitary time evolution. ECT is formulated on a Euclidean arena (P1), which provides an arena in which the OS hypotheses can be tested. Euclidean kinematics alone implies none of reflection positivity, OS-II growth/regularity, locality, clustering, an observable algebra or a physical state.

Reflection positivity for a supplied Euclidean measure. Reflection positivity is a condition on a complete Euclidean measure (or its Schwinger functions), not on a classical stationarity equation alone. Let Θ reverse the distinguished Euclidean coordinate, $\Theta w = -w$, while leaving x^i fixed. Write the supplied dimensionless measure exponent as $\mathcal{J}_E^\mu[\Phi]$:

$$Z_\mu = \int \mathcal{D}\Phi e^{-\mathcal{J}_E^\mu[\Phi]}. \quad (23.1)$$

For the bare P3 source model specifically,

$$\mathcal{J}_E^\mu = \kappa_{P3} \mathcal{J}_E^{P3} = \frac{S_E^{P3, \text{phys}}}{\Sigma_{PI, P3}}, \quad \kappa_{P3} > 0, \quad (23.2)$$

where A_{P3} normalises the physical classical action and $\Sigma_{PI, P3}$ independently normalises the path-integral weight. Other Gaussian, spinor or effective measures have their own exponent and normalisation; no equality to the P3 ratio, S_0 or $\hbar$ is implied.

The measure is reflection positive if, for every functional $F[\Phi]$ supported in $w > 0$,

$$\langle (\Theta F)^* F \rangle_E := \frac{1}{Z_\mu} \int \mathcal{D}\Phi (\Theta F[\Phi])^* F[\Phi] e^{-\mathcal{J}_E^\mu[\Phi]} \geq 0. \quad (23.3)$$

When this condition holds as part of the complete applicable OS package, the reconstruction theorem supplies the corresponding Hilbert-space structure and positive time-translation generator. Reflection positivity alone does not identify the physical ECT observables, the clock/metric map, a physical energy unit or the canonical S_0 .

Euclidean spinor conventions and the graded OS form. For the spinorial extension of the OS programme the following conventions are fixed once and used throughout (in particular in the spin–statistics analysis of §9.9 and Appendix AU). (i) *Gammas and Wick map.* Euclidean gamma matrices are hermitian and satisfy

$$\{\gamma_A^E, \gamma_B^E\} = 2\delta_{AB}, \quad (\gamma_A^E)^\dagger = \gamma_A^E, \quad A, B \in \{0, 1, 2, 3\}, \quad (23.4)$$

with γ_0^E associated to the distinguished Euclidean coordinate w_E . Within this purely Euclidean OS subsection and its test appendix we use the shorthand $\gamma_A \equiv \gamma_A^E$. For the mostly-plus Lorentzian house convention $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$, the compatible continuation is

$$x_L^0 = -iw_E, \quad \gamma_L^0 = \gamma_0^E, \quad \gamma_L^i = i\gamma_i^E, \quad (23.5)$$

which gives $\{\gamma_L^\mu, \gamma_L^\nu\} = -2\eta^{\mu\nu}$ and maps

$$D_L = i\gamma_L^\mu \partial_\mu^L - m \mapsto -(\gamma_A^E \partial_A^E + m) = -D_E.$$

Thus the Euclidean free kernel below is retained. This is a convention-matching statement, not a derivation of a physical continuation or spinor sector. (ii) *Kernel.* The free minimal Euclidean Dirac operator and two-point kernel are

$$D_E = \gamma_A \partial_A + m, \quad m > 0, \quad S_E = D_E^{-1}, \quad \tilde{S}_E(p) = \frac{m - i\gamma_A p_A}{p^2 + m^2}, \quad (23.6)$$

For the normalisation convention used below, the Berezin pairing is $\langle \psi_\alpha(x) \bar{\psi}_\beta(y) \rangle = S_{E,\alpha\beta}(x-y)$ for Grassmann fields $\psi, \bar{\psi}$. More generally a separately supplied dimensionless Gaussian coefficient $\kappa_F > 0$ in $\kappa_F \int \bar{\psi} D_E \psi$ gives covariance $\kappa_F^{-1} S_E$. This positive rescaling does not change the projector sign test, and κ_F is not identified with the P3 ratio, S_0 or $\hbar$. (iii) *Reflection*. The OS reflection is the same $\Theta : (w, \vec{x}) \mapsto (-w, \vec{x})$ already used above; on spinor fields it acts antilinearly, reverses operator order, and carries the spinorial reflection matrix γ_0 for the Dirac variables used here (if the spinor sector is later rewritten in Majorana or charge-conjugated variables, the reflection matrix and adjoint must be translated accordingly):

$$\Theta \psi(x) = \bar{\psi}(\Theta x) \gamma_0, \quad \Theta \bar{\psi}(x) = \gamma_0 \psi(\Theta x), \quad \Theta(AB) = \Theta(B) \Theta(A). \quad (23.7)$$

The Euclidean adjoint of a smeared field polynomial is defined through this Θ -operation together with complex conjugation of coefficients. (iv) *Graded OS form*. For elements F, G of the field algebra supported in the half-space $w > 0$, the OS sesquilinear form is

$$(F, G) \equiv \langle F \cdot \Theta G \rangle_E, \quad (23.8)$$

with the operator order fixed as written. Equation (23.8) is the order convention used in the spinorial OS test; its positivity properties are not assumed here, but are checked explicitly in the free-kernel calculation of Appendix AU. Equation (23.8) uses the same reflected-order convention as the bosonic formula (23.3). It is literally a restriction of one mixed measure only after that common boson–fermion measure and its independent normalisations are supplied. (v) *Scope and status*. These conventions lie in the standard Euclidean Fermi-field OS family, with γ_0 here playing the role usually denoted γ_4 . They are fixed at the level of the free minimal first-order kernel; Majorana/charge-conjugated reformulations, higher-derivative kernels, and the interacting sector would require separate treatment. The conventions themselves carry no derivational content (definition-level; no Level assignment); all statements proved with them inherit their own Levels.

Which OS inputs can and cannot be supplied by the ECT arena. The ECT postulate structure identifies a Euclidean setting for the OS programme, but the reconstruction hypotheses remain separate inputs or theorems:

- (i) P1 gives the Euclidean 4-manifold $\mathcal{M}^4$;
- (ii) P2 gives the $O(4)$ -invariant action $S_E[\Phi]$;
- (iii) P4 supplies a distinguished direction $n^A = \delta_w^A$ and BR1 supplies a separate coherent closure; together they provide a coordinate adaptation for the later Lorentzian interpretation; the reflection-positivity condition itself is imposed at the Euclidean level, independently of SSB;
- (iv) P5 gives the ordered branch a genuine medium interpretation: the Euclidean functional measure is read as the statistical weight of a physical Φ -medium configuration, not merely as a formal integration device. This interpretation does not prove reflection positivity or any other OS axiom;
- (v) the formal continuation (22.2) provides the Euclidean–Lorentzian amplitude bridge used in the coherent-sector reading; in this sense it is compatible with the OS route, while remaining distinct from the real parametrisation $x^0 = w$ used for the effective metric and hyperbolic dynamics.

Thus the Euclidean start makes the OS tests relevant but does not satisfy them. Only a supplied measure or Schwinger-function family that passes the complete named OS package supports the corresponding conditional reconstruction.

Status in explicitly supplied free Gaussian subsectors. For a supplied positive massive scalar Gaussian, define the dimensionless measure exponent

$$\mathcal{J}_{E,G}^{(2)}[\varphi] := \frac{\kappa_G}{2} \int d^4X [(\partial_A \varphi)^2 + m^2 \varphi^2], \quad \kappa_G > 0. \quad (23.9)$$

Its covariance is

$$C_G(p) = [\kappa_G(p^2 + m^2)]^{-1}, \quad (23.10)$$

and the formal one-loop Gaussian factor is proportional to $\det[\kappa_G(-\partial_E^2 + m^2)]^{-1/2}$ after the measure, regularisation and boundary domain are supplied. Thus κ_G controls fluctuation amplitudes but not the pole location or correlation length. For every $m^2 > 0$ and finite $\kappa_G > 0$, the corresponding Gaussian measure $e^{-\mathcal{J}_{E,G}^{(2)}} \mathcal{D}\varphi$ has the standard reflection-positive free-field OS package [148]. The normalisation convention $\kappa_G = 1$ may be chosen for this source model; it is not a derivation of equality between a physical action prefactor and a path-integral denominator.

The supplied covariance is Euclidean covariant and symmetric, has the standard regularity and local observable algebra, and clusters for $m > 0$. The P3 radial curvature satisfies $m_G^2 = 2\mu^2 > 0$ (Eq. (5.12)), so the standard free-field result applies to a declared P3 Gaussian only after its Hessian, measure ratio and domain are identified with this source model. This is Level A standard mathematics for the declared free Gaussian. Identifying it with the complete physical ECT radial sector or extending it to an interacting ordered measure remains Open.

Conditional same-channel OS–canonical normalisation gate. There is nevertheless a sharp consistency test once one and the same positive free mode has been completed in both languages. Let $m, \omega > 0$, use the infinite-Euclidean-time vacuum, and fix the same normalised observable q , the same calibrated m and pole residue, and no field/source renormalisation between the OS and canonical descriptions. If the supplied reflection-positive Euclidean channel has

$$S_{E,\text{ch}}^{\text{phys}}[q] = \int d\tau \frac{m}{2} [(\partial_\tau q)^2 + \omega^2 q^2], \quad d\mu_{\text{ch}} \propto \exp[-S_{E,\text{ch}}^{\text{phys}}/\Sigma_{\text{PI},\text{ch}}] \mathcal{D}q, \quad (23.11)$$

then its full normalised vacuum two-point function is

$$C_E(\tau) = \langle q(\tau)q(0) \rangle_E = \frac{\Sigma_{\text{PI},\text{ch}}}{2m\omega} e^{-\omega|\tau|}. \quad (23.12)$$

For the canonical oscillator with the same q, m, ω , Hamiltonian $H = p^2/(2m) + m\omega^2 q^2/2$ and $[\hat{q}, \hat{p}] = iS_{0,\text{ch}}$, the analytically continued ground-state correlator is

$$C_{\text{can}}(\tau) = \frac{S_{0,\text{ch}}}{2m\omega} e^{-\omega|\tau|}. \quad (23.13)$$

Equality of the full normalised correlators for this same completed channel therefore forces

$$\Sigma_{\text{PI},\text{ch}} = S_{0,\text{ch}}. \quad (23.14)$$

This is Level A free-mode mathematics under all printed owner-identification premises and a B/Open closure gate for an ECT channel. An isolated equal-time variance without those premises is insufficient: a field map $q_E = cq_{\text{can}}$, a mass or residue renormalisation, finite temperature, interactions, gauge/constraint reduction, or an open/non-vacuum state changes the comparison. If the Euclidean measure is obtained by Wick rotation from the canonical oscillator, the equality is true by construction rather than a new derivation. The gate derives neither a P3 identification nor $A_{\text{P3}} = \Sigma_{\text{PI},\text{P3}}$, no cross-sector universality, and no $S_0 = \hbar$; the last remains an empirical calibration. Applying the gate to P3 first requires a named physical Gaussian mode, its observable/source map, clock, state, residue and canonical completion.

Ordered-branch sign, state covariance and the reflection gate. The positive isotropic free Gaussian measure above must not be identified silently with the separately supplied real- w ordered-branch quadratic operator. For $n^A = \delta_w^A$ and $\alpha > \beta > 0$, the latter contains

$$S_{\text{ord}}^{(2)} \supset \frac{1}{2} \int d^4X (\beta - \alpha)(\partial_w \phi)^2. \quad (23.15)$$

As an ordinary real Euclidean Gaussian form this term is unbounded below at sufficiently large w -momentum and is not the positive measure used in the free OS proof. This sign is nevertheless the declared hyperbolic-signature feature of the effective real- w description; by itself it is not evidence that the fundamental Euclidean measure violates reflection positivity. Relating a positive branch measure, contour or analytic continuation to the physical hyperbolic operator remains Open.

There is a separate state-level gate. P2 supplies an $O(4)$ -covariant action, whereas P4 supplies a nonzero-vector chart with $O(3)$ stabiliser. The ordinary reflection $w \mapsto -w$ exchanges the $+u_0$ and $-u_0$ gradient branches. A paired-branch measure, combined internal/reflection operation, superselection reconstruction or reduced-symmetry replacement therefore requires a proof; averaging all orientations can erase the order parameter and fail clustering. The physical P4 measure and branch-compatible reflection remain Open.

Supplied free scalar and director-coordinate toy measures. The supplied nonzero-director orbit $O(4)/O(3) \simeq S^3$ has three local coordinates, but this is target-space kinematics rather than a physical Goldstone count. In the scalar exact-gradient ECT map, integrability leaves one longitudinal coordinate at linear order; no stationary P4 Hessian or physical soft-particle spectrum is presently owned.

Independently, a supplied free massless scalar in four Euclidean dimensions has the standard reflection-positive Gaussian measure, subject to its usual infrared/zero-mode domain qualifications [148]. Therefore each separately declared free scalar/director-coordinate Gaussian toy measure is locally reflection-positive and admits the corresponding sector-by-sector OS reconstruction once its domain is fixed. This mathematical statement does not promote three orbit coordinates to physical ECT particles, does not derive a P4 quadratic measure, and does not establish the OS axioms for the interacting ordered-sector completion.

Relation to the P3 correlation scale and Open P4 domain. The positive homogeneous P3 Gaussian has correlation length $m_{\Phi, \text{rad}}^{-1}$. It does not furnish a cutoff for the interacting ordered P4 branch or weaken the OS requirements there. A physical P4 measure, observable algebra, reflection, spectrum and validity domain remain Open and cannot be inherited from the P3 Gaussian by owner transfer.

Status beyond the quadratic sector (Level B/open). For the full interacting coherent-sector functional integral, the verification of reflection positivity is a nontrivial constructive task. In standard constructive QFT, such verification has been completed only for a small number of models (e.g. ϕ_2^4, ϕ_3^4). For ECT the analogous programme is:

- (i) construct the interacting Euclidean measure or Schwinger functions and the physical observable algebra beyond the Gaussian approximation;
- (ii) verify Euclidean covariance, the declared time-reflection map, reflection positivity, symmetry/regularity, and all applicable locality and clustering hypotheses for the reduced constrained sector;
- (iii) demonstrate that the resulting OS-reconstructed Hilbert space, time translations, and observables coincide with the physical coherent-branch clock/metric and wave-dynamical description.

This programme is Level B/open: it is structurally well-motivated and uses only ingredients already present in ECT, but the full nonperturbative completion has not yet been carried out. It belongs to the long-term constructive programme of the theory and is flagged as an open problem (OP-Q16).

Connection to graviton quantisation. If the orientation (spin-2) sector of ECT supplied a complete applicable OS package—not reflection positivity alone—then the same reconstruction theorem would provide a candidate Hilbert space for that sector without a separate canonical imposition of commutators on the metric variable. This would be a different reconstruction route from canonical quantum gravity and loop quantum gravity, which formulate geometric quantum kinematics and dynamics through their own quantisation programmes. The present work establishes this only as a programme target: the physical spin-2 mode, constrained observable algebra, Euclidean correlators, full OS hypotheses, and Lorentzian metric/clock identification remain open (OP-OS1).

Connection to the UV-threshold programme. The later UV-threshold programme (§8.3) is structurally relevant for the OS route. A future derived P4 pole could delimit the domain of the corresponding ordered-branch EFT, but no such threshold is presently owned. It therefore cannot solve or weaken the constructive reflection-positivity requirements for the interacting theory. The threshold and Hilbert-space programmes may later be linked only after the stationary P4 action and pole are supplied.

What a complete applicable corrected OS-II package would establish. If the full coherent sector supplies its Euclidean measure/correlators, observable algebra and time reflection, and verifies Euclidean covariance, reflection positivity, symmetry/regularity, and the applicable locality and clustering hypotheses, the OS theorem would deliver:

- (i) a physical Hilbert space $\mathcal{H}_{\text{phys}}$ with positive-definite inner product;
- (ii) a vacuum state $|\Omega\rangle$ with $\hat{H}|\Omega\rangle = 0$;
- (iii) a self-adjoint Hamiltonian $\hat{H} \geq 0$;
- (iv) a unitary time-translation group in the reconstructed parameter; writing it physically as $U(t) = e^{-i\hat{H}_{\text{phys}}t/S_0}$ additionally requires the clock, energy and canonical-action normalisation map;
- (v) a spectral representation for all correlation functions.

These are constructive consequences of the complete applicable corrected OS-II hypothesis set, not of reflection positivity alone. The full canonical quantisation programme of the Quantum Sector would then be grounded in this reconstruction rather than assumed independently.

Why this bridge matters downstream. The Hilbert space reconstructed by the OS route is the common arena for the rest of the Quantum Sector. Specifically: (i) the canonical commutator programme requires a positive-norm $\mathcal{H}_{\text{phys}}$ as its domain (Section 23.3); (ii) the Born-type probability reading requires positive inner-product structure (Section 30.4); (iii) the entanglement programme presupposes tensor-product structure inside a physical Hilbert space (Section 31); (iv) decoherence and the arrow-of-time programme require reduced-state descriptions built from an underlying globally unitary theory (Section 26.1); (v) the black-hole information programme likewise relies on global unitarity at the level of the full coherent description (Section 36). In this sense, reflection positivity is an indispensable falsifiable member of the OS bridge. The downstream quantum architecture also depends on the other reconstruction hypotheses, the physical observable/state map, and the separate probability and measurement owners.

Relation to the formal continuation used in ECT. The continuation (22.2) appears as an amplitude-language bridge and as the formal operation entering OS reconstruction. The relabelling $x^0 = w$ obtains neither a metric nor physical time. In Gaussian/free subsectors satisfying all OS axioms, the standard theorem reconstructs a positive Hilbert space and time-translation semigroup. For the full interacting coherent sector, the measure, reflection positivity, locality, cluster properties, physical observables and clock/metric map remain Open.

Discriminant: Euclidean-first versus Lorentzian-first readings. ECT postulates a primary Euclidean arena (P1) and treats OS reconstruction as a candidate route to a Lorentzian quantum description in verified subsectors. This ontological direction differs from an auxiliary Euclidean continuation, but it does not strengthen the theorem: a physical Hilbert space, metric, clock and interacting dynamics are outputs only where every OS and matching hypothesis is established.

Falsifier for the Hilbert-space programme. If the full interacting coherent-sector functional integral fails to satisfy reflection positivity, then the OS route cannot construct a physical Hilbert space with positive norm and globally unitary evolution. In that case the downstream coherent-branch programmes for canonical algebra, Born-type probabilities, entanglement, decoherence, and the black-hole-information reading would lose their common structural foundation. Reflection positivity is therefore a genuine architectural falsifier for the quantum reading of ECT, not a merely technical side condition.

Complementarity with the compact-phase action-scale programme. The reflection-positivity / OS line and the compact-phase / loop line developed in Section 21 address different aspects of the quantum programme (cf. the three-question decomposition in §20). The former concerns the *structural reconstruction* of a physical quantum theory from Euclidean data: Hilbert space, positive-definite norm, and unitary time evolution. The latter organises compact winding sectors, introduces two conditional loop-scale candidates, and leaves their map to the operational S_0 Open. The OS route can reconstruct the physical Hilbert-space and time-translation structure only after the complete applicable corrected OS-II package holds; reflection positivity alone is insufficient. Compact phase owns the integer winding but not the Euclidean measure exponent $\mathcal{J}_E^\mu$, the P3 ratio κ_{P3} , or a universal physical action scale. Conversely the Euclidean measure does not derive the canonical S_0 . The two are complementary Open closure programmes, not completed mechanisms.

Status summary. *Established (Level A mathematics in the stated free models):* (i) the explicitly supplied massive scalar Gaussian measure has the standard free-field OS package and reconstruction; (ii) the supplied free massless scalar coordinate measures have the same local result after their infrared/zero-mode domain is fixed; (iii) these mathematical reconstructions do not by themselves identify the complete physical ECT particle content or interacting observable algebra. *Structural programme (Level B/open):* the full interacting coherent-sector verification remains a constructive-QFT problem, but in ECT it is structurally tied to the same ordered-branch architecture that underlies the UV-threshold, canonical, Born, entanglement, decoherence, and black-hole-information programmes (OP-Q16).

23.2 Status of unitarity in the coherent branch

Status: Level A mathematics for global unitarity in the reconstructed free Gaussian models after their complete applicable corrected OS-II hypothesis set is satisfied; Level B/open for the full interacting coherent branch. Level A for the conceptual distinction between global unitarity of the full coherent branch and effective reduced-sector non-unitarity after tracing over unresolved modes. Connection: reflection positivity (Section 23.1), P2 ($O(4)$ -invariant Euclidean action), P5 (medium character of the ordered branch), and the downstream black-hole, decoherence, and entanglement programmes.

In each supplied free model where the complete applicable corrected OS-II hypotheses and reconstruction are actually established, unitarity of the reconstructed Lorentzian evolution is a mathematical consequence rather than an additional postulate. This is proved here only for the stated free Gaussian models. The interacting measure/correlators, observable algebra, full OS hypotheses, physical map, and unitarity remain Open, so no theory-wide consequence is asserted.

Global unitarity in explicitly positive Gaussian subsectors. For the supplied free Gaussian models satisfying the complete applicable corrected OS-II package, the reconstructed Hamiltonian is self-adjoint and bounded below, so the time-evolution operator $U(t) = e^{-i\hat{H}t/S_0}$ is unitary. This follows from the

standard OS theorem applied to those declared Gaussian measures and is a Level A mathematical result in those models, not yet a theory-wide physical ECT unitarity theorem. Extension to the full interacting coherent branch remains conditional on the completion of the OS programme (OP-Q16).

Global unitarity beyond the Gaussian regime (conditional/open). Beyond the Gaussian subsectors, global unitarity is a conditional structural expectation: it is the natural consequence of the OS programme applied to the full interacting action, but its verification requires the completion of OP-Q16. It should not be presented as an independently established result.

Two levels of unitarity in ECT. It is essential to distinguish two levels:

- (i) **Global unitarity target:** the total coherent-sector evolution on $\mathcal{H}_{\text{phys}}$ would be unitary after a positive interacting OS reconstruction. It is established here only in supplied free Gaussian models with their complete applicable corrected OS-II package and remains Open for the full branch. This is the target relevant for the black-hole information programme (Section 36).
- (ii) **Reduced-sector non-unitarity:** after tracing over unresolved condensate modes, the reduced density matrix of a subsystem evolves non-unitarily. This is the level relevant for decoherence and the arrow of time (Section 26.1).

The two levels are mutually consistent whenever global unitarity has been established, while the appearance of non-unitary evolution in reduced descriptions is a consequence of the system–environment split, not of fundamental information loss.

Medium reading of the two-level distinction (P5). Under the medium interpretation (P5), the two-level unitarity distinction acquires a physical rather than merely formal meaning. Conditional on completion of the interacting OS programme, global unitarity would mean that the total Φ -medium state evolves without fundamental information loss. Reduced-sector non-unitarity means that an observer with access only to a subsystem of the medium sees effective mixedness, not because the medium itself has lost information, but because unresolved condensate modes have been traced out. This is the same standard subsystem logic that governs decoherence in condensed-matter analogue systems: a total state can remain pure while a local probe exhibits effectively thermal or mixed behaviour. Its application to the complete interacting Φ medium is conditional, not an additional consequence of P5.

Why this matters. The distinction between global and reduced unitarity is an organising target of the Quantum Sector. It resolves the apparent tension between:

- the need for unitary evolution to prevent information loss (black-hole information problem, Section 36);
- the need for effective non-unitarity to produce decoherence, the arrow of time, and the quantum-to-classical transition (Section 26.1).

A completed positive theory would not need to choose between the two: standard partial tracing would then produce reduced non-unitarity without violating global unitarity. The present interacting ECT action has not yet completed the premise of that conditional statement.

Connection to the downstream quantum programmes. The two-level unitarity distinction is not an abstract point. It feeds directly into three already developed programmes: (i) the black-hole information programme (§36), which requires the still-open global-unitarity completion in order to exclude fundamental information destruction; (ii) the decoherence and arrow-of-time programme (§26), which requires effective reduced-sector non-unitarity in order to produce irreversibility and classicalisation; and (iii) the entanglement programme (§31), which, once a nonfactorising state/channel is supplied, must track how its correlations behave when reduced subsystems become mixed. In this sense, the unitarity programme is a prerequisite rather than an already established common owner for all three downstream programmes.

Discriminant with respect to collapse and fundamental non-unitary alternatives. Two familiar alternatives should be sharply distinguished from the ECT reading. In collapse-type models, microscopic evolution itself is modified by an intrinsically non-unitary law that produces classical outcomes. In fundamental-information-loss scenarios, the total state evolves from pure to mixed at a basic level. ECT investigates the globally unitary route, but P1–P6 do not yet prove it for the interacting theory. The checked statement is conditional: if the full coherent-branch description admits a positive OS reconstruction and unitary evolution, then apparent non-unitarity may arise in reduced subsystem descriptions after tracing over unresolved degrees of freedom. This is standard reduced-state logic, not an ECT-specific proof that the completed medium is globally unitary.

Falsifier for the unitarity programme. The ECT unitarity programme would fail if (i) the full interacting coherent sector failed reflection positivity or any other necessary OS reconstruction hypothesis, so that the Hilbert-space construction could not be completed; (ii) a completed analysis showed that reduced-sector non-unitarity is incompatible with global unitarity at the level of the full coherent theory, rather than arising as a standard subsystem effect; or (iii) the two-level distinction could only be maintained by introducing an additional collapse mechanism not derivable from the ordered-medium architecture. Any of these would force a fundamental revision of the downstream black-hole, decoherence, and entanglement programmes.

Relation to P2. Postulate P2 requires that the Euclidean action $S_E[\Phi]$ is $O(4)$ -invariant. This symmetry is a necessary but not sufficient ingredient for Lorentzian unitarity. In a complete OS reconstruction, Euclidean covariance contributes to Poincaré covariance, while the result also requires supplied correlators/measure, an observable algebra and time reflection, reflection positivity, symmetry/regularity, and the applicable locality and clustering conditions. P2 supplies only the action-level symmetry input; it does not establish the complete correlator axioms or physical ECT reconstruction.

Status summary. *Established (Level A mathematics in stated models):* (i) global unitarity in the standard reconstruction of the supplied free Gaussian measures after the complete applicable corrected OS-II hypothesis set is met; (ii) the distinction between global unitarity of the full coherent description and effective reduced-sector non-unitarity of traced subsystems is conceptually well-defined and structurally consistent. *Conditional programme (depends on OP-Q16):* global unitarity for the full interacting coherent branch; not yet independently established. *Structural consequence:* reduced-sector non-unitarity is compatible with global unitarity and underlies the decoherence, arrow-of-time, entanglement, and black-hole-information programmes without requiring a primitive collapse or information-destruction law.

23.3 Canonical structure and the status of commutation relations

Status: Level A mathematics for the canonical organisation of the supplied reduced phase dynamics; Level B/Open for using the supplied operational parameter S_0 as a candidate normalisation of the coherent-sector commutator bridge, with its loop-sector owner unproved; uniqueness, nonzero universality and the full operator algebra, multi-particle completion, and the final physical identification with the observed canonical quantum structure. Connection to previous subsections: Sections 21.1, 21.2, 22.1, 23.1.

Given a supplied operational action unit S_0 , its Open owner map, and a Schrödinger-type wave-dynamical reduction, the next question is whether its variables admit a canonical organisation. This question must be treated carefully. ECT does not begin by postulating canonical commutators. Rather, it asks whether the coherent branch naturally develops the kinematic structure that ordinary quantum theory usually assumes from the outset.

What is already fixed before operator quantisation. Before any operator algebra is postulated, the current closure supplies three ingredients: (i) a reduced phase dynamics with canonical variables, (ii) one operational action-unit slot S_0 reused by the conditional wave/canonical sector, whose loop owner, nonzero

value and universality are not fixed by winding, and (iii) a Hilbert-space route in the reflection-positivity programme (Section 23.1)—standard and rigorous for the supplied free Gaussian packages under all named OS hypotheses, and structurally prepared, but not yet completed, for the full interacting coherent branch. The canonical-commutator question in ECT is therefore not whether a commutator can be freely imposed, but whether these already established ingredients single out a unique normalisation candidate for the operator bridge.

Canonical organisation of the reduced coherent sector

The action-dimension factorisation must also be carried through the canonical variables. Let $\mathcal{L}_{\text{red}}$ denote the Lagrangian density of the dimensionless positive-energy Lorentzian functional $\mathcal{I}_{\theta,L}$, and let $\mathcal{L}_{\text{phys}} = \Sigma_{\theta} \mathcal{L}_{\text{red}}$ with the declared finite positive, constant and θ -independent $\Sigma_{\theta} > 0$. Define

$$\begin{aligned}\pi_{\theta}^{\text{red}}(x, t) &:= \frac{\partial \mathcal{L}_{\text{red}}}{\partial(\partial_t \theta)}, \\ \pi_{\theta}^{\text{phys}}(x, t) &:= \frac{\partial \mathcal{L}_{\text{phys}}}{\partial(\partial_t \theta)} = \Sigma_{\theta} \pi_{\theta}^{\text{red}}(x, t).\end{aligned}\tag{23.16}$$

The last equality is conditional on the exact regular same-coordinate first-order relation $\mathcal{L}_{\text{phys}} = \Sigma_{\theta} \mathcal{L}_{\text{red}}$ after all constraints and gauge reductions, with constant Σ_{θ} and no additional symplectic term. More generally the symplectic potentials may obey $\Theta_{\text{phys}} = \Sigma_{\theta} \Theta_{\text{red}} + dF + \Theta_{\text{top}}$. An independently normalised compact total derivative $a\dot{\theta}$ is locally exact, generically carries a nontrivial holonomy on S^1 , and gives $\pi_{\theta}^{\text{phys}} = \Sigma_{\theta} \pi_{\theta}^{\text{red}} + a$; the integrated generator shifts by the integrated coefficient and the flat holonomy/twist shifts with it. In the zero-mode convention printed below, removal leaves the periodic domain unchanged only for trivial quantum holonomy $\exp(i2\pi a/S_0) = 1$, equivalently $a/S_0 \in \mathbb{Z}$; otherwise it moves the effect into the quasiperiodic parameter v_{tw} . This absorption is allowed only after the action, convention and self-adjoint domain are declared, and the term must not then be counted a second time as an independent twist. Under the printed no-extra-term premises the scaling is Level A variational mathematics inside the supplied smooth reduced model. It neither derives the physical quantum operators nor fixes Σ_{θ} or S_0 .

Poisson brackets versus quantum commutators

The reduced and physical symplectic normalisations must not be mixed. With the bracket normalised to the reduced functional,

$$\{\theta(\mathbf{x}, t), \pi_{\theta}^{\text{red}}(\mathbf{y}, t)\}_{\text{red}} = \delta^{(3)}(\mathbf{x} - \mathbf{y}).\tag{23.17}$$

Equivalently, the symplectic bracket defined by the physical action uses $\{\theta, \pi_{\theta}^{\text{phys}}\}_{\text{phys}} = \delta^{(3)}$. These are the same classical organisation after the constant rescaling, not two independent canonical pairs.

In a local angular chart the conditional quantum bridge may be written formally on a declared common core for the physical momentum:

$$[\hat{\theta}(\mathbf{x}, t), \hat{\pi}_{\theta}^{\text{phys}}(\mathbf{y}, t)] = iS_0 \delta^{(3)}(\mathbf{x} - \mathbf{y}) \iff [\hat{\theta}(\mathbf{x}, t), \hat{\pi}_{\theta}^{\text{red}}(\mathbf{y}, t)] = i \frac{S_0}{\Sigma_{\theta}} \delta^{(3)}(\mathbf{x} - \mathbf{y}).\tag{23.18}$$

Thus the unscaled iS_0 commutator cannot be attached to $\pi_{\theta}^{\text{red}}$. Because the phase is compact, the displayed coordinate CCR is not asserted as a global identity of self-adjoint angle and momentum operators on one invariant domain; the global unitary/Weyl form is given below. Equation (23.18) remains Level B/Open: it requires a Hilbert/Weyl representation, operator domains and quantisation map, and it does not derive S_0 , Σ_{θ} , their owner map or the calibration $S_0 = \hbar$.

Status of the commutator-normalisation candidate. The physical phase-weight ansatz $\exp(i\Sigma_{\theta} \mathcal{I}_{\theta,L}/S_0)$, the physical-momentum commutator above and a Schrödinger-type reduction may reuse one supplied operational action unit. This is an economical consistency ansatz, not a proof that either loop route owns S_0 , that Σ_{θ} is universal, or that the operator representation exists.

Compact shift orbit and global generator. Integer winding of an S^1 -valued field configuration and the spectrum of a canonical global-shift generator are different statements. The rotor route exists only if the constant-shift orbit $[\theta] \mapsto [\theta + \alpha]$, $\alpha \in S^1$, survives the physical gauge quotient and all Gauss constraints, and if its zero-mode symplectic form is finite and nondegenerate (equivalently, the reduced rotor has finite nonzero inertia). A gauged or Stueckelberg completion can remove this orbit; for example, a constrained zero mode of $L = I(\dot{\theta} - A_0)^2/2$ may impose $Q = 0$. Compact field values alone do not guarantee a physical rotor. On a noncompact slice a normalised smearing or another finite zero-mode construction is also required.

In the trivial spatial-winding sector on a finite three-torus, a chosen lift allows the orbit coordinate α to be represented by the volume-average angle θ_0 modulo 2π . In a nonzero-winding sector no global real lift and no cut-independent volume average generally exists. One instead chooses a reference map θ_* and writes the constant-shift orbit as $\theta_\alpha = \theta_* + \alpha$; the global unitary is $U_0 = e^{i\alpha}$, not the exponential of a cut-dependent average. When the physical quotient, finite symplectic and no-extra-topological-term premises hold, define

$$Q_\theta^{\text{red}} := \int d^3x \pi_\theta^{\text{red}}, \quad Q_\theta^{\text{phys}} := \int d^3x \pi_\theta^{\text{phys}} = \Sigma_\theta Q_\theta^{\text{red}}. \quad (23.19)$$

In a local angular chart the formal CCR has the sign convention $[\hat{\alpha}, \hat{Q}_\theta^{\text{phys}}] = iS_0$ and hence $[\hat{Q}_\theta^{\text{phys}}, \hat{\alpha}] = -iS_0$. It is not a global identity between self-adjoint operators on the circle: multiplication by α does not preserve the quasiperiodic domain below.

The globally defined angular operator is $U_0 = e^{i\alpha}$. On $L^2(S^1, d\alpha/2\pi)$, more precisely on sections of the supplied flat line bundle that implements the chosen holonomy, the representation $\hat{Q}_\theta^{\text{phys}} = -iS_0\partial_\alpha$ requires a self-adjoint domain. For the quasiperiodic family $\psi(\alpha + 2\pi) = e^{iv_{\text{tw}}} \psi(\alpha)$, $0 \leq v_{\text{tw}} < 2\pi$, the domain-preserving Weyl relation and spectrum are

$$[\hat{Q}_\theta^{\text{phys}}, U_0] = S_0 U_0, \quad U_0^\dagger \hat{Q}_\theta^{\text{phys}} U_0 = \hat{Q}_\theta^{\text{phys}} + S_0, \quad \text{spec } \hat{Q}_\theta^{\text{phys}} = S_0 \left(\mathbb{Z} + \frac{v_{\text{tw}}}{2\pi} \right). \quad (23.20)$$

$\hat{Q}_\theta^{\text{red}} = \hat{Q}_\theta^{\text{phys}}/\Sigma_\theta$ therefore has the general spectrum

$$\text{spec } \hat{Q}_\theta^{\text{red}} = \frac{S_0}{\Sigma_\theta} \left(\mathbb{Z} + \frac{v_{\text{tw}}}{2\pi} \right).$$

The strictly periodic scalar-wavefunctional domain $v_{\text{tw}} = 0$ gives the special lattices $S_0\mathbb{Z}$ and $(S_0/\Sigma_\theta)\mathbb{Z}$. The supplied physical shift orbit, Weyl representation and self-adjoint domain yield this standard extension family; compactness alone selects neither the orbit's physical survival, $v_{\text{tw}} = 0$ nor the representation. A nonzero twist may equivalently encode a supplied flat holonomy. Spectral spacings determine S_0 for the physical generator, or S_0/Σ_θ for the reduced generator, only after the corresponding operator has been physically identified; spacings alone determine neither the twist offset nor Σ_θ . The topological winding label $n_{\text{wind}} \in \mathbb{Z}$ remains exact in the declared compact smooth-field domain and must not be identified automatically with a generator eigenvalue.

Local versus global content. The local CCR, survival of the constant-shift orbit after the physical quotient, its finite symplectic form, the physical/reduced momentum map, the absence or declared single absorption of extra topological symplectic terms, the zero-mode self-adjoint domain and the physical identification of the generator are separate owners. The displayed domain-dependent spectrum is Level A operator mathematics after those premises are supplied; the ECT representation and normalisation remain B/Open.

Structural triangle: winding – canonical – action scale. Integer winding, a candidate canonical algebra, Σ_θ and S_0 have different owners. Winding follows from the supplied compact topology. The canonical bridge requires a physical symplectic form, quantisation map and operator domain. The periodic spectrum additionally requires $v_{\text{tw}} = 0$. Therefore there is no closed theorem deriving the physical action quantum or its universality from winding.

Connection to the downstream quantum programmes. If completed, the physical canonical bridge supplies an input to the uncertainty, probability-representation, entanglement and decoherence programmes. None of those programmes converts the conditional commutator or the periodic-domain spectrum into a derived probability or outcome rule.

From phase variables to wavefunction kinematics

In the wave-dynamical representation (22.4), the phase functional $\mathcal{S}$ plays the role of the Hamilton principal function of the coherent branch. In this representation the natural wave-mechanical counterpart of the canonical bracket structure is a momentum operator of the schematic form

$$\hat{p} \sim -iS_0 \nabla_{\mathbf{x}} \quad (23.21)$$

(schematic 3D form; the 4D covariant version follows the same logic with $g_{\text{eff}}^{AB} \partial_B$ replacing $\nabla_{\mathbf{x}}$). Again, ECT has not proved full operator quantisation. Within the supplied candidate closures, the same parameter S_0 is reused in:

- loop-sector phase weights,
- Schrödinger-type evolution (22.7),
- and candidate canonical-operator normalisation.

This is an internal consistency ansatz, not evidence that S_0 is the physical universal action scale; its finite value, origin, and universal ownership remain Open. Once the operator bridge is admitted, the same single normalisation extends further: it links loop-sector phase weights, the Schrödinger-type wave reduction, the candidate canonical commutator, and the uncertainty bound of Section 23.5. If any one of these required a different action-scale normalisation, the compact-phase coherent architecture would fragment.

What is still missing

Despite this structural picture, the following steps are not yet complete:

- (i) a rigorous construction of the full operator algebra of the coherent sector;
- (ii) the derivation of canonical commutators beyond the phase variable itself (charge sectors, spin, multi-particle structure);
- (iii) the proof that the canonical structure survives the full decohering environment and defect sectors;
- (iv) the final empirical identification $S_0 = \hbar$ (21.7).

For this reason the Quantum Sector does not claim that canonical quantisation of ordinary quantum mechanics has been derived. The supplied closure identifies a possible operator-normalisation slot and one candidate coefficient; it does not establish that this slot is physically realised or that the coefficient is correct or unique.

Discriminant with respect to standard and hidden-variable readings. Standard quantum mechanics takes the canonical commutator as a primitive axiom. The ECT programme instead asks whether a medium completion can derive it. The present canonical bridge is only a conditional kinematic organisation of the supplied smooth coherent branch. Whether canonical, loop, and wave normalisations share one physical S_0 is an Open matching/closure problem, not a consequence of P1–P6.

External matching target: universality of the quantum of action. Observed physics uses one and the same $\hbar$ across atomic spectroscopy, molecular binding, solid-state band structure, matter-wave interferometry, and particle-physics cross sections. No sector of observed physics has ever required a different action scale for its canonical commutation structure. This universality is an external matching target and constraint, not evidence for the ECT single-medium architecture. The present coherent closures reuse one operational parameter S_0 ; they do not derive it from either loop candidate or prove that all physical sectors inherit a unique canonical normalisation. Only after a common ECT owner is derived would a sector-dependent commutator scale test that named completion; before then it constrains the external target rather than falsifies an existing ECT prediction.

Falsifier for the canonical-bridge programme. If different parts of the coherent quantum programme required incompatible action-scale normalisations for (i) loop weights, (ii) Schrödinger-type evolution, (iii) canonical commutators, and (iv) Born-type probability structure, then the single-scale coherent-branch architecture would fail. Likewise, if a completed operator derivation forced a canonical normalisation unrelated to S_0 , the ECT canonical-bridge programme would be structurally falsified. The canonical question is therefore a genuine cross-sector test of the one-medium architecture, not a merely technical completion problem.

Why this is already a substantial result

Most approaches that attempt to derive quantum mechanics from a deeper substrate struggle to explain why a canonical algebra should emerge at all, and why it should be normalised by a universal action scale. ECT turns this into a concrete matching programme: the reduced closure already contains a common parameter S_0 and a wave-dynamical structure. The compact phase quantises winding, but does not topologically prepare a nonzero universal action scale or derive the physical canonical algebra.

Bridge to uncertainty, probability, and decoherence

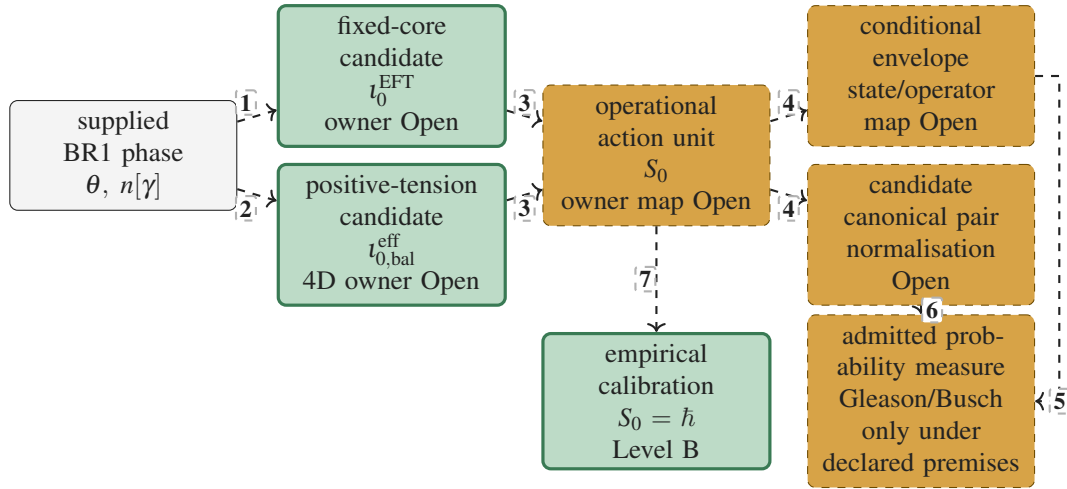
Canonical structure alone is not yet enough to recover the full operational content of quantum theory. One must still explain: (i) why the commutator bridge yields the uncertainty relation (Section 23.5), (ii) which physical probability measure is supplied and whether ordinary projector Gleason represents that normalised, noncontextual, countably additive projector measure on a complex Hilbert space with $\dim \mathcal{H} \geq 3$ (with the Busch effect/POVM theorem under separately declared C2q hypotheses required for a qubit) (Section 30.4), and (iii) why coherent amplitudes decohere in macroscopic environments (Section 26.1). Those questions belong to the subsequent sections of Part III.

Status summary. *Established (Level A mathematics under supplied premises):* (i) a supplied symplectic phase-space model has the standard Poisson algebra; (ii) a supplied compact phase has integer winding/representation labels. Neither premise is a derivation of the physical quantum operator algebra. *Conditional candidate (Level B/Open):* S_0 is the common parameter used for coherent phase weights, wave reduction and the candidate commutator normalisation; uniqueness and the physical match to $\hbar$ remain to be proved. *Open (Level B/open):* the full local operator algebra, many-body canonical completion, and final physical identification with the observed canonical quantum structure remain to be derived in full detail.

For the present loop-based calculation status and the open first-principles derivation of the action scale S_0 , see Appendix AS.

23.4 Conditional charge lattice in a supplied compact local $U(1)$ completion

Status: Level A mathematics for integer representations of an already supplied compact local $U(1)$ bundle. P3–P5 do not derive that bundle, its physical electromagnetic identification, matter representations, the elementary unit e_0 , an unbroken photon, or the observed charge lattice; these remain Open. The



Arrow key: 1 fixed length/profile; 2 positive tension; 3 still-Open owner map; 4 supplied action unit; 5 measure/state required; 6 does not derive probability; 7 separate physical match.

Figure 21: Status-separated dependency map for the coherent-sector quantisation programme. The phase-only fixed-core coefficient ι_0^{EFT} and the optional positive-tension coefficient $\iota_{0, \text{bal}}^{\text{eff}}$ are distinct reduced-model candidates. Neither is silently identified with the operational action-unit slot S_0 used by the conditional wave and canonical closures; each owner map is Open. The later $S_0 = \hbar$ identification is a second, explicit Level B empirical calibration. The fixed-length bound and the minimisation of the supplied positive-tension 1D functional are exact only inside their declared models. No arrow derives a probability measure, an outcome, a four-dimensional loop owner, or Planck's constant. Numbered arrow keys replace the former sloped micro-labels without changing the graph topology. Colour is redundant with literal status text and dashed borders.

commutator route is Level B/Open because its operator algebra and S_0 normalisation are not derived. The electroweak $\mathbb{Z}_2$ quotient is conditional Level B. Connection: compact phase sector (§21.1), conditional local $U(1)$ completion (§7.2), Dirac quantisation condition (23.20), electroweak structure (§49.14).

Problem statement: conservation versus quantisation versus the observed charge spectrum

Inside a supplied local $U(1)$ completion, Noether's theorem implies conservation of the corresponding model charge. This, however, is not yet the same as charge quantisation. Conservation states that a charge, once assigned, is preserved. Quantisation asks why the allowed charges belong to a discrete lattice rather than a continuum. Three levels must be kept sharply distinct:

- (i) *charge conservation*, $\partial_\mu J^\mu = 0$, which is conditional on a supplied local symmetry and current (§7.2);
- (ii) *Abelian charge quantisation*, $q \in \mathbb{Z} \cdot e_0$, i.e. why charges form a discrete lattice;
- (iii) *the full observed Standard-Model charge pattern*, $Q_{\text{em}}/e = T_3 + Y$ in the small- Y convention, with values $0, \pm 1/3, \pm 2/3, \pm 1$, which requires in addition electroweak embedding, the integer character $q_f = 6Y_f$, the colour–weak centre interlock, and a derivation of the fermion representation content.

The present subsection gives the conditional compact-group representation theorem behind (ii). It does not establish that the ECT compact phase is the physical electromagnetic bundle or that its representation labels are observed electric charges; both (ii) and (iii) remain Open as ECT identifications.

Conditional compact-phase route to an integer lattice

For a supplied compact $U(1)$, fixed generator normalisation and strongly continuous single-valued one-dimensional unitary characters, the representation labels are integers. The topology of a noncompact

$\mathbb{R}$ gauge group does not enforce such a lattice, although a particular noncompact theory may still have a discrete matter spectrum from separate dynamics or representation content. Compactness fixes the character labels under the stated premises; it fixes neither the elementary unit e_0 nor the observed electric-charge map. Which bundle and matter content nature realises is an empirical input.

In ECT the supplied BR1 phase is compact, $\theta \sim \theta + 2\pi$ (§7.2). This gives a compact global variable. It does not by itself derive a local gauge group, its matter representations, or the physical electromagnetic charge operator. If those completion data identify the local fibre with compact $U(1)$, the standard integer representation theorem below applies.

Theorem 23.1 (Compact- $U(1)$ charge lattice). *Let the condensate phase sector be compact, $\theta \sim \theta + 2\pi$, and suppose a supplied Abelian gauge completion makes local rephasings the redundancy of this compact phase. Then any matter excitation transforming in a strongly continuous, single-valued one-dimensional unitary representation of this compact $U(1)$ has an integer character label and, after a separately supplied generator normalisation e_0 , carries charge*

$$q_n = n e_0, \quad n \in \mathbb{Z}, \quad (23.22)$$

where e_0 is the elementary Abelian charge unit of the condensate sector.

Proof. A charged excitation transforms as $\psi \rightarrow e^{i\lambda\alpha(x)}\psi$ under a local gauge transformation with parameter α . Because the gauge parameter is angular ($\alpha \sim \alpha + 2\pi$), the transformation with $\alpha = 2\pi$ is the identity element of compact $U(1)$. Physical single-valuedness therefore requires $e^{i2\pi\lambda} = 1$, hence $\lambda \in \mathbb{Z}$. Restoring the elementary charge unit e_0 , the covariant derivative takes the form $D_\mu = \partial_\mu - i n e_0 A_\mu$, and the allowed charges are precisely $q_n = n e_0$. $\square$

Status. Theorem 23.1 is Level A mathematics conditional on a supplied compact local bundle, fixed generator normalisation and strongly continuous single-valued one-dimensional unitary representation. P5 does not derive those physical premises. The mapping of the integer label to electric charge, the elementary unit e_0 , fractional Standard Model charges, matter representations and anomaly constraints remain Open. The theorem does *not* require the canonical commutator candidate (23.18).

Canonical generator spectrum (independent conditional route)

A second route concerns the generator conjugate to the compact global-shift orbit; it does not by itself reproduce the electric-charge lattice of Theorem 23.1. This route first assumes that the constant-shift orbit survives the physical gauge quotient/Gauss constraints and has a finite nondegenerate symplectic form and finite nonzero inertia. A constrained gauge zero mode such as $L = I(\dot{\theta} - A_0)^2/2$ can instead impose $Q = 0$ and remove the rotor. In a local chart with orbit coordinate α the formal signs are $[\hat{\alpha}, \hat{Q}_\theta^{\text{phys}}] = iS_0$ and $[\hat{Q}_\theta^{\text{phys}}, \hat{\alpha}] = -iS_0$, while globally the unitary $U_0 = e^{i\alpha}$ obeys $[\hat{Q}_\theta^{\text{phys}}, U_0] = S_0 U_0$. The reduced-generator relation $\hat{Q}_\theta^{\text{red}} = \hat{Q}_\theta^{\text{phys}}/\Sigma_\theta$ additionally assumes the exact constant action scaling and absence of an independently normalised total-derivative/topological symplectic term. A term $a\dot{\theta}$ shifts the physical generator and flat holonomy; it may be absorbed once into the twist below only after the action/domain convention is fixed. On sections of the supplied flat line bundle over $L^2(S^1, d\alpha/2\pi)$, take the self-adjoint domain $\psi(\alpha + 2\pi) = e^{iv_{\text{tw}}} \psi(\alpha)$, Eq. (23.20) gives

$$\text{spec } \hat{Q}_\theta^{\text{phys}} = S_0 \left(\mathbb{Z} + \frac{v_{\text{tw}}}{2\pi} \right).$$

The reduced spectrum is $(S_0/\Sigma_\theta)(\mathbb{Z} + v_{\text{tw}}/2\pi)$; only the separately supplied periodic domain $v_{\text{tw}} = 0$ yields the unshifted lattices. The surviving physical shift orbit, Weyl representation and self-adjoint extension make the spectrum discrete, but compactness alone selects neither the twist, S_0 , Σ_θ nor the physical charge map. Level spacings identify S_0 or S_0/Σ_θ only after the physical or reduced generator is known; they do not determine the twist offset or Σ_θ by themselves.

This route is Level A operator mathematics under those premises and B/Open as an ECT physical identification. The compact- $U(1)$ representation theorem is logically independent: its integer character lattice uses the group, representation and elementary charge unit. Identifying the canonical rotor generator with observed electric charge requires an additional representation/unit map and cannot be inferred from compactness alone.

Loop/holonomy viewpoint

The same conclusion may be restated in the loop language natural to ECT. Transport of a charged excitation around a closed loop must return the same physical state up to an allowed $U(1)$ phase. Compatibility with the compact phase fibre forces the charge to lie on the discrete lattice $q/e_0 \in \mathbb{Z}$.

Electroweak conditional refinement

If a completion explicitly supplies $U(2) = [SU(2) \times U(1)_n]/\mathbb{Z}_2$, the $U(1)_n$ character must be an integer. With the full colour-compatible normalisation $n = q_f = 6Y_f$, a representation descends exactly when $d_f + q_f \equiv 0 \pmod{2}$ (49.2). The quark doublet $(d, q) = (1, 1)$ therefore is a valid standalone $U(2)$ representation. What standalone $U(2)$ lacks is the independent colour- $\mathbb{Z}_3$ triality interlock. For colour singlets only one may alternatively use $n_{\text{sing}} = q_f/3 = 2Y_f$, which gives $Q_{\text{em}}/e = T_3 + n_{\text{sing}}/2 = T_3 + Y$; this weight is not integral for quarks and must not be applied to them.

An even order-parameter potential establishes only $V(\Phi) = V(-\Phi)$ for that potential. It does not prove that the central element acts trivially on every physical field, operator and line operator, so it does not derive the physical $U(2)$ quotient. The refinement is Level A conditional group mathematics and Open as an ECT identification.

Anomaly cancellation as a self-consistency requirement

For any supplied chiral gauge completion, anomaly cancellation is not an external aesthetic criterion but a non-negotiable self-consistency requirement: if the chosen completion contains a chiral gauge sector, its supplied matter content *must* be anomaly-free; otherwise that completion is inconsistent. This does not derive the observed fermion representations or show that they emerge from the condensate. It constrains any acceptable completion to satisfy the standard anomaly-cancellation conditions. A systematic anomaly audit for a supplied chiral gauge completion, including the standard electroweak consistency constraints, the anomaly-transparency of the director coupling $\mathcal{L}_5$, and the open status of the colour-sector completion, is developed in §9.5.

Neutrino compatibility

Within the full-weight standalone- $U(2)$ condition $d_f + q_f \equiv 0 \pmod{2}$, a singlet with $T = 0, Y = 0$ and hence $(d, q) = (0, 0)$ has $Q_{\text{em}} = 0$. This is compatible with a sterile-neutrino-like state and the supplied seesaw route, but it is not an independent derivation of that state or its charge.

What remains open

The compact- $U(1)$ result does not yet derive the full observed Standard-Model electric-charge spectrum. In particular:

- (i) the appearance of quark charges $\pm e/3$ and $\pm 2e/3$ requires the colour sector (OP-GUT1) and the electroweak representation content (OP-GUT3b);
- (ii) the precise equality $|Q_e| = |Q_p|$ requires anomaly-consistent representations (OP-GUT6);
- (iii) the full observed charge pattern likely depends on the global structure of the complete low-energy gauge group, potentially of $(SU(3) \times SU(2) \times U(1)_Y)/\Gamma$ type with $\Gamma = \mathbb{Z}_6, \mathbb{Z}_3$, or $\mathbb{Z}_2$ [84];

(iv) three fermion generations remain an open problem (OP-GUT5).

Comparison with other routes to charge quantisation

Table 75: Comparison of routes to charge quantisation.

Aspect	Standard QED	GUT embedding	ECT
Why quantised	Not explained ($U(1)$ can be noncompact)	Group embedding forces it	Integer representations only after a compact local bundle is supplied
Status of derivation	Empirical input	GUT hypothesis required	Conditional mathematics A; ECT physical owner Open
Fractional charges	Empirical	Predicted by rep content	Requires EW+colour completion
Anomaly role	Empirical consistency check	Automatic within GUT	Mandatory consistency test of any supplied chiral completion

Discriminant: ECT charge quantisation versus standard approaches. The BR1 phase supplies a compact global variable, but a local compact $U(1)$ bundle does not follow from that fact alone. If a completion supplies the bundle and its matter representations, integer representation labels follow by standard group theory. The local bundle, unbroken photon, physical charge operator, elementary unit e_0 , representation map and observed spectrum remain part of the open gauge-matching programme (OP-GUT4).

Falsifier for the charge-quantisation programme. The programme would be falsified if (i) the condensate phase sector were shown to be topologically noncompact (contradicting the construction); (ii) a completed ECT gauge/matter sector produced anomaly-violating representations; or (iii) the electroweak/colour completion failed to reproduce the observed charge assignments.

Status summary. *Established (Level A mathematics, conditional premises):* (i) integer one-dimensional representations and $q \in \mathbb{Z} \cdot e_0$ once a compact local $U(1)$ bundle, matter representation and charge unit are supplied (Theorem 23.1);

Architectural / consistency requirement: (i) anomaly cancellation as a mandatory self-consistency condition on any emergent chiral sector (consistency target, not yet a derived representation theorem).

Conditional / Open: (i) the conditional canonical-generator spectrum of Eq. (23.20), after a physical Weyl representation and self-adjoint domain are supplied; the periodic $S_0\mathbb{Z}$ special case additionally requires $v_{tw} = 0$; (ii) standard $U(2)$ descent condition $d_f + q_f \equiv 0 \pmod{2}$ after an integer character and quotient are supplied. The physical quotient and charge map remain Open.

Open: (i) the local compact bundle, unbroken photon, physical charge operator, elementary unit and matter representation map; (ii) full observed electric-charge spectrum and hypercharge quantisation; (iii) full global gauge-group structure $(SU(3) \times SU(2) \times U(1)_Y)/\Gamma$; (iv) observed fermion representation content.

23.5 Uncertainty principle in the coherent branch

Status: Level A for the mathematical uncertainty inequality once the canonical commutator structure of the smooth coherent branch is admitted; Level B for the physical normalisation of that inequality by the observed $\hbar$, which depends on Eq. (21.7).

The uncertainty relation discussed below inherits its status directly from the canonical-bridge programme of Section 23.3: it is mathematically strict once the commutator bridge is granted, but physically matched only after the $S_0 = \hbar$ identification.

The uncertainty principle is not introduced in ECT as an independent axiom. If the coherent branch already admits (i) a wave-dynamical representation, (ii) a canonical organisation, and (iii) a commutator candidate normalised by the supplied coefficient S_0 (Section 23.3), then the uncertainty relation follows as a standard mathematical consequence. This conditional theorem does not own the coefficient or its physical matching.

Operator input

The Robertson–Schrödinger theorem applies to supplied self-adjoint operators on the common product domains required below. For a noncompact physical canonical pair (q, p_{phys}) , the conditional ECT bridge is

$$[\hat{q}, \hat{p}_{\text{phys}}] = iS_0. \quad (23.23)$$

After the separate empirical calibration $S_0 = \hbar$ (21.7), this becomes

$$[\hat{q}, \hat{p}_{\text{phys}}] = i\hbar. \quad (23.24)$$

The representation, domains and physical owner are B/Open; the theorem below is Level A mathematics once they are supplied.

Robertson–Schrödinger inequality

For self-adjoint $\hat{A}, \hat{B}$ and a normalised state in the domains of $\hat{A}\hat{B}$ and $\hat{B}\hat{A}$, define $\delta\hat{A} = \hat{A} - \langle\hat{A}\rangle$ and $\delta\hat{B} = \hat{B} - \langle\hat{B}\rangle$. Cauchy–Schwarz gives

$$(\Delta A)^2 (\Delta B)^2 \geq \frac{1}{4} |\langle[\hat{A}, \hat{B}]\rangle|^2 + \frac{1}{4} |\langle\{\delta\hat{A}, \delta\hat{B}\}\rangle|^2. \quad (23.25)$$

Dropping the nonnegative covariance term gives

$$\Delta A \Delta B \geq \frac{1}{2} |\langle[\hat{A}, \hat{B}]\rangle|. \quad (23.26)$$

For the supplied physical pair,

$$\Delta q \Delta p_{\text{phys}} \geq \frac{S_0}{2}, \quad (23.27)$$

and, after $S_0 = \hbar$,

$$\Delta q \Delta p_{\text{phys}} \geq \frac{\hbar}{2}. \quad (23.28)$$

If the reduced phase momentum is used on the corresponding local common core, $\pi_{\theta}^{\text{phys}} = \Sigma_{\theta} \pi_{\theta}^{\text{red}}$ instead gives $\Delta\theta \Delta\pi_{\theta}^{\text{red}} \geq S_0/(2\Sigma_{\theta})$. This local-chart statement does not create a global self-adjoint angle operator.

Compact phase: use the unitary relation, not a naive angle bound. Multiplication by a sawtooth representative of the global orbit coordinate α does not preserve the quasiperiodic self-adjoint domain of $Q_{\theta}^{\text{phys}} = -iS_0\partial_{\alpha}$. Therefore $\Delta\alpha \Delta Q_{\theta}^{\text{phys}} \geq S_0/2$ is not a theorem: indeed a rotor eigenstate has $\Delta Q_{\theta}^{\text{phys}} = 0$ while the sawtooth angle has finite variance. The missing hypothesis is precisely a common invariant product domain.

The globally defined unitary $U_0 = e^{i\alpha}$ obeys $[Q_{\theta}^{\text{phys}}, U_0] = S_0 U_0$. Under the corresponding domain assumptions, Cauchy–Schwarz yields the valid circular relation

$$\Delta Q_{\theta}^{\text{phys}} \sqrt{1 - |\langle U_0 \rangle|^2} \geq \frac{S_0}{2} |\langle U_0 \rangle|. \quad (23.29)$$

For a generator eigenstate $\langle U_0 \rangle = 0$, so the bound is consistent with $\Delta Q_{\theta}^{\text{phys}} = 0$. The twist v_{tw} , periodic special case and spectrum are those of Eq. (23.20). Integer field winding remains a separate topological statement and derives neither the self-adjoint domain nor a common physical S_0 .

Interpretation

The Robertson–Schrödinger inequality is a standard theorem once a Hilbert space, physical operators, domains and their commutator are supplied. ECT reproduces that mathematics conditionally; it has not yet derived the physical operators, commutator normalisation or universal S_0 from P1–P6.

Numerical anchor. After the identification $S_0 = \hbar$, the uncertainty bound takes its standard value $\hbar/2 \approx 5.27 \times 10^{-35}$ J s. For an electron confined to a hydrogen-like atomic scale $\Delta x \sim a_0 \approx 0.529 \text{ \AA}$, one obtains

$$\Delta p_{\min} \sim \frac{\hbar}{2a_0} \approx 1.0 \times 10^{-24} \text{ kg m/s}, \quad (23.30)$$

which is of the standard atomic order of magnitude. ECT adds no correction to this result at the current level of the coherent-branch closure: once $S_0 = \hbar$ is matched, the usual Heisenberg scale is reproduced exactly.

Discriminant: conditional theorem versus physical ownership. In standard axiomatic quantum mechanics, the uncertainty principle is a theorem of the already assumed Hilbert-space and operator framework. In ECT the same theorem holds in every supplied representation satisfying its hypotheses. The proposed compact-phase/canonical route is a programme for the missing representation and normalisation, not a derivation that traces uncertainty to winding or to a loop-generated action scale.

Falsifier for the uncertainty programme. The uncertainty programme would fail if a completed physical canonical pair required a normalisation incompatible with its wave-sector S_0 , or if no self-adjoint/Weyl representation with the declared domains could realise the compact generator relation (23.20). Integer winding alone is not such a compatibility condition. At the observational level, any reproducible violation of the bound $\Delta q \Delta p \geq \hbar/2$ in the regime where the standard canonical description is known to apply would directly challenge the canonical reading of the coherent branch. No such violation has ever been observed.

Status summary. *Conditional Level A mathematics:* given a Hilbert space, the required common product domains and the supplied physical-pair commutator (23.23), Eq. (23.25) is the Robertson–Schrödinger theorem and Eq. (23.27) its noncompact physical-pair corollary. The compact angle instead uses the unitary relation (23.29). Compact phase independently supplies an integer winding representation; it does not prove that its global generator uses the same S_0 normalisation. *Level B/Open:* the physical commutator and Heisenberg form $\Delta q \Delta p \geq \hbar/2$ require those owners and the separate matching $S_0 = \hbar$ (21.7). No experimental violation of this bound has been observed.

Bridge to the next chapter. With the Hilbert-space bridge, unitarity status, canonical organisation, and uncertainty structure in place, the coherent branch is ready for the next layer of analysis: Euclidean path-dependence, exchange sectors, and the route to Dirac structure. Those questions are the subject of the next chapter.

24 Euclidean Path-Dependence, Quantum Nonclassicality, Statistics, and Dirac Structure

Inputs: coherent-branch wave kinematics (Section 22), the Hilbert-space bridge (Section 23), canonical structure (Section 23.3), and compact-phase topology (BR1). Aim: to separate three conditional structures of the coherent-sector programme: Euclidean path-dependence in a supplied boundary-value problem; exchange topology in a separately supplied defect-free point-configuration space; and a possible Dirac sector after a physical Lorentzian metric, spinor bundle and irreducible Clifford-factorisation action

are supplied. These are parallel inputs and results, not the implication chain $O(4) \rightarrow O(3) \Rightarrow \mathbb{Z}_2 \Rightarrow \text{Spin}(3) \Rightarrow \mathfrak{so}(1,3) \Rightarrow \text{Clifford} \Rightarrow \text{Dirac}$. Status: Level A conditional mathematics for the elliptic two-sided propagator in its declared BVP; Level A topology for $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ in the declared defect-free $\mathbb{R}^3$ point model, without representation selection; and Level A standard representation theory only after the physical metric/tetrad, real form, spin structure and Clifford module have been supplied. Interference, tunnelling and Dirac dynamics are Level B conditional on their additional actions, states and observable maps. The physical spinorial spectrum, CAR/OS completion, indefinite-causal-order reconstruction and the interacting spin–statistics theorem remain Open.

24.1 Euclidean path-dependence: intermediate results from both past and future

A fundamental feature of the ECT framework is that the Euclidean equation of motion

$$(-\nabla_E^2 + m^2) \Phi(X) = 0, \quad \nabla_E^2 = \delta^{AB} \partial_A \partial_B, \quad (24.1)$$

is an *elliptic* partial differential equation. Consequently, the value of Φ at an interior point X_m depends simultaneously on boundary conditions at *both* $w < w_m$ (the “past”) and $w > w_m$ (the “future”). The Euclidean Green’s function $G_E(X, X')$ is fully symmetric under $w \leftrightarrow -w$, encoding this two-sided dependence.

Exact boundary-value fact and Open quantum bridge. Two-sided Euclidean boundary dependence is also present in classical elliptic systems, including electrostatics, and therefore supplies no nonclassical consequence by itself. The following items separate that exact PDE fact from the additional quantum owners.

1. *Quantum interference.* The four-dimensional Euclidean resolvent does *not* obey the transition-kernel composition law printed here previously. For $G_m = (-\nabla_E^2 + m^2)^{-1}$ its exact spacetime convolution is

$$\int d^4 X_m G_m(X_f, X_m) G_m(X_m, X_i) = -\frac{\partial}{\partial m^2} G_m(X_f, X_i), \quad (24.2)$$

as follows immediately in momentum space. A genuine composition law instead belongs to a time-sliced heat/transition kernel, $K_{T_1+T_2}(x_f, x_i) = \int d^3 x_m K_{T_2}(x_f, x_m) K_{T_1}(x_m, x_i)$, after a slicing, state space and measure have been supplied. Formal Euclidean–Lorentzian continuation can then connect the quadratic kernel to standard amplitude language; ellipticity alone does not derive the quantum measure or superposition rule.

2. *Quantum tunnelling.* A supplied Euclidean quantum source model assigns a dimensionless weight $e^{-\mathcal{J}_E^\mu}$ to its integration configurations. A tunnelling rate follows only after a named saddle or thimble, boundary conditions, determinant/prefactor and Euclidean–Lorentzian observable map are supplied. The bare P3 measure would use $\mathcal{J}_E^\mu = \kappa_{\mathbb{P}^3} \mathcal{J}_E^{\mathbb{P}^3}$; merely integrating over configurations does not derive an ECT tunnelling amplitude or identify its denominator with S_0 .
3. *Directionality and irreversibility.* The two-sided w -dependence of the Euclidean propagator does *not* by itself produce an arrow of time: the Euclidean Green’s function is symmetric under $w \leftrightarrow -w$ and treats past and future equally. Coarse-graining may yield dissipative reduced dynamics, but it does not by itself define a forward/reverse probability ratio. Such a ratio requires declared protocols, path measures, a time-reversal map, and local detailed balance. Deriving those ECT objects remains OP-Q17.

Structural significance. Two-sided Euclidean dependence is an exact property of the supplied elliptic boundary-value problem. It is neither sufficient nor discriminating for interference, tunnelling, complex amplitudes, Born probabilities or superposition as a quantum-state principle. A physical bridge would additionally require an admitted measure, complex/coherent structure, OS reconstruction, composition law, preparation/state and observable map. The present text therefore retains ellipticity as Level-A mathematics in the supplied model and the quantum interpretation as Open, not as an architectural explanation.

What this does *not* mean. The two-sided dependence along the Euclidean coordinate w must not be misread as controllable signalling from the future in Lorentzian time. At the fundamental Euclidean level, w is not yet the observed time coordinate. Only after a supplied scalar operator with $\beta(\beta - \alpha) < 0$ (or $\alpha > \beta > 0$ on the retained $\beta > 0$ component), orientation and initial-boundary prescription is there a conditional scalar notion of retarded propagation. A physical clock, cross-sector cone and operational no-signalling theorem remain completion gates. Thus ECT does not claim that laboratory agents can influence the past; Euclidean two-sided boundary data alone have no operational signalling interpretation. This clarification also applies to indefinite-causal-order protocols: ECT reads them as coherent manifestations of a pre-Lorentzian two-sided amplitude structure, not as operational permission to signal to the past.

Boundary-value compatibility only. Two-sided dependence in an elliptic Euclidean boundary-value problem does not derive physical reverse $O(3) \rightarrow O(4)$ segments, their action weight, or a laboratory quantum–classical threshold. Establishing such segments would require a saddle and backreaction calculation with declared boundary conditions. No causal-order, signalling, threshold-suppression, or metric-signature consequence follows from ordinary two-sided boundary dependence. The retained claim is only that the Euclidean-first boundary-value architecture is compatible with coherent amplitudes whose Lorentzian reconstruction can depend on data specified on more than one Euclidean boundary component.

Open comparison: Euclidean-first and Lorentzian formulations. The primary ECT arena is Euclidean and its supplied scalar equation is elliptic, so two-sided boundary dependence precedes any proposed Lorentzian reconstruction. This ordering is an interpretive choice, not a derivation of nonclassical amplitude structure: classical elliptic countermodels have the same boundary dependence without quantum states or probabilities. No ECT-specific discriminator exists until the missing quantum bridge is derived.

Falsifier for the path-dependence programme. The path-dependence programme would fail if the Euclidean two-sided propagation structure, in the ordered Lorentzian branch, proved structurally incompatible with the observed interference, tunnelling, or superposition phenomena. It would also be weakened if the elliptic Euclidean boundary-value character could not be made consistent with the retarded causal structure established elsewhere for the ordered branch. A quantitative test additionally requires a frozen Euclidean exponent $\mathcal{J}_E^\mu$, measure, analytic-continuation prescription and observable map; failure would reject that supplied bridge, not an undefined universal weight. No such physical ECT bridge is presently complete, so no data comparison is claimed here.

Indefinite causal order: compatibility, not an ECT prediction

Status: Open as an ECT-specific result. Quantum-switch and process-matrix predictions belong to standard quantum-information theory; the present Euclidean-first architecture is compatible with them but has not derived their process operators, causal witnesses, or platform dependence [149, 150].

The Procopio–Walther experiment [151] reports high interferometric visibility for a photonic quantum switch. In a separately declared pure-dephasing proxy one may define

$$\Phi_{\text{vis}} := -\ln V, \quad (24.3)$$

which gives $\Phi_{\text{vis}} = 0.0060 \pm 0.0020$ from $V = 0.994 \pm 0.002$. This is a transformation of a reported visibility, not a microscopic condensate calculation and not evidence for reverse Euclidean segments. A universal order-resource threshold at $\Phi = 1$ is not derived.

The Procopio–Walther quantum switch

Procopio et al. [151] realised coherent control of the order of two operations. ECT is compatible with that process but has not derived its process operator, causal witness, or apparatus kernel; the result is not evidence for physical reverse trajectories.

Programme-level expectations for indefinite-order experiments

Platform-level expectations require an ECT record operator and noise kernel. Without those owners, temperature, coupling, and visibility trends are ordinary open-system hypotheses rather than ECT predictions.

Expectation 1 (coherence-threshold degradation). Loss of a switch advantage must be computed from the platform's process matrix and declared record channel; no universal crossover follows from one scalar proxy.

Expectation 2 (environmental scaling). Increasing a specified environmental coupling can reduce a coherent-control witness, but the scaling depends on the coupled operator, spectrum, state, and protocol. An unresolved “bath size” alone is not a predictive variable.

Expectation 3 (temperature sensitivity). Temperature sensitivity is conditional on a thermal state and the platform spectral density. No universal ECT temperature law is asserted without a KMS state and the relevant coupling. ECT becomes testable in this sector only after it supplies a platform record operator and kernel that predicts a correction to the standard switch statistics. Until then the result is interpretive compatibility and provides no ECT-specific discriminator.

Quantitative dephasing estimate. The reported visibility $V = 0.994 \pm 0.002$ gives the source-model proxy $\Phi_{\text{vis}} := -\ln V = 0.0060 \pm 0.0020$. This arithmetic is retained, but it is neither a microscopic condensate calculation nor an ECT-specific fit [151].

Delayed choice, quantum erasure, and entanglement swapping

Interpretive conclusion

Indefinite-order experiments are compatible with a pre-Lorentzian two-sided amplitude architecture, while process-matrix theory already describes them operationally [150]. ECT becomes discriminating only after predicting a held-out correction. Delayed-choice and quantum-eraser experiments are described consistently by ordinary quantum amplitudes, conditioning, and later classical comparison; they do not enable signalling to the past. Delayed-choice entanglement swapping likewise reorganises conditioned correlations without changing an earlier locally available record [152, 153, 154, 155, 156, 157]. ECT presently reproduces no platform-specific statistic beyond that standard description.

The Euclidean boundary-value picture may be used as an interpretation, but it does not establish literal reverse $O(3) \rightarrow O(4)$ transitions, a universal coherence threshold, or a new retrocausal mechanism. Any ECT claim in this family requires derived apparatus couplings and a held-out quantitative deviation. The current status is Open as an ECT discriminator.

Wheeler's delayed-choice experiment. Wheeler/Jacques delayed choice changes the late measurement basis and hence conditioned statistics without signalling to the past [152, 153, 154]. ECT currently offers a boundary-value interpretation, not an experiment-specific correction.

Comparison with standard quantum theory. Standard superposition and the Born rule quantitatively own the delayed-choice outcome. ECT's present addition is only the statement that global Euclidean boundary data need not be organised by a primitive one-sided time before Lorentzian reconstruction.

Quantum eraser. In a quantum eraser, which-path marking removes interference in the unconditioned ensemble and coincidence conditioning restores it only in selected subensembles [155, 156]. No completed macroscopic collapse is reversed.

Comparison with standard quantum theory. The density-matrix and conditional-probability account is the quantitative owner of quantum erasure. A medium interpretation may identify the physical record degrees of freedom only after their coupling and trace are explicitly derived.

Delayed-choice entanglement swapping. Delayed-choice entanglement swapping changes the correlation class identified after Bell-state analysis and later classical comparison; no earlier local record is changed [157].

Comparison with standard quantum theory. Entangled projection and classical record comparison already account for the observations. The common-medium picture is at present an interpretation and supplies no new numerical prediction.

Summary of delayed-choice experiments. All three families preserve operational no-signalling. Global Euclidean boundary-value dependence must not be described as controllable retrocausation in any supplied Lorentzian observable completion.

Experiment family	Standard operational statement	ECT status
Wheeler/Jacques delayed choice	Late basis choice changes conditioned statistics without past signalling	Compatible; no ECT correction derived
Quantum eraser	Interference reappears only in conditioned subensembles after record erasure	Standard channel result; ECT Open
Delayed-choice entanglement swapping	Correlations are identified after Bell analysis and classical comparison	Standard result; ECT Open
Procopio quantum switch	High-visibility coherent control of operation order	$\Phi_{\text{vis}} = -\ln V$ is a source proxy only

Requirements for a future programme-level discriminator. A genuine ECT discriminator would be a platform-specific, pre-registered correction computed from the condensate coupling and record kernel. Generic visibility loss under added noise is not such a discriminator.

Wigner’s friend paradox

Status: interpretive / Level B. ECT does not modify the operational predictions but provides a structural reframing of the problem.

In the Wigner’s friend scenario, an observer (the “friend”) performs a measurement inside a sealed laboratory, while a superobserver (Wigner) treats the entire laboratory as a quantum system. In the standard presentation, the paradox is that the friend assigns a definite outcome while Wigner assigns a superposition.

In the ECT reading, the tension is reformulated structurally rather than eliminated by a new measurement postulate. The friend’s measurement device may form a robust reduced-state record in a specified channel. This does not derive one realised outcome; the update rule remains OP-Q19. Wigner, if treated as external to that record-generating process, may still assign a more weakly reduced description to the laboratory as a whole, but only insofar as the relevant lab–environment coarse graining has not yet been specified at his level of description. ECT therefore shifts the question from “who is observing” to “which degrees of freedom have already been locked by environmentally amplified record formation”.

Comparison with standard quantum theory. Standard quantum theory can accommodate both the friend-collapsed and Wigner-superposed descriptions within the relative-state formalism, but does not by

itself privilege one physical criterion for when a definite record should count as established. ECT proposes such a criterion only at closure level: large effective environmental locking of the record-generating degrees of freedom. This is a structural closure-level reading of the paradox, not yet a completed model-independent theorem of measurement.

24.2 Two-sided Euclidean paths and protocol-defined reverse weights

The Euclidean functional integral can contain both orientations of a path, but this kinematic fact does not define thermodynamic forward and reverse experiments. For a declared protocol and measurable time-reversal involution Θ , transport the normalised reverse measure to the forward path space as $Q = P_R^\Theta$, with $Q(A) = P_R(\Theta A)$. Assuming $P_F \ll Q$, define

$$\Sigma[\gamma] := \ln \left(\frac{dP_F}{dQ}(\gamma) \right), \quad \frac{dQ}{dP_F}(\gamma) = e^{-\Sigma[\gamma]} \quad \text{when } P_F \sim Q. \quad (24.4)$$

Thus one-way absolute continuity is enough to define Σ P_F -almost surely, whereas the reciprocal and the standard unit integral-fluctuation identity require mutual absolute continuity. Singular reverse weight belongs to the separate absolute-irreversibility case. This is a protocol definition, not a universal identification of a reverse/forward path ratio with the exponential of a dephasing scalar. A pairwise dephasing exponent measures loss of interference between alternatives; it is not a path-probability ratio. An ECT fluctuation relation remains Open pending derived forward/reverse measures, initial ensembles and local detailed balance.

24.3 Tunnelling from a supplied Euclidean saddle

Status: Level A external mathematics for the standard leading WKB exponent under its stated one-particle assumptions; Level B for using a specified Euclidean saddle in a supplied ECT functional with declared physical-action and measure owners; Open for a literal excursion through the fourth Euclidean coordinate or an ECT-specific correction to a measured tunnelling rate.

WKB exponent and Euclidean rederivation

Let a one-dimensional Hamiltonian $H = p^2/(2m) + V(x)$ have simple turning points $x_- < x_+$ at energy E , with $V(x) > E$ between them. At leading WKB order the positive physical under-barrier action of this supplied external one-particle source model is

$$S_{\text{WKB}}(E) := \int_{x_-}^{x_+} dx \sqrt{2m[V(x) - E]}, \quad T_{\text{WKB}}(E) \asymp e^{-2S_{\text{WKB}}(E)/\hbar}, \quad (24.5)$$

up to the standard connection prefactor and higher semiclassical orders. This formula follows without assigning ontological reality to an extra coordinate.

The same exponent has a controlled Euclidean-saddle derivation. For the energy-shifted physical Euclidean action of that external source model

$$S_E[x] - E\Delta\tau_E = \int d\tau_E \left[\frac{1}{2}m\dot{x}^2 + V(x) - E \right],$$

the zero-energy first integral of the shifted Euclidean problem is $m\dot{x}^2/2 = V - E$. Evaluating the shifted action on one monotone crossing therefore gives

$$S_E[x_{\text{sad}}] - E\Delta\tau_E = \int_{x_-}^{x_+} dx \sqrt{2m[V(x) - E]} = S_{\text{WKB}}(E).$$

For false-vacuum or metastable decay, by contrast, the standard quantum rate has the form $\Gamma \simeq Ae^{-B/\hbar}$ with a vacuum-subtracted *bounce* action B and a fluctuation/zero-mode prefactor A ; that bounce problem must not be identified with the one-crossing transmission exponent without its own boundary conditions.

The $\hbar$ in these displayed formulae belongs to the externally calibrated quantum problem. A prospective ECT Euclidean channel must instead supply a vacuum-subtracted physical bounce action $B_{\text{ch}}^{\text{phys}}$ and its own finite positive path-integral denominator $\Sigma_{\text{PI, ch}}$, giving $\exp[-B_{\text{ch}}^{\text{phys}}/\Sigma_{\text{PI, ch}}]$. The undeformed P3 quartic has degenerate minima, $\varepsilon = \Delta V = 0$, and owns no finite false-vacuum bounce by itself: its thin-wall R and B diverge. Only an explicitly biased/deformed P3 potential, finite boundary/protocol or another named channel could supply a positive finite $\Delta \mathcal{S}_E^{\text{P3}}$ and the conditional factor $\exp[-\kappa_{\text{P3}} \Delta \mathcal{S}_E^{\text{P3}}]$. Complex thimbles require their own contour interpretation. No map from a channel denominator to the canonical S_0 or external $\hbar$ is presently derived.

What a fourth-spatial bypass would require

P1 states that the microscopic manifold is four-dimensional and Euclidean; it does not specify the particle/worldline reduction or a barrier $V(\mathbf{x}, w)$. In particular, writing only $V(\mathbf{x})$ normally means that the potential is independent of w and hence extends uniformly along that coordinate. It does *not* imply an opening around the barrier. Conversely, a barrier localised in w would be a new operator and boundary assumption.

If one properly normalised ECT action and boundary-value problem actually supply a stationary four-coordinate saddle $X_{\text{sad}}^A(\sigma)$, its coordinate excursion may be defined by

$$\Delta w_{\text{sad}} := \max_{\sigma} w_{\text{sad}}(\sigma) - \min_{\sigma} w_{\text{sad}}(\sigma). \quad (24.6)$$

This definition predicts neither $\Delta w_{\text{sad}} \neq 0$ nor $\Delta w_{\text{sad}} \sim x_+ - x_-$. The former estimate $\mathcal{S}/p_w$ was only a dimensionally admissible ratio and cannot prove a geometric path, because no p_w , $V(\mathbf{x}, w)$, variational saddle or projection map owned it.

The preserved ECT route is therefore conditional: derive from one action (i) the physical tunnelling coordinate and its canonical kinetic term, (ii) the full w -dependent barrier/operator and boundary state, (iii) the constrained Euclidean saddle and determinant, and (iv) the analytic-continuation and detector map. Only then may one compare the saddle action with $S_{\text{WKB}}(E)$ and ask whether a nonzero Δw_{sad} changes the transmission exponent or prefactor. The gate is a preregistered barrier- and platform-specific deviation after the ordinary multidimensional instanton calculation; generic agreement with WKB is not an ECT discriminator.

Connection to PES-R and protocol reversal

The tunnelling exponent is owned by the supplied saddle and boundary-value problem. PES-R can subsequently classify a metastable state's physical pole width, finite-time survival and record formation for a declared channel; it does not derive the saddle. A coordinate derivative $\partial_w \Psi$, the translation susceptibility Γ_{tr} , a pairwise dephasing exponent and a decay rate are distinct objects.

Echo/control protocols may refocus phases or erase a specified record, but they neither implement a literal return to the bare $O(4)$ phase nor furnish a universal reverse-path law. Euclidean boundary correlations likewise do not become controllable Lorentzian retrocausation. Thus the standard semiclassical exponent survives, while the literal fourth-spatial bypass and any new numerical law remain Open behind the stated action/saddle/detector gate.

Status summary. *Established (Level A):* the Euclidean coherent-field equation is elliptic, so interior values depend on boundary data from both sides of the Euclidean ordering coordinate. *Level B route / Open physical ECT:* after an admitted quantum measure and coherent structure, or a named Euclidean saddle, state, continuation and observable map are supplied, standard Euclidean methods can organise interference or tunnelling. Two-sided elliptic dependence alone derives neither phenomenon, and no ECT-specific amplitude or rate is presently obtained. The quantitative reconstruction of operational indefinite-causal-order protocols, switch correlators, and platform-dependent degradation laws from condensate dynamics likewise remains incomplete.

24.4 Quantum statistics: exchange sectors and spinorial compatibility

Status: Level A topology inside the declared defect-free $\mathbb{R}^3$ point-excitation configuration space: $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$, giving two homotopy classes. Its application to physical ECT is conditional on a separately supplied physical three-dimensional slice and excitation map; P1 and P4 do not derive either. The characters ± 1 follow only after restricting to one-dimensional unitary representations. Level B is a route from those admitted characters and separately supplied tensor/spinor sectors to the familiar exchange assignments. Open (OP-Q11) are the physical slice and excitation map, representation selection, the treatment of higher-dimensional S_N sectors, the spin–statistics theorem, CAR and the many-excitation reconstruction. Connection: P4, P5, §9.9, §24.5, and §31.

Belonging to the same emergent mode sector is not by itself a proof that physical excitations have no unresolved core, orientation, internal or environmental labels. If two physical ECT excitations are independently shown to be identical unlabeled point excitations in defect-free $\mathbb{R}^3$, their supplied coordinate configuration space is the quotient by exchange used below. ECT has not yet derived that excitation/identity map or excluded additional discrete fibres and sectors. The quotient-space topology is therefore a conditional theorem for the declared point-excitation model, not a derivation of physical indistinguishability.

If the quantum sector of ECT is to reproduce many-particle physics, it must derive the physical spatial slice, relevant configuration space, state representation and observed exchange assignment. At present the declared defect-free $\mathbb{R}^3$ point model supplies only a conditional three-dimensional homotopy input. ECT has not derived the physical three-dimensional slice, point-excitation configuration space, representation or physical assignment; these are separate completion steps.

Exchange topology

For two identical excitations in the declared defect-free $\mathbb{R}^3$ point model, the reduced configuration space is

$$\mathcal{M}_2 = (\mathbb{R}^3 \times \mathbb{R}^3 \setminus \Delta) / S_2, \quad (24.7)$$

whose fundamental group is

$$\pi_1(\mathcal{M}_2) = \mathbb{Z}_2. \quad (24.8)$$

This yields exactly two homotopy classes of loops (Level A topology for the supplied configuration space). Calling them symmetric and antisymmetric already adds a representation: the one-dimensional unitary characters of $\mathbb{Z}_2$ are $+1$ and -1 . The topology does not choose one for a species or supply a quantum amplitude.

What the declared defect-free configuration-space model establishes. For identical point excitations in defect-free $\mathbb{R}^d$ with $d \geq 3$ and the collision diagonal removed, exchange loops reduce to the permutation-group topology; for two excitations this is $\mathbb{Z}_2$. This statement is not extended to arbitrary multiply connected ambient spaces, whose independent cycles can contribute additional loop classes. In $d = 2$, the braid group is richer and supports continuous Abelian anyonic phases [158, 159]. Conditional on P1 and on the point-excitation configuration-space model, P4 therefore blocks importing such a generic two-dimensional braid phase into the primary branch. It does not derive bosons, fermions, representation selection or physical statistics.

External QED reference. Observed particle physics in $3 + 1$ dimensions exhibits exactly the bosonic/fermionic dichotomy for fundamental excitations: photons and gauge bosons are bosonic, while electrons, neutrinos, and quarks are fermionic. By contrast, anyonic exchange has been observed only in effectively two-dimensional condensed-matter systems, where braid topology differs from the $3 + 1$ -dimensional particle-exchange problem. This empirical split is compatible with a future physical completion whose spatial slices are three-dimensional, while anyonic statistics remains an emergent lower-dimensional quasiparticle phenomenon rather than a fundamental exchange law of the coherent branch.

Representation-theoretic compatibility

Given the $O(3)$ stabiliser of the supplied P4 vector, a separately supplied excitation sector may be organised in its representations. If a one-dimensional exchange character is also selected, integer-spin/+1 and half-integer-spin/-1 are compatible with the standard assignment. This is an admissible pairing, not a theorem derived from the two group facts.

Medium reading of exchange (P5). P5 motivates treating same-species excitations as modes of one medium, but it does not determine their configuration space, state representation or exchange character. Those owners must be derived in the completed many-excitation sector; the full spin-statistics link remains Open.

Conditional canonical-normalisation no-go in a supplied fermionic EFT. A natural question is whether a *supplied* local fermionic/CAR completion allows small violations of the Pauli principle in the presence of a spatially varying ordered condensate, such as inside strong gravitational fields or in dense astrophysical regimes. The answer obtained in this conditional EFT is essentially the opposite of what a naive gradient estimate might suggest: smooth local inhomogeneity of the ordered condensate does not by itself furnish a mechanism for violating fermionic exclusion on the coherent ordered branch.

For a local first-order fermionic EFT with a smooth and strictly positive temporal kinetic coefficient $\mathcal{K}(x) > 0$ multiplying $i\psi^\dagger \partial_t \psi$, a general condensate-dependent temporal kinetic term of the form $i\mathcal{K}(x)\psi^\dagger \partial_t \psi + \dots$ can be brought to canonical form by the local field redefinition

$$\chi_\alpha(x) \equiv \mathcal{K}(x)^{1/2} \psi_\alpha(x), \quad (24.9)$$

after which the equal-time anticommutator takes the standard canonical form

$$\{\chi_\alpha(t, \mathbf{x}), \chi_\beta^\dagger(t, \mathbf{y})\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad (24.10)$$

and local interaction terms (including condensate-dependent mass $M(\phi, n)\bar{\psi}\psi$, the candidate preferred-direction operator $\mu_5(\phi, n)\bar{\psi}\gamma^A n_A \psi$, and local four-fermion operators $C_k(\phi, n)\mathcal{O}_k^{(4f)}$) modify the dynamics of fermionic modes without altering the underlying exclusion algebra. A full technical exposition of the canonical normalisation procedure and its explicitly conditional scope is given in Appendix AT.

Restricted conditional statement. If the eventual fermionic reconstruction closes to a local first-order Grassmann theory with a smooth positive scalar kinetic factor and standard CAR before the rescaling, the rescaling above preserves the CAR. This excludes only a statistics violation generated solely by that scalar normalisation. Any departure would require failure of at least one supplied premise. Non-exhaustive possibilities include:

- (i) a noncanonical kinetic or symplectic structure, a singular or sign-changing kinetic operator, or a failure of the assumed local CAR;
- (ii) *nonlocal branch mixing* — sector couplings that cannot be cast in the form of a local first-order fermionic EFT on the coherent branch.
- (iii) *topological or defect sectors of the ordered condensate* beyond the ordinary coherent branch (e.g. condensate-defect backgrounds with nontrivial π_k content; cf. the topological-sector catalogue in Section 8.1).
- (iv) *branch-transition / coherence-loss regions*, where the ordered-branch description itself fails and the fermionic reconstruction programme ceases to close.

This list is not a classification theorem. Physical fermion reconstruction, spin-statistics assumptions, the operator domain and any nonstandard completion remain Open under OP-Q11 and OP-Q21.

Scope: no astrophysical inference. The equivalence principle removes a connection at a point but does not fix the order or Wilson coefficients of curvature-dependent local operators. The present local

canonical-normalisation argument therefore implies no packet-mismatch scaling, compact-object estimate or astrophysical exclusion window. White-dwarf and neutron-star matter cannot be classified here without a physical ECT fermion action, CAR owner, curvature operators, state and observable map.

Conditional on the fermionic reconstruction programme. At the present stage, the no-go statement is therefore conditional not on the canonical-normalisation argument itself, which is technically complete within the local first-order EFT closure described above, but on the completion of the fermionic reconstruction from the exchange-topological and representation-theoretic sector (§24.4, §24.5) to the full operator-level canonical anticommutation structure (OP-Q11, OP-Q21).

What is and is not claimed

The exact topological input is the supplied configuration-space result $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ in the declared defect-free $\mathbb{R}^3$ point model. It gives two homotopy classes and, after an additional one-dimensional-unitary-sector assumption, the characters ± 1 . It does not select either character, construct CAR/Fock space, or identify spin with exchange. For $N \geq 3$, the group S_N also admits higher-dimensional representations, so excluding parastatistical descriptions requires the standard additional locality and relativistic quantum assumptions.

Connection to downstream programmes. This corrected input remains useful conditionally. It supplies the exchange-loop topology for the fermion, Dirac and entanglement programmes inside the declared defect-free $\mathbb{R}^3$ point model and rules out a generic two-dimensional braid phase only after a physical three-dimensional slice and point-excitation map are supplied. Each downstream programme must still specify its physical slice, state space, representation, observable algebra and local dynamics.

Discriminant and recovery route. In two spatial dimensions the braid group permits continuous Abelian anyonic phases. In three spatial dimensions the point-particle exchange group collapses to the permutation group; for two particles it is $\mathbb{Z}_2$. That dimensional contrast is Level-A topology for the separately supplied defect-free $\mathbb{R}^3$ point-configuration model. Its application to physical ECT is conditional on the physical spatial slice and excitation map, neither of which follows from P1/P4. It is not an ECT-specific prediction of bosons, fermions or the absence of every parastatistical description. A physical discriminant would require a completed excitation sector and an exchange-interferometry observable specified before data.

Status summary. *Established (Level A, conditional on the separately supplied defect-free $\mathbb{R}^3$ point-excitation configuration space):* $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$, and continuous Abelian braid phases of the two-dimensional anyon type are unavailable in that model. P1 and P4 do not by themselves derive the physical $\mathbb{R}^3$ slice, particle identity or excitation map.

Level B route: after selecting a one-dimensional unitary representation, the characters $+1$ and -1 are compatible with bosonic and fermionic exchange sectors.

Open (OP-Q11): the physical excitation space, representation selection, treatment or reduction of higher-dimensional S_N sectors, the spin–statistics theorem, CAR and the many-excitation reconstruction.

Table 77: Quantum statistics in ECT vs standard frameworks.

Aspect	Standard QM/QFT	ECT
Route to bosonic/fermionic split	Full spin-statistics theorem within an already assumed operator framework	Topological exchange dichotomy in $d = 3$ plus structural compatibility with ordered-branch representation content (Level B; not yet full PLZ; free-kernel Euclidean OS test Level A, Q11.1a)

Aspect	Standard QM/QFT	ECT
Operator-algebra basis	Bosonic/fermionic operator algebra fixed within the standard quantised-field framework	Topological and representation-theoretic skeleton established; full operator-algebraic completion remains open
Pauli exclusion	Follows for fermions once the fermionic operator algebra / spin-statistics structure is imposed	Structural Level B consequence once the spinorial sector and spin-statistics route are combined; not yet a full operator-level theorem
Full PLZ theorem	Yes	Open
Beyond present closure	Standard many-body framework available	Full many-body fermionic construction remains open

24.5 Representation-theoretic route to Dirac structure

Status: the supplied $O(4) \rightarrow O(3)$ director branch, the exchange quotient of identical points in defect-free $\mathbb{R}^3$, the double cover of a supplied connected rotation/Lorentz group, and the irreducible Clifford factorisation are distinct conditional constructions. Each standard mathematical statement is Level A under its own hypotheses; no one of them derives the next. Dirac-type dynamics is Level B conditional only after a physical Lorentzian metric/tetrad, spin structure, realised four-component irreducible Clifford module and action are supplied. The physical spinorial spectrum, masses, gauge couplings, CAR/Fock/OS completion and interacting spin-statistics theorem remain Open. Connection: exchange topology (§24.4), the scoped Clifford theorem (Eq. (9.3)), fermionic programme (§9.9), Open electroweak/chirality programme (§7.4), fifth-force coupling (§7.6), and entanglement (§31).

The exchange-sector analysis of Section 24.4 establishes two loop classes and two one-dimensional character choices in the declared configuration-space model. It does not select a species assignment and does not establish that the ECT broken phase physically contains either bosonic or fermionic particles. The present subsection shows that the symmetry structure of the broken condensate provides a natural representation-theoretic route to the Dirac structure.

Full structural chain from the ordered branch to Dirac form. The representation-theoretic route to Dirac structure should be read as the final part of a longer ordered-branch chain:

- (i) P4 supplies a nonzero vector with $O(3)$ stabiliser; a physical transition, branch and three-space interpretation remain Open;
- (ii) after a physical three-dimensional spatial slice is separately supplied, the exchange topology of identical excitations is $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ (Section 24.4);
- (iii) the residual rotational sector admits the spinorial double cover $\text{Spin}(3) \rightarrow SO(3)$;
- (iv) after a physical metric/clock and the standard complex generator continuation are supplied, the usual Lorentz algebra $\mathfrak{so}(1, 3)$ is available conditionally;
- (v) a supplied spinor representation of that algebra carries the standard Clifford structure (24.11);
- (vi) within a supplied minimal local linear first-order Lorentz-covariant fermion closure, the corresponding free dynamics is Dirac-type.

The exchange topology and Euclidean group facts are structural results. The physical Lorentz metric, fermion representation, Clifford action and Dirac dynamics are conditional completion data, not consequences of P4 alone.

Conditional continuation from $O(4)$ generators to the Lorentz algebra

The Lie algebra $\mathfrak{so}(4)$ has 6 generators J_{AB} . Under $O(4) \rightarrow O(3)$ with the condensate direction $n^A = \delta_4^A$:

- **Unbroken** ($O(3)$ subgroup): J_{ij} , $i, j \in \{1, 2, 3\}$ — spatial rotation generators.
- **Broken** (coset $O(4)/O(3)$): J_{i4} , $i \in \{1, 2, 3\}$.

At the algebraic level, one may perform the standard generator continuation that identifies the broken Euclidean generators with the Lorentzian boosts:

$$K_i \equiv -iJ_{i4}.$$

This representation-theoretic step is distinct from the real ordered-branch parametrisation $x^0 = w$ used in the hyperbolic field equations. A direct calculation then confirms that J_{ij} and K_i satisfy the Lorentz algebra $\mathfrak{so}(1, 3)$, with $[K_i, K_j] = -iJ_{ij}$. Thus the continued generators realise the Lorentz algebra exactly. This is Level-A representation theory after the complex continuation is supplied; P4 alone neither selects that continuation nor provides a physical metric, tetrad or Osterwalder–Schrader reconstruction.

Conditional representation-theoretic route to Clifford structure

The Dirac field representation is $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. To avoid conflating the abstract continued generators above with their spinor matrices, define

$$\Sigma_L^{\mu\nu} = \frac{i}{4}[\gamma_L^\mu, \gamma_L^\nu], \quad J_i^{\text{spin}} = \frac{1}{2}\epsilon_{ijk}\Sigma_L^{jk}, \quad K_i^{\text{spin}} = -\Sigma_L^{0i}.$$

Together with the supplied mostly-plus Clifford representation

$$\{\gamma_L^\mu, \gamma_L^\nu\} = -2g^{\mu\nu}\mathbf{1}_4, \quad (24.11)$$

these matrices realise the standard Lorentz commutators. This is Level A conditional representation theory after the metric, continuation and representation map are supplied. The Lorentz algebra alone does not force this physical ECT owner, and the scalar characteristic tensor supplies no spinor bundle or tetrad.

Conditional structural theorem: irreducible Clifford-factorisation class

This subsection uses exactly the scoped theorem of Eq. (9.3). Supply a physical Lorentzian metric/tetrad and spin structure, and one four-complex-component irreducible Clifford module carrying $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. Require local, linear, constant-coefficient Lorentz-covariant dynamics and the exact factorisation of the Klein–Gordon polynomial displayed there. Inside that class the irreducible complex Clifford module is unique up to similarity and convention choices, and the free equation can be written

$$(iS_0 \gamma_L^\mu \partial_\mu - m_\psi c_{\text{char}}) \psi = 0. \quad (24.12)$$

This is not uniqueness in the full class of first-order systems. Reducible direct sums (including Dirac modules with different masses), Duffin–Kemmer–Petiau or Maxwell systems, constrained/gauge systems, additional background tensors and non-Clifford first-order systems lie outside the hypotheses. The theorem is standard conditional mathematics; it neither supplies the physical ECT spinor sector nor derives its action.

Relation to the Part I Dirac equation. Part I uses the same metric, Clifford, adjoint and Hamiltonian package with the observed $\hbar$. Equation (24.12) instead uses the pre-identification action scale S_0 . The forms agree only after their metric, spinor, domain, action and normalisation inputs are admitted; this subsection does not derive $S_0 = \hbar$.

What this theorem does and does not establish. *It does establish:* scoped standard uniqueness inside the complete declared operator class. *It does not establish:* a physical ECT metric or spinor spectrum, CAR/Fock structure, mass generation, gauge couplings, the operator-class premises, or complete quantisation from bare P3.

Medium reading (P5). Postulate P5 does not by itself derive the Dirac operator. Its role here is interpretive. It says that, if spinorial excitations are realised, they are not abstract fields on a pre-given background but physical excitations of one common ordered medium. Accordingly, the Clifford and Dirac structures are read in ECT not as fundamental ingredients of spacetime, but as representation-theoretic organisations of coherent medium excitations on the ordered branch.

Discriminant: proposed medium descent versus supplied Dirac structure. Standard relativistic field theory starts from a Lorentzian metric, spinor bundle and Dirac action. ECT supplies an earlier Euclidean ordered-medium architecture, exact conditional orientation-group facts, and exact topology inside a separately declared defect-free $\mathbb{R}^3$ point-configuration model. It does not derive the physical spatial slice or rotation-group identification; the physical metric, generator continuation, spinor sector and restricted first-order operator class are still completion inputs. Once those inputs are supplied, standard conditional mathematics fixes the Dirac-type operator inside that class. The distinctive ECT programme is therefore a proposed descent of those missing owners from the medium, not a completed derivation of the Clifford or fermion sector.

Connection to downstream programmes. The Dirac route is not an endpoint. Once admitted, it provides the fermionic backbone for several later programmes: (i) the Open chiral-bundle and matter-projector problem in the electroweak completion (§7.4); (ii) the neutrino-sector discussion in Part I (§9.9), where the Dirac/Majorana question is posed within the same spinorial framework; (iii) the ECT-specific fifth-force coupling $\Delta\mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi$ (§7.6); and (iv) the spin-statistics and entanglement programmes (§24.4, §31), which use the same spinorial structure as part of their closure.

External target and reference case. Observed charged leptons and quarks are described by Dirac-type relativistic spinor dynamics to very high precision. Neutrinos also belong to the same relativistic spinorial framework, even though the detailed Dirac-versus-Majorana completion remains open. ECT does not yet derive the fermion mass spectrum from condensate microdynamics, but it does provide the structural statement that if the ordered branch supports spinorial excitations, then the minimal local linear first-order Lorentz-covariant free equation available to them is of Dirac type. The observed fermionic sector is therefore an external target for a future ECT spinor completion. Its agreement with standard Dirac theory does not support an ECT spinor owner; the physical bundle, state, vertex, representations, masses and mixings remain Open.

Falsifier for the Dirac-structure programme. The conditional Dirac-structure programme would fail if (i) no supplied physical metric/clock completion admitted a consistent Lorentz spinor representation; (ii) the corresponding Clifford structure were inconsistent with that supplied metric; (iii) within the same explicitly declared local, linear, constant-coefficient, nondegenerate, Hermitian, parity-even minimal first-order class and modulo invertible constant field redefinitions, a non-Dirac free dynamics were equally admissible; or (iv) the physical fermionic sector ultimately required a structure incompatible with the ordered-branch spinorial route. Any of these would force a substantial revision of the present representation-theoretic construction.

Pauli exclusion. Exchange topology alone supplies neither a fermionic character nor CAR. If a physical spinor kernel, local quantum reconstruction and antisymmetric CAR sector are independently supplied, the standard free-kernel reflection-positivity check selects the fermionic quantisation within its declared

class and CAR then implies Pauli exclusion. This is Level A conditional mathematics for that supplied kernel/algebra; deriving the physical ECT spinor sector, CAR and the interacting spin–statistics theorem remains Open (OP-Q2, OP-Q11).

Status summary. *Established and conditional components:* (i) Euclidean generator conservation, exchange topology and the relevant covering-group identities are Level A mathematics in their stated models; (ii) after a physical metric/clock and the standard complex continuation are supplied, the Lorentz algebra and its Clifford representation are standard conditional mathematics; (iii) the Φ -medium descent of the physical metric, spinor bundle and fermion action remains Open.

Level B (conditional physical route): If physical spinorial excitations are realised, then within the minimal local linear first-order closure their free massive dynamics is Dirac-type.

Level A conditional algebra / Open physical owner: With a supplied physical spinor kernel, quantum reconstruction and antisymmetric CAR, Pauli exclusion follows algebraically. This does not require $S_0 = \hbar$, a separate Level B calibration whose microscopic owner and universality remain Open.

Open (OP-Q3, OP-Q11): the physical spinorial spectrum, mass generation, gauge couplings, full second-quantised operator algebra, and the completed fermionic many-body construction.

Bridge to the vacuum-response programme. With Euclidean path-dependence, exchange sectors, and the structural route to Dirac dynamics in place, the coherent branch has acquired its basic nonclassical, statistical, and spinorial architecture. The next chapter turns from these kinematical and representation-level questions to conditional response calculations in supplied coherent-vacuum models under boundaries, acceleration, time dependence, and strong-field interfaces. Their common physical ECT owner remains Open.

25 Conditional quantum-vacuum models and the universal-response programme

Inputs: supplied ordered P4 chart and coherent BR1 closure; wave dynamics and action scale S_0 from Sections 21–22; formal continuation (22.2); medium character (P5). Aim: to test whether a future physical ordered-vacuum completion can relate boundary, acceleration, time-dependent and strong-field responses. The displayed Casimir, Unruh, particle-production and Hawking calculations currently have different supplied owners. Status: the standard scalar boundary-value, Rindler/KMS, and Bogoliubov calculations quoted below are Level A only inside their explicitly supplied QFT toy models. Their ownership by ECT is Open: the physical reduced soft bundle, its state, boundary and detector couplings, stress tensor, physical Lorentzian cone, and strong-field metric completion have not been obtained from the present Φ -medium action. In particular, neither a three-scalar Casimir coefficient nor an ECT Unruh/Hawking prediction is established. Connections: vacuum ontology (§25.1), unified-response logic (§25.2), action-scale universality (§5.6), UV-threshold programme (§8.3), analogue-laboratory programme (§37), and the downstream decoherence/Born/black-hole programmes.

Connection to PES-R. The vacuum or ground sector is fixed by the owning Hamiltonian/effective action, its domain, state, and boundary conditions. PES-R does not identify it by minimising a universal non-stationarity/decoherence scalar. After the physical vacuum modes are known, their poles, finite-time retention, and record distinguishability may be classified separately. Casimir response, Unruh response, and particle production remain owned by their correlation functions and backgrounds.

What is established, what is conditional, and what remains open in the vacuum-response programme. The ECT vacuum-response programme should be read in three layers. First, the supplied ordered-phase variables define candidate coherent correlators and a genuine-medium interpretation under P5. This does not yet establish their response to physical boundaries, detectors, or strong-field interfaces.

Second, on top of that structural basis, ECT defines Open routes to four families of phenomena: Casimir-type boundary response (§25.3), Unruh-type accelerated-vacuum response (§25.4), particle production in nonstationary ordered backgrounds (§25.5), and horizon thermality in the strong-field sector (§36). The standard QFT calculations used as targets are Level A within their own supplied assumptions; their ECT ownership remains Open. Third, what remains open is the completed reduced coherent EFT needed for definitive force coefficients, detector-response kernels, and production rates in realistic geometries and backgrounds. Thus the proposed one-medium unification is a research hypothesis, not a completed structural or quantitative result. The following subsection (§25.1) provides the ontological and stiffness-level basis for that unified response logic.

25.1 Quantum vacuum structure of the ordered condensate

Status: Level A only for algebraic correlators and stiffnesses inside each explicitly supplied reduced model. Open for the physical quantum state and for the hypothesis that Casimir response, accelerated-detector response, particle production, and horizon thermality are probes of one ECT vacuum operator. The common bundle, state, continuation, couplings, and microscopic parameter map are not derived. Connection: $O(4) \rightarrow O(3)$ ordering (P4), medium character (P5), the two-branch architecture (§6), the action-scale universality (§5.6), the UV-threshold programme (§8.3), and the downstream vacuum-response programme (§25).

The Quantum Sector explores the completion hypothesis that particles are excitations of a structured ordered medium rather than entities placed on an otherwise empty background. BR1 supplies a phase-sensitive calculational closure; it does not identify a physical vacuum state or prove that the closure is the complete vacuum theory.

Vacuum as an ordered medium

Inside the supplied ordered/coherent closure, the background is represented with the following structures:

- (i) a nontrivial order parameter $\Phi_{\text{eff}} = \rho e^{i\theta}$;
- (ii) a phase-sensitive coherent sector with winding structure;
- (iii) stiffness scales controlling deformations of the ordered state;
- (iv) topological sectors, two conditional loop-scale candidates, and loop/history weights by an operational S_0 only after an explicit owner map is supplied;
- (v) nontrivial vacuum correlators of coherent observables.

Promoting this inventory to a physical ECT vacuum requires a stationary action, state, admissible configuration space, continuation and observable response map; those owners remain Open.

Medium character of the vacuum (P5). Postulate P5 motivates the ontological programme in which a future physical vacuum is an ordered medium rather than an abstract lowest-energy state on a pre-existing Lorentzian background. P5 does not derive that state, physical correlators, stiffness, topology, response functions or particle spectrum. The statement that particles are coherent excitations *of* this medium is therefore a completion target, not a result of P5 or BR1 alone.

Unified vacuum viewpoint

This is a common-origin research programme of ECT, not an established unification theorem. The displayed scalar, phase, gauge, tensor and response sectors require distinct normalisations and physical owner maps; one bare order parameter does not by itself make their observed constants or response operators identical.

This unified reading is consistent with the broader one-medium architecture established earlier in the article: one physical-cone programme, one candidate coherent action scale, sector-specific variational backbones, and compatible conservation laws (§11.1). The present subsection specialises that architecture to supplied vacuum-EFT models. It investigates whether the same physical medium could underlie excitations, geometry and vacuum response; that common ownership is not derived.

Vacuum stiffness and Planck-scale interpretation

Within the declared models, an ordered-medium response is represented by stiffness coefficients. In the geometric branch this appears as a candidate gravitational stiffness (Section 13.4). In the coherent branch it appears as phase stiffness K_θ^{eff} , loop action cost S_0 , and the energetic structure of the supplied coherent-vacuum EFT. A full first-principles derivation relating all these stiffness notions to one common microscopic set of condensate parameters remains Open. Their placement in one two-branch diagram is organisational and does not establish a common physical owner.

Two microscopic scales and a separate fit length. The ordered-medium programme motivates comparisons among gravitational, coherent, and conditional electroweak stiffnesses, but a common microscopic derivation remains Open. The conditional HRC matching length L_{gal} is not a third stiffness or VEV: it is a Level-C phenomenological fit quantity pending the scalar/AQUAL bridge. The vacuum-response programme below is formulated most explicitly for the primary coherent-vacuum sector.

Connection to the vacuum-response programme. The manuscript proposes an Open common-response programme for four families of downstream phenomena: (i) boundary sensitivity and Casimir-type response (§25.3), (ii) accelerated-detector and Unruh-type response (§25.4), (iii) particle production in time-dependent ordered backgrounds (§25.5), and (iv) horizon thermality in the strong-field sector (§36). The conditional dependency logic is developed later in §25.2. The present subsection supplies no physical vacuum state, common correlator or response vertex, so it does not yet provide a common foundation.

External QED/QFT reference cases. Casimir forces between boundaries [160, 161], vacuum-sensitive atomic effects such as the Lamb shift, and spontaneous emission are established external QED/QFT phenomena. They neither require nor select an ECT ordered-medium ontology. ECT does not yet derive their magnitudes, boundary maps or detector observables from condensate microdynamics; they are targets for a future vacuum-response completion, not empirical support for one.

Open interpretive hypothesis: ordered-medium and ground-state vacuum descriptions. In standard quantum field theory, the vacuum is introduced as the lowest-energy state of quantised fields on a Lorentzian spacetime that is already assumed. ECT asks whether that ground-state description can emerge as an effective reading of a deeper ordered substrate. This is an ontology hypothesis, not a consequence of the presently supplied order parameter: the physical ECT vacuum state, its correlators, excitation algebra, boundary conditions and observable maps remain Open. Standard QED/QFT is an explicit successful description of the cited phenomena without selecting this hypothesis.

Completion test for the common-vacuum programme. After one ECT vacuum state, correlator, response vertex and joint observable map are derived, their failure to reproduce the named channels without sector-by-sector retuning would reject that common-kernel completion. It would not falsify the supplied scalar identities or establish that every possible ECT completion fails.

Status summary. *Established (Level A in supplied models):* the displayed order parameters, correlators and stiffness identities.

Open: the physical vacuum state and one derived response operator that owns the Casimir, Unruh-type, particle-production, and horizon-thermality channels and fixes their amplitudes.

25.2 Candidate common-response programme and analogue-model comparison

Status: Open research hypothesis: boundary response, accelerated-detector response, particle production, and horizon thermality may become probes of one ordered coherent vacuum only after one physical response bundle and state are derived. Analogue-gravity examples are external source models and do not supply an ECT owner or evidence for a common response. *Connection*: the vacuum ontology of Section 25.1, the action-scale universality (§5.6), the UV-threshold programme (§8.3), the decoherence programme (§26), and the analogue-laboratory programme (§37).

Conditional unified-response logic

The supplied vacuum-model effects below define a comparison programme. The claim that Casimir response, particle production, Unruh detector response and Hawking thermality are manifestations of one physical ECT operator remains Open; their displayed calculations currently use different supplied boundaries, states, continuations and geometries.

The logic can be organised as follows:

- (i) **Boundary sensitivity**: boundaries reshape the coherent vacuum mode spectrum, producing Casimir-type effects (Section 25.3).
- (ii) **Time-dependent background sensitivity**: nonadiabatic variation of the ordered background leads to Bogoliubov-type mode mixing and particle-production effects (Section 25.5).
- (iii) **Observer sensitivity**: accelerated detectors probe the coherent vacuum differently from inertial ones, giving rise to Unruh-type response (Section 25.4).
- (iv) **Horizon/interface sensitivity**: strong-field causal interfaces generate reduced-state thermality in black-hole settings (Section 36).

In all four supplied models the vacuum state has nontrivial correlators whose response depends on the declared probe. Identifying these models with one physical ordered ECT medium is an Open matching problem.

What is proposed and what remains open. The programme asks whether the same physical ordered coherent state can own four distinct physical channels: boundaries, time dependence, accelerated observers, and causal interfaces. What is proposed here is that these channels should be read as one unified response family rather than as unrelated effects. What remains open is the completed microscopic derivation in which one and the same physical correlator infrastructure quantitatively reproduces all four sectors at once.

Why this matters. This unified viewpoint prevents the Quantum Sector from fragmenting into separate phenomenological anecdotes. It is the vacuum-response analogue of the common-origin programmes stated earlier; physical ownership remains Open in each channel. Here the same philosophy is applied specifically to vacuum response: one candidate common owner, several observational channels.

Connection to the vacuum ontology of Section 25.1. Section 25.1 supplies a coherent-vacuum effective model. The present subsection goes one step further: it organises the principal observable consequences of that statement. Whether one physical correlator, metric and state own all four channels is Open. The current boundary, acceleration, time-dependent and horizon blocks use different supplied operators and assumptions.

Connection to decoherence and influence-function kernels. The unified-response logic is structurally parallel to the decoherence programme. In both cases, the central object is not a formal particle ontology but the correlator structure of the ordered medium. A future common kernel would have to reproduce the boundary, detector, particle-production and reduced-state channels after their physical bundles and states are specified (§26). No such equality is presently derived; vacuum response and decoherence remain separate Open programmes connected by a testable common-owner hypothesis.

Discriminant: unified ordered-vacuum response versus separate mechanisms. In standard presentations, Casimir forces, Unruh response, cosmological particle production, and Hawking thermality are often developed as conceptually separate effects tied to different calculational setups. ECT instead proposes that they are four observational windows into one common ordered-vacuum response structure. The one-medium diagram does not make that common origin physically natural or probable; it merely states the hypothesis to be tested after a common kernel, state and observable map are derived.

Analogue-gravity continuity of the QS vacuum picture

ECT does not introduce this vacuum-response logic in isolation. Analogue-gravity source models show that effective horizons, mode conversion, and thermal detector response can arise in emergent media without requiring spacetime geometry itself to be fundamental. This establishes a logical possibility in those source models. It neither supports ECT nor supplies its action, state or response kernel.

This does not mean ECT is merely an analogue model. The point is only that horizon thermality and vacuum response need not by themselves prove a fundamental microscopic-spacetime ontology. Whether the ECT coherent branch provides a physical instance remains Open.

No inference from analogue continuity. The analogue cases motivate asking whether Hawking-, Unruh-, Casimir-, and particle-production phenomena can be limits of one ECT response family. They do not strengthen that reading or make it more probable: no common ECT kernel has been derived.

External source-model and observational references. The four response channels have different external statuses: Casimir response in precision boundary-force experiments, Unruh-type logic in accelerated-detector analyses and analogue platforms, particle-production mechanisms in time-dependent quantum backgrounds, and Hawking-type horizon thermality in black-hole and analogue-horizon settings. These references constrain or motivate future calculations but are neither evidence for ECT nor tests of a common ECT kernel.

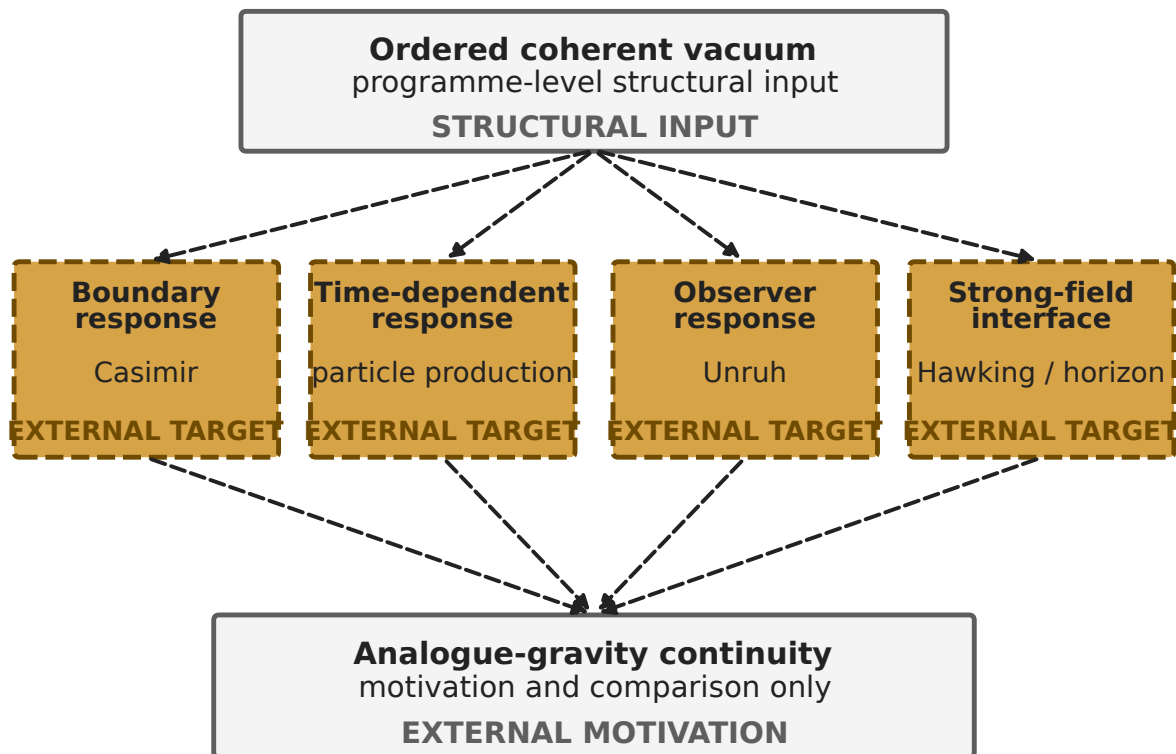
Completion test for the unified-response programme. Once a named common ECT kernel is derived, incompatible kernels or failed held-out channel predictions would reject that named unification. At present there is no physical one-kernel closure to test, and the absence of a known contradiction supplies no positive evidence.

Status summary. *Open programme:* derive one physical response bundle, state and set of couplings and then test whether boundary effects, accelerated-detector response, time-dependent particle production, and horizon thermality are its different limits. Analogue-gravity continuity is motivation, not an ECT derivation.

Open: the completed one-kernel microscopic derivation that quantitatively reproduces all four response families from one common coherent-vacuum EFT.

Open common-response programme

All four response targets remain Open.
Dashed links show conceptual continuity only.



No common physical bundle, state, continuation, or coupling has been derived.

Mechanism names and explicit status labels—not hue—
distinguish the four targets.

Figure 22: Open common-response programme for the quantum sector. Boundary, time-dependent, observer and strong-field responses remain four external targets. The shared gold fill marks the Open-status class. All links are dashed because no common physical ECT bundle, state, continuation and coupling owning all four responses has been derived; analogue-gravity continuity is motivation only.

For a more extended structural discussion of the unified vacuum-response logic and its analogue-gravity continuity, see Appendix AV.

25.3 Boundary sensitivity in a supplied vacuum model and the Casimir calculation

Status: Level A for the elementary Dirichlet boundary-value calculation inside a supplied set of independent massless scalar fields. Open (OP-Q12, OP-Q18) for an ECT Casimir effect: the current strict scalar map has one longitudinal coordinate rather than three independent coordinates, while the physical mode count and boundary coupling, mode multiplicity, renormalised stress tensor, force sign, and laboratory implementation are not derived. Connection: vacuum ontology (§25.1), unified-response logic (§25.2), medium character (P5), Goldstone counting from $O(4) \rightarrow O(3)$, analogue-laboratory programme (§37), and Appendix J.

An ordered medium can in principle respond to boundaries, but this general possibility is not yet an ECT prediction of a measurable Casimir force. Such a prediction requires a physical boundary operator that changes the spectrum of the reduced propagating bundle and a derived stress observable.

Why a Casimir-type effect is expected

If a supplied coherent-vacuum model supports fluctuation modes, then imposing boundary conditions changes the allowed mode spectrum. Schematically,

$$E_{\text{vac}}^{(\text{boundary})} \neq E_{\text{vac}}^{(\text{free})}. \quad (25.1)$$

In the supplied scalar toy problem the resulting energy difference induces a force. Whether a material boundary constrains the physical Φ -medium modes in this way is an additional, presently Open, question.

ECT interpretation

In ordinary QFT the Casimir force is often phrased as a consequence of zero-point mode subtraction. If a physical ECT boundary operator is derived, P5 would motivate interpreting the corresponding energy difference as response of an ordered medium rather than merely as formal mode subtraction. The present manuscript does not yet derive that operator.

Medium reading of boundary sensitivity (P5). Under the medium interpretation (P5), a future physical boundary map could be read as deformation of a genuine ordered medium. Only after that map is known may one infer that the condensate between boundaries has a modified physical spectrum or measurable stress. This medium-level reading is conceptually continuous with superfluid and analogue-vacuum Casimir phenomena, though it does not by itself prove the quantitative ECT coefficient in any given laboratory setup.

Status and future derivation

The present ECT action does not yet support a theorem asserting the existence or coefficient of a physical condensate-sector Casimir force. It only motivates the boundary-response programme and supplies a setting in which a future reduced action and boundary map can be tested (OP-Q12).

Status summary. *Established (Level A, supplied toy model only):* the standard Dirichlet calculation for N independent massless scalar fields gives the algebraic expression (25.3) below.

Open (OP-Q12, OP-Q18): the value of N in the physical reduced ECT bundle, the boundary coupling, the completed coherent-vacuum stress calculation in realistic geometries, and the identification of systems that realise the required boundary conditions.

25.3.1 Supplied scalar Dirichlet boundary-value problem

Scope and applicability (read before the formula). The calculation below is a supplied boundary-value toy model. No ordinary material, superfluid interface, BEC wall, or cosmological phase boundary has been shown to impose its Dirichlet condition on the physical ECT soft bundle. In particular, neither transparency of metals nor a purely gravitational coupling of that bundle follows from the present action.

25.3.2 Derivation inside the supplied scalar toy model

Boundary condition assumption. Dirichlet conditions and N mutually independent massless scalar fields are supplied as inputs. They are not consequences of the current ECT action.

For reference, the standard QED result for two photon polarisations between parallel plates separated by a is

$$\left. \frac{F}{A} \right|_{\text{QED}} = -\frac{\pi^2}{240} \frac{\hbar c}{a^4}. \quad (25.2)$$

For N supplied, independent, massless scalar fields with propagation speed c_{toy} and ideal Dirichlet plates, the fields contribute independently:

$$\boxed{\left. \frac{F}{A} \right|_{N \text{ scalars}} = -\frac{N\pi^2}{480} \frac{\hbar c_{\text{toy}}}{a^4}}, \quad (25.3)$$

which follows from zeta-function regularisation [162]:

$$E_0^{\text{ren}}(a) = -N \frac{\pi^2}{1440} \frac{\hbar c_{\text{toy}} A}{a^3} \implies \frac{F}{A} = -\frac{\partial E_0^{\text{ren}}}{\partial a} \bigg/ A. \quad (25.4)$$

Equation (25.3) is standard toy-model algebra, not an ECT force law. The once-used substitution $N = 3$ is not licensed by the current integrable soft sector and is therefore not retained as a prediction. If three independent Dirichlet scalar channels are *supplied by hand* and $c_{\text{toy}} = c$, the standard formula gives $(F/A)_{N=3}/(F/A)_{\text{QED}} = 3/2$. This arithmetic identity is retained as a check of the toy BVP; it is not an ECT polarisation count or laboratory prediction.

Soft-mode counting caveat. The supplied orientation target $O(4)/O(3) \simeq S^3$ carries three coordinate directions in the unconstrained sigma-model parametrisation. However, in the scalar-only integrable basis these do not automatically correspond to three independent propagating soft particles: the strict linear scalar map is parametrised by one longitudinal coordinate χ (Level A kinematics; Section 4.3, Appendix J). The physical soft-mode count requires a supplied reduced action, quotient or domain, and projector; $N_G^{\text{phys}} = 1$ is only a Level B possibility inside such a reduced EFT, not a consequence of integrability. Therefore the earlier substitution of a three-component coordinate count into a physical Casimir coefficient is invalid. Even the conditional physical mode represented by this coordinate cannot be inserted into (25.3) until its material boundary coupling and stress operator are derived. This caveat applies equally to any use of the three-component count in the dark-sector (§18.1) and particle-spectrum sections.

Summary statement and status caveat. Equation (25.3) records only the supplied scalar Dirichlet algebra. ECT presently predicts neither a correction to the electromagnetic metallic-plate force nor a distinct condensate-sector coefficient for any laboratory setup.

Discriminant: laboratory QED Casimir versus ECT condensate Casimir. Three cases must be kept sharply distinct.

- (i) **Standard metallic-plate Casimir experiments:** these test the electromagnetic QED vacuum and do not by themselves test an ECT condensate sector, because no derived map shows that they impose boundary conditions on the soft ordered-branch sector.

- (ii) **Condensate-reflecting boundaries:** the supplied scalar calculation applies only after the number of independent fields and their Dirichlet coupling have been posited. No such physical ECT boundary model has yet been identified.
- (iii) **Intermediate analogue systems:** superfluid, BEC, domain-wall, or phase-boundary systems may probe the same boundary-sensitivity logic without reproducing the exact idealised Dirichlet setup assumed in the toy estimate. They motivate future boundary-response models but do not yet test a quantitative ECT prediction.

The next theoretical gate is to derive a physical reduced boundary operator and stress tensor before proposing a discriminating experiment.

Empirical anchor. The standard electromagnetic Casimir force has been measured with high precision in metallic and related laboratory setups [160, 161]. These experiments do *not* directly test an ECT condensate-sector law, because no such law or material boundary map has yet been derived. They establish boundary-sensitive QED vacuum response in their owning theory. For ECT they are only an external matching target; they neither select ordered-medium ontology nor support an ECT boundary law.

Gate for the boundary-sensitivity programme. The programme advances only if a completed reduced ECT action yields a boundary-coupled propagating bundle and a renormalised stress response. Only then can a physical coefficient and falsifier be defined. Standard metallic-plate Casimir experiments currently test QED, not a derived ECT boundary law.

Connection to the analogue-laboratory and unified-response programmes. Boundary sensitivity is the most direct and experimentally accessible entry point into the unified vacuum-response logic developed in Section 25.2. It is also the vacuum-response sector that lies closest to the analogue-laboratory programme of Section 37, where phonon, superfluid, and BEC systems provide the most natural candidate platforms for testing ordered-medium boundary response beyond the purely electromagnetic Casimir effect.

Programme-level connection. A future calculation may relate the boundary, detector, and influence-functional correlators after their physical bundles, states, and couplings are specified. Their reduction to one universal ECT correlator is a proposed programme, not a present result.

25.4 Accelerated observers and the Unruh-type response

Status: Level A for the standard Rindler/KMS reconstruction inside a supplied Lorentzian QFT state with a massless correlator and an Unruh–DeWitt detector. Open (OP-Q12) as an ECT result: the physical Lorentzian state, reflection positivity/analytic continuation, detector coupling, physical cone, and the matching $S_0 = \hbar$ have not all been derived from the current Φ -medium action. Connection: vacuum ontology (§25.1), unified-response logic (§25.2), medium character (P5), Euclidean $O(4)$ symmetry (P2), Casimir boundary sensitivity (§25.3), strong-field horizon response (§36), and the analogue-laboratory programme (§37).

The standard Unruh calculation demonstrates what a completed ECT vacuum sector would have to reproduce. It does not establish that the current ordered-medium variables already define the required Lorentzian QFT state and detector observable.

Kinematic expectation

For uniform acceleration a , the standard QFT result gives

$$T_U = \frac{\hbar a}{2\pi c k_B}. \quad (25.5)$$

This formula uses the observed $\hbar$ and c . In ECT the identifications $S_0 = \hbar$ and of a physical characteristic cone with c are matching conditions, not consequences of P1–P6.

Medium reading of the Unruh response (P5). Under P5, a future completed ECT construction would interpret an Unruh response as detector sampling of a genuine medium state. The present action has not yet shown that its Euclidean correlator reconstructs the required physical Lorentzian state or that an accelerated detector couples to it in the standard way.

ECT status

An Unruh-type effect remains a target of the ordered-vacuum programme. What is not yet in place is its detector-response derivation from a completed ECT coherent EFT (OP-Q12).

Status summary. *Established (Level A, supplied QFT assumptions only):* the standard massless Wightman function restricted to a Rindler trajectory satisfies the KMS condition and gives (25.8).

Open (OP-Q12) for ECT: the physical state, continuation, cone and detector-response derivation from the completed coherent EFT. Direct experimental detection of the Unruh effect also remains unsettled.

Observational status and empirical anchor. At present there is no universally accepted direct laboratory detection of the Unruh effect itself. Accordingly, ECT should not claim observational confirmation here. Indirect discussions exist in the literature, for example in connection with electron depolarisation in storage rings [163], but such arguments are not generally regarded as a clean standalone detection of the Unruh effect. The calculation below is therefore a conditional target and an internal consistency test, not an ECT-specific empirical success.

25.4.1 Conditional correlator-based reconstruction of the Unruh response

25.4.2 Derivation

The derivation below is a conditional reconstruction using the standard Euclidean correlator of a massless scalar/Goldstone-coordinate mode in a supplied free Gaussian model, taken as a proxy for the completed soft coherent sector. It does not identify a physical ECT particle or derive $m_\chi = 0$. It shows structural compatibility and continuity with the standard result, not yet a full first-principles ECT derivation from the completed coherent EFT.

The Unruh effect [164, 165, 166] states that a uniformly accelerated observer detects the Minkowski vacuum as a thermal bath at $T_U = \hbar a / (2\pi k_B c)$. In ECT this structural reconstruction proceeds from the Euclidean correlator via analytic continuation.

The Euclidean two-point function of that supplied massless scalar proxy is [162]:

$$G_E(X, X') = \frac{1}{4\pi^2 |X - X'|^2}, \quad |X - X'|^2 = (x - x')^2 + (w - w')^2. \quad (25.6)$$

If the supplied Euclidean correlator satisfies the hypotheses needed for reconstruction of a physical Lorentzian state, its continuation is

$$G_M^+(x, x') = -\frac{1}{4\pi^2 c^2} \cdot \frac{1}{(t - t' - i\varepsilon)^2 - |x - x'|^2/c^2}, \quad (25.7)$$

the standard Wightman function. Invoking the Osterwalder–Schrader theorem [73] here is an assumption about the supplied correlator and state; it is not yet a proof that the present ECT Euclidean action satisfies all reconstruction axioms.

Along the Rindler trajectory with proper acceleration a , G_M^+ becomes periodic in imaginary proper time with period $\beta_U = 2\pi c/a$ — a thermal distribution. The Unruh–DeWitt detector response gives a Planck spectrum with:

$$T_U = \frac{\hbar a}{2\pi c k_B}, \quad (25.8)$$

which is the standard result. Replacing c by a model characteristic speed is not licensed until that speed is shown to be the physical Lorentzian detector cone. No accelerator number is quoted as an ECT prediction because an accelerated proton is not an Unruh–DeWitt thermometer and its response map has not been derived.

25.4.3 Physical interpretation

In a possible ECT completion, the Unruh effect would not be an additional postulate about accelerated observers. It would have to follow from the physical continuation and detector sampling of the ordered-medium state. That completion is Open. The Euclidean two-point function (25.6) is manifestly $O(4)$ -invariant. Under the supplied formal Euclidean–Lorentzian continuation, that object is accessed through different Lorentzian decompositions: inertial observers use the standard Minkowski mode split, whereas uniformly accelerated observers probe the same vacuum along Rindler trajectories. The thermal character of the response is therefore not a new dynamical heating of the vacuum, but a consequence of how the *supplied* Lorentzian state is sampled by the detector. Ownership of that state and detector by ECT remains Open.

This supplied-QFT calculation is mathematically analogous to the Casimir boundary problem and the near-horizon Rindler target. It does not prove that one ECT correlator owns all three responses. It is also consistent with the decoherence programme (§26), though the present Unruh effect is not itself a decoherence calculation and should not be reduced to environmental tracing alone.

Connection to the unified vacuum-response programme. The Unruh response is one target of the unified vacuum-response programme in Section 25.2. Establishing that it shares a physical ECT operator and state with the boundary, particle-production and black-hole sectors is an Open test of that programme.

Discriminant: emergent versus assumed Unruh response. In standard Lorentzian quantum field theory, the Unruh effect is derived after one has already assumed the quantum vacuum and its Lorentzian analyticity properties. ECT seeks to reverse that logical direction by deriving the physical state and continuation from an ordered Euclidean medium. The displayed calculation shows the required endpoint but does not yet perform that derivation.

Gate for the Unruh-response programme. Before an ECT-specific falsifier can be stated, the theory must derive a reflection-positive Euclidean state, its physical Lorentzian continuation and cone, and a detector coupling. The completed model must then reproduce the KMS temperature (25.8); failure would reject that completion.

25.5 Cosmological particle production and time-dependent ordered backgrounds

Status: A external/QFT mathematics for Bogoliubov mixing after a time-dependent quadratic action, in/out states and particle basis are supplied. Open ECT for the physical time-dependent action, transition history and clock, in/out vacua, particle observable, production rates $|\beta_k|^2$, and primordial fluctuation spectrum. Connection: vacuum ontology (§25.1), unified-response logic (§25.2), medium character (P5), MP cosmology (§14.1), the ordering-transition programme (§25.5, §13.8), and the analogue-laboratory programme (§37).

If the ordered condensate background changes in time, the notion of “particle” defined relative to one coherent vacuum slicing need not coincide with that defined relative to another. This is a standard

candidate setting for cosmological particle production, complementary to the macroscopic transition programme of Section 14.1.

Physical picture: particle creation as vacuum reconfiguration. If a completed ECT transition supplies the required time-dependent quantum action and states, particle creation can be read as a change of excitation content rather than “appearance from nothing”. This is entirely analogous to the physics of known media: rapidly cooling a superfluid produces phonons and rotons not because they “materialise from the void”, but because the background against which “silence” is defined has changed. Whether the resulting excitations are physical ECT particles, and their number and spectrum, depends on the missing action, state, detector and transition-history maps.

Time-dependent ordered branch

In a nonstationary background, the effective coherent vacuum is itself time-dependent. Mode decomposition at early and late times therefore differs. Schematically, the early-time and late-time mode operators are related by a Bogoliubov transformation,

$$a_k^{(\text{out})} = \alpha_k a_k^{(\text{in})} + \beta_k a_{-k}^{(\text{in})\dagger}, \quad (25.9)$$

with $|\beta_k|^2$ measuring the produced excitation density. This is A mathematics for a supplied quadratic QFT problem. It is not yet an ECT result: the actual α_k, β_k and their particle interpretation require the completed time-dependent action and in/out state construction.

ECT interpretation

Conditional on those owners, particle production would be a reorganisation of excitation content relative to a changing ordered background. This is an interpretation target, not a derived event in the present ECT history.

Medium reading of particle production (P5). Under the medium interpretation (P5), a future particle-production completion would be read as reorganisation of excitation content within an ordered medium whose physical background configuration changes in time. What standard curved-spacetime QFT describes as particle creation in a time-dependent geometry, the ECT programme would read as mode redistribution in one candidate ordered medium whose effective parameters evolve during a separately supplied physical history. No ordering transition is derived.

Empirical anchor. Time-dependent particle production is not a speculative mechanism. Bogoliubov-type mode mixing is standard in nonstationary quantum-field settings, and closely related laboratory logic has been observed in the dynamical Casimir effect [167]. ECT does not yet claim a completed first-principles derivation of cosmological production rates from condensate dynamics. The present text records compatibility with that external formalism and the precise missing owners; it does not claim a produced ECT population.

Relation to MP cosmology

Macroscopic Physics describes the large-scale evolution of the ordered branch (Section 14.1). The present QS subsection gives the coherent-vacuum interpretation of how such time dependence can populate excitation sectors. The two descriptions are complementary: MP governs the background history; QS governs excitation production against that history.

The most important application is the $O(4) \rightarrow O(3)$ ordering transition itself (§25.5). If that transition is non-adiabatic in a derived physical quadratic mode action, it can generate Bogoliubov mixing. The resulting coefficients are relevant in principle for (i) the primordial fluctuation spectrum, (ii) the initial

excitation content of the ordered branch, and (iii) the ordering-transition gravitational-wave programme (§13.8). The external Bogoliubov framework is stated here as the target; both its ECT ownership and the quantitative completion belong to the Open cosmological programme of Part II.

Status summary. *Established externally (A mathematics):* given a time-dependent quadratic action and two specified positive-frequency splittings, Eq. (25.9) is the standard canonical transformation.

Open ECT (OP-Q12, OP3): the time-dependent physical action, transition history and clock, in/out states, particle map, production rates $|\beta_k|^2$, and the primordial fluctuation spectrum.

Discriminant: time-dependent ordered-medium production versus postulated vacuum particle creation. In standard curved-spacetime QFT, cosmological particle production is derived once one has already assumed a quantum vacuum on a given time-dependent background. In ECT the ordered medium is primary, and the changing background is a change of the condensate state itself. The mathematics of Bogoliubov mixing is continuous with the standard treatment, but the ontological reading is reversed: the “vacuum” is not a passive background on which particles appear, but the physical medium whose nonstationary evolution reorganises its own excitation content.

Falsifier for the particle-production programme. The programme passes only if a completed ECT transition produces a positive, stable time-dependent mode action, well-defined in/out states and particle observables, and an excitation history compatible with the background and primordial data. Failure rejects that completion; the generic external Bogoliubov identity is not itself an ECT prediction.

25.5.1 Vacuum-response target table

Table 78: Supplied QFT response calculations used as targets for a future ECT vacuum completion. “A mathematics” refers only to the stated external/toy assumptions; all physical ECT owners listed here remain Open.

Effect	Supplied/external formula	ECT disposition	Standard result	Status
Supplied N -scalar Dirichlet BVP	$-\frac{N\pi^2\hbar c_{\text{toy}}}{480a^4}$	No ECT force coefficient	Standard scalar BVP	A mathematics / Open ECT
Standard Unruh target	$\frac{\hbar a}{2\pi c k_B}$	No ECT detector prediction	$\frac{\hbar a}{2\pi c k_B}$	A external / Open ECT
Particle production in time-dependent background	$a_k^{(\text{out})} = \alpha_k a_k^{(\text{in})} + \beta_k a_{-k}^{(\text{in})\dagger}$	No ECT particle rate or state map	Bogoliubov mixing	A external / Open ECT
Standard Hawking target	$T_H = \frac{\hbar \kappa}{2\pi c k_B}$	Physical ECT tensor geometry, state and strong-field response Open	Standard semiclassical-QFT form	A external / Open ECT

Status note on the horizon-thermality row. The Hawking row is a standard target, not an ECT derivation. A physical tensor metric, horizon state, continuation and strong-field response operator are all missing from the current completion.

Parker [168] showed how time-varying supplied backgrounds can produce particle pairs. The ECT $O(4) \rightarrow O(3)$ transition history, physical clock and duration are not derived; in particular no Planck-time transition estimate is retained.

For a separately supplied scalar quadratic mode v_k , the standard oscillator target is [168, 162]:

$$\ddot{v}_k + \Omega_k^2(t) v_k = 0, \quad \Omega_k^2(t) = c_{\text{char}}^2(t) k^2 + m_{\text{eff}}^2(t), \quad (25.10)$$

where $c_{\text{mode}}(t)$ and $m_{\text{mode}}^2(t)$ are supplied functions of the chosen background. Using the established equation label, we write them as $c_{\text{char}}(t)$ and $m_{\text{eff}}(t)$ in Eq. (25.10); neither function is derived for a physical ECT transition. Scalar integrability leaves one longitudinal coordinate in the strict map, not three independent π^i coordinates; the physical mode count remains conditional on the supplied action and projector.

Candidate time-dependent mode speed. Equation (25.10) allows a supplied mode speed $c_{\text{char}}(t)$ to enter as a time-dependent parameter. In standard particle-production treatments on a fixed Lorentzian background, the speed of light is constant and non-adiabaticity enters through the background geometry or effective masses alone. In a future ECT completion, the ordered medium may carry additional time dependence during the transition through parameters that set $c_{\text{char}}(t)$. This is a possible source of mode mixing, not a currently derived ECT-specific effect. Its existence and quantitative importance belong to the completed transition-history calculation.

The expected particle number per mode is $\mathcal{N}_k = |\beta_k|^2$, where β_k are Bogoliubov coefficients [162]. An ECT-native computation of n_s and r from the condensate fluctuation spectrum during the ordering transition remains open (OP3, §14.6).

Discriminant with respect to standard vacuum programmes. ECT differs from the standard presentation of vacuum phenomena in one central respect: Casimir response, Unruh response, particle production, decoherence, and horizon thermality are compared as candidate probes of one ordered medium. This is an Open hypothesis, not a derived equivalence. The theory seeks a single condensate correlator infrastructure behind boundary response, detector response, nonstationary production, and open-system decoherence. If this unified correlator logic fails, the vacuum-response programme fails as a whole rather than piecemeal.

Falsifier for the coherent-vacuum response programme. The present vacuum-response programme would be seriously weakened if the different sectors ultimately required mutually unrelated vacuum structures: for example, if Casimir-type boundary response required one underlying vacuum, Unruh response another, particle production a third, and decoherence kernels a fourth. It would also be weakened if a completed coherent EFT showed that no single ordered-vacuum correlator can jointly support boundary sensitivity, accelerated-detector response, nonstationary mode mixing, and the horizon thermality route. Conversely, a successful cross-link would show only that one specified source-model kernel can reproduce the named effects. It would support ECT only after an independently derived ECT correlator and observable bridge are identified.

External mechanisms, not ECT evidence. The following empirical or standard-theory facts belong to their own source models: the success of standard QED Casimir measurements for electromagnetic boundaries, the observed dynamical Casimir effect in driven effective boundary systems [167], the standard consistency of Bogoliubov-type particle production in nonstationary backgrounds, and the broad analogue-gravity continuity between detector response, horizon thermality, and effective-vacuum physics. They establish neither a common correlator nor an ECT bridge and therefore are not confirmations, support, or consistency evidence for the ordered-vacuum interpretation.

25.6 Summary and status of coherent-vacuum response

Status: summary subsection for the coherent-vacuum programme of Chapter 25. Collects the Level A structural claims, the Level B conditional or closure-dependent results, and the open quantitative programmes. The chapter-level discriminant, falsifier, and empirical anchors are developed in Section 25.5.

Table 79: Coherent-vacuum response map of ECT. Level A = exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. Level B = structural/conditional derivation or closure route with a declared open matching, input, or identification step. Open = not yet derived; OP-Q reference given.

Vacuum-sector statement	Status	Basis / comment
Supplied coherent-vacuum model has non-trivial correlators	Level A within the declared model / Open ECT owner	Model-internal state and correlator statement; Section 25.1
Phase-sensitive vacuum response in the supplied coherent sector	Level A within the declared model / Open ECT owner	Requires BR1 and the separately supplied state and action; Section 25.1
Physical medium ownership of the vacuum-response programme (P5)	Level B/Open	P5 motivates relational medium response but does not derive a common physical vacuum, state or correlator owner; Section 25.1
Supplied soft-sector boundary-value response	Level A mathematics / B–Open physics	Each declared operator and boundary condition has a calculable mode response; the physical condensate boundary and its coupling to ordinary matter are not derived; Section 25.3
Supplied scalar Dirichlet coefficient	Level A mathematics / Open ECT	Eq. (25.3); physical ECT mode count, boundary coupling and stress response are missing
Time-dependent ordered backgrounds: background-dependent particle notion	A external / Open ECT	Standard statement after a time-dependent action and in/out states are supplied; Section 25.5
Bogoliubov-type mode mixing in the coherent vacuum	A external / Open ECT	Eq. (25.9); ECT action, states, particle map and rates remain open; Section 25.5
Candidate time-dependent mode speed $c_{\text{char}}(t)$	Open	Supplied function in the displayed oscillator target; its physical ECT owner and quantitative importance are not derived; Section 25.5
Quantitative production rates $ \beta_k ^2$	Open (OP-Q12)	Requires completed time-dependent coherent-vacuum calculation
Euclidean–Rindler/KMS route	Level B/Open	The standard response follows only after supplying the physical metric, state/KMS condition, detector coupling and clock/action matchings; Section 25.4

(continued on next page)

Vacuum-sector statement	Status	Basis / comment
Unruh temperature formula in the supplied route	Level A external / Open ECT	Eq. (25.8); requires $S_0 = \hbar$, a physical cone, and the missing detector/state owners; Section 25.4
Common vacuum-response programme (Casimir/production/Unruh/horizon)	Open	Section 25.2; a proposed common architecture, not a single physical response operator derived for all four regimes
Analogue-gravity continuity of the ECT vacuum interpretation	Motivation / Open ECT	Section 25.2
Unified coherent-vacuum response map	Open programme map	Fig. 22; Section 25.2
Horizon thermality as a coherent-vacuum response channel	A external / Open ECT	Standard near-horizon target; physical ECT tensor geometry, state and strong-field response are missing; Section 36
Phase-action-quantization structural chain	Level B	Sections 21.1–30.4; Fig. 21
Unified vacuum/coherent/geometry interpretation	Level B	Shared MP–QS condensate foundation

Summary. The coherent-vacuum programme of Chapter 25 now has a clear three-layer structure.

First, supplied reduced models define calculable correlators and stiffnesses. Physical boundary sensitivity, detector response and strong-field ownership remain Open.

Second, treating Casimir, Unruh, particle-production and horizon thermality as regimes of one common ECT operator is an Open organising hypothesis.

Third, the chapter has identified which parts of this architecture are already structural, which are conditional on specific closures, and which remain open quantitative programmes.

In this sense, Section 25.1 presents an Open map linking four supplied calculations. Section 26 separately studies declared environmental reductions; equality of its kernel with the boundary, detector or horizon kernels is not assumed.

26 Open Quantum Systems, Decoherence, and Irreversibility

Inputs: coherent amplitude structure, a supplied causal retarded response, and the ordered-vacuum sector. Status: Level A conditional mathematics for retarded support and for a nonnegative spectral/noise quadratic only after the corresponding positive Hilbert/state/observable, residue or oscillator representation is supplied; Level B for generic influence-functional bookkeeping in a specified Gaussian model; Open for the ECT system–environment split, physical record operators, state-dependent spectral/noise owner, entropy-production theorem, and unique outcome rule.

Physical picture: reduced-state records. Tracing unresolved degrees of freedom can suppress interference and stabilise pointer records. This is a statement about a reduced density operator, not selection of one ontologically realised branch. The unresolved sector may ultimately be part of the same condensate, but its coupling, state, and coarse-graining must be derived rather than inferred from that ontology.

What is established, what is closure-dependent, and what remains open in the decoherence programme. Established results are restricted to the declared source model: causal retarded support and a positive pairwise noise quadratic. The ECT split, vertices, state, entropy theorem, and outcome update remain Open.

Recoherence. Finite or engineered environments can erase or refocus records and partially restore off-diagonal terms. The amount of recovery is computed from the specified influence functional or channel. No universal reverse/forward probability follows from the pairwise decoherence exponent.

Connection to PES-R. The causal retarded kernel determines admissible modes and their pole structure; the positive noise kernel determines pairwise record distinguishability. PES-R keeps these roles separate. A one-history translation susceptibility can be a declared robustness diagnostic, but it is not the generic Feynman–Vernon mean equation. Figure 23 summarises this channel-specific regime split without assigning an outcome.

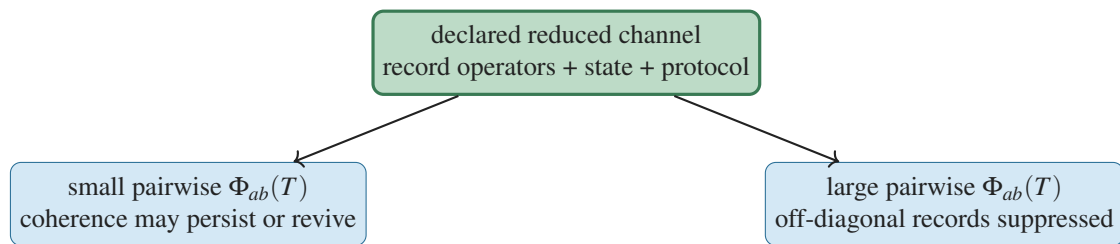


Figure 23: Conditional reduced-state regime split. The result is controlled by the physical record differences and noise kernel, not by the total number of environmental degrees of freedom alone. Neither branch changes metric signature or selects one realised outcome. Green marks the declared conditional channel and blue marks consequences inside it; labels and borders carry the same information in grayscale.

26.1 Influence functional, decoherence, and the arrow of time

Status: Level A conditional mathematics within a declared quadratic source model for causal retarded support and for spectral/noise positivity when a positive state/observable/residue or oscillator decomposition is supplied; Level B as generic influence-functional bookkeeping after a specified split; Open for the ECT split, vertices, state/spectral/noise owner, entropy theorem, and outcome update.

Position within the full argument. This programme follows the amplitude and causal-sector analysis but does not retroactively derive it. It classifies reduced records only after the physical modes and couplings are known.

Relation to ECT Foundations. The foundational ontology suggests that resolved and unresolved degrees of freedom may belong to one condensate. That statement does not identify the trace, state, record operator, or kernel required for a reduced dynamics.

Why decoherence belongs to the coherent branch

Decoherence is discussed in the coherent branch because it acts on interference between coherent alternatives. It suppresses selected off-diagonal records; it neither creates the branch nor chooses one actual outcome.

Discriminant with respect to standard decoherence and arrow-of-time programmes. The discriminating task is not to reproduce generic decoherence but to derive the ECT-specific record channel and a held-out deviation. An unspecified large environment is not an ECT prediction.

Medium reading of decoherence (P5). P5 motivates a medium interpretation of the unresolved sector. The quantitative content still comes from a declared interaction vertex, state, and reduced map.

Connection to the global/reduced unitarity distinction. Fine-grained unitary or reflection-positive reconstruction and reduced non-unitary evolution are compatible descriptions at different levels. Their compatibility does not furnish a unique-outcome rule.

Connection to the unified vacuum-response infrastructure. Vacuum-response calculations and reduced-state records may share physical modes, but this creates a double-counting obligation rather than an automatic identification of kernels.

Influence-functional form of the effective description

For a declared Gaussian split one must also specify the joint initial preparation and closed-time-path contour. After centring the unresolved one-point function, a compatible factorised preparation is sufficient for the usual bulk decomposition. Initial system–environment correlations instead require an explicit initial-boundary term S_{ic} , an equivalent extended-contour construction, or a proof that the extra contribution vanishes or has been absorbed consistently. The bulk influence functional then separates retarded response from a Hermitian noise quadratic. Their normalisations are fixed by Eqs. (29.8)–(29.10); they are not interchangeable.

Status. The bulk quadratic influence-functional algebra is Level A in the stated source model and Level B as generic bookkeeping, conditional on the declared initial preparation and contour. The correlated initial-boundary term and ownership of the complete functional by the bare ECT action remain Open.

Causal retarded kernel: the Level A result

The retarded kernel satisfies $D^R(\tau < 0) = 0$ in the owning hyperbolic construction. This is a causality statement, not an entropy sign or outcome-selection theorem.

Physical meaning. Retarded response fixes how a perturbation propagates, whereas the noise kernel fixes pairwise record loss. Even in equilibrium their relation requires the state and fluctuation–dissipation assumptions.

Beyond the quadratic sector. Beyond a Gaussian truncation, higher cumulants and non-linear vertices must be retained. Positivity of the quadratic term alone does not control the full non-Gaussian influence functional.

Low-entropy coherent initial sector

A coherent initial sector is not automatically a low-entropy sector. The initial density operator and coarse-graining must be supplied; P4 supplies the orientation datum, not a thermodynamic boundary condition.

Structure of the initial sector. The initial-sector structure must be stated through its state, constraints, and physical modes. Neither one realised orientation nor high coherence determines its statistical entropy.

Path-integral weighting and the initial coherent sector

A supplied Euclidean weight $e^{-\mathcal{J}_E^\mu}$ defines a measure only after its domain, boundary data and normalisation are supplied; for P3 the exponent is $\kappa_{P3}\mathcal{J}_E^{P3}$. It is neither fixed by S_0 nor a Boltzmann probability distribution for macroscopic entropy.

Status. The path-integral expression is retained as amplitude bookkeeping. Its conversion into a physical initial ensemble remains an Open reconstruction/state problem.

Conditional spectral positivity and dissipative kernel

A nonnegative spectral measure supports a positive pairwise noise quadratic only after its physical Hilbert space or oscillator decomposition, state, observable, normalisation and residues have been supplied. Causal dispersion relations additionally require the retarded analyticity, Fourier and subtraction conventions. A classical positive scalar Hamiltonian and retarded support do not derive those spectral owners. The conditional spectral algebra is Level A under the stated premises; its ECT identification is Open, and it implies neither Markovianity nor entropy monotonicity.

Effective ohmic regime and Markov approximation

An ohmic spectrum and short memory are model assumptions useful for a Markov approximation. The corresponding coefficients must be derived for each physical channel; they are not universal condensate constants.

Qualitative monotonicity of entropy

Finite environments can exhibit recoherence, and even Markovian channels need not increase von Neumann entropy unless the required channel property is stated.

Why this is enough. Causal support plus positive noise is enough for a consistent quadratic record model, not for a thermodynamic theorem. The stronger claim requires the missing reduced-map hypotheses.

Status. Accordingly, the entropy-growth programme remains Open for ECT. Only conditional channel theorems may be used before the physical reduction is derived.

Working explicit entropy formula

The applicable explicit relation is the contractive relative-entropy inequality for a named channel and stationary reference state; no universal scalar rate is retained.

Role of the explicit formula. The conditional formula is a diagnostic of an already specified channel, not a derivation of that channel from the condensate or a numerical ECT prediction.

Macroscopic irreversibility versus quantum exception

Macroscopic irreversibility and quantum recoherence can coexist because different environments and timescales define different reduced channels. No sharp universal boundary follows from an order-unity scalar.

Numerical illustration. Any numerical illustration of a Markov source model is labelled model-internal. It cannot be promoted to an ECT lifetime or entropy rate without the ECT vertex and state.

Interpretive consequence. The interpretation is therefore conditional: large pairwise record exponents suppress selected coherences, while actual outcomes and thermodynamic irreversibility remain separate questions.

Status. The quantitative example is retained only as a source-model illustration. It is not an experimental verification of an ECT kernel.

Bridge to probability and Born-type interpretation

Decoherence can make a reduced density matrix approximately diagonal but does not derive Born weights or select one realised outcome. Ordinary projector Gleason represents only an independently supplied normalised, noncontextual, countably additive projector measure on a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. It does not derive the physical state, preparation, event, probability origin, outcome or update.

Falsifier for the decoherence and thermodynamic-arrow programme. Falsification requires a derived ECT record operator and kernel: if their predicted channel conflicts with held-out interference or recoherence data, that channel fails. Generic agreement with standard decoherence is not discriminating.

External source-model anchors only. C_{60} , cavity, qubit, and photonic experiments are empirical anchors for their source models. Reproducing those models checks arithmetic and conventions, not ECT-specific ownership.

Conditional cross-protocol consistency test for a future ECT record channel. The conditional consistency requirement is this: once ECT fixes a vertex and state, the same kernel must account for multiple protocols without channel-by-channel retuning. Until then the statement is a test plan, not a prediction.

Connection to the structural universalities. Causal response, compact phase, and reduced records are distinct structural ingredients. Their coexistence does not license a universal selection functional.

Summary. The decoherence architecture is thus a reproducible framework for named channels, with ECT-specific vertices and states still Open. Outcome, Born, and entropy claims are not included in its established core.

Medium reading (P5). P5 can motivate that apparatus and environment are excitations of one medium, but it does not eliminate the need for a physical tensor product or algebraic subsystem split.

Relation to the general decoherence programme. The general programme retains finite-time recoherence, null spaces, and state dependence. A universal monotone scalar is neither required nor derived. For two alternatives a, b , a declared Gaussian environment, and a named joint initial preparation, a nonzero normalised scalar influence functional, when it exists and includes all initial-boundary terms, has the branch-free exact polar decomposition

$$\mathcal{F}[a, b] = U_{ab}^{\text{full}} \exp\{-\Xi_{ab}^{\text{full}}\}, \quad U_{ab}^{\text{full}} := \frac{\mathcal{F}[a, b]}{|\mathcal{F}[a, b]|} \in U(1), \quad \Xi_{ab}^{\text{full}} := -\ln |\mathcal{F}[a, b]|. \quad (26.1)$$

On a domain where U_{ab}^{full} admits a chosen continuous real phase lift one may write $U_{ab}^{\text{full}} = \exp(i\Theta_{\text{IF}}^{\text{full}})$, with the dimensionless $\Theta_{\text{IF}}^{\text{full}}$ defined modulo 2π ; no global lift is implied. If a separate reduction derives an action-valued real influence functional, one may then define $S_{\text{IF}}^R := S_0 \Theta_{\text{IF}}^{\text{full}}$ modulo $2\pi S_0$. That additional action owner and the cross-sector normalisation of S_0 are not assumed by the polar decomposition. This full quantity is a nonnegative suppression exponent only when the declared reduced construction also

establishes $|\mathcal{F}| \leq 1$; only then do we set $\Phi_{ab} := \Xi_{ab}^{\text{full}} \geq 0$. For a compatible factorised preparation the displayed exponent can reduce to the nonnegative bulk Gaussian noise quadratic of Result A5. For a correlated preparation it may also contain initial-boundary contributions and cannot be reconstructed, assigned a sign, or promoted to Φ_{ab} from the bulk kernel N alone. The causal response kernel satisfies

$$D^R(\tau < 0) = 0, \quad (26.2)$$

under the owinging hyperbolic/retarded construction. Positivity of the quadratic spectral measure,

$$\rho(\mu^2) \geq 0, \quad (26.3)$$

supports a positive noise quadratic; it does not imply entropy monotonicity.

The Euclidean weight

$$Z_E = \int \mathcal{D}\Phi e^{-\mathcal{J}_E^\mu[\Phi]}, \quad \mathcal{J}_E^\mu = \kappa_{P3} \mathcal{J}_E^{P3} \quad \text{only for the supplied bare P3 measure,} \quad (26.4)$$

does not by itself prove that the realised ordered branch is a low-entropy initial ensemble. That requires a measure, coarse graining, and state selection and remains Open.

For a declared three-dimensional derivative-coupling toy, the bare mode-density weight and the Weiss spectral density are related by

$$J_{\text{bare}}(\omega) \propto N_{\text{eff}} \omega^2 |g(\omega/c_{\text{char}})|^2 \sim N_{\text{eff}} \omega^4, \quad J_W(\omega) \equiv \frac{J_{\text{bare}}(\omega)}{2\omega} \sim N_{\text{eff}} \omega^3, \quad (26.5)$$

whereas the effective ansatz

$$J_W(\omega) \simeq N_{\text{eff}} \gamma \omega \quad (26.6)$$

and the short-memory condition

$$\tau_{\text{env}} \ll \tau_{\text{dyn}} \quad (26.7)$$

are model inputs, not consequences of the minimal ECT bath. Even together they do not fix the sign of von Neumann entropy. For a declared contractive channel with stationary state ρ_* ,

$$\frac{d}{d\omega} D(\rho_w \| \rho_*) \leq 0, \quad (26.8)$$

while for a general derived generator $\mathcal{L}_w$ one only has

$$\frac{dS_{\text{vN}}}{d\omega} = -\text{Tr}[\mathcal{L}_w(\rho_w) \ln \rho_w], \quad (26.9)$$

whose sign must be proved from the channel and state.

In a specified open-system model, macroscopic irreversibility can emerge when uncontrolled records make recoherence operationally inaccessible, while echo/control protocols show that the statement is finite-time and channel-dependent. A declared dephasing channel may establish reduced-state record formation inside that model; it does not give a unique outcome, Born weights, a metric-signature transition, or a Crooks relation. The physical ECT record operators, state, kernels, trace, entropy functional, and update rule remain OP-Q5/OP-Q17/OP-Q19 inputs.

26.2 Decoherence timescale in supplied external source models

Status: standard source-model definition for declared reduced channels; Open for an ECT-specific rate. A decoherence time is computable only after the physical alternatives, record operator, environmental state, noise kernel, protocol, and full-channel contractivity have fixed a nonnegative $\Phi_{ab}(T)$. The ECT resolved/unresolved split and those kernels have not been derived from the condensate action. Consequently there is no general ECT $N_{\text{eff}}\gamma$ timescale formula, and the numerical applications below must retain the status of their owinging external source models.

General record-channel definition

For two declared alternatives a, b , a decoherence time can be defined only after their nonnegative pairwise record exponent has been calculated from a physical record operator and environmental state and the required full-channel contractivity has been established:

$$\tau_{ab} := \inf\{T > 0 : \Phi_{ab}(T) \geq 1\}. \quad (26.10)$$

The threshold is a convention for order-unity visibility loss, not a universal quantum–classical boundary. There is no general ECT identity $\tau = (N_{\text{eff}}\gamma)^{-1}(\ell/\Delta q)^2$.

For dilute-gas collisional decoherence, the source-model collision rate has the dimensionally consistent form

$$\gamma_{\text{coll}} = n_{\text{gas}} \langle v_{\text{rel}} \sigma_{\text{eff}}(v_{\text{rel}}) \rangle, \quad (26.11)$$

with the actual decoherence function additionally depending on momentum transfer, path separation, molecular form factors, and the observation protocol. This is standard environmental-decoherence input, not an ECT kernel derived from the condensate action.

C₆₀ matter-wave interferometry: source-model status

The Arndt and Hornberger results [169, 170] establish matter-wave interference and its collisional degradation. The expression $k_B T \sigma / (S_0 v)$ has dimensions of length, not inverse time, and therefore cannot define an ECT decoherence rate. Reproducing the source model requires the pressure-dependent gas distribution, differential cross section, momentum-transfer kernel, path geometry, and flight-time protocol. No distinct ECT prediction is currently available for C₆₀.

Table 80: Current C₆₀ external source anchor. A quantitative prediction requires the complete apparatus inputs listed in the text.

Experimental system	Observation	ECT status
C ₆₀ interferometer [169]	Interference visible	External observation; exact pressure owner not identified; no ECT discriminator

Extension to massive-particle interferometry

Fein et al. 2019 and Pedalino et al. 2026 [171, 172].

Fein et al. [171] established the historical macromolecular interference anchor above 25 000 Da. Pedalino et al. demonstrated a clear quantum/classical separation around 172 kDa for sodium nanoparticles with more than 7 000 atoms [172]; their centre of mass was delocalised by more than an order of magnitude beyond the particle diameter. Higher-mass fringe records in the same apparatus, extending into the reported roughly 0.4–1 MDa tail, do not by themselves carry the same discriminating inference and are not promoted here to that status. The platform is described by standard quantum mechanics and constrains parameterised macrealistic modifications; it does not by itself identify an ECT record or decoherence channel.

The observed coherence in both platforms must be compared with standard collisional and thermal source models in which gas distribution, scattering, transit time, material response and apparatus geometry matter. No ECT-specific rate is evaluated here. Inserting apparatus parameters into an external collapse functional would be a separate source-model calculation, not an ECT gravitational regime, because the corresponding ECT vertex and noise kernel are absent. The larger interferometric frontier supplies a future chassis for a derived mass-, separation- and time-dependent ECT residual, not a presently predicted condensate rate [173, 174].

Comparison with standard quantum theory. At the Fein and Pedalino scales, standard quantum and environmental- decoherence models provide the operational comparison. The present manuscript retains these experiments as external source anchors; it neither reproduces their full apparatus likelihoods nor derives a distinct environmental or gravitational ECT rate. The Pedalino platform would become an ECT discriminator only after one action-owned residual kernel predicts, before the decisive data, a joint dependence on mass, delocalisation, duration, material and environment together with an independent noise/heating observable.

Origin of the mass-scale estimate. Any maximum-mass estimate for visible interference in this regime follows from the environmental condition $\tau_{\text{dec}} \gtrsim \tau_{\text{pass}}$ and is not a uniquely ECT-derived law. The Fein molecules remain a historical 25 kDa benchmark, while the Pedalino sodium nanoparticles establish clear quantum/classical separation around 172 kDa; the higher-mass fringe tail is not assigned the same inferential status. Any comparison with a mesoscopic gravity-related channel must use an explicitly declared imported benchmark or a future derived ECT spectrum, state and normalisation.

Case 2: Gravity-induced decoherence

Diósi 1987 [175]; Penrose 1996 [176].

Diósi and Penrose propose that quantum superpositions of massive objects collapse with a timescale $\tau_{\text{DP}} = \hbar/E_G$, where E_G is the self-gravitational energy of the difference between the two mass distributions ρ_1, ρ_2 of the superposed branches:

$$E_G = \frac{G_N}{2} \iint \frac{[\rho_1(\mathbf{r}) - \rho_2(\mathbf{r})][\rho_1(\mathbf{r}') - \rho_2(\mathbf{r}')] }{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r'. \quad (26.12)$$

The corresponding external source-model timescale is

$$\tau_{\text{DP}}^{\text{ext}} = \frac{\hbar}{E_G}. \quad (26.13)$$

Evaluating it requires a separately specified mass-density profile, regularisation and geometry. One narrow external orientation point can be evaluated without identifying an ECT channel. For two identical sharp, homogeneous continuum spheres of radius R and mass m , with d the centre-to-centre branch separation, $x = d/R$, the factor-1/2 convention in Eq. (26.12) and the explicitly adopted map $\tau_{\text{DP}}^{\text{ext}} = \hbar/E_G$ give

$$C_{\text{DP}}(x) := \frac{E_G}{G_N m^2 / R} = \frac{x^2}{2} - \frac{3x^3}{16} + \frac{x^5}{160}, \quad 0 \leq x \leq 2. \quad (26.14)$$

At the explicitly supplied nominal point $x = 1$, $R = 1 \mu\text{m}$, and $\rho = 2000 \text{ kg m}^{-3}$, so that $m = 4\pi\rho R^3/3 = 8.3776 \times 10^{-15} \text{ kg}$,

$$C_{\text{DP}}(1) = \frac{51}{160}, \quad E_G = 1.4931 \times 10^{-33} \text{ J}, \quad \tau_{\text{DP}}^{\text{ext}} = 70.6 \text{ ms}. \quad (26.15)$$

This is Level-A geometry/arithmetic inside the imported source model and a Level-C external benchmark, not a data test, ECT collapse time, prediction or uncertainty statement. For $0 < x \leq 2$, at fixed x it scales as $\tau \propto G_N^{-1} \rho^{-2} R^{-5} C_{\text{DP}}(x)^{-1}$; at $x = 0$ the spheres coincide, $E_G = 0$, and τ has only the $+\infty$ limiting interpretation. Profiles, microscopic smearing, separation, regularisation and normalisation conventions change it. The previous erroneous time payload and old numerical tables remain withdrawn. This new scoped point does not replace the separately supplied unit-coefficient guide and collisional-comparison laws below: inserting $C_{\text{DP}}(1)$ into those laws would change their coefficients and would define a different comparator.

More importantly, no ECT-specific source projector, vertex, Hadamard/noise kernel, state, finite-body charge or absolute normalisation has been shown to reproduce this functional. A positive or null result constrains only the fully declared external model unless that missing ECT owner chain is first supplied.

Status and closure test. The common influence-functional notation is a useful organisational language, but it does not show that environmental and gravitational channels are bath sectors of one identified ECT medium. Closing the ECT claim requires a physical source projector, canonical residue, state-dependent noise kernel, resolved/unresolved split, subtraction rule, and finite-window protocol.

Status summary. *Source model:* standard environmental-decoherence anchors; the cited apparatus rates were not independently reproduced here. *External comparison:* the symbolic Diósi–Penrose functional above. *Open:* distinct environmental and gravitational ECT record channels, their rates, corrections, and discriminants.

26.3 Diósi–Penrose benchmark and ECT gravitational record-channel status

Status: the Diósi–Penrose mass-density functional and $\tau_{\text{DP}}^{\text{ext}} = \hbar/E_G$ are an external symbolic comparison. The present ECT gravitational record channel is *Open*: its response vertex is at most parametric, while its state-dependent noise kernel and absolute rate are not identifiable from the displayed action.

Response is not a noise derivation

For a canonically normalised scalar ϕ , the most that follows from the current scalar–tensor ansatz is the parametric retarded response

$$\Delta\phi_{\text{resp}}(x) = g_{\phi T} \int d^4x' D_R(x, x') \Delta T(x'), \quad g_{\phi T} = -\frac{\beta_\phi}{2\bar{M}_{\text{Pl}} \sqrt{\omega_0 + 3\beta_\phi^2/2}}. \quad (26.16)$$

This is a trace response (PARAMETRIC ONLY), not a derivation of a Hadamard kernel or of branch distinguishability. It vanishes at tree level for a classically trace-free source in the minimal model, and the charge of a screened extended body is not yet identifiable.

For comparison only, the external Diósi–Penrose functional is Eqs. (26.12)–(26.13); Eqs. (26.14)–(26.15) add one declared geometry-resolved orientation point. No equality between either external object and an ECT decoherence rate is claimed. The separately retained unit-coefficient guide/crossover relations remain distinct approximations. The minimal free-vacuum scalar completion instead gives a translated equal-mass spectrum $J_\phi \propto \omega^3$ and a saturating finite-time exponent, so it is incompatible with a universal D–P linear-time law. Reusing the same scalar exchange both in the fitted gravitational response and as an unsubtracted bath contribution would be double counting.

Current bounded record-channel ledger. The authoritative ten-row terminal disposition for the presently named and owned candidate corpus is collected in Section 33.5, Table 91.

Bounded closure of the enumerated search. That authoritative table is terminal for the presently named and owned candidate universe: every listed route has a negative, parametric or non-identifiability disposition, and none supplies a derived physical record channel. This is a bounded classification result, not a no-go theorem for all future ECT completions. A new stationary action, nonlinear/tensor field content or explicit matter/record vertex re-enters the owner gates as a new candidate with a new scope. OP-grad is therefore neither a universal canonical “Gate 0” nor identical to a record-channel certificate: some negative terminal results require no positive OP-grad owner, whereas every positive rate calculation still requires its own action–projector–vertex– state/noise–subtraction chain. PES-R remains a Level-B organising taxonomy; physical/global PES remains Open.

Response–noise firewall. A static susceptibility, HRC compliance or fitted mean gravitational response does not determine the retarded bath kernel, the Hadamard/noise kernel or the environmental state. The same mode may not be counted simultaneously in the mean action and the unresolved bath without an explicit split, subtraction and counterterm ledger.

D–P-filter guard. Existing non-interferometric bounds on gravity-related collapse [177, 178, 179] constrain specific stochastic, Markovian, white-noise Lindblad implementations. They are not automatic passes or failures of a future ECT channel. Any proposed ECT $J_{\text{grav}}(\omega, k)$ must declare its spectrum, constituent-scale density resolution, state, passivity/energy balance, and finite-window protocol before radiation, force-noise, heating, or visibility bounds are compared.

26.4 Protocol-defined entropy production: open ECT construction

Medium reading (P5). The medium reading is maintained only after canonical normalisation and constraint reduction. Gauge or null directions must not be counted as physical environmental records.

Effective entropy production in the Gaussian–Markov reduction. A Gaussian–Markov reduction may possess an entropy-production functional after local detailed balance is imposed. That functional is not the pairwise dephasing exponent and is not presently derived for ECT.

Why w is traversed in one direction macroscopically. The macroscopic direction assigned to w follows from the ordered causal branch. Why a cosmological state has a thermodynamic arrow is a separate initial-state and coarse-graining problem.

Status: Open for ECT. A causal response and positive pairwise noise kernel are structural ingredients, but neither a Crooks theorem nor monotone von Neumann entropy follows from a decoherence exponent.

For a declared coarse graining, the reduced state is defined by a dynamical map

$$\rho_{\text{red}}(w) = \Lambda_{w,0}[\rho_{\text{red}}(0)], \quad (26.17)$$

and its von Neumann entropy is

$$S_{\text{ent}}(w) = -\text{Tr}[\rho_{\text{red}}(w) \ln \rho_{\text{red}}(w)]. \quad (26.18)$$

No identity $S_{\text{ent}} = \text{const} + \Gamma$ is assumed. Under a declared contractive channel with stationary state ρ_* one may instead use

$$\frac{d}{dw} D(\rho_{\text{red}}(w) \| \rho_*) \leq 0, \quad (26.19)$$

with the hypotheses stated explicitly.

For a forward protocol F , reversed protocol R , measurable time-reversal involution Θ , and normalised path measures, first transport the reverse measure to the common forward path space: $Q = P_R^\Theta$, $Q(A) = P_R(\Theta A)$. If $P_F \ll Q$, stochastic entropy production is defined P_F -almost surely by

$$\Sigma[\gamma] := \ln \left(\frac{dP_F}{dQ}(\gamma) \right). \quad (26.20)$$

The reciprocal relation $dQ/dP_F = e^{-\Sigma}$ and the standard identity $\langle e^{-\Sigma} \rangle_F = 1$ require the additional condition $Q \ll P_F$, hence $P_F \sim Q$. If a singular reverse component is present, the applicable statement contains the corresponding absolute-irreversibility correction. A Crooks-type distributional relation requires, in addition, microscopic reversibility or local detailed balance and the appropriate initial ensembles [180]; none of these physical inputs follows from the Radon–Nikodym definition. The pairwise record exponent Φ_{ab} is not Σ . Constructing these objects from the ordered-condensate coarse graining remains OP-Q17.

Discriminant: ECT Crooks-type relation versus standard fluctuation theorems. A Crooks relation is stronger than the definition $\Sigma = \ln[dP_F/dP_R^\Theta]$: it requires named protocols, reversed dynamics, initial ensembles, and microscopic reversibility or local detailed balance.

Falsifier for the Crooks programme. The Crooks programme is falsifiable only after those measures are derived. A protocol-free reverse/forward ratio is excluded from the theory.

Bridge to the probability and measurement programme. Probability and measurement are not closed by a fluctuation relation. Born weights, record formation, and outcome update retain separate owners and statuses.

27 Source-model anchors, operational proxies, and external benchmarks for a future ECT record channel

This section collects quantitative source-model estimates, order-of-magnitude consistency checks, operational proxies, and external benchmarks relevant to the PES-R programme (Appendix AX). They are not outputs of an action-owned ECT record channel: the ECT operator, vertex, environment split, state, and response/noise kernels remain Open.

Visibility-based dephasing proxies. In a specified pure-dephasing model, interferometric visibility can be written $V = e^{-\Phi}$, so the reported visibility defines the operational proxy $\Phi^{(\text{exp})} = -\ln V$. This does not determine an ECT noise kernel and does not define a reverse/forward path-probability ratio.

Anchor / marker	Observable or model	Φ proxy	Status
Procopio quantum switch (2015)	$V = 0.994 \pm 0.002$	0.006 ± 0.002	Visibility-derived
Jacques delayed choice (closed configuration)	measured $V \approx 0.94$ [154]	$-\ln V \approx 0.0619$	Manuscript transform; ECT kernel Open
Werner-visibility CHSH toy	$2\sqrt{2}e^{-\Phi} = 2$	$\ln \sqrt{2}$	Model threshold
Qubit information–dephasing toy	one-bit marker	model-dependent	Illustrative

These values are source-model or visibility proxies. They are not first-principles ECT calculations and do not provide an ECT-specific discriminator.

How to read the coherence–dephasing crossover. For a declared pair of alternatives a, b in the pure-dephasing closure, the operational coherence-retention ratio is $V_{ab}/V_0 = \exp[-\Phi_{ab}]$. Figure 24 therefore distinguishes the asymptotic regimes $\Phi_{ab} \ll 1$, $\Phi_{ab} \sim 1$, and $\Phi_{ab} \gg 1$ and makes the continuous transition between them visible. This is a protocol-dependent statement about one declared pairwise record exponent. It is not a universal quantum–classical boundary, a collapse or outcome-selection criterion, or an ECT-specific prediction until the physical record channel, environmental state, and noise/kernel owners are supplied.

27.1 Environmental decoherence: C_{60} and macromolecules

Status: standard collisional-decoherence source-model anchor; Open for a distinct ECT rate. The condensate action has not supplied the physical record operator, environmental state, differential cross section, or noise kernel needed for an ECT prediction.

For dilute residual gas, the source-model collision rate is

$$\gamma_{\text{coll}} = n_{\text{gas}} \langle v_{\text{rel}} \sigma_{\text{eff}}(v_{\text{rel}}) \rangle, \quad (27.1)$$

with visibility additionally controlled by the momentum-transfer distribution, path separation, molecular form factor, flight time, and apparatus protocol [169, 170]. A white-noise amplitude D_{eff} may parameterise a declared Markov source model, but its use does not turn that model into a rate derived from ECT.

Protocol-dependent coherence-retention regimes

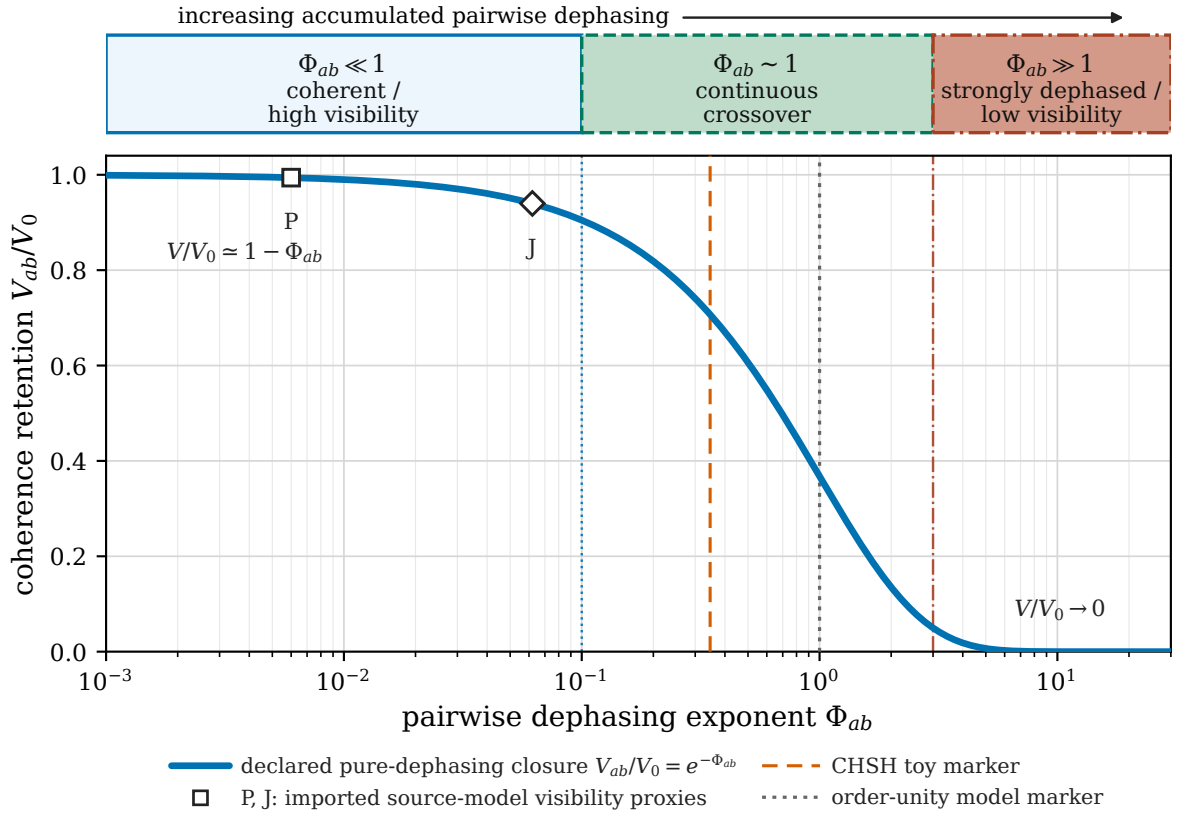


Figure 24: Protocol-dependent coherence retention in the declared pure-dephasing closure. The solid blue curve is $V_{ab}/V_0 = \exp(-\Phi_{ab})$. The three directly labelled bands show the asymptotic readings $\Phi_{ab} \ll 1$ (coherent/high visibility), $\Phi_{ab} \sim 1$ (continuous crossover), and $\Phi_{ab} \gg 1$ (strongly dephased/low visibility). The guide edges at 10^{-1} and 3 organise the picture only; they are not physical thresholds. The open black markers retain the imported source-model visibility proxies and the vertical lines retain the declared toy/model markers. Nothing in this figure defines a universal quantum–classical boundary, collapse or outcome selection, or an ECT-specific transition scale without the missing channel/state/kernel owners.

Table 82: C₆₀ and macromolecular environmental-decoherence status.

Experiment	Observable	Owning result	ECT status	Assessment
C ₆₀ interferometer [169]	Interference visible	Observed/source-model regime	No distinct ECT rate	Source anchor
Fein 2019 (25 000 amu)	Interference visible	Environmental source-model regime	No distinct ECT rate	Source anchor
Pedalino 2026 (~ 172 kDa)	Quantum–classical test	Standard-QM source model	No distinct ECT rate	Separation scale; tail qualified

Comparison with observations. The observed C₆₀ interference is an external experimental anchor [169]. Its environmental conditions must be quoted only from an identified primary owner; the previously repeated 10^{−6} Pa sub-number is therefore not retained. Agreement with standard collisional decoherence is not an ECT-specific discriminator.

27.2 Diósi–Penrose benchmark; ECT record channel Open

Status: external functional plus one declared hard-sphere orientation point. The geometry algebra is exact inside the supplied source model and the number is Level C externally; ECT has not derived the required physical vertex, state-dependent noise kernel, or absolute normalisation.

The primitive external D–P input is $\tau_{\text{DP}}^{\text{ext}} = \hbar/E_G$, with E_G defined by the mass-density functional in Eq. (26.12). The idealised point $C_{\text{DP}}(1) = 51/160$, $\tau_{\text{DP}}^{\text{ext}} = 70.6 \text{ ms}$ is fixed by Eqs. (26.14)–(26.15); it is neither an uncertainty statement nor a prediction that ECT decoherence persists in vacuum. A future ECT-specific test requires a derived kernel and a held-out feature not shared by the external functional.

27.3 External collisional–Diósi–Penrose equality scale

Status: exact algebra inside the explicitly supplied hard-sphere and unit-coefficient external comparison; Level C as a source-model benchmark; Open as an ECT result.

Two different mass constructions must not be conflated. Inverting an external D–P time at fixed radius defines the guide mass $m_{\text{guide}}^{\text{DP}}$ in Section 32.5; it is not a collisional crossover. By contrast, take a homogeneous sphere $m = 4\pi\rho R^3/3$, the declared hard-sphere collision rate $\lambda_{\text{coll}}^{\text{hs}} = n_{\text{gas}}v_{\text{th}}\pi R^2$, and the separately supplied unit-coefficient external comparison $\lambda_{\text{DP}}^{\text{ext}} = G_N m^2/(\hbar R)$. Equality of those two imported rates gives

$$R_{\text{cross}}^{\text{coll-DP}} \sim \left(\frac{9\hbar n_{\text{gas}} v_{\text{th}}}{16\pi G_N \rho^2} \right)^{1/3}, \quad m_{\text{cross}}^{\text{coll-DP}} \sim \frac{3\hbar n_{\text{gas}} v_{\text{th}}}{4G_N \rho}. \quad (27.2)$$

This follows algebraically and dimensionally from the displayed approximate rate laws. It is not universal: a different mass-density geometry, branch separation or D–P regularisation can change the gravitational coefficient and, outside the displayed unit-coefficient regime, the functional dependence; a differential cross section, molecular form factor or momentum-transfer kernel can likewise change the collisional rate law. No numerical value is published because the apparatus inputs are not fixed. Even after they are fixed, Eq. (27.2) compares two external source models; it is not an ECT crossover or a claim that an ECT channel dominates or persists in vacuum without the missing vertex, state and noise normalisation.

27.4 Photonic visibility bounds

Status: source-model/visibility consistency only. For a specified pure-dephasing description, a measured visibility defines $\Phi^{(\text{exp})} = -\ln V$. The Procopio value is therefore a small visibility-loss proxy, not a

microscopic ECT noise calculation. Delayed choice, eraser, swapping, and quantum-switch experiments retain their standard operational predictions. No inference that free photons inhabit a different metric-signature regime, or that detection is a literal signature transition, follows from these visibilities.

27.5 Cavity-cat decoherence: next quantitative target

Status: standard cavity-QED source-model target / ECT Open. The calculation below uses a declared Gaussian–Markov environmental model with parameters matched to the apparatus. No distinct environmental or gravitational ECT rate is derived; the physical vertices, state, noise kernel, and absolute normalisation remain Open.

The Haroche–Raimond cavity-QED experiments [181] directly measure decoherence of mesoscopic microwave-field cat states. With D_{eff} matched to the measured cavity lifetime, the declared Markov closure reproduces the standard cavity-QED scaling. This is a source-model consistency check, not evidence that a condensate-gravitational term is negligible. A distinct ECT contribution cannot be calculated until its vertex/state/noise chain is supplied.

27.6 Summary of anchors, proxies, benchmarks, and Open targets

Table 83 collects the quantitative estimates, source-model transforms, imported benchmarks, and Open targets relevant to a future ECT record channel.

Table 83: Summary of source-model estimates, operational proxies, imported benchmarks, and Open targets relevant to a future ECT record channel. No row is an ECT-specific held-out prediction.

System	Estimate / benchmark	Observation	Status
C_{60} [169]	Observed interference	Interference ✓	External source anchor; pressure sub-number unowned; ECT rate Open
Fein 25 000 amu	Standard environmental regime	Interference ✓	External source anchor; ECT rate Open
Pedalino ~ 172 kDa	Standard-QM/source-model regime	Quantum/classical separation	Demonstrated separation scale; higher-mass fringe tail not assigned the same status; ECT rate Open
Procopio switch	$V = 0.994 \pm 0.002 \Rightarrow \Phi_{\text{vis}} = 0.006 \pm 0.002$	ICO visible	Source visibility transform; ECT kernel Open
Diósi–Penrose functional	$\tau_{\text{DP}}^{\text{ext}} = \hbar/E_G$; $C_{\text{DP}}(1) = 51/160$, 70.6 ms at the declared sharp-sphere point	External formula plus scoped Level-C orientation point	ECT source projector, vertex, state, noise kernel and normalisation Open; not a data test or ECT rate
Environmental–D–P comparison	Eq. (27.2) under the stated hard-sphere/unit-coefficient assumptions	External conditional algebra; no numerical value	Not an ECT crossover; ECT channel Open
Cavity cat	Open target within present closure	Measured decoherence times (Brune 1996)	Programme-level target

28 ECT reduced-state programme for the hybrid-consistency problem

Status: structural framing for the gravitational, measurement, and black-hole sectors of Part III. No new derivations; this section fixes the two-level ontology invoked in Sections 26.3, 26.4, 30.2, 36, and the conditional named-completion tests of Section 38.4. The comparison with theories of Lorentzian emergence (Einstein–aether, Hořava–Lifshitz, CDT, Hartle–Hawking) given in Section 3.10 compares ECT along the signature-emergence axis; the present section instead compares architectures for the gravity–matter interface. The two comparison axes are logically distinct.

28.1 The hybrid-consistency problem

Reconciling a quantum description of matter with a classical or emergent description of the gravitational sector produces a characteristic set of structural obstructions known collectively as the hybrid-consistency problem. Any non-metric-fundamental approach to the gravity–matter interface must confront at least five demands: positivity of reduced states; operational no-signalling; controlled energy–momentum bookkeeping; relativistic compatibility; and gauge or diffeomorphism consistency of the emergent mediator. Historical semi-classical proposals of Møller and Rosenfeld [182, 183] already exhibited the relevant tensions. These were later sharpened by the Page–Geilker experiment [184] and by a broader literature of thought experiments and consistency critiques, including the influential but debated analysis of Eppley and Hannah [185]. Low-energy effective-field-theory treatments [186, 187] remain internally consistent in their domain, but do not by themselves resolve the ultraviolet side or the hybrid measurement problem. The post-quantum classical gravity programme of Oppenheim [188, 189, 190, 191] addresses the problem by postulating fundamentally stochastic dynamics for both metric and matter, at the cost of irreducible irreversibility and fundamental information loss.

The architectural claim of ECT is not that the five consistency demands disappear, but that they are displaced from the fundamental law to the reduction map between the full condensate theory and its effective descriptions.

28.2 Central thesis

Conditional programme. ECT asks whether irreversibility, decoherence, effective thermal-ity and apparent stochasticity can be realised as reduced descriptions of a future globally unitary Euclidean-condensate completion. This is a research programme, not a resolution: global unitarity, the physical reduction map, the gravitational mediator/channel and their joint consistency remain Open.

The conditional thesis has three components. First, global unitarity must be established at the level of the full theory. Second, every apparent cost of hybrid consistency enters only through the reduction to effective descriptions. Third, the reduction is not a collapse postulate and not an added stochastic law; it is the choice of which condensate degrees of freedom are tracked explicitly and which are integrated out.

Two qualifications are important at the outset. *First*, ECT does not identify “non-fundamental metric” with “non-quantum gravity”; what is denied is the need to quantise an independent metric field as the fundamental starting point. *Second*, in ECT classicality is not a separate ontological sector but a regime of reduced description.

28.3 Full and reduced descriptions: status guard

The full condensate description and a chosen reduced description must be kept distinct. Standard global unitary dynamics is established only in supplied subsectors for which the complete applicable corrected OS-II package and reconstruction are in place; reflection positivity alone is insufficient, and this is not yet a theorem of the full interacting ECT functional. A reduced channel may preserve coherence or form records depending on its physical vertices, state, and protocol. A small or large dephasing proxy does not define an ontological phase, quantise gravity, or decide whether a mediator can entangle. The

useful content of the Level IIa/IIb language is therefore operational and conditional, not a universal Γ_{loop} partition.

Level I — full condensate description. Level I is the full constrained condensate description before a resolved/unresolved split. Its physical observable algebra and state are the inputs from which any reduction must be derived.

Level IIa — reduced coherent-effective description. Level IIa is a coherent effective description after controlled mode reduction. Interference remains in the retained sector and no classical outcome is assumed.

Level IIb — reduced macroscopic classical-looking description. Level IIb is a reduced macroscopic description with robust records for a declared channel. Approximate diagonality is not ontological outcome selection.

Bridge between levels. A bridge between levels requires an explicit trace/coarse graining, record operator, state, and error estimate. The levels must not be identified by prose alone. Figure 25 displays the resulting status-separated map.

28.4 Costs of consistency across frameworks

Each candidate architecture for the gravity–matter interface pays a characteristic price for consistency. Table 84 summarises the main frameworks along four axes: the fundamental mediator picture, whether fundamental stochasticity or irreversibility is postulated, whether global unitarity is maintained at the full-theory level, and the principal consistency cost.

Table 84: Costs of consistency for gravity–matter architectures. The entries are qualitative structural summaries; detailed references are given in the surrounding text.

Framework	Fundamental mediator picture	Fundamental stochasticity / irreversibility	Global unitarity	Main consistency cost
Linearised quantum gravity / graviton EFT	Quantised gravitational mediator	No / No	Yes	Quantised metric sector; UV non-renormalisability
Semi-classical Møller–Rosenfeld	Classical expectation-value geometry	No / No	Problematic in superposition regimes	Hybrid inconsistency tensions (Eppley–Hannah, Page–Geilker)
Newton–Schrödinger class	Classical self-potential in non-linear QM	No / No	Modified	Non-linearity of QM; in the standard additive two-body form, no mediator-based entanglement generation
PCCG / Oppenheim	Classical metric with fundamental stochastic law	Yes / Yes	No	Fundamental stochasticity and information loss

Framework	Fundamental mediator picture	Fundamental stochasticity / irreversibility	Global unitarity	Main consistency cost
Measurement-feedback hybrids	Monitored classical channel	Yes / Yes	No	Explicit monitoring/feedback architecture
Phenomenological Diósi–Penrose collapse	Correlated classical noise / collapse law	Yes / Yes	No	Free collapse scale R_0 ; experimental constraints
ECT (this work)	Candidate reduced mediator programme; no physical mediator has yet been derived from the condensate	Open; neither fundamental stochasticity nor fundamental irreversibility is established or excluded by the present completion	Open	A future full theory must derive the mediator, reduced channel and consistency properties without assuming them

Read column-by-column, every candidate pays somewhere. The present ECT programme places a corresponding burden on a proposed ordered-medium architecture: a successful completion would have to derive, rather than presuppose, the physical mediator, the system/environment split, the reduced channel and any resulting irreversibility, decoherence, effective thermality or classical-looking gravity. None of those global properties follows from the current Level I→Level II bookkeeping alone. Whether this architectural choice succeeds depends on the open items in the status map. The discriminants in Section 38.4 become exclusion tests only for a future named completion that has first fixed the corresponding assumptions.

28.5 Bridges to the rest of Part III

To the Diósi–Penrose benchmark. The Diósi–Penrose timescale [175, 176] is used here as an external phenomenological benchmark, not as a structurally determined reduced-state limit of condensate distinguishability. The current ECT action does not fix it through ξ_{cond} and does not provide the required state/noise kernel. Any future comparison with X-ray or levitated-optomechanics bounds must begin from a declared ECT spectrum and density-resolution convention.

To the black-hole sector. One candidate architecture would extend the reduced-state programme to the strong-field sector. Whether reduced thermality or Page-curve-type behaviour arises from Level I→Level II reduction, from a different completion, or not at all is Open (Section 36).

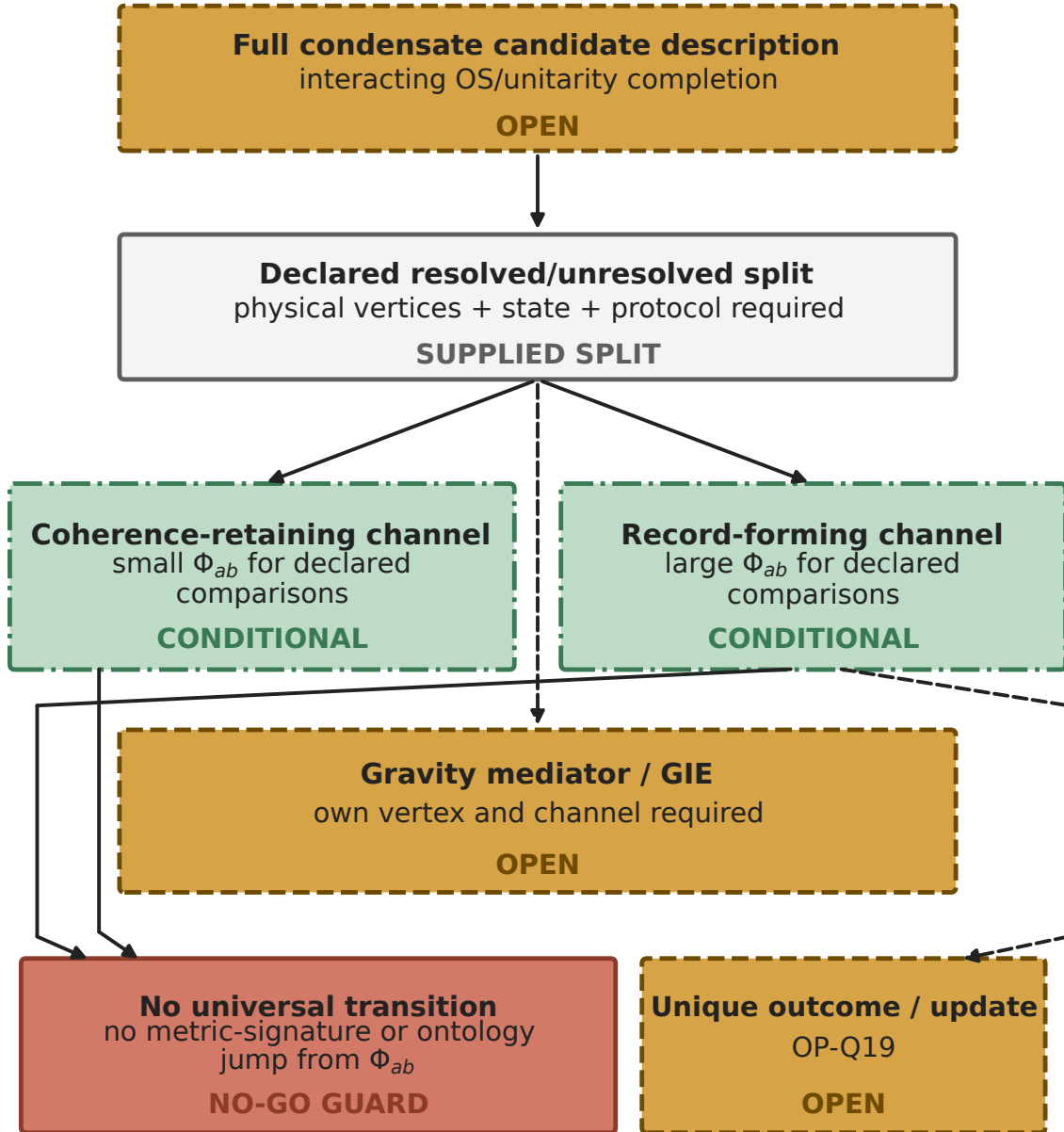
To conditional completion tests. Section 38.4 records how future observations could exclude named completions after they have committed to exclusively reduced-state stochasticity, exact global unitarity or a specific mediator protocol. These tests do not turn those presently Open commitments into properties of P1–P6.

28.6 Scope and non-claims

The present section does not claim: (i) a closed-form completion of the full non-linear Level I dynamics; (ii) a microscopic derivation of the Level I→Level II reduction for arbitrary coarse-grainings; (iii) a

Status-guarded full/reduced-description map

Status is explicit; border style and luminance remain redundant.



Solid arrows: declared reduction/guard. Dashed arrows: Open branch.

The map classifies supplied, conditional, Open, or ruled-out steps;
it does not add a metric-signature, gravity-mediator, or outcome theorem.

Figure 25: Status-guarded full/reduced-description map. The declared split can produce conditional coherence-retaining or record-forming reduced channels; the gravity-mediator and unique-outcome branches remain Open. The red node is the explicit no-go guard that a pairwise record exponent alone causes no universal metric-signature or ontology transition. Solid/dashed edges and literal status words carry the logic independently of colour.

strong Level-A prediction of BMV-type gravity-induced entanglement; (iv) a complete solution of the single-outcome measurement problem or an explicit Page-curve derivation from first principles. Its claim is narrower: ECT provides a single ontological architecture within which these programmes are jointly well-posed and can be assessed by common structural criteria.

Forward connection. The next section introduces the Principle of Euclidean Stationarity as one concrete organising principle through which the Level I→Level II transition becomes physically legible in the quantum, decoherence, and persistence sectors.

29 Principle of Euclidean Stationarity

Status: Level B as a response–persistence organising framework. For a declared ECT sector, admissible configurations are supplied by its causal retarded dynamics, Hamiltonian, Liouvillian, or Floquet problem and are then classified by physical widths, finite-time retention, and pairwise record exponents for a declared protocol. Compact phase and winding constraints apply only where the sector has that topology. The ECT-specific record operators, environment split, retarded/noise kernels, and any unique outcome rule remain Open (OP-PES-1/OP-Q19; physical record-channel owner). PES-R does not derive all quantisation, spectra, least action, or Born weights.

The preceding chapters supply several logically distinct inputs: sector dynamics and boundary data, the structural action scale in the sectors where it has been established, and reduced-state influence-functional tools for declared open systems. PES-R does not derive or merge those inputs into one variational law. It provides a status-separated classification of modes that have already passed their owning dynamical problem: dynamical admissibility, finite-time retention, and pairwise record distinguishability are tested separately (Fig. 26). Compact phase and winding are additional constraints only in sectors that possess that topology.

29.1 Statement of the principle

Short formulation.

Principle of Euclidean Stationarity, response–persistence form (PES-R). *Within a declared ECT sector, first solve the physical causal or constrained dynamical problem. Then classify the resulting modes by their finite-time retention and pairwise record distinguishability, subject to the sector’s boundary, charge, representation, and—where present—topological constraints. PES-R classifies finite-window persistence using preregistered diagnostics appropriate to the spectral type; it supplies no universal scalar or cross-type ranking. It does not generate every spectrum, derive Born weights, or select a unique outcome.*

Why this is not a new postulate. PES-R is not a new fundamental postulate and is not a universal variational law. It is a status taxonomy assembled from separately owned dynamics, retention, and record calculations.

Derivational status. PES-R assembles three distinct structures already used in the manuscript: (i) the sector’s causal or constrained dynamics; (ii) its spectral and boundary data; and (iii) a pairwise influence-functional record exponent for a declared resolved/unresolved split. The tolerances and observation window are protocol data. The ECT-specific split, record operators, kernels, and a universal one-history path measure have not been derived and remain OP-PES-1 physical-channel inputs.

PES-R/static-response firewall. PES-R does not derive S11, ERP binary recruitment, the minimum-energy continuation, HRC-0/HRC-3, or the gravity bridge. Conversely, the HRC static primitive does not supply the retarded/noise kernels or record operators needed by PES-R. Any common scalar/orientation

PES-R: response-persistence classification

Persistence is mode-type specific; none of these tests is a Born, spectrum or outcome theorem.

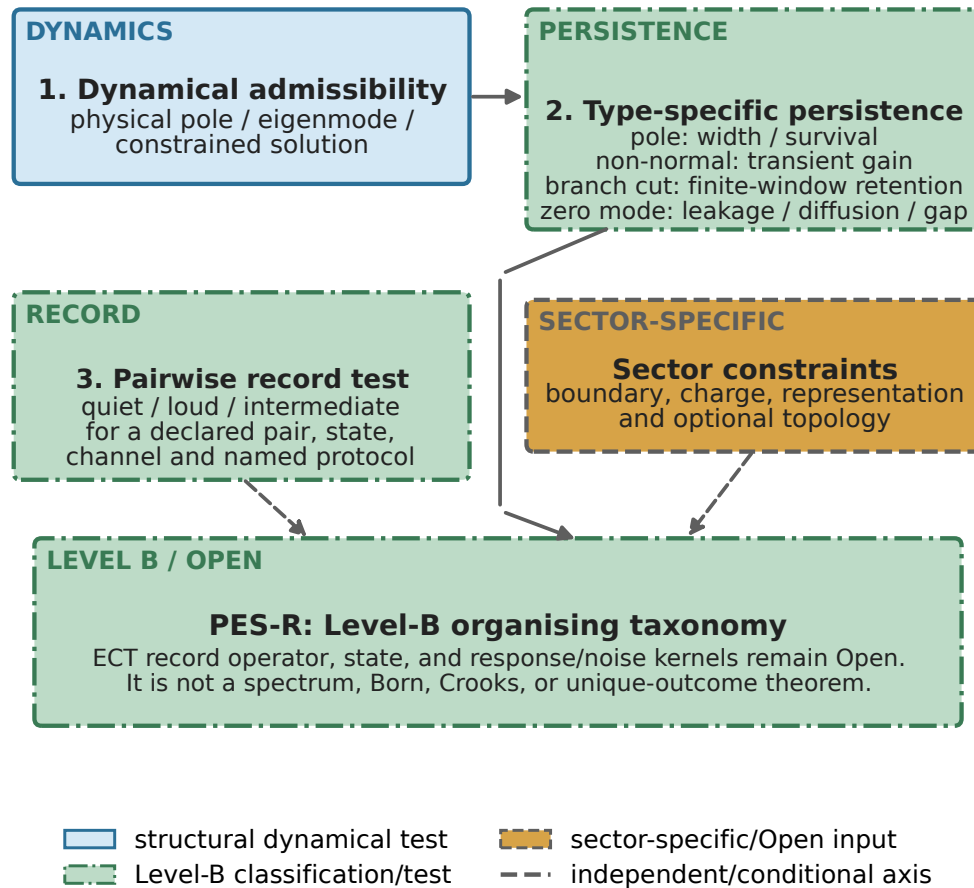


Figure 26: Response–persistence formulation of PES-R. A candidate must be dynamically admissible and receive a pairwise record classification for a named state, channel and protocol. Persistence is tested by a criterion appropriate to the spectral object: width/survival for an isolated pole, transient gain or finite-time singular values for non-normal dynamics, finite-window retention for a branch cut, and leakage, diffusion or a gap for a zero mode. Boundary, charge, representation and optional topology remain sector-specific constraints. PES-R is a Level-B organising taxonomy, not a spectrum, Born, Crooks or unique-outcome theorem; its ECT record operator, state and response/noise kernels remain Open.

modes require one resolved/unresolved split, canonical normalisation, and an explicit subtraction ledger. A finite readout coupling may shift a constitutive threshold; the protected-probe limit, a Ward identity, a counterterm, or the shifted law must therefore be stated rather than assumed. This paragraph is a no-conflict/double-counting guard, not a derivation of either framework.

Notation firewall: occupation versus noise. The HRC recruitment/occupation variable is reserved as $\mathcal{Q}$. The symbols $N(t - t')$ and $N^{(S)}(\omega)$ are reserved exclusively for the PES-R time-domain and spectral noise kernels, with $S_H(\omega) = S_0 N^{(S)}(\omega)$. No N appearing in the PES-R sector is an HRC occupation number, and no $\mathcal{Q}$ appearing in the HRC sector is a noise kernel.

29.2 Formal realisation

Two-history influence functional and causal response. Let $q_c = (q_+ + q_-)/2$ and $q_\Delta = q_+ - q_-$:

$$q_c = \frac{q_+ + q_-}{2}, \quad q_\Delta = q_+ - q_-. \quad (29.1)$$

Fix a joint initial preparation at t_0 and a closed-time-path contour, and centre the unresolved-sector one-point functions or absorb their contribution into S_{sys} . For a Gaussian unresolved sector coupled through physical record operators $\mathcal{O}_a[q]$, the bulk coarse-grained effective action has the schematic Feynman–Vernon/Schwinger–Keldysh form [192, 22]:

$$S_{\text{bulk}} = S_{\text{sys}}[q_+] - S_{\text{sys}}[q_-] + \int dX dX' \Delta \mathcal{O}_a(X) D_{ab}^R(X, X') (\mathcal{O}_b)_c(X') \quad (29.2)$$

$$+ \frac{i}{2} \int dX dX' \Delta \mathcal{O}_a(X) N_{ab}(X, X') \Delta \mathcal{O}_b(X'), \quad (29.3)$$

where D^R is causal and $N = N^\dagger \succeq 0$. A compatible factorised preparation is sufficient for this bulk form without an additional initial-correlation influence term. For a correlated preparation the full effective action is instead

$$S_{\text{CGEA}}^{\text{corr}}[q_+, q_-] = S_{\text{bulk}}[q_+, q_-] + S_{\text{ic}}[q_+, q_-; t_0], \quad (29.4)$$

or is represented equivalently on an extended contour. The term S_{ic} may be omitted only after it has been shown to vanish or to have been absorbed consistently. The non-negative bulk pairwise record exponent is

$$\Phi_{\text{dec}}^{\text{bulk}}[q_+, q_-] = \frac{1}{2S_0} \int dX dX' \Delta \mathcal{O}_a(X) N_{ab}(X, X') \Delta \mathcal{O}_b(X') \geq 0. \quad (29.5)$$

The full log-modulus diagnostic $\Xi_{ab}^{\text{full}} = -\ln |\mathcal{F}|$ may additionally contain preparation-dependent initial-boundary terms and need not be nonnegative without a contractivity premise.

Distinction: stationarity, disturbance, action. Stationarity of the system action, small physical decay width, and small pairwise record exponent are inequivalent. Each has its own units, kernel, and experimental owner.

Normalisation dictionary: N , S_H , and the finite record. The full log-modulus diagnostic is defined operationally before any kernel convention,

$$\Xi_{ab}^{\text{full}}(T) := -\ln |\mathcal{F}[a, b; T]|, \quad \Phi_{ab}(T) := \Xi_{ab}^{\text{full}}(T) \geq 0 \quad \text{only when } |\mathcal{F}[a, b; T]| \leq 1 \text{ has been established.} \quad (29.6)$$

After the spatial/projector integrations have produced a finite vector of record differences $\Delta \mathcal{O}_{ab}(t)$, assume only for this spectral rewrite that the bulk state and noise kernel are stationary on the observation window, so that $N(t, t') = N(t - t')$, and that initial-boundary/preparation transients are absent, negligible on that window, or explicitly separated. Without these premises the two-time kernel $N(t, t')$ must be retained

and no single-frequency kernel $N^{(S)}(\omega)$ is defined generically. Under the stated premises we use the unnormalised finite-window and Fourier conventions

$$d_{ab,T}(\omega) := \int_0^T dt e^{i\omega t} \Delta \mathcal{O}_{ab}(t), \quad (29.7)$$

$$N^{(S)}(\omega) := \tilde{N}(\omega) := \int_{-\infty}^{\infty} d\tau e^{i\omega\tau} N(\tau), \quad N(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega\tau} N^{(S)}(\omega). \quad (29.8)$$

The action-scaled spectral record kernel used in the PES-R robustness statements is, by definition,

$$S_H(\omega) := S_0 N^{(S)}(\omega), \quad S_H(\omega) = S_H^\dagger(\omega) \succeq 0. \quad (29.9)$$

The time-domain and spectral expressions are then the same bulk contribution:

$$\boxed{\Phi_{ab}^{\text{bulk}}(T) = \frac{1}{2S_0} \int_0^T dt \int_0^T dt' \Delta \mathcal{O}_{ab}^\dagger(t) N(t-t') \Delta \mathcal{O}_{ab}(t') = \frac{1}{2S_0^2} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} d_{ab,T}^\dagger(\omega) S_H(\omega) d_{ab,T}(\omega).} \quad (29.10)$$

If S_{ic} is absent or has no modulus contribution on the declared protocol, then $\Xi_{ab}^{\text{full}} = \Phi_{ab} = \Phi_{ab}^{\text{bulk}} \geq 0$. Otherwise Ξ_{ab}^{full} also contains the explicitly retained preparation term and cannot be reconstructed from $N(t-t')$ alone; a nonnegative full Φ_{ab} is NOT IDENTIFIABLE until full contractivity is proved. Thus the time quadratic has units $[S_0]$, the spectral integral has units $[S_0]^2$, and both displayed ratios are dimensionless. No factor of T is hidden in $d_{ab,T}$; a periodogram convention divided by T would require the compensating factor in its spectral kernel. S_H is an action-scaled record spectrum for the declared operator normalisation, not an independently identified raw detector covariance. D^R remains a distinct causal response kernel: stationarity, positivity, or an equilibrium fluctuation–dissipation relation does not identify response and noise without the required state and KMS assumptions. The mean equation follows by varying the complete coarse-grained action and taking the physical diagonal,

$$\left. \frac{\delta(S_{\text{bulk}} + S_{\text{ic}})}{\delta q_\Delta(X)} \right|_{q_\Delta=0} = 0, \quad S_{\text{ic}} = 0 \text{ when no such term is present}, \quad (29.11)$$

with the retarded force and nonlinear vertices understood. The first derivative of the bulk noise quadratic is proportional to Nq_Δ and vanishes on the diagonal. Noise therefore controls bulk fluctuations and branch records rather than supplying a generic additional mean potential. A correlated-preparation term S_{ic} can nevertheless contribute to the mean equation and must be varied rather than silently discarded.

Translation susceptibility: optional diagnostic. For a declared deformation $q_\epsilon = q + \epsilon\eta + O(\epsilon^2)$, keep the state, kernel, record protocol and interval fixed. A finite bulk susceptibility follows only if the record map is differentiable in the quadratic-form topology of N :

$$\Delta \mathcal{O}[q, q_\epsilon] = -\epsilon D \mathcal{O}_q[\eta] + r_\epsilon, \quad D \mathcal{O}_q[\eta] \in \text{Dom}(N^{1/2}), \quad \|N^{1/2} r_\epsilon\| = o(|\epsilon|). \quad (29.12)$$

Under these premises,

$$\Gamma_{\text{tr}}^{\text{bulk}}[q; \eta] := \lim_{\epsilon \rightarrow 0} \frac{\Phi_{\text{dec}}^{\text{bulk}}[q, q_\epsilon]}{\epsilon^2} = \frac{1}{2S_0} \|N^{1/2} D \mathcal{O}_q[\eta]\|^2 \geq 0. \quad (29.13)$$

If form-domain differentiability or finiteness has not been established, the only generally defined diagnostics are the extended quantities

$$\underline{\Gamma}_{\text{tr}}^{\text{bulk}} := \liminf_{\epsilon \rightarrow 0} \frac{\Phi_{\text{dec}}^{\text{bulk}}[q, q_\epsilon]}{\epsilon^2}, \quad \overline{\Gamma}_{\text{tr}}^{\text{bulk}} := \limsup_{\epsilon \rightarrow 0} \frac{\Phi_{\text{dec}}^{\text{bulk}}[q, q_\epsilon]}{\epsilon^2}, \quad \underline{\Gamma}_{\text{tr}}^{\text{bulk}}, \overline{\Gamma}_{\text{tr}}^{\text{bulk}} \in [0, \infty]. \quad (29.14)$$

A finite Γ_{tr} exists only when these two quantities coincide and are finite; otherwise the finite susceptibility is reported as NOT IDENTIFIABLE, rather than assumed. With a correlated initial preparation, S_{ic} may also

contribute to the full $-\ln|\mathcal{F}|$ and hence to a full translation susceptibility; the bulk formula above is not by itself that full result.

Its value depends on the direction normalisation, state, record operators, interval, and boundary prescription. Within one fixed choice of all those data it may order already admissible modes; it supplies no cross-protocol or cross-type ranking, but varying it together with the system action would introduce a force not derived from the influence functional. A one-history Onsager–Machlup route requires a separately derived path measure, drift, diffusion, Jacobian, stochastic convention, and boundary data; that route is Open.

Relation to extremal action. Closed-sector constrained extrema remain the starting point. In an open sector the retarded kernel shifts poles and widths, while the noise kernel governs pairwise records. PES-R therefore does not replace the closed-sector variational equation by an additive one-history noise-gradient equation.

Retention set and separate pairwise record classification. For a declared sector $\mathcal{C}_n$, observation window T , and tolerances, define the protocol-level retention set

$$\mathfrak{P}_\varepsilon^{\text{ret}}(T) = \bigcup_n \left\{ a \in \mathcal{C}_n : \begin{array}{l} a \text{ solves the declared physical response/spectral problem,} \\ \text{the retention diagnostics applicable to its spectral type} \\ \text{meet the preregistered finite-window tolerances,} \\ \text{sector boundary, charge, representation, and optional topology hold} \end{array} \right\}. \quad (29.15)$$

The relevant diagnostic set is type-dependent rather than one universal conjunction: an isolated pole uses its physical width and survival/residue; a non-normal or Jordan sector uses the singular-value gain in a declared norm together with full finite-window retention; a cut-dominated object uses the finite-window survival functional directly; and a zero mode uses leakage, diffusion, or gap diagnostics rather than a quality factor. At an exceptional point the defective Jordan block, not ill-defined coalescing residue labels, is the object to classify. All objects are defined after gauge/constraint reduction.

For a channel in which a nonnegative full Φ_{ab} has been established, the pairwise record exponent is an independent axis, not a property of one mode and not a necessary retention condition. With preregistered thresholds $\varepsilon_{\text{quiet}} < \varepsilon_{\text{loud}}$, define

$$\mathcal{Q}_\Phi(T) := \{(a, b) : \Phi_{ab}(T) \leq \varepsilon_{\text{quiet}}\}, \quad (29.16)$$

$$\mathcal{L}_\Phi(T) := \{(a, b) : \Phi_{ab}(T) \geq \varepsilon_{\text{loud}}\}. \quad (29.17)$$

Pairs between the thresholds are intermediate. Small Φ_{ab} means that the owning channel is record-quiet and the compared coherence can persist or revive; large Φ_{ab} means strong pairwise dephasing and suppression of the corresponding off-diagonal element in the named reduced channel. Calling that suppression an accessible physical record additionally requires an identified environmental or readout observable. Objective pointer redundancy is a still stronger, fragment-resolved condition and is not inferred from one scalar Φ_{ab} .

To combine the two axes without confusing an object with a pair, define

$$R_{ab}(T) := \mathbf{1}[a, b \in \mathfrak{P}_\varepsilon^{\text{ret}}(T)]. \quad (29.18)$$

The joint taxonomy then contains six cells, $R_{ab} \in \{0, 1\}$ times $\{\text{quiet}, \text{intermediate}, \text{loud}\}$. The four quiet/loud endpoint cells are only the extreme subtable obtained after the intermediate band has been excluded; retaining zero, one, or two members of the pair would instead give a finer nine-cell diagnostic. This is a Level-B protocol taxonomy, not a necessary-and-sufficient persistence, pointer, or outcome theorem.

Non-normal, higher-order-pole, and cut guard. Let $U_{\text{phys}}(t, 0)$ be the amplitude propagator after the declared constraint/gauge reduction, let $P_a = P_a^\ddagger = P_a^2$ be the orthogonal physical projector onto the block being classified, and let $\|\cdot\|_{\text{phys}}$ be a preregistered positive physical norm. The block gain is

$$\mathcal{G}_a(T) := \sup_{0 \leq t \leq T} \|P_a U_{\text{phys}}(t, 0) P_a\|_{\text{phys}, \text{op}}. \quad (29.19)$$

For density-operator or Liouvillian evolution, let $\mathcal{E}_t$ be the declared physical reduced map and let $0 \preceq E_a \preceq I$ be a positive effect representing survival in the declared sector. For a normalized initial density operator ρ_a prepared wholly in that sector, $\text{Tr}_{\text{phys}}(E_a \rho_a) = 1$, the survival probability is

$$\mathcal{S}_a(t; \rho_a) := \text{Tr}_{\text{phys}}[E_a \mathcal{E}_t(\rho_a)], \quad 0 \leq t \leq T, \quad (29.20)$$

where $\ddagger$ denotes the adjoint in the declared physical inner product. A non-normal Riesz projector is not silently substituted for the positive effect in this probability. If the initial preparation is not wholly in the sector, the ratio to its initial occupation is merely a normalized retention ratio and can exceed unity through inflow; it is not called a probability. The initial state or admissible state class, projector/effect, norm, observation window, and tolerance functional on the full curve must be listed by each application; neither equation is a universal scalar selection rule. For non-normal physical evolution the finite-time singular values of the reduced propagator, not the eigenvalue real parts alone, control transient gain. A Jordan block or higher-order pole generates polynomial prefactors multiplying its exponential contribution, while a branch cut can generate an algebraic long-time tail. Therefore $\mathcal{G}_a(T)$ and the full curve $\mathcal{S}_a(t; \rho_a)$ must include pole, Jordan and cut contributions over the declared finite window; a small modal width by itself is insufficient.

29.3 Immediate consequences

Spectral discreteness and persistence

A discrete spectrum is supplied by the sector operator, its domain, boundary conditions, and potential—for example by a self-adjoint operator with compact resolvent. Such spectra exist on topologically trivial configuration spaces. PES-R asks which supplied modes remain persistent; it does not create the spectrum from compact phase. Compact phase remains important in sectors where it adds winding, flux, circulation, holonomy, or charge labels.

Extremal action and persistence

Extremal action and open-system persistence answer different questions. Constrained extrema determine closed-sector solutions; a retarded environment shifts poles and widths; the noise kernel controls record distinguishability. No general bound relating $|\delta S_E|$ to a decoherence exponent has been derived. A survival derivation of least action remains OP-PES-2.

Stationary, oscillatory, and decoherence-free modes

Stationarity is spectral rather than the universal coordinate condition $\partial_w \Psi = 0$. An energy eigenprojector may be stationary while its state vector acquires phase; resonances have finite width; Liouvillian and Floquet modes may persist at nonzero frequency. Environmental silence for a pair requires the physical record difference to lie in the null space of the noise kernel. Widths must be calculated from vertices, retarded self-energies, form factors, channels, occupation factors, and phase space.

Connection to $S_0 = \hbar$: no PES-R derivation

The fixed-core and positive-tension loop constructions assign distinct sector-specific coefficients t_0^{EFT} and $t_{0, \text{bal}}^{\text{eff}}$, but PES-R selects neither owner map to the operational S_0 and proves no cross-sector universality. Identifying S_0 with $\hbar$ requires a separate canonical/experimental matching and remains Level B/Open. Persistence provides no independent upgrade of that status and no universal lower-action theorem.

Gravitational sector: conditional programme

ECT proposes that gravitational and matter excitations belong to one condensate architecture. It has not derived that they share one canonical action quantum, nor has PES-R quantised the physical tensor sector. Closing this claim requires the tensor modes, constraints, canonical normalisation, propagator, interacting RP/OS reconstruction, and comparison of the resulting action scale with the electromagnetic one. Cross-sector universality and “no separate quantisation step” therefore remain Open hypotheses (OP-PES-3), not predictions.

29.4 Relation to known selection mechanisms

PES did not appear in a vacuum: several of its special cases are classic results, and its quantum-information face has a developed formalism. Placing PES in this landscape both anchors it and makes precise what is new.

Physical analogies, not validations. The Landau critical-velocity condition, environment-stabilised molecular chirality, and measured resonance widths illustrate three different mechanisms—kinematic channel closure, record-induced dephasing, and retarded pole decay [193, 194]. PES-R offers a common vocabulary for comparing their retention, but it does not derive these results or identify them with one scalar “quietness” mechanism. Lifetimes and widths remain owned by their physical vertices, self-energies, states, and phase space.

Quantum-information anchors. Environment-induced superselection, predictability sieves, decoherence-free subspaces, noiseless subsystems, and quantum Darwinism are established open-system structures [195, 196, 197, 198, 199, 200, 201, 202]. In matrix notation exact quietness of a declared record difference $d(\omega)$ requires $S_H^{1/2}(\omega)d(\omega) = 0$ almost everywhere. Infrared superselection is a pairwise divergence of Φ_{ab} , not a one-history zero-noise or “quietness” condition.

If $S_H(\omega) \sim \omega^a$ and $|d(\omega)| \sim \omega^{-\sigma}$, the infrared integral scales as

$$\Phi_{\text{IR}} \sim \int_0^\Omega d\omega \omega^{a-2\sigma-\zeta}, \quad \zeta = 0 \ (T = 0), \quad \zeta = 1 \ (\text{thermal IR tail}), \quad (29.21)$$

and diverges when $a - 2\sigma - \zeta \leq -1$.

ECT-specific status. ECT hypothesises that the unresolved sector belongs to the same condensate, but the split, record operators, and kernels have not been derived from the bare action. PES-R organises possible ECT applications of einselection; it does not yet derive or extend einselection from first principles.

29.5 Stability, lifetimes, and decay widths

Long-lived excitations are real poles or poles/Liouillian modes with small damping. Their widths are determined by the retarded self-energy, interaction vertices, form factors, available channels, occupation factors, and phase space. A translation susceptibility may be tested as a correlate of selected form factors, but it is not a universal lifetime law. The former ansatz is retained only as an explicitly Open target,

$$\gamma_a \stackrel{?}{\propto} \Gamma_{\text{tr}}[q_a; \eta_a], \quad (29.22)$$

after OP-PES-4 supplies physical kernels and normalisation. No observed particle-lifetime hierarchy is claimed as an ECT prediction at present.

Instability from a residual broadening or decay channel. A residual coupling can destabilise a mode only through its self-energy, phase space, and state. Pairwise record loudness is a separate property and is not a universal scalar source of instability.

Qualitative hierarchy. Any qualitative lifetime hierarchy remains conditional on the physical vertices and densities of states. PES-R can compare computed widths; it cannot generate them from topology alone. For audit continuity, representative external lifetime landmarks are retained as inputs only [86, 203]:

Table 85: Representative particle-lifetime data. These values constrain a future width calculation; they do not validate PES-R or determine Γ_{tr} .

Particle	Lifetime / bound	ECT status
Proton	$> 10^{34}$ yr	External lower bound; microscopic ECT stability derivation Open
Electron	Stable within observations	External fact; owning symmetry/decay-channel argument required
Neutron (free)	$\simeq 879$ s	External input
Muon	$\simeq 2.2 \mu\text{s}$	External input
Charged pion	$\simeq 26$ ns	External input
W boson	$\sim 3 \times 10^{-25}$ s	Width-derived external scale

Rydberg states: external target only. Long Rydberg lifetimes do not test ECT while its electromagnetic and environmental channels are missing. They are external data for a future owner-complete calculation, not a qualitative consistency check.

29.6 Quality factors, persistent bands, and finite spectra

Distinguish the dimensionless pairwise exponent $\Phi_{ab}(T)$, decay rate γ_a , and energy width $\Gamma_{E,a} = S_0 \gamma_a$. For angular frequency Ω_a and $E_a = S_0 \Omega_a$,

$$Q_{\omega,a} = \frac{\Omega_a}{\gamma_a} = \frac{E_a}{\Gamma_{E,a}}, \quad N_{\text{coh},a} = \frac{Q_{\omega,a}}{2\pi}. \quad (29.23)$$

For the isolated-pole subclass with a declared exponential survival law $P_a(T) = e^{-\gamma_a T}$, the condition $P_a(T_{\text{obs}}) \geq P_{\min}$ gives $Q_{\omega,a} \geq \Omega_a T_{\text{obs}} / [-\ln P_{\min}]$. Within that subclass a declared high-quality-factor count is

$$N_{\text{persistent}} = \#\{a : Q_{\omega,a} > Q_{\text{sel}}\}, \quad (29.24)$$

with the line-shape and protocol convention declared. This is a coherent-cycle/high- Q count, not the general retention set of Eq. (29.15); when Ω_a varies, a common Q_{sel} is not a common finite-window survival threshold. A common survival threshold instead uses the mode-dependent bound above, equivalently $\gamma_a \leq -\ln P_{\min} / T_{\text{obs}}$. Branch-cut, non-normal, exceptional-point and zero-mode objects require their type-specific retention diagnostics.

In the following positive two-channel toy, the persistent sector can be a finite *band* rather than a prefix of low-lying modes. If, for asymptotic energies $E_n \sim A n^s$, the environmental width contains both an infrared-enhanced and an ultraviolet-enhanced broadening channel on top of the barrier one,

$$\Gamma_n = \Gamma_n^{\text{tunnel}} + \gamma_- n^{-r} + \gamma_+ n^p, \quad r > 0, \quad p > s, \quad (29.25)$$

then, in the high-barrier limit,

$$Q_n \simeq \frac{A n^s}{\gamma_- n^{-r} + \gamma_+ n^p}, \quad (29.26)$$

the exact high-barrier persistent set is

$$\mathcal{J}_Q = \left\{ n \in \mathbb{N} : \gamma_- n^{-(s+r)} + \gamma_+ n^{p-s} < \frac{A}{Q_{\text{sel}}} \right\}. \quad (29.27)$$

For $s + r > 0$ and $p > s$, convexity in $\ln n$ makes this set a single integer interval when it is nonempty. Its exact real endpoints are roots of the combined equality, not of the two terms separately. The separate-term bounds give only the necessary outer enclosure

$$\left(\frac{\gamma_- Q_{\text{sel}}}{A}\right)^{1/(s+r)} < n < \left(\frac{A}{\gamma_+ Q_{\text{sel}}}\right)^{1/(p-s)}. \quad (29.28)$$

A simple sufficient inner enclosure is obtained by requiring each positive term to be smaller than $A/(2Q_{\text{sel}})$:

$$\left(\frac{2\gamma_- Q_{\text{sel}}}{A}\right)^{1/(s+r)} < n < \left(\frac{A}{2\gamma_+ Q_{\text{sel}}}\right)^{1/(p-s)}. \quad (29.29)$$

The exact count is $N_{\text{persistent}} = \#\mathcal{J}_Q$, with maximal quality factor at $n_*^{p+r} = (s+r)\gamma_-/[(p-s)\gamma_+]$. Thus the positive two-channel toy can produce a finite window — neither the softest nor the hardest modes need survive — but either enclosure may contain no integer and the exact set may be empty. This toy parameterisation is not an ECT mass calculation; it only illustrates the mechanism by which condensate spectral data can produce a finite persistent sector. More generally, for $E_n \sim A n^s$ and multi-channel widths $\Gamma_n = \sum_i \gamma_i n^{b_i}$ the narrow-width condition reads $\sum_i \gamma_i n^{b_i-s} < A/Q_{\text{sel}}$; a channel with $b_i > s$ enforces an ultraviolet cap, a sufficiently strong channel with $b_i < s$ removes the lowest modes, and, since $\sum_i \gamma_i e^{(b_i-s)\ln n}$ is convex in $\ln n$, its sublevel set is one contiguous interval, possibly empty or unbounded, within this positive monomial asymptotic family. This conclusion is not generic for resonant or nonmonotone form factors, which can produce disjoint windows. When the ultraviolet channel arises from resonant sampling of a spectral density $J(\omega) \sim \omega^a$ with squared form-factor scaling n^u , its exponent is fixed by the dictionary $p = as + u$. Within this isolated-pole toy family, Eq. (29.24) converts the declared high- Q count into a spectral problem: determine (E_n, Γ_n) from the condensate defect spectrum — schematically $\Gamma_n \sim \int_0^\infty d\omega J(\omega, \mathbf{x}) |F_n(\omega)|^2$ with F_n the mode form factor — then count. It does not determine the number of generations, create the supplied spectrum, or classify general non-pole stability. *Status: Level B as a mathematical mechanism; the ECT microphysical input — the spectral data (E_n, Γ_n) , equivalently the condensate spectral density $J(\omega, \mathbf{x})$ and the mode form factors F_n — is Open and constitutes the common target of OP-PES-1, OP-PES-2, and OP-PES-4.*

29.7 Sector-specific quantisation and PES-R status

PES-R does not identify one universal origin of quantisation. The owning sector supplies each spectrum or representation; PES-R can only classify finite-time persistence afterwards.

Table 86: Sector-specific quantisation mechanisms and honest status.

Quantity	Owning mechanism	PES-R role	Status
$S_0 = \hbar$	Empirical calibration of the operational ECT action-unit slot	Calibration explicit; physical owner and absolute derivation Open	B calibration / Open owner
Bound-state levels	Hamiltonian/operator, domain, potential, boundaries	Retention after spectrum is solved	A/B
Angular momentum / spin	Rotation-group representations and topology	Sector constraints only	A/B

Quantity	Owning mechanism	PES-R role	Status
Electric charge / flux	Owning compact gauge sector	Sector constraints only	A/B/Open
Gravitational quanta	Physical tensor-mode reconstruction	Persistence test after kernel closure	Open
Area scale	Strong-field/entropy programme	No derivation from PES-R	Open

29.8 Measurement, information, and decoherence: summary

Specified reduced channels can suppress interference and form stable records. The qubit information–dephasing calculation is a model-specific trade-off, not a protocol-independent measurement bound. PES-R supplies no outcome update rule and no identity between photon proper time, coherence, and collapse.

29.9 Protocol reversal and tunnelling geometry: summary

Tunnelling retains its stated WKB/instanton and geometric status. A reverse/forward fluctuation relation requires explicit path measures and local detailed balance; no ratio is inferred from the pairwise decoherence exponent.

29.10 Geometric locking conjecture: Open

What happens at $\alpha \rightarrow \beta$. At $\alpha \rightarrow \beta$ the kinetic tensor becomes singular in the owning signature analysis. No universal decoherence transition or quantum–classical boundary follows from that singularity.

Conjectural bridge to decoherence. No signature-boundary record channel is defined without an explicit vertex, state, and constraint-reduced kernel. A connection between the signature boundary and reduced-state decoherence is therefore only an Open programme.

What this does not yet establish. The boundary analysis does not establish a measurement event, outcome rule, entropy law, or common action scale for all sectors. Those items remain separate Open tasks. No map between a subsystem dephasing exponent and the background coefficient $\alpha - \beta$ has been derived. Decoherence of a reduced state does not by itself change the signature of the effective metric. Any such connection requires a backreaction calculation from the ECT action and remains OP-PES-5.

Owner-separated benchmark scales near the ordering programme. The following quantities come from different owners and must not be read as scales set by one derived $O(4) \rightarrow O(3)$ transition:

Scale	Value / matching choice	Status	Physical meaning
Condensate VEV ϕ_0	$\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$	Level C/Open matching	Homogeneous-P3 minimum coordinate matched to the reduced Planck scale; not a P4 amplitude

Scale	Value / matching choice	Status	Physical meaning
Critical temperature T_c	$2\phi_0$	Level A inside a separately supplied one-real-scalar thermal toy	One-real-scalar Z_2 restoration; not the P4 gradient-ordering transition

This table is a summary device, not a new derivation. Its purpose is to collect in one place the characteristic scales that are otherwise derived or motivated across different parts of the article. For one real scalar with $V = -\mu^2\Phi^2/2 + \lambda\Phi^4/4$, the leading high-temperature term is $\Delta V_T = \lambda T^2\Phi^2/8$, so $m_T^2 = -\mu^2 + \lambda T^2/4$ and $T_c = 2\mu/\sqrt{\lambda} = 2\phi_0$. This standard one-loop toy calculation [204] describes Z_2 restoration of the bare scalar. It does not derive the P4 $O(4) \rightarrow O(3)$ gradient-ordering transition or a physical ECT thermal state. The two entries remain owner-separated: a P3 calibration and a standard finite-temperature toy for a different, explicitly supplied scalar model. Neither derives a stationary P4 action, pole, transition scale or thermal history. No Planck-length correlation scale or Planck-epoch transition energy is assigned.

29.11 Gravitational record channel: current status

The external Diósi–Penrose comparison consists of the symbolic functional $\tau_{\text{DP}}^{\text{ext}} = \hbar/E_G$ and one fully declared hard-sphere orientation point, $C_{\text{DP}}(1) = 51/160$ and $\tau_{\text{DP}}^{\text{ext}} = 70.6\text{ ms}$. The latter is deterministic arithmetic inside the imported model, not physical precision and not an ECT rate. The scalar trace response is parametric, while the ECT source projector, vertex, state-dependent noise kernel, screened-body charge, and absolute rate remain Open/Not Identifiable. See Section 26.3 for the gate ledger and owner firewall.

29.12 What PES-R changes in the quantum programme

PES-R changes bookkeeping rather than the microscopic equations. It prevents four conflations: spectrum versus persistence, retarded response versus noise, pairwise records versus one-history sensitivity, and decoherence versus unique outcome selection.

Table 88: Status-corrected role of PES-R.

Topic	Owning calculation	PES-R role
Spectrum	Sector operator, domain, boundary conditions	Classify retention after solving; no spectrum derivation
Mean dynamics	Retarded physical kernel	Dynamical admissibility; noise is not an extra mean force
Records	Pairwise noise kernel and protocol	Test distinguishability; no unique-outcome theorem
Stability	Pole/Liouvillian/Floquet analysis plus finite-time guards	Compare physical widths and retention
Topology	Owning compact/gauge sector	Add winding/charge constraints only where present

Topic	Owning calculation	PES-R role
Admitted-measure representation / physical weights	Independently supplied normalised, noncontextual, countably additive projector measure and ordinary Gleason representation in complex dimension ≥ 3 ; Busch effect/POVM theorem under separately declared C2q hypotheses for a qubit	No derivation edge from PES-R; state, preparation, event, probability origin, outcome and update remain Open

29.13 What PES-R does not prove

Discipline statement. The frozen P6 calculations retain their own assumptions, normalisations, and numerical owners. PES-R status edits must not alter their verified arithmetic. PES-R does not derive a universal one-history functional, Born weights, least action, spectra, particle lifetimes, a signature transition, or one realised measurement outcome. Its ECT-specific closure requires:

- (i) the physical resolved/unresolved split and record operators;
- (ii) canonical physical modes, retarded and noise kernels, state, and counterterms for at least one ECT sector;
- (iii) a finite-time protocol with gauge/constraint reduction;
- (iv) for any one-history large-deviation route, a derived stochastic process, path measure, diffusion, convention, Jacobian, and boundary data;
- (v) for measurement, an update/unique-outcome rule beyond decoherence.

What PES does and does not identify. PES-R identifies neither a universal origin of spectra nor an outcome. It can organise already-derived poles, lifetimes, finite-time records, and topology while preserving their distinct owners. The active gravitational record-channel programme may terminate as DERIVED, PARAMETRIC ONLY, NOT IDENTIFIABLE, INCOMPATIBLE, DOUBLE-COUNTED, or MISSING VERTEX; analogy to a Diósi–Penrose kernel is not closure.

Frozen spectral-record development (P6 appendices). The P6 record sector remains a Level B generic spectral-record framework with operational falsifiers. Its valid basis is matrix-positive noise, causal retarded response, physical record differences and form factors, line-plus-continuum spectra, Kramers–Kronig relations, and declared held-out protocols. It contains no universal additive action–noise stationarity equation. ECT-specific vertices, record operators, state, kernels, and counterterms remain Open. Existing numerical P6 results are unchanged; only their derivational status is corrected.

30 Quantum–Classical Boundary, Probability, Born Rule, and Measurement Status

Inputs: the complete applicable corrected OS-II Hilbert route, coherent amplitudes, and declared reduced channels. Status: Level A mathematics for the kinematic distinctions proved in their owning models and, independently, for ordinary projector Gleason’s representation of a supplied normalised, noncontextual, countably additive projector measure on a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. State/preparation, event, probability origin, ECT detector channel, pointer basis, unique outcome, and update remain Open.

Decoherence is not a premise of Gleason's theorem, is not a metric-signature field, and does not by itself define a universal quantum–classical boundary.

30.1 Quantum–classical boundary: reduced-state status

The ordered ECT branch supplies Lorentzian causal dynamics, while a specified reduced channel can suppress interference and stabilise records. A pairwise decoherence exponent is not a signature-changing field and does not prove a literal $O(3) \rightarrow O(4)$ laboratory transition. Echo and dynamical-decoupling protocols demonstrate finite-time recoherence inside reduced models, not a return to the microscopic symmetric phase. The universal geometric mapping $\Gamma_{\text{loop}} \leftrightarrow \alpha - \beta$ and any ECT-specific quantum–classical threshold remain Open.

Structural content of reverse transitions

A reverse transition is a physical process only after a saddle, boundary conditions, and transition amplitude are supplied. Two-sided Euclidean dependence alone does not create such a transition.

What reverse coherence does and does not mean

Reverse coherence means that a finite reduced record can be erased or refocused in a specified protocol. It is not a negative entropy event or an ontological return to an unrealised branch.

Relation to known recoherence protocols

Spin echo, dynamical decoupling, and quantum erasure are standard recoherence protocols. They constrain a proposed ECT channel only after its couplings predict the protocol dependence.

Status of the experimental programme

The experimental programme is Open as an ECT discriminator. Visibility and lifetime source data remain valid anchors but do not identify the condensate kernel.

30.2 Measurement records and environmental locking

Status: Level B for reduced-state dephasing and pointer-record stabilisation in declared models; Open for ECT record operators, a branch update rule, pointer-basis completion, and unique outcome selection.

Interaction as a Lorentzian concept

Interaction becomes a Lorentzian causal concept after ordered- branch reconstruction. Euclidean correlations are not automatically event vertices. A measurement couples a microscopic degree of freedom to an apparatus and unresolved modes. Pairwise exponents $\Phi_{ab}(T)$ suppress off-diagonal reduced-state terms and may stabilise a pointer algebra. The number of effective environmental modes is the number that actually couples through the protocol, not the total atom count.

A threshold $\Phi_{ab} = O(1)$ marks a model- and protocol-dependent reduced-state crossover. It neither generates Lorentzian signature nor selects one realised outcome. Persistent modes may be oscillatory poles, Liouvillian/Floquet modes, or decoherence-free record directions. Photon propagation and absorption require their physical vertices and channels; no identity between null proper time, a universal decoherence scalar, and collapse is asserted. Unique outcome selection remains OP-Q19.

Measurement as environmental locking

Figure 27 makes the distinction explicit: measurement is not defined as environmental locking. A declared channel may form a stable reduced record, while outcome and update remain Open.

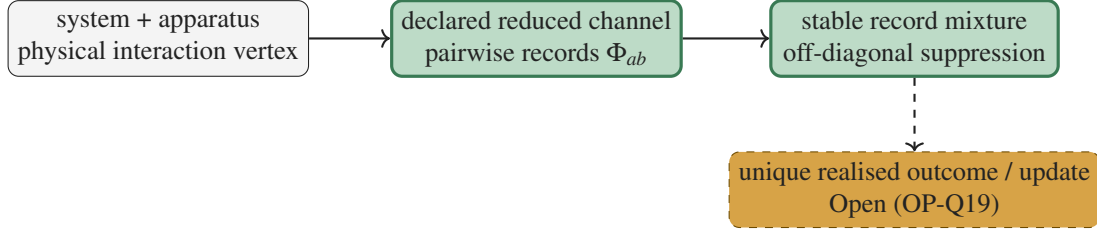


Figure 27: Status-separated measurement chain. A specified interaction and reduced channel can form stable records; decoherence alone does not supply the dashed unique-outcome update and does not create Lorentzian signature. Gray, green and orange-dashed roles denote supplied, conditional and Open nodes; literal labels and borders remain decisive in grayscale.

The persistent quantum sector as the sector of minimal environmental coupling

The persistent quantum sector is not universally the sector of minimal environmental coupling. Persistence can arise from symmetry, kinematic closure, topology, or a small computed width.

The photon as a Euclidean traveller

A photon is a physical gauge-sector excitation after constraint reduction; the Euclidean-first ontology does not by itself imply zero decoherence or a special traveller status.

30.3 Qubit dephasing: a model-specific correlation trade-off

Status: Level A for the algebra of the declared pure-dilation qubit model; Open for an ECT detector channel and for any protocol-independent information–disturbance bound.

Derivation for a two-level system

For a pure-dephasing qubit, the exact reduced-state eigenvalues and information measures are retained in the owning derivation. The calculation tests that source model and does not select an outcome. Take the initial qubit state $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and a pure system–environment dilation whose reduced channel multiplies the off-diagonal element by $e^{-\Phi}$, with $\Phi \geq 0$. Then

$$\rho'_S = \frac{1}{2} \begin{pmatrix} 1 & e^{-\Phi} \\ e^{-\Phi} & 1 \end{pmatrix}, \quad (30.1)$$

and its eigenvalues are $\lambda_{\pm} = (1 \pm e^{-\Phi})/2$. The reduced entropy is therefore

$$S(\rho'_S) = h_2\left(\frac{1+e^{-\Phi}}{2}\right), \quad h_2(p) = -p \ln p - (1-p) \ln(1-p), \quad (30.2)$$

and, because the joint state is pure in this particular dilation,

$$I(S:E) = 2S(\rho'_S) = 2h_2\left(\frac{1+e^{-\Phi}}{2}\right). \quad (30.3)$$

For $\Phi \ll 1$,

$$I(S:E) = \Phi \ln \frac{2e}{\Phi} + O(\Phi^2 |\ln \Phi|), \quad (30.4)$$

whereas $I(S:E) \rightarrow 2 \ln 2$ as $\Phi \rightarrow \infty$.

Equation (30.3) is a monotonic relation between total system–environment mutual information and residual qubit coherence $V = e^{-\Phi}$ under the assumptions just stated. It is not, by itself, a bound on accessible classical information, detector readout capacity, single-copy tomography, collapse, or selection

of one outcome. A mixed environment, a different dilation, feedback, sequential weak measurements, or a specified detector POVM changes the operational information question. No value such as $\Phi = 1$, the one-bit point, or the CHSH visibility marker is a universal quantum–classical boundary. Numerical evaluations of this same declared toy channel, including the one-bit and Werner-visibility markers with their scope guards, are tabulated in Appendix AY.

Sequential weak interactions

If independent pure-dephasing steps multiply the same off-diagonal element, their exponents add,

$$V_N = \prod_{i=1}^N e^{-\Phi_i} = e^{-\sum_i \Phi_i}. \quad (30.5)$$

The final-state mutual information then follows from Eq. (30.3) with $\Phi = \sum_i \Phi_i$, provided the total dilation remains pure. Correlated environments, feedback, adaptive measurements, and detector readout change this relation. Thus the additive statement is a conditional channel result, not a protocol-independent weak-measurement bound or a tomography theorem.

An ECT application must derive the physical record operator, environmental state, channel, and readout protocol. Until those inputs are supplied, this calculation is a source/model example and provides neither an ECT prediction nor an ECT-specific measurement theorem (OP-PES-6/OP-Q19).

Weak measurements

Weak measurements are described by the declared pre-selection, interaction, and post-selection. Their anomalous weak values do not provide a protocol-independent probe of an ECT selection scalar.

Can one observe without collapse? Observation without a projective update is possible through weak or nondemolition protocols, but the information–disturbance tradeoff remains owned by the channel. Reduced dephasing is not synonymous with collapse. Figure 28 displays the exact toy-channel relation and its explicitly non-universal markers.

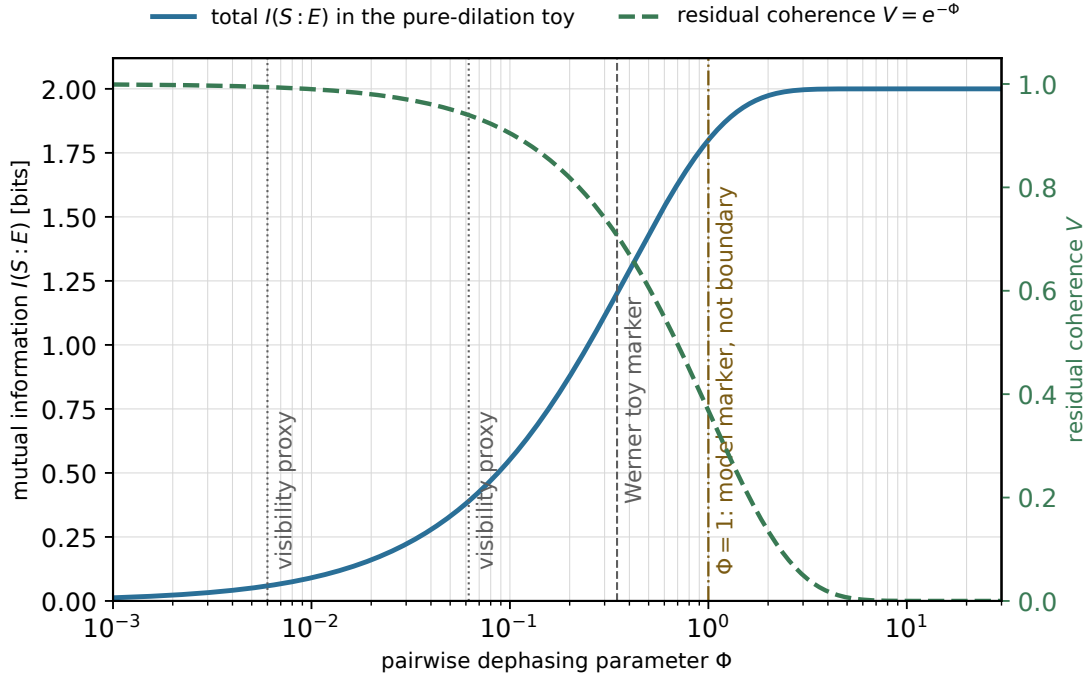


Figure 28: Pure-dilation qubit toy. The blue solid curve is the unchanged total mutual information and the green dashed curve is the unchanged residual coherence. Vertical lines are explicitly labelled proxy or toy markers, not collapse, outcome or metric-signature boundaries. *Display semantics: Both continuous traces are analytic relations of the declared pure-dilation toy; no sampled-data markers are implied.*

Scope and open status. The scope is limited to source-model record formation and finite-time retention. A universal measurement law, Born rule, and unique outcome remain Open.

Euclidean meaning of the reverse segments. Euclidean reverse segments are not inferred from re-coherence. Where reverse-oriented saddles exist, their weights belong to the system boundary-value problem, not to a dephasing exponent.

Medium reading (P5). P5 supplies an ontological motivation for common-medium correlations, while all operational measurement statements still require normalised physical modes and apparatus couplings.

Connection to decoherence and the arrow-of-time programme. Record formation can contribute to an operational arrow for a specified apparatus but does not prove a cosmological thermodynamic arrow. The two questions remain separated.

Discriminant: ECT boundary versus standard decoherence crossover. A quantum–classical crossover is channel- and protocol-dependent. ECT differs from a standard source model only after a condensate calculation predicts a held-out correction.

Falsifier for the quantum–classical boundary programme. The programme fails if a derived ECT channel contradicts interference, re-coherence, or no-signalling data. It is not falsified by the failure of an unowned universal threshold, which is no longer claimed.

Empirical anchor. C_{60} visibility, photonic coherence, and qubit/cavity data are retained as empirical source-model anchors. Their agreement is not evidence for ECT until the ECT channel is derived.

Bridge to the probability and measurement analysis. The bridge to probability requires a separately justified measure on alternatives. Pairwise dephasing changes accessibility of interference, not the origin of Born weights.

30.4 Probability, Born-type interpretation, and measurement status

Status: Level A mathematics for distinguishing amplitudes, reduced-state dephasing, diagonal weights, and a separately admitted probability measure in supplied models. A physical ECT decoherence channel is Level B/Open until its coupling, state, kernel, trace, and timescale are derived. The projector version of Gleason gives a conditional representation theorem only for a supplied complex Hilbert space of dimension at least three and a normalised, noncontextual, countably additive measure on orthogonal projector decompositions; dimension two instead requires the separately declared C2q effect-measure hypotheses stated below. The physical preparation/state, Born measure, event map, outcome/update rule, and measurement channel remain Open. Connections: Sections 22.1, 23.3, 23.1, 23.2, 26.1, and 30.1.

The previous subsections introduced supplied wave-dynamical and canonical templates and analysed dephasing inside declared reduced-state models. They did not derive a physical ECT macroscopic decoherence channel. Such a channel still needs an owned coupling, state, influence/noise kernel, trace, and finite timescale. Even when those inputs suppress specified coherences, dephasing neither creates a probability measure nor selects an outcome. The separate question below is what standard conditional mathematics says after Hilbert/projector and probability inputs are admitted.

Physical picture: the probability gap. In standard quantum mechanics the Born rule $P = |\langle \psi | \phi \rangle|^2$ is postulated. In the present ECT construction a declared environment can suppress interference between specified reduced alternatives, but that dephasing neither creates a probability measure nor chooses one outcome. A complete applicable corrected OS-II package can reconstruct a positive quadratic Hilbert norm in a supplied subsector; reflection positivity alone is insufficient, and neither result identifies diagonal matrix elements with physical frequencies. The physical state/preparation, event map, probability origin, outcome and update therefore remain Open reconstruction targets.

Inputs to the conditional measure representation. For clarity, the discussion separates ordinary C1/C2 projector hypotheses, the alternative C2q effect-measure hypotheses in dimension two, and the C3 physical-application input; OS-II reconstruction and dephasing are not Gleason premises:

- (C1) a complex Hilbert space and projector event algebra are supplied, with $\dim \mathcal{H} \geq 3$ for the ordinary projector theorem;
- (C2) a normalised, noncontextual, countably additive probability measure is supplied on every orthogonal projector decomposition in the ordinary theorem's domain;
- (C2q) for $\dim \mathcal{H} = 2$, a separately supplied generalised measure ν is defined on the full effect set $\mathcal{E}(\mathcal{H}) = \{E : 0 \leq E \leq I\}$, obeys $0 \leq \nu(E) \leq 1$, $\nu(I) = 1$, and is sigma-additive for every countable sequence of effects whose operator sum is at most I . Normalised POVM decompositions with sum I are the corresponding special case. These are the separately declared C2q hypotheses used by Busch's effect version of Gleason's theorem [205];
- (C3) the physical preparation/state is supplied, together with the identification of mathematical projectors (ordinary C2 route), or effects/POVM outcomes (C2q route), with laboratory events. The applicable theorem produces a representing density operator for the admitted measure, but does not derive which physical preparation it represents.

A different dimension-two representation theorem would require its own separately declared premises and is not used here. An independently specified record or dephasing channel may help identify an approximately orthogonal event algebra in an application, but it is not a premise of either representation

theorem. ECT does not yet derive (C1), (C2), (C2q), or (C3) from the bare postulates. Under (C1)–(C2), ordinary projector Gleason fixes the representation in complex dimension at least three. Under the separately declared (C1)+(C2q) hypotheses, the Busch effect route supplies the dimension-two representation. The supplied preparation plus projector/effect event map (C3) is still required for a physical application. None of these steps is a dynamical ECT derivation of probabilities.

Kochen–Specker boundary. The Kochen–Specker theorem separately obstructs a global noncontextual 0/1 assignment to projectors that assigns exactly one value 1 in every orthonormal resolution of the identity, in Hilbert-space dimension at least three [206]. It is not a premise of the Bell–CHSH or MDL bounds and neither supplies Born probabilities nor excludes contextual, nonlocal or measurement-dependent completions. Applying this gate to ECT would first require a physical observable algebra, compatible measurement contexts and an event/value map; those owners remain Open.

Medium reading (P5). P5 motivates seeking amplitudes, environment channels and detector records in one medium. It does not establish an amplitude-to-probability transition. Tracing unresolved modes can dephase a supplied reduced state, but the probability measure, event algebra and outcome rule remain separate owners.

Connection to the global/reduced unitarity distinction. The probability question is logically independent of the global/reduced unitarity distinction (§23.2); decoherence can make a supplied density operator approximately diagonal without creating the measure used to read its entries. Standard global unitary dynamics is established only in supplied subsectors for which the complete applicable corrected OS-II package and reconstruction are in place; reflection positivity alone is insufficient, and the interacting full-theory statement remains Open. The conditional measure-representation discussion begins only after a normalised, noncontextual, countably additive projector measure has been independently admitted. Approximate diagonality identifies a possible record basis in a specified channel, not the probability or outcome law.

Amplitude versus probability

Within the supplied coherent closure used in this section, the bookkeeping objects are amplitudes with weights of the form (22.1),

$$\mathcal{A}[\theta] \sim \exp\left(\frac{i\Sigma_{\theta}\mathcal{I}_{\theta,L}[\theta]}{S_0}\right),$$

and their declared superposition over admissible histories. This is an EFT input, not a derivation of the amplitude space, measure, or state from DP. In regimes where those supplied sectors interfere, their description remains amplitude-based; no probability or outcome interpretation follows.

One specified record protocol may proceed from a supplied amplitude/state to a reduced description in which selected coherences are suppressed. This gives diagonal weights, not probabilities. For that supplied protocol the bookkeeping is

$$\begin{aligned} \text{supplied amplitudes and state} &\longrightarrow \text{specified reduced-state dephasing} \\ &\longrightarrow \text{diagonal weights.} \end{aligned} \tag{30.6}$$

This is not a universal temporal or logical order. An additive probability measure is an independent premise and may be formulated without decoherence; the physical ECT channel, event interpretation, and outcome/update remain Open.

Why specified dephasing is not the Born rule

A declared system–environment channel can make selected off-diagonal reduced terms small under its stated assumptions. That calculation does not explain why diagonal weights are probabilities or why their

law is quadratic. An approximately diagonal supplied density matrix is therefore neither a derivation of Born's rule nor an ECT-owned probability mechanism.

This distinction must remain explicit. Many discussions conflate four conceptually separate steps:

- (i) suppression of interference;
- (ii) emergence of classical alternatives;
- (iii) assignment of probabilities;
- (iv) the specific quadratic weighting rule.

In ECT these four steps are kept conceptually distinct.

Ordinary projector-Gleason representation of an independently supplied measure

A Born-type probability law is not derived from first principles in ECT. The present comparison keeps four separately owned layers distinct:

- (i) a supplied coherent amplitude and action scale;
- (ii) a supplied norm-preserving complex dynamics;
- (iii) the ordinary projector hypotheses (C1)–(C2), or the dimension-two effect hypotheses (C1)+(C2q), plus the supplied preparation and projector/effect event map (C3);
- (iv) an optional specified reduced-state dephasing channel used only at the application/record layer, whose physical ECT owner remains Open.

The applicable representation theorem under (C1)–(C2), or under the separately declared (C1)+(C2q) hypotheses, is conditionally unique; the physical preparation, event map, probability premise, record channel, and outcome are not generated by the theorem. In the projector theorem's domain there is a unique positive trace-one operator ρ_μ representing the admitted measure. If (C3) supplies a normalised pure preparation $\rho_\mu = |\Psi\rangle\langle\Psi|$ and a rank-one event $P_\alpha = |\phi_\alpha\rangle\langle\phi_\alpha|$, then

$$\mu(P_\alpha) = \text{Tr}(\rho_\mu P_\alpha) = |\langle\phi_\alpha|\Psi\rangle|^2. \quad (30.7)$$

This is the standard representation of the admitted measure. It is not produced by dephasing and does not identify a physical branch or realised outcome without the missing event and update owners.

Status of Eq. (30.7): conditional Gleason representation. For a complex Hilbert space of dimension at least three, assumptions (C1)–(C2) place the admitted measure in the domain of Gleason's theorem [207]. The measure then has the unique representation $\mu(P) = \text{Tr}(\rho_\mu P)$ by a positive trace-one operator. The ordinary projector theorem does not cover a two-dimensional Hilbert space; a qubit statement requires the separately stronger Busch effect/POVM extension under separately declared C2q hypotheses. Under the complete chosen hypotheses there is no extra functional-form freedom in the representation.

What this establishes: Gleason gives a conditional representation theorem. Once the relevant Hilbert/projector structure, dimension, normalised noncontextual countable additivity, and theorem domain are supplied, the representing form is fixed. This is Level A mathematics inside the stated theorem, not the physical origin of the measure, state, events, or outcomes.

What this does not establish: ECT has not yet derived the full Hilbert/projector event structure, probability measure, or physical preparation map from bare P1–P6. The Gleason representation applies *conditional* on (C1)–(C2), and its physical reading additionally depends on (C3). The physical ECT Born rule therefore remains Open (OP-Q7). Only the conditional Gleason form, after its probability assumptions are supplied, is Level A mathematics inside that stated model.

OS-II Hilbert-norm layer and its separation from probability

The decision-theoretic routes of Deutsch [208] and Wallace [209] obtain quadratic weights from rationality axioms in a branching quantum world. Zurek's distinct envariance route [210] argues from entangled-state symmetries and its own probability assumptions. In ECT a quadratic *norm* can be reconstructed conditionally from a supplied Euclidean model only when its complete applicable corrected OS-II package holds. Reflection positivity is one necessary hypothesis, not the full reconstruction package or a Gleason premise. Calling the norm a probability still requires the independent probability assumptions listed above.

A supplied Euclidean condensate functional measure (23.1) is one input from which the quadratic Hilbert structure can be reconstructed under the complete applicable corrected OS-II package. For functionals F, G supported in the half-space $w > 0$, the OS route (§23.1) defines a physical inner product:

$$\langle F|G \rangle_{\text{OS}} = \frac{1}{Z_\mu} \int \mathcal{D}\Phi (\Theta F[\Phi])^* G[\Phi] e^{-\mathcal{I}_E^\mu[\Phi]}. \quad (30.8)$$

This is the OS form of the supplied measure. Calling its completion physical requires the remaining OS-II and observable/clock maps; the measure exponent does not contain the canonical S_0 unless a separate owner map is proved. This inner product is quadratic in the amplitudes by construction: it is a defining property of the Euclidean measure and the reflection operation, not an auxiliary assumption about probability. RP guarantees a positive semi-definite OS form. Quotienting its null states and completing gives a mathematical Hilbert space for that form; identifying it as the reconstructed physical Hilbert space $\mathcal{H}_{\text{phys}}$ with time-translation dynamics requires the complete applicable corrected OS-II package and the physical observable/clock map.

Once the OS-reconstructed Hilbert space is in place, the diagonal elements of a reduced density matrix are automatically of the form

$$\rho_{\alpha\alpha} = \langle \alpha | \hat{\rho} | \alpha \rangle = |\langle \alpha | \Psi \rangle|^2 \quad (\text{pure state}). \quad (30.9)$$

This is an algebraic pure-state diagonal identity in the RP-defined inner product, not a probabilistic theorem. The diagonal matrix elements inherit the quadratic structure of $\langle \cdot | \cdot \rangle_{\text{phys}}$, while their interpretation as normalised physical probabilities remains an additional step.

The conditional comparison chain is therefore:

$$\begin{array}{ccc}
 \underbrace{\mathcal{I}_E^\mu, d\mu, \Theta, \mathcal{A}_{\text{obs}}}_{\text{supplied Euclidean/OS inputs}} & \xrightarrow{\text{complete applicable corrected OS-II package}} & \underbrace{\mathcal{H}_{\text{phys}}, \langle \cdot | \cdot \rangle}_{\text{quadratic}} \\
 & & \text{quadratic} \\
 \underbrace{\mathcal{H}_{\text{phys}}, \langle \cdot | \cdot \rangle}_{\text{quadratic}} & \xrightarrow{\text{independent input}} & \underbrace{\text{projector events + normalised noncontextual measure?}}_{\text{(C1)–(C2)}} \\
 \underbrace{\text{projector events + normalised noncontextual measure}}_{\text{(C1)–(C2)}} & \xrightarrow{\text{Gleason representation}} & \mu(P) = \text{Tr}(\rho_\mu P), \\
 \underbrace{\rho_\mu, P_\alpha}_{\text{representation}} & \xrightarrow{\text{supplied preparation and projector/effect event map (C3)}} & \mu(P_\alpha), \\
 & & \mu(P_\alpha) = |\langle \phi_\alpha | \Psi \rangle|^2 \quad \text{for a supplied pure state.}
 \end{array} \quad (30.10)$$

For a two-dimensional sector, the projector-measure step is replaced by the full effect algebra and separately declared C2q hypotheses, giving $v(E) = \text{Tr}(\rho_v E)$; C3 then maps effects/POVM outcomes to laboratory events. This is a separate theorem domain, not a consequence of the projector chain. Dephasing remains a separate reduced-state record step, not a theorem premise. The quadratic Hilbert norm is inherited from a complete OS reconstruction only in the verified subsector; the physical state/event map, probability premise, and one realised outcome are not inherited from it.

Comparison with decision-theoretic routes. The Deutsch–Wallace programme starts from an assumed Hilbert space and derives the Born rule from rationality constraints on agents. In supplied Gaussian subsectors, ECT starts from a Euclidean functional measure and reconstructs a Hilbert norm only under the complete applicable corrected OS-II package. A Born probability does not follow without the separate probability-measure assumptions. The Deutsch–Wallace programme is therefore an external comparison, not a step derived by the present ECT construction.

Residual gap and honest status. The RP/OS route narrows the open gap, but it does not close it:

- (C1) is available only in explicitly supplied Gaussian OS-reconstructed subsectors; the interacting Hilbert/event algebra is a constructive target (OP-Q16);
- (C2) is an independent projector-measure premise in the ordinary dimension-at-least-three theorem domain. Its normalisation, noncontextuality, and countable additivity are not derived by RP, OS reconstruction, or decoherence;
- (C2q) is the separately supplied full-effect generalised measure premise in dimension two, with the positivity, normalisation and sigma-additivity hypotheses stated above; it too is not derived by RP, OS reconstruction, or decoherence;
- (C3) remains a physical owner problem: neither the prepared state nor the laboratory projector or effect/POVM-event identification is selected by the representation theorem;
- the **record/dephasing layer** remains independently Open as an ECT channel and is not a hypothesis of Gleason’s theorem.

Level summary for the separate OS-II and Gleason layers. *Level A mathematics under stated hypotheses:* the complete supplied free-Gaussian corrected OS-II packages and, independently, ordinary projector Gleason’s representation of the admitted measure under (C1)–(C2), with a qubit handled only by the Busch effect/POVM theorem under separately declared C2q hypotheses. *Level B:* conditional reduced-state dephasing and approximate record sectors for a declared system–environment split; diagonal identities remain kinematical. *Open:* ECT ownership of (C1), (C2) or (C2q), and (C3), the interacting OS completion, the physical state/preparation and projector/effect event map, the record channel, probability origin, and the outcome/update rule.

An independently supplied projector measure therefore has only the conditional Gleason trace representation after the exact theorem hypotheses are supplied. This is not a completed ECT Born derivation from P1–P6; the physical state/preparation, event map, probability origin, outcome and update remain Open.

Reduced density matrix viewpoint

In a supplied reduced description one may write the schematic matrix

$$\rho_{\text{red}}(q, q') = \sum_{\alpha, \beta} \Psi_{\alpha}(q) \Psi_{\beta}^*(q') \mathcal{D}_{\alpha\beta}, \quad (30.11)$$

where $\mathcal{D}_{\alpha\beta}$ is the channel-dependent dephasing factor of that model. If its declared coupling, state, kernel, trace, and timescale give

$$\mathcal{D}_{\alpha\beta} \approx 0 \quad (\alpha \neq \beta), \quad (30.12)$$

the matrix is approximately diagonal. Its diagonal entries remain weights, not probabilities, until an event algebra and probability measure are supplied.

What this does and does not show. The calculation shows only dephasing in the specified source model. It neither derives a physical ECT channel nor makes a probability interpretation natural or necessary. Gleason fixes a trace representation only after (C1)–(C2) are admitted in their precise theorem domain; its physical Born reading also requires (C3). Deriving these inputs, the physical channel, and the outcome/update rule remains Open.

Measurement status

ECT does not yet contain a completed first-principles measurement theory. The manuscript can organise the following supplied bookkeeping ingredients without claiming their physical derivation: coherent amplitudes, a specified reduced-state dephasing map, diagonal record weights, and a conditional Gleason representation after a probability measure is admitted. The physical coupling, state, kernel, trace, timescale, stable-record map, probability measure, and outcome/update rule all remain Open; no universal branching or Born mechanism follows from the displayed chain.

What remains open. A full measurement theory would require:

- (i) a precise branch-selection and state-update rule;
- (ii) a physical derivation of the state/preparation, event map and probability origin from the condensate dynamics;
- (iii) a treatment of pointer sectors and stable macroscopic records;
- (iv) control of the rôle of defect and topological sectors during measurement-like processes (Section 34).

Why the current result is still meaningful

The present construction usefully separates Hilbert kinematics, reduced-state dephasing, probability assignment, and outcome selection. It does not yet explain why physical probabilities exist or why one outcome is realised. Its positive result is a scoped conditional comparison route, not a solution of the Born or measurement problem.

Bridge to entanglement and Bell-type correlations

The probability and measurement discussion has so far used supplied reduced-state dephasing models and conditional record alternatives. The next step analyses the case of supplied coherent correlations that are non-factorisable and are therefore not exhausted by that conditional bookkeeping. Neither the dephasing model nor an effectively classical alternative is thereby promoted to an ECT-derived physical channel. That is the subject of the next section on entanglement and Bell-type correlations.

Discriminant: ECT conditional measure-representation programme versus standard probability postulates. Standard formulations of quantum theory usually take the Born rule as an axiom or attempt to justify it through auxiliary principles such as decision theory. ECT treats the origin of the physical probability measure as Open. The present text organises supplied coherent amplitudes and derives or reproduces reduced-state dephasing only inside specified source models; the corresponding physical ECT channel remains Open, and neither step creates a measure. Ordinary projector Gleason asks what trace form an independently supplied normalised, noncontextual, countably additive projector measure must take on a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. The ECT-specific programme is to derive the missing state/preparation, event, probability-origin, outcome and update structures from the medium. At present only the conditional trace representation is available after those inputs are supplied.

Falsifier for the Born-weight programme. The present programme would be weakened if a completed ECT derivation showed that the physical alternatives do not admit the Hilbert/projector structure assumed in (C1), or that the measure premise (C2) is inconsistent with the coherent-branch architecture. It would be more strongly challenged if a completed derivation forced a non-quadratic probability rule while retaining the same Hilbert-space structure. At present, no such incompatibility is known; the open point is not inconsistency, but incompleteness.

Empirical status. The quadratic Born weighting is extraordinarily well confirmed across atomic, molecular, optical, and solid-state quantum experiments. ECT does not claim a distinct experimental prediction at this stage. Its present contribution is architectural: it identifies precisely which inputs are needed before the conditional quadratic representation applies.

Status summary. *Established (Level A mathematics in stated models):* the logical distinction among amplitudes, reduced-state dephasing, diagonal weights, and an admitted probability measure; complete applicable corrected OS-II packages for supplied Gaussian subsectors reconstruct a positive quadratic Hilbert norm but not physical probabilities.

Conditional theorem: under (C1)–(C2), including a complex Hilbert space with $\dim \mathcal{H} \geq 3$ and normalised noncontextual countable additivity, ordinary projector Gleason’s theorem [207] fixes the trace representation of the independently admitted measure; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. The physical application still needs (C3). OS-II reconstruction is a separate layer and is not a Gleason premise.

Open: (i) full interacting RP verification (OP-Q16); (ii) the probability origin/premise (C2) and physical state, preparation plus projector/effect event map (C3); (iii) the system–environment coupling, state, kernel, trace, timescale, and projector/record channel at the independent application layer; (iv) the outcome/update rule.

Summary. ECT does not postulate probability at the foundation. A specified supplied dephasing model can suppress declared coherences, but no physical ECT channel has yet been derived and dephasing does not create a probability measure or select an outcome. Once (C1)–(C2) are independently admitted in the exact theorem domain, Gleason conditionally fixes the trace representation; a physical Born application still needs (C3). The channel, origin of the measure, physical preparation plus projector/effect event map, actual outcomes, and update rule remain Open. The next step is to analyse the case of a supplied coherent correlation structure that is non-factorisable before branch-based probability language is invoked.

Table 89: Comparison of Born-rule justification routes and the ECT conditional measure-representation programme.

Aspect	Deutsch (1999) [208]	Wallace (2012) [209]	ECT (this work)
Framework	MWI + decision theory	MWI + decoherence	Supplied coherent/dephasing templates + Hilbert norm from a complete applicable corrected Gaussian OS-II package + independent ordinary projector-Gleason representation of an admitted measure; physical ECT owners Open
Source of weights	Rationality axioms	Rationality + branch structure	Under (C1)–(C2), or under separately declared (C1)+(C2q) hypotheses in dimension two, the admitted measure has a unique trace representation; physical use additionally requires the preparation plus projector/effect event map (C3)

(continued on next page)

Aspect	Deutsch (1999) [208]	Wallace (2012) [209]	ECT (this work)
Origin of quadratic structure	Decision-theoretic axiom	Rationality + separability	Hilbert norm from a specified complete applicable corrected OS-II package plus an applicable independently supplied projector measure (C2), or full-effect generalised measure (C2q) in dimension two; OS-II is not a Gleason premise
Dephasing / decoherence	Not used	Postulated	Derived or reproduced only inside specified source models with declared state, split, coupling, kernel, trace and timescale; physical ECT owner Open and not a probability owner
Extra assumptions	Rationality	Rationality + separability	(C1)–(C2) for the ordinary projector theorem, or separately declared (C1)+(C2q) full-effect hypotheses in dimension two; (C3) supplies the preparation plus projector/effect event map
Known limitations	Circularity [211]	Albert (2010) objection	Conditional representation only; first-principles derivation of (C1), (C2) or (C2q), and (C3), the probability owner, and unique outcome selection still open (OP-Q19)

31 Quantum Entanglement and Bell-Type Correlations

Status: Level A standard mathematics for a supplied tensor-product state/channel and its reduced density operators; ECT ownership Open. A common medium alone does not force nonfactorisation. The physical subsystem/state, local measurement map, Bell correlators, Tsirelson-type bounds and no-signalling completion remain Open. Connection: the Hilbert-space bridge (§23.1), the global/reduced unitarity distinction (§23.2), the exchange-sector topology (§24.4), the decoherence programme (§26.1), the vacuum-response infrastructure (§25), the Born-weight route (§30.4), medium character (P5), and the quantum–classical boundary (§30.1).

Physical picture: entanglement as a trace of a common configuration. In the standard formulation, entanglement is introduced as mathematical non-factorisability in a tensor product of Hilbert spaces. ECT may host such a state in a common medium, but common support alone neither creates a tensor-product Hilbert space nor forces the joint state to be nonfactorising. The physical subsystem map, nonfactorising state/channel, local measurement operators and no-signalling/Bell correlators must be supplied or derived independently. A common-medium picture is therefore an interpretation of an already specified entangled state, not its Level B derivation.

Connection to PES-R. Shared-medium language is an interpretation of a previously specified joint state; it does not by itself derive entanglement, Bell correlators, or a no-signalling theorem. PES-R may classify the retention and record distinguishability of modes after the joint Hilbert-space dynamics and subsystem map have been established. Neither $\Phi_{ab} \ll 1$ nor $\Phi_{ab} \gg 1$ defines a metric signature, an interaction law, or a measurement update. The ECT derivation of Bell-inequality-violating correlators and the physical subsystem/record operators therefore remains Open (OP-Q20/OP-PES-1).

What is established, what is structurally suggested, and what remains open. The tensor-product and reduced-state language used below presupposes the physical Hilbert-space bridge discussed in Section 23.1.

The existing infrastructure provides only conditional ingredients: specified Gaussian correlators, influence functionals after a system–environment split, and standard reduced-state algebra. These tools can propagate or degrade a nonfactorising state once it is supplied. They do not prove that the physical ECT state is nonfactorising, derive a Born measure, or select local measurement operators. The full ECT state/channel and Bell map remain Open.

Medium reading (P5). P5 motivates looking for a common-medium owner, but factorising and nonfactorising states are both compatible with a common substrate. No generic nonfactorisation theorem follows without an interaction, boundary condition or state-selection rule that fixes the cross-correlators.

No-signalling and operational locality. The common-medium reading of entanglement does not imply operational superluminal signalling. For a supplied quantum channel, no-signalling and Bell inequalities can be checked by the standard operational rules. The present ECT action has not yet derived that channel, so it claims neither a signalling mechanism nor a medium-based derivation of Bell violation.

Connection to exchange sectors. For systems of indistinguishable excitations, the entanglement programme is constrained by the exchange-sector topology established in Section 24.4. The two exchange homotopy classes of a supplied identical-point configuration space constrain possible representations, but do not select the physical bosonic/fermionic representation or particle sector. This does not mean that all entanglement reduces to exchange structure. Rather, exchange topology provides part of the kinematical backbone for entanglement in the indistinguishable-sector case, while the broader non-factorisable correlation logic applies to distinguishable reduced subsystems as well.

Minimal structural statement. Given a supplied total state, $\hat{\rho}_{AB} = \text{Tr}_{\text{env}} \hat{\rho}_{\text{tot}}$ may factorise or not; the trace operation alone decides neither case. A nonzero connected cross-correlator or an explicit nonseparable preparation/channel is the required owner. Thus ECT currently supplies a place where an entanglement mechanism could live, but not a theorem that the medium forces nonseparability or Bell violation. There is likewise no amplitude $\rightarrow$ decoherence $\rightarrow$ probability derivation.

Discriminant with respect to axiomatic QM and local hidden-variable pictures. Three correlation pictures should be kept distinct. In axiomatic quantum mechanics, entanglement is treated as a primitive property of the tensor-product state space. In local hidden-variable models, Bell-violating correlations are forbidden once factorisability and measurement independence are imposed. ECT has not yet been placed in either class operationally because its physical joint state and local measurement map are missing. A viable completion must reproduce Bell-type correlations and no-signalling without reducing to an excluded Bell-local hidden-variable model.

CHSH premise ledger: the common-measure step. The standard CHSH conclusion is a theorem about a specified probability class, not about deterministic language by itself [212, 213, 214]. Let

$x, y \in \{0, 1\}$ denote settings and $a, b \in \{-1, +1\}$ denote outcomes. A Bell-factorisable model with measurement independence has

$$P(a, b | x, y) = \int_{\Lambda} d\rho(\lambda) p_A(a | x, \lambda) p_B(b | y, \lambda), \quad d\rho(\lambda | x, y) = d\rho(\lambda), \quad (31.1)$$

where $d\rho$ is one positive normalised measure for all four setting pairs. Define the bounded conditional response means

$$A_x(\lambda) = \sum_{a=\pm 1} a p_A(a | x, \lambda), \quad B_y(\lambda) = \sum_{b=\pm 1} b p_B(b | y, \lambda), \quad |A_x|, |B_y| \leq 1. \quad (31.2)$$

Factorisability gives $E_{xy} = \int d\rho A_x B_y$. With the convention

$$S = E_{00} + E_{01} + E_{10} - E_{11}, \quad (31.3)$$

all four terms may be combined under the same measure:

$$S = \int_{\Lambda} d\rho(\lambda) C(\lambda), \quad (31.4)$$

$$C(\lambda) = A_0(B_0 + B_1) + A_1(B_0 - B_1). \quad (31.5)$$

For each λ ,

$$|C(\lambda)| \leq |B_0 + B_1| + |B_0 - B_1| = 2 \max\{|B_0|, |B_1|\} \leq 2, \quad (31.6)$$

and therefore

$$|S| \leq 2. \quad (31.7)$$

For deterministic responses $A_x, B_y \in \{-1, +1\}$ the integrand is exactly $+2$ or -2 . Thus the proof uses, separately, (i) dichotomic bounded outcomes, (ii) Bell factorisability of the conditional responses, and (iii) one setting-independent probability measure. Dropping any one premise removes this particular inference; it does not by itself select a replacement theory or a value of S .

What a CHSH violation excludes. A statistically validated value $|S| > 2$ does not rule out hidden variables as such. It excludes the Bell-local common-measure class in which one fixed preparation is represented on a common latent space by one positive normalised setting-independent measure, bounded outcomes, and Bell- factorisable response functions used consistently in all four setting contexts. Nonlocal, contextual, measurement-dependent, retrocausal, atemporal global-constraint and selection-dependent hidden-variable completions are not excluded by this inequality alone. Conversely, dropping one premise neither derives the observed quantum correlation set nor proves operational no-signalling or the Tsirelson ceiling.

The present ECT source has not derived a physical preparation-level latent variable or quantum state, setting generators, event measure, detector instruments or complete conditional response law. It is therefore not presently classified as Bell-local, nonlocal, measurement-dependent, retrocausal or superdeterministic. Bell violation is neither a current confirmation nor a current falsification of ECT; it is a mandatory external gate for any completed operational model.

What changes for a setting-dependent measure. Suppose the local response maps are retained but the conditional hidden-state measure is allowed to depend on the joint setting:

$$P(a, b | x, y) = \int_{\Lambda} d\rho_{xy}(\lambda) p_A(a | x, \lambda) p_B(b | y, \lambda). \quad (31.8)$$

The four correlators are then integrals with four measures. They may be represented relative to a common dominating measure, but in general with four different Radon–Nikodym weights; the single common-weight pointwise CHSH integrand (31.5) is therefore unavailable. Hence $d\rho_{xy} \neq d\rho$ invalidates the common-measure proof of Eq. (31.7). This statement is only logical: setting dependence permits but does not guarantee a violation, and a statistical dependence $P(\lambda | x, y) \neq P(\lambda)$ does not identify whether its physical explanation, if any, is a common cause, retrocausal influence, global constraint, contextual preparation, or selection effect.

Explicit 16-state construction: a Tsirelson-value target. The following is a generic finite probability construction, not a model derived from ECT. Related measurement-dependent locally factorisable deterministic models are known independently of ECT [215]. Take

$$\Lambda = \{-1, +1\}^4, \quad \lambda = (\alpha_0, \alpha_1, \beta_0, \beta_1), \quad A_x(\lambda) = \alpha_x, \quad B_y(\lambda) = \beta_y. \quad (31.9)$$

The response at each wing is deterministic and depends only on the local setting once λ is fixed. Let $\eta_{00} = \eta_{01} = \eta_{10} = +1$, $\eta_{11} = -1$, and $c = 1/\sqrt{2}$. For each setting pair define

$$\rho_{xy}^{(T)}(\lambda) = \frac{1 + c \eta_{xy} \alpha_x \beta_y}{16}. \quad (31.10)$$

Because $|c| \leq 1$, every weight is non-negative. Summation over the 16 states gives

$$\sum_{\lambda} \rho_{xy}^{(T)} = 1, \quad \langle A_x \rangle_{xy} = \langle B_y \rangle_{xy} = 0, \quad E_{xy} = c \eta_{xy}. \quad (31.11)$$

Consequently

$$S = 4c = 2\sqrt{2}, \quad P(A = a | x, y) = P(B = b | x, y) = \frac{1}{2}. \quad (31.12)$$

The last identities verify operational no-signalling for these four finite distributions, but do not supply a causal explanation of them.

For the total-variation convention

$$M_{TV} = \max_{s,t} \text{TV}(\rho_s, \rho_t), \quad \text{TV}(p, q) = \frac{1}{2} \sum_{\lambda} |p(\lambda) - q(\lambda)|, \quad (31.13)$$

the signed products $\eta_{xy} \alpha_x \beta_y$ associated with two distinct setting pairs differ on eight of the 16 states. Therefore

$$M_{TV} = \frac{c}{2} = \frac{1}{2\sqrt{2}}. \quad (31.14)$$

This value characterises this construction under the declared TV convention; it is not claimed to be a minimal measurement-dependence cost.

Counterexample: setting dependence without Bell violation. On the same state space and with the same local deterministic responses, set

$$\tilde{\rho}_{xy}(\lambda) = \frac{1 + \frac{1}{2} \alpha_{1-x} \beta_{1-y}}{16}. \quad (31.15)$$

These four measures are setting dependent, non-negative and normalised, but they correlate only the response bits not sampled at the chosen pair. Direct summation gives

$$E_{00} = E_{01} = E_{10} = E_{11} = 0, \quad S = 0, \quad M_{TV} = \frac{1}{4}, \quad (31.16)$$

with uniform one-party marginals. Thus violation is not a consequence of setting dependence alone.

A TV-controlled sufficient bound. The failure of the common-measure proof can be quantified without assigning a physical mechanism. Choose ρ_{00} as a reference and add and subtract it in the other three terms:

$$S = \int d\rho_{00} [A_0 B_0 + A_0 B_1 + A_1 B_0 - A_1 B_1] \quad (31.17)$$

$$+ \int (d\rho_{01} - d\rho_{00}) A_0 B_1 + \int (d\rho_{10} - d\rho_{00}) A_1 B_0 - \int (d\rho_{11} - d\rho_{00}) A_1 B_1. \quad (31.18)$$

For $|f| \leq 1$, $|\int f d(p - q)| \leq 2\text{TV}(p, q)$. The first line of Eq. (31.18) has magnitude at most two, so

$$|S| \leq 2 + 2[\text{TV}(\rho_{01}, \rho_{00}) + \text{TV}(\rho_{10}, \rho_{00}) + \text{TV}(\rho_{11}, \rho_{00})] \leq 2 + 6M_{\text{TV}}. \quad (31.19)$$

Combining this with the algebraic ceiling gives the sufficient general bound

$$|S| \leq \min\{4, 2 + 6M_{\text{TV}}\}. \quad (31.20)$$

No tightness claim is made. The bound also holds for stochastic factorisable responses because the conditional response means remain in $[-1, 1]$.

Measurement-dependence metrics and the corrected Hall comparison. Hall's joint L^1 diameter M_{H} and the total-variation diameter used here obey $M_{\text{H}} = 2M_{\text{TV}}$. In Hall's deterministic no-signalling model class the relaxed CHSH relation is [216]

$$|S| \leq \min\{2 + 3M_{\text{H}}, 4\} = \min\{2 + 6M_{\text{TV}}, 4\}. \quad (31.21)$$

Consequently, within that declared class,

$$M_{\text{TV}} \geq \max\left\{0, \frac{|S| - 2}{6}\right\}. \quad (31.22)$$

The four-setting targets are

target	M_{H}	M_{TV}
$ S = 2\sqrt{2}$	$2(\sqrt{2} - 1)/3$	$(\sqrt{2} - 1)/3 \simeq 0.138071$,
$ S = 4$	$2/3$	$1/3$.

(31.23)

The $1/3$ value is therefore the cost of the algebraic maximum in this metric, not of the quantum maximum.

The published erratum to Hall's explicit model for the full continuum of projective singlet correlations [217] corrected its global diameter to $M_{\text{H}, \text{singlet}} \simeq 0.276434$, or

$$M_{\text{TV}, \text{singlet}} \simeq 0.138217, \quad F_{\text{singlet}} = 1 - M_{\text{TV}, \text{singlet}} \simeq 0.861783 \quad (31.24)$$

Thus the often quoted approximately 14% statement means $1 - F = M_{\text{TV}} \simeq 0.1382$ in Hall's convention. It is not a setting-predictability probability, mutual information or a universal fraction of free choice. The corrected full- singlet construction is close to, but does not exactly saturate, the four- setting floor above. The finite construction in Eq. (31.10), with $M_{\text{TV}} = 1/(2\sqrt{2})$, is nonminimal and is not ECT physics.

Mutual-information measures quantify a different resource. Barrett and Gisin constructed Bell-local correlated-setting models reproducing all projective singlet correlations while Bob's setting remains independent of the latent variable and with at most one bit of mutual information between Alice's setting and the latent variable [218]. This is a model-class existence upper bound, not a proved minimum, a conversion to M_{TV} or (ℓ, h) , or an ECT parameter.

Measurement-dependent-local inequalities as a future test. A bounded measurement-dependent-local (MDL) class may instead be defined by

$$\ell \leq P(x, y | \lambda) \leq h \quad (31.25)$$

for every allowed setting pair and latent state. The resulting behaviours form a convex polytope with Bell-like inequalities adapted to the declared bounds. For every strictly positive lower bound $\ell > 0$, suitable quantum correlations escape that MDL polytope [219]; this result does not cover unrestricted dependence with $\ell = 0$.

Two experiments illustrate that this bounded class is empirically testable. Using the additional observed-setting condition $P(x, y) = 1/4$, Aktas *et al.* violated the corresponding MDL inequality with a

partially entangled photon pair. Their data gave violations for $\ell_{\text{raw}} > 0.090(4)$ and, after accidental-count subtraction, $\ell_{\text{net}} > 0.074(6)$ [220]. The detection and locality loopholes remained open, and settings were changed only between successive run series. Liu *et al.* later used human-generated real-time settings and spacelike-separated measurement sites, reporting an MDL violation incompatible with the declared local models for $\ell > 0.10 \pm 0.05$ under their protocol assumptions [221]. These are external exclusions within specified bounded MDL classes: they do not measure a physical ℓ , cover unrestricted dependence with $\ell = 0$, or supply an ECT mechanism.

For ECT, Eq. (31.25) is a future falsification framework, not a present mechanism. The theory would first have to derive the preparation-level λ , setting generators, stable values or bounds for ℓ, h , the complete $P(a, b, x, y | p)$, detector/selection map and likelihood. There is no model-independent conversion between M_{TV} and (ℓ, h) without additional assumptions.

No-signalling does not supply the Tsirelson bound. Replacing $c = 1/\sqrt{2}$ by $c = 1$ in Eq. (31.10) preserves positivity, normalisation and exactly uniform one-party marginals, but yields $S = 4$. Thus a setting-dependent local-response construction may be operationally no-signalling and still exceed $2\sqrt{2}$. The value $2\sqrt{2}$ in Eq. (31.12) was selected as a target; it was not derived. In standard quantum mathematics the Tsirelson bound instead follows when one positive normalised state owns all four correlators and the Bell observables are bounded self-adjoint projective dichotomic observables (Hermitian unitaries),

$$\hat{A}_x^\dagger = \hat{A}_x, \quad \hat{B}_y^\dagger = \hat{B}_y, \quad \hat{A}_x^2 = \hat{B}_y^2 = \mathbb{I}, \quad [\hat{A}_x, \hat{B}_y] = 0 \quad \text{for all } x, y.$$

For

$$\widehat{\mathcal{S}} = \hat{A}_0(\hat{B}_0 + \hat{B}_1) + \hat{A}_1(\hat{B}_0 - \hat{B}_1), \quad (31.26)$$

$\widehat{\mathcal{S}}$ is self-adjoint and the standard identity

$$\widehat{\mathcal{S}}^2 = 4\mathbb{I} - [\hat{A}_0, \hat{A}_1][\hat{B}_0, \hat{B}_1]$$

holds. Hermitian unitarity gives $\|\hat{A}_0, \hat{A}_1\| \leq 2$ and $\|\hat{B}_0, \hat{B}_1\| \leq 2$, hence

$$\|\widehat{\mathcal{S}}\|^2 = \|\widehat{\mathcal{S}}^2\| \leq 8, \quad \|\widehat{\mathcal{S}}\| \leq 2\sqrt{2}.$$

Therefore every one common positive normalised state ω obeys $|\omega(\widehat{\mathcal{S}})| \leq 2\sqrt{2}$ [222, 223]. The involution identities $\hat{A}_x^2 = \hat{B}_y^2 = \mathbb{I}$ without self-adjointness are not sufficient for this norm proof. ECT would have to derive the physical state, operator algebra and locality map entering the stated hypotheses, or derive another comparably restrictive principle; the finite construction does neither.

Three CHSH ceilings and one common quantum state. The local Bell bound, the quantum Tsirelson bound and the algebraic bound,

$$|S| \leq 2, \quad |S| \leq 2\sqrt{2}, \quad |S| \leq 4, \quad (31.27)$$

answer different questions. The first follows from the Bell-local common-measure assumptions above. The second follows when one positive normalised state owns all four Bell-operator words and the bounded self-adjoint dichotomic and cross-party commutation hypotheses in the preceding paragraph hold. Four positive setting-conditioned states need not combine into that single functional. The third is the algebraic maximum and is compatible with no-signalling tables outside the standard quantum set.

The Osterwalder–Schrader programme is a conditional route to a positive Hilbert-space representation, not by itself a Bell completion. It requires the complete applicable reconstruction package and one physical preparation, local observable algebras, detector effects, event map and setting protocol. Reflection positivity of separate context-conditioned measures does not manufacture the common Bell state. Conversely, an OS/Hilbert representation and a contextual or setting-dependent hidden-state representation are not logical negations: the same operational table may admit more than one representation. A future ECT completion must derive why its full correlation family lies inside the quantum ceiling rather than merely selecting one value of S .

Conditional global-configuration template for ECT. The Euclidean boundary-value character of the bare model motivates a precise question but does not answer it. Let p denote a physical preparation, let $[\Gamma] \in \mathfrak{C}_{\text{phys}}$ denote a complete global configuration after all applicable constraints and gauge identifications, and let x, y denote physical apparatus settings. If a positive normalised conditional measure and gauge-invariant record maps were supplied, one could write

$$E_{xy}^{\text{glob}} = \int_{\mathfrak{C}_{\text{phys}}} d\mu_{p,xy}([\Gamma]) \mathcal{A}_x([\Gamma]) \mathcal{B}_y([\Gamma]), \quad |\mathcal{A}_x|, |\mathcal{B}_y| \leq 1. \quad (31.28)$$

For deterministic records the two maps take values ± 1 . A possible mathematical construction would be a push-forward

$$\mu_{p,xy} = (F_{p,xy})_{\#} \nu_p, \quad (31.29)$$

where $F_{p,xy}$ is a well-posed solution map from unresolved admissible boundary data to a global physical configuration and ν_p is a declared probability law on those data. Neither $F_{p,xy}$ nor ν_p is presently a physical ECT owner. In particular, a Euclidean action weight, an influence functional, a reduced state, practical ignorance, or PES-R bookkeeping does not by itself provide the positive experimental probability measure in Eq. (31.28).

The dependence of a global solution map on apparatus boundary data makes $\mu_{p,xy}$ a mathematically available conditional template. It does not establish that nature violates measurement independence, and it does not classify ECT as retrocausal or superdeterministic. Those labels refer to different causal explanations of the same possible statistical dependence and cannot be inferred before the physical setting generators, Lorentzian causal map and conditional independences are derived [224, 225].

Conditioning, interventions and the Bell latent variable. The setting-indexed measure $d\mu_{p,xy}$ above is only a context-indexed family. Without an operational construction it must not be identified either with the regular conditional measure $d\mu_p([\Gamma] \mid X = x, Y = y)$ or with an interventional ensemble $d\mu_p^{\mathcal{I}_{xy}}([\Gamma])$; these objects need not coincide. Standard Bell measurement independence is the statistical condition

$$d\rho_p(\lambda \mid X = x, Y = y) = d\rho_p(\lambda) \quad (31.30)$$

for a declared preparation-level latent variable in the realised setting protocol. The interventional condition

$$d\rho_p(\lambda \mid \text{do}(X = x), \text{do}(Y = y)) = d\rho_p(\lambda) \quad (31.31)$$

instead states invariance in a supplied causal model. These statistical and interventional conditions need not coincide. Causal-network analyses make the chosen relaxation and its observable implications explicit [226]; no equivalence between them is assumed here.

A complete post-setting history generally changes with the settings because it contains the apparatus settings and their descendants. That tautological dependence is not Bell measurement dependence. ECT first needs a physical projection

$$L_{\text{pre}} : \mathfrak{C}_{\text{phys}} \longrightarrow \Lambda_{\text{pre}} \quad (31.32)$$

and a preparation-preserving family of setting operations. The relevant question is whether the source-level push-forwards $(L_{\text{pre}})_{\#} \mu_p^{\mathcal{I}_{xy}}$ are common to all four settings. Varying admissible Euclidean boundary data is presently only a comparison of solution maps. Whether a local boundary/source modification realises a Pearl intervention rather than conditioning or a change of physical preparation remains Open until Lorentzian localisation, setting generators and a structural causal map are derived.

Failure of observational or interventional independence does not by itself select superdeterminism, retrocausality, a common cause, contextual preparation, selection or an atemporal global constraint. The current ECT causal class is therefore NOT IDENTIFIABLE.

Owner and closure map. Table 90 lists the additional owners required by the conditional global-configuration representation above. It does not make that representation mandatory for every possible ECT Bell completion.

Table 90: Owners required before the conditional global-configuration template can become an ECT Bell model. “Open” refers to ECT ownership, not to the standard mathematics used to formulate each gate.

Object or gate	Required mathematical/physical input	Present evidence	Status
Physical configuration space	Full field–apparatus configuration space with constraints imposed and gauge-equivalent descriptions quotiented	A bare real scalar and conditional reduced sectors exist; the physical matter, apparatus and interacting observable algebra are incomplete	Open
Preparation and settings	Operational maps from laboratory procedures to p, x, y boundary data or source insertions	The global-BVP language supplies an analogy, not these maps	Open
Probability measure	Positive, normalised $\mu_{p,xy}$ with regulator/continuum and preparation meaning fixed	No action-owned probability law; tracing and decoherence presuppose a state	Open: missing probability owner
Outcome/record maps	Gauge-invariant, detector-level $\mathcal{A}_x, \mathcal{B}_y$ and a rule for registered outcomes	The physical detector channel and record operators are not derived	Open: missing vertex/map
Causal classification	Physical Lorentzian localisation, setting generators and a causal graph fixing the relevant conditional independences	The Euclidean BVP does not determine a common-cause, retro-causal or superdeterministic arrow	NOT IDENTIFIABLE
No-signalling	Setting-independent local marginals for every admissible preparation and spacelike protocol	It holds in the engineered 16-state target only by its explicit symmetry	Open ECT theorem/test
Tsirelson ceiling	A derived positive normalised state and bounded self-adjoint projective dichotomic observables with commuting cross-party algebras, or an alternative principle excluding $2\sqrt{2} < S \leq 4$	The generic setting-dependent template permits the algebraic value four	Open (OP-Q20)
Full correlation law	Complete $P(a, b \mid x, y, p)$ across settings, preparations and detector contexts, not only one four-correlator target	No ECT Bell correlator has been calculated	Open
Robustness/fine tuning	No-signalling and the quantum boundary must follow from identities, symmetries or dynamically stable constraints under admissible perturbations	No microscopic parameter family or stability theorem has been supplied	Open

Operational gates for a future ECT completion. First, no-signalling must be proved rather than read from conditional locality at fixed $[\Gamma]$. With probabilities computed from Eq. (31.28), the required identities

are

$$P_A(a | x, y, p) = P_A(a | x, y', p), \quad (31.33)$$

$$P_B(b | x, y, p) = P_B(b | x', y, p) \quad (31.34)$$

for all admissible arguments. Local response functions alone do not imply these identities when the measure depends on both settings. In standard classical directed causal models, Bell-violating yet no-signalling independences generally require a failure of faithfulness/no-fine-tuning; that result is framework-dependent and is not a theorem against every all-at-once global-constraint model [227].

Second, a Tsirelson gate must exclude the no-signalling algebraic example $S = 4$ while retaining the derived quantum correlation family. Matching one engineered value $2\sqrt{2}$ is insufficient. Third, the equalities in Eq. (31.34) must be structurally stable. If θ denotes admissible microscopic couplings, boundary data and detector parameters, the no-signalling residuals must vanish by a proved identity or on the physically allowed family, rather than only at an isolated tuned θ_0 . Finally, the full construction must survive constraint and gauge reduction: a setting dependence that appears only through a coordinate choice, gauge representative or double-counted mode is not a physical Bell mechanism.

Experimental scope and the quantum ceiling. Modern event-ready and photonic tests reject their declared Bell-local null hypotheses while addressing the principal locality and detection loopholes under experiment-specific timing, setting-generation, detector and statistical assumptions [228, 229, 230]. Direct standard-CHSH values include $S = 2.42 \pm 0.20$ in the event-ready solid-state experiment of Hensen *et al.*; cosmic-setting experiments reported $S = 2.425$ and 2.502 with Milky-Way stars and $S = 2.65$ and 2.63 with high-redshift quasars [231, 232].

The high-precision photon-pair result

$$S = 2.82759 \pm 0.00051, \quad 2\sqrt{2} - S = 0.00084 \pm 0.00051 \quad (31.35)$$

is statistically compatible with the quantum ceiling, but it is not a loophole-free result because its inference uses the declared sampling and coincidence assumptions [233]. Among the explicitly cited flagship loophole-free experiments audited here, none reports a statistically significant standard unpostselected bipartite CHSH value above $2\sqrt{2}$; this bounded statement is not an exhaustive literature theorem. Claims called beyond Tsirelson that use postselection or a different operational scenario do not violate the hypotheses of the single-state standard-CHSH theorem merely by exceeding that numerical value.

Cosmic-setting tests move admissible ordinary common causes far into the past under their propagation and sampling assumptions. They do not exclude unrestricted initial-condition, retrocausal or atemporal global-constraint models, and they do not bound an ECT condensate parameter because ECT has no setting–source vertex or Bell likelihood.

Setting-generator provenance as a future ECT test. If a completed ECT model predicts a generator-class dependence or stable MDL bounds (ℓ, h) , the same Bell source should be tested with independently characterised quantum, physical, astronomical and algorithmic setting generators while all detector and selection models are held fixed. A preregistered absence of the predicted dependence would reject that completion; a reproducible dependence would still require exclusion of ordinary drift, memory, leakage and selection effects. Current ECT has no setting–source vertex or Bell forward law and therefore makes no definite prediction of generator-class dependence. This is a requirement for a future completed model, not present evidence.

Status summary. *Established (Level A standard/generic mathematics):* the common setting-independent measure is an explicit premise of the standard factorisable CHSH proof; the 16-state generic construction reproduces $S = 2\sqrt{2}$ with uniform marginals; a distinct nontrivially setting-dependent construction gives $S = 0$; and Eq. (31.20) is a sufficient TV bound under the stated convention. These statements are not ECT derivations.

Level B conditional organisation only: Eq. (31.28) states how a future gauge-reduced global-configuration model could be tested once its physical owners are supplied. It is not evidence that such an ECT measure exists.

Open: the physical ECT preparation and setting maps, positive probability measure, record operators, causal classification, no-signalling theorem, full Bell correlators, Tsirelson principle, robustness theorem and discriminating experimental prediction.

Connection to the coherent-sector infrastructure. The entanglement, vacuum-response, decoherence, probability and black-hole programmes use different mathematical owners: respectively a nonfactorising state/channel, boundary/KMS data, environment kernels, an additive measure, and a metric/state/surface completion. They may ultimately share microscopic medium inputs, but no common correlator or action has yet been derived. Their cross-consistency is a future matching test, not an established inheritance.

Connection to the probability and measurement route. Entanglement is not exhausted by the branch-based probability discussion of Section 30.4. There the main issue was the form of an independently supplied probability measure on projectors; decoherence does not confer those weights. Here the issue is earlier and different: how non-factorisable coherent correlations arise before or independently of any effective branch selection. The two programmes are therefore complementary rather than redundant.

Why entanglement is not equivalent to decoherence. Entanglement and decoherence are related but distinct phenomena in the coherent-branch framework. Decoherence describes the suppression of off-diagonal terms in a reduced description when unresolved environment modes are traced out. Entanglement describes the non-factorisability of the reduced bipartite state *before* environmental decoherence has acted. ECT must account for both: the persistence of non-classical correlations in isolated subsystems (entanglement) and their eventual suppression under environmental coupling (decoherence). An influence functional can describe degradation of a supplied entangled state for a specified coupling, but it does not generate every entangling channel and there is no universal common kernel. Entanglement concerns the non-factorisability of multipartite reduced states, whereas decoherence concerns the suppression of interference in a chosen reduced description under environmental tracing. In practice, increasing Γ_{irr} provides one natural route by which coherent entanglement signatures become fragile, but the two notions are not identical.

Scope of the remaining entanglement programme. The shared-medium picture supplies a research question, not a Bell mechanism: does one completed action and state generate the required nonfactorising channel, its open-system degradation and the apparatus records without double-counting degrees of freedom? The stronger expectation that Bell, decoherence and vacuum response must share one kernel is therefore a hypothesis to be tested after their separate operators and states are derived, not a present Level B result or prediction. Analogue ordered media may test specified correlation and degradation models, but agreement there would not by itself establish the physical ECT Bell channel.

Closure criterion and falsifier. A named ECT Bell completion must derive the full probability law $P(a, b \mid x, y, p)$, reproduce the observed quantum correlation family, prove operational no-signalling and the quantum boundary, and pass held-out changes of preparation, setting generator and detector context. After those objects are fixed, failure of any of these tests falsifies that completion. Before closure, a mismatch of a generic hidden-variable construction, a Werner visibility toy or an analogue platform does not falsify ECT as a whole, and success of such a proxy does not verify it.

Bridge to topological and defect sectors. The entanglement discussion, like the rest of the preceding quantum analysis, has been confined to the smooth coherent branch selected by BR1. The full coherent ontology of ECT is wider: it also includes nontrivial defect and topological configurations that may

contribute to dark-sector ontology and to non-perturbative coherent processes. Those wider sectors are the subject of the next section.

Bell-inequality violation and the decoherence crossover. Werner-state algebra and the CHSH visibility marker are retained as standard quantum-information results. A Bell crossover does not define a universal ECT decoherence threshold or a hidden-variable mechanism.

Entanglement, tunnelling, and a visibility toy. Entanglement and tunnelling are distinct results of their owning quantum dynamics. Calling both “Euclidean connectivity” is an interpretation, not a derivation of either from the other. Environmental coupling can alter tunnelling and entanglement witnesses in channel-dependent ways; no universal scalar suppression law follows.

For a Werner-state visibility toy only, $\text{CHSH}_{\text{eff}} = 2\sqrt{2}e^{-\Phi_{\text{vis}}}$ crosses the local bound at

$$\Phi_{\text{vis}}^{\text{Bell}} = \ln \sqrt{2} \approx 0.347. \quad (31.36)$$

This is model algebra, not an ECT derivation of Bell correlators, the Tsirelson bound, or a universal decoherence threshold.

Channel-specific Bell degradation. The Werner marker above is not universal. For a two-qubit state with correlation matrix T , the Horodecki criterion gives

$$S_{\text{max}} = 2\sqrt{u_1 + u_2}, \quad (31.37)$$

where u_1, u_2 are the two largest eigenvalues of $T^T T$ [234]. A Werner/depolarising family with $T = -e^{-\Phi} I_3$ gives $S_{\text{max}} = 2\sqrt{2}e^{-\Phi}$ and the marker $\Phi = \ln \sqrt{2}$. A phase-damped Bell state with $T = \text{diag}(e^{-\Phi}, -e^{-\Phi}, 1)$ instead gives

$$S_{\text{max}} = 2\sqrt{1 + e^{-2\Phi}} > 2 \quad (31.38)$$

for every finite $\Phi \geq 0$. A scalar dephasing exponent therefore does not define a universal Bell threshold.

For a separate conditional sensitivity benchmark, suppose that a single setting-independent isotropic residual multiplies the complete pair correlator as

$$S_{\text{obs}} = e^{-\Phi_{\text{res}}} S_{\text{pre}}, \quad 0 < S_{\text{pre}} \leq 2\sqrt{2}, \quad \Phi_{\text{res}} \geq 0. \quad (31.39)$$

Assuming $S_{\text{obs}} > 0$, the pre-channel ceiling implies the central upper envelope

$$\begin{aligned} \Phi_{\text{res}} &\leq \Phi_{\text{env}}(S_{\text{obs}}) \equiv \ln \frac{2\sqrt{2}}{S_{\text{obs}}}, & \Phi_{\text{env}}(2.82759) &= 2.96012 \times 10^{-4}, \\ \Phi_{\text{res}} &< 5.93 \times 10^{-4} & (\text{nominal one-sided Gaussian 95\%}). \end{aligned} \quad (31.40)$$

Equality in the central transform additionally assumes a saturating source, $S_{\text{pre}} = 2\sqrt{2}$; the quoted 95% number propagates the published symmetric uncertainty into the same upper envelope. This is arithmetic on an imported aggregate S , not a channel-resolved likelihood or ECT bound. ECT has not derived the photon–condensate vertex, state/channel/noise owner, detector map or pre-channel ceiling required by Eq. (31.39).

32 Gravity-induced entanglement: external criterion and ECT closure test

Status. The BMV/no-go/loophole statements below are external operational physics. The current ECT gravitational mediator, projected two-body vertex and state are NOT IDENTIFIABLE; the linear director candidate terminates at MISSING VERTEX.

32.1 Why this is a sharp completion test

BMV-class protocols [137, 138, 235] ask whether the interaction between two quantum masses implements a nonfactorising channel. This is sharper than generic medium coherence: it requires a physical mediator operator, two nonzero projected source vertices, a positive reduced state and a protocol-level phase/noise prediction.

32.2 External map of the debate

For strictly local factorising classical mediation the standard no-go logic applies. Locally mediated quantum gravity can remain causal, and explicit correlated-noise/common-medium models show that some broader nonfactorising classes evade the slogan-level inference [236, 237, 238]. These existence results do not assign ECT to a loophole class. The Aziz–Howl controversy and its critiques [239, 240, 241, 242] are retained as a warning that the reduced map, not its verbal label, must be computed.

32.3 Three status statements

(A) External implication. An observed entangling phase excludes a strictly factorising classical mediator under the protocol assumptions; it need not by itself select ECT.

(B) Current ECT result. No ECT reduced gravitational channel has yet been shown to be nonfactorising. A shared microscopic medium is not a proof of an entangling reduced map.

(C) Closure requirement. ECT must derive the mediator operator, two-body vertex, state, physical projector and protocol-level phase/noise pair before it can predict a BMV outcome.

Loophole availability is not loophole realisation. An external counterexample to a no-go theorem is not evidence that ECT realises the counterexample.

32.4 Operational criterion

The applicable BMV class is determined only after constructing the physical reduced channel. If it factorises, the classical no-go applies; if it is nonfactorising, an entangling phase is possible subject to decoherence. The present ECT channel is NOT IDENTIFIABLE, so neither branch may be asserted.

32.5 Imported D–P guide mass is not a crossover

For orientation only, choose the unit-coefficient external scaling $\tau_{\text{DP}}^{\text{ext}} \sim \hbar R / (G_N m^2)$ at a supplied radius and invert it at a declared experimental time:

$$m_{\text{guide}}^{\text{DP}} \sim \sqrt{\frac{\hbar R}{G_N \tau_{\text{exp}}}}. \quad (32.1)$$

This is exact algebra inside that external scaling, not an ECT decoherence time, entangling threshold or environmental crossover. In particular, $m_{\text{guide}}^{\text{DP}}$ must not be renamed $m_{\text{cross}}^{\text{coll-DP}}$: the latter is obtained by equating a declared collisional rate to the external D–P rate and depends on n_{gas} , v_{th} and ρ as in Eq. (27.2). Geometry, branch separation and regularisation can alter either source-model coefficient. No numerical guide or crossover is assigned here, and neither formula becomes ECT-specific without an action-owned vertex, state and noise kernel.

32.6 Falsifiers and null results

A completed ECT mediator is falsified by a protocol whose measured phase and noise cannot be produced by its declared reduced channel. Before completion, positive or null BMV results constrain external channel classes but do not constitute an ECT-specific pass or failure.

32.7 Scope and non-claims

No ECT entanglement rate, loophole-class membership or orientation-mediated phase is claimed here. The section defines the measurement and derivation gates that a future ECT gravity completion must pass.

32.8 Detailed external BMV map and ECT closure protocol

Standard BMV logic. In a BMV-class experiment two masses are placed in path superpositions. A branch-dependent interaction phase can entangle the path labels. Under the usual assumptions of local mediation, initially uncorrelated systems and a mediator carrying only classical commuting variables, observation of the appropriate entanglement witness excludes that restricted classical-mediator class. This is an external conditional theorem about the reduced channel; it does not identify which ECT field, if any, realises the mediator.

No-go assumptions and loopholes. Published no-go statements differ in locality, compositionality, mediator state space and admissible direct interactions. Constructive loopholes show that entanglement can arise in enlarged or nonstandard classical–quantum architectures when at least one assumption is relaxed. The existence of a logical loophole is not evidence that ECT occupies it. To make that claim one must derive the ECT reduced dynamics and demonstrate exactly which assumption fails while retaining positivity, no signalling and the observed weak-field limit.

Fixed director candidate. The previously proposed orientation mediator does not pass the first ECT gate. For a static nonrelativistic source the fixed map gives $T^{AB}h_{AB} = \rho h_{ww} = 0$. Hence the linear director channel has MISSING VERTEX; it cannot accumulate the required branch-dependent two-body phase. The spatial index on δn_i and its gapless wave equation do not repair this zero projection.

Scalar trace candidate. A supplied scalar–tensor ansatz can contain a trace coupling. Its canonical coefficient is parametric, and relativistic or classically trace-free sources may decouple at leading order. For finite composite bodies one must derive the scalar charge, form factor and possible screening. Without those inputs and without a state/noise kernel, the vertex is PARAMETRIC ONLY and the full GIE channel remains NOT IDENTIFIABLE.

Independent tensor candidate. If a healthy physical tensor with a universal $HT/2$ vertex is supplied, the standard quantum-mediator analysis may be applied. ECT must still specify the tensor state, constraints, branch-dependent propagator and detector protocol. The FP action by itself is a conditional target and does not establish the required reduced channel.

Operational phase calculation. For a specified two-body kernel $V_{ab}(t)$, the branch phase is

$$\Delta\varphi_{ab} = \frac{1}{S_0} \int_0^T dt V_{ab}(t),$$

with the experimentally relevant entangling combination formed from the four path branches. This equation becomes an ECT prediction only after the physical mediator, vertex, finite-body density profile and the match $S_0 \leftrightarrow \hbar$ are declared. Substituting a Newtonian potential without those derivations reproduces the external BMV source model, not an ECT-specific result.

No imported mediator threshold. Inverting an external Diósi–Penrose or Newtonian source model does not derive a mediator threshold and cannot decide between scalar, director and tensor candidates. No numerical mass guide is retained.

Null results. A null witness can arise from insufficient phase, ordinary environmental decoherence, imperfect visibility, an effectively factorising mediator, or a failure of the assumed source model. It does not falsify ECT as a whole until ECT predicts a nonzero witness for the exact protocol. Conversely, a positive witness excludes the tested classical-mediator assumptions but does not uniquely identify ECT.

Required ECT deliverables. An ECT-specific claim requires all of the following:

- (i) a physical reduced mediator operator and constrained state space;
- (ii) a nonzero projected vertex for both bodies and their finite-size form factors;
- (iii) a positive, trace-preserving reduced channel or unitary joint dynamics;
- (iv) the branch phase and accompanying noise/decoherence kernel;
- (v) a protocol-level witness including ordinary environmental channels;
- (vi) a comparison against classical, semiclassical and quantum-mediator alternatives using the same nuisance parameters.

Until this list is closed, BMV remains a sharp external completion test and the ECT mediator remains Open.

33 Reduced gravitational mediator: bounded terminal taxonomy

Status. Bounded terminal taxonomy for the currently enumerated candidate universe. No listed candidate supplies the complete ECT chain variable $\rightarrow$ constrained dynamics $\rightarrow$ physical vertex $\rightarrow$ state/noise. This is not a no-go theorem for a future named action/vertex completion and is not a derivation of physical PES.

33.1 Why a channel taxonomy is needed

Retarded response, symmetrised noise and a pairwise record operator are different objects. A static force law or compliance cannot be re-labelled as a noise kernel. The resolved/unresolved split and all counterterm subtractions must be fixed before the same mode is used in gravity and PES.

33.2 Scalar trace channel

The supplied scalar–tensor ansatz gives a parametric canonical trace vertex. Finite-body charge, the absolute Hadamard kernel and the ECT state are absent: vertex PARAMETRIC ONLY, full channel NOT IDENTIFIABLE. The minimal free vacuum has $J_\phi(\omega) \propto \omega^3$ and a saturating finite-time exponent, not a universal D–P law.

33.3 Linear director channel

Director source-normalisation guard. For

$$G^{AB} = \beta \delta^{AB} - \alpha n^A n^B, \quad n^A = (q^i, \sqrt{1 - q^2}), \quad (33.1)$$

the unit constraint makes the determinant independent of the tilt and gives the exact inverse-metric component

$$\delta g_{ww} = \frac{\alpha}{\beta(\alpha - \beta)} q^2. \quad (33.2)$$

On the untilted background a normalised fixed worldline has $(u^w)^2 = \alpha - \beta$ and $(u_w)^2 = (\alpha - \beta)^{-1}$. With that convention its exact matter-action change is

$$\frac{\Delta L_m}{d\tau_0} = m \left[1 - \sqrt{1 - \frac{\alpha}{\beta} q^2} \right] = m \left[\frac{\alpha}{2\beta} q^2 + \frac{\alpha^2}{8\beta^2} q^4 + \dots \right]. \quad (33.3)$$

For the supplied $\alpha > \beta > 0$ branch this fixed-coordinate line is timelike only for $0 \leq q^2 < \beta/\alpha$, becomes null at $q^2 = \beta/\alpha$, and has no physical-worldline interpretation beyond that point. The displayed Taylor series has the same convergence boundary. By contrast, a source comoving with the local director, $u^A = \sqrt{\alpha - \beta} n^A$, satisfies $g_{AB} u^A u^B = -1$ identically and has no tilt-action change. Hence the metric algebra is Level A, but the physical quadratic density vertex is PARAMETRIC ONLY until the current/worldline/source convention is derived. The substitution $q_i = \partial_i \chi / u_0$ produces a density-dependent gradient stiffness; it is not by itself a linear source, mass, thin shell or screening theorem. Theorem 8.1 gives $h_{ij}^{\text{TT}} = h_{ww} = 0$. Hence the current linear static-density route is MISSING VERTEX; δn_A is not an identified graviton or BMV mediator.

Qualification. A new nonlinear or independent tensor completion is not excluded, but it must be derived and re-enter Gates 0–3 as a new candidate rather than inherit the failed linear ownership.

33.4 Composite and mixed channels

For $O \sim \delta n_i \delta n_i = (\nabla \chi)^2 / u_0^2$, the strict massless Gaussian χ benchmark with linear dispersion has

$$G_O^>(\omega, \mathbf{k}) = \frac{\omega^4}{4\pi Z_\chi^2 u_0^4 c_\chi^7} B(c_\chi k / \omega) \Theta(\omega - c_\chi k). \quad (33.4)$$

Thus the ω^4 statement means $k = 0$ or homogeneous scaling at fixed $c_\chi k / \omega$; it is not a fixed-nonzero- k , $\omega \rightarrow 0$ limit. Only after the $d^3 k$ convolution with a smooth translated equal-mass source difference, $|\Delta \rho(\mathbf{k})|^2 = O(k^2)$, does the protocol-integrated record spectrum scale as $J_O(\omega) \propto \omega^9$. In the second-cumulant weak-coupling treatment, with a smooth ultraviolet regulator and the required contact-term subtraction, the resulting oscillatory approach to saturation has envelope $\Phi_\infty - \Phi(T) = O(T^{-8})$; a hard cutoff, threshold cusp, or massive mode can add different boundary tails.

For the additional free massless KMS continuum, the scattering contribution in the ordered regime $\omega \ll c_\chi k \ll T_b$ is

$$N_{O,\text{sc}}(\omega, k) \simeq \frac{2\pi^3}{15} \frac{T_b^5}{Z_\chi^2 u_0^4 c_\chi^8 k} \Theta(c_\chi k - |\omega|), \quad (33.5)$$

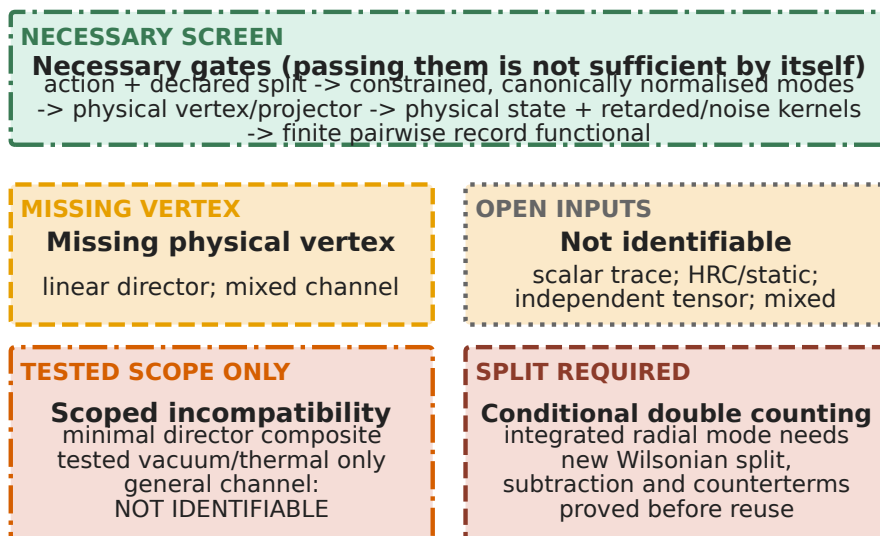
not a universal $1/k^2$ kernel. The $k = 0$ order of limits contains a delta-line, and finite volume produces discrete recurrences. The composite operator is non-Gaussian even when the underlying χ state is Gaussian. These minimal benchmark channels are INCOMPATIBLE with a universal D–P owner. General composites, mixed scalar–orientation operators, and a physical ECT record channel remain NOT IDENTIFIABLE.

33.5 Candidate-channel ledger

Figure 29 gives the reader-facing seven-family necessary-screen summary; passing a screen is not sufficient, and the complete ten-row dispositions remain in Table 91.

Record-channel necessary-screen summary

10 detailed ledger rows -> 7 reader-facing families -> 0 identified physical PES channels



The seven-family graphic aggregates - and does not replace - the detailed ten-row ledger.

Static HRC compliance is a response, not a PES noise kernel;
 imported D-P numbers are external Level C.

- | | |
|--|--|
| <p> missing physical vertex</p> <p> not identifiable / Open inputs</p> | <p> incompatible in tested scope</p> <p> double counting unless split proved</p> |
|--|--|

Figure 29: Reader-facing necessary-screen summary of the bounded record-channel audit. Seven reader-facing families aggregate the detailed ten-row ledger; the graphic does not replace that ledger. None of the tested rows supplies all necessary ingredients: an action and declared resolved/unresolved split, constrained and canonically normalised physical modes, a physical vertex/projector, a state with retarded and noise kernels, and a finite pairwise record functional. Passing this screen would be necessary, not by itself sufficient, for a physical PES channel. The minimal director composite is incompatible with a universal Diósi–Penrose owner only in the tested vacuum/thermal channels; the general case remains NOT IDENTIFIABLE. Reusing an integrated radial mode is DOUBLE-COUNTED only unless a new Wilsonian split, subtraction and counterterm ledger are proved. Static HRC compliance is a response rather than a PES noise kernel; the imported Diósi–Penrose functional remains external. Status colour is redundant with luminance-separated fill, border/arrow style and explicit terminal text; no hatch pattern encodes status. The following native table is the authoritative ten-row candidate/subcase ledger.

Table 91: Authoritative ten-row candidate/subcase ledger for the bounded mediator-channel audit.

Candidate	Vertex/projector	State/noise	Terminal status
Fixed-map linear director	$T^{ww}h_{ww} = 0$	absent	MISSING VERTEX
Scalar trace	parametric trace coupling	absent	vertex PARAMETRIC ONLY; channel NOT IDENTIFIABLE
Classically trace-free source	absent in minimal scalar model	absent	MISSING VERTEX
Finite composite body	charge not closed	absent	NOT IDENTIFIABLE
Minimal free scalar vacuum	trace vertex only	vacuum benchmark	INCOMPATIBLE with universal D–P
Quadratic director composite	coefficient/projector unclosed	tested minimal vacuum/thermal states	INCOMPATIBLE in tested channels; general case NOT IDENTIFIABLE
HRC/static response	static augmented-model response	no bath state	NOT IDENTIFIABLE as PES channel
Independent tensor completion	conditional $H_{\mu\nu}T^{\mu\nu}/2$	ECT state absent	NOT IDENTIFIABLE
Integrated radial mode reused as bath	possible trace term	split/subtraction/counterterms absent	DOUBLE-COUNTED only unless a new Wilsonian split, subtraction and counterterm ledger are proved
Mixed scalar–orientation	absent	absent	MISSING VERTEX / NOT IDENTIFIABLE

33.6 Implications for BMV

No imported collapse-model inversion chooses among these channels. Closure requires a nonzero projected two-body vertex, positive state, response and noise kernels, finite protocol and proof that the reduced map is nonfactorising where entanglement is claimed.

Open mediator tasks. Derive a physical constrained mode bundle and canonical residues; identify the matter and record operators; specify the state; compute D^R and N_H ; declare subtractions; then predict phase, retention and noise for a held-out protocol.

Static bridge and Wightman diagnostics. For this local static diagnostic define

$$g = |\nabla\Phi_{\text{gal}}|, \quad x = g/a_M \geq 0, \quad Y = x^2, \quad s = f(Y) = x^2 B(x),$$

where B is the dimensionless local bridge factor. The variables x, Y, s, B and

$$J_B = \frac{B'_+(0)}{B(0)^{3/2}}$$

are dimensionless; g and a_M have the same acceleration dimension. The diagnostic requires $B(0) > 0$ and a finite right derivative $B'_+(0)$. It is undefined if either condition fails or if the limiting derivative is direction-dependent. Its stated invariance is restricted to a positive horizontal acceleration-scale rescaling;

it is not a general coordinate, gauge or environment invariance. A value such as $J_B = -2$ fixes one local coefficient, not a unique bridge function, gravitational gauge invariant or ECT response owner. In the live static-action pullback a Jacobian is present, and a two-point static $D^R(0, \mathbf{k})$ cannot determine the first nonlinear bridge coefficient: that requires nonlinear response or a cubic vertex together with the physical source-to-metric map. M5 is thus a convention-declared local research diagnostic, not a record-channel closure milestone.

Likewise, the admissible M6 consistency test is the pair of physical Wightman inequalities in Eq. (BE.14), after the same projector, operator and filter have been applied to response and noise. The arbitrary-matrix absolute-value inequality is false, and complete positivity alone does not supply a frequency-local kernel theorem. The current ECT outcome for either diagnostic remains MISSING VERTEX / MISSING STATE / NOT IDENTIFIABLE; no positive ECT claim or numerical rate follows.

P28: primordial record-kernel programme (Open). CMB acoustic phase information can constrain a *fully specified* gauge-invariant primordial reduced channel if that channel predicts a phase-diffusion law, growing/decaying-mode mixing or isocurvature component that is propagated through recombination and the Boltzmann transfer map. Acoustic peaks do not directly measure the pairwise record exponent Φ_{ab} , the off-diagonal density matrix, state purity or a unique pointer basis. Strong dephasing in an amplitude basis can leave the diagonal two-point power unchanged, while a coherent Gaussian growing-mode seed can produce acoustic peaks without environment-induced dephasing.

For current ECT the gauge-reduced primordial alternatives, physical record operator and vertex are missing; the state and generally two-time nonequilibrium response/noise kernels at the ordering front are missing; and the curvature-variable, metric, recombination/transfer and likelihood maps are not identified. The terminal status is therefore MISSING VERTEX + MISSING STATE/KERNEL + NOT IDENTIFIABLE. P28 is a high-value Open PES–cosmology programme, not a claim that the CMB measures a Φ -field, a PES pointer or an ECT classicalisation event.

33.7 Scope and non-claims

The taxonomy records terminal outcomes. It does not promote an untested candidate to an ECT mediator.

33.8 Detailed candidate-channel separation

Why the channels cannot be merged. The scalar trace, linear director, quadratic director composite, independent tensor and mixed candidates have different operators, residues and source selection rules. A common long-distance mean force would not make their noise kernels or quantum channels identical. Each row of the terminal ledger must therefore be carried through independently.

Scalar branch. The scalar candidate couples parametrically to a trace. Its finite-body charge can depend on composition, binding energy and nonlinear screening. These data must be computed before a two-body phase, lensing response or PES kernel is assigned. A scalar channel that merely distinguishes the two path branches may still be factorising and need not entangle them.

Linear director branch. The fixed-map contraction with static density is zero, so this branch stops at MISSING VERTEX. A new dynamic-coefficient or nonlocal map would be a new owner and must be audited from Gate 0 rather than treated as a correction to the old channel.

Quadratic composite branch. The first nonzero director-density candidate may be quadratic in the soft field. Such a channel has a two-propagator spectrum and generally non-Gaussian statistics. In the verified strict massless Gaussian benchmark the local spectrum scales homogeneously as $\omega^4 B(c_\chi k / \omega) \Theta(\omega - c_\chi k)$; after spatial convolution with a smooth translated equal-mass $|\Delta\rho(\mathbf{k})|^2 = O(k^2)$ filter, the protocol-integrated record spectrum scales as ω^9 . With weak second-cumulant coupling, a smooth UV regulator,

and contact subtraction, its oscillatory saturation envelope is T^{-8} . Together with the thermal $1/k$ regime $\omega \ll c_\chi k \ll T_b$, this is incompatible with a universal DP kernel in the tested states. Other operators, filters, regulators, and states remain Open, not confirmed.

Independent tensor and mixed branches. An independent tensor can provide a standard universal vertex only after its action and state are derived. A mixed scalar–orientation branch requires an explicit operator and constrained Hessian; no such owner is present. Reusing a radial or HRC mode already included in the mean action as an unresolved bath without a Wilsonian split is DOUBLE-COUNTED.

Protocol handoff. For any surviving branch, the application file must list the operator, source projector, state, retarded kernel, Hadamard/noise kernel, finite-time window, detector coupling and counterterm ledger. Only then may one compute BMV phases, decoherence visibility, heating, radiation or force noise.

34 Topological Sectors, Defect Configurations, and the Dark Sector

Status: Level A only for winding and homotopy statements inside explicitly supplied smooth-phase or target models. Level B for a conditional inventory of nonsmooth or defect-like configurations once a wider configuration space, action, boundary class and admissibility rules are supplied. Open for their existence as physical ECT sectors, their dynamics, production, stability, interaction cross-sections, abundance and dark-sector role. Connection: the smooth coherent branch BR1, defect-sector material deferred from ECT Foundations (§6.7), the winding/conservation architecture (§22.3), the macroscopic dark-sector bridge (§18.1), vacuum-structure questions (§25.1), exchange-sector topology (§24.4), entanglement/common-medium correlations (§31), and medium character (P5).

So far the Quantum Sector has been developed inside the smooth coherent sector defined by the supplied closure BR1. This restriction was deliberate: it allowed the phase-sensitive sector to be constructed cleanly in terms of $\Phi_{\text{eff}} = \rho e^{i\theta}$, winding sectors, loop actions, wave dynamics, canonical structure, and decoherence. The present architecture does not establish that the physical ECT configuration space is wider than this smooth branch. The subsection instead records a conditional completion programme for such a wider sector.

34.1 Defect sectors and their relation to the smooth branch

Why defect sectors are not excluded by BR1

The smooth coherent-sector rule BR1 does not state that only smooth coherent configurations exist. Rather, it identifies the current constructive branch of the theory as the smooth coherent sector with well-defined phase and integer winding numbers. This leaves open whether an owner-complete theory admits a broader space of configurations outside the smooth branch, including singular cores, discontinuous ordering data, and nontrivial Euclidean topological transitions. Such a candidate state space was already flagged in the deferred defect-sector material (Section 6.7).

The rôle of the present subsection is therefore not to abandon BR1, but to state clearly:

- (i) which configurations could be admitted outside the smooth coherent branch after their action and admissibility rules are supplied;
- (ii) what would be required for physical relevance;
- (iii) how a future owner-complete sector could be tested as a dark-sector candidate.

Medium reading (P5). Under the medium interpretation (P5), defect and topological sectors are not external objects appended to the theory. They are candidate configurations of a proposed wider ordered-medium completion; their common physical condensate owner is not derived: regions where the modulus can vanish, the phase becomes ill-defined, the ordering data changes discontinuously, or the Euclidean configuration tunnels between distinct sectors. The dark-sector question is therefore not “what extra fundamental fields must be added?” but rather “what wider configurations does the same ordered medium admit beyond the smooth branch BR1?”

Connection to winding and conservation architecture. The conditional defect inventory developed here is a possible extension of the winding and conservation structure established in Section 22.3. In the smooth coherent branch, winding numbers are protected under smooth deformations on the declared domain. Thermal suppression of sector changes does not follow from topology; it additionally needs an action, state and barrier/rate calculation. If the wider configuration space is supplied, defect sectors can contain configurations through which topological protection fails: vortex cores, discontinuities, and instanton-like transitions are candidate places where topology can change. In this sense, Section 22.3 describes the protected regime, while the present section describes the wider regime in which protection can break down.

34.2 Conditional types of defect and topological configurations

If an owner-complete wider configuration space is supplied, the main structurally motivated candidate classes outside the smooth coherent sector are:

Vortex-core configurations. Configurations in which the condensate modulus vanishes on a line or axis, making the phase θ ill-defined there. They are the natural continuation of winding sectors to the regime where core singularities are allowed or regularised by $\rho \rightarrow 0$.

Domain-wall-like configurations. Sharp transitions or discontinuities in the ordering data, for example in orientation variables or in the local branch structure of the ordered phase.

Instanton-like Euclidean configurations. Topologically nontrivial Euclidean saddle configurations that interpolate between distinct vacuum sectors or coherent sectors. They lie outside the smooth ordered-sector configurations. If a Lorentzian/hyperbolic completion of that sector is supplied, they are not described by its perturbative coherent branch and are instead natural candidates for nonperturbative inter-sector transitions.

Composite defect configurations. Mixed objects in which phase winding, orientation mismatch, and local suppression of the order parameter coexist. At the present stage these should be regarded as plausible configurations of the wider ordered state space, not yet fully classified solutions.

Status of the defect ontology

The defect inventory is conditional (Level B/Open): if an owner-complete ordered-medium theory admits a phase field together with configurations where the modulus vanishes or the branch data cease to be smooth, defect-like sectors can be classified. P1–P6 and BR1 do not by themselves establish that physical configuration space or its dynamics. Core structure, stability, production channels, interaction cross-sections and relic abundance remain Open.

34.3 Dark-sector questions opened by the coherent sector

Status: Open programme. The Level-C HRC algebraic diagnostic, the late-time background closure, and possible particle, collective, defect, or topological sectors are separate objects. No common ECT dark-sector ontology or dynamical bridge among them has been derived.

The coherent sector permits candidate smooth excitations and, beyond the smooth EFT, candidate defect or topological configurations. Calling either class “dark-sector candidates” is presently a research classification, not a result that they are stable, produced in the required abundance, weakly coupled to visible matter, or gravitationally clustered.

The macroscopic galactic calculation supplies a Level-C HRC data diagnostic with no particle-halo or soft-mode term. It neither derives that fit from the condensate nor excludes additional particle or collective components. Likewise, the late-time background closure does not establish that a coherent or defect sector is the physical source of dark energy.

For a specified candidate to acquire a quantitative dark-sector status one must derive its action and spectrum, stability and production history, couplings to visible matter and gravity, relic abundance, structure-growth response, and discriminating observables. Failure of one such completion would constrain that named completion, not the whole ECT programme.

34.4 Measurement and observability status

Three observational levels can be distinguished:

- (i) **Macroscopic phenomenology (MP):** modified galactic and cosmological response, already developed in detail in Macroscopic Physics;
- (ii) **Indirect coherent signatures:** conditional relic populations, nonstandard clustering, or vacuum-transition effects after an action and production history are supplied (Level B/Open programme);
- (iii) **Direct microscopic detection:** currently open and likely model-dependent.

Only the first level is currently developed in substantial detail.

Status summary. *Established (Level A conditional mathematics):* (i) integer winding inside the explicitly supplied smooth BR1 closure; (ii) the stated homotopy groups of explicitly supplied target spaces and the corresponding absence or presence of protected map classes inside those targets.

Level B: (i) a conditional inventory of defect sectors after a wider configuration space, action, boundary class and admissibility rules are supplied; (ii) smooth collective excitations as the nearest dark-sector candidates within separately supplied coherent EFTs; (iii) the architectural split of the dark sector into a macroscopic effective layer and a coherent / topological layer.

Open: (i) the physical existence and full dynamical classification of defect sectors in ECT; (ii) their production, stability, and interaction mechanisms; (iii) a unique first-principles identification of the dominant dark-sector component; (iv) the quantitative rôle of defect ensembles in late-time cosmology.

Summary. BR1 supplies a smooth coherent-sector closure; it does not prove that this is the whole physical Quantum Sector or that a wider one exists. Defect and nonperturbative sectors beyond the smooth branch form a conditional completion inventory. They are conceptually useful candidate components of a coherent/topological dark-sector programme, not an established ECT ontology. This preserves the correct division of labour across the article: macroscopic dark phenomenology is developed in MP, whereas genuinely topological or defect-based dark ontology belongs to QS.

Bridge to black-hole thermodynamics

The conditional topological and defect inventory developed here records one possible extension of the Quantum Sector beyond the smooth coherent branch; it does not complete or establish its physical ontology.

The next section turns to the asymptotic infrared regime of the same ordered medium. Rather than defect and topological sectors beyond the smooth branch, it examines the long-wavelength ordered-branch structure relevant to asymptotic symmetries, gravitational memory, and soft graviton modes, before the subsequent transition to the strong-field black-hole sector.

35 Infrared and asymptotic completion targets

Status. BMS symmetries, gravitational memory and soft-graviton theorems are established results of asymptotically flat gravity. Their ECT ownership is Open because a physical tensor action, asymptotic charges, source, state and boundary conditions have not been derived.

35.1 Restriction to asymptotically flat backgrounds

Restriction. Null-infinity statements below apply only to an asymptotically flat completed metric theory and are not transferred automatically to an FRW background.

35.2 Why the infrared sector matters

BMS symmetry [243, 244], memory [142] and soft theorems [245, 246, 247] are mandatory comparison targets for any proposed ECT tensor completion.

35.3 BMS completion gate

One must derive the asymptotic phase space, constraints, charges and boundary conditions of the completed physical metric. The ordered data $(\phi_0, n_A^{(0)})$ alone do not imply the BMS group.

35.4 Memory completion gate

Gravitational memory is the standard permanent metric displacement sourced by radiation. Calling a persistent Δn_A “memory” requires a derived map from Δn_A to the physical radiative metric and its source. That map is currently NOT IDENTIFIABLE; no ECT memory amplitude is predicted.

35.5 Soft-graviton completion gate

Long-wavelength director modes may be soft orientation excitations, but the fixed-map theorem does not make them helicity-2 soft gravitons. The latter require the tensor action, physical projector, asymptotic charges and quantum state.

Caveat. Soft orientation data do not by themselves solve an information problem.

35.6 Soft-hair comparison

The soft-hair proposal [248] is retained as an external comparison. No ECT soft-hair storage or recovery mechanism follows before the tensor/asymptotic gates above are closed.

35.7 Scope and non-claims

This section defines external IR tests; it derives neither BMS charges, memory, soft-graviton theorems nor black-hole soft hair from P1–P6.

35.8 Detailed asymptotic-completion protocol

Asymptotic data required for BMS. The BMS group is defined for a physical asymptotically flat Lorentzian metric with specified fall-off and gauge conditions at null infinity. The scalar characteristic tensor supplies a cone in its own equations but not the nonlinear metric phase space, Bondi coordinates, symplectic form or surface charges. Therefore an ECT BMS statement requires a completed physical metric and boundary-value problem; residual $O(3)$ symmetry alone is insufficient.

Memory observable. Gravitational memory is a persistent change in a gauge-invariant detector observable produced by the integrated radiative flux. In a tensor completion it can be written as a change of asymptotic shear related to hard and soft charges. A persistent change Δn_A of the condensate orientation is a different variable. Establishing a map requires the tensor action, source, boundary conditions and detector metric. Without that map no memory amplitude or sign is predicted.

Soft theorem. The leading soft-graviton theorem factorises amplitudes when a physical massless helicity-2 quantum with universal coupling is added at vanishing energy. A long-wavelength gapless director disturbance is not automatically that particle: under integrability it is longitudinal and its fixed-map TT projection vanishes. The soft theorem remains an external target for a future tensor completion.

Relation among soft theorem, memory and asymptotic symmetry. In standard gravity these three descriptions are linked by a common physical metric phase space and its Ward identities. ECT may reproduce the triangle only after it derives the same common owner. It is inconsistent to import one corner—for example a long-wavelength mode—and infer the other two without the charges and radiative symplectic structure.

Soft hair and black holes. Soft-hair proposals concern asymptotic charges of a completed gravitational theory and do not by themselves solve the black-hole information problem. ECT defect or shell variables may provide additional boundary data, but their map to BMS charges, their conservation law and their coupling to Hawking radiation remain Open. No state-counting or information-recovery conclusion follows from the orientation analogy alone.

Finite-energy and detector gates. An observable prediction must specify a finite-frequency waveform, the source multipoles, propagation on the completed background, detector response and the limit connecting it to the asymptotic theorem. Boundary conditions must also distinguish a truly asymptotically flat solution from cosmological or finite-condensate geometries. These requirements are retained as the developed IR programme.

Status summary. BMS algebra, gravitational memory and soft-graviton factorisation are standard external Level-A results under their own assumptions. Their ECT realisation is Open. The fixed-map negative theorem rules out the former linear δn_A identification but leaves independent tensor, composite or nonlocal completions available for future testing.

36 Black-Hole Thermodynamics, Information Problem, and Holographic-Style Entropy Bounds

Inputs: coherent-vacuum response (§25), the Unruh-type response (§25.4), the global/reduced unitarity distinction (§23.2), the decoherence programme (§26.1), the Born-weight route (§30.4), the entanglement programme (§31), and the UV-threshold / singularity architecture (§8.3, §14.1). Aim: to reformulate black-hole thermodynamics within the ordered-condensate language, explain how apparent thermal mixedness can arise without fundamental information loss under supplied global unitarity, and assess a critical

ordered-phase shell as a C/Open strong-field candidate rather than an ECT result. Status: Level A only for the generic reduced-state fact that a thermal marginal can coexist with a pure global state. Exclusion of fundamental information destruction in ECT is conditional on a completed globally unitary interacting strong-field theory and remains Level B/Open. Level C/Open for the critical ordered-phase shell ansatz and for all constructive horizon-thermodynamic results built on it. Open for the explicit nonlinear strong-field profile of the shell, the constructive Page-curve mechanism, and the first-principles derivation of horizon thermodynamic coefficients from bare P3 (OP-Q13). Connection: medium character (P5), the coherent-vacuum response programme, the Unruh/horizon continuity, the global/reduced unitarity distinction, and the branch-breakdown logic of the ordered medium.

Candidate picture: the horizon as an emergent condensate interface. ECT investigates, but does not derive, a completion in which the black-hole horizon is represented by an emergent interface in the condensate medium. This provides a microscopic language for why semiclassical Hawking thermality may survive while allowing controlled corrections in dispersion, late-time correlations, and strong-curvature response. The present section organises the mechanism and its observable channels; it does not yet provide a complete first-principles microtheory of black holes. The shell model and Page-curve-like behaviour are organisational candidates. Absence of fundamental information loss additionally requires a supplied globally unitary strong-field completion.

What is established, what is closure-dependent, and what remains open in the black-hole programme. The black-hole and information programme in ECT should be read in three layers. First, the generic reduced-state identity fixes only that a mixed marginal can coexist with a pure global state. ECT has not yet derived the global strong-field Hilbert space, interacting unitary evolution, physical subsystem split, or the breakdown trajectory of the ordered branch. Second, the present article studies a C/Open strong-field candidate: the near-horizon critical shell. Within that closure, approximate horizon thermality, enlarged-state bookkeeping and reduced-state mixedness can be organised consistently. Preservation of the global information requires the additional hypothesis of a unitary interacting completion; it is not supplied by the shell ansatz. Third, what remains open is the completed nonlinear evaporation problem: the exact shell dynamics, the precise Page-curve mechanism, and the fully explicit bridge between the reduced thermal appearance and the globally unitary condensate evolution.

A consistent Quantum Sector cannot stop at wave dynamics, commutators, decoherence, and Born-type probabilities. It must also address the strongest known tension between quantum unitarity and horizon thermodynamics: the black-hole information problem. The corresponding accelerated-observer version of the same vacuum-response logic is developed earlier in Section 25.4. The purpose of the present subsection is twofold:

- (i) to reformulate black-hole thermodynamics within the ordered-condensate language of ECT;
- (ii) to show that apparent thermal mixedness is compatible with global purity, while keeping global ECT unitarity as an explicit Open owner.

Medium reading (P5). Under P5, a future strong-field completion may represent the black-hole horizon as an interface of an ordered condensate rather than as a primitive boundary. The present action has not derived that interface or its weak-to-strong-field metric map. The information problem is therefore reformulated in medium language: not “does spacetime destroy information?”, but “how does one ordered condensate reorganise accessible and inaccessible coherent sectors in the strong-field regime?” Apparent thermality, reduced-state mixedness, and limited observer access are all properties of one physical medium rather than of a separate ontological stage.

Connection to the Unruh/horizon continuity. The black-hole programme asks whether a future strong-field completion can connect to the standard Unruh/Rindler target of Section 25.4. Neither a common

ECT response operator nor a physical shell is presently derived. A Tolman temperature may be used as an input in the shell toy model, but it does not dynamically create or select that shell from P1–P6.

Discriminant with respect to standard black-hole resolutions. ECT investigates a completion in which global information is preserved, but this is a conditional programme rather than a consequence of P1–P6. It also does not solve the problem by simply postulating a bounce, a remnant, or a new microscopic interior law unrelated to the rest of the theory. Instead, it uses the same ordered medium, the same reduced-state logic, and the same branch structure already employed in decoherence, vacuum response, and singularity breakdown. The proposed closure reads apparent thermality as a reduced-state phenomenon and tests whether a condensate branch reorganisation can carry the missing correlations. Both the branch reorganisation and global unitarity remain to be constructed.

36.1 Effective horizon thermodynamics

Black-hole thermodynamics as the effective horizon response

At the effective macroscopic level, a black hole is characterised by its horizon area A_H and surface gravity κ_H . In standard semiclassical language (using the conventional $\hbar$ before the $S_0 = \hbar$ identification):

$$T_H = \frac{\hbar \kappa_H}{2\pi c k_B}, \quad (36.1)$$

$$S_{BH} = \frac{k_B A_H c^3}{4G_N \hbar}. \quad (36.2)$$

Within ECT these formulas are the effective thermodynamic target that the coherent branch must reproduce or reinterpret; they are not claimed as first-principles derivations from bare P3.

Once the identification (21.7) is completed, they can be written in the equivalent ECT form

$$T_H = \frac{S_0 \kappa_H}{2\pi c k_B}, \quad S_{BH} = \frac{k_B A_H c^3}{4G_N S_0} \quad (36.3)$$

This rewriting is Level B since it depends on the final $S_0 = \hbar$ identification (21.7).

First law and generalized entropy bookkeeping

At the effective thermodynamic level, the horizon sector should also satisfy the first-law structure

$$dM = T_H dS_{BH} + \Omega_H dJ + \Phi_H dQ, \quad (36.4)$$

where the angular-momentum and charge terms are included for generality. Within ECT this equation is an external effective consistency target, not a first-principles derivation from bare P3. Equation (36.4) should therefore be read as a Open effective thermodynamic target for the completed strong-field closure, not yet as a first-principles derivation from bare P3.

The important conceptual point is the entropy bookkeeping. The relevant entropy for the information problem is not the horizon entropy alone, but the *generalized* coarse-grained entropy of the external description:

$$S_{\text{gen}} = S_{BH} + S_{\text{ext}}^{\text{coarse}}, \quad (36.5)$$

where $S_{\text{ext}}^{\text{coarse}}$ is the coarse-grained entropy of the outside radiation sector. This is standard generalized-entropy bookkeeping after a subsystem split is supplied. ECT has not yet derived the required Hilbert-space factorisation or observer algebra.

Thus the correct thermodynamic question is not whether S_{BH} alone changes, but whether the external coarse-grained description evolves consistently once horizon, shell, and radiation sectors are treated together.

Status. Equation (36.4) is an external effective target. Equation (36.5) is supplied semiclassical book-keeping, with its physical ECT owner Open. Its exact dynamical evolution in the completed evaporation problem remains open (OP-Q13).

36.2 The information paradox in ECT

The paradox in brief

The information paradox arises from three apparently simultaneous ingredients:

- (i) pure-state gravitational collapse,
- (ii) thermal Hawking emission,
- (iii) complete evaporation of the hole.

If the outgoing radiation is exactly thermal and the hole disappears completely, the final state appears mixed even when the initial state was pure. Schematically,

$$|\Psi_{\text{in}}\rangle\langle\Psi_{\text{in}}| \longrightarrow \rho_{\text{rad}}^{\text{thermal}}, \quad (36.6)$$

which would violate unitarity if taken as the *fundamental* evolution law of the full system.

ECT reformulation of the paradox

ECT changes the conceptual framing. The relevant question is not: *does spacetime itself destroy information at the horizon?* The correct question inside ECT is: *how does the ordered condensate reorganise the distinction between accessible and inaccessible coherent sectors near a strong-field horizon?*

In a supplied globally pure coherent completion, mixedness may arise after coarse-graining or a partial trace. This is one consistent reading of Hawking thermality, not a proof that fundamental state destruction is excluded by the present ECT action.

Ontological status of the paradox in ECT

Before specifying the critical-shell mechanism, it is useful to state what the information-paradox question becomes once the primitive ontology of ECT is used consistently. In its sharpest semiclassical form the paradox relies on three assumptions that are usually left implicit:

- (i) Lorentzian spacetime is the fundamental arena;
- (ii) the horizon is a fundamental geometric locus rather than an emergent effective boundary;
- (iii) the inside/outside split defines an observer-independent decomposition of the fundamental degrees of freedom.

ECT does not treat these assumptions as primitive. The primitive object is the four-dimensional Euclidean condensate field Φ . Lorentzian spacetime, causal cones, horizons, and the inside/outside distinction belong to the ordered-branch effective description. This does not make them unreal or arbitrary: within the Lorentzian branch they are physically meaningful and operationally measurable. But their status is emergent and reduced, not fundamental.

This changes the bookkeeping but not the unitarity question. The fact that a horizon is emergent does not logically exclude non-unitary microscopic dynamics. It motivates treating the horizon as a boundary in a reduced ordered-branch description. The candidate split is therefore not between “fundamental inside” and “fundamental outside”, but between the full coherent/reconstructed condensate description and the exterior reduced state,

$$\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}} |\Psi\rangle\langle\Psi|,$$

or its appropriate reconstructed/coarse-grained analogue.

Three consequences follow.

First. Hawking thermality is naturally interpreted as reduced-state thermality. A thermal exterior density matrix does not by itself imply fundamental non-unitary evolution, just as a thermalised subsystem in ordinary quantum theory does not imply loss of the global state.

Second. The question “is information destroyed?” splits into two distinct questions: whether a globally unitary ECT completion exists, and whether observer-accessible asymptotic radiation recovers its correlations. Both depend on the completed strong-field dynamics; the second is the constructive Page-curve problem (OP-Q13).

Third. The critical-shell model is one possible C/Open completion to test. The present theory does not show that the ordered branch fails at a shell, that a pre-Lorentzian interior takes over, or that such a shell mediates between them.

Status. The generic conditional statement is that a globally unitary completed strong-field theory preserves global information by assumption. That premise is not supplied by P2 and remains Open. The constructive recovery of correlations in the exterior radiation, including a quantitative Page curve, remains Open. The critical ordered-phase shell is only one optional C/Open completion, not the owner of that problem.

The critical ordered-phase shell

One C/Open candidate is a strong-field transition region near the horizon, where the ordered Lorentzian branch becomes critically strained. In the supplied ansatz this is described as a *critical ordered-phase shell*: a finite region surrounding the effective horizon in which:

- (i) the macroscopic Lorentzian ordering becomes marginal,
- (ii) coherent and topological degrees of freedom become strongly coupled,
- (iii) the outside observer’s effective Hilbert-space description ceases to coincide with the full condensate state space.

This shell is a Level C/Open strong-field ansatz, not a result of bare P3. Its existence, location, stability, state space and coupling to radiation are all missing owners.

Enlarged Hilbert space and apparent mixedness

The key structural point is that the external observer does not have access to the full condensate state space. For a supplied shell toy model one may posit the decomposition

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{ext}} \otimes \mathcal{H}_{\text{shell}} \otimes \mathcal{H}_{\text{int}}, \quad (36.7)$$

where $\mathcal{H}_{\text{ext}}$ is the asymptotically accessible exterior sector, $\mathcal{H}_{\text{shell}}$ is the critical ordered-phase shell sector, and $\mathcal{H}_{\text{int}}$ is the interior or otherwise inaccessible strong-field sector. The decomposition (36.7) is Level C/Open: no physical ECT Hilbert space, tensor product, subsystem algebra or shell factor is derived from bare P3.

Figure 30 is schematic: it visualises the current Level C/Open ansatz and should not be read as a solved global metric profile. For the present strong-field bookkeeping, generalized-entropy structure, and the open shell-dynamics problem, see Appendices AZ, AZ.4, and AZ.5.

The physically observed radiation state is not the total state but a reduced state:

$$\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}}(|\Psi_{\text{total}}\rangle\langle\Psi_{\text{total}}|). \quad (36.8)$$

A reduced state of this form can be approximately thermal when the tensor-product split, pure total state and unitary evolution are all supplied. None of those black-hole owners follows from the shell ansatz itself.

HYPOTHETICAL COMPLETION · LEVEL C / OPEN

No ECT shell action, location, stability, coupling or dynamics has been derived.

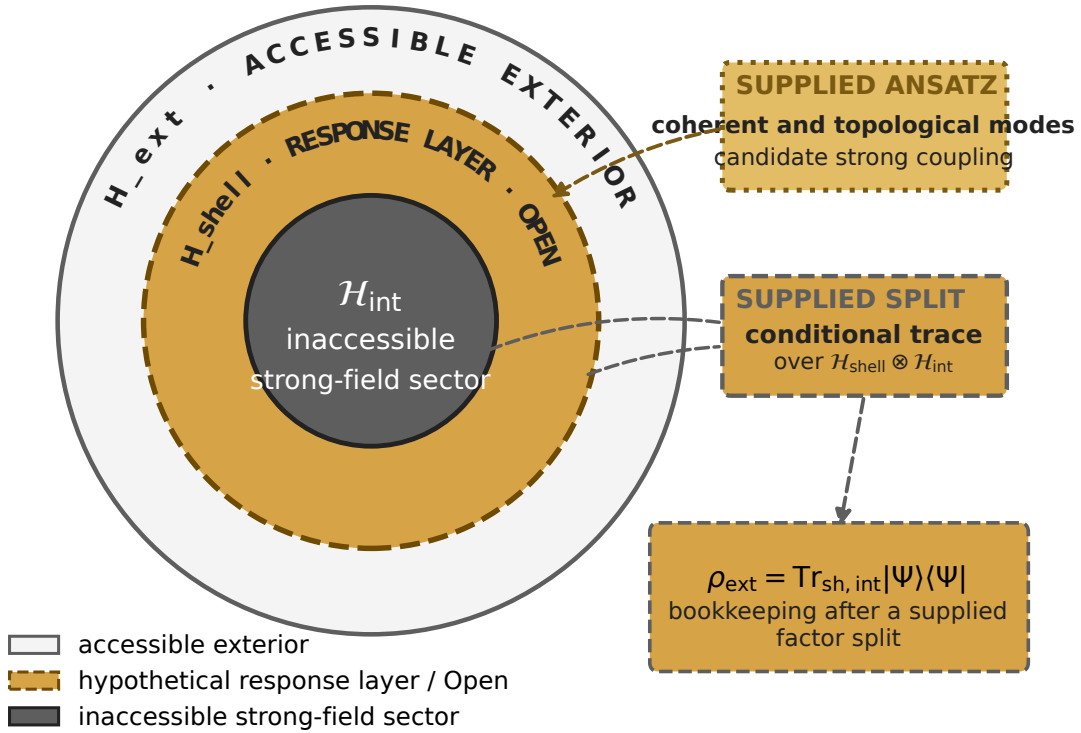


Figure 30: Level-C/Open schematic of a hypothetical near-horizon response layer. Curved labels follow the concentric regions: an accessible exterior, a hypothetical response layer, and an inaccessible strong-field sector. The amber callout marks the supplied ansatz that coherent and topological modes become strongly coupled in the shell. Two gray leader lines identify the shell and interior sectors as supplied factors of a conditional trace; they are not dynamical arrows. The sole gray arrow runs from that trace operation to the reduced exterior state ρ_{ext} . This bookkeeping requires a separately supplied Hilbert-factor split. No ECT shell action, location, stability, coupling, subsystem factorisation or dynamics is derived by this picture, and it is not a completed global solution of the evaporating black-hole problem.

Page-curve logic in ECT language

A useful way to formulate the information problem operationally is through the Page-curve question: does the entropy of the outgoing radiation continue to rise until the end of evaporation, or does it eventually turn over under a supplied complete-evaporation, no-remnant unitary channel with a physical time-dependent factorisation and final-radiation subsystem?

In ECT the sharpest current statement is structural. If the total ordered-condensate evolution remains unitary on $\mathcal{H}_{\text{total}}$ (36.7), then the radiation entropy seen by the external observer is the entanglement entropy of a reduced sector, not a fundamental entropy of the full state. Therefore a Page-like turnover is not forbidden in principle, but it is not selected by unitarity or subsystem bookkeeping alone (see (36.5)).

What is still missing is the explicit strong-field evolution law needed to compute a concrete Page curve from the ECT shell dynamics:

- a Page-compatible evolution is mathematically possible under a supplied globally unitary completion;
- the *explicit Page curve* is still open (OP-Q13).

Summary. The reduced-state formulation changes the logic of the Page problem: the radiation entropy is that of an observer sector, so apparent thermality does not by itself prove destruction of full-state information. It also does not prove that a globally pure interacting ECT completion exists.

Conditional reduced-state compatibility

The quantum-sector bookkeeping records the generic facts (Sections 26.1 and 30.4) that:

- (i) coherent amplitudes are primary;
- (ii) mixedness emerges from reduction or decoherence;
- (iii) reduced observers need not track the full coherent state space.

These facts do not weaken or solve the black-hole paradox on their own. The same reduced-state logic can be applied to a horizon only after a physical Hilbert-space factorisation and global dynamics are supplied. Observer-dependent mixedness is then a possible interpretation, not a default forced by P1–P6.

Conditional theorem: thermality without fundamental information loss

The strongest honest statement currently available is the following conditional result.

Conditional reduced-state identity. If a tensor-product split, globally pure state and global unitary evolution are supplied, then an approximately thermal reduced state ρ_{ext} (36.8) is compatible with preservation of the global state. This statement is generic quantum mechanics and does not derive a critical shell, Hawking spectrum, evaporation channel or Page curve.

Status:

- **Level A:** the general logic that thermal reduced states do not imply fundamental nonunitary evolution is a structural consequence of the global/reduced unitarity distinction together with the reduced-state formalism established in Sections 23.2, 26.1, and 30.4.
- **Level C/Open:** the critical ordered-phase shell ansatz; its physical state, vertex and dynamics are not derived.

Conditional reduced-state reframing

The black-hole information problem depends on a specific set of premises, including semiclassical exterior dynamics in its stated regime, a physical subsystem/algebra split, the evaporation channel, and a global information law. Declaring the Lorentzian metric emergent does not by itself remove or weaken the conflict. A completed ECT model would have to derive which premise is modified and then demonstrate a globally consistent evolution and the observable radiation channel. The present coherent/critical-shell picture supplies neither result and is therefore an Open strong-field programme, not a resolution.

36.3 Comparative mapping: a candidate shell and the islands/QES language

Status: Level C/Open analogy. The islands formula and the QES formalism are standard tools of the modern semiclassical/holographic black-hole programme. The present subsection translates the candidate ECT critical-shell picture of Section 36.2 into that language, without claiming a derivation of either framework from the other. Connection: the Hilbert-space decomposition (36.7); the reduced-state reading of Hawking thermality; the global/reduced unitarity distinction (§23.2); the IR/soft-hair bridge of Section 35.6.

Why a mapping is useful

Section 36.2 formulated the ECT reading of Hawking thermality through a critical ordered-phase shell and the Hilbert-space decomposition $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{ext}} \otimes \mathcal{H}_{\text{shell}} \otimes \mathcal{H}_{\text{int}}$ in (36.7). The modern holographic programme arrives at Page-curve-compatible radiation dynamics through a different language: that of quantum extremal surfaces and islands. The present subsection maps the two languages onto each other as far as current structure allows. *The point of the present mapping is compatibility of structural logic, not an ECT derivation of the Page curve.* Both pictures distinguish between globally unitary bookkeeping and the reduced thermal entropy seen by an exterior observer; that common logic is what makes the mapping possible.

Brief recap of the QES/islands programme

The modern holographic programme computes the entropy of Hawking radiation via the generalised entropy [249, 250]

$$S_{\text{gen}}(\Sigma) = \frac{\text{Area}(\partial\Sigma)}{4G_N} + S_{\text{bulk}}(\Sigma), \quad (36.9)$$

extremised over bulk regions Σ [251]. The inclusion of disconnected “island” regions inside the black hole proves decisive for radiation entropy [252, 253]: before the Page time the minimal QES is empty; after the Page time a non-trivial island dominates, yielding the Page-curve turnover.

Core structural mapping

Table 92 sets out the structural correspondence between the two languages.

Table 92: Structural mapping between the ECT critical-shell picture and the QES/islands language. Entries in the right column are structural analogies, not derived equivalences.

ECT side	Islands/QES side	Structural relation
$\mathcal{H}_{\text{int}}$ (interior / inaccessible sector)	island region	structurally analogous role: degrees of freedom outside the exterior reduced description but counted in the globally unitary picture

ECT side	Islands/QES side	Structural relation
boundary of critical ordered-phase shell	quantum extremal surface boundary $\partial\Sigma$	structurally analogous interface between accessible and inaccessible sectors
$\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}} \Psi_{\text{total}}\rangle \langle \Psi_{\text{total}} $	reduced radiation state whose entropy is given by S_{gen} at the dominant QES	both are reduced-state objects only after global unitarity and subsystem algebras are supplied
Page-curve possibility under supplied global isometry and factorisation	Page-curve turnover via island transition	qualitative Page-curve compatibility, not a derivation
long-wavelength δn_A modes near the critical shell	soft hair on or near the horizon	candidate long-wavelength book-keeping sector, independently formulated

Where the mapping is formal and where it is structural only

Formal identity and proposed analogy. The generic identity that a mixed reduced state may arise from a pure global state is common mathematical bookkeeping. No further equivalence has been derived. The proposed correspondence between an island/interior algebra and the supplied ECT $\mathcal{H}_{\text{int}}$ label, or between a shell boundary and $\partial\Sigma$, is an Open analogy until the ECT subsystem algebras, entropy functional and extremisation rule are constructed.

Structural only. ECT does not derive the generalised entropy formula (36.9) from first principles. The critical-shell / Hilbert-space decomposition is compatible with such a formula but does not imply it. ECT also does not provide a replica-wormhole-type computation; replica wormholes are a Euclidean gravitational path-integral device, and an analogous computation on the underlying Euclidean condensate path integral is an open item. The detailed time dependence of the ECT Page curve requires the explicit shell dynamics flagged as OP-Q13 in Section 36.2.

Shell is not a derived QES. In particular, the critical-shell boundary is not here identified with a derived quantum extremal surface. At present it is only a structurally analogous interface separating reduced exterior thermality from the globally unitary description.

Scope and non-claims

The present subsection does not claim (i) a first-principles derivation of the QES or islands formula from ECT; (ii) a replica-wormhole-type calculation in ECT; (iii) an explicit Page curve for a specific ECT evaporation scenario; (iv) that the critical-shell picture and the islands picture are formally equivalent. Its claim is narrower: the generic reduced-state identity is compatible with either bookkeeping scheme under separately supplied assumptions. No structural alignment, Page curve or common physical mechanism is inferred.

36.4 Black-hole entropy and observational signatures

Black-hole entropy in the ECT reading

In a supplied semiclassical black-hole model, S_{BH} is the standard Bekenstein–Hawking entropy. A future ECT completion might seek to represent it through a coarse-grained external description, but the required subsystem algebra, state, area coefficient and strong-field action have not been derived. The generic fact that reduced descriptions can be mixed does not identify S_{BH} with condensate entropy.

Relation to Hawking radiation and vacuum effects

ECT does not deny Hawking-like emission. Sections 25.5 and 25.4 record standard conditional targets; they do not derive horizon-induced excitation production from ECT. A reduced-state interpretation is compatible with a supplied globally pure completion, not forced by the present action.

Thermality versus exact thermality

The black-hole information problem becomes most severe if the outgoing radiation is assumed to be *exactly* thermal at all stages. ECT does not require such exact thermality. The shell ansatz does not itself derive approximate thermality of the reduced external state (36.8); that property must come from a supplied or derived evaporation channel.

This distinction matters. Approximate thermality is perfectly compatible with hidden correlations, late-time purification effects, and a Page-curve turnover. Exact thermality would be much harder to reconcile with global unitarity.

The correct conditional statement is: a thermal marginal can coexist with a pure global state when the physical factorisation and global isometry are supplied. This shows how a future unitary ECT completion could avoid the strongest form of the paradox, but neither that completion nor the quantitative evaporation problem has been derived (OP-Q13).

Missing strong-field forward model

No ECT strong-field observable follows from the shell label alone. A completion must first supply a physical action and state, a regular strong-field solution, matter/photon/tensor perturbation operators with boundary conditions, a source/detector vertex, a causal radiation channel, and an observable likelihood. Standard Hawking and exterior-metric results are external targets until that owner chain is complete.

No retained phenomenological response ansatz. No near-horizon dispersion law or energy-dependent gravitational response is derived from P1–P7. Arbitrary functions for those effects would parameterise generic deviations rather than ECT and are not retained.

Observable status. ECT currently predicts no nonzero ringdown shift, echo, reflectivity, primordial-black-hole correction or other shell signature. Future data can test only a completed model that derives one of these observables together with its nuisance and likelihood structure.

What ECT does not yet provide. ECT does not yet provide a full Level A dynamical solution of the black-hole information problem. The positive mechanism—how approximate horizon thermality, shell/interface dynamics, and global purity are jointly realised—remains to be constructed.

What is established and what remains conditional. The exact statement is the generic reduced-state identity: thermal marginals do not imply destruction of information if the global state evolves unitarily. ECT has verified this logic in reconstructed positive Gaussian subsectors. Applying it to an evaporating black hole requires a completed interacting Hilbert space, global unitary dynamics, strong-field metric/vertices and a physical subsystem factorisation; these are not consequences of P2 and remain B/Open. The relevant status distinction is developed in Sections 23.2, 26.1, and 30.4: reduced-state thermality does not imply fundamental loss of information *when* the total evolution remains globally unitary.

Status summary. *Established (Level A, conditional-domain mathematics):* (i) a thermal reduced state can be compatible with global purity; (ii) positive Gaussian subsectors admit the standard reconstructed unitary description under the named OS hypotheses.

Level C/Open shell ansatz: the critical shell, enlarged Hilbert-space bookkeeping, approximate Hawking channel, Page-like toy and branch-boundary interior picture. Global interacting unitarity and all physical subsystem/state owners remain Open.

Open (OP-Q13): (i) the explicit nonlinear strong-field profile of the critical shell; (ii) the constructive Page-curve mechanism; (iii) the first-principles derivation of horizon thermodynamic coefficients from bare P3; (iv) the completed strong-field evaporation dynamics.

In particular, the present shell analysis does not yet derive the full interior profile, first-principles horizon thermodynamic coefficients, or a constructive Page curve; the numerical coefficient in the area-law entropy $S_{\text{shell}} \sim A/\ell_*^2$ remains an open matching condition.

36.5 Conditional boundary-cell counting and the open area-law programme

Status: Level A mathematics only for the conditional finite-cell counting lemma below, after a subsystem split, coarse-graining, finite local entropy capacity, and an area-law state class have all been supplied. ECT ownership of those inputs, the ultraviolet completion, the black-hole coefficient, and the physical entropy bound is Open. Connection: shell entropy (§AZ.3), conditional theorem (§AZ.4), decoherence programme (§26.1), entanglement (§31), EFT breakdown scale (§8.3), medium character (P5).

Problem statement: area law, holographic bound, and full holography

In black-hole physics, entropy scales with area rather than volume. In the broader quantum-gravity literature this motivates several related but conceptually distinct ideas:

- (i) the Bekenstein–Hawking area law for black-hole entropy [254];
- (ii) more general holographic-style entropy bounds [255, 256, 257];
- (iii) the stronger holographic principle in the sense of bulk encoding or reconstruction by boundary data [258].

These statements should not be conflated. An area law does not by itself imply full holography, and a shell-counting argument near horizons does not by itself establish a covariant entropy bound for arbitrary dynamical geometries.

The ECT question is narrower than an assertion of holography:

Which additional microscopic and state assumptions would make an area-type entropy count compatible with the local ordered-medium picture, and which of those assumptions are actually derived?

The answer presently available is conditional. Locality and a mass gap motivate an interfacial description of low-energy correlations, but they do not by themselves regulate continuum entanglement or prove an area law for an arbitrary state.

Necessary distinctions

Three scales and statements must be kept separate.

- (i) The homogeneous P3 mass $m_{\Phi, \text{rad}}$ defines a P3 Gaussian infrared correlation length $\xi_{\text{P3}} = m_{\Phi, \text{rad}}^{-1}$. It does not define a P4 or gravitational EFT breakdown scale. A massive continuum field nevertheless contains modes of arbitrarily large wave number within the formal continuum theory. Therefore m_{σ}^{-1} is not, without a microscopic completion, a lattice spacing, a hard ultraviolet regulator, or a finite-dimensional local Hilbert-space cell.
- (ii) Exponential decay of selected two-point functions does not imply that all bipartite entropy is confined to a boundary layer. Arbitrary states of a local system may have volume-law or data-hiding entanglement despite short-ranged two-point correlations.

- (iii) A finite area count follows only after a physical split and a state class with an area-law property are specified together with a finite coarse-grained local capacity. Those inputs are not consequences of P1–P6 at present.

Conditional setup

Let Σ be a smooth closed two-surface separating exterior and interior subsystems. The reduced state of the exterior observer is

$$\rho_{\text{ext}} = \text{Tr}_{\text{int}} |\Psi\rangle\langle\Psi|, \quad S_{\text{red}}(\Sigma) = -\text{Tr}(\rho_{\text{ext}} \ln \rho_{\text{ext}}). \quad (36.10)$$

Equation (36.10) itself requires a supplied factorisation (or an appropriate algebraic split). Gauge and constrained sectors may require edge modes and need not admit the naive tensor-product factorisation.

Supporting lemma (exponential clustering). For the supplied homogeneous P3 Gaussian sector, with $m_\sigma \equiv m_{\Phi, \text{rad}} > 0$, the connected two-point correlator satisfies $|\langle \Phi(x)\Phi(y) \rangle_c| \leq C e^{-m_{\Phi, \text{rad}}|x-y|}$ (standard result for massive Euclidean fields [148]). Its cross-boundary *two-point* correlations are therefore exponentially suppressed over $\xi_{\text{P3}} = O(m_{\Phi, \text{rad}}^{-1})$. This is useful long-distance information inside that supplied sector, but it is neither a P4 pole nor a UV regulator or entropy-area theorem.

Supplied coarse-grained inputs. Let ℓ_{cg} be an explicitly chosen physical lattice spacing, UV cutoff, or coarse-graining resolution; let $s_{\text{cg}} < \infty$ be the maximum entropy assigned to one retained cell under a declared finite-dimensional truncation or energy restriction. Neither ℓ_{cg} nor s_{cg} is currently derived from m_σ or from P1–P6.

In addition, assume a state class for which the cross-interface entropy is supported, up to a declared remainder ΔS , in a layer of thickness ξ_{ent} . Examples of sufficient inputs would be a finite-depth local preparation from a product state, an appropriate quantum-Markov/conditional-mutual-information bound, or an independently proved area law for the selected gapped state. Finite correlation length alone is not substituted for this assumption.

General theorem

Theorem 36.1 (Conditional boundary-cell counting lemma). *Let Σ be a smooth closed two-surface of area $A(\Sigma)$ separating supplied interior and exterior subsystem algebras. Suppose that:*

- (i) *a lattice, UV completion, or coarse-graining of resolution ℓ_{cg} is supplied;*
- (ii) *each retained cell has entropy capacity at most $s_{\text{cg}} < \infty$ under the stated truncation or energy bound;*
- (iii) *the selected state class obeys a boundary-localisation hypothesis: all but a remainder ΔS of the entropy across the split is carried by cells within distance ξ_{ent} of Σ ;*
- (iv) *the geometry is regular on those scales, so the number of cells in that layer is bounded by $C_\Sigma A(\Sigma) \xi_{\text{ent}} / \ell_{\text{cg}}^3$.*

Then the coarse-grained reduced entropy satisfies

$$S_{\text{red}}(\Sigma) \leq C_\Sigma s_{\text{cg}} \frac{A(\Sigma) \xi_{\text{ent}}}{\ell_{\text{cg}}^3} + \Delta S. \quad (36.11)$$

If, as an additional matching assumption, $\xi_{\text{ent}} = O(\ell_{\text{cg}})$ and ΔS is sub-area, this is an area-type bound $S_{\text{red}} \lesssim c_0 A(\Sigma) / \ell_{\text{cg}}^2$.

Proof

Step 1: count the supplied boundary cells. The boundary-layer volume scales as $V_{\text{layer}} \leq C_{\Sigma} A(\Sigma) \xi_{\text{ent}}$. At resolution ℓ_{cg} , the number of retained cells is therefore bounded by

$$\mathcal{N}_{\Sigma} \leq C_{\Sigma} \frac{A(\Sigma) \xi_{\text{ent}}}{\ell_{\text{cg}}^3}. \quad (36.12)$$

Step 2: apply the supplied entropy-capacity and state-class bounds. Subadditivity and the finite capacity give at most $s_{\text{cg}} \mathcal{N}_{\Sigma}$ for the retained layer, while the boundary-localisation hypothesis contributes the declared remainder ΔS . Substitution of (36.12) proves Eq. (36.11). $\square$

Interpretation

The lemma is correct but conditional: the supplied state-class and finite-cell assumptions do the essential work. It is not an ECT area-law theorem and does not show that an arbitrary ECT state obeys an area law. It is retained because it identifies, without hiding them, the precise inputs that a microscopic completion would need to own.

Medium reading (P5). The medium interpretation (P5) makes an interfacial state class a natural programme target, but P5 does not supply a UV lattice, a finite entropy capacity, a subsystem factorisation, or the required state theorem. No gravity-specific statement is inserted into P5.

Relation to the black-hole shell result

The shell count introduced earlier (§AZ.3),

$$S_{\text{shell}}^{\text{toy}} \sim \frac{A \ln q}{\ell_{\text{cg}}^2}, \quad (36.13)$$

is an instance of the conditional cell count only if a physical shell, a factorisation, the cell scale, the local dimension q , and an appropriate state are separately supplied. It is not a concrete ECT black-hole realisation at present.

The exact Bekenstein–Hawking coefficient corresponds to the matching condition

$$\frac{\ln q}{\ell_{\text{cg}}^2} = \frac{1}{4\ell_{\text{Pl}}^2}. \quad (36.14)$$

This is a target matching condition inside the supplied toy, not a derivation of either ℓ_{cg} or q . In particular one may not set $\ell_{\text{cg}} = m_{\sigma}^{-1}$ without an independent microscopic argument.

Entanglement-entropy bridge

The above result also clarifies the relation between ECT and the standard area law of entanglement entropy in continuum QFT.

In continuum QFT, entanglement entropy across a surface is known to scale as area, but with a UV-divergent coefficient [259, 260]:

$$S_{\text{ent}}^{\text{QFT}} \sim \frac{A}{\varepsilon^2}, \quad (36.15)$$

where ε is a UV regulator. The current ECT continuum action does not remove this divergence. Its radial mass gap controls long-distance correlations, not the short-distance mode count. A physical UV completion or declared coarse-graining is therefore still required before a finite entropy can be assigned. Replacing ε by m_{σ}^{-1} is not an allowed derivation.

Role of the Euclidean elliptic structure

The Euclidean branch provides an additional architectural reason why boundary control is natural in ECT. Because the Euclidean condensate equation is elliptic, classical field configurations are naturally organised by boundary data.

This makes boundary-value problems conceptually native to ECT, but classical ellipticity neither specifies a quantum subsystem split nor implies an entropy bound. It is architectural motivation only.

Soft sectors and subleading corrections

Soft or constrained sectors can invalidate the naive factorisation or add edge contributions: Goldstone sectors, gauge-edge modes, long-range constrained sectors, and other nearly gapless collective excitations.

Their treatment has not been worked out within ECT. No assumption that their corrections are subleading is used as a result.

Relation to the Page-curve and information-capacity programmes

The lemma does not strengthen the physical Page programme, because a finite shell capacity is one of its supplied assumptions rather than an ECT output. A physical split, shell Hilbert space, global channel, information-return dynamics, and state evolution all remain Open.

Relation to the decoherence programme

The same interfacial logic clarifies the relation between entropy bounds and decoherence. The supplied influence-functional source models of Section 26.1 show how tracing declared inaccessible degrees of freedom can produce reduced-state evolution and dephasing. They are not yet an ECT-owned physical channel. Both questions may involve an interface after a reduced model is supplied, but the cell-counting lemma gives neither a coupling, state, decoherence kernel, trace, timescale, nor an entropy-production theorem. No quantitative PES or physical ECT decoherence result follows from it.

From area law to holography: what is and is not derived

It is essential to distinguish:

- (i) a *conditional finite-cell area count*, which is the only result established here;
- (ii) a physical *ECT area-law entropy bound*, which is Open;
- (iii) *full holography* in the strong reconstruction sense, which remains open.

The correct summary is:

ECT provides an interface-motivated programme and a conditional counting lemma. It does not yet derive a physical area law, a covariant entropy bound, or a bulk–boundary duality.

Unlike AdS/CFT, ECT currently has no analogue of a Ryu–Takayanagi extremal-surface prescription; the present result is only a conditional finite-cell count, not a geometrised entanglement functional. Modern quantum error correction reformulations of holography likewise remain outside the current scope.

Comparison with other approaches

Table 93: Comparison of area-law routes.

Aspect	AdS/CFT	LQG	ECT
Why area scaling	Duality; extremal surfaces; boundary CFT	Quantum-geometric state counting	Conditional boundary-cell count after supplied coarse-graining and state assumptions
UV scale	String / AdS completion	Planck discreteness	Open microscopic completion; m_σ^{-1} is not a UV cutoff
Exact coefficient	Model-specific, often controlled	Counting-dependent; Immirzi parameter	Not derived; toy matching condition (36.14)
Full bulk reconstruction	Central feature	Not standard	Open
Physical picture	Gravity/gauge duality	Quantum geometry	Interface-motivated conditional programme

Discriminant: ECT area bound versus standard approaches. The candidate route is architecturally distinct, but it is not yet a physical alternative to the established approaches. Its decisive test is whether an ECT microscopic completion supplies the split, finite local capacity and state-class bound without assuming the desired area law.

Falsifier for the area-law programme. The programme fails as a derivation if the completed physical ECT state has volume-law entropy, if no finite subsystem split/capacity exists, or if the proposed coarse-graining cannot be connected to an observable black-hole channel. A value of m_σ alone cannot pass or fail the UV test because it is not the regulator.

Status summary. *Established (Level A mathematics under supplied inputs):* the finite-cell counting implication in Theorem 36.1.

Open in ECT: (i) the microscopic UV scale ℓ_{cg} and finite local capacity s_{cg} ; (ii) a physical subsystem split and the relevant area-law/Markov or finite-depth state theorem; (iii) all soft, gauge-edge and constrained-sector contributions; (iv) the black-hole shell, coefficient, capacity and Page dynamics; (v) a covariant entropy bound and full bulk–boundary reconstruction.

36.6 Singularity, interior, and structural programme

Black-hole singularity as breakdown of the coarse-grained branch. The same branch-boundary logic developed for the cosmological sector (§14.1) applies to the black-hole interior. ECT does not yet provide a unique microscopic interior solution that replaces the classical GR singular core by a completely solved regular geometry. The present action does not decide whether the classical singularity is physically fundamental. It motivates an Open programme in which the ordered-branch description may cease to apply before the formal GR singularity, but neither a breakdown surface nor a replacement interior is derived. The shell construction below is one C/Open toy realisation of that possibility, not evidence that the singular core is an artefact.

Three-regime structure of the near-horizon condensate. The candidate ECT picture of the near-horizon region (Figure 31a) is parameterised by three proposed regimes:

- (i) *Exterior ordered Lorentzian regime* ($\rho \gg \rho_c$, $T_{loc} \ll T_c$): standard effective geometry is assumed valid outside the toy layer;

- (ii) *Critical shell / interface regime* ($\rho \sim \rho_c, T_{\text{loc}} \sim T_c$): described at the closure level by the Ginzburg–Landau / Tolman near-horizon construction (Appendix AZ.5); the toy ordered-branch amplitude is assumed to transition;
- (iii) *Pre-Lorentzian core regime* ($\rho \ll \rho_c, T_{\text{loc}} \gg T_c$): a non-geometric condensate completion is required but not derived; $\mathcal{H}_{\text{core}}$ in (AZ.9) is supplied bookkeeping, not an identified physical sector.

The Ginzburg–Landau mean-field description of Appendix AZ.5 applies only in the near-critical interface region (ii), not as a description of the full interior. A completed theory of the deep interior (iii) requires the pre-Lorentzian condensate dynamics, which is not yet available.

Semiclassical benchmark; ECT curve Open

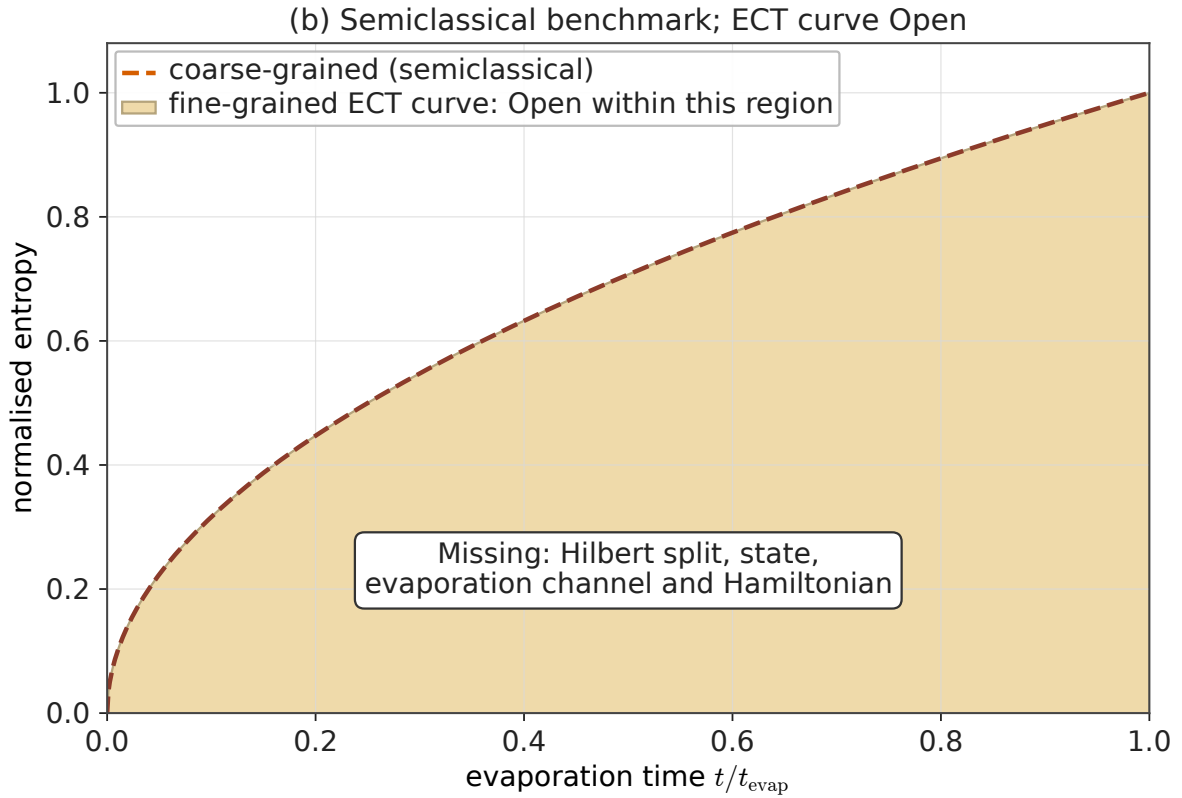


Figure 31: Imported semiclassical coarse-grained entropy benchmark and ECT status guard. This page shows only the Page-curve benchmark. The Open region is not a predicted curve: the physical Hilbert-space split, evaporation channel, state and Hamiltonian required for an ECT fine-grained entropy history remain Open. The Tolman near-horizon and Hawking-temperature benchmarks are retained as separate full-width atlas panels; ECT presently identifies no shell depth, critical surface or information-return law.

Appendix BO preserves the separate Tolman and Hawking benchmark panels with the same Open-owner guard. Figure 31 displays only external Tolman/Hawking kinematics and the missing-owner inventory. It is not a derived near-horizon ECT structure.

Topological consistency of the shell picture. The supplied orientation target $O(4)/O(3) \simeq S^3$ has $\pi_2(S^3) = 0$ (Section 8.10). If a candidate shell is a closed two-surface S^2 , the vanishing of π_2 shows only that the restricted n^A field carries no π_2 charge. It does not prove shell formation, stability, dynamics, or coupling. This is at most a consistency check: if π_2 were nontrivial, the shell could trap topological charge, potentially obstructing shell formation/dissolution and complicating any evaporation scenario based on a finite-lifetime shell interface. The nontrivial homotopy sector $\pi_3(S^3) = \mathbb{Z}$ (Section 8.10) admits

topological configurations of the exterior ordered-branch field; however, the present work does not derive a stabilising Skyrme-type higher-derivative term or a finite-energy gravitating topological black-hole solution.

Connection to the UV-threshold and breakdown programmes. The strong-field toy is compared with the entanglement and UV-threshold programmes, but equality of their states, kernels and physical subsystem maps is not derived. Whether a future action-owned P4 validity scale controls a physical near-horizon breakdown is Open; the P3 curvature does not supply that scale. Using one branch-breakdown mechanism in cosmology and black holes is an Open economy hypothesis, not a consequence of the common bare field.

Cross-sector consistency. The cosmological and black-hole singularity programmes explore a common branch-breakdown hypothesis. Neither replacement mechanism is yet derived in the strong-field regime. The completed theory may select a common mechanism or distinct state-dependent mechanisms; neither outcome is imposed here. Both illustrative programmes reuse the one-real-scalar toy temperature $T_c = 2\phi_0$ and the same bare order parameter; this does not identify the P4 gradient-ordering transition: using the same Z_2 toy in both settings does not show that a cosmological branch emerges thermally or that a black-hole branch dissolves. That proposed common mechanism is Open.

Discriminant with respect to alternative singularity programmes. The ECT singularity programme differs from two common alternatives:

- (i) **Classical-GR singularity realism:** the candidate programme tests whether the divergence marks branch failure; it does not establish that interpretation;
- (ii) **Bounce-by-postulate models:** ECT does not simply replace the singularity by a bounce imposed from outside; it asks whether loss of validity of the ordered geometric branch can be derived and leaves the microscopic completion as an open condensate-dynamics problem.

This is an Open alternative to both fully specified bounce models and unmodified extrapolation of classical GR to infinite curvature; its relative viability cannot be ranked before a strong-field completion is derived.

Falsifier for the singularity-breakdown programme. The present ECT singularity programme would be seriously weakened if a completed condensate analysis showed that the ordered Lorentzian branch remains fully valid all the way to the formal GR singularity with no critical-shell, no pre-ordered transition, and no condensate breakdown regime. It would also be weakened if the only consistent completion of the theory forced one back to a physically fundamental divergent singularity rather than to a branch-boundary interpretation. Conversely, an independently established critical regime before a classical singularity would motivate this class of completion, but would not identify the ECT action, state, or response mechanism.

36.7 Observational anchors and the Open black-hole programme

External observations and constraints. The following observations constrain future strong-field completions but currently provide neither compatibility evidence nor support for the ECT black-hole programme. The published Event Horizon Telescope comparison of the 2017 and 2018 M87* observations finds a persistent same-scale ring/shadow structure [261]. This owns the empirical persistence statement; it does not identify an ECT horizon response, interior, or surface mechanism. Gravitational-wave ringdown tests remain broadly consistent with Kerr-like exterior behaviour. The present ECT candidate supplies no owned strong-field action/state, exterior metric solution, interior/subsystem map, radiation response or likelihood, and therefore makes no derived statement about these data. They are external targets for a future exterior/interior completion, not consistency anchors for it.

Conditional geometric thinness of a supplied layer. For a non-extremal black hole and a supplied transition temperature T_* , the Tolman toy defines $\rho_* = \hbar c / (2\pi k_B T_*)$. Relative to the Schwarzschild radius $r_s = 2G_N M / c^2$, this gives

$$\frac{\rho_*}{r_s} = \frac{\hbar c}{2\pi k_B T_* r_s}, \quad (36.16)$$

so thinness is conditional on the still-unidentified physical T_* . No exterior-deviation bound follows from the present toy alone. Timing observables, late-time correlations and horizon reflectivity require a derived metric/transfer map. No exhaustive experimental channel can be selected before that map exists.

The present shell construction is developed for the non-rotating (Schwarzschild) case. For Kerr black holes the Tolman relation is modified by rotation and the critical-shell radius may depend on latitude; the analysis of this extension is left to future work.

Connection to the analogue-laboratory programme. The ordered-medium reading of horizon thermality finds its closest controlled analogue in superfluid and BEC experiments where sonic horizons generate phononic analogues of Hawking-like radiation (§37). These systems test their own ordered-medium horizon models. An analogue-to-ECT map is missing, so they currently provide motivation rather than a test of the optional ECT shell completion.

Acceptance criteria for the optional shell completion. The shell ansatz becomes physical only if a completed action derives its interface/control observable, subsystem algebra, state, radiation vertex and observable metric, while reproducing the semiclassical Hawking target where applicable. A Page curve then requires the full evaporation channel and state; the displayed tensor-product split does not select it.

Failure of any of these gates rejects this shell completion only. P1–P6 may admit a different strong-field completion, and no common mechanism with cosmological branch breakdown is imposed in advance.

Summary in one sentence. The paradox is not solved: reduced-state thermality is compatible with purity, while global ECT unitarity, the shell channel and information return remain Open.

For a more explicit derivational sketch of the reduced-thermality logic, the Hilbert-space factorisation, and the Page-curve discussion, see Appendix BA.

Summary. Black-hole thermodynamics in ECT belongs to the Quantum Sector because it probes the coherent-vacuum and reduced-state structure of the ordered condensate. The information paradox is decomposed into a generic reduced-state question and an ECT-specific global-dynamics question. Only the former is exact mathematics under its stated Hilbert-space assumptions. The globally unitary completion, concrete shell implementation, subsystem map and Page-curve-compatible evaporation route remain Open. The full nonlinear strong-field completion remains open (OP-Q13).

36.8 Diagnostic discriminants across the gravity–matter interface

Status: compact cross-programme diagnostic summary. No new physics is introduced; the table consolidates the status of three focal frameworks along axes that discriminate them structurally and observationally. Connection: the costs-of-consistency table of Section 28.4 compared frameworks by architectural cost; the present table instead compares them by diagnostic signatures and structural commitments.

Table 94 summarises three focal frameworks — linearised-graviton effective field theory, the post-quantum classical-gravity (PCCG) hybrid, and the ECT candidate common-medium programme — along six diagnostic axes relevant to the gravity–matter interface.

Table 94: Cross-programme diagnostic discriminants among three frameworks for the gravity–matter interface. Entries summarise current structural and experimental commitments; detailed references are in the surrounding text and in the cited sections.

Axis	Linearised-graviton EFT	PCCG / Oppenheim hybrid	ECT candidate programme
Independent metric quantisation	Yes (fundamental)	No (classical metric)	Open/not identified: the present scalar characteristic map is not a physical metric or tensor completion
Fundamental stochasticity	No	Yes (fundamental stochastic law)	Open: neither fundamental stochasticity nor an exclusively reduced-state origin has been established
Global unitarity	Yes	No (intrinsic information loss)	Open; not implied by P2 or by the present reduced-state constructions
GIE status in mesoscopic protocols	Expected in standard protocols via nonclassical mediator exchange	Framework-dependent; the standard stochastic reading is not built around unitary mediator-induced entanglement, while loop-hole constructions exist in neighbouring hybrid classes	No protocol-level ECT prediction: the physical mediator, state, channel and apparatus coupling remain Open (§32)
Evaporation / information stance	Outside the scope of linearised EFT; the answer depends on a separate strong-field/ultraviolet completion	Intrinsic information loss in the named stochastic model	A thermal marginal is compatible with a supplied global isometry; ECT factorisation, thermality and evaporation dynamics Open (§36.2)
Relation to islands/QES language	Outside linearised-EFT scope; islands belong to separate holographic strong-field constructions	No generic entry follows from the framework label alone	Analogy only: the ECT factorisation, generalized entropy and extremization rule are all Open (§36.3)

Reading the table. The rows are not logically independent: a framework’s stance on metric quantisation strongly constrains its stance on stochasticity and on global unitarity. No single row uniquely falsifies any framework; the table summarises the joint pattern of commitments or open completion requirements recorded here. In particular, an Open ECT entry is not a prediction of the present action.

Scope and non-claims. The table does not claim (i) that the three frameworks exhaust the space of gravity–matter architectures (Newton–Schrödinger-class, measurement-feedback hybrids, and others

remain); (ii) that any individual cell is beyond revision with further work in the respective framework; (iii) that current experiments have decided between the frameworks. It is a structural cross-programme summary, not a verdict.

Bridge to the analogue-laboratory programme. The black-hole programme, including its comparative mapping to the islands/QES language (§36.3) and the cross-programme diagnostic summary (§36.8), concerns the same ordered medium as the vacuum-response, decoherence, and entanglement sections, but in a different regime. The next section turns from the astrophysical strong-field regime to controlled laboratory analogues, where horizons, mode conversion, and effective thermal response can be studied in experimentally accessible ordered media.

37 Analogue Laboratory Programme: Named Source-Model Tests

Status: Level A only for mathematics and measurements inside each named laboratory source model; Level B for a fully specified conditional map between such a model and a declared ECT sector; Open for an ECT-specific discriminator. Existing analogue results are neither evidence for ECT nor an ECT falsifier. Connection: compact-phase winding (§22.3), supplied vacuum-response models (§25), open-system kernels (§26), the quantum–classical boundary (§30.1), horizon interfaces (§36), and medium character (P5).

Purpose and inference boundary. Laboratory media can realise mechanisms that also appear in ECT candidate constructions: compact phases, winding sectors, mode conversion, response kernels and environment-sensitive coherence. An experiment first tests the action, state, operators and observables of its own source model. Similarity of mechanism is a comparison, not evidence that the ECT vacuum has the same owner. Conversely, failure of an analogue implementation rejects that implementation, not ECT as a whole.

Minimum bridge contract. An analogue result becomes an ECT test only after one prespecified bridge supplies all of the following:

- (i) the laboratory source action and the candidate ECT action sector;
- (ii) the preparation/state and boundary or driving protocol on both sides;
- (iii) an operator or vertex map, including normalisation and domain;
- (iv) an observable map, calibration and finite-time measurement rule;
- (v) uncertainty, nuisance and detector-response models; and
- (vi) a quantitative discriminator against the ordinary laboratory source model and other non-ECT explanations.

Without these six owners there is no likelihood ratio or falsification statement about ECT. Shared words such as “medium”, “horizon” or “correlator” do not supply the bridge.

Four retained mechanism classes. The earlier analogue inventory survives after this correction, but with source-model ownership made explicit.

- (i) **Compact phase and winding.** Superfluid helium, atomic BEC and Josephson systems can have quantised circulation or flux and defect-mediated phase slips. The topological statement is exact when the named order parameter remains nonzero on the contour. BR1 has the same conditional topology only after its own complex field, domain and defect boundary are supplied.

- (ii) **Boundary, drive and horizon response.** Superconducting circuits exhibit dynamical-Casimir mode production [167], and acoustic BEC experiments realise source-model horizon/mode-mixing observables [262]. Each follows from its platform action, state and detector. Neither supplies an ECT Casimir, Unruh or Hawking owner.
- (iii) **Controlled decoherence.** Matter-wave and circuit platforms can vary bath occupation, spectrum, correlation time and coupling. This tests the declared influence-functional kernel. It does not identify an ECT record vertex, environment split or universal collapse boundary.
- (iv) **Background sensitivity.** Density, stiffness, condensate fraction and proximity to a transition can be controlled within one platform. Correlated changes of response and coherence test that platform's action. They do not establish the Open ECT dark-sector or galactic source/force map.

Conditional statements retained inside named source models. Several useful tests remain sharp. If a compact order parameter is nonzero on a contour, its winding cannot change smoothly; a change requires a defect or a crossing of the zero-amplitude set. If one specified action, state and operator generate several response channels, the corresponding kernels must obey that source model's common spectral and response identities. If a declared bath owns decoherence, controlled changes of its spectrum and occupation must change the measured coherence according to the same kernel. If a common background parameter enters several operators, their responses must follow the jointly propagated covariance. These are falsifiable source-model statements. They become ECT statements only after the bridge contract above is derived and frozen.

What existing experiments establish. Quantised circulation, phase slips, dynamical-Casimir radiation, acoustic mode mixing and tunable environmental decoherence establish that the corresponding mechanisms occur in their respective systems. They demonstrate physical possibility and provide test-bed technology. They neither confirm one universal condensate mechanism nor distinguish ECT from conventional descriptions of those platforms.

Falsification hierarchy. A discrepancy is classified in order: instrument/resolution, preparation or state, boundary/drive model, operator/vertex, kernel/counterterm convention, observable map, and finally the named source-model claim. An ECT claim is reached only if the frozen analogue-ECT bridge and its discriminator have survived every preceding check. This preserves the operational M3 hierarchy of Appendix BK.5: success is reported for the declared protocol and channel, while failure names the first failed guard.

For the source-model dictionary, see Appendix BB.

Status summary. *Established externally:* the listed laboratory mechanisms inside their named source models and protocols. *Conditional Level B route:* a future six-owner bridge can turn one of them into an ECT-sector test. *Open:* a current ECT-owned bridge, quantitative discriminator and corresponding likelihood.

Summary. The analogue programme is retained as an experimental design programme, not as empirical support already accumulated for ECT. Its positive contribution is concrete: it identifies controllable source models, guard variables and observables with which a future ECT operator-level completion could be tested.

Operational M3 protocols. The ordered-branch record programme now carries three operational falsification protocols, specified in Appendix BK: the intrinsic/external threshold contrast (M3-1), the zero-mode detector (M3-2), and the Airy-width/fingerprint protocol (M3-3). Their observable inputs are fixed by the width-sector interface of Appendix BI and the threshold package of Appendix BH, with

platform mapping through the analogue-observable dictionary of Appendix BB. A failed protocol is interpreted through the failure hierarchy stated there; it is not automatically a falsification of the wider framework.

Bridge to the status map. The analogue-laboratory programme completes the structural development of the Quantum Sector. The next section collects the results of the preceding Part III sections into a single status map, identifies the remaining open problems, and states the completion status of Part III as a whole.

38 Status Map, Open Problems, and Completion Status of Part III

This section collects the results of the preceding Part III sections into a single status map, identifies the irreducible structural core that ECT has already established, separates it from the closure-level and open fronts, and states the completion status of the Quantum Sector as a whole. Section 38.1 presents the detailed row-by-row status table and the open-problem list. Section 38.2 organises the minimal quantum core into a strict structural tier, a structurally singled-out backbone, and the nearest closure-level or open fronts. Section 38.5 states the overall completion status and provides the bridge from Part III to the summary, comparison, and outlook sections that follow.

38.1 Status table and open problems of the Quantum Sector

Purpose: to summarise what the Quantum Sector of ECT already establishes structurally, what is presently available only at closure level, and what remains open. This subsection plays for the Quantum Sector the same rôle that the final status table plays for Macroscopic Physics. The assignments below reflect the architectural strengthening of the preceding Part III sections, including medium character (P5), the global/reduced unitarity distinction, the unified vacuum-response programme, the decoherence/arrow-of-time architecture, the quantum–classical boundary, and the ordinary projector-Gleason representation of an independently supplied normalised, noncontextual, countably additive projector measure on a complex Hilbert space with $\dim \mathcal{H} \geq 3$; physical application input (C3) and the Busch effect/POVM theorem under separately declared C2q hypotheses for a qubit remain separate, followed by the entanglement programme, the wider topological/dark-sector ontology, the conditional black-hole information-preservation programme, and the analogue-laboratory instantiability programme.

Methodological note on status levels. The strengthened postulate and branch structure improves the Quantum Sector in an important but limited way. In several places it still does not promote familiar quantum laws to unconditional Level A theorems. Its real gain is subtler and more architectural: it turns earlier heuristic claims into conditional structural statements, removes overclaims, narrows the admissible logical possibilities, sharpens the distinction between global and reduced descriptions, while leaving the global information question explicitly Open. The table below reflects these deliberately honest levels.

The status map condenses the nineteen Part III sections into the following twelve reader-facing thematic steps (including the scope/inputs layer as the first step). Dedicated bridge modules are retained in their owning rows and dependency links rather than counted as additional themes:

- (i) scope, inputs, and logical programme of the Quantum Sector;
- (ii) coherent phase dynamics, loop sectors, and the pre-quantum action scale S_0 ;
- (iii) wave kinematics, common-length hyperbolic parametrisation, extremal-action structure, and conservation laws;
- (iv) Hilbert-space bridge: the complete applicable corrected OS-II package, including reflection positivity, followed by unitarity, canonical structure, and the uncertainty principle;
- (v) Euclidean path-dependence, exchange sectors, and Dirac structure;
- (vi) quantum vacuum structure and the unified response programme (Casimir, Unruh, particle production, horizon thermality);

- (vii) open quantum systems, decoherence, and the arrow of time;
- (viii) quantum–classical boundary, probability, Born-type interpretation, and measurement status;
- (ix) quantum entanglement and Bell-type correlations;
- (x) topological / defect sectors and the dark sector;
- (xi) black-hole thermodynamics, Hawking-type radiation, and the information problem;
- (xii) analogue laboratory programme.

The resulting picture is already substantial, but deliberately not overclaimed. Figure 32 and Table 95 distinguish sharply between what is structurally established, what is plausible within the current coherent-sector closure, and what remains open. The status table below should be read together with the open-problem list and the dependency map: the table records local status row by row, the map records logical dependence, and the open-problem list records what is still needed for programme closure.

Table 95: Status map of the Quantum Sector of ECT. Level A = exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. Level A/B = an explicit Level A algebraic or obstruction result paired with a Level B structural/conditional route whose physical identification or full closure remains open. Level B = structural/conditional derivation or closure route with a declared open matching, input, or identification step. Open = not yet derived; where available, an explicit open-problem reference is given.

Statement / claim	Status	Basis / comment
Statement / claim	Status	Basis / comment
Supplied coherent $U(1)$ phase-sensitive sector	Level B / supplied closure	Not implied by the real P3 scalar; Section 20.1
Phase variable θ inside that supplied compact sector	Level B / supplied closure	Section 21.1
Integer winding sectors exist	Level A mathematics conditional on the supplied compact phase	Single-valuedness of Φ_{eff} ; Eq. (20.2)
Phase-only fixed-core reduced-coefficient candidate $t_{\text{elem}}^{(\text{fixed core})} = 2\pi t_0^{\text{EFT}}$	A fixed- L mathematics / B-Open closure	Defined by evaluating the fixed-length phase-gradient functional at a supplied core scale; not a global minimum and not fixed by topology
Tension-balanced finite-loop minimum $\mathcal{J}_{1,\text{min}} = 2\pi t_{0,\text{bal}}^{\text{eff}}$	A mathematics / B-Open ECT route	Exact for the supplied positive-tension 1D functional; the tension, common 4D reduction and unwinding/backreaction completion are Open
First-principles physical action scale	Open (OP-Q1)	Derive or refute K_θ^{1D} and $\mathcal{J}_{\text{loop}}$ from one 4D owner, then test universality; Appendix AS
Operational S_0 and the identification $S_0 = \hbar$	Level B calibration / Open owner	Either loop candidate requires its own owner map to the operational slot; the calibration is explicit, while its absolute derivation and cross-sector universality are independent Open tests; Eq. (21.7)
Hamilton–Jacobi / continuity decomposition	A mathematics given a supplied complex evolution law; ECT owner B/Open	Polar split alone supplies no Hamiltonian, current, domain, or dynamics; Section 22.1

Statement / claim (cont.)	Status	Basis / comment
Probability-current reading of a supplied complex continuity law	B/Open	Requires the complex law, state, inner product, domain, clock and physical matching; it is not the compact-phase Noether current
Schrödinger-type complex recombination	Level B	Structural route; $S_0 = \hbar$ and $\hat{H}_{\text{coh}}$ still needed; Eq. (22.7)
Lorentzian stationary-phase asymptotic	A-math conditional / B/Open ECT owner	Requires a supplied oscillatory integral; the Euclidean extremum has distinct dominance conditions
Declared conservation laws in their owning actions	A within each supplied action	Real-scalar Noether currents and compact-phase EFT current are distinct; no common probability-current owner follows
Reflection positivity in massive Gaussian broken-phase subsectors	Level A mathematics in supplied free model	Section 23.1; RP plus the standard free-scalar covariance, symmetry, regularity, locality and clustering package
Reflection positivity in free massless Goldstone-coordinate Gaussian subsectors	Level A mathematics in supplied free model	Section 23.1; standard local massless-scalar OS package after the infrared/zero-mode domain is fixed; physical mode count remains Open
Reflection positivity for full interacting coherent sector	Open (OP-Q16)	Section 23.1; constructive QFT programme
Global unitarity in reconstructed positive Gaussian subsectors	Level A mathematics in supplied free models	Follows only after the complete applicable corrected Gaussian OS-II hypothesis set, not from reflection positivity alone; Section 23.2
Global unitarity beyond Gaussian regime	Level B/open	Depends on OP-Q16; not yet independently established; Section 23.2
Global vs. reduced unitarity distinction	Level A	Well-defined structural distinction between full coherent unitarity and reduced-sector effective non-unitarity; Section 23.2
Canonical organisation of smooth coherent branch	Level A mathematics in supplied reduced model	Section 23.3; $\pi_{\theta}^{\text{phys}} = \Sigma_{\theta} \pi_{\theta}^{\text{red}}$ for a finite positive constant Σ_{θ}
Poisson brackets of coherent canonical variables	Level A mathematics in supplied reduced/physical symplectic normalisations	Eq. (23.17); the reduced and physical brackets must not be mixed
Local-chart physical-momentum commutator candidate normalised by S_0	Level B/Open	Eq. (23.18); $[\theta, \pi_{\theta}^{\text{phys}}] = iS_0\delta$, equivalently $[\theta, \pi_{\theta}^{\text{red}}] = i(S_0/\Sigma_{\theta})\delta$ on a declared common core; no global self-adjoint angle CCR is claimed
Integer winding of the compact phase field	Level A topology in supplied smooth compact-field domain	Eq. (20.2); independent of a canonical-generator spectrum

Statement / claim (cont.)	Status	Basis / comment
Canonical compact-shift-orbit generator spectrum	Level A operator mathematics under supplied Weyl representation and self-adjoint domain / B/Open physical ECT owner	Eqs. (23.19)–(23.20); $[\hat{Q}_\theta^{\text{phys}}, U_0] = S_0 U_0$ and spectrum $S_0(\mathbb{Z} + v_{\text{tw}}/2\pi)$, reducing to $S_0\mathbb{Z}$ only for the supplied periodic domain $v_{\text{tw}} = 0$
Momentum-operator-type wave kinematics	Level B	Eq. (23.21); depends on the physical S_0 normalisation and representation
Full operator algebra and many-body canonical completion	Open (OP-Q4)	Section 23.3; operator-level derivation beyond the reduced coherent sector
Uncertainty inequality $\Delta q \Delta p_{\text{phys}} \geq S_0/2$	Level A operator mathematics conditional / B–Open physical owner	Eqs. (23.26), (23.27); applies to a supplied self-adjoint noncompact canonical pair with common product domains, not automatically to the compact angle and rotor
Heisenberg form $\Delta q \Delta p_{\text{phys}} \geq \hbar/2$	Level B calibration / Open owner	Requires the separate match $S_0 = \hbar$ (21.7)
Two-sided Euclidean boundary dependence	Level A	Elliptic Euclidean field equation; Section 24.1
Interference/tunnelling route from Euclidean methods	Level B route / Open physical ECT	Two-sided elliptic dependence is Level-A BVP mathematics but is neither sufficient nor discriminating. Interference additionally needs an admitted quantum measure/coherent structure, composable kernel, state/preparation and observable map; tunnelling needs a named action/saddle, state, boundary/continuation prescription, determinant/prefactor and detector map. No ECT-specific amplitude or rate is derived; Sections 24.1 and 24.3
Indefinite-causal-order compatibility route	Level B	Structural reading only; quantitative switch reconstruction open; Section 24.1
Visibility proxy for the quantum-switch source model	Source-model / ECT Open	$\Phi_{\text{vis}} = -\ln V$ is an observational transform, not a universal degradation law; Eq. (24.3)
Topological exchange-sector dichotomy ($\mathbb{Z}_2$)	Level A conditional topology / physical ECT map Open	$\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ for two supplied identical unlabeled points in defect-free $\mathbb{R}^3$; the physical spatial/excitation map and representation selection remain Open; Section 24.4
Spatial dimensionality $d_{\text{spatial}} = 3$ fixes the exchange topology	Level A conditional / physical slice Open	Given a separately supplied three-dimensional spatial slice, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$; P4 alone does not derive that slice; Section 24.4

Statement / claim (cont.)	Status	Basis / comment
Bosonic/fermionic structural compatibility	Level B	Exchange topology plus ordered-branch representation content; Section 24.4
Abstract $\mathfrak{so}(4) \rightarrow \mathfrak{so}(1,3)$ generator continuation	Level A conditional mathematics	Standard complex continuation after its map is supplied; not a physical metric derivation; Section 24.5
Clifford representation $\{\gamma_L^\mu, \gamma_L^\nu\} = -2g^{\mu\nu}$	Level A conditional mathematics / Open ECT owner	Requires supplied metric, tetrad and spinor representation
Dirac-type free dynamics for realised spinorial excitations	Level B (conditional)	Only inside the supplied four-complex-component irreducible Clifford-factorisation class: physical Lorentzian metric/tetrad and spin structure, $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ module, local linear constant-coefficient Lorentz covariance and exact KG factorisation; reducible, constrained/gauge and non-Clifford systems lie outside; Eq. (24.12); Section 24.5
Physical spinorial sector of ECT	Open (OP-Q11)	Requires metric/spinor descent, spectrum, CAR/Fock construction and state
Pauli exclusion	Level B (conditional)	From Dirac structure + spin-statistics argument; conditional on $S_0 = \hbar$ (OP-Q2)
Full spin-statistics theorem	Open (OP-Q11)	Requires operator completion beyond current closure; the free-kernel Euclidean OS test is now Level A (Q11.1a, Section 9.9)
Spin discreteness / Stern–Gerlach selection	A-math conditional / B / Open	Discreteness is standard compact- $SU(2)$ representation mathematics after a physical rotation/spinor sector is supplied; its ECT owner, apparatus channel, outcome and update remain Open; Section 24.5
Supplied ordered/coherent models have non-trivial correlator and stiffness structure	Level A inside the supplied reduced models / physical ECT vacuum Open	A physical common ECT vacuum still requires its state, operator/bundle, continuation, couplings and response ownership; Section 25.1
Supplied soft-sector boundary-value response	Level A mathematics / B–Open physics	Each declared operator and boundary condition has a calculable mode response; the physical condensate boundary and ordinary-matter coupling are not derived; Section 25.3
Supplied scalar Dirichlet coefficient	Level A mathematics / Open ECT	Eq. (25.3); physical mode count, material boundary map and stress response are missing

Statement / claim (cont.)	Status	Basis / comment
Euclidean–Rindler/KMS route	Level A external / Open ECT	The standard response follows only after supplying the physical metric, state/KMS condition, detector coupling and clock/action matchings; Section 25.4
Unruh temperature formula in the supplied route	Level A external / Open ECT	Eq. (25.8); requires $S_0 = \hbar$, a physical cone, and the missing detector/state owners; Section 25.4
Time-dependent ordered backgrounds: background-dependent particle notion	Level A external / Open ECT	Standard after a time-dependent action and in/out states are supplied; ECT ownership Open; Section 25.5
Bogoliubov-type mode mixing in ECT coherent vacuum	Level A external / Open ECT	Eq. (25.9); ECT action, states, particle map and quantitative rates open (OP-Q12); Section 25.5
Quantitative production rates $ \beta_k ^2$	Open (OP-Q12)	Requires completed time-dependent coherent-vacuum calculation
Horizon-thermality comparison	Level A external / Open ECT	Standard Hawking target; requires an independently completed physical metric, quantum state and strong-field sector; Sections 25.2, 36
Common vacuum-response programme (Casimir/production/Unruh/horizon)	Open	Section 25.2; a proposed common architecture, not one response operator derived for all four regimes
Analogue-gravity source-model comparison for coherent-vacuum mechanisms	External source-model results / ECT Open	Section 25.2; mechanism comparison is not evidence for ECT and no analogue result is a generic ECT falsifier
Reduced influence-functional organisation	Level B	Open-system EFT of coherent branch; Eq. (26.1)
Retarded propagation of ordered-branch scalar Φ	Level A (quadratic)	Appendix F
Nonnegative spectral measure $\rho(\mu^2) \geq 0$	Level A conditional / ECT owner Open	Follows only for a supplied positive Hilbert/state/observable and residue representation or explicit positive oscillator decomposition; it is not derived by the classical broken-phase Hamiltonian alone
Three-dimensional derivative-coupling bath toy gives $J_{\text{bare}} \propto \omega^4$ and $J_W \propto \omega^3$	Level A conditional mathematics within a supplied quadratic bath	Not a derived ECT record channel; vertex- and state-dependent; Eq. (26.5)
Effective ohmic regime	Model ansatz	Not derived from the minimal bath; Eq. (26.6)
Markov approximation	Model condition	Insufficient for monotone entropy; Eq. (26.7)
Low-entropy initial-branch interpretation	Open	Requires a measure, coarse graining, and state-selection rule; Section 26.1

Statement / claim (cont.)	Status	Basis / comment
Entropy monotonicity theorem for an ECT reduced channel	Open	Requires the normalised channel/state, entropy functional, and unitality/detailed-balance/contractivity hypotheses; OP-Q17
Protocol-defined Crooks fluctuation relation	Open	Requires forward/reverse path measures, time reversal, initial ensembles, and local detailed balance; Eq. (26.20) is a definition only
C_{60} collisional-decoherence source-model anchor	External observation / source model / ECT Open	Interference is retained from Arndt et al. [169]; the exact pressure sub-number lacks an identified primary owner and is not quoted; no distinct ECT rate is published; Section 26.2
Diósi–Penrose external comparison	External formula and Level-C hard-sphere point / ECT Open	$C_{DP}(1) = 51/160$ and 70.6 ms only for the declared continuum geometry; ECT projector, vertex, state, noise and absolute normalisation not identified; Section 26.3
Common influence-functional notation	Level B organisational / physical identification Open	Does not establish one bath or one kernel for environmental and gravitational channels
Physical reverse $O(3) \rightarrow O(4)$ segment mechanism	Open	Ordinary Euclidean boundary dependence does not establish physical reverse segments, their weight, or a laboratory threshold; Section 24.1
Reduced-state dephasing / record crossover	Level B/Open	Channel- and protocol-specific; does not change metric signature or select a unique outcome; Section 30.1
Pure-dilation qubit mutual-information toy	Level A model algebra / ECT Open	Total system–environment mutual information in the declared toy, not accessible detector information or a protocol-independent bound; Section 30.3
Wigner’s-friend reduced-state record reading	Interpretive / Open	A specified coarse graining can form records; pointer basis, unique outcome, and update remain Open; Section 30.1
Echo/control recoherence	Standard channel result / ECT Open	No Lorentzian-to-Euclidean signature reopening or physical $O(3) \rightarrow O(4)$ backreaction map is derived; Section 24.1
Amplitude / decoherence / probability distinction	Level A logic under supplied state/channel	Decoherence can suppress coherences but does not create a measure or outcome; Section 30.4; Eq. (30.6)

Statement / claim (cont.)	Status	Basis / comment
Admitted projector measure: conditional ordinary Gleason representation	Level A mathematics under theorem hypotheses / ECT Open	Unique trace representation only for an independently supplied normalised, noncontextual, countably additive projector measure on a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. State/preparation, event, probability origin, outcome and update remain Open (OP-Q7)
Independent inputs (C1), (C2) or (C2q), and (C3) for the conditional Gleason representation	Conditional / Open ECT ownership	(C1) supplied Hilbert/event algebra; (C2) supplied projector measure in complex dimension at least three, or separately declared C2q full-effect measure hypotheses in dimension two; (C3) supplied preparation plus projector/effect event map. A dephasing channel is a separate application layer; Section 30.4
Unique outcome selection beyond an admitted physical probability-weight map	Open (OP-Q19)	Relation between branch weights, decohered alternatives, and actually realised records remains incomplete; Section 30.4
Full Born rule as first-principles theorem	Open (OP-Q7)	Requires physical derivation of the state/preparation, event map, probability origin, outcome and update from coherent-sector dynamics
Completed measurement-update theory	Open (OP-Q8)	Pointer sectors, branch update, records not complete
Non-factorisable coherent correlations (entanglement route)	Standard conditional mathematics / ECT Open	Requires a supplied nonfactorising state/channel; a single medium does not force nonfactorisation; Section 31
No-signalling compatibility of common-medium entanglement	Conditional / ECT Open	Must be verified for the supplied state and local measurement channel; Section 31
Bell-correlator completion	Open (OP-Q20)	Must derive the physical preparation, state-or-configuration owner, settings, measurements and full joint outcome law, together with no-signalling and the Tsirelson restriction. If a hidden/configuration representation is claimed, it must additionally supply a positive measure, response maps and causal/intervention semantics and determine rather than assume measurement independence and Bell factorisability; Section 31
Werner-state CHSH visibility marker	Source-model toy / ECT Open	$\Phi_{\text{vis}}^{\text{Bell}} = \ln \sqrt{2}$ follows from the declared Werner proxy; it is not an ECT Bell-correlator derivation; Section 31

Statement / claim (cont.)	Status	Basis / comment
Entanglement–tunnelling relationship	Interpretive / ECT derivation Open	The phenomena retain separate dynamical owners; no common suppression scalar or derivation of one from the other is claimed; Section 31
Wider defect sector beyond smooth coherent branch	Level A	Section 34 + 6.7
Defect sectors: structural existence	Level A	Section 34
Defect sectors: full dynamics and classification	Open (OP-Q9)	Core structure, stability, production; not yet complete
Smooth collective dark-sector candidates	Level B	Bridge MP to QS; Section 18.1
Topological/defect dark-sector candidates	Level B	Section 34; detailed dynamics open
Quantitative dominant dark-matter identification	Open (OP-Q10)	No unique first-principles candidate established yet
Black-hole thermodynamics as an external effective target	A external / Open ECT	Eqs. (36.1)–(36.3); rewriting requires $S_0 = \hbar$
Reduced-state thermality $\neq$ fundamental information loss	Level A	General QS reduced-state logic; Eq. (36.8)
Global information preservation in an interacting ECT completion	Open / conditional	Reduced-state thermality is compatible with purity (A mathematics), but P2 does not construct a Hilbert space or prove global interacting unitarity; Section 36
Candidate ordered-phase shell near the horizon	Level C/Open	Section 36; supplied strong-field ansatz
Generalized entropy $S_{\text{gen}} = S_{BH} + S_{\text{ext}}^{\text{coarse}}$	A external book-keeping / Open ECT	Eq. (36.5); natural reduced-observer bookkeeping
Page-curve-compatible unitary evaporation logic	A mathematics under supplied isometry / Open ECT	Does not select a Page curve; Section 36
Explicit Page-curve computation	Open (OP-Q13)	Requires global nonlinear shell dynamics
Full strong-field evaporation completion (Page curve)	Open (OP-Q13)	Global nonlinear horizon solution not yet complete
Strong-field observable response	Open	No ECT dispersion, energy-response, ringdown, echo or primordial-black-hole correction is derived
Named analogue source-model mechanisms	External Level A only inside each supplied platform model	Compact phase, winding, response and decoherence mechanisms are experimentally accessible; not evidence for ECT and not an ECT falsifier
Analogue–ECT operator-level bridge	Open	Requires action, state, operator, observable, uncertainty and discriminator maps; Section 37

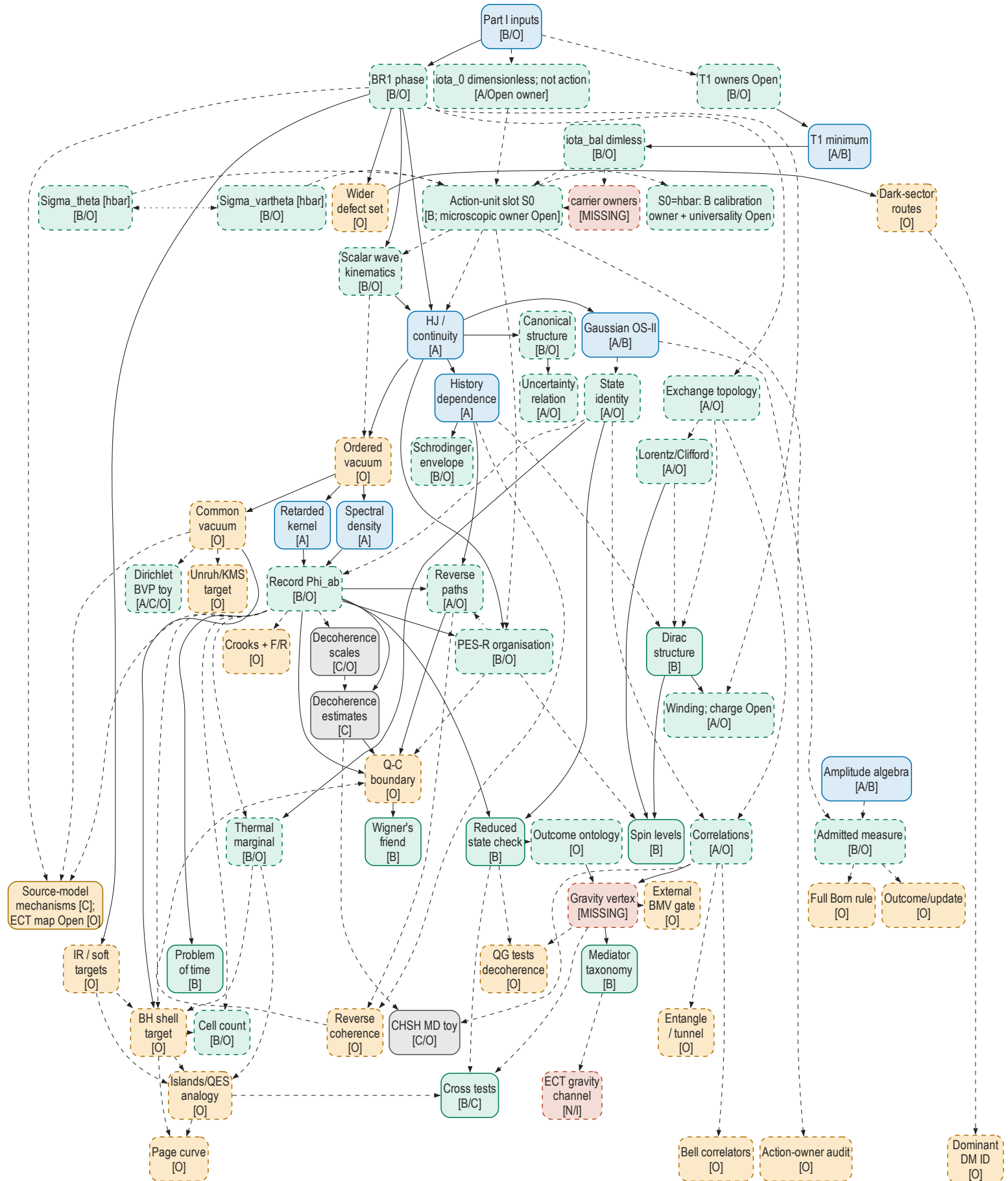
Statement / claim (cont.)	Status	Basis / comment
Visibility-derived dephasing proxies	Source-model / ECT Open	Procopio and Jacques values are $\Phi_{\text{vis}} = -\ln V$ transforms; no ECT record kernel is inferred; Section 27
Werner-state CHSH marker $\Phi_{\text{vis}} = \ln \sqrt{2}$	Source-model toy / ECT Open	Eq. (31.36); no universal boundary or ECT discriminator; Section 27
PES-R response–persistence taxonomy	Level B / ECT inputs Open	Dynamical admissibility, finite-time retention, and pairwise records are distinct; no universal intersection or quantisation/Born/outcome theorem; Section 29
Frozen P6 spectral-record framework (record kernels, dephasing classes, line/ribbon records, threshold protection, width interface, P26 over-closure)	Level B	Frozen with non-proliferation guard; Appendices BE–BL
M3 operational falsification protocols (M3-1/2/3)	Level B	Appendix BK; reporting packet and failure hierarchy specified
Analogue-platform execution of the M3 protocols	Open	Platform selection open (frozen-ledger item O4); Appendix BK
Problem of time in ECT: owner hierarchy	Level B/Open	P1 supplies no external time; P4 supplies a direction; scalar hyperbolicity, physical clock/lapse, record dynamics and thermodynamic arrow retain separate owners; Section 3.7
Conditional discriminants for named quantum-gravity/decoherence completions	Open premises / conditional tests	Exclusions apply only after a completion owns and preregisters the mediator, state, channel and global-dynamics commitments; Section 38.4
Reduced-state resolution of the hybrid-consistency problem	Level B (architectural)	Candidate reduced-state architecture, not a finished theory; Section 28, Table 84
Two-level ontology (Level I / IIa / IIb)	Level B/Open architecture	Conditional bookkeeping; the interacting observable algebra, state, universal partition and concrete reduction maps remain Open; Section 28.3
Gravity-induced-entanglement channel	Open / NOT IDENTIFIABLE	External BMV classes are known; ECT mediator/vertex/state are absent; Section 32
Operational criterion for BMV no-go vs loop-hole regimes	External conditional result	Applicable only after a physical reduced mediator is derived; Section 32.4
Mediator taxonomy: scalar trace, fixed director, composite and tensor candidates	Terminal bounded ledger	Scalar vertex parametric; director missing vertex; full ECT channel not identifiable; Section 33
Physical gravitational entangling mediator	Open	Requires nonzero two-body vertex, positive state and a proved nonfactorising reduced channel; Section 33.3

Statement / claim (cont.)	Status	Basis / comment
BMS, memory and soft-graviton sector	External completion tests	No ECT tensor charges, memory map or soft-graviton owner is identified; Section 35
Islands/QES structural mapping onto critical-shell picture	Level B/C	Structural alignment, not a derivation of S_{gen} or of replica wormholes from ECT; Section 36.3, Table 92
Cross-programme diagnostic discriminants	Summary-level	Six axes comparing ECT common-medium with linearised-graviton EFT and PCCG-hybrid; Section 36.8, Table 94

Selected status and dependency map. Figure 32 is a selected connected Part III dependency graph. It keeps coherent-phase dynamics, the two distinct action-scale candidates, the candidate T1 owner calculation, OS-II reconstruction, canonical organisation, response/decoherence, PES-R, admitted measures, Born-rule status, particle/statistics routes and quantum-gravity targets in one connected view. In particular, it contains no direct decoherence, quantum–classical-boundary or unitarity derivation edge into the Born node. The OS-II/Hilbert route supplies event structure only under its full stated hypotheses; an independently admitted measure and the physical state/preparation, probability, outcome and update owners remain Open.

Selected Part III status and dependency map

Status focus: dimensionless ι_0 is not an action; S0 is an action-unit slot; S0=hbar is a Level-B calibration with owner and universality Open.



P postulate/input | A exact in stated model | B conditional/calibration | C fit/toy | O Open | MISSING owner absent. Solid: exact declared step; dashed/dotted: conditional or Open dependency. Literal node status controls.

Reading discipline. The poster uses the same status semantics as the other part maps: solid arrows organise declared dependencies, dashed arrows show conditional/Open links, and literal node text controls status. T1 supplies neither $S_0 = \hbar$ nor a PES-selected occupation law. The connected A4 map preserves the full topology in a zoomable vector rendering; exact source, label and edge records remain in its deterministic companion key.

Principle of Euclidean Stationarity as organising output. PES-R is a Level-B/Open taxonomy with three non-equivalent layers: sector-specific dynamical admissibility, finite-time retention, and pairwise record distinguishability. It does not replace the sector equations, derive a universal scalar bath, or supply Born weights and measurement outcomes. The physical ECT kernels and record operators remain OP-PES-1 record-channel inputs.

Open problems of the Quantum Sector.

OP-Q1 Derive or refute, from one 4D coherent-sector action and profile, the reduced stiffness K_θ^{1D} , positive line tension $\mathcal{T}_{\text{loop}}$, unwinding barrier and backreaction entering the conditional balanced-loop route; then test whether its action scale is universal (Appendix AS).

OP-Q2 Final identification of the pre-quantum action scale with the observed $\hbar$ (21.7).

OP-Q3 Full derivation of the effective coherent Hamiltonian $\hat{H}_{\text{coh}}$ for the relevant physical sectors.

OP-Q4 Rigorous derivation of quantum commutators from the coherent branch beyond canonical plausibility arguments (23.18).

OP-Q5 Derive the physical resolved/unresolved split, record operators, state, retarded kernel, and noise kernel entering Eq. (26.1).

OP-Q6 Derivation of the effective ohmic regime and of the Markov limit from the condensate bath without auxiliary closure assumptions (26.6).

OP-Q7 Full first-principles emergence of the Hilbert-space projector and probability-measure structure from bare P1–P6. Complete applicable corrected OS-II packages close the standard Hilbert reconstruction only for supplied Gaussian broken-phase models (Level A mathematics); what remains includes the full interacting OS-II package (OP-Q16), a physical state/preparation and event algebra, probability origin, record/dephasing channel, outcome and update. Ordinary projector Gleason only represents an independently admitted measure in complex dimension at least three; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses.

OP-Q8 Completed measurement theory: branch update, pointer stability, and record formation.

OP-Q9 Dynamical classification, production, and stability of topological defect sectors outside the smooth coherent branch (Section 34).

OP-Q10 Quantitative link between coherent dark candidates and observed dark-sector phenomenology.

OP-Q11 Physical realisation of spinorial excitations in ECT and an explicit emergent many-excitation sector decomposition (Fock-like reconstruction) with variable excitation number from condensate dynamics. ECT does not require Fock space as a fundamental postulate, but the effective reconstruction—especially for the fermionic many-body sectors—remains to be constructed. Note: the abstract generator continuation and Clifford matrix algebra are Level A conditional mathematics only after their metric and representation inputs are supplied; the restricted Dirac theorem is Level B (Section 24.5). The physical metric/spinor descent, excitation spectrum and CAR/Fock sector remain Open.

- OP-Q12** Quantitative derivation of coherent-vacuum response effects (Casimir force, particle production rates, Unruh detector response) from the completed reduced coherent EFT (Sections 25.3–25.4).
- OP-Q12c** Thermal completion of the hot Big-Bang branch: entropy production, thermalisation, and radiation-bath formation after geometric branch onset; CMB acoustic-peak structure with cosmological matter content and matching left Open.
- OP-Q12b** Derive or falsify a physical ECT owner for the conditional boundary-cell count, then test a covariant Bousso-type formulation for dynamical horizons and null hypersurfaces; full bulk–boundary reconstruction in the strong holographic sense remains Open.
- OP-Q13** Global nonlinear completion of the black-hole evaporation problem in the presence of a critical ordered-phase shell, including (i) explicit shell dynamics and profile, (ii) generalized-entropy evolution (36.5), and (iii) a Page-curve-type computation for the reduced radiation sector (Section 36).
- OP-Q14** Construct an owner-complete strong-field action, state, solution, perturbation operators, radiation channel and likelihood; only then determine whether any nonzero compact-object or near-horizon observable differs from the external reference theory.
- OP-Q15** Derive the physical ECT resolved/unresolved split, record operators, state, and spectral kernels, then determine whether any weak/strong monitoring interpolation exists. Total mode count and an ohmic/Markov ansatz do not define a universal regime boundary.
- OP-Q16** Construct the interacting coherent-sector Euclidean measure or Schwinger functions, physical observable algebra, and time reflection; verify temperedness together with the applicable OS-II regularity/growth condition (for example E_0' linear growth), Euclidean covariance, reflection positivity, permutation/graded symmetry, and all applicable locality and clustering requirements; then complete the Osterwalder–Schrader reconstruction and identify its time generator, clock/metric map, and supplied S_0 with the physical coherent branch. Reflection positivity alone does not establish global unitarity beyond the quadratic regime.
- OP-Q17** Construct an ECT reduced channel, state, entropy functional, and conditional entropy theorem; separately define forward/reverse protocols, path measures, time reversal, initial ensembles, and local detailed balance before testing a Crooks relation (Section 26.4).
- OP-Q18** Full quantitative Casimir derivation for the condensate-reflecting sector and identification of laboratory systems implementing the required ordered-medium boundary conditions (Section 25.3).
- OP-Q19** Unique-outcome selection beyond conditional Born-rule weights, including the relation between branch weights, decohered alternatives, and actually realised measurement records (Section 30.4).
- OP-Q20** Construct the physical Bell experiment from condensate dynamics: derive the preparation, physical state-or-configuration owner, setting generators, measurement maps and full joint outcome law. If a hidden/configuration representation is proposed, also define its constrained space, positive measure, response/record maps and causal/intervention semantics, and determine rather than assume whether measurement independence or Bell factorisability fails. In every route, derive the angular correlation law, operational no-signalling and the Tsirelson restriction, and test robustness across detector, random-generator and cosmic-setting protocols (Section 31).
- OP-Q21** Smooth-branch exactness of the Pauli exclusion principle and its admissible departures. Determine whether the completed fermionic reconstruction of ECT enforces exact canonical anticommutation relations throughout the smooth coherent ordered branch, as provisionally indicated by the canonical-normalisation no-go argument of Section 24.4 and Appendix AT. If genuine departures from standard fermionic statistics exist in ECT, identify the precise nonlocal, topological-defect, or branch-transition mechanism responsible, rather than attributing them to smooth local condensate gradients.

PES-R-specific open targets (§29.13):

- OP-PES-1** Derive the physical resolved/unresolved split, record operators, and ECT retarded/noise kernels from the action.
- OP-PES-2** Derive, or rule out, a one-history large-deviation/path measure with its drift, diffusion, Jacobian, convention, and boundaries.
- OP-PES-3** Complete the physical graviton sector and UV closure; PES-R itself is not a quantisation prescription.
- OP-PES-4** Derive physical decay widths from vertices, self-energies, form factors, states, and phase space.
- OP-PES-5** Test whether any backreaction map connects a reduced-state record exponent to $\alpha - \beta$; no such map is currently known.
- OP-PES-6** Generalise the toy information–dephasing relation to a specified detector channel; no protocol-independent bound is claimed.
- OP-PES-7** Construct one action-owned, constraint-reduced physical channel in which retarded response and ordered noise are measured with the same operator, source vertex, projector, form factor, state and detector transfer. Apply the Wightman inequalities first and then test a preregistered single-temperature KMS model on held-out frequency bins.
- OP-PES-8** Derive a gauge-invariant primordial record operator, state and generally two-time ordering-front kernel; propagate its declared phase-diffusion or mode-mixing template through the Boltzmann/recombination map before using acoustic phase data. The CMB does not directly identify Φ_{ab} or a pointer basis.

Frozen P6 record-sector open targets (Appendices BE–BL).

- OP-P6-1** Per-protocol selection of the effective record-temperature source T_{eff} for the M3 implementations (frozen-ledger register O2).
- OP-P6-2** Unresolved record/line-ensemble entropy matching in the P6 sector. The frozen ledger uses the shorthand s_{unres}/μ for the open problem of defining whether unresolved record degeneracy should be represented by a coarse-grained entropy density s_{unres} , by a line/path multiplicity μ , or by a protocol-specific entropy factor. Any such term must be defined at a declared core scale and compared with the pair pressure σ_{dec} ; it is not an adjustable stabilising entropy.

The common technical object unifying OP-PES-1, OP-PES-2, OP-PES-4, and the quantitative side of OP-PES-6 is the condensate spectral density $J(\omega, \mathbf{x})$: one kernel fixes decoherence, dissipation, decay widths, lifetimes, and the quietness index.

38.2 Status-separated minimal quantum reconstruction core

The following results organise the minimal quantum core of ECT into three layers: a strict structural core, a structurally singled-out backbone, and the nearest closure-level or open fronts. They should be read together with the detailed status map in Section 38.1: the purpose of the present subsection is not to replace the full table, but to identify the irreducible backbone that already exists inside the theory and to separate it clearly from what still depends on closure or remains open.

Status-separated mathematical core.

- (i) Integer winding classification is Level A topology only within the supplied compact- $U(1)$ phase EFT.
- (ii) Conditional Madelung decomposition after a norm-preserving complex evolution law, Hamiltonian, domain and clock convention are supplied; ECT ownership of those inputs is Open.
- (iii) Poisson-bracket organisation is classical conditional mathematics; a physical canonical quantum representation remains Open.
- (iv) Discreteness of the integrated compact-phase generator follows only after its representation is supplied; the physical generator and its S_0 normalisation are Open.
- (v) The Robertson–Schrödinger inequality is Level A mathematics only after a Hilbert space, self-adjoint domains and commutator are supplied; the physical commutator and coefficient S_0 remain Level B/Open.
- (vi) Reflection positivity and standard OS reconstruction in the supplied free Gaussian scalar models under their complete declared hypotheses (massive radial and free massless Goldstone-coordinate measures); physical ECT mode identification remains Open.
- (vii) The structural distinction between global and reduced unitarity.
- (viii) Two-sided Euclidean boundary dependence from the elliptic field equation.
- (ix) Exchange-sector topology $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ in the separately supplied defect-free $\mathbb{R}^3$ point-configuration model; its physical slice and excitation map remain Open.
- (x) Standard Lorentz-generator and Clifford matrix algebra after the complex continuation, metric and spinor representation are supplied; their physical ECT owners remain Open.
- (xi) Retarded propagation of the supplied quadratic ordered-branch scalar, together with the exact conditional hyperbolicity gate for a declared $P(X)$ branch and nonnegative spectral density in the quadratic sector.
- (xii) Structural admissibility of reverse $O(3) \rightarrow O(4)$ Euclidean segments in the path integral.
- (xiii) The amplitude / decoherence / probability distinction.
- (xiv) A future globally unitary interacting completion would exclude fundamental information destruction; P1–P6 do not yet prove that completion.
- (xv) The ordered-condensate state space is not exhausted by the smooth coherent branch BR1.

Structurally singled-out backbone (Level A/B).

- (i) The presence of two supplied loop-scale candidates, ι_0^{EFT} and $\iota_{0,\text{bal}}^{\text{eff}}$, together with an Open programme for deriving their common four-dimensional owner, selecting a viable route and mapping the corresponding action-valued product $\Sigma_\theta \iota_0$ to operational S_0 , and testing universality.
- (ii) The unified vacuum-response logic relating Casimir, Unruh, particle production, and horizon thermality as four probing regimes of one coherent vacuum.
- (iii) Named laboratory source models realise compact phase, winding, response and decoherence mechanisms. These external results motivate controlled bridge-building, but are not evidence for ECT; an ECT test remains Open until the six-owner bridge and discriminator are supplied.

Closure-level results and open fronts (Level B / Open).

- (i) Generic influence-functional and reduced-state mathematics is available, but the ECT environment split, record operators, kernels, state, and entropy-production theorem remain Open.
- (ii) C_{60} , Fein and Pedalino observations are external environmental-decoherence anchors; the apparatus rates are not reproduced here. The Diósi–Penrose functional and declared hard-sphere orientation point are external comparisons, not an ECT rate.
- (iii) Visibility losses define dephasing proxies in declared simple models; they are neither ECT kernel determinations nor reverse-path probabilities.
- (iv) A common ordered medium supplies a candidate arena for correlated subsystems, but no nonfactorisation theorem, Bell measure or physical correlation law follows from common support; those owners remain Open.
- (v) The qubit information–dephasing and CHSH calculations are toy-model thresholds, not universal quantum–classical or measurement theorems.
- (vi) Born-type weights have a conditional trace representation only after (C1)–(C2), with (C3) needed for a physical application and an explicit extension needed in dimension two; the physical measure and unique outcome remain Open.
- (vii) PES-R organises modes already supplied by the sector dynamics; its one-history, signature-changing, lifetime, and universal-selection readings are not derived and remain Open.
- (viii) Full interacting OS closure, spin–statistics, Bell correlators, condensate-reflecting Casimir response, nonperturbative evaporation, defect dominance, and the microscopic value of S_0 remain their named Open tasks.

This distinction is essential. ECT does not yet claim to have derived all of standard quantum mechanics from its postulates. What has been identified is a dependency map: supplied scalar/coherent models host several exact mathematical substructures, while physical quantum states, operators, probability, outcomes, entanglement channels and strong-field owners remain explicitly separated Open tasks.

Mesoscopic gravity owner firewall. The external Diósi–Penrose functional supplies no ECT-specific rate or correction. An application requires its own density profile and regularisation, while an ECT claim additionally requires the record vertex, state, noise kernel, and finite-body charge.

Logical relocation of quantum mechanics. Part III reconstructs several kinematic ingredients and states explicit Open targets. A complete applicable corrected OS-II package governs the separate Hilbert reconstruction. Independently, ordinary projector Gleason represents only an already supplied normalised, noncontextual, countably additive projector measure in complex dimension at least three, with the Busch effect/POVM theorem under separately declared C2q hypotheses required for a qubit. State/preparation, event, probability origin, outcome and update remain Open; decoherence explains reduced-state interference suppression but does not create those owners, and PES-R is an organising taxonomy rather than their derivation.

38.3 ECT as a foundational framework for quantum mechanics: structural comparison with the standard formalism

Status: comparative and pedagogical synthesis of results established in the preceding sections of Part III. No new physical results are introduced here. The role of this subsection is to organise the structural comparison between ECT and the standard quantum formalism along three explicit axes: ontology of the primary object, type of mathematical problem, and origin of the statistical description. The formal

status ledger of the same material, organised by derivational level and keyed to the open-problem labels, is given separately in Section 38.2; current limitations of the formulation are collected in Section 49.10.

Standard quantum mechanics is an extraordinarily successful predictive formalism. Part III asks whether selected pieces of that formalism can be reconstructed conditionally from a Euclidean condensate model. The answer is status-separated: the bare Euclidean equation and some linear/Gaussian identities are Level A inside their stated models; the scalar Lorentzian and Schrödinger-type reductions are conditional; and the physical Hilbert-space map, operator algebra, universal action scale, probability measure, outcomes, particle statistics, and detector channels remain Open. The comparison below therefore organises a reconstruction programme. It does not claim that ECT already explains why the full quantum formalism is realised in nature.

Two different questions answered by two theories

Standard quantum mechanics supplies the tested operational rules. ECT treats some of those rules as possible reconstruction targets. Table 96 distinguishes what is part of the present ECT input, what has been obtained conditionally, and what still lacks an owner.

Table 96: Two questions, two frameworks. Each row records the fundamental ingredient on which the two theories differ at the architectural level; detailed cross-references to Part III are given in the concept-mapping Table 102 below.

Aspect	Standard quantum mechanics	Euclidean Condensate Theory
Primary question	How to compute observed outcomes efficiently	Can specified quantum structures be reconstructed from a supplied Euclidean-medium model?
Status of axioms	Foundational postulates (Hilbert space, Born rule, canonical commutators)	Reconstruction targets; most physical owners remain Open
Primary physical object	Complex amplitude $\psi(x, t)$	Real scalar field $\Phi(X)$ in the bare model; its physical state map is an additional matching problem
Arena	3+1 Lorentzian spacetime, assumed as background	4D Euclidean manifold; a scalar Poincaré-form principal operator is conditional on supplied P4 background/EFT data
Role of time	Fundamental external parameter	A candidate real parameter $x^0 = w$ exists on the supplied ordered branch; physical clocks and time observables remain Open
Role of causal speed	Speed of light postulated	The dimensionless scalar ratio $\hat{c}_*^2 = \beta/(\alpha - \beta)$ follows conditionally; $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration after the unit/lapse map, while photon, matter and tensor cone equality remains Open
Role of $\hbar$	Fundamental universal constant	Two supplied loop closures carry distinct candidates; neither owns the operational S_0 , and compactness alone fixes neither a nonzero universal microscopic owner; $S_0 = \hbar$ is instead the adopted Level-B calibration (see §21.2)
Role of probability	Primitive (Born rule)	Probability measure and outcome law remain Open; tracing and decoherence do not create either

Aspect	Standard quantum mechanics	Euclidean Condensate Theory
Causal structure	Built in from the start	Scalar characteristic cone conditional on P4/EFT inputs; physical causal structure remains a completion target
Type of mathematical problem	Forward-time Cauchy problem for amplitude	Elliptic boundary-value problem for the bare scalar; the quantum state, preparation and detector map remain Open

The table is a dependency map, not evidence that the reconstruction has been completed. Standard quantum mechanics remains the empirical calculational theory against which every proposed ECT owner must be tested.

Hierarchy of effective equations

Figure 33 and Table 97 show a *candidate reduction chain*. Its arrows do not all have the same status: Level 0 is the bare model, P4 supplies rather than derives the ordered branch used at Level 1, the scalar Klein–Gordon form is a conditional principal-operator result, and the Schrödinger endpoint additionally requires a state space, operator normalisation and a supplied S_0 .

Level 0: Euclidean condensate equation. At the most fundamental level one has the Euclidean field equation

$$\delta^{AB} \partial_A \partial_B \Phi - V'(\Phi) = 0, \quad (38.1)$$

which is the Euler–Lagrange equation of the bare Euclidean micro-action (2.3); it is written here only to display the logical hierarchy, not as a new independent postulate. It is a four-dimensional elliptic equation for the condensate field Φ . At this level there is no distinguished time coordinate, no $\hbar$, and no built-in probabilistic interpretation. All four coordinates are spatial from the Euclidean point of view. Boundary data specify the problem on the full four-dimensional domain, but determine a unique solution only after existence, uniqueness and stability/well-posedness are proved for the stated nonlinear equation, domain and boundary class; these global properties remain Open.

Level 1: adopted P4 branch plus a separately supplied scalar EFT. P4 postulates the physically relevant nonzero-gradient branch and supplies its directional chart; its $O(3)$ stabiliser is exact conditional mathematics. If a scalar kinetic EFT, the coefficients α, β and the effective mass input are additionally supplied, the relevant fluctuation equation takes the form

$$K^{AB} \partial_A \partial_B \chi + m_{\text{eff}}^2 \chi = 0. \quad (38.2)$$

Within the restricted parity-even, two-derivative scalar principal-symbol basis built only from δ^{AB} and the supplied director n^A , the general symmetric tensor has the form

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B. \quad (38.3)$$

This basis-level classification is conditional on that operator inventory; P4 neither derives the scalar EFT nor fixes α, β or m_{eff} (see Eq. (3.9) and the subsequent scoped uniqueness discussion). For a homogeneous background with $n^A = \delta_w^A$, this gives $K^{AB} = \text{diag}(\beta, \beta, \beta, \beta - \alpha)$, so that the Lorentzian regime is controlled by the sign of $\beta - \alpha$, equivalently by

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta}. \quad (38.4)$$

Thus, on the declared $\beta > 0$ branch, $\alpha > \beta > 0$ gives a Poincaré-form scalar principal operator, $\alpha < \beta$ gives an elliptic operator, and $\alpha = \beta$ is singular. Without $\beta > 0$, the full product criterion must be used. This is a conditional statement about the scalar characteristic operator, not a derivation of physical clocks, photons, rods, tensor gravity or quantum kinematics.

Level 2: relativistic Lorentzian wave equation. For the supplied ordered branch with $\alpha > \beta > 0$ on the retained $\beta > 0$ component, after the real parametrisation $x^0 = w$, the fluctuation equation takes the relativistic Klein–Gordon form (3.24) (see Section 22 for the derivation). A separate formal continuation $w_E \rightarrow \pm it$ may still be used when relating Euclidean coherent amplitudes to Lorentzian phase evolution; the first step is a scalar-coordinate reparametrisation in the ordered branch, while the second is only a translation of Euclidean coherent amplitudes into Lorentzian phase language.

Level 3: conditional nonrelativistic Schrödinger-type limit. Given a compatible complex state representation, a supplied action scale and the usual positive-frequency/nonrelativistic assumptions, extraction of the rest-energy phase $\psi = e^{-imc_{\text{char}}^2 t/S_0} \chi$ and neglect of terms of order v^2/c_{char}^2 give the Schrödinger-type coherent equation (22.7). Its identification with the standard Schrödinger equation of textbook quantum mechanics remains a Level-B/Open reconstruction step (see §22.1). The displayed equation is therefore a conditional endpoint, not an already-derived physical quantum sector.

Table 97: Hierarchy of effective equations in ECT, together with the type of problem they define, the data needed to solve them, and their standard-QM counterpart.

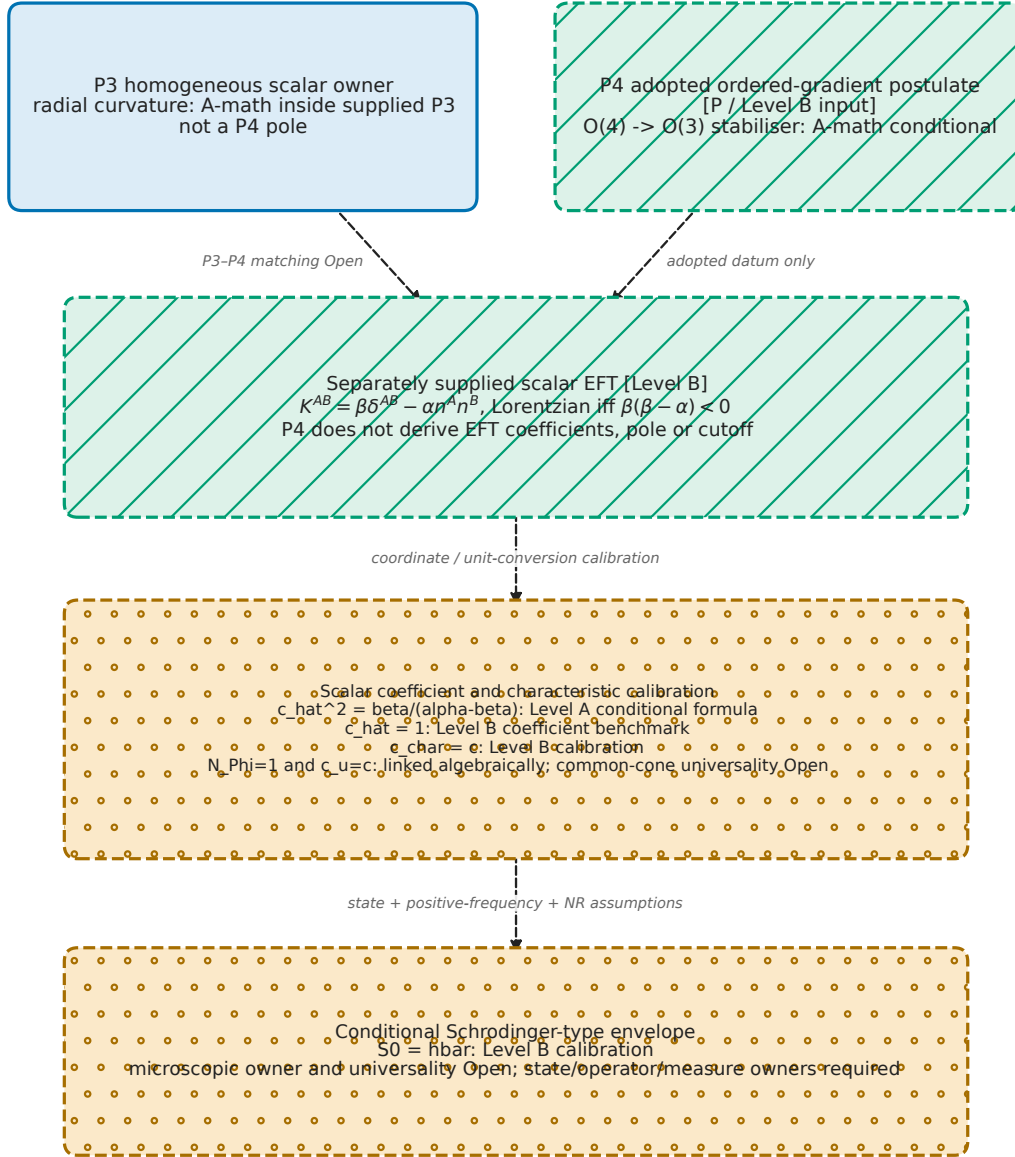
Lv.	Equation	Regime	Mathematical type	Data	Standard-QM analogue
0	$\delta^{AB} \partial_A \partial_B \Phi = V'(\Phi)$	P3 Euclidean regime before the supplied P4 datum	Elliptic	Global BVP on $\partial \mathcal{M}$	none
1	$K^{AB} \partial_A \partial_B \chi + m_{\text{eff}}^2 \chi = 0,$ $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$	P4 directional datum plus a separately supplied scalar EFT and coefficient branch	On the declared $\beta > 0$ branch: elliptic for $\alpha < \beta$, hyperbolic for $\alpha > \beta > 0$; in general the Lorentzian criterion is $\beta(\beta - \alpha) < 0$	BVP or IVP depending on the supplied coefficient branch	none
2	$(\square - m^2) \chi = 0, x^0 = w$	Conditional scalar Poincaré-form branch ($\alpha > \beta > 0$ on the retained $\beta > 0$ component)	Hyperbolic	Cauchy data on $t = \text{const}$	Klein–Gordon
3	$iS_0 \partial_t \psi = H \psi$	Conditional state/operator/ S_0 completion and nonrelativistic limit	Parabolic/effective	Cauchy data at $t = 0$	Schrödinger-type sector

What the ordered-branch equation contains beyond Schrödinger

The supplied ordered-branch equation (38.2) contains more scalar-EFT structure than the conditional Schrödinger-type endpoint (22.7). The comparison below separates algebraic facts from missing physical ownership.

- (i) *Both signatures in one equation.* On the declared $\beta > 0$ component, for $\alpha < \beta$, the effective equation remains elliptic and describes the Euclidean coherent sector, where there is no causal ordering and solutions depend on boundary data from the full domain. For $\alpha > \beta > 0$, the equation

Equation hierarchy — separated physical owners



Dashed arrows are conditional dependencies, not status upgrades.
Physical P4 formation, common cones, and action-scale universality remain Open.

Figure 33: Status-separated candidate hierarchy. Level 0 is the supplied bare Euclidean scalar model; P4 supplies the ordered background/EFT used at Level 1; Level 2 is a conditional scalar Klein–Gordon-form reduction; and Level 3 additionally assumes the state/operator/ S_0 data required for a Schrödinger-type equation. Layout arrows are organisational, not status upgrades. No arrow by itself derives a physical clock, Hilbert space, Born measure or detector channel.

becomes hyperbolic and supports causal propagation. The Schrödinger-type endpoint addresses only the selected hyperbolic scalar branch and only after additional nonrelativistic/state assumptions.

- (ii) *Relativistic parent.* Level 2 is manifestly a Klein–Gordon-type relativistic sector, not a nonrelativistic residue. The nonrelativistic limit appears only at Level 3.
- (iii) *Scalar slope and physical calibration.* The principal operator fixes the dimensionless $\hat{c}_*^2 = \beta/(\alpha - \beta)$ inside the supplied scalar EFT. It cannot be equated to a speed. The separate equality $c_{\text{char}} = c$ is the adopted Level-B scalar-clock calibration after the unit/lapse map; photon, matter and tensor cone equality remains Open.
- (iv) *Candidate loop scales and operational slot.* The supplied fixed-core phase-gradient functional defines ι_0^{EFT} by evaluation at L_{core} , while the supplied positive-tension model defines $\iota_{0,\text{bal}}^{\text{eff}}$ by minimisation (Section 21.2). Winding supplies the integer label only. A relation such as $\oint dw p_w = 2\pi n S_0$ additionally requires an independently derived canonical pair or action–angle map; it does not follow from compactness. Neither route maps itself to operational S_0 , the operator algebra, or $S_0 = \hbar$.
- (v) *Separate gravity completion.* The scalar ordered background is a possible common input to coherent and geometric completion programmes, but the current scalar action does not derive an independent helicity-2 mode, Newton’s constant, a physical metric, or their backreaction on the quantum state.
- (vi) *Status-separated structural programme.* The ordered-branch equation is compatible with two-sided Euclidean boundary dependence and with constructing declared reduced-state record channels. It does not derive a metric-signature transition, physical reverse $O(3) \rightarrow O(4)$ segments, one unique outcome, or a common dynamical mechanism for entanglement and tunnelling. The gravitational record projector, state, noise kernel, and rate remain Open.

The upshot is limited but useful: the supplied scalar EFT admits a conditional KG-to-Schrödinger reduction chain. Establishing that this chain is the physical quantum sector, and coupling it consistently to a completed metric sector, remain Open.

Ontological comparison: Φ versus ψ

Standard quantum mechanics formulates predictions with states and amplitudes. The ECT bare model instead begins with a real scalar $\Phi(X)$ on a Euclidean domain. Calling that scalar the physical medium is a defining ontology of the framework, whereas deriving the quantum state, probability and detector maps from it is an Open dynamical problem. Table 98 keeps those two statements separate.

Table 98: Ontological status of the primary object in the two frameworks.

Aspect	ψ in standard QM	Φ in ECT
Type of object	Complex-valued amplitude	Real scalar condensate field
Physical interpretation	Not a physical field on spacetime; a mathematical object used to compute probabilities	Candidate physical medium variable by ECT definition; quantum-state and observable ownership Open
Arena	3+1 Lorentzian spacetime, taken as a background	4D Euclidean manifold; scalar hyperbolic principal form is conditional, while a physical spacetime metric remains Open
Role of time	Explicit parameter in the evolution equation	Absent at Level 0; $x^0 = w$ is a conditional scalar coordinate and physical clocks remain Open
How it is fixed	Initial data at $t = 0$, evolved forward	Boundary data on $\partial\mathcal{M}$ in the full 4D domain

Aspect	ψ in standard QM	Φ in ECT
Role in measurement	Supports the standard measurement/update formalism	Reduced-state dephasing is available for supplied channels; pointer basis, outcome and update rule remain Open
Access by observer	Direct calculational object	Full condensate configuration is rarely accessible; subsystem access only

Formally, the key statement is:

Standard quantum mechanics uses a state/amplitude as its primary formal object. ECT posits a real Euclidean scalar medium as its primary model object. A derivation of the physical quantum state, probability measure and detector map from that scalar has not yet been completed.

The remaining question is therefore not whether one can rename standard quantum concepts in medium language, but whether explicit state, operator, probability and detector maps can be derived and tested.

Epistemological comparison: boundary-value versus Cauchy problem

The bare Euclidean scalar equation is naturally posed as a boundary-value problem, whereas standard Lorentzian quantum equations are normally posed with state/initial data. This mathematical contrast is exact at the level of the specified equations, but it does not prove uniqueness of the nonlinear Euclidean boundary-value problem or identify its boundary data with laboratory state preparation and readout. Table 99 summarises the conditional comparison.

Table 99: Boundary-value versus Cauchy description of quantum dynamics in ECT and standard quantum mechanics.

Aspect	Standard quantum mechanics	Euclidean Condensate Theory
Mathematical type	Parabolic Schrödinger equation; hyperbolic Klein–Gordon equation	Elliptic Euclidean condensate equation (Level 0); signature switches within Level 1
Problem type	Cauchy initial-value problem	Global boundary-value problem
Data	Initial amplitude $\psi(x, 0)$	Boundary data on $\partial\mathcal{M}$
Domain	3+1 spacetime, evolved forward in t	Full 4D Euclidean manifold, solved globally
Notion of determinism	Unitary forward evolution of a supplied state	A solution is fixed only when a well-posed/unique Euclidean BVP is supplied; uniqueness is not automatic
What counts as prediction	Probabilities from the standard state and Born rule	Requires an as-yet Open map from Φ and boundary data to physical preparations, observables and probabilities
Why probabilities appear	Postulated (Born rule) as an axiom	Not yet derived; coarse graining yields reduced states/dephasing but not an additive probability measure or outcome law
Role of future boundary data	None (initial-value problem)	Boundary data on all of $\partial\mathcal{M}$ enter a declared elliptic BVP; their physical preparation meaning is Open

Several consequences follow from this difference.

Probability remains a separate owner problem. For a declared system–environment split, integrating out untracked modes produces an influence functional and can suppress reduced-state coherences (Section 26.1). This operation presupposes a state and trace; it does not create an additive probability measure, the Born rule or one realised outcome. Those objects remain Open independently of the number of untracked modes.

What reduced dynamics establishes. The detector belongs to the full system. After a declared split and coarse graining, a specified reduced channel can suppress off-diagonal terms and form robust records. This does not fix one realised outcome, derive a pointer basis or update rule, or produce a Euclidean-to-Lorentzian metric-signature transition. Those detector-level objects remain Open (OP-Q19).

Conditional global boundary sensitivity; no causal classification. In a supplied well-posed elliptic BVP, a change of admissible boundary data can alter the global solution. In a Lorentzian slicing this may resemble two-sided conditioning, but it does not classify ECT as retrocausal or derive delayed-choice statistics. A physical conclusion would require the missing well-posed solution map, preparation and setting generators, state, detector map and Lorentzian causal structure. Until those owners are supplied, the causal class is NOT IDENTIFIABLE and no retrocausal signalling claim is made.

BVP determination is not a Born derivation. Even a unique deterministic solution of a fully specified Euclidean BVP would not by itself determine how an observer prepares a quantum state, assigns probabilities or records one outcome. Practical inaccessibility and coarse-graining may motivate statistical modelling, but epistemic ignorance is not a theorem establishing the Born measure. The BVP, reduced-state and probability questions must therefore remain distinct.

Candidate configuration-ensemble reading (Open). One possible but presently unclosed ECT interpretation is that a prepared subsystem state could encode coarse-grained information about an ensemble of admissible global condensate configurations. This is not an identification of the complex amplitude ψ with a positive density on configurations: the two objects have different mathematical types. To expose the missing phase data, suppose a Hilbert representation with orthonormal kets $|\Phi_1\rangle, |\Phi_2\rangle$ is independently supplied. A classical mixture retaining only equal positive weights on these two labels gives

$$\rho_{\text{mix}} = \frac{1}{2}(|\Phi_1\rangle\langle\Phi_1| + |\Phi_2\rangle\langle\Phi_2|), \quad (38.5)$$

whereas the coherent states

$$|\Psi_+\rangle = \frac{|\Phi_1\rangle + |\Phi_2\rangle}{\sqrt{2}}, \quad |\Psi_-\rangle = \frac{|\Phi_1\rangle - |\Phi_2\rangle}{\sqrt{2}}, \quad (38.6)$$

have density operators $\rho_{\pm} = |\Psi_{\pm}\rangle\langle\Psi_{\pm}|$ with opposite off-diagonal terms and therefore different interference in another basis. Thus a positive measure on only the two coarse labels does not encode their relative phase. A richer hidden/configuration space could carry additional phase-like variables; whether ECT supplies such variables and a valid reconstruction map is Open. A viable construction must separately define the admissible physical configuration space $\mathcal{C}_{\text{adm}}$, a positive normalised measure μ , a complex reconstruction/coarse-graining map $\mathcal{R}$, physical preparation and event maps $\mathcal{M}$, and an outcome/update rule $\mathcal{U}$. P6 supplies only configurational multiplicity and possible inhomogeneity; PES-R supplies neither these weights nor a selection theorem. Classical statistical mechanics also uses ensembles, so the contrast cannot be phrased as “classical physics describes one configuration whereas quantum physics describes an ensemble”. This candidate reading remains Open and is not a derivation of the wavefunction, Born weights, Bell correlators or one realised outcome.

Interpretational comparison without reclassification. The listed families allocate states, probabilities and outcomes to different owners; most are interpretations of the standard operational predictions, whereas GRW-type theories modify the dynamics and can differ experimentally. The families listed below are internally diverse; the table records only the distinctions relevant to the Bell and measurement discussion. It is not a ranking and does not identify ECT with any row.

Table 100: Interpretational owner map for states, outcomes and Bell–CHSH correlations. Empirical agreement with standard quantum mechanics is not an ECT-specific success.

Framework	Ontology, dynamics and outcomes	Bell correlations and settings	Relation to current ECT
Bohr/Copenhagen family [263]	Operational quantum state and a measurement context described with a classical apparatus vocabulary; the Born rule and registered outcomes are part of the standard formalism, with ontological readings varying across the family	Entangled states violate CHSH without a joint setting-independent assignment of pre-existing local values; operational no-signalling is retained	ECT has not derived the quantum state, Born measure, classical-apparatus cut or detector update, so shared operational language is not an identification
de Broglie–Bohm [264, 265]	Universal wave function plus actual configuration guided deterministically and nonlocally; outcomes are definite configurations; $ \psi ^2$ quantum equilibrium supplies the usual statistics	Bell violation is carried by explicit nonlocal guidance while ordinary setting independence is normally retained; equilibrium prevents operational signalling	ECT has no derived pilot wave, guidance law or quantum-equilibrium measure; a deterministic Euclidean BVP is not Bohmian mechanics
Everett/many worlds [266]	Universal quantum state with unitary dynamics and decohering branches; no fundamental single-outcome collapse; branch weights require a separate Born-weight justification	Bell correlations belong to the nonseparable global state; operational no-signalling and standard free-setting practice are retained	ECT has not derived a universal quantum state, branch ontology or branch measure; reduced-state decoherence alone does not make ECT Everettian
GRW-type objective collapse [267]	Stochastic nonlinear modification of state dynamics, with flash, mass-density or related ontologies; macroscopic definiteness is dynamical and collapse parameters are in principle testable	Bell correlations require non-local stochastic coordination, arranged so that the operational theory does not permit controllable superluminal signalling	ECT presently contains no GRW stochastic law, collapse parameters or collapse ontology; PES-R and dephasing are not objective collapse
QBism [268]	The quantum state encodes an agent’s personal probabilities and the Born rule is a normative consistency relation; an outcome is an agent’s experience, not the revelation of a pre-existing universal value	Bell data constrain an agent’s probability assignments rather than demanding an ontic hidden-variable mechanism; measurement independence is not introduced as a hidden-state resource	ECT posits an objective Euclidean field, so its ontology is different; it also lacks the probability and outcome rules needed for a substantive comparison

Framework	Ontology, dynamics and outcomes	Bell correlations and settings	Relation to current ECT
Retrocausal and global-constraint families [225]	Complete histories or boundary data may be constrained by both temporal directions or globally; models differ on whether causal influence or an atemporal constraint is fundamental	$P(\lambda x, y)$ may depend on settings while propagation remains locally formulated; Born weights, no-signalling and the Tsirelson ceiling remain model-specific obligations	ECT's elliptic BVP supplies only a mathematical family resemblance. No physical future-boundary, causal or Bell map has been derived
Superdeterministic families [224]	Hidden variables and setting generators share prior dynamical causes, so measurement independence fails without a backward causal arrow; concrete models must specify preparation and setting statistics	Local deterministic responses can evade the standard CHSH premise, but quantum probabilities, no-signalling and robustness under changes of setting generators are not automatic	Current ECT has no derived common-cause setting mechanism or conditional probability measure and therefore is not classified as superdeterministic
Current ECT	Bare ontology: a real scalar on a Euclidean domain with a global BVP; conditional reduced sectors exist, while the physical quantum state, probability owner, outcome and update remain Open	The physical subsystem map, local observables, Bell correlators, no-signalling theorem, Tsirelson principle and causal status of settings remain Open (OP-Q20)	Operationally unclassified. Global-conditioning language is a candidate question, not evidence for Bohmian, Everettian, collapse, retrocausal or superdeterministic completion

The comparison sharpens the present ECT boundary. A global Euclidean description may motivate a conditional-measure calculation, but the calculation becomes interpretationally and physically meaningful only after the probability, preparation, setting, record, constraint and causal owners are derived. Reproducing four target correlators by the generic construction above would establish neither those owners nor an ECT-specific prediction.

Present interpretive status of ECT. ECT is intended as an ontic medium and reconstruction programme rather than an agent-centred or purely operational reading, and it does not take a fundamental measurement cut or collapse postulate as a derived result. Nevertheless, until the state, probability, event, Bell and update owners are constructed, this ontological intention is not an empirically completed alternative to the interpretation families above and supplies no ranking among them.

Operational equivalence and the boundary of interpretation tests. An ontological reformulation is not by itself an experimental discriminator. Let $P_T(o | \mathcal{P})$ denote the complete normalised outcome law assigned by a theory T to a specified protocol $\mathcal{P}$. If

$$P_{\text{ECT}}(o | \mathcal{P}) = P_{\text{QM}}(o | \mathcal{P}) \quad \text{for every accessible } o, \mathcal{P}, \quad (38.7)$$

then the likelihood ratio is unity for every data set and no experiment can distinguish the two descriptions. Experimental discrimination requires a preregistered protocol for which ECT derives a normalised probability law outside the standard-QM predictive set after environmental, detector and selection nuisance parameters are included. Current ECT has not supplied that law. Agreement with experiments already predicted by standard quantum mechanics is compatibility, not ECT-specific confirmation. The Copenhagen family is also internally diverse rather than a unique competing dynamical law.

Experimental discriminator programme. The experiments below become tests of ECT only after the named forward owner is derived:

Table 101: Potential ECT discriminators. Every row is a future protocol, not a current ECT prediction or confirmation.

Protocol	Required precomputed ECT signature	What the experiment would decide	Present status
Massive-particle interference plus independent noise/heating	One action-owned correction with common mass, separation, time, material and environment scaling in visibility and in a second noise/heating observable	Detect that joint residual or exclude the parameterised completion; ordinary decoherence must be fitted simultaneously	MISSING VERTEX/KERNEL; Pedalino is an external frontier [172]
Bell settings with generator provance / MDL	A stable generator-class dependence or derived (ℓ, h) and complete $P(a, b, x, y)$	Exclude that bounded causal completion or detect a reproducible generator-linked residual	Probability and causal owners Open
Controlled record/decoherence protocol	A physical subsystem/environment split, record operator, vertex, environment state and response/noise kernels	Test the PES-R channel across controlled couplings and independently read records; failure rejects the channel, not all ECT	PES-R Level B organisation; physical PES Open
Noninterferometric radiation/heating	An ECT master equation with stochastic energy injection and a detector-level emission/noise spectrum	Recast or reanalyse existing null data using the derived ECT spectrum and detector response	No ECT stochastic law; current limits are model-indexed [269]
Higher-order multipath interference	A derived non-Born probability composition law and a complete source and detector model	Test a preregistered higher-order interference residual after controlling source, propagation and detector nonlinearities	No ECT non-Born prediction

No row presently confirms ECT and refutes Copenhagen. A statistically robust anomaly that survives systematic checks and independent replication would first establish failure of the tested standard-QM-plus-nuisance model, and would support ECT only if the ECT magnitude and cross-protocol scaling had been fixed before the decisive data. A null result could exclude a specified parameter region of a completed ECT model only when the experiment has demonstrated adequate sensitivity to its preregistered prediction.

Origin of the statistical description

The present theory has three logically separate statistical layers. First, a supplied state and system–environment split define reduced density operators and influence functionals; this conditional algebra can yield dephasing and record stability. Second, a normalised, noncontextual, countably additive projector measure must be independently supplied before ordinary Gleason’s theorem can represent it in complex dimension at least three; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. Third, a detector channel and update/selection law are needed to describe one registered outcome. None of the latter two layers follows from tracing out modes, coarse-graining or lack of access.

Accordingly, the information–decoherence bound, PES-R record taxonomy, conditional Gleason result and entanglement programme are related entries in a dependency graph, not different proofs of one universal reduction mechanism. This separation prevents three invalid inferences: decoherence $\nRightarrow$ Born weights, reduced state $\nRightarrow$ unique outcome, and common medium $\nRightarrow$ nonfactorising quantum state. Deriving the missing state, measure and detector owners from the ECT action remains an explicit Open programme.

Interpretive mapping of standard-QM concepts onto ECT

The content of the preceding parts of Part III can be organised as a mapping between the main conceptual primitives of standard quantum mechanics and their ECT counterparts. Table 102 records this mapping together with the present status of each item. It is intended as a reading guide, not as an independent source of results; the status column is consistent with the more formal ledger in Section 38.2.

Table 102: Concept mapping between standard quantum mechanics and ECT. The *Status* column uses the same Level-A / Level-B / Open language as Section 38.2.

Concept	Status in standard QM	ECT origin/route	Status in ECT
Superposition	Primitive axiom	Linear coherent state space in the subsectors where the Hilbert-space reconstruction and sector dynamics are established	Level A mathematics from explicit derivations in the declared linear/Gaussian subsectors, with reproducible checks; physical state map and interacting completion Open (§23.1)
Born rule	Primitive axiom	Ordinary projector Gleason represents an independently admitted normalised, noncontextual, countably additive projector measure for a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses, while OS-II and tracing/dephasing are not theorem premises	Level A mathematics under full theorem hypotheses; physical state, preparation plus projector/effect event map, probability origin, outcome, update and ECT ownership Open (§30.4)
State reduction	Primitive/interpretational	Reduced-state dephasing and record stabilisation in specified channels; no signature change or unique-outcome update is derived	Open beyond the reduced-state channel (§30.2, OP-Q19)
Uncertainty principle	Theorem from commutators	Robertson–Schrödinger inequality for supplied operators and commutator normalisation	Level A mathematics; physical operators and S_0 owner Open (§23.5)
Canonical commutators	Primitive	Candidate descent from a supplied symplectic/Poisson structure; compactness alone does not fix the operator algebra or normalisation	Level B/Open (§23.3)
Entanglement	Non-separable two-party states	Requires a supplied nonfactorising state and channel; a common medium alone is insufficient	Standard conditional mathematics; ECT state/channel and Bell map Open (§31, OP-Q20)
Tunnelling	Barrier penetration via WKB	Conditional Euclidean saddle/WKB interpretation after the state, barrier, continuation and detector map are supplied	Level B/Open interpretation (§24.3)

Concept	Status in standard QM	ECT origin/route	Status in ECT
Identical particles	Symmetrisation postulate	Exchange-space result $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ for supplied identical-point configuration space	Level A conditional topology; physical particle/statistics map Open (§24.4)
Spin-statistics	Spin-statistics theorem in QFT	Residual-symmetry and exchange-topology route in the ordered branch; Euclidean OS free-kernel test Level A (Q11.1a)	Open (§24.4, OP-Q11)
Action scale $\hbar$	Fundamental constant	Two supplied loop closures define distinct candidates; winding gives integer labels, while their map to operational S_0 remains Open	Level B calibration / Open microscopic owner (§21.2)
Charge quantisation	Topological/empirical	Integer representations after a compact local bundle, charge unit and matter representations are supplied	Level A conditional mathematics; physical ECT charge lattice Open (§23.4)
Decoherence	Added phenomenology	Influence-functional coupling of the coherent sector to untracked condensate modes	Level B conditional open-system algebra; ECT kernels, state and record operators Open (§26.1)

The table is intentionally shorter than the full status ledger: its purpose is to show that the main conceptual primitives of standard quantum mechanics acquire explicit structural counterparts in ECT, together with their derivational level.

Illustrative examples

The structural comparison above is best seen on several concrete examples. The examples below are not new derivations. They are interpretive reformulations of standard quantum phenomena in ECT language, with the present status of each reformulation stated explicitly. Each entry uses the same template so that the contrast can be read off directly.

Double-slit interference.

Standard QM. One specifies the incoming wave packet and evolves the amplitude through the two-slit apparatus as a Cauchy problem; the observed pattern is computed from $|\psi_1 + \psi_2|^2$, where ψ_1 and ψ_2 are the amplitudes associated with the two slits.

ECT. One solves a global boundary-value problem for the condensate configuration only after a source, slit, screen, state and detector map have been supplied. Two-sided Euclidean path dependence provides a possible geometric representation of the apparatus.

What is computed.

No new fringe law is derived here. Reproducing $|\psi_1 + \psi_2|^2$ requires the missing complex-state and Born/detector maps.

What is learned.

The elliptic BVP can encode dependence on both slits, but that fact alone does not explain quantum interference.

Status. Mathematical analogy / Open ECT ownership; not a first-principles laboratory calculation.

Quantum tunnelling.

Standard QM. One computes a transmission coefficient, typically by WKB matching across a classically forbidden region; the exponential suppression $T \sim \exp(-2 \int |p| dx / \hbar)$ follows.

ECT. The classically forbidden Lorentzian region is represented conditionally by an admitted Euclidean saddle after a physical state, barrier, continuation prescription and detector map are supplied.

What is computed.

Standard saddle/WKB algebra can reproduce a supplied semiclassical exponent; the present ECT action does not yet derive the full system-specific tunnelling rate.

What is learned.

Euclidean connectivity is a candidate geometrical language for an underbarrier saddle, not an explanation of tunnelling by itself.

Status. Level B/Open interpretation (§24.3); no common tunnelling–entanglement dynamics has been derived.

Measurement and outcome status.

Standard reduced-state result.

A specified system–apparatus–environment interaction can suppress interference and stabilise pointer records.

ECT status. The physical ECT record operators and detector channel are not yet derived. Decoherence yields an effective mixture, not one ontologically selected branch and not exact Born weights.

Open. Pointer-basis completion, unique outcome, and the update theorem remain OP-Q8/OP-Q19.

EPR-type correlations and entanglement.

Standard QM. Entangled states are non-separable and can violate Bell-type inequalities up to the Tsirelson bound $2\sqrt{2}$, excluding Bell-factorisable, measurement-independent models for the tested correlations while retaining operational no-signalling.

ECT. Correlated subsystems are understood as inheriting their reduced structure from a supplied nonfactorising state/channel. A common condensate may host such a channel but does not force it.

What is computed.

A separately declared Werner-state visibility toy gives $\Phi_{\text{vis}} = \ln \sqrt{2}$ at the CHSH local bound. This is standard model algebra, not a result derived from the ECT action (§38.2).

What is learned.

Common-medium language is compatible with no-signalling once an appropriate quantum channel is supplied, but compatibility is not a derivation of Bell-nonclassical correlations. The detailed premise audit in Section 31 provides a conditional template for testing a possible setting-dependent global-configuration measure, while proving that neither globality nor setting dependence alone supplies the observed correlators.

Status. Standard conditional state algebra / Open ECT state and channel owner (OP-Q20). The CHSH crossover is a toy proxy, not a derivation of the CHSH algebra or Tsirelson bound.

Identical particles and exchange statistics.

Standard QM. Particle statistics is imposed through symmetrisation (bosons) or antisymmetrisation (fermions) of many-body wave functions, with the boson/fermion dichotomy accepted as a primitive.

ECT. For a supplied configuration space of two identical points in three dimensions, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ classifies two exchange homotopy classes (§24.4).

What is computed.

The topology alone does not select a physical representation, define the particle sector, or prove symmetric versus antisymmetric many-body amplitudes.

What is learned.

The two-class topology is a necessary input to the usual 3D exchange construction, not an ECT derivation of physical statistics.

Status. Level A conditional topology; particle ownership, representation selection and the spin–statistics theorem remain Open (OP-Q11).

Hydrogen atom and spectral discreteness. Hydrogenic levels follow from the Coulomb Hamiltonian, its domain, boundary conditions, and the quantum kinematic structure. Compact phase supplies relevant representation/charge information but is not the universal origin of the bound-state spectrum. PES-R may classify level widths and retention after the spectral problem is solved; it predicts no new hydrogen spectrum at present.

Determinism at the fundamental level

The Level-0 Euclidean equation (38.1) is deterministic as a differential equation once a well-posed branch and complete boundary data are specified. Nonlinear BVP uniqueness is not automatic, and this mathematical determinism is not yet a physical hidden-variable completion of quantum mechanics. A supplied state and trace can yield an influence functional and reduced density matrix, but neither incomplete access nor coarse-graining derives exact Born weights or one outcome.

The classical-statistical-mechanics analogy is therefore only motivational. To make it a theorem ECT must supply a physical state-or-configuration space, preparation map, operator/measurement structure, joint probability law, detector/record channel and update rule. Its Bell completion must derive the operational probability model. If a hidden/configuration representation of Eq. (31.28) is claimed, it must determine whether Bell factorisability or measurement independence fails; a state/channel route must instead show why no factorisable common-measure representation applies. In every route ECT must derive the correlations, operational no-signalling and the quantum boundary rather than fitting them. Until those owners are derived, the manuscript makes no claim that quantum probability is merely epistemic or emergent from determinism.

What ECT already reproduces, and what remains open

It is useful to close with a reader-oriented inventory. Every item below is conditional on its stated inputs; it must not be read as a derivation of the full physical quantum formalism.

Part III currently establishes or formulates:

- (i) a conditional Schrödinger-type reduction after the P4 scalar EFT, state/operator representation, positive-frequency choice and S_0 are supplied;
- (ii) a conditional scalar Klein–Gordon-form principal operator under the supplied P4 branch/EFT, without yet identifying physical clocks or matter;

- (iii) uncertainty relations for supplied operator domains together with a conditional compact-rotor route: a physical constant-shift orbit must survive the gauge/Gauss quotient with finite nonzero symplectic form, and a Weyl representation plus self-adjoint quasiperiodic domain must be supplied. Compactness alone derives neither that rotor, its twist nor the operational S_0 ; the identification $S_0 = \hbar$ is the adopted empirical Level-B calibration; its owner remains Open;
- (iv) reduced-state dephasing and record formation in specified channels; the ECT split, record operators, state, and kernels remain Open;
- (v) reduced-state record formation without a metric-signature transition or unique-outcome theorem; pointer selection and update remain Open (OP-Q19);
- (vi) a structural route linking spin-sector discreteness to the $O(3)$ stabiliser and topology supplied with the P4 chart, rather than a physical $O(4) \rightarrow O(3)$ transition or a fully closed derivation of the complete spin sector;
- (vii) the PES-R status taxonomy, which separates dynamical admissibility, finite-time retention, and pairwise records without a universal selection theorem;
- (viii) an explicit programme for deriving a gravitational record channel from the condensate; the Diósi–Penrose rate is retained only as an external benchmark, not as a result of the present action;
- (ix) separate tunnelling and entanglement owners; Euclidean connectivity supplies at most a geometric analogy, while a common dynamical mechanism remains Open;
- (x) an operational-unit closure that reuses S_0 across several displayed effective sectors, with its loop owner, cross-sector universality and physical ownership left to the matching programme in Part III.

On the other hand, ECT does not yet fully derive:

- (i) the full Born rule from first principles beyond the present closures;
- (ii) a completed pointer basis, detector channel, unique-outcome update rule, and ECT record kernel;
- (iii) the physical Bell preparation, state-or-configuration, setting, measurement and joint-law owners, together with Bell correlators, no-signalling and the Tsirelson restriction directly from the condensate theory beyond generic countermodels and toy visibility proxies; a positive hidden measure and response maps are additionally required only if that representation is claimed;
- (iv) the detailed origin of three generations;
- (v) the exact value of α_{fs} ;
- (vi) the full Yukawa hierarchy.

These should be presented as open derivational targets, not as failures of the framework.

The list above is intended as a reader-oriented comparison between ECT and the standard quantum formalism. A stricter status map of the quantum sector, organised by derivational level (Level A / Level A-B / Level B / Open) and linked to the corresponding open-problem labels, is given separately in Section 38.2. The present limitations of the formulation are collected in Section 49.10. The two summaries serve different purposes: the present one is comparative and pedagogical, whereas Section 38.2 is the more formal status ledger of what is already structurally established and what remains open.

Having set out the structural comparison between ECT and the standard quantum formalism, it is natural to state explicitly what classes of future results could exclude particular, separately specified completion routes.

38.4 Conditional discriminants for named quantum-gravity and decoherence completions

Status: conditional future-test programme. Each exclusion applies only after a named completion has preregistered the corresponding mediator, state, channel and global-dynamics assumptions; it is not a falsifier of the present P1–P6 core.

The present formulation has not selected between fundamental and reduced-state stochasticity, has not established global unitarity, and has not derived a physical gravity-mediated entanglement protocol. Consequently, the following observations discriminate only among future ECT completion hypotheses whose commitments have first been made explicit.

First, compelling evidence for a genuinely fundamental stochastic metric law, with a noise spectrum that could not be reinterpreted as emergent from reduced matter–environment or condensate-sector dynamics, would exclude a named ECT completion that had assumed an exclusively emergent reduced-state origin. It would not contradict the present core, for which that choice is Open.

Second, robust evidence for fundamental black-hole information loss would exclude a named completion that had imposed exact global unitarity. The present work has not derived that global property.

Third, if a general no-go result established that gravity-induced entanglement in the relevant mesoscopic regime is possible only with a genuinely quantized mediator, even after allowing for correlated-noise, common-medium, or reduced-state routes of the Trillo–Navascúes type, then one of the main structural loopholes considered by a named common-medium completion would be removed, provided the theorem’s assumptions cover that completion’s exact mediator, state and apparatus channel [237, 240]. No such mediator completion is owned here.

Fourth, if a future named ECT mediator completion predicted decoherence that eliminated all mediator coherence throughout a specified mesoscopic protocol, that completion would fail to reproduce entanglement in that protocol. The claim requires the completed dynamics and cannot be applied to the current Open mediator programme.

Thus these are conditional completion tests, not present whole-theory falsifiers. They leave the already-derived structural mathematics unchanged and become ECT-specific only when the tested completion owns the full action-to-observable chain.

38.5 Completion status and bridge forward

The corrected quantum sector retains a strict structural core: declared wave-equation limits, canonical inequalities, exchange topology, and reflection-positive Gaussian subsectors, each under its stated assumptions. It also retains standard source-model open-system calculations and imported phenomenological benchmarks when labelled as such.

The following ECT-specific steps remain Open: interacting OS/Hilbert-space completion; the identification and cross-sector universality of $S_0 = \hbar$; physical record operators and state-dependent noise kernels; Born weights without importing the probability assumptions; pointer-basis completion and a unique-outcome update; Bell correlators; a Crooks/entropy production theorem; the gravitational record channel; and physical tensor quantisation. PES-R organises response, retention, and pairwise records but closes none of those owners by itself.

Consequently, visibility transforms, the Werner-state CHSH marker, C_{60} collisional decoherence, and Diósi–Penrose applications are respectively source-model proxies or external comparisons, not ECT confirmations. This status separation is the bridge to the remaining derivations and falsifiers.

Bridge to the rest of the paper. The bridge forward preserves the separation between derived quantum amplitudes, conditional open-system records, and Open measurement questions. Later applications may not strengthen these statuses.

What ECT does and does not establish in the Quantum Sector. ECT establishes a Euclidean-first amplitude architecture and several conditional/source-model calculations. It does not yet derive the full

quantum formalism, universal quantisation, Born weights, or outcomes.

Summary. The Quantum Sector is therefore complete as a status-audited research programme, not as a closed derivation. Every quantitative result retains its own assumptions, reproducibility owner, and falsifier.

Part IV

Reverse Analysis, the P7 $SU(3)$ Structural Postulate, and Physical-Colour Constraints

39 Methodology of reverse analysis

Status: Methodological. This section does not present new physics; it articulates the methodology of reverse analysis used throughout Part IV and its relation to the forward programme of Parts I–III.

Scope. Reverse analysis as used in Part IV is a closure/consistency layer on top of the forward derivations of Parts I–III. It is not a substitute for forward derivation, and must be used under explicit rules. As an ECT-native closure inference it reaches at most Level B (usually Level C); self-contained algebraic identities or obstruction theorems may still be Level A inside their declared assumptions.

Distinction from forward derivation. Forward derivation in ECT starts from P1–P6 and deduces consequences; this is how Parts I–III are structured, and it is the only route to Level A ECT-native descent claims. Reverse analysis starts from the partial success of the forward programme together with empirical constraints from observed physics, and identifies what minimal additional structure, beyond P1–P6, the medium must carry for consistency.

Legitimacy and scope of the inference. Reverse analysis may expose a scoped obstruction and then register candidate completion data, but it cannot turn the candidate into a derived physical sector. Here the dimension count proves only $\mathfrak{su}(3) \not\rightarrow \mathfrak{so}(4)$, while the VEV theorem says that a nonzero fundamental $SU(3)$ vector has stabiliser $SU(2)$. Neither statement excludes a reducible representation with a singlet VEV, an emergent/nonlinear construction, or a condensate-defect kernel whose Berry connection supplies an internal bundle. P7 adopts one separate $SU(3)$ -structured rank-three bundle as a Level-B structural postulate; it is not forced by those two facts.

Relation to the forward programme. Part IV is not a second derivation of gauge and fermion architecture. It compares the physical-completion implications of P7 with the conditional routes already developed in Sections 9.3, 9.5, 9.6, and 9.7. The physical colour, chirality, generation and matter owners remain Open after this comparison.

Rules of engagement. Five rules govern reverse analysis as applied in Part IV: (i) every reverse-engineered constraint must point back to a forward obstruction; (ii) new postulates must be geometric or structural, never numerical or phenomenological; (iii) outcomes are restricted to admissible classes—never derive desired results such as specific masses or couplings via reverse reasoning; (iv) any residual freedom in the completion must be catalogued as Open; (v) reverse-engineered postulates must be flagged as such (e.g. P7) and not confused with the primary postulates P1–P6.

What Part IV does within these rules. Section 40 defines and audits the supplied P7 bundle candidate while retaining alternative routes. Sections 41–42 display a leading truncation and restate standard matter-consistency conditions after further inputs are supplied. Section 43 is an Open-owner inventory, not a closure certificate. Section 44 keeps three conditional selection-rule patterns without deriving $U(1)_B$; Section 45 keeps the quark-mass phase in $\bar{\theta}$; and Section 47 separates the scoped Level-A mathematics from B/C/Open physical matching.

40 Physical-colour requirement and candidate non-Abelian completions

Observed requirement and current Open owner

Observed QCD is organised by an unbroken local $SU(3)_c$ gauge symmetry. A phenomenologically viable ECT completion must reproduce that physical sector, but P1–P6 and the present action do not derive it. Part IV therefore studies candidate completion data; it does not establish exact physical colour.

Scoped stabiliser and direct-embedding results

Invariant-VEV theorem and fundamental-triplet corollary. A VEV preserves precisely its stabiliser, $G_{\langle v \rangle} = \{g \in G \mid g\langle v \rangle = \langle v \rangle\}$. For a nonzero vector in the fundamental $\mathbf{3}$ of $SU(3)$ this stabiliser is $SU(2)$, so such a fundamental-triplet VEV would break $SU(3) \rightarrow SU(2)$. This is Level-A group theory inside the stated representation. It is not a universal statement about all condensate-internal realisations: a reducible representation can contain an invariant singlet, and a VEV in that summand preserves $SU(3)$.

Direct-embedding obstruction. Because $\dim \mathfrak{su}(3) = 8 > 6 = \dim \mathfrak{so}(4)$, there is no injective Lie-algebra embedding $\mathfrak{su}(3) \hookrightarrow \mathfrak{so}(4)$. This excludes only a direct-subalgebra origin inside the displayed $O(4)$ spacetime/order-parameter algebra. It does not exclude an emergent or nonlinear realisation, a reducible internal representation, an independently supplied bundle, or the defect-kernel/Berry route retained below.

P7 as an adopted rank-three structural completion

The following reverse-engineered postulate chooses one structural completion. Its singlet assignment for (ϕ, n^μ) is an assumption of P7, not a theorem about every possible ECT completion.

Postulate P7 (adopted $SU(3)$ -structured bundle). In addition to P1–P6, supply a global complex rank-three vector bundle $\mathcal{C} \rightarrow \mathcal{M}^4$ with transition cocycle, a positive-definite Hermitian form h , and a nowhere-vanishing complex volume form $\Omega \in \Gamma(\Lambda^3 \mathcal{C}^*)$ normalised by $|\Omega|_h = 1$. The ordered-branch fields ϕ and n^μ are singlets. The fibrewise stabiliser of (h, Ω) , equivalently the reduced structure group, is $SU(3)$. The global automorphisms preserving these data form the corresponding gauge group of sections and are not the finite-dimensional group $SU(3)$ itself.

Status of P7. P7 is an adopted reverse-engineered structural postulate (Level-B input), not a P1–P6 descent theorem. The reduction of the frame group to $SU(3)$ is Level-A mathematics conditional on $(\mathcal{C}, h, \Omega)$. On a suitable paracompact base a compatible mathematical connection exists, but P7 does not select its dynamics or identify it with physical colour. A physical colour connection, Yang–Mills action, matter representations, coupling normalisation, spectrum and QCD map remain Open. Reducible-singlet, emergent/nonlinear and kernel-bundle derivation routes remain available; a completed zero-mode programme could explain rather than merely postulate the P7 data.

Definition (the colour zero-mode problem). A first-principles derivation of P7 from condensate dynamics would require a zero-mode construction of the colour-completion datum. The problem data are: a local, condensate-equivariant elliptic operator $\mathcal{D}_{\text{defect}}$ built from the minimal condensate fields on stable defects or internal condensate sectors, for which the target statement is

$$\mathcal{C}_x \simeq \ker \mathcal{D}_{\text{defect}}(x), \quad \dim_{\mathbb{C}} \ker \mathcal{D}_{\text{defect}} = 3, \quad (40.1)$$

for a condensate-internal operator $\mathcal{D}_{\text{defect}}$ on stable defects or internal condensate sectors, with the Hermitian form h and the complex volume form Ω induced by the condensate inner product and orientation data rather than postulated. For a generic smooth rank-three Hermitian kernel bundle, an orthonormal zero-mode frame yields a unitary Berry–Wilczek–Zee connection whose holonomy is contained in $U(3)$. Rank three and the positive-definite Hermitian form alone do not reduce this connection to $SU(3)$. Such a reduction additionally requires a globally compatible determinant/volume owner and a volume-preserving connection,

$$\nabla h = 0, \quad \nabla \Omega = 0, \quad (40.2)$$

equivalently a vanishing trace- $U(1)$ connection in an Ω -compatible unitary frame, subject to the determinant-line and global trivialisation conditions stated below. No zero-mode-induced operator, smooth spectral projector, determinant-compatible selected connection or physical gauge/action map is constructed here. This does not negate the abstract existence of compatible connections on the P7 $SU(3)$ -reduced bundle; P7 does not select one as the physical dynamical connection.

Open problem (OP-GUT1, sharpened). Exhibit a condensate-owned operator and prove the smooth rank-three kernel statement (40.1), the induced (h, Ω) , and the compatibility condition (40.2). The current outcomes are MISSING DEFECT OPERATOR, MISSING SMOOTH KERNEL PROJECTOR, and MISSING DETERMINANT/TRACE-CONNECTION OWNER. Until these owners and the physical bundle/action map are supplied, the zero-mode derivation route remains Open; P7 itself remains the adopted structural postulate. Anomaly cancellation alone cannot substitute for rank or connection ownership.

Candidate zero-mode routes (all Open). Two structural routes are recorded, without commitment and without any existence claim. (i) *Index/charge route.* On θ -winding vortex-core backgrounds — which are not primary $O(4) \rightarrow O(3)$ strings: they rely on the compact phase fibre and require separate stability analysis — suitable Jackiw–Rossi/Callias-type settings can count zero modes in terms of winding/charge data, schematically $\dim \ker \sim |n|$; whether any ECT-native operator realises such a setting is open. This route would reduce the rank-three question to the stability/selection of charge-three cores; it would transform the rank-selection problem, not solve it. (ii) *Interface route.* Localised modes on ordered–Euclidean or inter-branch interfaces; counting data unclear; least developed. A tempting residual- $O(3)$ degeneracy argument is not used here: any $d_{\text{spatial}} = 3$ route must first pass an internalisation test, showing that the threefold degeneracy is an internal gaugeable fibre rather than an ordinary spatial vector multiplet.

The determinant-line problem for Ω . For a smooth complex rank-three kernel bundle $E = \ker \mathcal{D}_{\text{defect}}$, an $SU(3)$ -structure requires a positive-definite Hermitian metric and a nowhere-vanishing complex volume form $\Omega \in \Gamma(\det E^*)$, equivalently a trivialisation of the determinant line. The first Chern class $c_1(\det E)$ is the basic topological obstruction to such a trivialisation; its vanishing is necessary, and in the relevant geometric category may be part of the sufficient data, but it does not by itself choose a canonical Ω . A canonical Ω would require additional preferred trivialisation data, plausibly tied to condensate orientation data. The fibrewise L^2 inner product on the kernel supplies the positive-definite Hermitian metric h once the kernel bundle is smooth.

Connection and kinetics firewall. A Berry–Wilczek–Zee connection on a smooth kernel bundle gives a candidate gauge connection; it does not by itself derive the Yang–Mills action, the coupling g_s , or confinement. Possible failure modes include generic absence of zero modes, real rather than complex

kernels, unjustified charge-three selection, level crossing, nontrivial determinant-line obstruction, and failure to derive Yang–Mills kinetics.

Supplied bundle interpretation (Level B postulate; Level A conditional fibre algebra; C/Open physical status). P7 supplies an independent global complex rank-three bundle with transition cocycle, Hermitian form h and unit-norm complex volume form Ω ; it does not derive these data from the condensate. Pointwise h -preserving frame changes form $U(3)$, while the fibrewise stabiliser of both (h, Ω) is $SU(3)$. That structure-group reduction is exact mathematics for the supplied bundle data. Global structure-preserving automorphisms form an infinite-dimensional gauge group of sections.

P7 therefore does supply the abstract bundle, transition functions and $SU(3)$ reduction. It does not by itself identify them with physical colour, select a dynamical compatible connection satisfying $\nabla h = \nabla \Omega = 0$, supply triplet matter sections, a gauge action, gauge-volume reduction or observables. If triplet fields were independently supplied, the invariant ε_{abc} would permit an algebraic three-triplet singlet; this is only conditional representation theory and neither identifies quarks nor defines $U(1)_B$ or a baryonic state. The similarity to observed QCD is a Level C structural analogy. Physical colour and the strong sector remain Open.

The determinant/trace $U(1)$ removed when (h, Ω) reduces a unitary frame from $U(3)$ to $SU(3)$ is a determinant-line frame phase; it is not the physical electromagnetic $U(1)_{\text{em}}$. Conversely, the separately supplied compact phase does not derive an electromagnetic local bundle, connection, photon action or matter-charge map. Both identifications remain Open and must not be inferred from the fibre reduction.

Structural compatibility checks. A limited but nontrivial compatibility check concerns gauge-sector counting. If the colour completion proposed here is combined with the electroweak-like sector already discussed elsewhere in ECT, then the expected adjoint dimensions become structurally available: 8 for the colour sector, 3 for the weak sector, and 1 for the Abelian phase sector. This reproduces the Standard Model gauge-boson count at the level of symmetry dimensions only. It should not be confused with a derivation of the full electroweak and strong gauge dynamics, mass generation, or mixing structure.

A second compatibility check concerns colour neutrality outside the strong sector. In a separately supplied product-bundle completion, non-strong gauge fields, gravitational fields and declared condensate excitations can consistently be assigned singlet representations of the candidate $SU(3)$ factor. This is a conditional representation assignment, not a consequence of structural independence and not a derivation from P7. The physical representation map for every sector remains Open.

If the proposed colour module is combined with the weak-sector doublet structure discussed elsewhere in ECT, then quark-like and lepton-like sectors can be accommodated in a structurally compatible way. In particular, one may consistently describe excitations that transform nontrivially under both the colour and weak sectors, as well as excitations that remain colour singlets while still transforming under the weak sector. This should not be confused with a full derivation of the Standard Model matter content: chirality, hypercharge assignments, right-handed singlets, and generation structure remain open.

With supplied colour excitations in the fundamental representation $\mathbf{3}$, representation theory gives $\mathbf{3} \otimes \bar{\mathbf{3}} \supset \mathbf{1}$ and the invariant $\varepsilon_{abc} q^a q^b q^c$, so meson-like and baryon-like singlet operators are permitted. This is Level-A conditional $SU(3)$ algebra. It does not define a baryon-number group $U(1)_B$, assign baryon charges, produce a conserved current, or select baryon-violating vertices. Those require the full matter representation and action and remain Open, as do the bound-state dynamics.

What is not obtained. This completion does not yet derive full QCD. The matter content details, chirality pattern, coupling strength g_s , confinement mechanism, θ_{QCD} sector, Λ_{QCD} , and the full Standard Model representation structure (including hypercharge assignments and right-handed singlets) remain open.

Updated status under the P7 structural postulate

Table 103: Conditional consequences and Open physical owners under the adopted P7 structural postulate.

Item	Before P7	What the adopted structural postulate supplies	Still Open
Physical colour symmetry	Missing	An $SU(3)$ frame-redundancy candidate	Bundle, action and observable identification
Strong-sector gauge structure	Missing	A place where a supplied connection could act	Connection, kinetics and measure
Gauge-sector counting	Partial	Adjoint dimension eight after $SU(3)$ is supplied	Physical gauge-boson identification
Colour neutrality of non-strong sectors	Untested	Can consistently be assigned singlets in a separately supplied product-bundle completion	Physical representation map Open
Compatibility with EW sector	Partial	Product representations are algebraically allowed	Chiral matter embedding
Meson/baryon-like channels	No named colour owner	$SU(3)$ singlet operators are permitted	Bound-state dynamics and $U(1)_B$ owner
Strong-sector constants	Undefined without a supplied action	Become matching variables of a supplied action	ECT derivation
$\alpha_s(M_Z), \Lambda_{\text{QCD}}$	Undefined	Standard source-model quantities after action/matter input	ECT matching and RG
Proton/neutron mass	Not identified	A future gauge-invariant spectral target	Physical operator/dynamics map
Confinement and asymptotic freedom	No action owner	Standard questions can be posed after an action is supplied	ECT trajectory and proof
α_{fs} normalisation	Quark contribution undefined	P7 supplies no charged-matter data; observed charges are external benchmarks	Full electroweak/colour matter owner
Baryogenesis	Baryon current absent	A baryon-like colour singlet is permitted	$U(1)_B$, violating vertex, state and transport
Proton stability	Selection rules absent	Three conditional patterns can be catalogued	Pattern and operator ownership
Primary texture mass owner	Two-derivative $E_2(R) \propto R$ collapses	No change from P7 alone	Stabilising action and spectrum
Colour-sensitive laws/formulas	Blocked	Conditional source models can be written	Physical ECT derivation

Updated open questions under the primary completion

The primary completion improves consistency but does not end the programme. Its open questions are now sharper: why the internal module has rank three; why a distinguished antisymmetric trilinear form exists on the module; whether the non-Abelian connection acquires full Yang–Mills kinetics from ECT-native mechanisms; how quark representations and chirality arise; how hypercharge assignments and right-handed singlet structure emerge; whether confinement can be derived internally rather than imported from standard QCD intuition; whether strong-sector constants such as g_s and Λ_{QCD} can be constrained or derived; and what baryon-number selection rules are realised in the completed low-energy theory, which controls proton stability (§44). A deeper completion might eventually relate the number of colours to matter-representation content, but no such relation is derived here; conversely, the N_c -dependent anomaly classification of §9.5 shows that anomaly cancellation alone cannot supply this rank selection (it imposes only N_c odd), so the rank-3 input must come from the zero-mode programme below or from inherited charge data.

Alternative deeper geometric candidate: internal G_2 structure

Physical interpretation. A deeper geometric candidate would be an additional internal seven-dimensional real fibre $\mathcal{I}_x$, beyond the ordered branch, endowed with a metric and an antisymmetric composition law of octonionic type. Physically, this means that the medium possesses not merely internal degrees of local structure, but a specific non-associative geometry of how those internal channels combine locally. The group preserving this internal composition structure is G_2 .

Conditional mathematical kernel: G_2 stabiliser $SU(3)$. If such an internal G_2 structure exists, choosing a unit internal direction $u \in \mathcal{I}_x$ reduces the automorphism group to its stabilizer [270, 271]:

$$\text{Stab}_{G_2}(u) = SU(3). \quad (40.3)$$

On the orthogonal six-dimensional subspace $u^\perp$, the internal cross-product defines an almost-complex structure [271]:

$$J_u(v) = u \times v, \quad J_u^2 = -1, \quad (40.4)$$

so that $u^\perp \simeq \mathbb{C}^3$. The colour triplet then appears not as a postulated rank-3 module but as the complex three-space orthogonal to a distinguished internal calibration direction. These are standard mathematical facts about G_2 -structures; ECT does not derive the existence of such an internal octonionic fibre.

Relationship to the primary completion. The G_2 route is not merely an alternative to the primary completion. It can be viewed as a possible deeper origin of it: the $SU(3)$ -structure required by the primary completion arises naturally on $u^\perp$. The G_2 -compatible cross-product defines the complex structure J_u , while the induced positive-definite Hermitian form and complex volume form provide the $SU(3)$ -structure whose components appear as ε_{abc} in an adapted complex basis. In this sense, the G_2 completion implies the primary completion, but not conversely. The two candidates are nested, not parallel.

Why the G_2 route is not adopted as the primary completion. This route requires an exceptional internal geometry—a seven-dimensional real fibre with octonionic composition law—that is not otherwise forced by the current ECT postulates. It is physically less transparent and more speculative than the rank-3 module completion. The $SU(3)$ -structured rank-3 module completion is therefore adopted as the primary working hypothesis, while the G_2 route is recorded as a deeper candidate origin that may become preferable if future work uncovers independent evidence for an internal octonionic structure of the medium.

41 A leading colour-EFT truncation conditional on supplied P7 data

Status: Level B/Open leading truncation after a P7 bundle, a connection, a field content and locality/power counting are separately supplied. It is not a minimal or complete basis. The free health algebra is Level A only inside the stated constant-background branch.

Conventions and declared truncation. Use $\text{tr}(T^a T^b) = \delta^{ab}/2$ and

$$D_\mu = \partial_\mu - ig_s A_\mu^a T^a, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c, \quad qquad \tilde{F}^{a\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}^a. \quad (41.1)$$

With this convention F already contains g_s , so the correctly normalised gauge-theta density contains g_s^2 . One declared two-derivative truncation is

$$\mathcal{L}_{\text{col}}^{\text{lead}} = -\frac{1}{4} Z(\phi) F_{\mu\nu}^a F^{a\mu\nu} - \frac{1}{4} Y(\phi) (n^\mu F_{\mu\rho}^a) (n^\nu F_\nu^{a\rho}) + \frac{g_s^2 \theta_{\text{gauge}}(\phi)}{32\pi^2} F_{\mu\nu}^a \tilde{F}^{a\mu\nu} + \mathcal{L}_{\text{matter}}^{(3)} + \dots \quad (41.2)$$

Here $\phi = \phi_u$ is the dimensionless macroscopic coordinate. On a constant background define $Z_0 = Z(\bar{\phi})$, $Y_0 = Y(\bar{\phi})$ and $\theta_{\text{gauge},0} = \theta_{\text{gauge}}(\bar{\phi})$. The ellipsis is essential: a complete basis requires the field content, derivative counting, redundancies, director derivatives and constraint reduction.

Free constant-background health. For a constant aligned timelike director $n^\mu = (1, 0, 0, 0)$ in the declared mostly-plus convention, the transverse quadratic terms are

$$\mathcal{L}_{\text{col,quad}} = \frac{Z_0 - Y_0/2}{2} \mathbf{E}^a \cdot \mathbf{E}^a - \frac{Z_0}{2} \mathbf{B}^a \cdot \mathbf{B}^a.$$

Thus this free branch requires

$$Z_0 > 0, \quad Z_0 - \frac{Y_0}{2} > 0, \quad c_T^2 = \frac{Z_0}{Z_0 - Y_0/2}. \quad (41.3)$$

No statement about a generic background or interacting constraint surface follows from this quadratic test.

Three scoped non-identification results. (i) In a supplied exact gauge theory, a Proca term $m_g^2 A_\mu^a A^{a\mu}$ is not gauge invariant. This excludes that operator inside the declared truncation, not every mechanism for a gauge-invariant spectral scale. (ii) P7 assigns (ϕ, n^μ) to singlets and supplies no colour-charged scalar, so the displayed field content has no condensate-Higgs route. This does not exclude an added charged multiplet, a reducible singlet VEV, an emergent/nonlinear route or a kernel-bundle completion. (iii) A linearly spaced oscillator tower $m_n = n\omega$ is not automatically a physical colour spectrum. Identifying any tower requires a map to a gauge-invariant operator class, the gauge/constraint quotient, the physical action and state/measure, and the transfer-matrix or spectral dynamics. No universal no-go against a linear pattern follows.

Independent anisotropic coefficient. The $Y(\phi)(n \cdot F)^2$ operator is allowed in this truncation, but P7 neither requires Y to be nonzero nor makes it negligible. Sharing the spurion $n_\mu n_\nu$ with another sector transfers no Wilson coefficient or experimental bound. A relation would require one microscopic action and an explicit matching calculation; both are Open.

Separately declared one-coupling source-model branch. The ordinary pure- $SU(3)$ formula is used only after imposing all of

$$\mathcal{B}_{\text{src}} : \quad \partial \bar{\phi} = 0, \quad Z_0 > 0, \quad Y_0 = 0, \quad \theta_{\text{gauge},0} = 0, \quad \mathcal{L}_{\text{matter}} = 0, \quad \text{higher operators neglected.} \quad (41.4)$$

After canonical normalisation, $g_{\text{src}}^2 = g_s^2/Z_0$. Standard matter-free Yang–Mills then has $b_0 = 11$ and the model-internal scale

$$\Lambda_{\text{col}}^{\text{src}} = M_* \exp \left[-\frac{8\pi^2}{b_0 g_{\text{src}}^2(M_*)} \right] = M_* \exp \left[-\frac{8\pi^2 Z_0}{b_0 g_s^2(M_*)} \right]. \quad (41.5)$$

This is Level A algebra inside the supplied standard source model, not an ECT prediction or a transfer from the shared director. M_* , g_s and Z_0 remain matching inputs.

General anisotropic EFT. Away from $\mathcal{B}_{\text{src}}$, the running is a multi-coefficient Open RG problem for $(Z, Y, \theta_{\text{gauge}}, g_s, c_i, \dots)$ with operator mixing, matter thresholds and director-dependent counterterms. The coefficient $b_0 = 11$ and the one-dimensional flow for g_{src} cannot be copied to that system. In particular, no correction called “subleading in Y_0/Z_0 ” is justified before the coupled beta functions and physical matching are derived.

What is and is not established. Established conditionally are the displayed convention, the leading truncation, the free aligned-background health cone and the source-model reproduction. The physical colour connection/action, the value or sign of Y , the multi-coefficient RG, matter content, $\bar{\theta}$, confinement and the spectrum remain Open. The one-loop integration and illustrative source-model sensitivity arithmetic on $\mathcal{B}_{\text{src}}$ are collected in Appendix BD; they do not extend to the general anisotropic EFT.

42 Anomaly constraints and matter-sector consistency

Status: standard anomaly constraints, conditionally restated after a physical gauge and matter content are supplied. P7 alone does not establish that physical content; ECT representation ownership remains Open.

Setup. If the structural bundle postulated in P7 is endowed with a physical $SU(3)$ connection/action and chiral matter, the usual anomaly constraints must hold. The algebra below is established external gauge theory. It tests a supplied spectrum; it neither derives the spectrum nor upgrades P7 to physical colour.

$SU(3)^3$ anomaly. For coloured fermions in representations R_i , the cubic anomaly condition is

$$\sum_i \mathcal{A}(R_i) = 0, \quad \mathcal{A}(\mathbf{3}) = +1, \quad \mathcal{A}(\bar{\mathbf{3}}) = -1.$$

In the minimal realisation with matter restricted to fundamental and antifundamental colour representations, anomaly cancellation forces $N_3 = N_{\bar{3}}$, i.e. vector-like colour content. More general chiral spectra require their full cubic-anomaly sum. For example $\mathcal{A}(\mathbf{8}) = 0$ whereas $\mathcal{A}(\mathbf{6}) = 7$ in the declared convention, so an adjoint plus one rank-two symmetric representation is not anomaly-free. No exotic anomaly-free ECT spectrum is supplied here.

Mixed anomalies with additional gauge factors. Introducing additional gauge factors to accommodate chirality, the following mixed anomalies must cancel:

$$\begin{aligned} SU(3)^2 \cdot U(1) : \quad & \sum_i q_i T(R_i) = 0, \\ SU(2)^2 \cdot U(1) : \quad & \sum_{\text{doublets}} q_L = 0, \\ U(1)^3 : \quad & \sum_i q_i^3 = 0, \\ U(1) \cdot \text{grav}^2 : \quad & \sum_i q_i = 0. \end{aligned}$$

For the one-generation Standard-Model content $\{Q_L, u_R, d_R, L_L, e_R\}$, these constraints fix hypercharges up to normalisation, modulo the $u_R \leftrightarrow d_R$ relabelling and a degenerate lepton-neutral branch excluded

by the inherited charged-lepton assignment (solution structure: §9.5). With sterile-neutrino freedom a hypercharge-neutral singlet ($Y_V = 0$) can be added without disturbing anomaly balance; if instead the singlet hypercharge is left free, a $B-L$ flat direction opens and the anomaly system alone no longer fixes the assignment (ibid.); this is relevant for the neutrino discussion of §8.9 and for the proton stability discussion of §44.

Witten global anomaly. Any ECT completion with an $SU(2)$ weak factor must satisfy the Witten global anomaly constraint: the total number of left-handed $SU(2)$ doublets must be even. With four doublets per generation and three generations the count is twelve, and the anomaly is absent.

't Hooft anomaly matching. If the ECT completion admits distinct UV (medium microphysics) and IR (emergent SM-like EFT) descriptions, global anomalies must match between the two. This is a nontrivial structural constraint on the microphysical completion and is flagged as Open (§47).

Baryon-like singlets do not define $U(1)_B$. The invariant ϵ_{abc} permits a three-fundamental colour singlet, but colour representation theory alone supplies neither a baryon-number generator nor a conserved current. A $U(1)_B$ assignment requires the full matter representations, their charges, the action, anomalous/divergence equations and the allowed violating vertices. The three proton-stability patterns in Section 44 are therefore conditional completion hypotheses. If a future completion supplies a low-energy baryon selection rule, quantum-gravity no-global-symmetry arguments constrain how it can be realised; they do not derive it from P7. The unrelated determinant- line phase associated with Ω also has Open gauged/broken/protected status.

Local structure does not determine the global quotient. The local gauge structure discussed here does not determine the global quotient $(SU(3) \times SU(2) \times U(1))/\Gamma$, which remains an additional completion-level question tied to charge quantisation and global bundle structure. It is flagged as Open.

What this section claims and does not claim. Claimed: every supplied chiral gauge spectrum must pass the standard local and global anomaly tests; the displayed representation-theory arithmetic is conditional on the stated spectrum. Not claimed are a physical colour realisation, uniqueness of the Standard Model group or matter content, hypercharge normalisation, three generations, the global quotient, sterile- neutrino status, or a $U(1)_B$ current/selection rule. All are Open ECT matching inputs.

43 Inventory of $SU(3)$ -dependent questions from earlier parts

Status: Level-B structural input plus C/Open physical bookkeeping. The table records what becomes well-posed after the adopted P7 data are supplied and what remains physically Open; it is not a physical closure table.

Status convention. “Structural input” means adopted through P7 rather than derived from P1–P6. “Conditional” means exact only inside a declared action/representation/source model. “Sharpened” means that a missing owner is more precisely identified. “Open” means no physical ECT derivation or matching owner is available.

Table 104: Coverage of $SU(3)$ -dependent questions after the adopted P7 structural postulate.

Issue	Before Part IV	What Part IV actually supplies	Current status
Physical colour realisation (OP-GUT1)	Missing	Rank-three (h, Ω) frame candidate; direct-embedding and fundamental-stabiliser facts; alternative routes preserved	Supplied candidate / physical colour Open
Leading colour-EFT truncation	Not formulated	One convention-fixed truncation after connection/action input; not a complete basis	Level B/Open
Anomaly audit	Conditional EW audit	Standard $SU(3)^3$ and mixed conditions restated for a supplied spectrum	Conditional external algebra
Weak chirality and $SU(2)_L$	Open	P7 supplies no chiral projector	Open
Three fermion generations	Open	A supplied spectral toy can be tuned to three states, but no colour-rank, operator, index or selection map identifies them with physical families	Open
Matter representations	Open	Counterexamples show anomalies do not select a unique spectrum	Sharpened / Open
Charges and hypercharges	Conditional arithmetic	Supplied one-generation branches and global-quotient constraints remain model inputs	Conditional / Open
Strong-sector constants	Open	The $b_0 = 11$ formula is reproduced only on $\mathcal{B}_{\text{src}}$	Source-model reproduction / Open ECT matching
Confinement and glueball spectrum	Open	An oscillator tower is not identified without a gauge-invariant operator/dynamics map; no universal spectral no-go is proved	Open
Strong CP	Not owned	Correct coefficient $g_s^2 \theta_{\text{gauge}} F \tilde{F} / (32\pi^2)$ and invariant $\bar{\theta} \equiv \theta_{\text{gauge}} + \arg \det M_q \pmod{2\pi}$	Open; MISSING MASS-PHASE OWNER
Proton stability and baryon number	Conditional patterns	$SU(3)$ permits baryon-like singlets but does not define $U(1)_B$	Level C/Open
Neutrino/sterile status	Level B/Open	A neutral singlet can pass anomaly tests	Sharpened / Open
Global quotient	Implicit	Local Lie algebra does not determine $(SU(3) \times SU(2) \times U(1))/\Gamma$	Open
Yang–Mills existence and mass gap	Not owned	A corrected conditional transfer theorem and a finite-cutoff programme are stated	External conditional theorem / physical ECT Open

Phenomenological boundary. No bound on Y_0/Z_0 follows from a gravity-sector coefficient or from P7; the realised value could be zero or nonzero and requires its own observables. The neutron-EDM analysis constrains $\bar{\theta}$ through a hadronic map. It does not directly constrain $\theta_{\text{gauge}}(\bar{\phi})$ without an owner for $\arg \det M_q$. Thus the strong-CP row has outcome MISSING MASS-PHASE OWNER. M_* , Z_0 and g_s are likewise matching inputs; the displayed source-model scale has no ECT predictive leverage.

Coverage verdict. Part IV states the adopted P7 $SU(3)$ structural postulate and proves its exact scoped algebra. It does not close physical colour, a physical EFT basis, the coupled RG, baryon number, strong CP, confinement or the mass gap. Every topic from the prior coverage table is retained with a status no stronger than its owner supports.

44 Proton stability and baryon number as a UV discriminant

Status: standard conditional EFT algebra / Open ECT. This subsection is conditional on the unresolved gauge/flavour completion.

Proton stability is not part of the already established strict core of ECT; it is a discriminant of how the presently open gauge/flavour completion closes. The proton lifetime answers a structural question: what low-energy baryon-number selection rules are realised in the completed theory. The detailed selection rules depend on the completed gauge/flavour structure, an essential unresolved part of which is the colour sector (§40). Proton stability is therefore an Open ECT completion question; only the generic selection-rule taxonomy and supplied-EFT algebra are retained.

Three selection-rule patterns

The completed theory may realise several distinct baryon-number selection-rule patterns.

Pattern I: full low-energy protection. The completed gauge/flavour sector enforces a selection rule forbidding low-energy baryon-violating operators relevant for proton decay and for the leading $\Delta B = 2$ processes. In this case no low-energy proton-decay channel is generated within the completed effective theory. In a gravitationally complete setting such protection is more naturally realised through a gauged or discrete remnant than through an exact continuous global $U(1)_B$, in line with general quantum-gravity arguments suggesting the absence of exact continuous global symmetries [272, 273].

Pattern II: partial protection. $\Delta B = 1$ operators are forbidden, but $\Delta B = 2$ transitions remain allowed. Proton decay is absent—not because baryon number is exactly conserved, but because the completed low-energy selection rules forbid precisely the $\Delta B = 1$ channel. More generally, the completed theory may preserve or violate different combinations such as B , L , and $B - L$, which determine the allowed operator classes. In this pattern the relevant observational tests are $n-\bar{n}$ oscillations and dinucleon decay rather than proton decay.

Pattern III: conditional dimension-six benchmark. Assume all of the following additional inputs: a physical baryon current and charge assignment; a Standard-Model-like spectrum below a supplied matching scale Λ_B ; no lighter baryon-violating mediator; and a leading local $\Delta B = 1$ operator of the usual dimension-six form. Only in that source EFT may one write

$$\mathcal{O}_B \sim \frac{C_B}{\Lambda_B^2} qqql. \quad (44.1)$$

Dimensional analysis then gives, up to the channel-dependent hadronic matrix element, phase space, running and numerical normalisation,

$$\tau_p \sim \frac{\Lambda_B^4}{|C_B|^2 m_p^5}, \quad \Lambda_B > 0, \quad C_B \neq 0. \quad (44.2)$$

This scaling is Level A algebra inside the declared external EFT, not an ECT amplitude. The branch $C_B = 0$ has no decay through this operator and is not evaluated by the quotient. A named C/Open NDA reference branch may additionally set $\Lambda_B = c_B \phi_0$ with $c_B > 0$ and retain C_B as a supplied coefficient; the special choice $c_B = |C_B| = 1$ is only a unit-coefficient benchmark. On the further supplied point $\zeta_\phi = 1$, the bare natural-unit scaling, after restoring $\hbar$ and converting to Julian years, gives

$$\tau_{p,\text{NDA}}^{\text{unit}} \sim \frac{\bar{M}_{\text{Pl}}^4}{m_p^5} \simeq 1.01 \times 10^{42} \text{ yr.} \quad (44.3)$$

This is only a crude NDA reference scale: its displayed digits are not a physical precision, and channel-dependent hadronic, RG, flavour, phase-space and normalisation factors are omitted. ECT derives neither input nor those owners, so no proton-lifetime number is predicted. A protecting symmetry may instead give $C_B = 0$.

The three patterns are a generic conditional selection-rule taxonomy, not a structural ECT result. P7 supplies none of its physical owners. The current outcomes are MISSING BARYON-CURRENT OWNER, MISSING VIOLATING VERTEX, MISSING MATCHING-SCALE OWNER, and MISSING HADRONIC MAP.

Connection to baryogenesis

The present theory does not yet derive physical baryon number, a topological baryogenesis bias, or an electroweak sphaleron operator: the relevant compact bundle, chiral matter representations, violating vertex and state are missing. Such operators may occur in a supplied completion, but their existence cannot presently be used either to infer baryogenesis or to remove proton stability. The completed low-energy baryon-number selection rules relevant for proton decay remain Open.

Whether the same topological or sphaleron sector induces a local $\Delta B = 1$ nucleon-decay operator at zero temperature remains open. If it does, the present-day decay rate is controlled by a zero-temperature Euclidean action (instanton/bounce-type suppression) or by the induced effective operator, *not* by thermal Boltzmann suppression.

Experimental context and comparison

The 2020 Super-Kamiokande result in the $p \rightarrow e^+ \pi^0$ channel is $\tau_p/B > 2.4 \times 10^{34} \text{ yr}$ at 90% C.L. [203]. The Hyper-Kamiokande design report projects sensitivity of order 10^{35} yr [274]; this is a projected sensitivity, not an observed limit.

The present ECT framework, including the P7 structural postulate, does not determine the spectrum between the electroweak and ultraviolet scales, forbid an intermediate matching scale, or predict that proton decay lies outside experimental reach. A confirmed $\Delta B = 1$ proton-decay signal would exclude any named completion whose defining exact selection rule forbids $\Delta B = 1$ (Patterns I and II), but it would not by itself exclude ECT. A null result instead constrains only a violating completion whose baryon current, operator basis, mediator spectrum, matching scale, coefficients, running, hadronic matrix elements and branching fractions are specified; it cannot exclude the abstract Pattern III class.

Status summary. The dimensional scaling is Level A only inside the declared external dimension-six EFT. The displayed unit-coefficient NDA reference is conditional arithmetic, not a physical lifetime. The ECT baryon-number current, realised pattern, vertices, matching scale, decay amplitudes and every physical ECT lifetime prediction are Open.

Open problem (OP-proton). Derivation of baryon-number selection rules from the completed gauge/flavour sector; identification of which pattern (I/II/III) is realised; if Pattern III, identification of decay channels and coefficient structure.

45 Topological sector and strong CP

Status: standard-QCD convention and external constraint, with ECT matching Open. No strong-CP resolution is derived.

Normalisation. With $D_\mu = \partial_\mu - ig_s A_\mu^a T^a$ and F containing g_s as in Eq. (41.1), the topological term is

$$\mathcal{L}_\theta = \frac{g_s^2 \theta_{\text{gauge}}(\phi)}{32\pi^2} F_{\mu\nu}^a \tilde{F}^{a\mu\nu}. \quad (45.1)$$

Gauge invariance permits this operator but does not require a nonzero angle or derive its ϕ dependence.

What experiment constrains. In the declared sign convention the invariant is

$$\bar{\theta} \equiv \theta_{\text{gauge}}(\bar{\phi}) + \arg \det M_q \pmod{2\pi}, \quad \det M_q \neq 0. \quad (45.2)$$

For $\det M_q \neq 0$, the determinant phase is defined only modulo 2π . A massless-quark limit is a distinct case in which that phase is ill-defined/removable and cannot be analysed by the same shorthand. Combining the 90%-CL neutron-EDM limit [275] with the classic chiral-QCD estimate of Crewther et al., including its erratum [276], gives only the familiar order-of-magnitude inference $|\bar{\theta}| \lesssim 10^{-10}$. It is not a direct experimental inequality on θ_{gauge} , nor a precision hadronic determination. Until the quark mass matrix and its phase are owned, the correct ECT outcome is MISSING MASS-PHASE OWNER; no direct bound on $\theta_{\text{gauge}}(\bar{\phi})$ follows.

Possible resolution routes, all Open. *R1 — CP-symmetric branch.* A CP condition on θ_{gauge} is insufficient unless the same completion controls $\arg \det M_q$ and selects $\bar{\theta} = 0$ rather than π . *R2 — axion-like relaxation.* A physical pseudoscalar, anomaly vertex, potential and state would have to relax the invariant $\bar{\theta}$; none is derived from the present ordered-branch coordinates. *R3 — determinant-form geometry.* The supplied Ω might enter a future mass-phase/CP construction, but no theorem connects it to either term in Eq. (45.2).

Witten–Veneziano consistency target. In the convention $\langle 0 | J_{5\mu}^a | \pi^b(p) \rangle = i F_\pi p_\mu \delta^{ab}$ with $F_\pi \simeq 92$ MeV and $N_f = 3$, the leading large- N_c /chiral relation is

$$m_{\eta'}^2 + m_\eta^2 - 2m_K^2 \simeq \frac{2N_f}{F_\pi^2} \chi_{\text{YM}}. \quad (45.3)$$

This is an external QCD relation [277, 278], subject to its usual corrections and decay-constant convention. P7 supplies no correction or matching coefficient.

Final status. The gauge-theta operator is allowed with the convention-fixed g_s^2 factor; the observed small quantity is $\bar{\theta}$; and the missing quark-mass-phase owner prevents a direct ECT bound or resolution. R1–R3 remain Open.

46 ECT and the Yang–Mills mass-gap problem

Status: the standard pure-SU(3) one-loop and local heat-kernel identities below are conditional mathematics. P7 supplies an adopted Level-B internal SU(3)-structured bundle, not a physical colour connection, normalised action, reflection-positive measure, stable blocking trajectory or physical scale map. The conditional endpoint theorem therefore has no established ECT antecedent. HKLoc is retained as an incomplete attempted bookkeeping criterion; the physical/global ECT colour gap and the Clay problem remain Open.

This section addresses the Yang–Mills problem directly. Its purpose is neither to claim a solution of the Clay Millennium problem nor to dismiss it, but to separate the physical question from the mathematical one, to establish which of the two ECT engages, and to state precisely how far the engagement can presently be carried.

46.1 Separation of the physical and mathematical questions

The Clay formulation combines two logically independent statements, and ECT must keep them apart; conflating them produces the spurious dilemma of whether ECT “solves” the problem.

(K1) The physical question. Why does the observed strong sector exhibit confinement and a mass gap—no free, arbitrarily soft coloured excitations, and quarks permanently bound into colour-singlet states? This is a question about the real world.

(K2) The mathematical question. Does a constructively well-defined quantum *pure* Yang–Mills theory exist on $\mathbb{R}^4$ for a compact simple gauge group, with fundamental Poincaré invariance and satisfying the Wightman or Osterwalder–Schrader axioms, whose spectrum has a strictly positive gap $\Delta > 0$ [279]? For $SU(3)$ this means glueball states cannot be arbitrarily light. This is a question about a specific mathematical object: the continuum measure of pure Yang–Mills on flat $\mathbb{R}^4$.

The Clay prize is awarded for (K2), the construction; it is *motivated* by (K1), the physics. ECT relates differently to each, and we treat them separately.

46.2 Relation of the P7 structural postulate to the Clay object

The continuum Yang–Mills object in (K2) is a well-posed standard mathematical target. P7 neither proves that this object is non-fundamental in nature nor derives a physical finite-cutoff replacement.

Framework hypothesis H_{Clay} . ECT adopts as a separate Level-B/Open framework hypothesis that a future P7 physical colour completion may be emergent and finite-cutoff rather than the primitive continuum object appearing in (K2). This hypothesis is not a consequence of P7. It requires a colour action, measure, regulator and scale map, supplies no evidence against the continuum theory, and is neither a solution nor a dismissal of the Clay problem. Conditional on those owners, one may pose a distinct finite-cutoff spectral question for that model.

46.3 Domain comparison without an ontology upgrade

The Clay construction assumes the continuum and symmetry premises stated in its formulation. The current ECT manuscript has not established a physical colour sector with different premises. The optional anisotropic operator in Eq. (41.2) may have zero or nonzero coefficient; P7 does not decide. Therefore neither exclusion from nor identification with the Clay domain follows from the present action. The only justified statement is conditional: a supplied finite-cutoff anisotropic model would pose a different problem, while its $Y_0 = 0$ continuum source branch reproduces standard Yang–Mills premises.

46.4 Candidate finite-cutoff colour question and missing owners

After, and only after, a compact colour bundle, action, admitted measure, physical operator algebra, regulator and scale map are supplied, one may ask whether connected adjointed correlators of a transfer-cyclic/dense gauge-invariant class decay uniformly and hence whether the reconstructed Hamiltonian has a gap. A finite flux-tube or Wilson-loop question is a separate confinement diagnostic. None of these physical inputs is provided by P7 itself.

This finite-cutoff question is not a replacement for the Clay continuum construction. It lacks the $a \rightarrow 0$ requirement and therefore proves less even if solved. Conversely, a finite cutoff does not guarantee a gap; a massless phase remains possible. Pure gauge, gauge theory with matter, CP-even and CP-odd branches,

and zero- versus finite-temperature sectors must be treated separately. The source-model calculations below use only the constant-background $Y_0 = 0$, $\theta_{\text{gauge},0} = 0$, matter-free branch.

Centre symmetry and a Polyakov-loop expectation value are pure-gauge finite-temperature confinement diagnostics, not proofs of a zero-temperature global spectral gap. Dynamical fundamental matter explicitly breaks the centre symmetry, so the diagnostic cannot be transferred unchanged to a matter-coupled physical sector. The confinement and gap owners must therefore remain separate.

46.5 Source-model RG algebra and the Open anisotropic flow

The one-coupling calculation is restricted to the separately declared branch $\mathcal{B}_{\text{src}}$ of Eq. (41.4). On that constant-background, $Y_0 = 0$, CP-even, matter-free pure- $SU(3)$ source model, the standard beta function is

$$\mu \frac{dg_{\text{src}}}{d\mu} = -\frac{b_0 g_{\text{src}}^3}{16\pi^2}, \quad b_0 = 11, \quad (46.1)$$

For a separately supplied isotropic source model containing n_f Dirac flavours in the fundamental representation, the standard external one-loop coefficient is instead

$$b_0 = 11 - \frac{2}{3}n_f. \quad (46.2)$$

This is a source-model formula, not an ECT matter derivation, and it is not used on the matter-free branch $\mathcal{B}_{\text{src}}$. and hence

$$\frac{1}{g_{\text{src}}^2(\mu)} = \frac{1}{g_{\text{src}}^2(M_*)} + \frac{b_0}{8\pi^2} \ln \frac{\mu}{M_*}, \quad g_{\text{src}}^2(\mu) = \frac{8\pi^2}{b_0 \ln(\mu/\Lambda_{\text{col}}^{\text{src}})}. \quad (46.3)$$

Writing $x_{\text{src}} = g_{\text{src}}^2 = g_s^2/Z_0$, the formal pole and a chosen strong-coupling threshold obey

$$\Lambda_{\text{col}}^{\text{src}} = M_* \exp \left[-\frac{8\pi^2 Z_0}{b_0 g_s^2(M_*)} \right], \quad \mu_c = \Lambda_{\text{col}}^{\text{src}} \exp \left[\frac{8\pi^2}{b_0 x_c} \right]. \quad (46.4)$$

The inverse relation is

$$\frac{g_s^2(M_*)}{Z_0} = \frac{8\pi^2}{b_0 \ln(M_*/\Lambda_{\text{col}}^{\text{src}})}. \quad (46.5)$$

These are standard source-model identities. They show infrared growth only on $\mathcal{B}_{\text{src}}$ and neither identify the physical ECT trajectory nor prove entry into a constructive domain.

At M_* this source branch is in the weak-UV regime, so a strong-coupling or polymer expansion cannot be applied there. Such an endpoint theorem may be invoked only at a separately identified blocked scale after an action-owned RG/blocking trajectory, preservation of its locality/positivity class, and a quantitative polymer-entry bound have been proved. The one-loop running and formal pole do not establish any of those steps.

For $Y \neq 0$, running backgrounds, matter, CP-odd terms or generated higher operators, the action has a multi-coefficient Open RG problem. Its beta functions and operator mixing are not obtained by adding a “small Y ” correction to $b_0 = 11$. No shared-director coefficient transfer, nonzero- Y assumption, negligibility assumption or phenomenological Y_0/Z_0 bound is made. An oscillator tower likewise supplies no spectral conclusion without the physical gauge-invariant operator/dynamics map.

46.6 Conditional transfer-cyclic mass-gap endpoint and missing ECT owners

Theorem 46.1 (Conditional transfer-spectrum implication; not an ECT theorem). *Let a fully specified reflection-positive lattice colour theory reconstruct a Hilbert space $\mathcal{H}$, positive transfer operator $T = e^{-a_{\text{col}} H}$, self-adjoint $H \geq 0$, and unique vacuum Ω . Let $\mathfrak{A}_0$ be a physical gauge-invariant operator class for which*

$$\mathcal{D}_0 := \text{span}\{(\mathcal{O} - \langle \mathcal{O} \rangle)\Omega : \mathcal{O} \in \mathfrak{A}_0\} \text{ is dense in } \Omega^\perp. \quad (46.6)$$

Thus the class is transfer-cyclic/dense, not merely separating. Suppose there is one $m_{\text{lat}} > 0$ such that every $\mathcal{O} \in \mathfrak{A}_0$ has the positive adjointed reflected transfer correlator

$$C_{\mathcal{O}}(n) := \langle \Theta \mathcal{O}_0 T^n \mathcal{O}_0 \rangle = \langle \psi_{\mathcal{O}}, T^n \psi_{\mathcal{O}} \rangle \geq 0, \quad C_{\mathcal{O}}(n) \leq c_{\mathcal{O}} e^{-m_{\text{lat}} n} \quad (n \geq 0), \quad (46.7)$$

with a common exponent and the stated domain/uniformity. A stronger sufficient formulation is the uniform positive quadratic-form bound $\langle \psi, (T - P_0)^n \psi \rangle \leq e^{-m_{\text{lat}} n} \|\psi\|^2$ on a dense physical form domain, where $P_0 = |\Omega\rangle\langle\Omega|$. Then

$$\text{spec}(H) \cap (0, m_{\text{lat}}/a_{\text{col}}) = \emptyset, \quad \Delta_{\text{gap}} \geq m_{\text{lat}}/a_{\text{col}} > 0. \quad (46.8)$$

If, in addition, a separately derived positive two-sided physical scale map obeys

$$0 < c_- \Lambda_{\text{phys}} \leq a_{\text{col}}^{-1} \leq c_+ \Lambda_{\text{phys}} < \infty, \quad (46.9)$$

then $\Delta_{\text{gap}} \geq m_{\text{lat}} c_- \Lambda_{\text{phys}} > 0$ in physical units.

Why “separating” alone fails. Take $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ and $\mathfrak{M} = B(\mathcal{H}_1) \otimes I$. Choose Ω with full Schmidt rank on $\mathcal{H}_1$ but with Schmidt support a proper subspace of $\mathcal{H}_2$. Then Ω is separating for $\mathfrak{M}$, while $\overline{\mathfrak{M}\Omega}$ is not all of $\mathcal{H}$. A Hamiltonian can give unit gap on $\overline{\mathfrak{M}\Omega} \ominus \mathbb{C}\Omega$ and a gapless positive continuum on its orthogonal complement, with Ω the unique vacuum. All tested $\mathfrak{M}$ correlators decay with unit rate but the global gap is zero. Transfer-cyclicity/density is therefore essential.

P7 and HKLoc supply none of the action, admitted measure, positive adjointed reflection form, unique vacuum, dense physical class, uniform decay theorem or two-sided scale map. The physical/global ECT gap remains Open.

A positive transverse quadratic kernel, coercivity estimate or inserted mass parameter in a gauge-fixed/free truncation is not a surrogate for this nonperturbative gauge-invariant gap. Assuming a positive mass while using it to prove the gap is circular, while a Proca-like mass insertion is excluded by the exact gauge symmetry of the declared source theory.

46.7 Precise localisation of the residual: the A/B/C split and the polymer-entry criterion

The hypotheses of Theorem 46.1 dissect, in the ECT setting, into a known external theorem plus two ECT-specific bridges.

Part A — endpoint mathematics (external and conditional). For compact four-dimensional lattice $SU(3)$ in a stated strong-coupling/ polymer-convergence domain, constructive results establish clustering and confinement diagnostics [280, 281]. A global spectral-gap inference additionally requires the positivity, unique-vacuum, transfer-cyclic/dense-class and uniformity hypotheses of Theorem 46.1. These results are cited, not re-derived, and apply only after one fully specified action has entered their domain.

A finite cutoff is not a gap premise. A supplied finite cutoff removes the continuum-limit task from a candidate model, but it neither places the action in a strong-coupling domain nor excludes a massless phase. The present manuscript has not derived a physical cutoff or scale map for P7. Polymer entry, class preservation and the global transfer-spectrum hypotheses remain independent Open owners.

Part B — source-model RG and Open trajectory. On $\mathcal{B}_{\text{src}}$, the standard $b_0 = 11$ calculation reproduces infrared growth and the formal scale $\Lambda_{\text{col}}^{\text{src}}$. This is source-model algebra, not ECT-specific RG closure. Whether a physical P7 action exists, whether its generally anisotropic multi-coefficient flow enters the constructive domain, and whether it avoids an intervening phase are Open. No coefficient relation or small- Y expansion is supplied.

Part C — class preservation (Open). Even granting that the strong-coupling domain is reached, one must show that the blocked action at a_{col} remains in the local (or exponentially local), compact, transfer-matrix-positive class to which the endpoint theorem applies. Blocking from M_* to a_{col} may generate rectangular and chair loops, multi-plaquette interactions, long-range tails, anisotropic $n \cdot F$ terms, θ -dependent phases, matter-induced centre breaking, or loss of transfer positivity, any of which would take the system outside the theorem’s class. This is not automatic.

The residual as a checkable criterion. Rather than an abstract “the sector enters the confining basin,” the conjunction $B2 \wedge C$ is made concrete. Write $\mathcal{J}_{\text{eff}}$ and $\mathcal{J}_{\text{poly}}^{\text{nonplaq}}$ for the real dimensionless exponents of the supplied lattice source model. If a future physical colour action produces them, it must supply a finite positive action unit $\Sigma_{\text{col}}^{\text{act}} > 0$, for example $\mathcal{J}_{\text{eff}} = S_{\text{eff}}^{\text{phys}} / \Sigma_{\text{col}}^{\text{act}}$. No map from $\Sigma_{\text{col}}^{\text{act}}$ to either bare-P3 action owner A_{P3} or $\Sigma_{\text{PI},P3}$, their ratio κ_{P3} , the phase normalisations, S_0 or $\hbar$ is presently derived. Expand the blocked dimensionless exponent in $SU(3)$ characters,

$$e^{-\mathcal{J}_{\text{eff}}(U)} = \prod_p \left[1 + \sum_{R \neq 0} c_R(a_{\text{col}}) \chi_R(U_p) \right] \times e^{-\mathcal{J}_{\text{poly}}^{\text{nonplaq}}(U)}, \quad (46.10)$$

where R runs over nontrivial irreducible representations, χ_R are the characters, and c_R the activities. The displayed real logarithm is admitted only on a branch where the full supplied weight $W(U)$ on the right-hand side is pointwise positive, so that $\mathcal{J}_{\text{eff}} = -\ln W$ exists as a finite real exponent. At $W = 0$ the exponent diverges, while $W < 0$ requires a complex-log branch and lies outside this real source model. Conjugate pairing $c_{\bar{R}} = c_R^*$ together with real $\mathcal{J}_{\text{poly}}^{\text{nonplaq}}$ is necessary for reality but is not sufficient for $W > 0$. Even pointwise positivity is weaker than reflection or transfer-matrix positivity, which remain separate gates. For a Wilson-type seed $\mathcal{J}_W = -(\beta_{\text{eff}}/3) \sum_p \text{Re Tr } U_p$ at small β_{eff} the leading activities are $c_3 = c_{\bar{3}} = \beta_{\text{eff}}/6 + O(\beta_{\text{eff}}^2)$, with higher representations of order $O(\beta_{\text{eff}}^2)$. The polymer expansion converges if

$$Q(a_{\text{col}}) := \sum_{R \neq 0} d_R |c_R(a_{\text{col}})| e^{\alpha C_2(R)} + \|\mathcal{J}_{\text{poly}}^{\text{nonplaq}}(a_{\text{col}})\|_{\alpha} < Q_c(d=4, SU(3)), \quad (46.11)$$

with d_R the dimension of R , $C_2(R)$ its quadratic Casimir, α a norm parameter, and Q_c the convergence-domain constant. For a fully specified action, a proved bound $Q(a_{\text{col}}) < Q_c$ may yield exponential clustering in controlled channels. A global gap then requires the transfer-cyclic/dense physical class, unique vacuum, positive adjointed reflection form, common uniform decay exponent and positive two-sided scale map of Theorem 46.1. The present manuscript has not derived those premises for an ECT action.

The residual splits into two sharply scoped open problems.

(OP-YM-polymer-entry-I — class/positivity). Show that ECT Wilsonian blocking from M_* to a_{col} preserves compact $SU(3)$ link variables, locality (or exponential locality), and transfer-matrix positivity in the pure CP-even sector.

(OP-YM-polymer-entry-II — activity bound). Show that the blocked character/polymer activities satisfy $Q(a_{\text{col}}) < Q_c$, eq. (46.11)—equivalently, bound the blocked action in a polymer norm relative to the compact- $SU(3)$ Wilson strong-coupling action, $\|\mathcal{J}_{\text{eff}}(a_{\text{col}}) - \mathcal{J}_{\text{Wilson}}^{\text{strong}}\|_{\text{polymer}} < \epsilon_c$.

If OP-YM-polymer-entry-I and II were proved for one fully specified colour action, they would establish entry into the cited endpoint domain only. A global gap would still require the transfer-cyclic/dense-class, unique-vacuum, positive-adjointed-form, uniform-decay and two-sided-scale-map premises of Theorem 46.1. The RG input used here is only the standard $b_0 = 11$ algebra on $\mathcal{B}_{\text{src}}$; the physical anisotropic trajectory and class preservation remain Open.

Continuum bridge. If a fully specified physical colour theory admitted a continuum limit that met the Clay axioms and retained a positive gap, that separate construction would address (K2). The P7 structural postulate supplies none of these dynamical or continuum premises. The continuum question is therefore neither required nor excluded by ECT; it remains a standard external mathematical target.

The remaining bottlenecks. The numerical matching data (M_*, g_s, Z_0) are not ECT-derived, but they are not the only obstruction. Even a zero-mode origin for P7 would not by itself provide a normalised colour action, admitted measure, higher-cumulant bounds, reflection/transfer positivity, a stable RG trajectory or the physical scale map. Each is a separate completion owner. The exponential sensitivity of the standard transmutation formula reinforces the need for matching data but does not prove the existence of a gap.

Summary of the engagement. ECT neither solves nor dismisses the Clay problem. It proposes a distinct finite-cutoff colour-sector programme after a compact $SU(3)$ action and physical regulator are supplied. The standard $b_0 = 11$ running generates a formal transmutation scale only inside the supplied isotropic CP-even pure-YM source model; it does not prove that the actual blocked ECT action enters a constructive domain. The conditional theorem above yields a channel/global gap only after its full premises are established. The HKLoc calculation below preserves exact local convolution, Peter–Weyl damping and finite combinatorial arithmetic, but it leaves reflection positivity, repeated-leg cumulants, nonlinear stability, the actual RG trajectory and the physical scale map Open.

A heat-kernel/HKLoc bookkeeping attempt (incomplete). *Status: exact local heat-kernel identities and conditional arithmetic inside a fixed blocking skeleton; not a proof of reflection positivity, polymer entry, a colour gap or the Clay construction.* Standard heat-kernel conventions follow Refs. [282, 283, 284]; those references do not supply the ECT action or the missing global premises. Canonical $b = 2$ planar fusion gives the exact same-plane kernel K_{4s} . A two-leg fundamental non-singlet component has the formal Peter–Weyl envelope e^{-8s} , and the declared skeleton yields the arithmetic factor $16 \times 6 = 96$. These facts are retained below. They do not bound the initial cumulant amplitudes, higher repeated-leg singlets, nonlinear remainder transport or the complete shared-link four-dimensional measure. The anisotropy budgets are therefore conditional rehearsals, not the single remaining ECT input.

Derivation of the heat-kernel bookkeeping reduction. We record the finite calculation behind the preceding criterion, since it is part of the ECT-native polymer-entry problem rather than an external Clay–continuum construction. Throughout, $s > 0$ denotes the heat-kernel smoothing time of one blocking step, and

$$K_s(U) = \sum_R d_R e^{-sC_2(R)} \chi_R(U) \quad (46.12)$$

is the central heat kernel on $SU(3)$, with d_R the dimension, χ_R the character, and $C_2(R)$ the quadratic Casimir of the irreducible representation R ; the fundamental value is $C_F \equiv C_2(\mathbf{3}) = 4/3$ and the adjoint value $C_2(\mathbf{8}) = 3$.

(i) *Canonical planar fusion.* The kernels (46.12) form a convolution semigroup,

$$\int dV K_s(UV^{-1}) K_t(VW^{-1}) = K_{s+t}(UW^{-1}), \quad (46.13)$$

which follows termwise from the orthogonality relation $\int dV \chi_R(UV^{-1}) \chi_{R'}(VW^{-1}) = \delta_{RR'} \chi_R(UW^{-1})/d_R$. Iterating eq. (46.13), a 2×2 same-plane heat-kernel block fuses exactly to the coarse-plaquette kernel,

$$K_s^{*4} = K_{4s}. \quad (46.14)$$

Hence same-plane strips K_{2s} are intermediate objects in a canonical planar fusion tree; they are not independent polymer remainders. Distinct row/column fusion trees yield the same K_{4s} by associativity and commutativity of (46.13), so the choice of a canonical tree is a bookkeeping convention that avoids double-counting, not a physical anisotropy. With this convention the same-plane staple remainder vanishes identically, $R_{\text{staple}} = 0$. Each K_s is a strictly positive central class function and the local convolution is positivity preserving. This does not prove reflection positivity of the full shared-link blocked measure. Positive entries are not positive semidefiniteness: the symmetric kernel matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has positive entries

but eigenvalues -1 and 3 . The action-owned reflection form and its behaviour under the complete blocking map remain to be established.

(ii) *Cross-plane spectral contraction.* The genuine non-core source at this order is the cross-plane attachment sector. In a $b = 2, d = 4$ block the 2×2 coarse plaquette has four interior links, and at each such link there are four genuine cross-plane attachment pairs, so the combinatorial multiplicity is

$$N_{\text{cross}} = 4 \cdot 4 = 16 \quad (46.15)$$

(four interior links $\times$ four cross-plane attachment pairs per link, the latter being two directions transverse to the fixed coarse-plaquette plane times two orientations). The heat-kernel smoothing acts as the Peter-Weyl operator, diagonal on representation blocks,

$$T_t f_R = e^{-tC_2(R)} f_R. \quad (46.16)$$

Since the planar block advances $s \mapsto 4s$, the additional smoothing carried by each non-trivial leg is $\Delta s = 3s$. A leading cross cumulant has two fundamental non-singlet legs; after singlet (and normalisation) subtraction its spectral envelope is therefore

$$\exp[-2(3s)C_F] = \exp[-2(3s)\frac{4}{3}] = e^{-8s}, \quad (46.17)$$

which is the origin of the e^{-8s} contraction. A one-leg remainder, if present, would contract only as e^{-4s} ; this is precisely why the anisotropic Y -sector must be budgeted separately. That no non-trivial leg can integrate to a constant (which would escape the contraction (46.17)) follows from Haar orthogonality: a single fundamental leg has vanishing Haar average, so a local connected cumulant cannot shed a leg into the singlet without an explicit subtraction.

(iii) *Local cumulant norm and the constant 96.* For definiteness, let $V(X)$ be the lattice-site vertex set incident on the connected link support X , with coordinates measured in lattice units, and define the support metric used in this subcalculation by

$$\text{diam}_\infty(X) := \max_{x,y \in V(X)} \|x - y\|_\infty. \quad (46.18)$$

Control the remainders in the corresponding local connected-cumulant norm

$$\|R\|_{\text{HKLoc},\infty} = \sup_{X \text{ connected}} e^{L \text{diam}_\infty(X)} \sum_{\lambda} |r_X(\lambda)| \left(\prod_{\ell} d_{\lambda_\ell}^p \right) \exp[\alpha \sum_{\ell} C_2(\lambda_\ell)], \quad (46.19)$$

with the soft choice $p = 1$ and moderate $\alpha, L \in [\frac{1}{4}, \frac{1}{2}]$. Here λ are the representation labels on the legs and $r_X(\lambda)$ are the cumulant coefficients.

For the site projection used here, define $\pi(x)_j = \lfloor x_j/2 \rfloor$ and $V(\pi X) = \{\pi(x) : x \in V(X)\}$. Then, coordinate by coordinate, $|\lfloor x_j/2 \rfloor - \lfloor y_j/2 \rfloor| \leq \lceil |x_j - y_j|/2 \rceil$. Taking the maximum over coordinates and vertices gives the rigorous contraction

$$\text{diam}_\infty(\pi X) \leq \left\lceil \frac{\text{diam}_\infty(X)}{2} \right\rceil \leq \frac{1}{2} \text{diam}_\infty(X) + \frac{1}{2}. \quad (46.20)$$

This statement is only for ℓ^∞ diameter on site-projected supports; it is not asserted for graph distance or ℓ^1 diameter. If an actual link/plaquette reconstruction thickens the projected vertex set by r_∞ in ℓ^∞ , the only immediate bound is instead $\text{diam}_\infty(X_{\text{rec}}) \leq \lceil \text{diam}_\infty(X)/2 \rceil + 2r_\infty$. That reconstruction radius, preservation of the admitted connected-support class and the induced amplitude map are missing microscopic inputs.

For the declared site projection, the diameter-weight ratio is bounded by $e^{L/2}$. Let $C_{\text{loc}}^{(\infty)}(\alpha, L)$ denote the remaining dimensionless action-, amplitude-, representation- and ℓ^∞ -norm margin, including this factor. This constant must not be reused for a graph/ ℓ^1 norm or a thickened support rule without a new proof. With the separately supplied projection multiplicity $\mu_{\text{proj}} = 1$, coarse-plaquette valence $\mu_{\text{loc}} \leq 2(d-1) = 6$ and the leading count $N_{\text{cross}} = 16$, a future action-owned norm estimate would obey

$$K_{\text{cross}} \leq N_{\text{cross}} \mu_{\text{loc}} C_{\text{loc}}^{(\infty)}(\alpha, L) \leq 96 C_{\text{loc}}^{(\infty)}(\alpha, L). \quad (46.21)$$

If it also bounded the normalised two-leg cross-sector map by $96C_{\text{loc}}^{(\infty)}e^{-8s}$, the conditional threshold would be

$$96C_{\text{loc}}^{(\infty)}e^{-8s} < 1, \quad s > \frac{\ln(96C_{\text{loc}}^{(\infty)})}{8}. \quad (46.22)$$

The printed values at $C_{\text{loc}}^{(\infty)} = 1.95$ or 4 are reproducible arithmetic, not uniform bounds for the actual blocked action. Neither the initial two-leg amplitude nor nonlinear transport of this metric-specific norm is supplied.

(iv) *Higher cumulants remain uncontrolled.* For a component with m independently surviving fundamental non-singlet legs, isolated Peter–Weyl smoothing gives the formal factor e^{-4ms} . This does not bound the full m -leg cumulant, because $SU(3)$ admits singlet contractions already at three legs:

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} \supset \mathbf{1}, \quad \int_{SU(3)} dU U_{i_1 j_1} U_{i_2 j_2} U_{i_3 j_3} = \frac{1}{6} \epsilon_{i_1 i_2 i_3} \epsilon_{j_1 j_2 j_3}. \quad (46.23)$$

The count of transverse spatial directions therefore cannot reduce every three-leg source to the controlled two-leg sector. Three- and higher-leg connected cumulants require independent amplitude, support and HKLoc bounds.

(v) *Conditional plane-diagonal Y bookkeeping.* Colour-singlet coefficients and a director with no colour index do not by themselves make a generic discretised anisotropic action a one-plaquette central class function. For this subcalculation impose the additional assumptions that n is constant and aligned with one lattice axis, and that the discretisation is plane-diagonal at the one-plaquette core. Only under those assumptions does each plane contribution reduce to a c -number weight times a central one-plaquette function. Here ζ_Y is only a dimensionless plane-resolved lattice matching parameter; its map to $\kappa_n^{\text{col}} = Y_0/Z_0$ is action-, discretisation- and scheme-dependent and remains Open. The core weights may then be represented by $s_t = s/(1 + \zeta_Y)$ on planes containing the selected axis and $s_\sigma = s$ on transverse planes. For a nonconstant/non-axis-aligned director or a discretisation with mixed plane contractions, cross-plane terms need not be central and belong to an uncontrolled remainder.

Within the stated plane-diagonal skeleton, canonical fusion acts plane by plane and pointwise positivity holds for $s_t, s_\sigma > 0$; this still does not establish reflection positivity of the shared-link measure. The skeleton parity check rules out a single integrated-link plaquette remainder only in this declared discretisation. A frozen exhaustive enumeration on the periodic 4^4 patch gives the skeleton-link distribution $\{0:1152, 1:0, 2:384\}$. This is deterministic synthetic verification of the stated finite skeleton only, not an action-owned four-dimensional blocking theorem. It does not remove general cross-plane or higher-support operators.

The two-leg envelope is bounded by $e^{-8s_{\min}}$ with $s_{\min} = s/[1 + \max(0, \zeta_Y)]$. Because the contraction premise is strict, $96C_{\text{loc}}^{(\infty)}e^{-8s_{\min}} < 1$, assume $96C_{\text{loc}}^{(\infty)} > 1$ and define $s_{\text{entry},\infty} = \ln(96C_{\text{loc}}^{(\infty)})/8 > 0$. The matching budget is then also strict:

$$\max(0, \zeta_Y(a_{\text{col}})) < \frac{s}{s_{\text{entry},\infty}} - 1, \quad s > s_{\text{entry},\infty}, \quad 1 + \zeta_Y(a_{\text{col}}) > 0, \quad (46.24)$$

Rounded values such as $+0.53$ and $+0.34$ at $s = 1$ are strict upper boundaries, not admitted endpoints and not physical bounds on Y_0/Z_0 .

If a compatible blocking sequence and a nonlinear stability theorem are separately assumed, the same strict algebra at depth n reads

$$\exists n \geq 0 : \quad \max(0, \zeta_Y^{(n)}) < \frac{4^n s}{s_{\text{entry},\infty}} - 1, \quad 4^n s > s_{\text{entry},\infty}, \quad 1 + \zeta_Y^{(n)} > 0. \quad (46.25)$$

The transported norm, cross-plane remainder, $C_{\text{loc}}^{(\infty)}$ uniformity and actual RG trajectory remain uncontrolled. No conclusion is transferred to another support metric.

Consequence and status. What survives is the exact local convolution algebra, isolated Peter–Weyl damping, the stated $16 \times 6 = 96$ combinatorial arithmetic and the single-plaquette parity check inside

the declared skeleton. Pointwise positive kernels do not establish reflection positivity of the full blocked measure; initial activity smallness, all higher cumulants, nonlinear stability, the actual ECT RG trajectory and the physical scale map remain uncontrolled. Accordingly HKLoc is retained only as an incomplete attempted bookkeeping criterion. The physical/global ECT colour gap remains Open, and no Clay upgrade follows.

47 What Part IV establishes and what remains open

Status: P7 is an adopted Level-B structural input; its frame reduction and scoped group/source-model algebra are Level-A conditional. The leading physical truncation is Level-B/Open and physical colour remains Open.

Table 105: Scoped Part-IV results and their missing physical owners.

Item	What is actually demonstrated	Level / missing owner
Fundamental-vector stabiliser	A nonzero $\mathbf{3}$ vector has stabiliser $SU(2)$; reducible singlet VEVs are not excluded	Level A group theory / ECT representation Open
Direct embedding	$\mathfrak{su}(3) \not\hookrightarrow \mathfrak{so}(4)$ by dimension; only this route is excluded	Level A / emergent, reducible and kernel routes preserved
P7 structural postulate	Adopted $(\mathcal{C}, h, \Omega)$ has $SU(3)$ frame group	Level B input / Level A conditional; physical colour, connection dynamics and action Open
Leading EFT truncation	Eq. (41.2) is convention-fixed but not complete	Level B/Open
Free aligned-background health	$Z_0 > 0$, $Z_0 - Y_0/2 > 0$ and $c_T^2 = Z_0/(Z_0 - Y_0/2)$ in the stated free branch	Level A conditional algebra
Gauge-invariant mass operator	A Proca term is forbidden in the supplied exact gauge theory; P7 fields contain no charged Higgs	Scoped conditional result / other fields and dynamics Open
Oscillator tower	No physical spectrum is identified without a gauge-invariant operator/constraint/dynamics map	Non-identification only; no universal no-go
Coefficient ownership	A shared $n_\mu n_\nu$ spurion transfers no relation or bound to Y_0	Open matching
One-coupling transmutation	$b_0 = 11$ and $\Lambda_{\text{col}}^{\text{src}}$ follow only on $\mathcal{B}_{\text{src}}$	Level A source-model algebra / no ECT prediction
General anisotropic RG	Requires coupled beta functions and operator mixing	Open

Item	What is actually demonstrated	Level / missing owner
Anomaly arithmetic	Standard conditions test a supplied chiral spectrum	External conditional algebra / ECT spectrum Open
Baryon-like singlet	$\varepsilon_{abc}q^a q^b q^c$ is an $SU(3)$ singlet	Level A conditional / does not define $U(1)_B$
Strong CP	$\mathcal{L}_\theta = g_s^2 \theta_{\text{gauge}} F\tilde{F}/(32\pi^2)$ and $\bar{\theta} \equiv \theta_{\text{gauge}} + \arg \det M_q \pmod{2\pi}$	External convention / MISSING MASS-PHASE OWNER
Mass-gap endpoint	Theorem 46.1 needs a unique vacuum, positive adjointed form, transfer-cyclic/dense class, uniform decay and two-sided scale map	External conditional theorem / physical ECT Open
HKLoc	Local convolution, Peter–Weyl factors and $16 \times 6 = 96$ arithmetic survive	Conditional book-keeping / cross-plane and global owners Open

Non-replacement statement. Part IV does not close physical colour, weak chirality, generations, matter, baryon number, strong CP, confinement or a global gap. It supplies one candidate organisation and preserves alternative completion routes.

Remains Open. The items explicitly not closed by Part IV are collected in Table 106. For each we record the nature of the residual obstruction and the section that discusses it.

Table 106: Problems remaining open after Part IV.

ID	Item	Why still open	Locus
O1	Identification of the matching scale M_*	Multiple candidate scales can be supplied; no ECT-specific selector established	§41
O2	Derivation of $g_s(M_*)$	Colour coupling is matching data; microphysical origin not supplied by P7	§41
O3	Confinement (Wilson-loop area law)	Nonperturbative dynamics inaccessible from reverse analysis; inherited from standard YM	§41
O4	Full glueball spectrum	Bound-state dynamics; requires non-perturbative strong-sector computation not supplied by the admissible EFT class	§41
O5	Numerical value of $\Lambda_{\text{col}}^{\text{src}}$	M_* , g_s and Z_0 are source-model matching data; the general anisotropic RG is Open	§41

ID	Item	Why still open	Locus
O6	Strong CP resolution (R1/R2/R3)	Experiment constrains $\bar{\theta}$; the mass matrix, its determinant phase and any ECT relaxation mechanism are absent	§45
O7	Complete matter-content derivation	Anomaly cancellation narrows but does not uniquely fix the matter representations	§42
O8	Sterile-neutrino status in ECT	A singlet is admissible without disturbing anomaly balance, but no ECT postulate selects required/forbidden/permitted	§42
O9	Yukawa hierarchy as derivation	Structural routes in Part I remain Level B/C; not addressed by the completion layer	§9.6
O10	Three generations as a theorem	Spectral route of §9.6 remains the forward proposal; P7 does not convert it into a theorem	§9.7; App. AA
O11	CKM/PMNS mixing matrices	Flavour-closure problem; not addressed by the completion layer	§9.7
O12	Global quotient $(SU(3) \times SU(2) \times U(1))/\Gamma$	Local gauge structure does not determine the global quotient; identification of Γ requires bundle-level structure	§42
O13	Stiefel–Whitney and related topological obstructions on the colour bundle	Requires bundle-level topological analysis not performed here	§42
O14	't Hooft anomaly matching $UV \leftrightarrow IR$	UV medium microphysics not supplied; matching cannot be verified	§42
O15	Status of the global $U(1)$ carried by Ω	Whether gauged, broken, or quantum-mechanically protected is not fixed by P7	§42
O16	Yang–Mills existence, confinement and mass gap	HKLoc does not supply the full shared-link measure, repeated-leg bounds, reflection/transfer positivity, unique vacuum, transfer-cyclic/dense physical class, uniform decay, stable trajectory or two-sided scale map	§46
O17	Relation of P7 to chirality/flavour programmes	Whether the adopted P7 structural bundle coexists with or constrains the Open physical matter routes is not derived	§9.3–§9.7

Directional remarks (future work; not claimed here). The following remarks are conjectural pointers, collected in Table 107. None is a theorem; none is pursued quantitatively in this preprint.

Table 107: Directional remarks from Part IV.

ID	Direction	Content and status
D1	Weak-chirality closure test	The vector stabiliser and vector director operator select no chirality. Any P7-assisted route must exhibit an independent local weak bundle and a derived P_L matter vertex; without them no V–A claim is available.
D2	Colour rank and fermion-generation count	These are independent namespaces. Equality of the two observed integers has no operator, index or selection owner; no inference is drawn and the physical family count remains Open.
D3	Candidate Standard-Model compatibility class	Standard-Model representation data may be supplied as one completion, but P7 derives neither those data nor a prior or dynamical selection among alternatives. Physical selection remains Open.
D4	Swampland compatibility	Whether a completed ECT+P7 model satisfies any precise No-Global-Symmetries, Weak-Gravity, or Completeness conjecture is Open. Their premises and ECT operator/charge/spectrum maps have not been supplied, so no positive compatibility conclusion is drawn.

Three distinct questions in the strong sector. First, the ordinary $b_0 = 11$ running and formal transmutation scale are standard algebra only on $\mathcal{B}_{\text{src}}$. The general anisotropic multi-coefficient RG remains Open. Second, a physical global mass gap needs the full transfer-cyclic theorem premises and a two-sided scale map; HKLoc does not supply them. Third, confinement and the glueball/hadron spectrum need their own gauge-invariant operator and dynamics maps. The Clay continuum construction remains a distinct external problem.

Final note on claim levels. The stabiliser and direct-embedding statements are Level A mathematics inside their declared group assumptions. The one-coupling formula is Level A only inside the supplied source model. P7 is adopted Level-B structural data, while the physical-colour map is C/Open, and any claim that it describes physical colour or the fundamental ontology of nature is Open. No coverage row upgrades these distinctions.

48 Synthesis of Part IV

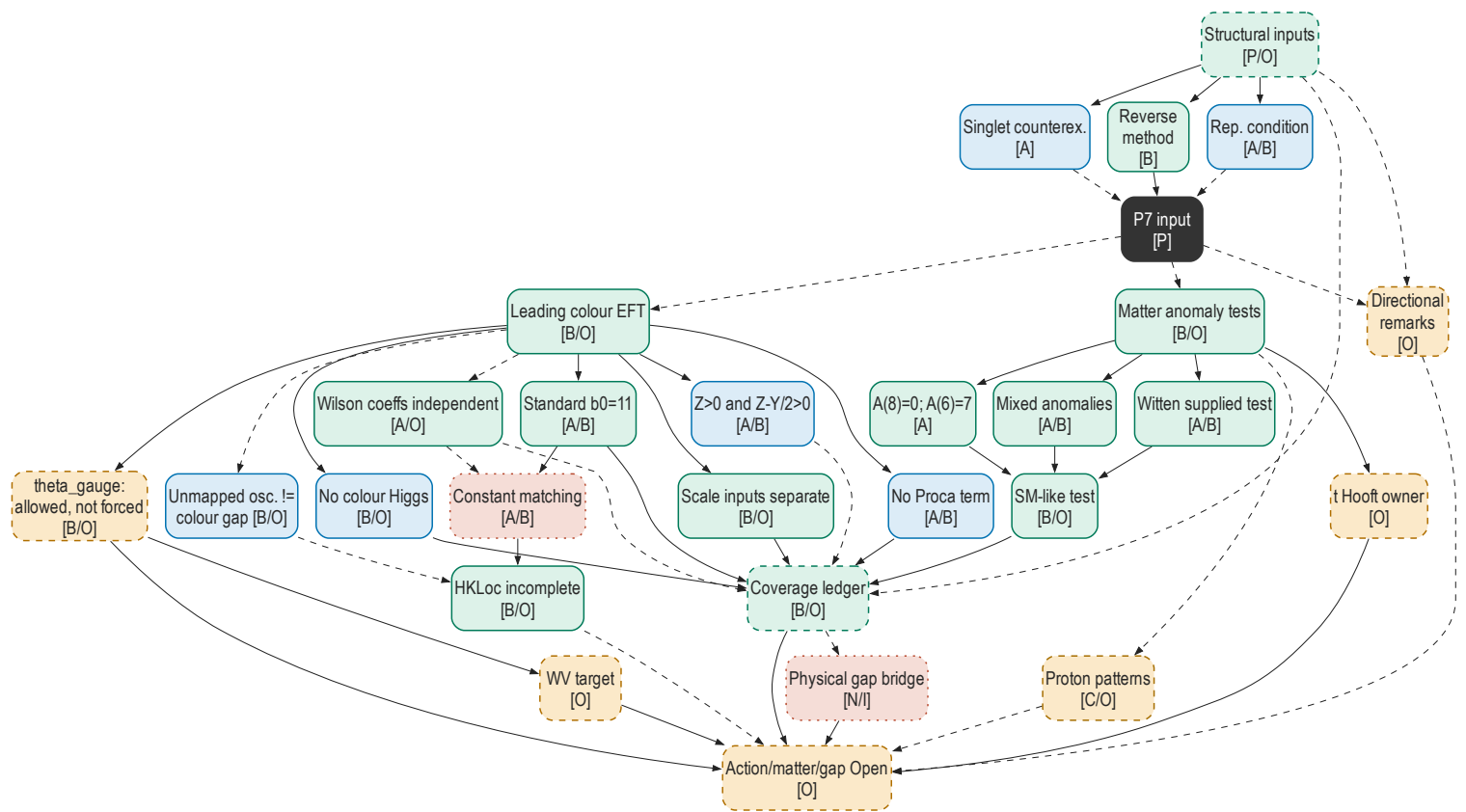
Status: Overview. No new claims are made in this section; it collects and depicts the structure of Part IV schematically.

Overall picture. Part IV is a reverse-analysis programme above Parts I–III. It formalises the supplied P7 rank-three internal module, proves scoped stabiliser and direct-embedding facts, displays one leading colour-EFT truncation with its free quadratic health cone, and restates standard anomaly constraints. It does not derive the P7 bundle, connection, action, Wilson coefficients, matter spectrum or nonperturbative vacuum. The coverage and status tables therefore organise conditional requirements and Open owners rather than declare a physical colour completion. Proton-stability patterns and the strong-CP routes remain conditional; HKLoc preserves local mathematics but does not establish a mass gap.

Status and dependency logic of Part IV. The logical structure of Part IV is shown in Figure 34.

Selected status and dependency map. Figure 34 displays a selected Part IV topology on one connected A4 reader map. It retains the forward inputs from Parts I–III, reverse-analysis obstructions and P7, the conditional colour completion, anomaly and proton-stability layers, and all still-Open strong sector targets without splitting them into locally readable but globally disconnected fragments.

Selected Part IV status and dependency map



P postulate/input | A exact in stated model | B conditional | C fit/toy | O Open | MISSING/N-I owner absent. Solid: exact declared step; dashed: conditional/Open dependency; dotted two-way: unproved compatibility.

Reading discipline. Solid arrows in Figure 34 denote declared logical dependence or organisation and never upgrade node status. Dashed arrows expose conditional or Open connections to the forward programme. The same colour/luminance and direct-label policy as in the other maps preserves the hierarchy under grayscale conversion.

Final statement for Part IV. Part IV preserves the adopted P7 $SU(3)$ structural postulate and a separate physical-colour completion programme; it does not claim that the postulate already supplies physical QCD. The scoped stabiliser and direct-embedding facts, the convention-fixed leading truncation, source-model RG algebra, standard anomaly identities and local heat-kernel calculations survive. The reducible, emergent/nonlinear and defect-kernel routes also survive. Physical colour, coefficient matching, the multi-coefficient RG, $U(1)_B$, the quark-mass phase, confinement, the global gap and the hadron spectrum remain Open.

Part V

Summary, Comparisons, and Outlook

49 Discussion

This section provides a consolidated discussion of the ECT framework: its conceptual logic, status-and-dependency architecture, current status-separated outputs, comparison with neighbouring programmes, observational reach, limitations, and future directions.

49.1 Conceptual interpretation and scope of the framework

ECT investigates whether Lorentzian spacetime, quantum dynamics, and gravitational interactions can all be obtained from one scalar medium in four-dimensional Euclidean space. P4 adopts the physically relevant nonzero-gradient branch and supplies its chart; a stationary action, physical formation/selection history and the cross-sector common owner remain Open. This is an organising hypothesis investigated through statements of varying status, as classified in Table 70; the conditional and Open items do not provide evidence that a common physical owner exists.

This perspective inverts the conventional logic of quantum gravity. Rather than attempting to quantise an existing spacetime geometry, ECT starts from a fundamentally Euclidean arena where no distinction between space and time exists, and recovers the Lorentzian structure as a candidate effective property of the supplied ordered-gradient chart. The displayed scalar action gives a Lorentzian principal tensor; a physical metric, clocks/rods and universal sector cone do not follow directly. The thermodynamic arrow of time is addressed at the level of the ordered-branch open-system description with clearly stated closure assumptions (Section 3.9). The same-medium hypothesis provides a common research framework for G_N , the EFT action scale later matched to $\hbar$, and the scalar speed map. Their microscopic ownership and mutual relations are not derived.

This framework should not be interpreted as claiming that time is “unreal”. It proposes that the distinction between temporal and spatial directions may be a collective property of a future physical ordered state. The present result establishes only the conditional scalar principal-symbol classification; a physical clock and universal metric remain to be derived.

The comparison with closely related Euclidean-to-Lorentzian constructions [3, 4] is useful in a second respect as well. It shows that the basic idea of obtaining Lorentzian characteristic operators from Euclidean field theory is not unique to ECT, but it also makes clearer which parts of ECT remain distinctive claims rather than generic consequences of that shared idea: the Open programme for an action-owned spontaneous origin of the preferred direction, the proposed gravitational completion, and the extension to quantum and phenomenological sectors.

49.2 Status-and-dependency architecture: from the Φ -medium to Open observable completions

The status architecture of ECT can be summarised as a directed graph from the foundational Φ -medium to derived supplied-model results, conditional reconstructions and Open completion targets (Figure 35). The graph is status-separated; it is not a claim that every major physical sector has emerged from P1–P6.

Step 1: the Φ -medium and the foundational postulates. The starting point is a scalar condensate field Φ living on a four-dimensional Euclidean manifold (M^4, δ_{AB}) with an $O(4)$ -invariant action $S_E[\Phi]$. No external time coordinate is postulated. P5 separately adopts a limited operational relational-response principle based on invariant comparisons with material and boundary probes; a complete relational quotient, physical observable algebra, clock/rod construction and relational dynamics have not been derived.

Step 2: adopted ordered-gradient branch and chart. Postulate P4 declares the physically relevant sector to have $\langle \partial_A \Phi \rangle \neq 0$ and supplies the representative $n_A = \delta_{Aw}$. The nonzero vector has $O(3)$ stabiliser inside $O(4)$. The complete stationary state/action, stability, spontaneous formation and dynamical selection from P1–P3, together with the descent of physical macroscopic/quantum sectors, remain Open.

Step 3: conditional scalar Lorentzian classification. Perturbations around the ordered background experience the effective ordered-branch kinetic tensor $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$. For a homogeneous ordered background $n^A = \delta_w^A$, the scalar principal tensor is Lorentzian exactly when $\beta(\beta - \alpha) < 0$. On the separately declared healthy $\beta > 0$ branch this is $\alpha > \beta > 0$, with $K^{AB} = \text{diag}(\beta, \beta, \beta, \beta - \alpha)$ and $\hat{c}_*^2 = \beta/(\alpha - \beta)$. This yields a scalar common-length tensor proportional to $\text{diag}(-1, 1, 1, 1)$ at $\hat{c}_* = 1$; it is not yet a physical light cone.

Step 4: macroscopic tensor branch (Part II). The macroscopic sector develops as follows:

- *General relativity.* A separately supplied healthy Fierz–Pauli seed could self-couple to an Einstein–Hilbert closure under the standard assumptions. At fixed α, β and tangent-director linear order about homogeneous $n = w$, the displayed map has no linear TT seed or static-density vertex, so the supplied tensor action is a completion target rather than a result of the scalar action; quadratic/nonlinear/composite routes remain Open. The scalar–tensor ansatz yields the corrected scalar metric and field equations. A preferred-direction curvature operator resembling an Einstein–aether term is a proposed separate extension [23]; it is not derived from the displayed scalar-only action and remains Open pending a covariant n^A sector and projection audit.
- *Fundamental constants.* $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ is a conditional physical-tensor matching, while $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$ requires a clock map and $c_{\text{char}} = c$ is the adopted scalar-clock calibration. None is derived from the condensate amplitude alone. Independently, $\phi_0 = \zeta_\phi \bar{M}_{\text{Pl}}$ with $\zeta_\phi > 0$, $\zeta_\phi = O(1)$ is the adopted Level-B scale ansatz; its microscopic explanation remains Open.
- *Scale inventory.* The energy/matching scales $\phi_0 \sim \bar{M}_{\text{Pl}}$ and $v_2 \approx 246 \text{ GeV}$ have distinct declared status. The kpc quantity L_{gal} is a matching length used in Level-C HRC diagnostics, not a third field scale; its ECT origin is Open (Section 5.1).
- *Cosmology.* The ordered branch supplies conditional background and perturbation constructions, not a completed numerical cosmology. The fixed orientation determinant does not derive vacuum-offset decoupling. A separately supplied trace-free completion is classically shift covariant through $V + C_I$, while a distinct ordinary two-slope completion gives a regular late-de-Sitter proof of class but is not additive-shift invariant. Their descent from the Φ medium, the physical matter/photon metric, global state and likelihood remain Open. The primordial spectrum (n_s, A_s, r) is likewise an open target.
- *Galactic dynamics.* HRC-0 and HRC-3 are the active augmented-model response candidates; the slope 4 is conditional algebra after a separate static gravity bridge. Existing SPARC/RAR/BTFR outputs are HRC-only algebraic diagnostics. P1–P6 ownership, source/metric map, and the matched $a_{M0} = cH_0/(2\pi)$ coefficient remain Open as stated.
- *Strong-field sector.* A critical-shell ansatz supplies only the conditional Tolman crossover $\rho_* = \hbar c/(2\pi k_B T_*)$ and a candidate route to black-hole thermodynamics (Level C/Open). No physical T_* or universal shell thickness is derived. Only the generic reduced-state identity is Level A mathematics; global interacting unitarity, purification and the Page curve remain Open.
- *Preferred-direction programme.* No coefficient or species law is derived, and the allowed vector operator does not generate a spin or body-force observable by itself. The homogeneous timelike vector term has no spin splitting, and the fixed spurion supplies no source-generated Eötvös force. The former numerical windows are withdrawn as outputs of this operator; external spin and WEP sensitivities constrain only a separately declared completion.

Step 5: quantum coherent branch (Part III). In parallel, the coherent sector develops:

- *Action-scale candidates and operational unit.* The supplied fixed-core and balanced loop closures define distinct coefficients; later conditional quantum formulae use an operational S_0 whose owner remains Open. Winding gives integer labels. The phase-only model has no nonzero global minimum. A supplied positive two-term 1D model has the exact conditional minimum $\mathcal{J}_{n,\min} = 2\pi|n|t_{0,\text{bal}}^{\text{eff}}$, but its common 4D owner, carrier/ensemble stability, cross-sector universality and any matching to $\hbar$ remain Open (Level B/Open for ECT ownership).
- *Schrödinger equation.* A supplied linear coherent sector admits a conditional envelope reduction and the usual uncertainty theorem for supplied operators. The physical operator/state map, CCR normalisation and universal S_0 are Open (Level B/Open).
- *Vacuum response.* Supplied scalar/QFT calculations compare a Dirichlet Casimir BVP, the standard Unruh/KMS target and time-dependent mode mixing. A common physical ECT response bundle, state and coupling are Open; no condensate-sector Casimir ratio is predicted.
- *Decoherence and arrow of time.* The influence-functional architecture yields environmental open-system closures and provides notation for candidate record channels. The Diósi–Penrose functional and its one declared hard-sphere orientation point are external comparisons; an ECT gravitational kernel remains Open. An entropy-growth theorem and a Crooks relation remain Open ECT constructions: the former requires a named channel/state/entropy functional and contractivity conditions, while the latter requires forward/reverse protocols and local detailed balance.
- *Born rule.* A complete applicable corrected OS-II package may reconstruct an inner product in a declared free model. Independently, ordinary projector Gleason represents an admitted normalised, noncontextual, countably additive projector measure only for a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. Neither result creates probabilities, selects an outcome, or derives the physical state/preparation, event, probability origin or update from P1–P6 (Level A mathematics under the stated theorem hypotheses; ECT ownership Open).
- *Exchange topology and spinors.* The $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ result from $d_{\text{spatial}} = 3$ is Level A conditional topology for a supplied identical-point configuration space. The physical particle map, spin-statistics hypotheses and Dirac matter action remain Open.
- *Entanglement.* Non-factorisable correlations follow only after a nonfactorising state and channel are supplied; a common ordered medium is insufficient (conditional mathematics / Open ECT).
- *Area-law entropy programme.* A conditional finite-cell counting lemma is exact after a coarse-graining, finite local capacity, physical split and area-law state class are supplied. ECT ownership of those inputs, a physical black-hole bound and holographic reconstruction remain Open (Theorem 36.1).
- *Charge quantisation.* Standard representation theory gives $q \in \mathbb{Z} \cdot e_0$ after a compact local $U(1)$ bundle, charge unit and matter representation are supplied (Level A conditional mathematics). ECT ownership of those premises, the unbroken photon and the observed Standard-Model charge spectrum remains Open.
- *Post-transition cosmological branch classification.* ECT does not yet derive the full nonequilibrium transition or the selection of the expanding branch from first principles, and a named supplied completion exhibits a regular expanding, early-decelerating, late-accelerating history. P1–P6 do not select that completion or history; thermal, recombination, perturbation and CMB closures remain Open (§14.2).
- *PES-R operational programme.* PES-R is a bounded Level-B organising taxonomy for dynamical admissibility, finite-time retention, and pairwise records. Visibility and qubit/CHSH numbers are declared source-model or toy proxies; they do not derive Born weights, measurement outcomes, spectra, or a universal quantum–classical boundary. ECT-specific kernels and record operators remain Open.

Step 6: gauge and matter sectors. BR1 supplies a compact global phase. A local $U(1)$ bundle, unbroken photon, charge map and matter representations do not follow from that fact; the non-Abelian

$SU(2)$ completion, $W^\pm/Z$ identification and radial-mode Higgs map are likewise conditional model-building routes (Level B/Open). The colour group $SU(3)$ and the fermion-generation structure remain Open. The adopted P7 $SU(3)$ -structured bundle is analysed in §40; its identification with physical colour and any baryon-number selection rule remain Open.

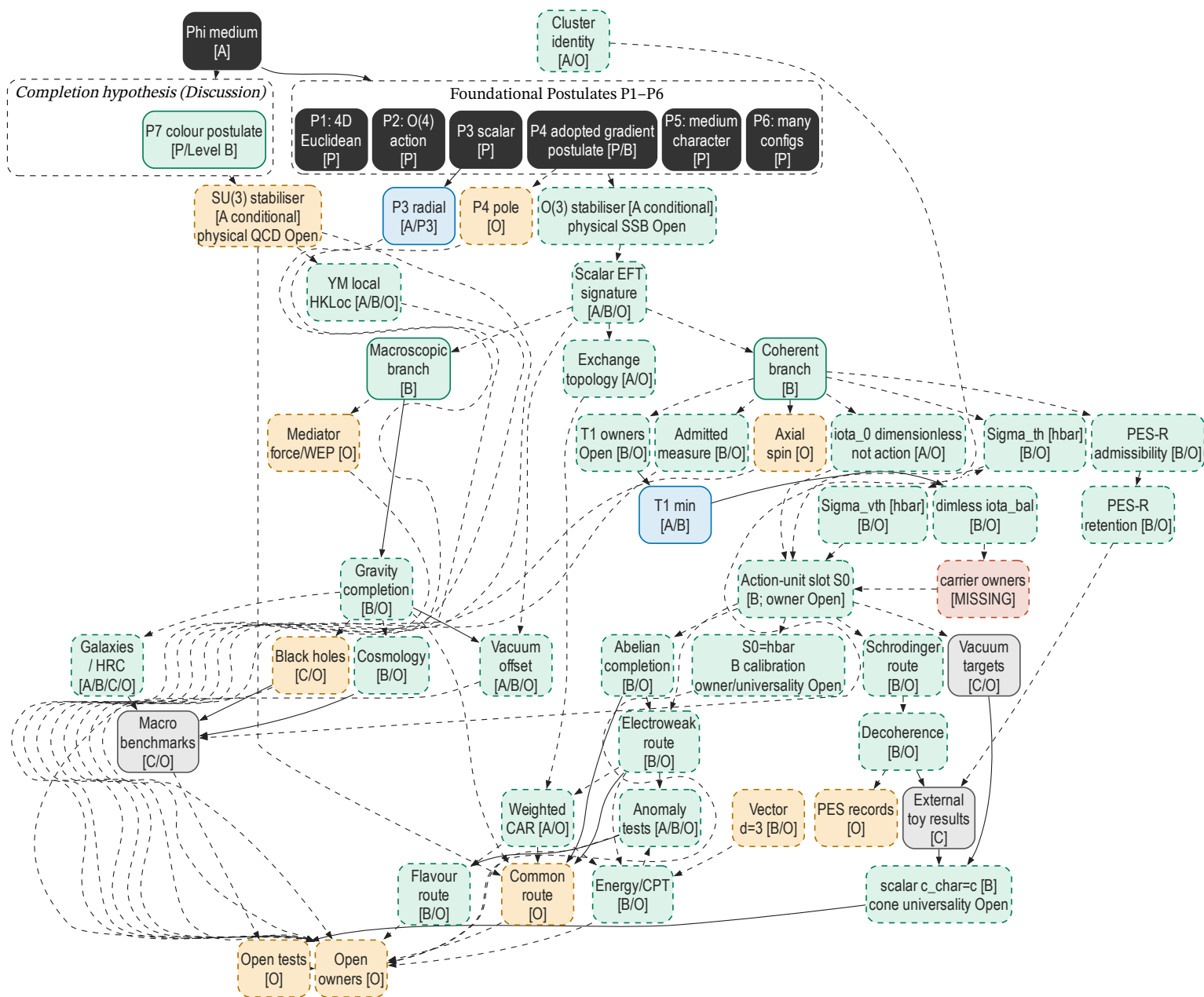
Particle spectrum. The condensate architecture organises the particle sector into different status layers. At the structural or partially established level this includes: a physical graviton only as an Open tensor-completion target (the fixed δn_A map has no linear TT/static-density seed only at fixed α, β and tangent-director order about homogeneous $n = w$; quadratic/nonlinear alternatives remain parametric/Open); the photon, $W^\pm/Z$, Higgs and fermions only as targets of supplied gauge/matter completions; their physical bundle, representation, mass and observable maps remain Open. The strict integrability-reduced orientation map instead retains one longitudinal scalar coordinate χ . A one-mode physical EFT is conditional; its existence, mass, abundance and possible cosmological or dark-sector role remain Open. For the primary $\pi_3(S^3)$ sector, topology supplies integer labels only and the two-derivative energy $E_2(R) \propto R$ collapses. Any finite particle and mass spectrum requires a separately derived stabilising action; a possible secondary-sector owner remains Open.

Step 7: results, benchmarks, tests, and Open gates. The map separates strict results from matched benchmarks and explicitly declared closures. In particular, a_M and the gravity bridge remain matched/Open; the deep-HRC slope 4 is Level A only inside the supplied functional, conditional after that bridge, and its data comparison is Level C. The no-halo statement is only a property of the stated diagnostic fit, and environment/orientation signals remain Open programmes. None predicts the absence of particle dark matter or of a laboratory signal. These are summarised in Figure 35 and Tables 69 and 70.

Complete programme map. Figure 35 is a selected ECT programme graph. It places postulate/structural inputs, conditional closures, Level-C benchmarks and Open owner maps on one topology-preserving A4 reader map. The map includes the corrected coherent-branch, T1, OS-II, admitted-measure and PES-R routes. It does not contain a direct Schrödinger-to-Born edge or an automatic T1-to- $\hbar$ edge.

Selected ECT status and dependency map

Status focus: P4 and P7 are adopted Level-B inputs; their stabiliser algebra is conditional; physical QCD, S0 ownership and cross-sector cone universality remain Open.



P postulate/input | A exact in stated model | B conditional/calibration | C fit/toy | O Open | MISSING owner absent. Solid: exact declared step; dashed/dotted: conditional or Open dependency. Literal node status controls.

Reading discipline. Solid arrows in Figure 35 organise declared dependencies; dashed arrows identify assumptions, matching steps or incomplete routes. Galactic nodes marked “matched” or “conditional” are not particle-ontology conclusions. Likewise, the generic fifth-force node refers only to a separately supplied dynamical mediator/profile, source and body charges, propagator, range and screening; the fixed homogeneous vector spurion is not such a force. The coherent-quantum gate still requires the applicable OS-II/Hilbert structure and an independently admitted measure; physical preparation, probability, outcome and update remain Open. The A4 reader map, its frozen source, deterministic exact key and the machine topology are jointly authoritative; the former fragmented reader pages and printed edge ledgers are not part of the current reader layer.

49.3 What ECT already establishes: emergence chains and current status

The programme claim of ECT is that several major physical frameworks can be organised within one ordered-condensate architecture as exact mathematical sectors under stated assumptions, conditional limits, or Open completion routes. These categories are kept distinct throughout; no single statement here claims their completed physical recovery.

General relativity. The supplied ordered scalar EFT conditionally yields a Lorentzian scalar principal/propagation sector. At fixed α, β and to linear order in a tangent director perturbation, the fixed director image generates neither a linear TT seed nor a linear static-density vertex. This scoped component theorem does not exclude nonlinear, composite, additional-field, or dynamical-parameter vertices; the general physical tensor/source/metric map remains Open. A Fierz–Pauli structure and Einstein closure are supplied completion targets. In a separately declared pure-metric completion the Einstein–Hilbert form is the unique two-derivative closure in four dimensions under its standard assumptions. *Status: Level A mathematics inside the supplied scalar and pure-metric assumptions; the negative fixed-map result is Level A only in the scoped linear fixed-parameter limit; physical metric/tensor/source completion Open.*

Quantum coherent dynamics. The corrected quantum programme has a strict separation of owners. Sector operators, domains, representations, and boundary conditions supply spectra and coherent dynamics; a complete applicable corrected OS-II package supplies the Hilbert-space reconstruction only in the declared subsectors where every hypothesis has been checked, with reflection positivity only one member of that package. Specified influence functionals supply reduced-state response and pairwise records. Independently, ordinary projector Gleason represents a supplied probability measure only after its dimension, projector-algebra, normalisation, noncontextuality and countable-additivity hypotheses are admitted. PES-R then classifies already admissible modes by finite-time retention and declared record comparisons. It derives neither $S_0 = \hbar$, Born weights, unique outcomes, universal compact-phase quantisation, nor a Crooks relation. The interacting Hilbert-space completion, ECT record operators and state, detector update rule, physical tensor quantisation, and forward/reverse path measures remain Open. This is a smaller but logically stronger programme than the former one-functional selection picture.

Decoherence and the macroscopic arrow of time. ECT contains a causal response backbone and can host specified reduced-state channels. Environmental estimates reproduce declared source models, while the gravitational values are external Diósi–Penrose benchmarks. A Crooks or entropy-production theorem requires explicit reduced dynamics, forward/reverse protocols, local detailed balance, and initial ensembles; these ECT-specific ingredients remain Open.

PES-R and quantum reconstruction. PES-R separates sector dynamics, finite-time retention, and pairwise record robustness. Spectra, least action, Born weights, and measurement outcomes remain owned by their separate derivations/Open programmes. Thus PES-R is a Level-B/Open organising taxonomy, not a universal selection mechanism.

Galactic weak-field phenomenology. HRC-0/HRC-3 are the active augmented-model response candidates. The BTFR/RAR and rotation-curve benchmarks have been recomputed as HRC-only Level-C algebraic diagnostics, not full field predictions. The live-action gravity map, environment law and particle content remain Open. *Status: augmented-model A/B; frozen fits C; live ECT Open.*

Gauge and matter sectors. ECT provides a condensate framework in which conditional gauge and matter completions can be formulated. Physical local bundles, unbroken gauge fields, representations, anomaly-complete chiral matter, and the coupling hierarchy are not derived from P1–P6. *Status: conditional structural programme with these physical owners Open.*

Interpretive conclusion. The correct claim is therefore not that every established theory has already been rederived as a completed Level A theorem. Rather, ECT organises general relativity, coherent quantum dynamics, the Hilbert-space bridge, decoherence, vacuum-response effects, entanglement, black-hole thermodynamics, the dark-sector ontology, and the analogue-laboratory programme inside one common ordered-medium picture while making explicit which parts are structural, which are structurally singled out but not fully closed, and which remain open.

49.4 Architectural comparison with neighbouring programmes

The purpose of this subsection is not to argue that ECT is “better” than all neighbouring frameworks. It is to locate ECT within the broader landscape of approaches to emergent spacetime and quantum gravity, highlighting where its architectural choices overlap with, differ from, and remain weaker or stronger than those of its neighbours.

Scope and methodology. The comparison is organised along eight structural axes: origin of time/Lorentzian structure, origin of gauge structure, origin/status of quantum axioms, gravity closure, dark-sector treatment, singularity logic, information/decoherence architecture, and falsifier structure. Table 108 summarises the comparison; an expanded version with additional axes is given in Appendix BC.

Comparison by test structure. A completed common-origin ECT model could have correlated cross-sector tests if one owner chain fixed several observables. No such global chain is derived here. At present, failure of one supplied cone, action-scale or ultraviolet completion rejects that completion, not P1–P6 as a whole. Standard particle, gravitational and cosmological theories likewise have tests at the level of their stated models and parameter sectors; no literature-wide superiority or uniqueness claim is made.

Comparison by number of independent primitives. Reducing the number of independent foundational primitives is an ECT design aim, not an established comparison metric. General relativity supplies a metric/diffeomorphism framework and the Standard Model supplies quantum gauge and matter sectors; neither theory by itself specifies a cosmological dark sector. Standard Λ CDM applications add cold dark matter and Λ as separately parameterised components. ECT asks whether some of these structures can descend from one ordered medium, but it has not derived the physical metric/tensor sector, quantum axioms, Standard-Model gauge and matter content, or a dark-sector completion. A quantitative like-for-like parameter census remains Open (Section 49.8).

Table 108: Architectural comparison of ECT with neighbouring programmes. Entries are necessarily compressed; for expanded discussion see Appendix BC.

Axis	GR + SM	Einstein-aether	Hořava–Lifshitz	Volovik analogue	ECT
Time / Lorentz	Fundamental	Fundamental; preferred frame added	Fundamental but anisotropic	Emergent from condensate	Conditional scalar Lorentzian principal tensor after supplied P4 and $\beta(\beta - \alpha) < 0$ (or $\alpha > \beta > 0$ on the retained $\beta > 0$ component); physical time, clock, and universal sector cones Open
Gauge structure	Postulated	Postulated	Postulated	Analogy only	Ordered-branch structural route to $U(1)$ and $SU(2)$; physical SM identification incomplete; $SU(3)$ open
Quantum axioms	Primitive	Primitive	Primitive	Primitive	Structural route via coherent branch; completion open
Gravity closure	Einstein eqs. exact	Einstein + aether	Modified dispersion	Acoustic metric only	Scalar characteristic geometry; physical metric/tensor/source Open
Dark sector	Extra particles + Λ	Extra particles + Λ	Extra particles + Λ	Not addressed	No-halo Level-C HRC data diagnostic; ECT particle content and matching Open
Singularities	Physical or censored	Physical or censored	Possibly resolved by UV	Not addressed	Candidate branch-boundary or condensate-breakdown interpretation; microscopic completion and replacement dynamics Open

Axis	GR + SM	Einstein-aether	Hořava–Lifshitz	Volovik analogue	ECT
BH information	Paradox open	Paradox open	Modified; details model-dependent	Not addressed	Reduced-state information reading; fundamental loss not required within the present programme, but completed evaporation closure remains open
Falsifier style	Sectorwise	Extra-mode / LIV	LIV / dispersion	Not systematic	Named-completion sectorwise; cross-sector only after a common owner

What ECT still does not derive. The comparison above should not be read as implying that ECT has already completed all structural derivations that are postulated in the standard framework. The main unresolved fronts are summarised more systematically in Section 49.10.

Interpretive note. The comparison above is architectural, not polemical. Each listed programme has independent strengths that ECT does not replicate. The point is not that ECT solves all problems, but that it pursues a distinctive architectural strategy—single ordered medium, emergent spacetime and quantum structure, cross-sector falsifiers—that is worth comparing on its own terms.

The remarks below complement the architectural comparison of this section and the extended discussion in Appendix BC by relating ECT to several further emergent-gravity programmes that emphasise different aspects of the same underlying question.

Sakharov’s induced gravity. Sakharov’s induced-gravity programme [5] treats the Einstein–Hilbert action as effectively induced by quantum fluctuations of matter fields propagating on a pre-existing Lorentzian metric. ECT shares the structural conclusion that the gravitational action need not be fundamental, but addresses a complementary level in the theoretical structure: in a supplied branch ECT constructs a scalar characteristic-tensor candidate from the condensate order parameter n_A on a Euclidean background; whether it is the physical metric remains Open. Sakharov’s mechanism addresses why $\int \sqrt{-g}R$ appears given a metric, whereas ECT poses a programme for deriving a physical metric from the medium.

Verlinde’s entropic gravity. Verlinde’s programme [285] interprets gravity as a thermodynamic, entropic effect arising from information associated with holographic screens. ECT shares the broad motivation that gravity need not be a fundamental interaction, but addresses a different aspect of the emergence problem: it provides a concrete medium-action candidate route through variation of a microscopic condensate action rather than an information-theoretic re-derivation; the physical metric/source/gravity map remains Open. An entropic reading, if valid, would be consistent with a thermodynamic shadow of the same candidate condensate dynamics.

Kleinert’s world-crystal picture. Kleinert [286] models curvature in terms of defect densities (disclinations and dislocations) in an effective elastic medium, with the gravitational field arising from elasticity theory on a world-crystal. ECT shares the philosophical position that gravity admits an underlying

medium, but the medium in ECT is a continuous condensate field Φ on a Euclidean manifold rather than a crystalline lattice. ECT therefore does not invoke an explicit discretisation; the role of geometric response is encoded in the gradients of the orientation field $\nabla_\mu n_\nu$, with no defect-density interpretation required at the level of the ordered-branch EFT.

49.5 Comparison with alternative quantum interpretations

Cramer’s transactional interpretation (TI). Cramer (1986) [287] models every quantum interaction as an exchange of “offer” (retarded) and “confirmation” (advanced) waves between emitter and absorber. ECT shares with TI the structural idea that the relevant coherent-sector description is not exhausted by a purely one-sided Lorentzian-causal picture. In TI this appears as the advanced–retarded offer/confirmation language. In ECT it appears through the two-sided boundary sensitivity of the Euclidean coherent branch. The similarity is therefore real but not an identity: TI is an interpretation of the standard quantum formalism, whereas ECT locates the two-sided structure in a specific Euclidean condensate ontology. The present comparison is conceptual and structural; it is not the claim that TI is fully derived as a special case of ECT. ECT therefore does not adopt the literal offer/confirmation-wave ontology of TI. The overlap lies only in the structural rejection of a primitively one-sided temporal description.

Page–Wootters mechanism. Page & Wootters (1983) [17] proposed that effective time can arise from entanglement between a “clock” subsystem and the rest of the universe within a globally stationary state. ECT shares the motivation of avoiding a fundamental external clock. Page–Wootters reconstructs relational time inside a supplied quantum state. P4, by contrast, adopts the physically relevant ordered-gradient sector and its preferred Euclidean direction. A temporal parametrisation of a supplied hyperbolic scalar operator is conditional, while a physical clock/lapse, quantum state and universal metric remain Open. The comparison therefore identifies two possible reconstruction programmes; ECT does not yet provide a deeper physical derivation of a time-bearing branch.

49.6 ECT as a foundational framework for quantum mechanics

The detailed structural comparison between ECT and the standard quantum formalism—covering the hierarchy of effective equations, the boundary-value versus Cauchy problem, the ontological distinction between Φ and ψ , the origin of the statistical description, the concept mapping onto the standard quantum primitives, and the illustrative examples of interference, tunnelling, measurement, entanglement, and identical particles—is developed as the capstone of Part III in Section 38.3. The role of the present subsection is not to duplicate that material, but to close two pan-structural threads of the comparison that naturally belong to the summary chapter: the common origin of the coherent and geometric branches of ECT, and a compact overall summary of the ECT-versus-QM relation that also serves as a bridge to the broader architectural comparison with neighbouring programmes.

Unified origin of the coherent and geometric branches

ECT proposes that gravity and quantum coherence may be two sectors of one ordered condensate completion. The present scalar action does not yet derive either a physical TT gravity sector or the full interacting quantum theory, so this is an architectural hypothesis rather than a closed result.

Candidate scalar kinematics in the two programmes may reuse the supplied principal-tensor template K^{AB} , but no common physical action has been derived. In the strict P4 scalar map, χ is the longitudinal coordinate defined by $\delta Q_A = \partial_A \chi$ and has the same field units as Φ ; in BR1, θ is a dimensionless compact phase with its own independently supplied stiffness K_θ^{eff} . Their variables, coefficients, action magnitudes and states are not identified. The long-wavelength geometric and shorter-wavelength coherent readings therefore remain candidate programmes of one ordered-medium architecture. ECT asks whether a completed ordered-condensate action can generate both; the displayed ordered functional does not yet establish that descent, and the common microscopic owner and back-reaction vertex remain Open.

The scalar mode decomposition is exact inside the supplied ansatz, but its identification with physical gravity and quantum mechanics is not. In this sense ECT proposes, rather than has yet demonstrated, a common substrate.

The supplied scalar Einstein sector contains the exact divided coefficients $G_{\text{div}}(X)$ and $\Lambda_{\text{div}}(X)$, and admits separately proposed tensor structures involving the ordered direction n_μ . Neither divided coefficient is by itself a physical response map. Conversely, in a completion that identifies the relevant amplitude variables and supplies a common action-owned vertex, coherent fluctuations could affect the effective amplitude sector and thereby back-react on the geometric side. No common action-owned backreaction vertex between the coherent and physical geometric/tensor sectors has yet been derived. Their possible common medium is an architectural hypothesis; a completed quantitative backreaction action, state and observable map remain Open.

Closing this common-origin map is one of the most important Open programmes of ECT.

Summary of the comparison

Standard quantum mechanics is an effective statistical theory of subsystems, formulated in terms of a probabilistic amplitude ψ and a forward-time Cauchy problem. ECT proposes a deeper level in which the fundamental object is the physical condensate field $\Phi(X)$, governed by a global four-dimensional boundary-value problem. Selected quantum mathematical structures can be reproduced conditionally after a coherent sector, state, continuation, operator normalisation and reduction map are supplied. Quantum mechanics as a complete physical theory is not yet recovered; the owner-by-owner status is given in Section 38.3.

ECT therefore sets a programme to reproduce and ultimately explain the effective quantum formalism (Fig. 36). Probability, outcomes/collapse, interacting superposition, universal $\hbar$, physical gravity and their common microscopic owner are explicit Open targets, not current explanations.

Standard quantum mechanics	Euclidean Condensate Theory
Cauchy (initial-value) problem $\psi(x, t_0)$ given $\rightarrow$ evolve forward	Specified BVP / ensemble Existence / uniqueness / selection: Open
$i\hbar \partial_t \psi = H\psi$ Schrodinger / Dirac equation	$\delta^{\wedge}(AB) \partial_A \partial_B \Phi - V(\Phi) = 0$ Euclidean condensate equation
$ \psi ^2 = \text{Born rule}$ Fundamental axiom	Admitted-measure representation dim ≥ 3 : C1+C2 projectors; dim 2: C1+C2q effects state / event / probability / outcome / update: Open
Time: fundamental coordinate Lorentzian spacetime a priori	P4 supplies an ordered direction Scalar signature needs supplied coefficients; selection / physical cones: Open
Measurement / update rule Textbook formulations; interpretations differ	Measurement / outcome map: Open Decoherence alone does not select outcome

Figure 36: Status-sensitive comparison of textbook quantum mechanics and the current ECT programme. The standard-QM column is an external comparison. Within ECT, the supplied Euclidean scalar equation is model-internal, the ordered-direction statement is conditional, and the BVP selection, Born weights/reconstruction and outcome map remain Open. Literal status wording is authoritative. Neutral, blue, green and amber luminance-ordered fills provide redundant external, model-internal, conditional and Open cues; solid, dash-dotted and dashed borders preserve the same distinctions without colour.

In this sense, ECT does not replace standard quantum mechanics as a laboratory calculation tool; it proposes a deeper level from which standard quantum mechanics emerges as an effective, statistically formulated subsystem theory.

49.7 Phenomenological reach and observational diagnostics

This subsection collects the main observational implications and detection strategies of ECT in one place, covering both dark-sector diagnostics and the most informative empirical tests.

Tests of the HRC no-halo algebraic diagnostic. The Level-C HRC data diagnostic is tested with held-out kinematics and prespecified transition/environment regressions. Lensing becomes a test only after a covariant two-potential metric completion; the current cluster proxy does not supply it. The diagnostic contains no particle-halo term, but this does not identify a condensate mechanism, exclude a particle flux, or predict a laboratory null result. The exact local-response theorem preserves input ordering and the set-valued argmax of the supplied total projected baryon proxy, even when the argmax sets are empty. Whenever maxima are attained, it forbids local displacement of a total-response maximizer relative to that input; it does not forbid an inherited gas/BCG offset or residual-map extrema and is not morphology evidence or a derived condensate redistribution.

Possible particle, soft, or topological excitations have Open spectra, couplings, abundance, and observables. Their absence from the minimal fit is not evidence that they are ultra-light, feebly interacting, or unnecessary.

Tests of the late-time scalar-background closure. The background ansatz may be compared with BAO, supernova, weak-lensing, redshift-space-distortion, and growth data through its declared $H(z)$ and

$w(z)$. Such comparisons test that closure. They do not establish that the microscopic condensate is the dark-energy source or that an independent component is absent. If the closure parameters are effectively frozen, its background becomes observationally close to Λ CDM and discrimination requires prespecified perturbation/growth observables.

Cluster-merger status. Equation (17.21) is the exact analytic result: a strictly increasing local response preserves all ordering and the set-valued argmax of the supplied total projected baryon proxy. Whenever maxima are attained, it cannot dynamically displace a total-response maximizer, but it may inherit an input gas/BCG separation and does not constrain the residual map. Separately, identifying $\tau_{\text{pole}} = \tau_{a_M}^{\text{match}}$ with one constant real Debye pole and ballistic transport is excluded by the three-order scale mismatch; this does not exclude general retarded kernels. A physical explanation requires the causal state evolution, matter and photon metrics, two lensing potentials, hydrodynamic/collisionless initial data and a mock-observation likelihood listed in Appendix AI.

The Great Attractor and large-scale peculiar flows. An ECT prediction of the Great Attractor or any large-scale peculiar-flow field requires a calibrated inhomogeneous cosmological solution, a derived matter-source and photon-metric map, survey selection and a velocity likelihood. These owners have not yet been supplied (OP21). In the restricted near-GR same-metric/quasistatic closure, the linear $aHfD$ carrier differs from its matched control by only 0.00282–0.00293% over $0 \leq z \leq 2$ [Eq. (AI.54)]. This gives no material large-flow enhancement and is not a survey bulk-flow prediction.

Tests of the HRC galactic programme. Void dwarfs, diffuse groups, and outer disks are useful pre-specified low-acceleration stress-test populations for HRC-0/HRC-3. The active registry defines the current algebraic diagnostics. Their placement on a canonical scalar-screening diagram has not been derived. Environment-tagged samples may test prespecified Ξ models and transition-shape correlations; neither an external-field effect nor a signed trend is presently an ECT prediction.

SPARC rotation-curve transition. The 165-galaxy statistics in Table 64, now report HRC-only held-out diagnostics with signed gas. They are Level C and do not replace the required finite-disk, nuisance-consistent publication likelihood; the representative Milky-Way sensitivity is reported in the same HRC registry but is not a modern joint posterior. The scale $a_{M0} = cH_0/(2\pi)$ remains matching, and no environment/history or external-field law is derived.

Lorentz-invariance violation gate. A viable ECT completion must make the physical gravitational, electromagnetic and any additional massless modes pass stringent relative-cone tests. The present scalar action does not predict their equality. The joint GW170817/GRB 170817A analysis constrains $-3 \times 10^{-15} \leq (c_{\text{GW}} - c_\gamma)/c_\gamma \leq +7 \times 10^{-16}$ under its source-association and emission-time assumptions [12]; this is an external kill gate for the missing tensor/photon completion, not a check satisfied by construction.

Late-time-source acceptance test. A physical ECT late-time result must be produced by the same varied action, state and matter/photon metric that determine $H(a)$, distances and the gauge-reduced perturbations. BAO, supernova, chronometer, growth and lensing data then test that completion jointly. At the present closure level no numerical equation-of-state trajectory or likelihood is identified.

Summary of observational strategy. Held-out kinematic and lensing data can test the conditional HRC galactic diagnostic and separate cluster benchmark, while prespecified regressions can test named Open environment extensions. Candidate soft or topological sectors acquire observable signatures only after their action, spectrum, abundance, couplings, and source projection are derived. These are closure-specific tests and Open programmes, not current universal ECT predictions.

Systematic role of the orientation proxy. The manuscript inventories how a candidate $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$ extension would enter background, galactic, growth, and cluster calculations (Table 51). This is a sensitivity study, not a derivation from the scalar action. Sub-horizon growth and cluster lensing appear most responsive in the proxy, but the orientation action, physical projection, sign, and coefficient must be established before any of the quoted corrections becomes an ECT prediction.

49.8 Parameter compression and comparison

ECT motivates a possible parameter-compression programme, but no complete owner/parameter census yet exists. At the benchmark level used in the present synthesis, the condensate-side description contains two kinds of ingredients: (i) a kinematic branch choice, encoded in the kinetic coefficients α, β , and (ii) non-kinematic condensate data, encoded in the gradient scale u_0 , the amplitude scale ϕ_0 , and the quartic coupling λ . Adopting the Level-B unit-slope coefficient benchmark $\hat{c}_* = 1$ gives $\alpha = 2\beta$; a common field rescaling cannot select this ratio. The numerical convention $\alpha = 2, \beta = 1$ fixes the remaining overall EFT normalisation but does not derive the photon or tensor speed map. The separate physical equality $c_{\text{char}} = c$ is the adopted clock calibration. The triplet (u_0, ϕ_0, λ) is only the displayed condensate-side subset. Additional branch coefficients, actions, states, response functionals, matching scales, operators and observable maps enter the macroscopic, gauge/matter and quantum completions.

These inputs do not by themselves derive the advertised low-energy sectors. The dimensionless scalar slope, characteristic cone and Klein–Gordon form are Level A inside the supplied scalar EFT. The physical clock/photon map, $S_0 = \hbar$ is the explicit Level B calibration and the physical owner and G_N completion remain Open; the Schrödinger envelope is a conditional coherent-sector reduction; a physical tensor completion and Einstein dynamics are Open. Horizon/flatness/relic statements and late-time backgrounds belong to named supplied completions, decoherence is conditional open-system algebra with ECT kernels/state Open, and black-hole entropy and information remain Open programmes. The galactic rotation-curve and BTFR results reported in this version have a different, explicitly layered status: they are conditional HRC algebraic diagnostics after a supplied identity gravity bridge. The HRC response algebra is Level A only inside the augmented response model; ownership of its source/force map by the live condensate action remains Open, and comparison with observations is Level C. At the more exploratory level, ECT also provides structural routes toward α_{fs} , generation number, and Yukawa hierarchy.

A numerical comparison with the Standard Model plus cosmology would not yet be like-for-like: ECT has not derived many parameters that those theories fit, and its own completion data have not been counted. Quantitative compression is therefore an Open falsifiable target: it may be claimed only after one owner-complete action/state/observable system reproduces the same data with a fully disclosed independent-parameter count.

49.9 Completion data and residual bifurcation points

The remaining non-closed sectors of ECT fall into three conceptually distinct classes, following the classification of incompleteness introduced in this section.

Type 1: Initial-condition-sensitive sectors. The arrow of time, thermodynamic irreversibility, and the second law depend not on a different “branch of laws” but on the realized initial state. This is not a special weakness of ECT: every fundamental theory (GR, QFT, string theory) requires initial conditions for the temporal asymmetry. ECT offers a candidate route: a future physical $O(4) \rightarrow O(3)$ transition could generate a non-equilibrium, coherent, low-entropy starting state. The present theory does not derive that transition or show that this outcome is generic; accordingly, ECT does not yet derive a strict second law from first principles.

Type 2: Open microphysics. This class includes: the CP-violating mechanism for baryogenesis (currently at the level of Sakharov-condition compatibility), the fine-structure constant (structural route via $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$, §7.2), three fermion generations (spectral route, §9.6), the Yukawa hierarchy

(overlap/tunnelling route, §9.6), the primordial perturbation spectrum (Lifshitz-window / transfer route, §14.6), and the origin of $SU(3)_{\text{colour}}$. For each, the structural mechanism is identified but the explicit computation remains an open calculational programme.

Type 3: Current condensate-side input data. The displayed non-kinematic inventory is current closure data, not a proved irreducible parameter set. In P3, (ϕ_0, λ) reparametrise the two potential couplings (μ^2, λ) through $\mu^2 = \lambda \phi_0^2$. P4 separately supplies u_0 , which has mass dimension two and whose relation to the P3 parameters remains Open. After the adopted α, β benchmark and normalisation convention, only ϕ_0 currently carries an explicit Level-B Planck-scale ansatz; λ and u_0 have no closed numerical observable map. Whether any of these inputs remain irreducible in a completed stationary ECT action is Open.

Epistemic vs. ontic distinction. It is important to note that if a parameter currently appears as closure data for a Level B result, this does not necessarily mean that the result is fundamentally underivable without it. It may mean only that the current version of the derivation does not close without this input. Future developments of ECT may reduce the number of independent closure data by deriving some of them from more fundamental condensate structures. The programme of strict classification of truly irreducible inputs versus current derivational gaps is itself an important open task.

49.10 Current limitations

Several important limitations should be noted.

(i) A supplied compact phase has exact global Noether algebra. Bundle covariance fixes the standard connection only after a local $U(1)$ bundle is independently supplied; physical $U(1)_{\text{em}}$, an unbroken photon, charges and matter vertices remain Open. The residual vector stabiliser is $SU(2)_{\text{diag}}$ and does not select a chiral weak factor. The inequality $\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$ excludes a direct injective Lie-subalgebra/residual-factor colour embedding in the displayed minimal $O(4) \rightarrow O(3)$ algebra (conditional Level A; see also §40). It does not exclude emergent/nonlinear, reducible/singlet, independently supplied bundle or defect-kernel routes. P7 is an adopted Level-B structural postulate whose physical-colour map remains C/Open; physical colour remains Open. The full Standard-Model matching—observed matter representations, charge assignments, anomaly cancellation, and the Weinberg angle—remains open (Section 49.14, Table 115).

(ii) The galactic weak-field sector is implemented through a declared phenomenological functional, not a boundary-value solution of the corrected scalar equation. Its deep-IR $Y^{3/2}$ form is motivated by scale invariance; the transition function and inner morphology remain closure choices.

(iii) The pipeline fits $a_{M,\text{eff}}$ per galaxy. The scatter is a data-interface output to be decomposed, not a predicted consequence of an ambient scalar background. A map $a_M = \mathcal{G}(q, \rho_{\text{env}}, \dots)$ remains Open.

(iv) Three fermion generations, the Yukawa coupling hierarchy, and the fine structure constant $\alpha_{\text{fs}} = 1/137$ are not yet derived from first principles, but structural routes have been identified for all three: $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$ inside a supplied gauge-normalisation completion (§7.2), family multiplicity through spectral counting of fermionic modes (§9.6), and Yukawa hierarchy through overlap/tunnelling suppression (§9.6). These are Level B/C routes, not completed derivations (Appendix AA).

(v) The homogeneous P3 Gaussian model has a correlation scale and may be studied with ordinary scalar power counting; the P4/gravitational validity threshold is Open. This does not solve quantum-gravity renormalisability: the physical tensor action, gauge constraints, interacting matter completion and their counterterms have not been derived. A remote Landau pole of the scalar subsector is not evidence for UV safety of a future gravitational completion. No quantum-gravity UV advantage is claimed. Open tasks: explicit loop power counting (OP-UV1), two-loop RG of the condensate–graviton system (OP-UV3), and OS proof for the spin-2 sector (OP-OS1).

(vi) The fixed-coefficient orientation tensor has the exact direction-independent determinant $\det K^{AB} = \beta^3(\beta - \alpha)$. This is Level A algebra but not an unconditional decoupling theorem. A separately supplied

trace-free/fixed-measure scalar–tensor completion reconstructs the metric equations with a global density C_I and is exactly covariant under $V \rightarrow V + C_0$, $C_I \rightarrow C_I - C_0$. The particular C_I may be boundary/state data of our condensate; the missing ECT result is the action-level descent and its matter/photon observable map. An ordinary two-slope completion supplies a stable, regular late-de-Sitter proof of class but is not invariant under an added constant. The observed late source, H_0 , and age therefore remain conditional on one selected action, state and metric map. The relation $\rho_{\text{late},0}^{(\text{GRref})} = 3\Omega_{\text{late},0}^{(\text{GRref})}\bar{M}_{\text{Pl}}^2 H_0^2$ is a reference convention after physical-metric and Planck calibration, not the generic scalar–tensor Friedmann equation or a determination of the source. A model-native present constraint also contains F_0 , $-3H_0\dot{F}_0$ and any separately supplied orientation stress.

(vii) Retarded support $K(\tau < 0) = 0$ is established at Level A for the supplied quadratic ordered-Lorentzian scalar operator (Appendices F, G). For a supplied $P(X)$ branch the appendix instead gives an exact sign test and conditional local domain-of-dependence theorem; a full nonlinear ECT response kernel is not established. The full open-system identification of the response kernel used in the influence-functional thermodynamic closure, and the associated ohmic/Markov/macrosopic bath assumptions, are not yet derived directly from Euclidean microdynamics. Gaussianity and Markovianity alone do not establish a second law. A conditional theorem requires a declared unital, detailed-balance, or contractive reduced channel together with its state and entropy functional; the ECT construction remains Open (OP-Q17).

(viii) The exact Bell-correlator reconstruction is still at the programme level, the full nonlinear gravitational closure is only partly theorem-level, and the black-hole evaporation/information programme is not yet quantitatively closed.

(ix) The electroweak scale v_2 is not yet derived from condensate dynamics; the physical identification $S_0 = \hbar$ is the explicit Level B calibration, while its absolute derivation and universal owner remain Open. The Yukawa hierarchy now has an identified structural route (overlap/tunnelling suppression, §9.6) but the explicit computation remains open.

49.11 Status-separated closure targets

Several central physical questions remain Open even where the manuscript provides exact conditional mathematics or an explicit list of missing owners. This subsection records those closure targets without treating a change of variables or a model-building option as progress toward a physical derivation.

Colour sector. The reduction supplies $SU(3)$ as the fibrewise stabiliser/structure group of a rank-three complex fibre with positive-definite Hermitian form h and unit-norm complex volume form Ω (§40, P7). Its identification with physical $SU(3)_c$ remains Open; the global automorphism group is the associated gauge group, not a single copy of $SU(3)$. The closure task is the zero-mode programme: construct a condensate-internal operator $\mathcal{D}_{\text{defect}}$ whose stable-defect or internal-sector kernel has $\dim_{\mathbb{C}} \ker \mathcal{D}_{\text{defect}} = 3$, derive (h, Ω) from the condensate inner product and orientation data, and identify the colour gauge field as the Berry–Wilczek–Zee connection on the zero-mode bundle. Completion of this programme would promote the colour-completion datum from an algebraic extension to a condensate-derived result.

Quantum of action. The identification $S_0 = \hbar$ is adopted as a Level-B empirical calibration; its absolute derivation, physical owner and universality remain Open. The supplied fixed-core route defines $t_{\text{elem}}^{(\text{fixed core})} = 2\pi t_0^{\text{EFT}}$, while the optional positive-tension route defines $\mathcal{S}_{1,\min} = 2\pi t_{0,\text{bal}}^{\text{eff}}$. The compact phase condition

$$\oint \partial_A \theta dX^A = 2\pi n$$

owns the integer winding, mathematically analogous to compact-phase winding in superfluid circulation [10] and superconducting flux [11]. It does not own the nonzero coefficient S_0 , the selected core representative, a positive tension, or a global action lower bound.

The phase-only unbounded-length class has infimum zero. T1 supplies one explicit sufficient stabilisation model, but its coefficients are not yet owned by ECT. The next Open tasks are therefore: (i) derive K_θ^{1D} and $\mathcal{J}_{\text{loop}}$ from one vacuum-subtracted four-dimensional profile, or derive a different stabiliser; (ii) exclude double counting and verify stability against shape, transverse, splitting, unwinding and backreaction variations; (iii) select and derive a route-to- S_0 owner map and test cross-sector normalisation; and (iv) derive the canonical commutators, Hilbert/projector structure and Born weights from an identified physical state and measure. Only that combined programme could promote a loop coefficient to a physical action-quantum candidate (OP-Q1, OP-Q2, OP-Q7).

Electroweak hierarchy. The electroweak scale is currently fixed by matching to the Fermi constant,

$$v_2 = (\sqrt{2} G_F)^{-1/2} \simeq 246 \text{ GeV},$$

and is used as the secondary ordering scale of the electroweak sector. ECT does not derive this scale from the minimal condensate dynamics. The identity

$$\mathcal{J}_{\text{EW}} = \ln(\phi_0/v_2) \simeq 36.8.$$

is only a matched logarithmic reparameterisation and has no explanatory content by itself. Closure requires a physical secondary operator and state, scale-generating dynamics, threshold matching and radiative stability. No alignment beta function or critical coupling is presently derived. A complete derivation would also have to connect this ordering scale to the chiral fermion/Yukawa sector, electroweak boson masses, neutrino-sector matching, and director-coupling suppression. Until that calculation is performed, the electroweak hierarchy remains Open; defining $\mathcal{J}_{\text{EW}}$ does not reduce it.

Fine-structure constant. The fine-structure constant is not derived in the present formulation. The identity in §7.2 and Appendix AA applies only after a local gauge completion, representations and charge normalisation have been supplied. Before electroweak breaking, one must compute the condensate-induced kinetic coefficients Z_B, Z_W and the corresponding charge normalisations q_Y, q_2 , with

$$\alpha_Y^{-1}(\mu) = \frac{4\pi Z_B(\mu)}{q_Y^2}, \quad \alpha_2^{-1}(\mu) = \frac{4\pi Z_W(\mu)}{q_2^2}.$$

At the electroweak matching scale,

$$e^{-2}(\mu_{\text{EW}}) = g_Y^{-2}(\mu_{\text{EW}}) + g_2^{-2}(\mu_{\text{EW}}), \quad \alpha_{\text{fs}}^{-1}(\mu_{\text{EW}}) = 4\pi e^{-2}(\mu_{\text{EW}}),$$

after which the QED coupling is run through the appropriate thresholds down to the electron scale. An ECT-specific result would have to derive the physical gauge bundle and matter representations, Z_B, Z_W , the charge-lattice normalisations (q_Y, q_2) , the Weinberg angle θ_W , and the threshold-corrected RG flow. No relation among these quantities is presently supplied by P1–P7. No simple geometric value such as $1/(4\pi^2)$ is claimed for α_{fs} in any limit; the relevant UV data are the pair $(Z_B^{(0)}, Z_W^{(0)})$ and θ_W , not a single geometric coefficient.

Ghost-freedom and unitarity. Within the retained scalar/coherent ordered-branch closures, the quadratic Hamiltonian positivity checks rule out the corresponding scalar ghost at that level. This does not yet constitute a complete unitarity theorem for the full metric/orientation sector. The spin-2 closure task comprises: (i) the complete quadratic expansion of an owner-complete condensate action around a derived stationary ordered state for all scalar, vector, and tensor perturbation modes; (ii) verification of the constraint algebra and projection onto exactly two transverse-traceless polarisations in the massless spin-2 sector; (iii) exclusion of Boulware–Deser-type scalar ghosts [288], or any analogous extra constraint-violating scalar mode, beyond the linearised approximation; (iv) a complete applicable Osterwalder–Schrader package [73, 74]: supplied Euclidean measure/correlators, temperedness together

with the applicable OS-II regularity/growth condition (for example E0' linear growth), covariance, a specified time reflection, reflection positivity, permutation/graded symmetry, the physical observable algebra, and the required locality/clustering properties, followed by the physical clock/metric and S_0 identification. Reflection positivity alone is not sufficient for reconstruction of the physical Lorentzian theory. Items (iii)–(iv) remain necessary before any nonperturbative unitarity claim can be made (OP-OS1, OP-UV3).

Born rule and measurement. Supplied influence-functional models can suppress selected coherences after a system–environment split, coupling, state, kernel, trace, and timescale are declared. The corresponding physical ECT channel is Open, and dephasing alone neither selects an outcome nor creates a probability measure or Born rule. The current mathematical representation is conditional (§38.1): for a supplied complex Hilbert/projector structure with $\dim \mathcal{H} \geq 3$ and an independently admitted normalised, noncontextual, countably additive projector measure, ordinary projector Gleason fixes its trace representation; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses. OS-II reconstruction and dephasing are not Gleason premises. The closure task is to derive the physical channel, state/preparation, event algebra, probability origin, pointer records, and outcome/update rule from condensate dynamics rather than to assume them. A complete account must also remain compatible with Bell correlations, no-signalling, and the Tsirelson bound (OP-Q7, OP-Q20).

Additional high-priority closure tasks. The broader list of uncovered structural problems is summarised in §49.10 and tracked in the central registry §38.1. For completeness, the most important cross-cutting targets that are not yet developed in the preceding paragraphs include: charge quantisation and the full Standard-Model hypercharge pattern (with the global discrete quotient); gauge and gravitational anomaly cancellation; flavour and mixing (three generations, CKM, PMNS, the CP phase, and the neutrino mass ordering); Bell–Tsirelson saturation and no-signalling closure; locality, cluster decomposition, analyticity, and microcausality of the reconstructed Lorentzian theory; black-hole entropy and Page-curve reconstruction from the condensate description of horizon regions; cosmological perturbations ($P(k), n_s, r$, non-Gaussianity) from the Euclidean ordering transition; the modified-gravity functions $\mu(k, z)$, $\eta(k, z)$, $\Sigma(k, z)$ for linear perturbations, separated from the local galactic-scale closure; and full UV closure including power counting, counterterm classification, loop closure, and reflection positivity.

49.12 Future directions and programme closure

The most important open problems for future work include:

(i) Deriving the missing $S \rightarrow G$ coordinate, source, force, and action map. Only after that bridge is fixed can a finite-body solver with environmental boundary conditions, realistic baryonic components (including SPARC gas profiles), and axisymmetric disk geometry test whether G follows from the scalar sector.

(ii) *Origin of the electroweak scale and locking structure (OP-EW).* The current formulation uses $v_2 \approx 246$ GeV as the secondary electroweak scale required by low-energy matching. A first-principles derivation remains open at several logically distinct levels: the compact $U(1)_\theta$ participant of the alignment sector (OP-EW-locking.0), the Hopf-fibered diagonal locking dynamics (OP-EW-locking.1), the non-polynomial scale-generation mechanism $v_2 = \phi_0 e^{-\mathcal{J}_{EW}}$ with $\mathcal{J}_{EW} \approx 36.8$ (OP-EW-scale), the local chiral $SU(2)_L \times U(1)_Y$ gauge completion including g, g' and the Weinberg angle (OP-EW-gauge), the protection of the secondary radial mode from additive $\mathcal{O}(\phi_0^2)$ corrections (OP-EW-naturalness), and the chiral fermion/Yukawa sector (OP-EW-matter). The reduced $O(3) \rightarrow O(2)$ analogue is described in Appendix M; the candidate Hopf-fibered phase–orientation geometry is described in Appendix N. Any complete derivation would feed simultaneously into the electroweak boson mass scale, fermion mass matching, the still-Open neutrino mass operators, and fifth-force suppression factors, all of which currently use v_2 as an input.

(iii) Numerical simulations of condensate evolution during cosmological phase transitions, including predictions for the primordial power spectrum beyond the slow-roll approximation.

(iv) Prespecified external validation of the HRC algebraic diagnostic on enlarged SPARC-like samples, including prespecified tests of the Open Ξ environment/history ansatz after observational systematics are controlled.

(v) Completion of the gauge-sector programme: explicit internal weak-gauge derivation beyond the present structural $SU(2)$ selection, derivation of the observed fermion representations and charge assignments, and a construction of the colour sector from extended internal condensate structure (see Section 49.14 and Table 115).

(vi) Derivation of baryon-number selection rules in the completed gauge/flavour sector, including whether low-energy $\Delta B = 1$ operators are absent, instanton-suppressed, or Planck-suppressed, what decay channels they induce, and whether $\Delta B = 2$ observables such as $n-\bar{n}$ oscillations and dinucleon decay provide an independent test (§44).

(vii) Completion of the structural routes to open microphysical targets: explicit computation of Z_A for the fine-structure constant (§7.2), spectral analysis of the condensate background for the generation number (§9.6), and overlap-integral computation for the Yukawa matrix (§9.6).

Highest-priority uncovered structural problems. Beyond the sector-specific items listed above, the following cross-cutting problems represent the largest remaining gaps between ECT and a complete physical theory, ordered by a combination of importance and proximity to existing ECT tools:

- (1) Full first-principles Born-rule closure (derive the owners of (C1), (C2) or (C2q), and (C3), including the physical probability and event map, from bare P1–P6; §30.4).
- (2) Charge quantisation: establish the local compact bundle, unbroken photon, charge unit and matter representations. Only the integer representation theorem for supplied premises is established; the physical ECT lattice, full SM pattern, hypercharge and global quotient remain Open (§23.4).
- (3) Gauge anomaly cancellation from admissible condensate excitations.
- (4) Full flavour closure: generations, textures, CKM, PMNS, CP phases, and mass ordering. The flavour architecture is now systematised in §9.7, but quantitative derivation remains open.
- (5) Strong CP problem (θ_{QCD}).
- (6) Why the fundamental arena is 4D Euclidean and why the emergent branch is specifically $3 + 1$.
- (7) Vacuum-offset and late-source completion (Section 15): derive the physical constrained Φ -medium action and Bianchi identity; decide whether its global density C_I is a boundary/state datum; derive the matter/photon metric; compute the full reduced Hessian and radiative counterterms; and fit one regular background and perturbation system. The orientation determinant identity, conditional trace-free shift covariance, and two-slope proof of class are distinct inputs to this programme (reformulated OP-Λ3).
- (8) Holography, area law, and Bekenstein bound as general medium properties; only a conditional finite-cell counting lemma is established after coarse-graining, finite capacity, physical split and an area-law state class are supplied (§36.5); physical ECT ownership, covariant and reconstruction extensions remain Open.
- (9) Quantitative black-hole information closure and Page curve beyond the current shell model.
- (10) Full low-energy QFT architecture: locality, cluster decomposition, analyticity, crossing, S-matrix structure.

(viii) Full derivation of the primordial perturbation spectrum (n_s, A_s, r) from the $O(4) \rightarrow O(3)$ ordering transition: derive a stationary regulated owner for $a_w, Z_u, \mathcal{C}_n, m_\chi^2$, test the exact anisotropic δ_a, δ_m, ρ finite-band bounds, construct the physical projector and RP-compatible state or explicitly classical statistical interpretation, and establish the curvature transfer $T_{\chi \rightarrow \zeta}$ and freeze-out / curvature-mode conversion (§14.6).

(ix) Construction of a covariant orientation action $S_n[g, n]$ and its on-shell stress tensor (OP-c1). The scalar action does not contain an independent n_μ , so the external Einstein–aether-like proxies used in the growth and cluster appendices do not reduce this problem to determining one coefficient c_1 . The vertex,

constraints, admissible operator basis, coefficients, signs, and physical projection all remain Open. Only after those steps could gravitational-wave, lensing, growth, or cluster observables be computed as ECT predictions.

(x) A complete first-principles derivation of cosmological linear perturbations on the Friedmann background, including the response functions $\mu(k, z)$, $\eta(k, z)$, and $\Sigma(k, z)$, remains an open technical step. A future covariant orientation sector might admit a mapping to a constrained Einstein–aether-like effective action. Whether such a mapping exists, which operators survive projection, and whether its coefficients are fixed by microscopic ECT parameters are Open questions. Existing Einstein–aether perturbation results may therefore be used only as external proxies, not applied directly as ECT predictions.

The ordered-branch amplitude variable as an organising coordinate. The coordinate $\phi_u = \beta_\phi^{-1} \ln(u/u_\infty)$ provides a common notation for several closure programmes. It does not make their dynamics identical. The corrected scalar ansatz, the cosmological background model, and the conditional HRC gravity response still require separate matching. In particular, density screening and the environment law Ξ do not follow from the coordinate definition. A future one-coordinate derivation may connect these layers, but at present the common symbol organises Open tasks rather than proving a unified mechanism.

Gauge-completion programmes and the vacuum-order hierarchy. The macroscopic amplitude variable does not derive electromagnetic, weak or strong interactions. The compact phase, orientation algebra and P7 module provide inputs to separate conditional completion programmes, while their physical local bundles, kinetic actions, states and matter maps remain Open. A future action may test whether the vacuum-order coordinate controls a transition in one of those completed sectors; no such cross-sector map is presently established.

Possible strengthening postulates. An important future question is whether some of the current Level B results can be promoted to Level A. In some cases this may require only further derivation from the present postulates; in other cases it may require one or more additional microscopic assumptions that constrain the condensate more tightly. Natural candidates for such strengthening assumptions could include:

- (i) a *spectral-normalisation postulate* fixing the absolute normalisation and stabilised finite representative of a coherent action cycle, thereby potentially producing a nonzero S_0 rather than assuming one;
- (ii) a *coherent-Hamiltonian closure postulate* selecting a unique real-time generator of the ordered branch compatible with the same action scale S_0 ;
- (iii) a *spinorial-sector postulate* specifying which condensate representation sector realises physical fermionic excitations;
- (iv) a *projector/probability completeness postulate* sharpening the route from decohered alternatives to a probability assignment;
- (v) a *strong-field shell regularity postulate* sufficient to derive near-horizon purification dynamics and Page-curve behaviour constructively.

Whether such additions would be elegant consequences of the ordered medium picture or merely ad hoc repairs must be judged carefully. But the important point is that the present framework now makes this question precise: it identifies not only which results remain below Level A, but also what kind of additional input would be needed to push them upward.

Candidate signatures. The architecture of ECT motivates several classes of empirical signatures, each of which becomes a quantitative prediction only after the corresponding response functional, coupling map, or closure task has been derived. These classes include: (i) modifications to gravitational-wave amplitude, friction, or lensing from the orientation sector, with any local propagation-speed deviation required to satisfy existing multimessenger bounds; (ii) matter-sector signals from separately completed director

couplings: composition-dependent forces require a source/body map, whereas spin or chirality signals require an independent spin-dependent Hamiltonian; the homogeneous vector-density term supplies neither, and MICROSCOPE, Eöt–Wash, clock-comparison and spin-precession data constrain only the corresponding declared completion; (iii) potential ultra-suppressed environmental corrections in the neutrino sector, conditional on first deriving an ECT neutrino mass operator, its coefficients and any dependence on condensate background variables. None of these is offered as a derived numerical prediction in the present formulation. The current state of discriminants and falsifiers, including those already developed at the quantitative level, is summarised in Table 72 (ND1–ND13).

49.13 Fundamental constants and present closure levels

ECT does not treat all entries traditionally listed among the “constants of Nature” as belonging to the same conceptual class. The following three tables separate native ECT structural outputs (Table 109), open gauge-matter closure parameters (Table 110), and derived combinations, conventions, and observables (Table 111).

Table 109: Native ECT structural outputs.

Quantity	ECT status	Comment
$D = 3$ (transverse coordinates of the supplied branch)	A-math conditional / physical map Open	P1 plus P4 leave three directions transverse to the selected w ; identifying them with all physical spatial dimensions requires the clock/metric/matter map
c (vacuum signal speed)	Level B scalar-clock calibration / photon and tensor cones Open	$\hat{c}_* = \sqrt{\beta/(\alpha - \beta)}$ is dimensionless; $c_{\text{char}} = c$ is the adopted clock calibration, while a physical photon/tensor owner and cone-universality theorem are still required
$\hbar$ (action scale)	Level B calibration / Open owner	Compact winding is integer-valued only after BR1 is supplied; fixed-core and balanced loop closures define distinct candidates, neither of which owns the operational S_0 . The calibration $S_0 = \hbar$ is explicit, but no route derives its absolute value, physical owner or cross-sector universality
G_N (gravitational coupling)	Matched / Open owner	Standard relation in a supplied tensor EFT; $\kappa_n \rightarrow M_G^2$ not derived
Massless photon	Open	Requires a local unbroken $U(1)$ bundle, gauge-volume quotient and physical transverse action
Physical tensor sector	A-math conditional negative result / Open physical spin-2 completion	No linear TT/static-density seed at fixed α, β and tangent-director order; quadratic source response is convention-dependent/parametric; Fierz–Pauli is supplied

Table 110: Open gauge-matter closure parameters.

Parameter	ECT status	Structural route
g_1, g_2, g_3 (at M_Z)	Open	Induced gauge kinetics
$\alpha_{\text{fs}} = 1/137$	Open	Conditional low-energy identity $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$ only; no ECT relation among Z_B, Z_W, θ_W or the physical charge normalisations (§7.2)
$\sin^2 \theta_W$	Open	Ratio of gauge normalisations
$v_{\text{EW}} = 246 \text{ GeV}$	Open	RG flow $\lambda(\mu)$ from Planck scale
λ_H	Open	Second condensate transition
y_f (Yukawa couplings)	Open (OP-Yukawa)	Fermion-condensate overlap integrals
CKM matrix (4 params)	Open	Multi-generation Yukawa
PMNS matrix (4–6 params)	Open	Supplied flavour/seesaw ansatz only; no physical matrix derived
$\theta_{\text{gauge}}, \bar{\theta}$	Open; MISSING MASS-PHASE OWNER	Convention fixed by $\bar{\theta} \equiv \theta_{\text{gauge}} + \arg \det M_q \pmod{2\pi}$; neutron-EDM matching constrains the invariant, not the gauge angle alone
$N_{\text{gen}} = 3$	Open (OP-generations)	Neither primary-vacuum homotopy nor anomaly cancellation selects three generations; bound-state and overlap constructions remain supplied model-building routes
$N_{\text{colour}} = 3$	Minimal closure hypothesis (C)	Minimal rank of the primary colour module compatible with an $SU(3)$ -structured completion; anomaly cancellation alone does not select the rank (N_c odd only; §9.5)
Λ_{div}	Level A coefficient inside Level-B ansatz / response Open	$U(\vec{\phi})/[\vec{M}_{\text{pl}}^2 F(\vec{\phi})]$ in the completely divided scalar equation; the scalar root, vacuum offset, selected branch, physical expansion response and observed value are not derived

Table 111: Combinations, conventions, observables, and Open targets.

Quantity	Status	Comment
e (elementary charge)	Follows from $\alpha + \hbar + c$	Not independent target
μ_0, ϵ_0	SI conventions	Fixed by c ; not physics targets
α_G	Follows from $G, \hbar, c, m$	Derived combination
m_f (fermion masses)	Follow from $y_f \cdot v_{\text{EW}}$	Requires Table 110 inputs
m_W, m_Z, m_H	Follow from $g_i, v_{\text{EW}}, \lambda_H$	Embedding (B/C)
H_0	Solution-dependent	Full cosmological model

Quantity	Status	Comment
n_s (spectral index)	Conditional strict-pullback route	A supplied pure-quartic kernel gives local $n_\chi = 1$ exactly when its combined κ_χ -kernel normalisation is scale-independent (separate constants suffice); positive constant lower-order terms give blue penalties fixed by δ_a, δ_m, ρ ; physical covariance, observed red tilt and amplitude require owned measure/coefficients/state/complete transfer/running/scale-map
η_B (baryon asymmetry)	No ECT value	External target from a named Planck base- Λ CDM fit and standard conversion [70]; physical current, vertex, state and transport owners are missing
$g - 2$	Precision observable	Requires full QED from ECT
Ω_{DM}	Open cosmological fraction	Not supplied by the Level-C HRC data diagnostic

Several entries in Table 110 remain blocked not by technical incompleteness alone, but by the absence of an exact colour sector in the current formulation. This motivates the following consistency analysis.

49.14 Candidate common-origin programme for the interaction sectors

Status: status-separated inventory from Sections 7–40. The hypothesis that all physical interactions share one condensate owner is not derived. The subsection separates exact algebra, supplied completion data and Open physical maps.

Definition of “unification” in ECT

In this paper, “unification” does not mean embedding into a single simple Lie group (as in conventional GUT programmes based on $SU(5)$, $SO(10)$, or E_6). Here it denotes the *hypothesis* that distinct physical sectors could descend from one pre-geometric condensate substrate. Three levels are distinguished:

Common-origin hypothesis: all interaction sectors are to emerge from one condensate.

Structural derivation: gauge groups are inevitable consequences of the condensate structure.

Quantitative unification: coupling constants, charges, and representations follow from condensate parameters.

The present work formulates candidate embedding routes; common origin, structural derivation and quantitative unification remain open at different levels.

Interaction-sector synthesis

Collecting the results of Sections 7–40, the proposed ECT architecture places five candidate interaction sectors on one ordered Φ -medium; their physical common origin is not yet derived:

Table 112: Status-separated interaction-sector inventory for a candidate common-origin programme.

Sector	Mechanism in ECT	Level	What is derived	What remains open
Gravity	Scalar characteristic tensor in a supplied ordered-branch EFT; separate scalar–tensor and HRC completion programmes	A-math conditional / B/Open	Scalar principal-symbol algebra in the declared model	Physical universal metric, source vertex, TT tensor action and nonlinear completion
EM ($U(1)$)	Supplied compact phase; a separately supplied local bundle admits a connection and the generic ordered-background basis $Z_1 I_1 + Z_2 I_2$ (Section 7.2)	A-math conditional / Open physical	Global Noether algebra and conditional two-invariant bundle mathematics	Local redundancy, gauge quotient, unbroken photon, charge/matter map and the additional Maxwell selection $Z_2 = 0$
Weak ($SU(2)$)	Residual carrier-orientation stabiliser $O(3)$ of $O(4) \rightarrow O(3)$ and the identity $\mathfrak{so}(3) \cong \mathfrak{su}(2)$; a distinct internal chiral bundle would have to be supplied (Section 7.3)	A algebra / B–Open completion	Lie-algebra identity only; no chiral factor is selected	Internal weak bundle, one-factor vertex; $\sin^2 \theta_W$; matter representations
Strong ($SU(3)_c$)	Adopted P7 rank-three (h, Ω) structural bundle, separate from the displayed condensate/order-parameter algebra (Section 40)	Level B postulate / Level A conditional frame reduction / C–Open physical	$SU(3)$ frame group is exact conditional mathematics; a physical EFT remains conditional on further owners	Physical bundle, determinant-compatible connection, action, matter map and ECT-native rank selection
Director coupling	Lowest-dimension symmetry-allowed fermion–condensate operator $\mu_5 \bar{\Psi} \gamma^A n_A \Psi$ (Section 7.6)	A-math conditional / physical Open	Operator-basis classification in the restricted EFT	Nonzero coefficient, species dependence, body response and observables

Scoped obstruction to direct colour embedding in the displayed $O(4)$ algebra

The Lie algebra $\mathfrak{su}(3)$ has dimension 8, while $\mathfrak{so}(4)$ has dimension 6. Since $\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$, the colour algebra cannot be a subalgebra of $\mathfrak{so}(4)$. Therefore $SU(3)_c$ cannot arise as a direct subgroup or residual gauge factor of the minimal $O(4) \rightarrow O(3)$ breaking pattern. This supplements the dynamical obstruction of Section 40 (colour-carrying condensate breaks $SU(3) \rightarrow SU(2)$) with an algebraic one: even setting dynamics aside, the algebra simply does not fit.

The dimension count excludes only a direct colour subalgebra or residual-factor construction inside the displayed ambient Euclidean $O(4)$ algebra. It does not derive an electroweak gauge group and does not exclude emergent/nonlinear, reducible/singlet, independently supplied bundle or defect-kernel colour routes. The compact phase and orientation algebra are only candidate inputs to distinct local bundle completions; the physical electroweak and colour sectors both remain Open ECT matching programmes.

Two-storey electroweak architecture

The candidate electroweak completion has two logically separate layers:

Geometric storey (Sections 7.3, 9.3): The supplied $O(4) \rightarrow O(3)$ orientation chart provides an $\mathfrak{so}(3)$ algebra, and a supplied compact phase provides global $U(1)$ algebra. Identifying the former with the physical spatial-rotation algebra requires the Open branch, clock/metric and matter maps. P5 motivates, but does not derive, their localisation as internal gauge redundancies; the vector stabiliser does not select chirality.

Phenomenological storey (Section 7.4): A separately supplied secondary condensate matched at $v_2 = 246$ GeV can reproduce the standard symmetry-breaking mass relations. Its action, gauge representations and matching are not derived from P1–P6.

Neither layer currently explains why the physical $SU(2)_L \times U(1)_Y$ theory exists; together they define a conditional completion route and its missing owners.

Director coupling: naming and status

The ECT-specific fermion–condensate coupling $\mu_5 \bar{\Psi} \gamma^A n_A \Psi$ (Section 7.6) is henceforth referred to as the *director coupling*, by analogy with the director field of nematic liquid crystals and the anchoring energy in those systems. The order parameter n_A plays the role of the director; the coupling describes how fermionic excitations interact with the orientation of the ordered medium.

Within the explicitly restricted parity-even, zeroth-derivative operator basis, the director term is an allowed dimension-four fermion bilinear. Operator allowance does not require a nonzero Wilson coefficient: field redefinitions, microscopic symmetries or matching cancellations may remove or alter it. Its coefficient and even its status as an observable single-species interaction therefore remain Open.

The homogeneous timelike vector term has no spin splitting [Eq. (7.41)] and therefore no associated ω_{dir} window. A precession signal requires an independently specified spin-dependent operator and detector Hamiltonian. Likewise, the fixed spurion contains no source equation, propagator, body-charge map or screening law, so it does not predict an Eötvös window. The MICROSCOPE result and spin-sensitive searches remain external constraints on future declared completions, not target values generated by the present operator.

Coleman–Mandula compatibility

The Coleman–Mandula theorem [51] constrains the possible exact symmetries of a nontrivial Poincaré-invariant, unitary Lorentzian S -matrix under the usual analyticity and mass-gap assumptions, with the graded extension provided by the Haag–Łopuszanski–Sohnius theorem [289].

ECT does not claim a Coleman–Mandula-forbidden mixing of Poincaré generators with internal gauge generators within the low-energy Lorentzian S -matrix. If a physical Lorentzian metric and an asymptotic scattering sector are supplied or derived in a completion of the Ordered Branch, the resulting low-energy theory is required to retain the standard Coleman–Mandula-compatible product structure between spacetime symmetry and internal gauge redundancy, up to the usual gauge identifications.

The common origin proposed in ECT lies instead at the pre-Lorentzian Euclidean condensate level. At that level there is no fundamental causal Lorentzian manifold, no fundamental Poincaré group, and no asymptotic Lorentzian S -matrix. The Coleman–Mandula restrictions therefore apply downstream, to the reconstructed Lorentzian scattering theory, not to the pre-geometric condensate data themselves. This is not a claim that ECT violates or algebraically circumvents the theorem; it is a statement that the completed low-energy EFT must remain compatible with it.

Global electroweak structure and charge-lattice constraint

If the global electroweak structure is supplied as

$$U(2) = \frac{SU(2) \times U(1)_n}{\mathbb{Z}_2}, \quad (49.1)$$

the Abelian character n must be integer. With the full colour-compatible weight $n = q_f = 6Y_f$, let $d_f = 2T_f \pmod{2}$. A representation descends exactly when

$$d_f + q_f \equiv 0 \pmod{2}. \quad (49.2)$$

The quark doublet $(d, q) = (1, 1)$ satisfies this standalone- $U(2)$ condition. Standalone $U(2)$ does not encode the colour- $\mathbb{Z}_3$ triality interlock; the full supplied $\mathbb{Z}_6$ condition is $2t_f + 3d_f + q_f \equiv 0 \pmod{6}$. In the small-hypercharge convention the electric operator is $Q_{\text{em}}/e = T_3 + Y$.

For colour singlets one may instead choose the singlet-adapted integer character $n_{\text{sing}} = q_f/3 = 2Y_f$, but it is fractional for quarks and cannot be their $U(1)$ weight. Finally, $V(\Phi) = V(-\Phi)$ is only a symmetry of the displayed potential. A physical gauge quotient requires the central element to act trivially on the complete field, operator and line-operator content. Therefore both the standalone $U(2)$ identification and the colour-completed global quotient remain Open ECT owners; the congruences are Level A conditional group arithmetic.

Hierarchical bosonic sectors

The bosonic content of ECT is best organised hierarchically, by structural level. The table below records the bosonic sectors retained in the current formulation at successive levels of condensate structure; it does not claim a final unique census of all possible emergent excitations. The Level-0 scalar content should be read as the retained low-energy condensate sector of the present closure, not as a theorem-level final census of all possible Goldstone-sector excitations.

Table 113: Hierarchical bosonic sectors in ECT.

Level	Field/sector	Spin	Origin	Status
<i>Level 0: supplied P4 orientation chart (nonzero vector with O(3) stabiliser; physical SSB Open)</i>				
0	Physical graviton	2 only in a supplied massless FP completion	Metric/tensor owner Open	Open
0	χ coordinate / conditional soft-mode sector	0	Scalar coordinate fixed by integrability; one-mode physical interpretation only in a supplied reduced EFT	A kinematics / B physical EFT; physical particle and mass Open
0	σ (radial)	0	Radial condensate oscillation	A/B
<i>Level 1: compact phase fibre (U(1) bundle)</i>				
1	Photon	1	Supplied unbroken local $U(1)_{\text{em}}$ completion	Open ECT / A conditional
<i>Level 2: electroweak condensate ($v_2 = 246 \text{ GeV}$)</i>				
2	$W^\pm$	1	Supplied $SU(2)$ doublet EFT	B/C/Open
2	Z^0	1	Supplied $SU(2) \times U(1)$ mixing	B/C/Open
2	Higgs (H)	0	Candidate radial interpretation of supplied doublet	B/C/Open

Level	Field/sector	Spin	Origin	Status
<i>Level 3: colour extension (scenario β)</i>				
3	8 gluons	1	Supplied $SU(3)_c$ completion target	Open

The present formulation does not derive the physical particle identity, mass, coupling, production history, lifetime, or detector response of either the χ coordinate or the radial variable σ , and therefore selects no search channel. A conditional decision tree can nevertheless be stated. If a physical χ pole is derived with an ultralight mass, a cosmologically populated state, a non-negligible energy fraction, a perturbation/response map and nonzero gravitational or matter coupling, background-evolution and large-scale-structure observables become candidate channels. A composition-dependent weak-equivalence-principle/Eötvös channel for ordinary test bodies additionally requires a non-universal or composition-dependent coupling with a relevant range; universal gravity alone does not select that channel. Strong-EP/Nordtvedt and PPN channels have separate self-gravity and metric owners. If a physical radial pole is heavy, collider, high-scale or early-Universe channels depend separately on its portal, production rate, lifetime and detector response. Mass alone selects neither branch; every future search statement still requires an action-owned spectrum and observable map.

Comparison with other unification programmes

Table 114: Comparison of interaction-sector scope.

Framework	Gauge sector	Gravity	Parity violation	Programme features or conditional outputs
Standard Model	Postulated	Absent	Postulated	—
GUT (SU(5), SO(10))	Simple group	Absent	From representations	Proton decay
String theory	Model- and compactification-dependent	Dynamical gravity in string constructions	Model-dependent	Spectra and moduli model-dependent
Loop quantum gravity	Matter/gauge sectors may be coupled; unification is not automatic	Quantum-geometric programme	Not a generic output	Kinematical area spectra under the supplied framework
Einstein–aether	External matter/gauge sector	Lorentzian metric plus dynamical aether	Model-dependent	Preferred-frame effects constrained observationally
ECT	Conditional bundle-completion programmes	Scalar characteristic tensor only; physical metric/tensor Open	Physical weak chirality Open	Candidate director operator and condensate-mode coordinates; their physical interactions and observables remain Open

ECT organises gravity, gauge, matter and director-completion questions in one pre-geometric research programme. This wider scope is not evidence that the minimal action has unified or derived those interactions.

The common-origin claim

ECT proposes one-medium completion programmes for gravity, Abelian and non-Abelian gauge structure, matter and a director coupling. The present minimal action has not derived any of the four physical interactions from a common microscopic owner: the tensor/source vertex, local gauge bundles, chiral matter representations, colour completion and nonzero director Wilson coefficient remain Open.

The present results inventory algebraic ingredients and consistency tests for possible gauge completions. They do not derive a physical gauge architecture, observed matter representations, charge assignments, anomaly cancellation, or the Weinberg angle. These belong to the open programme below.

Open problems toward quantitative unification

Table 115: Open problems toward quantitative unification.

ID	Problem	Status
OP-GUT1	Intrinsic derivation of physical $SU(3)_c$ from condensate structure. The dimension inequality $\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$ excludes only an injective direct Lie-subalgebra route inside the displayed $O(4)$ spacetime/order-parameter algebra. It does not exclude a reducible singlet-VEV representation, an emergent/nonlinear realisation, an independently supplied bundle, or a defect-kernel/Berry route. P7 supplies the adopted rank-three (h, Ω) structural bundle; its $SU(3)$ frame reduction is exact conditional mathematics, while the physical-colour identification, selected dynamical connection, action and matter map remain Open. Anomalies do not select the colour rank.	Direct route excluded / physical completion Open
OP-GUT2	Derive fermion representations from condensate. Bounded first-pass audit recorded in the P7-restricted test class (§9.5); anomalies alone shown not to select the matter pattern (explicit chiral quartet family); minimal-core selection remains convention-level.	Open
OP-GUT3a	Global EW structure $\rightarrow$ charge lattice; standard integer-representation theorem established only conditional on a supplied compact local $U(1)$ bundle; physical ECT charge lattice Open (Theorem 23.1); EW $\mathbb{Z}_2$ refinement conditional (B); colour-interlock congruence recorded (§9.5)	Open physically / conditional mathematics
OP-GUT3b	Derive observed charge assignments of quarks and leptons; sharpened to selection within an explicitly characterised discrete residual family (§9.5)	Open
OP-GUT4	Relate α_s , α_{em} , G_N to condensate parameters	Open

ID	Problem	Status
OP-GUT5	Three fermion generations; Yukawa hierarchy and flavour closure (CKM mixing, PMNS mixing, CP phases, absolute neutrino spectrum); systematised architecturally in §9.7 under OP-Yukawa / OP-generations local labels	Open
OP-GUT6a	Local triangle anomaly cancellation	Open (consistency test; explicit verification recorded for the inherited content, §9.5)
OP-GUT6b	Witten $SU(2)$ global $\mathbb{Z}_2$ -anomaly	Open (consistency test; doublet count verified for SM-like generation structure, §9.5)
OP-GUT6c	ECT-native interpretation of the mixed $U(1)_Y$ -gravitational consistency condition in the emergent-graviton setting (§9.5, Layer C)	Open (structural)
OP-GUT7	Weinberg angle $\sin^2\theta_W$ from condensate parameters	Open
OP-GUT8	Director coupling coefficient from microphysics	Open
OP-GRAV-WEAK	Independent spin-1 connection modes beyond metric sector	Open
OP-TORSION	Propagating torsion in a possible first-order reformulation	Open (speculative)
OP-Λ3-O1	Derive or refute a physical trace-free/fixed-measure action, its constraints and Bianchi identity from the Φ medium	Open
OP-Λ3-O2	Derive the universal matter/photon metric and the observable map for a global integration/state density C_I or dynamical late source	Open
OP-Λ3-O3	Compute the full reduced Hessian, measure and counterterms before making a quantum vacuum non-renormalization claim	Open
OP-Λ3-O4	Derive or reject the two-slope action and its parameters from the ordered Φ phase; preserve its regular background only as a conditional proof of class until then	Open
OP-Λ3-O5	Solve and fit one action-owned background and perturbation system, with C_I allowed as measured state/boundary data rather than required to be a universal law-predicted number	Open

In addition to the usual local triangle anomalies (OP-GUT6a), any chiral $SU(2)$ sector must also be checked for the global Witten $\mathbb{Z}_2$ anomaly [79], which excludes an odd number of left-handed $SU(2)$ doublets. This is logically distinct from the local triangle-anomaly constraints and therefore deserves a separate consistency check once the emergent fermion spectrum is fixed (OP-GUT6b).

Speculative routes (beyond present scope)

Two further routes merit recording as possible future directions, although they lie beyond the present metric–condensate formulation:

1. *Connection degrees of freedom beyond the metric.* A key technical question is whether the oriented

connection sector contains independent spin-1 modes beyond those fixed by the metric. If so, these modes could underlie the weak interaction, providing a geometric—rather than internal—derivation of the $SU(2)_L$ gauge sector (OP-GRAV-WEAK).

2. *Torsion-based reformulation.* A first-order (Palatini-type) reformulation of ECT with independent connection variables may render torsion components propagating. The self-dual torsion sector is a natural candidate for a geometric origin of weak bosons. However, the current metric–condensate formulation does not include independent torsion variables, and this route lies beyond the present scope (OP-TORSION).

50 Conclusion

50.1 Status-separated outputs and unresolved owners

Status-separated numerical and conditional inventory across Parts I–III. Table 116 collects the principal conditional outputs, matching inputs, external benchmarks, fit diagnostics, and Open quantities across Parts I–III. Its entries are not uniformly ECT predictions and must be read with the Assessment column.

Table 116: Conditional outputs, matching inputs, fit diagnostics, regime-level consistency checks, and external experimental anchors.

Item / statement	Value / conditional output	Observation	Assessment	Part
<i>Macroscopic sector (Part II)</i>				
HRC deep-branch BTFR slope	4 after declared bridge	3.98 ± 0.08	A model / B bridge / Open owner / C data	II
a_{M0} matching	$1.0824 \times 10^{-10} \text{ m s}^{-2}$	$1.2 \times 10^{-10} \text{ m s}^{-2}$ phenomenological scale	Matched input	II
MW representative HRC sensitivity	HRC-0: $\chi_v^2 = 1.813$; HRC-3: 2.287	Frozen Eilers-like points; no modern joint posterior	Level C	II
n_s, A_s, r	Conditional Lifshitz route; open	CMB scalar/tensor targets	open	II
Strict-pullback equal-slice scalar kernel	Exact covariance, positivity domain, deformation ratio, seed tilt and finite-band tolerances	Physical projector, RP/OS reconstruction, $\chi \rightarrow \zeta$ transfer and CMB likelihood required	A-math conditional; ECT owner Open	II
Photon–tensor speed ratio	Not predicted: no physical ECT photon/tensor principal symbols or common metric	Joint GW170817/GRB interval is an external constraint	Open completion gate; no ECT checkmark	II
Late-time expansion and source history	Not identified	Joint background, distance, growth and lensing likelihood required	Open	II

Item / statement	Value / output	Obs.	Assessment	Part
η_B	No ECT numerical result	External target from a named Planck base- Λ CDM fit plus standard conversion [70]; current physical owner chain is incomplete	Open	II
Orientation determinant / vacuum-offset action	$\det K$ is direction independent, but the physical trace-free action is not derived	Explicit constrained action, Bianchi relation and matter/photon map required	Algebra A; ECT descent Open	II
Conditional trace-free completion	$V + C_I$ is exactly shift covariant; C_I is a global state/boundary datum	Standard conditional completion	A inside named action / B–Open ECT	II
Two-slope late background	Stable de-Sitter root and regular calibrated history; ordinary action not shift invariant	Conditional proof of class, not observed-source determination	A in action / C benchmark / Open ECT	II
<i>Neutrino sector (Part I)</i>				
Seesaw normalisation	$M_V^{\text{light}} = -m_D M_R^{-1} m_D^T + \dots$	Standard imported algebra	Heavy spectrum, Yukawa matrix and ECT owner Open	I
Dimension-four neutrino operator class	$E C_{\text{osc}}(\hat{p})$	Traceless Hermitian contribution after identity projection	CPT-even SME operator inventory; coefficient, direction/flavour map and ECT vertex Open; no numerical benchmark retained	I
<i>Quantum sector: environmental decoherence (Part III)</i>				
Bounded gravitational record-channel audit	Enumerated routes terminal as negative, parametric or non-identifiable	No complete action–projector–vertex–state/ noise–subtraction chain	Bounded taxonomy closed; PES-R B, physical PES Open	III
C_{60} [169]	Interference	Interference	External source anchor; exact pressure unowned; ECT rate Open	III
Fein 25 kamu	Standard environmental regime	Interference	External source anchor; ECT rate Open	III

Item / statement	Value / output	Obs.	Assessment	Part
Pedalino ~ 172 kDa	Standard-QM/source-model regime	Quantum/classical separation	Demonstrated separation scale; higher-mass fringe tail not assigned the same status; ECT rate Open	III
Procopio switch	$V=0.994\pm0.002 \Rightarrow \Phi_{\text{vis}}=0.006\pm0.002$	ICO visible	Source visibility transform; ECT Open	III
Jacques delayed-choice	measured $V \approx 0.94$ [154]; manuscript $-\ln V \approx 0.0619$	Visibility retained	External measurement plus internal transform; ECT Open	III
Werner CHSH toy	$\Phi_{\text{vis}} = \ln \sqrt{2} \approx 0.347$	Model marker	Source-model toy; no ECT discriminator	III
Pure-dilation qubit toy	$I_{\text{bits}} = 1$ near $\Phi \approx 0.25$	Model evaluation	Total mutual information, not detector capacity	III
<i>Quantum sector: external symbolic comparison and Open ECT owner chain (Part III)</i>				
Diósi–Penrose functional and sphere point	$\tau_{\text{DP}}^{\text{ext}} = \hbar/E_G$; $C_{\text{DP}}(1) = 51/160$, 70.6ms at the declared nominal point	<i>External formula / Level-C orientation point</i>	Not a data test or ECT rate; source/vertex/state/noise chain Open	III
D–P guide / collisional comparison	$m_{\text{guide}}^{\text{DP}}$ and $m_{\text{cross}}^{\text{coll-DP}}$ kept distinct	<i>External conditional algebra</i>	No numerical value; apparatus inputs and ECT record channel missing	III
Casimir boundary-response programme	Supplied N -scalar Dirichlet algebra only; no ECT force coefficient or $3/2$ ratio	<i>Open</i>	Physical reduced bundle, material boundary map and stress response missing	II/III

Conditional benchmark registry with a public arithmetic owner. Table 117 is a scoped registry, not an exhaustive inventory of every number in the manuscript. Its six rows are reproduced by the standalone public verifier `scripts/verification/verify_conditional_benchmark_registry.py`; the machine-readable result is `data/verification/R190_CONDITIONAL_BENCHMARK_RESULTS_v1.json`. The verifier fixes all inputs, units, conventions and printed rounding, and fails closed if the scientific-status firewall is changed. Every row is *not a data test* and retains an Open or Not-Identifiable ECT owner. The sensitivity and matching columns prevent arithmetic reproducibility from being mistaken for physical closure.

Table 117: Scoped conditional benchmark registry with a public arithmetic owner.

Benchmark	Declared inputs	Printed output	Status	What changes the value	Physical closure / matching still required
Hubble soft-scalar orientation	$H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\zeta_H = \zeta_\phi = 1$, $\phi_0 = \bar{M}_{\text{Pl}}$	$1.493 \times 10^{-33} \text{ eV}$; $1.322 \times 10^{26} \text{ m}$; 6.131×10^{-61}	C; Open	H_0 , ζ_H , scale and reduced-Compton convention	physical pole, state, abundance and observable owner; no target match
Geometric seesaw norm and inverse fits	$c_R = \zeta_\phi = 1$; supplied 0.050 and 0.009 eV targets	$M_H = 2.4487 \times 10^{10} \text{ GeV}$; fitted $s_y = 6.355 \times 10^{-3}$ and 2.696×10^{-3}	C; Open	c_R , ζ_ϕ , scale identification, Y_ν , flavour and target; inverse problem	full Takagi matrix, RG and threshold owner; values are fits, not forward predictions
Weinberg unit point	$C_5 = c_\Lambda = \zeta_\phi = 1$ in the displayed normalisation	$1.2447 \times 10^{-5} \text{ eV}$	External-EFT A; C; ECT Open	$C_{5,\text{eff}}/(c_\Lambda \zeta_\phi)$, normalisation, RG	0.050 eV orientation needs $\simeq 4.02 \times 10^3$ in the relevant Takagi direction; relation to type-I is NOT IDENTIFIABLE without a common UV owner
Heavy-radial negative control	$\phi_\infty = \bar{M}_{\text{Pl}}$, $\lambda = 10^{-2}$, $\beta = 1$, unsourced compact exterior	$m_\sigma = 3.444 \times 10^{17} \text{ GeV}$; $\xi = 5.730 \times 10^{-34} \text{ m}$; 52.731 decades	C arithmetic; ECT Open	λ , β , boundary data, source and distinct P4 owner	excludes only the named unsourced exterior proxy, not driven or galactic response
Dimension-six proton NDA	$\Lambda_B = \bar{M}_{\text{Pl}}$, $ C_B = 1$, supplied external EFT	$\sim 10^{42} \text{ Julian yr}$	External-NDA A; C; ECT Open	C_B , scale convention, hadronic, RG, flavour and phase-space factors	operator and channel owner; $C_B = 0$ removes this decay branch
D–P hard sphere	factor-1/2 functional, $\tau = \hbar/E_G$, sharp spheres, centre separation $d = R = 1 \mu\text{m}$, $\rho = 2000 \text{ kg m}^{-3}$	$C_{\text{DP}} = 51/160$; 70.6 ms	External-model A; C; ECT Open	profile, smearing, d/R , ρ , R , regularisation and normalisation	no ECT projector, vertex, state, noise kernel or absolute rate; not the unit-coefficient crossover

Here an “External A” entry means exact algebra or geometry only inside the declared external/source model; it is not an ECT Level-A physical closure. Level C denotes the benchmark/application layer, while the ECT-specific owner remains Open or Not Identifiable in every row.

The table mixes four explicitly separated categories: (i) external observations, (ii) standard source-model or visibility transforms not independently reproduced here, (iii) imported phenomenological benchmarks, and (iv) ECT-specific conditional or untested targets. It is a status ledger, not a statistical scorecard, and the first three categories do not constitute evidence for ECT.

Two closure statements in the table must not be conflated. First, the fixed-coefficient mathematics of the displayed strict-pullback scalar kernel is closed after the exact coercivity, equal-slice, deformation and tolerance analysis; its physical ECT owner and observable transfer are Open. Second, the presently enumerated record-channel search is terminal only within its declared owner taxonomy; no listed route supplies a physical channel, and a genuinely new action, field content or vertex re-enters the gates as a new candidate. Neither result closes physical/global PES.

In this work we have developed Euclidean Condensate Theory (ECT) as a one-medium research architecture based on a scalar condensate field Φ in four-dimensional Euclidean space. P4 supplies a nonzero-vector chart whose stabiliser is $O(3)$; a stationary physical state and spontaneous $O(4) \rightarrow O(3)$ selection remain Open. The Lorentzian scalar characteristic and several conditional effective reconstructions follow within stated assumptions; physical gravity, gauge and matter completions, quantum outcome rules and the two dark-sector phenomena are not all derived from the minimal medium. The complete theoretical identity of the framework is captured by the foundational postulates, branch-selection rules, and signature relations summarised in Section 10; the status and dependency chain is mapped in Figure 35 and Table 69.

The central conceptual result is a unified research architecture—from one medium and six postulates—for testing whether structures that standard physics treats as independent inputs can share a common owner: the conditional scalar distinction between one timelike and three spacelike characteristic directions after P4 and independently supplied $\beta(\beta - \alpha) < 0$ (or $\alpha > \beta > 0$ on the retained $\beta > 0$ component), with physical clock and universal metric Open, conditional geometric and coherent reconstruction programmes (not yet a derivation of physical GR or of all quantum postulates), candidate gauge and matter-sector completion routes, whose local bundles, physical fields, charges, couplings and chiral representations remain conditional/Open, and an allowed preferred-direction fermion operator whose Wilson coefficient and observable map remain Open, a Level-C HRC no-halo algebraic diagnostic and a projected cluster benchmark, with condensate matching and particle content Open, a conditional late-de-Sitter proof of class in the two-slope scalar completion, with the physical ECT source and state still Open, the retarded scalar problem after a supplied hyperbolic operator and orientation, with an ECT thermodynamic entropy theorem still Open, and PES-R as a Level-B taxonomy separating dynamical admissibility, finite-time retention, and pairwise record distinguishability. ECT thereby provides structural or closure-level routes into several long-standing foundational problems, including the quantum-gravity interface, the origin of time, the cosmological-constant problem, dark-matter phenomenology, the black-hole information problem, and the measurement problem. A dedicated upgrade layer of Part III further develops, on the gravity–matter interface, a reduced-state architecture in which hybrid-consistency demands are displaced from fundamental law to the reduction map between the full condensate theory and its effective descriptions (Section 28); an architectural operational criterion for gravity-induced-entanglement tests (Section 32); a channel taxonomy of the effective gravitational mediator separating scalar distinguishability from an Open gravitational-mediator completion (Section 33); an infrared framing compatible with BMS-type asymptotic symmetries, gravitational memory, and soft-graviton bookkeeping (Section 35); and a structural mapping of the critical-shell black-hole picture onto the modern islands and quantum-extremal-surface language (Section 36.3), complemented by a compact cross-programme diagnostic summary (Section 36.8). The Level-C HRC algebraic diagnostic contains no particle-halo term, and the late-time background is represented by a separate ordered-branch closure. Neither choice is presently derived as the unique ECT reduction. In particular, the galactic fit does not exclude particle dark matter and predicts no absence of a laboratory particle signal; a confirmed particle explanation of the galactic discrepancy would disfavor this minimal fit interpretation, not ECT. Likewise, the background closure does not by itself establish the absence of every independent dark-energy sector. The relevant tests must be stated against each specified closure rather than against unresolved ECT particle content.

Several results are already genuinely nontrivial.

More precisely, the exact result is limited to reduced-state logic: a thermal marginal can coexist with global purity. Excluding fundamental information destruction in ECT additionally requires a completed globally unitary interacting strong-field theory, which is not yet derived. The shell mechanism, subsystem split, Page curve and horizon thermodynamics remain C/Open.

Among the strongest structural results are: the Lorentzian principal classification of the supplied P4 branch, with $\hat{c}_*$ fixed by the scalar coefficients while G_N and the EFT action scale remain conditional matching programmes with Open microscopic ownership; the pure-metric Fierz–Pauli to Einstein–Hilbert completion under the stated assumptions, together with a corrected Level-B scalar–tensor ansatz; candidate anisotropic tensors remain proxies; an orientation stress is Open because its covariant action and physical projection are missing; integer winding topology inside supplied BR1, two distinct supplied loop-scale candidates, and conditional canonical/Schrödinger/uncertainty routes using an operational S_0 whose owner, physical operators, normalisation and universality remain Open; reflection positivity in Gaussian subsectors and the structural global/reduced unitarity distinction; the conditional topology of the separately supplied defect-free $\mathbb{R}^3$ point-excitation model, $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$, which blocks continuous two-dimensional braid phases inside that model but does not derive the physical slice or excitation map, select a statistics representation, or derive fermions; the environmental influence-functional bookkeeping, positive pairwise record quadratic in declared Gaussian models, and the still-Open ECT entropy and Crooks programmes; candidate gravitational record channels remain Open and the Diósi–Penrose functional is an external comparison; same-channel KMS/FDT and primordial acoustic-phase routes are conditional/Open diagnostics until their physical operators, states, common filters and transfer maps are derived; the Open unified vacuum-response programme (supplied Casimir, Unruh/KMS and particle-production targets, without a common physical ECT owner); the conditional Gleason trace representation under its complete theorem hypotheses, with the physical Born owner still Open; the entanglement programme, which requires a supplied nonfactorising state/channel and has no common-medium derivation at present; the exact reduced-state identities, while global Hilbert-space unitarity and a strong-field evaporation channel remain Open; the Level A orientation determinant identity, which does not by itself derive vacuum decoupling; the exact classical shift covariance of the separately supplied trace-free completion through $V + C_j$; and the regular late-de-Sitter proof of class supplied by the distinct ordinary two-slope completion. The trace-free action descent, global state datum, physical metric and radiative completion remain Open, and $\rho_{\text{late},0}^{(\text{GRref})} = 3\Omega_{\text{late},0}^{(\text{GRref})} \bar{M}_{\text{pl}}^2 H_0^2$ is only a reference convention, not a generic scalar–tensor identity or microscopic determination; and an analogue source-model comparison programme whose existing mechanisms are not evidence for ECT and whose failure is not an ECT falsifier until a quantitative six-owner bridge is supplied.

Structural organisation and Open conceptual programmes. The present framework makes several organising choices and poses several reconstruction programmes. They are not asserted here as experimentally established novelties or physical predictions: the scalar characteristic tensor is conditional rather than fundamental; the physical metric and tensor gravity remain Open; three logically distinct classes of excitations replace the uniform field-postulate structure of QFT; the axioms of standard QM are relocated from postulates to reconstruction targets; the possibility that event-like interaction and quantum correlation have medium owners is an Open reconstruction programme; common support alone does not derive interactions or nonfactorising states; the ordered geometry supplies orientation data, but no physical weak chirality or parity-violating matter map; and Fock space is a reconstruction target, not a foundational axiom.

Mesoscopic owner firewall. The external Diósi–Penrose functional supplies one numerical source-model orientation point but no numerical ECT benchmark. ECT has not derived its record-channel projector, vertex, state, noise kernel, finite-body charge, absolute rate, or a distinctive correction. No universal reverse/forward path law is presently predicted from decoherence.

Neutrino sector. ECT does not yet close the flavour or neutrino programme. No physical chirality selection, sterile state, mass operator or PMNS matrix follows from P1–P6. Standard Weinberg and seesaw formulae may be supplied, but their coefficients and mass scales are not selected. No ECT-selected or predictive numerical mass, Yukawa, mixing or leptogenesis claim follows; the declared orientation points and inverse-fit arithmetic are nonpredictive Level-C benchmarks. PMNS/MSW phenomenology and neutrinoless double-beta decay belong to the chosen Dirac/Majorana completion and do not yet discriminate ECT. Full flavour closure—generations, textures, PMNS angles, phases, mass ordering—remains open.

Foundational questions organised within ECT. Beyond its technical statements, ECT organises several long-standing foundational questions into conditional routes and Open owner problems; this does not by itself solve those questions: (i) *Why 3 + 1 dimensions?*—given the P1 postulate of a four-dimensional Euclidean arena, a supplied nonzero vector has an $O(3)$ stabiliser and distinguishes one direction from three (Level A conditional group theory). Its physical interpretation as a temporal seed, stationary state and spontaneously selected branch remains Open; the deeper origin of four-dimensionality itself remains OP0. (ii) *What does the supplied S^3 target say about monopoles?*— $\pi_2(S^3) = 0$, so maps into that target carry no π_2 charge (Level A conditional mathematics). Identification with the full physical vacuum/configuration space, its action, transitions and relic history remains Open; no physical monopole exclusion follows yet. (iii) *Vacuum stability*—the homogeneous P3 minima satisfy $V''(\phi_0) > 0$ in the printed tree-level potential; this is not a stability theorem for the P4 gradient datum. In the strict one-real-scalar one-loop leading-log truncation, $\beta_\lambda > 0$ excludes the Standard-Model-like negative-running route at that order (§8.4.5; Level B conditional). It does not exclude additional extrema of the completed effective potential. The tree-level symmetric point $\Phi = 0$ is a hilltop rather than a Coleman false vacuum; full primary-branch and electroweak-secondary stability remain Open. (iv) *Quantum gravity without primitive metric quantisation*—gravity is treated as a condensate response rather than as a fundamental geometric field; the graviton is a candidate condensate excitation on a fixed Euclidean background. This reframes the usual perturbative non-renormalisability problem of quantised GR, but does not yet prove UV finiteness or perturbative renormalisability of the full ECT quantum sector (architecture; UV completion open). (v) *Cosmological coincidence*—the current action and state do not identify a physical late-time source or its density history. A genuine dynamical solution requires one globally regular background and its matter/photon observable map (Open). (vi) *Tensor sector*—at fixed α, β and tangent-director linear order about homogeneous $n = w$, the displayed fixed-map perturbation has no linear TT/static-density seed. Quadratic, nonlinear, composite and additional-field routes remain separate parametric/Open candidates. A massless Fierz–Pauli action remains an Open completion target; the LIGO graviton-mass bound is an external gate for that future action.

Table 118 collects these structural positions, conditional statements, and foundational questions in one place.

Table 118: Structural positions, conditional statements, and foundational questions organised within ECT.

Question / structural position	ECT route	Level	Scope / disposition
<i>Structural organisation and research programmes</i>			
Conditional scalar characteristic tensor, not a physical metric	A supplied scalar operator has a secondary characteristic geometry; the physical gravitational metric completion is Open	A/B/Open	Supplied-model statement
Three logical classes of particles	Required / embedded / topological tiers, unlike QFT	A/B	Classification proposal

Question / structural position	ECT route	Level	Scope / disposition
QM axioms $\rightarrow$ reconstruction targets	Conditional Hilbert/RP structures; Born weights, unique outcomes and update rule remain Open	A/B/Open	Reconstruction programme
Interaction/correlation ownership	Common-medium support is compatible with interactions and correlations but does not derive their vertices or nonfactorising states	Open	Programme
Weak-chirality programme	$O(4) \rightarrow O(3)$ supplies orientation algebra but does not select a local chiral weak bundle or physical representations	Open	Programme
Discrete-symmetry architecture	P, T, C , conditional CPT route via OS reconstruction; branch-pair layer on S^3 ; $\mathcal{L}_5$ re-classified as CPT-odd phenomenological operator	A/B	Conditional architecture
Anomaly architecture	EW anomaly constraints inherited; $\mathcal{L}_5$ anomaly-transparent; full SM completion open	A/B/Open	Consistency bridge to OP-GUT6
Flavour architecture	Standard basis-misalignment and phase bookkeeping; supplied overlap and seesaw matrices are Open model-building routes; near-rank-one structure alone does not imply alignment or FCNC suppression; anomaly coefficients omit flavour values once representations are supplied	A standard algebra / Open ECT	OP-Yukawa and OP-generations
Fock space as reconstruction target	Variable N = condensate reconfiguration	Open	Interpretive programme

Foundational questions organised

Why a 1+3 directional split on the supplied chart?	$O(4) \rightarrow O(3)$ leaves three transverse Euclidean directions plus one ordered direction. A scalar Lorentzian interpretation additionally requires $\beta(\beta - \alpha) < 0$; identifying those directions with physical space and time, and deriving a common matter/photon/tensor metric, remain Open. The deeper origin of four-dimensionality is OP0	A-math conditional / Open physical	Conditional group/principal-symbol statement
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Question / structural position	ECT route	Level	Scope / disposition
Why no π_2 class in the supplied orientation target?	$\pi_2(S^3) = 0$ exactly; physical vacuum identification, later gauged cosets and production histories remain Open	A-math conditional / Open ECT	Conditional target-space statement
Vacuum stability (primary sector only)	Tree-level local minimum plus positive quartic running in the strict one-loop scalar truncation; no Coleman false vacuum in the minimal tree-level potential; full effective-potential, branch and EW-inheritance analyses Open	B conditional / Open	No
Quantum gravity	Scalar characteristic geometry derived; physical graviton/metric Open	Open	Programme only
Physical tensor sector	$h_{ij}^{\text{TT}} = h_{ww} = 0$ only at fixed α, β and tangent-director linear order about homogeneous $n = w$; quadratic/nonlinear routes remain parametric/Open; Fierz–Pauli not derived	A-math conditional negative result / Open physical spin-2 completion	Completion gate
Late-time source and coincidence	A two-slope scalar action supplies a regular late-de-Sitter proof of class; the observed source and coincidence history require the selected ECT action and state	A conditional / C benchmark / Open ECT	Reframing only
Vacuum-offset problem	Fixed orientation determinant is algebra only; a separate trace-free action is shift covariant through $V + C_I$	A inside supplied completion; ECT action map Open	Conditional mechanism
Observed late source in GR-reference units	$\rho_{\text{late},0}^{(\text{GRref})} = 3\Omega_{\text{late},0}^{(\text{GRref})} \tilde{M}_{\text{Pl}}^2 H_0^2$ by definition after metric/Planck calibration; model-native scalar–tensor bookkeeping also contains $F_0, \dot{F}_0$; the pseudo-Goldstone operator, mass, state and abundance remain Open and no scaling is selected	State/matching datum; Open ECT owner	Not an independent prediction
<i>Benchmarks, conditional outputs, external gates, and Open audits</i>			
Diósi–Penrose functional	External functional plus the scoped 51/160, 70.6 ms sharp-sphere orientation point; ECT source/state/noise chain Open	External A/C; ECT Open	No ECT coefficient or rate; the point is not a data test

Question / structural position	ECT route	Level	Scope / disposition
Protocol-defined fluctuation relation	Requires forward/reverse path measures and local detailed balance; no universal decoherence-exponent identity	Open	No ECT-specific law yet
No particle halo term in the HRC algebraic diagnostic	Property of the supplied conditional static test; identity-bridge ownership and cosmological particle content Open	C/Open	Diagnostic scope, not an ECT particle prediction
Late-time cosmological source	Requires an action-owned globally regular background, physical metric and joint distance–growth likelihood	Open	Test each named completion with BAO, supernova, chronometer, RSD and lensing data
Director-force completion	The fixed vector spurion supplies no source-generated force or Eötvös window; field/profile, body-charge, range and screening owners are Open	Open	MICROSCOPE constrains only a declared completion
External collisional–D–P equality scale	External hard-sphere/unit-coefficient formula retained conditionally; no numerical value or ECT identification	C/Open	Complete apparatus inputs plus ECT vertex/state/noise required

On the macroscopic side, HRC-0/HRC-3 are the active augmented-model response candidates; after the separate static gravity bridge they yield the BTFR slope 4 conditionally. Existing rotation-curve/RAR outputs are current HRC-only algebraic diagnostics governed by $\mu_{g,k}(g^2/a_M^2)g = g_N$. The matching $a_{M0} = cH_0/(2\pi)$, live P1–P6 ownership and particle content remain Open. The cosmological and strong-field programmes retain their separately declared status, while the cluster result is an exact restricted no-go: a strictly increasing local response cannot move an attained total maximizer away from the supplied total-baryon maximizers; collision morphology and residual-map extrema remain outside that theorem. None of these statements excludes collisionless or particle dark matter. Baryogenesis and late-time expansion remain separate owner-complete calculation programmes; neither currently supplies an ECT numerical likelihood or prediction. The preferred-direction operator is retained without a derived or predictive numerical coefficient, suppression scale or cross-species law. The optional H_{5m} free- ζ_{5f} diagonal ansatz is C/Open and non-identifiable. The formerly quoted spin-precession and Eötvös scales are withdrawn as outputs of the homogeneous vector term: it has no spin splitting and no source-generated force. External spin/WEP sensitivities become relevant only after a separate spin-dependent or body-force completion is declared. The multimessenger interval $-3 \times 10^{-15} \leq (c_{GW} - c_\gamma)/c_\gamma \leq +7 \times 10^{-16}$ from GW170817/GRB 170817A [12] is an external kill gate, not satisfied by construction: the present fixed director map supplies no physical TT principal symbol, photon metric or predicted speed ratio. The SPARC and representative Milky-Way statistics are HRC-only algebraic diagnostics; the complete field-level publication likelihood remains pending. The scale inventory separates the energy/matching quantities $\phi_0 \sim \bar{M}_{P1}$ and $v_2 \approx 246$ GeV from the conditional HRC matching length L_{gal} ; the latter is not a scale of one field and its ECT derivation remains Open. At the exploratory level it also suggests a structured particle sector including a longitudinal soft scalar coordinate χ . A derivative-only supplied truncation is massless, but the physical ECT mass operator is not identified and no physical or predicted pseudo-Goldstone mass is derived. The retained Hubble-frequency number is only a C/Open orientation

benchmark. The primary texture sector has integer homotopy labels but no derived finite particle or mass; a stabilising primary or secondary action remains Open. Named analogue source models (superfluid helium, BEC and nematic liquid crystals) realise selected ordered-medium mechanisms. They provide platforms on which a future action/state/operator/observable bridge could be tested, but presently neither support nor falsify the ECT $\text{SSB} \rightarrow \text{scalar-hyperbolicity} \rightarrow \text{coherent-branch}$ programme. Table 51 records external sensitivity proxies for a possible orientation extension. Cluster lensing and JWST growth do not currently probe an ECT orientation-stress coefficient because the action, projection, sign, and coefficient are Open (OP-c1).

Candidate organisation of interaction-sector programmes. The manuscript organises the interaction-sector programmes (Section 49.14): conditional Abelian bundle completion from a supplied compact phase; an Open chiral $SU(2)$ completion not selected by the vector stabiliser; and a colour sector with a restricted direct-injective embedding no-go inside the displayed minimal $O(4)$ algebra ($\dim \mathfrak{su}(3) > \dim \mathfrak{so}(4)$). This no-go does not exclude emergent/nonlinear, reducible/singlet, independent-bundle or defect-kernel routes; P7 is the adopted Level-B structural bundle rather than a theorem that this is the necessary or unique microscopic origin of physical colour. Its physical-colour completion remains C/Open. Together with the Open tensor/source and director-vertex programmes, these form a common architectural hypothesis rather than a derived pre-geometric origin of the known interactions. The full Standard-Model matching—matter representations, charge assignments, anomaly cancellation, Weinberg angle—remains an open programme (Table 115).

The only exclusions are the explicitly scoped mathematical no-go statements, such as the absence of a direct injective $\mathfrak{su}(3) \hookrightarrow \mathfrak{so}(4)$ map in the displayed minimal algebra. No broader class of physical completion is excluded.

50.2 What remains open

The following central outputs remain Level B or Open: the physical identification $S_0 = \hbar$ (OP-Q2); the completed observational Schrödinger equation; the full physical Dirac sector beyond the structural spinor route (OP-Q11); the full spin–statistics/Pauli theorem (OP-Q11); the Born rule as an unconditional bare-postulate theorem (OP-Q7); the full Bell-correlator derivation (OP-Q20); the protocol-independent information–decoherence bound beyond the present qubit closure (OP-PES-6); a channel-specific echo/recoherence programme; no metric-signature reopening or physical $O(3) \rightarrow O(4)$ backreaction map has been derived; the full operational Bell programme beyond the present toy visibility threshold; the constructive strong-field shell/interface mechanism needed for an explicit Page-curve computation (OP-Q13); the colour group $SU(3)$ and the electroweak scale v_2 (the adopted P7 $SU(3)$ -structured bundle is analysed in §40, while physical colour remains Open); the Yukawa hierarchy, three fermion generations, and $\alpha_{\text{fs}} = 1/137$ (structural routes identified—overlap/tunnelling suppression, spectral bound-state counting, and a conditional gauge-normalisation completion respectively—but explicit computations remain open; see §9.6, §9.6, §7.2, and Appendix AA); the primordial perturbation spectrum (n_s, A_s, r) without requiring a separate inflationary mechanism (the supplied pure-quartic strict-pullback kernel has an exact flat equal-slice limit when its combined κ_χ -kernel normalisation is scale-independent (separate constants suffice), and the full fixed-coefficient anisotropic tolerances are known; their stationary action/measure owner, physical covariance, complete transfer and observable n_s, A_s, r remain Open; §14.6); and the full nonlinear gravitational closure at the theorem level.

These directions should be interpreted as identified calculational programmes with concrete numerical targets, not as completed derivations. These limitations are real, mark the current frontier, and are catalogued in Section 49.10. At the present foundational benchmark level, one may organise a restricted scalar toy slice around ϕ_0 and λ after imposing the unit-slope coefficient benchmark $\hat{c}_* = 1$, i.e. $\alpha = 2\beta$ at $\beta = 1$. This count excludes the still-open clock/lapse, photon, tensor and gravity-bridge owners and must not be read as a count of all physical ECT inputs. This statement refers only to the foundational benchmark architecture: several higher-level closures and sector-specific implementations remain Level B

or programme-level. No quantitative compression claim is made until an owner-complete, like-for-like parameter census exists (see §49.8).

Proton stability as a conditional completion test. After a completion supplies a baryon current and operator selection rules, one may classify it by exact low-energy protection, protection only of $\Delta B = 1$, or allowed $\Delta B = 1$ violation. This is a generic logical taxonomy, not an ECT derivation. Under the additional Standard-Model-like, no-lighter-mediator and dimension-six assumptions, and on the branch $\Lambda_B > 0$, $C_B \neq 0$, the symbolic scaling $\tau_p \propto \Lambda_B^4 / (|C_B|^2 m_p^5)$ follows inside that external EFT. The $C_B = 0$ branch has no decay through this operator; ECT derives neither input nor the hadronic amplitude, so it supplies no predicted physical lifetime. The retained unit-coefficient NDA reference is only nonpredictive conditional arithmetic. The 2020 Super-Kamiokande bound $\tau_p/B > 2.4 \times 10^{34}$ yr at 90% C.L. [203] constrains specified violating completions. A confirmed $\Delta B = 1$ signal would exclude any named exact-protection completion in Patterns I or II, but not ECT; a null result tests only an owner-complete Pattern-III spectrum/operator/ matching choice and not that abstract class. Physical baryon number, the realised pattern, matching and branching fractions remain Open.

50.3 Final status

ECT is not a finished physical theory. The manuscript contains exact mathematics inside stated models, conditional reconstruction routes and a large set of Open physical owner chains; these categories must not be combined into a claim of achieved unification.

The status-controlled programme asks whether Euclidean microscopic data can be connected to an ordered branch, Lorentzian structure, gravitational dynamics, coherent phase kinematics, physical open-system channels, vacuum response, dark-sector logic, and horizon thermality. These are not yet one derived continuous chain: several links are independently supplied actions, states, or observable maps. The missing links are now localised, named, and technically testable.

50.4 Comprehensive status map: standard structures, conditional reconstructions, and open ECT programmes

One defining feature of the ECT programme is the breadth of the questions posed within one medium architecture. Only a limited subset of the listed structures is presently derived; many others are conditional reconstruction routes, supplied closure models, phenomenological diagnostics, or named Open programmes. The scalar condensate Φ , the postulates P1–P6 and the supplied $O(4) \rightarrow O(3)$ ordered branch therefore organise the comparison; they do not by themselves turn the entries below into consequences.

Table 119 collects the central cases of this derivational scope. It is organised into seven thematic groups: foundations and spacetime, quantum mechanics postulates, fields and particle statistics, the gauge sector, vacuum response and thermodynamics, macroscopic physics and cosmology, and long-standing mysteries and programme-level targets. Entries that are not reinterpretations of standard postulates or open problems while candidate-completion tests, benchmarks and observational handles are collected separately in Table 120. For each item the table records its conventional status in standard physics, the ECT origin, and the current Level A/B/C classification maintained throughout the paper. Cross-references point to the sections or appendices where the corresponding derivation, source-model calculation, benchmark audit, or missing-owner analysis appears.

Three reading guidelines apply throughout: (i) The table is deliberately not a list of ECT’s “solutions”: many entries are still at Level B or Level C, and this is preserved honestly rather than glossed. The value of the table lies in exposing the dependencies and missing owners of a single research programme, not in uniform strength of the individual entries. (ii) A Level A entry is exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. A Level B entry means a derivation that requires an additional matching step, closure assumption, or effective-field-theory truncation that is explicitly identified. Level B/C and Level C entries are conditional

on still-open completions of the gauge, flavour, or CAR programmes, and their inclusion reflects the existence of a structural route, not its full completion. (iii) The separate table mixes conditional results, external constraints, observational handles and Open ansatz tests. In particular, environment/history modulation of the galactic critical scale is an Open Ξ regression family, not an ECT-specific prediction. Each row must be interpreted at its stated level.

Table 119: Comprehensive status map of standard structures, conditional reconstructions and Open ECT programmes. This mixed-status table is not a claim of uniform strength, consequence or derivation. It is grouped by thematic sector. Status column records the conventional treatment in standard physics; ECT origin gives the structural mechanism in the present framework; Level records the current Level A/B/C classification; § gives the main reference within this paper. Entries at Level B/C and Level C are conditional on still-open completions explicitly identified in the text. The table is not a claim of uniform strength; its value is the range of items organised within a single status-and-dependency research programme. Candidate-completion tests, benchmarks and Open observational handles are collected separately in Table 120; their inclusion does not make them uniformly ECT-specific predictions.

Standard physics	Status there	ECT origin	Level	§
Group 1 — Foundations and spacetime				
Lorentzian signature $(-,+,+,+)$	Fundamental input	Hyperbolic classification of the supplied scalar principal tensor iff $\beta(\beta - \alpha) < 0$; healthy shorthand $\alpha > \beta > 0$ only for $\beta > 0$	A-math conditional on P4; physical metric Open	3
Time as a distinguished direction	Fundamental	P4 adopts an ordered coordinate w ; a physical clock and dynamical orientation-selection owner remain Open	postulate / physical map Open	3
Arrow of time	Open / separate completion	Conditional scalar retarded support and $\Gamma_{\text{irr}} \geq 0$ in a declared influence-functional model do not select one physical or thermodynamic arrow	Open	3.8
Second law / entropy production	Open / empirical	Conditional only for a named reduced channel and entropy functional under unitality/detailed balance/contractivity; ECT channel Open	Open	3.9
Unique kinetic tensor K^{AB} of the broken phase	Postulated form	Uniqueness theorem under the residual $O(3)$ symmetry: $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$	A	3
Speed of light c	Fundamental constant	$\hat{c}_*^2 = \beta/(\alpha - \beta)$ is the dimensionless scalar slope; $c_{\text{char}} = c_u \hat{c}_* / \mathcal{N}_\Phi$ and $c_{\text{char}} = c$ is the adopted scalar-clock calibration; physical photon/tensor cone equality remains Open	A-math / Level-B calibration / Open universality	3, 5.1

Standard (cont.)	Status	ECT origin	Level	§
Gravitational constant G_N	Fundamental constant	Standard tensor-EFT normalisation $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$; identifying $M_G = \bar{M}_{\text{Pl}}$ is external calibration, while $\kappa_n \rightarrow M_G^2$ is Open	A algebra / physical matching Open	13.4
Reduced Planck constant $\hbar$	Fundamental constant	Two supplied loop closures define distinct candidates; neither is the operational S_0 without an owner map, while $S_0 = \hbar$ is the adopted Level-B calibration	B calibration / Open microscopic owner and universality	5.6, 21.2
Universal causal cone	Postulate	One scalar characteristic cone is derived in its supplied model; matter, photon, fermion and tensor cones require independent actions and a common observable metric	Open beyond scalar subsector	12.6
Propagation corrections	Not guaranteed	No physical ECT photon or tensor operator, coefficient, suppression scale or common clock map is derived; no time-delay estimate follows	Open	3.6, 13.8
Group 2 — Quantum mechanics postulates				
Superposition principle	Axiom	Linearity inside a supplied linear scalar/coherent sector; physical state and measurement map Open	A mathematics / Open ECT	22.1
Complex Hilbert space structure	Axiom	Standard OS reconstruction for explicitly supplied free Gaussian measures after covariance, time reflection, RP, symmetry/regularity, locality/clustering and observable-algebra hypotheses; interacting ECT closure Open	A mathematics in stated free models / Open ECT	23.1
Imaginary unit i in Schrödinger eq.	Axiom	Formal complex-envelope recombination after the clock, branch and phase closures are supplied	B/Open	22.1

Standard (cont.)	Status	ECT origin	Level	§
Schrödinger equation	Axiom / field-theoretic postulate	Nonrelativistic recombination of a supplied scalar KG sector plus state, clock and S_0 matching	B/Open	22.1
Planck–Einstein relation $E = \hbar\omega$	Quantum postulate / empirical energy–frequency identification	In the Schrödinger-type coherent-branch reconstruction (Eq. (22.7)), coherent stationary eigenmodes with $\Psi \sim e^{-i\omega t}$ and $\hat{H}_{\text{coh}}\Psi = E\Psi$ give $E = S_0\omega$ up to the controlled corrections denoted by $\dots$; the Level-B matching $S_0 = \hbar$ then yields $E = \hbar\omega$	B	5.6, 21.2, 22.1
Stationary action $\delta S = 0$	Variational axiom	Closed-sector constrained extremum; open-sector mean dynamics from the retarded kernel; PES-R does not derive least action	A/B / Open	29
Canonical commutator $[\hat{q}, \hat{p}] = i\hbar$	Postulate	A compact global generator is discrete only after a physical shift orbit survives the gauge/Gauss quotient with finite nonzero symplectic form and a Weyl representation/self-adjoint quasiperiodic domain is supplied; compactness alone derives neither that rotor, its twist nor the S_0 normalisation	A operator mathematics under premises / B/Open ECT owner	23.3
Uncertainty principle $\Delta q \Delta p \geq \hbar/2$	From CCR axioms	Robertson inequality is standard A mathematics after operators are supplied; the ECT operator/ S_0 owner is Open	A mathematics / Open ECT	23.5

Standard (cont.)	Status	ECT origin	Level	§
Reflection positivity / OS reconstruction	Imposed on field theory	RP is verified for supplied free scalar Gaussian measures; the standard reconstruction applies only with their complete applicable corrected OS-II hypothesis package, while physical interacting ECT identification remains Open	A mathematics in stated free models / Open ECT	23.1
Global unitarity (Gaussian regime)	Postulate	Standard consequence only for those supplied free Gaussian measures satisfying the complete applicable corrected OS-II package; not a consequence of RP alone and not theory-wide ECT unitarity	A mathematics in stated free models / Open ECT	23.2
Wavefunction collapse	Postulate	Reduced-state dephasing and record stabilisation in specified channels; pointer basis, unique outcome, and update rule remain Open	Open	30.1
Born rule $P = \psi ^2$	Postulate	A complete applicable corrected OS-II package for a supplied Gaussian model reconstructs a positive quadratic norm. Independently, ordinary projector Gleason gives a unique trace representation only for a supplied normalised, noncontextual, countably additive projector measure on a complex Hilbert space with $\dim \mathcal{H} \geq 3$; a qubit requires the Busch effect/POVM theorem under separately declared C2q hypotheses, and (C3) is required for physical application. OS-II and dephasing are not Gleason premises	Level A mathematics under full theorem hypotheses; Open for ECT state/preparation, event owner, probability origin, outcome and update	30.4

Standard (cont.)	Status	ECT origin	Level	§
Quantum tunnelling / interference	Wave-mechanical postulate	Two-sided dependence is exact for a supplied elliptic BVP but is neither sufficient nor discriminating. Interference additionally requires an admitted quantum measure/coherent structure, composable kernel, state/preparation and observable map; tunnelling requires a named action/saddle, state, continuation, prefactor and detector map. No ECT-specific amplitude or rate is derived	A BVP mathematics / A external WKB / B route / Open physical ECT	24.1, 24.3
Quantum entanglement	Interpretational puzzle	Non-factorisable correlations follow only after a non-factorisable state/channel is supplied; a common medium alone is insufficient	A conditional mathematics / Open ECT	31
Discrete energy levels	Boundary-condition consequence	Standard spectrum of a supplied sector operator/domain; ECT operator and physical map Open, and PES-R only classifies retention	A conditional mathematics / Open	29
Group 3 — Fields, particles, and statistics				
Klein–Gordon equation	Supplied field-theoretic EFT	Standard KG form of the supplied scalar operator after the declared real common-length parametrisation and clock/lapse map; the ordered-phase action, mass/domain and physical mode owner are not derived from P1–P6	A-math conditional / Open ECT owner	3, 22.1
Clifford algebra $\{\gamma_L^\mu, \gamma_L^\nu\} = -2g^{\mu\nu}$	Supplied for fermions	Standard matrix representation after a physical metric/tetrad and spinor bundle are supplied	A-math conditional / Open ECT	24.5
Spinor representation	Supplied for fermionic fields	The double-cover algebra admits spinors; physical ECT realisation remains Open	A algebra / Open physical	9, 24.5

Standard (cont.)	Status	ECT origin	Level	§
Dirac equation	Supplied fermionic dynamics	Unique only in the declared four-component irreducible Clifford-factorisation class; reducible, gauge/constrained, DKP/Maxwell and non-Clifford first-order systems are outside the theorem	A-math conditional / Open physical owner	9.2, 24.5
Exchange topology $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ in supplied defect-free $\mathbb{R}^3$	Configuration-space theorem	For two supplied identical unlabeled point excitations the theorem gives loop classes, not a physical P4 spatial slice or statistics selection	A topology conditional / Open physical ECT	24.4
Pauli exclusion principle	Postulate	Conditional canonical-normalisation no-go only after spinor fields and the CAR algebra are supplied; ECT origin of physical fermions/CAR/Pauli remains Open	Open for ECT / A-math conditional	24.4, AT
Fermi–Dirac / Bose–Einstein statistics split	Spin–statistics theorem with assumptions	Exchange topology and a supplied free spinor/CAR/OS kernel give a conditional consistency check; physical interacting spin–statistics theorem remains Open	Open / conditional	24.4, 9.9
Spin discreteness / Stern–Gerlach records	From $SU(2)$ representation theory	Compact $SU(2)$ representation content after a physical spatial-rotation group and spinor sector are supplied; their ECT ownership remains Open; apparatus dephasing and record formation require a declared channel, while unique-outcome selection remains Open	A-math conditional / B / Open	9, 24.5
Physical graviton	Fundamental/quantised in GR treatments	Open ECT tensor completion; no linear TT/static-density vertex at fixed α, β and tangent-director order; quadratic/nonlinear routes remain parametric/Open	A-math conditional negative result / Open physical spin-2 completion	8.5

Standard (cont.)	Status	ECT origin	Level	§
Absence of continuous Abelian braid phases in the supplied $d_{\text{spatial}} = 3$ point model	Not formulated	$\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ for two supplied identical unlabeled points; the physical spatial/excitation map, representation selection, CAR and higher-dimensional S_N sectors remain Open	A topology conditional / Open physical statistics	24.4
Group 4 — Gauge sector				
Photon as $U(1)$ gauge boson	Postulated gauge principle	Compact phase gives global Noether algebra; a local bundle, positive reduced Hessian, unbroken transverse photon, charge/matter map and metric remain Open	conditional/Open	7.2
Local $U(1)$ gauge invariance	Postulate	Global phase/winding algebra is exact after BR1; local bundle, gauge quotient, connection, photon and charge map remain Open	Open beyond global algebra	7.2
$SU(2)_L$ electroweak gauge sector	Postulated	$\mathfrak{so}(3) \cong \mathfrak{su}(2)$ is algebra only; local weak bundle and chirality Open	A algebra / Open physical	7.4
$W^\pm, Z^0$ gauge bosons	Postulated as EW gauge fields	Supplied electroweak completion; no P1–P6 descent	C/Open	7.4
Higgs boson	Postulated scalar	Radial mode of the $SU(2)$ condensate in the electroweak embedding	C	7.4
Yukawa couplings	Phenomenological inputs	Structural route via fermion–condensate couplings; hierarchy open	B–C	9
Group 5 — Vacuum response and thermodynamics				
Scalar Dirichlet Casimir BVP	Standard scalar QFT calculation	Exact only inside the supplied N -scalar toy model; ECT bundle, boundary map and force Open	A mathematics / Open ECT	25.3
ECT Casimir coefficient	Not present	No coefficient or $3/2$ ratio is predicted	Open	25.3
Unruh temperature $T_U = \hbar a / (2\pi c k_B)$	Standard Lorentzian QFT	Standard Rindler/KMS target; physical ECT state, cone and detector map Open	A external / Open ECT	25.4

Standard (cont.)	Status	ECT origin	Level	§
Hawking radiation	Semi-classical GR + QFT	Standard target; physical ECT tensor geometry, state and strong-field response Open	A external / Open ECT	36
Bogoliubov mode mixing on curved backgrounds	Standard technique	Standard transformation after a time-dependent quadratic action and in/out states are supplied; ECT transition history, state, particle map and rates Open	A external / Open ECT	25.5
Information preservation in black holes	Open problem	Reduced-state thermal-ity is compatible with global purity; a globally unitary interacting strong-field completion and Page curve are not derived	A mathematics / Open ECT	36
Crooks fluctuation theorem	Classical stat-mech result	Open ECT construction requiring forward/reverse path measures and local detailed balance; not a decoherence-exponent identity	Open	26.4
Decoherence as environment-induced effect	Effective description	Standard reduced-state result for a supplied split, vertex, state and kernel; ECT owners remain Open	B mathematics / Open ECT	26.1
Group 6 — Macroscopic physics and cosmology				
Einstein equations $G_{\mu\nu} = 8\pi GT_{\mu\nu}$	Fundamental dynamics in GR	Conditional scalar-tensor/Einstein completion; physical ECT tensor and universal source Open	B/Open	11
Candidate anisotropic curvature extension $\mathcal{X}_{\mu\nu}^{\text{cand}}[n]$	Absent in GR	Requires a separate covariant orientation action and physical projection	Open / proxy	AC
Spatial-flatness branch $k=0$	Observational input	The physical cosmological metric/embedding remains Open	B / Open	14.1
Primordial monopole question	Requires inflationary dilution in standard histories	$\pi_2(S^3) = 0$ only for the supplied target; physical vacuum/relic conclusion Open	A-math conditional / Open ECT	8.10
Primordial wall/string question	Requires absence of suitable physical π_0, π_1	Supplied S^3 target has $\pi_0 = \pi_1 = 0$; physical configuration owner Open	A-math conditional / Open ECT	8.10

Standard (cont.)	Status	ECT origin	Level	§
Bare scalar-ratio bound	Not an observable bound	$w_{\phi}^{\text{bare}} \geq -1$ for the supplied positive-kinetic bare pieces; non-minimal terms leave physical $w_{\text{eff}}(a)$ unconstrained	A in model / Open ECT	16.1
Late-time accelerated expansion	Cosmological constant Λ postulated	A chosen scalar potential supplies a proof-of-class only; ECT source, state and solution Open	Open / conditional	16.1
Dark matter as a separate particle substance (galactic scale)	Postulated halo with particle content	HRC candidates contain no particle-halo term; particle content and gravity ownership Open	B/C/Open	17, 18.2
Galactic rotation curves	Require particle dark matter halo	HRC-0/HRC-3 active candidates with re-computed algebraic diagnostics; full disk solve pending	B/C/Open	17
BTFR slope = 4	Empirical fit	Exact conditional algebra for either HRC deep branch after the separate gravity bridge	B / bridge Open	17.1
RAR and baseline acceleration $a_{M0} \approx cH_0/(2\pi)$	Empirical regularity	HRC augmented-model response plus matched present scale; live ownership and environment law Open	C/Open	17.1
Baryon asymmetry η_B	Sakharov conditions + specific model	Conditional owner inventory only; no ECT baryon vertex, transition state, transport solution or numerical result	Open	16.3
Local cluster-response no-go	Requires physical collision/lensing modelling	A strictly increasing local transform preserves the set-valued argmax and, when attained, cannot move a total-response maximizer relative to the supplied total-baryon maximizers; inherited gas/BCG offsets and residual extrema remain outside this theorem	A algebra / physical Open	17.2
Lensing–growth sector (S_8 , cosmic shear, CMB lensing)	Probe-dependent discrepancy / modified-gravity test sector	Gauge-reduced perturbation and photometric owners required; the current quasistatic two-slope diagnostic is conditional only	C/Open	13.5, Appendix AI, OP- c_1

Standard (cont.)	Status	ECT origin	Level	§
Group 7 — Long-standing mysteries and programme-level targets				
Proton stability	Symmetry-protected empirical fact	Generic selection-rule taxonomy only; the dimension-six $\tau_p \propto \Lambda_B^4/(C_B ^2 m_p^5)$ scaling is conditional; no ECT matching scale, coefficient, hadronic amplitude or lifetime is derived	Open	44
Three fermion generations	Empirical input	A supplied spectral well can be tuned to three states; the fermion operator, background, boundary data and physical family map are Open	C/Open	9
Fine-structure constant $\alpha_{\text{fs}} = 1/137$	Fundamental constant	Conditional identity $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$ for a supplied normalised photon sector; ECT derives neither that sector nor an electroweak/colour matching relation	Open	7.2
Neutrino mass operators and scales	Empirical completion target	Standard Weinberg/seesaw algebra may be supplied; coefficients, heavy scale and ECT owner are Open. Combinations of (ϕ_0, ν_2) remain unselected ansätze; fully declared unit-coefficient orientation points and inverse fits are Level-C benchmarks, not inferred masses	C/Open	8.9
Quantum gravity (UV completion)	Open problem	Candidate condensate orientation/tensor route; physical mode reduction, canonical normalisation, RP/OS reconstruction, and equality of action scales remain Open; PES-R supplies no quantisation	Open	8.5, 29

Standard (cont.)	Status	ECT origin	Level	§
Measurement problem	Open problem	Reduced-state dephasing is separate from the ordinary Gleason representation of an independently supplied projector measure; state/preparation, event, probability origin, pointer basis, unique outcome and update remain Open	Open	30.1, 29, 30.4
Particle stability (generic)	Empirical / symmetry-protected	Owned by sector symmetries, vertices, self-energies, phase space, and physical widths; PES-R only classifies retention after those inputs exist	Open programme	29
Cosmological-constant problem (10^{120} UV–IR mismatch)	Catastrophic fine-tuning in the standard framework	The orientation tensor has an exact fixed determinant but does not derive vacuum decoupling. An explicit trace-free completion is classically shift covariant through $V + C_I$, where C_I may be a measured global state/boundary datum; its ECT descent and radiative completion are Open. A separate two-slope action gives a regular late background but is not additive-shift invariant	Orientation algebra: A; supplied completions: A; ECT action/metric map: Open	15
Constancy of dimensionless constants (α, μ) over cosmic time	Empirical requirement	Requires the complete matter/gauge coupling, cosmological state and protected observable map; no such protection is presently derived	Open	16.5

Summary and reading of the table. From a Euclidean scalar architecture plus supplied ordered-branch and completion inputs, ECT derives several scoped algebraic and conditional results. It does not yet derive a physical clock, universal Lorentzian metric, Einstein tensor sector, gauge/matter sector or probability law. The candidate geometric programme includes a possible preferred-direction extension whose covariant action and stress remain Open, the Klein–Gordon and Schrödinger limits of the supplied scalar closure, a candidate canonical-commutator programme, standard uncertainty and linear-superposition mathematics after their operators are supplied, a Gaussian OS reconstruction route, and a conditional probability-measure representation via Gleason’s theorem, the conditional point-excitation topology $\pi_1(\mathcal{M}_2) = \mathbb{Z}_2$ and its two loop classes inside the declared defect-free $\mathbb{R}^3$ point model, without a derived physical slice, excitation map or statistics representation, and a conditional canonical-normalisation no-go after

a CAR algebra is supplied, not a derivation of Pauli exclusion, the representation-theoretic route to the Dirac structure, the Open Casimir/Unruh/particle-production/horizon common-response programme, the conditional global-isometry identity with physical black-hole unitarity still Open, the ϕ -first late-time closure with its dimensionally normalised bare scalar-ratio identity but no generic bound on observable w_{eff} , a Level-C HRC no-halo algebraic diagnostic for selected rotation-curve/BTFR/RAR benchmarks with ECT matching and particle content Open, and an exact peak-preservation identity inside the supplied monotone map, with the physical cluster response and morphology Open. It also provides structural routes into long-standing open problems (proton stability, three fermion generations, the fine-structure constant, neutrino mass ownership, quantum gravity, and the measurement problem) without claiming that any of those routes is fully closed at the present stage. The inclusion of quantum gravity in this map should be read in the same disciplined sense as the other B/C entries: ECT supplies a unified structural route and a common condensate architecture, not a completed ultraviolet closure.

The value of Table 119 is not in uniform strength of the entries—several are explicitly listed as Level B or Level C, and this is preserved honestly—but in the explicit map of claims that the one-medium organising hypothesis asks a future physical completion to connect. The present manuscript does not establish one common action/state/observable owner for those structures.

Candidate-completion tests, benchmarks, and observational handles

Table 119 collects the cases in which ECT reinterprets structures already present in standard physics. A complementary question is: what candidate-completion tests, benchmarks and observational handles remain once the mixed-status structures of Table 119 are set aside? They are collected in Table 120. They have mixed status and are not uniformly ECT predictions. A row becomes an ECT falsifiability channel only after its named completion owns the action, state, observable and uncertainty model. Some rows would directly exclude the corresponding ECT structural closure if observationally refuted; others constrain a closure parameter, an EFT matching coefficient, or an ansatz-level completion. The status of each entry should therefore be read together with the Level column and the explanatory paragraph below the table.

A more detailed observational test map, with target experiments and verdict-on-falsification entries, was already given in Table 71 of Part II. The present table provides a compact, status-tagged summary organised by physical sector rather than by observational channel; inclusion does not establish that an entry is ECT-specific.

Table 120: Status-tagged statements, benchmarks, candidate routes, and observational handles beyond the re-interpretive map of Table 119. The rows do not have uniform predictive status. The logical status of each entry is indicated by the Level column and should be read together with the explanatory paragraph below the table. Detailed observational targets and falsifier verdicts are recorded in Table 71.

Statement / benchmark / handle	Content	Observable channel	Level
Preferred-direction vector operator	Allowed restricted-basis vector-current operator; its coefficient may vanish, and the homogeneous timelike term has neither spin splitting nor a source-generated body force. Axial/spin and inhomogeneous force completions are independent and Open	Relative multi-species or gradient observables only after owner closure; not a MICROSCOPE or spin prediction by itself	A-math conditional / Open observable
Environment/history dependence of $a_{M,\text{eff}}$	Exploratory law $a_{M,\text{eff}} = a_{M0}\Xi(\rho_{\text{env}}, \bar{q}_{\text{env}}, \dots)$; no sign or amplitude derived	SPARC $\times$ 2MRS/SDSS cross-correlation	Open

Statement / benchmark / handle (cont.)	Content	Channel	Level
Primary texture stability and mass	$\pi_3(S^3)$ supplies integer labels, while $E_2(R) \propto R$ collapses; finite objects require a derived stabilising action	Soliton/collider searches only after an action-owned spectrum exists	A negative result / mass Open
Graviton polarisation: two TT modes in the completed linear closure	If the induced effective-metric sector closes exactly to the Fierz–Pauli form, only the two transverse-traceless tensor polarisations propagate; exact TT projection and ghost-freedom remain open verification tasks	Polarisation decomposition of stochastic GW background	B/Open
Photon/tensor propagation corrections	No ECT photon or tensor LIV operator, coefficient, suppression scale or common clock map is derived; no time-delay estimate is retained	Multi-messenger photon/GW/neutrino timing as an external completion gate	Open
Complete-final-state distinguishability under a supplied isometry	Only a state-erasing complete final channel is excluded; thermal one-body marginals and Page-curve shape are not fixed	Full final radiation channel (conceptual)	A mathematics / Open ECT
ECT Hubble inference	Requires one background, recombination, photon metric, sound horizon and late-distance likelihood; no shift currently derived	CMB, standard sirens and distance ladder	Open
<i>Additional structural and observational handles</i>			
External collisional–D–P equality scale	Conditional $R_{\text{cross}}^{\text{coll-DP}}$ and $m_{\text{cross}}^{\text{coll-DP}}$ formulae retained for the explicitly supplied external source models; no numerical application	Complete apparatus inputs and an ECT record channel required	External algebra / ECT Open
Divided matter coefficient $G_{\text{div}}(\phi)$	$G_{\text{div}} = G_* e^{-\beta_\phi \phi}$ is exact after complete division inside the supplied ansatz; a measured local or cosmological gravitational response requires additional transfer data	Cavendish/PPN/LLR, growth and distance observables after the response map is supplied	A algebra in B ansatz / response Open
Force-law dimensionality diagnostic $d_{\text{force}} : 3 \rightarrow 7/3 \rightarrow 2$	Point-source algebra only inside the conditional HRC-0 point-source response $\mu_{g,0}$	Extended-galaxy extraction requires measured $p_N(r)$ and is not performed	C / Open test
Redshift dependence of fitted $a_M(z)$	Exploratory test of whether one matched baseline remains adequate; no microscopic $\phi \rightarrow a_M$ transfer law derived	JWST, SKA high- z kinematic surveys	Open
Conditional constant-background GR-form equation / scalar-screening test	The metric background is algebraically GR-form if the scalar is constant; physical GR additionally requires a scalar root and a decoupled, short-range or screened finite-body response	Cassini, MICROSCOPE, LLR	Open
Recoherence under control protocols	Echo and record-erasure protocols can restore reduced-state coherence in specified channels; no physical $O(3) \rightarrow O(4)$ backreaction map has been derived	Echo and recoherence protocols in quantum optics	Standard channel result / ECT Open

Statement / benchmark / handle (cont.)	Content	Channel	Level
Indefinite causal order compatibility	Quantum-switch/process-matrix protocols can be represented within coherent quantum dynamics; no ECT-specific derivation or universal Φ degradation law is supplied	Quantum-switch experiments (Procopio–Walther)	Open as an ECT discriminator
JWST abundance and morphology	Require one action-owned background, perturbation, halo, baryonic and survey-selection forward model; no active ECT abundance or morphology prediction	JWST spectroscopic and imaging surveys at $z > 6$	Open
Candidate orientation-sector Poisson proxy	A separately specified n^A action may generate slip and modified-growth functions; the displayed scalar action does not fix this term	Euclid, Rubin, DESI, and CMB lensing after action/projection closure	Open / proxy

Reading the mixed candidate-test table. The entries in Table 120 are not of uniform maturity. Some entries are structural routes rather than coefficient-level predictions. In particular, photon-sector LIV remains Open: Fermi–LAT constrains declared external dispersion and source-lag models, and becomes an ECT gate only after an ECT photon operator, coefficient, suppression scale, clock and likelihood map are derived. The absence of extra graviton polarisations is tied to completion of the linear Fierz–Pauli closure. Other entries, such as the preferred-direction fermion operator are Open coefficient/observable classes that can be constrained only after the relevant physical channels are derived. The environment/history dependence of $a_{M,\text{eff}}$ is an Open test of the exploratory law Ξ . The combination $\gamma 4\pi\rho/U_0''$ does not follow from the scalar equation. A null result constrains the tested Ξ ansatz, while a detected correlation does not by itself establish the scalar source/screening chain or ECT. No numerical topological mass ansatz remains active: the physical texture spectrum is Open until a stabilising action, state, mode analysis and observable map are supplied. The table is therefore a status-separated exposure map. A row becomes an ECT test only after its named action, state, observable and uncertainty owners are closed.

Closing remark on the two tables. Taken together, Tables 119 and 120 summarise the two complementary aspects of the ECT programme: the derivational scope that relocates a broad range of standard postulates and open problems into a single condensate-based framework, and a mixed set of candidate tests, benchmarks and Open observational handles through which completed sectors could become empirically answerable. Neither is a claim of a finished theory: several entries in both tables are still explicitly at Level B, Level B/C, or Level C, and the corresponding open problems are tracked in Section 38.1 and in the Part-specific status tables. What the two tables record is how this manuscript uses one underlying object—the scalar condensate Φ on a four-dimensional Euclidean manifold, together with its supplied nonzero-gradient chart—to organise a wide conditional status-and-dependency map and a status-separated experimental-exposure programme. This organisation is not evidence that the Open physical owners exist.

Appendices

A Scalar-closure constraints and conditional response programmes

Status: current constraint audit. The generic scalar variation identities are exact inside the displayed ansatz. Heavy-amplitude no-go conditions, the provisional Δ_K benchmark, the physical coordinate, the critical fixed point, the AQUAL source/force map and the normalisation have separate statuses stated below. No route is selected by another route's failure.

A.1 Purpose and scope of this appendix

This appendix separates four current logical objects: a heavy-amplitude profile constraint, the anisotropy coordinate ψ_K , a coordinate-neutral scalar audit q , and a separately supplied conditional HRC/AQUAL candidate. In particular, $\phi_u = \beta_\phi^{-1} \ln(u/u_\infty)$ and $\psi_K = \ln(\Delta_K/\Delta_{K,\text{vac}})$ are not interchangeable; no $u \leftrightarrow \Delta_K$ map is known.

At each stage exact ansatz algebra, provisional benchmarks, Level-C phenomenology and Open ECT matching are kept distinct.

A.2 Basic physical premises

The ECT framework begins from two physical ideas that together set it apart from other approaches to emergent gravity.

Premise 1: conditional scalar Lorentzianness. P4 supplies an ordered direction, while the scalar principal tensor is Lorentzian/hyperbolic only for independently supplied coefficients $\beta(\beta - \alpha) < 0$; on the retained $\beta > 0$ component this is $\alpha > \beta > 0$. Defining $\Delta_K = \alpha - \beta$ is coefficient bookkeeping; it does not show that symmetry breaking selects $\Delta_K > 0$, that $\Delta_K = 0$ is the physical symmetric phase, or that the scalar tensor is the observable spacetime metric. Those branch, clock and metric maps remain Open.

Open matter-to-ordering map. The scalar calculation proves only a conditional monotonic response of the coupling function $f(q)$ at a stable homogeneous root. It neither identifies q with Δ_K nor fixes the sign of a physical Δ_K response. The matter-to-Lorentzian-order map is therefore Open, not an additional result of the present action. Any proposed completion must derive this map before it can be used for galactic dynamics.

A.3 Heavy scalar-amplitude exterior-tail constraint

This subsection tests a mass-dimension radial proxy denoted $u_{\text{prox}}(r)$. The displayed linear equation fixes the transfer function and, for a compact source, the exterior Yukawa range. It does not forbid a smooth profile driven by a spatially extended source and does not select ϕ_u , ψ_K , or the supplied HRC/AQUAL functional.

Throughout this Branch 1 discussion $u_{\text{prox}}(r)$ denotes a generic ordered-chart scalar-amplitude proxy with dimension of mass, used only in the conditional no-go calculation below. The notation is deliberately distinct from the global gradient datum $u_0 = |\langle \partial_A \Phi \rangle|$, which has dimension GeV^2 , and from the homogeneous-P3 scalar coordinate ϕ_0 .

In this appendix it is important to distinguish the asymptotic proxy amplitude $u_{\text{prox},\infty}$ from the bare-potential vacuum value $\phi_\infty \equiv \mu/\sqrt{\lambda}$. The tested route varies $u_{\text{prox}}(r)$; the quartic-potential scale ϕ_∞ enters only as the microscopic mass-setting scale in the simplest realisation.

A.3.1 No-go setup and scope

The local u_{prox} is a supplied radial proxy with dimensions different from the main gradient amplitude. Its compact-source exterior range can be stated symbolically, but ECT supplies no owned matching from the homogeneous-P3 curvature to a physical galactic mediator. The calculation also cannot establish that χ represents a physical soft ECT mode, identify it with ϕ_u , or derive the S→G/HRC gravity bridge. The decisive Yukawa-range calculation and supporting consistency checks are stated directly below.

A.3.2 Argument 1 (conditional): compact-source exterior-range theorem

For this named test, $u_{\text{prox}}(r)$ is a supplied mass-dimension amplitude proxy; it is not thereby the P4 radial mode or a derived condensate particle. The theorem leaves its mass symbolic; one separately declared unit benchmark is quoted below only as an exterior-range negative control. In the simplest quartic parametrisation, the underlying bare scalar potential $V(\Phi) = -\mu^2\Phi^2/2 + \lambda\Phi^4/4$ has homogeneous minimum coordinate

$$\phi_\infty \equiv \frac{\mu}{\sqrt{\lambda}}. \quad (\text{A.1})$$

To test the amplitude-route hypothesis, write the ordered-branch amplitude as

$$u_{\text{prox}}(r) = u_{\text{prox},\infty} + \delta u_{\text{prox}}(r), \quad (\text{A.2})$$

where $u_{\text{prox},\infty}$ denotes the asymptotic value of the proxy. Supply a static three-dimensional physical-space reduction with constant $\beta > 0$. Linearisation about the reference state then gives

$$-\beta \nabla^2(\delta u_{\text{prox}}) + m_\sigma^2 \delta u_{\text{prox}} = J(r), \quad m_\sigma^2 = 2\lambda \phi_\infty^2, \quad (\text{A.3})$$

where $J(r)$ is the still-unowned matter-to-amplitude source. This static three-dimensional Green function must not be confused with the separate four-dimensional Euclidean free correlator of Appendix P.

Assume $\beta > 0$ and define

$$\kappa_\sigma \equiv \frac{m_\sigma}{\sqrt{\beta}}, \quad \xi_\sigma \equiv \kappa_\sigma^{-1} = \frac{\sqrt{\beta}}{m_\sigma}.$$

Fourier transformation gives the exact linear transfer function

$$\delta u_{\text{prox}}(\mathbf{k}) = \frac{J(\mathbf{k})}{m_\sigma^2 + \beta \mathbf{k}^2}. \quad (\text{A.4})$$

Equivalently, the static three-dimensional solution with decaying boundary condition is

$$\delta u_{\text{prox}}(\mathbf{x}) = \int d^3x' \frac{e^{-\kappa_\sigma |\mathbf{x}-\mathbf{x}'|}}{4\pi\beta |\mathbf{x}-\mathbf{x}'|} J(\mathbf{x}').$$

Consequently an extended source can drive an extended smooth profile. In the long-wavelength regime $|\mathbf{k}| \ll \kappa_\sigma$ the response is local and suppressed,

$$\delta u_{\text{prox}}(\mathbf{k}) = \frac{J(\mathbf{k})}{m_\sigma^2} \left[1 - \frac{\beta \mathbf{k}^2}{m_\sigma^2} + \dots \right],$$

rather than being absent.

An exact negative control makes the source dependence explicit: for any smooth profile $h(\mathbf{x})$ with the chosen boundary behaviour, selecting $J = (-\beta \nabla^2 + m_\sigma^2)h$ gives the exact solution $\delta u_{\text{prox}} = h$. The field equation alone therefore cannot prohibit the spatial scale of an arbitrary source-driven profile.

A universal Yukawa statement follows only outside the support of a compact source. There $J = 0$, and the spherically symmetric exterior mode is

$$\delta u_{\text{prox,ext}}(r) = A \frac{e^{-\kappa_\sigma r}}{r}, \quad (\text{A.5})$$

with amplitude A fixed by the source and boundary data. Higher multipoles have the same exponential scale. Thus $\xi_\sigma = \sqrt{\beta}/m_\sigma$ is the range of the unsourced exterior tail, not a theorem about the spatial support of an arbitrary driven solution.

An exterior range of at least L therefore requires

$$\frac{m_\sigma}{\sqrt{\beta}} \lesssim L^{-1}.$$

This is a necessary symbolic gate, not a numerical ECT verdict. A useful negative-control point can nevertheless be declared by supplying

$$\phi_\infty^{\text{bench}} = \bar{M}_{\text{Pl}} = 2.4353 \times 10^{18} \text{ GeV}, \quad \lambda^{\text{bench}} = 10^{-2}, \quad \beta^{\text{bench}} = 1. \quad (\text{A.6})$$

Then

$$\begin{aligned} m_\sigma^{\text{bench}} &= \sqrt{2\lambda^{\text{bench}}} \phi_\infty^{\text{bench}} = 3.444 \times 10^{17} \text{ GeV}, \\ \xi_\sigma^{\text{bench}} &= \frac{\sqrt{\beta^{\text{bench}}} \hbar c}{m_\sigma^{\text{bench}}} = 5.730 \times 10^{-34} \text{ m}, \\ E_{\text{kpc}} &:= \frac{\hbar c}{1 \text{ kpc}} = 6.3949 \times 10^{-36} \text{ GeV}, \quad \frac{m_\sigma^{\text{bench}}/\sqrt{\beta^{\text{bench}}}}{E_{\text{kpc}}} = 5.386 \times 10^{52}. \end{aligned} \quad (\text{A.7})$$

The last ratio spans 52.731 decades. This Level C arithmetic excludes only the displayed P3-quartic/unit-gradient proxy point as an *unsourced exterior* kpc-range mediator. It is not a P4-pole identification, a galactic fit or an ECT prediction. It does not exclude an extended response driven by $J(r)$, another value of λ or β , or a distinct action-owned P4 mode. Applying the gate physically still requires an owned identification of m_σ , β , the length, source vertex and metric map.

A.3.3 Argument 2: a profile is not yet a gravity law

If one separately identifies this amplitude with the scalar–tensor coordinate and supplies $F = u/u_\infty$, division of the complete metric equation gives

$$G_{\text{div}} = \frac{G_*}{F} = G_* \frac{u_\infty}{u}, \quad \frac{\delta G_{\text{div}}}{G_{\text{div}}} = -\frac{\delta u}{u_\infty} + O(\delta u^2).$$

The older u^{-2} scaling is not the current closure. The corrected linear relation is only divided-equation bookkeeping and does not determine a Cavendish, finite-body or galactic law: Eq. (A.4) still needs an action-owned J , boundary data, canonical normalisation and the solved physical metric potentials. A hand-selected profile cannot replace those inputs, but the field equation also does not rule out every extended sourced profile.

A.3.4 Argument 3: conditional fifth-force gate

If the exterior pole is made light and the same field is also identified with a scalar–tensor mediator, its force strength is controlled by the canonical matter coupling, not by its mass alone. For the generic closure of Branch 3,

$$g_{\phi T} = -\frac{a(q)}{2\sqrt{f_0}\sqrt{K_E(q)}}, \quad K_E = \frac{K}{f} + \frac{3}{2}a^2.$$

Thus an order-unity fifth force is not automatic: it depends on a , the kinetic normalisation, the matter coupling, the scalar mass and any nonlinear finite-body screening. Cassini constrains an unscreened sufficiently long-range universal conformal coupling, while MICROSCOPE constrains relevant composition-dependent couplings [95, 49]. Neither experiment supplies a bound on an unowned ECT vertex.

The recovered result is therefore a pair of conditional gates. A proposed proxy completion must first satisfy the exterior-range condition $m_\sigma/\sqrt{\beta} \lesssim L^{-1}$ and must then pass a canonical finite-body coupling/PPN/equivalence-principle calculation. No current ECT parameter matching evaluates the first gate, and a proposed light completion cannot be evaluated without the second. This test derives neither a P4 pole nor a P4 ultraviolet threshold.

A.3.5 Auxiliary sub-routes and their actual status

Five nearby ideas can be retained as explicit tests, but the old numerical rejections did not follow from displayed owners:

(a) *Soft orientation sector*. In the scalar-only basis $Q_A = \partial_A \Phi$, integrability reduces the linear soft parametrisation to one longitudinal coordinate χ . A separately supplied massless local scalar with a point-source vertex gives a $1/r$ potential and a $1/r^2$ force, the same radial power as Newtonian gravity; it does not by itself yield the $1/r$ acceleration needed for flat rotation curves. Neither the physical ECT mode nor its source vertex follows from integrability.

(b) *Brans–Dicke limit*. The response is governed by the canonically normalised coupling and boundary solution, not by a universal estimate $\delta G/G \sim r_S/r$. The supplied minimal benchmark is tested separately in Branch 2, and the generic coupling is displayed in Branch 3.

(c) *RG running of $\lambda(\mu)$* . No numerical change in λ can be inferred without the complete field content, beta function, thresholds and matching conditions. Whether running can alter the pole is Open.

(d) *Running of $\beta(\mu)$* . The coefficient and its renormalisation owner are likewise unspecified; the former quantitative estimate is withdrawn rather than used as evidence.

(e) *Nonlinear response*. A nonlinear completion can screen, amplify or reshape a sourced profile. Its effect requires the full nonlinear boundary-value problem and cannot be excluded from the linear amplitude estimate alone.

A.3.6 Conclusion of the supplied heavy-proxy test

For the supplied linear equation, the exact proxy response is $\delta u_{\text{prox}}(\mathbf{k}) = J(\mathbf{k})/(m_\sigma^2 + \beta \mathbf{k}^2)$ and a compact source has an exterior range $\xi_\sigma = \sqrt{\beta}/m_\sigma$. This is a symbolic necessary range gate, not a numerical ECT exclusion or a derived P4 radial-particle or UV-threshold result. An extended source can nevertheless drive an extended, locally suppressed profile; its magnitude and gravitational observability remain Open until the source, normalisation and metric map are owned. No amplitude coordinate, Δ_K branch or HRC functional is selected by this test. The interior-response status is therefore MISSING VERTEX / PARAMETRIC ONLY.

A.3.7 No response completion is selected by this constraint

The supplied heavy-proxy test, the amplitude coordinate $\phi_u = \beta_\phi^{-1} \ln(u/u_\infty)$, and the anisotropy coordinate $\psi_K = \ln(\Delta_K/\Delta_{K,\text{vac}})$ are distinct objects. Freezing the heavy radial proxy would not identify Δ_K with u , identify either with the strict soft coordinate χ , or select either coordinate as the galactic field. No $u \leftrightarrow \Delta_K$ bridge or kinetic Jacobian has been derived.

A.4 Branch 2: provisional linear scalar–tensor benchmark for Δ_K

For a benchmark only, promote the anisotropy variable Δ_K in

$$S = \int d^4x \sqrt{-g} \left[\frac{M_*^2 \Delta_K}{2} R - \frac{Z_K(\Delta_K)}{2} (\partial \Delta_K)^2 - U_K(\Delta_K) \right] + S_m[g_{\mu\nu}]. \quad (\text{A.8})$$

Then

$$\omega_{\text{BD}} = \frac{Z_{K,\infty} \Delta_{K,\infty}}{M_*^2}. \quad (\text{A.9})$$

If one additionally assumes $Z_{K,\infty} \sim M_*^2/\Delta_{K,\infty}$, the benchmark gives $\omega_{\text{BD}} \sim 1$ and $\alpha_{\text{PPN}}^2 = 1/(2\omega_{\text{BD}} + 3) \sim 1/5$, in conflict with the Cassini bound [95]. Thus the minimal unscreened linear benchmark with that input is excluded. The normalisation $Z_{K,\infty}$, the identification of a physical Δ_K mode, and any nonlinear body solution are not fixed by ECT; the result does not exclude all Δ_K completions or force the AQUAL branch.

A.5 Branch 3: scalar–tensor response audit

The preceding branches motivate testing a nonlinear scalar response, but do not select a unique ECT degree of freedom. In particular, u and $\Delta_K = \alpha - \beta$ are distinct variables and no map between them has been derived. This subsection therefore uses a generic scalar–tensor coordinate q and states explicitly which results are coordinate invariant.

A.5.1 Canonical matter vertex

For

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} f(q) R - \frac{1}{2} K(q) (\partial q)^2 - V(q) \right] + S_m[g, \Psi], \quad f(q) > 0,$$

define $a(q) = d \ln f / dq$ and transform locally to the Einstein frame. With

$$K_E(q) = \frac{K(q)}{f(q)} + \frac{3}{2} a(q)^2, \quad \frac{d\varphi}{dq} = \sqrt{f_0 K_E(q)}, \quad (\text{A.10})$$

the canonical scalar couples parametrically to the trace of the matter stress tensor as

$$\mathcal{L}_{\varphi T} = g_{\varphi T}(q) \varphi T + \dots, \quad g_{\varphi T}(q) = -\frac{a(q)}{2\sqrt{f_0} \sqrt{K_E(q)}}. \quad (\text{A.11})$$

For the exponential closure at a reference point with $F = 1$ this becomes $g_{\varphi T} = -\beta_\phi / [2\bar{M}_{\text{Pl}} \sqrt{\omega_0 + 3\beta_\phi^2/2}]$. This is a parametric trace vertex, not a coupling to T^{00} alone and not by itself a derived non-relativistic force law. The Jordan-frame force must be obtained from the solved metric potentials.

A.5.2 Exact scalar equation and trace elimination

Variation with respect to q gives

$$K \square q + \frac{1}{2} K_{,q} (\partial q)^2 + \frac{1}{2} f_{,q} R - V_{,q} = 0. \quad (\text{A.12})$$

The scalar-only metric equation has trace

$$fR = -T + K(\partial q)^2 + 4V + 3\square f. \quad (\text{A.13})$$

The sign of the potential is positive in this solved trace equation. The normalisation is internally uniform: if f carries the Planck coefficient, the matter term is $-T$ rather than a second, independent factor of $8\pi/c^4$.

Using $\square f = f_{,q} \square q + f_{,qq} (\partial q)^2$ gives the exact combined scalar-only equation

$$\mathcal{A}(q) \square q + \mathcal{C}_X(q) (\partial q)^2 - \mathcal{E}(q, \rho_m) = 0, \quad T \simeq -\rho_m, \quad (\text{A.14})$$

where

$$\mathcal{A}(q) = K + \frac{3f_{,q}^2}{2f}, \quad (\text{A.15})$$

$$\mathcal{C}_X(q) = \frac{1}{2} K_{,q} + \frac{f_{,q} K}{2f} + \frac{3f_{,q} f_{,qq}}{2f}, \quad (\text{A.16})$$

$$\mathcal{E}(q, \rho_m) = V_{,q} - 2aV - \frac{a}{2} \rho_m. \quad (\text{A.17})$$

This equation contains the complete terms of the displayed scalar variation and has no fixed 4π source. It is exact within that closure before the additional quasistatic/weak-gradient approximation.

A.5.3 Homogeneous equilibrium and invariant response

A homogeneous equilibrium satisfies

$$V_{,q}(\bar{q}) - 2a(\bar{q})V(\bar{q}) = \frac{a(\bar{q})}{2}\rho_m. \quad (\text{A.18})$$

Equivalently,

$$\mathcal{E}(\bar{q}, \rho_m) = 0. \quad (\text{A.19})$$

An integrating factor gives a useful local potential for arbitrary $a(q)$:

$$U_{\text{loc}}(q; \rho_m) \equiv \frac{V(q) + \rho_m/4}{f(q)^2}, \quad \frac{\partial U_{\text{loc}}}{\partial q} = \frac{\mathcal{E}(q, \rho_m)}{f(q)^2}. \quad (\text{A.20})$$

Thus a root of $\mathcal{E} = 0$ is a stationary point of U_{loc} . The factor f^{-2} is essential; the former expression $V - 4\pi\rho_m q$ is not obtained from the displayed action.

Here and below

$$\mathcal{E}_{,q}|_{\rho_m} = V_{,qq} - 2a_{,q}V - 2aV_{,q} - \frac{a_{,q}}{2}\rho_m; \quad (\text{A.21})$$

the derivative is partial at fixed Jordan-frame matter density. On a fixed branch with smooth $f, K, V, f > 0$ and $\mathcal{E}_{,q}(\bar{q}, \rho_m) \neq 0$, the implicit-function theorem gives

$$\frac{d\bar{q}}{d\rho_m} = \frac{a(\bar{q})}{2\mathcal{E}_{,q}}, \quad \frac{d\ln \bar{f}}{d\rho_m} = \frac{a(\bar{q})^2}{2\mathcal{E}_{,q}}. \quad (\text{A.22})$$

The separate stable-root condition $\mathcal{E}_{,q} > 0$ makes the logarithmic response non-negative. The condition $\mathcal{A} > 0$ is instead a kinetic stability guard for the local scalar equation and is not an assumption of the implicit-function theorem. The logarithmic statement is invariant under a regular scalar reparameterisation; $d\bar{q}/d\rho_m$ is not. Terms proportional to $a_{,q}$ must not be dropped. At $\mathcal{E}_{,q} = 0$ the implicit-function theorem fails and the response is left Open rather than continued across a possible fold. Relativistic-trace, scalar-dependent-mass and correlated-state completions require a new matter derivative. The theorem concerns f only; it does not identify u, Δ_K or χ , and derives no finite-body screening.

A.5.4 Local response mass; screening still open

Linearising about a homogeneous root gives

$$(\nabla^2 - m_{\text{loc}}^2) \delta q = -\frac{a(\bar{q})}{2\mathcal{A}(\bar{q})} \delta \rho_m, \quad m_{\text{loc}}^2 = \frac{\mathcal{E}_{,q}(\bar{q}, \rho_m)}{\mathcal{A}(\bar{q})}. \quad (\text{A.23})$$

This establishes a locally stable Yukawa benchmark when $\mathcal{A} > 0$ and $\mathcal{E}_{,q} > 0$. It does not imply $dm_{\text{loc}}/d\rho_m > 0$, nor does it determine a finite body's scalar charge or post-Newtonian metric. Density screening and the previous AU-scale statement therefore remain Open.

Scope of this appendix block. The calculations collected in the present appendix explore candidate nonlinear weak-field branches and their EFT consequences. They should be read as analytical branch diagnostics and consistency tests, not as part of the strict structural core already established in the main text.

A.6 Branch 4: what does one loop actually provide?

Status: the cancellation of a constant Gaussian normalisation is exact. The derivative expansion on the $O(3)$ background contains distinct transverse and ordered-direction coefficients; their renormalised values are PARAMETRIC ONLY until the external-momentum expansion, operator ordering, regulator and counterterms are supplied. In particular, this block does not presently derive $K_K(\psi_K) \propto e^{-\psi_K/2}$ or a result for ϕ_u .

A.6.1 Setup

The three orientation coordinates δn_i ($i = 1, 2, 3$) of the separately supplied minimal EFT enter on the anisotropic chart with inverse propagator

$$G^{-1}(k) = \kappa_n D(k; \Delta_K), \quad D(k; \Delta_K) = \beta k_\perp^2 + \Delta_K k_w^2, \quad \kappa_n = \mathcal{C}_n u_0^2, \quad (\text{A.24})$$

where $k_\perp^2 = k_x^2 + k_y^2 + k_z^2$ and κ_n is held constant on the background used for this calculation. If κ_n or u_0 varies, their vertices must be included as additional fields; that is a different problem. For slowly varying $\Delta_K(X)$ the Gaussian determinant is

$$\Gamma_1[\Delta_K] = \frac{N_G}{2} \text{Tr} \ln[\kappa_n D(\Delta_K)] = \frac{N_G}{2} \text{Tr} \ln D(\Delta_K) + \frac{N_G}{2} (\text{Tr} 1) \ln \kappa_n. \quad (\text{A.25})$$

The last term is independent of Δ_K . Hence every derivative coefficient of Δ_K is exactly independent of the constant κ_n : rescaling u_0 cannot multiply such a coefficient by u_0^4 . $N_G = 3$ is only the naive coordinate multiplicity of this supplied Gaussian model, not a count of physical asymptotic modes.

A.6.2 Allowed derivative expansion and the missing coefficient owner

The background preserves $O(3)$ rather than $O(4)$. Therefore the most general quadratic two-derivative term has two coefficients,

$$\Gamma_{\text{der}} = \frac{1}{2} \int d^4 X [Z_\perp(\Delta_K)(\partial_i \Delta_K)^2 + Z_w(\Delta_K)(\partial_w \Delta_K)^2] + \dots \quad (\text{A.26})$$

Equality $Z_\perp = Z_w$ does not follow from the residual symmetry. With a cutoff regulator, power counting permits the owner-separated form

$$Z_a^{\text{ren}}(\Delta_K) = Z_a^{\text{ct}}(\Delta_K) + C_{a,\text{reg}} \Lambda^2 f_a(\beta, \Delta_K), \quad a \in \{\perp, w\}, \quad [Z_a] = \text{mass}^2, \quad (\text{A.27})$$

where the functions f_a , constants $C_{a,\text{reg}}$ and counterterms are not fixed by the displayed zero-external-momentum integral. They require an explicit expansion in the direction of the external momentum and a declared subtraction prescription.

The integral used in the former single-coefficient estimate remains a useful diagnostic, but not a completed kinetic calculation. For $k_w = \sqrt{\beta/\Delta_K} p_w$, a spherical hard cutoff $p^2 \leq \Lambda^2$, and $\beta, \Delta_K > 0$, it is

$$I_3(\beta, \Delta_K) := \int \frac{d^4 k}{(2\pi)^4} \frac{k_w^4}{[\beta k_\perp^2 + \Delta_K k_w^2]^3} = \frac{\Lambda^2}{128\pi^2 \beta^{1/2} \Delta_K^{5/2}}. \quad (\text{A.28})$$

Here $\langle p_w^4/p^4 \rangle_{S^3} = 1/8$ and $\int d^4 p/(2\pi)^4 p^{-2} = \Lambda^2/(16\pi^2)$. This diagnostic integral has the required mass dimension and contains no u_0^4 . However, a true two-point external-momentum expansion also expands the vertices and both propagators, distinguishes transverse from w momentum, and admits local counterterms. Thus I_3 alone proves neither $Z_\perp \propto \Delta_K^{-5/2}$ nor $Z_\perp = Z_w$. The quadratic divergence and its coefficient are regulator/subtraction dependent, and the gradient expansion also fails near a critical point (Branch 5).

A.6.3 Transformation to the anisotropy coordinate ψ_K

Define $\Delta_K = \alpha - \beta$ and $\psi_K = \ln(\Delta_K/\Delta_{K,\text{vac}})$, so that $\partial_a \Delta_K = \Delta_K \partial_a \psi_K$. Direction by direction,

$$K_a(\psi_K) = Z_a(\Delta_K) \Delta_K^2, \quad a \in \{\perp, w\}. \quad (\text{A.29})$$

If a future completed transverse calculation were to yield $Z_\perp \propto \Delta_K^{-5/2}$, then and only then

$$Z_\perp \propto \Delta_K^{-5/2} \implies K_\perp(\psi_K) \propto \Delta_K^{-1/2} = \Delta_{K,\text{vac}}^{-1/2} e^{-\psi_K/2}. \quad (\text{A.30})$$

This is a conditional implication, not the result of the present determinant. No corresponding K_w scaling has been derived. Neither coefficient can be assigned to ϕ_u without the missing $u \leftrightarrow \Delta_K$ map.

A.6.4 Conditional static exterior of the former single-coefficient surrogate

To retain the associated negative control, separately supply the old physical-space surrogate $K_\perp \propto e^{-\psi_K/2}$ and omit sources. Its source-free spherical equation is

$$\frac{1}{r^2} \frac{d}{dr} \left[r^2 e^{-\psi_K/2} \psi'_K \right] = 0, \quad (\text{A.31})$$

the first integral is $r^2 e^{-\psi_K/2} \psi'_K = C$. It can be integrated exactly:

$$e^{-\psi_K/2} = C_0 + \frac{C_1}{r}, \quad \psi_K(r) = -2 \ln \left(C_0 + \frac{C_1}{r} \right). \quad (\text{A.32})$$

For $C_0 \neq 0$ the asymptotic perturbation and its gradient behave as $\delta\psi_K \propto r^{-1}$ and $\delta\psi'_K \propto r^{-2}$. No $r^{2/3}$ solution exists for the displayed equation. Moreover, a force law cannot be read off from this profile without the missing metric and matter-coupling match.

A.6.5 Conclusion of the Gaussian determinant audit

The constant normalisation $\kappa_n = \mathcal{C}_n u_0^2$ cancels from the Δ_K -dependent Gaussian determinant.

The two allowed derivative coefficients $Z_\perp$ and Z_w remain regulator-, subtraction-, external-momentum- and mode-counting-dependent. If the former $K_\perp \propto e^{-\psi_K/2}$ surrogate is separately supplied, its exact source-free exterior has the usual $1/r$ asymptotic perturbation; neither $\psi \propto r^{2/3}$ nor $v^2 = c^2/3$ follows.

The recovered result is narrower but exact where stated: constant Gaussian normalisation cannot generate an u_0^4 kinetic coefficient, and a field-dependent quadratic kinetic surrogate without a derived source, renormalised coefficients, metric map and nonlinear infrared completion is insufficient to obtain the observed galactic scaling laws.

A.7 Branch 5: candidate nonlinear infrared completion

Status: Level C candidate / Open ECT derivation. The following power counting states sufficient conditions for a nonlinear regime; it does not derive those conditions from the ECT action.

A.7.1 A possible breakdown of the gradient expansion

Consider the static derivative expansion

$$\Gamma[q] = \int d^3x \left[\frac{K(q)}{2} (\nabla q)^2 + \frac{L(q)}{3} |\nabla q|^3 + \frac{M(q)}{4} (\nabla q)^4 + \dots \right]. \quad (\text{A.33})$$

The ratio of the cubic and quadratic terms is

$$\frac{L|\nabla q|^3}{K|\nabla q|^2} = \frac{L}{K} |\nabla q|. \quad (\text{A.34})$$

If a derived infrared flow were to satisfy $L/K \rightarrow \infty$ in a specified critical limit, then the crossover gradient $|\nabla q|_{\text{trans}} = K/L$ would approach zero and weak gradients could enter a nonlinear regime. This is a sufficient conditional statement.

The present one-loop calculation fixes neither $L(q)$ nor the ratio L/K . The often-used ansatz

$$K \propto e^{-aq}, \quad L \propto e^{-bq}, \quad b > a > 0,$$

therefore remains a hypothesis. In addition, the loop result was obtained for $q = \psi_K$, whereas the separate scalar ansatz S uses $q = \phi_u$. The HRC internal control coordinate and its map to either scalar variable remain Open; none of the three can be identified without the missing coordinate map.

A.7.2 Regime ordering as a target

A viable completion must reproduce Newtonian behaviour in dense/large- acceleration systems and a nonlinear branch in dilute galactic outskirts. A field-dependent threshold could realise this ordering if a solved ECT flow showed both

$$\begin{aligned} |\nabla q|_{\text{trans}} &\text{ large in the high-acceleration Newtonian regime,} \\ |\nabla q|_{\text{trans}} &\rightarrow 0 \text{ toward the infrared critical regime.} \end{aligned}$$

Neither inequality is established by the current action. Consequently the observed regime ordering is a design constraint and potential falsifier of the future closure, not a guaranteed consequence of “approach to criticality.” Screening and MOND-like dynamics may ultimately share a mechanism, but that identification is presently Open.

A.8 Candidate critical scaling: the conditional $Y_q^{3/2}$ form

Let $Y_q = (\nabla q)^2$. If the deep-infrared static theory is (i) local in three spatial dimensions, (ii) controlled by a fixed point, (iii) a function only of Y_q , and (iv) has a dimensionless field $[q] = 0$ with negligible anomalous dimension, scale counting gives

$$[Y_q] = L^{-2}, \quad [\mathcal{F}] = L^{-3}, \quad \mathcal{F} \propto Y_q^p \implies p = \frac{3}{2}. \quad (\text{A.35})$$

This implication is exact under the four assumptions; the assumptions are not derived by the present one-loop calculation. With anomalous dimension Δ_q , the same counting gives $p = 3/[2(1 + \Delta_q)]$. Nonlocal terms, logarithms, and additional derivative invariants can also change the fixed-point action.

The $Y_q^{3/2}$ operator is therefore a useful critical/AQUAL candidate. It produces a $1/r$ gradient and BTFR-like mass exponent once an appropriate matter source and metric-force map are separately assumed. It does not by itself establish an ECT fixed point, its normalisation, or its connection to the u -based amplitude closure.

A.9 Conditional BTFR algebra of an AQUAL benchmark

Status: Level C benchmark. The algebra below follows from the displayed nonlinear equation; ECT has not yet derived its source normalisation, force map, or fixed-point origin.

Let q be a dimensionless scalar coordinate and consider

$$\nabla \cdot \left[\mu_q \left(\frac{|\nabla q|}{q_*} \right) \nabla q \right] = \mathcal{S}_q \rho, \quad \mu_q(x) \rightarrow 1 \ (x \gg 1), \quad \mu_q(x) \rightarrow x \ (x \ll 1). \quad (\text{A.36})$$

Here ρ is baryonic mass density, while $\mathcal{S}_q$, q_* , and the map from q to the physical metric are matching inputs. They are not fixed by the scalar trace equation in Branch 3.

For $\mu_q(x) = x$, spherical integration around mass M gives

$$|q'|^2 = \frac{\mathcal{S}_q M q_*}{4\pi r^2}, \quad |q'| = \sqrt{\frac{\mathcal{S}_q M q_*}{4\pi}} \frac{1}{r}. \quad (\text{A.37})$$

If the solved weak-field metric additionally yields $|a_q| = \mathcal{C}_q c^2 |q'|$, then

$$v^4 = G_N M a_M, \quad a_M \equiv \frac{\mathcal{C}_q^2 c^4 \mathcal{S}_q q_*}{4\pi G_N}. \quad (\text{A.38})$$

Thus the BTFR exponent is an exact consequence of the assumed deep-AQUAL equation and force map. Its coefficient is not an ECT prediction until $\mathcal{S}_q$, $\mathcal{C}_q$, and q_* are derived. The former special choice $\mathcal{S}_q = 4\pi G_N/c^2$, $\mathcal{C}_q = 1/2$ gives $a_M = c^2 q_*/4$, but that normalisation is one benchmark rather than a consequence of Eq. (13.35).

A.9.1 Comparison with the quadratic branch

	Conditional quadratic surrogate	Critical/AQUAL candidate
Kinetic form	Separately supplied $K_{\perp}(\psi_K)(\nabla\psi_K)^2$	$Y_q^{3/2}$
Exact exterior	$\delta\psi_K \propto 1/r$	$ q' \propto \sqrt{M}/r$ if sourced as in Eq. (A.36)
Flat curves / BTFR	Not derived	Algebraically yes, conditional on force/source matching
ECT status	Conditional negative-control surrogate; the Gaussian loop coefficients are parametric only	Level C benchmark; fixed-point and matching Open

A.10 Environmental dependence as an operational hypothesis

No function $a_M = a_M(q)$ has yet been derived. If a future matching supplies $a_M = \mathcal{G}(q, \dots)$, define $\gamma_q \equiv \partial \ln \mathcal{G} / \partial q$. Combining that hypothesis with a fixed regular Branch 3 root gives the following response only where $\mathcal{E}_{,q} \neq 0$. A stable-root sign additionally requires $\mathcal{E}_{,q} > 0$, while at a fold the linear response fails closed:

$$\delta \ln a_M = \gamma_q \delta \bar{q} + \delta_{\text{other}}, \quad \delta \bar{q} = \frac{a(\bar{q})}{2\mathcal{E}_{,q}} \delta \rho_{\text{m,env}}. \quad (\text{A.39})$$

where δ_{other} denotes dependence on boundary conditions, galaxy structure, and any additional closure parameters. A fixed 4π coefficient and a numerical 10% modulation do not follow from this scalar equation.

Proposed observational test. Environment-split RAR fits remain useful as an empirical test of a specified closure. A correlation would motivate fitting $\mathcal{G}$ and the response combination $\gamma_q a / (2\mathcal{E}_{,q})$; a null result would constrain that mapping. Until the mapping is derived, either outcome tests the phenomenological environment ansatz rather than the full ECT framework.

A.11 Status of all results

Table 122: Status of results in the ordered-branch scalar-response programme.

Statement	Status	Depends on
$\phi_u = \beta_{\phi}^{-1} \ln(u/u_{\infty})$	Definition	Ordered-branch amplitude coordinate
$\psi_K = \ln(\Delta_K/\Delta_{K,\text{vac}})$	Definition	Anisotropy coordinate
$u \leftrightarrow \Delta_K$	Open	Missing EFT matching
Scalar equation and trace, Eqs. (A.12)–(A.13)	Derived in model	Scalar-only IR action
Regular-root susceptibility $d \ln \bar{f} / d \rho_{\text{m}} = a^2 / (2\mathcal{E}_{,q})$	Conditional model result	Fixed smooth branch, $f > 0$, $\mathcal{E}_{,q} \neq 0$; non-negative only for $\mathcal{E}_{,q} > 0$
Local $m_{\text{loc}}^2 = \mathcal{E}_{,q} / \mathcal{A}$	Conditional model result	Weak-gradient local linearisation; positive Yukawa mass requires the separate guards $\mathcal{A} > 0$ and $\mathcal{E}_{,q} > 0$

(continued on next page)

Statement	Status	Depends on
Density-growing mass and finite-body screening $K_{\perp} \propto e^{-\psi_K/2}$	Open Separately supplied conditional surrogate	Complete f, K, V , body charge and boundary solution Not derived by the Gaussian determinant; $Z_{\perp}, Z_w$, regulator, subtraction, external momentum and mode ownership remain open
$L/K \rightarrow \infty$ and critical flow $\mathcal{F} \propto Y_q^{3/2}$	Open Conditional scaling candidate	Non-perturbative RG 3D local fixed point, $[q] = 0$
BTFR exponent from Eq. (A.36)	Conditional algebra	Assumed AQUAL source and force map
ECT normalisation of a_M	Open	$\mathcal{S}_q, \mathcal{C}_q, q_*$ matching
Environmental modulation of a_M	Open test hypothesis	Missing $a_M = \mathcal{G}(q, \dots)$

A.12 Methodological summary: the audited branch tree

Table 123: Audited status of candidate galactic routes.

#	Route	Current verdict
1	Supplied radial amplitude proxy	Compact-source exterior range is $\sqrt{\beta}/m_{\sigma}$ and requires $m_{\sigma}/\sqrt{\beta} \lesssim L^{-1}$ for a target range L ; physical parameter matching and extended sourced response remain Open
2	Unscreened linear Δ_K scalar	Benchmark with $\omega_{\text{BD}} \sim 1$ conflicts with Cassini; the value is model-input dependent
3	Generic non-minimal scalar	Trace response derived inside the closure; coordinate identity and screening remain Open
4	Quadratic one-loop K_K	Exact exterior is $1/r$; it supplies neither an $r^{2/3}$ profile nor flat curves
5	Critical $Y_q^{3/2}$ /AQUAL	BTFR algebra conditional; ECT fixed point, source, force map and normalisation Open

This audit is not a proof that the fifth route is the unique endpoint. It is a record of which candidate calculations survive and which missing bridges must be supplied before a galactic claim can be upgraded.

A.13 Final perspective

The present galactic sector is a structured research programme, not yet a first-principles ECT derivation. What is established is the existence of an ordered branch, the exact scalar–tensor variational identities inside the chosen IR action, and the conditional regular-root response $d \ln \bar{f} / d\rho_m = a^2 / (2\mathcal{E}_q)$, which is non-negative only on a branch with $\mathcal{E}_{,q} > 0$; $\mathcal{A} > 0$ is the separate kinetic guard. The following remain Open: identification of the scalar with the soft ECT degree of freedom; the u – Δ_K bridge; density-enhanced finite-body screening; the non-perturbative critical fixed point; and the ECT source/metric normalisation of the AQUAL benchmark.

The AQUAL candidate reproduces flat-curve and BTFR scaling algebraically, but those are properties of the assumed closure, not yet ECT-specific predictions. Environmental modulation is an operational hypothesis whose interpretation requires a derived $a_M = \mathcal{G}(q, \dots)$ map. The next decisive calculations are

therefore (i) a covariant ECT-to-scalar matching, (ii) finite-body solutions and post-Newtonian observables, and (iii) a non-perturbative derivation or falsification of the proposed infrared fixed point (OP-new-1).

B Background Configuration and Fluctuation Expansion

Reference from main text: Eq. (2.7) and Section 4.5.

Background solution

Condition P4 requires $\langle \partial_A \Phi \rangle = u_0 \delta_{Aw}$. The simplest solution is:

$$\Phi_{\text{bg}}(X) = u_0 w + \Phi_*, \quad \Phi_* = \text{const.} \quad (\text{B.1})$$

Check: $\partial_A \Phi_{\text{bg}} = u_0 \delta_{Aw}$ for all X^A . ✓

Important note. This background is linear in w and therefore is not a minimum of the potential $V(\Phi_{\text{bg}})$ which grows without bound for $|w| \rightarrow \infty$. It does not claim to be a stationary point of the bare micro-action P3; it serves only as the supplied local ordered-sector analysis profile of P4, not as an established physical vacuum. Physically this is analogous to a gradient condensate in superfluid systems [10].

Conditional quadratic form of the broken-phase EFT

The effective broken-phase functional expanded around the background has the schematic form

$$S_{\text{eff}}^{\text{broken}}[\Phi_{\text{bg}} + \varphi] = S_{\text{eff}}^{\text{broken}}[\Phi_{\text{bg}}] + \underbrace{\int d^4X \frac{\delta S_{\text{eff}}^{\text{broken}}}{\delta \Phi} \Big|_{\text{bg}} \varphi}_{=0 \text{ only for a stationary completion}} + O(\varphi^2). \quad (\text{B.2})$$

P4 by itself does not make the linear term vanish: it specifies an ordered branch but not an action for which that branch is a saddle. Conditional on a completed stationary ordered-phase action, the supplied quadratic object is therefore parametrised by the same anisotropic characteristic form as in Eq. (3.12),

$$\mathcal{Q}_{2,\varphi} = \int d^4X \left[\frac{1}{2} K^{AB} \partial_A \varphi \partial_B \varphi + \frac{m_{\text{eff}}^2}{2} \varphi^2 \right], \quad K^{AB} = \beta \delta^{AB} - \alpha n^A n^B. \quad (\text{B.3})$$

It is neither the P3 Hessian about P4 nor a positive Euclidean statistical weight. On the retained $\beta > 0$, $\alpha > \beta > 0$ component, the positive-energy reduced Lorentzian functional and physical action are

$$\mathcal{J}_{L,\varphi}^{(2)} = -\mathcal{Q}_{2,\varphi}, \quad S_{L,\varphi}^{(2),\text{phys}} = A_{L,\varphi} \mathcal{J}_{L,\varphi}^{(2)}, \quad A_{L,\varphi} > 0.$$

The coefficients, mass, spatial domain, action magnitude and their P4 matching remain Open.

C Uniqueness of the Kinetic Tensor K^{AB}

Reference from main text: Theorem 3.2 in Section 3.3.

Setup

We seek the most general symmetric rank-2 tensor K^{AB} satisfying:

- (i) $K^{AB} = K^{BA}$;
- (ii) invariance under $O(3)$ rotations in the hyperplane $\{1, 2, 3\}$;
- (iii) built locally and algebraically from tensors $\{\delta^{AB}, n^A\}$, without additional derivatives, where $n^A = \delta^A_w$ is the fixed unit vector along the preferred direction.

Basis decomposition

The space of symmetric rank-2 tensors in $\mathbb{R}^4$ has dimension 10. Under $O(3)$ the available background objects generate two $O(3)$ -invariant symmetric rank-2 structures: δ^{AB} and $n^A n^B$. Without additional derivatives or pseudo-tensors (Levi-Civita is antisymmetric and excluded), these two span the space of admissible tensors. Hence the most general tensor is:

$$K^{AB} = a \delta^{AB} + b n^A n^B, \quad a, b \in \mathbb{R}.$$

Setting $a = \beta$, $b = -\alpha$ gives (3.11).

Connection to the quadratic form and physical action

The tensor fixes the kinetic part of the characteristic form:

$$\frac{1}{2} K^{AB} \partial_A \varphi \partial_B \varphi = \frac{\beta}{2} (\partial_A \varphi)^2 - \frac{\alpha}{2} (n^A \partial_A \varphi)^2. \quad (\text{C.1})$$

Equation (3.12) then introduces the separate overall Lorentzian orientation and positive action-valued magnitude. On the retained $\beta > 0$, $\alpha > \beta > 0$ component, positive energy fixes $\mathcal{S}_{L,\varphi}^{(2)} = -\mathcal{Q}_{2,\varphi}$ and $S_{L,\varphi}^{(2),\text{phys}} = A_{L,\varphi} \mathcal{S}_{L,\varphi}^{(2)}$ with $A_{L,\varphi} > 0$; the tensor theorem fixes neither that relative sign, the magnitude, nor a positive Euclidean measure. ■

See also [290, Ch. 2] and [291].

D Derivation of the Lorentzian Window

Reference from main text: Section 3.5.

Full signature classification and the healthy lower condition

From the quadratic action (3.12), choose an orthonormal frame whose last basis vector is the supplied ordered direction n^A . The principal tensor then has eigenvalues

$$\underbrace{\beta, \beta, \beta}_{n^\perp}, \quad \beta - \alpha \quad (n).$$

Consequently,

$$\begin{aligned} \beta(\beta - \alpha) > 0 & : \text{nondegenerate elliptic,} \\ \beta(\beta - \alpha) < 0 & : \text{Lorentzian/hyperbolic,} \\ \beta = 0 \text{ or } \alpha = \beta & : \text{degenerate.} \end{aligned}$$

The Lorentzian set has the two mathematical components

$$\{\beta > 0, \alpha > \beta\} \quad \text{and} \quad \{\beta < 0, \alpha < \beta\}.$$

In the first, the three transverse eigenvalues are positive and the ordered one is negative. In the second all signs are reversed. The manuscript separately retains $\beta > 0$ and fixes the physical Lorentzian orientation $\zeta_L = -1$ by the positive-energy Legendre test in Eq. (3.13); on that component the exact lower condition is $\alpha > \beta > 0$. For $\beta > 0$ and $\alpha < \beta$ the tensor is positive definite; for $\beta < 0$ and $\alpha > \beta$ it is negative definite. The mass term does not affect this principal-symbol classification.

Arithmetic isotropic-average benchmark: $\alpha < 4\beta$

Supply an averaging ensemble or cell of declared scale ℓ_{iso} whose orientation distribution has the $O(4)$ -isotropic second moment $\langle n^A n^B \rangle_{\text{arith}} = \delta^{AB}/4$. The pointwise arithmetic coefficient average is then

$$\langle K^{AB} \rangle_{\text{arith}} = \left(\beta - \frac{\alpha}{4} \right) \delta^{AB}.$$

Positive-definiteness of this arithmetic averaged form requires $\beta - \alpha/4 > 0$, i.e. $\alpha < 4\beta$. In d dimensions the analogous conditional inequality is $\alpha < d\beta$. Combining this extra assumption with the healthy hyperbolic component gives

$$\boxed{0 < \beta < \alpha < 4\beta} \quad (d = 4).$$

Status. The product-sign classification is Level A mathematics inside the supplied scalar EFT. The reduction to $\alpha > \beta > 0$ additionally uses the supplied healthy transverse-kinetic branch. The upper bound $\alpha < 4\beta$ is only the positivity condition for the declared arithmetic isotropic average. It requires an ensemble/cell and scale ℓ_{iso} that P1–P4 do not supply. Moreover, the arithmetic coefficient average need not equal the homogenised principal operator, which generally requires a cell problem and can depend on correlations and boundary data. Thus the upper bound is neither a necessary physical stability condition nor a derived ECT ultraviolet window. None of these statements selects the realised coefficients, a physical clock/metric, or common matter/photon/tensor cones.

E Photon LIV ownership and external constraint gate

The present ECT action supplies no physical photon kinetic operator, higher-dimension LIV term, suppression scale, coefficient, source-lag model or detector observable. Equation (3.21) records only the schematic dispersion of a future named operator completion. A unit-coefficient time-delay table would therefore be a hand-set phenomenological insertion and is not retained.

Published high-energy timing analyses, including the Fermi-LAT GRB 090510 study [292], constrain their own declared dispersion and source models. They become an ECT test only after a future ECT photon completion derives the corresponding operator and forward map.

F Retarded Green function for the supplied scalar operator: support and causal boundary

Reference from main text: Section 3.8, Result A3. Status: Level A operator mathematics for retarded support under the declared clock and initial–boundary problem; the displayed real-frequency pole proof additionally requires the nonnegative retained spectrum, while physical response normalisation remains Open.

Setup

After passing to the effective Lorentzian description (Section 3.6), small fluctuations satisfy the equation below. Here J is equation-normalised. The inverse operator, pole positions and causal support depend on the differential operator and retarded initial–boundary prescription, not on a finite nonzero constant overall action multiplier. They do not, however, fix a physical response residue. For the convention $S_{L,\varphi}^{\text{phys}} = A_{L,\varphi} \mathcal{J}_{L,\varphi}^{(2)} + \int d^4x j_{\text{phys}} \varphi$, variation gives $J = j_{\text{phys}}/A_{L,\varphi}$ and hence $\delta\varphi/\delta j_{\text{phys}} = G_{\text{ret}}/A_{L,\varphi}$. The positive Hamiltonian separately uses $\zeta_L = -1$ and the finite constant $A_{L,\varphi} > 0$ in Eq. (3.13); another source-sign convention changes the corresponding overall sign, not this normalisation firewall.

$$(\partial_t^2 - c_{\text{char}}^2 \nabla^2 + M^2) \varphi(t, \mathbf{x}) = J(t, \mathbf{x}). \quad (\text{F.1})$$

The retarded Green’s function:

$$(\partial_t^2 - c_{\text{char}}^2 \nabla^2 + M^2) G_{\text{ret}}(t, \mathbf{x}) = \delta(t) \delta^{(3)}(\mathbf{x}). \quad (\text{F.2})$$

Fourier representation and pole structure

For the real-frequency pole representation below assume

$$\Omega^2(\mathbf{k}) \equiv c_{\text{char}}^2 |\mathbf{k}|^2 + M^2 \geq 0 \quad \text{for every retained spatial mode.} \quad (\text{F.3})$$

On $\mathbb{R}^3$ this is ensured by $M^2 \geq 0$. If a negative spectral mode is admitted, the hyperbolic initial-value problem still has a retarded solution, proportional to $\theta(t) \sinh(|\Omega|t)/|\Omega|$ for that mode, but it grows and requires a Laplace/Bromwich contour; the following real-axis pole proof does not apply.

$$G_{\text{ret}}(\omega, \mathbf{k}) = \frac{1}{\omega_k^2 - (\omega + i0^+)^2}, \quad \omega_k = \sqrt{\Omega^2(\mathbf{k})} \in \mathbb{R}_{\geq 0}. \quad (\text{F.4})$$

For $\omega_k > 0$ both poles $\omega = \pm\omega_k - i0^+$ lie in the lower half-plane. The zero-mode formula is the continuous limit $\theta(t) \sin(\omega_k t)/\omega_k \rightarrow \theta(t)t$. Thus a semidefinite Hamiltonian may still admit linearly growing zero-mode data; strict bounded oscillatory stability requires a strictly positive spectral bottom or a separately constrained/removed zero mode.

Proof that $G_{\text{ret}}(t, \mathbf{x}) = 0$ for $t < 0$

For $t < 0$ the integrand $e^{-i\omega t}$ decays for $\text{Im } \omega \rightarrow +\infty$, so the ω -contour closes in the upper half-plane. Since both poles lie in the lower half-plane, the Cauchy theorem gives:

$$\boxed{G_{\text{ret}}(t, \mathbf{x}) = 0 \quad (t < 0).} \quad (\text{F.5})$$

Explicit form for $t > 0$

$$G_{\text{ret}}(t, \mathbf{x}) = \theta(t) \int \frac{d^3k}{(2\pi)^3} \frac{\sin(\omega_k t)}{\omega_k} e^{i\mathbf{k} \cdot \mathbf{x}}. \quad (\text{F.6})$$

The factor $\theta(t)$ explicitly confirms (F.5). For $M > 0$, let $\sigma = t^2 - r^2/c_{\text{char}}^2$ with $r = |\mathbf{x}|$. In flat 3 + 1 dimensions, evaluating (F.6) distributionally gives

$$G_{\text{ret}}(t, \mathbf{x}) = \theta(t) \left[\frac{\delta(\sigma)}{2\pi c_{\text{char}}^3} - \theta(\sigma) \frac{MJ_1(M\sqrt{\sigma})}{4\pi c_{\text{char}}^3 \sqrt{\sigma}} \right]. \quad (\text{F.7})$$

The first term of Eq. (F.7) is supported on $r = c_{\text{char}}t$ and is present for arbitrarily small positive M ; the second is the massive interior tail. Thus the massive kernel is supported on the closed cone, not strictly in its interior. In the continuous $M \rightarrow 0$ limit the tail vanishes and the flat-3 + 1 massless kernel is supported on the cone.

What follows and what does not

Follows (Level A under the displayed spectral and boundary hypotheses): the temporal support property $G_{\text{ret}}(t, \mathbf{x}) = 0$ for $t < 0$. Together with the symmetric-hyperbolic reduction below this gives finite domain of dependence for the supplied scalar operator in the owned clock patch. It is not by itself a universal physical-causality, arrow-of-time or all-sector statement.

Relation to the nonlinear-owner gate: Appendix G separates this exact quadratic retarded-kernel result from the conditional hyperbolicity test for a supplied quasilinear $P(X)$ operator. It does not promote the fixed-coefficient calculation to the full nonlinear medium.

Does not follow from this appendix alone: retardedness in defect-core regions or signature-changing domains (OP5); Markovianity, ohmicity, Born rule, Pauli principle, OS reconstruction.

G Conditional hyperbolicity of supplied scalar operators and the nonlinear-owner gate

Status. The constant-coefficient scalar equation obtained from Eq. (3.12) admits the explicit symmetric-hyperbolic first-order reduction displayed below and therefore has retarded support for the declared initial–boundary problem (Level A operator mathematics inside that supplied clock patch). This support statement is invariant under a finite nonzero constant action multiplier; positive energy and physical response magnitude additionally require the separate orientation, action scale and source vertex. For a separately supplied smooth $P(X)$ functional, the nonlinear principal tensor and its Lorentzian sign test are exact, but local well-posedness is conditional on uniform hyperbolicity, a timelike foliation and regular initial–boundary data. Neither result derives the operative nonlinear ordered-phase action from P1–P6, a universal matter/photon cone, or a nonlinear Green function.

Three different claims that must not be conflated

The ordered scalar coordinate is one constrained degree of freedom; this removes a spurious independent (σ, χ) two-field count, but it does not determine its nonlinear equation. Three statements remain distinct:

1. the fixed-coefficient quadratic EFT supplied in Eq. (3.12);
2. the quasilinear equation of a separately supplied $P(X)$ completion;
3. a physical causal theorem for the complete medium, including its clock, matter/photon metric, defects, boundaries and other sectors.

Only the first is closed without further dynamical input.

Exact principal tensor of a supplied $P(X)$ completion

For the dimensionless reduced functional and physical action

$$\mathcal{J}_P = \int d^4X [P(X) + V(\Phi)], \quad S_P^{\text{phys}} = A_P \zeta_P \mathcal{J}_P, \quad 0 < A_P < \infty, \quad (\text{G.1})$$

with $X = \partial_A \Phi \partial_A \Phi$ and constant field-independent A_P, ζ_P , the scalar equation has principal part

$$G_P^{AB}(\partial\Phi) \partial_A \partial_B \Phi + \text{lower-order terms} = 0, \quad G_P^{AB} = P_X \delta^{AB} + 2P_{XX} \partial^A \Phi \partial^B \Phi. \quad (\text{G.2})$$

The finite nonzero overall multiplier affects neither this equation nor its principal cone. In a local frame with $\partial_A \Phi = \sqrt{X} n_A$, the three transverse eigenvalues are P_X , while the longitudinal eigenvalue is

$$Q'(X) = P_X + 2XP_{XX}, \quad Q(X) = 2XP_X - P. \quad (\text{G.3})$$

Consequently the exact nondegenerate Lorentzian condition is

$$\boxed{P_X Q' < 0}, \quad (\text{G.4})$$

with two components,

$$(P_X > 0, Q' < 0) \quad \text{or} \quad (P_X < 0, Q' > 0). \quad (\text{G.5})$$

Both eigenvalues must additionally be bounded away from zero on the Cauchy domain. The first component is retained for direct matching to $\beta = P_X > 0$ and $\beta - \alpha = Q' < 0$; it implies the sharper condition $P_{XX} < -P_X/(2X)$ for $X > 0$. On either component local positive quadratic kinetic and gradient energy fixes $\zeta_P = -\text{sgn } P_X = \text{sgn } Q'$. This sign choice does not establish a globally bounded nonlinear Hamiltonian, the mass sign, or an ECT owner for A_P .

The second component is not empty. For any fixed $M_X > 0$, take $P(X) = -X + X^2/M_X^4$ at $X = M_X^4/3$. Then $P_X = -1/3$, $P_{XX} = 2/M_X^4$, $Q' = 1$ and $P_X Q' = -1/3 < 0$. This counterexample excludes the old

identification of the first component with the full Lorentzian domain. The recorded stationary-density point $P_X = 0$, $P_{XX} > 0$ remains degenerate and does *not* satisfy Eq. (G.4). Thus P4 and the bare $P_X = 0$ motivation alone do not own a nonlinear Lorentzian equation.

For smooth coefficients, a foliation whose normal covector is uniformly timelike with respect to G_p^{AB} , compatible regular initial–boundary data and controlled lower-order terms, the standard quasilinear symmetric-hyperbolic reduction gives local existence, uniqueness and a finite domain of dependence [87, 88]. These are explicit hypotheses, not consequences of the residual $O(3)$ tensor classification.

Explicit fixed-coefficient reduction that is actually closed

For the supplied quadratic EFT, choose a locally aligned constant-lapse clock patch. Including the lower-order field itself gives the closed state

$$U := \begin{pmatrix} \varphi \\ c_{\text{char}}^{-1} \partial_\tau \varphi \\ \partial_1 \varphi \\ \partial_2 \varphi \\ \partial_3 \varphi \end{pmatrix}. \quad (\text{G.6})$$

Here c_{char} is supplied by Eq. (3.23). For $(\partial_\tau^2 - c_{\text{char}}^2 \nabla^2 + M^2)\varphi = J$, the complete first-order system is

$$\partial_\tau U^0 = c_{\text{char}} U^1, \quad (\text{G.7})$$

$$\partial_\tau U^1 = c_{\text{char}} \sum_{i=1}^3 \partial_i U^{i+1} - \frac{M^2}{c_{\text{char}}} U^0 + \frac{J}{c_{\text{char}}}, \quad (\text{G.8})$$

$$\partial_\tau U^{i+1} = c_{\text{char}} \partial_i U^1, \quad i = 1, 2, 3. \quad (\text{G.9})$$

Writing $\partial_\tau U + \sum_k A^k \partial_k U = F(U, J)$,

$$A_{ab}^k = -c_{\text{char}}(\delta_{a1} \delta_{b,k+1} + \delta_{b1} \delta_{a,k+1}), \quad (A^k)^\text{T} = A^k, \quad (\text{G.10})$$

for $a, b = 0, \dots, 4$, with

$$F = \begin{pmatrix} c_{\text{char}} U^1 \\ -M^2 U^0 / c_{\text{char}} + J / c_{\text{char}} \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (\text{G.11})$$

The time matrix is $A^0 = I_5 > 0$; the principal determinant contains the propagating roots $\omega^2 = c_{\text{char}}^2 |\xi|^2$. The compatibility constraints

$$C_i = U^{i+1} - \partial_i U^0 = 0, \quad C_{ij} = \partial_i U^{j+1} - \partial_j U^{i+1} = 0 \quad (\text{G.12})$$

propagate because $\partial_\tau C_i = 0$ and $\partial_\tau C_{ij} = 0$. Thus the augmented system is equivalent to the supplied Klein–Gordon equation on compatible initial data; the mass and source are lower-order terms and do not change its characteristic cone. Hence the fixed scalar operator has a well-posed local Cauchy problem and retarded fundamental solution. Equivalently,

$K_\Phi^{\text{ret}}(\tau < 0) = 0 \quad \text{for the linearised supplied scalar operator in an owned Lorentzian clock patch.}$

(G.13)

This is Level A operator mathematics under the displayed assumptions. It is not a universal physical-cone claim.

Why there is no “nonlinear Green function” theorem here

A Green function is the inverse of a *linear* operator. Around any declared nonlinear background solution $\bar{\Phi}$, one may form the Fréchet derivative $L_{\bar{\Phi}}$ and, if the hypotheses above hold, its retarded inverse $G_{\bar{\Phi}}^{\text{ret}}$. Different backgrounds generally give different linearised kernels. The notation “retarded Green function of the full nonlinear ECT dynamics” is therefore not well defined and is not used.

Terminal scope

Established: the supplied fixed quadratic scalar EFT admits the displayed symmetric-hyperbolic first-order reduction; Eq. (G.2) and the full sign test (G.4) are exact for a supplied $P(X)$ functional. The retained $P_X > 0, Q' < 0$ component is a matching choice, not the full Lorentzian domain.

Conditional: local nonlinear well-posedness and domain of dependence for a declared solution on a uniformly hyperbolic branch with an owned foliation and regular initial–boundary data.

Open: derivation of that operative branch from P1–P6; crossing or evolution through $P_X = 0, Q' = 0$, defect cores or signature change; the physical matter/photon clock and metric; global-in-time existence; gauge, tensor and P7 sectors; and the open-system response/noise kernels.

Appendix F is therefore not a special case of a proved full-nonlinear ECT theorem. It is the explicit retarded solution of the same supplied quadratic scalar operator. A completed nonlinear action must pass the additional branch, foliation, boundary and stability gates listed above.

H Directed Gaussian dephasing: proof that $\Phi_{\text{irr}}^{\text{bulk}}[q_+, q_-] \geq 0$

Reference from main text: Section 3.8, Result A5. Status: Level A conditional mathematics for the bulk noise term of the supplied Gaussian oscillator bath, with a named closed-time-path contour; centred bath one-point function, and a compatible factorised initial preparation, or with every correlated initial-boundary term treated explicitly. The positivity proof does not require ohmicity or a Markov approximation. Causal support of the response kernel is a separate property; neither the physical subsystem/environment split nor the bath operator, initial preparation, state, spectral density or response/noise kernels are derived from the live ECT action.

Setup

For two closed-time-path histories $q_+(t)$ and $q_-(t)$, define

$$q_{\Delta}(t) = q_+(t) - q_-(t), \quad q_c(t) = \frac{q_+(t) + q_-(t)}{2}. \quad (\text{H.1})$$

Fix an initial time t_0 , a closed-time-path contour and a centred bath one-point function. If a coherent subsystem coordinate is coupled linearly to a supplied Gaussian environment with a compatible factorised preparation, the bulk reduced description takes the Feynman–Vernon form [192, 22]

$$\mathcal{F}_{\text{bulk}}[q_+, q_-] = \exp \left\{ \frac{i}{S_0} \int_0^T dt \int_0^T dt' q_{\Delta}(t) D_{\text{self}}^R(t-t') q_c(t') - \frac{1}{2S_0^2} \int_0^T dt \int_0^T dt' q_{\Delta}(t) C_H(t-t') q_{\Delta}(t') \right\}, \quad (\text{H.2})$$

where D_{self}^R is the *signed* retarded self-kernel in the bulk coarse-grained effective action, C_H is the physical symmetrised force correlator, and S_0 is the pre-quantum action scale. The sign relating D_{self}^R to a separately defined positive damping kernel depends on the interaction and equation-of-motion convention; no sign is inferred from retarded support alone. The dimensionless bulk pairwise dephasing exponent is

$$\Phi_{\text{irr}}^{\text{bulk}}[q_+, q_-] = \frac{1}{2S_0^2} \int_0^T dt \int_0^T dt' q_{\Delta}(t) C_H(t-t') q_{\Delta}(t'). \quad (\text{H.3})$$

For a correlated joint preparation the complete action additionally contains S_{ic} , or its extended-contour equivalent. The full mean equation and the full $-\ln|\mathcal{F}|$ may then receive initial-boundary contributions;

the displayed bulk formula is not complete until those terms are included or proved absent. If preparation transients are not absent or explicitly separated, one must retain $C_H(t, t')$ rather than $C_H(t - t')$, and the single-frequency representation below does not apply.

Spectral representation of the symmetrised force kernel

Let $B = \sum_k c_k x_k$ be the supplied bath force operator. In the declared stationary thermal/KMS oscillator state, after the preceding initial-transient separation,

$$C_H(t) \equiv \frac{1}{2} \langle \{B(t), B(0)\} \rangle = \frac{S_0}{\pi} \int_0^\infty d\omega J_W(\omega) \coth\left(\frac{S_0 \omega}{2T_{\text{eff}}}\right) \cos(\omega t), \quad (\text{H.4})$$

and the action-normalised kernel used in the canonical PES dictionary is

$$N(t) \equiv \frac{C_H(t)}{S_0} = \frac{1}{\pi} \int_0^\infty d\omega J_W(\omega) \coth\left(\frac{S_0 \omega}{2T_{\text{eff}}}\right) \cos(\omega t). \quad (\text{H.5})$$

Here T_{eff} is the temperature parameter of the supplied environment, not a temperature derived for the ECT condensate. The density is the Weiss/Feynman–Vernon density J_W , not the bare mode-density weight. In the zero-temperature limit, $\coth \rightarrow 1$ and

$$C_H(t) = \frac{S_0}{\pi} \int_0^\infty d\omega J_W(\omega) \cos(\omega t). \quad (\text{H.6})$$

For a stationary non-thermal Gaussian state, the $\coth$ factor must instead be replaced by its symmetrised occupation spectrum.

Positivity of $J_W(\omega)$

For oscillator masses m_k , the Weiss density is defined by the mode decomposition

$$J_W(\omega) = \pi \sum_k \frac{c_k^2}{2m_k \omega_k} \delta(\omega - \omega_k), \quad (\text{H.7})$$

which reduces to the unit-mass normal-mode convention after absorbing $m_k^{-1/2}$ into the coupling. Each term is nonnegative, so

$$J_W(\omega) \geq 0 \quad \forall \omega \geq 0. \quad (\text{H.8})$$

Inside the supplied bath model this requires a non-tachyonic spectrum ($\omega_k^2 > 0$ for all retained environment modes) and a well-defined coupling decomposition. Those physical bath modes and couplings are not identified by the live ECT action. The result holds for *any* spectral shape—ohmic, super-ohmic, or otherwise.

Positivity in this appendix follows from the positive oscillator-mode decomposition (H.7). Retarded support and the Kramers–Kronig relation alone do not fix the dissipative sign. A passive same-channel relation between J_W and an imaginary retarded response requires a named physical operator, state and an explicit normalisation; the canonical matrix-valued version is stated separately in Appendix BE.6.

Positive-definiteness of C_H

In the full-line Fourier convention, Eq. (H.4) gives

$$\tilde{C}_H(\omega) = S_0 J_W(|\omega|) \coth\left(\frac{S_0 |\omega|}{2T_{\text{eff}}}\right) \geq 0. \quad (\text{H.9})$$

By Bochner’s theorem, a function whose Fourier transform is nonnegative is a positive-definite kernel:

$$\int dt \int dt' f(t) C_H(t - t') f(t') \geq 0 \quad \text{for all real } f. \quad (\text{H.10})$$

Main result

Choosing $f = q_\Delta$ in (H.10) gives

$$\Phi_{\text{irr}}^{\text{bulk}}[q_+, q_-] = \frac{1}{2S_0^2} \int dt \int dt' q_\Delta(t) C_H(t-t') q_\Delta(t') \geq 0. \quad (\text{H.11})$$

Equality means that the physical record difference lies in the null space of the positive noise kernel. This need not imply $q = q'$ when the channel has decoherence-free or unmonitored directions.

Consequently:

$$|\mathcal{F}_{\text{bulk}}[q_+, q_-]| = e^{-\Phi_{\text{irr}}^{\text{bulk}}} \leq 1, \quad |\mathcal{F}_{\text{bulk}}[q_+, q_-]| = 1 \iff C_H^{1/2} q_\Delta = 0 \quad \text{almost everywhere.} \quad (\text{H.12})$$

For a correlated preparation this is a theorem about the bulk factor. The full influence functional satisfies the same displayed identification only after every S_{ic} modulus contribution has been included or proved absent.

What follows and what does not

Follows (quadratic sector):

- $\Phi_{\text{irr}}^{\text{bulk}}[q_+, q_-] \geq 0$ for the supplied Gaussian bath whenever $J_W(\omega) \geq 0$.
- $|\mathcal{F}_{\text{bulk}}| \leq 1$: record differences are not amplified by the modulus, and are suppressed only outside the noise-kernel null space.

Does not follow from this theorem alone:

- Monotonicity of $S_{\text{ent}}(w)$ in time: Markovianity is insufficient; the reduced channel, state, entropy functional, and unitality/detailed-balance/contractivity hypotheses must be declared.
- Impossibility of partial recoherence in non-Markovian environments.
- The former scalar entropy-rate formula: it is not derived from ohmicity plus Markovianity and requires a normalised reduced dynamics, state, and explicit entropy calculation.
- Full H-theorem for the complete nonlinear branch (OP1, OP5, OP-Q17).
- Extension beyond the quadratic sector of the ordered branch.
- A first-principles identification of the effective open-system bath-response kernel from ECT microdynamics, beyond the already established retarded support of the supplied quadratic scalar operator (OP1, OP5).

Relation to other appendices

Appendix F establishes retarded propagation in the quadratic ordered-branch sector, i.e. the vanishing of the retarded Green function for negative time separation. Appendix G rewrites the same fixed operator as a symmetric-hyperbolic system and separately records the exact sign and regularity gates for a supplied quasilinear $P(X)$ equation. It does not define or derive a Green function for the full nonlinear medium. In the open-system reduction used in the present appendix, this retarded structure is encoded by a causal response kernel $K(\tau < 0) = 0$. What remains open is the full first-principles identification of this effective bath-response kernel directly from Euclidean microdynamics, especially beyond the scalar smooth ordered branch and without auxiliary closure assumptions (OP1, OP5). Appendix I provides the scaling analysis of the spectral density; it explains why the declared derivative-coupling benchmark is super-ohmic and what other supplied vertices or coarse-graining mechanisms can give ohmic behaviour.

I Scaling Analysis of the Bath Spectral Density

Reference from main text: Sections 3.8 and 3.9. Status: scaling analysis; strict derivation deferred to Section 26.1.

Spectral density in 3D with linear dispersion

For a declared benchmark bath of linearly dispersing soft modes with $\omega_k = c_{\text{char}}k$:

$$J_{\text{bare}}(\omega) \propto N_{\text{eff}} \omega^2 |g(\omega/c_{\text{char}})|^2 \quad (\text{up to powers of } c_{\text{char}}). \quad (\text{I.1})$$

The scalings are listed in both conventions. The Weiss-normalised spectral density entering the noise kernel of Appendix AX carries one power of ω less, $J_W(\omega) = J_{\text{bare}}(\omega)/(2\omega)$. The boundary scaling $J_W \propto \omega^0$ corresponds to $s = 0$, not to a member of the regular $s > 0$ family used below: its damping and finite-temperature noise integrals require a gap, finite-volume cutoff or another explicit infrared regulator. Scaling regimes:

Coupling	Bare mode-density scaling	Weiss scaling	Standard CL regime
$ g ^2 = \text{const}$	$J_{\text{bare}} \propto \omega^2$	$J_W \propto \omega$	ohmic
$ g ^2 \propto k^2$	$J_{\text{bare}} \propto \omega^4$	$J_W \propto \omega^3$	super-ohmic
$ g ^2 \propto k^{-1}$	$J_{\text{bare}} \propto \omega$	$J_W \propto \omega^0$	marginal IR end-point; regulator required

Standard 3D derivative coupling gives the super-ohmic Weiss contribution $J_W(\omega) \propto \omega^3$ (equivalently $J_{\text{bare}} \propto \omega^4$), not an ohmic one. Within this supplied normalisation a constant mode-space coupling gives $J_W \propto \omega$; whether ECT owns such a physical vertex remains Open. Starting from a derivative vertex, an effective ohmic law would require additional coarse-graining, reduced dimensionality or another declared mechanism.

Markovian approximation: heuristics

A declared source model may assign a memory scale τ_{env} and admit a white-noise approximation only when that scale is short compared with every resolved protocol/dynamical timescale. Neither $\tau_{\text{env}} \sim m_{\sigma}^{-1}$ nor a universal $\tau_{\text{dec}} = 1/(N_{\text{eff}}\gamma)$ follows from the present ECT action.

Ohmic behaviour and Markovianity are phenomenological model inputs. Deriving the physical split, state, vertices, memory kernel, and applicable timescale hierarchy from ECT remains Open (OP-Q5/OP-Q15).

J Goldstone sector in scalar-only ECT: integrability, inverse-Higgs reduction, and EFT status

Status: mixed. Scalar-only microscopic statements are Level A. Reduced soft-orientation EFT statements are Level B. No claim is made that the present minimal P1–P6 scalar basis proves three independent physical Goldstone particles.

A. Microscopic starting point

In the present minimal P1–P6 scalar basis the fundamental field is a single real scalar $\Phi(X)$ on $\mathcal{M}^4$. The ordered quantity is therefore not an independent vector multiplet but the exact one-form

$$Q_A \equiv \partial_A \Phi. \quad (\text{J.1})$$

In the ordered branch one writes $Q_A = u n_A$ with $u = |Q|$, $n_A n^A = 1$. However, u and n_A are *not independent microscopic variables* because Q_A is an exact gradient.

B. Integrability constraint

Exactness of $Q_A = \partial_A \Phi$ gives identically

$$\partial_A Q_B - \partial_B Q_A = 0. \quad (\text{J.2})$$

Substituting the decomposition:

$$\partial_A (u n_B) - \partial_B (u n_A) = 0. \quad (\text{J.3})$$

This is the integrability condition constraining the orientation sector. The ordered branch in scalar-only ECT is *not* equivalent to a generic unconstrained $O(4)/O(3)$ sigma model.

C. Linearised broken phase

Take uniform ordered background $Q_A^{(0)} = u_0 \delta_{Aw}$ and write $\Phi(X) = u_0 w + \chi(X)$, so $\delta Q_A = \partial_A \chi$. In the $u n_A$ language with $u = u_0 + \sigma$, $n_A = (\pi_i, 1) + \mathcal{O}(\pi^2)$:

$$\delta Q_i = u_0 \pi_i, \quad \delta Q_w = \sigma. \quad (\text{J.4})$$

Comparing with $\delta Q_A = \partial_A \chi$ gives the exact linear relations

$$u_0 \pi_i = \partial_i \chi, \quad \sigma = \partial_w \chi. \quad (\text{J.5})$$

Hence the apparent orientation triplet is determined by a single scalar χ .

D. Immediate consequences

Equation (J.5) implies

$$\pi_i = \frac{1}{u_0} \partial_i \chi, \quad (\text{J.6})$$

so the linearised orientation fluctuation is purely longitudinal:

$$\varepsilon_{ijk} \partial_j \pi_k = 0. \quad (\text{J.7})$$

At linear order, one longitudinal scalar coordinate χ parametrises the strict scalar-only map. Whether it is one independent physical soft mode is a Level-B statement requiring the supplied reduced action, domain and projector.

E. Relation to inverse-Higgs reduction

This reduction is the inverse-Higgs phenomenon [29, 30] for spacetime symmetry breaking. The broken object $Q_A = \partial_A \Phi$ is a geometric gradient field, not an internal multiplet. The broken-generator counting for $O(4) \rightarrow O(3)$ gives three, but the integrability condition (J.3) removes two as independent coordinates of the strict scalar map. It does not by itself perform the constrained physical mode count.

F. Minimal soft-orientation EFT

One may still organise the broken branch through a minimal orientation EFT on $O(4)/O(3) \simeq S^3$. In that effective description the effective stiffness is

$$\kappa_n = \mathcal{C}_n u_0^2 \quad (\text{Level B, EFT identity}), \quad (\text{J.8})$$

where $\mathcal{C}_n$ is an effective coefficient with dimension mass^{-2} , not yet derived from P1–P6 (Theorem 3.1; OP-Planck). After the reduction $\pi_i = \partial_i \chi / u_0$, a supplied derivative-only reduced EFT gives the conditional dispersion

$$\omega^2 = c_\chi^2 k^2, \quad c_\chi^2 = \frac{K_s}{K_w} c_{\text{char}}^2 \quad (\text{A-math conditional; Level B as an ECT mode}). \quad (\text{J.9})$$

The statement $c_\chi = c_{\text{char}}$ remains open. The derivative-only equation does not establish $m_\chi = 0$ in the physical constrained Hessian; integrability supplies no owned shift/Ward protection, and the mass operator remains Open.

G. Status summary

Statement	Status	Comment
Supplied P4 vector has $O(3)$ stabiliser and S^3 orbit	A-math conditional / physical SSB Open	Group geometry
$Q_A = \partial_A \Phi$ implies integrability	A	Exact scalar-only
$\pi_i = \partial_i \chi / u_0$ at linear order	A	Longitudinal reduction
One scalar coordinate χ parametrises the strict linear map	A	Kinematic reduction
One-mode physical interpretation of χ within a supplied reduced EFT	B	Requires action, quotient/domain and projector
Minimal 3-component orientation EFT	B	Effective parametrisation
$\kappa_n = \mathcal{C}_n u_0^2$; $[\mathcal{C}_n] = \text{mass}^{-2}$	B	NLO/induced EFT identity
$\mathcal{C}_n$ from bare P1–P6	Open	OP-Planck
$\omega^2 = c_\chi^2 k^2$	A-math conditional / B	Supplied derivative-only reduced EFT
m_χ^2 and a protecting Ward identity	Open	Stationary owner, projector and radiative closure missing
$c_\chi = c_{\text{char}}$	Open	Matching not proved
Three physical Goldstone particles	Not established	Not represented by the strict scalar map; a larger physical bundle is not derived
Dark-matter role of soft sector	Open	HRC diagnostic omits it; abundance and physical role not derived

K Conditional principal-cone result for the supplied second-order scalar operator and comparison with competing frameworks

Reference from main text: Universality Corollary (§4.3), cone inheritance (§4.7), and comparison with other theories (§3.10). Status: Level A mathematics for the explicitly supplied reduced scalar quadratic operator; Level B as an ECT physical mode; separate gauge, matter, photon and tensor sectors remain Open.

K.1 Theorem: one principal cone of the supplied linear scalar operator

Theorem. *Let Φ be the single scalar field and suppose that the supplied ordered-branch reduced quadratic EFT has no additional independent physical fields or principal operators and uses $K^{AB} = \beta \delta^{AB} - \alpha n^A n^B$ as its scalar principal tensor (Theorem 3.2). Then the characteristic surface $K^{AB} \xi_A \xi_B = 0$ defines the one principal cone of that scalar operator. The derivative coordinates σ and π_i inherit it inside this truncation; no all-sector or P1–P4-only cone theorem follows.*

Proof.

Step 1. From the uniqueness theorem (Theorem 3.2, §3.3), the allowed two-tensor form of the explicitly supplied K^{AB} is fixed up to two parameters (α, β) by the $O(4) \rightarrow O(3)$ symmetry data. This algebraic form does not by itself supply the physical action or its coefficients.

Step 2. From integrability (Appendix J), $Q_A = \partial_A \Phi$ is an exact one-form. At the linear level, $\delta Q_A = \partial_A \chi$, so that $\sigma = \partial_w \chi$ and $\pi_i = \partial_i \chi / u_0$.

Step 3. By the stated reduced-EFT assumption, the quadratic characteristic form is

$$\mathcal{Q}_{2,\chi} = \frac{1}{2} \int d^4 X K^{AB} \partial_A \chi \partial_B \chi = \frac{1}{2} \int d^4 X [\beta (\partial_i \chi)^2 - (\alpha - \beta) (\partial_w \chi)^2]. \quad (\text{K.1})$$

It yields one equation for one field. The positive-energy reduced Lorentzian functional on the retained branch is $\mathcal{J}_{L,\chi}^{(2)} = -\mathcal{Q}_{2,\chi}$; its overall sign leaves the characteristic proof unchanged. A physical action also needs an independent $A_{L,\chi} > 0$, whose relation to other sector normalisations is Open.

Step 4. The equation of motion $\beta \nabla^2 \chi - (\alpha - \beta) \partial_w^2 \chi = 0$ has the characteristic polynomial $\beta \mathbf{k}^2 - (\alpha - \beta) k_w^2 = 0$, defining a single cone with speed

$$\hat{c}_*^2 = \frac{\beta}{\alpha - \beta}. \quad (\text{K.2})$$

Since σ and π_i are derivatives of χ , no second principal symbol exists within this supplied scalar operator. Additional fields, constraints, nonlocalities or higher-derivative operators are outside the theorem. $\square$

K.2 Comparison with Einstein–aether theory

In Einstein–aether theory [23, 26] the preferred direction is carried by a unit timelike vector u^μ with its own kinetic action:

$$S_{\text{ae}} = -\frac{M^2}{2} \int d^4 x \sqrt{-g} [c_1 (\nabla_\mu u_\nu) (\nabla^\mu u^\nu) + c_2 (\nabla_\mu u^\mu)^2 + c_3 (\nabla_\mu u_\nu) (\nabla^\nu u^\mu) + c_4 u^\mu u^\nu (\nabla_\mu u_\alpha) (\nabla_\nu u^\alpha)]. \quad (\text{K.3})$$

The four free coefficients $c_1 \dots c_4$ give rise to three distinct propagation speeds for spin-0, spin-1, and spin-2 perturbations. Their equality $c_0 = c_1 = c_2$ requires two fine-tuning relations among the c_i [26].

In ECT, $n_A = \partial_A \Phi / |\partial \Phi|$ is a derived object, not an independent dynamical field. There are no free $c_1 \dots c_4$ in the scalar-only principal tensor: the unique K^{AB} has two parameters (α, β) and yields one scalar slope $\hat{c}_*$. This does not establish universal-cone equality for photon, matter, fermion or tensor sectors; those require their own actions and matching.

K.3 Comparison with the Standard Model Extension

The Standard Model Extension (SME) [27, 28] parametrises all possible Lorentz-violating operators for each sector independently. For the photon sector alone the CPT-even term involves the rank-4 coefficient tensor $(k_F)^{\kappa\lambda\mu\nu}$ with 19 independent components. The electron sector carries a separate tensor $(c_e)^{\mu\nu}$ with 9 components. These coefficients are phenomenologically independent: the SME does not relate them.

In the supplied linear scalar ordered branch, the only principal tensor is K^{AB} with parameters (α, β) . This removes independent Lorentz-violation coefficients only inside that scalar sector. Photon, fermion, matter and tensor actions can contain independent allowed operators; their coefficients and cones must be derived or bounded sector by sector.

Nagel et al. [293] report the rotating-cavity result

$$\Delta v/v = (9.2 \pm 10.7) \times 10^{-19}.$$

This representative 10^{-18} -level electrodynamics test constrains convention-dependent combinations of photon-sector coefficients together with matter and electron coefficients; it is not a standalone bound on $\delta c_\gamma/c$. Gravitational–electromagnetic nonuniversality is bounded by $-3 \times 10^{-15} \leq (c_{\text{GW}} - c_\gamma)/c_\gamma \leq +7 \times 10^{-16}$ (GW170817/GRB 170817A [12]). These bounds are external kill gates. The present ECT action neither derives the relevant physical-sector coefficients nor proves Planck suppression.

K.4 Scope and limitations

The theorem covers only the explicitly supplied reduced linear scalar operator; it neither follows from P1–P4 alone nor proves that this coordinate is the complete physical scalar spectrum. Gauge and fermion routes are conditional programmes (§4.7, §9); a physical tensor principal symbol is absent. No all-sector universality follows until those actions and their radiative stability are established.

L EFT Motivation via Integrating Out the Radial Mode

Reference from main text: Section 4.5.

Status. This appendix gives a schematic EFT template about the supplied P4 ordered-branch chart, not a derivation of a vacuum sector or of the broken-phase action from P3. The affine background Φ_{bg} is not a solution of the P3 equations, and no stationary P4 action or state is presently supplied.

Phase action and heavy-mode integration

Assume a separately supplied P4 completion has a positive isolated heavy pole M_{P4} and a controlled low-energy expansion. Only then may that pole be integrated out to define an effective action for a retained soft fluctuation φ . The isotropic kinetic term in the broken phase may acquire the $O(3)$ -invariant deformation parametrised by (α, β) :

$$\mathcal{Q}_{2,\varphi}^{(2)} = \int d^4X \left[\frac{1}{2} K^{AB} \partial_A \varphi \partial_B \varphi + \frac{m_{\text{eff}}^2}{2} \varphi^2 \right], \quad K^{AB} = \beta \delta^{AB} - \alpha n^A n^B.$$

On the retained positive-energy hyperbolic branch the reduced quadratic Lorentzian functional is $\mathcal{S}_{L,\varphi}^{(2)} = -\mathcal{Q}_{2,\varphi}^{(2)}$ and the physical action is $S_{L,\varphi}^{(2),\text{phys}} = A_{L,\varphi} \mathcal{S}_{L,\varphi}^{(2)}$ with $A_{L,\varphi} > 0$. Higher-derivative operators require their own continuation, pole/constraint analysis and relative coefficients; the two-derivative sign test does not orient them by an ellipsis. Neither (α, β) , $A_{L,\varphi}$ nor these relative Lorentzian data are derived by the schematic heavy-mode template; all require the missing stationary P4 matching (OP-grad).

M Structural candidate mechanism for the second condensate transition $O(3) \rightarrow O(2)$

Status: structural Level B/C. This appendix describes a structural candidate mechanism, not a derived result. It does not prove that the second transition occurs, does not derive v_2 from first principles, and does not establish the electroweak gauge sector from ECT postulates P1–P6.

A. Conditional inputs and exact group algebra

After P4 supplies a nonzero vector and BR1 separately supplies a coherent complex-field closure, the architecture contains two declared analysis sectors:

- a coherent phase sector with compact $U(1)$ -like structure in the effective loop/phase description (the winding variable θ ; its role as an electroweak-locking participant is not derived from P1–P6);
- an orientation sector with $O(3)$ stabiliser for the supplied P4 vector.

The Lie algebras satisfy the exact isomorphism

$$\mathfrak{so}(3) \cong \mathfrak{su}(2). \quad (\text{M.1})$$

Furthermore, the connected rotation subgroup admits an $SO(2) \cong U(1)$ subgroup as the stabiliser of a chosen axis. Equivalently, a further supplied axis within the $O(3)$ stabiliser has an $O(2)$ (or connected $SO(2)$) stabiliser. These are Level-A group facts conditional on the supplied data; physical transitions and selections are not derived.

B. Structural mechanism candidate (Level B/C)

The following scenario is structurally admissible within ECT but requires two additional assumptions not yet derived from P1–P6.

Assumption A1. At some intermediate scale $v_2 \ll \phi_0$, a coupling between the coherent phase sector and the residual orientation sector becomes operative.

Assumption A2. The minimum of the condensate functional at scale v_2 aligns the $U(1)$ axis of the phase sector with a preferred direction in $O(3)$.

Under A1 and A2, the residual $O(3)$ is broken to its $O(2)$ stabiliser:

$$O(3) \xrightarrow{\text{phase-orientation alignment}} O(2). \quad (\text{M.2})$$

This breaking pattern is *formally analogous* to a reduced weak-like symmetry-breaking pattern. It should not be identified directly with the full Standard Model electroweak transition. The analogy is structural; it does not constitute a derivation of the electroweak sector.

C. Formal analogy with a weak-like sector

Under the breaking (M.2), the candidate second transition provides

- two broken directions;
- one unbroken residual $O(2) \cong U(1)_{\text{res}}$ direction;
- and, in an EFT embedding with a second condensate potential, one radial excitation at the intermediate scale.

This does *not* reproduce the full Standard Model electroweak generator counting, since $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$ contains three broken generators, whereas $O(3) \rightarrow O(2)$ contains only two. Accordingly, the present construction should be read as a *reduced structural analogue* or candidate weak-like sector, not as a completed derivation of the physical electroweak spectrum.

Only if a more complete local gauge structure were shown to emerge, together with the missing generator content, could one begin to relate this sector to the physical $W^\pm$, Z^0 , Higgs, and photon degrees of freedom. At the present stage the analogy is intentionally weaker.

D. Scale v_2 : matching constraint, not derivation

If A1 and A2 hold, the scale of the second transition is fixed by low-energy matching to the observed Fermi interaction:

$$v_2 = \left(\sqrt{2} G_F \right)^{-1/2} \approx 246.22 \text{ GeV}, \quad (\text{M.3})$$

where the current PDG-2026 web review gives $G_F = 1.1663785(6) \times 10^{-5} \text{ GeV}^{-2}$ [32]. This value is a standard electroweak *matching constraint*, not a first-principles ECT prediction.

As an order-of-magnitude estimate, dimensional transmutation gives

$$v_2 \sim \phi_0 e^{-\mathcal{C}}, \quad \mathcal{C} \approx \ln \frac{\phi_0}{v_2} \approx 37. \quad (\text{M.4})$$

For a schematic one-loop alignment running with coefficient b_a and dimensionless locking coupling g_a , this would correspond to $b_a g_a^2(\phi_0) \approx 8\pi^2/37 \approx 2.1$ in the strong-critical limit. This is numerically plausible but does not constitute a derivation.

E. What remains open

1. Does assumption A1 hold in ECT? Is there a phase–orientation coupling in the condensate action?
2. Is the $O(3)$ orientation symmetry local or global at scale v_2 ? Locality is required for the massive vector boson interpretation.
3. What is the explicit dynamical mechanism generating the scale v_2 (RG running, topological instanton, or other)?
4. How do SM fermion charges and Yukawa couplings arise?

Until these are answered, the second transition $O(3) \rightarrow O(2)$ remains a *structural candidate mechanism* of ECT at Level B/C, not an established result.

Generator counting

For $O(3) \rightarrow O(2)$, the dimension of the broken-generator space is

$$\dim O(3) - \dim O(2) = 3 - 1 = 2. \quad (\text{M.5})$$

Thus the candidate second transition yields two broken and one unbroken generator. This is the exact group-theoretic content of the analogy used in Section 7.4. No stronger physical identification follows from this counting alone.

What this appendix establishes and what it does not

This appendix does not derive the physical electroweak sector from the minimal ECT postulates. What it establishes is only the following structural chain:

- (i) if a second effective ordering step $O(3) \rightarrow O(2)$ is admitted, then it provides a reduced orientation-sector weak-like analogue with two broken directions, but not the full electroweak generator counting; the Hopf-fibered phase–orientation extension of Appendix N is required for the candidate three-broken-direction geometry;
- (ii) if a condensate doublet EFT is admitted, then a Higgs-like radial mode and the standard electroweak matching relations can be written in the usual low-energy form;
- (iii) this provides a structural embedding of the electroweak sector into the condensate architecture of ECT.

What remains open is:

- (a) the dynamical origin of the second transition,
- (b) the derivation of the doublet structure,
- (c) the derivation of the gauge couplings and Yukawa sector,
- (d) the derivation of the hierarchy v_2/ϕ_0 .

N Hopf-fibered phase–orientation locking as a candidate electroweak geometry

This appendix presents a structural geometric proposal addressing the generator-counting limitation of the reduced $O(3) \rightarrow O(2)$ analogue of Appendix M. The proposal identifies the *pre-gauge* vacuum manifold of a phase–orientation diagonal locking with the fixed-radius Higgs-doublet vacuum manifold of the Standard Model. It does not derive the dynamics of locking, the chiral electroweak gauge sector, the gauge couplings, or the scale of the transition.

N.1 Generator-counting limitation of $O(3) \rightarrow O(2)$

Appendix M analyses the ordering pattern $O(3) \rightarrow O(2)$ as a reduced analogue of electroweak symmetry breaking. The number of broken generators is

$$\dim O(3) - \dim O(2) = 3 - 1 = 2. \quad (\text{N.1})$$

This is insufficient for $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$, which requires three broken directions to provide longitudinal modes for $W^\pm$ and Z via the Higgs mechanism. The reduced analogue therefore captures only the orientation sector and cannot, by simple identification, be the full electroweak transition; a complete description requires the full phase-plus-orientation structure.

N.2 Pre-locking symmetry: $SU(2)_{\text{orient}} \times U(1)_\theta$

Two structures are present in the effective ECT alignment sector.

Orientation lift to $SU(2)_{\text{orient}}$. Since the connected orientation group has Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, its spinorial lift is naturally described by $SU(2)_{\text{orient}}$. This is the natural double-cover language for fermionic couplings (cf. the orientation operator entering the 5th-force coupling $L_5 = \beta \bar{\Psi} \gamma^A n_A \Psi$ in Section 7), but it does not by itself identify the group with the chiral weak group $SU(2)_L$.

Coherent phase $U(1)_\theta$. The phase variable θ enters the supplied coherent phase-sector EFT. Its compactness gives the winding condition $\oint \partial_A \theta dX^A = 2\pi n$ and supports a $U(1)_\theta$ structure. It does not select a minimal loop, either reduced-loop coefficient, the operational S_0 , or the matching to $\hbar$. The fixed-core and tension-balanced coefficients are separate conditional candidates with Open owner maps.

Caveat (OP-EW-locking.0). The fundamental ECT field Φ is a real scalar (P4); the appearance of a compact coherent phase variable θ in the alignment sector is a modelling assumption of the effective theory, not a derived consequence of P1–P6. Whether $U(1)_\theta$ admits a derivation from purely real- Φ dynamics, and whether it plays the role of a participant in an electroweak-like locking sector distinct from any post-breaking phase, is itself open.

N.3 Diagonal locking: pre-gauge geometry

Suppose the alignment dynamics is energetically minimised by diagonal locking of phase and orientation:

$$SU(2)_{\text{orient}} \times U(1)_\theta \longrightarrow U(1)_{\text{diag}}. \quad (\text{N.2})$$

The number of broken directions is

$$\dim[SU(2) \times U(1)] - \dim U(1)_{\text{diag}} = (3 + 1) - 1 = 3, \quad (\text{N.3})$$

matching the number required by $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$.

The vacuum manifold is the coset

$$\mathcal{M}_{\text{locked}} = \frac{SU(2)_{\text{orient}} \times U(1)_\theta}{U(1)_{\text{diag}}} \simeq SU(2) \simeq S^3, \quad (\text{N.4})$$

where the diagonal $U(1)$ acts freely. This is the same topology as the fixed-radius Higgs-doublet vacuum manifold $|H|^2 = v^2/2$.

Hopf fibration realisation. The coset S^3 admits the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$, with S^2 the orientation base and S^1 the locked phase fibre. This is structurally distinct from a trivial product $S^2 \times S^1$: the phase circle is non-trivially fibered over the orientation sphere. When field configurations are mapped into the S^2 base, their topological sectors can be classified by the Hopf invariant.

Pre-gauge versus post-gauge caveat. The coset identification $\mathcal{M}_{\text{locked}} \simeq S^3$ holds at the pre-gauge / global-symmetry level. In gauged electroweak theory, points of S^3 related by gauge transformations are physically identified, so the physical vacuum after modding out gauge redundancy is not the same object. Hopf-locking therefore reproduces the pre-gauge Higgs-doublet vacuum geometry and generator counting; it does *not* by itself establish that this geometry is gauged identically to the Standard Model.

N.4 Hopf-locking operator and Berry connection

The natural alignment operator compatible with locking dynamics is schematically of the form

$$\mathcal{L}_{\text{lock,L}}^{\text{schem}} = -\frac{1}{2} K_\theta g_{\text{eff}}^{AB} (\partial_A \theta + \mathcal{A}_A[n]) (\partial_B \theta + \mathcal{A}_B[n]) - \frac{1}{2} \kappa_n g_{\text{eff}}^{AB} \partial_A n \cdot \partial_B n + \dots, \quad (\text{N.5})$$

where $\mathcal{A}_A[n]$ is the connection one-form induced by the orientation field. In a local coordinate patch on the S^2 base,

$$\mathcal{A}_A[n] = \frac{1}{2} (1 - \cos \beta) \partial_A \alpha, \quad (\text{N.6})$$

with (α, β) the standard angles on S^2 . Geometrically, $\mathcal{A}_A[n]$ is the Berry / Dirac-monopole connection on orientation states; as a monopole connection it is not globally defined in a single patch.

Schematic, not derived. Equation (N.5) is a candidate effective density written as a locking term, not a derived ECT action. Its coefficients K_θ and κ_n are not specified by P1–P6; κ_n specifically depends on closure of OP-Planck/OP- C_n through the NLO/induced route discussed in Section 4.4 and paragraph 5.1.

Composite versus independent connection (OP-EW-gauge-field). A connection $\mathcal{A}_A[n]$ determined entirely by n is a *composite* connection and does not carry independent propagating degrees of freedom. In the Standard Model, by contrast, electroweak gauge bosons $W^\pm$, Z , γ are independent local fields with their own kinetic terms. Two distinct routes are available:

- (a) *Composite gauge route.* Standard Model gauge bosons are interpreted as collective excitations of the orientation/phase sector, with $\mathcal{A}_A[n]$ reproducing their leading behaviour. Closing this route requires deriving Yang–Mills-type kinetic terms F_{AB}^2 from ECT dynamics and showing that the propagating modes have the correct spin and mass content.
- (b) *Independent connection route.* A separately supplied local bundle structure, motivated but not generated by P5, is gauge-fixed by introducing an independent compensating connection; $\mathcal{A}_A[n]$ is then a background/Berry connection separate from the dynamical gauge field. This is closer to standard electroweak gauge theory but loses the structural unity of the composite picture.

Neither route is established here. The choice is itself open (OP-EW-gauge-field).

N.5 Hopfion structure: mathematical template, not closure

The closest developed mathematical template for the topological locking scenario is the Faddeev–Niemi model [41, 42, 43, 44]. In that setting, static three-dimensional Hopfion configurations obey a Vakulenko–Kapitanski-type lower bound on their finite static energy,

$$E_{\text{Hopfion}}[Q_H] \geq c|Q_H|^{3/4}, \quad (\text{N.7})$$

where $Q_H \in \pi_3(S^2) = \mathbb{Z}$ is the Hopf invariant and c is a model-dependent constant set by the stiffness and Skyrme-type coefficients of the effective model.

Critical caveat. Faddeev–Niemi solitons are static configurations in three spatial dimensions, with finite static energy. A four-dimensional Euclidean semiclassical suppression factor instead requires a localised bounce with a finite vacuum-subtracted physical action B_H . Only after supplying a finite positive action unit $\Sigma_H^{\text{act}} > 0$ is the exponent dimensionless: $\mathcal{J}_H := B_H/\Sigma_H^{\text{act}}$ and the weight is $e^{-\mathcal{J}_H}$. It is a genuine suppression factor only when $\text{Re } \mathcal{J}_H > 0$ (for a real Euclidean bounce, $B_H > 0$); $\text{Re } \mathcal{J}_H = 0$, a negative real bounce difference, and a complex thimble require separate interpretations. Neither Σ_H^{act} nor a map to the bare-P3 owners A_{P3} and $\Sigma_{P1,P3}$, their ratio κ_{P3} , Σ_θ , Σ_ϑ , S_0 or $\hbar$ is derived. A decay interpretation additionally needs the bounce state and boundary data, a contour, the fluctuation determinant with its appropriate negative and zero modes, and a normalised prefactor. A static three-dimensional soliton extended trivially in Euclidean time has action $E_{\text{static}}T_E$, which diverges for infinite T_E . Translation from a Hopfion energy bound to an ECT scale formula therefore requires either

- a genuine 4D Euclidean Hopf-locking instanton (finite-action localised configuration), not yet constructed within ECT; or
- an alternative interpretation in which Hopfion energy sets a threshold mass scale rather than a tunnelling exponent.

We record the Faddeev–Niemi structure as a promising mathematical template, not as a closure of the topological scale-generation route of equation (5.9).

N.6 What this appendix establishes and what it does not

What this construction achieves.

- (1) Hopf-fibered diagonal locking provides the correct number of broken generators (three) for an electroweak-like pattern, structurally improving on the reduced $O(3) \rightarrow O(2)$ analogue.
- (2) The resulting pre-gauge vacuum manifold is topologically S^3 , matching the fixed-radius global Higgs-doublet manifold of the Standard Model.
- (3) The minimal alignment operator takes covariant Hopf form, and the connection $\mathcal{A}_A[n]$ is identified geometrically as a Berry / Dirac-monopole connection on the S^2 base.

What this construction does not achieve.

- (a) *Pre-gauge vs post-gauge.* The match S^3 holds at the global-symmetry level only. Identification with the gauged electroweak vacuum requires a derivation of local gauge redundancy in the locking sector (OP-EW-gauge-field).
- (b) *Chirality.* Left-handed Standard-Model fermions occur in $SU(2)_L$ doublets [32]; this is an external representation fact. Any action of $SU(2)_{\text{orient}}$ on aligned matter through a double-cover lift is a distinct ECT-specific hypothesis and is not a priori chiral. The reduction $SU(2)_{\text{orient}} \rightarrow SU(2)_L$ is a separate open problem (OP-EW-chiral).
- (c) *Hypercharge.* $U(1)_\theta$ is a compact phase; $U(1)_Y$ is the hypercharge group with specific anomaly structure and matter-charge assignments. The identification $U(1)_\theta \rightarrow U(1)_Y$ is non-trivial and not established here (OP-EW-hypercharge).
- (d) *Couplings.* The geometric construction does not derive g , g' , or the Weinberg angle. A derivation of these from κ_n and Hopf geometry is an open possibility, not a result (OP-EW-couplings).
- (e) *Scale.* Locking geometry alone does not generate v_2 . Combination with a non-polynomial scale-generation mechanism (Section 5.1, equation (5.9)) is required (OP-EW-scale).
- (f) *Naturalness.* Protection of the secondary radial mode σ_2 from additive $\mathcal{O}(\phi_0^2)$ corrections is not addressed by the locking geometry: the supplied sigma-model symmetry can protect tangent orientation coordinates inside that model, not the radial mode or a separately owned physical spectrum (OP-EW-naturalness).
- (g) *Matter.* Chiral fermion representations and Yukawa hierarchy are not derived (OP-EW-matter).

N.7 Conditional layered closure theorem

We collect the chain of dependencies as a layered conditional statement.

Conditional OP-EW Reduction Theorem (layered). Assume:

- *Layer 0.* NLO/induced ECT dynamics generate the orientation stiffness $\kappa_n = \mathcal{C}_n u_0^2$ (OP-Planck / OP- C_n subproblem).
- *Layer 1.* A compact $U(1)_\theta$ is established as a genuine symmetry of the effective alignment sector, distinct from any post-breaking phase (OP-EW-locking.0).
- *Layer 2.* Diagonal locking $SU(2)_{\text{orient}} \times U(1)_\theta \rightarrow U(1)_{\text{diag}}$ is the energetically preferred minimum, giving Hopf-fibered S^3 pre-gauge vacuum manifold (OP-EW-locking.1).
- *Layer 3.* A physical scale-generating mechanism, threshold match and radiative closure derive v_2 rather than merely rewrite v_2/ϕ_0 logarithmically (OP-EW-scale).

- *Layer 4.* The pre-gauge structure is promoted to a local chiral $SU(2)_L \times U(1)_Y$ gauge symmetry, with couplings g, g' either derived from the ECT alignment/gauge sector or introduced as low-energy matching data (OP-EW-chiral, OP-EW-hypercharge, OP-EW-couplings, OP-EW-gauge-field).
- *Layer 5.* The radial excitation σ_2 is a collective critical mode protected from additive $\mathcal{O}(\phi_0^2)$ corrections (OP-EW-naturalness); chiral fermions and Yukawa hierarchy follow from the locking structure (OP-EW-matter).

If Layers 0–5 close, the resulting theory must calculate v_2 and the electroweak structure from the derived action and state and then compare them with the external Standard-Model values. The matched number $\mathcal{J}_{EW} \approx 36.8$ is not an order-one microscopic input.

Currently: Layer 0 partially established (NLO/induced route identified, κ_n derivation incomplete); Layer 2 structurally identified by Hopf-locking but dynamics not derived; Layers 1, 3, 4, 5 fully open.

This is a closure programme, not a solution. It converts the diffuse statement “the energy hierarchy and separate galactic matching are not derived” into a sharply defined sequence of intermediate problems with explicit dependencies.

O Orientation-stiffness dimension audit and open tensor matching

O.1 Primary condensate quantities

The dimensions of u_0, m_σ, ϕ_0 and the scalar coefficients remain those derived in the owning amplitude/scalar sectors.

O.2 Orientation-stiffness sector

The NLO operator

$$\frac{\mathcal{C}_n}{2} u^2 (\partial_B n_A) (\partial^B n^A), \quad \kappa_n = \mathcal{C}_n u_0^2,$$

has $[\mathcal{C}_n] = M^{-2}$ and $[\kappa_n] = M^2$. This is a valid orientation-EFT statement, not a tensor-gravity normalisation.

O.3 Candidate tensor-scale matching

Equal dimensions permit only the Open hypothesis

$$M_G^2 \stackrel{?}{=} c_M \kappa_n,$$

where M_G is defined in a separately supplied physical tensor action. Weak-field calibration of that action does not determine c_M or create the missing fixed- α, β tangent-director linear TT/static-density vertex; quadratic/nonlinear alternatives remain parametric/Open.

O.4 Parametric orientation-stiffness matching template

The exact-gradient decomposition obeys the integrability identity

$$\partial_A (u n_B) - \partial_B (u n_A) = 0. \quad (\text{O.1})$$

It does not supply an independent radial pole or a vertex coupling that pole to $(\partial n)^2$. A future stationary completion might contain a canonically normalised orientation-sector heavy field H_n with operator

$$\mathcal{O}_{H_n} = -\partial^2 + M_{n,\text{heavy}}^2 + \dots \quad (\text{O.2})$$

and an explicit $H_n(\partial n)^2$ interaction. Only under those supplied assumptions, with a declared measure, domain, regulator, subtraction and complete operator basis, could a determinant generate the power-counting template

$$\mathcal{C}_n = \frac{\hat{a}_n}{16\pi^2 M_{n,\text{heavy}}^2}, \quad (\text{O.3})$$

$$\hat{a}_n = \gamma_{\text{geom}} \mathcal{N}_{\text{match}}. \quad (\text{O.4})$$

At present this is PARAMETRIC ONLY / MISSING VERTEX. The homogeneous P3 curvature $m_{\Phi,\text{rad}}$ is not an owner of $M_{n,\text{heavy}}$, and no equality to another sector's heavy scale is implied.

For a supplied $\mathcal{C}_n$, the definition $\kappa_n = \mathcal{C}_n u_0^2$ is dimensionally consistent. It still does not create a helicity-2 field or determine a gravitational residue:

$$M_G^2 \stackrel{?}{=} c_M \kappa_n, \quad c_M, \text{ tensor action, projector, pole, residue and source Open.} \quad (\text{O.5})$$

Lemma O.1 (Scoped heavy-field matching template). *If one supplied stationary completion contains the positive heavy operator, the required heavy-orientation vertex and a valid local expansion, then its renormalised determinant may contribute an orientation Wilson coefficient with the dimensional form of Eq. (O.3). Neither its sign nor finite value follows without the full matching calculation, and no physical tensor or gravity coefficient follows from it.*

Tensor-normalisation bridge remains Open

The established chain ends at κ_n ; dimensional equality supplies neither helicity-2 dynamics nor a static-density vertex.

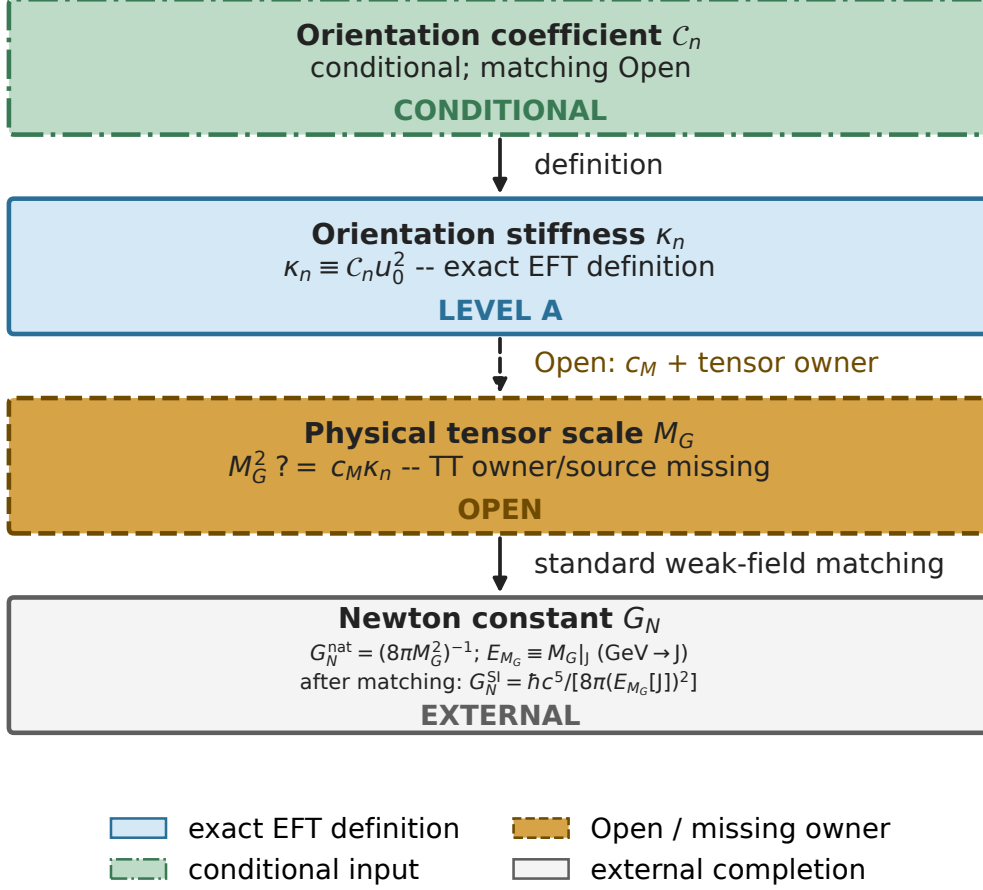


Figure 37: Corrected ownership chain from the ordered variables through the conditional NLO orientation coefficient and the exact EFT definition $\kappa_n \equiv \mathcal{C}_n u_0^2$. The dashed step $M_G^2 \stackrel{?}{=} c_M \kappa_n$ is Open: the physical helicity-2 action, source normalisation, residue and the coefficient c_M are not supplied. The natural-unit tensor normalisation $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ is standard only inside a supplied tensor completion. An SI Newton constant is defined only after physical tensor, source, clock, light and scalar-response matching, as $G_N^{\text{SI}} = \hbar c^5 / [8\pi (E_{M_G}[J])^2]$, where $E_{M_G} \equiv M_G|_{\text{joule}}$ denotes the manuscript energy scale converted from GeV to joules. Colour is redundant with fill luminance, border style, arrow style and literal status text. The diagram therefore does not derive gravity from κ_n . Complementary full-width panels are retained in Appendix BP.

O.5 Phase sector and action scale

The action scale S_0 and its matching to $\hbar$ are independent of the Open tensor-normalisation bridge.

Three distinct response levels. The scalar principal tensor, an orientation stiffness and a physical tensor coefficient are different operator owners. A dimensionless derivative expansion diagnostic, after normalising the leading operator, is for example

$$\varepsilon_n(k) = |\mathcal{C}_n| k^2 \ll 1. \quad (\text{O.6})$$

The former quantity $c_M \mathcal{C}_n u_0^2 / c_{\text{char}}^2$ had mass dimension two and was not a suppression factor; it is withdrawn.

O.6 Owner chain

The established chain ends at the orientation operator:

$$\partial_A \Phi = u n_A \longrightarrow \mathcal{C}_n \longrightarrow \kappa_n = \mathcal{C}_n u_0^2.$$

The continuation $\kappa_n \rightarrow M_G^2 \rightarrow G_N$ is Open at its first arrow; the second is standard weak-field matching inside a supplied tensor/Einstein EFT.

Detailed derivation and matching inventory

Canonical dimensions. In four Euclidean dimensions a canonically normalised scalar has mass dimension one, its gradient has dimension two, and the ordered-gradient amplitude u therefore has dimension two. The unit director n_A is dimensionless. The operator $u^2(\partial n)^2$ has dimension six, so its Wilson coefficient $\mathcal{C}_n$ has dimension minus two and $\kappa_n = \mathcal{C}_n u_0^2$ has dimension two. This audit establishes the orientation stiffness and nothing about its particle representation.

Why leading $P(X)$ is insufficient. For a one-scalar pullback $Q_A = \partial_A \Phi = u n_A$, the exact integrability condition $\partial_{[A} Q_{B]} = 0$ relates director gradients to amplitude gradients. A pure local $P(X)$ action depends algebraically on $Q_A Q^A$ and does not generate an independent two-derivative director elasticity at tree level. Such an operator must arise from higher derivatives, additional fields, loops, boundaries or a declared nonlocal reduction. This identifies the derivative order but not a tensor-gravity owner.

Conditional heavy-amplitude determinant. Expanding $u = u_0 + \sigma$ around the ordered background gives a massive radial operator of Laplace type only after the background, measure and constraint reduction are fixed. Formally integrating the radial field produces $\Gamma_\sigma = \frac{1}{2} \text{Tr} \ln \mathcal{O}_\sigma$. A proper-time or heat-kernel expansion then organises local operators in powers of $1/m_\sigma$. The coefficient multiplying a two-derivative orientation invariant is scheme- and basis-dependent; power divergences, logarithms and finite counterterms must be declared. P4 supplies neither this operator nor its mass; identifying m_σ with the homogeneous-P3 curvature or a Planck-scale pole is a separate matching ansatz.

Foliation geometry. When the director defines codimension-one leaves, its projected symmetric derivative can be organised as an extrinsic-curvature tensor only after the normal, projector and integrability assumptions are supplied. The exact pointwise relation is Eq. (14.3): the leaf Ricci scalar equals the projected ambient Riemann contraction plus $\varepsilon(K^2 - K_{ab}K^{ab})$. In the Euclidean convention $\varepsilon = +1$, and flat ambient geometry removes only the projected ambient term. Boundary divergences enter a scalar-curvature action decomposition, not this identity. Neither $K^2 - K_{ab}K^{ab}$ nor the Ricci scalar is identically $(\partial n)^2$; the general local basis contains independent trace, traceless, acceleration and boundary operators. Reduction to one coefficient $\mathcal{C}_n$ therefore requires a trace constraint, integration-by-parts identity, boundary condition or matching prescription. This ambiguity is physical and cannot be hidden in a notation change.

Conditional leaf-conformal kernel. If one deliberately chooses a scalar Laplace-type operator on three-dimensional leaves with conformal coefficient $\xi_{\text{leaf}} = 1/8$, the standard curvature coefficient contains $1/6 - \xi_{\text{leaf}} = 1/24$. That arithmetic is exact under the declared closure. It fixes only a geometric factor in one curvature channel. It does not fix the finite Wilson coefficient, the reduction to $(\partial n)^2$, or the normalisation of a physical tensor pole. The factor $1/24$ must therefore remain conditional rather than being presented as a Planck-scale prediction.

Renormalised coefficient. The parametrisation

$$\mathcal{C}_n = \frac{\hat{a}_{\text{eff}}}{16\pi^2 m_\sigma^2}, \quad \hat{a}_{\text{eff}} = \gamma_{\text{fol}} \mathcal{N}_{\text{match}},$$

separates the loop scale from a finite matched normalisation. The symbol $\mathcal{N}_{\text{match}}$ includes regulator-dependent pieces, operator-basis reduction and finite counterterms. It must not include a hidden spin-2 identification: the latter is a separate map from the orientation Hessian to a physical constrained tensor residue.

What the one-loop route actually establishes. Under its assumptions, the radial determinant makes an NLO orientation operator plausible and supplies a reproducible power-counting form. It does not establish the sign or numerical size of the complete renormalised coefficient without a UV prescription. More importantly, it changes a coefficient of the orientation EFT; it does not change the fixed- α, β tangent-director linear theorem $h_{ij}^{\text{TT}} = h_{ww} = 0$ or create a linear static-density vertex. Quadratic/nonlinear response remains a separate parametric/Open question.

Open tensor map. To relate κ_n to a tensor normalisation, one must enlarge or alter the field map, derive the constrained quadratic operator, isolate a symmetric physical pole and compute its residue. Only then is a coefficient c_M in $M_G^2 = c_M \kappa_n$ meaningful. Setting $c_M = 1$ by convention before that calculation is not weak-field matching; it is an unsupported identification.

Weak-field calibration. Given a supplied Einstein/FP tensor action, the observed G_N calibrates its coefficient M_G . This calibration may be used to infer what value of a microscopic combination would be required, but the inference is not a prediction of that combination. A large value of $\mathcal{N}_{\text{match}}$ or c_M is a naturalness diagnostic, not evidence that the missing tensor map has been derived.

Phase action scale remains separate. The phase stiffness, fixed-core coefficient l_0^{EFT} and positive-tension coefficient $l_{0,\text{bal}}^{\text{eff}}$ belong to supplied compact-phase loop models. Their dimensional reduction, either map to the operational S_0 , and the separate matching $S_0 \leftrightarrow \hbar$ do not determine M_G . Conversely, calibrating M_G with G_N does not derive S_0 . Common origin in one condensate does not collapse independent Wilson coefficients into one parameter.

Three response levels. The manuscript therefore distinguishes: (i) the leading scalar principal tensor and its cone; (ii) an NLO orientation stiffness; and (iii) a possible physical tensor coefficient. These are different operator owners. The diagnostic ratio in Eq. (O.6) is dimensionally valid, but it becomes gravitational only after the Open coefficient and tensor map are derived.

Reproducibility requirements. A future closure of this appendix must provide the expanded radial operator, constraint Jacobian, boundary conditions, regularisation and subtraction scheme, complete foliation-invariant operator basis, finite Wilson coefficients, constrained pole decomposition, tensor projector and matter vertex. The calculation should be reproducible both in the Euclidean parent description and in the reduced Lorentzian EFT. Until then the complete established chain ends at κ_n .

P Correlation Length from the Radial-Mode Propagator

Reference from main text: Section 5.5. Status: standard EFT estimate from the massive Euclidean propagator.

Euclidean propagator

Quadratic action of the radial mode: $S_\sigma^{(2)} = \frac{1}{2} \int d^4X [(\partial_A \sigma)^2 + m_\sigma^2 \sigma^2]$. Fourier propagator: $G_\sigma(k) = 1/(k^2 + m_\sigma^2)$.

Coordinate representation at large distances

In $d = 4$ Euclidean space the exact free propagator and its large-distance asymptotic are

$$G_\sigma(r) = \frac{m_\sigma}{4\pi^2 r} K_1(m_\sigma r) \xrightarrow{m_\sigma r \rightarrow \infty} \frac{\sqrt{m_\sigma}}{4\sqrt{2}\pi^{3/2}} \frac{e^{-m_\sigma r}}{r^{3/2}}. \quad (\text{P.1})$$

The exponential factor $e^{-m_\sigma r}$ defines the P3 homogeneous radial decay length

$$\boxed{\xi_{\text{P3}} = m_{\Phi, \text{rad}}^{-1}} \quad (\text{P.2})$$

This is standard P3 free-field mathematics. It is not a P4 correlation length or ordered-branch threshold; both that owner map and any Planck-length matching remain Open.

Q Phase Periodicity, Winding Sectors, and the Fixed-Core Coefficient $\mathfrak{l}_0^{\text{EFT}}$

Reference from main text: Section 5.6. Status: Level A mathematics for the supplied dimensionless fixed-core evaluation; Level B route / Open physical owner and matching. The compact phase and its periodicity belong to the $U(1)$ -extended phase EFT (modelling step beyond bare P3). The independent positive action factor Σ_θ , its owner, and any map $\Sigma_\theta \mathfrak{l}_0^{\text{EFT}} \rightarrow S_0 \leftrightarrow \hbar$ are not derived.

This appendix makes explicit the chain behind the candidate map $\Sigma_\theta \mathfrak{l}_0^{\text{EFT}} \rightarrow \hbar$. Its purpose is not to derive the full quantum formalism: the supplied $U(1)$ /fixed-core extension produces a dimensionless reduced coefficient, while an independent finite positive Σ_θ is required before an action-valued product exists. It should be noted that the real scalar field Φ of P3 does not by itself carry a compact $U(1)$ phase variable; the extension to $\Phi_{\text{eff}} = \rho e^{i\theta}$ is an additional modelling step for the coherent sector, not a direct topological consequence of the supplied orientation orbit $O(4)/O(3) \simeq S^3$ (which has $\pi_1(S^3) = 0$).

A. Phase variable on the coherent branch

Let the coherent collective state be described by a phase-like variable θ . Because physically equivalent coherent states are identified after a full winding, the phase lives on a circle:

$$\theta \in S^1, \quad \theta \sim \theta + 2\pi. \quad (\text{Q.1})$$

This is the minimal topological statement behind the coherent winding sector (BR1).

B. Optional action–angle parametrisation

A phase is dimensionless. If one separately supplies a multivalued action–angle coordinate $\mathcal{S}$ with an action scale S_{ph} , one may write the Level B/Open map

$$\theta = \frac{\mathcal{S}}{S_{\text{ph}}}, \quad (\text{Q.2})$$

and hence

$$e^{i\theta} = \exp\left(i \frac{\mathcal{S}}{S_{\text{ph}}}\right). \quad (\text{Q.3})$$

This is not an identity between the angular coordinate and the single-valued Euclidean phase-gradient functional $\mathcal{S}_{\text{loop}}[\theta]$. The canonical pair, monodromy and owner of S_{ph} must be derived or supplied.

C. Winding and the fixed-core evaluation

For a supplied closed one-dimensional carrier, single-valuedness requires

$$\oint d\theta = 2\pi n, \quad n \in \mathbb{Z}. \quad (\text{Q.4})$$

Only under the separate action–angle assumption (Q.2) does this imply the monodromy

$$\oint d\mathcal{S} = 2\pi n S_{\text{ph}}. \quad (\text{Q.5})$$

It does not identify the value of the phase-gradient functional: for a uniform winding, $\mathcal{S}_{\text{loop}}$ is constant after a circuit even though $\oint d\theta = 2\pi n$. The supplied fixed-core representative is instead defined by evaluating the fixed- L functional of Eq. (5.22) at $n = 1$ and $L = L_{\text{core}}$:

$$\iota_{\text{elem}}^{(\text{fixed core})} := \mathcal{S}_{\text{loop}}[\theta_{\text{unif}}; L_{\text{core}}] = \frac{2\pi^2 K_{\theta}^{\text{1D}}}{L_{\text{core}}} = 2\pi \iota_0^{\text{EFT}}. \quad (\text{Q.6})$$

The last equality is the fixed-core coefficient definition, not a topological quantisation of the full action.

D. Status of the identification with $\hbar$

The correspondence $\hbar \leftrightarrow \Sigma_{\theta} \iota_0^{\text{EFT}}$ should be understood as follows. What is established is only that a supplied compact-phase/fixed-core EFT may contain the dimensionless reduced coefficient ι_0^{EFT} obtained from its phase-gradient functional, while winding organises integer sectors. The candidate physical coefficient is the independent product $\Sigma_{\theta} \iota_0^{\text{EFT}}$. A separate action–angle map may use an action scale, but topology does not fix that scale, a nonzero physical value, or its universality. The candidate-to-operational map $\Sigma_{\theta} \iota_0^{\text{EFT}} \rightarrow S_0$ remains Open because the physical operator/state map and cross-sector normalisation are not derived. The final equality $S_0 = \hbar$ is separately adopted as a Level-B physical calibration; its microscopic explanation and universality remain Open.

E. Summary

The coherent branch provides:

1. a periodic phase variable (Q.1);
2. integer winding sectors (Q.4);
3. a fixed-core representative obtained by evaluating the supplied phase-gradient functional after the core length and EFT coefficient are supplied, not from winding alone;
4. an optional action–angle analogy whose canonical owner is Open;
5. an Open owner programme for $\Sigma_{\theta} \iota_0^{\text{EFT}} \rightarrow S_0$, followed by the separately adopted Level-B calibration $S_0 = \hbar$; neither step is a derivation of the calibration’s microscopic origin.

R Noether Currents: Translational and Rotational Sectors

Reference from main text: Section 6.6. Status: classical field theory; no quantisation required.

A. Translational invariance and T^{AB}

Scalar bare sector (P3):

$$\tau_{P3}^{AB} = \partial^A \Phi \partial^B \Phi - \delta^{AB} \left[\frac{1}{2} (\partial_C \Phi)^2 + V(\Phi) \right], \quad \partial_A \tau_{P3}^{AB} = - \frac{\delta \mathcal{J}_E^{P3}}{\delta \Phi} \partial^B \Phi, \quad T_{P3, \text{phys}}^{AB} = A_{P3} \tau_{P3}^{AB}. \quad (\text{R.1})$$

The reduced and physical currents have the same on-shell conservation law for constant $A_{P3} > 0$. The path-integral denominator $\Sigma_{PI, P3}$ and ratio κ_{P3} control the supplied Euclidean measure, not this physical-current prefactor.

Orientation sigma-model (AB.1):

$$\begin{aligned} T_{(n)}^{AB} &= \kappa_n \partial^A n_C \partial^B n^C - \delta^{AB} \frac{\kappa_n}{2} (\partial_D n_E)^2, \\ \partial_A T_{(n)}^{AB} &= - \frac{\delta S_n}{\delta n_C} \partial^B n_C, \\ \partial_A T_{(n)}^{AB} &= 0 \quad \text{on the sigma-model shell.} \end{aligned} \quad (\text{R.2})$$

These local on-shell identities follow from translation invariance of their respective owning actions, not from homogeneity or $O(4)$ rotations. Boundary assumptions enter differentiable generators and conserved integrated charges.

B. $O(4)$ -invariance and angular currents

For the P3 $O(4)$ -invariant scalar action, Noether's theorem [47] for $\delta X^A = \omega^A_B X^B$ gives

$$j_{P3}^{C, AB} = X^A \tau_{P3}^{CB} - X^B \tau_{P3}^{CA}, \quad J_{P3, \text{phys}}^{C, AB} = A_{P3} j_{P3}^{C, AB}, \quad \partial_C j_{P3}^{C, AB} = \partial_C J_{P3, \text{phys}}^{C, AB} = 0 \quad (\text{R.3})$$

only on the P3 equations of motion. Off shell its divergence is the corresponding Euler–Lagrange combination. Admissible boundaries are additionally required for a differentiable generator and conserved integrated angular charge. For the orientation field n_A (a vector under $O(4)$), a spin contribution is added:

$$J_{(n)}^{C, AB} = X^A T_{(n)}^{CB} - X^B T_{(n)}^{CA} + \kappa_n (n^A \partial^C n^B - n^B \partial^C n^A). \quad (\text{R.4})$$

C. Currents under a declared branch restriction

The translational and rotational currents above belong to the bare real-scalar P3 action. A branch-restricted action inherits them only when the restriction, boundary data, and allowed variations preserve the corresponding symmetry. Separately supplied branch EFTs own their own currents; their existence does not follow from the bare-current formulas alone.

S Variational Stationarity and Reduced Extremal Dynamics

Reference from main text: Section 6.6. Status: schematic reduction programme at Level B/Open. Variation inside an explicitly supplied reduced action is exact for that owner, while descent from DP remains Open until a variation-compatible restriction or controlled elimination is derived.

This appendix records the conditions under which medium-level stationarity may descend to a branch effective action. It does not assume a universal chain: an explicit variation-compatible restriction or controlled elimination is required in each case.

Step 1. Medium extremality. By DP, every physically realised configuration must satisfy the necessary stationarity condition

$$\delta S_E[\Phi] = 0. \quad (\text{S.1})$$

Step 2. Branch decomposition. In the ordered branch, decompose the full field into background and fluctuations:

$$\Phi = \Phi_{\text{branch}} + \delta\Phi, \quad (\text{S.2})$$

where Φ_{branch} specifies the ordered background (P4) and $\delta\Phi$ denotes fluctuations over it.

Step 3. Conditional construction of a reduced functional. A branch decomposition does not by itself define an effective action. One must either impose a consistent restriction of S_E or integrate out declared modes with the measure, boundary data, constraints, and induced terms tracked. Only for such a variation-compatible map R may one prove

$$\delta S_E[\Phi]|_R = 0 \implies \delta S_{\text{eff}}[\psi_i] = 0 \quad \text{for the variations preserved by } R. \quad (\text{S.3})$$

An EFT action introduced independently is instead its own conditional owner; matching its coefficients does not prove the implication above.

Step 4. Interpretation and status. Equation (S.3) is therefore a Level A conditional variational statement once a particular compatible map R is derived. The existence of such maps for all phase, gauge, fermion, gravity, cosmology, HRC, and open-system actions displayed in this manuscript is Open. Lorentzian stationary-phase dominance additionally requires its own oscillatory measure, contour, state, Hessian and boundary data, and does not follow from Eq. (S.3). No universal DP-to-EFT-to-macroscopic descent chain is asserted.

T Noether current and local $U(1)$ completion in the compact phase EFT

This appendix collects the explicit derivations underlying the $U(1)$ gauge construction of Section 7.2.

A. Global phase symmetry and Noether current

Let $\Phi_{\text{eff}}(X) = \rho(X)e^{i\theta(X)}$, with effective Lagrangian

$$\mathcal{L} = \partial_A \Phi_{\text{eff}}^* \partial^A \Phi_{\text{eff}} - V(|\Phi_{\text{eff}}|). \quad (\text{T.1})$$

Under the infinitesimal global transformation $\delta\Phi_{\text{eff}} = i\alpha\Phi_{\text{eff}}$, $\delta\Phi_{\text{eff}}^* = -i\alpha\Phi_{\text{eff}}^*$, Noether's theorem gives

$$j^A = \frac{\partial \mathcal{L}}{\partial(\partial_A \Phi_{\text{eff}})} \delta\Phi_{\text{eff}} + \frac{\partial \mathcal{L}}{\partial(\partial_A \Phi_{\text{eff}}^*)} \delta\Phi_{\text{eff}}^*. \quad (\text{T.2})$$

Since $\partial \mathcal{L} / \partial(\partial_A \Phi_{\text{eff}}) = \partial^A \Phi_{\text{eff}}^*$ and $\partial \mathcal{L} / \partial(\partial_A \Phi_{\text{eff}}^*) = \partial^A \Phi_{\text{eff}}$, one obtains

$$j^A = i\alpha (\partial^A \Phi_{\text{eff}}^* \Phi_{\text{eff}} - \partial^A \Phi_{\text{eff}} \Phi_{\text{eff}}^*). \quad (\text{T.3})$$

Removing the overall infinitesimal parameter and adopting the standard normalisation yields

$$j^A = \frac{-i}{2} (\Phi_{\text{eff}}^* \partial^A \Phi_{\text{eff}} - \Phi_{\text{eff}} \partial^A \Phi_{\text{eff}}^*). \quad (\text{T.4})$$

Substituting $\Phi_{\text{eff}} = \rho e^{i\theta}$:

$$j^A = \rho^2 \partial^A \theta. \quad (\text{T.5})$$

B. Necessity of the compensating connection under local phase redefinition

Under $\Phi_{\text{eff}}(X) \rightarrow e^{i\alpha(X)} \Phi_{\text{eff}}(X)$, the derivative transforms as

$$\partial_A \Phi_{\text{eff}} \rightarrow e^{i\alpha(X)} (\partial_A \Phi_{\text{eff}} + i(\partial_A \alpha) \Phi_{\text{eff}}). \quad (\text{T.6})$$

The inhomogeneous term $i(\partial_A \alpha) \Phi_{\text{eff}}$ prevents covariance. Introduce a field A_A with transformation $A_A \rightarrow A_A + \frac{1}{e} \partial_A \alpha$. Then

$$D_A \Phi_{\text{eff}} \equiv (\partial_A - ieA_A) \Phi_{\text{eff}} \quad (\text{T.7})$$

transforms covariantly: $D_A \Phi_{\text{eff}} \rightarrow e^{i\alpha(X)} D_A \Phi_{\text{eff}}$.

C. Leading kinetic term

The antisymmetric combination $F_{AB} = \partial_A A_B - \partial_B A_A$ is gauge invariant. On a supplied ordered background the leading local parity-even quadratic basis contains two invariants, $I_1 = F_{AB} F^{AB}$ and $I_2 = (n^A F_{AB})(n_C F^{CB})$, up to normalisation and total derivatives. Thus the general leading action is

$$S_{\text{gauge}} = -\frac{1}{4} \int d^4 X [Z_1 I_1 + 2Z_2 I_2]. \quad (\text{T.8})$$

The Maxwell form is the special isotropic choice $Z_2 = 0$; P5 alone does not impose it. Coefficients, signs, reduced Hessian and physical photon matching remain Open.

D. Compact phase, local gauge completion, and holonomy

For compact $\theta \sim \theta + 2\pi$, the gauge-invariant phase gradient is $\mathcal{D}_A \theta = \partial_A \theta - e A_A$. Single-valuedness around a closed loop γ implies only

$$\oint_{\gamma} \partial_A \theta dX^A = 2\pi n, \quad n \in \mathbb{Z}. \quad (\text{T.9})$$

Consequently the covariant circulation is

$$\oint_{\gamma} \mathcal{D}_A \theta dX^A = 2\pi n - e \oint_{\gamma} A_A dX^A. \quad (\text{T.10})$$

Flux quantisation follows only after the additional equilibrium or zero-covariant-circulation condition $\oint_{\gamma} \mathcal{D}_A \theta dX^A = 0$ is imposed. It is therefore a conditional property of the supplied local completion, not a consequence of single-valuedness alone.

U Non-Abelian local completion and the $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ bridge

This appendix records the explicit structural steps behind Section 7.3.

A. Local multiplet completion

Let Φ be an N -component effective condensate multiplet. Suppose the effective action is invariant under global $SU(N)$ transformations

$$\Phi \rightarrow U\Phi, \quad U \in SU(N). \quad (\text{U.1})$$

If one promotes U to a local transformation $U(X)$, covariance of derivatives requires the introduction of a Lie-algebra-valued connection

$$A_A = A_A^a T^a, \quad (\text{U.2})$$

with covariant derivative

$$D_A = \partial_A - ig A_A. \quad (\text{U.3})$$

This is the standard local completion of a global multiplet symmetry.

B. Field strength from commutator closure

Demanding that the covariant derivatives close under commutation gives

$$[D_A, D_B]\Phi = -ig F_{AB}\Phi, \quad (\text{U.4})$$

with

$$F_{AB} = \partial_A A_B - \partial_B A_A - ig[A_A, A_B]. \quad (\text{U.5})$$

In component form this yields

$$F_{AB}^a = \partial_A A_B^a - \partial_B A_A^a + g f^{abc} A_A^b A_B^c. \quad (\text{U.6})$$

Thus the commutator term is forced by local non-Abelian covariance.

C. Leading ordered-background kinetic basis

At leading derivative order, a supplied ordered director allows two local parity-even quadratic invariants,

$$I_1^{\text{NA}} = F_{AB}^a F^{aAB}, \quad I_2^{\text{NA}} = (n^A F_{AB}^a)(n_C F^{aCB}), \quad (\text{U.7})$$

up to normalisation and total derivatives. The general residual- $O(3)$ action therefore has independent coefficients multiplying both invariants. The canonical isotropic Yang–Mills action is recovered only if a stronger symmetry or matching condition sets the second coefficient to zero; neither P5 nor local gauge covariance alone does so.

D. The $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ bridge

The Lie algebra $\mathfrak{so}(3)$ has generators J_i obeying

$$[J_i, J_j] = \varepsilon_{ijk} J_k. \quad (\text{U.8})$$

The Lie algebra $\mathfrak{su}(2)$ has generators T_i obeying

$$[T_i, T_j] = \varepsilon_{ijk} T_k \quad (\text{U.9})$$

up to conventional normalisation. Hence

$$\mathfrak{so}(3) \cong \mathfrak{su}(2). \quad (\text{U.10})$$

This is an isomorphism of Lie algebras only. It does not imply that the global groups $SO(3)$ and $SU(2)$ are identical, nor does it by itself determine the physical matter representation content.

E. Limit of the argument

This appendix derives the standard connection, curvature and two-invariant kinetic-basis algebra only after a local multiplet redundancy, bundle and action have been supplied. It also records the exact Lie-algebra isomorphism $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$. It does not derive a physical gauge bundle from P1–P6, select canonical Yang–Mills/Maxwell ($Z_2 = 0$), or derive Standard-Model representations, couplings, confinement or electroweak breaking.

V Conditional operator classification and missing matching owner for the fermion–director coupling

This appendix records the explicit EFT logic behind Section 7.6.

A. Bilinear basis

The standard local fermion bilinears are

$$\bar{\Psi}\Psi, \quad \bar{\Psi}i\gamma^5\Psi, \quad \bar{\Psi}\gamma^A\Psi, \quad \bar{\Psi}\gamma^A\gamma^5\Psi, \quad \bar{\Psi}\sigma^{AB}\Psi.$$

To form a scalar interaction linear in the preferred direction n_A , the natural zeroth-derivative candidates are

$$n_A \bar{\Psi}\gamma^A\Psi, \quad n_A \bar{\Psi}\gamma^A\gamma^5\Psi.$$

If one restricts attention to the parity-even EFT sector, the axial candidate is excluded and the vector operator remains as the leading allowed term.

B. Why the vector operator is leading

Derivative couplings such as

$$n_A \bar{\Psi} \sigma^{AB} \partial_B \Psi, \quad (\partial_A n_B) \bar{\Psi} \sigma^{AB} \Psi$$

are higher-order in the derivative expansion and are therefore subleading in the low-energy EFT. Accordingly, within the parity-even, zeroth-derivative basis, the leading ECT-specific operator is

$$n_A \bar{\Psi} \gamma^A \Psi.$$

C. Dimensional analysis

In four spacetime dimensions the fermion field has mass dimension

$$[\Psi] = \frac{3}{2},$$

so the bilinear

$$\bar{\Psi} \gamma^A n_A \Psi$$

has dimension three. Therefore the coefficient multiplying it in the Lagrangian density must have mass dimension one:

$$[\mu_5] = 1.$$

This is why the coefficient should not be treated as purely dimensionless at the Lagrangian level.

D. Dimensionless parameterisation and missing scale

A possible parameterisation is

$$\mu_5 = \beta_5 m_f,$$

where β_5 is dimensionless. Dimensions alone do not select β_5 , a nonzero coefficient, its sign, mass dependence, suppression scale or species dependence. No numerical benchmark is retained.

E. Lorentzian reduction and observability caveat

In the Lorentzian ordered branch, the preferred Euclidean direction induces a timelike vector u_μ , and the operator becomes

$$\mu_5 u_\mu \bar{\Psi} \gamma^\mu \Psi.$$

For a constant background and a single fermion species, such a term can be partly removed by field redefinitions. Hence the physical content of the coupling is tied to relative effects, species dependence, gradients, or environmental variation of the ordered branch. This explains why the structural admissibility of the operator is stronger than the current proof of its directly observable low-energy effects.

The distinction among this microscopic operator, the Level-B scalar ansatz, and the supplied conditional HRC gravity functional is recorded in Appendix W.

W Microscopic operator, scalar ansatz, and galactic functional

This appendix records three objects rather than two versions of one mechanism.

M — microscopic preferred-direction operator

$$M: \quad \Delta \mathcal{L}_5 = \mu_5 \bar{\Psi} \gamma^A n_A \Psi, \quad [\mu_5] = 1.$$

Its operator classification is A/B conditional; its coefficient and observable multi-species reduction are Open.

S — scalar response ansatz

$$S: \quad \phi_u = \beta_\phi^{-1} \ln(u/u_\infty), \quad F(\phi_u) = u/u_\infty.$$

This Level-B ansatz yields conditional local scalar identities. It does not yet yield a body charge, screening profile, or galactic force law.

G — conditional HRC gravity functional

$$G: \quad \mu_{g,k}(g^2/a_M^2)g = g_N.$$

Its response algebra is Level A inside the supplied augmented model; the identity/microscopic bridge is Level B/Open and the fits are Level C. The fit arithmetic does not identify its physical scalar coordinate or derive its source/force normalisation.

The strict result is $M \neq S$ by variable/operator content. The relation $S \rightarrow G$ is an Open matching problem and must be represented by a dashed edge. No statement here derives all three from a common branch-level response.

X Dimensional scaling and open condensate-scale assignments for the preferred-direction coefficient

This appendix records the scale argument behind Section 7.9.

A. Operator dimension. The operator

$$\bar{\Psi} \gamma^A n_A \Psi$$

has mass dimension three in four dimensions. Its coefficient must therefore have mass dimension one.

B. Available scales. In the present ECT architecture, the preferred direction n_A is a normalised-gradient variable on the adopted P4 chart. P4 postulates the physically relevant ordered branch; its stationary action, transition and dynamical formation/selection mechanism remain Open. Several scales are nevertheless present: the gradient order parameter u_0 , the radial/Planck-scale parameter ϕ_0 associated with the symmetry-breaking amplitude of the bare potential, and the later electroweak-like scale v_2 .

C. Consequence. Dimensional analysis establishes only $[\mu_5] = 1$. The available scales do not select a nonzero coefficient or a monomial dependence on m_f , ϕ_0 , u_0 or v_2 . The choices $\mu_5 = 0$ and arbitrary dimension-one matching functions are direct counterexamples to any unique species-scaling inference.

D. Open owner. A value or species law for μ_5 requires a physical fermion action, representations, matching calculation and radiative closure. None is derived, so no preferred suppression scale is assigned.

Y ϕ -first nonlinear gravity: complete derivation inside the supplied IR closure

This appendix gives the complete derivation internal to the supplied ϕ -first scalar–tensor IR closure, states its conditional theorems precisely, and compares its equations with analogous equations in related theories. It does not derive that closure from P1–P6 or establish a physical ECT metric completion.

A.1 Theorem A: Allowed quadratic scalar tensor and its conditional principal cone

Statement. On the ordered branch, the most general leading quadratic $O(3)$ -invariant symmetric rank-2 scalar kinetic tensor has the two-parameter form

$$K^{AB} = \beta \delta^{AB} - \alpha n^A n^B. \quad (\text{Y.1})$$

Once this second-order scalar operator is supplied, it has one scalar principal polynomial and, in the hyperbolic window, one conditional scalar principal cone. The symmetry classification does not by itself supply a physical action, exclude higher-order principal factors, or derive universal metric coupling. The latter is assumption A4 (IR closure assumption; not proved beyond linear order).

Proof. Any $O(3)$ -invariant symmetric rank-2 tensor built from $\{\delta^{AB}, n^A\}$ has the form $T^{AB} = a \delta^{AB} + b n^A n^B$ with two scalars a, b . Matching the displayed operator gives $a = \beta$, $b = -\alpha$; hence $K^{ij} = \beta \delta^{ij}$ and $K^{ww} = \beta - \alpha = -\Delta_K$ on $n^A = \delta_w^A$. No other independent structure exists. $\square$

A.2 Lemma B: conditional logarithmic parameterisation

For any scalar–tensor closure with positive non-minimal factor $f(q)$,

$$\psi_f(q) \equiv \ln \frac{f(q)}{f_0}, \quad f(q) = f_0 e^{\psi_f(q)}, \quad (\text{Y.2})$$

is an exact algebraic reparameterisation. In the amplitude-dominated closure one further assumes $f/\bar{M}_{\text{Pl}}^2 = u/u_\infty = e^{\beta_\phi \phi_u}$, so that $\psi_f = \beta_\phi \phi_u$. This latter identification is Level B matching, not an algebraic theorem from bare P3.

The anisotropy variable $\Delta_K = \alpha - \beta$ is distinct. Defining $\psi_K = \ln(\Delta_K/\Delta_{K,\text{vac}})$ is separately algebraic, but no $u \leftrightarrow \Delta_K$ map has been derived. The two logarithmic variables must not be identified without that missing bridge.

A.3 Proposition C: scalar-only IR action (effective ansatz)

S_{IR} in Eq. (13.35) is a local two-derivative scalar–tensor ansatz consistent with A1–A4 after the amplitude-dominated matching is assumed. It is not derived from S_E without additional matching. Its functions $\omega(\phi)$ and $U(\phi)$ remain closure inputs. The displayed action contains no independent n_μ sector.

A.4 Proposition D: field equations and conditional extension

Variation of the displayed action gives the scalar-only metric equation (13.38) and the scalar equation recorded in Appendix A.5.2. The non-minimal terms $(g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)F$ are mandatory. The natural-unit matter coefficient is $8\pi G_*$ before the full equation is divided by F , where $G_* \equiv (8\pi \bar{M}_{\text{Pl}}^2)^{-1}$. Its separate SI conversion and match to observed G_N require the local scalar response and are not fixed by division.

An orientation contribution is not obtained by varying Eq. (13.35). It requires a separately specified diffeomorphism-invariant $S_n[g, n]$, including its constraint and equation of motion, and enters through

$$T_{\mu\nu}^{(n)} = -\frac{2}{\sqrt{-g}} \frac{\delta S_n}{\delta g^{\mu\nu}}.$$

The covariant completion and its matching to scalar-only ECT remain Level B/Open.

A.5 Corollary E: constant-scalar limit

For $\phi = \bar{\phi} = \text{const}$, the scalar-only metric equation reduces to

$$G_{\mu\nu} + \Lambda_{\text{div}}(\bar{\phi}) g_{\mu\nu} = 8\pi G_{\text{div}}(\bar{\phi}) T_{\mu\nu}^{\text{mat}}, \quad (\text{Y.3})$$

where

$$G_{\text{div}}(\bar{\phi}) = \frac{G_*}{F(\bar{\phi})} = G_* e^{-\beta_\phi \bar{\phi}}, \quad \Lambda_{\text{div}}(\bar{\phi}) = \frac{U(\bar{\phi})}{\bar{M}_{\text{Pl}}^2 F(\bar{\phi})}. \quad (\text{Y.4})$$

On the declared reference branch $\bar{\phi} = 0$ and $F = 1$, so the divided coefficient equals the action normalisation G_* , not automatically the observed G_N . This algebraic GR-form background limit is exact inside the scalar closure. It does not prove that the constant branch solves the scalar equation, is stable and dynamically accessible, or has a decoupled finite-body scalar response. In vacuum a constant branch must in particular satisfy $d[U/F^2]/d\phi = 0$; with matter its action-owned root equation must be used. Even at constant background F , a light scalar perturbation can give $G_{\text{Cav}} \neq G_{\text{div}}$ and $\gamma_{\text{PPN}} \neq 1$. An orientation vacuum contribution may be added only after an S_n is specified and its on-shell stress is evaluated.

Note. This subsection does not derive the galactic/AQUAL branch, which requires the additional open matching and nonlinear assumptions listed in Section 17.1.

A.6 Theorem F: Lovelock uniqueness of the pure-metric IR limit

Additional assumption A5. ϕ and n^A are frozen or integrated out; the remaining pure-metric IR theory is local, diffeomorphism-invariant, and two-derivative at leading order in 4D.

Lovelock theorem (1971) [92]. Under A5, the leading pure-metric two-derivative IR field equations are uniquely of Einstein form:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (\text{Y.5})$$

Limitations. After integrating out ϕ and n^A , corrections $\sim R^2, R_{\mu\nu}R^{\mu\nu}$ are generated but suppressed as $(E/m_\phi)^2$ in the IR. Theorem F is a statement about leading IR uniqueness, not a completed microscopic derivation from S_E . $\square$

A.7 Comparison with related theories

Table 126 places the ECT ϕ -first closure in context with Einstein-aether theory, scalar-tensor gravity, and standard GR.

Table 126: ECT ϕ -first IR closure compared to related theories. ϕ = logarithmic macroscopic amplitude variable (local stiffness variable of the ordered branch); n^A = orientation director; $G_{\text{div}} = G_*/F$ = divided matter coefficient, distinct from the measured gravitational response. Entries in **bold** indicate ECT-specific features.

Feature	GR	Brans-Dicke / ST	Einstein-aether	ECT (ϕ -first)
Primary object	$g_{\mu\nu}$ (postulated)	$g_{\mu\nu}, \phi$	$g_{\mu\nu}, u^\mu$	ϕ (order field); g^{eff} emergent
Divided matter coefficient	constant	$G \propto 1/\phi$	constant	$G_{\text{div}} = G_* e^{-\beta_\phi \phi(X)}$ in the supplied matching; measured response Open
Divided potential coefficient	constant	scalar-potential dependent	constant	$\Lambda_{\text{div}}(\phi)$; cosmological response Open
Metric equation	$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$	$G_{\mu\nu} = \dots + \nabla\nabla\phi$	$G_{\mu\nu} + \Theta_{\mu\nu} = 8\pi G T_{\mu\nu}$	eq. (13.38) with $\nabla\nabla M_{\text{eff}}^2$
Einstein response limit	exact	constant scalar plus vanishing scalar response	$u^\mu = \delta_0^\mu$ with a decoupled aether response	constant-background eq. (Y.3) plus a decoupled, short-range or screened scalar response

(continued on next page)

Feature	GR	Brans-Dicke / ST	Einstein-aether	ECT (ϕ -first)
MOND-like sector	absent	absent	absent	conditional HRC static functional; scalar map Open
Preferred direction	no	no	yes (u^μ)	yes (π^A, supplied by P4)
Origin of direction	postulated	postulated	postulated	P4 adopted ordered-gradient postulate; stationary action and dynamical formation/selection Open
Lovelock uniqueness	yes	yes (BD)	yes (aether)	yes, in pure-metric limit (Thm. F)
Full derivation from condensate	N/A	N/A	N/A	open (OP2/OP3)

Component glossary for eq. (13.38).

$F(\phi)G_{\mu\nu}$ Einstein tensor weighted by the dimensionless non-minimal factor. After the complete equation is divided by F , its matter coefficient is $G_{\text{div}} = G_*/F$. This is a complete local response only when the scalar perturbation and finite-body charge are absent or sufficiently suppressed.

$T_{\mu\nu}^{\text{mat}}$ Stress tensor of matter minimally coupled to the Jordan-frame metric (universality is closure assumption A4); it obeys $\nabla^\mu T_{\mu\nu}^{\text{mat}} = 0$ on the matter equations.

$(g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)F$ Non-minimal derivative terms required by variation of $F(\phi)R$.

$-\omega(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2}g_{\mu\nu}(\partial\phi)^2)$ Scalar kinetic contribution written on the left-hand side.

$(U/\bar{M}_{\text{Pl}}^2)g_{\mu\nu}$ Scalar potential contribution. Its divided coefficient is $\Lambda_{\text{div}} = U/(\bar{M}_{\text{Pl}}^2 F)$; the observed cosmological response requires the full solved background, branch, vacuum offset and physical metric/clock matching.

$T_{\mu\nu}^{(n)}$ Not part of Eq. (13.38) or of the displayed action. It may be added only from a separately specified covariant $S_n[g, n]$; its ECT matching is Open.

This appendix records a conditional tensor-EFT normalisation, not a microscopic ECT derivation of Newton's constant.

Y.1 Supplied quadratic tensor target

If a healthy physical tensor $H_{\mu\nu}$ with coefficient M_G^2 is supplied, its Fierz–Pauli action is Eq. (8.27). The current fixed director map does not provide this field or action.

Y.2 Weak-field matching

Comparison of that supplied action with linearised Einstein gravity gives

$$G_N^{\text{nat}} = \frac{1}{8\pi M_G^2}. \quad (\text{Y.6})$$

The observed G_N then calibrates M_G in the supplied completion.

Y.3 Open microscopic bridge

The chain

$$(u_0, \alpha, \beta, \dots) \rightarrow \kappa_n \not\rightarrow M_G \rightarrow G_N$$

is interrupted at the tensor owner: the displayed director response has no linear TT/static-density vertex. Closure requires a new physical tensor or reduction map, its constrained Hessian, residue and universal source.

Y.4 Interpretive limit

Medium “stiffness” is only an analogy until the coefficient of the physical gravity operator and its source are derived. No numerical ECT prediction of G_N is claimed.

Z Numerical algorithms

This appendix summarises the computational procedures used to generate the figures in this paper. Versioned source and validation materials for released revisions are provided at <https://github.com/chufelo/ECT-preprint-code>. The exact owner of a released PDF is its tagged commit and frozen release manifest; the moving default branch is not an exact revision identifier.

Z.1 Galaxy rotation curves

The current manuscript algorithm registry contains only HRC-0 and HRC-3. For each radius and trial a_M , it forms the signed baryonic acceleration in Eq. (AK.7) and solves

$$\mu_{g,k}(g^2/a_M^2)g - g_N = 0, \quad k \in \{0, 3\}, \quad (\text{Z.1})$$

by deterministic monotone bracketing. HRC-0 also has the explicit solution in Eq. (AK.6). Whole galaxies, not radial points, define the held-out folds. Signed gas, split seeds, fit modes, omitted nuisance parameters and the disk-curl limitation are recorded in Appendix AL.

All current galactic calculations, tables and figures are generated by the HRC-only registry described above.

Z.2 RG running of $\lambda(\mu)$

The one-loop β -function (Section 5.1) is integrated numerically from $\mu = M_{\text{Pl}}$ downward using a standard adaptive Runge–Kutta method, with boundary condition $\lambda(M_{\text{Pl}}) \approx 10^{-3}$. This yields the dimensionless scalar diagnostic $g_{\text{sc}}(M_{\text{Pl}}) = \lambda(M_{\text{Pl}})/(8\pi) \approx 4 \times 10^{-5}$ (Eq. (8.29)). Since $\beta_\lambda > 0$, $\lambda(\mu)$ decreases monotonically toward the infrared and $g_{\text{sc}}(\mu)$ remains small across the displayed sub-Planckian scalar-EFT range. This is not a running Newton coupling. The formal Landau pole of the strict one-loop scalar closure lies at $\mu_{\text{LP}} \approx M_{\text{Pl}} e^{8.77 \times 10^3}$ (Eq. (8.30)). This extrapolation is far above the chosen Planck reference scale and cannot define the physical ultraviolet validity of ECT. The P3 mass m_σ is neither a P4 EFT threshold nor a microscopic regulator; both the P4 threshold and ultraviolet completion remain Open.

Z.3 Figure production

All figures use publication-quality settings: serif font (Times New Roman), 9 pt labels, inward tick marks, 220 dpi output, with `bbox_inches='tight'`.

Regime diagram (Fig. 11). The figure is a two-band phenomenological schematic organised by g/a_M . It contains no calibrated $u \leftrightarrow \Delta_K$ coordinate, density-screening boundary, or fixed placement of Solar-System, galactic, and void environments. The high-acceleration Newtonian asymptote and the low-acceleration HRC branch are properties of the supplied response model after the identity bridge; canonical screening mass, composite-body charge, metric completion and PPN reachability are marked Open. The script `scripts/fig_regime_diagram.py` produces the schematic.

AA Structural routes to open microphysical targets

This appendix separates standard external calculations, supplied model examples and the physical ECT owners that remain missing for the fine-structure constant, generation number, Yukawa hierarchy and primordial perturbation spectrum. Numerical examples below do not establish naturalness, selection or an ECT derivation.

A. Fine-structure constant: electroweak gauge normalisation

From (7.16), $\alpha_{\text{fs}} = q^2/(4\pi Z_A)$. The observational target is $Z_A^{\text{IR}} \approx 10.9$ in the stated normalisation. This algebra does not derive a gauge field, charge normalisation or ECT ultraviolet boundary condition.

A.1 Why the ultraviolet problem is electroweak

Below the electroweak scale M_W , the photon A_μ is a mass eigenstate and $\alpha_{\text{fs}}^{-1} = 4\pi Z_A/q^2$ is a well-defined running quantity. Above M_W , however, the photon is not a propagating eigenstate. The relevant ultraviolet gauge basis is $U(1)_Y \times SU(2)_L$, with independent gauge couplings g' and g (or equivalently α_Y and α_2) and independent gauge-kinetic normalisations Z_B and Z_W . The electromagnetic coupling emerges only after electroweak symmetry breaking through

$$\alpha_{\text{fs}}^{-1} = \alpha_Y^{-1} + \alpha_2^{-1}.$$

Inside an already supplied Standard-Model electroweak completion the ultraviolet bookkeeping uses (Z_B, Z_W) and θ_W . ECT does not derive that completion or identify these quantities with condensate data.

A.2 External one-loop electroweak benchmark

Using the observed Standard-Model electroweak data only as an external benchmark (not as an ECT derivation), one proceeds as follows. Using the conventional LEP line-shape input $M_Z = 91.1876 \pm 0.0021$ GeV [32], the rounded-scale external inputs are $\alpha_{\text{em}}^{-1}(M_Z) \approx 127.95$ and $\sin^2 \theta_W(M_Z) \approx 0.2312$. The standard electroweak decomposition gives

$$\alpha_Y^{-1}(M_Z) = \alpha_{\text{em}}^{-1}(M_Z) \cos^2 \theta_W \approx 98.4, \quad \alpha_2^{-1}(M_Z) = \alpha_{\text{em}}^{-1}(M_Z) \sin^2 \theta_W \approx 29.6.$$

Employing the standard hypercharge normalisation (not the GUT-normalised convention), the one-loop SM beta coefficients are $b_Y = 41/6$ and $b_2 = -19/6$. The renormalisation group equations

$$\alpha_i^{-1}(\phi_0) = \alpha_i^{-1}(M_Z) - \frac{b_i}{2\pi} \ln \frac{\phi_0}{M_Z}$$

with $\ln(\phi_0/M_Z) \approx 37.8$ yield

$$\alpha_Y^{-1}(\phi_0) \approx 57, \quad \alpha_2^{-1}(\phi_0) \approx 49.$$

These should be read as benchmark ultraviolet kinetic targets inferred from the external one-loop electroweak running, not yet as microscopic coefficients derived from the ECT action. The corresponding benchmark coefficients are

$$Z_B^{\text{bench}} \approx 4.6, \quad Z_W^{\text{bench}} \approx 3.9.$$

For the observed three-generation Standard-Model matter benchmark, externally hand-setting the measured colour multiplicities and electric charges gives

$$C_{\text{eff}} = \sum_f N_{c,f} Q_f^2 = 3 \times 1^2 + 3 \times 3 \times (2/3)^2 + 3 \times 3 \times (1/3)^2 = 8.$$

This is exact arithmetic conditional on those external inputs and a Level C benchmark, independent of P7. The P7 rank-three module supplies no quark or lepton sections, generations, colour multiplicities, hypercharges or electric charges, so $C_{\text{eff}} = 8$ is not obtained from the candidate colour structure and is not an ECT prediction.

A.3 Conditional matching interpretation

The coefficients $Z_B^{(0)}$ and $Z_W^{(0)}$ are matching inputs of a supplied electroweak completion. The compact global phase and the orientation algebra do not derive a local $U(1)_Y \times SU(2)_L$ bundle, its gauge-reduced Hessian or its kinetic normalisations. A future microphysical completion could test whether those coefficients are related to phase and orientation response kernels, but the identifications $Z_B^{(0)} \leftrightarrow K_\theta^{\text{eff}}$ and $Z_W^{(0)} \leftrightarrow \kappa_n$ are not established by P1–P6. No relation between the fine-structure constant, the gravitational coupling or S_0 follows from the current action.

At the benchmark level, the ultraviolet mixing angle can be expressed directly in terms of the two electroweak inverse couplings, or equivalently in terms of the corresponding benchmark kinetic targets:

$$\sin^2 \theta_W(\phi_0) = \frac{\alpha_2^{-1}(\phi_0)}{\alpha_Y^{-1}(\phi_0) + \alpha_2^{-1}(\phi_0)} = \frac{Z_W^{\text{bench}}}{Z_B^{\text{bench}} + Z_W^{\text{bench}}}.$$

The external benchmark value $\sin^2 \theta_W(\phi_0) \approx 0.46$ is a target for such a completion. Without a derived map from the condensate response kernels to $(Z_B^{(0)}, Z_W^{(0)})$, neither proximity to $1/2$ nor the deviation from it is an ECT prediction.

A.4 Limitations and honest status

The external benchmark used above is subject to several limitations: (i) two-loop and higher-order corrections to the beta coefficients have been ignored; (ii) threshold corrections at intermediate scales (m_t , M_W , M_Z , M_H) have been treated as step functions rather than properly matched; (iii) below M_Z , hadronic vacuum-polarisation contributions cannot be honestly reduced to free quark thresholds; the low-energy running between the Thomson limit and M_Z involves nonperturbative QCD effects; (iv) the observed SM three-generation matter content, hypercharge assignments, and right-handed singlet structure have been used as external inputs, not as quantities derived within ECT; (v) the colour completion itself is a candidate route (Level Open/C), not a proven theorem.

The robust conclusion at the present stage is narrower: the external one-loop Standard-Model extrapolation defines benchmark electroweak kinetic targets. It does not determine whether a future microscopic model generates them through induced terms, bare matching terms, or a mixture. That classification remains Open until the UV charged spectrum and multiplicities, its couplings, regulator and renormalisation scheme, threshold matching and an explicit induced- Z calculation are supplied. The $C_{\text{eff}} = 8$ arithmetic and $\sin^2 \theta_W^{\text{bench}}(\phi_0) \approx 0.46$ are external benchmarks, independent of P7; neither is evidence for an ECT condensate-response map. The first-principles derivation of $(Z_B^{(0)}, Z_W^{(0)})$ therefore remains Open (OP- α_{fs}), with outcome PARAMETRIC ONLY and MISSING ELECTROWEAK ACTION/RESPONSE OWNER.

B. Three fermion generations: spectral counting

In the conventional attractive Pöschl–Teller toy operator $H_v = -\partial_x^2 - v(v+1) \text{sech}^2 x$ on $L^2(\mathbb{R}, dx)$, with dimensionless x and self-adjoint domain $H^2(\mathbb{R})$ (the Friedrichs realisation of its $C_c^\infty(\mathbb{R})$ restriction), the negative normalisable levels have integer labels $n \geq 0$ with $n < v$. Their exact count is $N_- = \lceil v \rceil$ for $v > 0$ (the integer- v threshold state is not negative). Here the declared toy-profile map is $v = \kappa_{\text{PT}}|y|/\sqrt{2\lambda_{\text{eff}}}$ (Eq. (9.29)), with $\kappa_{\text{PT}} > 0$ and $\lambda_{\text{eff}} > 0$, so $v \geq 0$; the count above applies for $v > 0$, equivalently $|y| > 0$, on the nontrivial attractive self-adjoint branch. The physical operator/profile map and the coefficient κ_{PT} are supplied rather than derived. In this toy well, the exact three-level count is $2 < v \leq 3$. It cannot be converted into a numerical physical Yukawa window without an action-owned fermion operator, profile and normalisation. The parameter λ_{eff} refers to the local effective profile entering the toy spectral problem; it is not yet identified with the fundamental quartic parameter of the full ECT action.

The complementary topological route is only a future specified-completion programme. One must first provide an elliptic chiral Dirac operator, bundle and connection, matter representation, domain/boundary conditions and an admissible background. An applicable index theorem could then constrain the chiral

difference $N_L - N_R$ for that completed problem. The homotopy statement $\pi_3(S^3) = \mathbb{Z}$ alone fixes neither the index nor the number of zero modes of either chirality. The physical identification of three Standard-Model families therefore remains Open.

Status: Level A conditional mathematics for the exact negative-level count and $2 < \nu \leq 3$ wedge inside the declared self-adjoint toy operator; Level C/Open for matching its operator, profile and levels to physical fermions. Boundary data, representations and exactly three ECT families are not derived (OP-generations).

C. Yukawa hierarchy: overlap and tunnelling suppression

If families correspond to three quasi-localised fermionic modes, then a supplied overlap model may parametrise Yukawa eigenvalues as $y_i \sim y_{0i} e^{-\Xi_i^{\text{ov}}}$. The exponent is dimensionless: $\Xi_i^{\text{ov}} = \kappa_i d_i$ in a simple localisation-tail model with an explicit penetration length d_i , or $\Xi_i^{\text{ov}} = B_i / \Sigma_{\text{ov}}^{\text{act}}$ in a WKB model with a finite positive action unit. Magnitude suppression requires $\text{Re } \Xi_i^{\text{ov}} \geq 0$. For species-dependent prefactors the exact identity is $\ln |y_a/y_b| = \ln |y_{0a}/y_{0b}| - \text{Re } \Xi_a^{\text{ov}} + \text{Re } \Xi_b^{\text{ov}}$. Consequently external Yukawa data can fix total log ratios, not the exponent differences separately; the latter coincide only under a declared common-prefactor-magnitude convention. Numerical ratios require a declared renormalisation scale, scheme and external data source and are not retained in this ECT derivation. The identity is a symbolic parametrisation, not evidence that the required profiles or barriers are statistically or dynamically natural; that claim would require a derived prefactor distribution and action owner.

Status: Level C/Open. Overlap/tunnelling suppression is a standard conditional mechanism, while the profiles, boundary conditions, d_i or B_i , $\Sigma_{\text{ov}}^{\text{act}}$ and the explicit computation of Y_{ij} from condensate microphysics remain Open (OP-Yukawa).

D. Primordial perturbations: projection diagnostic and Lifshitz-window route

The earlier logarithmic $d=4$ estimate is no longer treated as an ECT-derived prediction. Within the declared infinite- w , $k > 0$ convergence domain, the projection diagnostic of Section 14.6 shows that an ordinary Euclidean $1/p^2$ critical correlator projects to $\Delta^2 \propto k^2$, i.e. a blue spectrum; logarithmic corrections cannot remove this leading power. The current strict-pullback scale-invariance route is instead conditional. The simplest isotropic idealisation is a supplied Euclidean Lifshitz window with $G_\chi(p) \simeq \kappa_\chi^{-1} / (\kappa_4 p^4)$, whose equal-slice projection gives the dimensionless curvature spectrum $\Delta_\xi^2(k) \propto \kappa_\chi^{-1}(k) \mathcal{T}_{\chi \rightarrow \xi}^2(k) / \kappa_4(k)$. In the strict pure-quartic limit the dimensionless spectrum is exactly flat only when the logarithmic derivative of $\mathcal{T}_{\chi \rightarrow \xi}^2 / (\kappa_\chi \kappa_4)$ vanishes; separate constant $\mathcal{T}_{\chi \rightarrow \xi}$, κ_χ , κ_4 are a sufficient special case. In a finite Lifshitz window this statement is only the leading approximation; the exact lower-order formula below supplies the residual tilt. The strict-pullback lemma below generalises this result: in the pure-quartic limit the recorded $Z_u, \mathcal{C}_n$ second-gradient operators give the same local seed tilt $n_\chi = 1$ exactly when $\kappa_\chi \sqrt{\mathcal{C}_n} (\sqrt{Z_u} + \sqrt{\mathcal{C}_n})$ is scale-independent over the band; separate constant $\kappa_\chi, Z_u, \mathcal{C}_n$ are a sufficient special case, while the positive stiffness ratio still changes the amplitude prefactor. Lower-order and mass deformations require the exact controls derived below.

The amplitude is not predicted parameter-free; up to convention-dependent factors it fixes the combination

$$A_s \propto \frac{\kappa_\chi^{-1} T_{\chi \rightarrow \xi}^2}{u_0^2 \sqrt{\mathcal{C}_n} (\sqrt{Z_u} + \sqrt{\mathcal{C}_n})}, \quad Z_u = \mathcal{C}_n = 2\kappa_4 \Rightarrow A_s \propto \frac{\kappa_\chi^{-1} T_{\chi \rightarrow \xi}^2}{4u_0^2 \kappa_4},$$

whose isotropic limit is the scaling $A_s \propto \kappa_\chi^{-1} T_{\chi \rightarrow \xi}^2 / (u_0^2 \kappa_4)$. The observed red tilt, the tensor-to-scalar ratio r , the non-Gaussianity parameters $f_{\text{NL}}, g_{\text{NL}}$, and possible statistical anisotropy g_*^{aniso} remain open completion targets. The fixed-coefficient equal-slice calculation is closed below; the central physical task is to derive the stationary regulated owner and its $a_w, Z_u, \mathcal{C}_n, m_\chi^2$ running, then test the exact finite-band inequalities rather than an averaged-stiffness proxy. Integrability alone does not own that hierarchy.

Status: Level A for the declared-domain projection diagnostic; Level B/C for the conditional Lifshitz-window route; Open for the reduced-action/measure owner, $A_\chi, \Sigma_{\text{PI}, \chi}$, coefficient ownership, physical

reflection-positive/projected covariance, the transfer map, the observed tilt, tensors, non-Gaussianity and Lorentzian projection.

Lemma (strict-pullback Lifshitz seed from $Z_u, \mathcal{C}_n$). In the strict integrable single-field pullback (cf. the pure- $P(X)$ fluctuation analysis of §3.1 and the ordered-branch EFT (3.9)), the recorded second-gradient operators $\frac{Z_u}{2}(\partial_A \sigma)^2$ and $\frac{\mathcal{C}_n}{2}u_0^2(\partial_A \pi_i)^2$, with $\sigma = \partial_w \chi$ and $\pi_i = \partial_i \chi / u_0$, give the fourth-order kernel

$$K_4(p_w, k) = \frac{1}{2}(p_w^2 + k^2)(Z_u p_w^2 + \mathcal{C}_n k^2), \quad k^2 = \sum_{i=1}^3 p_i^2.$$

The exact four-dimensional $O(4)$ bi-Laplacian form $K_4 = \kappa_4(p_w^2 + k^2)^2$ holds iff $Z_u = \mathcal{C}_n$, with $\kappa_4 = Z_u/2 = \mathcal{C}_n/2$; for $Z_u \neq \mathcal{C}_n$ one has $\kappa_4^{\text{eff}} = (Z_u + \mathcal{C}_n)/4$ and a residual w -versus-spatial term $\frac{1}{4}(Z_u - \mathcal{C}_n)[p_w^4 - (k^2)^2]$. For $Z_u, \mathcal{C}_n > 0$, at fixed $k > 0$ and with the massless zero mode regulated or removed, the positive Fourier kernel defines the formal equal- w covariance

$$P_\chi(k) = \int \frac{dp_w}{2\pi} \frac{\kappa_\chi^{-1}}{K_4(p_w, k)} = \frac{\kappa_\chi^{-1}}{k^3 \sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})},$$

which reduces to $\kappa_\chi^{-1}/(4\kappa_4 k^3)$ when $Z_u = \mathcal{C}_n$. Hence the seed power per logarithmic interval, $k^3 P_\chi(k)/2\pi^2$, is scale-invariant exactly when $\kappa_\chi \sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})$ is scale-independent; separate constant $\kappa_\chi, Z_u, \mathcal{C}_n$ are a sufficient special case, giving a local seed tilt $n_\chi = 1$ (equivalently $n_s = 1$ only after the complete transfer $\mathcal{T}_{\chi \rightarrow \zeta} = T_{\chi \rightarrow \zeta}/u_0$ is scale-independent), at the strict Euclidean seed level. The mismatch $Z_u \neq \mathcal{C}_n$ changes the amplitude prefactor but not the tilt; because the spatial $O(3)$ is intact, no ordinary three-dimensional quadrupolar g_*^{aniso} arises in the equal- w scalar power spectrum (the w -versus-spatial anisotropy affects amplitudes and transfer-sector correlators, not the angular dependence of $P_\chi(|k|)$).

The algebraic average κ_4^{eff} above is not a sharp dominance constant. The exact two-sided global quartic bound is

$$\frac{1}{2} \min(Z_u, \mathcal{C}_n)(p_w^2 + k^2)^2 \leq K_4(p_w, k) \leq \frac{1}{2} \max(Z_u, \mathcal{C}_n)(p_w^2 + k^2)^2, \quad (\text{AA.1})$$

and both constants are sharp. Fixed- k mass dominance and projected observable tolerances are different questions, treated below.

Spectral tilt from the strict-pullback kernel (derivation). The local seed power from the lemma is $P_\chi(k) = \kappa_\chi^{-1}(k)/[k^3 g(Z_u, \mathcal{C}_n)]$, with

$$g(Z_u, \mathcal{C}_n) \equiv \sqrt{\mathcal{C}_n}(\sqrt{Z_u} + \sqrt{\mathcal{C}_n}) = \mathcal{C}_n(\sqrt{r} + 1), \quad r \equiv Z_u/\mathcal{C}_n.$$

The second equality is exact. The pure-quartic seed tilt is therefore

$$n_\chi - 1 = -\frac{d \ln[\kappa_\chi g(Z_u, \mathcal{C}_n)]}{d \ln k}.$$

Thus $n_\chi = 1$ if and only if the combined normalisation $\kappa_\chi g$ is scale-independent over the band. Separate constancy of $\kappa_\chi, Z_u, \mathcal{C}_n$ is sufficient but not necessary, because their runnings can cancel.

Define the complete field-to-curvature transfer

$$\mathcal{T}_{\chi \rightarrow \zeta}(k) \equiv \frac{T_{\chi \rightarrow \zeta}(k)}{u_0(k)}, \quad \zeta_k = \mathcal{T}_{\chi \rightarrow \zeta}(k) \chi_k.$$

The curvature variance is

$$\Delta_\zeta^2(k) \propto k^3 P_\zeta(k) = \frac{\kappa_\chi^{-1}(k) \mathcal{T}_{\chi \rightarrow \zeta}^2(k)}{g(Z_u(k), \mathcal{C}_n(k))} = \frac{\kappa_\chi^{-1}(k) T_{\chi \rightarrow \zeta}^2(k)}{u_0^2(k) \mathcal{C}_n(k) (\sqrt{r(k)} + 1)}.$$

Hence

$$n_s - 1 = -\frac{d \ln \kappa_\chi}{d \ln k} + 2 \frac{d \ln \mathcal{T}_{\chi \rightarrow \zeta}}{d \ln k} - \frac{d \ln \mathcal{C}_n}{d \ln k} - \frac{d \ln(\sqrt{r} + 1)}{d \ln k} = n_\chi - 1 + 2 \frac{d \ln \mathcal{T}_{\chi \rightarrow \zeta}}{d \ln k}.$$

Equivalently, resolve $\mathcal{T}_{\chi \rightarrow \zeta} = T_{\chi \rightarrow \zeta}/u_0$ and define $Q_n(k) \equiv u_0(k) \sqrt{\mathcal{C}_n(k)}$. The exact algebraic rewrite is

$$n_s - 1 = -\frac{d \ln \kappa_\chi}{d \ln k} + 2 \frac{d \ln T_{\chi \rightarrow \zeta}}{d \ln k} - 2 \frac{d \ln Q_n}{d \ln k} - \frac{d \ln(\sqrt{r} + 1)}{d \ln k}, \quad r = Z_u/\mathcal{C}_n. \quad (\text{AA.2})$$

The historical label is retained, but Q_n is not M_G : the bridge to a physical tensor normalisation is Open. The notation $\kappa_\chi(k)$ is conditional on a supplied RG-improved prescription $\kappa_\chi(\mu)|_{\mu=k}$; at a fixed matching scale it is constant. Its running is not a derived cross-sector action unit. If a freeze-out map is supplied, one may write $d \ln Q_n/d \ln k = s_* E_Q$ with $E_Q = d \ln Q_n/d \ln a$. The boxed terms are measure-normalisation, numerator-transfer, orientation-normalisation and anisotropy-ratio drift; the preceding form groups the physical transfer as the complete quantity $\mathcal{T}_{\chi \rightarrow \zeta}$.

Status of the zero-deformation tilt budget (Open). Every term in (AA.2) is an open completion target: the κ_χ running requires its action/measure owner; E_Q requires the closure-level ordering/Hubble derivation from one complete ECT-background action; s_* requires the freeze-out map; the r -running requires the anisotropy sector; and $T_{\chi \rightarrow \zeta}$ and u_0 require the emergence map defining the complete transfer. Consequently $n_s \simeq 0.965$ is not a derived prediction. In the supplied pure-quartic limit $a_w = m_\chi^2 = 0$, $n_\chi = 1$ holds exactly when $d \ln(\kappa_\chi g)/d \ln k = 0$; separate constant $\kappa_\chi, Z_u, \mathcal{C}_n$ are a sufficient special case. Under that seed condition, $n_s = 1$ additionally requires scale-independent $\mathcal{T}_{\chi \rightarrow \zeta}$. The exact zero is not protected by a derived Ward identity and this conditional limit is not a theorem of genericity in ECT. Finite constant deformations and their running are treated next.

Exact anisotropic lower-order deformation. In the displayed stationary-density parametrisation, restoring the ordinary w -derivative coefficient and allowing a constant quadratic term gives

$$K(p_w, k) = a_w p_w^2 + \frac{1}{2}(p_w^2 + k^2)(Z_u p_w^2 + \mathcal{C}_n k^2) + m_\chi^2, \quad k > 0, \quad (\text{AA.3})$$

with

$$K = A p_w^4 + B p_w^2 + D, \quad A = \frac{Z_u}{2}, \quad B = a_w + \frac{Z_u + \mathcal{C}_n}{2} k^2, \quad D = m_\chi^2 + \frac{\mathcal{C}_n}{2} k^4. \quad (\text{AA.4})$$

For fixed real coefficients and $Z_u > 0$, $K(p_w, k)$ is strictly positive for every real p_w and its equal-slice integral is finite if and only if

$$D > 0, \quad B + 2\sqrt{AD} > 0. \quad (\text{AA.5})$$

After $p_w = (D/A)^{1/4} x$, the identity $\int_{-\infty}^{\infty} dx/(x^4 + qx^2 + 1) = \pi/\sqrt{2+q}$ for $q > -2$ gives

$$P_\chi(k) = \kappa_\chi^{-1} \int \frac{dp_w}{2\pi} \frac{1}{K(p_w, k)} = \frac{\kappa_\chi^{-1}}{2\sqrt{D} \sqrt{B + 2\sqrt{AD}}}. \quad (\text{AA.6})$$

No factorisation into two real positive roots of the polynomial in p_w^2 is required.

Equation (AA.5) is a fixed- k theorem, not by itself global positivity over the full (p_w, k) domain. On the continuous domain $p_w \in \mathbb{R}, k \in [0, \infty)$ (or a spectrum accumulating at $k = 0$), and for $Z_u, \mathcal{C}_n > 0$, the global minimum occurs at $k = 0$. A negative m_χ^2 is globally inadmissible; for $m_\chi^2 = 0$ one needs $a_w \geq 0$ and retains the zero mode at the origin; and for $m_\chi^2 > 0$ global strict positivity is equivalent to

$$a_w > -\sqrt{2Z_u m_\chi^2}. \quad (\text{AA.7})$$

The nonnegative branch $a_w, m_\chi^2 \geq 0$ used below is therefore sufficient, but not the complete signed fixed- k domain. On a compact/discrete spatial domain with the zero mode removed and $k \geq k_{\min} > 0$, the fixed- k conditions are instead authoritative and may admit some negative m_χ^2 ; boundary data must then be declared.

Reflection-positivity, zero-mode and boundary guards. Positivity of the displayed Fourier kernel and its formal Gaussian covariance does not imply Osterwalder–Schrader reflection positivity. In dimensionless variables with the reference frequency set to $\Omega = 1$, the isotropic member $K_{\text{dp}}(p_w) = (p_w^2 + 1)^2$ has

$$C_{\text{dp}}(t) = \kappa_\chi^{-1} \int \frac{dp_w}{2\pi} \frac{e^{ip_w t}}{(p_w^2 + 1)^2} = \frac{\kappa_\chi^{-1}}{4} (1 + |t|) e^{-|t|},$$

but for distinct $t_1, t_2 > 0$

$$\det[C_{\text{dp}}(t_i + t_j)]_{i,j=1}^2 = -\frac{\kappa_\chi^{-2} e^{-2(t_1+t_2)}}{16} (t_1 - t_2)^2 < 0.$$

Thus the bare constant-numerator χ covariance already contains an explicit reflection-positivity counterexample. This does not exclude a different constrained physical observable or projector; it proves that the equal-slice calculation alone establishes neither an OS Hilbert space nor a positive Lorentzian spectral measure or ghost-free unitary continuation. In particular, a coincident formal cone from the double characteristic does not remove this double-pole obstruction.

Unequal-pole reflection-positivity no-go for the named bare covariance. The double-pole witness above is the degenerate member of a stronger fixed- k result. For the bare strict-pullback kernel

$$K_4(p_w, k) = \frac{1}{2} (p_w^2 + k^2) (Z_u p_w^2 + \mathcal{C}_n k^2), \quad Z_u, \mathcal{C}_n > 0, \quad (\text{AA.8})$$

take the constant-numerator covariance κ_χ^{-1}/K_4 and define $a = k$, $b = k\sqrt{\mathcal{C}_n/Z_u}$. If $a \neq b$, the fixed- k time kernel is

$$C_k(t) = A e^{-a|t|} + B e^{-b|t|}, \quad A = \frac{\kappa_\chi^{-1}}{Z_u a (b^2 - a^2)}, \quad B = -\frac{\kappa_\chi^{-1}}{Z_u b (b^2 - a^2)}, \quad AB < 0. \quad (\text{AA.9})$$

For unequal positive times,

$$\det[C_k(t_i + t_j)]_{i,j=1}^2 = AB \left(e^{-at_1 - bt_2} - e^{-at_2 - bt_1} \right)^2 < 0. \quad (\text{AA.10})$$

Together with the double-pole limit, this proves that the bare constant-numerator χ covariance fails this two-time reflection-positivity gate throughout the declared positive $(Z_u, \mathcal{C}_n)$ range. It is a Level-A no-go for that named translation-invariant coordinate covariance, not for every physical observable or UV completion.

Polynomial-numerator reflection-positivity gate. At fixed k , the factorisation has a Pais–Uhlenbeck-type higher-derivative form. Calling either formal branch a physical ECT ghost or a Källén–Lehmann particle would additionally require a stationary branch measure, physical local observable, Lorentzian continuation, domain, star, inner product, constraint quotient and source-coupled projector. Those owners are absent. Integrability does not itself remove a pole: the relations $\sigma = \partial_w \chi$ and $u_0 \pi_i = \partial_i \chi$ were already used to derive K_4 , and the natural pullback covariances retain both factors.

For a future real polynomial even numerator $N(z, k)$, with $z = p_w^2$, write at fixed $k > 0$

$$\frac{\kappa_\chi^{-1} N(z, k)}{K_4} = Q(z, k) + \frac{r_a}{z + a^2} + \frac{r_b}{z + b^2}, \quad r_a = \frac{2\kappa_\chi^{-1} N(-a^2, k)}{Z_u (b^2 - a^2)}, \quad r_b = -\frac{2\kappa_\chi^{-1} N(-b^2, k)}{Z_u (b^2 - a^2)}. \quad (\text{AA.11})$$

The corresponding time-kernel weights are $w_a = r_a/(2a)$ and $w_b = r_b/(2b)$. For the standard even scalar reflection and test functions supported strictly in the positive-time half-space, Q contributes only reflection-plane contact terms and the non-contact covariance is reflection positive if and only if $r_a, r_b \geq 0$; equivalently, its non-contact rational part is a Stieltjes function of z . For $b > a$ this becomes $N(-a^2, k) \geq 0$

and $N(-b^2, k) \leq 0$. If boundary-supported test functions are admitted, the contact polynomial requires a separate reflection-plane prescription. If a more general numerator introduces additional poles or cuts, the full non-contact part must admit a nonnegative Stieltjes measure; the two displayed pole signs are then necessary for those poles but are not by themselves sufficient.

Pointwise Euclidean positivity $N(z, k) \geq 0$ for $z \geq 0$ is not the Stieltjes criterion. For $0 < a < c < b$, the Euclidean-positive numerator $N(z) = z + c^2$ gives two positive residues; $N(z) = z + b^2$ cancels the b pole and leaves a positive one-pole covariance. Conversely, a nonzero p_w -independent numerator gives opposite residues and retains the no-go above. In the double-pole limit $a = b$, the non-contact scalar gate requires $N(-a^2) = 0$, so that one pole cancels, and $(2\kappa_\chi^{-1}/Z_u)N'(-a^2) \geq 0$ for the remaining simple pole. A genuine double pole fails the two-time determinant test.

A positive momentum-independent scalar rescaling leaves reflection-positivity determinant signs unchanged. A momentum-dependent numerator, even when positive on the real Euclidean axis, generally changes the pole residues, can cancel poles, and can add reflection-plane contacts. Its physical use requires an action-owned observable, reflection parity, constraint quotient, projector, normalisation and boundary prescription. The current outcome remains MISSING ACTION / MISSING OBSERVABLE / MISSING PROJECTOR. Neither passing nor failing this gate is equivalent to having exactly one characteristic cone; compare the conditional second-order scalar corollary, whose single-principal-polynomial assumption is not made for K_4 .

All massless baseline and ratio statements below require $k \geq k_{\min} > 0$. The pure quartic kernel has $K_4(0, 0) = 0$, while its four-dimensional covariance is logarithmically singular at coincidence. A full Gaussian measure, determinant or action normalisation therefore needs declared ultraviolet and zero-mode prescriptions. A finite or compact w direction replaces the continuous integral by a boundary-condition-dependent sum. Equations below use translation-invariant coefficients, the covariance convention κ_χ^{-1}/K , the Fourier measure $dp_w/(2\pi)$, and a formal or regulated infinite- w covariance. The polynomial-numerator gate above applies only to its declared fixed- k scalar reflection and positive-time test-function domain.

Exact ratio and finite-deformation tilt. For the positive massless reference define

$$\begin{aligned} H &\equiv (\sqrt{Z_u} + \sqrt{\mathcal{C}_n})^2, & \delta_a(k) &\equiv \frac{2a_w}{Hk^2}, & \delta_m(k) &\equiv \frac{2m_\chi^2}{\mathcal{C}_n k^4}, \\ \rho &\equiv \frac{2\sqrt{Z_u \mathcal{C}_n}}{H}, & 0 < \rho &\leq \frac{1}{2}. \end{aligned} \quad (\text{AA.12})$$

Relative to

$$P_0(k) = \frac{\kappa_\chi^{-1}}{k^3 \sqrt{\mathcal{C}_n} (\sqrt{Z_u} + \sqrt{\mathcal{C}_n})}, \quad (\text{AA.13})$$

the exact result is

$$\boxed{\frac{P_\chi}{P_0} = \frac{1}{\sqrt{1 + \delta_m} \sqrt{1 + \delta_a + \rho(\sqrt{1 + \delta_m} - 1)}}}. \quad (\text{AA.14})$$

The fixed- k positivity domain becomes $\delta_m > -1$ and $1 + \delta_a + \rho(\sqrt{1 + \delta_m} - 1) > 0$. For small deformations,

$$\begin{aligned} \frac{P_\chi}{P_0} &= 1 - \frac{1}{2}\delta_a - \frac{2+\rho}{4}\delta_m + \frac{3}{8}\delta_a^2 + \frac{2+3\rho}{8}\delta_a\delta_m \\ &\quad + \frac{3(\rho^2 + 2\rho + 4)}{32}\delta_m^2 + O((|\delta_a| + |\delta_m|)^3). \end{aligned} \quad (\text{AA.15})$$

For $Z_u = \mathcal{C}_n = 2\kappa_4$ and $a_w = 0$, this returns the R165 pure-mass coefficients $-5/8$ and $63/128$.

The live operator $a_w p_w^2$ is not the isotropic R165 operator $\kappa_2(p_w^2 + k^2)$. At one fixed k , the latter maps into Eqs. (AA.3)–(AA.4) only through the paired substitutions

$$a_w = \kappa_2, \quad m_{\text{here}}^2 = m_{\text{R165}}^2 + \kappa_2 k^2. \quad (\text{AA.16})$$

This is a fixed- k algebraic map, not an action-level identification of constant mass parameters. The R165 two-derivative coefficients therefore do not transfer to $a_w p_w^2$ alone.

For scale-independent $\kappa_\chi, Z_u, \mathcal{C}_n, a_w, m_\chi^2$, the exact local seed tilt is

$$n_\chi - 1 \equiv \frac{d \ln(k^3 P_\chi)}{d \ln k} = \frac{2\delta_m}{1 + \delta_m} + \frac{\delta_a + \rho \delta_m / \sqrt{1 + \delta_m}}{1 + \delta_a + \rho(\sqrt{1 + \delta_m} - 1)}. \quad (\text{AA.17})$$

Thus

$$n_\chi - 1 = \delta_a + (2 + \rho)\delta_m + O((|\delta_a| + |\delta_m|)^2). \quad (\text{AA.18})$$

Positive constant a_w and m_χ^2 give blue seed contributions. If the coefficients run while remaining diagonal functions of k , the pointwise power formula survives but the full derivative is

$$n_\chi - 1 = -\frac{d \ln \kappa_\chi}{d \ln k} - \frac{1}{2} \frac{d \ln \mathcal{C}_n}{d \ln k} - \frac{d \ln(\sqrt{Z_u} + \sqrt{\mathcal{C}_n})}{d \ln k} + \frac{d \ln(P_\chi/P_0)}{d \ln k}. \quad (\text{AA.19})$$

The last term includes the running of δ_a , δ_m and ρ . Frequency-dependent, coordinate-dependent or nonlocal coefficients require a fresh operator calculation. The observable curvature tilt additionally contains the u_0 and $\chi \rightarrow \zeta$ transfer terms of Eq. (AA.2).

Pointwise dominance and observable tolerances. Global coercivity (AA.1), fixed- k operator dominance and integrated tolerance are distinct. For $a_w, m_\chi^2 \geq 0$,

$$\min_{p_w} K_4(p_w, k) = \frac{\mathcal{C}_n}{2} k^4, \quad \sup_{p_w} \frac{m_\chi^2}{K_4} = \delta_m, \quad (\text{AA.20})$$

whereas

$$\sup_{p_w} \frac{a_w p_w^2}{K_4} = \delta_a, \quad p_w^2 = k^2 \sqrt{\frac{\mathcal{C}_n}{Z_u}} \quad \text{at the maximum.} \quad (\text{AA.21})$$

For the constant-coefficient band used below define explicitly

$$\delta_{a,\min} \equiv \delta_a(k_{\min}), \quad \delta_{m,\min} \equiv \delta_m(k_{\min}).$$

Here the subscript labels the lower-momentum endpoint; because the nonnegative constant-coefficient ratios decrease with k , these endpoint values are their band maxima, not minima of the ratios. Thus $m_\chi^2 \leq \varepsilon(\mathcal{C}_n/2)k_{\min}^4$ is necessary and sufficient for a uniform mass-only pointwise bound on a constant-coefficient band $k \geq k_{\min}$. It is independent of Z_u , but is not the sharp equal-slice amplitude tolerance. For the combined nonnegative deformation, the exact uniform pointwise condition is

$$\begin{aligned} \delta_{m,\min} &\leq \varepsilon, \\ \delta_{a,\min} H &\leq \varepsilon(Z_u + \mathcal{C}_n) + 2\sqrt{\varepsilon Z_u \mathcal{C}_n (\varepsilon - \delta_{m,\min})}. \end{aligned} \quad (\text{AA.22})$$

For nonnegative constant coefficients the exact power ratio is smallest at $k_{\min}$. A fractional amplitude tolerance $0 < A_{\text{tol}} < 1$ is therefore equivalent to

$$(1 + \delta_{m,\min}) \left[1 + \delta_{a,\min} + \rho(\sqrt{1 + \delta_{m,\min}} - 1) \right] \leq (1 - A_{\text{tol}})^{-2}. \quad (\text{AA.23})$$

The Z_u dependence is operationally material. For example, at $\mathcal{C}_n = k = 1$, $a_w = 0$ and $m_\chi^2 = 0.05$, the same $\delta_m = 0.1$ gives $P/P_0 = 0.942036943592800$ for $Z_u = 1$ but 0.953462542708278 for $Z_u = 10^{12}$; a 5% tolerance therefore fails in the first case and passes in the second. The leading amplitude budget is

$$\frac{1}{2} \delta_{a,\min} + \frac{2 + \rho}{4} \delta_{m,\min} \lesssim A_{\text{tol}}. \quad (\text{AA.24})$$

For a tilt tolerance no endpoint reduction is assumed; the exact bandwise condition is

$$\sup_{k \in [k_{\min}, k_{\max}]} |n_\chi(k) - 1| \leq n_{\text{tol}}, \quad (\text{AA.25})$$

using Eq. (AA.17) for fixed coefficients and the full logarithmic derivative when they run.

At one momentum the normalised power constrains only one combination of $(\delta_a, \delta_m, \rho)$; moreover, ρ is invariant under $Z_u/\mathcal{C}_n \leftrightarrow \mathcal{C}_n/Z_u$. Absolute normalisation is also degenerate with κ_χ^{-1}, u_0 , transfer normalisation and the ordering-scale map. The exact formulas are therefore constraints on supplied combinations, not an inverse identification of microscopic ECT coefficients.

Status of the strict-pullback result. The pure- K_4 projection and Eqs. (AA.6)–(AA.25) are Level A conditional mathematics of the displayed translation-invariant scalar kernel. This bounded fixed-coefficient mathematical branch is closed. Its ECT ownership is not: the stationary action, regulator-compatible measure, constrained physical projector, $a_w, Z_u, \mathcal{C}_n, m_\chi^2$ values and running, mass-protecting identity, and radiative stability remain Open. The number of small operator ratios is not by itself a theorem about the number of independent microscopic tunings. The bare χ covariance does not establish reflection positivity or a unitary Lorentzian theory. The scale map, $\chi \rightarrow \zeta$ transfer, A_s , observed red tilt, tensors and non-Gaussianity remain Open. No PES-R record vertex, response/noise kernel, state, subtraction or physical/global PES result follows from this equal-slice calculation.

AB Effective Action for Orientation Modes: Nonlinear Sigma-Model

Reference from main text: Sections 4.4 and 4.6. Status: minimal orientation EFT on the broken branch; Level B. This appendix does not claim three independent physical Goldstone particles; see Appendix J for the integrability analysis.

Status note. Any three-component orientation language used in this appendix refers only to the minimal reduced EFT description of the broken branch. In the scalar-only microscopic ECT basis, integrability of $Q_A = \partial_A \Phi$ reduces the strict linear scalar map to one longitudinal scalar coordinate; a physical one-mode interpretation remains conditional on the reduced action, domain and projector. See Appendix J.

Setup and caution

In the ordered branch one may introduce the orientation field n_A with $n_A n^A = 1$ and write the following supplied dimensionless reduced Euclidean broken-branch functional:

$$\mathcal{J}_{n,E} = \frac{\kappa_n}{2} \int d^4 X \partial_B n_A \partial^B n^A + \int d^4 X \Lambda(X) (n_A n^A - 1), \quad (\text{AB.1})$$

$$S_{n,E}^{\text{phys}} = A_{n,E} \mathcal{J}_{n,E}, \quad 0 < A_{n,E} < \infty. \quad (\text{AB.2})$$

Here $\kappa_n > 0$ is the effective stiffness, Λ enforces the unit constraint, and $A_{n,E}$ is a constant field-independent action magnitude. This is a nonlinear sigma model with target $S^3 \simeq O(4)/O(3)$. The notation is deliberately distinct from the still-missing generally covariant orientation action $S_n[g, n]$ and from the phase-loop functional $\mathcal{J}_n(L)$, whose subscript is an integer winding label.

Important caveat. In the scalar-only ECT basis $n_A = Q_A/|Q| = \partial_A \Phi/|\partial \Phi|$, so the decomposition $Q_A = u n_A$ is subject to the integrability constraint $\partial_A Q_B - \partial_B Q_A = 0$. This means the three “transverse” components δn_i are not fully independent in the underlying scalar-only theory. The minimal orientation EFT is a useful effective description, but must not be interpreted as a theorem about three independent propagating physical particles.

Linearisation and reduced content

Setting $n_A = \bar{n}_A + \delta n_A$, $\bar{n}_A = \delta_{Aw}$ (P4). From $n_A n^A = 1$: $\delta n_w = 0$ at linear order. The quadratic reduced functional is

$$\mathcal{J}_{n,E}^{(2)} = \frac{\kappa_n}{2} \int d^4 X \partial_B \delta n_i \partial^B \delta n_i, \quad S_{n,E}^{(2),\text{phys}} = A_{n,E} \mathcal{J}_{n,E}^{(2)}. \quad (\text{AB.3})$$

After imposing the integrability reduction $\delta n_i = \partial_i \chi / u_0$ (Appendix J), this supplied reduced EFT becomes an effective action for one longitudinal scalar mode χ ; the physical projector and domain remain part of its Level-B assumptions.

Stiffness coefficient κ_n

From the ordered-branch EFT (3.9) at uniform background $u = u_0$:

$$\frac{\mathcal{C}_n}{2} u_0^2 (\partial_A n_B) (\partial^A n^B) \rightarrow \frac{\kappa_n}{2} (\partial_A \delta n_i) (\partial^A \delta n_i), \quad \kappa_n = \mathcal{C}_n u_0^2 \quad (\text{Level B}). \quad (\text{AB.4})$$

The coefficient $\mathcal{C}_n$ has dimension mass^{-2} and is not yet derived from bare parameters (μ, λ) of P3 (OP-Planck).

Dispersion in the supplied reduced soft-mode EFT

After the integrability reduction and within the supplied derivative-only reduced EFT, the equation of motion $\partial_B \partial^B \delta n_i = 0$ becomes an equation for χ :

$$(\partial_w^2 + c_\chi^2 \nabla^2) \nabla^2 \chi = 0, \quad c_\chi^2 = \frac{K_s}{K_w} c_{\text{char}}^2. \quad (\text{AB.5})$$

For non-trivial modes ($k \neq 0$):

$$\omega^2 = c_\chi^2 k^2 \quad (\text{Level B}). \quad (\text{AB.6})$$

The identification $c_\chi = c_{\text{char}}$ would require $K_s = K_w$; this is not yet derived (open).

Coherent branch selection and the horizon problem

Conditionally, the horizon problem may be reformulated as asking whether a candidate progenitor of the observable ordered branch can be coherently ordered on a sufficient macroscopic scale. The chronology, physical clock/metric and transfer to observables remain Open.

The functional used below is a separately supplied positive sigma-model completion; it is not derived from P1–P6 or from the strict scalar-gradient Hessian. The following result is therefore Level A mathematics inside that completion and Open as a statement about the physical ECT branch.

Let Ω_{br} be a connected progenitor domain of the future observable branch, and let $n_A(x)$ be an orientation field valued in the supplied $S^3 \simeq O(4)/O(3)$ target. The relevant Euclidean ordering functional is

$$S_{\text{ord}}[n] = \frac{\kappa_n}{2} \int_{\Omega_{\text{br}}} d^d x (\partial_i n_A) (\partial^i n^A), \quad \kappa_n > 0.$$

Since the integrand is non-negative, one has $S_{\text{ord}}[n] \geq 0$, with equality if and only if $\partial_i n_A = 0$ on Ω_{br} , that is,

$$n_A(x) = \text{const} \quad \text{on } \Omega_{\text{br}}.$$

Thus, on a connected progenitor domain, the exact global minimum of the orientation ordering problem is a uniform branch orientation.

This proves only the conditional minimum lemma for the displayed functional. It neither derives that functional from the frozen ECT action nor proves that a realised cosmological branch occupies the minimum. A macro-coherent $O(4) \rightarrow O(3)$ transition is therefore a candidate state of the supplied completion, not an established ECT horizon mechanism.

To translate this into a cosmological horizon statement one further assumes that the coherence scale of the progenitor branch satisfies

$$L_{\text{br}} \frac{a_0}{a_{\text{br}}} \gtrsim L_{\text{hom}},$$

where L_{hom} is the present homogeneity scale to be explained. At present this is a working hypothesis, not a first-principles theorem.

The preference for coherent branches over strongly fragmented ones can be motivated at the level of classical action comparison: a macroscopic $O(1)$ twist over a scale L contributes $S_{\text{ord}} \sim \kappa_n (\Delta\theta)^2 L^{d-2}$, so large-scale branch fragmentation is not action-free. However, establishing branch preference rigorously would require a full semiclassical treatment including the entropy of the configuration space; the strict result in this appendix is therefore the existence of the coherent exact minimum, while the stronger branch-preference claim in the main text remains a Level B statement.

No quantitative support follows from assigning a soft-mode mass by hand. The identity $\xi_G = \hbar/(m_G c)$ becomes physical only after an action-owned mass operator, state and coherence observable are derived. Those inputs and the transfer to an observed homogeneous region remain Open.

Conditional defect classification for the supplied S^3 target

For reference, the topological defect types generated by a symmetry breaking $G \rightarrow H$ are classified by the homotopy groups $\pi_n(G/H)$ [9]:

$$\begin{aligned} \pi_0(G/H) \neq 0 &\Rightarrow \text{domain walls;} & \pi_1(G/H) \neq 0 &\Rightarrow \text{strings;} \\ \pi_2(G/H) \neq 0 &\Rightarrow \text{monopoles;} & \pi_3(G/H) \neq 0 &\Rightarrow \text{textures.} \end{aligned}$$

For the supplied P4 orientation target $\mathcal{T}_{\text{P4}} \simeq S^3$:

Homotopy group	Value	Defect type
$\pi_0(S^3)$	0	No π_0 protection within this target
$\pi_1(S^3)$	0	No π_1 protection within this target
$\pi_2(S^3)$	0	No π_2 protection within this target
$\pi_3(S^3)$	$\mathbb{Z}$	Integer target-map classes

The only non-trivial class $\pi_3(S^3) = \mathbb{Z}$ corresponds to texture-type target maps. Physical defect and relic conclusions require a complete vacuum/configuration space, exact-gradient admissibility, action, boundary class and transition history; those owners remain Open.

AC Audit of the candidate anisotropic curvature coupling

The scalar-only action (13.35) contains no independent orientation field and therefore cannot yield an n_μ stress tensor by variation. This appendix records the exact covariant identities that a future orientation completion must respect and identifies the step that prevented the previous argument from being a derivation.

AC.1 Required variational starting point

A consistent calculation must begin with a specified covariant action $S_n[g, n, \lambda_n]$. Its stress tensor is defined by

$$T_{\mu\nu}^{(n)} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta S_n}{\delta g^{\mu\nu}}. \quad (\text{AC.1})$$

This definition automatically includes connection variations, constraint terms, integrations by parts, and every term required by the chosen operator basis. The schematic tensor in Eq. (13.22) is not a substitute for this full variation.

AC.2 What the Ricci identity actually implies

With convention

$$[\nabla_\alpha, \nabla_\mu]n_\nu = R_{\alpha\mu\nu}{}^\beta n_\beta, \quad (\text{AC.2})$$

and acceleration $a_\nu \equiv n^\alpha \nabla_\alpha n_\nu$, one has

$$n^\alpha \nabla_\alpha \nabla_\mu n_\nu = \nabla_\mu a_\nu - (\nabla_\mu n^\alpha)(\nabla_\alpha n_\nu) + n^\alpha R_{\alpha\mu\nu}{}^\beta n_\beta. \quad (\text{AC.3})$$

This identity can reorganise terms already produced by a variation. It does not determine their coefficients and does not replace the metric variation of S_n .

The former reduction relied on the non-identity

$$n^\alpha n^\beta R_{\alpha\mu\nu\beta} \neq n_\mu n^\alpha R_{\alpha\nu} \quad \text{for a generic metric and vector field.} \quad (\text{AC.4})$$

The left-hand side retains a projected Riemann component; a Ricci tensor is obtained by a metric contraction over an independent index pair, not by replacing one metric leg with n_μ . Consequently the advertised Ricci structure does not follow from the commutator and Leibniz rule alone.

AC.3 Status of the candidate operator

For bookkeeping, define

$$\mathcal{X}_{\mu\nu}^{\text{cand}}[n] \equiv c_1 \left(n_\mu n^\alpha R_{\alpha\nu} + n_\nu n^\alpha R_{\alpha\mu} - \frac{1}{2} g_{\mu\nu} n^\alpha n^\beta R_{\alpha\beta} \right). \quad (\text{AC.5})$$

Equation (AC.5) is a candidate tensor structure, not the stress tensor derived from Eq. (13.35). To promote it to a field-equation contribution one must:

- (i) specify the complete generally covariant S_n and its unit and scalar-integrability constraints;
- (ii) vary that action with respect to the metric and n_μ ;
- (iii) show that the full on-shell tensor, not an isolated curvature term, is compatible with the Bianchi identity;
- (iv) match its coefficients and physical modes to the scalar-only ECT basis.

Einstein–aether theory provides a useful external comparison [23], but importing its stress tensor is not an ECT derivation. This appendix does not close OP2; the orientation completion and the coefficient c_1 remain Open.

AD Technical Details for the Conditional Vacuum-Offset Completions

This appendix separates three calculations that were previously conflated: the exact rank-one determinant identity, the standard trace-free scalar–tensor completion, and the ordinary two-slope background completion. Only the first follows from the fixed-coefficient orientation ansatz. The other two are explicitly supplied action-level completions whose relation to the microscopic Φ medium remains to be derived.

AD.1 Rank-one determinant identity and the non-converse

For an invertible matrix M , the determinant lemma gives

$$\det(M + c n n^\top) = (1 + c n^\top M^{-1} n) \det M. \quad (\text{AD.1})$$

Taking $M = \beta I$, $c = -\alpha$, and $n^\top n = 1$ yields

$$\det(\beta I - \alpha n n^\top) = \beta^4 \left(1 - \frac{\alpha}{\beta} n^\top n\right), \quad (\text{AD.2})$$

$$\det K = \beta^3(\beta - \alpha). \quad (\text{AD.3})$$

For a tangent variation, $n \cdot \delta n = 0$, and therefore

$$\delta_n \det K = -2\alpha\beta^3 n \cdot \delta n = 0. \quad (\text{AD.4})$$

If the additional densitized-metric identification $K^{AB} = C_K \sqrt{-g} g^{AB}$ is imposed with constant C_K , then

$$\sqrt{-g} \propto \sqrt{-\det K} = \sqrt{\beta^3(\alpha - \beta)}. \quad (\text{AD.5})$$

This conclusion concerns the orientation submanifold at fixed amplitude and fixed α, β . It is not a derivation of a constrained metric action. The non-converse follows both from the degree-of-freedom mismatch and from the explicit determinant-preserving family

$$g_{\mu\nu}(s) = \text{diag}(-e^{3s}, e^{-s}, e^{-s}, e^{-s}), \quad \det g = -1. \quad (\text{AD.6})$$

All members have the same determinant, while their lapse and spatial response differ. Similarly, a fixed determinant does not forbid FLRW expansion; in unimodular time,

$$ds^2 = -a^{-6}(\tau) d\tau^2 + a^2(\tau) d\mathbf{x}^2, \quad \det g = -1, \quad (\text{AD.7})$$

but the determinant constraint supplies no equation for $a(\tau)$.

AD.2 Explicit constrained action and Bianchi reconstruction

Consider the separately supplied action

$$S_{\text{TF}} = \int d^4x \left\{ \sqrt{-g} \left[\frac{1}{2} f(q) R - \frac{1}{2} K(q) (\partial q)^2 - V(q) + \mathcal{L}_m \right] + \lambda (\sqrt{-g} - \varepsilon_0) \right\}. \quad (\text{AD.8})$$

The constraint is $\sqrt{-g} = \varepsilon_0$. Metric variation gives

$$\begin{aligned} f G_{\mu\nu} &= T_{\mu\nu}^m + K \left(\partial_\mu q \partial_\nu q - \frac{1}{2} g_{\mu\nu} (\partial q)^2 \right) - V g_{\mu\nu} \\ &\quad + \nabla_\mu \nabla_\nu f - g_{\mu\nu} \square f + \lambda g_{\mu\nu}. \end{aligned} \quad (\text{AD.9})$$

The scalar equation is

$$K \square q + \frac{1}{2} K_{,q} (\partial q)^2 + \frac{1}{2} f_{,q} R - V_{,q} = 0. \quad (\text{AD.10})$$

Using this equation, $\nabla^\mu T_{\mu\nu}^m = 0$, and the contracted Bianchi identity in the divergence of Eq. (AD.9) gives $\partial_\nu \lambda = 0$. Writing $\lambda = -C_I$ reconstructs

$$\begin{aligned} f G_{\mu\nu} &= T_{\mu\nu}^m + K \left(\partial_\mu q \partial_\nu q - \frac{1}{2} g_{\mu\nu} (\partial q)^2 \right) - [V + C_I] g_{\mu\nu} \\ &\quad + \nabla_\mu \nabla_\nu f - g_{\mu\nu} \square f, \quad \partial_\mu C_I = 0. \end{aligned} \quad (\text{AD.11})$$

Under $V \rightarrow V + C_0$ and $C_I \rightarrow C_I - C_0$, both Eqs. (AD.10) and (AD.11) are invariant. Equivalently, in the Lagrange-multiplier form one shifts $\lambda \rightarrow \lambda + C_0$. This is a classical change of representative in the pair (V, C_I) , not a calculation of the global datum C_I and not yet a quantum non-renormalization theorem.

For a constant scalar in vacuum, the trace and scalar equation give

$$R = \frac{4(V + C_I)}{f}, \quad V_{,q} - 2\frac{f_{,q}}{f}(V + C_I) = 0. \quad (\text{AD.12})$$

For $f = f_* e^{aq}$ and $V = V_1 e^{bq}$,

$$e^{bq_0} = \frac{2aC_I}{(b-2a)V_1}, \quad \frac{aC_I}{(b-2a)V_1} > 0. \quad (\text{AD.13})$$

The displayed inequality, together with a nonzero denominator, is the general sign/existence condition. The frequently used branch $a, C_I, V_1 > 0$ and $b > 2a$ is sufficient but not necessary. Thus the fixed-curvature root is the on-shell form of an integration-density floor. The local field equations admit a continuum of C_I values. This is compatible with treating C_I as a boundary/state datum of a particular condensate; ECT still has to derive the action and its observable coupling.

AD.3 Exact FLRW equations of the declared scalar–tensor action

For the Jordan action

$$S_J = \int d^4x \sqrt{-g} \left[\frac{1}{2} f(q) R - \frac{1}{2} K(q) (\partial q)^2 - V(q) - C_I \right] + S_m[g, \psi] + S_r[g, A_\mu], \quad (\text{AD.14})$$

the flat-FLRW equations, with $v = \dot{q}$, are

$$3fH^2 = \rho_m + \rho_r + \frac{1}{2}Kv^2 + V + C_I - 3Hf_{,q}v, \quad (\text{AD.15})$$

$$-2f\dot{H} = \rho_m + \frac{4}{3}\rho_r + Kv^2 + \ddot{f} - H\dot{f}, \quad (\text{AD.16})$$

$$K(\dot{v} + 3Hv) + \frac{1}{2}K_{,q}v^2 + V_{,q} - 3f_{,q}(2H^2 + \dot{H}) = 0. \quad (\text{AD.17})$$

The last two equations form

$$\begin{pmatrix} 2f & f_{,q} \\ -3f_{,q} & K \end{pmatrix} \begin{pmatrix} \dot{H} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} -[\rho_m + \frac{4}{3}\rho_r + Kv^2 + f_{,qq}v^2 - Hf_{,q}v] \\ -3KHv - \frac{1}{2}K_{,q}v^2 - V_{,q} + 6f_{,q}H^2 \end{pmatrix}. \quad (\text{AD.18})$$

For $f = f_* e^{aq}$ and $K = \kappa f$ the determinant is

$$\Delta_{Hv} = 2fK + 3f_{,q}^2 = 2f^2 \left(\kappa + \frac{3a^2}{2} \right). \quad (\text{AD.19})$$

These equations belong to the declared completion. They are not generated by Eq. (AD.3).

Here C_I is simply a declared constant term in an ordinary Jordan action. It must not be identified with the integration datum of the trace-free completion unless a single constrained action deriving that identification is supplied.

AD.4 Two-slope ordinary completion

Let

$$f = f_* e^{aq}, \quad K = \kappa f, \quad V = V_- e^{cq} + V_+ e^{bq}, \quad C_I = 0, \quad 0 < c < 2a < b. \quad (\text{AD.20})$$

For $V_{\pm} > 0$, the constant-field balance has the unique root

$$e^{(b-c)q_{\text{vac}}} = \frac{(2a-c)V_-}{(b-2a)V_+}, \quad (\text{AD.21})$$

and its restoring derivative is

$$\mathcal{E}_{,q}(q_{\text{vac}}) = (2a-c)(b-c)V_-e^{cq_{\text{vac}}} > 0. \quad (\text{AD.22})$$

The Einstein-frame potential

$$\mathcal{U}_E \propto V_-e^{(c-2a)q} + V_+e^{(b-2a)q} \quad (\text{AD.23})$$

diverges at both ends and is strictly convex. If $f > 0$, $\kappa + 3a^2/2 > 0$, and the solution is on an expanding branch with nonnegative dust and radiation, it has one global minimum, confines each finite-energy future trajectory in that branch, and supplies a regular de-Sitter endpoint.

In this normalisation $\kappa + 3a^2/2 > 0$ is equivalently the statement that the Einstein-frame scalar kinetic coefficient is positive.

This result regularizes the ordinary scalar dynamics without a literal constant floor. It does not possess the shift covariance of the trace-free completion: adding C_0 to V changes both the Friedmann equation and the scalar trace balance.

The frozen near-GR benchmark quoted in the main text directly integrates Eqs. (AD.15)–(AD.17) and gives

$$\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.96368677864. \quad (\text{AD.24})$$

The dimensionless interval is owned by that calibrated action and state. Converting it to Gyr with a measured H_0 does not import a Λ CDM expansion history; it supplies the single dimensional scale of the already fixed orbit. It also does not make the number a unique ECT prediction, because the action coefficients, state and physical metric have not been selected by the microscopic theory.

AD.5 Why $\tilde{M}_{\text{Pl}}^2 H_0^2$ is not a determination

Choose a physical metric and a fixed reference Planck calibration, and define

$$\rho_{c,0}^{\text{GRref}} \equiv 3\tilde{M}_{\text{Pl}}^2 H_0^2, \quad \Omega_{\text{late},0}^{\text{GRref}} \equiv \frac{\rho_{\text{late},0}^{(\text{GRref})}}{\rho_{c,0}^{\text{GRref}}}. \quad (\text{AD.25})$$

Then $\rho_{\text{late},0}^{(\text{GRref})} = 3\Omega_{\text{late},0}^{\text{GRref}}\tilde{M}_{\text{Pl}}^2 H_0^2$ is a definition, not an identity of every physical Friedmann model. For the supplied scalar–tensor action, the present model-native bookkeeping instead obeys

$$3\tilde{M}_{\text{Pl}}^2 F_0 H_0^2 = \rho_{m0} + \rho_{r0} + U_0 + \frac{\tilde{M}_{\text{Pl}}^2}{2} \omega_0 \dot{\phi}_0^2 - 3\tilde{M}_{\text{Pl}}^2 H_0 \dot{F}_0 + \rho_{n0}, \quad (\text{AD.26})$$

where ρ_{n0} is present only after a separate orientation action is supplied. Dimensional analysis can motivate candidate operators but cannot select the coefficient, sign, state, microscopic carrier, physical metric or Cavendish/Planck matching. In particular, the current ECT action derives no soft-mode operator, mass, state or abundance that can be assigned to the observed late source. Writing $H_{0,E} := \hbar H_0^{(\text{freq})}$ to distinguish frequency and natural-unit energy conventions, the separately named C/Open orientation hypothesis $m_{\chi,E} = \zeta_H H_{0,E}$ is not selected by this action and does not identify a carrier of that source. No Planck-scale field excursion, numerical infrared-density scaling, abundance or cosmological-role inference is retained.

External-Euclidean guard. No spherical boundary is assumed. A local determinant constraint does not fix a global integration datum, and determinant-preserving metrics are not unique. Any proposed external owner must state the condensate domain, boundary conditions, state and response operator and reproduce the same local limit for nonspherical domains. The internal Lorentzian age and an external Euclidean formation duration also remain distinct until an explicit clock map relates them.

AE Conditional two-slope ECT cosmology: action, state, and computed observables

AE.1 Supplied action and regularity conditions

A fully specified conditional scalar–tensor completion is obtained by supplying the Jordan-frame action

$$S_{2s} = \int d^4x \sqrt{-g} \left[\frac{1}{2} F_s(\sigma) R - \frac{1}{2} K_s(\sigma) (\nabla \sigma)^2 - V_s(\sigma) \right] + S_m[g, \Psi] + S_r[g, A_\mu], \quad (\text{AE.1})$$

with

$$\boxed{F_s = F_* e^{a_s \sigma}, \quad K_s = \kappa_s F_s, \quad V_s = V_- e^{c_s \sigma} + V_+ e^{b_s \sigma}}, \quad 0 < c_s < 2a_s < b_s, \quad V_\pm > 0. \quad (\text{AE.2})$$

Here F_s and K_s have mass dimension two. Thus the exact coefficient of the matter source after the complete two-slope metric equation is divided by F_s is

$$G_{\text{div},s}(N) \equiv \frac{1}{8\pi F_s(N)}, \quad \frac{G_{\text{div},s}(N)}{G_{\text{div},s}(0)} = \frac{F_s(0)}{F_s(N)} \quad (c = \hbar = 1). \quad (\text{AE.3})$$

This dimensionful- F_s convention must not be replaced by G_N/F_s unless an additional matching $F_s = \tilde{M}_{\text{Pl}}^2 F$ and the action normalisation are stated explicitly. The coefficient $G_{\text{div},s}$ is not by itself a Cavendish or cosmological response. The subscript s keeps this supplied completion distinct from the amplitude coordinate and the N -velocity $q = d\phi/dN$ used above. Equation (AE.1) is a complete model for the calculations in this subsection; its derivation from the microscopic Φ -medium is Open. The two-slope potential supplies a stabilising wall without inserting a literal constant floor, but an ordinary scalar–tensor action of this form is not invariant under $V_s \mapsto V_s + C$. The construction therefore does not replace the separate trace-free/fixed-measure vacuum-offset mechanism; adding a constant defines a different completion and changes its metric trace.

Exact field equations and response determinant. For a homogeneous field $\sigma(t)$, $v = \dot{\sigma}$, dust and radiation, the flat-FLRW equations are

$$3F_s H^2 = \rho_m + \rho_r + \frac{1}{2} K_s v^2 + V_s - 3H F_{s,\sigma} v, \quad (\text{AE.4})$$

$$-2F_s \dot{H} = \rho_m + \frac{4}{3} \rho_r + K_s v^2 + \ddot{F}_s - H \dot{F}_s, \quad (\text{AE.5})$$

$$K_s (\dot{v} + 3Hv) + \frac{1}{2} K_{s,\sigma} v^2 + V_{s,\sigma} - 3F_{s,\sigma} (2H^2 + \dot{H}) = 0. \quad (\text{AE.6})$$

The last two equations form a linear system for $(\dot{H}, \dot{v})$ whose determinant is

$$\boxed{\Delta_{Hv} = 2F_s K_s + 3F_{s,\sigma}^2 = 2F_s^2 D_s}, \quad D_s \equiv \kappa_s + \frac{3}{2} a_s^2. \quad (\text{AE.7})$$

Thus $F_s > 0$ and $D_s > 0$ give local existence and uniqueness; the Friedmann root, the physical kinetic form and the gradient eigenvalues remain separate gates.

Vacuum root, stability and future regularity. For a constant field in vacuum the balance $V_{s,\sigma} - 2(F_{s,\sigma}/F_s)V_s = 0$ has the unique root

$$\boxed{e^{(b_s - c_s)\sigma_0} = \frac{(2a_s - c_s)V_-}{(b_s - 2a_s)V_+}}. \quad (\text{AE.8})$$

At this root,

$$\mathcal{E}_{,\sigma}(\sigma_0) = (2a_s - c_s)(b_s - c_s)V_- e^{c_s\sigma_0} > 0, \quad (\text{AE.9})$$

$$m_0^2 = \frac{\mathcal{E}_{,\sigma}(\sigma_0)}{F_s(\sigma_0)D_s} > 0. \quad (\text{AE.10})$$

The Einstein-frame potential is proportional to

$$\mathcal{U}_E(\sigma) \propto V_- e^{(c_s - 2a_s)\sigma} + V_+ e^{(b_s - 2a_s)\sigma}. \quad (\text{AE.11})$$

It diverges at both ends, is strictly convex and has one global minimum. For an expanding solution of the supplied action with non-negative dust and radiation, finite initial energy confines σ to a compact interval, keeps F_s bounded above and away from zero, and prevents a finite future endpoint. This proves future regularity inside the supplied completion from any finite regular ordered-branch slice. It does not derive the ordering event and does not remove a possible past singularity.

Analytic polynomial special case. For $c_s = a_s$, $b_s = 3a_s$, and $y = e^{a_s\sigma} = F_s/F_*$,

$$V_s = V_- y + V_+ y^3, \quad y_0 = \sqrt{V_-/V_+}, \quad H_{\text{vac}}^2 = \frac{2V_-}{3F_*}, \quad \frac{m_0^2}{H_{\text{vac}}^2} = \frac{3a_s^2}{D_s}. \quad (\text{AE.12})$$

This is an analytic polynomial in the positive amplitude ratio; no fractional-power response has been inserted.

Exact radiation and dust asymptotics. In the leading radiation regime, writing $r = d \ln F_s / dN = a_s d\sigma / dN$ and neglecting the trace-free potential correction, the scalar equation factorises as

$$\frac{r}{a_s} \left(\kappa_s + \frac{3}{2} a_s^2 \right) = 0. \quad (\text{AE.13})$$

For $D_s > 0$ the healthy solution is therefore $d\sigma/dN = 0$, $F_s = \text{const}$ and $H^2 \propto a^{-4}$; linearisation gives $(d\sigma/dN)' \simeq -d\sigma/dN$ in the forward expanding direction.

When dust and the upper exponential dominate, define

$$s_d \equiv \frac{3a_s}{b_s}, \quad \frac{d\sigma}{dN} = -\frac{3}{b_s}, \quad F_s \propto a^{-s_d}, \quad H^2 \propto a^{-(3-s_d)}. \quad (\text{AE.14})$$

All three equations (AE.4)–(AE.6) have zero residual provided

$$\frac{\rho_m}{F_s H^2} = 3 - \frac{7}{2} s_d - \frac{1}{2} s_d^2 - \frac{9\kappa_s}{b_s^2} > 0, \quad (\text{AE.15})$$

$$\frac{V_+ e^{b_s\sigma}}{F_s H^2} = \frac{1}{2} s_d (1 + s_d) + \frac{9\kappa_s}{2b_s^2}. \quad (\text{AE.16})$$

The local clock relation is $a(t) \propto t^{2/(3-s_d)}$ and $t = 2/[(3-s_d)H]$. It is an asymptotic tracker relation, not the total age.

Planck-scale and local-gravity guard. On the high-density stationary branch,

$$e^{b_s\bar{\sigma}} \sim \frac{a_s \rho_m}{2(b_s - 2a_s)V_+}, \quad F_s(\bar{\sigma}) \propto \rho_m^{a_s/b_s}, \quad \Delta \ln G_{\text{div},s} = s_d \Delta N \quad (\text{AE.17})$$

on the exact dust tracker. Hence the same construction that removes the zero-density degeneration can produce unacceptable divided Planck-coefficient drift. A frozen-field orbit can suppress that drift, but then its initial state and relaxation time are physical inputs. Tracker and frozen branches may not be selected separately for different observables. The finite-body scalar charge, range, physical metric and PPN parameters must be derived before $G_{\text{div},s}$ can be transferred to a measured local response or this completion can be regarded as viable.

Two-method global existence benchmark. A frozen reproducibility run used

$$a_s = 0.01, \quad b_s = 1, \quad c_s = 0.005, \quad \kappa_s = 1, \quad \Omega_{m0}^{\text{in}} = 0.315, \quad \Omega_{r0}^{\text{in}} = 9.19 \times 10^{-5}. \quad (\text{AE.18})$$

Two logically distinct inputs are supplied explicitly: the early frozen scalar state and the overall potential scale, so that $\sigma(0) = 0$ and $H(0) = H_*$ in dimensionless units. DOP853 and Radau both traverse $-20 \leq N \leq 10$. The determinant factor is $2\kappa_s + 3a_s^2 = 2.0003 > 0$, while F_s , H^2 and the algebraic Friedmann factor remain positive on 3001 sampled points. The internal interval is

$$\boxed{\int_{-20}^0 \frac{dN}{H(N)} = 0.95060544 H_*^{-1}.} \quad (\text{AE.19})$$

This dimensionless interval is not converted to gigayears: H_* , the density inputs, action parameters, initial state and ordering event were not predicted by ECT, and no external-to-internal clock map has been derived.

Conditional age of a second, near-GR member of the supplied two-slope family. The age question can nevertheless be answered conditionally for a named completion, rather than by importing a Λ CDM background. A second frozen member of Eq. (AE.1) uses

$$a_s = c_s = 0.01, \quad b_s = 0.03, \quad \kappa_s = 10, \quad V_- = 0.82485, \quad V_+ = 1.27485, \quad \rho_{m0} = 0.9, \quad \rho_{r0} = 3 \times 10^{-4} \quad (\text{AE.20})$$

in units $F_* = H_* = 1$. Evolving the selected regular radiation branch from $N = -60$ and integrating the same scalar action to $N = 0$ gives

$$\boxed{\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.9636867786415146.} \quad (\text{AE.21})$$

DOP853 and Radau agree in the displayed digits; moving the initial cutoff from $N = -40$ to -60 changes the result by less than 5×10^{-15} . The direct asymptotic tail before the diagnostic slice $N = -20$ is 2.12×10^{-16} in Hubble-time units; it is not a tail from $N = -60$. Starting instead at $N = -20$ with $q = p = 0$ shifts the age by approximately 0.5619 yr under the 67.4 calibration. Here $N = -60$ is an assumed sufficiently early slice of an already ordered radiation branch. Cutoff convergence proves that the integral has reached that branch's asymptotic clock limit; it does not derive the formation-front crossing Σ_{ord} or identify $N = -60$ with a microscopic ordering event. The displayed interval is therefore exact for the named action/state and its early-ordered boundary condition, while the duration from a dynamically derived ordering front remains Open.

If—and only if—one now supplies an external dimensional Hubble normalisation, the conversion

$$t_{0,2s}[\text{Gyr}] = \tau_{2s} \frac{977.7922216807891}{H_0[\text{km s}^{-1} \text{Mpc}^{-1}]} \quad (\text{AE.22})$$

gives

$H_0 [\text{km s}^{-1} \text{Mpc}^{-1}]$	$t_{0,2s} [\text{Gyr}]$	
67.40	13.9804960874	
73.04	12.9009506612	(AE.23)

Conversely, supplying $t_{0,2s} = 13.8$ Gyr for this same orbit fixes $H_0^{\text{cal}} = 68.2815533545 \text{ km s}^{-1} \text{Mpc}^{-1}$. This inverse example is not a third trajectory: it makes explicit that one absolute datum calibrates the common dimensional scale, after which H_0 and age cannot be retuned independently. The second calibration is the local SH0ES value reported by Riess et al. [294]; it is a disclosed conditional single-scale calibration, not an ECT prediction. Likewise, globular-cluster age determinations provide an external comparison target only [295]; they do not supply the ordered-branch dynamics or its formation-front

boundary condition. In the explicitly restricted outside-view special case $\mathcal{N}_\Phi = 1$, $c_u = c$ and an aligned planar congruence, these clock intervals correspond to Euclidean normal depths

$$L_w = (4.28644, 3.95545) \text{ Gpc} = (13.98050, 12.90095) \text{ Gly}, \quad (\text{AE.24})$$

respectively. This is depth along the selected external direction, not a radius of the Universe and not evidence for a spherical boundary. A non-trivial lapse, curved formation front or non-aligned matter congruence changes the map according to Eqs. (16.14) and (16.15). These numbers are calculated from the ECT-candidate two-slope dynamics, not from a Λ CDM expansion function. Their status is nevertheless conditional: H_0 , the action coefficients, density normalisation, initial state, physical metric, Planck matching and external depth/lapse map are inputs rather than predictions. The two Gyr values are the expected dimensional readings of one and the same dimensionless orbit under two externally supplied H_0 calibrations. They demonstrate state and scale dependence and the fact that P1–P6 do not yet select H_0 ; they are not evidence of a numerical defect in the supplied two-slope dynamics. They are conditional estimates for the named candidate completion, not a unique P1–P6 prediction, an independent age measurement, or a resolution of the cosmic-age question by themselves.

Covariance-aware chronometer calibration of the same orbit. The preceding conditional conversion can be calibrated against an observed clock channel without pretending that the universal postulates predict the dimensional scale. We use the official 15-point BC03 subset and the “suggested” covariance prescription distributed for the Moresco cosmic-chronometer analysis [296, 297]. For the frozen two-slope shape

$$H_{\text{model}}(z) = H_0 E_{2s}(z), \quad C = \text{diag}(\sigma_{H,i}^2) + u_{\text{IMF}} u_{\text{IMF}}^T + u_{\text{SPS},000} u_{\text{SPS},000}^T, \quad (\text{AE.25})$$

where $(u_X)_i = H_i f_X(z_i)$, the one-scale generalised least-squares profile is analytic,

$$\hat{H}_0 = \frac{E^T C^{-1} H}{E^T C^{-1} E}, \quad \sigma_{H_0} = \left(E^T C^{-1} E \right)^{-1/2}. \quad (\text{AE.26})$$

Direct solves, Cholesky whitening and a rank-two Woodbury inversion agree and give

$$H_0 = 68.5860 \pm 3.9491 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad t_{0,2s} = 13.7387 \pm 0.7911 \text{ Gyr}, \quad (\text{AE.27})$$

with $\chi_{\min}^2 = 6.2806$ for 14 nominal degrees of freedom. The age uncertainty shown is the linearised Gaussian propagation of only the chronometer scale uncertainty through Eq. (AE.22); it excludes uncertainty in the action coefficients, selected state, formation front, lapse, physical metric, redshift map and stellar-population prescription. The low reduced $\chi^2 = 0.4486$ is therefore not evidence for ECT; the small subset and its correlated systematic model are conservative. Directly transforming $H_0 \pm \sigma_{H_0}$ through the reciprocal map gives the asymmetric interval $t_{0,2s} = 13.7387^{+0.8394}_{-0.7480} \text{ Gyr}$. In the explicitly restricted aligned planar outside-view limit $\mathcal{N}_\Phi = 1$ and $c_u = c$, the symmetric linearised result corresponds to $L_w = 4.2123 \pm 0.2425 \text{ Gpc}$. This is a lapse-weighted depth along the selected congruence, not a universal radius or evidence for a spherical boundary.

Correlation of the dimensional clock and depth outputs. For a fixed dimensionless orbit and the restricted aligned planar map,

$$t_0 = \frac{\tau_{2s}}{H_0}, \quad L_w = c t_0, \quad \frac{\delta t_0}{t_0} = \frac{\delta L_w}{L_w} = -\frac{\delta H_0}{H_0}. \quad (\text{AE.28})$$

Consequently the displayed $t_0 = 13.7387 \pm 0.7911 \text{ Gyr}$ and $L_w = 4.2123 \pm 0.2425 \text{ Gpc}$ inherit the same chronometer scale uncertainty. With all action/state and lapse inputs held fixed, the first-order propagated covariance is rank one: within this linearisation, t_0 and L_w are perfectly positively correlated and both are perfectly anticorrelated with H_0 . The exact reciprocal map $t_0 \propto H_0^{-1}$ is nonlinear, so a finite non-Gaussian posterior need not have Pearson correlation exactly -1 . The three displayed quantities remain derived

readings of one calibration, not three independent constraints. Uncertainties in the orbit, formation front, lapse or physical metric would add new covariance directions and are not included in the quoted pair.

The scale sensitivity to the other covariance combinations displayed in the same official notebook is

Covariance prescription	H_0 [km s ⁻¹ Mpc ⁻¹]	$t_{0,2s}$ [Gyr]
suggested: diagonal + IMF + out-of-overlap SPS	68.586 ± 3.949	13.739 ± 0.791
suggested + stellar-library covariance	67.335 ± 6.077	13.994 ± 1.263
diagonal + IMF + stellar-library + full SPS	67.445 ± 7.689	13.971 ± 1.593

These rows are a systematic-prescription sensitivity test, not three independent measurements. A matched flat radiation–matter–constant-source control gives $\Delta H_0 = 1.03 \times 10^{-4}$ km s⁻¹ Mpc⁻¹ and $\Delta \chi^2 = 1.47 \times 10^{-5}$ relative to the selected ECT orbit. Thus this calculation is a reproducible Level C calibration of one realised ECT-candidate history, but neither ECT-specific discrimination nor a solution of the Hubble tension. The subset is BC03-only and is not the later complete 32-point chronometer likelihood; obtaining the latter with its exact row-ordered covariance is the next data-level gate.

Direct expansion history and distances of the same completion. No independently chosen interpolation function is used. Evaluating the same integrated orbit gives the following dimensionless expansion history and radial comoving distance. The comparison column is a flat radiation–matter–constant-source control with the same present derived density fractions; it is not inserted into the ECT solve.

Table 128: Conditional two-slope background observables. All entries are dimensionless; $E = H/H(0)$ and $\chi H(0)/c$ assumes the Jordan photon metric equals the background metric.

z	E_{2s}	$\Delta E/E_{\text{ctl}}$ [%]	$H(0)t_{2s}(z)$	$H(0)\chi_{2s}/c$
0	1.000000	0	0.96368678	0
0.5	1.308748	-1.884×10^{-4}	0.60284226	0.44096584
1	1.761039	-2.009×10^{-4}	0.41138834	0.77136045
2	2.967681	-6.434×10^{-5}	0.23067504	1.20927441
4.9	7.901164	$+2.795 \times 10^{-4}$	0.08454036	1.80243046
7	12.437605	$+4.363 \times 10^{-4}$	0.05355261	2.01370397
10	20.035592	$+5.982 \times 10^{-4}$	0.03318519	2.20316721
14.32	32.936279	$+7.649 \times 10^{-4}$	0.02015576	2.37077216
20	52.897950	$+9.226 \times 10^{-4}$	0.01252924	2.50643624

The complementary conditional background and scope panels are retained in Appendix BQ.

The corresponding distance differences from the control stay below $1.5 \times 10^{-4}\%$ on $0 < z \leq 20$. Thus this particular near-GR state is a constructive viable-background benchmark, not a mechanism for a large late-distance shift.

Conditional high-redshift time budgets. At the externally supplied rounded reference calibration $H_0 = 67.4$ km s⁻¹ Mpc⁻¹, corresponding to the named Planck base- Λ CDM inference 67.36 ± 0.54 km s⁻¹ Mpc⁻¹ [70], the same ECT-candidate orbit gives

$$\begin{array}{c|ccccc}
 z & 4.9 & 7 & 10 & 14.32 & 20 \\
 \hline
 t_{2s}(z) \text{ [Gyr]} & 1.22645 & 0.77690 & 0.48143 & 0.29241 & 0.18177
 \end{array} \quad (\text{AE.29})$$

and the intervals $(z_{\text{form}} \rightarrow z_{\text{obs}}) = (10 \rightarrow 4.9), (14.32 \rightarrow 7), (20 \rightarrow 10)$ and $(20 \rightarrow 14.32)$ are respectively 0.74502, 0.48450, 0.29966 and 0.11064 Gyr. With $H_0 = 73.04$ the corresponding intervals are 0.68750, 0.44709, 0.27652 and 0.10210 Gyr. These are genuine clock outputs of the supplied two-slope dynamics

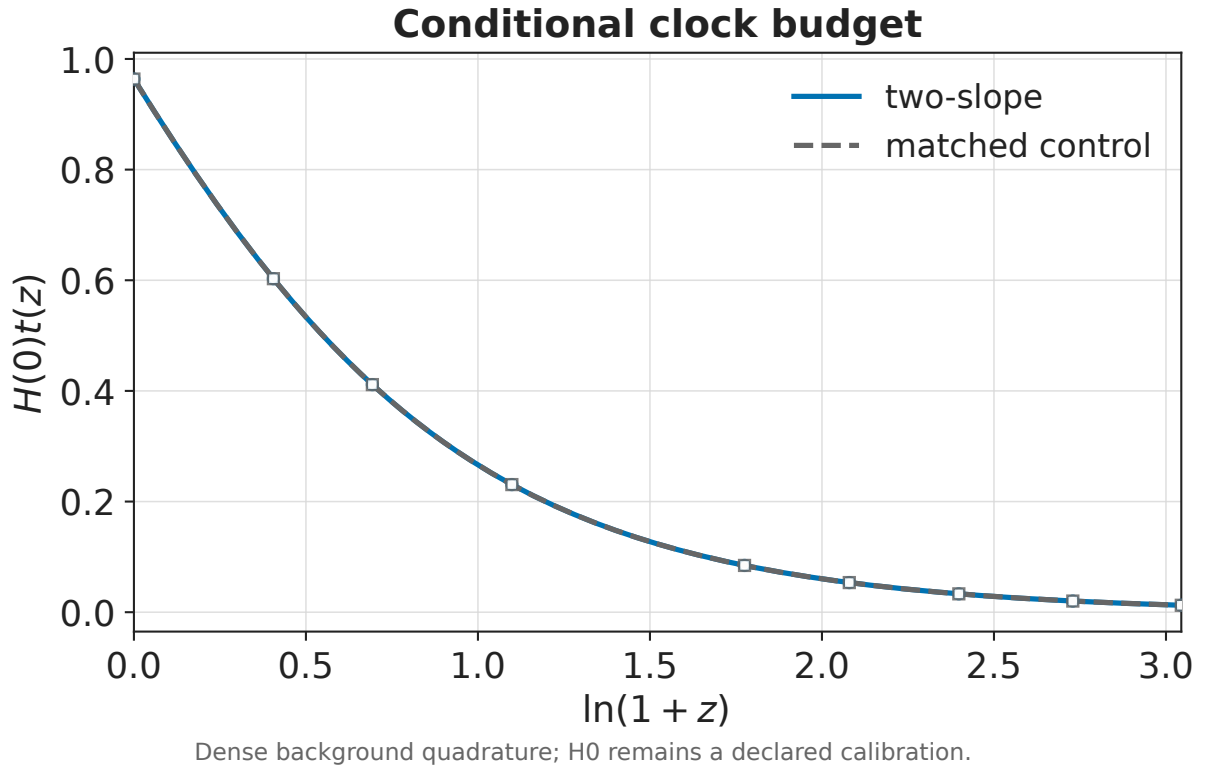


Figure 38: Conditional dimensionless clock budget $H(0)t(z)$ for the supplied two-slope action/state and its same-present-fraction reference control. Named expansion, comoving distance, total kinematic w_{eff} , the calibrated family and the explicit scope statement are separate full-width atlas panels. The continuous traces are replayed from the frozen 3001-point background owner, while the nine published redshift nodes are retained as separate markers. The dense traces and published nodes are direct outputs of the declared orbit, not a unique P1–P6 cosmology or a dark-energy equation-of-state fit. Colour is redundant with marker and line style. *Display semantics:* The continuous clock traces are replayed from the frozen 3001-point background owner; the nine published nodes are shown separately.

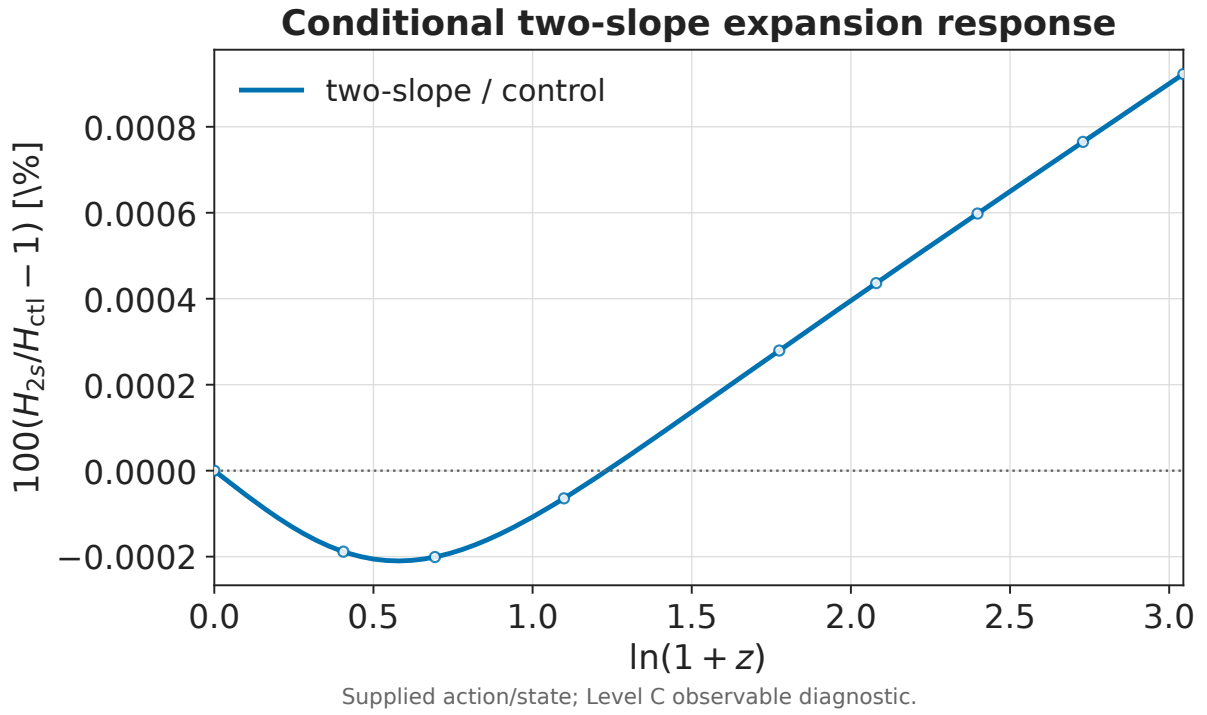


Figure 39: Focused status-safe fractional expansion response of the supplied two-slope background relative to the matched control. The total kinematic $w_{\text{eff}} = -1 - 2H'/(3H)$ is a separate full-width atlas panel; the inverse- F background proxy follows as a continuation and is not a local Newton constant. The curves contain no common- ε law. They are Level A only inside the supplied action and state, and a Level-C observable diagnostic rather than a unique P1–P6 cosmology. Colour is redundant with line style, marker shape and marker fill. The continuous expansion trace is replayed from the frozen 3001-point background owner; the nine published redshift nodes are retained as separate markers rather than used as vertices of a polygonal guide. *Display semantics: The continuous expansion trace is replayed from the frozen 3001-point background owner; the nine published nodes are shown separately.*

after dimensional matching. They are not a JWST abundance prediction: stellar-population inference, primordial power, scale-dependent growth, halo collapse, baryonic assembly and survey selection still have to be propagated through the same completion.

Conditional quasistatic growth diagnostic. As a deliberately restricted first perturbative estimate, impose conserved pressureless Jordan matter, a subhorizon quasistatic limit, negligible scalar mass and mixing, scale-independent response and $\mu_{\text{QS}} = F_s(0)/F_s(N)$. Then

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \frac{\rho_m}{3F_s(0)H^2} \mu_{\text{QS}} D = 0. \quad (\text{AE.30})$$

Normalising $D(0) = 1$, the fractional difference from the matched control is

z	0.5	1	2	4.9	(AE.31)
$10^4 \Delta D / D_{\text{ctl}} [\%]$	-1.04	-2.51	-5.17	-9.95	
z	7	10	14.32	20	
$10^4 \Delta D / D_{\text{ctl}} [\%]$	-12.09	-14.31	-16.60	-18.74	

This near-GR result is numerically small. Its status is Level C conditional because the complete gauge-reduced scalar/vector/tensor action, physical metric, scalar mass, scale dependence, initial spectrum, non-linear evolution and bias likelihood have not been derived.

Owner-specific early-response envelope (separate from the two-slope orbit). The following calculation is not another trajectory of the named two-slope action and does not restore a universal common- ε cosmology. It is a deliberately restricted response-envelope calculation in which a constant channel slope ζ_{ER} dresses the matter and constant-source terms, but not radiation:

$$E_{\text{ER}}^2(N) = \Omega_r e^{-4N} + \Omega_m e^{-(3+2\zeta_{\text{ER}})N} + \Omega_\Lambda e^{-2\zeta_{\text{ER}}N}, \quad (\text{AE.32})$$

$$\Omega_m^{\text{grav}}(N) = \frac{\Omega_m e^{-(3+2\zeta_{\text{ER}})N}}{E_{\text{ER}}^2(N)}, \quad \mu_{\text{ER}}(N) = e^{-2\zeta_{\text{ER}}N}. \quad (\text{AE.33})$$

The factor μ_{ER} already appears once in Ω_m^{grav} . The single-count restricted growth equation is therefore

$$D_{NN} + \left(2 + \frac{E_{\text{ER},N}}{E_{\text{ER}}}\right) D_N - \frac{3}{2} \Omega_m^{\text{grav}} D = 0, \quad (\text{AE.34})$$

without a second factor of μ_{ER} in the source. This closure differs from the near-GR two-slope calculation in Eq. (AE.30); their outputs must not be merged or described as one action/state prediction.

Initial-mode and boundary semantics. Write $N_{\text{on}} = -\ln(1 + z_{\text{on}})$ and $B_{\text{on}} = 2 + (E_{\text{ER},N}/E_{\text{ER}})_{N_{\text{on}}}$. The common-normalisation boundary is fixed by

$$D(N_{\text{on}}) = a_{\text{on}} = e^{N_{\text{on}}}, \quad D_N(N_{\text{on}}) = p_0 D(N_{\text{on}}), \quad p_0 = \frac{-B_{\text{on}} + \sqrt{B_{\text{on}}^2 + 6\Omega_m^{\text{grav}}(N_{\text{on}})}}{2}. \quad (\text{AE.35})$$

The first condition fixes the source calculation's scale-factor normalisation; equivalently, the rescaled variable D/a_{on} has unit value at the boundary. The second condition selects the local growing eigenmode. In this sensitivity calculation z_{on} is the boundary at which the model and control are assigned the same primordial normalisation. It is not a derived activation redshift, a phase-transition epoch, or a switch in the action.

Exact equality identity inside the same envelope. With only the matter term in Eq. (AE.32) tilted relative to radiation, matter–radiation equality obeys

$$1 + z_{\text{eq}}(\zeta_{\text{ER}}) = \left(\frac{\Omega_m}{\Omega_r}\right)^{1/(1-2\zeta_{\text{ER}})}, \quad \frac{1 + z_{\text{eq}}(\zeta_{\text{ER}})}{1 + z_{\text{eq}}(0)} = \left(\frac{\Omega_m}{\Omega_r}\right)^{2\zeta_{\text{ER}}/(1-2\zeta_{\text{ER}})}. \quad (\text{AE.36})$$

This is an exact identity of the declared envelope, not a CMB or BAO prediction. A radiation tilt, an independently derived transfer sector, or a different physical metric changes the interpretation and can be degenerate with it.

Numerical growth and collapse sensitivity. For $z_{\text{on}} = 1000$, $z = 10$, $\Omega_m = 0.315$ and $\Omega_r = 9.2 \times 10^{-5}$, flat closure fixes $\Omega_\Lambda = 1 - \Omega_m - \Omega_r = 0.684908$. Independent solvers reproduce the following single-count results. The five ζ_{ER} values are frozen stress-test coordinates inherited from the audited calculation chain; they are not posterior samples, an allowed corridor, or predictions of ECT. The exact inputs underlying the first three displayed rows are $1.5612556036955692 \times 10^{-6}$, 0.007500442556492628 and 0.029588344765651916 ; the first table column is rounded only for layout. The conditional barrier is obtained from a pressureless spherical top-hat in the same envelope,

$$\delta_{NN} + \left(2 + \frac{E_{\text{ER},N}}{E_{\text{ER}}}\right) \delta_N - \frac{4}{3} \frac{\delta_N^2}{1 + \delta} - \frac{3}{2} \Omega_m^{\text{grav}} \delta(1 + \delta) = 0, \quad (\text{AE.37})$$

with the same growing-mode initial slope and a frozen numerical collapse criterion $\delta = 10^7$. The reported barrier is the corresponding linear solution extrapolated to the observation slice; the control value is $\delta_{c,0} = 1.7267600460$. For an external reference peak height $v = 5$, the fixed-barrier and top-hat cumulative sensitivities are

$$R_{\text{PS}}^{\text{fixed}} = \frac{\text{erfc}\left[v/(\sqrt{2}D/D_0)\right]}{\text{erfc}(v/\sqrt{2})}, \quad (\text{AE.38})$$

$$R_{\text{PS}}^{\text{top-hat}} = \frac{\text{erfc}\left[v(\delta_c/\delta_{c,0})/(\sqrt{2}D/D_0)\right]}{\text{erfc}(v/\sqrt{2})}. \quad (\text{AE.39})$$

Table 129: Owner-specific ζ_{ER} sensitivity envelope. None of the rows is the named two-slope orbit or a joint JWST/CMB likelihood.

ζ_{ER}	D/D_0	$(1 + z_{\text{eq}})/(1 + z_{\text{eq},0})$	δ_c	$\delta_c/\delta_{c,0}$	$R_{\text{PS}}^{\text{fixed}}$	$R_{\text{PS}}^{\text{top-hat}}$
1.5612556×10^{-6}	1.00000521	1.00002541	1.72676032	1.00000016	1.00013505	1.00013091
0.00750044	1.02494986	1.13195313	1.72810966	1.00078159	1.86656763	1.83082689
0.02958834	1.09767905	1.66846599	1.73234129	1.00323220	9.13465297	8.51585875
0.03300000	1.10882221	1.77730862	1.73302549	1.00362844	11.34568870	10.50228040
0.03755500	1.12366682	1.93658864	1.73395051	1.00416414	14.99787480	13.75594876

The first numerical column is Level A inside Eqs. (AE.32)–(AE.35); the equality column is exact algebra in that envelope. The top-hat barrier and Press–Schechter ratios are Level C sensitivity estimates. They omit a derived ECT primordial spectrum, scale-dependent transfer, baryons, calibrated halo multiplicity, mass inference and selection. They therefore show that an early response can produce a large rare-tail sensitivity while simultaneously shifting equality; they do not constitute a JWST abundance prediction or re-establish a common ε theory. In particular, no displayed row has yet closed equality, early growth or JWST abundance, transfer physics and Solar-System constraints in one action/state solution: the near-zero row produces negligible rare-tail enhancement, whereas the boosted rows shift the declared equality coordinate by approximately 13–94%. This is a cross-sector tension to be resolved by a derived radiation/transfer and metric completion, not evidence that any one row is observationally viable.

Conditional sound-horizon diagnostic. If one additionally imports a standard photon/recombination layer with $\Omega_b = 0.0493$, $\Omega_\gamma = 5.45 \times 10^{-5}$, $z_{\text{drag}} = 1059.94$, $c_\gamma = c$ and $c_s/c = [3(1 + 3\rho_b/4\rho_\gamma)]^{-1/2}$, the same orbit gives

$$\frac{H(0)r_s}{c} = 0.03272615, \quad \frac{\Delta r_s}{r_{s,\text{ctl}}} = -0.003157\%. \quad (\text{AE.40})$$

This equals 145.565 Mpc for the 67.4 calibration and 134.324 Mpc for the 73.04 calibration. Recombination and the dimensional normalisation are external inputs here, so these numbers neither infer H_0 nor resolve its early/late discrepancy. They instead show quantitatively that the frozen near-GR member leaves the control sound horizon essentially unchanged.

Numerical validation and immediate falsifiers. The background was integrated independently with DOP853 and Radau at relative tolerance 10^{-11} ; τ_{2s} agrees below 5×10^{-13} . On 3001 points, F_s , H^2 , the algebraic Friedmann denominator and Δ_{Hv} remain positive. The present dimensionless Planck drift is $d \ln F_s / dN = 3.1225 \times 10^{-6}$ for this frozen state. A publication claim still requires a finite-body PPN solution, a direct comparison of the dimensional drift with local bounds, and the full perturbation/recombination owners just listed.

Calibrated family scan and the early-time/drift trade-off. To test whether the preceding near-GR point hides a useful high-redshift effect, a second frozen scan used $c_s = a_s$, $b_s = 3a_s$ and imposed the vacuum root $\sigma_{\text{vac}} = 0$ by

$$\frac{V_-}{V_+} = \frac{b_s - 2a_s}{2a_s - c_s} = 1. \quad (\text{AE.41})$$

For each (a_s, κ_s) , the early frozen value and total potential normalisation were shot to $\sigma(0) = 0$, $H(0) = 1$, with $\Omega_{m0} = 0.315$ and $\Omega_{r0} = 9.19 \times 10^{-5}$. DOP853 and Radau agree in τ_{2s} better than 10^{-15} for all nine points, and every sampled F_s , Friedmann denominator and response determinant is positive.

Table 130: Technical solver replay of the calibrated two-slope scan. This table records background age, early-clock, local-limit and drift diagnostics; the main cross-channel result including sound horizon, early coupling and the rough acoustic rescaling is Table 60. The comparison is the flat radiation–matter–constant-source control with the same present density fractions. The fifth column is the exact unscreened massless scalar limit; the last is the homogeneous-background quantity $|d \ln G_{\text{div},s} / dN|_0 = |d \ln F_s^{-1} / dN|_0$. Neither that divided coefficient nor the fifth-column unscreened proxy substitutes for a screened finite-body solution.

a_s	κ_s	Δt_0 [%]	$\Delta t(z=10)$ [%]	$ \gamma_{\text{PPN}} - 1 _{\text{uns}}$	$ d \ln G_{\text{div},s} / dN _0$
0.01	0.05	+0.00332	−0.1930	1.99×10^{-3}	8.96×10^{-4}
0.03	0.05	+0.02926	−1.6994	1.74×10^{-2}	7.90×10^{-3}
0.08	0.05	+0.18126	−10.4912	1.02×10^{-1}	4.93×10^{-2}
0.01	1	+0.00017	−0.0097	1.00×10^{-4}	4.49×10^{-5}
0.03	1	+0.00150	−0.0870	8.98×10^{-4}	4.04×10^{-4}
0.08	1	+0.01056	−0.6140	6.32×10^{-3}	2.85×10^{-3}
0.01	10	+0.00002	−0.0010	1.00×10^{-5}	4.49×10^{-6}
0.03	10	+0.00015	−0.0087	9.00×10^{-5}	4.04×10^{-5}
0.08	10	+0.00106	−0.0619	6.39×10^{-4}	2.87×10^{-4}

Indeed, with $\Phi_{\text{BD}} = F_s$ the action has constant $\omega_{\text{BD}} = \kappa_s / a_s^2$, so in the unscreened light-scalar limit

$$\gamma_{\text{PPN}} = \frac{\omega_{\text{BD}} + 1}{\omega_{\text{BD}} + 2} = \frac{\kappa_s + a_s^2}{\kappa_s + 2a_s^2}, \quad \beta_{\text{PPN}} = 1. \quad (\text{AE.42})$$

The corresponding weak-body coupling and Cavendish coefficient are given by Eq. (16.28); they are distinct from $G_{\text{div},s}$ before that response limit is imposed. Under the same explicitly imported standard

photon/recombination inputs used in Eq. (AE.40), the scan gives

$$\frac{\Delta r_s}{r_{s,\text{ctl}}} [\%] = \begin{array}{c|ccc} & a_s = 0.01 & a_s = 0.03 & a_s = 0.08 \\ \hline \kappa_s = 0.05 & -0.642 & -5.546 & -30.646 \\ \kappa_s = 1 & -0.0323 & -0.290 & -2.031 \\ \kappa_s = 10 & -0.00323 & -0.0290 & -0.206 \end{array} . \quad (\text{AE.43})$$

This is sound-horizon leverage, not an inferred H_0 : the distance to last scattering, recombination perturbations and late likelihood have not been fitted. The deliberately crude fixed-acoustic-angle proxy $H_{0,\theta}^{\text{proxy}} = 67.4 r_{s,\text{ctl}}/r_s$ gives $H_{0,\theta}^{\text{proxy}} = (67.84, 71.36, 97.18) \text{ km s}^{-1} \text{ Mpc}^{-1}$ along the three $\kappa_s = 0.05$ rows, and $(67.60, 68.80)$ for $(a_s, \kappa_s) = (0.03, 1), (0.08, 1)$. These numbers expose the possible acoustic leverage only; they are not a CMB or distance-ladder likelihood. The row $(a_s, \kappa_s) = (0.03, 0.05)$ illustrates the central trade-off: it changes r_s by -5.55% but has the unscreened $|\gamma_{\text{PPN}} - 1| = 0.0174$. The only scanned row satisfying the displayed absolute γ proxy gate without screening, $(0.01, 10)$, changes r_s by merely -0.00323% . The corresponding early-to-present divided-coefficient ratios $G_{\text{div},s}^{\text{early}}/G_{\text{div},s}^0 = F_s(0)/F_{s,\text{early}}$ are 1.1411 and 1.000075, respectively (the strongest scanned -30.65% sound-horizon row reaches 2.3255). Any large acoustic effect is therefore correlated with divided Planck-coefficient drift, but its physical gravitational calibration and BBN viability are determined only by the complete early-universe response pipeline.

The scan establishes a concrete negative result for the nine evaluated nodes: the points that most change the early time budget make the Universe *younger* at fixed high redshift and simultaneously increase the homogeneous divided Planck-coefficient drift. The last column is $|d \ln G_{\text{div},s}/dN|_0$ and cannot be compared directly with lunar ranging [298, 299]; LLR constrains an orbital/Cavendish response after the scalar range and body charges are specified. The primary Cassini result $\gamma_{\text{PPN}} - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [95] is compared here only with the explicitly unscreened massless-scalar proxy. In that limit one of the nine nodes, $(a_s, \kappa_s) = (0.01, 10)$, passes the displayed absolute γ gate; it is 1.35σ from the measured central value and changes $t(z=10)$ by only $-9.7 \times 10^{-4}\%$. Each sampled node with a large early-time effect therefore requires a separately derived finite-body or metric completion. The finite-body mechanism analysed for the named calibration supplies no mass screening, but neither that result nor failure to find another completion rejects unsampled members of the continuous two-parameter family. The current result is a reproducible nine-node action/state scan, not a fitted effective one-parameter law or an exhaustive family no-go.

The exact radiation limit, exact dust tracker, stable vacuum and global two-solver orbit establish a Level A existence result inside the supplied action and a Level B proof-of-class for an ECT completion. They do *not* select a unique physical $H_{\text{ECT}}(a)$ or $t_{0,\text{ECT}}$ from P1–P6. They do identify the conditional curves and clock outputs of the explicitly supplied completion above. The next gates are: derive Eq. (AE.1) and its parameters from the ordered Φ -medium; identify the physical matter/photon metric and ordering event; derive the gauge-reduced perturbation action; and pass BBN, recombination, varying- G , growth, lensing and finite-body PPN tests. No spherical boundary or spherical Universe is assumed in this construction.

AE.2 Relation to observables

The two-slope example demonstrates the distinction between an action-owned conditional observable and a unique ECT prediction. Its background clocks and distances are computed directly; its growth and sound horizon require the disclosed extra assumptions. For any future completion, recombination and photon propagation precede the Hubble likelihood; the quadratic perturbation action precedes growth, ISW and lensing; a non-linear halo and baryonic forward model precedes JWST. The complete owner matrix and acceptance gates are given in §16.5 and Appendix AH.

AF Conditional tensor-EFT scaffold for gravitational waves associated with ordering

This appendix provides a minimal dimensional scaffold for a possible stochastic gravitational-wave background associated with inhomogeneous ordering. It applies only after a physical Lorentzian tensor EFT, a clock/metric, a TT source vertex and a matching interface from ordering inhomogeneities to that tensor have been supplied. It therefore parametrises a late-stage or post-ordering transfer; it is not a first-principles calculation of the pre-Lorentzian ordering process and does not assert that ordering produces a nonzero tensor source. Its purpose is to make the conditional programme of §13.8 quantitatively traceable without promoting it to a prediction.

Source equation. For the constant- (M_T, c_T) tensor benchmark on an emergent FRW background, define the dimensionless physical transverse–traceless metric perturbation by $g_{ij} = a^2(\delta_{ij} + H_{ij})$, with $\partial_i H_{ij} = H_{ii} = 0$. In conformal time η and units $c = 1$ it satisfies

$$H_{ij}'' + 2\mathcal{H}H_{ij}' + c_T^2 k^2 H_{ij} = \frac{2a^2}{M_T^2} \Pi_{ij}^{\text{TT,phys}} \equiv \frac{2}{M_T^2} \Pi_{ij}^{\text{TT,conf}}, \quad \Pi_{ij}^{\text{TT,conf}} := a^2 \Pi_{ij}^{\text{TT,phys}}, \quad (\text{AF.1})$$

where $\Pi_{ij}^{\text{TT,phys}}$ is the TT projection of the physical orthonormal spatial anisotropic stress (equivalently the mixed spatial stress on this background, with displayed spatial indices moved by δ_{ij}), and the second symbol is its explicitly defined conformal rescaling. Only in the GR-normalised specialization $c_T = 1$ and $M_T^2 = (8\pi G_N^{\text{nat}})^{-1}$ does the right-hand side become $16\pi G_N^{\text{nat}} a^2 \Pi_{ij}^{\text{TT,phys}} = 16\pi G_N^{\text{nat}} \Pi_{ij}^{\text{TT,conf}}$. A time-dependent M_T changes the damping term and lies outside this minimal benchmark. Specifically, allowing $M_T = M_T(\eta)$ in the same reduced action replaces $2\mathcal{H}$ by $2\mathcal{H} + (M_T^2)'/M_T^2$. Neither the TT stress nor its coupling is supplied by P1–P6.

Dimensional estimate. Up to convention-dependent order-one factors, the characteristic tensor amplitude at production scales in natural units as

$$H_{\text{prod}} \sim \frac{2\Delta\rho L_{\text{corr}}^2}{M_T^2 c_T^2} \longrightarrow \frac{16\pi G_N^{\text{SI}} \Delta\rho L_{\text{corr}}^2}{c^4} \quad \text{in the GR-normalised SI benchmark,} \quad (\text{AF.2})$$

where $\Delta\rho$ is a physical TT stress/energy-density scale, not the full ordering energy density unless its TT projection is derived, and L_{corr} is the gradient-controlled correlation length of the matched source. The displayed coefficient follows from the chosen tensor-action normalisation; source geometry, TT projection and time-profile factors remain parametric. The peak frequency at production is parametrically set by

$$f_{\text{prod}} \sim \frac{c_T}{2\pi L_{\text{corr}}} \quad (c = 1) \longrightarrow \frac{c}{2\pi L_{\text{corr}}} \quad \text{in the GR-normalised SI benchmark.} \quad (\text{AF.3})$$

Redshift to today. If the ordering transition occurs at an effective redshift z_{ord} , the peak frequency today is approximately

$$f_0 \sim \frac{f_{\text{prod}}}{1 + z_{\text{ord}}}, \quad (\text{AF.4})$$

while the present-day gravitational-wave energy fraction is related to its production value by the standard redshifting logic

$$\Omega_{\text{GW},0}(f_0) \sim \Omega_{\text{GW,prod}}(f_{\text{prod}}) \left(\frac{a_{\text{ord}}}{a_0} \right)^4 \left(\frac{H_{\text{ord}}}{H_0} \right)^2. \quad (\text{AF.5})$$

Thus the present signal depends not only on the source strength at production but also on the subsequent expansion history.

What this does and does not provide. Equations (AF.2)–(AF.5) provide a dimensional scaffold, not a closed prediction. The quantity $\Omega_{\text{GW,prod}}$ must itself be obtained from the ordering-source efficiency and cannot yet be computed from first principles within the present closure. The quantities $\Delta\rho$, L_{corr} , and z_{ord} are transition parameters that must be derived from the ordering dynamics (duration, latent heat analogue, correlation length, domain structure). Until this microphysics is closed, $\Omega_{\text{GW}}(f)$ remains parametrically open.

Distinction from standard first-order transitions. The ECT ordering transition is not a thermal bubble-nucleation event in a pre-existing Lorentzian spacetime. Standard thermal-transition formulae (e.g. based on α , β/H , v_w) therefore cannot be applied directly. The scaffold above is formulated in terms of general dimensional quantities applicable to any causal source of anisotropic stress, without assuming a thermal equation of state.

AG Status of sign-level ordered-branch cosmology statements

This appendix separates exact statements inside a supplied scalar model from claims that require a cosmological completion.

C1. Conditional bare scalar-ratio inequality. For the action normalisation of Eq. (13.35), $K_\phi^{\text{phys}} = \bar{M}_{\text{Pl}}^2 \omega \dot{\phi}^2 / 2 \geq 0$ and $K_\phi^{\text{phys}} + U > 0$ imply $w_\phi^{\text{bare}} = (K_\phi^{\text{phys}} - U) / (K_\phi^{\text{phys}} + U) \geq -1$. This is Level A algebra for the bare Jordan-bookkeeping pieces. It is not a bound on the observable effective history, which also contains $\dot{F}$ and $\ddot{F}$.

C2. Conditional frozen-scalar limit. If ϕ is constant, the orientation stress vanishes or is separately fixed, and the remaining potential is constant, the supplied background equations reduce algebraically to an Einstein–Friedmann form. Reachability and stability of that branch are separate questions.

C3. No model-independent Hubble-shift sign theorem. A change in an effective gravitational response can alter both the sound horizon and distance kernels. The inferred H_0 shift depends on the full recombination-to-late-time likelihood and is not fixed by the sign of a single late-time function. A sign claim is therefore Open until the complete ECT background, photon metric and acoustic calibration are solved jointly.

C4. No model-independent JWST enhancement theorem. Early-galaxy counts depend on the background volume and age, the primordial spectrum, transfer and growth, non-linear collapse, baryonic evolution and selection. A stronger fitted response in one diagnostic equation is not a theorem about abundance or morphology.

C5. Conditional correlation only after common ownership. Age, distance, growth and potential evolution are correlated once one complete action and state determine them. Their common dependence on an assumed background does not demonstrate that ECT has derived that background.

AH ECT cosmological identifiability and action-to-observable closure

This appendix gives the equations and gates required for a genuine ECT cosmology. Each observable is derived from its physical owner in one common action and state.

AH.1 Conditional-estimate and replacement policy

Approximation is not itself a reason to discard a calculation. A numerical estimate is admissible as a Level C conditional result when it is generated by one named action or response closure, one admissible state and boundary-data record, one consistent dimensional calibration, and a stated approximation domain. The same owners and calibration must then be propagated to every correlated clock, distance, growth, lensing and local-gravity output. Universal postulates need not select the integration constants or the absolute age and Hubble scale of our particular condensate realisation; those may be measured state descriptors. What the theory must determine is the lawful relation between them for the selected solution.

A formula is rejected only when its algebra is false, its dimensions or normalisations are inconsistent, or it assigns one unowned parameter to physically inequivalent operators. Retirement of that formula does not retire the physical question. Each affected observable is retained and replaced by one of three explicit objects: a conditional calculation from a named action/state, a theorem or no-go result that bounds the answer, or a forward equation with its missing owners listed. A superseded empirical interpolation is not kept as an active “legacy” law in the theory exposition, and an empty deletion is not an acceptable correction.

AH.2 Frozen action hierarchy

The bare microscopic scalar action owns local $O(4)$ dynamics, but its P4 gradient background is off shell. The ordered Euclidean derivative expansion owns a useful field inventory, while its coefficients and gravity matching remain Open. The scalar-only covariant action owns exact field equations inside its declared Jordan-frame model, but not the orientation stress. The supplied two-slope scalar–tensor action owns the conditional background and clock calculations reported here, while its coefficients, state and relation to P1–P6 remain inputs. These layers must not be merged into one silently complete action.

AH.3 Background boundary/initial-value problem

A candidate completion must vary one action to obtain a Hamiltonian constraint, an evolution equation for each independent condensate degree of freedom, the orientation constraints and matter/radiation conservation. In a homogeneous branch, the solve must determine

$$\mathcal{B}_{\text{ECT}} = \{a(t), H(t), u(t), n_A(t), \rho_m(t), \rho_r(t), \dots\} \quad (\text{AH.1})$$

from one physical state and one set of boundary/initial data. The Planck normalisation and H_0 may be matched to observations only when explicitly identified as matching inputs; they cannot simultaneously be advertised as predictions.

AH.4 Identifiability theorem

For the current scalar-only equations, the functions in Eqs. (16.20) and (16.21) reconstruct an admissible formal closure for arbitrary sufficiently smooth $H(t)$ and $F(t) > 0$. Hence one background cannot be inferred from those equations until the closure functions and state are independently fixed. Positivity and stability can reject members of the family but do not select a unique member without additional dynamics.

AH.5 Physical age

The only active age statement is Eq. (16.13). Its lower endpoint is the derived ordering event, not automatically $a = 0$. External reference integrations validate a numerical quadrature but do not constitute ECT ages. An accepted calculation must publish the full derived $H_{\text{ECT}}(a)$, the ordering criterion, input ownership and independent integration checks.

AH.6 Distance and acoustic owners

For a spatially flat physical photon metric, the formal distance and sound horizon are

$$\chi(z) = \int_0^z \frac{c_\gamma(z')}{H_{\text{ECT}}(z')} dz', \quad (\text{AH.2})$$

$$D_A(z) = \frac{\chi(z)}{1+z}, \quad (\text{AH.3})$$

$$r_s(z_*) = \int_{z_*}^\infty \frac{c_s(z)}{H_{\text{ECT}}(z)} dz. \quad (\text{AH.4})$$

ECT must derive c_γ , c_s , z_* , ionisation history and the metric seen by photons. A late-time change cannot be propagated through these integrals without the high-redshift solution.

AH.7 Perturbation and growth owners

Let $\mathbf{v}_k$ denote the constrained physical perturbations after gauge and non-dynamical fields are removed. A complete quadratic action has the form

$$S^{(2)} = \frac{1}{2} \int d\eta d^3k \left[\mathbf{v}_k'^\dagger \mathbf{K} \mathbf{v}_k' - \mathbf{v}_k'^\dagger \mathbf{v}_k \right]. \quad (\text{AH.5})$$

Ghost freedom requires positive kinetic form on the physical subspace; gradient stability requires the relevant eigenvalues of the principal symbol to be non-negative. The same reduction must derive the matter Poisson kernel, slip, sound speeds and anisotropic stress. Only its solutions can own a growth factor, $f\sigma_8$, ISW source or lensing kernel.

AH.8 Early-object forward model

A JWST prediction is a predicted distribution, not a shifted scalar number. It requires

$$\frac{dN}{dz d\mathcal{O}} = \frac{dV_{\text{ECT}}}{dz} \int dM \frac{dn_{\text{ECT}}}{dM}(M, z) P(\mathcal{O} | M, z, \mathcal{B}_{\text{baryon}}) \mathcal{S}(\mathcal{O}, z), \quad (\text{AH.6})$$

where the volume, halo mass function, baryonic map and selection function are all declared. Age and linear growth alone do not fix this observable.

AH.9 Lensing, clusters and mergers

Photon lensing depends on the Weyl potential of the physical photon metric. A strictly increasing local transformation preserves the possibly empty set-valued argmax of the supplied total projected baryon proxy. Whenever maximizers are attained, it cannot by itself displace a total-response maximizer away from the supplied total-baryon maximizers. It can inherit a gas/BCG separation already encoded in the input, and it says nothing about extrema of the residual map. Explaining collision-generated morphology therefore requires a time-dependent non-local response, its transport state and a hydrodynamic/collision forward model. HRC by itself supplies none of these owners.

AH.10 Local gravity and PPN gate

The cosmological completion and the weak-field finite-source completion must be limits of the same theory. The PPN parameters are read from the physical metric after solving its constraints. A scalar propagation cone or a galactic interpolation does not determine them. Solar-System constraints are therefore an early kill gate for proposed completions, not an output that can be appended after a cosmological fit.

AH.11 External Euclidean geometry

For a nonspherical interface the response depends on intrinsic momentum and the full extrinsic-curvature tensor. The admissible external construction must specify the actual interface, boundary conditions and state and must derive its contribution to the on-shell Lorentzian stress tensor. Flat-front translation and a codimension-one spherical-radius substitution are excluded as generic Hubble owners by the isometry and Gauss obstructions stated in §16.4.

AH.12 Acceptance matrix

Output	Full ECT-native acceptance evidence	Named supplied result now	Two-layer status
Background $H(a)$	One derived action, state, physical clock/congruence and stable solve	Globally regular two-slope trajectories; exact radiation/dust/vacuum limits	Full owner Open; A in supplied action / C as ECT completion
Age	Same $H(a)$ plus congruence depth/lapse map and one declared unit calibration	$\tau_{2s} = 0.96368678$ and disclosed conditional Gyr/high- z clocks; covariance-calibrated $t_0 = 13.74 \pm 0.79$ Gyr	Full map Open; A in state / C dimensional comparison
Hubble inference	ECT recombination, spectra and one early+late likelihood	Official 15-point BC03 covariance calibration of the named orbit plus a conditional imported-recombination r_s scan and rough fixed-angle diagnostic	Full inference Open; C calibration/diagnostic
JWST	End-to-end abundance and morphology forward model	Conditional time/volume budgets at nine prespecified nodes; none enlarges the sampled $t(z = 10)$ age budget, while the continuous family is not exhausted	Full prediction Open; C finite-grid background diagnostic
Growth/ISW/lensing	Gauge-reduced perturbation action and photon/matter metric map	Restricted quasistatic growth plus instantaneous Weyl and conformal-derivative carriers computed; no projected spectra	Full owner Open; C limiting diagnostics
Clusters	Retarded/non-local metric response plus collision model	Exact local-map peak no-go only	Full collision/lensing Open; A conditional theorem / physical Open

PPN	Finite-source metric and Solar-System comparison	Exact unscreened massless-family γ, β diagnostic; one point satisfies the displayed absolute-deviation proxy and lies 1.35σ from the Cassini central value	Not a full finite-body/PPN fit; full owner Open
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The right two columns prevent an Open full-theory owner from erasing a valid conditional result, and prevent a conditional result from being promoted into an ECT-native prediction. No fitted cross-channel interval replaces either layer.

AI Protocol for a future observable-complete ECT cosmology

This section defines the calculation that an ECT completion must perform. The formulae below are exact definitions or standard consequences of a supplied physical metric, background and perturbation state. They are not numerical ECT predictions until those owners have been obtained from one varied ordered-medium action. The purpose of writing the chain explicitly is twofold: no missing operator can be hidden inside a fitted scalar function, and every future result has a finite set of reproducibility and falsification gates.

AI.1 Minimal action and state record

A candidate cosmological completion starts from a covariant action record

$$\mathfrak{R}_{\text{act}} = \{S_\Phi, S_n, S_{\text{constr}}, S_{\text{grav}}, S_m, S_\gamma, \mathcal{B}, \Psi_{\text{state}}\}, \quad (\text{AI.1})$$

where $\mathcal{B}$ denotes all boundary terms and interface conditions. This record must identify which variables are independent before variation. In particular, a director constrained by $n_A n^A = 1$ and an amplitude related to a gradient potential cannot subsequently be varied as independent fields. The same reduced variable count must be used in the background, perturbation and finite-source calculations.

The homogeneous state record is

$$\mathfrak{R}_{\text{bg}} = \{a, a_{\text{ord}}, u, n_A, \rho_i, s_i; \mathcal{C}_0 = 0, \mathcal{C}_I = 0; \mathcal{J}_{\text{ord}}\}, \quad (\text{AI.2})$$

where $\mathcal{C}_0$ is the Hamiltonian constraint, $\mathcal{C}_I$ are all additional constraints, and $\mathcal{J}_{\text{ord}} = 0$ is the relational event that starts the Lorentzian clock. A coordinate choice such as $a(t_0) = 1$ is not an initial condition, and a present-day normalisation is not a derivation of the state.

Background closure gate. For evolved variables X^A the numerical system must be written as

$$\mathbf{M}_{AB}(X) \frac{dX^B}{dN} = \mathbf{F}_A(X, N), \quad N \equiv \ln a, \quad (\text{AI.3})$$

after all algebraic constraints have been reduced. A physical trajectory requires

$$\det \mathbf{M}_{\text{phys}} \neq 0, \quad H^2 > 0, \quad F_{\text{grav}} > 0, \quad \lambda_{\min}(\mathbf{K}_{\text{phys}}) > 0, \quad c_{s,I}^2 \geq 0 \quad (\text{AI.4})$$

throughout the domain from $\mathcal{J}_{\text{ord}} = 0$ to the present. A field that diverges while an algebraic Friedmann denominator remains finite is still a physical failure; conversely, a coordinate singularity may be crossed only after an independent regular-variable formulation demonstrates regularity.

AI.2 Age, lookback time, and chronometers

Once the background, physical photon-redshift law $1+z=1/a$ with $a_0=1$, and a monotone expanding segment are owned, the ordered-branch age and lookback time are

$$t_{0,\text{ECT}} = \int_{a_{\text{ord}}}^1 \frac{da}{aH(a)}, \quad t_{\text{b}}(z) = \int_{1/(1+z)}^1 \frac{da}{aH(a)}, \quad (\text{AI.5})$$

$$\Delta t(z_1, z_2) = \int_{1/(1+z_2)}^{1/(1+z_1)} \frac{da}{aH(a)}, \quad z_2 > z_1. \quad (\text{AI.6})$$

Cosmic chronometers constrain the local derivative of the same clock,

$$H(z) = -\frac{1}{1+z} \frac{dz}{dt_{\text{pop}}}, \quad (\text{AI.7})$$

only after the population-age calibration and its covariance are declared. The chronometer methodology and its principal systematics are reviewed, for example, in Ref. [297]; that external calibration remains a nuisance owner rather than an ECT dynamical input. Chronometer points cannot select an ECT completion if the stellar clock model is changed simultaneously with the background.

For the named two-slope orbit, the current highest-integrity approximate calibration uses the official 15-point BC03 subset and suggested covariance of Eq. (AE.25). It gives $H_0 = 68.59 \pm 3.95 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and the conditional age $t_0 = 13.74 \pm 0.79 \text{ Gyr}$, with only the scale uncertainty propagated. This is a dimensional scale calibration of the already fixed dimensionless orbit, not a postulate-level derivation of either number. The matched flat control is indistinguishable and the 15-point BC03 subset is not a full current chronometer likelihood.

An accepted age result must publish: the ordering criterion, the entire $H(a)$ array, the integration measure, endpoint treatment, and two independent quadratures. It must also distinguish a dimensionless interval $H_* \Delta t$ from a duration in gigayears. The latter requires an explicitly declared physical matching of H_* ; if that matching is taken from data, the resulting age is conditional on it.

General P1–P6 age-identifiability limit. The inverse reconstruction identities (16.20)–(16.21) produce inequivalent smooth background histories within the unowned scalar closure. Therefore neither Eq. (AI.5) nor a precise quadrature package can select one unique law-level age directly from P1–P6. Numerical precision does not cure that dynamical non-identifiability. This limit does not forbid a numerical age for a fixed named action, state, formation-front prescription and dimensional calibration: such an age is a conditional output of that realised condensate history, as in Section AE, rather than a universal constant of the postulates.

AI.3 Photon metric, redshift, and distances

The physical photon line element must first be derived. For a homogeneous isotropic metric written as

$$ds_{\gamma}^2 = -N_{\gamma}^2(t) c^2 dt^2 + a_{\gamma}^2(t) \gamma_{ij} dx^i dx^j, \quad (\text{AI.8})$$

the observed redshift is the ratio of photon energies measured by physical matter clocks. Only when N_{γ} and a_{γ} are shown to coincide with the background clock and scale factor may one set $1+z=a^{-1}$. This is a metric/clock bridge, not notation.

After that bridge, define

$$\chi(z) = \int_0^z \frac{c_{\gamma}(z')}{H_{\gamma}(z')} dz', \quad (\text{AI.9})$$

$$D_M(z) = S_K[\chi(z)], \quad D_A(z) = \frac{D_M(z)}{1+z}, \quad D_L(z) = (1+z)D_M(z), \quad (\text{AI.10})$$

where $S_K(\chi)$ is the curvature-dependent transverse-distance function. Distance duality in Eq. (AI.10) additionally requires photon-number conservation and null geodesic propagation in one metric. Violating those assumptions changes the supernova and BAO likelihood and must be derived, not absorbed in a nuisance offset.

The differential volume entering number counts is

$$\frac{dV_c}{dz d\Omega} = \frac{c_\gamma(z) D_M^2(z)}{H_\gamma(z)}. \quad (\text{AI.11})$$

This owner is essential for JWST: the same observed catalogue implies a different comoving density if the ECT distance/volume map changes.

AI.4 Recombination, acoustic scale, and CMB inference

The baryon–photon sound speed is conditional on the physical matter and photon sectors,

$$c_s^2(z) = \frac{c_\gamma^2(z)}{3[1 + R_b(z)]}, \quad R_b(z) = \frac{3\rho_b(z)}{4\rho_\gamma(z)}. \quad (\text{AI.12})$$

The sound horizon and photon diffusion scale are then

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H_\gamma(z)} dz, \quad (\text{AI.13})$$

$$r_{\text{diff}}^2(z_*) = \int_{z_*}^{\infty} \frac{c_\gamma^2(z) \mathcal{D}_\gamma(z)}{H_\gamma(z)} dz, \quad (\text{AI.14})$$

with $\mathcal{D}_\gamma$ fixed by the Thomson opacity and ionisation history. The last-scattering redshift is not an input label: it follows from the visibility function

$$g_{\text{vis}}(\eta) = -\dot{\tau}(\eta) e^{-\tau(\eta)}, \quad \tau(\eta) = \int_{\eta}^{\eta_0} an_e \sigma_T d\eta'. \quad (\text{AI.15})$$

With this future-directed optical-depth convention, $\dot{\tau} = -an_e \sigma_T \leq 0$, so $g_{\text{vis}} \geq 0$ as required for a visibility density.

The primary geometric observable is approximately

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}, \quad (\text{AI.16})$$

but a physical inference of H_0 uses the full temperature, polarisation and lensing spectra, foregrounds and covariance. Changing r_s while holding a reference D_M , or vice versa, does not define an ECT Hubble shift. An ECT CMB pipeline must therefore freeze the same action/state hash used by the background and by the photon metric.

Acceptance gates. The CMB/Hubble channel is calculable only after the completion provides

- (a) a regular solution through the radiation and recombination eras;
- (b) physical baryon, photon and neutrino couplings and conservation laws;
- (c) a recombination/opacity module consistent with those couplings;
- (d) the complete scalar, vector and tensor transfer system;
- (e) an early+late joint likelihood with no duplicated calibration input.

The sign and magnitude of any inferred H_0 displacement remain Open until all five conditions are met.

AI.5 BAO, supernovae, sirens, and the distance ladder

BAO constrain combinations such as

$$\frac{D_M(z)}{r_{\text{drag}}}, \quad H_\gamma(z)r_{\text{drag}}, \quad \frac{D_V(z)}{r_{\text{drag}}}, \quad D_V(z) = \left[\frac{c_\gamma(z)zD_M^2(z)}{H_\gamma(z)} \right]^{1/3}. \quad (\text{AI.17})$$

Here $r_{\text{drag}} = r_s(z_{\text{drag}})$ is the sound horizon at baryon drag; it is distinct from the photon diffusion scale r_{diff} . The ruler r_{drag} must be generated by the ECT pre-drag plasma; it cannot be borrowed from a reference cosmology while using an ECT late-time distance. Type-Ia supernovae constrain [300]

$$\mu_{\text{SN}} = 5 \log_{10} \left(\frac{D_L}{10 \text{ pc}} \right) + \Delta \mathcal{M}, \quad (\text{AI.18})$$

where the calibration and population model are nuisance owners rather than new gravitational dynamics.

For a standard siren, the waveform propagation equation must determine a gravitational-wave luminosity distance D_L^{GW} . It may differ from the electromagnetic distance only if the tensor kinetic normalisation and damping are derived. Since the present ECT action does not own a physical tensor sector, D_L^{GW} and a siren-based ECT H_0 are Open. No scalar-cone result may be substituted for this tensor calculation.

The local distance ladder additionally depends on the matter metric, stellar structure and possible variation of dimensionless couplings. A complete ECT likelihood must propagate any predicted coupling change through calibrator and host physics rather than treating it as a free shift of H_0 .

Dimensionless constants and clock/rod calibration. The Euclidean medium amplitude, the homogeneous cosmological scalar and the locally relaxed finite-body state are distinct objects until a matching map is derived. A common rescaling of all dimensional masses is not observable as a variation of a dimensionless constant, whereas non-universal Yukawa, QCD or gauge-kinetic dressing is. For an action containing

$$-\frac{1}{4}B_{\text{em}}(\sigma)F_{\mu\nu}F^{\mu\nu} - \sum_f m_f(\sigma)\bar{\psi}_f\psi_f, \quad (\text{AI.19})$$

the physical sensitivities include

$$\frac{d \ln \alpha_{\text{em}}}{d\sigma} = -\frac{d \ln B_{\text{em}}}{d\sigma}, \quad \frac{d \ln \mu}{d\sigma} = \frac{d \ln m_p}{d\sigma} - \frac{d \ln m_e}{d\sigma}, \quad \mu = \frac{m_p}{m_e}. \quad (\text{AI.20})$$

Neither function is supplied by the scalar background action used here. Selective Φ -dressing, gauge-kinetic inheritance, or a Planck-gapped matching mode may protect or vary these ratios only after the matter/gauge vertices, threshold corrections and coarse-grained renormalisation are derived. A heavy matching mode alone is not a non-renormalisation theorem.

The observable programme combines quasar atomic spectra, including direct fine-structure-constant bounds [301], molecular constraints on the proton–electron mass ratio [302, 303], local atomic-clock comparisons [304], geological/nuclear constraints and equivalence-principle tests with the same cosmological and finite-body scalar solution. These are external empirical anchors, not evidence for a nonzero ECT coupling variation. The calculation must propagate the derived sensitivities through recombination, standard-candle luminosities and clock/redshift calibration; it must not identify a homogeneous background value with the screened laboratory value by notation. Present status: no nonzero variation and no protection mechanism is an ECT prediction. The acceptance gate is one action-owned set of $B_i(\sigma)$, masses/Yukawas, local transfer functions and a joint likelihood satisfying all cosmological and laboratory bounds.

AI.6 Gauge-reduced perturbation action

Let q^I denote all scalar perturbations before constraints. After lapse, shift and other nondynamical fields are integrated out, one must obtain

$$S_{\text{sc}}^{(2)} = \frac{1}{2} \int d\eta \frac{d^3 k}{(2\pi)^3} [\mathbf{v}'^\dagger \mathbf{K}_s \mathbf{v}' - \mathbf{v}^\dagger (k^2 \mathbf{G}_s + \mathbf{M}_s^2) \mathbf{v}]. \quad (\text{AI.21})$$

The rank of $\mathbf{K}_s$ is the physical scalar degree count. Stability requires positive nonzero eigenvalues of $\mathbf{K}_s$ and nonnegative high- k eigenvalues of $\mathbf{K}_s^{-1}\mathbf{G}_s$. A zero eigenvalue may represent a constraint, a strong-coupling boundary, or a gauge direction; that classification must be performed before discarding it.

The matter and metric variables then define, in a declared gauge or through gauge-invariant combinations,

$$-k^2\Psi = 4\pi G_N a^2 \mu_{\text{cos}}(k, a) \rho_m \Delta_m, \quad (\text{AI.22})$$

$$\Phi = \eta_{\text{slip}}(k, a) \Psi, \quad \Phi_{\text{W}} = \frac{\Phi + \Psi}{2}. \quad (\text{AI.23})$$

The functions μ_{cos} and η_{slip} are outputs of the quadratic action and matter/photon metric. They are not the galactic HRC function and cannot be assigned its form by analogy.

Tensor and vector guard. Each physical tensor/vector mode requires its own kinetic term, source vertex and characteristic cone. The absence of a transverse–traceless seed in a particular linear director map is a negative result for that map, not a derivation of a massless graviton. CMB tensor spectra, gravitational-wave speed/damping and standard-siren propagation remain Open until an explicit tensor completion passes ghost, gradient and source tests.

AI.7 Linear growth, $f\sigma_8$, and redshift-space distortions

For pressureless matter whose physical conservation law has been derived, the sub-horizon density contrast may be written schematically as

$$D'' + \left(2 + \frac{H'}{H} + \Gamma_m\right) D' - \frac{3}{2} \Omega_m(a) \mu_{\text{cos}}(k, a) D = \mathcal{S}_{\text{mix}}, \quad (\text{AI.24})$$

where primes denote $d/d \ln a$. Both an exchange term Γ_m and a mixed source $\mathcal{S}_{\text{mix}}$ must follow from the matter coupling; setting them to zero is an assumption, not a generic ECT result.

The logarithmic growth rate is

$$f(k, z) = \frac{d \ln D(k, z)}{d \ln a}. \quad (\text{AI.25})$$

If growth is scale dependent, σ_8 is not obtained by multiplying one $D(k, z)$ value. It must be recomputed from the evolved spectrum,

$$\sigma_8^2(z) = \int_0^\infty \frac{dk}{k} \Delta_m^2(k, z) \left[\frac{3j_1(kR_8)}{kR_8} \right]^2, \quad R_8 = 8h^{-1} \text{Mpc}. \quad (\text{AI.26})$$

The familiar scalar $f\sigma_8(z)$ is recovered only in the scale-independent limit. In the general case the owned observable is the window-convolved redshift-space power or its multipoles, schematically

$$P_s(k, \mu, z) = [b(k, z) + f(k, z)\mu]^2 P_m(k, z) + P_{\text{stoch}}, \quad (\text{AI.27})$$

with bias, velocity dispersion, Alcock–Paczynski mapping and survey window included in the likelihood. Thus a growth prediction requires the primordial spectrum and its normalisation. A small improvement after multiplying a reference growth curve by a free response is not a prediction of Eq. (AI.24).

The required publication products are the matrices $\mathbf{K}_s, \mathbf{G}_s, \mathbf{M}_s^2$, the transfer functions, initial conditions, the exact survey window convolution and the covariance-weighted likelihood. Stability must be checked on the entire k – a grid used by the likelihood, not at one fiducial point.

AI.8 ISW, weak lensing, and CMB lensing

The late integrated Sachs–Wolfe temperature source is

$$\left. \frac{\Delta T}{T} \right|_{\text{ISW}}(\hat{n}) = 2 \int_{\eta_*}^{\eta_0} d\eta \Phi'_W[\chi(\eta)\hat{n}, \eta], \quad (\text{AI.28})$$

subject to the sign convention for the metric potentials. A galaxy–ISW cross-spectrum requires tracer bias, redshift distribution and covariance. Representative observational analyses combine large-scale-structure tracers with CMB maps precisely at this likelihood level [305]; they are external targets rather than a substitute for the ECT transfer system. Its sign cannot be inferred from a background coupling alone because Φ'_W depends jointly on growth and slip.

For a source distribution $n_s(\chi)$, the weak-lensing convergence is

$$\kappa(\hat{n}) = \int_0^{\chi_H} d\chi W_\kappa(\chi) \nabla_\perp^2 \Phi_W(\chi\hat{n}, \eta_0 - \chi/c), \quad (\text{AI.29})$$

where W_κ follows from the physical photon distance and source distribution. Prediction of shear two-point functions requires nonlinear matter power, intrinsic-alignment treatment, baryonic feedback and survey covariance. The current ECT theory supplies none of these as a closed chain.

For tomographic bins i, j , the corresponding convergence spectrum has the schematic Limber form

$$C_\ell^{\kappa_i \kappa_j} = \int_0^{\chi_H} \frac{d\chi}{\chi^2} W_i(\chi) W_j(\chi) P_{\Phi_W}\left(\frac{\ell + 1/2}{\chi}, z(\chi)\right), \quad (\text{AI.30})$$

with all metric-to-density normalisations derived from the same perturbation action. The commonly reported compression $S_8 \equiv \sigma_8(\Omega_m/0.3)^{1/2}$ is not itself an ECT equation; an ECT test requires the survey data vector and covariance,

$$-2 \ln \mathcal{L}_{\text{shear}} = (\mathbf{d}_{\text{sh}} - \mathbf{m}_{\text{sh}})^T \mathbf{C}_{\text{sh}}^{-1} (\mathbf{d}_{\text{sh}} - \mathbf{m}_{\text{sh}}) + \text{const}, \quad (\text{AI.31})$$

including calibration, photometric-redshift, intrinsic-alignment and baryonic nuisance owners.

CMB lensing uses the line-of-sight potential

$$\varphi_{\text{CMB}}(\hat{n}) = -2 \int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi} \Phi_W(\chi\hat{n}, \eta_0 - \chi/c), \quad (\text{AI.32})$$

and compares its reconstructed spectrum $C_L^{\varphi\varphi}$, including normalisation bias and covariance, with the prediction of the same transfer system. A fitted multiplier on a reference $C_L^{\varphi\varphi}$ is not a derived ECT lensing prediction.

Exact negative result. A static scalar response that is a positive local monotone function of a baryonic proxy preserves the ordering and positions of isolated maxima. Therefore it cannot, by itself, produce a displaced lensing peak in a merging cluster. This no-go does not say that every ECT completion fails; it states that the completion must add a physical photon metric and time-dependent, non-local or transported degrees of freedom.

AI.9 JWST number counts, masses, and morphologies

A survey prediction begins with the halo abundance

$$\frac{dn}{dM}(M, z) = \frac{\bar{\rho}_m}{M} f_{\text{coll}}(\nu, z) \left| \frac{d \ln \sigma^{-1}(M, z)}{dM} \right|, \quad \nu = \frac{\delta_c(M, z)}{\sigma(M, z)}. \quad (\text{AI.33})$$

The collapse barrier δ_c , variance σ , and multiplicity function f_{coll} must be recomputed if the ECT force, slip or expansion differs; a reference calibration cannot automatically be reused.

There is one useful restricted diagnostic for the named near-GR two-slope state. Its same-metric quasistatic calculation gives, at $z = 10$,

$$\left. \frac{\Delta D}{D} \right|_{z=10} = -1.430830934 \times 10^{-5}. \quad (\text{AI.34})$$

This number is owned by the named conditional ECT action/state and the restricted growth approximation. To expose its possible rare-halo impact without pretending to possess a halo forward model, hold the transfer function, collapse threshold, present-day variance shape and normalisation, volume and baryonic selection fixed, and import only the local *differential fixed-mass* Press–Schechter sensitivity

$$\Delta \ln n \simeq (v^2 - 1) \Delta \ln D, \quad (\text{AI.35})$$

where $v = \delta_c / [D\sigma(M)]$. The interval $v = 3, 4, 5, 6$ is an external sensitivity bracket, not an ECT-derived assignment of a JWST sample to peak height. Linearised and exact finite Press–Schechter ratios are

v	$(\Delta n/n)_{\text{lin}}$	$(\Delta n/n)_{\text{exact}}$
3	−0.01144665%	−0.01144626%
4	−0.02146246%	−0.02146064%
5	−0.03433994%	−0.03433480%
6	−0.05007908%	−0.05006764%

Thus this near-GR background plus restricted-growth route produces no material rare-halo enhancement in the declared hybrid diagnostic. This is a Level C sensitivity calculation, not a JWST abundance prediction or a universal no-go: the ECT-native primordial spectrum, transfer function, mass-dependent variance, calibrated differential and cumulative mass functions, nonlinear/baryonic assembly, mass-to-light inference, survey volume, completeness and selection likelihood remain missing owners.

The predicted catalogue intensity is

$$\frac{dN_{\text{pred}}}{dz d\mathcal{O}} = \Delta\Omega \frac{dV_c}{dz d\Omega} \int dM_h \frac{dn}{dM_h} \int d\vartheta_b P(\mathcal{O} | M_h, z, \vartheta_b) P(\vartheta_b | M_h, z) \mathcal{S}(\mathcal{O}, z), \quad (\text{AI.36})$$

$$\mathcal{L}_{\text{JWST}} = \text{PoissonPointProcess}[\{\mathcal{O}_i, z_i\} | dN_{\text{pred}}/(dz d\mathcal{O})]. \quad (\text{AI.37})$$

Here ϑ_b contains star-formation efficiency, duty cycle, dust, metallicity, IMF and black-hole/AGN components. Photometric-redshift and stellar-mass posteriors must be integrated, not replaced by central values. Morphology additionally needs the point-spread function, surface-brightness selection and an image-level forward model.

Status-neutral observational target. JWST has reported very-high-redshift spectroscopic candidates in the JADES programme [306], candidates for overmassive black holes relative to their stellar hosts [307, 308], luminous or massive systems at very high redshift including spectroscopically confirmed objects at $z \gtrsim 14$ [309], and massive quiescent systems at high redshift [310, 311]. These are external empirical targets with selection, redshift, stellar-population and mass-inference uncertainties. They motivate Eqs. (AI.36)–(AI.37) but do not by themselves establish a failure of a reference cosmology, an ECT enhancement, or a specific formation mechanism.

No statement about “extra formation time” is sufficient to own Eqs. (AI.36)–(AI.37). The same ECT completion must first pass the background, CMB, growth, lensing and local-gravity gates; otherwise the early-object channel is not a closed test of the theory.

AI.10 Cluster dynamics and collision lensing

For a merging system the required state is at least

$$\mathfrak{R}_{\text{cl}} = \{\rho_{\text{gas}}, f_{\text{gal}}, \rho_*, \Phi, \Psi, \Xi_{\text{resp}}, \mathcal{K}_{\text{ret}}; \mathbf{b}, \mathbf{v}_{\text{rel}}, t_{\text{obs}}\}, \quad (\text{AI.38})$$

where Ξ_{resp} is any independent medium response and $\mathcal{K}_{\text{ret}}$ its causal/Euclidean-to-Lorentzian response kernel. The gas evolves hydrodynamically, collisionless components obey their kinetic equations, and the physical photon metric produces the convergence map.

A minimal observable comparison uses the reduced shear

$$g_{\text{red}} = \frac{\gamma}{1 - \kappa}, \quad \kappa = \frac{\Sigma_{\text{lens}}}{\Sigma_{\text{crit}}}, \quad (\text{AI.39})$$

with the survey source-redshift distribution and mass-sheet degeneracy treated in the likelihood. A local algebraic orientation term or a rescaling of a baryonic map is not a solution of Eq. (AI.38).

The calculability gates are: derived response vertex; causal and stable kernel; common matter/photon metric; initial collision state; hydrodynamic and collisionless numerical convergence; mock-observation pipeline; and a blinded likelihood against convergence, shear, X-ray, SZ and galaxy maps. Until these are satisfied, residual cluster mass and Bullet-like morphology remain Open.

Cluster abundance and counts. The collision problem is distinct from the abundance problem. For a survey selection $S_{\text{cl}}(M, z, \mathcal{O})$, the latter requires

$$N_{\text{cl}} = \Delta\Omega \int dz \frac{dV_c}{dz d\Omega} \int dM \frac{dn}{dM}(M, z) \int d\mathcal{O} P(\mathcal{O} | M, z) S_{\text{cl}}(M, z, \mathcal{O}). \quad (\text{AI.40})$$

The restricted near-GR growth proxy in Eq. (AI.35) changes the growth-only high-peak sensitivity by less than 0.046% for $v \leq 5$. It therefore does not materially alter cluster counts through this one channel. This is not an ECT cluster-count likelihood: the volume, transfer function, collapse barrier, mass calibration, observable scatter and selection remain uncomputed owners of the same action and state.

AI.11 Finite-body weak field and PPN

PPN is the parametrised post-Newtonian expansion of a physical metric around a slowly moving, weakly gravitating finite source. In a standard isotropic bookkeeping convention,

$$g_{00} = -1 + \frac{2U}{c^2} - \frac{2\beta_{\text{PPN}}U^2}{c^4} + O(c^{-6}), \quad (\text{AI.41})$$

$$g_{ij} = \left(1 + \frac{2\gamma_{\text{PPN}}U}{c^2}\right) \delta_{ij} + O(c^{-4}), \quad (\text{AI.42})$$

$$g_{0i} = O(c^{-3}), \quad (\text{AI.43})$$

with additional preferred-frame, nonconservation and nonlinear parameters when the completion permits them. These coefficients are defined only after the full constrained field equations are solved with interior/exterior matching and asymptotic boundary conditions.

The finite-body pipeline must determine: the matter source vertex; scalar, director and tensor charges; possible thin-shell or finite-size suppression; the two metric potentials; preferred-frame terms; and the motion of the source and observer relative to the ordered medium. A homogeneous algebraic stationary point is not a screening calculation, and a scalar characteristic cone is not a PPN metric.

Finite-body scalar gate for the same two-slope state. The homogeneous action can be tested around a prescribed finite density profile before a full metric solution is available. For the named near-GR slice $c_s = a_s$, $b_s = 3a_s$, set

$$y = e^{a_s \sigma} = \frac{F_s}{F_*}, \quad D_s = \kappa_s + \frac{3a_s^2}{2}.$$

On a static flat background with prescribed pressureless density, the exact field substitution gives

$$\boxed{\nabla^2 y = \frac{a_s^2}{F_* D_s} \left(V_+ y^3 - V_- y - \frac{\rho}{2} \right)}. \quad (\text{AI.44})$$

Unlike the one-exponential toy family, this action has the positive zero-density root

$$y_{\text{vac}} = \sqrt{V_-/V_+} = 0.80437385.$$

For the cosmological exterior state $\rho_{\text{out}} = 0.9$, $y_{\text{out}} = 1$, define $u = y/y_{\text{out}}$, $x = m_{\text{out}}r$, $s = \rho/\rho_{\text{out}}$, $A = V_+y_{\text{out}}^2$ and $B = V_-$. The spherical equation is

$$u'' + \frac{2}{x}u' = \frac{Au^3 - Bu - (A - B)s(x)}{3A - B}, \quad u'(0) = 0, \quad u(\infty) = 1. \quad (\text{AI.45})$$

At the deliberately artificial scale $m_{\text{out}}R = 1$, a collocation solve and an independent sparse finite-difference Newton solve give

Table 132: Conditional two-slope finite-body scalar proxy at $m_{\text{out}}R = 1$. The last column compares the nonlinear far amplitude with the exact linear response of the same smooth profile.

$\rho_{\text{in}}/\rho_{\text{out}}$	surface flux / linear volume source	$A_{\text{far}}/A_{\text{lin}}$
10	0.761419	0.888498
100	0.537185	0.451047
10^3	0.253502	0.101308
10^4	0.102100	0.0147107

The two methods agree within 2.13×10^{-4} , and the collocation surface flux agrees with the volume identity within 4.5×10^{-7} . This is genuine model-internal surface localisation when a body is already large in exterior Compton units.

Estimator registry for the strongest finite-body proxy row. The finite-body reduction admits several nearby but non-identical tail statistics. With

$$r = \frac{B}{A}, \quad \eta = \frac{(1-r)(s_{\text{in}} - 1)}{3 - r}, \quad s_{\text{in}} = \frac{\rho_{\text{in}}}{\rho_{\text{out}}}, \quad (\text{AI.46})$$

where $A = V_+y_{\text{out}}^2$ and $B = V_-$ are the coefficients already used in Eq. (AI.45); this is the named $(m, n) = (1, 3)$ slice. The frozen dimensionless row $(r, \eta, m_{\text{out}}R) = (0.99, 3333, 1)$, and for its regular vacuum limit, the central values are

Declared statistic	nonlinear/linear ratio
Finite-window mean of $xe^x[u(x) - 1]$ on $5 \leq x \leq 9$, $r = 0.99$	0.006399711753
Green-identity / finite-difference asymptotic coefficient, $r = 0.99$	0.006393895852
Asymptotic coefficient in the regular $r \rightarrow 1$ limit at fixed η	0.006372442154

These are three declared estimators or boundary limits, not three measurements to be averaged. The $r \rightarrow 1$ entry is the regular limit of the asymptotic coefficient; it is not a direct evaluation of the singular coordinate $s = \rho/\rho_{\text{out}}$ at $\rho_{\text{out}} = 0$. A further independent discretisation differs by only 0.0689–0.0717% and is retained as a cross-method validation, not as a replacement central value. The result establishes nonlinear far-tail suppression in the fixed-metric scalar BVP. It does not identify the ratio with the physical sensitivity α_A/α_∞ , does not supply a coupled metric or fixed-baryon mass derivative, and does not imply a Cassini, WEP or ranged-PPN pass. The versioned calculation owner is `data/cosmology_r114/MANIFEST_SHA256_v1.json`; it hashes the producer, independent red-team verifier, inputs and outputs. Local presence does not imply a public release, archival deposit or DOI.

The same calculation supplies a decisive dimensional gate for the named cosmological state. Interpreting the displayed potential coefficients in $F_*H_0^2$ units with the one declared $H_0 = 67.4$ dimensional scale calibration gives $m_{\text{out}}/H_0 = 0.0054769$. In the small-body regime the controlling linear source parameter is

$$\begin{aligned}\Xi_{\text{body}} &\equiv \frac{A-B}{3A-B}(s-1)(m_{\text{out}}R)^2, \\ \delta u(0) &= \frac{\Xi_{\text{body}}}{2} + O(m_{\text{out}}R\Xi_{\text{body}}), \\ \delta u(R) &= \frac{\Xi_{\text{body}}}{3} + O(m_{\text{out}}R\Xi_{\text{body}}).\end{aligned}\tag{AI.47}$$

The last two expressions follow by matching the regular uniform-sphere solution of the linearised massive equation to its exterior Yukawa tail. Crude spherical mean-density estimates are

Table 133: Dimensional finite-body gate for the named two-slope state. The formal equilibrium roots are counterfactual comparators: because $\Xi_{\text{body}} \ll 1$, the finite bodies remain near $u = 1$ and do not reach them. These are mean-density linear estimates, not coupled metric body solutions.

object	$m_{\text{out}}R$	Ξ_{body}	formal $F_{\text{in}}^{\text{eq}}/F_{\text{out}}$
Earth	2.54×10^{-22}	2.09×10^{-14}	9.13×10^9
Sun	2.78×10^{-20}	6.36×10^{-11}	5.79×10^9
Jupiter	2.79×10^{-21}	6.05×10^{-13}	5.68×10^9
Milky-Way mean within 15 kpc	1.85×10^{-8}	5.74×10^{-12}	34.1
cluster mean within 1 Mpc	1.23×10^{-6}	1.51×10^{-10}	6.21

The fixed-metric tail-estimator detail is retained in Appendix BR.

Figure 41 groups the three already frozen finite-body gates without combining their estimators or upgrading their physical status.

All Ξ_{body} values are tiny. The formal high-density equilibrium roots and their large Planck-factor contrasts are therefore not reached; the actual linear proxy has $u = 1 + O(\Xi_{\text{body}})$. Consequently this named near-GR slice does *not* obtain a thin-shell or mass-screening suppression from the displayed scalar mechanism. Its separately computed unscreened estimate $\gamma_{\text{PPN}} - 1 = -0.99998 \times 10^{-5}$ satisfies only the declared absolute-deviation proxy gate 2.3×10^{-5} . Relative to the primary Cassini measurement $(2.1 \pm 2.3) \times 10^{-5}$ [95], the manuscript's Gaussian standard-error proxy is 1.35σ from the central value, outside the literal one-sigma interval and inside two sigma. That number is internal arithmetic, not a significance reported by Cassini, a full likelihood, or a complete ECT PPN fit. The finite-body result is therefore not by itself a local-gravity failure of this near-GR point. It also cannot rescue stronger scan points whose unscreened PPN estimates fail. Physical body charge, metric backreaction, β_{PPN} , preferred-frame coefficients, composition dependence/WEP, the two metric potentials and lensing still require the full coupled finite-body action and remain Open.

This non-uniqueness is constructive rather than merely verbal. Metric completions sharing the same tested scalar cone yield $\gamma_{\text{PPN}} = 1, 0$, or $1/3$ depending on the physical metric map, and the naive direct nonlinear continuation of the scalar completion yields $\beta_{\text{PPN}} = 2$. The latter is excluded at order unity by Solar-System bounds. These are kill tests for those named completions, not predictions that ECT has $\beta_{\text{PPN}} = 2$ or any one value of γ_{PPN} . They prove that the cone and HRC response do not select a Solar-System metric.

The first local acceptance tests are

$$|\gamma_{\text{PPN}} - 1| \ll 1, \quad |\beta_{\text{PPN}} - 1| \ll 1, \quad |\alpha_{1,2,3}| \ll 1,\tag{AI.48}$$

with the numerical bounds and covariance taken from the chosen current data release. A candidate giving an order-unity deviation is killed before it is tuned to galaxies or cosmology. Passing PPN does not by itself validate the cosmological completion; it is a necessary cross-scale gate.

AI.12 External Euclidean response and the outside view

The outside description is formulated on an interface Σ with induced metric h_{ij} , second fundamental form K_{ij} and a specified exterior operator. For a massive scalar in a locally flat half-space, the full flat-boundary Fourier symbol is

$$\sigma_{\text{flat}}(D_E)(x, \xi) = \sqrt{M^2 + h^{ij}(x)\xi_i\xi_j}. \quad (\text{AI.49})$$

In ordinary classical pseudodifferential calculus its homogeneous principal symbol is $\sigma_{\text{pr}}(D_E) = |\xi|_h$; the mass enters at lower order. Curvature contributes at subprincipal and lower order through invariants of K_{ij} , derivatives and boundary conditions. It cannot be inserted into the flat full symbol as a universal replacement without deriving the operator.

Two exact guards follow. First, translating a flat interface in ambient Euclidean space is an isometry, so its DtN spectrum is unchanged; translation speed is not a Hubble law. Second, an umbilic embedding of a spatially flat three-slice in flat E^4 with $K_{ij} = (H/c)h_{ij}$ violates the Gauss equation unless $H = 0$. Non-umbilic embeddings can evade the scalar conclusion only by introducing directional structure that must be reconciled with observed isotropy.

The outside view can nevertheless provide a valid owner if a concrete interface, state and response map are derived from the ordered phase. It must then reproduce the internal stress tensor and perturbation operator on shell. Agreement of the two descriptions is an equality of effective actions or response kernels, not a visual analogy between a boundary radius and c/H .

AI.13 Unified numerical workflow

For each candidate completion the following workflow is mandatory.

- G1.** Freeze the action, independent variables, constraints, state, boundary conditions, conventions and all matching inputs.
- G2.** Derive the background equations twice: directly by variation and by an independent symbolic or Hamiltonian formulation.
- G3.** Integrate from the derived ordering event; monitor every constraint, denominator, response determinant and stability eigenvalue.
- G4.** Repeat with at least two integration methods and a refinement sequence; classify every endpoint before interpreting a time interval.
- G5.** Derive and reduce the complete quadratic perturbation action on the accepted background; scan ghost and gradient eigenvalues over the publication k - a domain.
- G6.** Generate recombination, distances, transfer functions and nonlinear structure from that same frozen background and physical metric.
- G7.** Apply observable-specific forward models and likelihoods with nuisance parameters, selection functions and covariance.
- G8.** Publish machine-readable inputs, outputs, hashes, convergence records, failure flags and an owner/status ledger.
- G9.** Run a cross-sector gate: the same completion must pass age, CMB/BAO, growth/lensing, cluster and PPN tests without assigning incompatible meanings to one parameter.

Two-method background residual. For a numerical solution $X(N)$ define

$$\mathcal{R}_{\text{bg}} = \max_N \left\{ \frac{|\mathcal{C}_0|}{1 + \mathcal{S}_0}, \frac{\|\mathcal{E}_A\|}{1 + \mathcal{S}_A}, \frac{\|X_{(1)} - X_{(2)}\|}{1 + \|X_{(1)}\|} \right\}, \quad (\text{AI.50})$$

where $\mathcal{S}$ are declared physical scales and (1), (2) label independent methods. The acceptance tolerance must be chosen before looking at observable residuals. A stable PDF curve without a constraint residual is not a reproducible cosmological solution.

Frozen scientific payload versus runtime provenance. Every numerical row added above is accompanied by a machine-readable input record, output table, verification script and checksum manifest. Scientific payload fields are frozen separately from runtime metadata such as the Python, NumPy and SciPy version strings. A cross-version replay that changes only those metadata strings is a provenance delta, not a scientific disagreement; conversely, a change in a physical value requires a newly named refreeze and cannot be hidden by serialization. Displayed growth ratios are rounded to no more than thirteen significant digits, below the independent solver agreement. The finite-window, asymptotic and regular- $r \rightarrow 1$ -limit finite-body statistics retain separate field names and definitions and must not overwrite one another.

Frozen calculation owners added with this candidate. The owner-specific response envelope is generated by `scripts/cosmology/compute_r113_early_response_growth_collapse_envelope.py`; the restricted one-real-pole cluster no-go by `scripts/cosmology/verify_r113_one_pole_cluster_no_go.py`; and the director-source convention audit by `scripts/verification/r113/verify_r113_director_source_normalisation.py`. Their frozen outputs and checksum manifest reside in `data/cosmology_r113/MANIFEST_SHA256_v3.json`. The two-slope clock, including $t(z = 10) = 0.4814276482$ Gyr under the stated $H_0 = 67.4$ calibration, is owned by `scripts/cosmology/compute_r103_two_slope_conditional_observables.py` and `data/cosmology_r103/R103_TWO_SLOPE_CONDITIONAL_OBSERVABLES_v1.json`. These paths become immutable public provenance only when the scripts, data and manifest are included in the same reviewed repository release as the manuscript; local file existence alone is not a publication guarantee.

AI.14 Publication products and status transitions

Table 134: Required publication products for an ECT cosmological completion.

Channel	Machine-readable product	Scientific acceptance statement	Status now
Action/ background	Action record, equations, initial data, $H(a)$, constraints, response and stability spectra	One globally regular ordered-branch solution with a physical clock map	A in supplied two-slope action; full ECT owner Open

Channel	Machine-readable product	Scientific acceptance statement	Status now
Age	Formation-front congruence/lapse, $H(a)$, dual quadratures and one dimensional scale calibration	Same physical solution and outside/inside clock agreement	Conditional τ_{2s} , Gyr clocks and covariance-aware 15-point BC03 calibration computed; full map Open
CMB/Hubble	Recombination, transfer functions, spectra, joint covariance	One early+late likelihood; no duplicated ruler or calibration	Conditional r_s background diagnostic and one chronometer scale calibration computed; full inference Open
BAO/SN/sirens	$D_M, D_A, D_L, r_{\text{drag}}, D_L^{\text{GW}}$ and nuisance posterior	Physical photon/tensor metrics and declared distance-duality assumptions	Conditional background distances computed; full likelihood/tensor owner Open
Growth	Reduced quadratic matrices, transfer grid, $f\sigma_8$ likelihood	Stable physical modes and derived matter coupling on full grid	Restricted quasistatic diagnostic computed; full result Open
Owner-specific ζ_{ER} envelope	Frozen background, equality, growth, collapse and rare-tail tables with dual-solver replay	Algebra and linear growth are Level A inside the declared envelope; collapse and Press–Schechter tails are Level C; no universal ECT parameter is inferred	

Channel	Machine-readable product	Scientific acceptance state- ment	Status now
ISW/lensing	Φ, Ψ , kernels, spectra and covariance	Derived Weyl potential of the physical photon metric	Restricted Weyl/ derivative proxy computed; physical spectra Open
JWST	Catalogue intensity, selection, baryonic/halo posterior	End-to-end counts, masses and morphologies from one completion	Conditional time/ volume clocks computed; abundance/ morphology Open
Clusters	Time-dependent state, response kernel, mock maps and likelihood	Hydrodynamic/collisionless evolution plus physical lensing map	Level A theorem in supplied local map / physical map Open
PPN	Finite-body solution and PPN coefficients	Same completion passes local metric and preferred-frame bounds	A unscreened- family diagnostic; full finite-body result Open

A row moves from Open to Level B when the physical owner and a complete calculation route are derived but an external matching input remains. It moves to Level C when a declared phenomenological completion is fitted. It moves to Level A only when it is exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; this does not by itself establish descent from P1–P6 or physical ECT ownership. No agreement in one row promotes any other row automatically.

AI.15 Current terminal conclusions

The exact results presently available are:

- (i) the scalar background equations are internally consistent inside their declared action;
- (ii) the current unowned scalar closure is non-identifiable by inverse reconstruction, so it does not predict one $H(a)$ or age;
- (iii) the supplied two-slope action/state has a globally regular conditional background, but its coefficients and ordering state are not selected by P1–P6;
- (iv) a strictly increasing local cluster map preserves the set-valued argmax and, whenever maxima are attained, cannot displace a total response maximizer relative to the supplied total projected baryon

maximizers; it neither generates the hand-set gas/BCG separation nor constrains residual-map extrema;

- (v) one constant real Debye pole cannot simultaneously own the declared static a_M matching and ballistic 70–100 kpc merger offsets; multipole, dispersive and separately owned response kernels remain Open;
- (vi) a restricted same-metric, zero-slip, sub-horizon closure gives a reproducible Weyl-amplitude/derivative proxy for one named state, but not a physical ISW or lensing spectrum;
- (vii) HRC is a quasistatic galactic response candidate and does not replace the homogeneous stress, perturbation, photon-metric or PPN owners.

Taken together, corrections A1–A9 close a manuscript-ready audit of the current cosmological sector: valid conditional results, null results, tensions, restricted no-go statements and Open owners are all retained. They do not close a unique or observationally complete ECT cosmology. The full likelihoods and physical-owner results in the Hubble, age, JWST, growth, ISW/lensing, cluster-merger and PPN programme remain explicitly enumerated calculations beyond the conditional diagnostics above. This conclusion is stronger and more useful than a common fitted interval: it states exactly which equations must be derived, which counterexamples already apply, and which observations can falsify each future completion.

AI.16 Hubble and acoustic calculation

Physics of the channel. The early acoustic inference constrains the angular and damping scales of a coupled photon–baryon plasma, while the late ladder constrains calibrated luminosity distances. The ECT question is whether one action/state changes these observables consistently. It is not whether one background parameter can be assigned an interval.

Required inputs. The calculation requires the physical background and photon metric; baryon, photon and neutrino densities; the electron mass and binding energies; Thomson opacity; the primordial state; scalar/vector/tensor transfer operators; foreground model; late calibrator physics; and the complete early+late covariance. Every input is classified as derived, matched, measured nuisance or Open.

Conditional background calculation now available. For each calibrated two-slope orbit, the present bundle computes

$$r_s^{\text{cond}}(z_{\text{drag}}) = \int_{z_{\text{drag}}}^{z_{\text{max}}} \frac{c_{\text{std}} dz}{\sqrt{3[1 + R_b^{\text{std}}(z)]} H_{2s}(z)} + r_s^{\text{rad.tail}}, \quad (\text{AI.51})$$

using a declared standard photon/recombination layer. This isolates the effect of the action-owned background and is reproducible, but it is not an ECT recombination prediction.

Current numerical result. Across the nine points of the prespecified 3×3 Cartesian grid, $\Delta r_s/r_s$ ranges from -0.00323% to -30.646% . The largest sampled changes occur where the early long-range response changes by factors 1.14 to 2.33 and the unscreened PPN diagnostic fails badly. The one sampled point satisfying the displayed absolute γ -proxy gate changes r_s by only -0.00323% . Thus none of these nine nodes simultaneously gives a large acoustic shift and satisfies that local proxy; no statement about unsampled parameter values follows.

Current model-class comparison. The distinction is not merely numerical. In baseline Λ CDM the early ruler is owned by the standard radiation/matter expansion and recombination sector; Early Dark Energy [312] changes it through an additional early component whose perturbations and decay must be evolved; generic modified-gravity models, including the scalar-tensor/ $f(R)$ model class reviewed in Ref. [313], change it only after their background, photon sector and gauge-reduced perturbations are specified. The conditional two-slope ECT family changes the action-owned background but still imports the standard photon/recombination layer. Therefore its displayed r_s shift is a background-sensitivity diagnostic, not a completed alternative to any of those model classes and not evidence of an observational advantage.

Robustness and missing calculation. The result must be repeated after deriving the photon metric, recombination microphysics, finite-body scalar charge and full transfer system. The fixed- θ_* rescaling in Table 60 is deliberately not used as evidence for a resolved Hubble tension. The final statistic is one joint likelihood, not an intersection of separately constructed bands.

Falsifiers. A supplied completion fails this channel if it cannot reach recombination regularly, violates BBN or opacity constraints, produces unacceptable damping/peak phases, fails the late distance likelihood, or requires different parameter meanings in the early and late calculations.

AI.17 Chronometer and age calculation

Physics of the channel. The observable chronometer relation measures a derivative of population age, while ECT age is the proper duration of the ordered branch. Both use the same matter-clock lapse. An arbitrary external Euclidean depth is not a duration; it becomes one only through Eq. (16.14).

Current numerical result. The named near-GR two-slope completion gives $\tau_{2s} \equiv H_{2s}(0)\Delta t = 0.9636867786415146$. Under two disclosed conditional dimensional scale calibrations of that same dimensionless orbit, it corresponds to 13.9804960874 or 12.9009506612 Gyr. These are not merely arbitrary unit rescalings unless all physical matter, clock, Planck and recombination scales are co-rescaled. The absolute-second sound horizon is therefore only a unit-reading diagnostic until the physical scale map is owned; neither calibration is a second dynamical orbit.

High-redshift clock outputs. Using the externally supplied reference dimensional calibration $H_0 = 67.4$, the supplied orbit gives $t(z) = 1.22645, 0.77690, 0.48143, 0.29241, 0.18177$ Gyr at $z = 4.9, 7, 10, 14.32, 20$. These numbers are calculated from the ECT-candidate orbit rather than a reference expansion function. Their scientific status is conditional on the action/state, photon/matter clock map and scale calibration.

Required observational comparison. The chronometer likelihood must propagate stellar population synthesis, metallicity, star-formation history and covariance. Absolute-age tests must propagate object dating and formation priors. No object older than the conditional branch age is allowed, but passing this inequality does not select the action.

Falsifiers. A completion fails if a robust object age exceeds its ECT branch age, if the chronometer derivative is inconsistent with the same $H(z)$ used for BAO and supernovae, or if the external-depth/lapse map becomes singular or state-dependent in a way incompatible with physical clocks.

AI.18 JWST maturity and number-count calculation

Physics of the channel. Early massive objects test a product of available time, comoving volume, initial fluctuations, scale-dependent growth, nonlinear collapse, baryonic conversion and selection. A

time-budget factor is only one term in that product. The current correction therefore replaces the former one-number extraction by a forward intensity in catalogue space.

Background inputs now available. The conditional two-slope orbit supplies $t(z)$, $H(z)$, and—under the declared photon-metric assumption— $D_M(z)$ and $dV_c/dz d\Omega$. The nine-node calibrated grid tests whether stronger sampled scalar-background response helps. It does not at those nodes: the largest sampled departure reduces $t(z=10)$ by 10.491167%, so none of the nine background nodes enlarges the sampled maturity interval. Unsourced regions and the continuous family remain untested.

Perturbative and nonlinear inputs still required. The action must predict the primordial covariance and the transfer function $T_m(k, z)$. A validated collapse calculation then generates

$$\frac{dn_h}{dM_h}(M_h, z) = \mathfrak{F}_{\text{coll}}[P_m(k, z), \delta_c(z), \mathcal{E}_{\text{env}}], \quad (\text{AI.52})$$

where $\mathcal{E}_{\text{env}}$ denotes any derived environmental response. The stellar relation $P(M_*, L, \mathcal{M} | M_h, z)$ and survey selection must be propagated rather than replaced by a fixed efficiency.

Catalogue statistic. For bins B_i in redshift, mass, luminosity, size and morphology, the expected counts are

$$\lambda_i^{\text{ECT}} = \int_{B_i} dz dM_h d\mathbf{y} \frac{dV_c}{dz d\Omega} \frac{dn_h}{dM_h} P(\mathbf{y} | M_h, z) \mathcal{S}(\mathbf{y}, z), \quad (\text{AI.53})$$

and the likelihood includes photometric-redshift, lensing-magnification, stellar-population and cosmic-variance covariances. Posterior stellar masses cannot be treated as model-independent data when the distance map changes.

Morphology and assembly clocks. Sizes, Sérsic indices, colours, line widths and merger indicators constrain assembly histories beyond the scalar age inequality. The simulation or semi-analytic model must output these observables at the image level. A completion that produces enough halos but cannot produce the observed morphology remains incomplete.

Current verdict and falsifiers. The present ECT result is a direct conditional clock/volume calculation at nine prespecified nodes; all nine reduce the high-redshift interval relative to the declared control. This is not a negative theorem for the continuous scalar-background family. Abundance and morphology remain Open. A separately declared owner-complete model is falsified by joint failure of its catalogue likelihood, by an impossible formation time, or by an early-growth enhancement that violates CMB, lensing or local-gravity constraints.

AI.19 Growth and redshift-space calculation

Physics of the channel. Growth is controlled by the full gauge-reduced perturbation action and matter coupling, not by the background Planck factor alone. The observable is a window-convolved anisotropic clustering statistic. A scalar $f\sigma_8(z)$ summary is exact only in the scale-independent limit.

Restricted diagnostic now available. For the near-GR named completion, imposing conserved pressureless Jordan matter, negligible scalar mass/mixing and $\mu_{\text{QS}} = F_s(0)/F_s(N)$ gives Eq. (AE.30). The difference from its matched control remains below $1.9 \times 10^{-3}\%$ through $z=20$. This is a Level C limiting calculation, not the physical ECT growth result.

Restricted peculiar-velocity and bulk-flow carrier. With Fourier convention $e^{i\mathbf{k}\cdot\mathbf{x}}$ and $\theta_m \equiv i\mathbf{k} \cdot \mathbf{v}_m$, conserved dust in this same closure obeys

$$\theta_m = -aHf\delta_m, \quad \mathbf{v}_m(\mathbf{k}) = i aHf \frac{\mathbf{k}}{k^2} \delta_m(\mathbf{k}). \quad (\text{AI.54})$$

For an isotropic real-space window W_R , the linear bulk-flow variance is

$$\langle |\mathbf{V}_R|^2 \rangle = \frac{(aHf)^2}{2\pi^2} \int_0^\infty dk P_{\delta\delta}(k, a) |W_R(k)|^2, \quad (\text{AI.55})$$

before survey geometry, nonlinear velocities and covariance are added. Using the same primordial normalisation as the restricted growth/Weyl replay, the carrier $aHfD$ differs from the matched control by (0.002894, 0.002933, 0.002905, 0.002821)% at $z = (0, 0.5, 1, 2)$; its squared-amplitude carrier differs by (0.005789, 0.005866, 0.005809, 0.005642)%. Thus this near-GR closure gives no material enhancement of a Great-Attractor or large-scale-flow signal. This is a Level C linear proxy, not a survey bulk-flow prediction: the primordial spectrum, scale dependence, nonlinear reconstruction, observer window and velocity likelihood remain Open. A static galactic HRC map may not be substituted for the cosmological velocity transfer. The versioned replay owners are `scripts/cosmology/compute_r103_restricted_isw_lensing_proxy.py` and `scripts/cosmology/compute_r103_restricted_large_flow_proxy.py`.

Full calculation. After constraint reduction, the matter contrast obeys a generally scale-dependent system

$$D'' + \left(2 + \frac{H'}{H} + \Gamma_m\right) D' - \frac{3}{2} \Omega_m \mu_{\cos}(k, a) D = \mathcal{S}_{\text{mix}}, \quad (\text{AI.56})$$

$$-k^2 \Phi_W = 4\pi G_N a^2 \Sigma_{\cos}(k, a) \rho_m \Delta_m. \quad (\text{AI.57})$$

The functions $\Gamma_m, \mu_{\cos}, \Sigma_{\cos}$ and $\mathcal{S}_{\text{mix}}$ are outputs of the physical matrices and vertices. They cannot be assigned the HRC form.

Likelihood inputs. The solver evolves $P_{\delta\delta}, P_{\delta\theta}, P_{\theta\theta}$, constructs the redshift-space multipoles including bias and velocity dispersion, applies Alcock–Paczynski remapping and the survey window, and uses the full covariance. Weak-lensing and CMB-lensing spectra are fitted jointly because they constrain the same Weyl response.

Current verdict and falsifiers. No ECT-native S_8 or $f\sigma_8$ prediction is claimed. A completion fails if it has a ghost/gradient instability, unacceptable scale dependence, wrong RSD/lensing spectra, or requires a primordial normalisation inconsistent with its CMB state. A small background-only or restricted Weyl-carrier diagnostic cannot rescue such a failure.

AI.20 ISW and lensing calculation

Physics of the channel. ISW measures the line-of-sight time derivative of the Weyl potential and lensing measures its transverse gradients. The background expansion enters, but does not determine, either response. The photon metric and gravitational slip are indispensable owners.

Transfer calculation. For a primordial mode $\mathcal{R}(\mathbf{k})$, define

$$\Phi_W(\mathbf{k}, z) = T_W(k, z) \mathcal{R}(\mathbf{k}), \quad \Phi'_W(\mathbf{k}, z) = T'_W(k, z) \mathcal{R}(\mathbf{k}). \quad (\text{AI.58})$$

The same T_W produces CMB lensing, galaxy weak lensing and cluster lensing. The derivative T'_W produces the ISW source. Separate normalisations for those channels would violate common ownership.

Projected spectra. For tracer kernels W_X , the observable spectra have the schematic form

$$C_\ell^{XY} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\mathcal{R}}(k) \Delta_\ell^X(k) \Delta_\ell^Y(k), \quad (\text{AI.59})$$

where the transfer kernels include the physical distances and survey selection. The ISW–galaxy sign depends on the full time evolution of T_W , tracer bias and redshift distribution; it cannot be inferred from the sign of a background coupling.

Restricted limiting proxy now available. For the named near-GR two-slope state, impose conserved pressureless Jordan dust, a common matter/photon metric, the sub-horizon quasistatic limit, negligible scalar mass, mixing and scale dependence, zero slip $\Phi = \Psi$, the response $\Sigma_{QS} = F_s(0)/F_s(N)$, and the same primordial growth normalisation in model and control. With $\mathcal{F} = F_s/F_s(0)$ define

$$\begin{aligned} Q_W^{\text{proxy}} &= \frac{D}{a\mathcal{F}}, & S_{\text{ISW}}^{\text{proxy}} &= \frac{aH}{H(0)} \frac{dQ_W^{\text{proxy}}}{dN}, \\ \frac{d \ln Q_W^{\text{proxy}}}{dN} &= f - 1 - \frac{d \ln F_s}{dN}. \end{aligned} \quad (\text{AI.60})$$

Relative to the matched control, the fractional changes are

$$\begin{aligned} Q_W &: (0.002674, 0.002763, 0.002825, 0.002915)\%, \\ S_{\text{ISW}}^{\text{proxy}} &: (0.003083, 0.003459, 0.004256, 0.008127)\%, \quad z = (0, 0.5, 1, 2). \end{aligned}$$

DOP853/Radau and analytic/finite-difference derivatives agree at relative levels below 4.3×10^{-11} and 1.4×10^{-10} , respectively. This is a Level C restricted-completion diagnostic. It is not a prediction of C_ℓ , weak or CMB lensing, an ISW angular spectrum, or an ISW–galaxy correlation. In particular, the quasistatic assumption removes the horizon-scale dynamics central to ISW. The versioned publication reproducibility owners are `scripts/cosmology/compute_r103_restricted_isw_lensing_proxy.py` and `data/cosmology_r103/R103_RESTRICTED_ISW_LENSING_PROXY_v1.json`.

Figure 42 displays only the four evaluated nodes; it does not supply a continuous perturbation history.

Full result still open. The supplied two-slope background gives the expansion entering the projection integrals, but its gauge-reduced perturbation and photon metric are not derived. The restricted proxy cannot be upgraded by reusing a reference transfer function: the physical kernels, spectra and likelihood remain Open.

Falsifiers. A completion fails if its Weyl transfer produces the wrong sign or amplitude of cross-correlations, violates weak/CMB-lensing spectra, or is inconsistent with the same growth and cluster metric. Passing background distances alone does not weaken these tests.

AI.21 Joint multi-observable acceptance

The programme combines channels at the action/state level. Let ϑ_{act} be genuine action and state parameters and $\mathbf{v}_i$ observational nuisances. The joint likelihood is

$$-2 \ln \mathcal{L}_{\text{joint}}(\vartheta_{\text{act}}, \{\mathbf{v}_i\}) = \sum_{ij} [d_i - m_i(\vartheta_{\text{act}}, \mathbf{v}_i)] C_{ij}^{-1} [d_j - m_j(\vartheta_{\text{act}}, \mathbf{v}_j)] \quad (\text{AI.61})$$

with cross-covariances included. There is no universal empirical cosmological drift coordinate and no intersection of probe-specific intervals in the active theory.

Acceptance hierarchy. The order of gates is:

- (1) action/constraint consistency and external-depth/lapse ownership;
- (2) globally regular background and physical perturbations;
- (3) BBN, recombination, photon and local-gravity consistency;
- (4) observable-specific forward models;
- (5) a joint likelihood and posterior predictive checks.

Failure at an earlier gate is not repaired by a later fit. Passing one channel does not promote the status of another.

Figures and tables. Publication figures are generated only from action-owned arrays: background $H(z)$ and age; distance ratios; recombination and sound horizon; transfer and growth matrices; Weyl/ISW kernels; JWST forward counts; cluster retarded-response maps; and finite-body metric residuals. Until a listed owner exists, the manuscript provides equations, conditional tables and explicit empty slots rather than a decorative proxy curve.

Current cross-channel conclusion. The supplied two-slope action demonstrates a regular conditional background, and the nine-node Cartesian grid exposes a sampled early-acoustic/local-gravity tradeoff. None of those nine nodes solves the Hubble or early-clock target; growth, ISW and JWST remain owner-incomplete rather than family-wide negative results. The local-HRC theorem separately preserves the possibly empty set-valued argmax and, whenever maximizers are attained, excludes static pointwise displacement of a total-response maximizer relative to the supplied total-baryon maximizers; it does not decide residual morphology or a gas/BCG separation already present in the input. These finite-protocol diagnostics and the separate exact local theorem define the present scope without asserting a continuous-family no-go.

AJ Conditional force-law effective-dimensionality identity in the HRC-0 point-source diagnostic

Status: the response algebra is Level A inside the supplied HRC-0 augmented model; the identity/source bridge is Level B/Open and this point-source application is Level C. The resulting curve and its mass independence follow exactly after the HRC-0 response and point-source map are imposed. They are not a derivation of G from the microscopic action and are not closure-independent ECT predictions; other functions share the limiting behaviour only if the same asymptotes are assumed.

AJ.1 Definition

For a gravitational force $g(r)$ obeying $g \propto 1/r^{d-1}$ in d effective dimensions, the force-law dimensionality is

$$d_{\text{force}}(r) = 1 - \frac{d \ln g_{\text{obs}}}{d \ln r}. \quad (\text{AJ.1})$$

AJ.2 General algebra inside the conditional HRC-0 point-source diagnostic

Starting from

$$\mu_{g,0}(x^2) g_{\text{obs}} = g_N(r), \quad x \equiv g_{\text{obs}}/a_M, \quad (\text{AJ.2})$$

take the logarithmic derivative with respect to $\ln r$:

$$\frac{d \ln g_N}{d \ln r} = \left(1 + \frac{d \ln \mu_{g,0}}{d \ln x} \right) \frac{d \ln g_{\text{obs}}}{d \ln r}. \quad (\text{AJ.3})$$

Therefore

$$\frac{d \ln g_{\text{obs}}}{d \ln r} = \frac{d \ln g_N / d \ln r}{1 + d \ln \mu_{g,0} / d \ln x}, \quad (\text{AJ.4})$$

and hence

$$d_{\text{force}}(r) = 1 + \frac{p_N(r)}{1 + d \ln \mu_{g,0} / d \ln x}, \quad (\text{AJ.5})$$

where $p_N(r) \equiv -d \ln g_N / d \ln r$.

For a point mass, $g_N = G_N M / r^2$ gives $p_N = 2$.

AJ.3 Point-source result for $\mu_{g,0}(x^2) = x / \sqrt{1 + x^2}$

Compute:

$$\ln \mu_{g,0} = \ln x - \frac{1}{2} \ln(1 + x^2), \quad \frac{d \ln \mu_{g,0}}{d \ln x} = 1 - \frac{x^2}{1 + x^2} = \frac{1}{1 + x^2}. \quad (\text{AJ.6})$$

Substituting into (AJ.5):

$$d_{\text{force}}(x) = 1 + \frac{2(1 + x^2)}{2 + x^2}. \quad (\text{AJ.7})$$

AJ.4 Limiting cases

Regime	x	d_{force}	Physics
High acceleration	$\gg 1$	$\rightarrow 3$	$g \propto 1/r^2$, Newtonian
Conditional HRC-0 crossover	$= 1$	$= 7/3 \approx 2.33$	point-source value imposed by $\mu_{g,0}$
Conditional HRC-0 low-acceleration limit	$\ll 1$	$\rightarrow 2$	$g \propto 1/r$, $v_{\text{flat}} = \text{const}$

Since $v^2 = rg$, the exact local identities are

$$\frac{d \ln v^2}{d \ln r} = 2 - d_{\text{force}}, \quad \frac{d \ln v}{d \ln r} = \frac{2 - d_{\text{force}}}{2}.$$

The familiar power laws (Keplerian at $d_{\text{force}} = 3$ and flat at $d_{\text{force}} = 2$) apply only where d_{force} is locally or approximately constant; they are not a global power-law solution through the varying crossover.

AJ.5 Characteristic radius and the true $x = 1$ radius

The reference length

$$r_* = \sqrt{\frac{G_N M}{a_M}} \quad (\text{AJ.8})$$

is defined by $g_N = a_M$ for the matched conditional HRC-0 scale. For a general monotone response,

$$r_{x=1} = \frac{r_*}{\sqrt{\mu(1)}}, \quad r_{x=1} = 2^{1/4} r_* \quad \text{for } \mu_{g,0}(x^2) = \frac{x}{\sqrt{1 + x^2}}. \quad (\text{AJ.9})$$

The conditional HRC-0 script uses the declared matching value $a_{M0} = 1.0824 \times 10^{-10} \text{ ms}^{-2}$, giving $r_*(10^8 M_\odot) \approx 0.36 \text{ kpc}$, $r_*(5 \times 10^{10} M_\odot) \approx 8.0 \text{ kpc}$, and $r_*(5 \times 10^{11} M_\odot) \approx 25.4 \text{ kpc}$. This is a bridge-level matched benchmark, not a derivation of a_M .

AJ.6 Distinction from morphological dimensionality

The force-law diagnostic d_{force} describes radial scaling inside the conditional HRC-0 point-source diagnostic and a declared source map. Morphological dimensionality instead describes sheets, filaments, and other spatial matter distributions. No map from either the conditional HRC-0 point-source response $\mu_{g,0}$ or the full HRC field problem to a cosmic-web tidal tensor has been derived, so the latter is a separate Open problem and is not inferred from d_{force} .

AJ.7 Numerical implementation

Equation (AJ.7) is evaluated analytically for a point source. The mapping $x(r)$ uses the exact solution of the conditional HRC-0 point-source relation:

$$g_{\text{obs}}^2 = \frac{g_N^2 + \sqrt{g_N^4 + 4g_N^2(a_M)^2}}{2}, \quad x = g_{\text{obs}}/a_M. \quad (\text{AJ.10})$$

The versioned implementation is included in the released source package. Exact reproduction uses the tagged commit and frozen release manifest associated with the revision, not the moving default branch.

AK HRC galactic functional and algebraic numerical diagnostic

This appendix records the current HRC functional and the reproducible algebraic SPARC diagnostic. It does not supply the missing map from the live Φ action to the gravitational potential and does not replace the full three-dimensional disk problem.

AK.1 Variational structure

After the separate matching in Eqs. (17.55) and (17.57), the static functional is

$$S_{\text{HRC}}^{\text{stat}} = - \int dt d^3x \left[\frac{a_M^2}{8\pi G_N} W_g(Y_g) + \rho_b \Phi_{\text{gal}} \right], \quad Y_g = \frac{|\nabla \Phi_{\text{gal}}|^2}{a_M^2}. \quad (\text{AK.1})$$

Variation gives

$$\nabla \cdot [\mu_g(Y_g) \nabla \Phi_{\text{gal}}] = 4\pi G_N \rho_b, \quad \mu_g = W_{g,Y_g}. \quad (\text{AK.2})$$

The two registered response laws are

$$\mu_{g,0}(Y_g) = \sqrt{\frac{Y_g}{1+Y_g}}, \quad \mu_{g,3}(Y_g) = \mu_{g,0}(Y_g) \left[1 - \frac{4}{3} \frac{Y_g}{(1+Y_g)^2} \right], \quad (\text{AK.3})$$

$$W_{g,0}(Y_g) = \sqrt{Y_g(1+Y_g)} - \text{arsinh} \sqrt{Y_g},$$

$$W_{g,3}(Y_g) = W_{g,0}(Y_g) - \frac{8}{3} \left[\text{artanh} m - m - \frac{m^3}{3} \right], \quad m = \sqrt{\frac{Y_g}{1+Y_g}}. \quad (\text{AK.4})$$

Both have the required limits $\mu_g \sim \sqrt{Y_g}$ for $Y_g \rightarrow 0$ and $\mu_g \rightarrow 1$ for $Y_g \rightarrow \infty$. HRC-0 is exact in the augmented inverse-DtN/recruitment model; HRC-3 additionally assumes the supplied equal-three-channel rank-one spectral vertex. The equalities in this registry also assume $s = Y_g$ and $A_g = 1$; ERP itself does not derive that bridge.

AK.2 Symmetry-reduced algebra and the disk guard

For an isolated spherical configuration,

$$\mu_{g,k}(g^2/a_M^2)g = g_N, \quad k \in \{0, 3\}. \quad (\text{AK.5})$$

The positive solution is unique because both registered responses are positive and monotone. HRC-0 can be solved explicitly:

$$g^2 = \frac{g_N^2 + \sqrt{g_N^4 + 4g_N^2 a_M^2}}{2}. \quad (\text{AK.6})$$

Equation (AK.6) is an HRC-0 identity. The frozen held-out HRC-3 path uses deterministic monotone bracketing; the per-galaxy sensitivity path uses a 60-step safeguarded Newton iteration in $z = g^2/a_M^2$, initialised by the HRC-0 positive root and protected by a positive half-step fallback. Their forward residuals are checked on a logarithmic stress grid. For a generic disk, the algebraic relation does not determine the divergence-free curl field and is used only as a Level-C diagnostic.

AK.3 Baryonic SPARC input

At each radius the diagnostic uses

$$V_{\text{bar}}^2 = V_{\text{gas}}|V_{\text{gas}}| + \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2, \quad g_N = V_{\text{bar}}^2/R. \quad (\text{AK.7})$$

The sign of the gas component is retained. Negative total V_{bar}^2 rows are excluded from the algebraic inversion and must be handled at the field/source level in a future solver.

AK.4 Deep branch and BTFR

For $g \ll a_M$,

$$g^2 \simeq g_N a_M, \quad v_\infty^4 = G_N M_b a_M. \quad (\text{AK.8})$$

This derives the exponent conditionally. It does not derive a_M , a redshift law, or the finite-disk normalisation.

AK.5 Possible orientation-sector degeneracy

No orientation source is included. The live integrability reduction leaves one longitudinal scalar rather than three independent galactic response modes. The three-channel HRC-3 fibre is therefore a supplied response bundle, not a counted set of live propagating Goldstone modes. A future covariant orientation action must be varied and projected before it can alter any curve or fitted scale.

AL Computational procedure for the HRC SPARC diagnostic

AL.1 Frozen inputs and sample

The source table is `MassModels_Lelli2016c.mrt`. The diagnostic retains 165 galaxies. The fixed- M/L mode has 3343 usable physical points after the declared parsing and positive-source gates. Whole galaxies, rather than individual radial points, are assigned to five held-out folds. Five fixed seeds test split sensitivity.

AL.2 Fit modes

The fixed- M/L mode uses $\Upsilon_d = 0.5$ and $\Upsilon_b = 0.7$. The common- M/L mode fits a shared disk mass-to-light ratio on the training galaxies and transfers it to the held-out fold. Its bound is $0.3 \leq \Upsilon_d \leq 1.0$ and its parameter-independent validity mask is frozen at the lower bound; this leaves 3342 test points per seed (three excluded from the 3345 parsed rows), rather than the 3343 points of the fixed- M/L mode. Neither

mode marginalises distance, inclination, correlated uncertainties, intrinsic scatter or stellar-population hierarchy. The velocity objective uses the disclosed floor $\sigma_{V,i}^{\text{use}} = \max(\sigma_{V,i}^{\text{table}}, 2 \text{ km s}^{-1})$; it must not be described as using the raw table errors without modification.

The separate per-galaxy M/L sensitivity diagnostic uses $0.1 \leq \Upsilon_d \leq 2.5$ and $10^{-2} \leq a_M/a_{M0} \leq 10^2$, with $\Upsilon_b = 0.7$ fixed. It uses a deterministic grid-seeded bounded local optimiser and an independent differential-evolution audit for every row. If the local and first global solutions disagree, two additional fixed seeds are run; verification requires at least two independently seeded global searches to agree at the lowest objective within relative tolerance 10^{-7} . Boundary hits are reported rather than interpreted as physical stellar-population estimates.

AL.3 Law registry and inversion

Only HRC-0 and HRC-3 are registered in the manuscript calculation. For every trial a_M , the unique positive root of

$$f(g) = \mu_{g,k}(g^2/a_M^2)g - g_N = 0 \quad (\text{AL.1})$$

is obtained either by monotone bracketing (held-out point registry) or by the safeguarded z -Newton solver above (per-galaxy sensitivity registry). The two implementations share the same monotone response and are guarded by a direct forward-residual test. The model velocity is $V_{\text{mod}} = \sqrt{gR}$. The training objective is the declared sum of squared velocity residuals with the error-floor convention above; the held-out objective is never used to choose the response law or its fixed HRC-3 coefficient.

AL.4 Reproducibility and interpretation

The HRC-0/HRC-3 point predictions, per-galaxy contributions, regime decomposition, fold table and machine-readable result JSON belong to the versioned HRC numerical reproducibility package. For a released manuscript revision, the exact package owner is the corresponding tagged commit and frozen release manifest; the main-text figures are deterministic renderings of the files identified there. HRC-3 was designed after inspection of the crossover tension, so the present dataset cannot serve as its independent confirmation. A prespecified external-galaxy sample is a mandatory next test.

The current calculation is exact only as an algebraic diagnostic. A publication-grade observational result requires the density-aware axisymmetric PDE, scale heights and boundary conditions, the external field in the same operator, covariance and hierarchical nuisance marginalisation.

AM Local-HRC argmax theorem and physical cluster requirements

Status: the strictly increasing-map order/argmax theorem is Level A inside the supplied local map. The physical cluster source, response, photon metric, merger dynamics and observational map remain Open. No hand-set cluster mass, width, offset, critical scale or synthetic replay is used as ECT phenomenology.

AM.1 Exact monotonicity theorem

Let D be the projected domain, let $s(\mathbf{x}) = \Sigma_b(\mathbf{x}) > 0$, and define $\arg \max_D g = \{\mathbf{x} \in D : g(\mathbf{x}) = \sup_D g\}$, which may be empty. Let the proposed static response be $F(\mathbf{x}) = f[s(\mathbf{x})]$ with f strictly increasing on $s(D)$. A sufficient smooth condition is $f \in C^1$ with $f'(u) > 0$ for every u in the interval hull of $s(D)$. Strict monotonicity gives

$$s(\mathbf{x}_1) > s(\mathbf{x}_2) \iff F(\mathbf{x}_1) > F(\mathbf{x}_2), \quad \arg \max_D F = \arg \max_D s. \quad (\text{AM.1})$$

The set identity remains valid when both argmax sets are empty, and the same ordering argument preserves every isolated local maximum. Whenever a peak is attained, no choice of parameters in this strictly increasing positive local map can move it away from the baryonic proxy that sources it. A prescribed input map can only encode an offset already present in its source.

For the explicit local HRC-0 map used in the main text, Eq. (17.22) proves the required positivity, while Eq. (17.23) shows that the map compresses contrast. These exact statements require no synthetic cluster construction or hand-set numerical input.

AM.2 Required physical calculation

The physical merger problem is the time-dependent state $\{\rho_{\text{gas}}, f_{\text{gal}}, \rho_*, \Phi, \Psi, \Xi_{\text{resp}}, \mathcal{K}_{\text{ret}}; \mathbf{b}, \mathbf{v}_{\text{rel}}, t_{\text{obs}}\}$. It requires a derived source vertex and stable causal kernel, a common matter/photon metric, converged hydrodynamic and collisionless evolution, and mock-observation likelihoods for shear, convergence, X-ray, SZ and galaxy maps. These owners are Open. Equation (AM.1) prevents a local static rescaling from being mistaken for that calculation.

AM.3 What would count as closure

A quantitative ECT cluster result must evolve the same action through the collision, derive the causal response kernel and both matter/photon metrics, project a mock convergence map, and compare it jointly with shear, X-ray, SZ and galaxy data including nuisance parameters and covariance. The local HRC theorem remains useful as a kill gate: if a proposed implementation is only a strictly increasing pointwise remapping, it preserves the set-valued argmax and, whenever maxima are attained, cannot dynamically move a total-response maximizer relative to its supplied total-baryon maximizers, no matter how one tunes its scalar scale. This does not test an inherited gas/BCG separation or residual-map extrema and is not a no-go for a general causal merger kernel.

AN Open calculation of scalar response lengths across environments

For a declared scalar coordinate q , a homogeneous local linearisation defines only the conditional response scale

$$m_{\text{loc}}^2 = \frac{\mathcal{E}_{,q}(\bar{q}, \rho_{\text{m}})}{\mathcal{A}(\bar{q})}, \quad \lambda_{\text{loc}} = m_{\text{loc}}^{-1},$$

where $\mathcal{A}$ and $\mathcal{E}$ are defined in eq. (17.2). A real positive local Yukawa length requires the separate conditions $\mathcal{A} > 0$ and $\mathcal{E}_{,q} > 0$; the formula must not be continued through $\mathcal{E}_{,q} = 0$. Even when these local guards hold, this is not yet the canonical finite-body mass or a screening length measured by a Solar-System experiment.

A quantitative environment calculation must specify:

- (i) the functions f, K, V and asymptotic background;
- (ii) canonical Einstein-frame normalisation and Hessian;
- (iii) regular interior/exterior solutions for each body;
- (iv) composite-body charge and matching conditions; and
- (v) PPN, fifth-force, and equivalence-principle observables.

Until these steps are completed, the density trend and every numerical response length remain Open.

AO Operational test of the exploratory environment/history law

This appendix defines a regression protocol for the Open hypothesis Ξ . It does not assume that $a_{M,\text{eff}}$ is non-universal for physical reasons or that ECT predicts a correlation.

Covariates. Environment may be represented by nearest-neighbour density from 2MRS [314], group membership or host-halo mass from SDSS [315], tidal-field measures, and explicitly declared history proxies. These variables probe different scales and must not be collapsed into one density without testing that reduction.

Protocol.

- (i) fit the conditional HRC rotation-curve diagnostic to obtain $a_{M,\text{eff}}$ per galaxy;
- (ii) cross-match the sample to the declared covariates;
- (iii) regress residuals after controlling for baryonic mass, morphology, distance, inclination, and stellar M/L ;
- (iv) compare a universal-scale model with prespecified Ξ models using held-out or information-criterion tests; and
- (v) repeat on gas-dominated and morphology-stratified subsamples.

Interpretation. A threshold such as $p < 0.01$ may be declared in advance for one analysis, but it is a statistical decision rule rather than an ECT prediction. A null result rejects or constrains the tested Ξ model. A robust correlation is an empirical regularity that still requires a microphysical scalar and screening derivation before it can be attributed to ECT.

AP Status, ownership, and falsifiers of the HRC programme

This appendix prevents three logically different statements from being collapsed into one: an exact identity inside a supplied response model, a conditional medium-to-gravity bridge, and an observational diagnostic. The current ownership chain is

$$\begin{aligned}
 & \text{S11a local deformability} + \text{S11b branch reconfigurability} \not\Rightarrow \text{ERP-}\Phi, \\
 & \text{S11b branch reconfigurability} \supset \text{ERP-}\Phi \text{ ideal regime} \implies \hat{\mu}_0(s) \text{ or } \hat{\mu}_3(s), \\
 & \hat{\mu}_k(s) \xrightarrow[\text{Open}]{s=f(Y_g), A_g} \mu_{g,k}(Y_g), \quad k \in \{0, 3\}. \quad (\text{AP.1})
 \end{aligned}$$

S11a says that sufficiently small guarded variations exist; S11b says only that selected finite transition paths may exist. Neither derives ERP’s equal channels, sharp threshold, no-hysteresis limit or bounded occupation. The final arrow is a separate gravitational matching problem and is not part of any property or postulate of the pre-geometric Φ -medium.

AP.1 Owner-by-owner status

Table 136: Claim ownership in the HRC candidate. “Exact” always means exact inside the model named in the middle column.

Claim	Current owner/status	Missing closure or falsifier
Infinitesimal local deformability	Conditional Level A tangent-space result for regular scalar configurations	Requires fixed admissible boundary data, regularity margin and topology; it implies neither finite connectivity nor relaxation
Finite branch reconfigurability	Candidate S11b, gravity-free property	Derive which branches are joined by finite-action paths and whether the dynamics can traverse them; no universal connectivity is implied

Claim	Current owner/status	Missing closure or falsifier
Stationary ordered response action	Open / MISSING ACTION; P4 is not a P3 saddle	Close OP-grad or supply one explicit stable P4-supporting action before identifying a physical constrained Hessian
ERP- Φ	Author-proposed constitutive principle for an ideal symmetric, quasistatic, additive and non-hysteretic regime	Derive the physical constrained bundle, equal shift, bounded two-branch occupation and trace normalisation from a microscopic ordered-phase model; derive a factorised gap, protected Kubo limit or other exact finite-coupling mechanism
Inverse-DtN compliance	Level A for the stated positive massive Euclidean half-space problem	Identify the physical interface, boundary condition, finite-depth and curvature corrections in the live Φ action
Finite-interface infrared limit	Level A compact periodic-cube counterexample; smooth HRC density conditional on the continuum-first order of limits	Identify a noncompact/gapless internal fibre or derive $m_R L$ and quantify the actual low-spectrum corrections for the physical nonspherical interface
HRC-0 primitive and response	Level A in the augmented inverse-DtN/recruitment model; Level B mechanism candidate	Failure of the three-dimensional spectral density, a residual radial weight, gain/ghost modes, unavoidable hysteresis or an unprotected finite-probe threshold shift refutes this completion
HRC-3 crossover factor	Level A in the supplied equal-three-channel rank-one matrix model; Level B proof of class	The live scalar-only action does not provide the required three-channel fibre or equal weights; the coefficient was selected after the crossover tension was inspected
Internal-to-gravity map	Open; represented by $s = f(Y_g)$ and A_g	Exact HRC gravity additionally assumes $f(Y_g) = Y_g$ and $A_g = 1$; a derived nonidentity map changes the response law
Static HRC field equation	Level A after the supplied gravity functional	Matter vertex, covariant completion, both metric potentials and double-counting control remain Open
BTFR slope and spherical RAR asymptotes	Exact conditional algebra in the deep, isolated, spherical branch	The finite-disk normalisation, scatter, environment dependence and scale owner are not thereby derived
$a_{M0} = cH_0/(2\pi)$	Explicit present-epoch matching condition	The factor $1/(2\pi)$ and any redshift or memory law remain Open
SPARC figures and tables	Level C held-out algebraic diagnostics	Require a density-aware axisymmetric PDE, curl field, covariance and hierarchical nuisance model plus a prespecified independent sample
Lensing, clusters, cosmology and PPN	Open in HRC	Require the covariant metric response and additional tensor/shear, finite-body and dynamical owners; they are not consequences of the scalar algebraic law alone

AP.2 Exterior Euclidean check for an arbitrary local interface

Let Σ be a smooth local interface patch, with no assumption about the global shape of the condensate domain. For a positive quadratic exterior operator, minimum-energy continuation of boundary data φ_Σ into the adjacent Euclidean region gives

$$S_{E,\text{on shell}}^{(2)} = \frac{1}{2} \langle \varphi_\Sigma, D_E \varphi_\Sigma \rangle, \quad D_E = \sqrt{\mathcal{L}_\Sigma} \quad (\text{AP.2})$$

at principal-symbol order. The static compliance is D_E^{-1} . In a local flat chart with $\mathcal{L}_\Sigma = -\Delta_\Sigma + m_R^2$, normalising by the zero-momentum value gives

$$C(\Lambda) = (1 + \Lambda)^{-1/2}. \quad (\text{AP.3})$$

This supplies the compliance factor used by HRC without assuming a spherical Universe. It does not supply the released-channel projector, the physical interface or the map to Y_g . Extrinsic curvature, finite depth, mixed boundary conditions and anisotropic principal symbols are finite tests, not optional decorations: a residual spectral weight that destroys the Newtonian limit falsifies the corresponding interface realisation.

AP.3 Why the current numerical calculation is not yet the field solution

For spherical symmetry, integrating the divergence equation gives

$$\mu_g(g^2/a_M^2)g = g_N. \quad (\text{AP.4})$$

For a disk, the relation instead has the Helmholtz form

$$\mu_g g = g_N + \nabla \times H, \quad \nabla \cdot (\mu_g g) = 4\pi G_N \rho_b. \quad (\text{AP.5})$$

The curl field depends on the full three-dimensional density, thickness and boundary data. Mid-plane component velocities alone do not determine it. Consequently the present radius-by-radius SPARC computation is an auditable diagnostic of the response shape, not a completed prediction of disk galaxies.

AQ HRC stress test with ultra-diffuse galaxies

This appendix gives an HRC-only central-value UDG diagnostic. It is deliberately weaker than a spherical or axisymmetric Jeans posterior, but it exposes the inverse problem and the missing dynamical owners explicitly.

For a pressure-supported system, use the Wolf-type central proxy

$$R_{1/2} = \frac{4}{3}R_e, \quad M_b(< R_{1/2}) \simeq \frac{1}{2}M_\star, \quad g_{\text{obs}} = \frac{3\sigma_{\text{los}}^2}{R_{1/2}}, \quad g_N = \frac{G_N M_\star}{2R_{1/2}^2}. \quad (\text{AQ.1})$$

The central inputs and their provenance are

Table 137: Inputs of the HRC UDG proxy. The quoted dispersions are heterogeneous measurements or literature brackets, not a joint-confidence sample.

Object	$M_*/10^8 M_\odot$	R_e/kpc	$\sigma_{\text{los}}/\text{km s}^{-1}$	$\mathcal{R}_{\text{dyn}}$	Role/status
NGC 1052–DF4	1.50	1.60	$6.3^{+2.5}_{-1.6}$	0.787	Low-dispersion stress case [119, 118]
FCC 224	1.74	1.89	7.8 ± 2.0	1.229	Stellar-kinematics low-dispersion case [120]
NGC 1052–DF2	1.30	2.20	8.6–14.9	2.328–6.988	Ambiguous bracket [122, 123, 118]
NGC 5846-UDG1	1.10	2.10	17 ± 2	10.262	Matched-mass control [124]
Dragonfly 44 [125]	3.00	4.70	33 ± 3	31.734	High-dispersion control
AGC 114905	Gas-rich rotating disk; spherical proxy not applied				Axisymmetric solver required [126]

Define $\mathcal{R}_{\text{dyn}} = g_{\text{obs}}/g_N$. A finite positive- a_M HRC enhancement solution requires $\mathcal{R}_{\text{dyn}} > 1$; equality is the Newtonian boundary $a_M = 0$. Under HRC-0,

$$\mu_{g,0}(Y_g) = \frac{1}{\mathcal{R}_{\text{dyn}}} \implies \frac{a_M^{(0)}}{g_{\text{obs}}} = \sqrt{\mathcal{R}_{\text{dyn}}^2 - 1}. \quad (\text{AQ.2})$$

For HRC-3, monotonicity in the retained rank-one class gives one positive root x_3 of

$$\mu_{g,3}(x_3^2) = \frac{1}{\mathcal{R}_{\text{dyn}}}, \quad a_M^{(3)} = \frac{g_{\text{obs}}}{x_3}. \quad (\text{AQ.3})$$

The resulting values are reported in Table 65. They show that a single universal present-epoch scale does not reproduce the central proxies of all listed systems: one object lies outside the positive enhancement domain, while the others span effective scales from below one per cent to nearly five times the matched value. This is a tension and a target for a derived environment/history operator, not evidence that such an operator has already been identified.

The calculation must not be read more strongly than its inputs permit. Anisotropy, inclination, binary contamination, stellar population, distance, non-sphericity and selection effects can change the inverse scale. A complete UDG result requires posterior propagation through a Jeans or distribution function model, the same covariant HRC field solver used for ordinary galaxies, and a predicted incidence distribution rather than a separately fitted a_M for every object.

AQ.1 Jeans upgrade path

The Wolf proxy suppresses, but does not remove, tracer and anisotropy dependence. A forward model must solve

$$\frac{d(v\sigma_r^2)}{dr} + \frac{2\beta(r)}{r} v\sigma_r^2 = -v(r)g(r), \quad (\text{AQ.4})$$

with $g(r)$ obtained from the same HRC field equation used for ordinary galaxies. The projected and aperture-weighted observables are

$$\sigma_{\text{los}}^2(R) = \frac{2}{I(R)} \int_R^\infty \left(1 - \beta \frac{R^2}{r^2}\right) \frac{v(r)\sigma_r^2(r)r dr}{\sqrt{r^2 - R^2}}, \quad (\text{AQ.5})$$

$$\langle \sigma_{\text{los}}^2 \rangle_{\text{ap}} = \frac{\int_0^{R_{\text{ap}}} W I \sigma_{\text{los}}^2 R dR}{\int_0^{R_{\text{ap}}} W I R dR}. \quad (\text{AQ.6})$$

Here v , I , β , W and the interloper/binary model are nuisance inputs. Equations (AQ.4)–(AQ.6) define the required upgrade; no posterior from them is claimed here.

AQ.2 Boundary-aware dispersion-only intervals

At fixed central (M_*, R_e) , propagating only the quoted dispersion endpoints through the HRC inverses gives Table 138. These are not common-confidence intervals. A dash at the low endpoint means $\mathcal{R}_{\text{dyn}} < 1$ and hence no positive-enhancement HRC solution.

Table 138: Dispersion-only HRC boundary audit. Scale ranges are in units of a_{M0} .

Object	σ range	$\mathcal{R}_{\text{dyn}}$ range	HRC-0 a_M/a_{M0}	HRC-3 a_M/a_{M0}
NGC 1052-DF4	4.7–8.8	0.438–1.536	— to 0.0380	— to 0.0188
FCC 224	3.4–14.5	0.234–4.248	— to 0.309	— to 0.282
NGC 1052-DF2	8.6–14.9	2.328–6.988	0.0476–0.470	0.0278–0.457
NGC 5846-UDG1	15–19	7.990–12.819	0.572–1.480	0.560–1.468
Dragonfly 44	30–36	26.226– 37.766	3.381–7.013	3.374–7.006

The large ranges show why the central inverse is a stress diagnostic rather than a population verdict. Distance, M/L , anisotropy, shape, aperture and membership must be sampled jointly with the physical domain constraint.

AQ.3 External field, shape, controls and selection

HRC currently has no derived external-field operator. Rough host-field estimates for DF4, FCC 224, UDG1 and Dragonfly 44 are of the same order as their internal accelerations in some cases, so an EFE cannot be declared negligible from g_{ext}/a_{M0} alone. Its sign and angular structure must come from the same nonspherical response operator as the disk curl field; inserting a scalar suppression factor would be another fit law.

FCC 224 has apparent ellipticity $\varepsilon \simeq 0.36$ [120]; its spherical row is therefore only an order-unity proxy. The juxtaposition of low-dispersion DF4/FCC 224 with the similarly diffuse but high-dispersion NGC 5846-UDG1 is a matched-class challenge: a completion must predict which additional state, history or boundary datum separates them. It may not assign an independent a_M after observing the dispersion.

The known sample is small and selection-biased toward unusual globular cluster populations. No selection-corrected observational parent sample is supplied here, so no population percentage is asserted. Collision/tidal channels are external astrophysical alternatives [144, 146]. A viable ECT completion must predict both the conditional kinematics and the incidence/morphology, not only accommodate individually selected objects. AGC 114905 supplies a complementary disk-domain test once the axisymmetric HRC solver exists.

AR Numerical algorithm for a physical ECT cosmological completion

The active numerical target is one constraint-preserving integration of a complete varied ECT action. A reference background, an independently tuned growth multiplier, or a manually attached high-redshift tail is not an ECT solver.

AR.1 Required state and equations

After gauge reduction the solver state must contain every independent homogeneous field and its conjugate datum, together with conserved matter and radiation variables. At each step it must solve the Hamiltonian constraint, all evolution equations and all algebraic orientation constraints. The minimal diagnostics are

$$\mathcal{G} = 0, \quad \Delta_F > 0, \quad \lambda_{\min}(\mathbf{K}_{\text{phys}}) > 0, \quad c_{s,i}^2 \geq 0, \quad (\text{AR.1})$$

where $\mathcal{G}$ is the background constraint, Δ_F denotes every algebraic denominator and $\mathbf{K}_{\text{phys}}$ is the constrained kinetic matrix. Any crossing is a terminal physical failure, not an invitation to splice in another background.

AR.2 Ordering-to-present integration

The initial surface is the relational ordering event obtained through the external-depth/lapse map. Its state and boundary conditions must be supplied by the ordering dynamics. The solver then advances the same equations to the present, determining $H(a)$ and the physical normalisation together. The age is accumulated as an auxiliary quadrature of Eq. (16.13); it is not an input.

AR.3 Constraint reduction and regular variables

Write the homogeneous equations before elimination as the differential algebraic system

$$\mathcal{E}_A(X, \dot{X}, H) = 0, \quad \mathcal{C}_\alpha(X, \dot{X}, H) = 0, \quad (\text{AR.2})$$

where $\mathcal{C}_\alpha$ contains the Hamiltonian, unit-director and any response constraints. The code first computes the rank of the Jacobian with respect to the algebraic variables. On a regular chart,

$$\det \frac{\partial \mathcal{C}_\alpha}{\partial Y^\beta} \neq 0, \quad Y^\beta = \{H, \lambda_{\text{constr}}, \dots\}, \quad (\text{AR.3})$$

these variables are solved and substituted into a first-order system in $N = \ln a$. A vanishing determinant triggers a chart change or termination; it is never regularised by clipping the denominator.

The two-slope scalar implementation uses the state

$$X = (\sigma, p, \tau, \chi), \quad p = \frac{d\sigma}{dN}, \quad \frac{d\tau}{dN} = \frac{1}{H}, \quad \frac{d\chi}{dN} = -\frac{e^{-N}}{H}, \quad (\text{AR.4})$$

with H obtained from the positive Friedmann root at every evaluation. The age and distance quadratures are therefore integrated by the same adaptive solver and at the same accepted steps as the fields.

AR.4 Shooting and calibration record

A named conditional completion may impose terminal calibration targets, but the shooting variables and targets must be reported separately from theory inputs and observables. For the calibrated family the residual is

$$\mathbf{R}_{\text{shoot}}(\sigma_i, V_{\text{tot}}) = \begin{pmatrix} \sigma(N=0) \\ H(N=0) - H_* \end{pmatrix}, \quad (\text{AR.5})$$

with $p(N_i) = 0$. A two-dimensional root solve determines $(\sigma_i, V_{\text{tot}})$. The residual tolerance, Jacobian method, iteration count and final values are stored in the result file. A target such as $H(0) = H_*$ defines dimensionless units; it does not predict the observed value of H_0 .

The vacuum-ratio condition is checked independently:

$$\frac{V_-}{V_+} = \frac{b_s - 2a_s}{2a_s - c_s}, \quad \partial_\sigma \frac{V_s}{F_s^2} \Big|_{\sigma=0} = 0. \quad (\text{AR.6})$$

Failure of either equality invalidates the declared vacuum calibration even when the shooting residual is small.

AR.5 Two-integrator and cutoff validation

Every production orbit is evaluated with an explicit high-order non-stiff method and an implicit stiff method. For a quantity Q ,

$$\Delta_Q^{\text{method}} = \frac{|Q_{\text{DOP853}} - Q_{\text{Radau}}|}{\max(1, |Q_{\text{DOP853}}|, |Q_{\text{Radau}}|)} \quad (\text{AR.7})$$

is published for the age, terminal state, $H(0)$, distances and any integrated acoustic quantity. Agreement of two methods is a numerical check, not a physical validation of the action.

The initial-cutoff test repeats the integration at $N_i = -20, -30, -40, -60$ where supported. The missing radiation-era tail is bounded analytically from the leading radiation solution,

$$0 < \Delta\tau_{\text{tail}} \leq \frac{e^{2N_i}}{2H_0\sqrt{\Omega_{r0}}} [1 + \mathcal{O}(e^{N_i})]. \quad (\text{AR.8})$$

The bound and the observed cutoff drift are reported separately.

AR.6 Perturbation and observable stages

Only after the background passes every gate may the same completion generate the quadratic perturbation operators. Recombination, distances, transfer functions, growth, lensing, cluster transport and early-object forward models are separate downstream stages with frozen inputs and likelihoods. Their outputs must carry hashes of the background and action that generated them.

Stage P0: background publication

The frozen background file contains $N, H, \sigma, \sigma', F, V, \rho_i$, all algebraic denominators, constraint residuals, age and distance quadratures, plus the external-depth/lapse variables. It also carries the action, parameter, initial-state and source hashes. Derived interpolation objects are constructed from this file and never from a separately fitted curve.

Stage P1: physical clocks, photons and distances

The matter-clock and photon metrics are evaluated on the background. Redshift is computed from the photon energy measured by the matter clock; the simplification $1 + z = a^{-1}$ is used only after that equality is verified. The stage publishes

$$\{z(N), H_\gamma(z), D_M, D_A, D_L, dV_c/dz d\Omega\} \quad (\text{AR.9})$$

and tests distance duality under its stated assumptions. This stage is the owner of chronometer, BAO, supernova and catalogue-volume inputs.

Stage P2: thermal and recombination history

The baryon, photon and neutrino sectors are evolved with their physical couplings. The ionisation fraction $x_e(z)$, optical depth, visibility function, sound speed, r_s , r_{drag} and diffusion length are computed without borrowing a reference expansion. The imported standard photon layer used for the conditional two-slope diagnostic is a controlled test of the background only and is labelled separately in every table.

Stage P3: gauge-reduced linear perturbations

The full quadratic action is reduced before integration. At every (k, N) the code records the rank and eigenvalues of the kinetic matrix, the characteristic speeds, the mass matrix and the source vertices. Adiabatic and all physical isocurvature modes are initialised in a declared state. The transfer file contains

$$\{T_{\mathcal{R}}, T_m, T_\theta, T_\Phi, T_\Psi, T_{\text{vec}}, T_{\text{tens}}\}(k, N) \quad (\text{AR.10})$$

with gauge-invariant definitions. A transfer function is not emitted for a mode whose kinetic term or source vertex is missing.

Stage P4: spectra and linear likelihoods

The transfer functions are convolved with the action-owned primordial state to obtain matter, velocity, Weyl, CMB and tensor spectra. Survey windows, bias parameters and covariance matrices are applied at this stage. The pipeline reports both raw theory arrays and residuals in data space so that normalisation, window and likelihood errors can be distinguished.

Stage P5: nonlinear objects and clusters

Halo collapse and baryonic assembly are simulated or calibrated only after the linear spectrum is fixed. JWST number counts and morphologies use the same distances, volume and time budget as the background. Cluster mergers add the retarded response kernel, gas dynamics, collision geometry and photon metric; the local monotone HRC limit is kept as a negative control.

Stage P6: finite-body and local gravity

The cosmological solution supplies the asymptotic state for an extended-body boundary-value problem. The resulting scalar charge and metric are used to calculate PPN, orbital, time-delay and equivalence-principle observables. This stage decides whether a cosmologically interesting completion is viable; screening may not be inserted as a binary flag.

AR.7 Failure taxonomy

Every failed trajectory receives one primary machine-readable reason:

Code	Meaning
BG-CONSTRAINT	Hamiltonian or algebraic constraint exceeds tolerance
BG-SINGULAR	physical reduced denominator or response determinant vanishes
BG-GHOST	a physical kinetic eigenvalue becomes non-positive
BG-GRADIENT	a physical squared characteristic speed becomes negative
MAP-MISSING	matter, photon, tensor or depth/lapse map is not supplied
VERTEX-MISSING	a required perturbation or source vertex is absent
REC-INCONSISTENT	recombination inputs are not generated by the same state
PPN-FAIL	the finite-body completion fails a declared local-gravity gate
LIKELIHOOD-OPEN	physical predictions exist but no frozen data likelihood has been run

The first four are failures of the supplied physical completion. The next three identify missing ownership. PPN-FAIL rejects that completion in its declared local limit. LIKELIHOOD-OPEN forbids a data-agreement claim but does not by itself falsify the dynamics.

AR.8 Verification requirements

An accepted implementation must provide: two independent formulations of the background equations; constraint residuals; convergence under step and precision refinement; continuation tests near every apparent singularity; positive-kinetic and gradient checks; a complete input-ownership table; and reproduction

from a frozen environment. Control solutions may test the code, but their ages or parameters must never be reported as ECT predictions.

Independent formulation gate. The first formulation evolves the reduced first-order system. The second evaluates the unreduced Euler–Lagrange and constraint equations on the interpolated orbit. Let $\mathcal{E}_A^{\text{full}}$ denote the latter residuals. The report provides

$$R_A = \frac{\max_i |\mathcal{E}_A^{\text{full}}(N_i)|}{\max_i [1, |\mathcal{T}_{A,1}|, \dots, |\mathcal{T}_{A,n}|]}, \quad (\text{AR.11})$$

where $\mathcal{T}_{A,j}$ are the individual equation terms. A small reduced-solver error with a large R_A exposes an elimination or sign mistake.

Precision and grid gate. The integration is repeated with at least two relative/absolute tolerances, two output grids and two dense-output interpolation orders. Integrated quantities are compared before rounding. A result is quoted to no more digits than remain stable under all variations and between the two integrators.

Limit gate. The implementation reproduces analytic radiation, matter-tracker and vacuum limits whenever they exist. It also reproduces the constant- F control after the scalar coupling is disabled. These tests validate code paths; they do not make the control cosmology an ECT prediction.

Dependency gate. Every downstream file includes the SHA-256 of the exact background input. A change of action, state, scale calibration, external-depth/lapse map, photon metric or perturbation operator invalidates all dependent ages, distances, spectra, figures and likelihoods until they are regenerated.

Counterexample gate. The solver suite keeps known negative cases in its private verification corpus: the local-HRC cluster offset no-go, an unscreened point that fails Cassini, and a background point with a vanishing response determinant. A verification system that accepts any of these as a complete viable cosmology has failed.

AR.9 Publication artefacts

Figures and tables are generated only from a background that passes the exact solver gates. The supplied two-slope completion passes the background gates and therefore owns the conditional tables in Section AE. Its restricted growth and imported-recombination layers remain conditional, while a full Hubble likelihood, JWST abundance/morphology and gauge-complete growth curve still await their listed owners.

The minimum frozen release bundle is:

- (i) action, boundary terms, state and unit/depth-lapse records;
- (ii) source code and machine-readable environment;
- (iii) raw background and constraint arrays;
- (iv) shooting, convergence and independent-residual reports;
- (v) photon/recombination and perturbation arrays when present;
- (vi) every figure source and the exact data array plotted;
- (vii) likelihood configuration, covariance and residual vectors;
- (viii) a claim ledger separating Level A internal results, Level B proof-of-class statements, Level C conditional comparisons and Open owners.

Versioned reproducible artefacts. The versioned reproducibility owners contain three independent calculation products: the named near-GR two-slope orbit and its observables; the nine-point Cartesian action/state grid with sound-horizon and unscreened-PPN diagnostics; and the exact local-HRC monotone-map theorem. None uses a universal empirical cosmological drift parameter. The publication source contains the numerical inputs, equations, assumptions and status of each product; their machine-readable JSON/CSV and scripts are identified by the frozen release manifest and tagged repository state.

AS Conditional loop-scale closures and the remaining first-principles task

Status: Level A mathematics for the fixed-length bound and for the minimum of the explicitly supplied positive-tension 1D functional. Level B/Open as an ECT route because the compact phase, transverse profile, line tension, unwinding sector and 4D-to-1D reduction are not derived from P1–P6. CDE- Φ is only a post-order candidate constitutive hypothesis and acceptance criterion for a completed branch EFT. The owner remains the explicit 4D action, state, profile, vacuum subtraction and physical quotient; CDE- Φ is not a theorem or owner of the minimal action. No first-principles value of $\hbar$ or universal occupation law is claimed. Reference from the main text: Section 5.6.

AS.1 Two different reduced-loop closures

The phase-only Euclidean construction starts from $\mathcal{J}_{\theta,E} \sim \int d^4X K_\theta \delta^{AB} \partial_A \theta \partial_B \theta / 2$ and introduces the transverse three-volume weight $K_\theta^{1D} = K_\theta \mathcal{A}_\perp$. At fixed loop length L and winding n , Cauchy–Schwarz gives

$$\mathcal{J}_{\text{loop}}(L, n) \geq \frac{2\pi^2 K_\theta^{1D} n^2}{L}, \quad (\text{AS.1})$$

with equality for uniform winding. Selecting a core-sized length by hand defines the phase-only coefficient ι_0^{EFT} used in the main text. Because Eq. (AS.1) tends to zero as $L \rightarrow \infty$, topology and phase stiffness alone do not select a nonzero global reduced minimum; a physical action additionally requires Σ_θ .

The tension-balanced route adds a physically different term. Suppose a declared one-dimensional reduced model contains

$$\mathcal{J}_n(L) = \mathcal{J}_{\text{loop}} L + \frac{2\pi^2 K_\theta^{1D} n^2}{L}, \quad K_\theta^{1D} > 0, \quad \mathcal{J}_{\text{loop}} > 0, \quad n \neq 0. \quad (\text{AS.2})$$

Here $\mathcal{J}_{\text{loop}}$ is the reduced inverse-length coefficient and is not the chemical potential, a particle mass, or a fitted infrared cutoff. In ordinary units $[K_\theta^{1D}] = L$ and $[\mathcal{J}_{\text{loop}}] = L^{-1}$, so both $\sqrt{K_\theta^{1D} \mathcal{J}_{\text{loop}}}$ and $\mathcal{J}_n$ are dimensionless. The phase-route physical coefficients are $\Sigma_\theta K_\theta^{1D}$ and $\Sigma_\theta \mathcal{J}_{\text{loop}}$, with dimensions action times length and action per length. The natural-unit entries are $[K_\theta^{1D}] = \text{GeV}^{-1}$ and $[\mathcal{J}_{\text{loop}}] = \text{GeV}$; they do not themselves supply an action unit. Interpreting the physical line coefficient as a worldline rest-action cost or as an ordinary Lorentzian string energy per unit length requires, respectively, a separately supplied proper-time/state map or a clock/worldsheet reduction. Differentiation gives

$$\frac{d\mathcal{J}_n}{dL} = \mathcal{J}_{\text{loop}} - \frac{2\pi^2 K_\theta^{1D} n^2}{L^2}, \quad \frac{d^2\mathcal{J}_n}{dL^2} = \frac{4\pi^2 K_\theta^{1D} n^2}{L^3} > 0. \quad (\text{AS.3})$$

Together with $\mathcal{J}_n \rightarrow \infty$ for $L \rightarrow 0^+$ and $L \rightarrow \infty$, this proves that Eq. (5.28) is the unique global minimum with respect to L , at fixed nonzero winding, inside the supplied one-coordinate ansatz. At $L_{*,n}$,

$$\mathcal{J}_{\text{loop}} L_{*,n} = \frac{2\pi^2 K_\theta^{1D} n^2}{L_{*,n}} = \pi |n| \iota_{0,\text{bal}}^{\text{eff}}. \quad (\text{AS.4})$$

This equality of the two contributions is why the total balanced coefficient is twice the phase-only coefficient evaluated at the same selected length, as stated for the elementary $n = 1$ sector in Eq. (5.30).

For $n = 0$, however, $\mathcal{J}_{n=0}(L) = \mathcal{J}_{\text{loop}}L$ has infimum zero as $L \rightarrow 0^+$ and no finite interior minimum. Moreover $\mathcal{J}_{n,\min} = |n|\mathcal{J}_{1,\min}$, so this two-term model alone leaves one multiwinding loop marginally degenerate with $|n|$ unit-winding loops; splitting and binding require additional dynamics.

AS.2 Ownership, non-double-counting and falsifiers

The added term makes a finite conditional minimum possible in the supplied model. It is not yet an ECT derivation. A valid descent must satisfy all of the following conditions:

1. derive the compact $U(1)$ collective coordinate and its admissible winding sectors rather than inserting them into the real P3 scalar;
2. derive K_θ^{1D} and $\mathcal{J}_{\text{loop}}$ from the same vacuum-subtracted transverse profile, normalisation and coarse-graining map, with a non-overlapping reduction of the form $\mathcal{J}_{4\text{D},\text{red}}^{\text{sub}} \rightarrow \int ds [\mathcal{J}_{\text{loop}} + K_\theta^{\text{1D}}(\partial_s \theta)^2/2 + \dots]$, with a separate positive physical-action factor, so that the longitudinal winding energy is not counted again inside the tension;
3. derive the correct worldline or worldsheet embedding and its declared ensemble, while retaining shape modes, curvature, collapse, splitting, self-intersection, radial relaxation and backreaction at the order needed to test the full constrained Hessian and whether the minimum survives;
4. test fragmentation and multi-loop channels explicitly: because $\mathcal{J}_{n,\min} = |n|\mathcal{J}_{1,\min}$, a single $|n| > 1$ loop is marginally degenerate in the two-term model with $|n|$ separated unit-winding loops; the interactions that decide splitting are absent from Eq. (AS.2);
5. show that the $n \neq 0$ loop cannot unwind through a finite-action path excluded by the reduced ansatz;
6. demonstrate that the resulting scale is independent of arbitrary parametrisation, infrared box size and subtraction convention; and
7. establish cross-sector universality before comparing the candidate with the measured $\hbar$.

AS.3 Four-dimensional owner forks

Codimension-three worldline route. The d^3y reduction used in the main text is geometrically consistent with a localised object on a selected three-dimensional slice whose history is a curve in the four-dimensional Euclidean carrier. If its stationary profile is $\Psi_*^I(y; \vartheta)$ and its physical moduli space contains $\vartheta \in S^1$, insertion of the slowly varying ansatz into one vacuum-subtracted action gives Eq. (5.31). In this route, $\mathcal{J}_{\text{loop}}$ is the reduced inverse-length coefficient and K_ϑ^{1D} is the positive reduced moduli-space metric. Only $\Sigma_\vartheta \mathcal{J}_{\text{loop}}$ is the Euclidean physical action per worldline length, and only $\Sigma_\vartheta K_\vartheta^{\text{1D}}$ has dimensions action times length. The proper length L is varied only in a declared closed-worldline/proper-time ensemble; it is not the texture radius and is external for propagation over a prescribed duration. A reparametrisation-invariant representative makes this ensemble guard explicit. With $\tau \in [0, 1]$, einbein $e(\tau) > 0$, $L = \int_0^1 e d\tau$ and $\vartheta(1) - \vartheta(0) = 2\pi n$, the leading dimensionless modulus functional is

$$\mathcal{J}_{\text{wl}}[e, \vartheta] = \int_0^1 d\tau \left[\mathcal{J}_{\text{loop}} e + \frac{K_\vartheta^{\text{1D}}}{2e} \left(\frac{d\vartheta}{d\tau} \right)^2 \right], \quad S_{\text{wl,phys}} = \Sigma_\vartheta \mathcal{J}_{\text{wl}}. \quad (\text{AS.5})$$

The ϑ equation gives $K_\vartheta^{\text{1D}} \dot{\vartheta}/e = \text{const}$ and therefore $\dot{\vartheta} = 2\pi n e/L$; substitution yields exactly Eq. (5.35). This proves that the two-term functional is compatible with worldline reparametrisation invariance. It does not make its stationary L physical: the proper-time zero mode must actually belong to the declared measure or variational ensemble rather than being gauge or externally fixed. Any cross-term must be retained unless its vanishing is proved after the profile equations, constraints, zero-mode quotient and relaxation are imposed; an informal “background–fluctuation orthogonality” is not a substitute for that calculation.

A three-dimensional texture with nonzero $\pi_3(S^3)$ is the natural topological candidate for the carrier, while a compact normalisable collective coordinate can supply the longitudinal phase. This is not yet a solution of the live theory. The minimal two-derivative texture collapses, and in the static exact-gradient branch $Q_A = \partial_A \Phi$ with fixed $Q_w = u_0$ the normalised orientation remains in one contractible hemisphere. A nontrivial carrier therefore requires either w -dependence, access to $Q = 0$, a stabilising higher-gradient term, or an independent orientation completion. The carrier charge and the longitudinal winding must not be identified.

External proof-of-class toy, not an ECT route. Standard Skyrme-type models show that an *independent* unit-orientation field with a separately supplied positive four-derivative operator can evade the two-derivative Derrick collapse [40]. This is only an external generic-EFT existence analogy. Before it could enter an ECT owner chain, the independent field and coefficient would have to be derived or explicitly supplied, and the actual profile, physical quotient, compact modulus and constrained Hessian would still have to be solved. No explicit Skyrme completion is adopted in the present T1 route.

Codimension-two vortex-worldsheet route. A compact $U(1)$ vortex has codimension two. In four Euclidean dimensions it is a two-dimensional worldsheet. A spatial vortex loop that persists for a w -interval Δw reduces instead as

$$S_E = \Delta w \left[\tau L + \frac{2\pi^2 K_{ws} n^2}{L} + \dots \right]. \quad (\text{AS.6})$$

The bracket is an energy-like functional or action per unit w -duration. Turning it into the worldline action used in Eq. (5.27) requires an explicit clock, worldsheet state and dimensional reduction. Consequently an ordinary vortex ring cannot be used as an unqualified owner of the existing $\mathcal{A}_\perp \propto \xi^3$ formula. Superconducting-string models are an external example of the additional carrier/internal-current split required for such a worldsheet route [39].

Why constrained orientation does not itself produce a compact modulus. For the minimal scalar, $Q = d\Phi$ leaves a longitudinal real fluctuation rather than a compact angle. Rotations that leave one chosen director fixed rotate an auxiliary transverse frame; without an additional material order parameter this is basis freedom, not a physical S^1 . Moreover a Hopf circle embedded in S^3 is contractible inside the full S^3 target. Therefore compactness must come from the supplied BR1 phase, a physical collective coordinate of a derived defect, or another explicit completion. It is not a consequence of integrability alone. No new foundational compactness postulate is adopted here: the least invasive conditional test is whether $\theta|_\Gamma$, or another modulus of an explicit owner solution, survives as a normalisable physical S^1 after the constraint, gauge, frame and zero-mode quotient. The longitudinal scalar χ remains real and is not identified with that modulus.

AS.4 Elastic completion, shape and splitting

At fixed L , the tension and uniform-winding terms depend only on length; they do not select shape. If the same owner yields the positive bending term in Eq. (5.31), Fenchel's theorem and Cauchy–Schwarz give

$$\oint \mathcal{K}^2 ds \geq \frac{4\pi^2}{L}, \quad (\text{AS.7})$$

with equality for a once-covered planar circle. This proves the circular minimum inside the positive-bending elastic model and yields Eq. (5.39). It also shows why the original two-term value cannot be retained when κ_b is material. The same equation gives a conditional finite $n = 0$ minimum and makes the minimum nonlinear in $|n|$, thereby lifting the bare splitting marginality even before profile interactions are restored.

The relation $\mathcal{J}_{n,\min} = |n|\mathcal{J}_{1,\min}$ is an exact marginal degeneracy, not a stability theorem. Attraction, repulsion and a BPS-like marginal regime are all possible in admissible completions. Even repulsion may favour separated unit-winding carriers rather than selecting one loop. The sign of the coincident-core binding term, the allowed splitting channel and the phase-slip barrier must be computed in the same reduced model.

AS.5 Measure, nucleation and the limited role of PES-R

The stationary loop of the two-term functional is not automatically a nucleation bounce. A semiclassical channel weight has the schematic form

$$\Gamma_n \sim \mathcal{P}_n \exp\left[-\frac{B_n}{S_{\text{measure}}}\right], \quad (\text{AS.8})$$

where B_n and S_{measure} are independently owned physical actions, the former has the required bounce and negative mode, and $\mathcal{P}_n$ contains determinant and zero-mode factors. The expression $e^{-2\pi|n|}$ follows only after imposing $B_n = \Sigma_{\vartheta}\mathcal{J}_{n,\min}$, $S_{\text{measure}} = \Sigma_{\vartheta}\iota_{0,\text{bal}}^{(\vartheta)}$ and an n -independent prefactor. None of those identifications is derived by the two-term minimum.

Likewise, a loop-gas proliferation threshold balances physical action against configurational entropy. Let b be a declared lattice/coarse-graining length, S_{measure} the action denominator, and μ the connective factor of the specified asymptotic ensemble. Its marginality condition is

$$\frac{\Sigma_{\vartheta}\mathcal{T}_{\text{loop}}b}{S_{\text{measure}}} = \ln \mu. \quad (\text{AS.9})$$

Only after independently imposing $\Sigma_{\vartheta}\mathcal{J}_{1,\min}/S_{\text{measure}} = 2\pi$ does one obtain

$$\frac{L_*}{b} = \frac{\pi}{\ln \mu}. \quad (\text{AS.10})$$

For representative $\mu = 6.774\text{--}8$, the latter gives $L_*/b = 1.51\text{--}1.64$ and therefore cannot be evaluated consistently with the large-loop asymptotics that produced μ ; it is also outside a controlled thin-curve regime when the core width is of order b . Equation (AS.10) is thus a UV-control failure of the combined assumptions, not a continuum prediction or a derivation of a_{measure} , S_0 or $\hbar$. The regulator, self-avoidance, minimum allowed polygon, curvature and interactions all remain model data. PES-R can test an already supplied mode for admissibility, finite-time retention and record formation. It neither supplies the Euclidean measure nor selects exact marginality, and non-normal dynamics can make an almost marginal mode transiently amplified or noise dominated rather than robust.

AS.6 Finite acceptance ladder for the T1 programme

Table 140: Owner gates for the tension-balanced T1 route.

Gate	Required calculation	Fail condition
Carrier	Derive a localised finite-action profile and its codimension from one stationary ordered-phase action	No stationary profile, Derrick collapse, or mismatch between worldline and worldsheet reduction

Gate	Required calculation	Fail condition
Length/ensemble	Define worldline proper time versus spatial circumference, the einbein or parametrisation convention, boundary period, state and measure; derive the proper-time zero mode and prove why L is varied	L is an externally prescribed duration, or the einbein zero mode is gauge or absent from the physical measure
Compact modulus	Identify a physical S^1 collective coordinate after quotienting gauge/frame zero modes	The angle is contractible, pure frame choice, or absent from the physical reduced space
Common owner	Compute $\mathcal{T}_{\text{loop}}$ and $K_{\vartheta}^{\text{1D}}$ from the same vacuum subtraction and profile	Either coefficient is nonpositive, convention dependent, or double counted
Projected stability	Evaluate the full constrained Hessian, including profile relaxation and zero modes	A negative physical mode survives or the Schur complement is nonpositive
Shape	Derive bending and higher-gradient terms	Shape is unstable or the induced correction invalidates the two-term scale
Topology/barrier	Compute phase-slip and carrier-unwinding paths, barrier and lifetime in a declared state and clock	Any finite path defeats <i>absolute</i> protection; metastability fails only if the calculated lifetime is below the declared observation window/tolerance
Splitting	Compute coincident-core binding and multi-carrier interactions	The chosen elementary sector fragments or has no isolated state
Measure	Derive a bounce, determinant, zero-mode measure and initial state	The stationary loop is not the relevant saddle or weights are normalisation dependent
PES-R	Supply Lorentzian dynamics, state, width/noise and record protocol	The mode is inadmissible, too short lived or record incompatible
Universality	Compare the same operational action-scale map across at least two independently normalised sectors	Sector-dependent values or an uncontrolled matching convention

AS.7 Future calculations conditional on closing the owner, profile, Hessian and barrier gates

Only after the explicit owner, stationary profile, physical quotient, projected Hessian and phase-slip/carrier barrier have passed the table above would the T1 route support applications. The profile calculation would then provide a rest-action density $\mathcal{T}_{\text{loop}}$, an internal inertia $K_{\vartheta}^{\text{1D}}$, a bending modulus and a barrier. Those outputs could support the following conditional future tests:

1. a Lorentzian Hamiltonian/eigenmode spectrum only after a supplied continuation, symplectic structure, operator/domain and state; this is not the rejected $e^{-2\pi|n|}$ sector-weight template;

2. a quantitative phase-slip or unwinding lifetime, rather than an assumed topological permanence;
3. a binding/fragmentation phase diagram comparing a multiwinding carrier with separated unit carriers;
4. a cross-sector universality test of $\sqrt{2K_{\theta}^{\text{1D}} \mathcal{T}_{\text{loop}}}$, using one operational normalisation protocol in at least two independently derived sectors;
5. a possible action-scale matching test only if an independently normalised source, measure or canonical datum breaks the common rescaling degeneracy and the same value is recovered in held-out sectors; and
6. a PES-R application to the derived carrier's poles, widths, transient gain, environmental records and survival time.

Failure of the final two items would not erase a sector-specific stable texture; it would only reject its interpretation as the universal quantum of action or as a persistent record carrier.

Non-positive coefficients or non-negligible L -dependence invalidate the closed-form minimum above and require the derived functional to be re-minimised. Loss of every stable minimum under profile and shape relaxation, or an admissible finite-action unwinding path, falsifies the stabilised-loop route. Sector-dependent values of $\sqrt{2K_{\theta}^{\text{1D}} \mathcal{T}_{\text{loop}}}$ falsify a universal action scale, but do not by themselves falsify sector-specific loop stabilisation. Conversely, a successful 4D solution would upgrade the ownership of the balanced scale but would still require an independent physical normalisation and quantum-sector map before $\Sigma_{\theta} \iota_{0,\text{bal}}^{\text{eff}} = \hbar$ could be claimed.

AT Conditional canonical-normalisation result for a supplied local-CAR fermionic EFT

Status: Level A mathematics conditional on a supplied local first-order fermionic EFT, positive kinetic coefficient and CAR algebra. ECT has not derived that fermionic completion from P1–P6. No universal curvature-controlled mismatch estimate or astrophysical conclusion is identified. Open (OP-Q21) for the full theorem, contingent on the completion of the fermionic reconstruction programme (OP-Q11). Connections: §24.4 (exchange sectors and spinor compatibility), §24.5 (representation-theoretic route to Dirac structure), §7.6 (ECT fermion–condensate coupling μ_5).

This appendix collects the technical material underlying the provisional no-go statement of Section 24.4: smooth local inhomogeneity does not deform an already supplied local CAR algebra merely through a positive kinetic normalisation. This is not a derivation of Pauli exclusion. The appendix proceeds in three steps. First, we write a representative local first-order fermionic EFT compatible with the presently identified ECT structures (§AT.1). Second, we show that a condensate-dependent kinetic coefficient is absorbed into a local canonical normalisation, leaving the equal-time anticommutator in standard form (§AT.2). Third, we show why the equivalence principle alone fixes neither curvature-dependent fermion operators nor a packet-level scaling, so numerical laboratory or compact-object conclusions are not identifiable without additional owners (§AT.3).

AT.1 General local fermionic EFT on the coherent ordered branch

On the smooth coherent ordered branch, a representative local first-order fermionic EFT compatible with the presently identified ECT structures—the conditionally supplied Dirac dynamics (§24.5), medium-ontological identification of fermionic excitations with spinor modes of the ordered condensate (§9.9),

and the ECT-specific preferred-direction coupling (§7.6)—can be written schematically as

$$\begin{aligned}
S_\psi^L = \int d^4x e(x) & \left[\frac{i}{2} Z(\phi, n) \bar{\psi}(x) \gamma_L^\mu e_a^\mu(x) \overleftrightarrow{D}_\mu \psi(x) \right. \\
& - M(\phi, n) \bar{\psi}(x) \psi(x) \\
& + \mu_5(\phi, n) u_\mu(x) \bar{\psi}(x) \gamma_L^\mu \psi(x) \\
& \left. + \sum_k C_k(\phi, n) \mathcal{O}_k^{(4f)}(x) + \dots \right],
\end{aligned} \tag{AT.1}$$

where $e(x) = \sqrt{-\det g_L(x)}$ belongs to a separately supplied mostly-plus Lorentzian metric/tetrad completion. Its curved matrices are $\Gamma^\mu = e_a^\mu \gamma_L^a$ with $\{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}$. The function $Z(\phi, n)$ is smooth and strictly positive, $M(\phi, n)$ is an effective mass, and $\mu_5 u_\mu \bar{\psi} \gamma_L^\mu \psi$ is the Lorentzian preferred-direction operator discussed in Section 7.6. For the declared future-oriented rest vector it equals $-\mu_5 j^0$; the coefficient sign remains Open. The $\mathcal{O}_k^{(4f)}$ are local four-fermion composite operators admitted by the supplied symmetries.

The working assumption of this appendix is that the background fields are smooth (all spatial and temporal derivatives are finite and of order the characteristic condensate scales), the kinetic coefficient is nondegenerate ($Z(\phi, n) > Z_{\min} > 0$ and $e(x) > 0$ everywhere in the region of interest), there are no higher-time-derivative fermionic kinetic pathologies, and no nonlocal sector-mixing kernels coupling the ordered-branch fermionic sector to its exterior. These are precisely the conditions defining “smooth coherent ordered branch” for the purposes of the no-go statement.

AT.2 Canonical normalisation and the equal-time anticommutator

Extracting the temporal part of the kinetic term in (AT.1) yields, up to the choice of local frame and temporal foliation,

$$\mathcal{L}_{\text{kin}}^{(t)}(x) \sim i \mathcal{K}(x) \psi^\dagger(x) \partial_t \psi(x) + \dots, \quad \mathcal{K}(x) > 0, \tag{AT.2}$$

where $\mathcal{K}(x)$ is the resulting smooth and strictly positive temporal kinetic coefficient, and the ellipsis denotes terms involving spatial derivatives and mass/interaction pieces that do not contribute to the equal-time canonical momentum. In the schematic ECT EFT of Eq. (AT.1) one has $\mathcal{K}(x) \sim e(x) Z(\phi(x), n(x))$ up to the choice of local frame and temporal foliation, so that all statements below apply to this EFT as a particular case. The canonical momentum conjugate to ψ is then

$$\Pi_\alpha(x) = \frac{\partial \mathcal{L}}{\partial(\partial_t \psi_\alpha(x))} = i \mathcal{K}(x) \psi_\alpha^\dagger(x), \tag{AT.3}$$

and the equal-time canonical anticommutation relation

$$\{\psi_\alpha(t, \mathbf{x}), \Pi_\beta(t, \mathbf{y})\} = i \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}), \tag{AT.4}$$

which in the present normalisation takes the background-weighted form

$$\{\psi_\alpha(t, \mathbf{x}), \psi_\beta^\dagger(t, \mathbf{y})\} = \mathcal{K}(x)^{-1} \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \tag{AT.5}$$

Equation (AT.5) is the technical object that a naive gradient-based argument would interpret as a modification of the exclusion algebra in the presence of a spatially varying condensate. The correct reading, however, is that ψ is simply not the canonically normalised field. The local field redefinition

$$\chi_\alpha(x) \equiv \mathcal{K}(x)^{1/2} \psi_\alpha(x) \tag{AT.6}$$

brings the kinetic term to canonical form and yields, at equal times,

$$\{\chi_\alpha(t, \mathbf{x}), \chi_\beta^\dagger(t, \mathbf{y})\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}), \tag{AT.7}$$

which is the standard canonical anticommutation relation in its unmodified form. Creation operators built from χ then satisfy $(\hat{a}_i^\dagger)^2 = 0$ mode by mode, and two-fermion states constructed from them are exactly antisymmetric.

Three comments are in order.

(i) Condensate-dependent mass and preferred-direction coupling do not alter the exclusion algebra.

The condensate-dependent mass term $M(\phi, n)\bar{\psi}\psi$ and the candidate Lorentzian preferred-direction operator $\mu_5(\phi, n)u_\mu\bar{\psi}\gamma_L^\mu\psi$ do not contribute to the temporal kinetic term (AT.2) and hence do not modify the canonical momentum or the equal-time anticommutator. They modify the Dirac operator and the local mode functions, shifting energies, effective masses, and transport coefficients, but they do not act on the exclusion algebra itself.

(ii) Local four-fermion operators modify dynamics, not statistics. Local four-fermion operators $C_k(\phi, n)\mathcal{O}_k^{(4f)}$ are also absent from the temporal kinetic term at leading order in the first-order formulation, and therefore do not modify the canonical momentum conjugate to ψ . They can shift spectra, drive pairing or condensation phenomena, and modify effective equations of state, but on the operator-algebra level the fermionic exclusion structure is fixed by the kinetic term through the canonical anticommutation relation ((AT.7)), not by the interaction sector. This is a standard feature of canonical fermionic field theory.

(iii) Linear gradient contributions cancel for symmetric packets. A naive gradient-based argument would proceed by expanding the background around a point $\mathbf{x}_0$ and retaining the linear term $\partial_i \ln \mathcal{K}(x)$. For a symmetrically localised one-particle wave packet, however, the linear expansion contributes with vanishing first moment, $\langle y^i \rangle \equiv \langle (x - x_0)^i \rangle = 0$, and the leading physical correction begins only at second order in the background inhomogeneity. Furthermore, even this second-order correction is a packet-level mismatch of mode overlap integrals, not a violation of the equal-time anticommutator (AT.7).

These three observations close the canonical-normalisation argument at the level of the present local first-order EFT.

AT.3 Scope of curvature-dependent terms

The equivalence principle permits removal of the connection at a point but does not determine the allowed curvature-dependent operators in a physical fermion action. Such operators may enter at different orders with independent Wilson coefficients. Therefore neither a universal packet mismatch proportional to $(\mathcal{E}L^2/c_{\text{char}}^2)^2$ nor any compact-object numerical conclusion follows from the conditional canonical-normalisation argument.

The result established here is narrower: after a smooth, local, invertible first-order fermionic EFT with the standard CAR algebra is supplied, canonical normalisation preserves that algebra. Physical ECT fermions, the CAR owner, curvature operators and coefficients, states and observable maps remain Open. No white-dwarf, neutron-star, VIP-2, Borexino or Ramberg–Snow comparison is made without those owners.

Relation to the existing fermionic programme

The no-go statement developed here complements the existing fermionic programme of the main text in three ways. First, it combines the exchange-topology classification of Section 24.4 with a conditional no-go inside a supplied local CAR theory. It neither derives the CAR algebra nor physical fermions from P1–P6. Second, it is consistent with a supplied spinorial/Dirac completion, on which the present canonical-normalisation argument operates. Third, it makes explicit the connection between the ECT preferred-direction coupling, written as $\mu_5 u_\mu \bar{\psi} \gamma_L^\mu \psi$ in the supplied Lorentzian completion, and the statistics sector of Section 7.6: μ_5 contributes to dynamics, not to the exclusion algebra. The full theorem—unconditional exactness of Pauli exclusion throughout the coherent branch, or an explicit identification of the nonlocal, topological-defect, or branch-transition mechanism responsible for any departures—remains a distinct target (OP-Q21) contingent on the completion of the fermionic reconstruction programme (OP-Q11).

AU Osterwalder–Schrader spin–statistics test for the free minimal Euclidean Dirac kernel

Status: Level A mathematical result (Q11.1a) for the free minimal Euclidean Dirac kernel, in the conventions of §23.1 (eqs. 23.4–23.8); the ECT application remains Level B conditional (Q11.1b). Connection: §9.9, §23.1, §9.2.

Throughout, $m > 0$, $\omega(\vec{p}) = \sqrt{\vec{p}^2 + m^2}$, and the reordering rule

$$\langle \bar{\psi}_\beta(y) \psi_\alpha(x) \rangle = \sigma S_{E,\alpha\beta}(x-y), \quad \sigma = \begin{cases} -1 & \text{Grassmann,} \\ +1 & \text{commuting,} \end{cases} \quad (\text{AU.1})$$

is the only point at which the statistics assignment enters (σ is local notation of this appendix). Define the hermitian matrices $h_\pm(\vec{p}) = m\gamma_0 \pm i\gamma_0 \vec{\gamma} \cdot \vec{p}$; hermiticity follows from $(i\gamma_0 \vec{\gamma})^\dagger = -i\vec{\gamma}\gamma_0 = +i\gamma_0 \vec{\gamma}$, and $h_\pm^2 = \omega^2$ (cross terms cancel by $\{\gamma_0, \vec{\gamma}\} = 0$), so $(\omega \pm h_\pm)/2\omega$ are orthogonal rank-2 projectors. At the quadratic level the alternative order assignment $(F, G) = \langle \Theta F \cdot G \rangle$ differs from (23.8) by the global reordering sign σ on the elementary families considered below. The convention (23.8) is the one for which both elementary Grassmann families are reflection-positive; the final identification with standard Euclidean Fermi-field OS conventions is part of the convention-reconciliation step recorded at the end of this appendix.

Contour lemma

For $W > 0$ and either sign $s = \pm$:

$$\int \frac{dp_0}{2\pi} \frac{e^{is p_0 W} (m - i\gamma_0 p_0 - i\vec{\gamma} \cdot \vec{p})}{p_0^2 + \omega^2} = \frac{e^{-\omega W}}{2\omega} (m + s\omega\gamma_0 - i\vec{\gamma} \cdot \vec{p}), \quad (\text{AU.2})$$

by closing in the upper ($s = +$, pole $p_0 = i\omega$) or lower ($s = -$, pole $p_0 = -i\omega$) half-plane.

Grassmann case: both elementary families are reflection positive

Family I. $F = \int_{w>0} d^4x f^\dagger(x) \psi(x)$, so $\Theta F = \int_{w>0} d^4y \bar{\psi}(\Theta y) \gamma_0 f(y)$ and

$$\mathcal{Q}^I[f] = \langle F \Theta F \rangle = \int \int d^4x d^4y f^\dagger(x) S_E(x - \Theta y) \gamma_0 f(y), \quad (\text{AU.3})$$

with no reordering (fields already ordered as in the pairing). Since $x - \Theta y = (w_x + w_y, \vec{x} - \vec{y})$, the contour lemma (AU.2) with $s = +$ factorises the w -dependence through $h_f(\vec{p}) = \int_0^\infty dw e^{-\omega w} \tilde{f}(w, \vec{p})$:

$$\mathcal{Q}_{\text{Gr}}^I[f] = \int \frac{d^3p}{(2\pi)^3} h_f^\dagger(\vec{p}) \mathcal{M}_1(\vec{p}) h_f(\vec{p}), \quad \mathcal{M}_1 = \frac{(m + \omega\gamma_0 - i\vec{\gamma} \cdot \vec{p})\gamma_0}{2\omega} = \frac{\omega + h_+(\vec{p})}{2\omega} \geq 0. \quad (\text{AU.4})$$

Family II. $G = \int_{w>0} d^4x \bar{\psi}(x) g(x)$, so $\Theta G = \int_{w>0} d^4y g^\dagger(y) \gamma_0 \psi(\Theta y)$. In components,

$$G \cdot \Theta G = \int \int d^4x d^4y g_\alpha(x) g_{\sigma'}^*(y) (\gamma_0)_{\sigma'\beta} \bar{\psi}_\alpha(x) \psi_\beta(\Theta y), \quad (\text{AU.5})$$

and the expectation with the reordering rule (AU.1), $\langle \bar{\psi}_\alpha(x) \psi_\beta(\Theta y) \rangle = \sigma S_{E,\beta\alpha}(\Theta y - x)$, gives

$$\mathcal{Q}^{\text{II}}[g] = \sigma \int \int d^4x d^4y g^\dagger(y) \gamma_0 S_E(\Theta y - x) g(x). \quad (\text{AU.6})$$

Here $\Theta y - x = (-(w_x + w_y), \vec{y} - \vec{x})$, so (AU.2) applies with $s = -$ and momentum $\vec{q}$ conjugate to $\vec{y} - \vec{x}$:

$$\gamma_0 S_E(\Theta y - x) \Big|_{\vec{q}} = e^{-\omega W} \frac{m\gamma_0 - \omega - i\gamma_0 \vec{\gamma} \cdot \vec{q}}{2\omega} = e^{-\omega W} \frac{h_-(\vec{q}) - \omega}{2\omega}, \quad (\text{AU.7})$$

and after the Fourier transform and relabelling of dummy variables (the $e^{-\omega w_x}$, $e^{-\omega w_y}$ factors attach to $g(x)$ and $g^\dagger(y)$ respectively, producing the same half-space profile h_g):

$$\mathcal{Q}^{\text{II}}[g] = \sigma \int \frac{d^3 q}{(2\pi)^3} h_g^\dagger(\vec{q}) \frac{h_-(\vec{q}) - \omega}{2\omega} h_g(\vec{q}). \quad (\text{AU.8})$$

For Grassmann fields ($\sigma = -1$):

$$\mathcal{M}_{\text{II}}^{\text{Gr}}(\vec{q}) = \frac{\omega - h_-(\vec{q})}{2\omega} \geq 0 \quad (\text{orthogonal projector}). \quad (\text{AU.9})$$

The spatial-momentum orientation differs between the families (h_+ vs h_-); both conclusions are orientation-independent: positivity is the projector property, and the converse below uses only $\text{spec } h_- = \{\pm\omega\}$. Extension from the two elementary quadratic families to the full graded polynomial algebra is the standard OS fermionic construction and is cited, not re-derived, here.

Commuting case: explicit negative modes

The strict negative-mode test assumes $m > 0$; the massless case requires separate infrared treatment and is not part of Q11.1a. Family II with $\sigma = +1$ gives

$$\mathcal{M}_{\text{II}}^{\text{com}}(\vec{q}) = \frac{h_-(\vec{q}) - \omega}{2\omega}, \quad \text{spec } \mathcal{M}_{\text{II}}^{\text{com}} = \{0, -1\} \text{ (each twice)}. \quad (\text{AU.10})$$

Explicit negative-norm test function. Take $\vec{q} \approx 0$, a spinor u_- with $\gamma_0 u_- = -u_-$ (so $h_-(0)u_- = -\omega u_-$), and $g(w, \vec{x}) = \chi(w)\rho(\vec{x})u_-$ with χ supported in $w > 0$ and ρ a broad spatial envelope; then $\mathcal{Q}_{\text{com}}[g] \rightarrow -|h_g(0)|^2 < 0$. Family I remains positive for $\sigma = +1$ (no reordering occurs there); reflection positivity nevertheless fails because the OS form must be non-negative on *all* test functionals. *Mechanism.* The statistics assignment enters in exactly one place: the reordering sign σ of (AU.1). Grassmann statistics converts the Family-II kernel into the projector $(\omega - h_-)/2\omega$; commuting statistics leaves $(h_- - \omega)/2\omega$ with spectrum $\{0, -1\}$. *Diagnostic remark (not part of the proof).* A separate consistency route would require constructing a positive commuting Gaussian covariance with spinor indices; if enforcing ordinary symmetric covariance removes the first-order Dirac numerator, the resulting object is no longer the same minimal kernel.

Convention reconciliation and scope

The conventions of §23.1 lie in the standard Euclidean Fermi-field OS family [316, 317, 318], with γ_0 playing the role usually denoted γ_4 . The reconciliation has been performed at the level of the defining data and is recorded here. (a) *Reflection action.* The trial action (23.7) matches the standard form of fermionic time reflection used in Euclidean/lattice constructions, $(\theta\bar{\psi})(x) = \gamma_0\psi(\theta x)$, $(\theta\psi)(x) = \bar{\psi}(\theta x)\gamma_0$, antilinear and order-reversing, up to notation and the identification of the Euclidean time gamma (γ_0 here, often denoted γ_4); see, e.g., [319] for an explicit lattice Euclidean convention of this reflection type. (b) *Order and pairing.* Standard formulations state reflection positivity as $\langle\langle\Theta F\rangle F\rangle \geq 0$, with action-sign, measure-ordering, and pairing conventions that vary between sources, whereas the present free-kernel test uses $\langle F\Theta G\rangle$ with a supplied normalised Berezin Gaussian ($\kappa_F = 1$ in the printed covariance convention) and pairing $\langle\psi\bar{\psi}\rangle = +S_E$, where here $S_E = D_E^{-1}$ is the Euclidean Dirac kernel, not an action. A general finite $\kappa_F > 0$ rescales the covariance and both OS forms by positive factors without changing the sign verdict; it is a separate source-model owner. At the quadratic level these two assignments differ by one global sign from the pairing convention and one from the reordering. With the order convention (23.8), these signs combine to the same projector-positive branch on the two elementary Grassmann families analysed above. This is the global- σ equivalence noted at the start of this appendix. The page-level comparison with the original OS Fermi-field constructions, which completes this convention check, is recorded in item (c) below. (c) *Page-level verification (completed).* The conventions

have been verified directly against the original constructions. In [317]: hermitian Euclidean gamma matrices, $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$, $\gamma_\mu^\dagger = \gamma_\mu$ (footnote 5, p. 286, with an explicit chiral representation in his eq. (32)); the reflected-first form $(\mathcal{O}_1, \mathcal{O}_2) = \langle \theta(\mathcal{O}_1^+) \mathcal{O}_2 \rangle$ with the conjugation example $\psi^+ = \gamma_0 \bar{\psi}$ (eqs. (4)–(5), p. 284) and Berezin weight e^{+A} (eqs. (1),(6)); strict positivity for the Grassmann Wilson-fermion sector (Propositions 1–2, pp. 287–289). In [316]: the doubled-field construction (§3.2, pp. 284–285), hermitian Euclidean gammas $\gamma_{E,\text{src}}^0 = \gamma_L^0$, $\gamma_{E,\text{src}}^j = +i\gamma_L^j$ (p. 282), the time-inversion involution exchanging the doubled species with the matrix γ_0 (eqs. (4.1)–(4.3), pp. 290–291), the reflected-first form $\langle X, Y \rangle = (\Theta X, Y)$ with positivity (Lemma 4.1, p. 292, and §6), and the explicit remark that the adjoint of the non-hermitian fermionic action is implemented by Euclidean time inversion (Introduction, item III, p. 278). Relative to the house map $\gamma_L^j = i\gamma_{E,\text{house}}^j$, the source uses $\gamma_{E,\text{src}}^j = -\gamma_{E,\text{house}}^j$ with the same time gamma. The two Euclidean representations are unitarily equivalent: conjugation by γ_0^E leaves γ_0^E fixed and flips all spatial gammas. The order/action-sign differences relative to the present text reduce to the two global signs documented in (b). This verification concerns the Euclidean spinor/OS conventions and the Grassmann reflection-positive branch. The negative-mode exclusion of the commuting/symmetric alternative is the explicit free-kernel calculation given above. The ECT application remains conditional on deriving or verifying the ECT spinor kernel and spinorial OS structure. Scope: minimal first-order kernel; $m > 0$; pointlike primary-branch excitations; no higher-derivative kernels; Majorana/charge-conjugated reformulations would require translating the reflection matrix and adjoint. The ECT application (Q11.1b) and the ordered-branch descent (Q11.2) remain Level B conditional (§9.9).

AV Unified vacuum-response logic: boundaries, acceleration, time dependence, and horizons

Status: Open (OP-Q12) for the unified ECT response claim. Each displayed standard/toy calculation can be exact inside its own supplied assumptions, but a common physical bundle, state, continuation and coupling that derive all four effects from the reduced ECT action are missing.

The four effects discussed in Sections 25.3–25.4 and 36—Casimir-type boundary response, particle production in time-dependent backgrounds, Unruh-type thermality, and Hawking-type horizon thermality. These are supplied comparison targets. It is Open whether they are probes of one physical ECT response operator.

The unifying logic

In each supplied calculation the mathematical question is how a stated state and operator respond when accessibility conditions change:

- (i) **Boundary problems:** part of the mode spectrum is excluded geometrically; the vacuum response appears as a boundary-dependent shift in effective energy (Casimir-type).
- (ii) **Time-dependent backgrounds:** mode definitions at early and late times no longer coincide; the mismatch appears as Bogoliubov-type particle production.
- (iii) **Accelerated observers:** kinematic access to the coherent vacuum is restricted observer-dependently; the reduced state takes an approximately thermal form (Unruh-type).
- (iv) **Horizons:** division into exterior, shell, and inaccessible sectors yields a reduced state after a physical split is supplied. Thermality requires an additional state and Hawking-channel owner; a partial trace alone does not imply it.

What is and is not claimed

ECT proposes, but does not yet prove, that these supplied effects can be obtained from one ordered medium under different restrictions on mode accessibility and coarse-graining. Presently they have

different owners: a scalar boundary BVP, a time-dependent mode action, a Lorentzian QFT state/detector, and a semiclassical strong-field metric. No stronger common-kernel conclusion follows until those owners descend from one reduced ECT action.

Analogue-gravity comparison

Analogue-gravity source models show that effective thermality and horizon response can arise in ordered media. This is external mechanism evidence inside those named models, not a Level-B ECT closure. The ECT common-response hypothesis remains Open until one ECT action, state, boundary/detector operators, observable maps and uncertainty model reproduce the channels and supply a discriminator against their separate standard owners.

AW Bath regimes, effective spectral structure, and the limits of the ohmic/Markov closure

Status: standard reduced-state bookkeeping for a declared channel; Level A only for algebra within the specified quadratic source model; Open for the ECT split, vertices, state, and quantitative bath interpolation (OP-Q15).

This appendix clarifies the status of the effective bath description used in Section 26.1. The main text employs a reduced influence-functional language in which a subsystem of coherent variables interacts with an effective environment. Such a split may be supplied after coarse-graining over selected coherent and topological degrees of freedom, but the physical subsystem/environment map and detailed bath model remain an effective closure rather than a first-principles theorem from bare P3.

Effective spectral density

The central quantity is the Weiss-normalised effective spectral density $J_W(\omega)$, which determines dissipation and noise kernels in the reduced dynamics. It is distinct from the bare mode-density weight by $J_W(\omega) = J_{\text{bare}}(\omega)/(2\omega)$ in the convention fixed in Appendix BE. In the simplest closure one assumes a smooth positive spectral density and, in the many-mode regime, an effectively Markovian coarse-grained behaviour. For a supplied three-dimensional soft-coordinate bath with derivative coupling, mode counting gives $J_{\text{bare}}(\omega) \propto \omega^4$ and therefore $J_W(\omega) \propto \omega^3$. This is Level A conditional mathematics inside that supplied bath model, not a derivation of a physical ECT bath, vertex or record channel. The effective ohmic/Markov approximation is a model ansatz. The general entropy-rate identity (26.9) has no fixed sign until the reduced generator and state are supplied.

Two regimes

Two regimes must be kept distinct:

Strong declared record channel. When the physical record differences and noise kernel make $\Phi_{ab}(T)$ large, off-diagonal reduced-state terms are strongly suppressed and records can become robust. The total mode count alone does not ensure this, and entropy monotonicity still requires a channel theorem.

Weak bath: $N_{\text{eff}} \sim 1$. When only very few effective environmental channels are active, coherent recurrences, partial reversibility, and incomplete decoherence remain possible. This is the regime in which the subsystem retains recognisably quantum behaviour (Section 26.1).

Table 141: Declared reduced-channel regime summary.

Regime	Effective condition	Behaviour	Status
Weak declared record channel	Small pairwise $\Phi_{ab}(T)$	Coherence may persist or revive; mode count alone is insufficient	Source-model / ECT Open
Strong declared record channel	Large pairwise $\Phi_{ab}(T)$	Selected off-diagonal terms are suppressed; entropy sign is not fixed	Source-model / ECT Open
Conditional entropy theorem	Declared unital, detailed-balance, or contractive channel plus state and entropy functional	Monotonicity only for the named functional under its hypotheses	Standard conditional theorem / ECT Open

Open problem

Generic subsystem–environment bookkeeping exists for a declared source model. What remains Open in ECT is the physical split, interaction vertices, record operators, state, noise kernel, reduced map, and entropy functional. Only after these are derived can a bath hierarchy or monotone entropy theorem be tested (OP-Q15/OP-Q17).

AX Mathematical apparatus for conditional decoherence calculations and future ECT record-channel tests

Status: Level A conditional mathematics / Level B calculational apparatus for a supplied subsystem–bath model; physical ECT ownership is Open. The generic Weiss spectral-density family, the noise and dissipation kernels, and the decoherence exponent are constructed only after a split, free bath, vertex and state are supplied. The specific exponent is parameterised rather than fixed by the live ECT action.

This appendix should be read as an operational extension of Appendix AW. The earlier appendix classifies the admissible bath regimes and the status of the ohmic/Markov closure; the present appendix supplies a generic calculational toolkit within that closure language. It does not upgrade the bath model to a first-principles theorem from bare P3.

AX.1 Generic subsystem–bath coupling class

As a generic reference calculation, suppose a supplied three-dimensional free bath has $N_G = 3$ independent linearly dispersing coordinate fields, $\omega_k = c_{\text{char}}|\mathbf{k}|$. This is the naive unconstrained orientation-bath benchmark, not the integrability-reduced physical spectrum of the live ECT action (Section 4.3). For a localised subsystem with generalised coordinate q , a separately supplied coupling to these modes takes the generic form

$$H_{\text{int}} = q \sum_{\alpha} g_{\alpha}(\mathbf{k}_{\alpha}) \chi_{\alpha}(\mathbf{x}_q), \quad (\text{AX.1})$$

where $g_{\alpha}(\mathbf{k})$ is the coupling vertex. No such physical ECT record vertex, resolved/unresolved split or bath state is identified here.

The bare mode-density weight and the Weiss/Feynman–Vernon spectral density must be kept distinct:

$$J_{\text{bare}}(\omega) = \mathcal{A}_b \omega^2 |g(\omega/c_{\text{char}})|^2 e^{-\omega/\omega_c}, \quad J_W(\omega) = \frac{J_{\text{bare}}(\omega)}{2\omega}, \quad \omega_c \sim c_{\text{char}}/\xi_{\text{cond}}. \quad (\text{AX.2})$$

The estimate $\omega_c \sim c_{\text{char}}/\xi_{\text{cond}}$ is only a Level C/parametric matching ansatz for this supplied source model. Neither the physical cutoff nor the dimensionful c_{char} is owned by the live ECT action. Here ξ_{cond} denotes

a separately declared bath cutoff length; it is not identified with ξ_{P3} , an Open P4 pole length, or a universal condensate scale. Within this fixed $d = 3$, $z = 1$, canonically normalised coordinate-field benchmark, the remaining exponent shift is controlled by the declared coupling vertex. In general it also depends on the physical mode projector, dispersion, dimensionality, canonical normalisation, degeneracy and infrared structure:

- density coupling, $g(k) \sim \text{const}$: $J_{\text{bare}}(\omega) \propto \omega^2$ and $J_W(\omega) \propto \omega$ (ohmic in the standard CL classification);
- derivative coupling, $g(k) \propto k$: $J_{\text{bare}}(\omega) \propto \omega^4$ and $J_W(\omega) \propto \omega^3$ (super-ohmic in the supplied three-dimensional coordinate-field bath with gradient interaction — this is the regime identified in the main text, eq. (26.5));
- an effective ohmic law $J_W(\omega) \propto \omega$ can arise from the constant-coupling benchmark in this declared normalisation, or from additional coarse-graining/reduced dimensionality applied to another vertex. No such physical ECT vertex is selected by eq. (26.6).

To maintain generality, the subsequent formulae are written for the generic Weiss power-law family

$$J_s(\omega) \equiv J_W(\omega) = A_s \omega^s e^{-\omega/\omega_c}, \quad (\text{AX.3})$$

with $A_s \geq 0$, $\omega_c > 0$ and $s > 0$. The direct examples are $s = 1$ (ohmic constant-coupling benchmark) and $s = 3$ (super-ohmic derivative-coupling benchmark) as the two direct cases above; other $s > 0$ remain admissible supplied effective families. None is thereby selected as the physical ECT record channel.

AX.2 Symmetrised force kernel $C_H(\tau)$ and action-normalised noise $N(\tau)$

Given the Weiss spectral density (AX.3), suppose first that the stationary Gaussian oscillator bath is in a declared thermal/KMS state at T_{eff} (with $k_B = 1$). Its physical symmetrised bath-force correlator and the action-normalised noise kernel are

$$C_H(\tau) = \frac{S_0}{\pi} \int_0^\infty d\omega J_s(\omega) \coth\left(\frac{S_0\omega}{2T_{\text{eff}}}\right) \cos(\omega\tau), \quad N(\tau) \equiv \frac{C_H(\tau)}{S_0}. \quad (\text{AX.4})$$

Thus C_H is a force–force covariance, whereas N is the kernel in the canonical PES normalisation of Eq. (29.10). Treating the spectral integral simultaneously as both quantities would introduce a spurious factor of S_0 . For a stationary non-thermal Gaussian state, the coth factor must be replaced by the corresponding symmetrised occupation spectrum; assigning a scalar “effective temperature” does not by itself establish the KMS condition.

Vacuum limit ($T_{\text{eff}} \rightarrow 0$)

Setting $\coth \rightarrow 1$, the integral evaluates in closed form:

$$C_{H,0}(\tau) = S_0 N_0(\tau) = \frac{S_0 A_s \Gamma(s+1) \omega_c^{s+1}}{\pi} \frac{\cos[(s+1) \arctan(\omega_c \tau)]}{(1 + \omega_c^2 \tau^2)^{(s+1)/2}}. \quad (\text{AX.5})$$

At $\tau = 0$: $C_{H,0}(0) = S_0 A_s \Gamma(s+1) \omega_c^{s+1} / \pi$ and $N_0(0) = C_{H,0}(0) / S_0$.

High-temperature limit ($T_{\text{eff}} \gg S_0 \omega_c$)

Using $\coth(x) \approx 1/x$ for $x \ll 1$:

$$C_{H,T}(\tau) = S_0 N_T(\tau) = \frac{2T_{\text{eff}} A_s \Gamma(s) \omega_c^s}{\pi} \frac{\cos[s \arctan(\omega_c \tau)]}{(1 + \omega_c^2 \tau^2)^{s/2}}. \quad (\text{AX.6})$$

Both formulae exhibit the supplied bath memory time $\tau_{\text{env}} \sim \omega_c^{-1}$. The further estimate $\tau_{\text{env}} \sim \xi_{\text{cond}}/c_{\text{char}}$ inherits the same Level C/parametric status as the cutoff ansatz above.

AX.3 Causal damping and dissipation kernels

For the velocity-memory form of the generalised Langevin equation, define the causal damping kernel

$$\gamma_R(\tau) = \frac{2}{\pi} \Theta(\tau) \int_0^\infty d\omega \frac{J_s(\omega)}{\omega} \cos(\omega\tau). \quad (\text{AX.7})$$

For the generic family (AX.3), this gives

$$\gamma_R(\tau) = \frac{2A_s \Gamma(s) \omega_c^s}{\pi} \Theta(\tau) \frac{\cos[s \arctan(\omega_c \tau)]}{(1 + \omega_c^2 \tau^2)^{s/2}}. \quad (\text{AX.8})$$

The coordinate–coordinate dissipation kernel is instead the sine transform of J_s itself,

$$\mu_R(\tau) = \frac{2}{\pi} \Theta(\tau) \int_0^\infty d\omega J_s(\omega) \sin(\omega\tau) = -\partial_\tau \gamma_R(\tau) \quad (\tau > 0), \quad (\text{AX.9})$$

for the displayed kernel definitions. The dictionary from μ_R to the signed D_{self}^R in Eq. (H.2) must additionally state the interaction and closed-time-path sign convention. Passing from the $\mu_R * q$ form to the $\gamma_R * \dot{q}$ form by integration by parts produces both a local contact/frequency-renormalisation term and an initial-slip term proportional to $\gamma_R(t)q(0)$. The contact term may be absorbed into a declared frequency counterterm. The slip term must instead be retained, incorporated into the initial data/noise, or removed by a separately declared correlated stationary preparation; a frequency counterterm alone does not remove it. In the reduced effective description one imposes the retarded support $\gamma_R(\tau < 0) = \mu_R(\tau < 0) = 0$ through the $\Theta(\tau)$ prefactor, consistently with the Level A retarded-kernel structure of the supplied quadratic ordered-Lorentzian scalar operator. A full first-principles derivation of this identification from ECT microdynamics remains open.

The damping and dissipation kernels govern memory/friction effects, Lamb-shift-like renormalisation, and the response part of the reduced dynamics. In one standard interaction-sign convention, after the contact term has been absorbed into Ω_R , the supplied oscillator model gives

$$M\ddot{q} + M\Omega_R^2 q + \int_0^t d\tau' \gamma_R(t-\tau') \dot{q}(\tau') = \xi_B(t) - \gamma_R(t)q(0), \quad (\text{AX.10})$$

for a factorised preparation. A declared correlated stationary preparation can remove the displayed slip term; it cannot be removed by a frequency counterterm. The equivalent real Hubbard–Stratonovich force and the operator-valued bath force obey, respectively,

$$\langle \xi_B(t) \xi_B(t') \rangle_{\text{stoch}} = \frac{1}{2} \langle \{ \hat{\xi}_B(t), \hat{\xi}_B(t') \} \rangle = C_H(t-t') = S_0 N(t-t'), \quad (\text{AX.11})$$

where the operator equality concerns the symmetrised, not the unordered, correlator. If Eq. (AX.10) is divided by M , then $\gamma_R \mapsto \gamma_R/M$ and the covariance of the acceleration source is C_H/M^2 .

AX.4 Decoherence exponent for spatial superpositions

For a fixed-separation spatial superposition in the Gaussian reduced description, and neglecting drift/relaxation relative to pure dephasing, the dimensionless pairwise dephasing exponent accumulated over time t is

$$\Phi_{\Delta q}(t) = \frac{(\Delta q)^2}{2S_0^2} \int_0^t dt_1 \int_0^t dt_2 C_H(t_1 - t_2) = \frac{(\Delta q)^2}{S_0^2} \int_0^t d\tau (t - \tau) C_H(\tau) = \frac{(\Delta q)^2}{S_0} \int_0^t d\tau (t - \tau) N(\tau). \quad (\text{AX.12})$$

This is the central bridge from the apparatus to observable quantities: interference fringe visibility decays as $e^{-\Phi_{\Delta q}(t)}$. For the declared thermal power-law bath it also has the exact spectral form

$$\Phi_{\Delta q}(t) = \frac{(\Delta q)^2}{\pi S_0} \int_0^\infty d\omega J_s(\omega) \coth\left(\frac{S_0 \omega}{2T_{\text{eff}}}\right) \frac{1 - \cos(\omega t)}{\omega^2}. \quad (\text{AX.13})$$

The nonnegative integrand proves $\Phi_{\Delta q}(t) \geq 0$, but it does not by itself prove monotonicity in t . With the exponential ultraviolet cutoff shown in Eq. (AX.3), the large- t infrared scaling is

	growing power	logarithmic	saturating	
$T_{\text{eff}} = 0$	$0 < s < 1 : t^{1-s}$	$s = 1 : \ln t$	$s > 1$	(AX.14)
$T_{\text{eff}} > 0$ (thermal IR)	$0 < s < 2 : t^{2-s}$	$s = 2 : \ln t$	$s > 2$	

up to model-dependent positive coefficients and transient terms. In particular, finite-temperature ohmic $s = 1$ permits linear growth, whereas the direct derivative-coupling $s = 3$ benchmark saturates rather than approaching a universal linear law.

AX.5 Short-time and Markov limits

Short-time (non-Markov) regime: $t \ll \tau_{\text{env}}$

For $t \ll \omega_c^{-1}$, the kernel is approximately constant, $C_H(\tau) \approx C_H(0)$, and

$$\Phi_{\Delta q}(t) \approx \frac{(\Delta q)^2 C_H(0)}{2S_0^2} t^2 = \frac{(\Delta q)^2 N(0)}{2S_0} t^2. \quad (\text{AX.15})$$

This is a general result: for analytic kernels with finite $N(0)$, the early-time decoherence is quadratic, as illustrated here for the supplied derivative-coupling super-ohmic benchmark.

Markov (white-noise) regime

Short memory, $t \gg \tau_{\text{env}}$, is necessary but not sufficient for a white-noise reduction. The symmetrised spectrum

$$\tilde{C}_H(\omega) = S_0 J_s(|\omega|) \coth\left(\frac{S_0 |\omega|}{2T_{\text{eff}}}\right) \quad (\text{AX.16})$$

must also be approximately flat over the resolved system/protocol band, and the initial-preparation/slip conditions must be controlled. For example, at finite temperature its low-frequency behaviour is

$$\tilde{C}_H(\omega) \sim 2T_{\text{eff}} A_s |\omega|^{s-1}. \quad (\text{AX.17})$$

Thus the supplied $s = 1$ benchmark has a white low-frequency plateau $2T_{\text{eff}} A_1$, whereas the supplied $s = 3$ derivative benchmark behaves as $2T_{\text{eff}} A_3 \omega^2$ and does not become white merely because $t \gg \omega_c^{-1}$.

If the flat-band conditions do hold and the physical force kernel is written in the common full-line convention $C_H(\tau) \approx 2D_H \delta(\tau)$, the symmetric endpoint prescription gives

$$\Phi_{\Delta q}(t) \approx \frac{D_H}{S_0^2} (\Delta q)^2 t. \quad (\text{AX.18})$$

Equivalently, $N(\tau) = 2D_N \delta(\tau)$ with $D_N = D_H/S_0$ gives $\Phi_{\Delta q} = D_N (\Delta q)^2 t / S_0$. This gives a linear-in-time pairwise dephasing exponent in the declared white-noise model; it is not the entropy-rate identity (26.9).

Thus a specified Gaussian source model can interpolate between quadratic short-time and linear Markov dephasing. Establishing that kernel and its state from ECT remains Open.

AX.6 Conditional external collisional–D–P equality scale

No ECT environmental–gravitational crossover can be computed from the current action. The following calculation is retained only because it is a valid conditional comparison between two separately supplied external source models.

For a homogeneous sphere of density ρ ,

$$m = \frac{4\pi}{3} \rho R^3. \quad (\text{AX.19})$$

Supply the hard-sphere collision-rate approximation

$$\lambda_{\text{coll}}^{\text{hs}} = n_{\text{gas}} v_{\text{th}} \pi R^2 \quad (\text{AX.20})$$

and, independently, the unit-coefficient D–P comparison rate

$$\lambda_{\text{DP}}^{\text{ext}} = \frac{G_N m^2}{\hbar R} = \frac{16\pi^2}{9} \frac{G_N \rho^2 R^5}{\hbar}. \quad (\text{AX.21})$$

Equating these two supplied rates gives

$$R_{\text{cross}}^{\text{coll-DP}} \sim \left(\frac{9\hbar n_{\text{gas}} v_{\text{th}}}{16\pi G_N \rho^2} \right)^{1/3} \quad (\text{AX.22})$$

and hence

$$m_{\text{cross}}^{\text{coll-DP}} = \frac{4\pi}{3} \rho (R_{\text{cross}}^{\text{coll-DP}})^3 \sim \frac{3\hbar n_{\text{gas}} v_{\text{th}}}{4G_N \rho}. \quad (\text{AX.23})$$

Equations (AX.22)–(AX.23) are Level-A algebra conditional on the displayed rate laws, not a universal collapse calculation. The coefficients change with the D–P mass-density geometry, separation and regularisation and with the collisional cross section, form factor and momentum-transfer kernel. They are also distinct from the fixed-radius guide mass $m_{\text{guide}}^{\text{DP}}$ in Eq. (32.1). A numerical comparison additionally requires a complete apparatus specification. No inference that an ECT channel dominates or persists in vacuum follows; that claim still requires an action-owned source projector, physical vertex, state-dependent noise kernel and absolute normalisation.

AX.7 Domain of applicability

What this apparatus can compute:

- (i) Standard environmental-decoherence source-model calculations for matter-wave interferometry, once all apparatus inputs are supplied. The C_{60} /Fein/Pedalino observations are retained as external anchors; no distinct ECT rate is currently computed: Section 26.2.
- (ii) Conditional external D–P guide and collisional-comparison algebra: Eq. (32.1) at fixed supplied radius/time and Eqs. (AX.22)–(AX.23) for the separately supplied hard-sphere/unit-coefficient rate laws. These are not ECT rates. The separate $C_{\text{DP}}(1) = 51/160$, 70.6ms sphere point fixes a complete idealised geometry only; a physical apparatus comparison still requires its full inputs and does not inherit the unit-coefficient guide/crossover coefficients.
- (iii) Cavity/cat-state decoherence in Haroche–Raimond-type microwave CQED experiments (an open quantitative target).
- (iv) Source-visibility transforms for photonic delayed-choice, eraser, entanglement-swapping, and quantum-switch experiments. These are declared reduced-model proxies, not ECT weak-locking bounds; deriving the detector-specific ECT channel remains Open.

What this apparatus cannot yet compute:

- (i) Diósi–Penrose object times or an environmental–collapse crossover as an ECT result; the external symbolic functional and conditional source-model algebra do not supply the missing ECT owner chain.
- (ii) Lifetimes or decay widths of elementary particles (requires the full particle sector, not the open-system appendix).
- (iii) Precision spectroscopy (requires explicit interaction vertices beyond the generic coupling family).

(iv) The fine-structure constant or fermion mass ratios (requires the gauge and Yukawa sectors).

The boundary is sharp: the present apparatus is an *open-system / influence-functional* framework; it governs decoherence, measurement, and quantum–classical boundary phenomena. Full particle-physics predictions require the separate ECT particle programme, which is not the subject of this appendix.

AY Qubit dephasing numerics: declared toy channel

For the pure-dilation channel of Section 30.3, define $V = e^{-\Phi}$ and

$$I_{\text{bits}}(S:E) = 2h_2\left(\frac{1+e^{-\Phi}}{2}\right) / \ln 2. \quad (\text{AY.1})$$

Direct evaluation gives

Φ	$I_{\text{bits}}(S:E)$	V	Status of the displayed point
0.006	0.0588	0.9940	Model evaluation at a reported-visibility proxy
0.062	0.3894	0.9399	Model evaluation at a reported-visibility proxy
0.25	1.0034	0.7788	Near the one-bit point of this model
$\ln \sqrt{2}$	1.2018	$1/\sqrt{2}$	Visibility marker used in a separate Werner-state CHSH toy
1	1.8001	0.3679	Order-unity model parameter; no universal boundary
3	1.9964	0.0498	Near-complete dephasing in this model

These are evaluations of one declared channel, not “ECT-calibrated” measurements. In particular, mutual information includes quantum and classical correlations and is not generally equal to the accessible detector record. The earlier approximate entries 0.10, 0.50, and 1.69 bits at $\Phi = 0.006$, 0.062, and 1 are corrected here. A detector-level ECT calculation remains Open.

AZ Candidate critical-shell toy and black-hole information programme

This appendix develops a *C/Open* shell toy and the conditional reduced-state identities relevant to a future ECT black-hole completion. We consistently distinguish what is structural within the present ECT framework, what depends on the current shell/interface closure, and what remains open.

AZ.1 Microscopic unitarity and apparent information loss

The underlying condensate is specified by a real Euclidean variational functional $S_E[\Phi]$ (postulate P2). This fact alone does not construct a Lorentzian Hilbert space, a self-adjoint Hamiltonian, reflection-positive interacting measure or globally unitary strong-field evolution. Global information preservation is therefore a completion requirement, not a theorem derived from P2.

The standard Hawking calculation is semiclassical. Interpreting its radiation state as a partial trace over shell/core sectors is possible only after the physical ECT Hilbert space and factorisation in Eq. (AZ.9) are supplied; they are not derived here. The standard reduced-state statement is conditional: a pure global state can have a mixed marginal when it is entangled across a supplied factorisation [197]. A product pure state remains pure under the corresponding partial trace, and coarse graining by itself does not establish entanglement, an ECT state, or information loss.

AZ.2 Conditional Tolman–thermal shell toy

Rindler near-horizon geometry. Near any non-extremal stationary horizon the proper-distance metric takes the local Rindler form

$$ds^2 \approx -\left(\frac{\kappa\rho}{c}\right)^2 c^2 dt^2 + d\rho^2 + r_h^2 d\Omega^2, \quad (\text{AZ.1})$$

where κ is the surface gravity and r_h is the horizon radius. The Tolman temperature for a static observer at proper distance ρ is

$$T_{\text{loc}}(\rho) = \frac{T_H}{\sqrt{-g_{tt}}} = \frac{\hbar c}{2\pi k_B \rho}, \quad (\text{AZ.2})$$

which diverges as $\rho \rightarrow 0$. This is a standard result [164, 162].

ECT phase transition temperature. The condensate effective potential at finite temperature is

$$V_{\text{eff}}(\Phi, T) = -\frac{\mu^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4 + \frac{\lambda T^2}{8}\Phi^2, \quad (\text{AZ.3})$$

from the Dolan–Jackiw one-loop correction [204]. The ordered phase ($\langle\Phi\rangle \neq 0$) exists while $-\mu^2 + \lambda T^2/4 < 0$, i.e. for $T < T_c$ with

$$T_c = 2\phi_0, \quad (\text{AZ.4})$$

where $\phi_0 = \sqrt{\mu^2/\lambda}$ is the bare field VEV. This is only the one-real-scalar Z_2 thermal toy; it is not a derivation of the gradient-ordered P4 transition.

Conditional crossover surface. Setting $T_{\text{loc}}(\rho_c) = T_c$ and substituting Eq. (AZ.2):

$$\rho_c = \frac{\hbar c}{4\pi k_B \phi_0}. \quad (\text{AZ.5})$$

The additional identifications formerly used to replace ϕ_0 by a Planck scale are not derived and are not used to claim a physical number. The toy result is retained only in its conditional form

$$\rho_* = \frac{\hbar c}{2\pi k_B T_*}, \quad (\text{AZ.6})$$

where T_* must be derived as the control temperature of the physical P4 ordering transition before ρ_* has an ECT interpretation. Figure 31(a) illustrates the Tolman temperature profile and marks a supplied toy crossover.

No numerical proximity to a membrane-paradigm stretched horizon is claimed: the relevant control observable, T_* , physical metric and ordered-phase transition owner are all Open.

AZ.3 Shell entropy and the area law

If the shell of area A and thickness ρ_c is decomposed into effective condensate cells of linear scale ℓ_* , the microstate count is

$$N \sim A/\ell_*^2, \quad S_{\text{shell}} \sim N \ln q = \frac{A \ln q}{\ell_*^2}. \quad (\text{AZ.7})$$

Matching the Bekenstein–Hawking entropy $S_{BH} = A/(4\ell_{\text{Pl}}^2)$ [254] requires

$$\frac{\ln q}{\ell_*^2} = \frac{1}{4\ell_{\text{Pl}}^2}. \quad (\text{AZ.8})$$

We do not derive the exact coefficient from first principles here; the matching condition (AZ.8) shows what is required, but the microscopic counting remains an open problem (Table 144). The cell count only shows what a supplied boundary-layer model must match. It does not derive a shell, a Planck-scale cell size, a state count, or the 1/4 coefficient from ECT.

AZ.4 Conditional isometry theorem and its limits

The mere presence of a critical shell or additional condensate degrees of freedom does not resolve the information problem. The exact statement that can be made without specifying the dynamics is a conditional theorem about a separately supplied unitary map.

Conditional isometry theorem. Assume that (i) a completed strong-field theory supplies a Hilbert space, (ii) its evolution from a space of distinguishable initial microstates to the complete final state is an isometry, and (iii) evaporation is complete with no unobserved final subsystem. Write the intermediate bookkeeping as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_{\text{rad}} \otimes \mathcal{H}_{\text{shell}} \otimes \mathcal{H}_{\text{core}}. \quad (\text{AZ.9})$$

Then inner products of initial states are preserved in the *complete* final radiation state. In particular, a channel that maps two orthogonal initial states to the same complete final density operator is incompatible with the assumed isometry.

Proof. For an isometry U , $U^\dagger U = \mathbf{1}$, hence $\langle \psi_1 | \psi_2 \rangle = \langle U \psi_1 | U \psi_2 \rangle$. Orthogonal inputs therefore remain orthogonal outputs. At intermediate times a reduced radiation state

$$\rho_{\text{rad}}(t) = \text{Tr}_{\text{shell,core}} |\Psi(t)\rangle \langle \Psi(t)| \quad (\text{AZ.10})$$

may nevertheless be mixed. This proves compatibility of a thermal marginal with global purity; it does not prove that ECT owns the assumed isometry.

For reference, a general emission step can be expanded as

$$|s\rangle|0\rangle \longrightarrow \sum_{\omega,s'} A_{\omega s' s} |s'\rangle |\omega\rangle. \quad (\text{AZ.11})$$

A state-independent *complete* final channel would violate the theorem, but identical thermal one-particle spectra need not: distinguishability may be stored in phases and higher-order or multipartite correlations. The theorem therefore neither forces every observable to be shell-state dependent nor derives a transfer factor. The optional H_{sh} transfer model introduced below is a separately supplied C/Open completion and is not a consequence of this theorem.

Entropy statement and Page-curve guard. For a pure bipartite state at a fixed time,

$$S_{\text{rad}}(t) = S_{\text{rem}}(t) \quad (\text{AZ.12})$$

with S_{rem} the entropy of the complementary subsystem [320]. If the complement truly becomes one-dimensional at the end, the final complete radiation state is pure. These endpoint facts do not determine the time-dependent Page curve. Its shape requires a defined time-dependent tensor factorisation, initial state, changing subsystem dimensions and dynamical or typicality/scrambling assumptions.

ECT status. The isometry and reduced-state statements are Level A mathematics under their explicit assumptions. P2 is a Euclidean variational postulate and does not derive reflection positivity, a Lorentzian Hilbert space, a self-adjoint strong-field Hamiltonian, the factorisation above, complete evaporation or a shell/radiation vertex. All of those ECT owners, and any quantitative Page curve, remain Level B/Open. The Mathur small-corrections obstruction remains an external constraint on future completions [321].

AZ.5 Near-horizon χ_{sh} -field dynamics: an effective mean-field construction

The conditional theorem above is purely structural. To go beyond it and exhibit an explicit mechanism, we construct an effective mean-field model of the shell, valid in the Ginzburg–Landau regime. This is *not* a full microscopic derivation from the underlying condensate action, and the results below have a weaker

status than the theorem. For this optional submodel only, use $\hbar = c = 1$, let t and ρ have dimension E^{-1} , and choose the shell order parameter dimensionless: $[\chi_{\text{sh}}] = 1$. The angular integral is suppressed in $\mathcal{F}_{\text{GL}}$ (energy per unit solid angle), so $[a_{\text{sh}}] = E^2$ and $[\lambda_{\text{GL}}] = E^4$. These symbols are local to H_{sh} and are not the P3 gradient or quartic coefficients. We mark the status of each step explicitly.

Step 1: local outer asymptotics of the χ_{sh} field. Near the critical point $T_{\text{loc}}(\rho_c) = T_c$, the reduced temperature is

$$\varepsilon(\rho) = 1 - \frac{T_{\text{loc}}(\rho)}{T_c} = \frac{\rho - \rho_c}{\rho} \approx \frac{\rho - \rho_c}{\rho_c} \quad (\rho \downarrow \rho_c^+). \quad (\text{AZ.13})$$

The Ginzburg–Landau effective action for the shell order parameter χ_{sh} , with Tolman temperature as control parameter, is

$$\mathcal{F}_{\text{GL}}[\chi_{\text{sh}}] = \int d\rho r_h^2 \left[a_{\text{sh}} (\partial_\rho \chi_{\text{sh}})^2 - \frac{\lambda_{\text{GL}}}{6} \varepsilon(\rho) \chi_{\text{sh}}^2 + \frac{\lambda_{\text{GL}}}{4} \chi_{\text{sh}}^4 \right], \quad (\text{AZ.14})$$

where the potential terms use one supplied normalisation of the one-real-scalar Dolan–Jackiw toy [204]. The local coefficients satisfy $a_{\text{sh}} > 0$ and $\lambda_{\text{GL}} > 0$ in this branch. Their values, the normalisation and physical identity of χ_{sh} , the Tolman/control map, and the descent from a strong-field ECT action are not fixed by that potential and remain Open.

In the mean-field approximation (gradient term subdominant), the local outer asymptotic scaling just outside the shell is

$$\chi_{\text{sh}}(\rho) \propto \left(\frac{\rho - \rho_c}{\rho_c} \right)^{1/2} \quad (\rho \downarrow \rho_c^+). \quad (\text{AZ.15})$$

This is the leading outer asymptotic behaviour of the order parameter, not a complete radial profile. The full profile, including the interface region and the inner branch $\rho < \rho_c$, requires solving the Euler–Lagrange equation of $\mathcal{F}_{\text{GL}}$ numerically. The mean-field exponent $1/2$ is the leading result inside this supplied one-real-scalar Z_2 equilibrium toy. If that toy is separately assigned to the 3D Ising universality class, fluctuations modify it toward $\beta \approx 0.326$; this is not a result for the P4 transition or a physical black-hole shell.

The GL description (AZ.14) is a near-critical mean-field closure, valid in the outer-shell interface region $\rho \gtrsim \rho_c$ where the condensate amplitude transitions between its vacuum value and zero. It is not a derived description of the deep interior $\rho \ll \rho_c$, which belongs to the pre-Lorentzian core sector (Appendix AZ.7). A complete interior profile requires the pre-Lorentzian condensate dynamics, which is outside the scope of the present mean-field construction.

Step 2: separately supplied radial operator inside H_{sh} . One optional near-horizon completion supplies the lapse/redshift ansatz

$$g_{tt}^{\text{eff}}(\rho) = -Z_t \left(\frac{\chi_{\text{sh}}(\rho)}{\chi_{\text{sh,vac}}} \right) (\kappa\rho)^2, \quad Z_t(1) = 1, \quad (\text{AZ.16})$$

with the benchmark $Z_t(\chi_{\text{sh}}/\chi_{\text{sh,vac}}) = \chi_{\text{sh}}/\chi_{\text{sh,vac}}$. The component g_{tt} alone does not determine a wave operator: a radial metric coefficient, measure and probe action are also required. Therefore H_{sh} separately defines the positive radial coefficient

$$c_{\text{eff}}^2(\rho) := c^2 \frac{\chi_{\text{sh}}(\rho)}{\chi_{\text{sh,vac}}} \quad (\text{AZ.17})$$

and, as an independent Sturm–Liouville input on a declared interval $I_\rho \subset (\rho_c, \infty)$, the s-wave equation

$$-\partial_\rho [p(\rho) \partial_\rho f] = \omega^2 r(\rho) f, \quad p(\rho) := c_{\text{eff}}^2(\rho) > 0, \quad r(\rho) := \frac{1}{(\kappa\rho)^2} > 0, \quad \varphi = e^{-i\omega t} f(\rho). \quad (\text{AZ.18})$$

The Hilbert-space weight is $r(\rho)d\rho$, and the boundary conditions are chosen so that this Sturm–Liouville operator is self-adjoint. This radial operator is not derived from Eq. (AZ.16); both are supplied C/Open data of the same optional completion. A covariant realisation must instead specify the full metric, scalar action and measure and show that their Euler–Lagrange equation reduces to Eq. (AZ.18).

Step 3: finiteness of the shell correction. In the WKB language the shell correction to the outgoing channel is encoded in

$$\ln |\mathcal{T}_\omega^{(0)}| \sim - \int_{\text{shell}} d\rho \Im k_\omega(\rho). \quad (\text{AZ.19})$$

For the supplied radial operator, the local WKB magnitude obeys $|k_\omega(\rho)| = O[(\rho - \rho_c)^{-1/4}]$ when $c_{\text{eff}}(\rho) \propto (\rho - \rho_c)^{1/4}$ and $\rho_c > 0$. This does not imply that k_ω is imaginary: a propagating branch may have $\Im k_\omega = 0$. On any chosen complex branch satisfying $|\Im k_\omega| \leq |k_\omega|$, the endpoint contribution to Eq. (AZ.19) is bounded by an integrable $(\rho - \rho_c)^{-1/4}$ envelope and is therefore finite. This establishes only a local conditional integrability bound inside the supplied WKB toy. It does not establish a nonzero transmission correction, its sign or magnitude, a Hawking channel, or Planck suppression. Those claims require a physical metric, radiation vertex, state and complete scattering problem.

Step 4: optional shell–radiation completion H_{sh} . The present action does not establish that a shell coordinate exists as a physical mode, encodes a black-hole microstate, or couples to radiation. For a declared stationary background $\bar{\chi}_{\text{sh}}$, canonically normalised probe, boundary conditions and passive mode-diagonal scattering prescription, one may define a *flux-normalised* C/Open channel amplitude

$$\mathcal{T}_{\omega\ell}[\bar{\chi}_{\text{sh}}] = \exp(i\delta_{\omega\ell} - \Gamma_{\omega\ell}), \quad \delta_{\omega\ell} \in \mathbb{R}, \quad \Gamma_{\omega\ell} \geq 0, \quad 0 < |\mathcal{T}_{\omega\ell}| < \infty. \quad (\text{AZ.20})$$

Passivity then gives $|\mathcal{T}_{\omega\ell}| \leq 1$ only in this declared flux convention. After fixing a logarithm branch and a radial projection of the stationary (ω, ℓ) mode, define the Fréchet kernel by

$$K_{\omega\ell}(\rho) := \left. \frac{\delta \ln \mathcal{T}_{\omega\ell}[\chi]}{\delta \chi(\rho)} \right|_{\chi=\bar{\chi}_{\text{sh}}}, \quad \delta \ln \mathcal{T}_{\omega\ell} = \int_{I_p} d\rho K_{\omega\ell}(\rho) \delta \chi_{\text{sh}}(\rho). \quad (\text{AZ.21})$$

With $[\chi_{\text{sh}}] = 1$, $[K_{\omega\ell}] = E$. The state, Green function, flux normalisation, interval, boundary data and kernel remain supplied completion data. Mode mixing, resonances, superradiance or nonstationary scattering require a different representation. Nothing here derives a Hawking spectrum, Page curve, echo or ringdown forecast.

Effective shell Hamiltonian. Using the shell parametrisation $\chi_{\text{sh}}(\rho, \Omega, t) = \bar{\chi}_{\text{sh}}(\rho) + \delta \chi_{\text{sh}}(\rho, \Omega, t)$, one may define the dimensionless shell coordinate $\psi_{\text{sh}} = \ln(\chi_{\text{sh}}/\chi_{\text{sh,vac}})$. This toy coordinate is not identified with either the amplitude coordinate $\phi_u = \beta_\phi^{-1} \ln(u/u_\infty)$ or the strict soft scalar coordinate χ of Section 4.3; no such bridge has been derived. The GL free energy Eq. (AZ.14) gives a *static* energy functional for the declared χ_{sh} variable. Supplying a local time-kinetic term, angular stiffness and potential gives the dimensionally closed candidate Hamiltonian

$$H_{\text{shell}} = \int d\rho d\Omega r_h^2 \left[\frac{\pi_{\text{sh}}^2}{2Z_{\text{sh}}} + a_{\text{sh}} (\partial_\rho \delta \chi_{\text{sh}})^2 + \frac{W_{\text{sh}}}{2r_h^2} (\nabla_\Omega \delta \chi_{\text{sh}})^2 + U_{\text{eff}}(\delta \chi_{\text{sh}}; \rho) \right], \quad (\text{AZ.22})$$

where $\pi_{\text{sh}} = Z_{\text{sh}} \partial_t \delta \chi_{\text{sh}}$, $[Z_{\text{sh}}] = [a_{\text{sh}}] = [W_{\text{sh}}] = E^2$ and $[\pi_{\text{sh}}] = E^3$. The radial term now uses exactly the GL coefficient and measure of Eq. (AZ.14). The supplied effective potential is

$$U_{\text{eff}}(\delta \chi_{\text{sh}}; \rho) = \frac{1}{2} A_{\text{sh}}(\rho) \delta \chi_{\text{sh}}^2 + \frac{\lambda_{\text{sh}}}{4} \delta \chi_{\text{sh}}^4 + \dots, \quad A_{\text{sh}}(\rho) = A_0 \frac{\rho_c - \rho}{\rho_c}, \quad [A_0] = [A_{\text{sh}}] = [\lambda_{\text{sh}}] = E^4. \quad (\text{AZ.23})$$

All coefficients are effective inputs not derived from the full condensate action.

Status of Eq. (AZ.22). Equation (AZ.22) is one supplied C/Open ansatz. The GL potential, Rindler background and a zero at ρ_c do not uniquely fix the kinetic, angular, higher-operator or constraint structure. Neither the field χ_{sh} nor the coefficients Z_{sh} , W_{sh} , λ_{sh} have been descended from a physical strong-field ECT action.

Supplied vertex inside H_{sh} . No physical shell mode, radiation vertex or shell Green function follows from the present action. The optional completion H_{sh} supplies, as one explicit local choice,

$$\mathcal{L}_{\text{int}}^{H_{\text{sh}}} = -\frac{g}{2} \delta\chi_{\text{sh}} (\partial\varphi)^2, \quad (\text{AZ.24})$$

where the probe is a canonically normalised four-dimensional scalar, $[\varphi] = E$. Since $[\chi_{\text{sh}}] = 1$, the coefficient g is dimensionless in the declared natural-unit convention; it is independent and may vanish. This operator defines a new conditional model rather than derived ECT phenomenology; other operators are not excluded, and no observable follows until the background, state, propagators, boundary conditions and detector map are supplied.

AZ.6 Toy finite-memory shell model and the Page mechanism

The conditional isometry theorem fixes only complete-state distinguishability and endpoint purity under its assumptions; it does not determine an entropy curve. The following *illustrative* finite-memory model adds explicit Page-like typicality assumptions. It is not derived from the condensate action and is not a prediction of the shell closure.

Setup. Discretise the shell into $N \sim A/\ell_*^2$ local cells, where ℓ_* is an effective UV cutoff. Give each cell a q -dimensional Hilbert space. The shell's total Hilbert space is $\mathcal{H}_{\text{shell}} = (\mathbb{C}^q)^{\otimes N}$ with

$$S_{\text{shell}}^{\text{max}} = N \ln q \sim \frac{A}{\ell_*^2} \ln q. \quad (\text{AZ.25})$$

Neither ℓ_* nor q is supplied by ECT. Only the symbolic matching condition in Eq. (AZ.8) is retained; no hand-set numerical cell model is inferred.

Unitary emission map. Each emission step is a unitary map on the full Hilbert space:

$$U_n : \mathcal{H}_{\text{shell}}^{(n)} \otimes \mathcal{H}_{\text{vac}}^{(n)} \longrightarrow \mathcal{H}_{\text{shell}}^{(n+1)} \otimes \mathcal{H}_{\text{rad}}^{(n)}, \quad (\text{AZ.26})$$

The toy additionally assumes that one q -dimensional factor is transferred from the remaining subsystem to the radiation at each step and that the resulting pure state is typical on the declared bipartition. Unitarity alone does not impose either assumption or state dependence of every marginal.

Resulting entropy curve. Under those declared assumptions the leading Page estimate is

$$S_{\text{rad}}(n) \approx \min(n, N-n) \ln q, \quad 0 \leq n \leq N, \quad (\text{AZ.27})$$

with a smooth turnaround at the Page time $n_P \sim N/2$.

Interpretation. The early and late branches describe the typical entanglement entropy of the chosen finite bipartition. They do not reproduce a Hawking spectrum or rate without a separate emission Hamiltonian and state.

This model demonstrates one consistent Page-like construction, not an automatic consequence of a finite shell and unitary map. An ECT Page curve requires the microscopic H_{shell} or an explicitly supplied H_{sh} Hamiltonian/vertex (including g), subsystem factorisation, initial state, transfer law and correlation observables. The optional completion above supplies none of these from P1–P7.

AZ.7 Interior and the singularity

One C/Open picture assumes that a Tolman-controlled order parameter crosses a supplied T_* and that the Lorentzian effective description then fails before the classical singularity. The current theory does not derive this control map, transition, breakdown surface, physical interior, or a core Hilbert-space sector. Thus $\mathcal{H}_{\text{core}}$ in Eq. (AZ.9) is only bookkeeping for the toy completion; it does not replace the singular interior as a theorem of ECT. The relation to any cosmological pre-Lorentzian phase is likewise an Open common-mechanism hypothesis.

AZ.8 Observable status

The present theory derives no nonzero shell observable, sign, magnitude or scaling. The Tolman toy is not a physical shell solution, and analogue horizons test their own platform models unless an action-, state- and observable-level ECT map is constructed. Any future comparison with ringdown, radiation thermality or other strong-field data must first derive the physical shell mode, metric and radiation vertices, boundary conditions, causal response and likelihood. Until then these data are external completion tests only.

AZ.9 Requirements for a candidate unitary ECT black-hole completion

The conditional isometry theorem (Section AZ.4) gives a necessary endpoint constraint, not a complete resolution of the paradox. The following items are requirements for one candidate shell-based unitary completion; satisfying them still does not derive the dynamics, information channel or Page curve.

The four genuine conditions. The theorem's input conditions, and what they require of the underlying ECT microcompletion, are:

- (C1) **Self-adjoint microscopic dynamics.** The condensate Hamiltonian must be self-adjoint: $H_{\text{micro}} = H_{\text{micro}}^\dagger$. P2 does not supply this Hamiltonian or its domain. Reflection positivity, Lorentzian reconstruction and interacting strong-field self-adjointness must be established independently.
- (C2) **Finite critical temperature and near-horizon ordered-phase shell.** The condensate must have a finite critical temperature T_c above which the ordered macroscopic phase ceases to exist. Moreover, the local Tolman temperature must be the correct thermodynamic control parameter for this phase transition near the horizon. These two sub-requirements would permit a finite-temperature interface at $\rho_c = \hbar c / (2\pi k_B T_c)$. The temperature in the displayed one-real-scalar thermal toy does not derive the P4 gradient-ordering transition, and the Tolman control map remains Open.
- (C3) **Complete evaporation with no infinite-capacity remnant.** Evaporation must proceed to completion without leaving a remnant carrying an unbounded amount of hidden information. This is an assumption about the final state of evaporation not derived from the current formulation; the Planck-remnant case requires separate treatment.
- (C4) **No additional non-unitary dissipation.** The interaction between the shell and the radiation sector must not introduce intrinsic non-unitarity (e.g. an open-system environment outside the ECT degrees of freedom). The absence of an additional non-unitary environment is itself an assumption until the complete state space and dynamics are identified.

If C1–C4 hold, the *complete final radiation state* must retain initial state distinguishability. This does not force state dependence of each one-particle spectrum, exclude thermal marginals, or determine late-time entropy evolution; information may reside in higher correlations and phases.

Condition violated	Implication for the information paradox
(C1) violated	ECT itself admits fundamental non-unitarity; the theory requires substantial revision before the paradox can be addressed.
(C2) unavailable (no physical T_* owner)	This particular Tolman–thermal shell route is unavailable. Other strong-field completions are neither excluded nor selected.
(C2) unavailable (Tolman not the control parameter)	This shell route must be rebuilt from the actual control observable; no location or interface follows from the present toy.
(C3) violated (infinite remnant)	The paradox is formally evaded by hiding information indefinitely, but at the cost of pathological remnant physics (infinite pair-production rate in external fields, etc.).
(C4) violated	The ECT condensate is open to an external non-unitary environment; additional quantum gravity degrees of freedom must be specified.

Classification of microcompletions.

Current status within ECT. Condition (C1) is Open: the Euclidean action P2 is not a self-adjoint Lorentzian Hamiltonian. Condition (C2a) has only a supplied one-real-scalar thermal toy (Section 25.1), not a derivation of P4 ordering. Condition (C2b) (Tolman as control parameter) is the central remaining structural assumption; deriving it from the black-hole branch of the full condensate action is the sharpest open problem in the ECT black-hole sector. Conditions (C3) and (C4) are natural but not derived.

Thus none of C1–C4 is presently owned as a complete strong-field ECT result. The scalar thermal toy evaluates one conditional ingredient; all remaining links require a physical tensor geometry, state, interface action and evaporation map.

AZ.10 Summary table

Table 144: Status of components of the ECT black-hole information scenario. Level A = exact in the declared model and assumptions, with an explicit reproducible derivation; reproducible checks verify that derivation but do not replace it; it does not by itself establish descent from P1–P6 or physical ECT ownership. Level B = structural/conditional derivation or closure route with a declared open matching, input, or identification step in the present shell/interface construction; Open = problem not yet completed.

Statement	Status	Reference
<i>Completion inputs and conditional toys</i>		
Global interacting unitarity	Open	Not implied by P2
One-real-scalar thermal benchmark $T_c = 2\phi_0$	C / not P4 transition	Eq. (AZ.4)
Critical-shell location from supplied Tolman-control assumption	C/Open	Eq. (AZ.6)
<i>Conditional theorem (T)</i>		
Conditional boundary-cell counting lemma (Theorem 36.1)	A mathematics under supplied inputs / Open ECT	§36.5

(continued on next page)

Statement	Status	Reference
Complete final radiation must preserve initial distinguishability under a supplied isometry	A mathematics / Open ECT	§AZ.4
Thermal one-body spectra and Page-curve shape	Open	Not fixed by the isometry theorem
<i>Structural / effective model (S)</i>		
Area-scaling shell entropy toy	C/Open	§AZ.3
Conditional boundary-cell counting lemma (Theorem 36.1)	A mathematics under supplied inputs / Open ECT	§36.5
Physical UV regulator and finite local entropy capacity	Open	§36.5
GL outer asymptotics of a supplied χ toy	A within toy / Open ECT	Eq. (AZ.15)
Candidate near-horizon closure and separately supplied effective mode equation	C/Open	Eqs. (AZ.16) and (AZ.18)
Finiteness of a supplied shell correction integral	A within toy / Open ECT	Eq. (AZ.19)
$\pi_2(S^3) = 0$ for a supplied $S^2 \rightarrow S^3$ shell map	A topology / C–Open shell	Shows only absence of π_2 charge; formation, stability and dissolution are not derived
Conditional layer ratio: $\rho_*/r_s = \hbar c / (2\pi k_B T_* r_s)$	C/Open	Eq. (36.16)
Shell–radiation response kernel	Open	No physical mode, vertex or scattering owner
<i>Open problems (O)</i>		
Effective shell Hamiltonian ansatz	C/Open	Eq. (AZ.22)
Shell–radiation coupling	Open	No operator or coefficient derived
Toy finite-memory Page model (illustrative)	C/Open	§AZ.6
Shell–radiation coupling coefficient g	Open	—
Shell Hamiltonian coefficients from microdynamics	Open	—
Exact coefficient 1/4 in $S_{BH} = A / (4\ell_{Pl}^2)$	Open	—
Quantitative ECT Page curve	Open	—
Explicit radiation response from first principles	Open	—
Tolman heating as ordered-phase control parameter (C2b)	Open	—
Firewall vs. smooth-horizon consistency	Open	—

BA Reduced thermality, generalized entropy, and the black-hole information problem

Status: Level A for the generic reduced-state identity only. Global interacting unitarity and exclusion of information destruction in ECT are B/Open. Level C/Open for the shell/interface ansatz and the generalized entropy bookkeeping. Open (OP-Q13): explicit shell dynamics, Page curve, and purification mechanism.

This appendix clarifies the precise status of the black-hole discussion in Section 36. The central point is not that the full black-hole information problem has been solved dynamically. The central point is more limited and more secure: reduced-state thermality does not imply fundamental destruction of information.

Hilbert-space factorisation

In the ECT framework the relevant Hilbert-space decomposition is organised schematically as

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{ext}} \otimes \mathcal{H}_{\text{shell}} \otimes \mathcal{H}_{\text{int}}, \quad (\text{BA.1})$$

the same decomposition as (36.7). The state seen by an exterior observer is obtained by tracing over inaccessible sectors:

$$\rho_{\text{ext}} = \text{Tr}_{\text{shell,int}}(|\Psi_{\text{total}}\rangle\langle\Psi_{\text{total}}|). \quad (\text{BA.2})$$

Such a reduced state may be approximately thermal even when the global state remains pure.

Conditional negative route

ECT would strengthen this observation if its full interacting coherent sector admitted the required positive OS reconstruction and globally unitary strong-field continuation. P2 asserts $O(4)$ invariance, not unitarity, so genuine non-unitary destruction of information cannot yet be excluded from P1–P6 alone.

Generalized entropy and the Page problem

The appropriate entropy variable for the exterior observer is the generalised entropy (36.5):

$$S_{\text{gen}} = S_{\text{BH}} + S_{\text{ext}}^{\text{coarse}},$$

rather than the Bekenstein–Hawking term alone.

Generic reduced-state mathematics permits approximate thermality and hidden correlations to coexist, but this supplies no ECT Page-curve result or positive evidence for one. An explicit Page curve remains Open: it requires a completed strong-field shell closure and a quantitative evolution law for S_{gen} (OP-Q13).

Summary of what is established

- (i) Apparent exterior thermality is structurally compatible with global purity (Level A; reduced-state logic).
- (ii) P1–P6 do not prove a globally unitary interacting completion; information preservation is therefore Open rather than a consequence of P2.
- (iii) The strong-field dynamics, subsystem map, purification mechanism and Page curve remain Open (OP-Q13).

The reduced-state framework shows why thermality alone does not decide the information problem. A completed ECT theory must still show whether and how approximate exterior thermality and global purity are jointly realised.

BB Source-model dictionary for ordered-medium comparisons

This appendix records mechanisms and observables in named laboratory source models. It is not an identification dictionary: no row equates a laboratory degree of freedom with an ECT field. An ECT inference additionally requires the action/state/operator/observable/uncertainty/discriminator bridge of Section 37. Thus a source-model success is not evidence for ECT and a source-model failure is not an ECT falsifier.

Table 145: Named source-model mechanisms and candidate ECT comparison targets. The last column is an inference firewall, not a derived map.

Source-model mechanism	Observable	Example platform	ECT inference status
Compact order-parameter phase	Phase difference	Superfluid He, atomic BEC	BR1 field/domain map Open
Integer winding	Quantised circulation or flux	Superfluid vortex, Josephson junction	Same homotopy form only after BR1 is supplied
Defect-mediated sector change	Phase slip or vortex nucleation	BEC, superconducting nanowire	ECT defect/action map Open
Boundary-sensitive spectrum	Boundary force or mode shift	Superconducting waveguide	ECT field, boundary operator and stress Open
Driven mode mixing	Produced quanta/correlations	Superconducting circuit	ECT time-dependent action/state Open
Acoustic horizon	Bogoliubov mode correlations	BEC acoustic black hole	Quantitative analogue–ECT horizon map Open
Accelerated detector model	Detector transition spectrum	Named detector/source model	ECT detector, metric and state Open
Environmental kernel	Fringe visibility/reduced coherence	Matter-wave interferometry	ECT subsystem split, vertex and record kernel Open
Tunable ordered background	Density/stiffness dependence	Variable-density BEC, thin-film He	No ECT galactic/dark-sector transfer law
Critical or zero-amplitude set	Threshold/phase-transition response	Quench experiment	No universal ECT branch threshold derived

BC Expanded architectural comparison of ECT and neighbouring programmes

This appendix provides an expanded version of the architectural comparison given in Section 49.4, using the same set of programmes as the main table but with additional comparison axes.

BC.1 Closely related Euclidean-to-Lorentzian constructions

The idea that Lorentzian physics can emerge as an effective property of a fundamentally Euclidean field theory has been explored in several independent constructions.²

Girelli, Liberati and Sindoni [3] showed that a scalar field with a nonvanishing gradient on a Riemannian manifold generates an effective Lorentzian metric for perturbations. Their gravitational sector, however, reduces to Nordström (scalar) gravity rather than a tensor theory.

Mukohyama and Uzan [4] developed a more complete classical construction. They introduced a “clock field” ϕ whose gradient $\partial_\mu \phi = M^2 n_\mu$ selects a preferred direction on a 4-dimensional Riemannian manifold and allows scalar, vector, and Dirac-spinor sectors to propagate in an effective Minkowski metric within a region $\mathcal{M}_0$ where $\partial_\mu \phi \neq 0$. Their gravitational sector is not general relativity but a covariant Galileon/Horndeski-type theory.

ECT shares the mechanism-class idea that a supplied scalar-gradient background can support a Lorentzian principal tensor. In ECT this result is presently restricted to the supplied scalar EFT with $\beta(\beta - \alpha) < 0$; on its declared healthy $\beta > 0$ branch this is $\alpha > \beta > 0$; physical spacetime, clock and cross-sector cones remain Open. P4 supplies, but P1–P3 do not dynamically derive, the preferred direction. The two constructions differ in three substantial respects.

- **Origin of the preferred direction.** Mukohyama–Uzan impose a nonvanishing clock-field gradient as a background condition. In ECT, P4 supplies a preferred direction through an $O(4) \rightarrow O(3)$ ordered condensate-gradient branch; its dynamical formation from P1–P3 is Open.
- **Gravity sector.** Mukohyama–Uzan obtain a covariant Galileon/Horndeski-type gravitational theory. ECT currently supplies a scalar–tensor ansatz plus a kinematic preferred direction. A dynamical covariant orientation stress is an Open extension and is not part of the displayed field equation.
- **Scope of development.** Mukohyama–Uzan restrict their analysis to classical field theory. ECT carries the construction further into a quantum/coherent branch, vacuum-response questions, and astrophysical/cosmological phenomenology.

A further important contrast concerns cone universality. In the clock-field construction, different matter sectors can in principle see different effective metrics, and a matching condition between sector-dependent couplings must be imposed to obtain a universal lightcone. In ECT, universality is intended to follow instead from the single-medium origin of the broken-phase kinetic structure. Whether this remains exact beyond the minimal EFT truncation is itself part of the open consistency programme.

The comparison is informative in both directions. On the one hand, the existence of these closely related constructions shows that Lorentzian principal operators can arise conditionally in Euclidean field models. It does not establish that the scalar operator is the observed physical spacetime metric. On the other hand, it sharpens which parts of ECT remain genuine open problems: the full closure of the gravitational sector, the complete fermion and gauge reconstruction, and the final status of cross-sector universality beyond the benchmark EFT.

Table 146: Closely related Euclidean-to-Lorentzian constructions.

Framework	Preferred direction	Lorentzian sector	Gravity sector	Quantum sector	Main limitation
Girelli–Liberati–Sindoni (2009)	Imposed scalar gradient	Emergent for perturbations	Nordström (scalar)	No	No tensor gravity
Mukohyama–Uzan (2013)	Imposed clock-field gradient	Emergent for scalar/vector/spinor sectors	Covariant Galileon / Horndeski	No (classical only)	Sector matching required for universal cone

²An early discussion of dynamical Lorentzian-signature emergence appears in Ref. [322].

Framework	Preferred direction	Lorentzian sector	Gravity sector	Quantum sector	Main limitation
ECT	P4 adopted nonzero-gradient sector with $O(3)$ stabiliser; stationary action/state and dynamical formation/selection Open	Scalar cone conditional on supplied-chart coefficients	Scalar–tensor ansatz; orientation stress Open	Partial / coherent branch developed	Open closures in gravity, fermions, and full universality

BC.2 Comparison with further neighbouring programmes

Table 147: Expanded architectural comparison of ECT with neighbouring programmes.

Axis	GR + SM	Einstein-aether	Hořava–Lifshitz	Volovik analogue	ECT
Starting ontology	Lorentzian manifold	Lorentzian manifold + unit vector	Foliated manifold	Condensate / superfluid	Euclidean $\mathbb{R}^4$ + scalar Φ
Time status	Fundamental	Fundamental	Fundamental but anisotropic	Emergent	P4 supplies an ordered direction datum; $\beta(\beta - \alpha) < 0$ conditionally makes the scalar principal tensor Lorentzian; physical clock Open
Lorentzian structure	Postulated	Postulated; preferred frame added	Postulated; foliation-dependent	Emergent acoustic metric	Scalar Lorentzian slope conditional on P4; physical speed/metric Open
Gauge status	SM group postulated	SM group postulated	SM group postulated	Analogy only	Global phase / Lie-algebra comparisons only; physical $U(1) \times SU(2) \times SU(3)$ Open
Quantum axioms	Primitive	Primitive	Primitive	Primitive	Structural route via coherent branch; completion open
Gravity closure	Einstein eqs. exact	Einstein + aether sector	Modified UV dispersion	Acoustic metric only	Scalar characteristic geometry; physical metric/tensor/source Open
Dark matter	Extra particles	Extra particles	Extra particles	Not addressed	Level-C HRC no-halo algebraic diagnostic; ECT matching and particle content Open
Dynamic DE	Λ (rigid)	Λ (rigid)	Λ or modified	Not addressed	Conditional divided coefficient $\Lambda_{\text{div}}(X)$ and $w_\phi^{\text{bare}} \geq -1$ only for the supplied bare positive-kinetic scalar pieces; non-minimal terms leave the physical expansion response, w_{eff} and the late history Open

Axis	GR + SM	Einstein-aether	Hořava–Lifshitz	Volovik analogue	ECT
Singularities	Physical or censored	Physical or censored	Possibly UV-resolved	Not addressed	Candidate branch-boundary or condensate-breakdown interpretation; microscopic completion and replacement dynamics Open
BH information	Paradox open	Paradox open	Modified; model-dependent	Not addressed	Reduced-state reading; fundamental loss not required; final evaporation reconstruction open
Decoherence / arrow	External quantum postulate	External quantum postulate	External quantum postulate	Not addressed	Influence-functional structural route; quantitative entropy closure depends on bath assumptions
Entanglement	Primitive axiom	Primitive axiom	Primitive axiom	Not addressed	A common medium may host a supplied nonfactorising state/channel but does not derive one; Bell completion Open
UV / EFT validity scale	None (SM); Planck scale (GR)	None built-in	Lifshitz scaling	Phonon cutoff	P3 homogeneous radial correlation scale m_σ only; P4 pole/cutoff and physical UV completion Open
Falsifier style	Sectorwise	Extra-mode / LIV	LIV / dispersion	Not systematic	Named-completion sectorwise; cross-sector only after a common owner
Independent primitives	~ 6	~ 7	~ 5	$\sim 2-3$	Programme target: compress toward 1 medium + P1–P6; completion still open

CDT / causal dynamical triangulations. CDT [24] shares with ECT the goal of obtaining Lorentzian geometry from a more fundamental starting point, but does so through discrete simplicial complexes with a built-in causal ordering. ECT starts from a continuum Euclidean arena and, after a supplied ordered branch and independently supplied coefficients, classifies a Lorentzian scalar characteristic sector. It does not yet derive observed physical time, a universal metric, or the complete 3 + 1 matter/photon/tensor structure from SSB alone. CDT instead imposes causality at the lattice level and recovers 4D geometry in the continuum limit. Neither approach addresses the gauge or dark sectors in detail.

Barbour timeless mechanics. Barbour’s relational programme [323] shares ECT’s goal of eliminating external time but works in 3D configuration space rather than 4D Euclidean space. ECT and Barbour’s programme are therefore architecturally distinct: ECT starts from a 4D arena and conditionally classifies one scalar Lorentzian sector after ordering and coefficient input, whereas Barbour reconstructs dynamics from instantaneous configurations.

Why CDT and Barbour are treated separately. CDT and Barbour’s timeless mechanics are included here because they share with ECT an interest in emergent temporal structure, but they are not the closest neighbours of ECT along the gauge, medium, and condensate-response axes. For that reason, the main architectural table focuses on GR + SM, Einstein-aether, Hořava–Lifshitz, Volovik-type analogue emergence, and ECT, while CDT and Barbour are discussed in the expanded appendix as additional comparison points.

Scope note. This comparison is necessarily incomplete. Many other approaches (loop quantum gravity, string theory, asymptotic safety, causal sets, group field theory) have rich internal structures that cannot be compressed into a single table row. The purpose here is to locate ECT among its closest architectural neighbours, not to survey the full landscape.

BD Source-model RG algebra conditional on the P7 structural data

This appendix reproduces standard one-loop arithmetic only on the separately declared branch $\mathcal{B}_{\text{src}}$: constant background, $Y_0 = 0$, $\theta_{\text{gauge},0} = 0$, matter-free pure $SU(3)$ and neglected higher operators. It does not describe the general anisotropic EFT and makes no ECT physical claim.

One-loop integration on $\mathcal{B}_{\text{src}}$. For $g_{\text{src}}^2 = g_s^2/Z_0$ and $b_0 = 11$, $\mu dg_{\text{src}}/d\mu = -b_0 g_{\text{src}}^3/(16\pi^2)$ gives

$$\frac{1}{g_{\text{src}}^2(\mu)} = \frac{1}{g_{\text{src}}^2(M_*)} + \frac{b_0}{8\pi^2} \ln \frac{\mu}{M_*}, \quad (\text{BD.1})$$

and $\Lambda_{\text{col}}^{\text{src}} = M_* \exp[-8\pi^2/(b_0 g_{\text{src}}^2(M_*))]$. For a chosen threshold $x_c = g_{\text{src}}^2(\mu_c)$,

$$\mu_c = \Lambda_{\text{col}}^{\text{src}} \exp\left[\frac{8\pi^2}{b_0 x_c}\right]. \quad (\text{BD.2})$$

This is formal source-model running, not proof that a physical P7 action follows this trajectory or enters a constructive domain.

For the illustrative source-model inputs $\Lambda_{\text{col}}^{\text{src}} = 0.2$ GeV and $b_0 = 11$, the same formula gives $x_c = 2 \Rightarrow \mu_c \approx 7$ GeV and $x_c = 4 \Rightarrow \mu_c \approx 1.2$ GeV. These are Level-C source-model examples only; they neither locate a physical ECT blocking scale nor prove polymer entry.

Illustrative matching arithmetic. Inverting the formula gives $g_s^2(M_*)/Z_0 = 8\pi^2/[b_0 \ln(M_*/\Lambda_{\text{col}}^{\text{src}})]$. The following values merely reproduce the source model for the illustrative choice $\Lambda_{\text{col}}^{\text{src}} = 0.2$ GeV.

Table 148: Illustrative $\mathcal{B}_{\text{src}}$ matching arithmetic, $\Lambda_{\text{col}}^{\text{src}} = 0.2$ GeV and $b_0 = 11$; not an ECT prediction.

M_*	$\ln(M_*/\Lambda_{\text{col}}^{\text{src}})$	$x_{\text{src}} = g_s^2/Z_0$	$\alpha_{\text{src}} = x_{\text{src}}/(4\pi)$
$v_2 = 246$ GeV	7.11	1.009	0.080
10^{16} GeV	38.45	0.187	0.015
$\bar{M}_{\text{Pl}} = 2.435 \times 10^{18}$ GeV	43.95	0.163	0.013

At $M_* = \bar{M}_{\text{Pl}}$, changing the model input $x_{\text{src}} = 0.1633$ by ± 0.01 changes the formal scale from about 0.011 to 2.5 GeV around the 0.20 GeV reference. This reproducible sensitivity shows why matching inputs cannot yield a parameter-free scale.

General branch firewall. When Y is present, backgrounds run, matter is included or higher operators mix, the RG is multi-coefficient and Open. The table neither assumes Y is nonzero nor negligible, and no gravity/director coefficient supplies a bound. The $b_0 = 11$ source-model coefficient and table values must not be transferred to that general EFT.

BE P6-A: Conventions, record kernels, and normalisation dictionary

Status and scope. This appendix begins the technical P6/PES block. The P6/PES branch is frozen here as a *Level-B spectral-record framework with operational falsifiers*, not as a completed first-principles proof of quantum mechanics from the bare postulates. The purpose of the present appendix is narrower: it

fixes the conventions, record kernels, normalisations, counterterm conventions, and protocol guards used by the later P6 appendices. Throughout the P6 appendices the symbol c denotes the signal speed of the record medium in the channel under discussion; for condensate perturbations about the homogeneous ordered background it coincides with the effective propagation speed c_{char} of Section 5.3. No formula in this appendix should be read as a derivation of $S_0 = \hbar$, of the Born rule, of C_n or OP-Planck, of the colour postulate P7, of confinement, or of a universal scalar bath for all record channels.

BE.1 Record data and the Weiss convention

A certified P6 record channel is specified not by a bath exponent alone, but by the record data

$$(J_W(\omega), F_n(\omega), \Delta F_{\text{record}}(\omega), S_{\pm}(\omega) \text{ or } T_{\text{eff}}(\omega), \delta_{ct}), \quad (\text{BE.1})$$

where J_W is the Weiss-convention spectral density, F_n is the mode form factor, ΔF_{record} is the record operator difference relevant to the protocol, $S_{\pm}$ encode occupation/noise, and δ_{ct} denotes the subtraction or counterterm convention used in the self-energy. In the multi-channel case J_W is replaced by a positive spectral matrix

$$J_{ab}^W(\omega) \succeq 0. \quad (\text{BE.2})$$

A scalar one-kernel description is therefore a channel-selected or diagonalised limit, not the generic form of the theory.

All P6 exponent statements use the Weiss convention

$$a_{\text{bare}} = a_W + 1. \quad (\text{BE.3})$$

This matches the convention $J_W(\omega) = J_{\text{bare}}(\omega)/(2\omega)$ already used in Appendix AX. Thus an apparently identical power law quoted in a bare density and in a Weiss density differs by one power unless the convention is declared explicitly.

BE.2 Canonical C8 width-channel kernels

For the certified width-channel record operator, the canonical soft-edge kernels are

$$J_{\text{temp}}^{\text{can}}(\omega, k_z) = \frac{\omega^2 - c^2 k_z^2}{16\pi\omega^3} \Theta(\omega - c|k_z|), \quad (\text{BE.4})$$

$$J_{\text{spat}}^{\text{can}}(\omega, k_z) = \frac{\omega^2 - c^2 k_z^2}{16\pi c^2 \omega} \Theta(\omega - c|k_z|). \quad (\text{BE.5})$$

They obey the convention-independent invariant

$$\frac{J_{\text{spat}}^{\text{can}}}{J_{\text{temp}}^{\text{can}}} = \frac{\omega^2}{c^2}. \quad (\text{BE.6})$$

This invariant is the preferred check on component normalisation, because it is insensitive to the overall record-source strength.

The noncanonical reduced columns used in the P6 audit differ from the canonical columns by

$$J_{\text{red}} = 4J_{\text{can}}. \quad (\text{BE.7})$$

This factor belongs only to the noncanonical J -column audit. It must not be applied to native Φ -measure quantities such as the line tension coefficient $c_{\text{line}} = 1$, the exact cusp kernel, or the record-normalisation functional below.

BE.3 Pair normalisation and the value of ζ_0

The canonical line-overlap functional for a pair-built record is

$$\Phi_{\text{pair}}(R, L_z) = 2c_{\text{line}} \frac{W^2 K_\theta c_\chi}{S_0} L_z R + O(\xi_c L_z), \quad c_{\text{line}} \equiv \int_0^\infty \frac{1 - J_0(x)}{x^2} dx = 1. \quad (\text{BE.8})$$

The default P6 convention is therefore

$$\zeta_0^{\text{pair}} = 2, \quad \sigma_{\text{dec}} \equiv \sigma_{\text{pair}}, \quad \sigma_{\text{pair}} = 2\sigma_{\text{face}}. \quad (\text{BE.9})$$

The alternative $\zeta_0^{\text{face}} = 1$ is reserved for genuine one-sheet objects. No such one-sheet record construction is certified in the current P6 ledger. The winding factor $2\pi W$ is already part of the normalised source convention; it must not be inserted again as an extra $4\pi^2$ multiplier.

BE.4 Bosonic Kramers–Kronig mirror and counterterms

For oscillator-like bosonic record channels, the real self-energy shift must use the mirror-symmetric Kramers–Kronig kernel

$$\Sigma'(\omega) = \frac{2}{\pi} \mathcal{P} \int_E^\infty \frac{\omega' \Sigma''(\omega')}{\omega'^2 - \omega^2} d\omega'. \quad (\text{BE.10})$$

The one-sided principal-value kernel $1/(\omega' - \omega)$ is not the convention used in the frozen P6 branch. Any threshold-protection statement must also declare whether Ω_0 is a bare frequency, a physical $g \rightarrow 0$ calibrated frequency, or a counterterm-subtracted frequency. Without this declaration the absolute binding below the edge is not an invariant statement, although the vanishing of the width for a pole below the soft edge is robust within the stated channel.

BE.5 Effective temperature and occupation data

The operational detailed-balance thermometer is

$$T_{\text{eff}}(\omega) = \frac{S_0 \omega}{\ln[S_+(\omega)/S_-(\omega)]}. \quad (\text{BE.11})$$

Here $S_+(\omega)$ and $S_-(\omega)$ are ordered so that a thermal channel satisfies $S_+(\omega)/S_-(\omega) = \exp(S_0 \omega/T)$ for $\omega > 0$; if the opposite positive/negative-frequency convention is used, the ratio must be inverted. In the ledger units $S_0 = 1$ used in the frozen P6 derivation files this reduces to $\omega/\ln[S_+/S_-]$; the explicit S_0 restores the $\coth(S_0 \omega/2T_{\text{eff}})$ convention of Appendix AX. A scalar T_{eff} exists only if this expression is constant across the relevant frequency band. Otherwise the protocol must use $S_\pm(\omega)$, or equivalently the frequency-dependent occupation, directly. For an intrinsic ground-state threshold channel, $T_{\text{eff}} = 0$ is an occupation statement; it is not a licence to replace the full soft-edge kernel by an ohmic Markov coefficient.

BE.6 Same-channel response/noise and KMS–FDT diagnostic

Status and scope. The identities in this subsection are Level A standard spectral mathematics for a supplied stationary physical channel. Their use as a measurement architecture is a Level B conditional protocol. The present ECT gravitational application remains Open: the physical constrained operator, source vertex, state, canonical normalisation and common response/noise transfer have not been derived. No statement below is an ECT-specific experimental prediction.

Let $F_a = F_a^\dagger$ be centred environment force or record operators after constraint/gauge reduction and after the spatial smearing of the protocol has been frozen. With $F_a(t) = e^{+iH_E t/S_0} F_a e^{-iH_E t/S_0}$ and the Fourier convention $\tilde{K}(\omega) = \int dt e^{+i\omega t} K(t)$, define

$$C_{ab}^>(t) := \text{Tr}[\rho_E F_a(t) F_b(0)], \quad C_{ab}^<(t) := \text{Tr}[\rho_E F_b(0) F_a(t)], \quad (\text{BE.12})$$

$$D_{ab}^R(t) := -\frac{i}{S_0} \theta(t) \langle [F_a(t), F_b(0)] \rangle, \quad N_{ab}(t) := \frac{1}{2S_0} \langle \{F_a(t), F_b(0)\} \rangle. \quad (\text{BE.13})$$

These definitions use the passive convention $\text{Im}_H D^R(\omega) \preceq 0$ at positive frequency, where $\text{Im}_H D^R = (D^R - D^{R\dagger})/(2i)$. They reproduce the live dictionary $S_H = S_0 N^{(S)}$ and give

$$\boxed{C^> = S_H - S_0 \text{Im}_H D^R \succeq 0, \quad C^< = S_H + S_0 \text{Im}_H D^R \succeq 0.} \quad (\text{BE.14})$$

This pair is stationary physical Wightman positivity. It is not a theorem of abstract complete positivity alone: a reduced completely positive map does not uniquely determine a stationary dilation, D^R , or S_H . For a fixed scalar projection $F_v = v_a F_a$ it implies $h_v \geq |\mathcal{D}_v|$, with

$$h_v := v^\dagger S_H v, \quad \mathcal{D}_v := -S_0 v^\dagger \text{Im}_H D^R v. \quad (\text{BE.15})$$

The spectral data must also pass the negative-frequency and causal ledgers. For Hermitian channels, $C_{ab}^<(\omega) = C_{ba}^>(-\omega)$, $D_{ab}^R(-\omega) = D_{ab}^R(\omega)^*$ componentwise and $D^A(\omega) = D^R(\omega)^\dagger$; in a declared reciprocal matrix basis the last two relations combine to $D^R(-\omega) = D^R(\omega)^\dagger$. The retarded kernel must be analytic in the upper half-plane and its real and dissipative parts must obey the appropriate Kramers–Kronig relation. Any required subtraction, unresolved high-frequency tail, equal-time contact term and source Ward term belongs to that relation and must be fitted or fixed rather than silently discarded. For noncommuting matrix channels the stronger-looking expression $S_H \succeq |S_0 \text{Im}_H D^R|$ is false in general and is not used.

If the same stationary channel is in a canonical KMS state with respect to the named generator H_E at energy temperature T , then for $\omega > 0$ $C^<(\omega) = e^{-S_0 \omega/T} C^>(\omega)$ and hence

$$\boxed{S_H(\omega) = -S_0 \coth\left(\frac{S_0 \omega}{2T}\right) \text{Im}_H D^R(\omega).} \quad (\text{BE.16})$$

For a passive scalar projection on nonzero positive-frequency support, with $0 < \mathcal{D}_v < h_v$, the equivalent response/noise thermometer is

$$r_v(\omega) := \frac{\mathcal{D}_v(\omega)}{h_v(\omega)}, \quad \boxed{T_{\text{eff},v}(\omega) = \frac{S_0 \omega}{2 \text{artanh } r_v(\omega)}} = \frac{S_0 \omega}{\ln[S_{+,v}(\omega)/S_{-,v}(\omega)]}, \quad (\text{BE.17})$$

where $S_{+,v} = v^\dagger C^> v$ and $S_{-,v} = v^\dagger C^< v$ in the positive-frequency convention. If the ordered spectra are measured independently, their directly observable ratio fixes T/S_0 ; kelvin then requires the separate S_0 matching and division by k_B . If instead $S_\pm$ are reconstructed from S_H and D^R , that reconstruction already contains S_0 . In that case S_0 must be fixed by an independent matching or fitted jointly with T , and the resulting T – S_0 degeneracy must be reported rather than counted as a second thermometer. At a passive nondegenerate ground-state positive-frequency channel, $S_- = 0$ and the scalar inequality saturates by a limiting statement. This does not extend automatically to driven, squeezed, inverted, degenerate or nonstationary states.

The comparison is admissible only when response and noise use the same Hermitian physical operator, source vertex, canonical normalisation, projector, spatial form factor, two-sided angular-frequency convention, window/binning and detector transfer. The same congruence filter must act on both operator legs. Detector imprecision, backaction and their cross-spectrum must be calibrated, and the drive must be extrapolated to the undisturbed state. The matrix test must either use the full matrix or a preregistered informationally complete set of complex projections, including relative-phase directions such as $e_a + e^{i\phi} e_b$; an eigenbasis selected after looking at the same data is not a valid independent test.

Identical finite windows are necessary but not sufficient. For a positive leakage kernel L_B the measured ordered spectra are

$$\bar{C}_B^{\geq}(\omega_0) = \int d\omega L_B(\omega_0, \omega) C^{\geq}(\omega), \quad L_B \geq 0. \quad (\text{BE.18})$$

A thermal hypothesis must therefore be forward-convolved through the actual L_B : one compares $\bar{S}_{H,B}$ with the convolution of $-S_0 \coth[S_0 \omega/(2T)] \text{Im}_H D^R(\omega)$. Pulling the coth factor outside a broad bin and evaluating it at ω_0 is invalid unless a controlled narrow-bin error bound is supplied. Poles, cuts and zeros

are likewise compared only after this common positive convolution; a 0/0 bin carries no temperature information, and the strict inequalities used in Eq. (BE.17) are not asserted off nonzero positive-frequency support.

The state preparation is part of the channel definition. The exact global KMS theorem does not require a factorised system–environment state. A bath-only reduced implementation, however, must declare its autonomous-bath and product/weak-coupling domain or derive a compatible correlated influence functional. Generic initial probe–environment correlations or preparation transients can add inhomogeneous response/noise terms; source contact terms and Ward identities must be retained before testing stationarity, causality or KMS. Gaussianity is not required for the two-point KMS identity, but it is an additional assumption if these kernels are to close the entire influence functional.

A frequency-dependent T_{eff} after all these guards rejects only a canonical two-point KMS model for the named operator, state, band and generator. It does not by itself establish a generic “nonthermal state”: a generalised Gibbs/KMS state may involve declared conserved charges, chemical potentials or a rotating generator, with correspondingly shifted detailed balance. Ordinary multiple baths, feedback, active reservoirs, driven occupations and transfer mismatch can produce the same outcome. Conversely, a constant value in one projection does not prove a global KMS state. In particular, the currently available ω^3 and ω^9 laws belong to differently filtered or composite candidates and cannot be inserted as a same-channel response/noise pair without an absolute common operator and filter. Their present ECT comparison is therefore NOT IDENTIFIABLE.

Figure 43 visualises the protocol mechanics and its principal false-positive controls.

BE.7 Exact cusp kernel and intrinsic resolution floor

The exact C10 cusp kernel is

$$\Phi(\Delta) = 4\pi \left[\sqrt{4\xi_c^2 + \Delta^2} - 2\xi_c - 2\xi_c \ln \left(\frac{2\xi_c + \sqrt{4\xi_c^2 + \Delta^2}}{4\xi_c} \right) \right]. \quad (\text{BE.19})$$

It implies the intrinsic resolution floor

$$\xi_{\text{res}}^{\text{intr}} = 4\xi_c. \quad (\text{BE.20})$$

The large-separation logarithmic part contains the shifted logarithm

$$-8\pi\xi_c \ln(\Delta/4\xi_c), \quad (\text{BE.21})$$

which should be used when comparing finite-window fits with the asymptotic law.

BE.8 Core-profile fingerprint constants

For a regulated core profile $f(u)$, the approach to the universal asymptotic line slope contains a profile-dependent fingerprint

$$a_1[f] = \int_0^\infty \frac{1-f(u)}{u^2} du. \quad (\text{BE.22})$$

The certified examples are

$$a_1^{\text{Gauss}} = \sqrt{\pi}, \quad a_1^{\text{Lorentz}} = \frac{\pi}{2}. \quad (\text{BE.23})$$

For the exponential/C10 class the approach is logarithmic rather than finite- a_1 :

$$D(R) = \frac{\xi}{R} \left[\ln \frac{R}{2\xi} + 1 \right] + \dots. \quad (\text{BE.24})$$

Thus fitted exponents below unity in finite windows should not be promoted to new physical powers until this logarithmic law has been excluded.

BE.9 Airy-width stiffness convention

The width-sector tilt parameter is the physical stiffness

$$\kappa_b = \frac{T_{\text{line}}}{2S_0}. \quad (\text{BE.25})$$

With the transfer Hamiltonian

$$H = -\frac{1}{2\kappa_b} \partial_x^2 + \sigma_{\text{dec}} x, \quad (\text{BE.26})$$

the Airy ground-state scale obeys

$$E_0 = |a_1| \left(\frac{\sigma_{\text{dec}}^2}{2\kappa_b} \right)^{1/3}, \quad \ell_w \propto (\kappa_b \sigma_{\text{dec}})^{-1/3}, \quad (\text{BE.27})$$

and the coherence-window scaling is

$$L_z^{\text{coh}} = \frac{-\ln P_{\text{min}}}{\sigma_{\text{dec}} \ell_w} \propto \kappa_b^{1/3} \sigma_{\text{dec}}^{-2/3}. \quad (\text{BE.28})$$

A width law with $\kappa_b^{+1/3}$ is valid only after introducing a separate flexibility $D_b = 1/\kappa_b$. The same symbol must never be used for stiffness and flexibility.

BE.10 Protocol split and matrix-kernel guard

Two P6 protocol classes must not be mixed. External or detuned monitors use the golden-rule form

$$\Gamma_n^{\text{ext}} = 2\pi J_W(\Omega_n) |F_n(\Omega_n)|^2 [1 + n_{\text{eff}}(\Omega_n)]. \quad (\text{BE.29})$$

Intrinsic threshold channels instead use the dressed soft-edge pole problem and may produce a sub-edge line plus a continuum shoulder. In such channels a below-edge pole satisfies $\Gamma(\omega_b) = 0$ within the ideal soft-edge kernel, while the spectral weight is shared between the bound line and the continuum shoulder.

For multiple record operators the spectral object is a positive matrix, not a single scalar density:

$$J_{ab}(\omega) \succeq 0, \quad |J_{ab}(\omega)|^2 \leq J_{aa}(\omega) J_{bb}(\omega). \quad (\text{BE.30})$$

Scalar one-kernel closure is a further channel-selection or diagonalisation step. This guard is required before applying the P6-S7 triangle or the P26 over-closure test.

BE.11 Correction index local to this appendix

The conventions above incorporate the frozen P6 correction history: the finite- k_z C8 correction, the spatial factor-of-two correction caught by $J_{\text{spat}}/J_{\text{temp}} = \omega^2/c^2$, the $4\times$ reduced/canonical dictionary, the bosonic KK mirror correction, the finite-window/log artefact corrections in P25/P18, and the P11 stiffness/flexibility transcription correction. These are current notes: they are part of the reproducibility conditions for the P6/PES appendices.

BF P6-B: Core spectral laws, dephasing classes, and tomography

Status and scope. This appendix records the frozen P6/PES core spectral laws. It uses the conventions fixed in Appendix BE: Weiss spectral densities, explicit S_0 in thermal factors, canonical versus reduced record kernels, and the matrix-kernel guard. The results below are Level-B spectral-record statements. They organise how a certified record channel produces dephasing, induced coefficients, tomography data, and sign constraints. They do not prove the full PES principle from the bare postulates, do not derive $S_0 = \hbar$, and do not assert that every ECT channel is ohmic or scalar.

BF.1 L12': dispersion–dimension law for the Weiss exponent

For a channel with effective dimension d_{eff} , dispersion $\omega \propto k^z$, vertex derivative order ℓ , and possible infrared matrix-element singularity exponent ν , the Weiss low-frequency spectral density

$$J_W(\omega) \sim C\omega^{a_W} \quad (\text{BF.1})$$

has exponent

$$a_W = \frac{d_{\text{eff}} + 2\ell}{z} + 2\nu - 2. \quad (\text{BF.2})$$

This is a counting law for a declared record operator and channel. It is not a rule that turns every boundary, line, or bulk channel into an ohmic bath. In particular, a quoted power must always specify whether it is a bare density or a Weiss density, since the P6 convention is $a_{\text{bare}} = a_W + 1$.

Equation (BF.2) is the guard against exponent-only reasoning. Persistent spectra are not determined by a_W alone; they are determined by the pair (J_W, F_n) , together with the protocol occupation data and counterterm convention. This is the spectral version of the pairwise record-kernel quietness criterion: robustness is determined by the actual record difference, kernel, and form factor, not by a slogan such as “ohmic bath” or by a universal one-history PES scalar.

BF.2 L16 and L40: dephasing trichotomy

For a stationary Gaussian record channel one may write the dephasing part in the spectral form

$$\Phi(t) = \int_0^\infty d\omega S_{\text{rec}}(\omega) \frac{1 - \cos(\omega t)}{\omega^2}, \quad (\text{BF.3})$$

where S_{rec} includes the Weiss spectral density, the relevant record-operator difference, form factors, and the occupation/noise factor. (The symbol S_{rec} is used for this record-noise spectrum to avoid any confusion with the effective action S_{eff} of the main text.) Equivalently, for a thermal scalar channel one may absorb the factor $\coth(S_0\omega/2T_{\text{eff}})$ into S_{rec} . Formula (BF.3) is a compact representation of the influence-functional class; outside the Gaussian/stationary setting the cumulant form in Section BF.4 is the safer definition.

If $J_W(\omega) \sim C\omega^{a_W}$ and the channel is in its vacuum occupation, the large-time behaviour falls into the trichotomy

$$\Phi(t) \sim \begin{cases} t^{1-a_W}, & a_W < 1, \\ \log t, & a_W = 1, \\ \Phi_\infty - O(t^{1-a_W}), & a_W > 1. \end{cases} \quad (\text{BF.4})$$

For an actually occupied thermal channel, the additional infrared factor from $\coth(S_0\omega/2T_{\text{eff}})$ shifts the thresholds by one power:

$$\Phi_T(t) \sim \begin{cases} t^{2-a_W}, & a_W < 2, \\ \log t, & a_W = 2, \\ \Phi_{T,\infty} - O(t^{2-a_W}), & a_W > 2. \end{cases} \quad (\text{BF.5})$$

Thus an occupied ohmic channel gives Markov-like linear growth, while a vacuum ohmic channel gives a logarithmic dephasing law. The word “temperature” in this statement refers to the occupation of the actual record channel. If the detailed-balance thermometer of Appendix BE.5 is not constant across the relevant band, the protocol must use $S_\pm(\omega)$ directly rather than fitting a scalar T_{eff} .

BF.3 L19: induced-coefficient sum rule

The same positive spectral kernel that measures record loudness also induces the corresponding quadratic coefficient when the record channel is integrated out. In the Weiss convention the induced coefficient is

$$\kappa_{\text{ind}} = \frac{2}{\pi} \int_0^\infty \frac{J_W(\omega)}{\omega} d\omega. \quad (\text{BF.6})$$

For the regulated power-law model

$$J_W(\omega) = C \omega^{a_W} \exp(-\omega/\omega_c), \quad (\text{BF.7})$$

one obtains

$$\kappa_{\text{ind}} = \frac{2C}{\pi} \Gamma(a_W) \omega_c^{a_W}. \quad (\text{BF.8})$$

This regulated family has the same form as $J_s(\omega) = A_s \omega^s e^{-\omega/\omega_c}$ of Appendix AX, eq. (AX.3), with $C = A_s$ and $a_W = s$ when J_s is read in the Weiss convention; whether a given tabulated exponent is bare or Weiss must be declared explicitly, per Appendix BE. Consequently, once the induced coefficient and cutoff are fixed by a sector calculation, the spectral amplitude C is no longer a free loudness parameter. In the gravitational matching route this is the bridge from the induced coefficient to the M_G^2 normalisation; in the P6 record route it is the same object appearing in dephasing and line-shift observables. This is the first form of the one-kernel closure later sharpened in the P6-S7/P26 appendix.

BF.4 L23, L29, L29a, and L30: non-Markov windows and finite-bath guards

Some P6 protocols use early-time or finite-window information rather than an asymptotic Markov rate. A certified example is the square-root branch

$$\Phi(t) + 2\sqrt{\pi} \propto t^{1/2}, \quad (\text{BF.9})$$

which is used later as part of the guarded zero-mode/Berry detector. This law is a channel-specific signature, not a generic substitute for the trichotomy of Section BF.2.

For non-Gaussian records the Gaussian variance formula must be replaced by the cumulant definition

$$\Gamma = -\ln |\langle e^{iX} \rangle|. \quad (\text{BF.10})$$

Here X denotes the accumulated record variable of the channel over the protocol window. The Gaussian expression is recovered as the leading second-cumulant approximation, but the exact characteristic function is the object that decides whether a record is erased or superselected.

A useful corollary is the quiet-lattice rule. If the environment records only quantised kicks on a lattice, the dephasing functional is bounded and periodic; in particular it vanishes at integer multiples of the environment's own quantum. Such a bath cannot superselect states separated only by multiples of its own kick quantum. Winding superselection therefore requires the continuum or non-commensurate record channel, not merely a quantised finite lattice of kicks.

For finite baths two questions must be distinguished. Coherence is preserved throughout a window when

$$\sup_{0 \leq t \leq T_{\text{obs}}} \Phi(t) \leq -\ln V_{\text{min}}, \quad (\text{BF.11})$$

whereas persistence of an already formed record requires

$$\inf_{t_{\text{rec}} \leq t \leq T_{\text{obs}}} \Phi(t) \geq -\ln V_{\text{rec}}, \quad t_{\text{rec}} > 0. \quad (\text{BF.12})$$

The infimum over $[0, T_{\text{obs}}]$ is trivial because $\Phi(0) = 0$.

BF.5 L31–L35b: dephasing tomography package

When $\Phi(t)$ is known over a sufficient window and has the stationary spectral representation (BF.3), the effective spectral kernel may be reconstructed by the second-derivative cosine inversion

$$S_{\text{rec}}(\omega) = \frac{2}{\pi} \int_0^\infty \Phi''(t) \cos(\omega t) dt. \quad (\text{BF.13})$$

This inversion supplies the basic tomography route. Its first guard is positivity:

$$S_{\text{rec}}(\omega) \geq 0. \quad (\text{BF.14})$$

A negative reconstructed kernel is a diagnostic of noise, window artefacts, an incorrect subtraction, or an invalid Gaussian/stationary model.

The second guard is detailed balance. With explicit S_0 convention, a thermal channel satisfies

$$S_{-}(\omega) = \exp(-S_0\omega/T)S_{+}(\omega), \quad \omega > 0. \quad (\text{BF.15})$$

Failure of this relation is not automatically a failure of PES; it means that the channel is athermal, driven, coarse-grained, or otherwise outside the scalar thermal closure.

For kernels with an asymptotic plateau Φ_{∞} , a derivative-free inversion is often better conditioned:

$$S_{\text{rec}}(\omega) = \frac{2}{\pi} \omega^2 \int_0^{\infty} [\Phi_{\infty} - \Phi(t)] \cos(\omega t) dt. \quad (\text{BF.16})$$

The frozen P6 audit found this route to be far better conditioned than direct differentiation in noisy windows. The methodological lesson is binding: sub-ohmic and near-ohmic classes are window-hungry, and a finite observation time can mimic a shifted exponent unless the window law is included.

The two-regime window law is

$$\text{IR bias} \sim \begin{cases} T_{\text{obs}}^{-a_W}, & \omega T_{\text{obs}} \lesssim 1, \\ T_{\text{obs}}^{-(a_W+1)}, & \omega T_{\text{obs}} \gg 1, \end{cases} \quad (\text{BF.17})$$

up to channel-dependent prefactors. Apparent exponents extracted from short windows must therefore be reported together with the window, cutoff, and inversion route.

BF.6 L33 and P6-S7 preview: triangle consistency before full closure

Already at the tomography level, dephasing, widths, and shifts are not independent fit parameters. For a fixed channel, the reconstructed J_W that explains $\Phi(t)$ must also predict the golden-rule widths and Kramers–Kronig shifts once the form factors and counterterm convention are supplied. This is the preliminary form of the one-kernel triangle. The full P6-S7/P26 statement is deferred to the dedicated P6-F appendix because the generic object is a positive spectral matrix $J_{ab}(\omega)$, not a universal scalar bath.

BF.7 L36: sign-protection chain

For a certified passive record channel,

$$J_W(\omega) \geq 0 \quad (\text{BF.18})$$

implies

$$\kappa_{\text{ind}} = \frac{2}{\pi} \int_0^{\infty} \frac{J_W(\omega)}{\omega} d\omega \geq 0. \quad (\text{BF.19})$$

Thus the induced contribution generated by tracing out a positive record kernel is non-negative. This is a sign-protection statement for the induced contribution, not a claim that every renormalised low-energy coefficient in the full theory must be positive after all counterterms, competing sectors, and matching conditions are included. Whenever a negative effective coefficient is inferred, the audit question is therefore: which subtraction, active channel, or non-passive sector has been introduced?

BF.8 Handoff to the ribbon and threshold appendices

Appendix BE fixes the conventions. The present appendix records the core spectral laws and tomography tools. The next P6 appendices specialise these tools to the ribbon/line sector, the certified record operators and form factors, the intrinsic threshold-protection mechanism, and the M3 falsification protocols. The frozen ordering is important: later claims must inherit the convention and tomography guards above rather than redefining the bath exponent, the occupation convention, or the counterterm convention locally.

BG P6-C: Ribbon/line sector, record classes, and form factors

Status and scope. This appendix specialises the spectral-record machinery of Appendices BE and BF to the certified P6 ribbon/line sector. The status is Level B throughout, except where an open input is explicitly named. The claims below are structural record-sector statements: they identify the line tension, endpoint guards, core and tilt inputs, Magnus fork, record operator, record-class hierarchy, and external-monitor form-factor laws used by the later threshold and M3 appendices. They do not derive the microscopic value of S_0 , do not prove the colour postulate P7, and do not turn analogue-line falsification into a falsification of the whole ECT framework.

A notational warning is needed. Appendix BE.3 denotes the P6 Bessel-integral line coefficient by $c_{\text{line}} = 1$. This notation is used throughout the P6 appendices to avoid collision with the orientation-stress $\text{OP-}c_1$ used elsewhere in the manuscript. Equivalently,

$$c_{\text{line}} \equiv \int_0^\infty \frac{1 - J_0(x)}{x^2} dx = 1. \quad (\text{BG.1})$$

(This is the same dimensionless number written c_1 in the P6 derivation files.)

BG.1 C1–C2: pair tension and ribbon area law

For a pair-built winding record of transverse separation R and length L_z , the P6 pair convention of Appendix BE.3 gives

$$\Phi_{\text{pair}}(R, L_z) = \sigma_{\text{dec}} L_z R + O(\xi_c L_z), \quad \sigma_{\text{dec}} \equiv \sigma_{\text{pair}} = 2 \frac{W^2 K_\theta c_\chi}{S_0}. \quad (\text{BG.2})$$

In the ledger units this is the statement that the Bessel-integral slope is $2c_{\text{line}} = 2$. The factor of two is not an optional monitor convention: all certified P6 line, width, and cusp constructions are pair constructions. A one-face coefficient $\sigma_{\text{face}} = \sigma_{\text{pair}}/2$ may be introduced only for a genuine one-sheet object; none is currently certified in the P6 ledger.

Because the pair can be extended along z without changing the transverse kernel, the line law is also a ribbon area law,

$$\Phi_{\text{ribbon}} = \sigma_{\text{dec}} A_{\text{ribbon}} + O(\xi_c L_z), \quad A_{\text{ribbon}} = L_z R. \quad (\text{BG.3})$$

The excluded alternative is an isolated unscreened winding sector, whose record cost grows with the transverse system size rather than with a finite pair separation. Thus the finite P6 object is a bound record pair/ribbon, not a freely isolated line charge.

BG.2 C4: endpoint mass, screening, and line-breaking guard

The endpoint contribution is the core/perimeter part left after the extensive pair tension has been separated. Without screening, an endpoint carries an infrared-linear self-loudness and cannot be used as a finite line-breaking defect. With a screening length ξ_{scr} supplied by a specified carrier, the certified endpoint estimate is

$$M_{\text{end}} = \frac{1}{2} \sigma_{\text{dec}} \xi_{\text{scr}}. \quad (\text{BG.4})$$

If breaking a ribbon changes the effective tension by $\sigma_{\text{br}} > 0$, the thin-wall estimates are

$$R_{\text{br}} = \frac{2M_{\text{end}}}{\sigma_{\text{br}}} = \frac{\sigma_{\text{dec}}}{\sigma_{\text{br}}} \xi_{\text{scr}}, \quad B = \frac{\pi M_{\text{end}}^2}{\sigma_{\text{br}}} = \frac{\pi}{4} \frac{\sigma_{\text{dec}}^2}{\sigma_{\text{br}}} \xi_{\text{scr}}^2. \quad (\text{BG.5})$$

The thick-wall prefactor is not fixed by this argument. The screening carrier and ξ_{scr} are open sector inputs, not consequences of the line tension alone.

If an unresolved line ensemble has entropy per unit length $s_{\text{line}} = \ln \mu_{\text{step}} / \xi_{\text{line}}$, compare it with the record cost per unit length. For $\Phi = \sigma_{\text{dec}} L_z R$,

$$\sigma_{\text{dec}} R > \frac{\ln \mu_{\text{step}}}{\xi_{\text{line}}}, \quad R \sim \xi_{\text{line}} : \quad \sigma_{\text{dec}} \xi_{\text{line}}^2 > \ln \mu_{\text{step}}. \quad (\text{BG.6})$$

The ensemble, core scale, and meaning of μ_{step} are inputs.

BG.3 S1 and C5: core profile and tilt stiffness

The reduced Ginzburg–Pitaevskii core calculation fixes the line-core profile and the dimensionless core constant used in the ribbon sector. In the frozen ledger the profile audit reproduces the GP value

$$\frac{c_{\text{GP}}}{\pi} = 0.3811 \quad (\text{BG.7})$$

within the quoted numerical tolerances. (This constant is written c_0 in the P6 derivation files; it is renamed c_{GP} here because c_0 denotes an entropy-capacity coefficient elsewhere in the manuscript.) The window-independent gradient length is

$$\xi_{\text{grad}} = 1.477 \xi_{\text{cond}}, \quad (\text{BG.8})$$

where ξ_{cond} is the core/healing-length parameter of this supplied GP profile only; it is not ξ_{P3} , an Open P4 pole length, or the bath cutoff length unless a separate matching is supplied. Any fitted ξ_{eff} must state the finite window used to extract it.

The width-sector tilt coefficient is an independent stiffness. In the stiffness convention fixed in Appendix BE.9,

$$\kappa_b = \frac{T_{\text{line}}}{2S_0}, \quad (\text{BG.9})$$

where T_{line} is the line’s energetic tilt cost, including the appropriate core/log-running input for the width scale under study. Tension does not by itself determine the Airy width spectrum; an independent calibration of κ_b is required. The loudness-gradient part of the stiffness remains an explicitly named open input until a sector calculation supplies it.

BG.4 C6–C7: Magnus fork and the minimal ohmic branch

The minimal ribbon action inherited from the strictly second-order phase sector has the schematic form

$$\frac{S_{\text{line}}^{(2)}}{S_0} = \frac{1}{2} \int dt dz [M_{\perp} (\partial_t X)^2 + T_{\text{line}} (\partial_z X)^2] + \dots \quad (\text{BG.10})$$

It gives a linear branch $\omega \propto |k_z|$, hence dynamical exponent $z = 1$. In the P6 count this is the minimal ohmic ribbon branch,

$$n_0 = 0, \quad \kappa_M^{\text{bulk}} = 0, \quad z = 1, \quad a_W = 1. \quad (\text{BG.11})$$

The word “ohmic” here is a statement about the minimal Weiss exponent of this certified branch, not a licence to replace every subsequent protocol by a Markov bath.

A first-order Magnus/Berry term would instead have the schematic form

$$\frac{S_M}{S_0} = \frac{i\kappa_M}{2} \int dt dz (X \partial_t Y - Y \partial_t X), \quad \kappa_M \propto \frac{W n_0}{S_0}, \quad (\text{BG.12})$$

and would generate a quadratic Kelvin-like branch in the infrared. The frozen S2b verdict for the minimal bulk ribbon is $n_0 = 0$; the $n_0 \neq 0$ route is a separate core-zero-mode/Berry branch and is tested by the M3 zero-mode protocol, not assumed inside the minimal ribbon sector.

BG.5 C8: certified bulk record operator and soft-edge kernels

The certified bulk record operator is dictated by the pair geometry. For a pair separated by d in the transverse plane its Fourier kernel is

$$G(\mathbf{k}; d) = -2\pi i W \frac{k_y \cos(k_x d/2)}{k_x^2 + k_y^2 + k_z^2}. \quad (\text{BG.13})$$

Thus $|G|^2$ contains the same inverse-square kernel that appears in the C1 line-tension integral. The record operator is not selected from a menu after the fact: it is the particular combination of width coordinate and time-derivative records induced by this vertex.

After the canonical C8 normalisation of Appendix BE.2, the temporal and spatial record kernels are

$$J_{\text{temp}}^{\text{can}}(\omega, k_z) = \frac{\omega^2 - c^2 k_z^2}{16\pi\omega^3} \Theta(\omega - c|k_z|), \quad (\text{BG.14})$$

$$J_{\text{spat}}^{\text{can}}(\omega, k_z) = \frac{\omega^2 - c^2 k_z^2}{16\pi c^2 \omega} \Theta(\omega - c|k_z|). \quad (\text{BG.15})$$

They are repeated here only to make the line-sector handoff explicit. The ratio $J_{\text{spat}}^{\text{can}}/J_{\text{temp}}^{\text{can}} = \omega^2/c^2$ is the preferred component-normalisation check. The noncanonical reduced columns are $4J^{\text{can}}$ and should not be mixed into the native Φ -measure line tension or cusp formulas.

BG.6 L42–L45 and C10: record classes and cusp linkage

Record classes are organised by the infrared measure of the record difference. In the G -kernel measure, the critical exponent is

$$\rho_{G,\text{crit}} = 1, \quad (\text{BG.16})$$

corresponding to $\rho_{\theta,\text{crit}} = 3/2$ in the certified Φ -measure dictionary. The frozen class hierarchy is

record class	infrared status	interpretation
endpoint	hard, power-divergent	the only hard class, $\rho = 2$
pair separation	finite tension	the C1/C2 pair/ribbon law
width difference	finite but linearly recorded	effective pair of size $ \Delta d $
translation	soft	no hard record by itself

The width-difference law, in the pair convention, is

$$\frac{\Phi_{\text{width}}(\Delta d)}{L_z} = \sigma_{\text{dec}} |\Delta d| + O(\xi_c). \quad (\text{BG.17})$$

This statement is one reason the default assignment $\sigma_{\text{dec}} = \sigma_{\text{pair}}$ is used uniformly throughout P6.

The exact C10 cusp kernel and the intrinsic resolution floor are given in Appendix BE.7. The important line-sector interpretation is that separated width sectors are not local delta-function monitors; they are cusp-kernel objects with intrinsic floor

$$\xi_{\text{res}}^{\text{intr}} = 4\xi_c. \quad (\text{BG.18})$$

The Gaussian monitor introduced below is a resolved-window approximation to this cusp object, not a replacement for it at intrinsic resolution.

BG.7 C11: Gaussian width-monitor anchor

For a Gaussian resolved width monitor with resolution ξ_{res} , the certified anchor is

$$\lambda_n^{\text{rec}} = \frac{\sigma_{\text{dec}}}{2\xi_{\text{res}}}. \quad (\text{BG.19})$$

It is mode-independent at the level of the resolved Gaussian anchor. This formula is the bridge from the exact C10 cusp kernel to the form-factor taxonomy of the external-monitor calculation. In intrinsic width-sector work one should use $\xi_{\text{res}} = 4\xi_c$ only when the protocol really probes the intrinsic floor; an external detector may have a larger experimental resolution.

BG.8 C12 and L46: form factors and external-monitor hierarchy

External or detuned monitors use the golden-rule form already fixed in Appendix BE.10,

$$\Gamma_n^{\text{ext}} = 2\pi J_W(\Omega_n) |F_n(\Omega_n)|^2 [1 + n_{\text{eff}}(\Omega_n)]. \quad (\text{BG.20})$$

The key C12 lesson is that the bath exponent alone does not determine the persistent spectrum. After dividing out the same J_W and the same occupation factor, the certified form-factor hierarchy is

$$|F_n|^2 \sim n^0, \quad |F_n|^2 \sim n^2, \quad |F_n|^2 \sim n^2, \quad |F_n|^2 \sim n^{-2} \quad \text{on the odd-parity subsector.} \quad (\text{BG.21})$$

The two n^2 entries correspond to distinct record operators with the same large- n power. They should not be collapsed unless the underlying operator has also been identified. Conversely, an observed width hierarchy that does not match any certified F_n class after division by J_W and occupation data is a record-operator discriminator: it challenges the proposed monitor, not the entire P6/PES framework.

BG.9 Handoff to threshold protection and M3 protocols

This appendix has established the line/ribbon object and the external-monitor form-factor hierarchy. The intrinsic width channel is different: its mode sits at its own soft radiation edge and must be treated by a dressed pole plus continuum calculation. That threshold-protection mechanism, its sum rule, and its five-case detuning map are deferred to the next P6 appendix. The M3 protocols then use the separation between external C12 monitoring and intrinsic C13' threshold behaviour as the primary falsifier. Mixing the two protocols is the main interpretive error this appendix is designed to prevent.

BH P6-D: Intrinsic threshold protection and edge-plus-line spectra

Status and scope. This appendix records the conditional C13' threshold-protection mechanism of the frozen P6 ledger. It uses the conventions of Appendix BE, the core spectral laws of Appendix BF, and the certified line/width record operator of Appendix BG. The status is Level B conditional. The result is a statement about an intrinsic width-channel self-record sitting at its own soft radiation edge. It is not a generic statement about every external monitor, every oscillator coupled to a bath, or every line in the analogue laboratory. In particular, the external-monitor golden-rule hierarchy remains the C12 result of Appendix BG.8; the present appendix treats the intrinsic dressed-pole problem.

The robustness claim below is the vanishing of the width for a pole below the edge, together with the line-plus-shoulder spectral form and the associated sum rule. The absolute binding energy below the edge is not invariant until the counterterm and detuning convention have been declared.

BH.1 Soft edge inherited from C8

For fixed longitudinal momentum k_z , the certified C8 kernels have the threshold

$$E \equiv E_{\text{edge}} = c|k_z|, \quad (\text{BH.1})$$

and vanish below it. Near the edge, with $\varepsilon = \omega - E > 0$, Appendix BE.2 gives

$$J_{\text{temp}}^{\text{can}}(\omega, k_z) = \frac{\varepsilon}{8\pi E^2} + O(\varepsilon^2), \quad (\text{BH.2})$$

$$J_{\text{spat}}^{\text{can}}(\omega, k_z) = \frac{\varepsilon}{8\pi c^2} + O(\varepsilon^2). \quad (\text{BH.3})$$

Thus the intrinsic width channel sees a soft linear continuum onset rather than a hard threshold singularity. For the threshold analysis it is useful to write the effective edge kernel in the form

$$J_{\text{edge}}(\omega) = g^2 A_E (\omega - E) \Theta(\omega - E) f_{\text{UV}}(\omega), \quad A_E > 0, \quad f_{\text{UV}}(E) = 1, \quad (\text{BH.4})$$

where g is the channel amplitude and A_E includes the selected temporal/spatial component, source normalisation, and any declared projection. Equation (BH.4) is a local threshold model, not a replacement for the full C8 kernel away from the edge.

BH.2 Dressed intrinsic response and bosonic mirror dispersion

The intrinsic width mode is represented by the retarded dressed response

$$G_R(z) = \frac{1}{D(z)}, \quad D(z) = z^2 - \Omega_0^2 + \Sigma_R(z) - \delta_{ct}. \quad (\text{BH.5})$$

Here Ω_0 is the frequency in the declared bare or calibrated convention, δ_{ct} is the subtraction convention, and Σ_R is computed with the bosonic mirror kernel of Appendix BE.4. Recalling that convention, with a positive continuum width above the edge, the real part is

$$\Sigma'(\omega) = \frac{2}{\pi} \mathcal{P} \int_E^\infty \frac{\omega' \Sigma''(\omega')}{\omega'^2 - \omega^2} d\omega'. \quad (\text{BH.6})$$

The one-sided kernel $1/(\omega' - \omega)$ is not used in the frozen P6 branch. It fails the oscillator sum-rule audit and is included in the correction history rather than in the certified convention.

A real pole below the edge satisfies

$$D(\omega_b) = 0, \quad \omega_b < E. \quad (\text{BH.7})$$

Because $J_{\text{edge}}(\omega_b) = 0$, the intrinsic continuum gives

$$\Gamma(\omega_b) = 0 \quad (\text{BH.8})$$

before any external readout convolution or environmental leakage is added. This is the core of threshold protection. It should not be confused with a weak but finite Lorentzian width above the edge.

BH.3 C13': the conditional threshold-protection statement

In the self-consistent intrinsic width channel, the undressed width mode is edge-aligned in the ideal minimal geometry,

$$\Omega_0 = E_{\text{edge}}, \quad (\text{BH.9})$$

and is dressed by its own soft edge. In a bare-edge convention this produces a subedge pole,

$$\omega_b < E_{\text{edge}}, \quad E_{\text{edge}} - \omega_b = O(g^2), \quad \Gamma(\omega_b) = 0, \quad (\text{BH.10})$$

with a soft continuum shoulder above the edge. The $O(g^2)$ statement is made with g the channel amplitude; if the scan variable is already g^2 , the fitted exponent is unity.

The intrinsic spectral form is therefore

$$\rho_{\text{intr}}(\omega) = \frac{Z_b}{2\omega_b} \delta(\omega - \omega_b) + \rho_{\text{cont}}(\omega) \Theta(\omega - E_{\text{edge}}). \quad (\text{BH.11})$$

This is not the C12 external-monitor spectrum. The C12 form is recovered only when an external or detuned monitor reads the width family through the golden-rule expression in Appendix BG.8. The primary M3 discriminator is precisely this two-protocol contrast.

BH.4 Spectral-weight sum rule

The subedge line and the continuum shoulder are not independent fit components. In the oscillator normalisation used in the P6 audits they obey

$$Z_b + \int_E^\infty 2\omega \rho_{\text{cont}}(\omega) d\omega = 1. \quad (\text{BH.12})$$

Equivalently, Z_b is the oscillator-normalised residue of G_R at the subedge pole,

$$Z_b = \frac{2\omega_b}{D'(\omega_b)} = \left[1 + \frac{1}{2\omega_b} \frac{d}{d\omega} (\Sigma'(\omega) - \delta_{ct}) \Big|_{\omega=\omega_b} \right]^{-1}, \quad (\text{BH.13})$$

when the counterterm is frequency-independent. If a different response normalisation is used, the sum rule must be transformed with it; the invariant content is that line weight lost from Z_b appears in the same channel's continuum shoulder. This guard eliminates the false positive in which an unrelated trapped mode is fitted as a protected intrinsic line.

BH.5 Five-case detuning and counterterm map

The detuning is

$$\Delta = \Omega_0 - E_{\text{edge}}. \quad (\text{BH.14})$$

Throughout this appendix Δ denotes the edge detuning in Eq. (BH.14); the transverse-separation variable used in the cusp-kernel discussion of Appendix BE.7 is local to that discussion and is not used below. The following cases define the frozen C13' map.

case	condition	outcome and guard
bare edge	$\Delta = 0$ with a declared bare-edge convention	A subedge pole detaches with binding $O(g^2)$ and zero intrinsic width. The numerical binding is convention-dependent.
physical edge	δ_{ct} chosen so that the $g \rightarrow 0$ physical edge is fixed	The pole may sit at the edge in the calibrated convention, but any pole below the edge still has $\Gamma = 0$. Do not quote an absolute detachment without the subtraction convention.
positive detuning	$\Delta > 0$	Below a critical coupling the mode is an over-edge resonance. Above it the pole rebinds below the edge. The transition defines P24.
negative detuning	$\Delta < 0$	The pole is already subedge in the calibrated limit; protection is then kinematic rather than a nontrivial rebinding.
finite system or finite readout	edge levels, disorder, external leakage, finite T_{obs}	The mathematical pole is observed with residual width set by the largest non-intrinsic broadening scale in Eq. (BH.17).

For the positive-detuning branch, the soft-edge model gives the leading scaling

$$g_{\text{crit}}^2(\Delta) = \frac{\Delta}{\chi_E} + O(\Delta^2), \quad \chi_E > 0, \quad (\text{BH.15})$$

Here g_{crit}^2 is the critical channel amplitude of this detuning map; it is unrelated to the alignment coupling g_{crit} of the electroweak sector of the main text. It follows that

$$\frac{d \log g_{\text{crit}}^2}{d \log \Delta} = 1 + O(\Delta). \quad (\text{BH.16})$$

The coefficient χ_E is not universal: it depends on the edge normalisation, cutoff, projection, and counterterm convention. The unit exponent is the P24 prediction. If the measured continuum-onset exponent is not the C8 soft-edge exponent, the P24 exponent must be rederived before interpreting a failure as a failure of threshold protection.

BH.6 Finite-resolution and observed width

Below-edge zero width is an intrinsic-channel statement. The measured line shape is a convolution of intrinsic, external, finite-size, thermal, disorder, finite-window, and instrumental responses. Independent Lorentzian rates add, Gaussian variances add in quadrature, and mixed broadening is generally Voigt-like. Only as a conservative scale estimate,

$$\Gamma_{\text{obs}}^{(\text{scale})} \sim \max(\Gamma_{\text{ext}}, \Gamma_{L_z}, \Gamma_T, \Gamma_{\text{disorder}}, T_{\text{obs}}^{-1}, \delta\omega_{\text{res}}), \quad (\text{BH.17})$$

while a quantitative fit must convolve the declared response functions.

Resolving the line as a subedge line rather than as an edge shoulder requires

$$E_{\text{edge}} - \omega_b \gtrsim 3\delta\omega_{\text{res}}, \quad (\text{BH.18})$$

while resolving the shoulder as a continuum rather than as a few finite-size levels requires the opposite hierarchy for the edge-level spacing,

$$\delta\omega_{\text{level}} \ll \delta\omega_{\text{res}}. \quad (\text{BH.19})$$

Both inequalities must be stated in any proposed M3 implementation. A failure caused by finite resolution is a protocol failure, not a falsification of C13'.

BH.7 P23/P24 two-protocol falsifier

The intrinsic protocol fits the spectrum, after convolution with the experimental resolution kernel R_{res} , as

$$\rho_{\text{intr}}^{\text{obs}}(\omega) = \left[\frac{Z_b}{2\omega_b} \delta(\omega - \omega_b) + \rho_{\text{cont}}(\omega) \Theta(\omega - E) \right] * R_{\text{res}}(\omega), \quad (\text{BH.20})$$

with $\omega_b < E$, $\Gamma(\omega_b) = 0$ before convolution, and the sum rule (BH.12). The external protocol, recalling the C12 monitor law of Appendix BG.8, fits the same width-mode family with

$$\Gamma_n^{\text{ext}} = 2\pi J_W(\Omega_n) |F_n(\Omega_n)|^2 [1 + n_{\text{eff}}(\Omega_n)]. \quad (\text{BH.21})$$

After division by the same J_W and occupation data, the external hierarchy must reduce to the C12 form-factor classes of Appendix BG.8. P23 is the contrast between these two protocols. P24 is the positive-detuning transition

$$\text{over-edge resonance} \longrightarrow \text{subedge line plus soft shoulder}, \quad g_{\text{crit}}^2(\Delta) \propto \Delta. \quad (\text{BH.22})$$

The falsification hierarchy is deliberately local. A failed P23/P24 implementation first challenges the proposed C8/C13' intrinsic record operator, then the edge-alignment assumption, then the soft-edge threshold mechanism, and only after the protocol guards are controlled does it challenge the wider P6/PES spectral-record framework. It does not by itself falsify all of ECT.

External band-edge sanity checks. The line-plus-continuum component of the C13' package is the same open-system structure that appears in standard band-edge reservoir physics: a bound component below or outside a continuum edge, together with non-Markovian continuum decay. Existing superconducting-metamaterial and matter-wave experiments provide external consistency checks on the P6 conventions. In Mirhosseini et al. [324], the same engineered reservoir accounts for both the lifetime anomaly and the Lamb-shift anomaly of a transmon near a metamaterial band edge; this is a source-model-level pass of the one-reservoir width/shift pattern, not yet an independent quantitative P26 residual test. In the matter-wave experiments of Krinner et al. [325] and Stewart et al. [326], bound states and non-Markovian continuum dynamics realise the C13' line-plus-continuum pattern in the singular one-dimensional edge class. In all of these published platforms the accessible edge is of van-Hove type, effectively $m = -1/2$, or more singular, so the P24 soft-edge prediction $g_{\text{crit}}^2 \propto \Delta$ is not tested there. These comparisons show that the frozen P6 conventions reproduce known band-edge open-system physics; they are not claimed as evidence that PES is superior to standard quantum open-system theory. A direct P24 test requires an engineered convergent one-sided reservoir, for example a gapped Ohmic edge $J(\omega) = \eta(\omega - E)\Theta(\omega - E)$.

BH.8 Operational guards

The following guards must accompany any use of C13' or P23/P24.

1. *Backaction guard.* Vary readout strength and extrapolate to the weak-reading intercept; a strong readout can turn an intrinsic channel into an external C12 monitor.
2. *Trapped-mode guard.* A subedge line without transfer of weight to the soft shoulder is not a C13' line. The sum rule (BH.12) is mandatory.
3. *Detuning guard.* Calibrate Δ by a declared $g \rightarrow 0$ or counterterm-subtracted procedure before fitting $g_{\text{crit}}^2(\Delta)$.
4. *Edge-exponent guard.* Measure the continuum-onset exponent independently. The P24 unit exponent assumes the C8 soft linear edge.
5. *Finite-size guard.* Check Eqs. (BH.18) and (BH.19) before interpreting line/shoulder visibility.
6. *Matrix-channel guard.* If intrinsic and external readouts project different eigenchannels of J_{ab}^W , scalar closure can fail without falsifying the matrix P6-S7 framework.

These guards are operational rather than decorative: they determine whether a laboratory outcome is a C13' result, a record-operator failure, a protocol failure, or an ambiguous mixed-channel measurement.

BH.9 Handoff

Appendix BG established the certified external-monitor hierarchy. The present appendix establishes the intrinsic edge-plus-line mechanism and its detuning transition. Appendix BI supplies the width-sector observable interface, Appendix BJ supplies the P6-S7/P26 over-closure test, and Appendix BK assembles M3-1, M3-2, and M3-3 as explicit falsification protocols. The handoff rule is simple: C12 governs external or detuned monitoring, C13' governs the intrinsic self-channel, and a calculation that mixes the two without stating the protocol is not a valid P6 prediction.

BI P6-E: Width-sector observable interface and M3-3 over-closure inputs

Status and scope. This appendix is an interface layer. It uses the convention dictionary of Appendix BE, the line/width record operators of Appendix BG, and the intrinsic threshold-protection test of Appendix BH. It does not rederive the pair normalisation $\zeta_0^{\text{pair}} = 2$, the C10 cusp kernel, the P25/P25' fingerprint constants,

the Airy transfer Hamiltonian, or C13' threshold protection. Its role is to state the observable chain used by the P6-S7/P26 over-closure test of Appendix BJ and by the M3 protocols of Appendix BK, without duplicating the derivations already owned by Appendices BE–BH.

BI.1 Observable chain

The width-sector chain is

$$\begin{aligned} \text{static core-profile fit} &\longrightarrow \widehat{\xi}_c \longrightarrow \widehat{\xi}_{\text{res}} = 4\widehat{\xi}_c \longrightarrow \lambda_n^{\text{rec}} \\ &\longrightarrow \left(\widehat{\ell}_w, \widehat{E}_0, \widehat{L}_z^{\text{coh}}\right) \longrightarrow \text{C13' protection check.} \end{aligned} \quad (\text{BI.1})$$

The first arrow uses the P25/P25' profile fingerprints of Appendix BE.8. The second uses the intrinsic resolution floor of Appendix BE.7. The monitor anchor λ_n^{rec} is the Gaussian record coefficient of Appendix BG.7. The Airy quantities use the stiffness convention of Appendix BE.9. The final arrow is not optional: a width distribution is a persistent quantum width candidate only if the intrinsic line also passes the C13' edge-plus-line test of Appendix BH.

BI.2 Independent tilt calibration

The width stiffness must be calibrated independently from a tilt-cost measurement. For a small imposed tilt v , the operational calibration is

$$\frac{\Delta\Phi_{\text{tilt}}(v)}{L_z} = \frac{\widehat{\kappa}_b}{2} v^2 + O(v^4). \quad (\text{BI.2})$$

The hat emphasises that this is a measured stiffness. It must not be inferred from the Airy width fit itself and then reused to claim the $\kappa_b^{-1/3}$ width scaling. That circular procedure would make the stiffness exponent tautological even for a false width model. The Airy-width test is therefore meaningful only after Eq. (BI.2), or an equivalent independent stiffness calibration, has fixed $\widehat{\kappa}_b$.

BI.3 Resolution and deconvolution guard

Let $P_{\text{obs}}(x)$ be the measured width distribution and $R_{\text{res}}(x)$ the independently measured instrumental resolution. The model comparison should use

$$P_{\text{obs}}(x) = P_{\text{wid}}(x) * R_{\text{res}}(x), \quad (\text{BI.3})$$

not an unsmeared Airy profile. For an approximately Gaussian resolution with variance σ_{res}^2 , the variance-level deconvolution is

$$m_{2,\text{dec}} = m_{2,\text{obs}} - \sigma_{\text{res}}^2. \quad (\text{BI.4})$$

This formula is only a variance diagnostic; the full profile comparison should convolve the candidate distribution with the measured R_{res} . The certified operational requirement from the frozen P6 M3-3 audit is

$$\sigma_{\text{res}} \lesssim 0.3 \widehat{\ell}_w. \quad (\text{BI.5})$$

Above this scale, deconvolution errors can mimic or erase the Airy tail diagnostics. (In the frozen P6 audit ledger this bound is recorded as $\sigma_{\text{res}} \lesssim 0.3 \ell_A$, where ℓ_A is the ledger name for the Airy width scale denoted ℓ_w in Appendix BE.9.)

BI.4 Shape collapse and non-circular Airy test

Using the independent $\widehat{\kappa}_b$ and the already fixed σ_{dec} convention, the Airy prediction gives a width scale $\widehat{\ell}_w$ by Appendix BE.9. The shape-collapse test compares

$$\widehat{\ell}_w P_{\text{wid}}(\widehat{\ell}_w u) \quad \text{against the universal half-line Airy profile in } u. \quad (\text{BI.6})$$

The scale collapse is not a substitute for the exponent test: both the profile shape and the independent scalings with σ_{dec} and $\widehat{\kappa}_b$ must be checked. Conversely, a cube-root scaling alone is not sufficient evidence for a persistent quantum width, because thermal broadening, trapping, or resolution convolution can generate similar apparent powers over finite windows.

BI.5 C13' protection check

The width-sector observable chain must be paired with the intrinsic threshold test. The required C13' signature is the package

$$\text{subedge line} + \text{soft continuum shoulder} + \text{same-channel sum rule} + \text{P24 detuning transition}. \quad (\text{BI.7})$$

Here the sum rule and detuning map are those of Appendix BH.4 and Appendix BH.5. A width profile without this intrinsic protection check should be reported as a width-profile observation, not as a persistent Airy-width record.

BI.6 M3-3 decision table

The following table is an operational interface for M3-3 in Appendix BK. It is not a new derivation of the Airy or fingerprint laws.

Table 150: Operational M3-3 width, fingerprint, and intrinsic-protection decision interface.

Observation	Required companion check	Interpretation
Fingerprint fit gives stable $\widehat{\xi}_c$	Consistent $\widehat{\xi}_{\text{res}} = 4\widehat{\xi}_c$ in the record monitor	Core profile scale is usable as width-sector input
Airy-like width profile	Independent tilt calibration from Eq. (BI.2)	Non-circular candidate for quantum width
Apparent cube-root scaling only	Resolution bound Eq. (BI.5) and full convolution test	Ambiguous unless deconvolution and shape collapse pass
Width profile passes, C13' package fails	Threshold package Eq. (BI.7)	Width is not yet a persistent intrinsic record
All checks pass	Same-channel P23/P24 contrast in Appendix BH.7	Candidate M3-3 success case

BI.7 Handoff

Appendix BF.6 previews the idea that dephasing, shifts, and widths should over-close when they are projections of the same positive record kernel, and Appendix BJ gives the full P6-S7/P26 test. The present appendix supplies the width-sector observable inputs for that over-closure test and for the M3-3 protocol of Appendix BK. The protocol should call the quantities defined here rather than repeating the normalisation, fingerprint, and Airy derivations of Appendix BE.

BJ P6-F: Matrix one-kernel triangle and P26 over-closure

Status and scope. This appendix gives the full P6-S7/P26 closure test. The label P6-S7 refers to the spectral-record milestone of the frozen P6 ledger, not to the foundational S7 admissibility postulate of the main axiomatic list. The preview in Appendix BF.6 is upgraded here to an operational over-closure test. The generic object is a positive record spectral matrix; a scalar one-kernel triangle is valid only after a channel-selection or diagonalisation step.

The synthetic replay is Level A only inside its supplied identifiable two-parameter kernel family. P26 as a preregistered cross-arm protocol is Level B. A model-independent reconstruction of an arbitrary continuum from a finite set of widths is not established, and a physical ECT implementation remains Open until its record operator, state, form factors, kernel family and calibrated measurement map are supplied.

This appendix does not rederive the dephasing classes of Appendix BF.2, the C12 form-factor hierarchy of Appendix BG.8, the C13' threshold mechanism of Appendix BH, or the width-observable chain of Appendix BI. It uses those results as inputs.

BJ.1 Matrix record kernel

For a set of record operators labelled by a, b , the P6 record environment is represented by a Weiss spectral matrix

$$J_{ab}^W(\omega) \succeq 0, \quad \omega > 0. \quad (\text{BJ.1})$$

Recalling the matrix guard of Appendix BE.10, positivity implies the pointwise Cauchy–Schwarz test

$$|J_{ab}^W(\omega)|^2 \leq J_{aa}^W(\omega) J_{bb}^W(\omega). \quad (\text{BJ.2})$$

A violation of Eq. (BJ.2) is not a subtle model-selection issue: it means the proposed cross-channel spectral data cannot be a passive positive record kernel in the declared convention.

At a fixed frequency one may diagonalise

$$J_{ab}^W(\omega) = \sum_r j_r(\omega) e_a^{(r)}(\omega) e_b^{(r)}(\omega)^*, \quad j_r(\omega) \geq 0. \quad (\text{BJ.3})$$

A scalar triangle is therefore a selected-channel statement. It is justified when the protocol excites a single eigenchannel r , or when a declared projection Π_r is stable over the frequency window being tested. If the eigenvectors $e^{(r)}(\omega)$ rotate significantly across the window, scalar closure can fail even though the matrix P6-S7 statement remains viable.

BJ.2 P6-S7 triangle for one selected channel

For a selected channel r , define the projected density

$$J_r^W(\omega) = \Pi_r^\dagger J^W(\omega) \Pi_r. \quad (\text{BJ.4})$$

The P6-S7 triangle is the statement that, once the protocol data are declared, the three observed quantities are not independent fits:

$$J_r^W(\omega) \implies \{\Phi_r(t), \Gamma_{n,r}, \delta\Omega_{n,r}\}. \quad (\text{BJ.5})$$

The dephasing arm uses the spectral representation of Appendix BF.2. The width arm uses the C12 external-monitor form factors of Appendix BG.8, or the intrinsic threshold objects of Appendix BH when the channel is a C13' self-channel. The shift arm uses the bosonic mirror convention of Appendix BH.2 and the counterterm convention declared for the protocol.

For an external monitor, the width arm is

$$\Gamma_{n,r}^{\text{ext}} = 2\pi J_r^W(\Omega_n) |F_{n,r}(\Omega_n)|^2 [1 + n_{\text{eff}}(\Omega_n)]. \quad (\text{BJ.6})$$

The corresponding counterterm-relative leading shift is obtained from the same projected kernel through the bosonic mirror map,

$$\delta\Omega_{n,r} = -\frac{\Sigma'_{n,r}(\Omega_n) - \delta_{ct,n}}{2\Omega_n} + O(\Sigma^2), \quad (\text{BJ.7})$$

where the sign follows the dressed-response convention $D(z) = z^2 - \Omega_n^2 + \Sigma_R(z) - \delta_{ct,n}$ used in Appendix BH.2. If a different subtraction convention is used in a concrete protocol, Eq. (BJ.7) must be translated before comparison with data.

BJ.3 P26 reconstruction loop

P26 is an over-closure loop that fits or identifies a preregistered kernel family from one arm, when that family is identifiable, and tests it against the other two. In an external C12 protocol the width arm first supplies sampled projected values:

$$\hat{J}_r^W(\Omega_n) = \frac{\Gamma_{n,r}^{\text{ext}}}{2\pi|F_{n,r}(\Omega_n)|^2[1 + n_{\text{eff}}(\Omega_n)]}. \quad (\text{BJ.8})$$

A finite set of widths does not determine the continuum kernel between the Ω_n , its unmeasured bands, ultraviolet tail, subtraction, or counterterm. A positive interpolation or kernel family is therefore an additional, preregistered model input. Conditional on its identifiability, the fitted $\hat{J}_r^W(\omega)$ then predicts

$$\hat{\Phi}_r(t_i) = \int_0^\infty d\omega \hat{S}_{\text{rec},r}(\omega) \frac{1 - \cos \omega t_i}{\omega^2}, \quad (\text{BJ.9})$$

using the same record-spectrum convention as Appendix BF.2, and predicts

$$\widehat{\delta\Omega}_{n,r} = -\frac{\widehat{\Sigma}'_{n,r}(\Omega_n) - \delta_{ct,n}}{2\Omega_n}. \quad (\text{BJ.10})$$

The hats mark predictions from the same fitted/identified kernel family, not independent fits and not a model-independent reconstruction of an arbitrary continuum spectrum. The uncertainties and covariance of $F_{n,r}(\Omega_n)$ and $n_{\text{eff}}(\Omega_n)$ must be propagated through Eq. (BJ.8); divisions by a poorly known or near-zero form factor are not informative. If the family has parameters θ , the sampled forward map $\theta \mapsto \{\Gamma_{n,r}^{\text{ext}}\}_n$ must be injective on the preregistered domain before the width arm is called *globally* identifiable. Full Jacobian rank at a point supports only local identifiability in a neighbourhood of that point and must not be reported as global reconstruction.

The over-closure residuals may be reported as

$$R_\Phi^{\text{oc}} = \max_i \left| \frac{\Phi_r^{\text{obs}}(t_i) - \hat{\Phi}_r(t_i)}{\sigma_{\Phi,i}} \right|, \quad R_\Omega^{\text{oc}} = \max_n \left| \frac{\delta\Omega_{n,r}^{\text{obs}} - \widehat{\delta\Omega}_{n,r}}{\sigma_{\Omega,n}} \right|. \quad (\text{BJ.11})$$

The choice of norm is protocol-dependent. The denominators must include the observational covariance and the propagated uncertainty of the width fit, interpolation/family parameters, leakage model, extrapolated bands, ultraviolet tail and counterterm; a correlated analysis should use the corresponding full covariance rather than pointwise errors alone. What matters is that the same fitted positive $\hat{J}_r^W$ and the same uncertainty model are used in all arms. Continuum or informationally complete spectroscopy can weaken the family dependence, but the present synthetic replay closes only its supplied two-parameter family.

BJ.4 Matrix-channel diagnostics

When multiple record channels are measured, the scalar reconstruction should be supplemented by matrix diagnostics. A useful positivity ratio is

$$\eta_{ab}^{\text{rec}}(\omega) = \frac{|J_{ab}^W(\omega)|^2}{J_{aa}^W(\omega)J_{bb}^W(\omega)}, \quad 0 \leq \eta_{ab}^{\text{rec}}(\omega) \leq 1. \quad (\text{BJ.12})$$

Values above unity flag an inconsistent cross-spectrum, a convention mismatch, or an uncontrolled background subtraction. Values near unity indicate approximate rank-one behaviour in the measured two-channel block, while smaller values indicate genuinely multi-channel record structure.

A second diagnostic is the eigenvalue leakage ratio

$$\eta_r^{\text{leak}}(\omega) = \frac{\sum_{s \neq r} j_s(\omega)}{j_r(\omega)} \quad (\text{BJ.13})$$

computed after the diagonalisation in Eq. (BJ.3). Scalar P26 closure requires η_r^{leak} to be small over the tested window, or requires the leakage to be explicitly modelled. Intermediate apparent power laws are not new P6 exponents until this matrix-channel leakage has been controlled.

BJ.5 Intrinsic threshold channel

For an intrinsic C13' channel the width arm is not Eq. (BJ.6). The relevant observables are instead the bound-line position, the line weight, and the continuum shoulder:

$$\{\omega_b, Z_b, \rho_{\text{cont}}(\omega)\}_r \quad \text{with} \quad Z_b + \int_E^\infty 2\omega \rho_{\text{cont}}(\omega) d\omega = 1, \quad (\text{BJ.14})$$

using the sum rule of Appendix BH.4. The same projected kernel must also account for the P24 detuning transition and for the dephasing arm after the declared occupation data are supplied. Applying a golden-rule Lorentzian width to the intrinsic subedge line would mix protocols and would fail the guard of Appendix BH.7.

BJ.6 Failure interpretation

A failed scalar over-closure does not immediately falsify P6/PES. The failure hierarchy is

$$\begin{aligned} &\text{wrong form factor} \rightarrow \text{wrong record operator} \rightarrow \text{wrong occupation/counterterm convention} \\ &\rightarrow \text{uncontrolled matrix-channel leakage} \rightarrow \text{failure of the selected-channel P6-S7 closure.} \end{aligned} \quad (\text{BJ.15})$$

Only after these guards are controlled does the result challenge the wider spectral-record framework. In particular, failure of a scalar fit in a multi-channel setting may simply mean that Eq. (BJ.3) has not reduced to a stable single channel.

BJ.7 Handoff to M3 protocols

The P26 loop is a laboratory analysis pattern, not yet a platform specification. Appendix BI provides the width-sector inputs used by the M3-3 protocol, while Appendix BH.7 provides the P23/P24 intrinsic-threshold contrast. Appendix BK assembles these inputs into explicit falsification protocols without rederiving the kernel, width, or threshold laws.

BK P6-G: M3 operational falsification protocols

Status and scope. This appendix collects the three M3 protocols attached to the frozen P6 record framework. It is a protocol appendix, not a new derivation appendix. The threshold physics of P23/P24 is owned by Appendix BH.7; the width-sector observable interface is owned by Appendix BI; and the one-kernel over-closure logic is owned by Appendix BJ. The analogue-observable dictionary is Appendix BB. The role of the present appendix is to specify what must be measured, how the guards are sequenced, and how a pass or fail should be interpreted.

A failure of an analogue M3 protocol is not by itself a falsification of all ECT. It first tests the declared record operator, channel map, calibration, and platform assumptions. Only after those guards are controlled does a failure bear on the corresponding P6 record claim.

BK.1 Common M3 reporting packet

Every M3 implementation should begin with a platform packet

$$P_{M3} = \{\text{platform, ordered variable, record channel, } J_{ab}^W, F_n, S_{\pm} \text{ or } T_{\text{eff}}, R_{\text{res}}, \delta_{ct}, \text{ calibrations}\}. \quad (\text{BK.1})$$

The packet is deliberately redundant: it records the conventions needed to decide whether a failed protocol is a physical failure or a convention mismatch. The symbols J_{ab}^W , F_n , $S_{\pm}$, T_{eff} , R_{res} , and δ_{ct} are the same objects used in Appendices BE–BJ. They should not be redefined locally.

The minimal reporting standard is

$$P_{M3} + \text{raw spectra} + \text{resolution kernel} \\ + \text{readout-strength sweep} + \text{counterterm or calibration convention}. \quad (\text{BK.2})$$

Without this packet, the M3 outcome is not interpretable as a P6 test.

BK.2 M3-1: intrinsic/external threshold contrast

M3-1 operationalises the P23/P24 contrast of Appendix BH.7. The same width-mode family is read in two protocols:

- (i) an intrinsic self-channel readout, which should show the subedge line, soft shoulder, same-channel sum rule, and positive-detuning transition of Appendix BH.7;
- (ii) an external detuned monitor, which should reduce to the C12 form-factor hierarchy of Appendix BG.8.

The first operational guard is weak-readout extrapolation. Let ϵ_{rd} denote the readout strength. The measured line width is fitted as

$$\Gamma_b^{\text{obs}}(\epsilon_{\text{rd}}) = \Gamma_0 + A_{\text{rd}}\epsilon_{\text{rd}} + O(\epsilon_{\text{rd}}^2). \quad (\text{BK.3})$$

For an intrinsic C13' line the intercept should be consistent with zero within the declared error model. A nonzero intercept is evidence for an external broadening channel, a trapped mode, or an uncontrolled environmental width, not for the intrinsic zero-width statement.

The second guard is spectral-weight transfer. Using the same-channel normalisation of Appendix BH.4, report

$$Z_b + Z_{\text{cont}} \simeq 1, \quad Z_{\text{cont}} \equiv \int_E^{\infty} 2\omega \rho_{\text{cont}}(\omega) d\omega. \quad (\text{BK.4})$$

A trapped-mode false positive can produce a narrow line, but it will not generically show the correlated line-to-shoulder weight transfer required by the C13' channel.

The third guard is detuning calibration. When a positive-detuning resonance exists below the rebinding threshold, define

$$\Omega_{0,\text{cal}} = \lim_{g^2 \rightarrow 0} \omega_{\text{res}}(g^2), \quad \Delta_{\text{cal}} = \Omega_{0,\text{cal}} - E_{\text{edge}}. \quad (\text{BK.5})$$

The P24 test is then the calibrated relation

$$g_{\text{crit}}^2(\Delta_{\text{cal}}) \propto \Delta_{\text{cal}}, \quad (\text{BK.6})$$

with the notation and disambiguation of Appendix BH.7. A detuning map without the $g^2 \rightarrow 0$ calibration is not a quantitative P24 test.

BK.3 M3-2: zero-mode detector through $t_*(R)$

M3-2 tests the P6 zero-mode/Berry alternative. The notation P7' used here is a P6/M3 detector label; it is not the colour postulate P7 of the main text. The minimal line/ribbon branch has $n_0 = 0$ and a linear crossover scale, while a core-Berry branch gives a log-corrected quadratic scale. The two flatness transforms are

$$Q_{\min}(R) = \frac{t_*(R)}{R + R_{\text{off}}}, \quad Q_{\text{core}}(R) = \frac{t_*(R) \ln(R/\xi)}{R^2}. \quad (\text{BK.7})$$

The flatness transforms $Q_{\min}$ and Q_{core} are M3-2 detector statistics; they are distinct from the quietness index $Q[\Psi]$ of Section 29.6. The offset R_{off} was denoted R_0 in some P6 derivation files; the present notation avoids collision with the unrelated collapse-scale R_0 used elsewhere in the manuscript. The raw slope of the core-Berry branch over a finite window is not exactly two:

$$p_{\text{raw}} = 2 - \frac{1}{\ln(R_{\text{geo}}/\xi)} + \dots. \quad (\text{BK.8})$$

Therefore the protocol must test the transformed flatness, not merely fit a raw power law.

A positive zero-mode/Berry claim requires three correlated signatures:

$$Q_{\text{core}}(R) \simeq \text{const}, \quad \Phi(t) + 2\sqrt{\pi} \propto t^{1/2}, \quad \Delta\omega_{\text{odd}}(W) \propto W n_0. \quad (\text{BK.9})$$

The middle signature recalls the square-root branch of Appendix BF.4. The odd frequency shift is extracted by

$$\Delta\omega_{\text{odd}}(W) = \frac{\Delta\omega(+W) - \Delta\omega(-W)}{2}. \quad (\text{BK.10})$$

This removes even backgrounds such as pinning shifts. A raw R^2 scaling without log-flatness and odd Berry response is a trap/diffusion false positive.

The scale ξ used in Eq. (BK.7) should be supplied by an independent core fingerprint or profile measurement. If ξ is floated freely in the same fit, the corrected slope can be biased and should not be read as new physics.

BK.4 M3-3: Airy-width and fingerprint protocol

M3-3 uses the observable interface of Appendix BI. The protocol should report the data vector

$$\mathbf{D}_{\text{M3-3}} = \left(\widehat{\xi}_c, \widehat{\xi}_{\text{res}}, \widehat{\kappa}_b, \widehat{\ell}_w, \widehat{E}_0, \widehat{L}_z^{\text{coh}}, Z_b, \rho_{\text{cont}} \right), \quad (\text{BK.11})$$

with the hatted quantities obtained by the non-circular steps of Appendix BI.1. The decision logic is Table 150 in Appendix BI.6; it should not be repeated as a second table here.

The core warning is circularity. Extracting κ_b from the Airy width and then claiming the κ_b exponent is a tautology. The stiffness must instead come from the independent tilt calibration of Appendix BI.2. The resolution guard is likewise inherited from Appendix BI.3. A successful width profile without the C13' package of Appendix BI.5 is a width-profile observation, not a persistent intrinsic record claim.

BK.5 Shared interpretation rules

The shared M3 failure hierarchy is

$$\begin{aligned} & \text{instrumental/resolution failure} \rightarrow \text{wrong channel map} \\ & \rightarrow \text{wrong occupation/counterterm convention} \\ & \rightarrow \text{wrong record operator or form factor} \\ & \rightarrow \text{failure of the corresponding P6 claim.} \end{aligned} \quad (\text{BK.12})$$

This hierarchy is the operational counterpart of the scalar-channel failure discussion in Appendix BJ.6. It prevents two opposite overclaims: treating every analogue discrepancy as a falsification of ECT, and treating every discrepancy as a harmless apparatus problem.

A successful M3 result should be reported as a success of the declared protocol and channel, not as a proof of all P6/PES. Conversely, a controlled failure should name the first failed guard in Eq. (BK.12).

BK.6 Handoff to the correction index

The next appendix records correction history and convention reconciliation. Protocol outcomes should be interpreted through that index: factor-of-two, counterterm, and spectral-density convention mistakes can mimic physical failures if the audit trail is not explicit. The already flagged $J_s/J_W/J_{\text{bare}}$ convention reconciliation belongs there, not in the M3 protocol definitions.

BL P6-H: Honesty log, correction index, and convention reconciliation

Status and scope. This appendix is an audit index for the P6/PES appendices. It records the correction events and convention reconciliations that are part of the frozen P6 transfer into the preprint. It is not a new physics appendix: the convention dictionary is owned by Appendix BE, the core spectral laws by Appendix BF, the line/record sector by Appendix BG, threshold protection by Appendix BH, the width-observable interface by Appendix BI, the P26 over-closure by Appendix BJ, and the M3 protocols by Appendix BK. The role of this appendix is to make the audit trail visible and reproducible without repeating the derivations.

BL.1 Audit principles

The P6 transfer uses the following audit rules.

1. A correction is not a weakening of a test. When an error is found, the corrected convention or formula becomes part of the reproducibility conditions.
2. A repeated formula must be explicitly marked as recalled or used from its owner appendix. Silent redefinition is forbidden.
3. Level marks are part of the result. The P6/PES branch is frozen as a Level-B spectral-record framework with M3 falsifiers, not as a completed first-principles proof of quantum mechanics.
4. Analogue-laboratory failure is interpreted through the protocol failure hierarchy of Appendix BK.5; it is not automatically a falsification of the full ECT programme.
5. The public manuscript should expose the corrections that affect reproducibility. Purely local typography events are listed only when they affect future compilation or audit practice.

BL.2 Correction index

The following index gives the public correction map for the P6 appendices. The owner column identifies where the corrected convention or result is defined; this appendix only records the audit trail.

Table 151: P6 correction and convention index. The entries are audit pointers, not rederivations.

Event	Corrected issue	Operational meaning	Owner/status
C8 finite- k_z correction	Early finite- k_z record kernels inherited a $k_z = 0$ face formula too broadly.	Use the canonical soft-edge C8 kernels and edge-onset rules; do not use the older singular finite- k_z forms.	Appendix BE.2; Appendix BG.5; corrected.
C8 normalisation and factor-of-two guard	The canonical spatial line is fixed by the convention-free invariant.	$J_{\text{spat}}/J_{\text{temp}} = \omega^2/c^2$ fixes the canonical pair; noncanonical reduced audit columns are $4\times$ canonical and must not enter the native measure.	Appendix BE.2.
Pair normalisation	The line-overlap normalisation must distinguish pair and face conventions.	The P6 appendices use $\sigma_{\text{dec}} \equiv \sigma_{\text{pair}}$ and $\zeta_0^{\text{pair}} = 2$; single-face quantities require an explicit declaration.	Appendix BE.3; closed convention.
Bosonic KK mirror	A one-sided dispersion kernel misses the oscillator mirror term.	Self-energy and shift formulas use the bosonic mirror convention; threshold residues and sum rules are checked in that convention.	Appendix BE.4; Appendix BH.2; corrected.
Detailed-balance thermometer	A ledger-unit thermometer without explicit S_0 is dimensionally incomplete in the preprint convention.	The preprint uses the explicit S_0 detailed-balance thermometer and states the ordering convention for S_+/S_- .	Appendix BE.5; corrected during transfer.
Same-channel response/noise test	A detailed-balance ratio is meaningful only for one physical operator, state, projector and common filter.	Use the Wightman pair $S_H \mp S_0 \text{Im}_H D^R \succeq 0$ before a KMS fit; do not replace it by an arbitrary-matrix absolute-value inequality or infer it from complete positivity alone.	Appendix BE.6; Level A conditional mathematics / Level B protocol / ECT Open.
Static bridge versus PES response	A local static response-coordinate invariant was proposed as a retarded-kernel owner.	The invariant fixes at most one convention-dependent nonlinear coefficient; one static two-point D^R does not own the nonlinear bridge or the gravity map.	Paragraph 33.6; terminal diagnostic only.
Line coefficient notation	The P6 Bessel line coefficient is distinct from OP- c_1 .	The P6 appendices use $c_{\text{line}} = 1$ for the line coefficient.	Appendix BG.1.
P11 stiffness sign	A ledger summary line once wrote the Airy width as if κ_b were a flexibility.	With κ_b the physical tilt stiffness, $\ell_w \propto (\kappa_b \sigma_{\text{dec}})^{-1/3}$ and $L_z^{\text{coh}} \propto \kappa_b^{1/3} \sigma_{\text{dec}}^{-2/3}$.	Appendix BE.9; Appendix BI.4; corrected.
Finite-window/log artefacts	Raw slopes near 0.97 or 1.94 can be intercept or logarithmic-window artefacts.	P25' and M3-2 use transformed variables and explicit logarithmic guards before interpreting exponents.	Appendix BE.8; Appendix BK.3; guarded.

Event	Corrected issue	Operational meaning	Owner/status
Matrix/scalar ambiguity	A scalar one-kernel statement can be over-read as a universal scalar bath.	The active P6 statement is matrix-positive; scalar P6-S7 is a selected-channel or diagonalised closure.	Appendix BJ.1; guarded.
Protocol mixing	Intrinsic C13' lines and external C12 monitors can be accidentally fitted with the same Lorentzian logic.	External protocols use C12 golden-rule widths; intrinsic threshold channels use edge-plus-line observables and the same-channel sum rule.	Appendix BH.7; Appendix BK.2; guarded.

BL.3 Bath-density convention reconciliation

The remaining convention issue connecting the older generic decoherence apparatus to the frozen P6 notation is the meaning of the exponent in a generic family $J_s(\omega)$. Earlier parts of the manuscript used $J_s(\omega) = A_s \omega^s \exp(-\omega/\omega_c)$ in the noise-kernel formula while also tabulating some bath powers as bare mode-density exponents. The frozen P6 convention, recalling the declaration already made in Appendix AX and Appendix BE.1, is that the spectral density entering the Feynman–Vernon/Weiss noise kernel is the Weiss density,

$$J_W(\omega) = \frac{J_{\text{bare}}(\omega)}{2\omega}, \quad a_W = a_{\text{bare}} - 1. \quad (\text{BL.1})$$

Hence any use of the generic family J_s must state whether s is a Weiss exponent or a bare exponent. If s is bare, the noise kernel must use the corresponding J_W ; if s is Weiss, comparisons with a bare-exponent table must shift the power by one. Appendix AX now declares $J_s \equiv J_W$ at the point of introduction and lists bare and Weiss powers separately; the generic-bath prose and the main-text derivative-coupling benchmark use the same reconciled convention.

For the already integrated P6 appendices the local protection is in place: Appendix BF.1 uses the conditional dictionary only when J_s is read in the Weiss convention, while the C8/C12/C13' record-channel statements use the explicitly declared P6 convention of Appendix BE.1.

BL.4 Transfer and typography events

The following transfer events do not change the P6 physics, but they are included to make the public manuscript reproducible.

Table 152: Transfer-process events relevant to reproducibility.

Event	Resolution	Status
AP6-C mini-table typography	A short local record-class table briefly used <code>longtable</code> and produced a new overfull vbox. It was returned to non-floating center/tabular form.	Local exception.
AP6-E pagination ripple	AP6-E added no block-local overfills, but the global count changed by a tiny front-matter pagination ripple. The distinction between block-local and global box counts is now tracked explicitly.	Logged.
AP6-G line-break targeting	A first typography break targeted the wrong line of a long hierarchy; the offending line was then correctly split.	Fixed.

Event	Resolution	Status
iCloud/dataless incident	Project files in an iCloud-synchronised area were temporarily evicted under disk pressure. The working preprint, book file, backups, and logs were rehydrated and verified.	Infrastructure.

BL.5 How to read the correction record

The correction index is part of the scientific claim. A corrected entry should not be read as evidence that the final transferred formula is optional; it identifies the path by which the final convention was selected and the tests that would catch a regression. Conversely, the existence of a correction record does not upgrade P6/PES beyond its stated status. The branch remains a Level-B spectral-record framework with operational M3 falsifiers and explicit open programmes.

The no-claim discipline is therefore unchanged. The P6 appendices do not prove the Born rule from bare P1–P6, do not derive $S_0 = \hbar$ beyond the matched Level-B role already stated in the manuscript, do not establish OP-Planck or OP-GUT1, and do not claim that an analogue-platform failure automatically falsifies the full ECT framework.

BL.6 Handoff

After this correction index, no separate P6-I appendix is introduced in the present manuscript. The residual P6 open items are folded into the existing status-map section rather than into a duplicate appendix. The PES, analogue-laboratory, and status-map sections have been updated in place using the stable anchors of Appendices BE–BL.

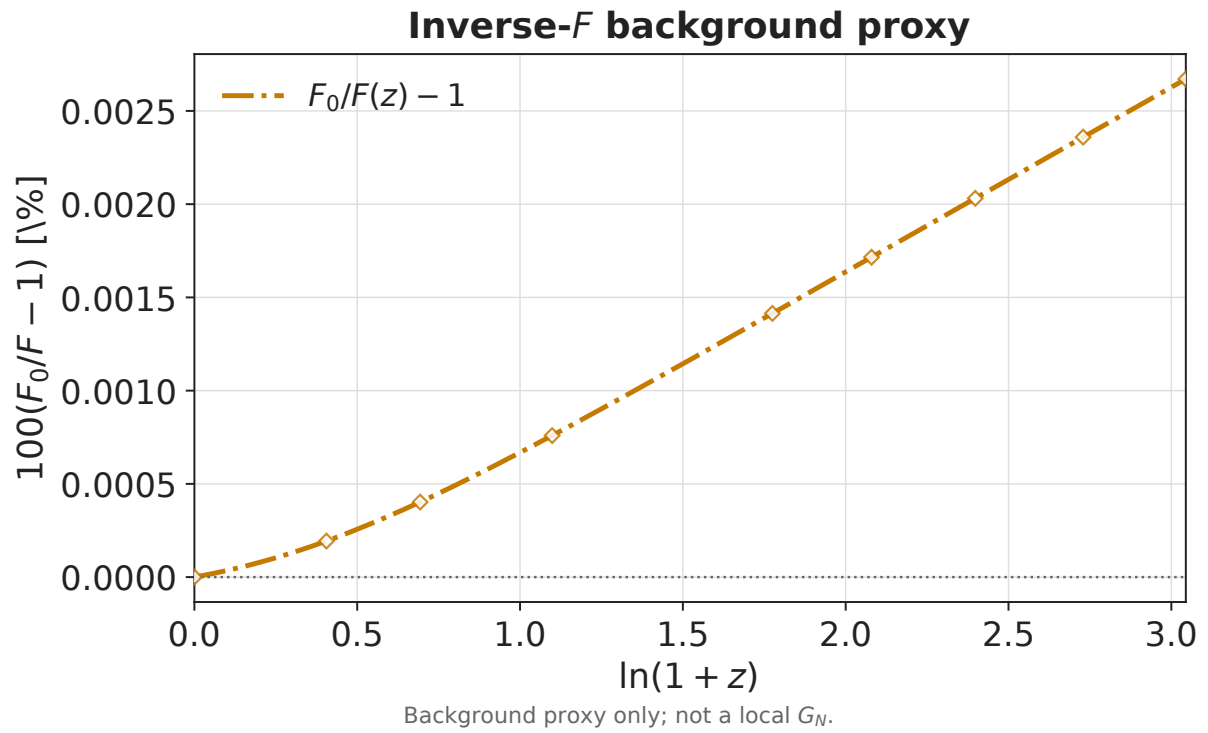


Figure 39, continued. Inverse- F background proxy. It is not a local Newton constant; a finite-body metric and PPN map remain Open. The intermediate w panel is retained in the visual atlas. *Display semantics:* The continuous inverse- F trace is evaluated directly on the frozen 3001-point background owner; the nine published nodes are shown separately.

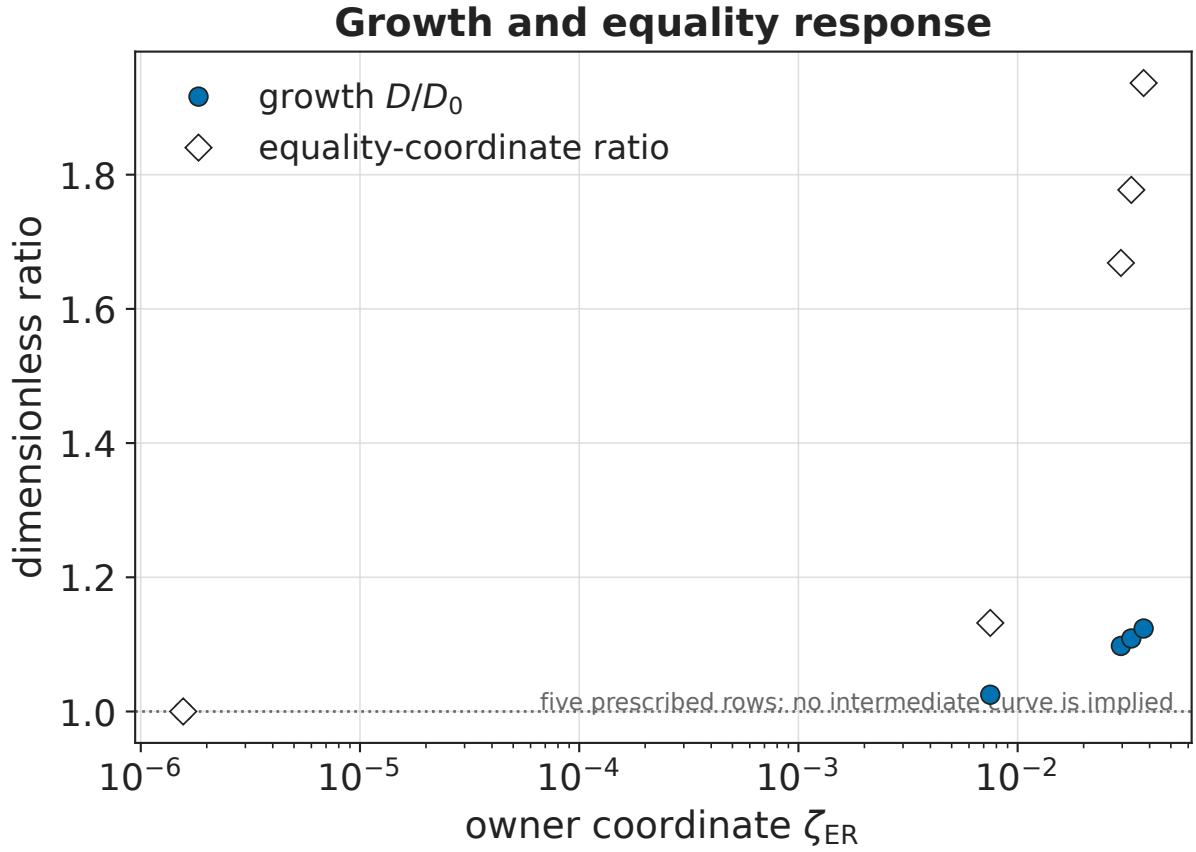


Figure 40: Owner-specific early-response sensitivity envelope at $z = 10$ for the five prescribed ζ_{ER} stress-test coordinates of Table 129. Linear growth and the equality shift are algebra inside the declared envelope; the top-hat barrier and Press–Schechter rare-tail ratios are Level-C sensitivity estimates. The five prescribed source rows are rendered as unconnected markers; no interpolant, continuous family, posterior, JWST abundance prediction or likelihood is inferred. Colour is redundant with marker shape, marker fill and direct legend text, so the panel remains interpretable in grayscale. The remaining owner panel follows as a full-width continuation. *Display semantics: The five prescribed owner rows are shown as unconnected markers; no intermediate interpolation is implied.*

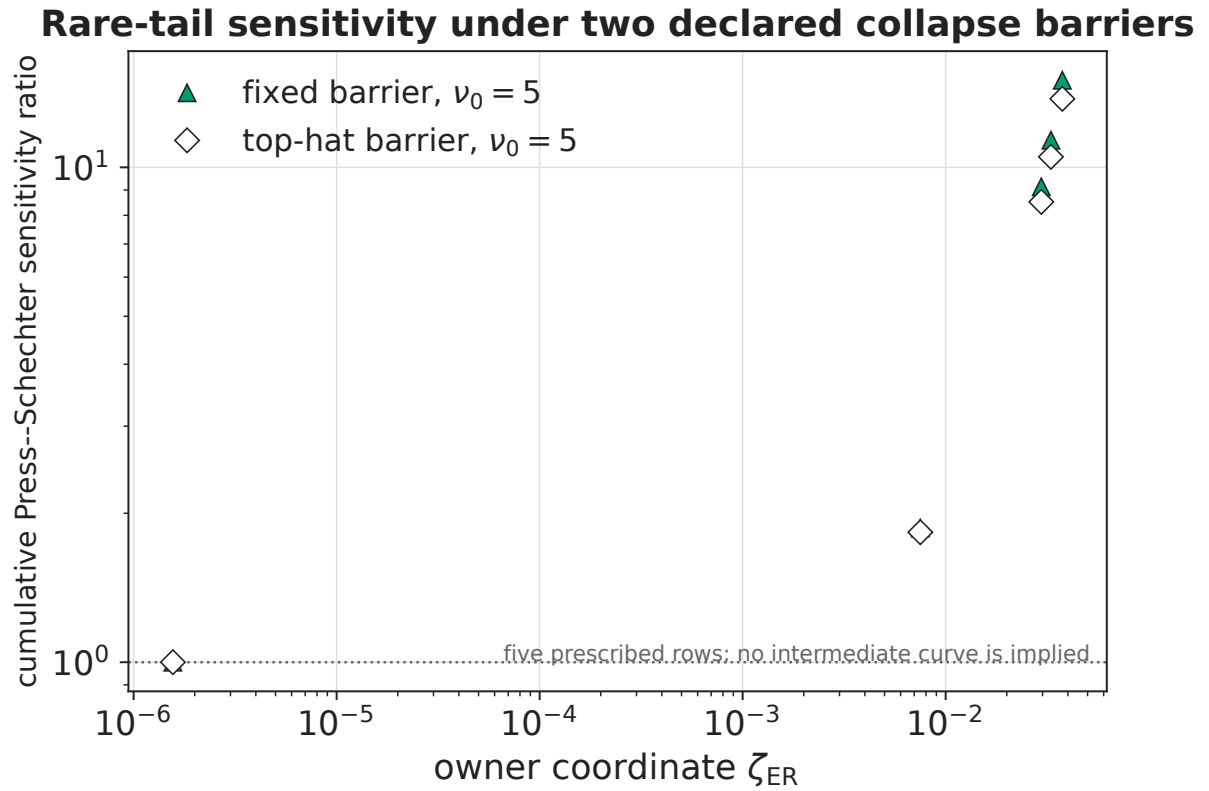


Figure 40, continued. Conditional top-hat and Press–Schechter rare-tail sensitivity. The same five discrete stress-test rows are unconnected markers and do not define a continuous response family. This is Level C and is not a JWST abundance prediction or likelihood. *Display semantics:* The five prescribed rare-tail rows are shown as unconnected markers; no intermediate interpolation is implied.

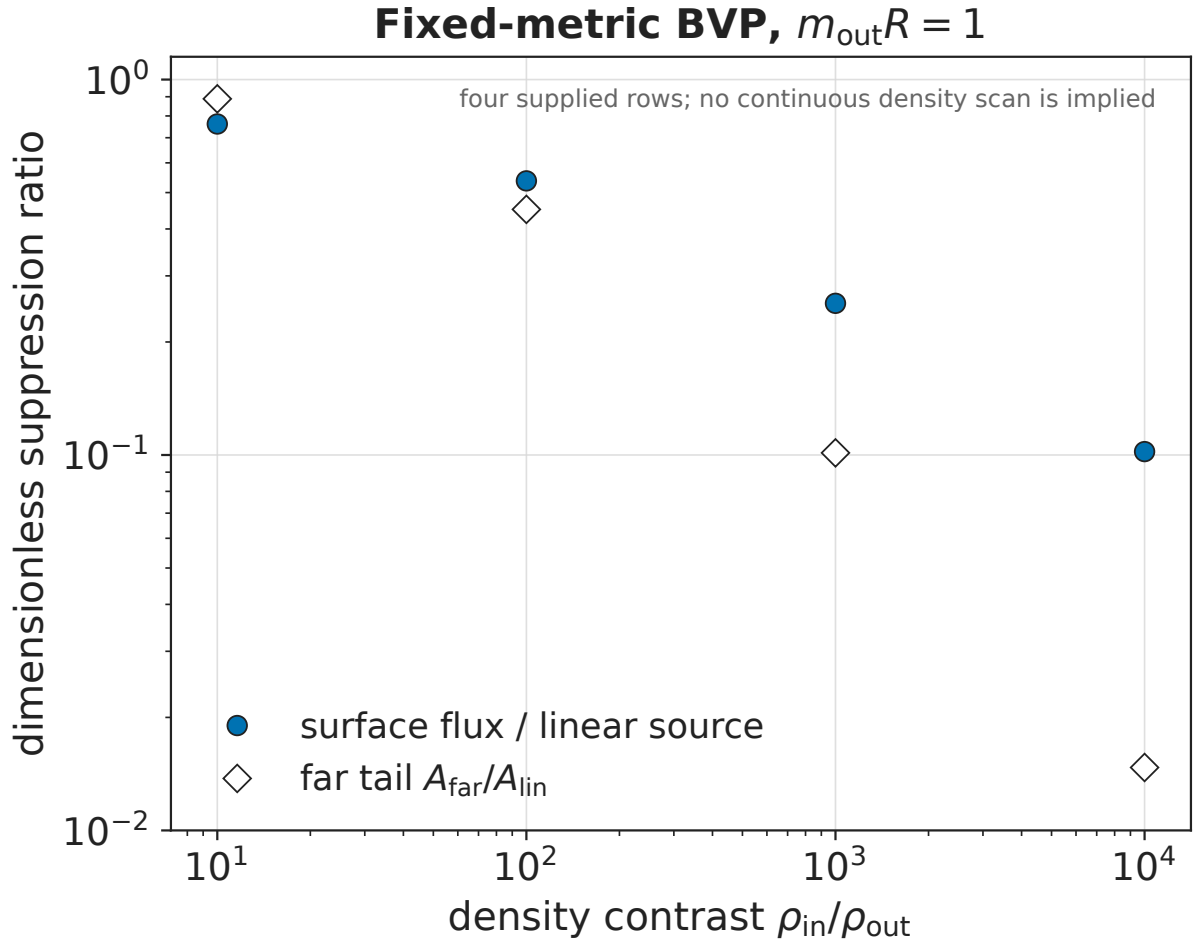
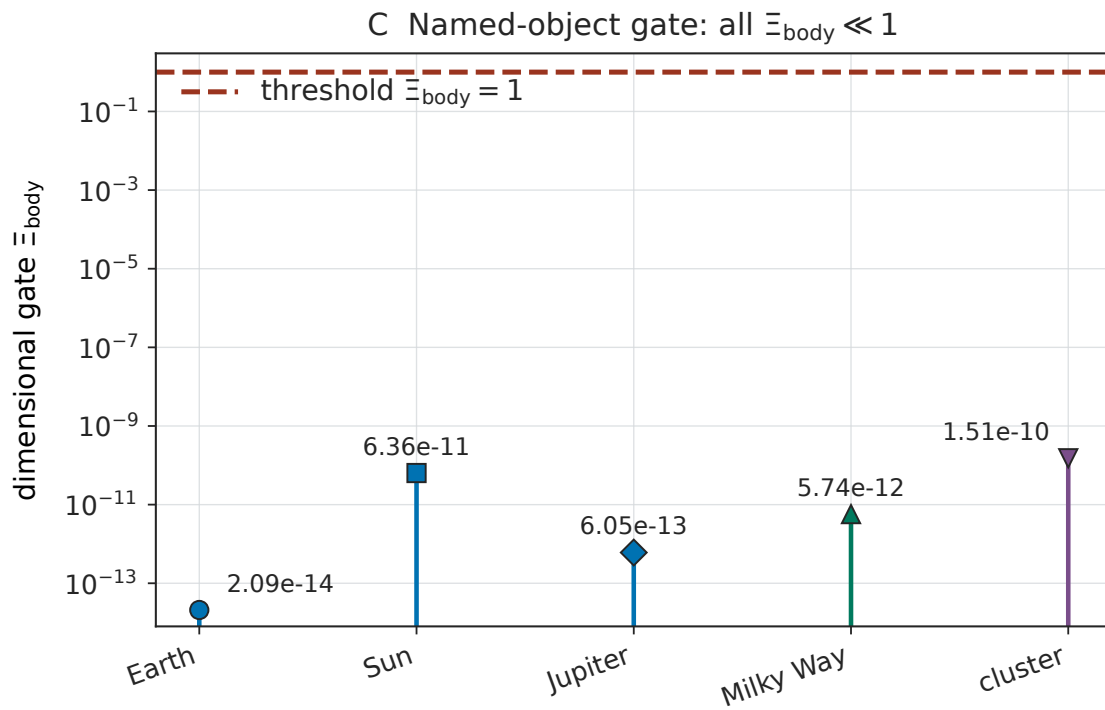


Figure 41: Conditional fixed-metric scalar-BVP gate for the four supplied rows of the named two-slope state at the deliberately artificial scale $m_{\text{out}}R = 1$; the four supplied rows are unconnected markers and no interpolant is implied. Tail-estimator definitions are retained as a separate full-width atlas panel, and the mean-density dimensional gate follows as a continuation. This is not a physical body sensitivity, coupled metric, Cassini, WEP or full-PPN prediction. Colour is redundant with marker shape, marker fill, luminance contrast and direct labels; no decorative hatching is used. *Display semantics: The four supplied fixed-metric BVP rows are shown as unconnected markers; no continuous density scan is implied.*

Finite-body scalar dimensional gates



Dimensional gate only; not physical body sensitivity or full PPN.

Figure 41, continued. Mean-density dimensional gate. The named state does not reach its formal dense roots; this is not a Cassini, WEP or full-PPN prediction. Estimator detail remains in the visual atlas.

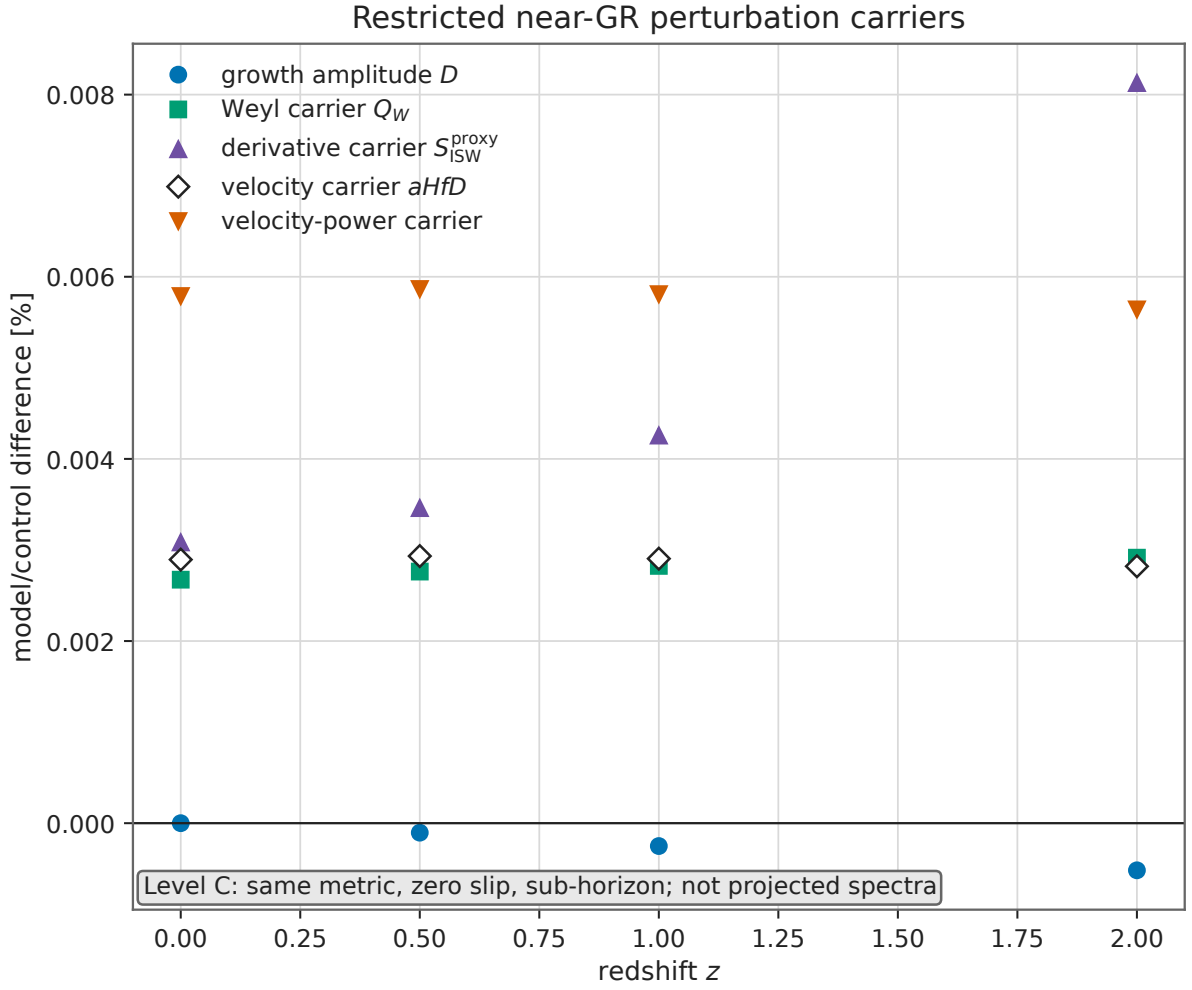
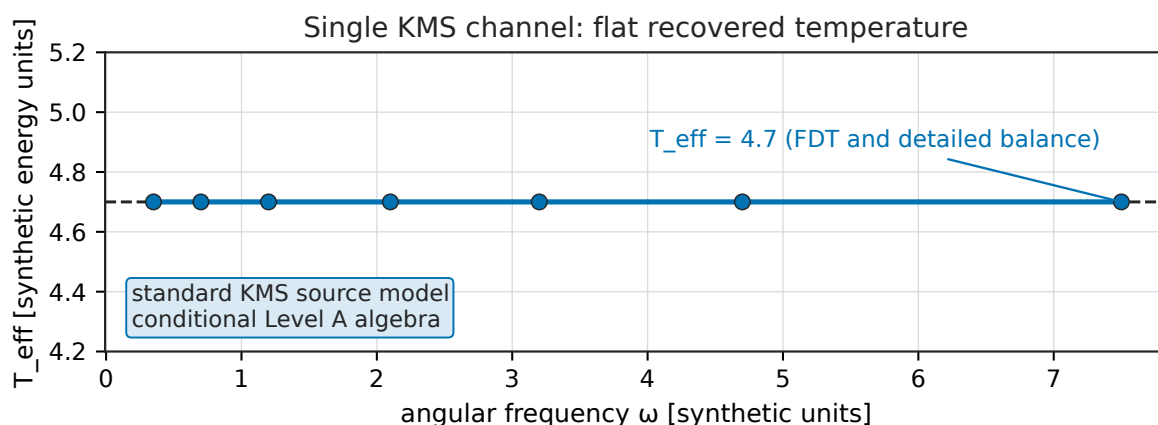


Figure 42: Restricted perturbation carriers for the named near-GR two-slope state relative to its reference control. The calculation assumes one matter/photon metric, zero slip, the sub-horizon quasistatic limit, negligible scalar mass/mixing/scale dependence and a common primordial normalisation. It is a Level C limiting diagnostic, not projected ISW, weak/CMB-lensing or survey bulk-flow spectra. The four evaluated redshift nodes are unconnected; no continuous perturbation history or interpolation is implied. Colour and marker shape provide redundant encodings.

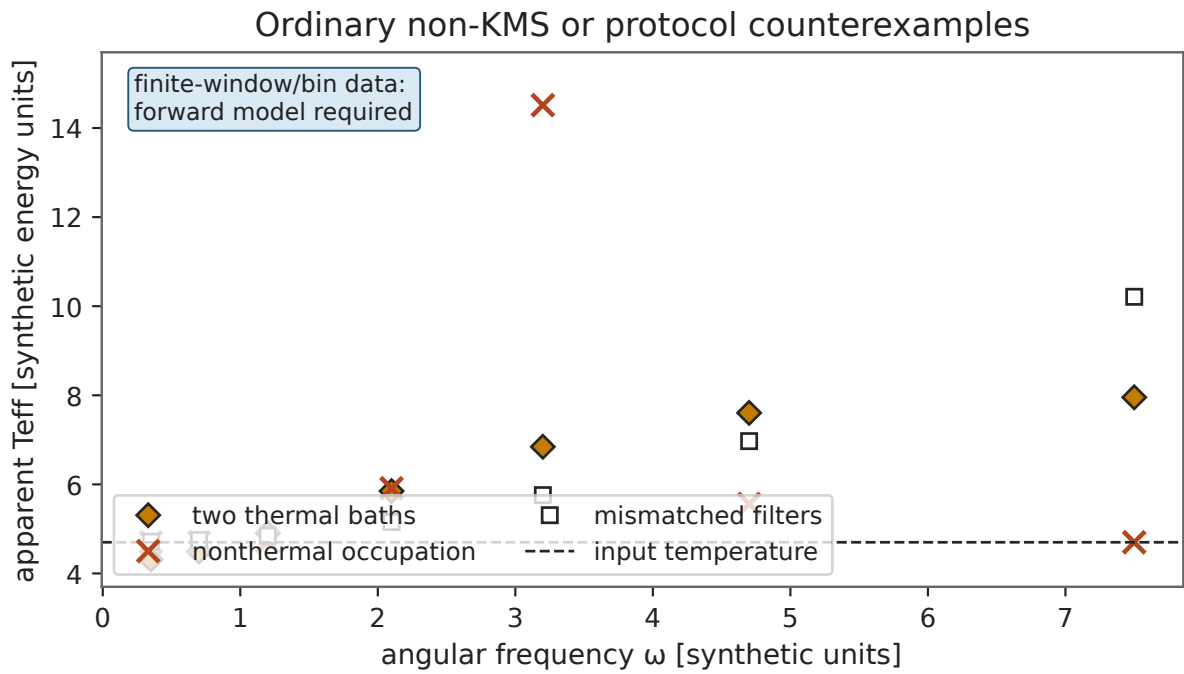
M1 same-channel FDT: single KMS channel



Deterministic standard-source-model replay — not experimental data or an ECT signal.
One-channel agreement is not evidence for global KMS.

Figure 43: M1 same-channel KMS/FDT protocol: canonical-channel replay. This page shows the flat temperature recovered in one declared canonical KMS channel. It is a deterministic standard-source-model algebraic check, not experimental data, an ECT signal or evidence for global KMS thermality. Two-bath, nonthermal-occupation and mismatched-filter counterexamples follow on a separate full-width continuation. A physical test must forward-convolve the signed-frequency leakage/bin kernel, freeze or fit S_0 and inter-arm gain, use an informationally complete matrix or preregistered projections, specify generator, charges, state and preparation, and pass causality, analyticity, Kramers–Kronig, tail, subtraction, contact and Ward-term gates. Current ECT evaluation remains Open because the owning operator, vertex, state and transfer function are missing or not identifiable.

M1 same-channel FDT protocol



Ordinary non-KMS/protocol counterexamples; mismatch is not by itself ECT evidence.

Figure 43, continued. Two-bath, nonthermal-occupation and mismatched-filter counterexamples. The markers are independent finite-window/bin values and are not joined into a forward curve. ECT evaluation remains Open until the physical operator, vertex, state and transfer function are identified.

Reference-ordered visual reproducibility atlas: reading guide

The main narrative uses one owner panel per local argument, three two-galaxy external-comparison pages as its rotation-curve owner, and five complete, connected A4 reader maps for the Part I–IV and whole-programme logic. The six appendix entries that follow preserve complementary scientific panels and earlier HRC-only galaxy diagnostics in first-reference order. The logic maps are not duplicated or fragmented here: each is printed once, at its owning synthesis section, with the full topology visible on one page.

The former multi-page reader routes and directed-edge ledgers were audit products, not adequate replacements for the connected scientific graphs. They are removed from the sequential print document but not discarded. Each ledger row recorded one canonical edge—source node and identifier, target node, literal edge label, line style and reading rule—so it remains useful for machine reproducibility rather than exposition. For the current reader maps, `FIGURE_REGISTRY.csv` and `FIGURE_REGISTRY.json` pin the rendered assets, current source-owner hashes, machine-readable semantic/topology keys, QA records and the required files under `provenance/figures/r190/`. Superseded intermediary map packages are not current scientific owners and are excluded from the public current-file surface. For a released manuscript revision, its tagged commit and frozen release manifest identify the exact applicable registry and provenance set. Exact rendered-PDF inspection remains required for every reader-facing map and scientific page; a detached route, clipped label or unreadable type size is a build failure. Across those pages, colour is redundant through luminance separation, line/marker/border form and direct labels, never through hatch patterns. The release semantic cascade preserves the corrected Part III and full programme-map dependencies: RP-to-unitarity is conditional on the complete applicable corrected OS-II package, no decoherence, quantum–classical-boundary or unitarity edge derives Born weights, and the full map replaces the direct Schrödinger-to-Born edge by a dashed coherent-branch gate requiring supplied OS-II/Hilbert structure and an independently supplied measure. The canonical sources and repository machine-topology records preserve those exact changes. No numerical array, fit, sample choice or nuisance parameter is changed. The older galaxy panels are provenance and stress diagnostics; they are not independent HRC–MOND– Λ CDM comparisons.

Table 153: Visual-atlas disposition. “Main owner” identifies the concise narrative figure; the atlas column identifies material retained here rather than deleted.

Family	Main owner	Atlas content	Status/owner guard
Part I logic maps	Fig. 4	one complete connected A4 reader map in the Part I conclusion; legacy atlas anchor 4 points to the same page	whole topology retained; canonical source and repository machine topology govern exact edges, and no arrow upgrades status
Acoustic/cluster gates	Figs. 7 and 8	Appendix BM: complementary fixed-angle and ballistic-distance panels	conditional diagnostics in their named models
HRC diagnostics and rotation comparisons	Figs. 12, 13, 14, 15–17	Appendix BN: nuisance decompositions, SPARC regime census, all 12 older rotation panels and the Milky-Way sensitivity curve	Level-C local algebraic diagnostics and provenance/stress only; NFW is a fitted Λ CDM-motivated benchmark
Part II logic	Fig. 20	one complete connected A4 reader map; the former panels 20–20 are retained only as aliases to that poster	literal node text is authoritative; exact edge data and hashes remain in the repository machine-topology package
Strong-field	Fig. 31	Appendix BO: Tolman and Hawking external benchmarks	standard benchmarks; ECT completion Open

Family	Main owner	Atlas content	Status/owner guard
Part III logic	Fig. 32	one complete connected A4 reader map in the Part III conclusion; legacy atlas anchor 32 points to the same page	corrected T1, OS-II and admitted-measure topology; exact machine edges remain repository-side and no arrow upgrades status
Part IV logic	Fig. 34	one complete connected A4 reader map in the Part IV synthesis; legacy atlas anchor 34 points to the same page	whole topology retained; canonical source and repository machine topology govern exact edges
Full status and dependency map	Fig. 35	one complete connected A4 programme map; legacy atlas anchor 35 points to the same page	corrected T1 and coherent-branch-to-admitted-measure gates; exact machine edges remain repository-side and physical Born owners are Open
Orientation/tensor	Fig. 37	Appendix BP: upstream orientation stiffness	conditional upstream relation; tensor bridge Open
Conditional background/cosmology	Figs. 38 and 39	Appendix BQ: distance, w , calibration and scope panels	conditional supplied state; no unique P1–P6 cosmology
Finite-body scalar	Fig. 41	Appendix BR: tail-estimator detail	fixed-metric numerical detail; not a PPN pass

BM Visual atlas: acoustic and cluster gates

These complementary panels retain the fixed-acoustic-angle proxy and the ballistic-distance mismatch without changing their arrays or ownership. They remain conditional diagnostics in their explicitly named models, not a CMB/distance-ladder likelihood or a universal cluster-response no-go. In the fixed-angle panel, the three separately computed nodes for each κ_s are unconnected; they are neither fits nor evaluated continuations.

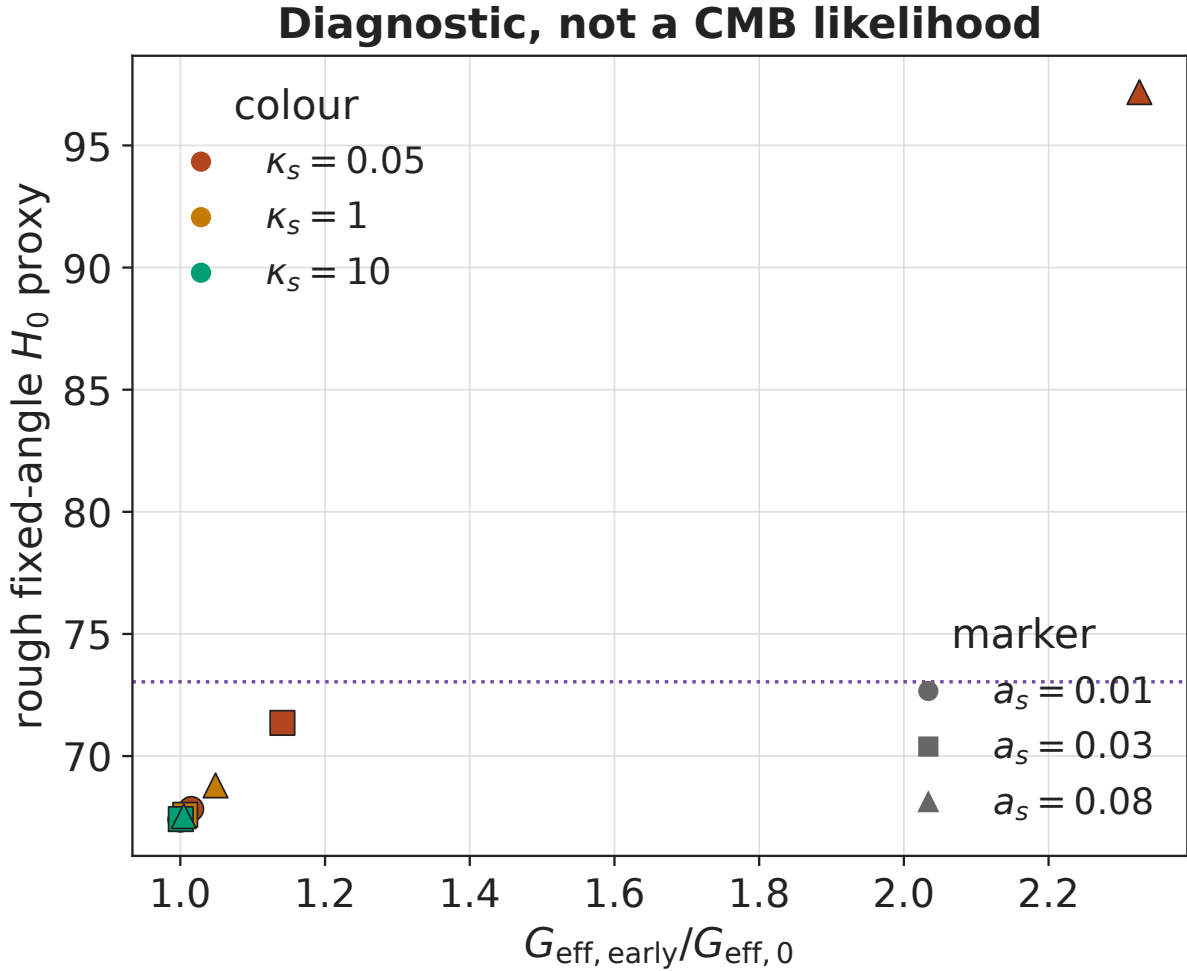
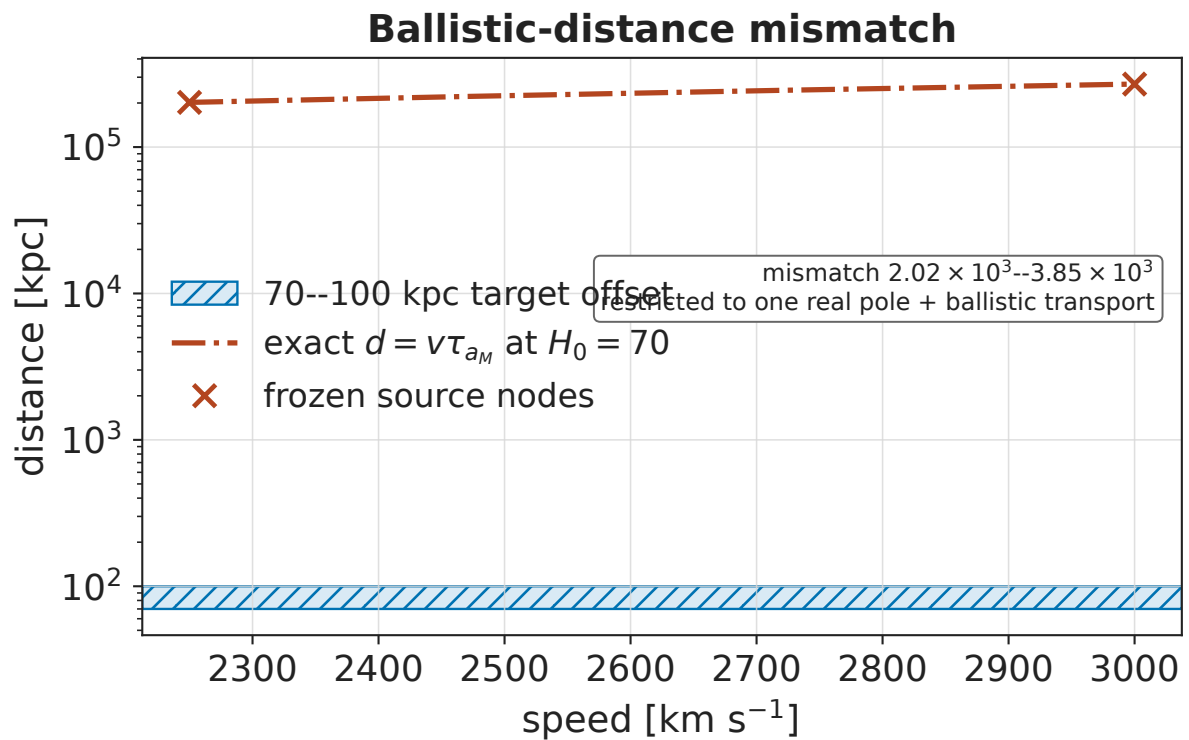


Figure 44: Fixed-acoustic-angle proxy. Complementary full-width panel for Figure 7. **Status retained:** Level C; three unconnected computed nodes per κ_s ; not a fit, continuation, CMB or distance-ladder likelihood; *Display semantics:* Each kappa family consists of three independently solved parameter nodes shown without connecting segments.. Numerical arrays, selection and owner are unchanged.



Conditional no-go for the named one-real-pole + ballistic model; general causal kernels remain Open.

Figure 45: Ballistic distance mismatch. Complementary full-width panel for Figure 8. **Status retained:** conditional one-pole no-go; *Display semantics:* The continuous trace is the densely evaluated exact relation $d=v\tau$; crosses mark the frozen source nodes.. Numerical arrays, selection and owner are unchanged.

BN Visual atlas: HRC diagnostics and rotation comparisons

This family retains the Level-C nuisance decompositions, frozen SPARC regime census, held-out examples, acceleration-stratified galleries, post-hoc residual stress cases and the HRC-only Milky-Way sensitivity curve. The galaxy panels remain local algebraic diagnostics. Their NFW curves are separately fitted Λ CDM-motivated benchmarks, not unique Λ CDM predictions or a population model-selection result. SPARC measurements are unconnected error-bar markers, and source model ordinates are faint, unconnected nodes. Smooth traces are residual-gated, shape-preserving display approximations through curvature-regularised display nodes; their maximum node-displacement tolerance is $0.10\text{--}0.25\text{ km s}^{-1}$. All fit and diagnostic metrics use the original nodes, and no continuous physical owner is implied. In the held-out panels the five split ranges are unconnected per-radius whiskers, not a smoothed envelope.

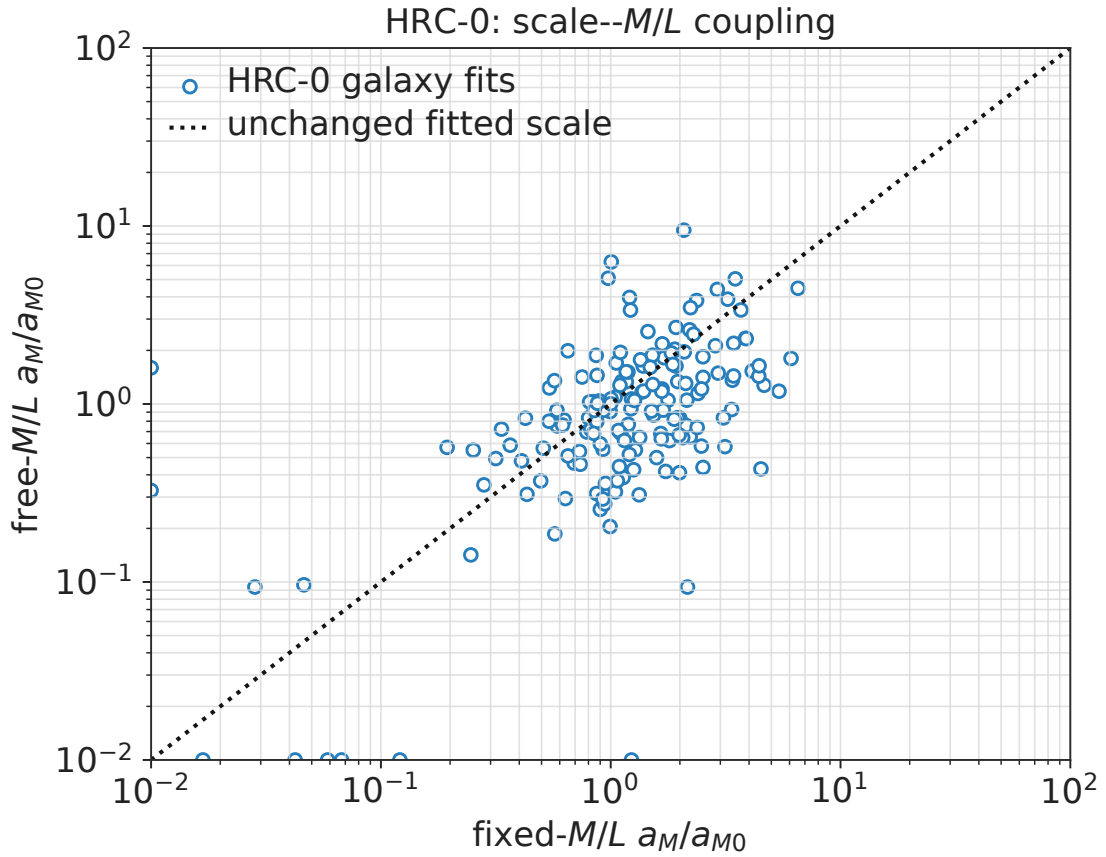


Figure 46: HRC-0 stellar- M/L coupling. Complementary full-width panel for Figure 13. **Status retained:** Level-C nuisance diagnostic. Numerical arrays, selection and owner are unchanged.

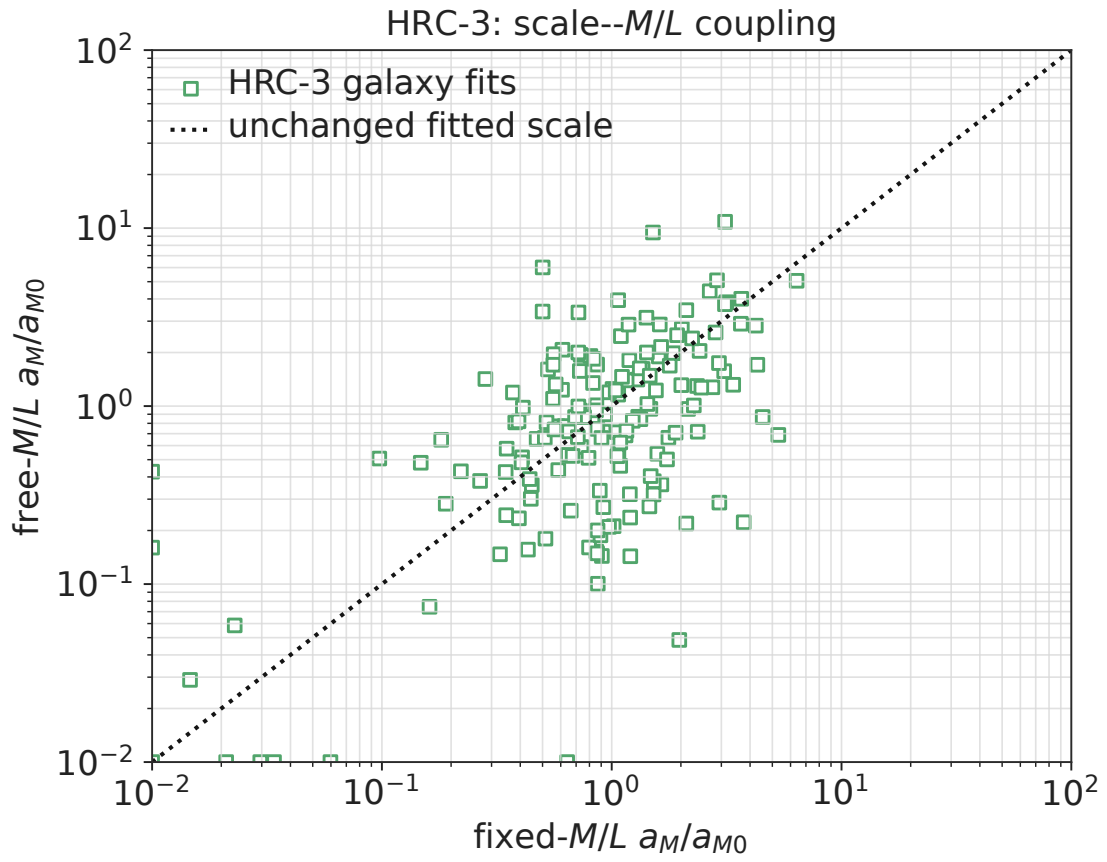


Figure 47: HRC-3 stellar- M/L coupling. Complementary full-width panel for Figure 13. **Status retained:** Level-C nuisance diagnostic. Numerical arrays, selection and owner are unchanged.

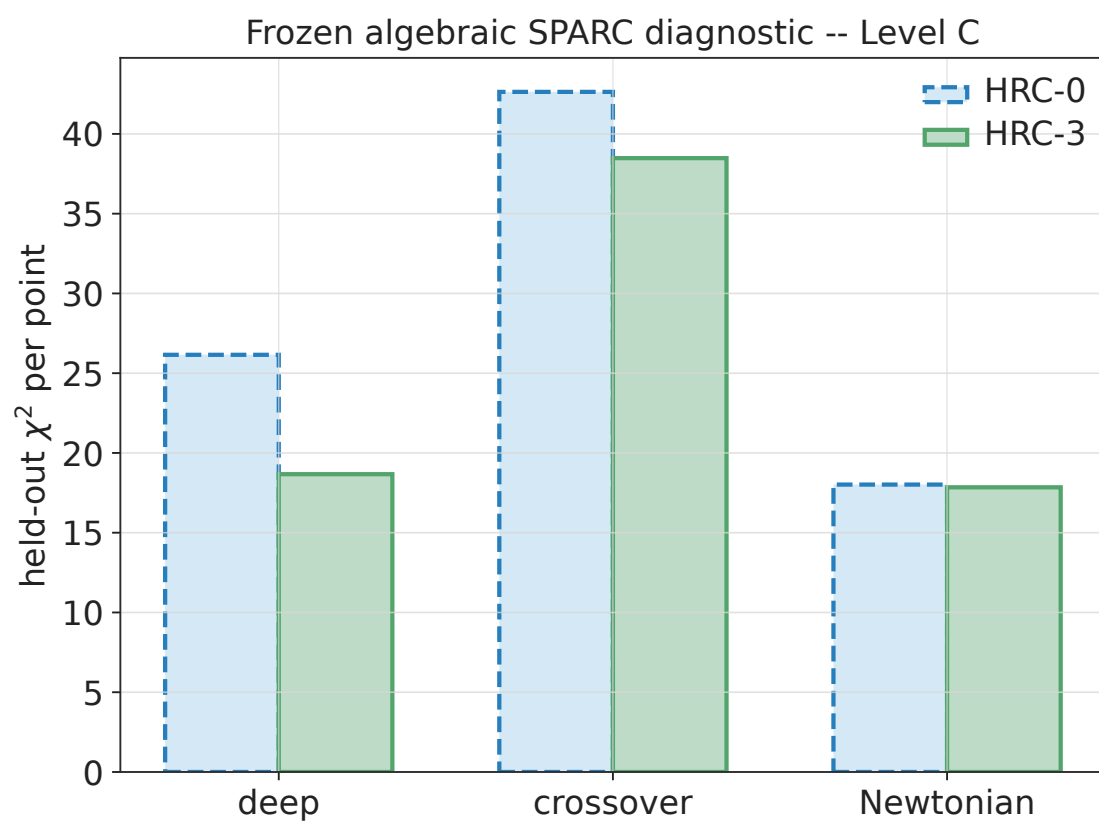


Figure 48: SPARC regime census. Complementary full-width panel for Figure 14. **Status retained:** frozen Level-C sample diagnostic. Numerical arrays, selection and owner are unchanged.

fig18_a: held-out SPARC external-comparison atlas

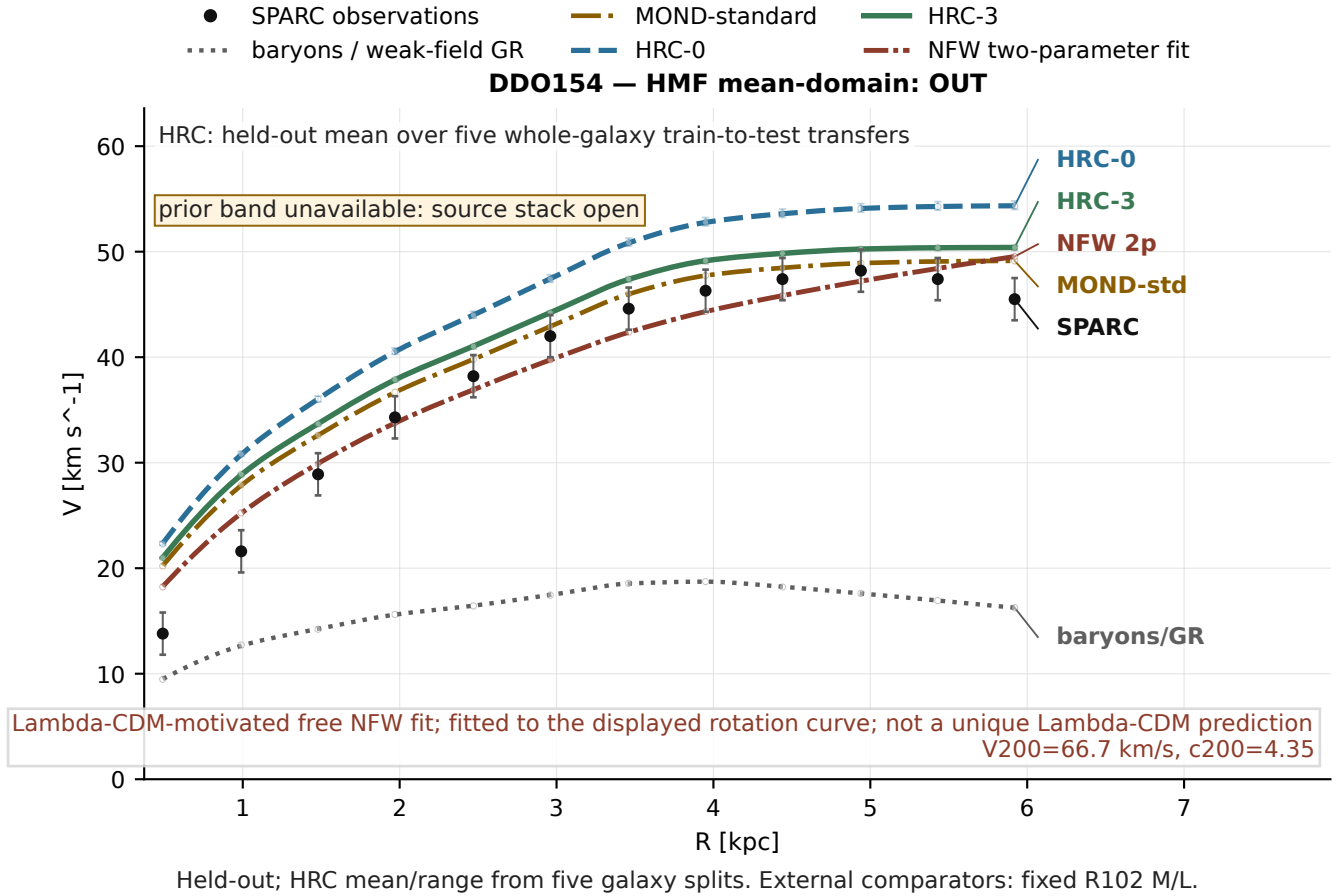


Figure 49: Held-out rotation-curve comparisons: DDO 154. Points are SPARC velocities; directly labelled curves show baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW halo. The galaxies and whole-galaxy split assignment were frozen before this rendering pass. The calculation is a local algebraic disk diagnostic, not a finite-thickness field solution or population likelihood. The NFW curve is Λ CDM-motivated and is not a unique Λ CDM prediction. The HRC centre is the pointwise mean of five held-out split predictions at the tabulated radii; the split ranges are unconnected per-radius whiskers, not a smoothed envelope.

fig18_b: held-out SPARC external-comparison atlas

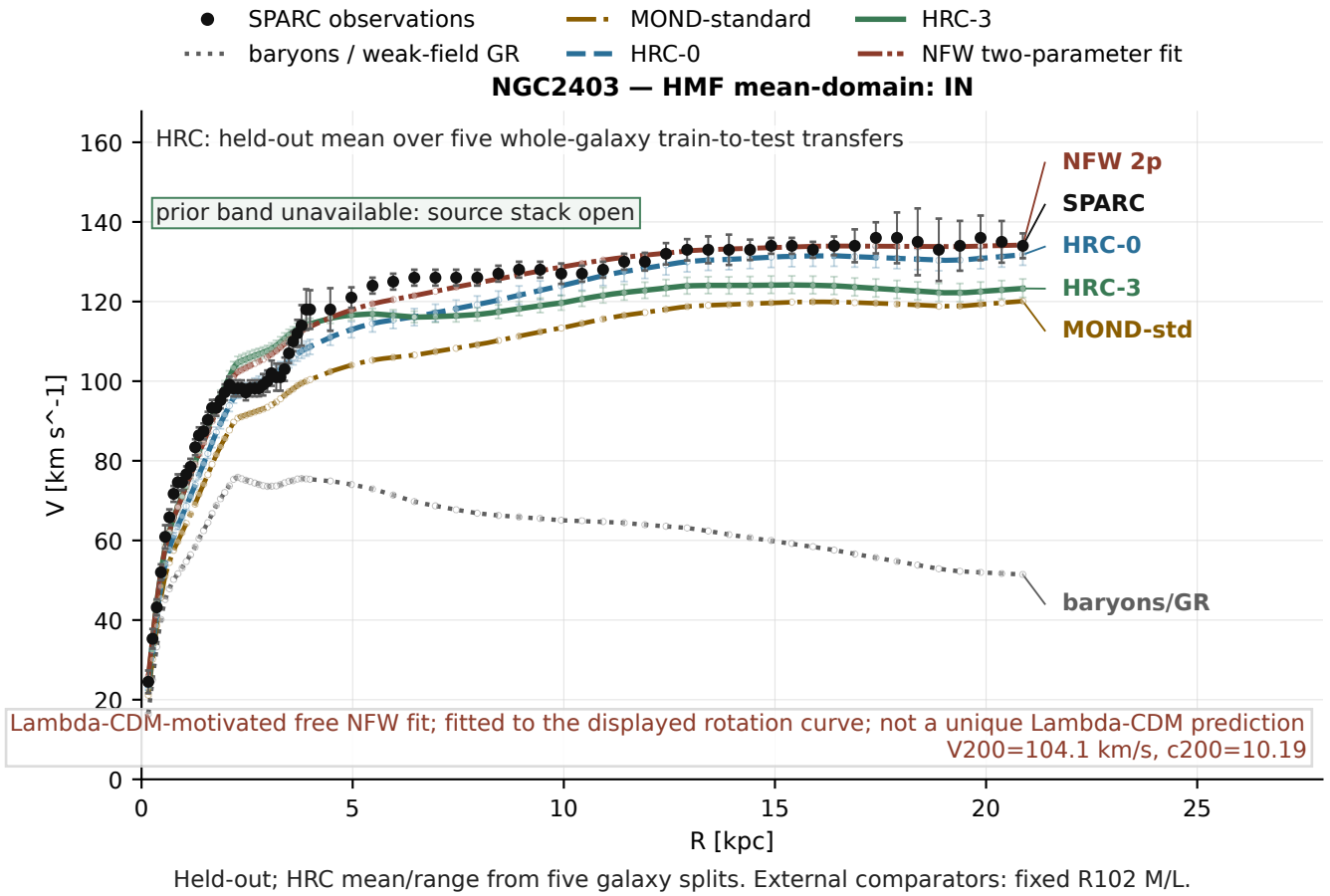


Figure 50: Held-out comparison: NGC 2403. Complementary full-width panel for Figure 15. **Status retained:** held-out Level-C external comparison; not population model selection. Numerical arrays, selection and owner are unchanged.

fig18_c: held-out SPARC external-comparison atlas

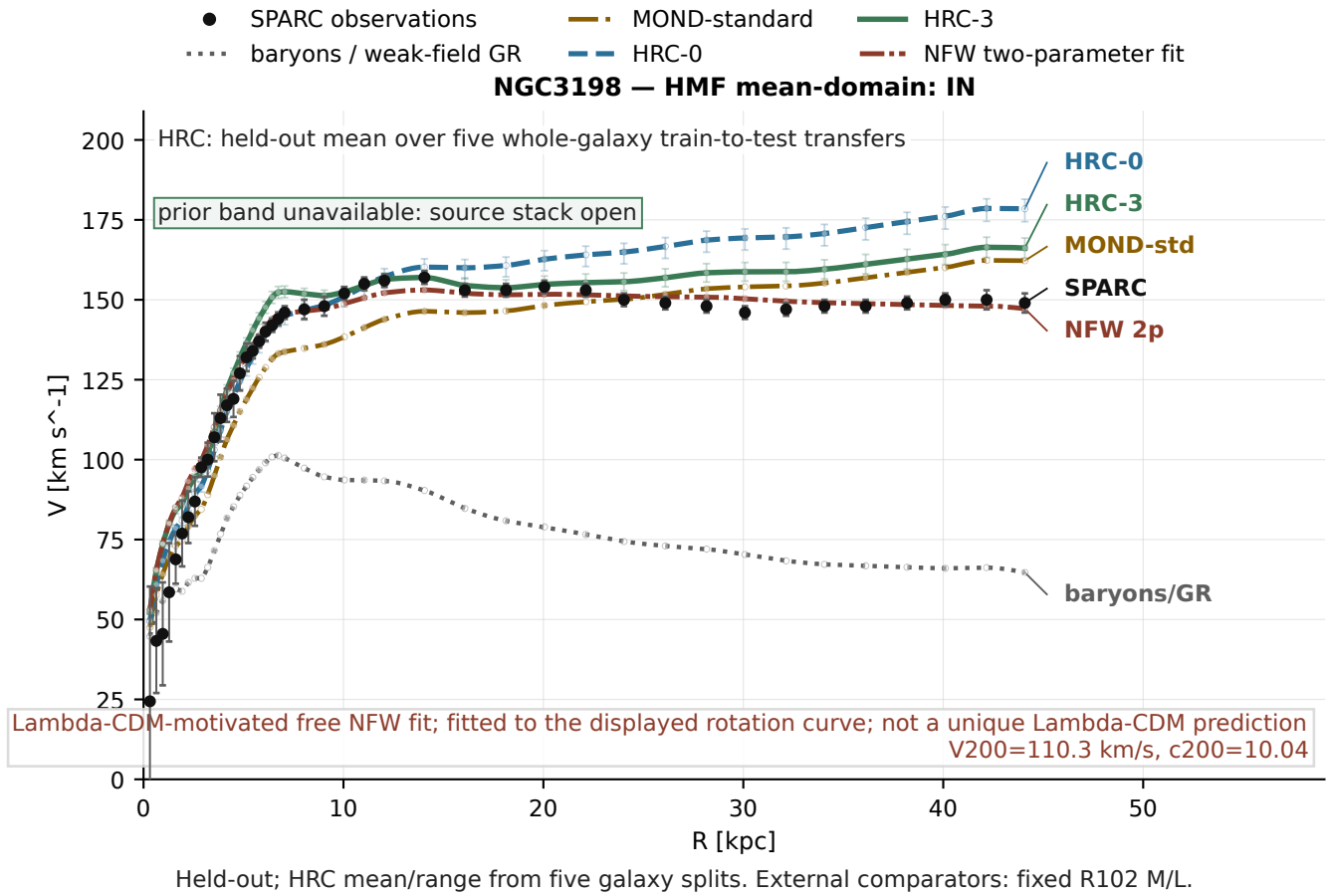


Figure 51: Held-out comparison: NGC 3198. Complementary full-width panel for Figure 16. **Status retained:** held-out Level-C external comparison; not population model selection. Numerical arrays, selection and owner are unchanged.

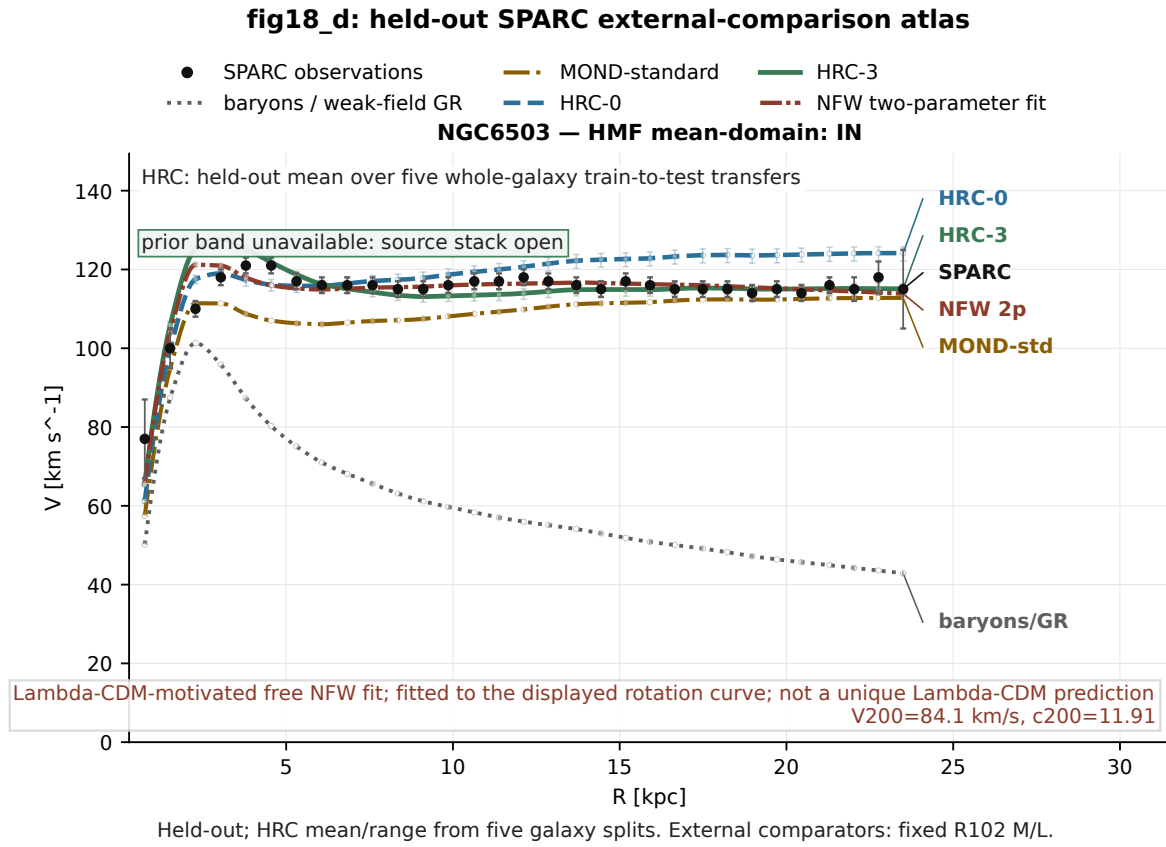


Figure 52: Held-out comparison: NGC 6503. Complementary full-width panel for Figure 16. **Status retained:** held-out Level-C external comparison; not population model selection. Numerical arrays, selection and owner are unchanged.

fig19_a: frozen acceleration-stratified SPARC gallery

Selection unchanged; HRC uses per-galaxy fitted aM with fixed M/L=0.5/0.7.

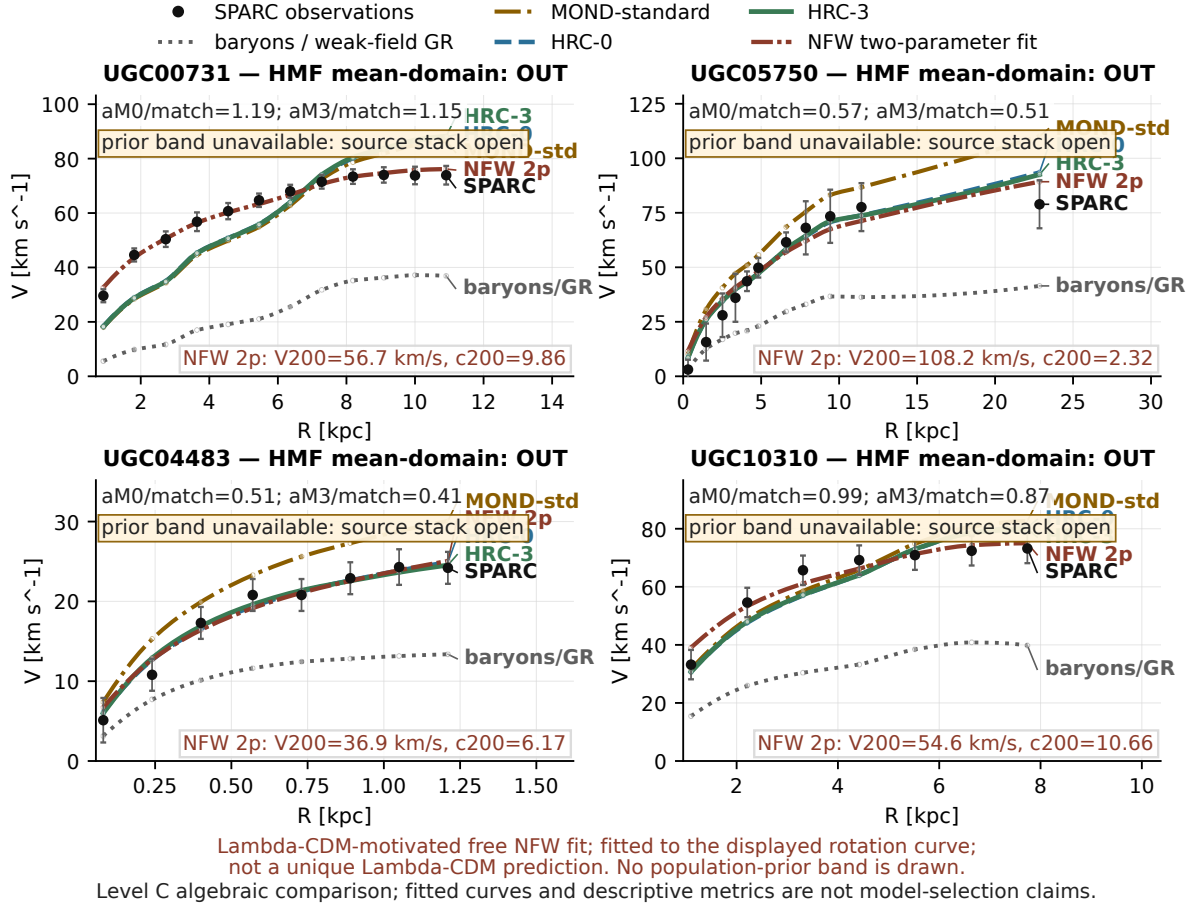
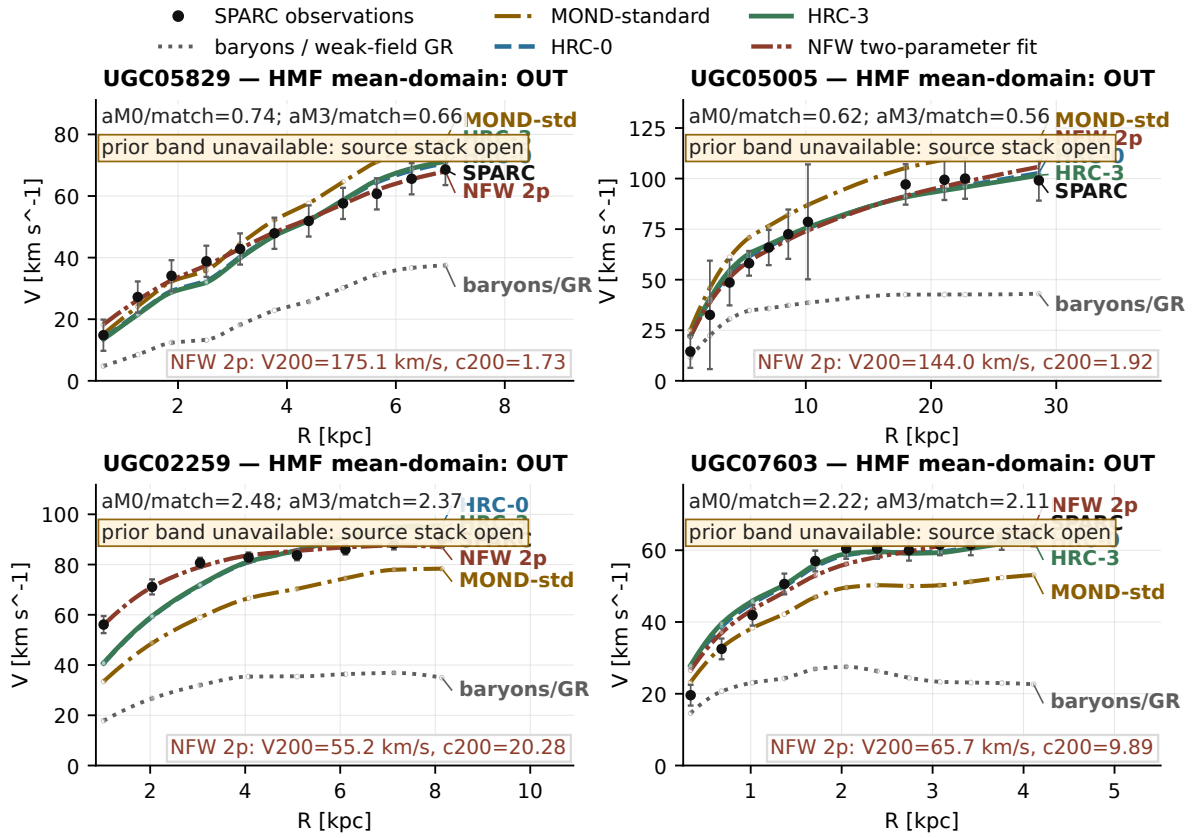


Figure 53: Frozen acceleration-stratified rotation-curve gallery. Selection is frozen without visual tuning: four quantiles are taken from each of four strata in median baryonic acceleration. Each panel shows SPARC data, baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW benchmark. Per-galaxy fitted response scales and NFW parameters make this a Level-C shape/residual diagnostic, not a universal-scale or population model-selection result. Smooth traces are display-only approximations through residual-gated display nodes; source ordinates remain faint and unconnected, and all metrics retain the original nodes.

fig19_b: frozen acceleration-stratified SPARC gallery

Selection unchanged; HRC uses per-galaxy fitted aM with fixed $M/L=0.5/0.7$.



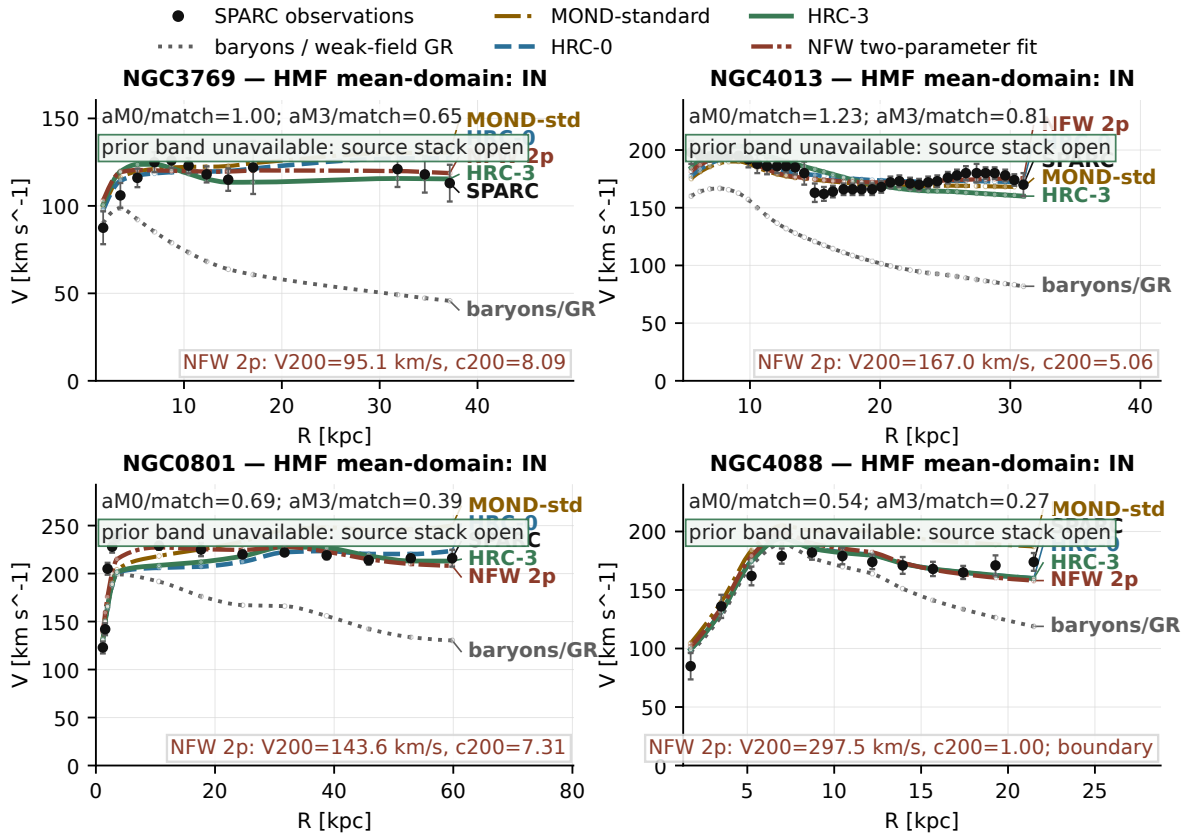
Lambda-CDM-motivated free NFW fit; fitted to the displayed rotation curve;
 not a unique Lambda-CDM prediction. No population-prior band is drawn.

Level C algebraic comparison; fitted curves and descriptive metrics are not model-selection claims.

Figure 54: Frozen comparison gallery, low-acceleration page II. Complementary full-width panel for Figure 53. **Status retained:** frozen Level-C external comparison; not population model selection. Numerical arrays, selection and owner are unchanged.

fig19_c: frozen acceleration-stratified SPARC gallery

Selection unchanged; HRC uses per-galaxy fitted aM with fixed M/L=0.5/0.7.



Lambda-CDM-motivated free NFW fit; fitted to the displayed rotation curve;
not a unique Lambda-CDM prediction. No population-prior band is drawn.

Level C algebraic comparison; fitted curves and descriptive metrics are not model-selection claims.

Figure 55: Frozen comparison gallery, high-acceleration page I. Complementary full-width panel for Figure 53.
Status retained: frozen Level-C external comparison; not population model selection. Numerical arrays, selection and owner are unchanged.

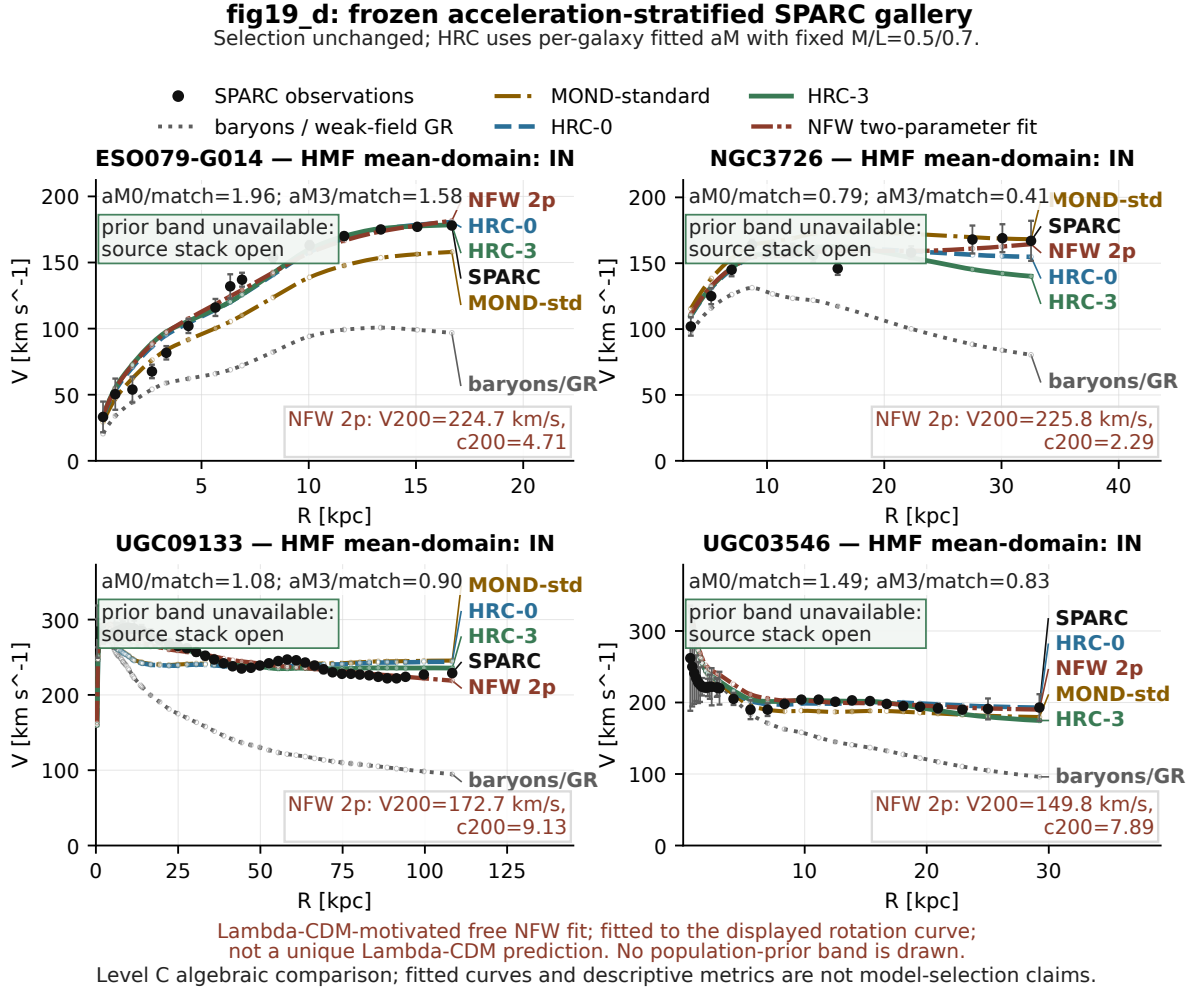


Figure 56: Frozen comparison gallery, high-acceleration page II. Complementary full-width panel for Figure 53. **Status retained:** frozen Level-C external comparison; not population model selection. Numerical arrays, selection and owner are unchanged.

fig20_a: post-hoc residual-stress external-comparison atlas

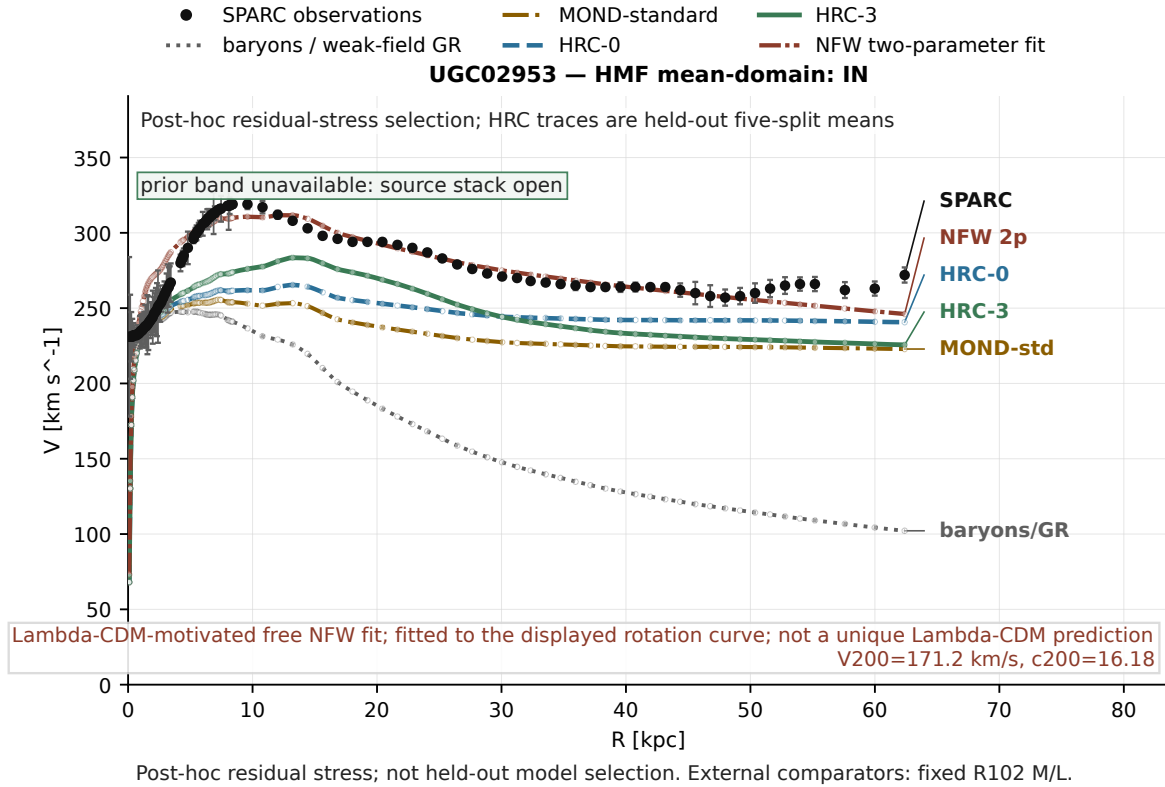


Figure 57: Post-hoc residual-stress comparisons. Two systems maximise the HRC-0 minus HRC-3 fit difference and two maximise the reverse difference in the frozen point registry. Every panel now includes SPARC data, baryon-only weak-field GR, MOND-standard, HRC-0, conditional HRC-3 and a separately fitted two-parameter NFW benchmark. Because the objects were selected after viewing the residual statistic, this remains an explanatory Level-C red-team diagnostic, not a held-out model-selection result. Model traces use residual-gated, shape-preserving display approximations through curvature-regularised display nodes; source ordinates remain faint and unconnected and all metrics use the original nodes.

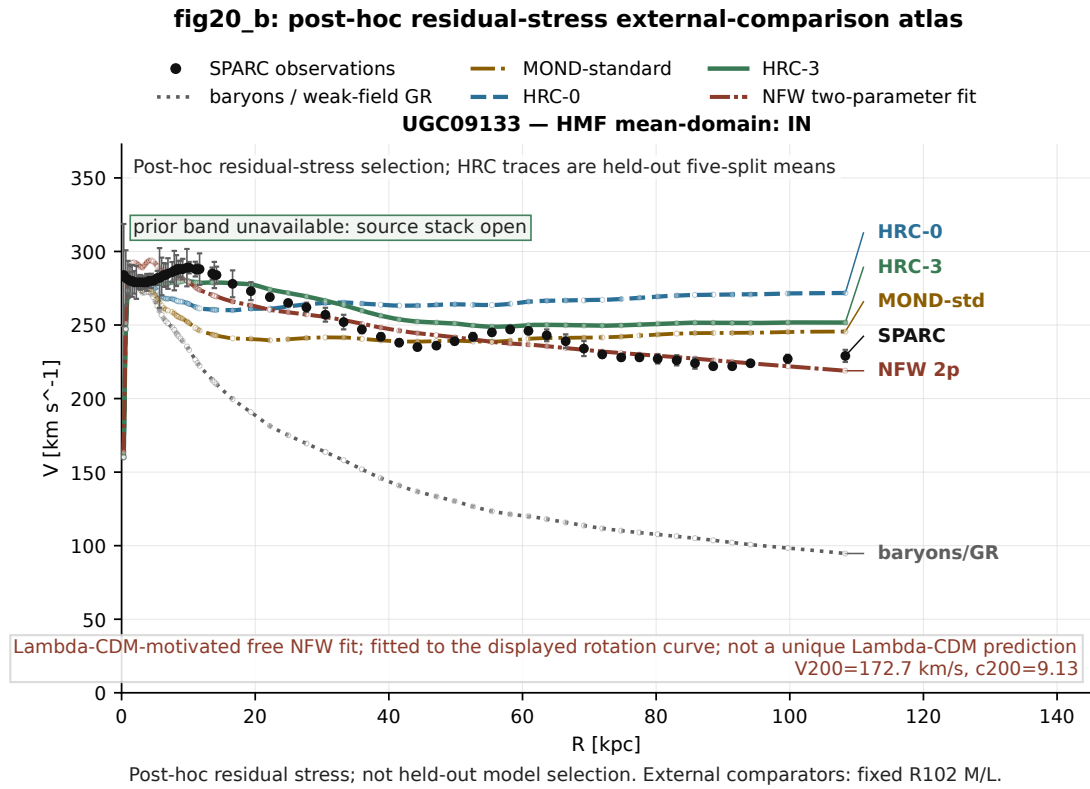


Figure 58: Post-hoc residual stress: UGC 09133. Complementary full-width panel for Figure 57. **Status retained:** Level-C red-team diagnostic. Numerical arrays, selection and owner are unchanged.

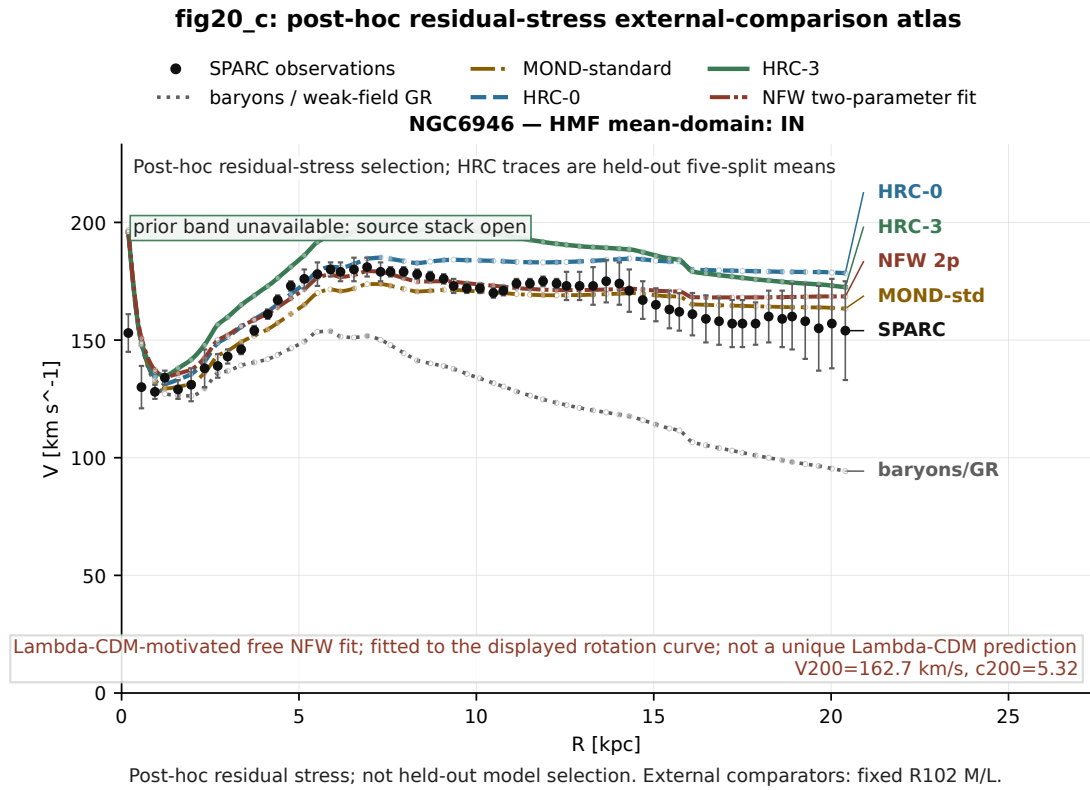


Figure 59: Post-hoc residual stress: NGC 6946. Complementary full-width panel for Figure 57. **Status retained:** Level-C red-team diagnostic. Numerical arrays, selection and owner are unchanged.

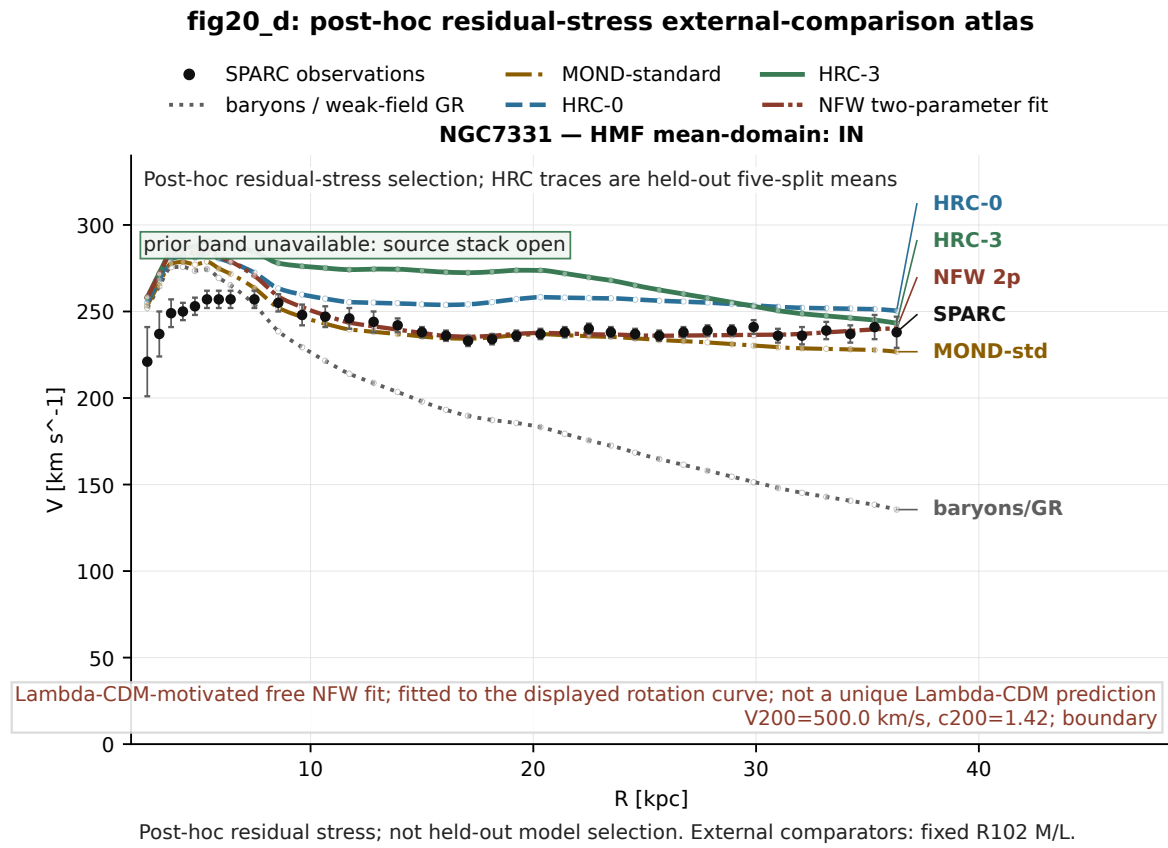


Figure 60: Post-hoc residual stress: NGC 7331. Complementary full-width panel for Figure 57. **Status retained:** Level-C red-team diagnostic. Numerical arrays, selection and owner are unchanged.

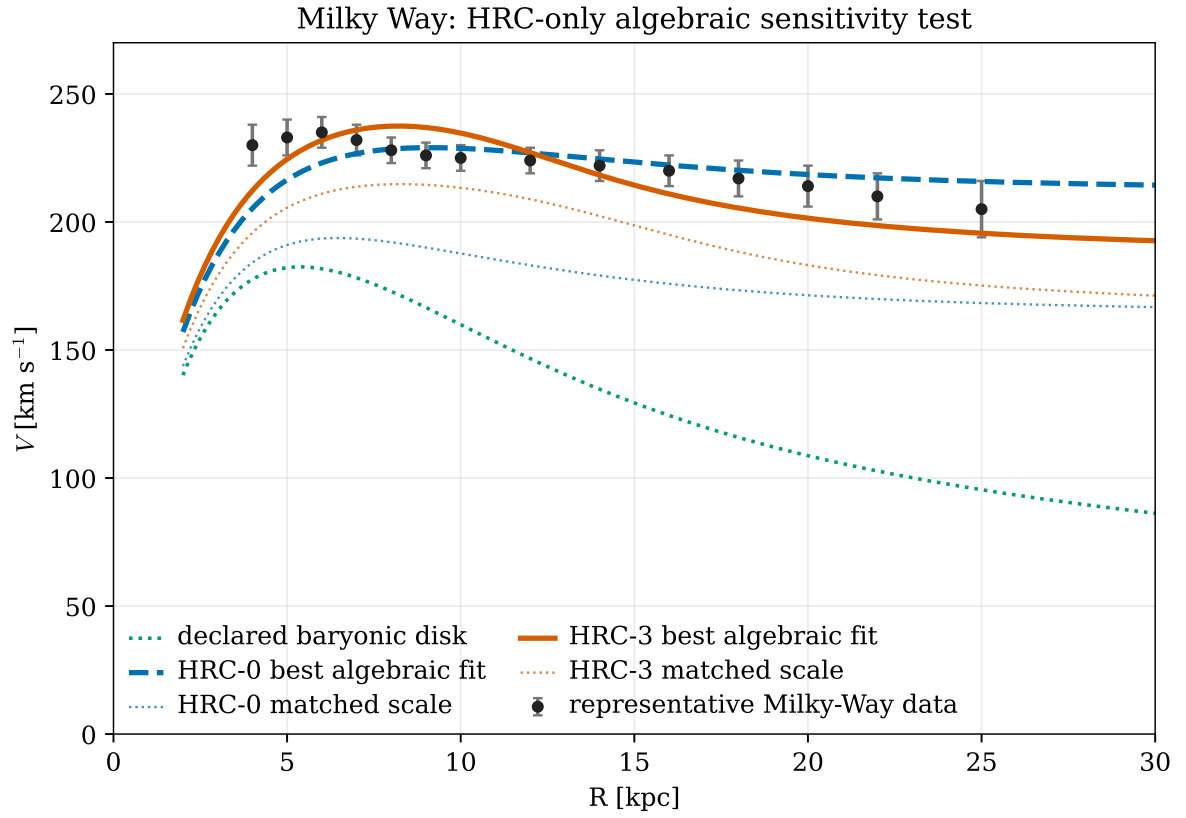


Figure 61: Representative Milky-Way HRC sensitivity test. Dotted HRC curves use the matched scale; solid/dashed curves use the algebraic best-fit scale for the declared exponential disk. The figure does not replace a full axisymmetric baryonic posterior. The curve is relocated here as readable provenance so that no HRC-only galaxy rotation plot remains a main-text model-comparison owner.

BO Visual atlas: strong-field benchmarks

The Tolman-kinematic and Hawking-temperature pages are retained as external standard benchmarks. They do not identify an ECT shell depth, critical surface, physical Hilbert-space split or information-return law; the ECT completion remains Open.

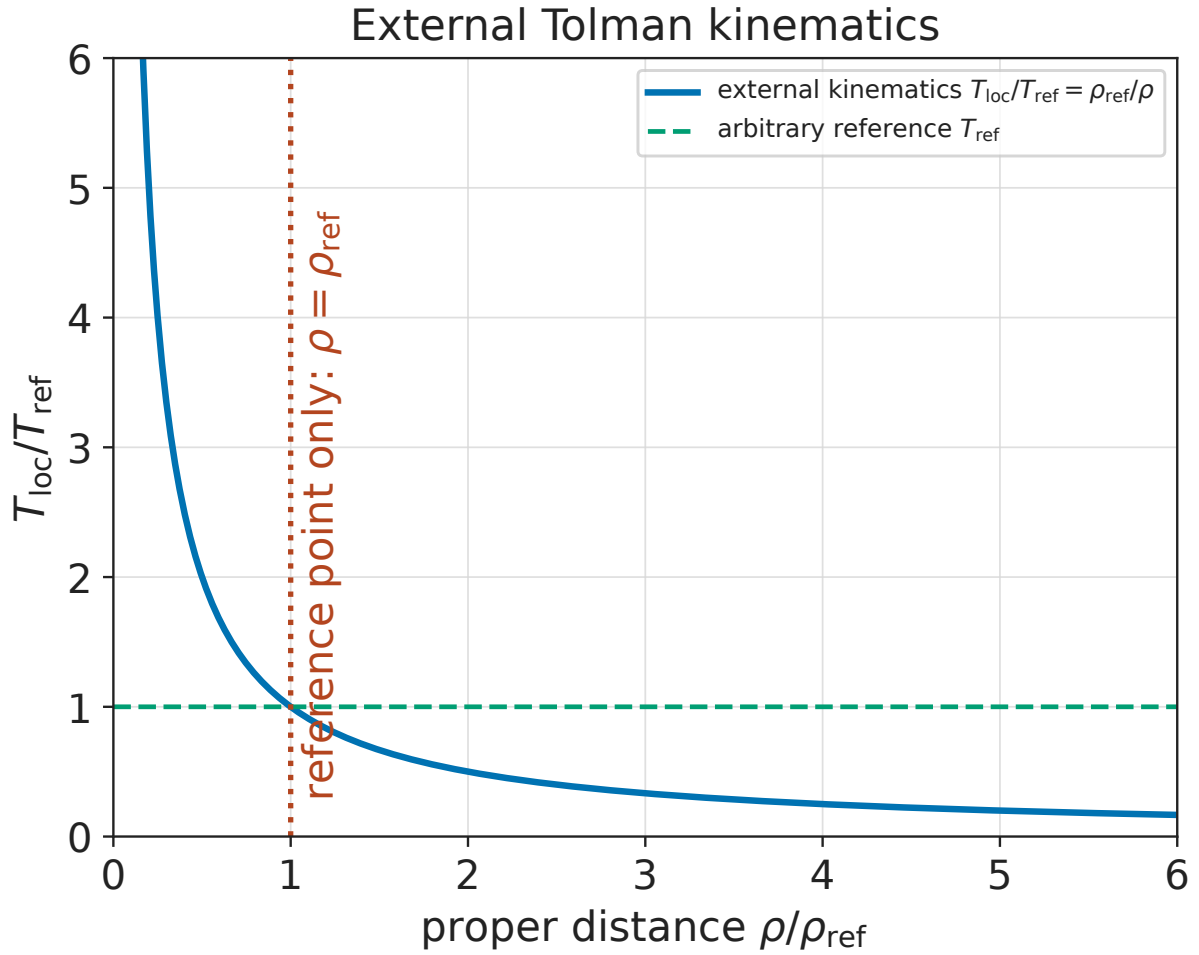


Figure 62: Tolman kinematic benchmark. Complementary full-width panel for Figure 31. **Status retained:** external standard benchmark; *Display semantics:* The smooth line is the densely evaluated exact external Tolman relation, not a fit to sampled data.. Numerical arrays, selection and owner are unchanged.

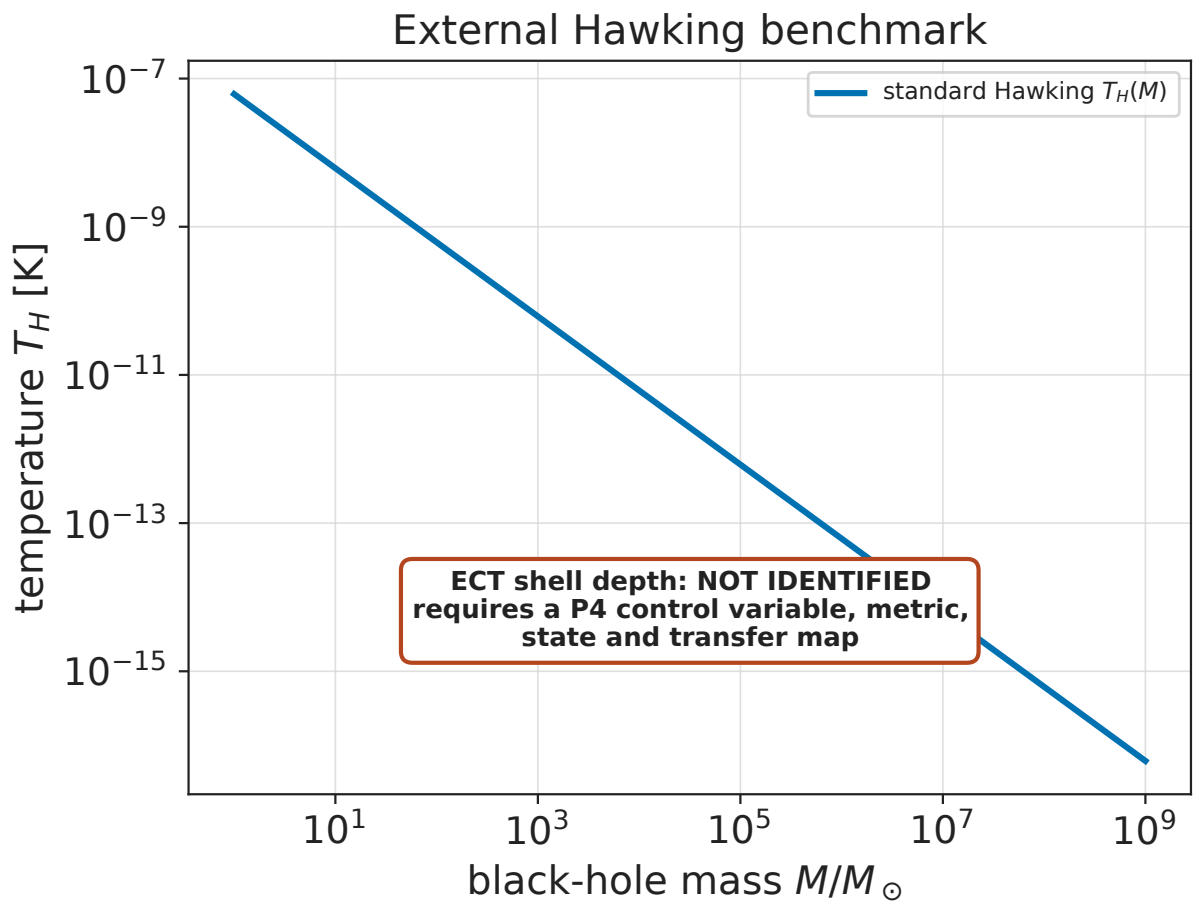


Figure 63: Hawking benchmark. Complementary full-width panel for Figure 31. **Status retained:** standard benchmark; ECT completion Open; *Display semantics:* The smooth line is the densely evaluated exact external Hawking benchmark, not a fit to sampled data.. Numerical arrays, selection and owner are unchanged.

BP Visual atlas: orientation and tensor bridge

This page preserves the conditional upstream orientation-stiffness relation. The continuation from that coefficient to a physical tensor normalisation remains an Open ownership bridge.

Orientation stiffness — conditional owner map

The EFT identity is retained; the microscopic heavy-field route is not promoted to a derivation.

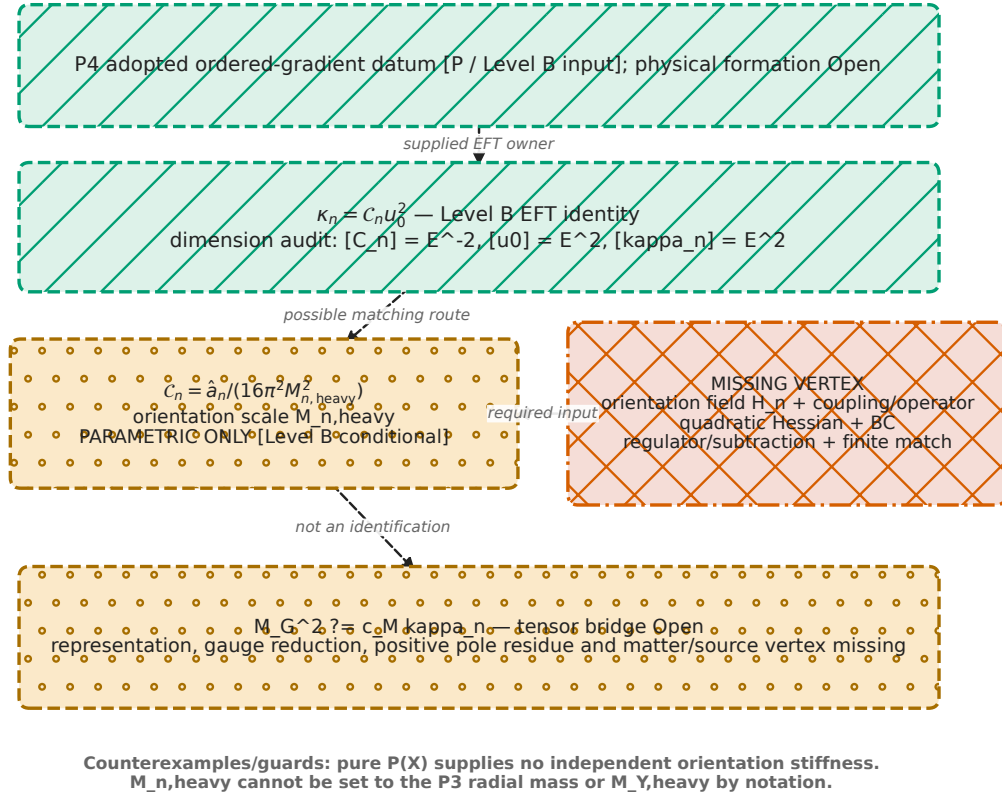


Figure 64: Upstream orientation stiffness. Complementary full-width panel for Figure 37. **Status retained:** conditional upstream relation; tensor bridge Open. Numerical arrays, selection and owner are unchanged.

BQ Visual atlas: conditional background and cosmology

These panels retain the named supplied expansion, distance and kinematic- w diagnostics, calibrated family and explicit scope statement. They remain outputs of the conditional supplied state and do not constitute a unique P1–P6 cosmology or a completed observable likelihood. Continuous traces are evaluated from the frozen dense background owner (with distance obtained by deterministic quadrature); the eight or nine published diagnostic nodes, as applicable, are shown separately.

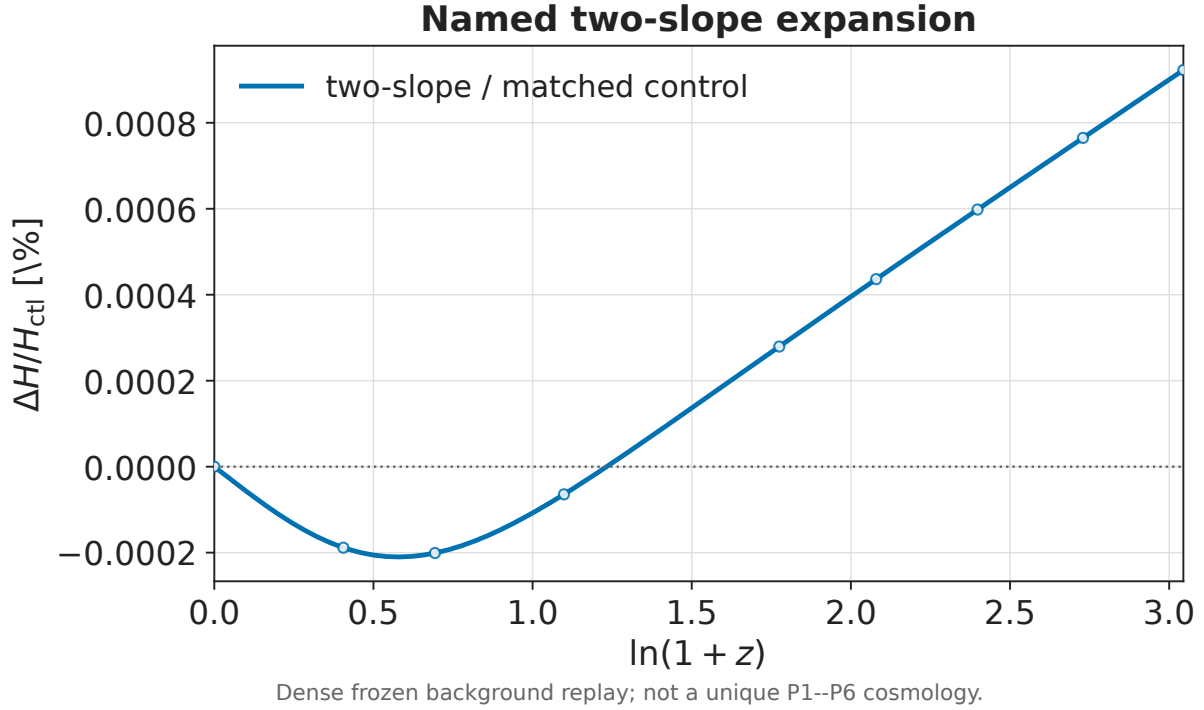


Figure 65: Named supplied expansion. Complementary full-width panel for Figure 38. **Status retained:** conditional supplied background; *Display semantics:* The continuous expansion trace is evaluated on the frozen 3001-point background owner; the nine published nodes are shown separately.. Numerical arrays, selection and owner are unchanged.

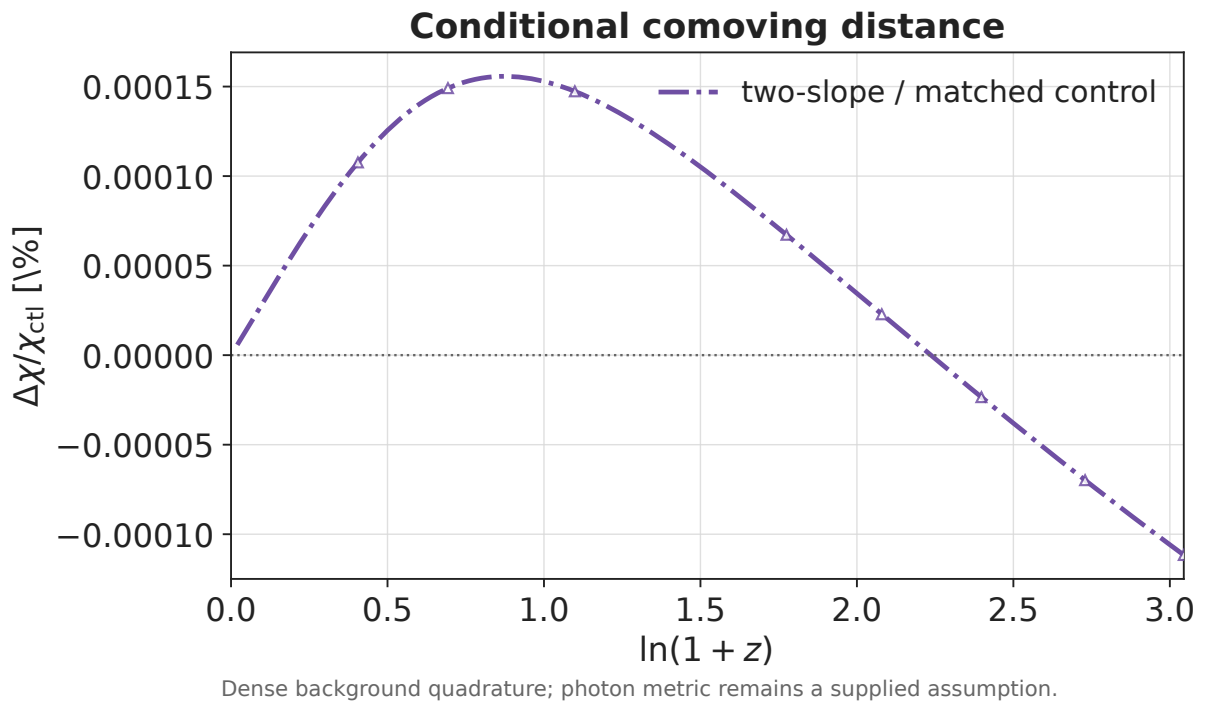


Figure 66: Comoving-distance diagnostic. Complementary full-width panel for Figure 38. **Status retained:** conditional supplied background; *Display semantics:* The continuous distance trace is obtained by deterministic quadrature of the frozen dense background; the eight nonzero- z published nodes are shown separately.. Numerical arrays, selection and owner are unchanged.

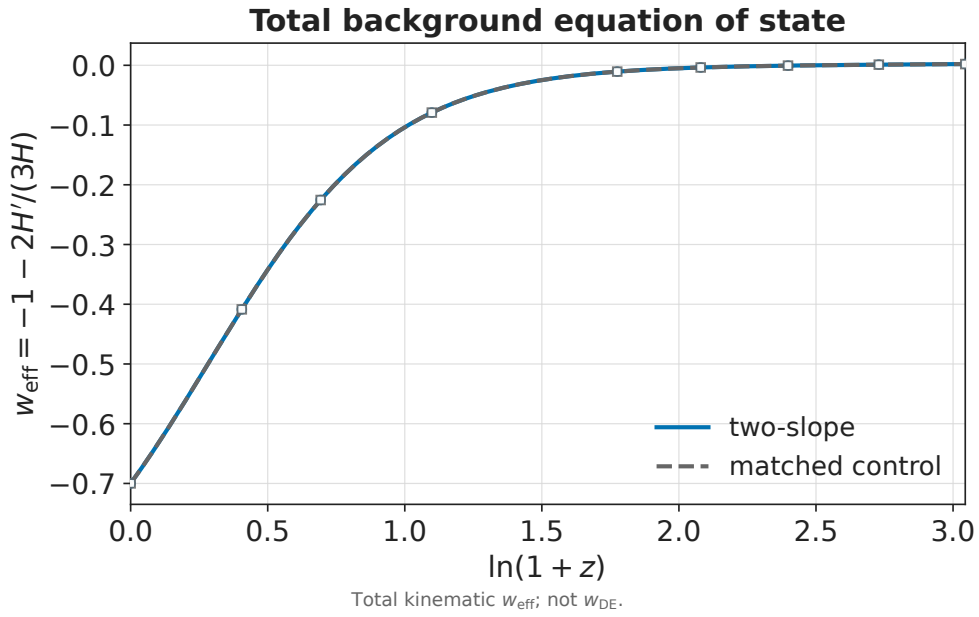


Figure 67: Background equation-of-state proxy. Complementary full-width panel for Figure 38. **Status retained:** conditional supplied background; *Display semantics:* Both continuous effective-equation-of-state traces are evaluated on the frozen dense background and declared matched control; the nine published nodes are shown separately.. Numerical arrays, selection and owner are unchanged.

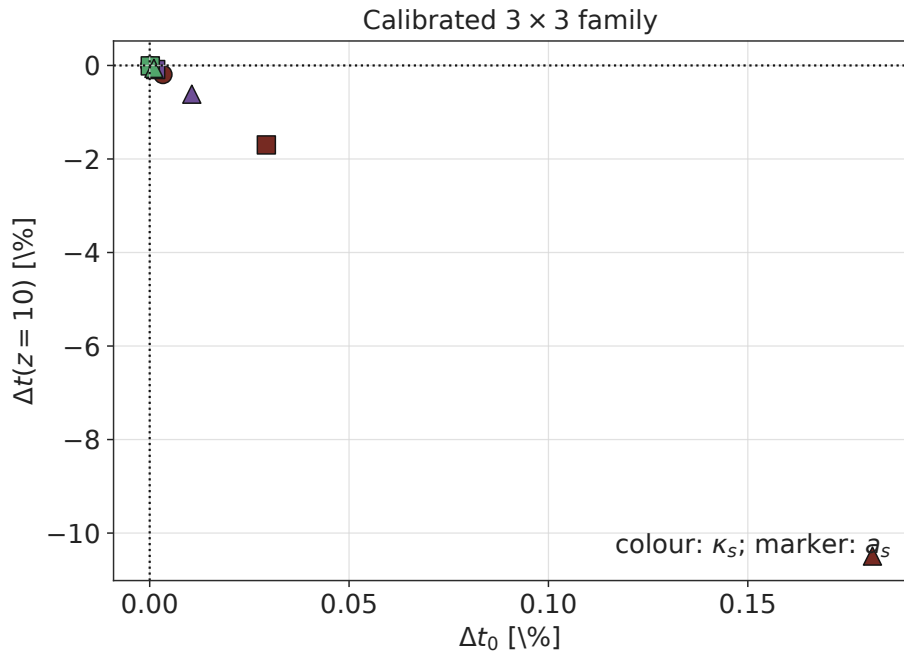


Figure 68: Calibrated background family. Complementary full-width panel for Figure 38. **Status retained:** Level-C conditional family. Numerical arrays, selection and owner are unchanged.

Conditional supplied-action outputs

- not a unique P1--P6 cosmology
- H_0 is one declared unit/state calibration
- w_{eff} is total background kinematics, not w_{DE}
- photon/perturbation likelihood owners remain Open

Figure 69: Background scope and ownership. Complementary full-width panel for Figure 38. **Status retained:** explicit Open-owner statement. Numerical arrays, selection and owner are unchanged.

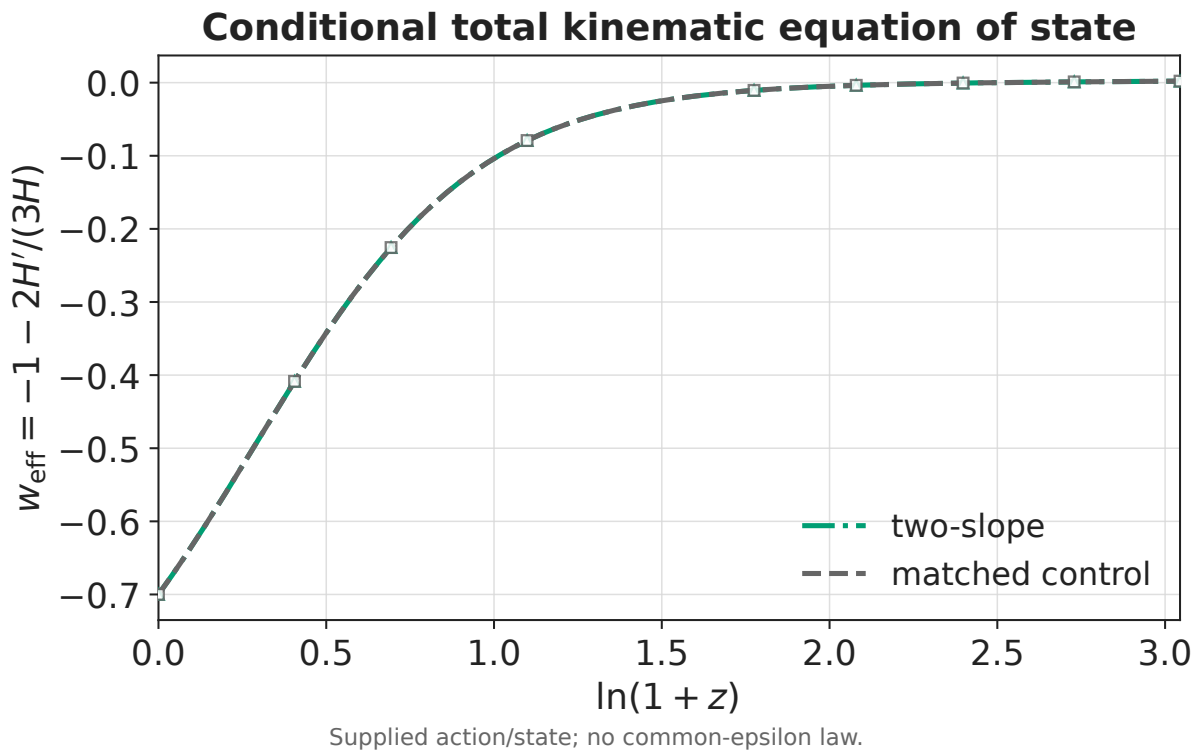


Figure 70: Two-slope w diagnostic. Complementary full-width panel for Figure 39. **Status retained:** conditional supplied background; *Display semantics:* Both continuous effective-equation-of-state traces are evaluated on the frozen dense background and declared matched control; the nine published nodes are shown separately.. Numerical arrays, selection and owner are unchanged.

BR Visual atlas: finite-body scalar gates

The retained tail-estimator page is fixed-metric numerical detail for the named scalar boundary-value diagnostic. It is not a coupled metric body solution, Cassini likelihood, WEP result or full-PPN pass.

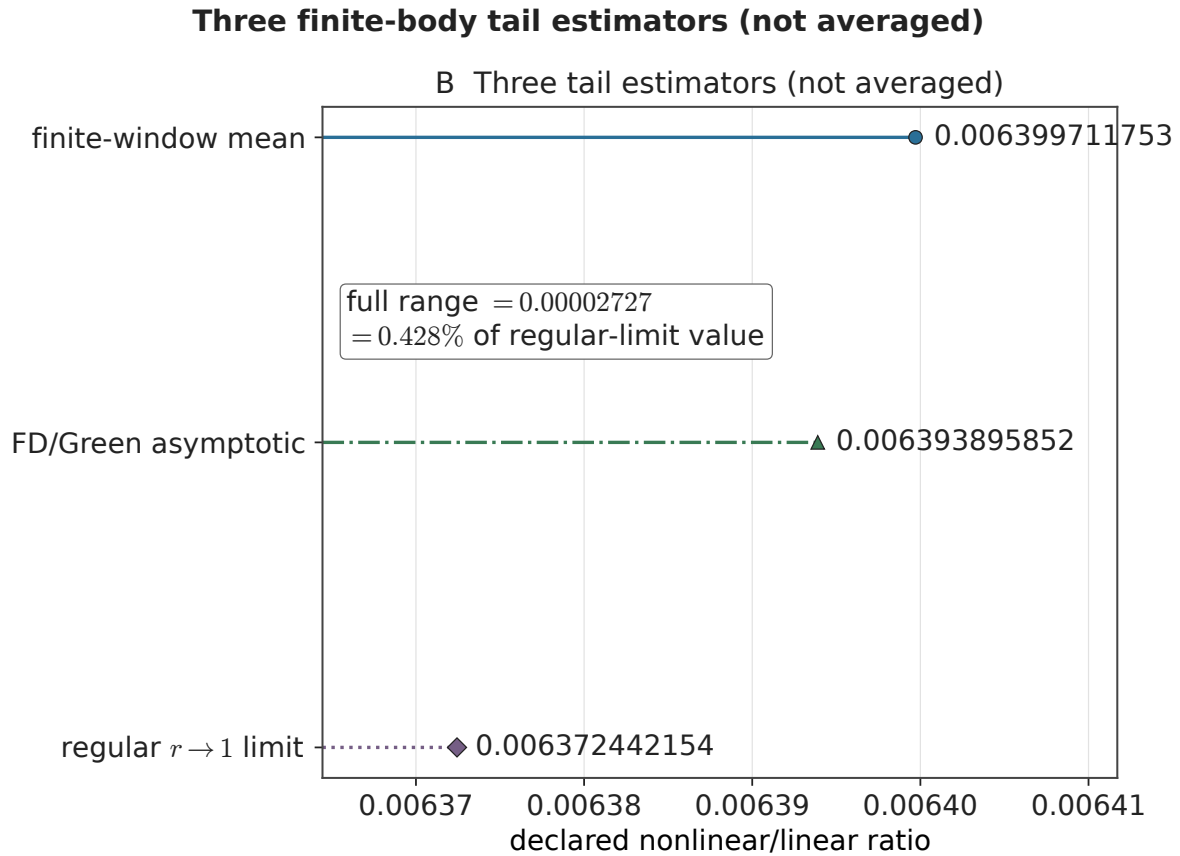


Figure 71: Finite-body tail estimators. Complementary full-width panel for Figure 41. **Status retained:** fixed-metric numerical detail; not a PPN pass. Numerical arrays, selection and owner are unchanged.

This atlas records visual preservation and the outcome of exact-page readability gates only. It does not close any Open action, state, metric, likelihood or experimental-owner gate identified by the corresponding main-text section.

Data and Code Availability

Galaxy rotation-curve data are drawn from the public SPARC database [327] (<http://astroweb.cwru.edu/SPARC/>). Versioned source and reproducibility materials for released revisions are available at:

<https://github.com/chufelo/ECT-preprint-code>

For an exact revision, use the tagged commit and frozen release manifest associated with its version-specific DOI. The moving default branch may later change and is not an exact citation owner.

The versioned validation package contains the P6 record-sector scripts for kernel-exponent, threshold-residue, sum-rule and M3 guard checks, together with the synthetic two-parameter-family P26 over-closure demonstration. It also contains the exact conditional two-slope cosmology equations, dual-integrator background and ordered-branch clock outputs, restricted growth/Weyl/flow diagnostics, and the owner/identifiability gates. These are conditional outputs of the declared action and state, not a unique P1–P6 cosmology or a full Hubble, JWST, lensing or growth likelihood. A physical numerical release must retain the continuous exact-action protocol of Appendix AR and must freeze every action, state, boundary datum and observable likelihood.

The same versioned package contains the HRC-0 analytic inversion, deterministic residual-checked HRC-3 solvers, signed-gas handling, the 165-galaxy HRC catalogue, fixed/free M/L sensitivity, representative Milky-Way and UDG stress tests, all nine HRC figures, and machine-readable verification outputs. The frozen HRC layer does not claim a finite-thickness disk PDE, covariance-complete likelihood or metric/lensing completion. The main narrative adds a separately frozen, qualified six-galaxy comparison with MOND-standard and a fitted two-parameter NFW halo benchmark. This is a Level-C algebraic diagnostic; NFW is Λ CDM-motivated and is not a unique Λ CDM prediction. The package also contains the action-scale, fifth-force, baryogenesis, dimensionality and P6 verification calculations described in their owning sections. The release manifest identifies the exact files and hashes belonging to a published revision.

The computational procedures used to generate the individual figures are summarised in Appendix Z.

The public repository can also be accessed by scanning the QR code (right). Original project code is MIT-licensed under the repository's declared licence scope; third-party data and assets retain their own terms. Each exact release is identified there by its tagged commit and frozen manifest.



[github.com/chufelo/
ECT-preprint-code](https://github.com/chufelo/ECT-preprint-code)

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